

If the tracer contributes vanishingly little to the denominator, then

$$SA_A' = \text{mol } A^*/(\text{mol } A + \text{mol } A_x)$$

Rewriting each of the above equations in terms of mol A^* , and we obtain the following

$$SA_A(\text{mol } A) = \text{mol } A^*$$

$$SA_A'(\text{mol } A + \text{mol } A_x) = \text{mol } A^*$$

Setting these equations equal to each other, we get

$$SA_A(\text{mol } A) = SA_A'(\text{mol } A + \text{mol } A_x)$$

or

$$SA_A/SA_A' = (\text{mol } A + \text{mol } A_x)/(\text{mol } A)$$

Dividing the right-hand numerator and denominator by (mol A), we arrive at the following expression:

$$SA_A'/SA_A = 1 + (\text{mol } A_x/\text{mol } A)$$

Thus, accurate determination of the specific activities SA_A and SA_A' immediately provides the value of the unknown A_x . Note: Whenever $A_x < A$, the isotope dilution method suffers from any statistical variation in the measurements of quantities that are nearly identical. Should this prove to be true for your measurements, use a lesser amount of isotopic probe, so that $A_x \ll A$.

ISOTOPE EXCHANGE

1. A reaction in which the reactants and products exchange atoms or groups of atoms that are chemically identical and differ only in isotopic composition. For example, the exchange of deuterium in D_2O with the labile hydrogens of glucose in solution represents this type of isotope exchange. 2. The exchange of an isotope from one isotopically substituted substrate in a multisubstrate enzyme-catalyzed reaction, with at least one of the other substrates.

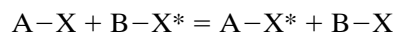
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Abortive complex interaction, **64**, 39-45; enzyme stability and, **64**, 23, 24; equilibrium exchange [equations, **63**, 76-79; **64**, 7-18; experimental protocol, **64**, 19-27, 31; initial rate method, **64**, 32, 33; ionic strength effect, **64**, 27, 32; kinetic parameter, **64**, 17, 18, 25, 31, 32; pH effect, **64**, 23; profile, **64**, 13, 17; quenching exchange reactions, **64**, 25; rapid equilibrium model, **64**, 6, 7, 17, 18; side reaction effect, **64**, 24; substrate inhibition, **64**, 34-36; substrate-product ratio, **64**, 22, 23; substrate stability, **64**, 24; theory, **64**, 7-18, 30, 32, 33, 44; types, **64**, 4, 5]; intermediate, **64**, 25, 26, 34, 36-38; irreversible enzyme reactions, **64**, 36-38; substrate synergism, **64**, 38, 39; equations for isotope exchange, **64**,

7-18 [derivation, **63**, 76-83; Cleland method, **63**, 80-83; equilibrium assumption, **63**, 76-79; steady state assumption, **63**, 79-83].

ISOTOPE EXCHANGE (Rate Law)

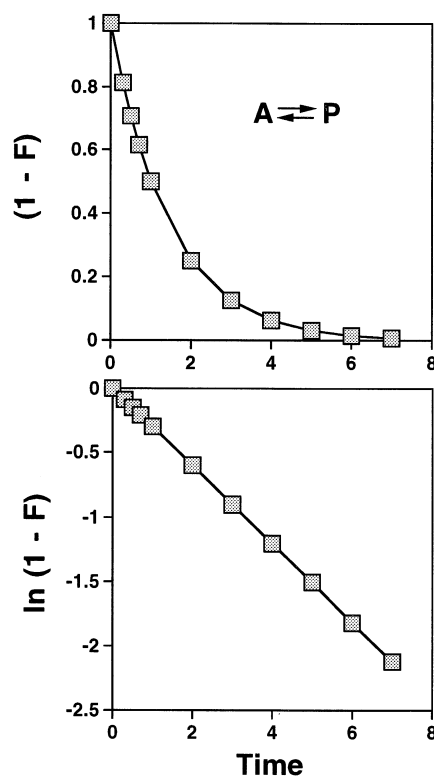
Duffield & Calvin¹ were among the first to derive the expression for an exchange reaction involving the transfer of isotope X^* between two reactants:



The gross rate of exchange R depends on the fraction of BX total that is present as BX^* and the fraction of AX_{total} present as AX .

$$-\ln(1 - F) = R[(a + b)/ab]t$$

where F is the fractional attainment of isotopic equilibrium (i.e., $F = [A-X^*]_t/[A-X^*]_{\text{eq}} = [B-X^*]_t/[B-X^*]_{\text{eq}}$), a is the concentration of $A-X$, and b is the concentration of $B-X$. R has units of molarity per unit time (i.e., M/s).



The expression² given above for R indicates that the exchange process is first order, and it will remain so provided tracer quantities are employed. It is also possible to roughly measure R by single-point determinations, but it is preferable to use several sampling times and to confirm the first-order nature of the measured exchange

rate. In enzyme-catalyzed reactions, R depends upon the fraction of enzyme capable of reacting with the labeled substrate (or product), and all substrates and products affect R . This is true because the concentrations of substrate(s) and product(s) control the concentration of all Michaelis complexes as well as any "mixed" substrate-product-enzyme complexes resulting from abortive complex formation. In some cases, this equation may be further simplified if the level of one exchange partner is considerably greater than the other. If a is much greater than b , the rate equation for exchange will become:

$$R = -\frac{[\ln(1-F)][X][H_2O]}{([X] + [H_2O])t} = -\frac{[\ln(1-F)][X]}{t}$$

This rate law, for example, applies to the tritium exchange from labeled dihydroxyacetone phosphate to water in the aldolase reaction or the exchange from labeled malate to water in the fumarase reaction.

¹R. B. Duffield & M. Calvin (1946) *J. Am. Chem. Soc.* **68**, 557.

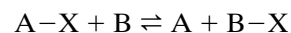
²A. A. Frost & R. G. Pearson (1961) *Kinetics and Mechanism*, 2nd ed., Wiley, New York.

ISOTOPE EXCHANGE (Reactions At Equilibrium)

Even at dynamic equilibrium, uncatalyzed and catalyzed reactions relentlessly shuttle reactants and products forth and back, and the flux in each direction is essentially constant on the time scale used in most kinetic experiments. Inasmuch as the rate of enzymic catalysis depends upon the concentration of enzyme-substrate(s) and enzyme-product(s) complexes, the enzyme's behavior at equilibrium is a complex composite of the availability of ligand binding sites and the magnitude of rate constants governing steps in the forward and reverse directions. It is also true that the equilibrium flux in each direction depends upon enzyme concentration, and one may adjust the concentration of enzyme to suit the needs of the particular experiment. The cardinal feature of all exchange processes is that they obey simple first-order kinetics, and despite the complexity of enzyme interactions with substrate(s) and effector(s), the experimental determination of rate behavior is therefore relatively straightforward. Introduction of a labeled substrate or product allows one to trace quantitatively the course of exchange reactions. The most common applications of equilibrium exchange kinetics are the determination of substrate binding mechanisms (see Tables I and II) and the influ-

ence of regulatory molecules on enzymic catalysis. On the other hand, equilibrium exchange reactions can provide evidence for certain chemical intermediates as well as positional isotope exchange.

Depending on the position of the isotopic atom(s) in the labeled substrate or product, various exchanges may be examined¹⁻³. The types of exchanges subject to measurement can be illustrated by considering a bisubstrate reaction:



For exchange to occur, atom(s) (or functional groupings of atoms, such as phosphoryl groups) must be transferred between two substrate-product partners. Thus, only three exchanges can occur in this example: A-X with A; A-X with B-X; B with B-X; but *not* B with A. In the hexokinase reaction, for example, one can observe exchange reactions between glucose and glucose 6-phosphate, ATP and ADP, and ATP and glucose 6-phosphate. But, there is no exchange of atoms between glucose and ADP. Likewise, NAD⁺-dependent dehydrogenases never exhibit exchange between oxidized substrate and oxidized coenzyme. More complicated enzyme systems may have a number of exchange processes associated with catalysis, and the glutamine synthetase reaction provides an excellent example of the types of possible exchange reactions: [¹⁴C]ATP with ADP; [¹⁴C]glutamate with glutamine; [γ -³²P]ATP with orthophosphate; [¹⁵N]ammonia with glutamine; [γ -¹⁸O]ATP with orthophosphate; [¹⁸O]glutamate with orthophosphate; [¹⁸O]glutamine with orthophosphate; [¹⁸O]ATP with glutamate; [¹⁸O]ATP with glutamine; and [¹⁸O]glutamate with glutamine. These do not include so-called positional isotope exchange (PIX) reactions which tell us about stereochemical properties and the reversibility of certain exchange reactions.

To maintain equilibrium, attention must be given to the mass-action ratio, which is the multiplicand of product concentrations (that is, [P], [Q], *etc.*) divided by the product of the substrate concentrations (that is, [A], [B], *etc.*):

$$K_{eq} = \{[P][Q] \dots\} / \{[A][B] \dots\}$$

Typically, the system can be treated in terms of pairwise concentrations of substrate and product (*e.g.*, let $\alpha = [P]/[A]$ and $\beta = [Q]/[B]$, where the product $\alpha\beta$ equals

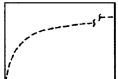
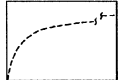
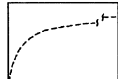
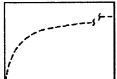
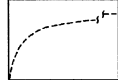
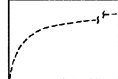
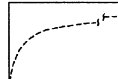
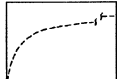
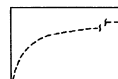
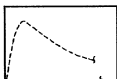
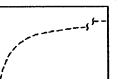
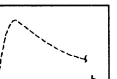
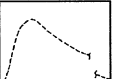
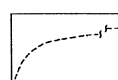
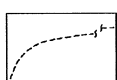
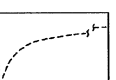
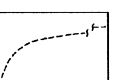
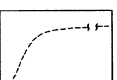
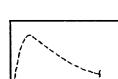
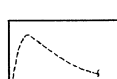
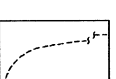
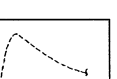
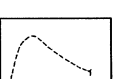
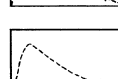
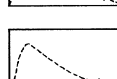
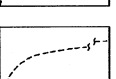
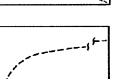
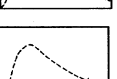
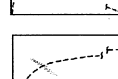
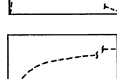
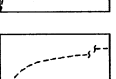
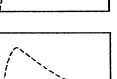
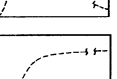
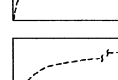
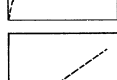
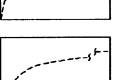
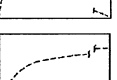
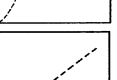
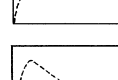
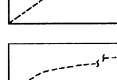
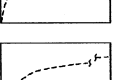
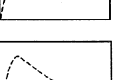
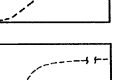
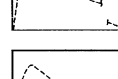
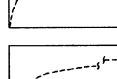
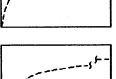
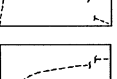
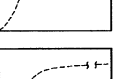
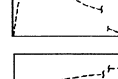
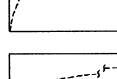
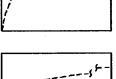
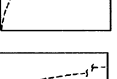
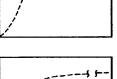
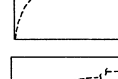
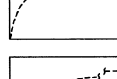
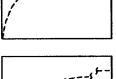
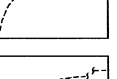
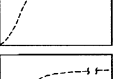
Table I Isotope Exchange-Rate Expressions for Selected Uni and Bi Substrate Systems^a

1. Uni Uni	$R_{A \leftrightarrow P} = \frac{V_1[A]}{K_a[1 + [A]/K_a + [P]/K_p]}$	
2. Ordered Bi Uni	$R_{A \leftrightarrow P} = \frac{V_1[A][B]}{(K_a K_b + K_a[B])[1 + [A]/K_a + [P]/K_p]}$	$R_{B \leftrightarrow P} = \frac{V_1[B]}{K_b[1 + K_b/[A] + K_a + [P]/K_p[A]]}$
3. Random Bi Uni (Rapid Equilibrium)	$R_{all} = \frac{V_1}{1 + K_a/[A] + K_b/[B] + (K_a K_b/[A][B])(1 + [P]/K_p)}$	
4. Random Bi Uni	$R_{A \leftrightarrow P} = \frac{[k_1 k_3 k_9 [A][B] + k_7 k_9 [A](k_2 + k_3[B])[B]/K_b]E_0}{[k_2(k_4 + k_8 + k_9) + k_3[B](k_8 + k_9)][1 + [A]/K_a + [B]/K_b + [A][B]/K_a K_b]}$ $R_{B \leftrightarrow P} = \frac{[k_5 k_7 k_9 [A][B] + k_3 k_9 [B](k_6 + k_7[A])([A]/K_a)]E_0}{[k_6(k_4 + k_8 + k_9) + k_7[A](k_4 + k_9)][1 + [A]/K_a + [B]/K_b + [A][B]/K_a K_b]}$	
5. Ordered Bi Bi	$R_{A \leftrightarrow Q} = \frac{V_1[A][B]}{[K_a K_b + K_a[B] + K_a K_b K_q [P]/K_q K_p][1 + ([A]/K_a)(1 + [B]/K_{ab}) + [Q]/K_{iq}]}$ $R_{A \leftrightarrow P} = \frac{V_1[A][B]}{[K_a K_b + K_a[B]][1 + ([A]/K_a)(1 + [B]/K_{ab}) + [Q]/K_{iq}]}$ $R_{B \leftrightarrow P} = \frac{V_1[B]}{K_b[1 + K_b/[A] + [B]/K_{ab} + K_a[Q]/K_{iq}[A]]}$ $R_{B \leftrightarrow Q} = \frac{V_1[B]}{[K_b + K_b K_q [P]/K_q K_p][1 + K_b/[A] + [B]/K_{ab} + K_a[Q]/K_{iq}[A]]}$	
6. Theorell–Chance Bi Bi	$R_{A \leftrightarrow Q} = \frac{V_1[A][B]}{[K_a[B] + K_a K_b + K_{iq}[P]/K_{eq}][1 + [A]/K_a + [Q]/K_{iq}]}$ $R_{A \leftrightarrow P} = \frac{V_1[A][B]}{[K_a K_b + K_a[B]][1 + [A]/K_a + [Q]/K_{iq}]}$ $R_{B \leftrightarrow Q} = \frac{V_1[B]}{[K_b + K_b [P]/K_{eq}][1 + K_b/[A] + K_a[Q]/K_{iq}[A]]}$ $R_{B \leftrightarrow P} = \frac{V_1[B]}{K_b[1 + K_b/[A] + K_a[Q]/K_{iq}[A]]}$	
7. Random Bi Bi (Rapid Equilibrium)	$R_{all} = \frac{V_1}{1 + K_a/[A] + K_b/[B] + K_a K_b/[A][B](1 + [P]/K_p + [Q]/K_{iq} + [P][Q]/K_p K_q)}$	
8. Random Bi Bi	$R_{A \leftrightarrow Q} = \frac{[k_9 k_{11} + k_{13}(k_{10}[P] + k_{11})][k_1 k_3 [A][B] + k_7([A][B]/K_{ib})(k_2 + k_3[B])]E_0}{(Z)\{k_2[k_{10}[P](k_4 + k_8 + k_{13}) + k_{11}(k_4 + k_8 + k_9 + k_{13})] + k_3[B][k_{10}[P](k_8 + k_{13}) + k_{11}(k_8 + k_9 + k_{13})]\}}$ $R_{A \leftrightarrow P} = \frac{[k_{13}k_{15} + k_9(k_{14}[Q] + k_{15})][k_1 k_3 [A][B] + k_7([A][B]/K_{ib})(k_2 + k_3[B])]E_0}{(Z)\{k_2[k_{14}[Q](k_4 + k_8 + k_9) + k_{15}(k_4 + k_8 + k_9 + k_{13})] + k_3[B][k_{14}[Q](k_8 + k_9) + k_{15}(k_8 + k_9 + k_{13})]\}}$ $R_{B \leftrightarrow Q} = \frac{[k_9 k_{11} + k_{13}(k_{10}[P] + k_{11})][k_5 k_7 [A][B] + k_3([A][B]/K_{ab})(k_6 + k_7[A])]E_0}{(Z)\{k_6[k_{10}[P](k_4 + k_8 + k_{13}) + k_{11}(k_4 + k_8 + k_9 + k_{13})] + k_7[A][k_{10}[P](k_4 + k_{13}) + k_{11}(k_4 + k_9 + k_{13})]\}}$ $R_{B \leftrightarrow P} = \frac{[k_{13}k_{15} + k_9(k_{14}[Q] + k_{15})][k_5 k_7 [A][B] + k_3([A][B]/K_{ab})(k_6 + k_7[A])]E_0}{(Z)\{k_6[k_{14}[Q](k_4 + k_8 + k_9) + k_{15}(k_4 + k_8 + k_9 + k_{13})] + k_7[A][k_{14}[Q](k_4 + k_9) + k_{15}(k_4 + k_9 + k_{13})]\}}$ where $Z = \{1 + [A]/K_a(1 + [B]/K_b) + [B]/K_{ib} + [P]/K_p + [Q]/K_{iq}\}$	
9. Random On–Ordered Off Bi Bi	$R_{A \leftrightarrow Q} = \frac{k_9 k_{11}[k_1 k_3 [A][B] + k_7([A][B]/K_{ab})(k_2 + k_3[B])]E_0}{\{k_2[k_{10}[P](k_4 + k_9) + k_{11}(k_4 + k_8 + k_9)] + k_3[B][k_9 k_{10}[P] + k_{11}(k_8 + k_9)]\}(Y)}$ $R_{A \leftrightarrow P} = \frac{k_9[k_1 k_3 [A][B] + k_7([A][B]/K_{ab})(k_2 + k_3[B])]E_0}{\{k_3[B](k_8 + k_9) + k_2(k_4 + k_8 + k_9)\}(Y)}$ $R_{B \leftrightarrow Q} = \frac{k_9 k_{11}[k_5 k_7 [A][B] + k_3([A][B]/K_{ab})(k_6 + k_7[A])]E_0}{\{k_6[k_{10}[P](k_4 + k_9) + k_{11}(k_4 + k_8 + k_9)] + k_7[A][k_9 k_{10}[P] + k_{11}(k_4 + k_9)]\}(Y)}$ $R_{B \leftrightarrow P} = \frac{k_9[k_5 k_7 [A][B] + k_3([A][B]/K_{ab})(k_6 + k_7[A])]E_0}{\{k_7[A](k_4 + k_9) + k_6(k_4 + k_8 + k_9)\}(Y)}$ where $Y = \{1 + ([A]/K_a)(1 + [B]/K_b) + [B]/K_{ib} + [Q]/K_{iq}\}$	
10. Ping Pong Bi Bi	$R_{A \leftrightarrow P} = \frac{V_1[A]}{K_a[1 + [A]/K_a + K_p[A]/K_p[P] + [Q]/K_{iq}]}$ $R_{B \leftrightarrow Q} = \frac{V_1[B]}{K_b[1 + [B]/K_b + [P]/K_p + K_q[B]/K_q[Q]]}$ $R_{A \leftrightarrow Q} = \frac{V_1[A][B]}{[K_a[B] + (K_a K_b[P]/K_p)][1 + [A]/K_a + [Q]/K_{iq} + K_p[A]/K_a[P]]}$	

^a The expressions are derived for systems at chemical equilibrium and in the absence of abortive complexes. All expressions, except for the Uni Uni and Rapid Equilibrium cases, were derived assuming only one central catalytic complex.

Table II

Exchange Profiles for Enzyme Kinetic Mechanisms^a

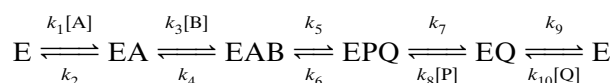
Mechanism	Exchange	Varied substrate-product pair(s)				
		A-P	B-P	A-Q	B-Q	A-B-P-Q ^b
1. Uni Uni	$A \leftrightarrow P$					
2. Ordered Bi Uni	$A \leftrightarrow P$					
	$B \leftrightarrow P$					
3. Random Bi Uni (Rapid Equilibrium)	Any					
4. Random Bi Uni	$A \leftrightarrow P$					
	$B \leftrightarrow P$					
5. Ordered Bi Bi	$A \leftrightarrow P$					
	$B \leftrightarrow P$					
	$A \leftrightarrow Q$					
	$B \leftrightarrow Q$					
6. Theorell-Chance Bi Bi	$A \leftrightarrow P$					
	$B \leftrightarrow P$					
	$A \leftrightarrow Q$					
	$B \leftrightarrow Q$					
7. Random Bi Bi (Rapid Equilibrium)	Any					
8. Random Bi Bi	Any					

^aSee Table I for a parallel listing of rate equations.^bA-B-P-Q refers to the Wedler-Boyer protocol of varying all substrates and products in a constant ratio.

the mass-action ratio). In this way, each substrate-product pair can be held constant with respect to the ratio of the absolute concentrations of substrate and product comprising that pair. In the case of the hexokinase reaction, for which the apparent equilibrium constant is about 500 at pH 6.5, we could choose $\alpha = [\text{glucose 6-phosphate}]/[\text{glucose}] = 25$ and $\beta = [\text{ADP}]/[\text{ATP}] = 20$, such that $\alpha\beta$ equals 500. The sugar pair could be held at any set of absolute concentrations, say for instance 10 mM and 0.4 mM, while one can establish a number of nucleotide pairs (e.g., $\beta_1 = ([\text{ADP}] = 20 \text{ mM})/([\text{ATP}] = 1 \text{ mM})$; $\beta_2 = ([\text{ADP}] = 16 \text{ mM})/([\text{ATP}] = 0.8 \text{ mM})$; $\beta_3 = ([\text{ADP}] = 12 \text{ mM})/([\text{ATP}] = 0.6 \text{ mM})$; $\beta_4 = ([\text{ADP}] = 8 \text{ mM})/([\text{ATP}] = 0.4 \text{ mM})$; $\beta_5 = ([\text{ADP}] = 4 \text{ mM})/([\text{ATP}] = 0.2 \text{ mM})$). Any $\alpha\beta$ product still satisfies the mass-action ratio, but this procedure allows one to vary the degree of saturation of hexokinase, in this case, with respect to the nucleotide pair. The symmetrical alteration of the sugar pair affords the opportunity to characterize the effects of increasing saturation of hexokinase by glucose and glucose 6-phosphate. Moreover, the nonvaried substrate-product pair can be prepared to contain buffer, metal ions, salt, and modifiers (if desired), and enzyme can be added to completely equilibrate the system. Because care has been exercised in choosing the above conditions, the reaction should be at or very near equilibrium, even before addition of enzyme. At this point, one may now introduce a small aliquot of stable or radioactive isotopically labeled reactant (where reactant is taken either to be substrate or product); the progress of the equilibrium exchange reaction may then be monitored by quenching an aliquot of the complete reaction sample, followed by chromatographic separation of labeled substrate and product. A useful check on systematic and/or consistent errors is that the progress curves for all exchange reactions are strictly first-order; moreover, the maximal rate of exchange should be linearly proportional to enzyme concentration.

Let us now examine the behavior of enzymes operating by way of ordered and random kinetic bisubstrate mechanisms:

1. Ordered Bi Bi Ternary Complex Mechanism



The A $\leftrightarrow$ Q exchange velocity as a function of the absolute concentrations of the [B]/[P] pair is the diagnostic test for compulsory ordered mechanisms wherein one substrate A must precede the binding of the other substrate, here B, and one product P must precede the release of the second product, here designated as Q. As the concentrations of [B] and [P] are raised, while maintaining the [B]/[P] ratio, a great deal of the enzyme will be drawn into the central ternary complexes, and the fraction of total enzyme present as uncomplexed E will become progressively lower. For A $\leftrightarrow$ Q exchange to occur, labeled A* or Q* must bind to free enzyme because the rate of all exchange processes reflects the fraction of reactant X that can combine with the fraction of another reactant Y. Since uncomplexed enzyme is required to form EA* (or EQ* in the reverse reaction), the A $\leftrightarrow$ Q exchange will increase as we proceed from low to moderate absolute concentrations of the [B]/[P] pair. However, the A $\leftrightarrow$ Q exchange will not be favored as we continue to raise the absolute concentrations of the [B]/[P] pair. Note that the A $\leftrightarrow$ Q exchange will not be depressed at high [A]/[Q] absolute levels, because increasing the A and Q concentrations actually favors the binding of A* (or Q*). Interestingly, the B $\leftrightarrow$ P exchange requires sufficient concentrations of EA and EQ for combination with B* or P*, respectively. Thus, raising the absolute concentrations of components of the [A]/[Q] or [B]/[P] pair will increase the B $\leftrightarrow$ P exchange rate, and a plot of the exchange rate *versus* absolute concentrations of either pair will be hyperbolic. This behavior is fully consistent with saturation kinetics. In summary, the depression of the A $\leftrightarrow$ Q exchange at high absolute levels of B and P serves to categorize an enzyme as a ordered ternary complex type.

The now classical example is lactate dehydrogenase. Silverstein and Boyer⁴ were the first to determine the rates of exchange between cognate pairs of reactants (*i.e.*, lactate and pyruvate as well as NAD⁺ and NADH). Convenient [NADH]/[NAD⁺] and [pyruvate]/[lactate] ratios were chosen such that when combined they satisfied the apparent equilibrium constant for the LDH reaction. These investigators first established that each exchange rate was directly proportional to the duration of exchange and likewise directly proportional to enzyme concentration. As an additional control, they also demonstrated the equality of the pyruvate $\leftrightarrow$ lactate exchange

rates to within ten percent by starting with either $[1-^{14}\text{C}]$ pyruvate or $[1-^{14}\text{C}]$ lactate. Then, to examine the order of substrate binding/release, Silverstein and Boyer examined the effect of raising the absolute concentrations of lactate and pyruvate on the rate of equilibrium exchange between pyruvate and lactate and between NAD^+ and NADH . As shown in Fig. 1, the exchange between pyruvate and lactate rises to what is an essentially stable plateau, whereas the exchange between coenzymes is markedly depressed over this concentration range of pyruvate and lactate. These observations exclude a random mechanism of substrate addition (see next paragraph) and were consistent with an ordered mechanism in which the coenzymes form the "outer" Michaelis complexes and each combines with its respective three-carbon acid to form "inner" ternary complexes.

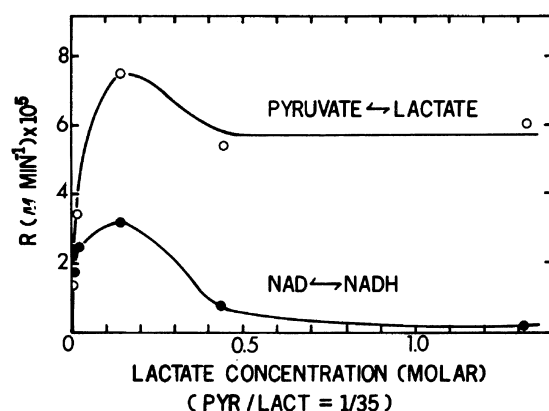
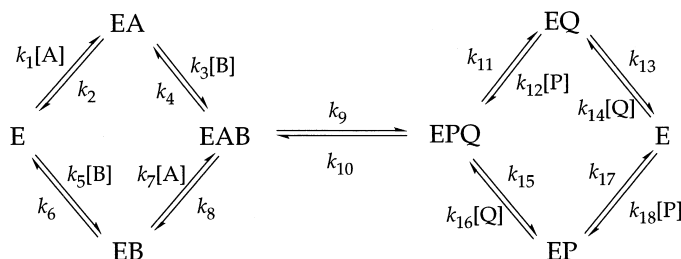


Figure 1. Effect of lactate and pyruvate concentrations on the equilibrium exchange rates of rabbit muscle lactate dehydrogenase at pH 7.9.⁴ Reaction mixtures contained 1.68–1.70 mM NAD^+ , 32.2–45.5 mM NADH , and the indicated concentrations of pyruvate and lactate.

2. Random Bi Bi Ternary Complex Mechanism

The random or noncompulsory ordered mechanism is noticeably symmetrical, with two different paths for producing the EAB complex from free enzyme E and its substrates A and B, as well as two different pathways for producing the EPQ complex from free enzyme E and its products P and Q:



For this reason, these alternative routes for isotope combination with enzyme-substrate and/or enzyme-product complexes ensures that raising the $[A]/[Q]$ or $[B]/[P]$ pair will not depress either the $A \leftrightarrow Q$ or the $B \leftrightarrow P$ exchanges. Fromm, Silverstein, and Boyer⁵ conducted a thorough analysis of the equilibrium exchange kinetic behavior of yeast hexokinase, and the data shown in Fig. 2 indicate that there is a random mechanism of substrate addition and product release.

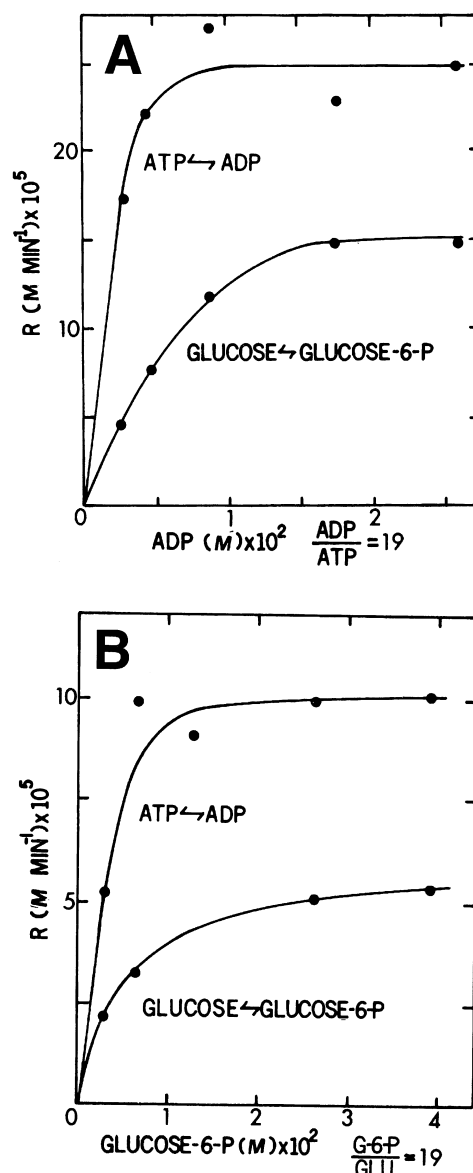


Figure 2. (A) Effect of simultaneously raising the absolute concentrations of ATP and ADP on the indicated equilibrium exchange reactions catalyzed by yeast hexokinase. (B) Effect of simultaneously raising the absolute concentrations of glucose and glucose 6-phosphate on the indicated equilibrium exchange reactions of yeast hexokinase.

These qualitative conclusions can be justified in terms of rate equations derived for each exchange catalyzed by ordered and random ternary complex mechanisms^{1,3}.

¹P. D. Boyer (1959) *Arch. Biochem. Biophys.* **82**, 387.

²P. D. Boyer (1978) *Acc. Chem. Res.* **11**, 218.

³D. L. Purich & R. D. Allison (1982) *Meth. Enzymol.* **64**, 3.

⁴E. Silverstein & P. D. Boyer (1964) *J. Biol. Chem.* **239**, 3901 and 3908.

⁵H. J. Fromm, E. Silverstein & P. D. Boyer (1964) *J. Biol. Chem.* **239**, 3645.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Abortive complex interaction, **64**, 39-45; enzyme stability and, **64**, 23, 24; equilibrium exchange [equations, **63**, 76-79; **64**, 7-18; experimental protocol, **64**, 19-27, 31; initial rate method, **64**, 32, 33; ionic strength effect, **64**, 27, 32; kinetic parameter, **64**, 17, 18, 25, 31, 32; pH effect, **64**, 23; profile, **64**, 13, 17; quenching exchange reactions, **64**, 25; rapid equilibrium model, **64**, 6, 7, 17, 18; side reaction effect, **64**, 24; substrate inhibition, **64**, 34-36; substrate-product ratio, **64**, 22, 23; substrate stability, **64**, 24; theory, **64**, 7-18, 30, 32, 33, 44; types, **64**, 4, 5; intermediate, **64**, 25, 26, 34, 36-38; irreversible enzyme reactions, **64**, 36-38; modifier effects, **87**, 647; substrate synergism, **64**, 38, 39; equations for isotope exchange, **64**, 7-18 [derivation, **63**, 76-83; Cleland method, **63**, 80-83; equilibrium assumption, **63**, 76-79; steady state assumption, **63**, 79-83].

ISOTOPE EXCHANGE (Reactions Away From Equilibrium)

Although most enzyme exchange studies have been investigated at equilibrium, the back exchange of labeled product while the reaction is proceeding in the forward direction can provide valuable information about enzymic catalysis. Under favorable conditions, one may utilize such isotope exchange data to learn about the order of product release and the presence of covalent enzyme-substrate compounds. One of the first systems to be characterized in this way was glucose-6-phosphatase^{1,2}.

During the course of glucose 6-phosphate hydrolysis, radiolabeled glucose was added and the back exchange to form labeled glucose 6-phosphate was examined. (Likewise, the possibility of incorporation of labeled phosphate into glucose 6-phosphate was examined under identical conditions, but none was observed.) The rate of the glucose $\leftrightarrow$ glucose 6-phosphate exchange correlated rather well with the amount of glucose inhibition of the phosphatase. These findings indicated that glucose is the first product to leave and that the negative free energy of hydrolysis is preserved as a phosphoryl-enzyme compound. It is also interesting to note that Zatman, Kaplan, and Colowick³ used such exchange phenomena away from equilibrium to synthesize [¹⁴C]NAD⁺ from [¹⁴C]-

nicotinamide and NAD⁺ with bovine spleen NADase. They also observed that labeled adenosine diphosphoribosyl pyrophosphate (ADPR) failed to exchange in a similar fashion.

These mechanisms involve ordered release of products, and they are formally of the ping pong Bi Bi type with H₂O presumably entering the catalytic process after release of the first product. Labeled product, P, can exchange back to substrate A in the absence of the other product Q only when there is a significant level of EQ in the steady state and P* is sufficiently high (*i.e.*, equal or greater than the respective K_i value for P). An exchange from Q* back to A may only occur when there is a significant level of P present. If Q* does exchange without P present, one must conclude that the order is noncompulsory (random) with an adequate level of the EP complex in the steady state to support significant exchange. In this respect, observation of a P* $\leftrightarrow$ A exchange is not strict proof of ordered release, and one may argue that a random mechanism with slow EQ breakdown to enzyme and Q is operative⁴.

¹L. F. Hass & W. L. Byrne (1960) *J. Am. Chem. Soc.* **82**, 947.

²R. C. Nordlie (1982) *Meth. Enzymol.* **87**, 319.

³L. J. Zatman, N. O. Kaplan & S. P. Colowick (1953) *J. Biol. Chem.* **200**, 197.

⁴D. L. Purich & R. D. Allison (1982) *Meth. Enzymol.* **64**, 1.

ISOTOPE RATIO MASS SPECTROMETER

An instrument that measures the isotopic mass ratio of a gas by bombarding the sample in an electron beam, such that the molecular ions generated can be deflected in their trajectories through a magnetic field in accordance to their charge/mass ratios. These devices are extremely accurate and reliable, and many stable isotope experiments can be analyzed by converting the isotopically substituted metabolite into carbon dioxide, water, or molecular nitrogen prior to IRMS measurements.

ISOTOPE SCRAMBLING

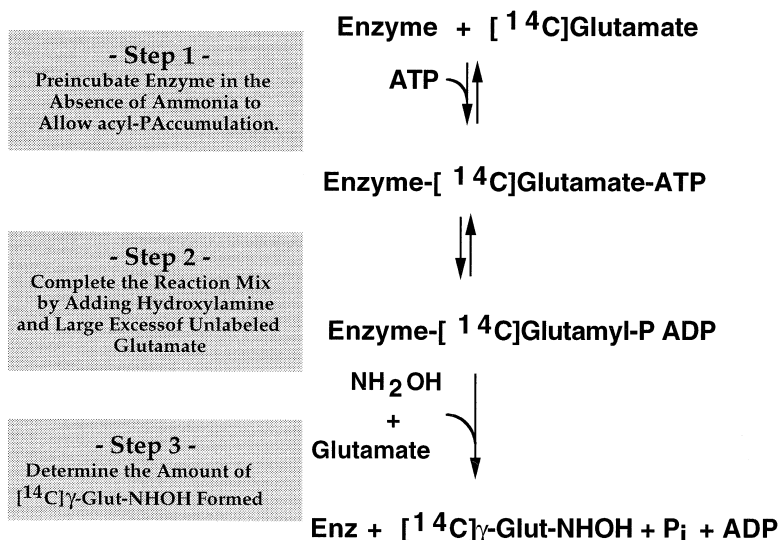
A change in the distribution of isotopes within a specified set of atoms of a particular molecular entity or entities. **See** *Positional Isotope Exchange; Bridge-to-Nonbridge Oxygen Exchange*

ISOTOPE TRAPPING

A technique for evaluating the rate (and rate constant) for the dissociation of an enzyme-substrate covalent

compound or noncovalent complex, as compared to its rate (and rate constant) for interconversion into product. While the method was first applied by Meister's group¹ in 1962, its general utility remained unappreciated until the studies of Rose and co-workers². (Among many practicing kineticists, the experimental strategy is now frequently called the Rose experiment.)

Rose and co-workers² first demonstrated that a proteolyzed form of hexokinase forms a "sticky" (or sluggishly dissociable) complex with glucose. The generalized application of this approach to the kinetic characterization of multisubstrate enzymes has been treated in detail³. **See also** *Partition Coefficient; Radiospecific Activity; Stickiness*



To adduce evidence for a γ-glutamyl phosphate intermediate in the sheep brain glutamine synthetase reaction, Meister's group incubated that enzyme with [¹⁴C]glutamate and ATP [in strict absence of the third substrate ammonia (or hydroxylamine)]. This solution was subsequently mixed with a second solution containing a great molar excess of unlabeled glutamate and the third substrate to initiate the overall reaction to form glutamine. The basic idea was that the formation of the γ-glutamyl phosphate intermediate in the absence of the third substrate would prime the enzymatic reaction to favor radioactive glutamine formation *versus* desorption of substrates. The reasoning went that if no acyl phosphate intermediate was formed, the radiolabeled glutamate would probably dissociate and become diluted with the excess of unlabeled glutamate, such that the amount of [¹⁴C]glutamine formed enzymatically would be miniscule. Rose offered the more rigorous interpretation that whether or not a covalent intermediary is in fact formed, the amount of [¹⁴C]glutamine depended only on the partition coefficient [$k_{\text{onward}}/(k_{\text{onward}} + k_{\text{dissoc}})$], where k_{onward} is the onward rate constant for product formation and k_{dissoc} is the rate constant for dissociation of the complex of enzyme and radioactive substrate.

¹P. R. Krishnaswamy, V. Pamljans & A. Meister (1962) *J. Biol. Chem.* **237**, 2932.

²I. A. Rose, E. L. O'Connell, S. Litwin & J. Bar-Tana (1974) *J. Biol. Chem.* **249**, 5163.

³I. A. Rose (1980) *Meth. Enzymol.* **64**, 47.

Selected entries from *Methods in Enzymology* [vol, page(s)]:
 Enzyme-substrate complex formation, **64**, 53; data analysis, **64**, 56, 57; extensions of technique, **64**, 57-59; as evidence for occurrence of intermediate, **64**, 47-59; kinetic equation, **64**, 49-52; limitations, **64**, 57-59; mixing procedure, **64**, 53-56; reaction condition, **64**, 56, 57; termination, **64**, 56, 57.

ISOTOPICALLY SENSITIVE STEP

An elementary reaction or step (in a chemical process) for which the rate constants are altered by an isotopic substitution in substrate, product, or solvent. **See** *Kinetic Isotope Effect; Solvent Isotope Effect*

ISOTOPIC PERTURBATION

Any observable effect of isotopic substitution on the rate or extent of a chemical/physical process. Equilibrium isotopic perturbation measurements can provide valuable information about kinetic isotope effects on enzymic catalysis¹. NMR shift difference measurements are also useful in detecting the effects of isotopic substitution on a fast (degenerate) equilibrium between two species differing only in their specific isotopic substitution^{2,3}. The

latter technique allows one to differentiate a rapidly equilibrating mixture with time-averaged symmetry from that of a single structure with higher symmetry^{2,3}.

¹W. W. Cleland (1980) *Meth. Enzymol.* **64**, 104.

²IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

³H. U. Siehl (1987) *Adv. Phys. Org. Chem.* **23**, 63.

ISOTOPOLOGUES

Molecules having the same type and number of isotopic atoms, albeit in different positions. CH₂DCH(NH₃⁺)COO⁻ and CH₃CD(NH₃⁺)COO⁻ are examples of isotopomers. Certain stereoisomers and *cis-trans* isomers can also be regarded as isotopomers: for example, (*R*)- and (*S*)-1-deuteroethanol, CH₃CHDOH.

ISOTROPIC

A term used with respect to a medium or substance or body having physical properties which are independent of direction. **See also** *Anisotropic*

ISOTROPIC MOTION

In the standard mathematical expressions for the contribution of quadrupolar relaxation to the relaxation rates of the quadrupolar nucleus (in nuclear magnetic resonance), rapid isotropic motion is assumed to occur. This behavior, in most cases, will not be true in the solid or liquid crystalline state¹⁻⁴.

¹A. Abragam (1961) *The Principles of Nuclear Magnetism*, Oxford Univ. Press, London.

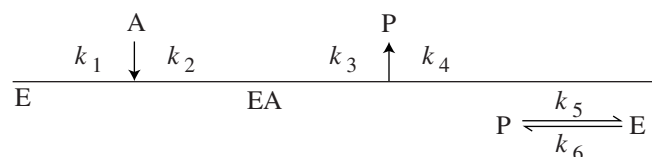
²T. L. James (1975) *Nuclear Magnetic Resonance in Biochemistry*, Academic Press, New York.

³K. Wuthrich (1976) *NMR In Biological Reactions*, Academic Press, New York.

⁴H. H. Mantsch, H. Saito & I. C. P. Smith (1977) *Prog. Nucl. Magn. Res. Spect.* **11**, 211.

ISO UNI UNI MECHANISM

The general reaction scheme for this mechanism can be drawn as shown below using Cleland's notation.



The complete steady-state rate expression for this scheme is

$$v = \frac{V_{\max,f} V_{\max,r} ([A] - \{[P]/K_{\text{eq}}\})}{V_{\max,r} K_a + V_{\max,r} [A] + (V_{\max,f} [P]/K_{\text{eq}}) + (V_{\max,r} [A][P]/K_{\text{iip}})}$$

where $V_{\max,f}$ and $V_{\max,r}$ are the forward and reverse maximum velocities, respectively (in which $V_{\max,f} = k_3 k_5 [E_{\text{total}}]/(k_3 + k_5)$ and $V_{\max,r} = k_2 k_6 [E_{\text{total}}]/(k_2 + k_6)$), K_a is the Michaelis constant for A (equal to $(k_2 + k_3)(k_5 + k_6)/\{k_1(k_3 + k_5)\}$), K_{eq} is the equilibrium constant (equal to $V_{\max,f} K_p/V_{\max,r} K_a$ in which K_p is the Michaelis constant for P equal to $(k_2 + k_3)(k_5 + k_6)/\{k_4(k_2 + k_6)\}$), and K_{iip} is equal to $(k_3 + k_5)/k_4$. Note that this general expression reduces to the same equation as is seen for the Uni Uni mechanism in the absence of the product P^{1,2}.

¹H. J. Fromm (1979) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

ISOVALERYL-CoA DEHYDROGENASE

This FAD-dependent enzyme [EC 1.3.99.10] catalyzes the reaction of 3-methylbutanoyl-CoA with an electron-transferring flavoprotein to produce 3-methylbut-2-enoyl-CoA and the reduced electron-transferring flavoprotein. Together with the electron-transferring flavoprotein and the electron-transferring-flavoprotein dehydrogenase [EC 1.5.5.1], this enzyme forms a multi-enzyme system that can reduce ubiquinone and other acceptors. Isovaleryl-CoA dehydrogenase is not identical with butyryl-CoA dehydrogenase, acyl-CoA dehydrogenase, or 2-methylacyl-CoA dehydrogenase.

Y. Ikeda & K. Tanaka (1988) *Meth. Enzymol.* **166**, 155 and 374.

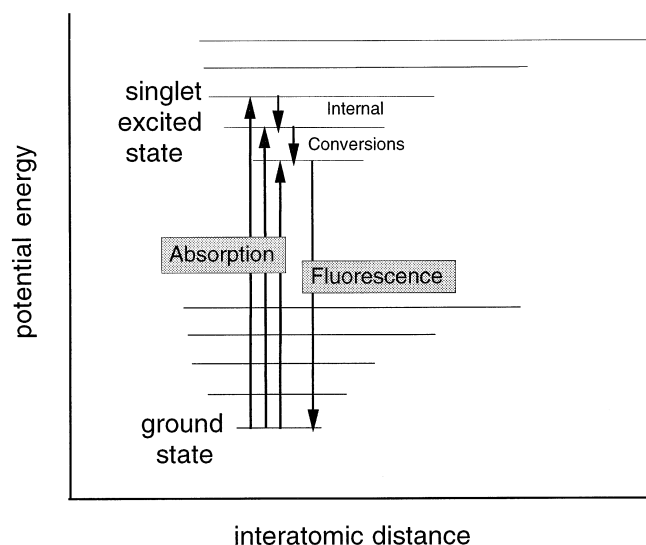
ISOZYMES

Enzymes catalyzing the same reaction, but differing from each other with respect to their primary structure. Such amino acid substitutions are often detected as differences in one or more physical and/or chemical properties, most notably electrophoretic mobility. Isoenzymes, although they catalyze the same reaction, may have different kinetic parameters, different regulatory characteristics, and/or different stabilities. Enzymes that catalyze the same reaction and have the same primary structure, while still differing by some measurable property, are often called pseudoisozymes (or, pseudoisoenzymes). Examples would be proteins that have undergone different degrees or types of covalent modification (including the cases of different enzymes derived from the same zymogen via varying levels of proteolysis). Other examples of pseudoisozymes are oligomeric proteins that form different degrees of aggregation.

J

JABLONSKI DIAGRAM

A diagram (plotting potential energy on the vertical axis *versus* interatomic distance for a bond on the horizontal axis) for illustrating how the excited state of a molecular entity undergoes an allowed transition between ground and excited states. The energy wells for the ground and excited states (often drawn as a Leonard-Jones potential energy diagram) have different interatomic distances representing the greater equilibrium bond length in the excited vibrational and electronic states. Vibrational states are depicted as a series of horizontal lines starting with the zero-point energy level, and the spacing between the vibrational state reflects the quantized nature of energy differences between successive vibrational states. In the electronic transition, one of the two electrons from a chemical bond is promoted to a higher energy state, and this will be a singlet excited state if the excited electron has a spin that is opposite that of the other electron in the electron-pair bond.



Electronic transitions occur much more rapidly than do the motions of nuclei (Franck-Condon principle), and this results in a vertical transition (*i.e.*, changes in the bond length cannot immediately attend photon absorption or emission).

If the promoted electron in the excited singlet state undergoes a change to a parallel spin relative to its partner, then it is said to have undergone an intersystem transfer from a singlet excited state to a triplet excited state. Longer-lived phosphorescent states are depicted as metastable states positioned at lower energy relative to the initial excited state, and the first excited triplet state is reached via radiationless transfer that delays photon emission in phosphorescence relative to the promptness of fluorescence emissions. Moreover, phosphorescence occurs with a lifetime ($\tau \approx 10^{-3}$ seconds or greater) that is much slower than the time scale of nuclear motions; hence this transition to the singlet ground state can be regarded as a nonvertical transition.

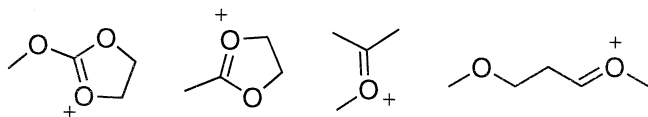
The name of this diagram (or plot) honors the pioneering contributions of Jablonski, who recognized that phosphorescence involves emission from a metastable triplet state lying at lower energy relative to the lowest lying excited singlet state.

J. R. Lakowicz (1983) *Principles of Fluorescence Spectroscopy*, Chapter 1, Plenum, New York.

JENCKS' CLOCK

A semiquantitative procedure used to estimate the lifetimes of carbocations and oxocarbenium ions by using diffusion-controlled trapping of the cations by nucleophiles^{1,2}. Ions of intermediate stability react with azide ions at a constant, diffusion-controlled rate and react with water by an activated process. The ratio of the products obtained from the azide path and the water path is dependent on the electronic characteristics of the cation.

The Jencks' clock, as well as pulse radiolysis experiments, indicate that oxocarbenium cations often can be quite stabilized, with a half-life on the order of milliseconds. For example, at room temperature, the structures shown here (left to right) have respective lifetimes of



0.25 millisecond, 0.7 microsecond, 1 nanosecond, and 0.5 nanosecond^{2,3}. The estimated lifetime of glucosyl cation in aqueous solution is about 1 picosecond⁴. While longer than the period required for a bond vibration, the magnitude of this value suggests that a discrete cationic intermediate is probably not meaningful in enzyme-catalyzed glycosyl transfer reactions, because the intermediates in those reactions typically persist for periods on the order of milliseconds. Nonetheless, a longer lived resonance-stabilized oxocarbenium ion may still form in the transition state of enzymic reactions.

¹P. R. Young & W. P. Jencks (1977) *J. Amer. Chem. Soc.* **99**, 8238.

²J. P. Richard, M. E. Rothenberg & W. P. Jencks (1984) *J. Amer. Chem. Soc.* **106**, 1361.

³M. Steenken & R. A. McClelland (1989) *J. Amer. Chem. Soc.* **111**, 4967.

⁴T. L. Amyes & W. P. Jencks (1989) *J. Amer. Chem. Soc.* **111**, 7888.

JENCKS MECHANISM

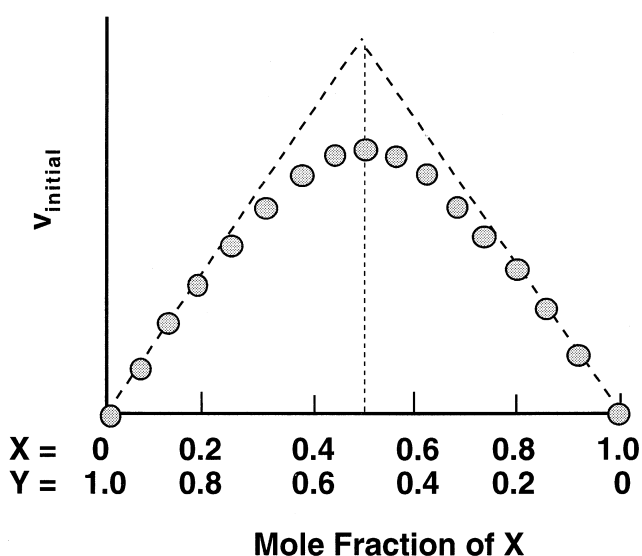
A hypothetical reaction scheme for enzyme action and cooperativity¹. In this ligand-substitution model, dissociation steps (for example, $EP \rightleftharpoons E + P$) are not treated as unimolecular steps; rather, dissociation is bimolecular, requiring that an incoming ligand displace the bound ligand (thus, $EP + X \rightleftharpoons EX + P$). A ligand X, referred to as the surrogate, is displaced by the incoming substrate or displaces the outgoing product. Once the enzyme-product binary complex (EP) is formed in the scheme above, EP can only undergo ligand displacement by a new S molecule or by the surrogate X. In other words, the catalytic conversion of ES to EP is unidirectional (*i.e.*, there is no k_{-3} term or direction; such a rate constant is present in the Briggs-Haldane type model). This would suggest an untenable outcome, namely that in the S to P conversion the reaction essentially proceeds unidirectionally through the transition state. Derivation of an

expression for a scheme containing a k_{-3} step yields a rate equation containing both $[X]^2$ and $[S]^2$ terms (note that the scheme above should also yield an expression having $[S]^2$ terms since S can bind to two different forms of the enzyme (namely EX and EP)). At high substrate concentrations, nonlinearity should be observed in double-reciprocal plots.

¹W. T. Jenkins (1982) *Adv. Enzymol.* **53**, 307.

JOB PLOT

A plotting protocol (also known as the method of continuous variation) that provides useful information about protein-ligand and protein-protein interactions. Mole fractions X_A and X_B of two interacting substances, say A and B, are varied such that the total molarity remains constant. Note that $X_A = \{[A]/([A] + [B])\}$ and $X_B = \{[B]/([A] + [B])\}$, such that $(X_A + X_B) = 1$. If an enzyme prefers to bind AB as a one-to-one complex, then the enzymatic activity will be maximal at a mole fraction X_A of 0.5 (*i.e.*, the point at which A and B are present in a one-to-one stoichiometry). Similarly, if AB_2 is the active species, then the enzyme will be most active at a X_A value of 0.33. In this manner, the stoichiometry of binding can be readily determined, and the technique can yield information concerning the affinity of the enzyme for the active species.



Huang¹ has provided an excellent account of the application of this method of continuous variation in studies

of enzymic catalysis. The technique can also assist in distinguishing between different cooperative models.

¹C. Y. Huang (1982) *Meth. Enzymol.* **87**, 509.

JOULE

A unit of energy (symbolized by J) equal to 10^7 ergs. In the SI system, the joule replaces the calorie, which is equivalent to 4.184 J. Formally, the joule equals the heat generated by an ampere flowing through a 1 ohm resistor over a one second interval.

JUGLONE MONOOXYGENASE

This enzyme [EC 1.14.99.27] catalyzes the reaction of 5-hydroxy-1,4-naphthoquinone with AH_2 and dioxygen to produce 3,5-dihydroxy-1,4-naphthoquinone, A, and water. The enzyme can also use 1,4-naphthoquinone, naphthazarin, and 2-chloro-1,4-naphthoquinone as substrates, but not other related compounds.

H. Rettenmaier & F. Lingens (1985) *Biol. Chem. Hoppe Seyler* **366**, 637.

K

K_a

1. Symbol for the association constant. Thus, for the reaction $A \rightleftharpoons P + Q$, $K_a = [A_{eq}]/([P_{eq}][Q_{eq}])$. This constant is the reciprocal of the dissociation constant and its negative logarithm is pK_a . **See** *Association Constant*.
2. The symbol for the dissociation constant for an acid. Thus, for the reaction $HA \rightleftharpoons H^+ + A^-$, $K_a = [H^+][A^-]/[HA]$. The negative logarithm of this constant is pK_a .
3. Symbol (also K_A) for the Michaelis constant for substrate A in an enzyme-catalyzed reaction.

KALLIKREIN

Plasma kallikrein [EC 3.4.21.34], also known as kininogenin and serum kallikrein, catalyzes the hydrolysis of Arg—Xaa and Lys—Xaa bonds in polypeptides. This includes the Lys—Arg and Arg—Ser bonds in human kininogen, thus producing bradykinin. Tissue kallikrein [EC 3.4.21.35] catalyzes the hydrolysis of peptide bonds, preferentially Arg—Xaa, in small-molecule substrates. It catalyzes the breaking of the appropriate bonds in kininogen resulting in the release of lysyl-bradykinin.

C. W. Wharton (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 345.

J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

KAMLET-TAFT SOLVENT PARAMETERS

Parameters of the Kamlet-Taft solvatochromic relationship. These parameters measure the contributions to overall solvent polarity of the hydrogen bond donor, the hydrogen bond acceptor, and the dipolarity/polarizability properties of solvents.

M. J. Kamlet, J. L. M. Abboud & R. W. Taft (1981) *Progr. Phys. Org. Chem.* **13**, 485.

KAPPA (κ)

1. Symbol for conductivity. 2. Symbol for compressibility (κ_T for isothermal compressibility $[-(1/V)(\partial V/\partial P)_T]$ and

κ_s for isentropic compressibility $[-(1/V)(\partial V/\partial P)_s]$). 3. Symbol for reciprocal radius of ionic atmosphere.

KATAL

A unit of enzyme catalytic activity equal to the conversion of one mole of substrate per second in a specified assay system. The katal (kat) is more commonly used in clinical enzymology. One unit of enzyme activity (*i.e.*, one micromole per minute) corresponds to 16.67 nkat.

K_b

1. Symbol for the dissociation constant of a base. Thus, for the reaction $BOH \rightleftharpoons B^+ + OH^-$, $K_b = [B^+][OH^-]/[BOH]$.
2. Symbol (also, K_B) for the Michaelis constant for substrate B in an enzyme-catalyzed reaction.

k_{cat}

See *Turnover Number*

k_{cat}/K_m

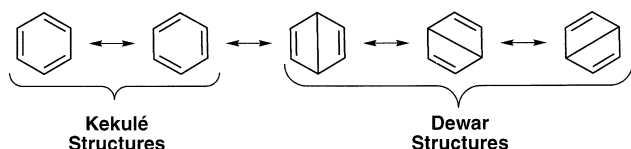
The ratio of the turnover number (*i.e.*, $V_{max}/[E_{total}]$) to the K_m value of a substrate in a particular enzyme-catalyzed reaction. When k_{cat} and K_m are the true steady-state parameters, this ratio (or the ratio V_{max}/K_m) is an excellent gauge of the specificity of the enzyme for that substrate. The larger the ratio, the more effective that substrate is used by the enzyme under study. In addition, the effects of a number of mechanistic probes of enzyme action on this ratio (for example, pH effects, isotope effects, temperature effects, the influence of various modifiers, *etc.*) can provide much information on the catalytic and binding mechanism. **See** V_{max}/K_m

K_d

Symbol for dissociation constant, equal to the reciprocal of the association constant. The negative logarithm of this constant is the pK_d . **See** *Dissociation Constant*; K_s

KEKULÉ STRUCTURES

Canonical forms of benzene with alternating double bonds (*i.e.*, “cyclohexatriene”), structures which contribute equally and predominantly to benzene’s resonance hybrid structure.



They are estimated by calculation to account for about 75% of the resonance stabilization of benzene with Dewar structures accounting for most of the remaining stabilization. However, such resonance structures have no separate physical reality or independent existence. A dotted circle is often used to indicate the resonance-stabilized bonding of benzene. Nonetheless, the most frequently appearing structures of benzene are the two Kekulé structures.

KELVIN

A unit of thermodynamic temperature (symbolized by K) that is one of the seven SI base units. The kelvin is equal to $1/(273.16)$ of the thermodynamic temperature of the triple point of water. (Note: Absolute zero is 0 K, and not 0°K.)

K_{eq}

Symbol for the thermodynamic equilibrium constant for a reaction.

3-KETOACID CoA-TRANSFERASE

This enzyme [EC 2.8.3.5], also known as succinyl-CoA:3-ketoacid CoA-transferase and 3-oxoacid CoA-transferase, catalyzes the reversible reaction of succinyl-CoA with a 3-oxo acid to produce succinate and a 3-oxo-acyl-CoA derivative.

W. P. Jencks (1973) *The Enzymes*, 3rd ed., **9**, 483.

β -KETOACYL-ACP REDUCTASE

This enzyme [EC 1.1.1.100], also known as 3-oxoacyl-[acyl-carrier protein] reductase, catalyzes the reaction of a (3*R*)-3-hydroxyacyl-[acyl-carrier protein] with NADP⁺ to produce 3-oxoacyl-[acyl-carrier protein] and NADPH. The enzyme prefers acyl-carrier protein derivatives over simple CoA derivatives as substrates.

J. Browse & C. Somerville (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 467.

S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.

β -KETOACYL-ACP SYNTHASE

This enzyme [EC 2.3.1.41], also known as 3-oxoacyl-[acyl-carrier protein] synthase, catalyzes the reaction of an acyl-[acyl-carrier protein] with malonyl-[acyl-carrier protein] to produce a 3-oxoacyl-[acyl-carrier protein], carbon dioxide, and the [acyl-carrier protein]. **See also** *Fatty Acid Synthetase*

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

J. Browse & C. Somerville (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 467.

S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.

P. R. Vagelos (1973) *The Enzymes*, 3rd ed., **8**, 155.

α -KETOBUTYRATE SYNTHASE

This enzyme [EC 1.2.7.2], also known as 2-oxobutyrate synthase, catalyzes the reaction of α -ketobutyrate (or, 2-oxobutanoate) with coenzyme A, and oxidized ferredoxin to produce propanoyl-CoA, carbon dioxide, and reduced ferredoxin.

B. B. Buchanan (1972) *The Enzymes*, 3rd ed., **6**, 193.

2-KETO-3-DEOXY-L-ARABINONATE DEHYDRATASE

This enzyme [EC 4.2.1.43], also referred to as 2-dehydro-3-deoxy-L-arabinonate dehydratase, catalyzes the conversion of 2-dehydro-3-deoxy-L-arabinonate to 2,5-dioxopentanoate and water.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

2-KETO-3-DEOXY-D-GLUCARATE ALDOLASE

This enzyme [EC 4.1.2.20], also referred to as 2-dehydro-3-deoxyglucarate aldolase, catalyzes the reversible reaction of 2-dehydro-3-deoxy-D-glucarate to pyruvate and tartronate semialdehyde.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

5-KETO-4-DEOXYGLUCARATE DEHYDRATASE

This enzyme [EC 4.2.1.41], also referred to as 5-dehydro-4-deoxyglucarate dehydratase, catalyzes the conversion of 5-dehydro-4-deoxy-D-glucarate to 2,5-dioxopentanoate, water, and carbon dioxide.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

2-KETO-3-DEOXY-L-PENTONATE ALDOLASE

This enzyme [EC 4.1.2.18], also referred to as 2-dehydro-3-deoxy-L-pentonate aldolase or 2-keto-3-deoxyarabonate aldolase, catalyzes the reversible conversion of 2-dehydro-3-deoxy-L-pentonate to pyruvate and glycolaldehyde.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

2-KETO-3-DEOXY-6-PHOSPHOGALACTONATE ALDOLASE

This enzyme [EC 4.1.2.21], also referred to as 6-phospho-2-dehydro-3-deoxygalactonate aldolase, catalyzes the reversible conversion of 2-dehydro-3-deoxy-D-galactonate 6-phosphate to pyruvate and D-glyceraldehyde 3-phosphate.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

2-KETO-3-DEOXY-6-PHOSPHOGLUCONATE ALDOLASE

This enzyme [EC 4.1.2.14], also referred to as phospho-2-dehydro-3-deoxygluconate aldolase, catalyzes the reversible conversion of 2-dehydro-3-deoxy-D-gluconate 6-phosphate to pyruvate and D-glyceraldehyde 3-phosphate. 2-Oxobutanoate can also serve as a substrate for this enzyme.

K. N. Allen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 135.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

α -KETOGLUTARATE DEHYDROGENASE COMPLEX

This multiprotein complex contains as its central enzyme α -ketoglutarate dehydrogenase (lipoamide) [EC 1.2.4.2], also referred to as oxoglutarate dehydrogenase (lipoamide) and oxoglutarate decarboxylase. This enzyme catalyzes the reaction of α -ketoglutarate (or, 2-oxoglutarate) with lipoamide to produce S-succinyl-dihydrolipoamide and carbon dioxide, using thiamin pyrophosphate as a cofactor.

R. N. Perlman (1991) *Biochemistry* **30**, 8501.

M. Hamada & H. Takenaka (1991) *A Study of Enzymes* **2**, 227.

S. J. Yeaman (1986) *Trends Biochem. Sci.* **11**, 293.

α -KETOGLUTARATE SEMIALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.26], also known as 2,5-dioxovalerate dehydrogenase, catalyzes the reaction of 2,5-dioxopentanoate with NADP⁺ and water to produce α -ketoglutarate (or, 2-oxoglutarate) and NADPH.

α -KETOGLUTARATE SYNTHASE

This enzyme [EC 1.2.7.3], also called 2-oxoglutarate synthase, catalyzes the reversible reaction of α -ketoglutarate (or, 2-oxoglutarate) with coenzyme A and oxidized ferredoxin to produce succinyl-CoA, carbon dioxide, and reduced ferredoxin.

B. B. Buchanan (1972) *The Enzymes*, 3rd ed., **6**, 193.

KETONIZATION

The process of converting an enol to a ketone. Pyruvate kinase catalyzes a ketonization reaction in the conversion of the enolpyruvate intermediate to pyruvate. **See Tautomerization**

I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.

3-KETO-5 β -STEROID Δ^4 -DEHYDROGENASE

This enzyme [EC 1.3.99.6], also known as 3-oxo-5 β -steroid 4-dehydrogenase and Δ^4 -3-ketosteroid 5 β -reductase, catalyzes the reaction of a 3-oxo-5 β -steroid and an acceptor to produce a 3-oxo- Δ^4 -steroid and the reduced acceptor.

S. J. Davidson (1969) *Meth. Enzymol.* **15**, 656.

3-KETOSTEROID Δ^5 -ISOMERASE

This enzyme [EC 5.3.3.1], also called steroid Δ -isomerase and Δ^5 -3-ketosteroid isomerase, catalyzes the interconversion of a 3-oxo- Δ^5 -steroid and a 3-oxo- Δ^4 -steroid.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

P. Talalay & A. M. Benson (1972) *The Enzymes*, 3rd ed., **6**, 591.

KEXIN

This calcium-ion-activated enzyme [EC 3.4.21.61] catalyzes the hydrolysis of peptide bonds at LysArg—Xaa and ArgArg—Xaa to process yeast α -factor pheromone and killer toxin precursors.

C. Brenner, A. Bevan & R. S. Fuller (1994) *Meth. Enzymol.* **244**, 152.

K_i (or, K_i)

Symbol for the dissociation constant of an inhibitor.

 K_{ia} , K_{ib} , and K_{ic}

Symbol for the dissociation constant of substrate A, B, and C.

 K_{ii}

Symbol for the dissociation constant of an inhibitor with respect to a particular form of the enzyme. This dissociation constant is associated with the intercept term in the double-reciprocal form of the initial-rate equation. For example, consider an inhibitor that can bind to either the free enzyme, E, or the binary central complex, EX, of a Uni Uni mechanism. K_{ii} would be the dissociation constant for the $EX + I \rightleftharpoons EXI$ step and is equal to $[EX][I]/[EXI]$. The binding of I to the free enzyme (*i.e.*, $E + I \rightleftharpoons EI$) is governed by K_{is} (equal to $[E][I]/[EI]$).

KINEMATIC VISCOSITY

The ratio (symbolized by ν) of the dynamic viscosity, or simply viscosity η , to the fluid density, ρ . Kinematic viscosity has SI units of meters squared per second.

KINESIN

The following are recent reviews dealing with this ATP-dependent motor protein that is responsible for vesicle locomotion along microtubules in eukaryotes. **See** *Force Effects on Molecular Motors*

S. Khan & M. Sheetz (1997) *Ann. Rev. Biochem.* **66**, 785.

R. A. Walker & M. P. Sheetz (1993) *Ann. Rev. Biochem.* **62**, 429.

J. R. McIntosh & M. E. Porter (1989) *J. Biol. Chem.* **264**, 6001.

KINETIC AMBIGUITY

Any condition of kinetic equivalence characterized by the absence of unique properties permitting one to choose among rival mechanisms. **See** *Kinetic Equivalence; Black Box*

KINETIC CONTROL OF PRODUCT COMPOSITION

When one or more molecular entity(ies) participates in two or more parallel and irreversible reactions in which different products are formed, the faster-forming product will accumulate by the reaction having the lowest activation energy. Thus, kinetically controlled processes are those whose proportion of products is governed by the relative rates of the competing reactions. If the reac-

tions are reversible and the process is stopped prior to the establishment of equilibrium, the overall process will also be kinetically controlled since the faster-formed product is still present in greatest abundance. **See** *Thermodynamic Control*

G. W. Klumpp (1982) *Reactivity in Organic Chemistry*, pp. 36-89, Wiley, New York.

KINETIC ELECTROLYTE EFFECT

The effect of added electrolyte and/or ionic strength on the rate constant(s) of a reaction in solution, exclusive of any role of that the electrolyte as a reactant or catalyst.

Recall from transition state theory that the rate of a reaction depends on k_o (the catalytic rate constant at infinite dilution in the given solvent), the activity of the reactants, and the activity of the activated complex. If one or more of the reactants is a charged species, then the activity coefficient of any ion can be expressed in terms of the Debye-Hückel theory. The latter treats the behavior of dilute solutions of ions in terms of electrical charge, the distance of closest approach of another ion, ionic strength, absolute temperature, as well as other constants that are characteristic of each solvent. If any other factor alters the effect of ionic strength on reaction rates, then one must look beyond Debye-Hückel theory for an appropriate treatment.

Ionic strength also has a great influence on protein structure, thus influencing enzyme-catalyzed rates above and beyond the effects on individual rate constants.

Although these effects are often collectively referred to as salt effects, IUPAC regards that term as too restrictive. If the effect observed is due solely to the influence of ionic strength on the activity coefficients of reactants and transition states, then the effect is referred to as a primary kinetic electrolyte effect or a primary salt effect. If the observed effect arises from the influence of ionic strength on pre-equilibrium concentrations of ionic species prior to any rate-determining step, then the effect is termed a secondary kinetic electrolyte effect or a secondary salt effect. An example of such a phenomenon would be the influence of ionic strength on the dissociation of weak acids and bases. **See** *Ionic Strength*

J. N. Brønsted (1922) *Z. Phys. Chem.* **102**, 169.

N. Bjerrum (1924) *Z. Phys. Chem.* **108**, 82.

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

KINETIC EQUIVALENCE

Any two reaction schemes are considered to be “kinetically equivalent” if they imply the same orders of reaction for all chemical species involved. For example, assume that the rate expression for a particular scheme for a chemical reaction is given by $d[P]/dt = v = k_1 k_3 [A][B]/(k_2 + k_3[B])$. For a second scheme, the expression is found to be $k_1' k_3' [A][B]/(k_2' + k_3'[B])$. Both expressions have the same general form and have the same dependence on reactant concentrations. Thus, the two schemes are kinetically equivalent. **See** *Kinetic Ambiguity*

KINETIC HALDANE RELATIONSHIPS

Haldane relationships based on the ratio of apparent rate constants for both the forward and reverse directions. Every enzyme mechanism has at least one kinetic Haldane relationship. For a Uni Uni mechanism¹, the kinetic Haldane relationship is: $K_{eq} = (V_{max,f}/K_{m,A})/(V_{max,r}/K_{m,P})$. **See** *Haldane Relationships; Thermodynamic Haldane Relationships*

¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 366.

KINETIC ISOTOPE EFFECTS

GENERAL COMMENTS. Isotopically substituted molecules having identical geometrical, optical, and conformational properties are called isotopic isomers (or isotopomers), and depending on the position of isotopic substitution, two isotopic isomers may differ in their reactivity (*i.e.*, their rate of reaction). Any observed rate difference is called a kinetic isotope effect, and the occurrence and magnitude of the isotope effect can provide valuable information and insight regarding reaction mechanisms. One can also employ isotope effect studies to ascertain whether the bond-breaking step is either a rate-limiting, or at least a rate-contributing, feature of catalysis. In this section, we discuss the essential properties of isotopic isomers that yield kinetic isotope effects, and we briefly analyze their utility in mechanistic studies of enzymic catalysis. The interested reader will obtain more advanced treatments of isotope effects elsewhere¹⁻⁴.

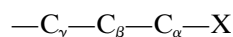
A kinetic isotope effect (or KIE) is usually written as a ratio (*e.g.*, v_H/v_D or k_H/k_D) to indicate the magnitude of the rate (or rate constant) obtained for one isotopomer (in this case H, or ¹H) divided by the observed rate or rate constant achieved with the heavier isotopically

labeled reactant (in this case D, or ²H). For simplicity, some kineticists use ^D*k* as the shorthand version of k_H/k_D . In enzyme-catalyzed kinetics, one must necessarily deal with the behavior of a multistep reaction scheme. For initial rate enzyme processes, one typically deals with collections of rate constants which appear in the form of the maximal velocity V_m (shortened to V) or the specificity constant V_m/K_m (shortened to V/K). Accordingly, enzyme kineticists will use ^D V and ^D V/K as an easy way to indicate the respective isotope effects $[(V_m)_H/(V_m)_D]$ and $[(V_m/K_m)_H/(V_m/K_m)_D]$, respectively.

TYPES OF ISOTOPE EFFECTS. Kinetic isotope effects provide information regarding the configuration of reactants as they approach the transition state of a reaction¹. These inferences are accessible, because the transition state, in large measure, determines the type and magnitude of observed changes in chemical reactivity. Early work on reaction mechanisms relied on the introduction of various substituents (*e.g.*, methyl, ethyl, propyl, *etc.*) near the reaction center, but such “homologues” often introduced confounding variables, such as differences in their solvation, acid/base chemistry, and conformation. Thus, on the basis of studies on structural homologues, one could not unambiguously assign differences in reaction rate to changes in the configuration of atoms within the transition state. Moreover, for enzymes, there is always the problem that substrate selectivity/specificity limits the extent to which one may introduce substituents into reactant’s structure without sacrificing the ability to bind at the active site. Isotopic substitution, by contrast, leads to more subtle and often far more predictable changes in active site geometry²⁻⁴.

When the reaction center (*i.e.*, the site of bond-breaking/making) is itself the site of isotopic substitution, any observable rate change is termed a primary isotope effect. Thus, substitution of deuterium or tritium for the transferred hydrogen atom of NADH can give rise to a primary kinetic isotope effect, and when observed, one can infer that hydrogen transfer is at least a rate-contributing step in a particular dehydrogenase reaction. Isotopic substitution outside the reaction center most often will have considerably less influence on reaction rate; such rate changes are termed secondary kinetic isotope effects. These isotope effects typically decrease in magnitude as the point of substitution lies farther from the

reaction center, and α , β , and γ are used to designate the positions of isotopic substitution:



For the purposes of this introductory treatment, we will confine our discussion to primary kinetic isotope effects as well as secondary α isotope effects.

PRIMARY ISOTOPE EFFECT. Without engaging a detailed statistical mechanical argument¹ about the origin of primary isotope effects, we can list the basic assumptions of the Bigeleisen-Mayer treatment of isotope effects: (a) the Born-Oppenheimer approximation holds, such that the electronic potential energy does not change upon isotopic substitution; (b) one can parse the partition functions into the independent contributions attributable to translational, vibrational, and rotational motions; (c) classical mechanics is sufficient to describe translational and rotational motion, and vibrational motion is harmonic; (d) isotopic substitution is without effect on κ , the transition-state transmission coefficient; and (e) no quantum mechanical tunneling occurs. These approximations are adequate to describe most kinetic isotope effects, with the notable exception of hydrogen transfer, for which tunneling can make a significant contribution.

Let us now examine an approximate method⁵ for comprehending their nature and magnitude. Consider the diagram in Fig. 1 where A—B and A—B' bonds of two

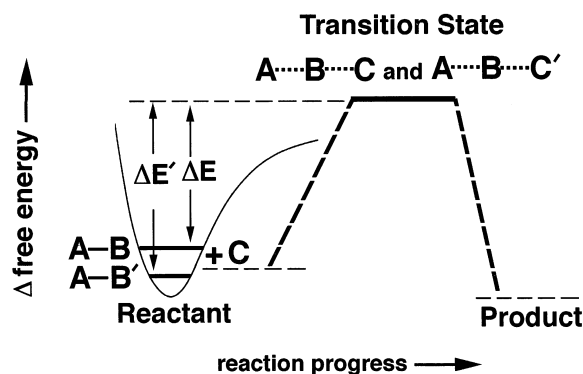


Figure 1. Reaction coordinate diagram for the reaction of A—B and A—B' with another agent, designated here as C. Note that the heavier isotopomer will lie at lower energy, as would be expected on the basis of its shorter equilibrium bond length and its consequentially lower zero point energy. Because the energy levels of the transition states for both reactions will be approximately the same, the slight difference in ground state zero point energy means that a greater fraction of A—B will reach the transition state, compared to A—B'.

isotopic isomers are cleaved. During reaction, the A—B and A—B' stretching frequencies differ only in the mass of the nucleus and are identical with respect to their essential electronic interactions.

During reaction, the A—B and A—B' stretching frequency is lost in the transition state, where B and B' are located at essentially identical positions. (In fact, vibrational frequencies other than the A—B and A—B' stretching mode may be likewise partly lost, but this represents what may be regarded as a minor refinement in most cases.) In the diagram above, the reader will recognize that there will always be a difference in the vibrational energies of the isotopic isomers A—B and A—B' reflecting the energy differences in the lowest lying vibrational level which has a finite zero-point energy (Z.P.E.). That a molecule has a finite Z.P.E. is a requirement of the Uncertainty Principle; any quantized harmonic oscillator will still experience unavoidable disturbances (hence, energy), even when that oscillator is dampened to its lowest energy state. The vibrational frequency of a bond stretching mode can be estimated by using an equation that accounts for the simple harmonic motion of a spring (**See Hooke's Law Spring**):

$$\nu_{\text{A-B}} = (1/2 \pi) [k/\text{mass}_{\text{reduced}}]^{1/2}$$

where k is the so-called force constant (in this case, a property of the electron density of the bond undergoing scission) and $\text{mass}_{\text{reduced}}$ is the reduced mass of the A—B molecule [in this case, $m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$]. We already indicated that isotopic substitution has no effect on the electron density of the bond, and for this reason, the value of k is the same for A—B and A—B'. Thus, for two isotopic isomers having essentially identical vibrational force constants, we can divide the equation for A—B above by its counterpart for A—B', and this yields:

$$\frac{\nu_{\text{A-B}}}{\nu_{\text{A-B'}}} = \left\{ \frac{m_{\text{B}}(m_{\text{A}} + m_{\text{B'}})}{m_{\text{B'}}(m_{\text{A}} + m_{\text{B}})} \right\}^{1/2} = M$$

By using this equation, one can estimate how the change in isotopic substitution can result in a small but experimentally significant change in the zero-point energy:

$$\Delta E = E_{\text{A-B}} - E_{\text{A-B'}} = 0.5 h \nu_{\text{A-B}} (1 - M)$$

where h is Planck's constant. The lighter isotopic isomer actually lies closer to the transition state by the amount given in this last equation.

This energy difference ΔE can be estimated reasonably accurately by using infrared absorption data to determine the magnitude of ν_{A-B} . If the A—B vibration is completely lost in going to the transition state during a chemical reaction, then the difference in zero-point energies for A—B and A—B' stretching will lead to a corresponding rate decrease for A—B' relative to A—B. This can be expressed as follows:

$$k_{A-B}/k_{A-B'} = \exp \{(-h\nu_{A-B}/2RT)[1 - M]\}$$

where R is the gas constant, h is Planck's constant, and T is the absolute temperature. For example, a C—H bond has a value of 3000 cm^{-1} for ν_{C-H} , whereas ν_{C-D} for a C—D bond is 3000 cm^{-1} . In this case, ΔE is about 1.2 kcal/mol, and if this difference is fully expressed in the primary kinetic isotope effect, k_H/k_D will be about seven at room temperature. In some cases, although the primary kinetic isotope probably occurs in the isotopically sensitive step within a reaction mechanism, the experimenter is left with little indication thereof, because the particular step has no effect on reaction rate (*i.e.*, it is faster than the rate-determining step). Nonetheless, one may use the above expression to estimate the following primary isotope effects, which are taken from Hine⁵:

k_{C-H}/k_{C-D}	8.2 (at 0°C)
k_{C-H}/k_{C-D}	6.9 (at 25°C)
k_{N-H}/k_{N-D}	9.2 (at 25°C)
k_{C-H}/k_{C-T}	16 (at 25°C)
$k_{12C-12C}/k_{12C-13C}$	1.03 (at 0°C)
$k_{12C-12C}/k_{12C-14C}$	1.09 (at 0°C)

These estimations engage the assumption that any additional effects due to bending vibrations will tend to contribute relatively weakly to the isotope effect. In fact, however, the above estimates are probably only “ball park” estimates, in that coupling of molecular vibrations of multiatomic molecules is ignored¹. Another key point is that a covalent bond need not be completely broken in the transition state, and two isotopic isomers may behave slightly differently in the transition state. Jencks⁶ discusses this matter as well as the problem of nonlinear transition states, a condition that takes into account the fact that bending frequencies often lessen the developed magnitude of isotope effects.

Examination of one real-life case may benefit the reader's understanding. Strittmatter⁷ studied the primary kinetic isotope effects arising in the NADH-dependent cytochrome b_5 reductase (EC 1.6.2.2). The oxidation of NADH and subsequent reduction of cytochrome b_5 is facilitated by the enzyme-bound FAD group, and the kinetics of the direct transfer of a hydrogen from the A-face (or *pro-R*) of NADH to the flavin can be monitored by the loss of the 340 nm absorbance of the NADH's dihydropyridine ring. Using deuterated isotopic isomers of NADH and several related compounds, Strittmatter⁷ obtained the primary kinetic isotope effect data compiled in the table below.

Rates of Oxidation of NADH and Several Pyridine Ring Analogues by Cytochrome b_5 Reductase^a

Nucleotide	Observed Turnover Number	
	(mol product/mol enzyme/min)	k_H/k_D
NADH	29,600	—
NADD (<i>pro-R</i> Deuterium)	8,100	3.66
NADD (<i>pro-S</i> Deuterium)	29,700	1.00
AcPyADH ^b	3,560	—
AcPyADD (<i>pro-R</i> Deuterium)	340	10.45
AcPyADD (<i>pro-S</i> Deuterium)	3,530	0.99
PyAlADH ^c	414	—
PyAlADD (<i>pro-R</i> Deuterium)	48	8.7

^aThese kinetic experiments were carried out in microcuvettes at 25°C, which contained 25 nmol reduced pyridine nucleotide, 50 nmol potassium ferricyanide (added at zero time as the electron acceptor), 1 nmol reductase in 0.2 mL 0.1 M Tris-acetate (pH 8.1) containing 1 mM EDTA.

^bAcPyADH = 3-Acetylpyridine analogue of NADH.

^cPyAlADH = 3-Pyridinealdehyde analogue of NADH.

The values of k_H/k_D were quite large for coenzymes deuterated in the *pro-R* position, and the authors carefully conducted control experiments with coenzymes labeled at the *pro-S* position. The experiments clearly demonstrated that the overall transfer of electrons to ferricyanide involves NAD—H_A bond-breaking in the rate-limiting step of the catalyzed reaction. Moreover, the isotope effect is greatest with alternative substrates, suggesting that rate constants defining the reaction with NADD/H may mask the full extent of the kinetic isotope effect. The reader will also note that substitution at the *pro-S* position was without effect on the rates of hydrogen transfer. Had there been such an effect, we would call it a secondary kinetic isotope effect, because the isotope is not at an atom directly undergoing bond scission.

Classical mechanical treatments of primary kinetic isotope effects require the reactants to surmount an energy barrier in order for them to form products. By contrast, quantum mechanics allows for the phenomenon of tunneling, wherein reactants of insufficient energy to pass over the top of an energy barrier can still be converted to product, provided that the width of the barrier is small. Only particles having de Broglie wavelengths ($\lambda = h/mv$) that are on the same scale as the barrier thickness can “penetrate” the barrier by quantum mechanical tunneling. Protons and hydride ions have 1–2 Å de Broglie wavelengths, and several enzymic reactions have been demonstrated to exhibit tunneling. Tell-tale features of tunneling are isotope effects that exceed the magnitude predicted from barrier heights obtained by applying the Bigeleisen-Mayer treatment as well as the temperature dependence of the observed isotope effect. Tunneling is favored by (a) high, thin potential energy barriers; (b) low temperature; and (c) steric hindrance which appears to influence solvent exclusion.

SECONDARY ISOTOPE EFFECTS. Changes in reaction may also result from isotopic substitutions at positions that are immediately adjacent to the reaction center (*i.e.*, the bond broken/made in the chemical reaction under investigation). We deal here only with so-called secondary α -isotope effects, and we will limit the scope further by considering only α deuterium and α tritium isotope effects on carbon. Isotopic substitution by heavier nuclides will also give rise to α isotope effects, but they are quite small. The magnitudes of the α isotope effects for C— $^1\text{H}_\alpha$ compared to C— $^2\text{H}_\alpha$ as well as for C— $^1\text{H}_\alpha$ compared to C— $^3\text{H}_\alpha$ are also relatively small, frequently necessitating the use of special techniques.

One of the principal applications of secondary isotope effects is to distinguish $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$ reactions. The α kinetic isotope effect for $\text{S}_\text{N}1$ reactions is typically significantly larger than that observed in $\text{S}_\text{N}2$ reactions. Such behavior reflects changes in the geometry of the carbon atom undergoing nucleophilic substitution (Fig. 2). If a nucleophilic substitution reaction is first-order ($\text{S}_\text{N}1$), the carbon atom changes from a tetrahedral sp^3 arrangement to a flattened sp^2 configuration prior to nucleophile attack.

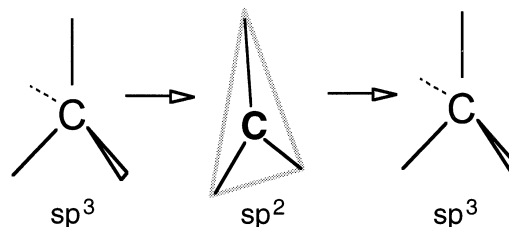


Figure 2. Rearrangement in molecular orbital hybridization and geometry of the carbon atom undergoing a first-order nucleophilic substitution (or $\text{S}_\text{N}1$) reaction.

In a second-order ($\text{S}_\text{N}2$) nucleophilic substitution reaction, the carbon atom simultaneously experiences the effects of the attacking nucleophile (N) and leaving group or exiphile (E). These mutual effects (shown in Fig. 3) serve to diminish energy differences between each isotopic substrate and its corresponding transition state.

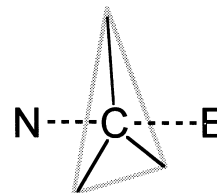


Figure 3. Structural rearrangements during a second-order nucleophilic substitution or $\text{S}_\text{N}2$ reaction.

Why does an $\text{S}_\text{N}1$ reaction have a significant secondary isotope effect? Detailed analysis^{1,2,5} of secondary kinetic isotope effects suggests that it is difficult to describe their origin explicitly in quantitative terms. One qualitative approach is to separate the electronic, vibrational, rotational, and translational energies of a molecule, and then to deal with each to explore any differences, say between C— $^1\text{H}_\alpha$ and C— $^2\text{H}_\alpha$. Considerable theoretical treatment has led to the conclusion that the potential energy surfaces of hydrogen- and deuterium-containing isotopic isomers are essentially identical⁸, suggesting that the secondary isotope effect results from different energies of nuclear motions (*i.e.*, vibrations, rotations, translations) and not from electronic differences. Of those involving nuclear motion, the vibrational energy dominates most, and the deviation from the equilibrium bond length (r_e) of the C— H_α and C— D_α can be taken as the dominant factor in the isotope effect. Because D is heavier, its mean-square amplitude will be closer to r_e which is the potential energy minimum. When steric repulsions between H_α (or D_α) and the other groups are relieved, then the H_α isotopic isomer will undergo a more favorable transition by dropping through a greater change of po-

tential energy than will its D_α isotopic isomer. The phenomenon can be qualitatively described by stating that a deuteration decreases the steric repulsion, and thus the formation of a flat (sp^2 -like) geometry from tetrahedral (or sp^3) bonding is less favorable for the heavier isotopic isomer⁹. Another way of dealing with the origin of α -KIE's is to view the primary contributing factors as the out-of-plane vibrations in the ground state of the reactant. The α effect is dependent on the reduction in the size of the hydrogen-carbon-exophile bending force constant in passing from reactant to its activated intermediate state (*i.e.*, $f_{\text{HCX}} \rightarrow f_{\text{HCX}}^*$), where as usual we use “ $\ddagger$ ” to denote the transition state. An independent method of correlating the $\text{H}-\text{C}-\text{X}$ bending constant with geometry is needed to permit a fuller analysis of geometry inferred on the basis of an observed kinetic α isotope effect.

Let us consider the classical secondary kinetic isotope experiments¹⁰ on lysozyme-catalyzed hydrolysis of glycosides. Earlier X-ray work¹¹ indicated that the enzyme

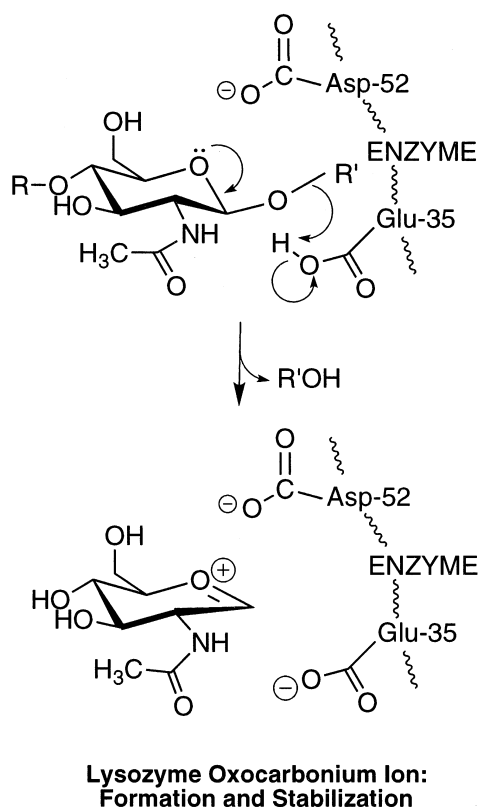


Figure 4. Essential features of the hydrolytic reaction catalyzed by hen egg white lysozyme. Notice that two nearby carboxyl groups should be able to stabilize the oppositely charged oxocarbenium ion intermediate.

binds oligosaccharides, such that the enzyme interacts with five monosaccharide units (designated alphabetically as subsites A, B, C, D, and E) stretching across the enzyme's adsorption pocket. Cleavage occurs between sites D and E, and the enzyme appeared to distort substrate analogues into a half-chair conformation, suggestive of a flattened oxocarbenium ion intermediate as illustrated in Fig. 4.

Chemical studies also demonstrated that lysozyme hydrolysis of the glycosidic bond proceeds with retention of configuration to at least 99.7%. Retention of configuration provides strong evidence against an in-line S_N2 -type mechanism, but the authors were particularly interested in determining the extent to which a secondary kinetic isotope effect is expressed during lysozyme catalysis. Using a phenyl glycoside with the following structure (Fig. 5),

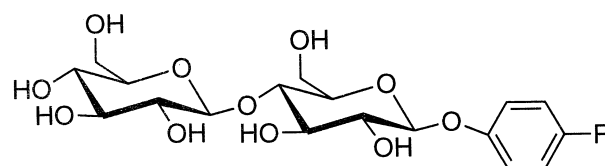


Figure 5. Phenylglycoside substrate substituted at C1 with either ^1H and ^2H .

the effect of deuteration at the C1 carbon of the glucosyl residue on the rate of phenol formation was measured. Actually, these investigators used either [^3H -phenyl]- and [^{14}C -phenyl]substrate to prepare the respective C1- ^1H and C1- ^2H isotopic isomers. The two isotopic isomers could then be combined with each other prior to enzyme addition, and the isotope effect ($k_{\text{H}}/k_{\text{D}}$) could be measured by determining changes in the $^3\text{H}/^{14}\text{C}$ ratio for the isolated phenol products¹². This strategy permits one to attain high sensitivity and precision, something that is difficult to achieve with separate kinetic runs using isotopic isomers separately. Their experimental findings are given below. The first tabulated value agrees with the oxocarbenium ion mechanism for acid-catalyzed glycoside hydrolysis¹³. The second value for base-catalyzed hydrolysis likewise agrees with that expected for an in-line S_N2 -type mechanism. Finally, the $k_{\text{H}}/k_{\text{D}}$ of 1.11 at all three pH values fits best with the idea of an oxocarbenium mechanism, and the observed retention of stereochemical configuration at C1 suggests that the reaction involves restricted entry of the incoming nucleophile

(i.e., water). If a fully developed oxocarbenium ion in 2M HCl yields a k_H/k_D of 1.13, then the value of 1.11 is really quite close to this limit. However, as noted below, the magnitude of an observed kinetic isotope effect may be observed by other kinetic considerations.

Observed Isotope Effects for Hydrolysis of Phenyl-4-O-(2-acetamido-2-deoxy- β -D-glucopyranosyl)- β -D-glucopyranoside^a

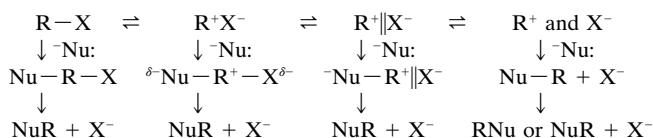
Conditions	Observed k_H/k_D
Acid (2 M HCl)	1.13
Base (3 M NaOCH ₃)	1.03
Lysozyme (pH 5.5)	1.11
Lysozyme (pH 3.1)	1.11
Lysozyme (pH 8.3)	1.11

^aFrom Dahlquist et al.¹⁰ with permission of the authors and the American Chemical Society. Reaction was permitted to proceed to 2–5% completion and was then adjusted with acid or base to pH 5.6. The mixture was extracted with ether, and the extract was washed twice with saturated NaCl, then dried with anhydrous sodium sulfate, and added to scintillation fluid for counting of radiolabeled phenol.

Why do secondary isotope effects in S_N2 reactions vary from about 0.95 to 1.06 upon deuterium substitution in the α position? The most basic explanation for this behavior relates to the fact that the repulsion lost as the leaving group departs is partly balanced by the repulsions developed by the incoming nucleophile. The presence of both nucleophile and exiphile within the transition state maintains the zero point energy difference throughout the course of the S_N2 reaction. The magnitude of k_H/k_D depends upon differences between the corresponding force constants (specifically $f_{HCX} \rightarrow f_{HCX}^\ddagger$ and $f_{HCX} \rightarrow f_{HCN}^\ddagger$, where the subscripted X and N indicate the exiphile and nucleophile, respectively). This shows that bonding by both the nucleophile and exiphile to the carbon atom undergoing substitution is important in S_N2 reactions. The degree to which the entry and removal of the nucleophile and exiphile are well “choreographed” will affect the magnitude of the observed secondary isotope effects, which tend to be 1.00 ± 0.06 . Thus, if an enzyme-catalyzed reaction yields a k_H/k_D isotope effect that is less than 1.07, the observed kinetic isotope effect is of dubious value in diagnosing the actual type of mechanism.

Are S_N1 and S_N2 mechanisms the whole story? For solution-phase nucleophilic substitution reactions, S_N1 and

S_N2 mechanisms are limiting cases that lie along a continuum of possible interactions¹⁴:



where $[R-X]$ is the covalently bonded form, $[R^+X^-]$ is the tight ion pair, $[R^+\|X^-]$ is the loosely associated ion pair, $[R^+ + X^-]$ represents the fully dissociated ions, Nu: is the nucleophile, and RN and NR are the two optical isomers formed by attack on either face of the electrophilic center. These intermediate types of mechanisms obviously should have geometries that lie on a continuum; so also must the secondary kinetic isotope effects associated with these hybrid mechanisms. Other factors (e.g., nucleophilic strength, solvent nucleophilicity, other solvation effects, the detailed nature of the leaving group, and temperature) can greatly influence the mechanistic course of a reaction. Even examples from physical organic chemistry (where reactant adsorption and release from an enzyme active site is not a factor) often fail to allow one to discriminate between these possible mechanisms. In this sense, it is not difficult to imagine that isotope effects often serve only as a guide for limiting the number of most likely mechanisms. In fact, the formation of tight-ion pairs, or even covalent intermediates, can introduce further complexity in enzymic reactions. Taking the case of lysozyme as an example, there still remains the possibility for transient formation of an acylal intermediate; this species could form in a post-rate-determining step, wherein the oxocarbenium ion and the active site carboxylate collapse to form a covalent bond. Such a pathway would lead to two Walden inversions: one forming the acylal, and a second inversion during hydrolysis of the acylal. This mechanistic route would therefore give the same apparent overall retention of stereochemical configuration predicted for the S_N1 case.

INFLUENCE OF OTHER STEPS ON THE MAGNITUDE OF OBSERVED ISOTOPE EFFECTS. As noted earlier, nonenzymatic reaction mechanisms do not involve those complexities imposed by substrate binding order, rates of substrate binding/release, as well as conformational changes that attend enzyme catalysis. As a result, the opportunity for detecting isotope effects is

cally sensitive step is rate-limiting), we get

$$(V_H/V_D) \approx k_{3H}/k_{3D}$$

(See Footnote¹⁶ for corresponding V/K expression.) We can see that in this case the isotope effect will be fully expressed in terms of (V_H/V_D) when the isotopically sensitive step is the rate-limiting step. One can now ask what happens if k_{5H} is not much larger than k_{3H} ? Let us assess the hypothetical case where the primary isotope is assumed to be 10:

$$(1) \text{ If } k_{3H}/k_{5H} = 1, \text{ then } (V_H/V_D) = \{10 + 1\}/(1 + 1) = 5.5$$

$$(2) \text{ If } k_{3H}/k_{5H} = 10, \text{ then } (V_H/V_D) = \{10 + 10\}/(1 + 10) = 1.8$$

These simple calculations indicate how the other rate constants in a kinetic equation collectively act to greatly suppress the magnitude of the observed primary isotope effect¹⁷.

With this background, let us examine how Northrop¹⁵ was able to develop a way to extract the true (or as we more commonly say, *intrinsic*) isotope effect. Recognizing that an underdetermined system can often be solved if an additional quantitative relation is considered, Northrop proposed to measure two isotope effects, namely $(V/K)_H/(V/K)_D$ and $(V/K)_H/(V/K)_T$ and to take advantage of the following relationship:

$$(k_{3H}/k_{3T}) = (k_{3H}/k_{3D})^{1.442}$$

This exponential relationship is called the Swain relationship in honor of the senior author of a report¹⁸ on how one can calculate the magnitude of primary isotope effects. Use of this relationship allowed Northrop to obtain the following expression¹⁹:

$$\frac{[(V/K)_H/(V/K)_D] - 1}{[(V/K)_H/(V/K)_T] - 1} = \frac{[(k_{3H}/k_{3D}) - 1]}{[(k_{3H}/k_{3D})^{1.442} - 1]}$$

OTHER EXAMPLES OF KINETIC ISOTOPE EFFECTS. The power of kinetic isotope effects in enzymology is well illustrated in the work of Rose²⁰ and Knowles²¹ dealing with hydrogen effects in proton transfer to and from carbon. Abstraction of a proton from a tetrahedral carbon is a fundamental step in many enzyme-catalyzed reactions. Intramolecular proton transfer as well as partial loss (wash-out) migrating protons have provided important clues in mechanistic investigations. Enol and enediolate formation constitute several

of the best examples of how isotope effects have become a major tool for unraveling subtle properties of enzyme catalysis and control. The prototypical enzyme system is the triose-phosphate isomerase reaction where a *cis*-enediol intermediate is formed upon removal of the *pro-R* proton from C-1 of dihydroxyacetone phosphate or the lone proton of C-2 in D-glyceraldehyde 3-phosphate. This proton abstraction reaction is facilitated by the γ -carboxylate of E165 in the isomerase^{22,23}. Rose's group demonstrated that a proton must also be transferred between the oxygen of C-1 and the oxygen of C-2 to achieve isomerization of dihydroxyacetone to D-glyceraldehyde 3-phosphate. Evidence for these important steps in the isomerase mechanism were provided by a combination of tritium isotope-exchange experiments and kinetic isotope-effect measurements²⁴.

Isotope effects have also been applied extensively to studies of $\text{NAD}^+/\text{NADP}^+$ -linked dehydrogenases. We typically treat these enzymes as systems whose catalytic rates are limited by product release. Nonetheless, Palm²⁵ clearly demonstrated a primary tritium kinetic isotope effect on lactate dehydrogenase catalysis, a finding that indicated that the hydride transfer step is rate-contributing. Plapp's laboratory²⁶ later demonstrated that liver alcohol dehydrogenase has an intrinsic k_H/k_D isotope effect of 5.2 with ethanol and an intrinsic k_H/k_D isotope effect of 3-6-4.3 with benzyl alcohol. Moreover, Klinman²⁷ reported the following intrinsic isotope effects in the reduction of *p*-substituted benzaldehydes by yeast alcohol dehydrogenase: k_H/k_D for *p*-Br-benzaldehyde = 3.5; k_H/k_D for *p*-Cl-benzaldehyde = 3.3; k_H/k_D for *p*-H-benzaldehyde = 3.0; k_H/k_D for *p*-CH₃-benzaldehyde = 5.4; and k_H/k_D for *p*-CH₃O-benzaldehyde = 3.4.

Bell²⁸ described how quantum mechanical tunneling can be an important factor in the rate of chemical reactions, and Bahnson and Klinman²⁹ have extended these considerations to define how isotope effects can be utilized to infer important features of hydrogen transfer reactions. They point out that the tunneling probability depends on the degree of solvent reorganization during catalysis³⁰, thermodynamic relationships between substrates and products^{24,31}, and the height and width of the energy barrier along the reaction coordinate³². Bahnson and Klinman²⁹ describe how the temperature dependence of enzyme kinetic isotope effects is an important criterion

for demonstrating tunneling. They provide examples from studies on alcohol dehydrogenase, bovine serum amine oxidase, and monamine oxidase B.

Although there are literally hundreds of reports (see reviews^{3,4,33-35}) dealing with kinetic isotope effects, the recent study of Scheuring & Schramm³⁶ on the ADP-ribosylase activity of *Bordetella pertussis* toxin provides an excellent example of the combined power of primary and secondary kinetic isotope effect experiments as well as solvent isotope effects. For the ribosyl-acceptor substrate, these investigators used a synthetic cysteine-containing peptide corresponding to the last twenty amino acids of the α -subunit of the G-protein G_i3. A family of kinetic isotope effects (KIE) were evaluated by using ³H-, ¹⁴C-, and ¹⁵N-labeled NAD⁺ as substrates. Primary kinetic isotope effects were 1.050 ± 0.006 for [1'N—¹⁴C] and 1.021 ± 0.002 for [1N—¹⁵N]. The double primary [1'N—¹⁴C, 1N—¹⁵N] effect was found to be 1.064 ± 0.002 . Secondary kinetic isotope effects were also evaluated: [1'N—³H]secondary-KIE = 1.208 ± 0.014 ; [2'N—³H]secondary-KIE = 1.104 ± 0.010 ; [4'N—³H]secondary-KIE = 0.989 ± 0.001 ; and [5'N—³H]secondary-KIE = 1.014 ± 0.002 . The measured commitment factor of 0.01, based on isotope trapping experiments, indicated that the observed isotope effects are intrinsic. The observed inverse solvent D₂O kinetic isotope effect suggested possible deprotonation of the attacking cysteinyl residue prior to reaching the transition state. The transition state structure was determined by a normal mode bond vibrational analysis. The transition state is characterized by a nicotinamide leaving group bond order of 0.14 (corresponding to a bond length of 2.06 Å). Their analysis also suggests a bond order of 0.11 for the incoming thiolate nucleophile, corresponding to a distance of 2.47 Å. The ribose ring appears to have strong oxocarbenium ion character. By comparing the likely transition states for the ADP-ribosylase activity and the inherent NAD⁺ hydrolase, Scheuring & Schramm³⁶ inferred that the cysteinyl sulfur is likely to be a more nucleophilic in the transferase reaction than is H₂O in the hydrolytic reaction. **See also** Born-Oppenheimer Approximation; Equilibrium Isotope Effect; Hydrogen Tunneling; Intramolecular Kinetic Isotope Effect; Inverse Kinetic Isotope Effect; Quantum Mechanical Tunneling; Swain Relationship; Enzyme Energetics (The Case of Proline Racemase); Isotopic Perturbation

- ¹L. Melander (1960) *Isotope Effects on Reaction Rates*, p.181, Raven Press, New York.
- ²E. K. Thornton & E. R. Thornton (1978) in *Transition States of Biochemical Processes* (R. D. Gandour & R. L. Schowen, eds.), pp. 3-76, Plenum Press, New York.
- ³W. W. Cleland, M. H. O'Leary & D. B. Northrop, eds. (1977) *Isotope Effects on Enzyme-Catalyzed Reactions*, University Park Press, Baltimore.
- ⁴R. D. Gandour & R. L. Schowen, eds. (1978) *Transition States of Biochemical Processes*, Plenum Press, New York.
- ⁵J. Hine (1962) *Physical Organic Chemistry*, McGraw-Hill, New York.
- ⁶W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York]
- ⁷P. Strittmatter (1964) *J. Biol. Chem.* **240**, 4481.
- ⁸This is sometimes confused since C—H_a and C—H_b have slightly different dipole moments. As noted by Thornton & Thornton², the different dipoles arise because of different average geometries due to the stretching vibration. Thus, the dipole moment is not strictly an electronic property.
- ⁹A corollary of this argument is that a change from *sp*² to *sp*³ hybridization should result in an inverse secondary isotope effect (i.e., $k_H/k_D < 1$).
- ¹⁰F. W. Dahlquist, T. Rand-Meir & M. A. Raftery (1969) *Biochemistry* **8**, 4214.
- ¹¹D. C. Phillips (1967) *Sci. Am.* **215** (Nov.), 78; (1964) *Proc. Natl. Acad. Sci. U.S.A.* **57**, 484.
- ¹²Dahlquist et al.¹⁰ also provided a useful appendix for converting radioactivity counting data into observed isotope effects.
- ¹³C. A. Bunton (1969) *Nucleophilic Substitution at the Saturated Carbon Atom*, McGraw-Hill, New York.
- ¹⁴S. Winstein, E. Clippinger, A. H. Fainberg, R. Heck & G. C. Robinson (1956) *J. Amer. Chem. Soc.* **78**, 328.
- ¹⁵D. B. Northrop (1977) *Biochemistry* **14**, 2644.
- ¹⁶We can also rewrite the expression for V/K (from the list above) in terms of k_{3H} and k_{3D} to obtain:

$$(V/K)_H = k_{1H}k_{3H}[E_{\text{total}}]/(k_{2H} + k_{3H})$$

$$(V/K)_D = k_{1H}k_{3D}[E_{\text{total}}]/(k_{2H} + k_{3D})$$

Upon dividing the first equation by the second, we arrive at:

$$(V/K)_H/(V/K)_D = \{k_{3H}/(k_{2H} + k_{3H})\}/\{k_{3D}/(k_{2H} + k_{3D})\}$$

which upon rearrangement becomes

$$[(V/K)_H/(V/K)_D] - 1 = \{(k_{3H}/k_{3D}) - 1\}/\{(k_{3H}/k_{2H}) + 1\}$$

If $k_{3H} \ll k_{2H}$, then we get the desired final expression:

$$[(V/K)_H/(V/K)_D] - 1 = \{(k_{3H}/k_{3D}) - 1\}$$

- ¹⁷The situation would, of course, be far worse for secondary isotope effects. For example, if we were to take 1.22 as the limiting 2° isotope effect for a S_N1 substitution reaction and 1.06 as the largest secondary isotope effect for a S_N2 substitution reaction, then we could ask what value of k_{3H}/k_{4H} would yield a value of 1.06 (i.e., lower k_{3H}/k_{3D} to the point where one could not distinguish the S_N1 and S_N2 cases.) In this case, $1.06 = \{1.22 + k_{3H}/k_{4H}\}/(1 + k_{3H}/k_{4H})$, or $k_{3H}/k_{4H} = 2.66$. The result reminds us that secondary isotope effects are exquisitely sensitive to the relative magnitudes of rate constants.
- ¹⁸C. G. Swain, E. C. Stivers, J. F. Reuwer, Jr., & L. J. Schaad (1958) *J. Am. Chem. Soc.* **80**, 5885.
- ¹⁹This treatment is based on the following assumptions¹⁸: (a) all differences, except for zero point vibrational energy are negligible; (b) the hydrogen isotope is attached to a heavy polyatomic molecule, such that isotopic substitution affects only the vibrations of

the bond holding the hydrogen; (c) the bond may be treated as a harmonic oscillator; and (d) no tunneling through the potential energy barrier occurs. This limits the value of k_H/k_T and k_H/k_D to mass effects only, such that

$$\frac{k_H}{k_T} = \left(\frac{k_H}{k_T} \right)^{\left[\frac{1/\sqrt{m_H} - 1/\sqrt{m_D}}{1/\sqrt{m_H} - 1/\sqrt{m_T}} \right]} = \left(\frac{k_H}{k_T} \right)^{1.442}$$

where m_H is 1.0081, m_D is 2.0147, and m_T is 3.0170. Although beyond the scope of this description, some dispute exists regarding the best value for the exponent, which may range from 1.33 to 1.58. In this respect, use of four significant figures may not be justified.

- ²⁰I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.
²¹J. R. Knowles (1980) *Ann. Rev. Biochem.* **49**, 877.
²²S. G. Waley, J. C. Mill, I. A. Rose & E. L. O'Connell (1970) *Nature* **227**, 181.
²³E. A. Komives, L. CX. Chang, E. Lolis, R. F. Tilton, G. A. Petsko & J. R. Knowles (1991) *Biochemistry* **30**, 3011.
²⁴W. J. Albery & J. R. Knowles (1976) *Biochemistry* **15**, 5631.
²⁵D. Palm (1968) *Eur. J. Biochem.* **5**, 270.
²⁶B. V. Plapp (1970) *J. Biol. Chem.* **245**, 1725; (1973) *J. Biol. Chem.* **248**, 3470; *Meth. Enzymol.* **249**, 91.
²⁷J. P. Klinman (1977) in *Isotope Effects on Enzyme-Catalyzed Reactions* (W. W. Cleland, M. H. O'Leary & D. B. Northrop, eds.), p. 176, Univ. Park Press, Baltimore.
²⁸R. P. Bell (1980) *The Tunnel Effect in Chemistry*, Chapman & Hall, New York.
²⁹B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.
³⁰P. A. Bartlett & C. K. Marlowe (1987) *Biochemistry* **26**, 8553.
³¹K. P. Nambiar, D. M. Stauffer, P. A. Kolodziej & S. A. Benner (1983) *J. Amer. Chem. Soc.* **105**, 5886.
³²J. Rodgers, D. A. Femec & R. L. Schowen (1982) *J. Amer. Chem. Soc.* **104**, 3263.
³³D. L. Purich, ed. (1980) *Meth. Enzymol.* vol. **64**; (1982) *Meth. Enzymol.* vol. **87**; and (1995) *Meth. Enzymol.* vol. **249**, Academic Press, New York.
³⁴P. F. Cook, ed. (1991) *Enzyme Mechanism from Isotope Effects*, CRC Press, Boca Raton, Florida.
³⁵D. Devault (1984) *Quantum Mechanical Tunneling in Biological Processes*, 2nd ed., Cambridge University Press, Cambridge.
³⁶J. Scheuring & V. I. Schramm (1997) *Biochemistry* **36**, 8215.

KINETIC PARAMETERS

1. The K_m , $V_{\max}$ (or, k_{cat}), K_{is} , K_{ii} , etc., values of an enzyme-catalyzed reaction. 2. The value(s) of the individual rate constant(s) of a reaction or process.

KINETIC RESOLUTION

Partial or complete separation of enantiomers within a racemic mixture as a result of unequal rates of reaction with another agent. The latter reagent, catalyst, solvent, or micelle must itself exhibit chirality, resulting in its stereoselective or stereospecific action on the racemic compound.

KING-ALTMAN METHOD

A method to assist the investigator in the derivation of enzyme-rate expressions¹. This method is schematic in nature, involving the drawing of all possible patterns of a geometric figure minus a loop. It is a very convenient

method for straightforward enzyme schemes. However, for reactions which contain three or more substrates or products, or a number of abortive complexes, or isomerization steps, etc., other methods are available that are easier to follow and aren't as prone to mistakes (e.g., it is a common mistake in the King-Altman method to miss one or more of the geometrical forms with complex mechanisms)². See *Enzyme Rate Equations* (2. Derivation of Initial Velocity Equations); *Fromm's Method for Deriving Enzyme Rate Equations*; *Net Rate Constant Method* (applies only to unbranched mechanisms)

¹E. L. King & C. Altman (1956) *J. Chem. Phys.* **60**, 1375.

²C. Y. Huang (1979) *Meth. Enzymol.* **63**, 54.

KINSIM/FITSIM

A computer software package for numerical integration of rate equations, first developed for use on VAX computers by Barshop, Wrenn and Frieden¹ under the name KINSIM to utilize kinetic information acquired during the full time course of a reaction to analyze the properties of candidate mechanisms. The second program, designated FITSIM, was developed by Zimmerle and Frieden² to carry out statistical fitting of rate data to kinetic simulations. These programs are now supported on the following platforms³: VMS, Silicon Graphics Iris (IRIX), IBM PC (in MS-DOS, Windows, and OS/2 versions), and Apple Macintosh.

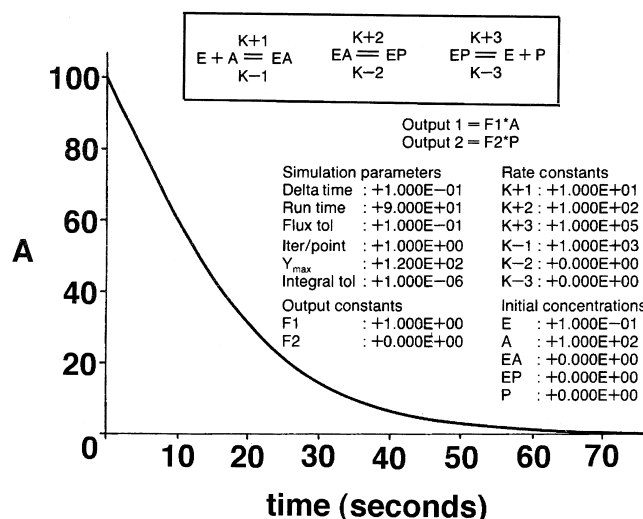


Figure 1. Uni Uni enzyme reaction scheme along with rate constants and a typical kinetic simulation.

Frieden^{4,5} has presented an overview of the KINSIM program, and a typical printed display for the two-intermediate Uni Uni enzyme kinetic mechanism is shown in Fig. 1.

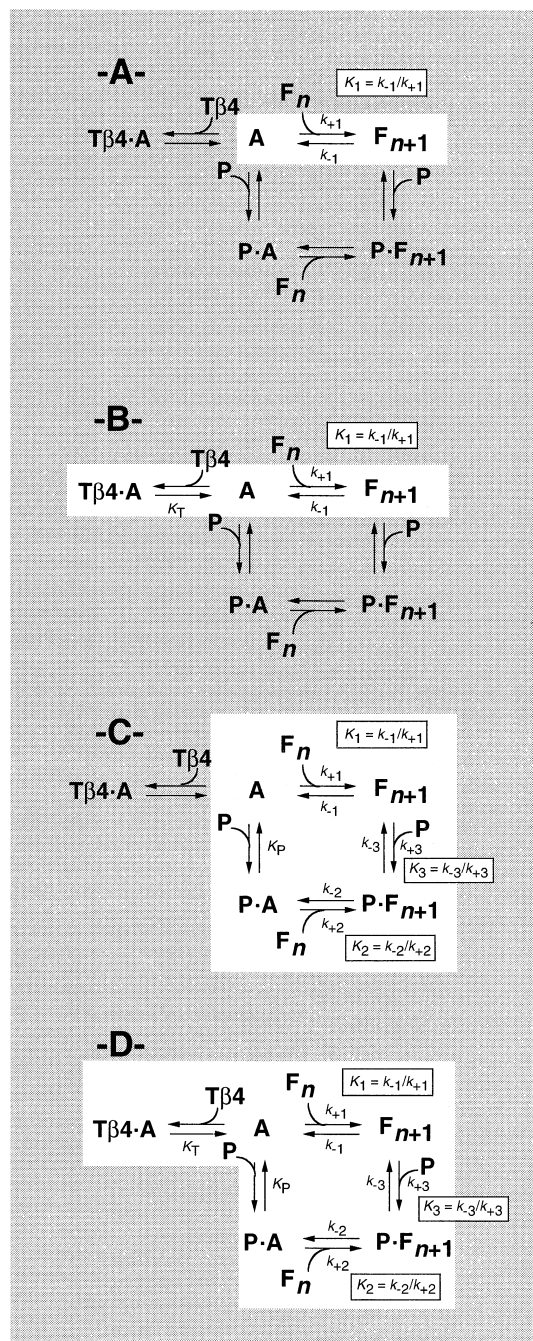


Figure 2. A diagrammatic representation of those segments of an overall reaction mechanism for barbed-end actin polymerization in the absence and presence of profilin and/or thymosin- $\beta 4$. For clarity the overall scheme is indicated. Experimental data were obtained for individual segments (indicated in black type), and those parts of the scheme that were not evaluated in a particular experiment are shown in gray.

Kang et al.⁶ recently used the KINSIM/FITSIM software to model barbed-end actin polymerization (*See Actin Assembly Kinetics*) in the absence and presence of profilin and/or thymosin- $\beta 4$. This data analysis protocol permitted them to derive rate constants from a series of

experiments where the rates and extents of polymerization were first determined by right-angle light scattering. Fig. 2 illustrates the segments of the overall scheme investigated in any single set of experiments, and KINSIM/FITSIM was used to achieve global data analysis on a series of runs at various protein concentrations (see one illustrative set of polymerization runs in Fig. 3).

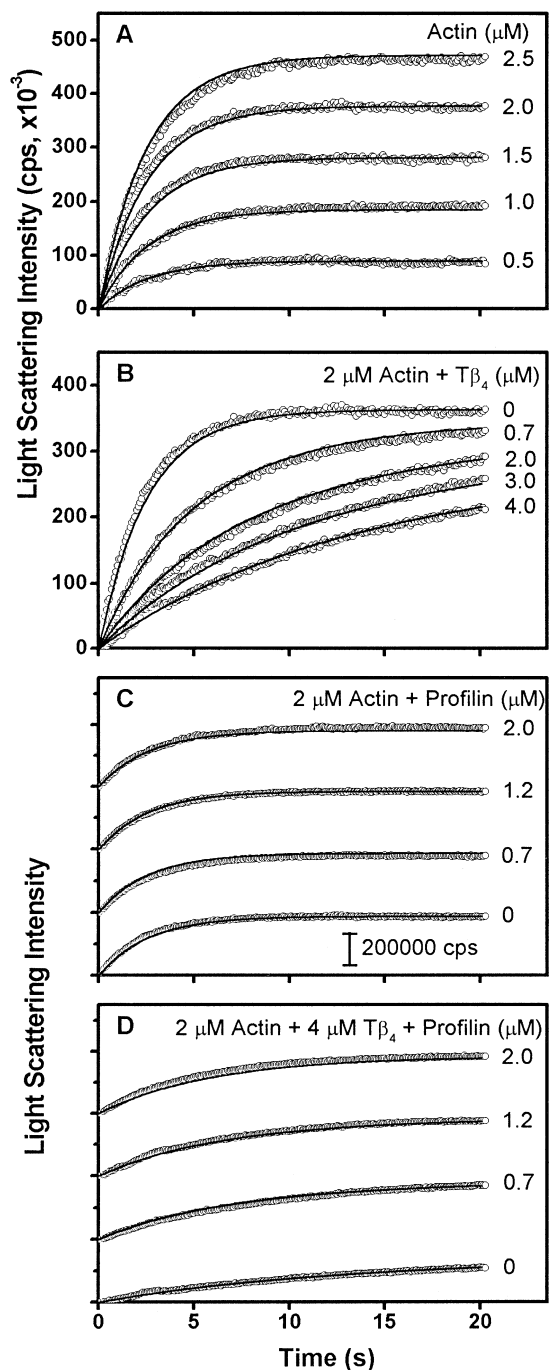


Figure 3. KINSIM/FITSIM applied to analysis of actin polymerization, showing actual light scattering data points and kinetic simulations (solid lines).

These experiments indicated (a) that profilin·actin·ATP can participate in actin filament, (b) that the derived rate and equilibrium constants define a model which satisfies a closed thermodynamic cycle, and (c) that contrary to earlier suggestions, there is no absolutely need to invoke any special property of profilin in promoting irreversible ATP hydrolysis during actin polymerization.

¹B. A. Barshop, R. F. Wrenn & C. Frieden (1983) *Anal. Biochem.* **130**, 134.

²C. T. Zimmerle & C. Frieden (1989) *Biochem. J.* **258**, 381.

³Those interested in obtaining information about the KINSIM/FIT-SIM software (KFSIM40.EXE), should consult the kinsim archive over the internet at: "http://wuarchive.wustl.edu."

⁴C. Frieden (1993) *Trends in Biochem. Sci.* **18**, 58.

⁵C. Frieden (1997) *Trends in Biochem. Sci.* **22**, 317.

⁶F. Kang, F.S. Southwick & D. L. Purich, unpublished findings.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Data analysis flow chart, **240**, 314-315; data point number requirements, **240**, 314; determination of enzyme kinetic parameters multisubstrate, **240**, 316-319; single substrate, **240**, 314-316; enzyme mechanism testing, **240**, 322; evaluation of binding processes, **240**, 319-321; instructions for use, **240**, 312-313.

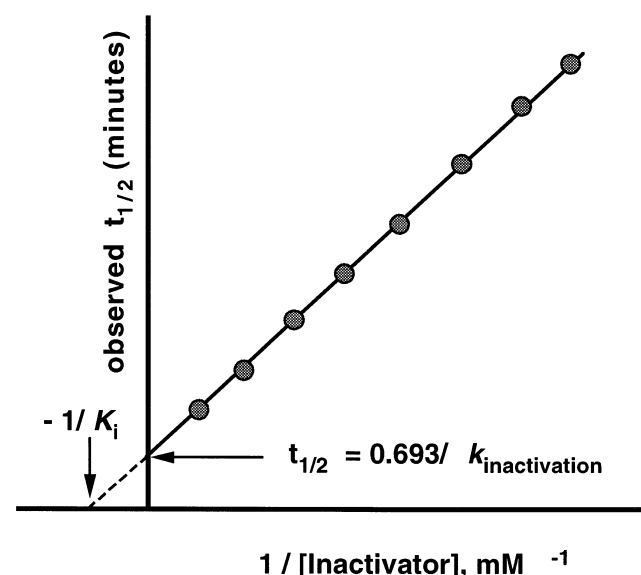
K_{ip} , K_{eq} , K_{ir} , . . .

Symbol for the dissociation constant for product P, Q, R, . . .

K_{is}

Symbol for the dissociation constant of an inhibitor with respect to the free enzyme, E. In the double-reciprocal form of the initial-rate equation, this term is associated with the slope portion of the expression. See K_{ii}

KITZ AND WILSON REPLOT



A graphical procedure¹ used in the study of irreversible inhibitors, particularly in mechanism-based inactiva-

tion², and with affinity labels³. In this method, the half-life ($t_{1/2}$) is measured at a number of different inhibitor concentrations (at constant enzyme concentration). The Kitz and Wilson plot is a replot of these $t_{1/2}$ values as a function of $1/[\text{inactivator}]$. From this plot, k_{inact} and K_i can be obtained.

¹R. Kitz & I. B. Wilson (1962) *J. Biol. Chem.* **237**, 3245.

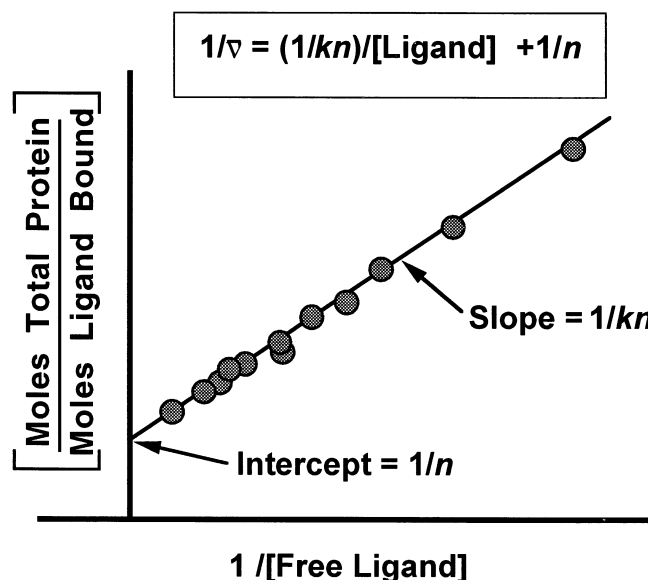
²R. B. Silverman (1995) *Meth. Enzymol.* **249**, 240.

³B. V. Plapp (1982) *Meth. Enzymol.* **87**, 469.

KLOTZ PLOT

A ligand binding plot ($1/v$ versus $1/[A]$) that permits one to determine the number of ligand binding sites and the dissociation constant for the protein-ligand complex. The typical binding equation for n -independent sites is

$$v = (n[\text{Ligand}]/K_d) / \{1 + [\text{Ligand}]/K_d\}$$



For the Klotz plot, this expression is transformed to yield:

$$1/v = \{(K_d/n)/[\text{Ligand}]\} + 1/n$$

where n is the fractional saturation of n sites with equivalent binding. A plot of $1/v$ versus $1/[A]$ gives a straight line with slope equal to K_d/n and an intercept of $1/n$. In the case of equilibrium dialysis or Hummel-Dreyer ligand binding experiments, the net accrual of ligand as macromolecule-ligand complex is obtained directly. In the case of an observed fluorescence change upon ligand binding, then the fraction of ligand bound at any given concentration of ligand is given as

$$X = (F_{\text{observed}} - F_{\text{ligand}}) / (F_{\text{maximum}} - F_{\text{ligand}})$$

where F_{maximum} is the fluorescence at saturating ligand concentration. Then a plot of the reciprocal of ν , which is again the number of moles of bound ligand per mole protein, *versus* the reciprocal of the free ligand complex relies on the following equation:

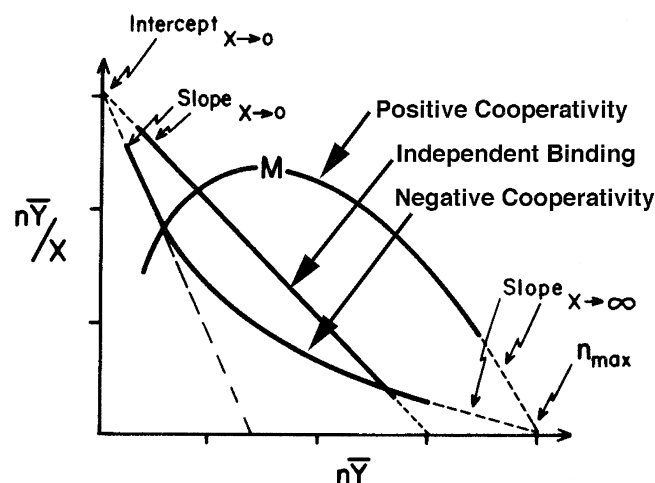
$$1/\nu = K_d/\{n(1-X)[\text{Ligand}]_{\text{total}}\} + 1/n$$

I. M. Klotz, F. M. Walker & R. G. Bivan (1946) *J. Am. Chem. Soc.* **68**, 1486.

K_m (or, K_M)

Symbol for the Michaelis constant (or, Michaelis-Menten constant). **See** *Michaelis Constant*

KOSHLAND-NÉMETHY-FILMER COOPERATIVITY MODEL



Scatchard plots for ligand saturation behavior of an oligomeric protein exhibiting positive cooperativity, negative cooperativity, or independent binding behavior. For addition details, **See** *Monod-Wyman-Changeux Cooperativity Model; Scatchard Plot*.

A model used to explain cooperativity on the basis of ligand-induced changes in conformation that may or may not alter the subunit-subunit interfaces of oligomeric enzymes and receptors¹⁻³. This model has also been referred to as the Adair-Koshland-Némethy-Filmer model (AKNF model), the induced-fit model, and the sequential model.

The model assumes: (1) that the protein is oligomeric (or that the protein contains several topologically distinct ligand binding sites on a single polypeptide chain); (2) that in the absence of the binding ligand (or substrate), the protein has only one conformational form; (3) that upon binding, the ligand induces a conformational

change in the subunit to which it is bound (hence, there is no symmetry requirement as is essential for the Monod-Wyman-Changeux model), (4) that this conformational change affects the interface to one or more of the adjacent subunits and this conformational change alters the affinity of the new ligand binding to that adjacent subunit, and (5) that binding of the ligand(s) is a rapid equilibrium step(s).

The Koshland model can account for positive and negative cooperativity as well as independent ligand binding. The key is how much binding energy is released as each ligand interacts with protein as saturation of sites is achieved. For energy diagrams explaining these features, **See** *Monod-Wyman Changeux Model*.

The KNF model is an appealing model that builds on the well-known ability of ligands to disturb protein structure beyond the binding site *per se*. Moreover, many proteins exhibit negative cooperativity, and the Monod model simply cannot accommodate the possibility of negative cooperativity. **See** *Induced Fit Model; Cooperativity; Allostery; Monod-Wyman-Changeux Model; Negative Cooperativity; Positive Cooperativity; Independent Binding; Linked Functions; Adair Equation*

¹D. E. Koshland, Jr., G. Némethy & D. Filmer (1966) *Biochemistry* **5**, 365.

²K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

³K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

KOSMOTROPES

Small ions of high charge density that bind water molecules more strongly than the strength of water-water interactions in bulk solvent¹. The major intracellular anions (*e.g.*, phosphates and carboxylates) are kosmotropes. These ions differ from chaotropes, which are large ions of low charge density and only weakly bind water molecules. **See** *Chaotropic Agents; Hofmeister Series*

¹K. D. Collins (1997) *Biophys. J.* **72**, 65.

$K_p, K_q, K_r, \dots$

Symbol for the Michaelis constant for product P, Q, R, ...

K_S

1. Symbol for the dissociation constant of a substrate, S, for a particular enzyme. Note that, when this symbol is

used, the investigator is assuming that the substrate binds in a rapid equilibrium fashion. The K_s value is not necessarily equal to the Michaelis constant. 2. Symbol for the Michaelis constant for product S. **See** *Michaelis-Menten Equation*; *Uni Uni Mechanism*

K SYSTEMS

A subset in allosteric models of cooperativity. If an allosteric effector, upon binding to a cooperative enzyme, alters the Michaelis or dissociation constants (or $[S_{0.5}]$ value) for the substrate(s) (but not the V_{max} values), then that protein is a *K* system enzyme. **See** *Monod-Wyman-Changeux Model*

K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

KUHN-ROTH OXIDATION

A method for the determination of carbon-linked methyl groups (and, by extension, used in the analysis of chiral methyl groups in stereochemical studies of enzyme-catalyzed reactions acting on methyl groups). In this procedure, the methyl groups are converted to acetic acid by oxidation of the metabolite with chromic and sulfuric acids. Milder versions are also available.

H. Simon & H. G. Floss (1967) *Bestimmung der Isotopenverteilung in markierten Verbindungen*, Springer-Verlag, Berlin.

J. W. Cornforth, R. H. Cornforth, A. Pelter, M. G. Horning & G. Popjak (1959) *Tetrahedron* **5**, 311.

KURGANOV METHOD FOR ANALYZING COOPERATIVITY

A procedure^{1,2} for estimating the cooperativity of a protein by using the initial rates at only three different concentrations of substrate to estimate the shape of the cooperative curve.

¹B. I. Kurganov (1973) *Acta Biol. Med. Germ.* **31**, 181.

²B. I. Kurganov (1982) *Allosteric Enzymes: Kinetic Behavior*, Wiley, New York.

K_w

Symbol for the product of the H^+ concentration (or, H_3O^+ concentration) and the OH^- concentration of an aqueous solution; the autoprotolysis constant. **See** *Water, Temperature Effects of pK_w of*

KYNURENINASE

This pyridoxal-phosphate-dependent enzyme [EC 3.7.1.3] catalyzes the hydrolysis of kynurenine to produce anthranilate and alanine. 3'-Hydroxykynurenine and some other (3-arylcarbonyl)alanines can also be acted upon by this enzyme.

A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.

K. Soda & K. Tanizawa (1979) *Adv. Enzymol.* **49**, 1.

KYNURENINE AMINOTRANSFERASE

Kynurenine:oxoglutarate aminotransferase [EC 2.6.1.7] catalyzes the pyridoxal-phosphate-dependent reaction of kynurenine and α -ketoglutarate (or, 2-oxoglutarate) to reversibly produce 4-(2-aminophenyl)-2,4-dioxobutanoate and glutamate. Kynurenine:glyoxylate aminotransferase [EC 2.6.1.63] catalyzes the reversible reaction of kynurenine with glyoxylate to produce 4-(2-aminophenyl)-2,4-dioxobutanoate and glycine.

M. C. Tobes (1987) *Meth. Enzymol.* **142**, 217.

K. Tanizawa, Y. Asada & K. Soda (1985) *Meth. Enzymol.* **113**, 90.

R. A. Hartline (1985) *Meth. Enzymol.* **113**, 664.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

KYNURENINE 3-HYDROXYLASE

This FAD-dependent enzyme [EC 1.14.13.9], also called kynurenine 3-monooxygenase, catalyzes the reaction of kynurenine with NADPH and dioxygen to produce 3-hydroxykynurenine, $NADP^+$, and water.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

L

LABILE

1. Referring to an unstable and/or transient chemical species. 2. Referring to a substituent, atom, or group of a molecular entity that is easily removable; *e.g.*, the labile proton of the carboxyl group in trichloroacetic acid. 3. In reference to coordinator complexes, referring to ligands that can be readily replaced by other ligands. 4. On occasions, the term is also used with respect to stable, yet reactive, chemical species.

LACCASE

This copper-dependent enzyme [EC 1.10.3.2] catalyzes the reaction of four benzenediol and dioxygen to produce four benzosemiquinone and two water.

- A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.

 β -LACTAMASES

Zinc-dependent enzymes [EC 3.5.2.6], including penicillinase and cephalosporinase, with varying specificity in their catalysis of β -lactam hydrolysis. Some act more readily on penicillins, whereas the catalysis of others is more efficient with cephalosporins.

- C. W. Wharton (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 345.
 M. Jamin, J.-M. Wilkin & J.-M. Frère (1995) *Essays in Biochem.* **29**, 1.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 N. Citri (1971) *The Enzymes*, 3rd ed., **4**, 23.

LACTASE

β -Galactosidase [EC 3.2.1.23], also called lactase, catalyzes the hydrolysis of terminal, nonreducing β -D-galac-

tose residues in β -D-galactosides. Lactase [EC 3.2.1.108] catalyzes the hydrolysis of lactose to produce glucose and galactose.

- M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.
 E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, **197**.

LACTATE DEHYDROGENASE

This NAD⁺-dependent oxidoreductase [EC 1.1.1.27] catalyzes the following reversible reaction:



with hydride transfer occurring to or from the A-side of the pyridine ring. The kinetic reaction mechanism is best described as an ordered Bi Bi ternary complex scheme, and formation of Enz·NAD⁺·Pyruvate or Enz·NADH·Lactate abortive complexes can lead to substrate inhibition under favorable conditions. **See also** Abortive Complexes; Isotope Exchange at Equilibrium

- R. Jaenicke, H. Schurig, N. Beaucamp & R. Ostendorp (1996) *Adv. Protein Chem.* **48**, 181.
 N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 H. F. Fisher (1988) *Adv. Enzymol.* **61**, 1.
 J. J. Holbrook, A. Liljas, S. J. Steindel & M. J. Rossmann (1975) *The Enzymes*, 3rd ed., **11**, 191.
 K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.
 M. G. Rossmann, A. Liljas, C.-I. Brändén & L. J. Banaszak (1975) *The Enzymes*, 3rd ed., **11**, 61.
 J. Everse & N. O. Kaplan (1973) *Adv. Enzymol.* **37**, 61.

Selected entries from Methods in Enzymology [vol, page(s)]:

Enzyme-substrate complex formation, **64**, 53; data analysis, **64**, 56, 57; extensions of technique, **64**, 57-59; as evidence for occurrence of intermediate, **64**, 47-59; kinetic equation, **64**, 49-52; limitations, **64**, 57-59; mixing procedure, **64**, 53-56; reaction condition, **64**, 56, 57; termination, **64**, 56, 57.

D-LACTATE DEHYDROGENASE (CYTOCHROME)

This FAD-dependent enzyme [EC 1.1.2.4] catalyzes the reaction of (*R*)-lactate with two ferricytochrome *c* to produce pyruvate and two ferrocytochrome *c*.

T. A. Hansen (1994) *Meth. Enzymol.* **243**, 21.
Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

L-LACTATE DEHYDROGENASE (CYTOCHROME)

This FMN- and protoheme IX-dependent enzyme [EC 1.1.2.3] catalyzes the reaction of (*S*)-lactate with two ferricytochrome *c* to produce pyruvate and two ferrocytochrome *c*.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference*, **3**, 83.
T. A. Hansen (1994) *Meth. Enzymol.* **243**, 21.
Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

LACTATE 2-MONOOXYGENASE

This FMN-dependent enzyme [EC 1.13.12.4] catalyzes the reaction of (*S*)-lactate with dioxygen to produce acetate, carbon dioxide, and water.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

LACTATE RACEMASE

This enzyme [EC 5.1.2.1], also referred to as lacticoracemase and hydroxyacid racemase, catalyzes the interconversion of (*S*)-lactate and (*R*)-lactate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
E. Adams (1976) *Adv. Enzymol.* **44**, 69.
L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

LACTOPEROXIDASE

This enzyme [EC 1.11.1.7] catalyzes the reaction of a donor with hydrogen peroxide to produce the oxidized donor and two water. **See also Peroxidase**

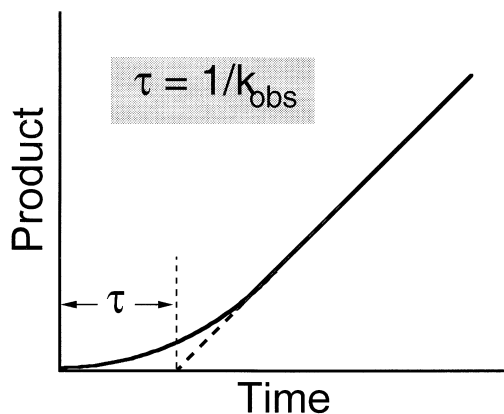
H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.
E. Cadenas (1989) *Ann. Rev. Biochem.* **58**, 79.
A. Naqui, B. Chance & E. Cadenas (1986) *Ann. Rev. Biochem.* **55**, 137.

LACTOSE SYNTHASE

This enzyme [EC 2.4.1.22] is a protein complex of two proteins (designated A and B) and catalyzes the reaction of UDP-galactose with D-glucose to generate UDP and lactose. In the absence of the α -lactalbumin (protein B), the enzyme catalyzes the transfer of galactose from UDP-galactose to *N*-acetylglucosamine (*i.e.*, the activity of *N*-acetylglucosamine synthase, EC 2.4.1.90).

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.
R. L. Hill & K. Brew (1975) *Adv. Enzymol.* **43**, 411.
K. E. Ebner (1973) *The Enzymes*, 3rd ed., **9**, 363.
K. Brew (1970) *Essays in Biochem.* **6**, 93.

LAG TIME



1. The pre-steady-state portion of progress curve, during which there is an exponential increase in the slope of the product concentration *versus* time plot. 2. The slow increase in the slope of the product concentration *vs.* time progress curve often observed when using coupled-enzyme assay protocols. Investigators strive to minimize such lag times¹. 3. A slow increase in the slope of the product concentration *versus* time or progress curve. Such systems respond slowly to rapid changes in substrate concentrations and are often associated with reaction schemes containing a slow isomerization step^{2,3}. **See also Hysteresis; Pre-Steady-State; Autocatalysis; Microtubule Assembly Kinetics**

¹F. B. Rudolph, B. W. Baugher & R. S. Beissner (1979) *Meth. Enzymol.* **63**, 22.

²K. E. Neet & G. R. Ainslie (1980) *Meth. Enzymol.* **64**, 192.

³K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

LAMBDA (Λ , λ)

Λ 1. Symbol for molar conductivity of an electrolyte. 2. Symbol for Ostwald solubility coefficient.

λ 1. Symbol for Avogadro's number (rarely). 2. Symbol for wavelength. 3. Symbol for radioactive constant (decay rate constant). 4. Symbol for absolute chemical activity. 5. Symbol for thermal conductivity. 6. Symbol for mean free path.

LAMBDA (or, Λ -) ISOMERS OF METAL ION-NUCLEOTIDE COMPLEXES

Coordination isomers of metal ion-nucleotide complexes. For most metal ions, for example Mg^{2+} , the metal

ion exchange times are fast, and it is difficult to separate and characterize particular isomers. However, when the metal ion is Cr^{3+} or Co^{3+} , exchange times are very slow and the corresponding nucleotides can be used as inhibitors, chirality probes, and in distance measurements in NMR experiments¹. If the terminal phosphate of a nucleotide triphosphate or a nucleotide diphosphate is chelated to the metal ion, two possible isomers can exist, Λ and Δ , which are defined by the screw sense of the coordination. If the Cr^{3+} or Co^{3+} is bound to both the β and γ phosphate groups of NTP, there are a total of four coordination isomers, two Λ and two Δ . These subsets of isomers are designated δ (or, pseudoaxial) and λ (or, pseudoequatorial). All of these isomers have proved to be quite useful in probes of an enzyme's active site¹. **See** *Exchange-Inert Metal-Nucleotide Complexes*

¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

LANGMUIR ISOTHERM

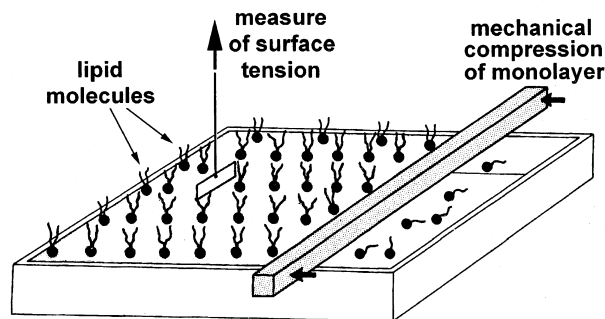
A theoretical curve that accounts for the adsorption of gases onto a surface. Because the strength of adsorption depends on the absolute temperature, binding must be measured at a single temperature (hence the term "isotherm"). The treatment assumes (a) that each binding site can accommodate a single adsorbed molecule; (b) that the affinity is the same at all sites; (c) that binding at each site is completely independent of binding at any other site; and (d) that the total number of sites per unit area of the surface is a constant. The fraction of occupied sites is represented by θ , and the fraction of unoccupied sites is given as $(1 - \theta)$. At equilibrium, the Langmuir isotherm can be derived from the balancing of the rate of adsorption ($k_{\text{adsorption}}(1 - \theta)P$, where P is the partial pressure of the adsorbing gas) and the rate of desorption ($k_{\text{desorption}}\theta$). The resulting equation can be rearranged to yield: $\theta = bP/[1 + bP]$, where b is the adsorption coefficient, the ratio $k_{\text{adsorption}}/k_{\text{desorption}}$. Because both ligand-receptor and substrate-enzyme saturation curves have the same mathematical form, they are sometimes said to be Langmuir isotherms. **See** *Biominingalization; Micelle*

LANGMUIR TROUGH AND BALANCE

A device for direct measurement of the surface pressure exerted by a surface film on liquids. The basic features of the trough include a horizontally positioned tray with a

mechanically positioned compressing barrier that retains the surface film on one end of the trough and a second float typically constructed of a mica strip to which is attached a torsion spring balance.

Langmuir Trough



In a typical experiment, a small volume of an insoluble surface-active material (dissolved in a water-insoluble solvent such as benzene) is placed atop a clean water surface. As the solvent evaporates away, a film remains and the moving barrier can be adjusted so that the surface film exerts pressure on the mica float. A calibrated torsion balance is used to measure the force that the film exerts on the float. That force divided by the length of the float is the force per unit length or the surface pressure. For studies of lipolysis kinetics^{1,2}, a Langmuir trough can be constructed so that one can measure lipase action under first-order and zero-order conditions.

¹R. Verger & G. H. de Haas (1973) *Chem. Phys. Lipids* **10**, 127.

²R. Verger (1980) *Meth. Enzymol.* **64**, 372.

LAPLACE TRANSFORM

A linear operator that permits one to solve the initial value problem

$$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy = f(t)$$

[where $y(0) = y_0$ and $y'(0) = y'_0$ for the continuous (or piecewise continuous) function $f(t)$, itself defined over the interval $0 \leq t < \infty$] in a manner analogous to that required for solving algebraic equations. The Laplace transform $\mathcal{L}\{f(t)\}$ is written as:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt = \lim_{A \rightarrow \infty} \int_0^A e^{-st} f(t) dt$$

where s is of sufficiently large magnitude and the parameter A lies in another interval. This transform is a linear operator, such that

$$\begin{aligned}\mathcal{L}\{r_1 f_1(t) + r_2 f_2(t)\} &= \int_0^\infty e^{-st} [r_1 f_1(t) + r_2 f_2(t)] dt \\ &= \mathcal{L}\{r_1 f_1(t)\} + \mathcal{L}\{r_2 f_2(t)\}\end{aligned}$$

The method of Laplace transformation represents a powerful tool for practicing kineticists seeking to solve linear differential equations. The three-step process includes: (a) transforming a differential equation (or group of differential equations) into a subsidiary algebraic equation(s); (b) solving this(these) subsidiary equation(s) using algebraic manipulations; and (c) retransforming the resultant equation(s) to obtain the sought after solution to the original equation(s). Many third-semester calculus textbooks, especially those applying differential equations to engineering problems, provide lucid accounts of the formalism of using Laplace transform methods.

Laplace transformation is particularly useful in pharmacokinetics where a number of series first-order reactions are used to model the kinetics of drug absorption, distribution, metabolism, and excretion. Likewise, the relaxation kinetics of certain multistep chemical and physical processes are well suited for the use of Laplace transforms.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Claverie approach to Padé-Laplace algorithm for sums of exponentials, **210**, 59; Taylor series expansion in analysis of sums of exponentials, **210**, 56.

LARMOR PRECESSION

The precession observed in a plane of an electron orbit of an atom when that atom lies within a uniform magnetic field. The plane will precess about the direction of the field such that a cone is traced out having an axis normal to the direction of the field. The frequency of this precession is usually symbolized by ν_L and is equal to $eB/(4\pi mc)$ where e is the electron charge, B is the field strength, m is the mass, and c is the velocity of the electron. Larmor precession is also important in NMR studies of enzymes.

LASER-FLASH KINETIC ANALYSIS

A rapid reaction technique for analyzing the kinetics of photo-sensitive or photo-responding biochemical reactions. One application deals with the determination¹ of

the rate constants and likely intermediate steps involved in the fast electron transfer from metalloproteins such as plastocyanin or cytochrome c_6 to photosystem I (PSI). P700 in PSI can be rapidly photo-oxidized by a saturating laser flash, thereby inducing almost instantaneously an 820 nm absorbing species that can then react with reductant (in this case the metalloproteins above). The time course of this reaction can be monitored by loss of the 820 nm absorbance, and the oscilloscope tracings of the reaction's time course can be analyzed to obtain valuable kinetic information about the mechanism.

¹M. Hervas, J. Navarro, A. Diaz, H. Bottin & M. De la Rosa (1995) *Biochemistry* **34**, 11321.

LATENT ACTIVITY

The amount of total enzymatic activity that becomes manifest only after disruption of membranous barriers between enzyme and substrate or upon removal of some otherwise inhibitory factor. Membrane disruption is often achieved by treatment with detergent to solubilize the enzyme. One example is the so-called *microsomal* glucose-6-phosphatase¹, an enzymatic activity that is located in the lumen of the endoplasmic reticulum but becomes trapped as a latent activity in microsome vesicles upon mechanical disruption of cells.

¹R. C. Nordlie (1982) *Meth. Enzymol.* **87**, 319.

LATENT HEAT

Heat released or absorbed in a process other than a change in temperature. For example, latent heat of fusion and latent heat of vaporization refer to the heat associated with the change of state with no temperature change.

LAW OF CONSTANT HEAT SUMMATION

See Hess' Law; Additivity Principle

LEAST-CONDUCTIVE STEP

A kinetic term introduced by Ray¹ to designate the bottleneck in a steady-state enzyme kinetic pathway as that step in which the enzyme form accumulates in highest concentration at saturating substrate concentration. Highest accumulation of this enzyme reflects the fact that this species faces the highest barrier that precedes an irreversible step in the forward direction.

¹W. J. Ray, Jr. (1983) *Biochemistry* **22**, 4452.

LEAST MOTION, PRINCIPLE OF

The hypothesis that, for a given set of reactants, those elementary reactions having the least change in nuclear positions will also have the lowest energies of activation.

J. Hine (1966) *J. Org. Chem.* **31**, 1236.

J. Hine (1977) *Adv. Phys. Org. Chem.* **51**, 1.

LEAST SQUARES METHOD

A statistical algorithm, also known as linear regression analysis, for systems where Y (the random variable) is linearly dependent on another quantity X (the ordinary or controlled variable). The procedure allows one to fit a straight line through points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , ..., (x_n, y_n) where the values x_i are defined before the experiment and y_i values are obtained experimentally and are subject to random error. The best fit line through such a series of points is called a "least squares fit", and the protocol provides measures of the reliability of the data and quality of the fit.

For such a series of ordered pairs, the vertical displacement of a sample value, say (x_j, y_j) , from the straight line $y = a + bx$ is $|y_j - a - bx_j|$, and the sum of these squares is given as:

$$Q = \sum_{j=1}^n (y_j - a - bx_j)^2$$

The values of a and b are then chosen to minimize Q which is "the sum of the squares of $(y_j - a - bx_j)$ ", hence the name *Least Squares*. Accordingly, the equation for the best fit least squares line is:

$$[y = a + bx]_{Q \min}$$

This line is frequently referred to as a regression line, and its slope b is the regression coefficient. This procedure can be extended to polynomial functions wherein segments can be treated as linear or nearly linear functions of Y on X .

Cleland¹ and Mannervik² have described least squares programs and procedures for treating enzyme kinetic data. The interested reader will also wish to consult numerous articles in vols. 210 and 259 in *Methods in Enzymology* (L. Brand & M. L. Johnson, eds.) dealing with numerical computer methods for statistical treatment of kinetic and equilibrium data.

¹W. W. Cleland (1979) *Meth. Enzymol.* **63**, 103.

²B. Mannervik (1982) *Meth. Enzymol.* **87**, 370.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Computer program, **63**, 111-138; evaluation, **63**, 109-111; evaluation of oxygen equilibria data, **76**, 478-481, 483; iterative fitting, **63**, 106; for numerical analysis of ligand binding kinetics, **76**, 661-662; nonlinear, **63**, 104, 105; residual, **63**, 110-112; surface, **63**, 106-109; weighting, **63**, 105; χ^2 method [goodness-of-fit, **210**, 102; parabolic extrapolation, **210**, 10; residual value frequencies, **210**, 93]; nonlinear [global analysis packages, construction, **210**, 44; in iterative deconvolution program, **210**, 511]; parameter estimation from experimental data, **210**, 1 {assumptions, **210**, 4; χ^2 method, parabolic extrapolation, **210**, 10; confidence intervals, **210**, 20 [asymptotic standard errors, **210**, 24; grid search method, **210**, 22; linear joint confidence intervals, **210**, 25; Monte Carlo method, **210**, 21; propagation, **210**, 28]; Gauss-Newton method, **210**, 11; Marquardt method, **210**, 16; Nelder-Mead simplex method, **210**, 18; performance methods, **210**, 9; sample analysis, **210**, 29; steepest descent method, **210**, 15]; simultaneous [free energy of site-specific DNA-protein interactions, **210**, 471; for model testing, **210**, 463; for parameter estimation, **210**, 463; separate analysis of individual experiments, **210**, 475; for testing linear extrapolation model for protein unfolding, **210**, 465.

LEAVING GROUP

An atom, group, or moiety (either charged or neutral) that detaches from the main portion of the substrate or reactant in a given reaction. If the leaving group detaches and carries away the electron pair that formerly constituted the bond to the central atom, the leaving group is said to be nucleofugal. If the electron pair is not removed, the leaving group is said to be electrofugal. A leaving group may also be termed an exiphile. **See also** *Electrofuge*; *Nucleofuge*; *Entering Group*; *Nucleophilic Substitution Reaction*; *Nucleophile*

LE CHATELIER'S PRINCIPLE

The tendency for a system at equilibrium to adjust in a manner that minimizes the effect of a change imposed by external factors (such as changes in temperature, pressure or electric field strength).

All homeostatic mechanisms can be regarded as biological feedback mechanisms designed to take maximal advantage of the self-compensating nature described as Le Chatelier's Principle¹. An example is the linked binding functions associated with oxygen binding in the Monod-Wyman-Changeux model for hemoglobin; an increase in oxygen partial pressure is compensated by a change in the distribution of hemoglobin from the T-state to the R-state.

¹H. L. Le Chatelier (1884) *Compt. Rend.* **99**, 786; (1888) *Compt. Rend.* **106**, 355, 598, 687, and 1008.

LECITHIN-CHOLESTEROL ACYLTRANSFERASE

This enzyme [EC 2.3.1.43] catalyzes the reaction of a phosphatidylcholine with a sterol to produce a sterol ester and a 1-acylglycerophosphocholine. The acyl moiety transferred can be a palmitoyl, oleoyl, or linoleoyl group. A number of sterols, including cholesterol, can act as the acyl acceptor.

C. J. Fielding & X. Collet (1991) *Meth. Enzymol.* **197**, 420.

LEE-WILSON TREATMENT

A treatment suggested by Lee and Wilson¹ for analyzing enzyme kinetic data in which a significant proportion of the initial substrate concentration has undergone conversion (thus, not under true initial rate conditions). It was suggested that the $[\bar{S}]$, that is, the arithmetic mean substrate concentration ($([S_0] + [S])/2$) be used in the double-reciprocal form of the normal kinetic expression. Note that this approach is improved if the product(s) is(are) continuously removed from the reaction system (for example, by a coupling system), that substrate inhibition is absent, that all other cosubstrates are saturating, and that the reverse reaction is poor. If products cannot be removed, then the procedure should not be used if product inhibition is significant.

¹H.-J. Lee & I. B. Wilson (1971) *Biochim. Biophys. Acta* **242**, 519.

LEFFLER'S ASSUMPTION

The transition state of a reaction bears greater resemblance to the less stable species (of reactant or reaction intermediate or product). **See** *Hammond Postulate*

J. E. Leffler (1953) *Science* **117**, 340.

LEGENDRE TRANSFORMATION

A mathematical tool of proven utility in dealing with certain differential equations. Consider a function f (a dependent variable) having $x_1, x_2, x_3, \dots, x_n$ as independent variables. Then $df = f_1 dx_1 + f_2 dx_2 + \dots + f_n dx_n$ where $f_1 = (\partial f / \partial x_1)_{x_2 \dots x_n}$, $f_2 = (\partial f / \partial x_2)_{x_1 x_3 \dots x_n}$, etc. Let the new function g be equal to $f - f_1 x_1$. It can then be shown that $dg = -x_1 df_1 + f_2 dx_2 + \dots + f_n dx_n$. This tool has been utilized to generate expressions for enthalpy, Helmholtz energy, and Gibbs free energy from the differential expression for internal energy.

LEGUMAIN

This enzyme [EC 3.4.22.34] catalyzes the hydrolysis of peptide bonds in proteins (for example, in azocasein) and polypeptides at Asn-Xaa, the preferential bond.

S.-i. Ishii (1994) *Meth. Enzymol.* **244**, 604.

LEISHMANOLYSIN

This zinc- and calcium-dependent enzyme [EC 3.4.24.36] catalyzes the hydrolysis of peptide bonds, exhibiting a preference for hydrophobic amino acid residues at positions P1 and P1', and basic amino acid residues at P2 and P3'.

J. Bouvier, P. Schneider & R. Etges (1995) *Meth. Enzymol.* **248**, 614.

LEUCINE 2,3-AMINOMUTASE

This cobalamin-dependent enzyme [EC 5.4.3.7] catalyzes the interconversion of (2*S*)- α -leucine and (3*R*)- β -leucine.

J. M. Poston (1988) *Meth. Enzymol.* **166**, 130.

O. W. Griffith (1986) *Ann. Rev. Biochem.* **55**, 855.

J. M. Poston (1986) *Adv. Enzymol.* **58**, 173.

B. M. Babior & J. S. Krouwer (1979) *Crit. Rev. Biochem.* **6**, 35.

LEUCINE AMINOPEPTIDASE

This zinc-dependent enzyme [EC 3.4.11.1], also referred to as cytosol aminopeptidase, leucyl aminopeptidase, and peptidase S, catalyzes the hydrolysis of a terminal peptide bond such that there is a release of an N-terminal amino acid, Xaa-Xbb-, in which Xaa is preferably a leucyl residue, but may be other aminoacyl residues including prolyl (although not arginyl or lysyl). Xbb may be prolyl. In addition, amino acid amides and methyl esters are also readily hydrolyzed, but the rates with arylamides are exceedingly slow. The enzyme is activated by heavy metal ions.

W. L. Mock (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 425.

H. Kim & W. N. Lipscomb (1994) *Adv. Enzymol.* **68**, 153.

B. L. Vallee & A. Galles (1984) *Adv. Enzymol.* **56**, 283.

R. J. DeLange & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 81.

LEUCINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.6] catalyzes the reversible reaction of leucine with α -ketoglutarate (or, 2-oxoglutarate) to produce 4-methyl-2-oxopentanoate and glutamate. **See also** *Branched-Chain Amino Acid Aminotransferase*

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

LEUCINE DEHYDROGENASE

This enzyme [EC 1.4.1.9] catalyzes the reaction of leucine with NAD⁺ and water to produce 4-methyl-2-oxopentanoate, ammonia, and NADH. Isoleucine, valine, norvaline, and norleucine can serve as substrates for this enzyme.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

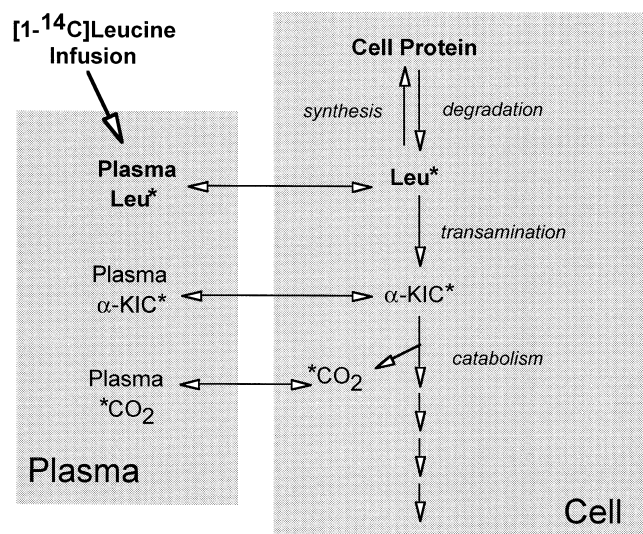
N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

G. Livesey & P. Lund (1988) *Meth. Enzymol.* **166**, 282.

LEUCINE KINETICS

The biosynthetic incorporation of amino acids into proteins makes these metabolites valuable endogenous tracers for the characterization of protein turnover¹. Of the naturally occurring amino acids, administration of a bolus dose of [³H]leucine is widely used as a tracer in kinetic investigations of protein synthesis and secretion.

Leucine as a Tracer in Protein Turnover



After intravenous injection, the specific radioactivity of plasma leucine decays over several orders of magnitude within the first 12 hours, followed by a slow decay lasting a number of weeks. This “tail” in the decay curve is the consequence of recycling of the leucine tracer as proteins are degraded, allowing the liberated [³H]leucine to reenter the plasma pool. Kinetic data can be analyzed by mathematical compartmental modeling (See SAAM), and plasma leucine activity is used as a forcing function to drive the input of [³H]leucine into the protein synthesis pathway. Leucine has many advantages as a protein

tracer: (a) it is an essential amino acid, and (b) the by-products of its degradation are not reincorporated into other amino acids. The initial step in leucine catabolism is transamination and formation of α-ketoisocaproic acid. This reaction is readily reversible. A subsequent series of oxidative reactions generate hydroxymethylglutaryl coenzyme A and acetate. In the oxidation of one of the intermediates, isovaleryl-coenzyme A, the tritium on position C-4 is lost, and the C-5 tritium can appear in steroids or fatty acids. Typically, any such tracer that becomes incorporated into lipid is greatly diluted. See *Protein Turnover Kinetics*; SAAM

¹W. R. Fisher, L. A. Zech, L. L. Kilgore & P. W. Stacpoole (1991) *J. Lipid Res.* **32**, 1823.

LEUCOLYSIN

This zinc- and calcium-dependent venom endopeptidase [EC 3.4.24.6] hydrolyzes the following peptide bonds in the insulin B chain: Phe¹-Val², His⁵-Leu⁶, Ala¹⁴-Leu¹⁵, Gly²⁰-Glu²¹, Gly²³-Phe²⁴, and Phe²⁴-Phe²⁵. It will also act on N-blocked dipeptides.

J. B. Bjarnason & J. W. Fox (1995) *Meth. Enzymol.* **248**, 345.

LEUCYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.4], also called leucine:tRNA ligase, catalyzes the reaction of leucine with tRNA^{Leu} and ATP to produce leucyl-tRNA^{Leu}, AMP, and pyrophosphate.

E. A. First (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 573.

A. Hampel & R. Tritz (1988) *Meth. Enzymol.* **166**, 260.

LEUKOTRIENE A₄ HYDROLASE

This enzyme [EC 3.3.2.6] catalyzes the hydrolysis of (7*E*,9*E*,11*Z*,14*Z*) - (5*S*,6*S*) - 5,6 - epoxycosa - 7,9,11,14 - tetraenoate to form (6*Z*,8*E*,10*E*,14*Z*)-(5*S*,12*R*)-5,12-dihydroxyicosa-6,8,10,14-tetraenoate. This enzyme is highly specific yet it also converts 4,5-leukotriene A₄ into leukotriene B₄. Note also that epoxide hydrolase [EC 3.3.2.3] is not identical with this enzyme.

N. Ohishi, T. Izumi, Y. Seyama & T. Shimizu (1990) *Meth. Enzymol.* **187**, 286.

J. Z. Haeggström (1990) *Meth. Enzymol.* **190**, 324.

LEUKOTRIENE C₄ SYNTHASE

This enzyme [EC 2.5.1.37] catalyzes the reaction of leukotriene A₄ with glutathione to generate leukotriene C₄ and water. This enzyme is not identical with glutathione

transferase [EC 2.5.1.18]. **See also** *Glutathione S-Transferase*

M. Söderström, B. Mannervik & S. Hammarström (1990) *Meth. Enzymol.* **187**, 306.

R. J. Soberman (1990) *Meth. Enzymol.* **187**, 335.

LEVANASE

This enzyme [EC 3.2.1.65], also known as 2,6- β -D-fructan fructanohydrolase, catalyzes the random hydrolysis of 2,6- β -D-fructofuranosidic linkages in 2,6- β -D-fructans (levans) containing more than three fructosyl units.

G. Avigad & S. Bauer (1966) *Meth. Enzymol.* **8**, 621.

LEVANSUCRASE

This enzyme [EC 2.4.1.10], also known as sucrose 6-fructosyltransferase, catalyzes the reaction of sucrose with [(2,6)- β -D-fructosyl]_n to produce glucose and [(2,6)- β -D-fructosyl]_(n+1). A few other fructosyl-containing sugars can act as substrates.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

LEVELING EFFECT

A property of solvents affecting the apparent acidity of Brønsted acids. This effect reflects the tendency of a solvent to make all Brønsted acids with acidities exceeding a certain value to appear equally acidic. In such cases, there is a transfer of a proton to the protophilic solvent from dissolved acid (which is a stronger acid than the conjugate acid of the solvent). Hence, the only acid present to any significant extent is the lyonium ion. For example, water has a leveling effect on the acidities of HClO₄, HCl, and HI. In this respect, the apparent strength of an acid depends on the basicity of the solvent: the more basic the solvent, the more the reactants will become ionized. Hence, to compare relative acidities, the solvent must be selected to prevent complete ionization of the acids being studied. A similar phenomenon occurs with strong bases except, in these cases, the solvent is protogenic.

LEWIS ACID

Any chemical species having a vacant orbital and thus acting as an electron-pair acceptor from a Lewis base. Examples of Lewis acids include: BF₃, Fe³⁺, Na⁺, Ca²⁺, and SO₃. **See** *Lewis Base; Acidity; Lewis Acidity*

LEWIS ACIDITY

The thermodynamic tendency of a substance to act as a Lewis acid. The strength of a Lewis acid depends on the nature of the base with which the Lewis acid forms a Lewis adduct. Hence, comparative measures of Lewis acidities are given by equilibrium constants for the formation of the adducts by a common reference base. **See** *Lewis Acid; Electrophilicity; Hard Acids; Soft Acids; Acceptor Number*

LEWIS BASE

Any chemical species that can donate a pair of electrons to a Lewis acid to form a Lewis adduct. Examples of Lewis bases include: NH₃, RNH₂, ROH, CO, CO₃²⁻, and RSH. **See** *Lewis Acid*

LEWIS BASICITY

The thermodynamic tendency of a substance to act as a Lewis base. The strength of a Lewis base depends on the nature of the acid with which the Lewis base forms a Lewis adduct. Hence, comparative measures of Lewis basicities are given by equilibrium constants for the formation of the adducts by a common reference acid. **See** *Lewis Base; Nucleophilicity; Hard Bases; Soft Bases; Donor Number*

L'HÔPITAL'S RULE (or, L'HOSPITAL'S RULE)

A differential technique¹ in mathematics used to obtain information about the composite function $f(x)/g(x)$ where $f(x)$ and $g(x)$ are functions of x . One version of the method requires that both functions are differentiable, symbolized by $f'(x)$ and $g'(x)$, at every point x greater than a certain fixed $M > 0$ and that $g'(x) \neq 0$. If $\lim_{x \rightarrow \infty} (f'(x)/g'(x))$ exists and has a certain value, then $\lim_{x \rightarrow \infty} (f(x)/g(x))$ also exists and has that same value.

This rule has found a number of uses in enzymology. A simple example applies to the standard Michaelis-Menten equation, $v = V_{\max}[S]/(K_m + [S])$. Here, $f(x) = V_{\max}[S]$ and $g(x) = K_m + [S]$. Hence, $\lim_{[S] \rightarrow \infty} (f[S]/g([S])) = \lim_{[S] \rightarrow \infty} (f'([S])/g'([S])) = \lim_{[S] \rightarrow \infty} (V_{\max}/1) = V_{\max}$. Thus, at $[S] = \infty$, $v = V_{\max}$. L'Hôpital's rule has also proved useful in determining expressions for the maximum rate of exchange in equilibrium exchange studies².

¹G. F. A. de l'Hôpital (1696) *Analyse des infiniment petits*, Paris.

²D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **64**, 3.

LIFETIME

The time (symbolized by τ) needed for a concentration of a molecular entity to decrease, in a first-order decay process to e^{-1} of its initial value. In this case, the lifetime (sometimes called mean lifetime) is equal to the reciprocal of the sum of rate constants for all concurrent first-order decompositions. If the process is not first-order, the term *apparent lifetime* should be used, and the initial concentration of the molecular entity should be provided. The terms lifetime and half-life should not be confused. **See** *Half-Life; Fluorescence*

LIGAND

[from the Latin, *ligare*, to bind] 1. A chemical group, ion, or molecule coordinately bound to the central atom in a complex; for example, the cyano moiety bound to the cobalt in cyanocobalamin or the porphyrin portion of heme. 2. An ion or molecule that binds noncovalently to a recognition site on a macromolecule (for example, D-glucose binding to hexokinase or a hormone binding to a receptor protein). 3. The analyte in a competitive binding assay (e.g., in radioimmunoassays). **See also** *Ligand Diffusion to Receptor; Metal Ion Coordination Reactions; Linked Functions; Scatchard Plot; Hill Equation and Plot*

LIGAND BINDING ANALYSIS

Ligand binding experiments are essential for determining the number of binding sites, the dissociation constant(s) for ligand binding site(s), and the most likely mechanism of binding interactions. Winzor and Sawyer¹ and Klotz² present an excellent account of the repertoire of available techniques, including equilibrium dialysis, ultrafiltration, partition equilibrium methods, gel chromatography, UV/visible and fluorescence methods, NMR and ESR measurements, ion-specific/selective electrodes, biosensor methods as well as other competitive binding assays. The choice of technique obviously depends on the ability to detect and quantify binding site occupancy, and one must consider (a) the amount of ligand and macromolecule required for a given technique, (b) the concentration range over which saturation of binding sites is to be assessed, (c) linearity between a technique's signal and the extent of site occupancy,

and (d) stability of the components under the conditions required to make a specific type of measurement.

There are four basic methods for graphically representing ligand binding data: (a) fractional saturation versus ligand concentration; (b) Scatchard plots of $\nu/[L]$ versus ν ; (c) double-reciprocal plots involving (fractional saturation)⁻¹ versus $[L]^{-1}$; and the Klotz plot. These are illustrated here by using different symbols for low and high ligand ranges to permit the reader the opportunity to assess the nature of each plot's representation of binding data. **See also** *Adair Equation; Linked Functions; Scatchard Plot; Hill Equation and Plot; Womack-Colowick Dialysis Method*

¹D. J. Winzor & W. H. Sawyer (1995) *Quantitative Characterization of Ligand Binding*, pp. 168, Wiley-Liss, New York.

²I. M. Klotz (1997) *Ligand-Receptor Energetics: A Guide For the Perplexed*, Wiley, New York.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Ligand Binding Measurements: By equilibrium gel penetration, **117**, 342; by gel chromatography, **117**, 346; by ultrafiltration, **117**, 354; using hydroxyapatite columns, **117**, 370; by electrophoretic and spectroscopic techniques, **117**, 381; by fluorescence spectroscopy, **117**, 400; by sedimentation velocity to investigate ligand-induced protein self-association, **117**, 459; linked to protein self-associations, **117**, 496; the meaning of Scatchard and Hill plots, **48**, 270; ligand-binding by associating systems, **48**, 299; measurement of protein dissociation constants by tritium exchange, **48**, 321; fluorescence methods for measuring reaction equilibria and kinetics, **48**, 380; theoretical calculations of relative affinities of binding, **202**, 497; quantitative structure-activity relationships and molecular graphics in evaluation of enzyme-ligand interactions, **202**, 512; theoretical calculations of relative affinities of binding, **202**, 497; quantitative structure-activity relationships and molecular graphics in evaluation of enzyme-ligand interactions, **202**, 512; ligand binding density functions in analysis of multiple ligand binding, **208**, 270; nucleic acid binding, **212**, 427, 447; theoretical behavior for data collection by reverse titration, **208**, 272; determination of binding constants using linear dichroism, **226**, 243; equilibrium ligand-nucleic acid interactions [analysis, **212**, 402; dependence on salt concentration, **212**, 411]; lanthanide ions, spectroscopic determination, **226**, 506; protein-nucleic acid interaction, measurement, **208**, 226; site-specific protein-DNA interactions [electrolyte effects, **208**, 308; equilibria, **208**, 295; solute effects, **208**, 306; temperature effects, **208**, 301; thermodynamics, **208**, 295]; numerical analysis of binding data, **210**, 481.

Hemoglobin: Assignment of rate constants for O₂ and CO binding to α and β subunits within R- and T-state human hemoglobin, **232**, 363; hemoglobin ligand binding and conformational changes measured by time-resolved absorption spectroscopy, **232**, 387; femtosecond measurements of geminate recombination in heme proteins, **232**, 416; double mixing methods for kinetic studies of ligand binding in partially liganded intermediates of hemoglobin, **232**, 430; hemoglobin-liganded intermediates, **232**, 445; hemoglobin-oxygen equilibrium binding: rapid-scanning spectrophotometry and singular value decomposition, **232**, 460; oxygen equilibrium curve of concen-

trated hemoglobin, **232**, 486; Adair fitting to oxygen equilibrium curves of hemoglobin, **232**, 559; weighted nonlinear regression analysis of highly cooperative oxygen equilibrium curves, **232**, 576; dimer-tetramer equilibrium in Adair fitting, **232**, 597; effects of wavelength on fitting Adair constants for binding of oxygen to human hemoglobin, **232**, 606; Adair equation: rederiving oxygenation parameters, **232**, 632; linkage thermodynamics, **232**, 655.

LIGAND DIFFUSION TO RECEPTOR

The random motion process by which a ligand approaches and penetrates a tissue surface: first, by passing down a concentration gradient by three-dimensional Brownian motion in the bulk solution; then, by reaching the surface where the dimensionality of random excursions reduces to a two-dimensional process; and finally, by docking at the receptor site¹⁻³. Once even weakly associated within the tissue surface, receptor electrostatics and mobility probably work together to guide the ligand to its binding site. Moreover, the kinetics of ligand diffusion to its receptor(s) is likely to greatly influence the time dependence of metabolite and drug action.

¹H. C. Berg & E. M. Purcell (1977) *Biophys. J.* **20**, 193.

²J. Crank (1975) *The Mathematics of Diffusion*, 2nd ed., Clarendon Press, Oxford.

³T. Kenakin (1993) *Pharmacologic Analysis of Drug-Receptor Interaction*, 2nd ed., pp. 20-24, Raven Press, New York.

LIGAND EXCLUSION MODEL

A multiligand binding mechanism observed in some multisubstrate enzyme-catalyzed reactions^{1,2}. Consider the case of a two-substrate enzyme in which substrate A binds deep within a cleft of the protein's active site. The second substrate, B, then binds over A to form the productive ternary complex, EAB. This is a specific type of ordered binding reaction. Binding of substrate A may also induce a conformational change in the enzyme, such that B binds more easily. In some ligand exclusion mechanisms, a competitive inhibitor of substrate B will also prevent the binding of A, resulting in nonhyperbolic behavior.

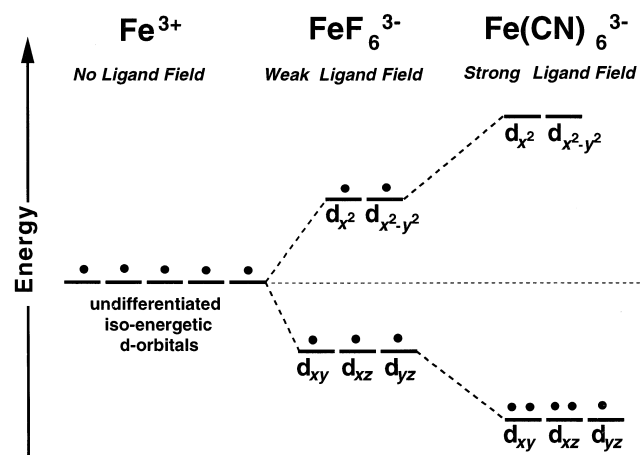
¹D. G. Cross & H. F. Fisher (1970) *J. Biol. Chem.* **245**, 2612.

²H. F. Fisher, R. E. Gates & D. G. Cross (1970) *Nature* **228**, 247.

LIGAND FIELD SPLITTING

The loss of degeneracy of atomic or molecular levels in a molecular entity with a given symmetry by the attachment or removal of ligands to produce reduced symmetries. Ligand field theory treats metal ligand complexation as a consequence of molecular orbital formation, whereas crystal field splitting considers ligands as point

charges that interact electrostatically with a central electropositive metal ion.



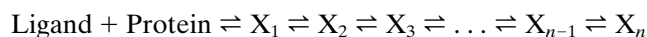
An example of ligand field theory is shown for the case of uncomplexed and complexed forms of Fe(III). Uncomplexed Fe^{3+} has 5 electrons occupying the 3d electron shell, one in each of the five isoenergetic orbitals. Upon complexation, each of the six negatively charged or electron-rich ligands are drawn toward the overall positively charged Fe(III) metal ion, leading to repulsion between the metal ion's 3d electrons and the electrons of the ligand. Recall that two of the 3d orbitals are geometrically disposed so that electrons in these orbitals will experience greater repulsion by ligands occupying their respective positions in a hexacoordinate complex. If the electron density of the ligands can repulsively interact with electrons in Fe(III), the 3d orbitals will split into high- and low-lying orbitals of different potential energy. For the FeF_6^{3-} complex, the extent to which the d orbitals split is relatively small, and the electrons remain unpaired by occupying each of the five d orbitals. With the $\text{Fe}(\text{CN})_6^{3-}$ complex, however, the degree of splitting is considerably greater. The interpretation is that the energy difference is of sufficient magnitude that all five d-orbital electrons now occupy the three lower energy orbitals. With an uneven number of d-orbital electrons, one electron remains unpaired. **See Crystal Field Splitting**

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Energy level diagram, **246**, 85-86, 109; experimental determination of parameters, **246**, 84-85, 87, 108; free ion single-electron spin-orbit coupling parameters, **246**, 102; molecular orbital theory, **246**, 86-87; orbital angular momentum operator, **246**, 94, 101; origin of spin Hamiltonian parameters, **246**, 100-104; resource materials, **246**, 16; strength of transitions, **246**, 76, 109; $^5\text{T}_{2g}$ ground state, **246**, 104-106; zero field splitting, **246**, 101, 103.

LIGAND-INDUCED CONFORMATIONAL CHANGE

Although Fischer's Lock-and-Key hypothesis for ligand binding strongly influenced scientific thought about enzyme-substrate and ligand-receptor interactions, there is abundant evidence that many ligands induce conformational changes, however large or subtle, within their macromolecular host. The mechanism of ligand binding can generally be written as a two-step process, although additional isomerizations may intervene:



The intermediate states X_1 , X_2 , X_3 , ..., X_{n-1} , and X_n can often be directly observed by fast reaction kinetic approaches, or their existence can be inferred from the magnitude of the observed equilibrium constants and reasonable limits on rate constants for ligand binding and release. Koshland¹ first proposed the "induced-fit" model to account for substrate-induced conformational changes that bring catalyzing groups into closer contact with reacting groups on the substrate. As originally propounded, the model did not consider that such conformational changes may serve either to introduce strain within the substrate or to stabilize the transition state configuration of a reactant. However, one can reasonably argue that the concept of a ligand-induced conformational change logically anticipates these related features of enzymic catalysis. **See also** *Linked Functions; Lock-and-Key Model of Enzyme Action; Mapping Substrate Interactions Using Substrate Data*

¹D. E. Koshland, Jr., (1958) *Proc. Natl. Acad. Sci. U.S.A.* **44**, 98.

LIGAND-INDUCED PROTEIN SELF-ASSOCIATION

When a binding protein exists in various oligomeric states linked by reversible reactions, differences in the strength of ligand-binding affinity by monomer and the higher oligomer species are often manifested as changes in the degree of oligomerization as the ligand concentration is altered. Such behavior constitutes an example of how the favorable Gibbs free energy of ligand binding ($\Delta G_{\text{binding}}$) creates a thermodynamic "pull" that alters the poise of the pre-existing the monomer $\rightleftharpoons$ oligomer equilibrium in a manner that leads to the accumulation of the latter.

Nichol and Winzor¹ described the binding equations that apply to such ligand-induced changes in receptor oligomerization. They also presented the following equation to describe the joint operation of allosteric ligand binding cooperativity and receptor self-association.

$$r = \frac{\sum_j \left\{ m_j \sum_{i=1}^{i=\tau_j} i \left\{ \prod_{l=1}^{l=i} K_{j,l} \right\} m_s^i \right\}}{\sum_j \left\{ j m_j \left(1 + \sum_{i=1}^{i=\tau_j} i \left\{ \prod_{l=1}^{l=i} K_{j,l} \right\} m_s^i \right) \right\}}$$

In this expression, j denotes the acceptor states (monomer, dimer, etc.), τ_j values describe the numbers of binding sites per molecule of the acceptor in each of its oligomeric states, the K 's are ligand binding equilibrium constants, and S is the ligand.

An excellent example is the binding of 5'-AMP to glycogen phosphorylase *b*, an event that is attended by conversion of inactive dimeric enzyme into a catalytically active tetramer resembling phosphorylase *a*. In what by now should be regarded as a classic paper in enzymology, Huang and Graves² used specific activity measurements and light scattering experiments to determine how AMP binding is coupled to tetramer formation and catalytic activity.

Another example is the ability of taxol to promote tubulin polymerization: taxol exhibits no detectable interaction with the $\alpha\beta$ -tubulin dimer; however taxol binds extremely tightly to microtubules. Howard and Timasheff³ examined tubulin polymerization into microtubules as well as tubulin-colchicine aggregation. The linkage free energy provided by taxol binding is approximately -3.0 kcal/mol $\alpha\beta$ -tubulin dimer, whereas this quantity is reduced to approximately -0.5 kcal/mol tubulin-colchicine, indicating that polymerization of the latter requires the expenditure of much more of the $\Delta G_{\text{binding}}$ to overcome likely steric hindrance and geometric strain.

¹L. W. Nichol & D. J. Winzor (1981) in *Protein-Protein Interactions*, (C. Frieden and L. W. Nichol, eds.) Wiley-Interscience, New York.

²C. Y. Huang & D. J. Graves (1970) *Biochemistry* **9**, 660.

³W. D. Howard & S. N. Timasheff (1988) *J. Biol. Chem.* **263**, 1342.

LIGASES

A major class of enzymes, also referred to as synthetases, that catalyze the joining of two entities with the concomitant hydrolysis of molecules such as ATP or GTP.

LIGHT POLARIZATION

A condition that is said to occur when the end point of the electric vector of the light moves along a certain path. Light is *linearly* polarized when the end point of the electric vector moves along a straight line in the direction of light propagation. Light is (a) *circularly* polarized if the end point traces a circle or (b) *elliptically* polarized if the electric vector moves in an elliptical path as the light propagates.

This method was the first accurate spectroscopic method for determining chemical reaction rates. In the mid-eighteenth century, kinetic measurements of changes in the rotation of plane polarized light upon acid-catalyzed hydrolysis of sucrose led to the concept of a dynamic equilibrium.

LIGHT SCATTERING

Classical elastic light scattering^{1,2} arises when light impinges on a molecule (the dimensions of which are much smaller than the wavelength of the light) and induces an oscillating dipole μ equal to the electric field strength E multiplied by the polarizability α . This oscillating dipole radiates light, and Rayleigh showed that for N identical particles the intensity I of the scattered light is given by:

$$I = I_0 N \{ (8\pi^2 \alpha^2 / \lambda^4 r^2) (1 + \cos^2 2\theta) \}$$

where I_0 is the intensity of the light beam per square centimeter, λ is the wavelength of light, r is the distance in centimeters from the sample to detector, and 2θ is the angle between the incident and the scattered light rays. The inverse fourth power dependence of I on wavelength (or in other words, its fourth power dependence on frequency) explains why shorter wavelength light is scattered far more effectively than light of longer wavelength.

In practice, depending on the angle between the incident and scattered light paths, one can make several types of light scattering measurements:

(a) Turbidity is a parameter relating the intensity I of the light passing through the sample to a photomultiplier tube at $\theta = 0$ (*i.e.*, along its original path between the light source and the light trap) to the incident light intensity I_0 . The following equation for turbidity is mathematically analogous to that describing absorbance:

$$I/I_0 = \exp(-\tau x)$$

where τ is the turbidity and x is the path length of the sample cuvette. Turbidity is related to the so-called Rayleigh ratio ($R_\theta' = I_\theta x^2/I_0$) at 90° as follows:

$$R_\theta' = (16\pi/3)R_{90}'$$

Turbidity has proven to be especially useful in studies of protein polymerization, where one can demonstrate that the turbidity is directly proportional to the polymer mass concentration. This is illustrated in the following plot (Fig. 1) obtained for assembled microtubules.

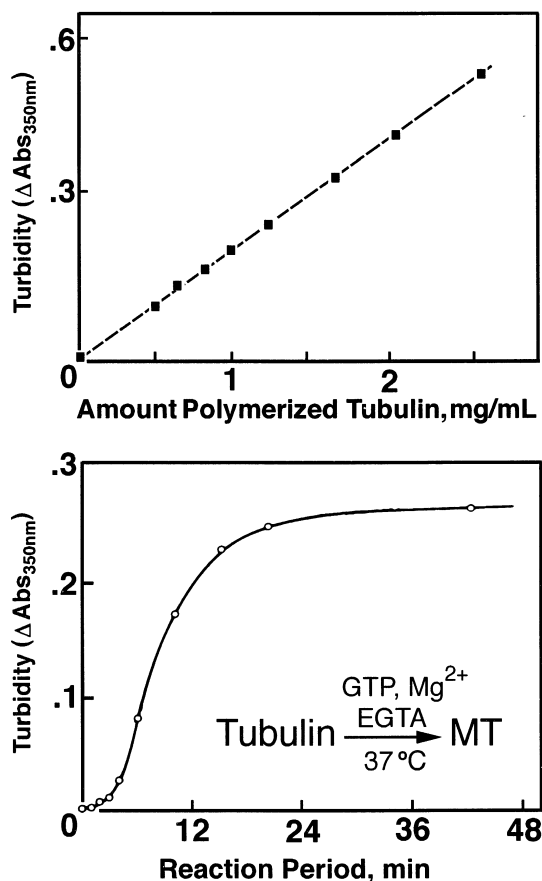


Figure 1. Top: Turbidity, measured at 350 nm, as a function of microtubule polymer mass concentration (expressed as mg/mL polymerized tubulin). Tubulin solutions of varying concentrations were polymerized until they reached stable plateau values in a Cary 118C spectrophotometer. Each sample was then transferred to an ultracentrifuge tube, and microtubules were pelleted, separated from the unpolymerized tubulin in the supernatant fraction, and then resuspended for protein concentration determination. The corresponding turbidity and polymer mass concentrations are plotted here. Bottom: Time-course of tubulin polymerization assayed by turbidity. Reproduced from MacNeal and Purich³ with permission from the American Society for Biochemistry and Molecular Biology.

(b) One can also measure the intensity of light scattering when the photomultiplier is positioned at an angle other

than 0° , and the detector is set at 90° as shown in Fig. 2. The angular dependence of scattered light intensity can provide valuable information about the hydrodynamic and thermodynamic properties of macromolecules.

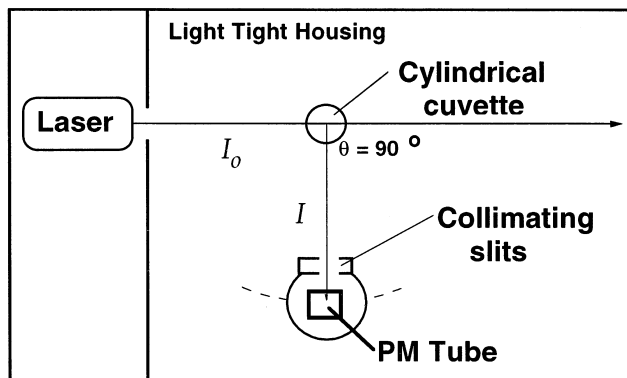


Figure 2. General layout of a light scattering apparatus. Light emitted from a light source impinges on a sample located in a cylindrical cell, and one can adjust the position of the photomultiplier (here shown at 90° relative to the transmitted light beam) along an arc centered at the midpoint of the sample cell.

Zimm⁴ demonstrated that when the scattering particle dimensions are not small relative to the wavelength of the incident light, one can use light scattering to infer important information about the size and shape of a macromolecule as well as the interactions of macromolecules. Accurate determination of molecular weight using light scattering requires the use of a double-extrapolation method known as a Zimm plot. Because this Handbook focuses on kinetic behavior, interested readers will wish to consult excellent discussions by Tanford¹ and Cantor and Schimmel². See *Actin Assembly Assays; Microtubule Assembly Kinetics; Fluorescence*

¹C. Tanford (1961) *Physical Chemistry of Macromolecules*, pp. 275-316, Wiley, New York.

²C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, Part II, pp. 838-842, Freeman, San Francisco.

³R. K. MacNeal & D. L. Purich (1978) *J. Biol. Chem.* **253**, 4683.

⁴B. H. Zimm (1948) *J. Chem. Phys.* **16**, 1093.

LIGHT SCATTERING (DYNAMIC)

Time-dependent fluctuations in the intensity and spectrum of light scattered as a consequence of fluctuations in the number of scattering molecules within a small aperture. Doppler shifts resulting from molecular motions also alter the wavelength of scattered radiation. The technique for measuring various dynamic light scattering is widely known as quasi-elastic light scattering (or QELS), because the motions of the scattering centers are not totally elastic. The scattered light spectrum produced by

Brownian motion of a single molecular species results in a Lorentzian spectrum defined by the relationship:

$$I(S, n) = \langle N \rangle \pi^{-1} \{ S^2 D / (4\pi^2 \nu^2 + (S^2 D)^2) \}$$

where I is the intensity, S is $2 \sin \theta / \lambda$, ν is the frequency of incident light, $\langle N \rangle$ is the number average of molecules in the sampled volume, and D is the diffusion coefficient. The half-width of the Lorentzian-shaped frequency distribution (or spectrum) of the scattered light equals $S^2 D$, and the diffusion constant can be used to determine molecular weight. Considerable effort has been devoted to understanding how macromolecules in real solutions affect the intensity and spectrum of scattered light.

C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, Part II, pp. 839-843, Freeman, San Francisco.

LIGNAN PEROXIDASE

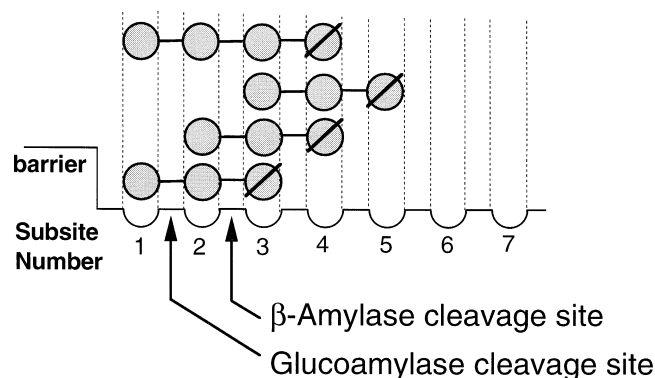
This heme-dependent enzyme [EC 1.11.1.14], also known as diarylpropane peroxidase, diarylpropane oxygenase, and ligninase I, catalyzes the reaction of 1,2-bis(3,4-dimethoxyphenyl)propane-1,3-diol with hydrogen peroxide to produce veratraldehyde, 1-(3,4-dimethylphenyl)ethane-1,2-diol, and four water molecules. The enzyme brings about the oxidative cleavage of C—C bonds in a number of model compounds and also oxidizes benzyl alcohols to aldehydes or ketones.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

F. P. Guengerich (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 97.

T. K. Kirk, M. Tien, P. J. Kersten, B. Kalyanaraman, K. E. Hammel & R. L. Farrell (1990) *Meth. Enzymol.* **188**, 154.

LIMIT DEXTRAN



Diagrammatic representation of the individual saccharide unit sub-sites and cleavage sites in β -amylase and glucoamylase.

A oligosaccharide that cannot be degraded further by a polysaccharide hydrolase or polysaccharide phosphoryl-

ase. The types of limit dextrans formed by various amylases and phosphorylases can provide useful information about ligand binding at the active site, and the structure of a limit dextran is also useful for recognizing and identifying new enzyme activities.

LIMITING ISOTOPE EFFECT

A term used in the study of isotope effects in enzyme-catalyzed reactions, namely $^D V_{\max}/K_m$ (equivalent to $^H V_{\max}/^H K_m / (^D V_{\max}/^D K_m)$) and $^D V_{\max}$ (equivalent to $^H V_{\max}/^D V_{\max}$).

D. B. Northrop (1982) *Meth. Enzymol.* **87**, 607.

LINEAR COMBINATION OF ATOMIC ORBITALS

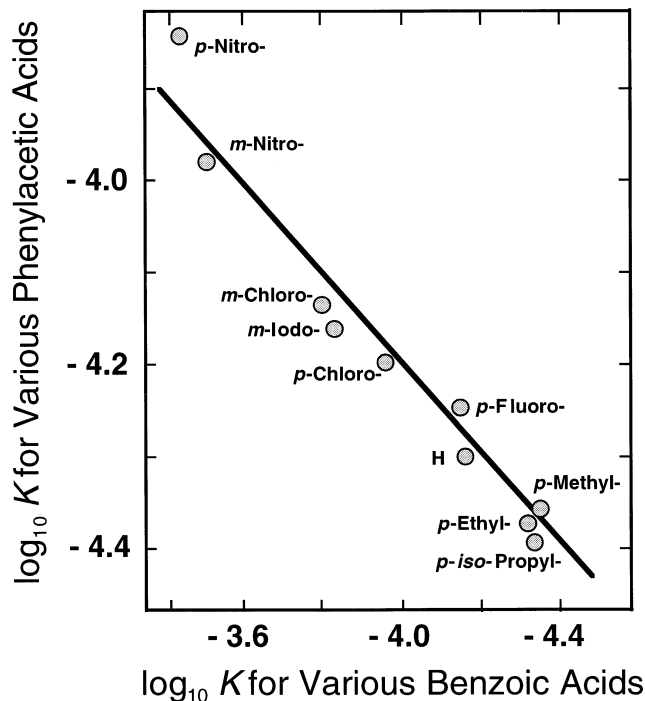
A theoretical approach in molecular orbital studies to formulate an expression for the wave function of a molecular orbital (both for bonding and antibonding orbitals) by linear combinations of the overlapping atomic orbitals with appropriate weighting factors.

LINEAR FREE-ENERGY CORRELATIONS

A plot of the logarithm of a rate constant (or an equilibrium constant) for one series of reactions *versus* the logarithm of the rate constant (or the equilibrium constant) for a related series of reactions. (Recall that at constant temperature and pressure the logarithm of an equilibrium constant is proportional to ΔG° , and the logarithm of a rate constant is proportional to $\Delta G^\ddagger$). An example of a linear free energy relationship is provided by the Hammett $\sigma\rho$ -equation. With equilibrium constants, this relationship is given by the expression:

$$\log(K_X/K_H) = \sigma_X \rho$$

where K_H is the equilibrium constant where the substituent is H, K_X is the equilibrium constant where the substituent is X, σ_X is a constant characteristic of the substituent X, and ρ is a constant for a given reaction under a set of conditions (**See Rho Value**). Recall that $\Delta G_X^\circ = -RT \ln K_X$ and $\Delta G_H^\circ = -RT \ln K_H$. Thus, the Hammett equation can be rewritten as $[-\Delta G_X/(2.3RT)] + [\Delta G_H/(2.3RT)] = \sigma\rho$; or, $-\Delta G_X = \sigma\rho 2.3RT - \Delta G_H$. Hence, for a given set of conditions (*i.e.*, ΔG_H , ρ , R , T are all constant) σ is linear with respect to ΔG_X .



Other examples of linear free energy relationships include the Brønsted relation, the Grunwald-Winstein relationship, and the Taft equation. **See Rho Value**

LINEAR SOLVATION ENERGY RELATIONSHIPS

Equations containing a number of solvent parameters in linear or multiple linear regression and expressing the effect of the solvent on the rate of the reaction or the thermodynamic equilibrium constant. **See E_T Values**; Kamlet-Taft Solvent Parameter; Koppel-Palm Solvent Parameter; Z Value

LINEAR STRAIN

A dimensionless quantity, symbolized by either ϵ or e , for the change in length (l) due to some force or interaction per unit length: thus, $\epsilon = \Delta l/l$. Linear strain is also known as longitudinal strain. Relative elongation is also measured by $\Delta l/l$.

LINE-SHAPE ANALYSIS

Determination of chemical exchange rate constants based on the shapes of spectroscopic lines (frequently NMR resonances) of dynamic processes.

LINEWEAVER-BURK PLOT

See Double-Reciprocal Plot

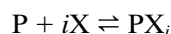
LINKED FUNCTION

If a macromolecule engages in two different association reactions with ligands X and Y at nearby (or structurally linked) binding sites, there arises the likelihood that the binding functions for these ligands will exhibit an interdependence that stems from the interaction between these sites. Such sites are said to be “linked,” and the saturation function describing their binding interactions is called a linked function.

The following treatment of linked function behavior closely follows Wyman’s approach^{1,2} where one considers a macromolecule P having q sites, each able to combine with one molecule of ligand X. One can denote X as the total concentration of ligand bound by the macromolecule:

$$X = [P_0] \sum_{i=0}^q i K_i x^i$$

where $[P_0]$ is the concentration of the uncomplexed macromolecule, x is the activity of the ligand, and K_i is the apparent macroscopic association constant for the reaction:



Note also that $K_0 = 1$. The total concentration of macromolecule in all forms is:

$$P = [P_0] + [P_1] + [P_2] + \cdots + [P_q]$$

$$P = [P_0] \sum_{i=0}^q K_i x^i$$

Using $\bar{X}$ to designate the amount of ligand bound per mole of macromolecule, we get:

$$\bar{X} = \sum_i i K_i x^i / \sum_i K_i x^i$$

If the K_i ’s are independent of the total extent of the association reaction, then the above expression can be written in the following form:

$$\bar{X} = \frac{\partial \ln \sum K_i x^i}{\partial \ln x}$$

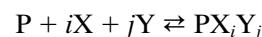
The fractional saturation of the macromolecule (written here as $\bar{x}$) is then:

$$\bar{x} = \bar{X}/q = \{d \ln \sum K_i x^i / d \ln x\} / q$$

A second ligand binding to another nearby site, then the total concentration of the macromolecule in all its forms will be:

$$P = [P_0] \sum_{i=0}^q \sum_{j=0}^r K_{ij} x^i y^j$$

where K_{ij} is the apparent equilibrium constant for the reaction:



This means that we can now write separate expressions for the amounts of ligands X and Y bound per mole of macromolecule:

$$\bar{X} = \frac{\partial \ln \sum_{i=0}^q \sum_{j=0}^r K_{ij} x^i y^j}{\partial \ln x} \quad \text{and} \quad \bar{Y} = \frac{\partial \ln \sum_{i=0}^q \sum_{j=0}^r K_{ij} x^i y^j}{\partial \ln y}$$

Letting $\Sigma\Sigma$ represent the double sum, we take advantage of the fact that $d \ln \Sigma\Sigma$ is a perfect differential that yields

$$\left(\frac{\partial \bar{X}}{\partial \ln y} \right)_x = \left(\frac{\partial \bar{Y}}{\partial \ln x} \right)_y$$

This expression is called a linked function and indicates how the binding of ligands at nearby sites can influence each other. **See also** Basic Regulatory Kinetics; Cooperativity; Allosterism; Feedback Effectors; Bohr Effect; Hemoglobin; Le Chatelier’s Principle; Adair Equation

¹J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, pp. 654-660, Academic Press, New York.

²J. Wyman & S. J. Gill (1990) *Binding and Linkage*, Univ. Sci. Books, Mill Valley, CA.

LINOLEATE ISOMERASE

This enzyme [EC 5.2.1.5] catalyzes the interconversion of 9-*cis*,12-*cis*-octadecadienoate and 9-*cis*,11-*trans*-octadecadienoate.

S. Seltzer (1972) *The Enzymes*, 3rd ed., **6**, 381.

C. R. Kepler & S. B. Tove (1969) *Meth. Enzymol.* **14**, 105.

LIPASES

Triacylglycerol lipases [EC 3.1.1.3] (also known as triglyceride lipases, tributyrases, or simply as lipases) catalyze the hydrolysis of a triacylglycerol to produce a diacylglycerol and a fatty acid anion. The pancreatic enzyme acts only on an ester-water interface; the outer ester links in the substrate are the ones which are preferentially

hydrolyzed. **See** *Lipoprotein Lipase; Micelles; and specific lipase.*

- D. M. Quinn & S. R. Feaster (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 455.
 M. E. Lowe (1997) *Ann. Rev. Nutr.* **17**, 141.
 Z. S. Derewenda (1994) *Adv. Protein Chem.* **45**, 1.
 L. C. Smith, F. Faustinella & L. Chan (1992) *Curr. Opin. Struct. Biol.* **2**, 490.
 H. Okuda (1991) in *A Study of Enzymes* (S. A. Kuby, ed.) **2**, 579.
 P. Strålfars, H. Olsson & P. Belfrage (1987) *The Enzymes*, 3rd ed., **18**, 147.
 M. A. Wells & N. A. DiRenzo (1983) *The Enzymes*, 3rd ed., **16**, 113.
 R. L. Jackson (1983) *The Enzymes*, 3rd ed., **16**, 141.
 J. C. Khoo & D. Steinberg (1983) *The Enzymes*, 3rd ed., **16**, 183.
 R. Verger (1980) *Meth. Enzymol.* **64**, 340.
 M. Sémériva & P. Desnuelle (1979) *Adv. Enzymol.* **48**, 319.
 P. Desnuelle (1972) *The Enzymes*, 3rd ed., **7**, 575.

LIPID ACTIVATION

Stimulation in the activity of enzymes that are normally associated with biomembranes by general or specific lipids within that membrane. If this activation phenomenon is relatively nonspecific, the term most often applied is *interface activation*¹.

¹R. Verger (1980) *Meth. Enzymol.* **64**, 340.

LIPID PHASE TRANSITION KINETICS

An emergent field dealing with the physical/chemical processes that underlie the changes of lipid phase state during such cellular events as membrane fusion, vesicle trafficking, and cell disjunction.

M. Caffrey (1989) *Ann. Rev. Biophys. Biophys. Chem.* **18**, 159.

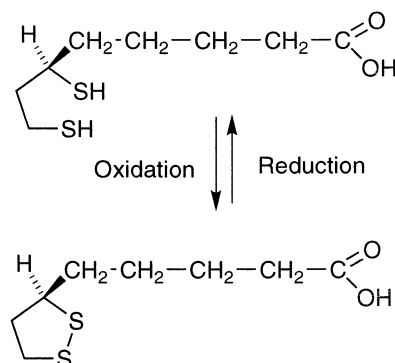
LIPID TRACER KINETICS

Wolfe¹ has presented an excellent description of the systematic application of stable and radioactive isotope tracers in determining the kinetics of intestinal fat absorption, hepatic triglyceride synthesis, lipid mobilization, triglyceride-fatty acid recycling, and cholesterol turnover.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, pp. 317-340, Wiley-Liss, New York.

LIPOIC ACID AND DERIVATIVES

An acyl-transfer and redox coenzyme containing two sulfhydryl groups that form a dithiolane ring in the oxidized (disulfide) form. The redox potential at pH 7 is -0.29 volts. Lipoic acid is attached to the ϵ -amino group of lysyl residues of transacetylases (subunit of α -ketoacid dehydrogenase complexes), thereby permitting acyl



transfer as a swinging arm akin to that of carboxybiotin.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Turbidimetric and polarographic assays for lipoic acid using mutants of *Escherichia coli*, **18A**, 269; chemical syntheses of ¹⁴C- α -lipoic acid, turbidimetric and polarographic assays for lipoic acid using mutants of *Escherichia coli*, **18A**, 269; lipoic acid transport, **18A**, 276; lipoic acid activation, **18A**, 282; purification and properties of lipoamidase, **18A**, 292; purification and properties of lipoamide dehydrogenases from pig heart α -keto acid dehydrogenase complexes, **18A**, 298; chromatographic and spectral properties of lipoic acid and its metabolites, **62**, 129; thin-layer chromatography of lipoic acid, lipoamide, and their persulfides, **62**, 135; preparation and separation of S-oxides of α -lipoic acid, **62**, 137; photochemical preparation of acyl lipoic acids, **62**, 145; lipoyl disulfide reducing polymers, **62**, 147; biosynthesis of lipoic acid via unsaturated fatty acids, **62**, 152; turnover of protein-bound lipoic acid, **62**, 159; the radial diffusion assay of lipoamide dehydrogenase, **62**, 166; asparaguate dehydrogenase and lipoyl dehydrogenase from asparagus, **62**, 172; preparation and determination of asparagusic acid, **62**, 18; methodology employed for anaerobic spectrophotometric titrations and for computer-assisted data analysis, **62**, 185.

LIPHILIC

Referring to bodies, substances, molecular entities, or portions of molecular entities that have a tendency to be soluble in apolar solvents. **See** *Hydrophobic Interaction*

LIPOLYPSACCHARIDE

N-ACETYLGLUCOSAMINYLTRANSFERASE

This enzyme [EC 2.4.1.56] catalyzes the reaction of UDP-*N*-acetyl-D-glucosamine with a lipopolysaccharide to produce UDP and an *N*-acetyl-D-glucosaminyl-lipopolysaccharide. Thus, this enzyme transfers *N*-acetylglucosaminyl residues to a D-galactose residue in the partially completed lipopolysaccharide core.

M. J. Osborn & L. I. Rothfield (1966) *Meth. Enzymol.* **8**, 456.

LIPOLYPSACCHARIDE GALACTOSYLTRANSFERASE

This enzyme [EC 2.4.1.44] catalyzes the reaction of UDP-galactose with a lipopolysaccharide to produce UDP and

a D-galactosyl-lipopolysaccharide. Thus, the enzyme transfers D-galactosyl residues to D-glucose in the partially completed core of a lipopolysaccharide.

M. J. Osborn & L. I. Rothfield (1966) *Meth. Enzymol.* **8**, 456.

LIPOPOLYSACCHARIDE GLUCOSYLTRANSFERASE

Lipopolysaccharide glucosyltransferase I [EC 2.4.1.58] catalyzes the reaction of UDP-glucose with a lipopolysaccharide to produce UDP and a D-glucosyl-lipopolysaccharide. Thus, this enzyme transfers glucosyl residues to the backbone portion of a lipopolysaccharide. Lipopolysaccharide glucosyltransferase II [EC 2.4.1.73] catalyzes the reaction of UDP-glucose with a lipopolysaccharide to produce UDP and a D-glucosyl-lipopolysaccharide. Thus, this enzyme transfers glucosyl residues to the D-galactosyl-D-glucosyl side chains in the partially completed core of a lipopolysaccharide.

M. J. Osborn & L. I. Rothfield (1966) *Meth. Enzymol.* **8**, 456.

LIPOPROTEIN LIPASE

This enzyme [EC 3.1.1.34] (also called clearing factor lipase, diglyceride lipase, and diacylglycerol lipase) catalyzes the hydrolysis of a triacylglycerol to produce a diacylglycerol and a fatty acid anion. This enzyme hydrolyzes triacylglycerols in chylomicrons and in low-density lipoproteins and also acts on diacylglycerols. **See also** *Lipases*

R. Potenz, J.-Y. Lo, E. Zsigmond, L. C. Smith & L. Chan (1996) *Meth. Enzymol.* **263**, 319.

H. Okuda (1991) *A Study of Enzymes* **2**, 579.

A. Bensadoun (1991) *Ann. Rev. Nutr.* **11**, 217.

R. L. Jackson (1983) *The Enzymes*, 3rd ed., **16**, 141.

P. Desnuelle (1972) *The Enzymes*, 3rd ed., **7**, 575.

LIPOXYGENASE

Lipoxygenase [EC 1.13.11.12] catalyzes the reaction of linoleate with dioxygen to produce (9Z,11E)-(13S)-13-hydroperoxyoctadeca-9,11-dienoate. This iron-dependent enzyme can also oxidize other methylene-interrupted polyunsaturated fatty acids. **See also** *specific enzyme*

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

B. J. Gaffney (1996) *Ann. Rev. Biophys. Biomol. Struct.* **25**, 431.

A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.

J. N. Siedow (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 145.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

T. Schewe, S. M. Rapoport & H. Kühn (1986) *Adv. Enzymol.* **58**, 191.

H. Kühn, T. Schewe & S. M. Rapoport (1986) *Adv. Enzymol.* **58**, 273.

O. Hayaishi, M. Nozaki & M.T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

5-LIPOXYGENASE

This iron-dependent enzyme [EC 1.13.11.34], better known as arachidonate 5-lipoxygenase and occasionally referred to as leukotriene A₄ synthase, catalyzes the reaction of arachidonate with dioxygen to produce (6E,8Z,11Z,14Z)-(5S)-5-hydroperoxyicos-6,8,11,14-tetraenoate, which rapidly converts to leukotriene A₄.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

A. W. Ford-Hutchinson, M. Gresser & R. N. Young (1994) *Ann. Rev. Biochem.* **63**, 383.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

T. Schewe, S. M. Rapoport & H. Kühn (1986) *Adv. Enzymol.* **58**, 191.

H. Kühn, T. Schewe & S. M. Rapoport (1986) *Adv. Enzymol.* **58**, 273.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

8-LIPOXYGENASE

This iron-dependent enzyme [EC 1.13.11.40], better known as arachidonate 8-lipoxygenase, catalyzes the reaction of arachidonate with dioxygen to produce (5Z,9E,11Z,14Z)-(8R)-8-hydroperoxyicos-5,9,11,14-tetraenoate.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

12-LIPOXYGENASE

This iron-dependent enzyme [EC 1.13.11.31], better known as arachidonate 12-lipoxygenase, catalyzes the reaction of arachidonate with dioxygen to produce (5Z,8Z,10E,14Z)-(12S)-12-hydroperoxyicos-5,8,10,14-tetraenoate, which then converts rapidly to the corresponding 12S-hydroxy compound.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

T. Schewe, S. M. Rapoport & H. Kühn (1986) *Adv. Enzymol.* **58**, 191.

H. Kühn, T. Schewe & S. M. Rapoport (1986) *Adv. Enzymol.* **58**, 273.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

15-LIPOXYGENASE

This iron-dependent enzyme [EC 1.13.11.33], better known as arachidonate 15-lipoxygenase or arachidonate ω -6-lipoxygenase, catalyzes the reaction of arachidonate with dioxygen to produce (5Z,8Z,11Z,13E)-(15S)-15-hydroperoxyicosa-5,8,11,13-tetraenoate, which rapidly converts to the corresponding 15S-hydroxy compound.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

LOCK-AND-KEY MODEL OF ENZYME ACTION

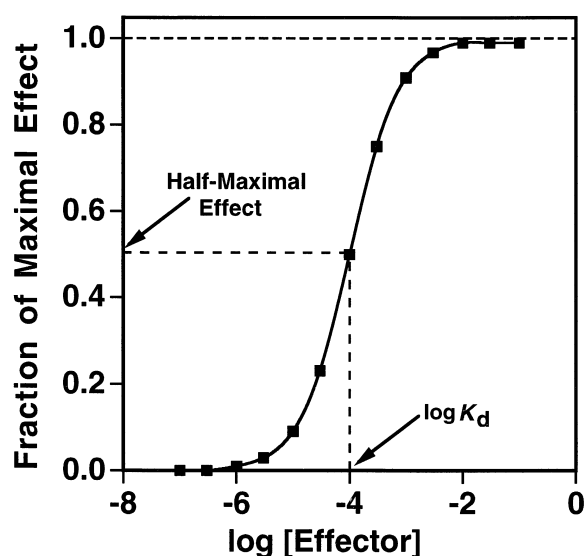
The original concept¹ offered to explain why enzymes exhibit such a high degree of substrate specificity. The active site of an enzyme was viewed as a topological template for a particular reactant; hence, there is an enzyme-substrate complementarity^{2,3}. **See** *Induced-Fit Model; Ligand-Induced Conformational Change*

¹E. Fischer (1894) *Ber.* **27**, 2985.

²A. Fersht (1977) *Enzyme Structure and Mechanism*, 2nd. ed., Freeman, San Francisco.

³A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

LOG-DOSE RESPONSE CURVE FOR ENZYME COOPERATIVITY



A method for graphically representing cooperativity in which the velocity (or percent response) is plotted as a

function of the logarithm of the substrate concentration (or concentration of the binding ligand)^{1,2}. Such curves are very commonly used in pharmacological studies.

A log dose-response curve affords an investigator an easy method for assessing cooperativity. By comparing the concentration of the ligand that yields a 90% response (or 90% $V_{\max}$) to the concentration that generates a 10% response, the ratio obtained (the R_s value) will provide information on the degree of cooperativity ($R_s < 81$ is positive cooperativity and $R_s > 81$ for negative cooperativity). The R_s value and the Hill coefficient are mathematically related: $R_s = 81^{1/n_H}$.

¹K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

²K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

LOG VELOCITY VERSUS LOG[SUBSTRATE] PLOT

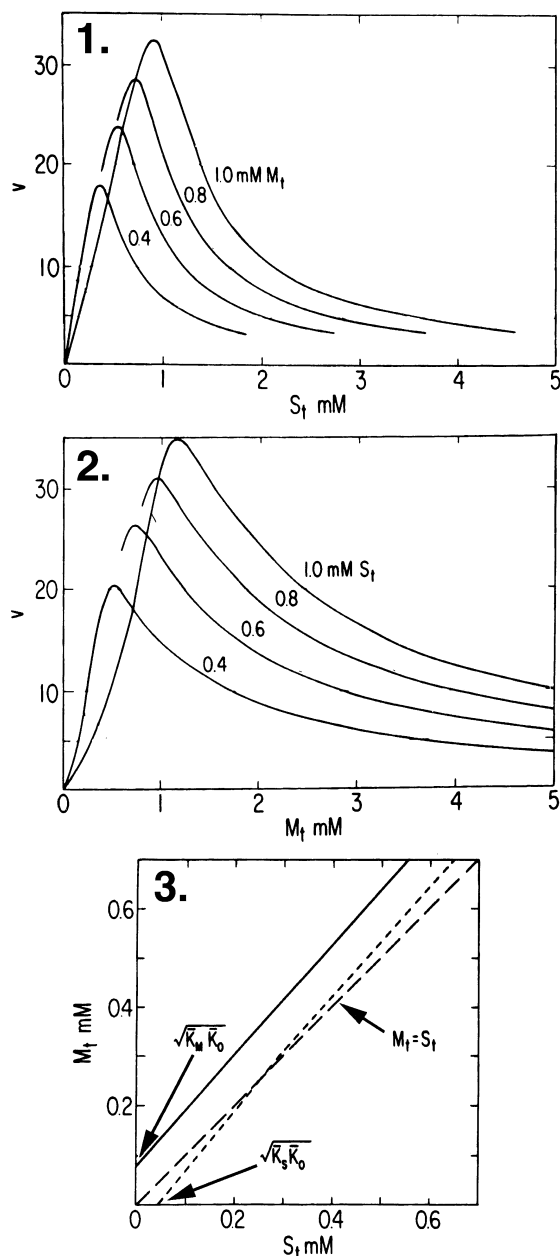
A plot that permits the characterization of enzyme rate behavior over a large range of substrate concentrations: the logarithm of the initial steady-state velocity is plotted as a function of the logarithm of the substrate concentration. If the enzyme obeys Michaelis-Menten kinetics, then a single S-shaped curve will be obtained, and the slope at the mid-point will be equal to one. If a sample contains two enzymes displaying markedly different Michaelis constants (*i.e.*, differing by a factor of seven or more), and if there are comparable amounts of each enzymic activity, then a log velocity-substrate plot may exhibit two S-shaped curves. Enzymes that display negative or positive cooperativity will give slopes other than unity.

LONDON FORCES

Attractive forces (sometimes also referred to as dispersion forces) between apolar molecules arising from the mutual polarizability of the interacting molecules. London forces also contribute to the interactive forces between polar molecules. **See also** *van der Waals Forces; Noncovalent Interactions*

LONDON-STECK PLOT

A graphical method¹ for analyzing enzyme activation (a metal ion for the case shown above). Initial velocities are plotted in Fig. 1 as a function of the total substrate (S) concentration at different, constant concentrations of the activator (A). An analogous plot (Fig. 2) is made



versus the activator concentration at different fixed concentrations of the substrate. If the ES complex cannot proceed to form products and the true substrate is the SA complex (for example, the binding of a required metal ion to a nucleotide), then the two plots will consist of curves in which the velocity increases to a maximum value, then decreases. If these maximum velocities are replotted (Fig. 3) as a function of the corresponding substrate concentration (or activator concentration), a new curve is generated which has a horizontal intercept that is a function of the dissociation constant for the SA complex and the dissociation constant for either the EA

complex or the ES complex. This analysis assumes rapid equilibrium binding.

¹W. P. London & T. L. Steck (1969) *Biochemistry* **8**, 1767.

LONE PAIR

A pair of nonbonding electrons localized in the valence shell on a single atom. Examples are the lone electron pair on the nitrogen atom in ammonia and the two lone pairs on the oxygen atom in water. In these cases, the lone pairs participate in hydrogen bonding interactions.

LONG-CHAIN ACYL-CoA DEHYDROGENASE

This enzyme [EC 1.3.99.13] catalyzes the reaction of a long-chain acyl-CoA with an electron-transferring flavoprotein to produce a 2,3-dehydroacyl-CoA and the reduced electron-transferring flavoprotein.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

LONG-CHAIN ALDEHYDE DEHYDROGENASE

This enzyme [EC 1.2.1.48] catalyzes the reaction of a long-chain aldehyde with NAD^+ to produce a long-chain acid anion and NADH. The best substrate is reported to be dodecylaldehyde.

M. Ueda & A. Tanaka (1990) *Meth. Enzymol.* **188**, 176.

LONG-CHAIN FATTY ACYL-CoA SYNTHETASE

Long-chain fatty acyl-CoA synthetase [EC 6.2.1.3] catalyzes the reaction of ATP with a long-chain carboxylic acid and coenzyme A to produce an acyl-CoA, AMP, and pyrophosphate. While utilizing a wide range of long-chain saturated and unsaturated fatty acids as substrates, enzymes from different tissues vary in their specificity.

J. C. Londesborough & L. T. Webster, Jr. (1974) *The Enzymes*, 3rd ed., **10**, 469.

LONG-RANGE LOOPS

Initial protein folding intermediates formed as a consequence of nonlocal interactions on the reaction path from unfolded to partially folded states. These loops are likely to determine the trajectory of later folding steps. The character of nonlocal interactions is likely to be much different than structures inferred from the results

of studies on unfolded states formed by perturbing an already well-folded protein. In the latter case, one implicitly assumes that the structure and reactivity of the species formed by the action of a pertubant on a folded protein are in fact related to structures formed in reactions that commence from a globally unfolded state. Ittah and Haas¹ provide an excellent account of how nonlocal interactions may act to stabilize long-range loops in the initial folding of reduced bovine pancreatic trypsin inhibitor.

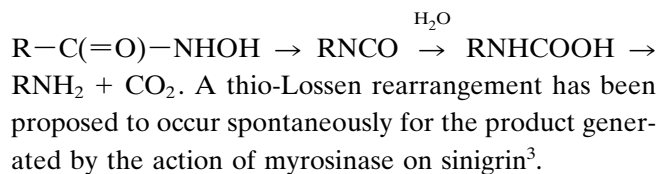
¹V. Ittah & E. Haas (1995) *Biochemistry* **34**, 4493.

LOOSE ION PAIR

An ion pair in which the constituent ions are separated by one or more solvent (or other neutral) molecules. If X^+ and Y^- represent the constituent ions, a loose ion pair is usually symbolized by $X^+||Y^-$. The constituent ions of a loose ion pair can readily exchange with other ions in solution; this provides an experimental means for distinguishing loose ion pairs from tight ion pairs. In addition, there are at least two types of loose ion pairs: solvent-shared and solvent-separated. **See** *Ion Pair*; *Tight Ion Pair*; *Solvent-Shared Ion Pair*; *Solvent-Separated Ion Pair*

LOSSEN REARRANGEMENT

Originally, the Lossen rearrangement specifically referred to the conversion of a hydroxamic acid to an amine with the loss of carbon dioxide^{1,2}:



¹W. Lossen (1872) *Ann.* **161**, 347.

²W. Lossen (1874) *Ann.* **175**, 271 and 313.

³G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

LOW-BARRIER HYDROGEN BONDS (Potential Role In Catalysis)

Low-barrier hydrogen bonds (abbreviated: LBHBs) are short, very strong hydrogen bonds formed when the electron-pair donors (most commonly oxygen or nitrogen atoms in proteins) sharing the hydrogen have comparable pK_a values.

Hydrogen bonds involve the sharing of a hydrogen atom by two electronegative partners by way of electrostatic and dispersion energy interactions. Pauling identified the properties of typical hydrogen bonds in polypeptides: (a) they are formed between electronegative atoms (most often amide hydrogens and amide carbonyls); (b) the strongest hydrogen bonds are linear, and any bending can severely lower their bond strength; and (c) they have bond lengths between 2.8 and 3.4 Å. When the bond lengths are compared to the sum of the van der Waals radii, one often observes a 10–25% reduction in distance upon hydrogen bond formation. There is an activation barrier for hydrogen “transfer” from one of the electronegative bonding partners to the other within a hydrogen bond. This barrier decreases as the difference in the pK_a values of the electron pair donors become comparable, hence the designation low-barrier hydrogen bonds. LBHBs have bond lengths of less than 2.5 Å. By contrast, weaker hydrogen bonds are formed when the pK_a values do not match.

The presence of LBHBs within enzyme active sites may partly explain the origin of enzymic rate enhancements^{1,2}. Frey *et al.*¹ obtained evidence with chymotrypsin and model compounds suggesting that a low-barrier hydrogen bond may be a mechanistic feature in serine protease catalysis. Cleland & Kreevoy³ have suggested that formation of low-barrier hydrogen bonds may supply 10–20 kilocalories per mole (or, 40–84 kJ/mol), thus facilitating difficult reactions such as enolization of carboxylate groups. The idea is that strong, low-barrier hydrogen bonds may greatly influence the transition state configuration. Induced conformational changes that more closely match the pK_a values may give rise to a strong, low-barrier bond: this is especially important when the pK_a values refer to those of a transition state or activated enzyme intermediate.

More recently, Cassidy *et al.*⁴ conducted additional ¹H NMR experiments to evaluate the basicities of the dyad H57-D102 in the tetrahedral complexes of chymotrypsin with the peptidyl trifluoromethyl ketones (TFKs): *N*-acetyl-L-Leu-DL-Phe-CF₃ and *N*-acetyl-DL-Phe-CF₃. The proton bridging His-57 and Asp-102 is part of a low-barrier hydrogen bond (LBHBs). In ¹H NMR spectra at pH 7.0, these protons appear at δ 18.9 and 18.6 ppm,

and the pK_a values of the dyads are 12.0 ± 0.2 and 10.8 ± 0.1 , respectively. The available structural data on chymotrypsin and other serine proteases, along with the high pK_a values of the hemiketals formed with TFKs, suggest LBHB formation in catalysis arises through a substrate-induced conformational transition leading to steric compression between His-57 and Asp-102. The LBHB's N-to-O distance is shorter than the van der Waals contact distance, and the LBHB may stabilize the tetrahedral intermediate. In their mechanistic proposal, substrate-induced steric compression within the dyad increases the basicity of N(ϵ)2 in His-57, rendering it more effective as a base to abstract a proton from Ser-195 in tetrahedral intermediate formation. In this model, the energy for conformational compression may be provided by binding interactions remote from the scissile bond.

Warshel *et al.*⁵ offer the argument that energy considerations suggest that low-barrier hydrogen bonds do not offer any special catalytic advantage as compared to ordinary hydrogen bonds. They evaluated the extent of transition state (TS) stabilization that might be achieved by low-barrier hydrogen bonds within enzyme active sites. They focused on three issues: (a) low-barrier hydrogen bonds are less stable than regular hydrogen bonds in water; (b) transition states are more stable in the enzyme active sites than in water, and (c) a nonpolar active site would destabilize the TS relative to its energy in water. By considering these points and related observations in a physically consistent frame-work, Warshel *et al.*⁵ are drawn to conclude that the occurrence of low-barrier hydrogen bonds cannot stabilize the TS more effectively than ordinary hydrogen bonds. **See also** *Hydrogen Bond; Water*

¹P. A. Frey, S. A. Whitt & J. B. Tobin (1994) *Science* **264**, 1927.

²J. A. Gerlt, M. M. Kreevoy, W. W. Cleland & P. A. Frey (1997) *Chem. Biol.* **4**, 259.

³W. W. Cleland & M. M. Kreevoy (1994) *Science* **264**, 1887.

⁴C. S. Cassidy, J. Lin & P. A. Frey (1997) *Biochemistry* **36**, 4576.

⁵A. Warshel & A. Papazyan (1996) *Proc. Natl. Acad. Sci. U.S.A.* **93**, 13665.

LUCIFERASE

The firefly enzyme (EC 1.13.12.7) catalyzes the intermediate formation of D(–)-luciferyl adenylate and pyrophosphate from D(–)-luciferin and ATP, followed by the oxidative reaction of the acyl adenylate with molecular oxygen to form an enzyme-bound product in the excited

state^{1–3}. Decay of this species to the ground state is attended by photon emission at a wavelength of 562 nm. This emission maximum is for *Photinus pyralis*, and other fireflies have slightly different emission properties.

The enzyme shows total specificity for ATP, but other nucleotides are poor competitive inhibitors of ATP. Those attempting to measure ATP exclusively must exercise care to block transphosphorylases (*e.g.*, nucleoside 5'-diphosphate kinase) that may indirectly synthesize ATP from ADP and another nucleoside 5'-triphosphate. Luciferase assays^{1,2} can be carried out using a kit available from Sigma Chemical Co. (St. Louis, MO; product numbers L1759, L5256, and L9009). Although the enzyme has a low turnover number, sensitive ATP assays down to the femtomole range can be conveniently achieved using a luminometer.

The bacterial luciferase^{4,5} [EC 1.14.14.3], also called alkaline monooxygenase (FMN-linked) and aldehyde monooxygenase, catalyzes the reaction of an aldehyde (RCHO) with dioxygen and FMNH₂ to produce a carboxylic acid (RCOOH), FMN, water, and light. The enzyme reaction mechanism involves incorporation of a molecule of dioxygen into reduced FMN and the subsequent reaction with the aldehyde to form an activated FMN·H₂O complex which breaks down with the emission of light. The enzyme is highly specific for reduced FMN and for long-chain aliphatic aldehydes with eight carbons or more.

¹F. R. Leach & J. J. Webster (1986) *Meth. Enzymol.* **133**, 51.

²S. Lin & H. P. Cohen (1968) *Anal. Biochem.* **24**, 531.

³M. DeLuca (1976) *Adv. Enzymol.* **44**, 37.

⁴J. W. Hastings (1978) *Crit. Rev. Biochem.* **5**, 163.

⁵V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

LUMEN

1. The SI unit of luminous flux, abbreviated lm. One lumen is the amount of luminous flux emitted in a unit solid angle of one steradian by a uniform point source having a luminous intensity of one candela. 2. The interior space of a tubular structure such as an intestine or artery.

LUMINESCENCE

An emission of radiation occurring spontaneously from an electronically or vibrationally excited species that is not in thermal equilibrium with its environment.

LYASES

A major class of enzymes that catalyze the cleavage of bonds by means other than hydrolysis or oxidation. When acting on a single substrate, at least one of the products formed by lyases is unsaturated.

LYATE ION

An anion produced as a result of the action of a Brønsted base on a protogenic solvent or of the action of autoprotolysis (loss of a hydron) of a solvent. Thus, OH^- is the lyate ion of water.

LYONIUM ION

A cation produced by the protonation of a solvent molecule. **See also** *Leveling Effect*

LYSINE 2,3-AMINOMUTASE

This enzyme [EC 5.4.3.2] catalyzes the interconversion of lysine and (3*S*)-3,6-diaminohexanoate. The enzyme is stimulated by *S*-adenosylmethionine and pyridoxal phosphate.

- G. H. Reed & M. D. Ballinger (1995) *Meth. Enzymol.* **258**, 362.
 P. A. Frey & G. H. Reed (1993) *Adv. Enzymol.* **66**, 1.
 E. J. Brush & J. W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.
 T. C. Stadtman (1973) *Adv. Enzymol.* **38**, 413.

 β -LYSINE 5,6-AMINOMUTASE

This cobalamin-dependent enzyme [EC 5.4.3.3] catalyzes the interconversion of (3*S*)-3,6-diaminohexanoate and (3*S*,5*S*)-3,5-diaminohexanoate.

- B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.
 P. A. Frey & G. H. Reed (1993) *Adv. Enzymol.* **66**, 1.
 T. C. Stadtman (1973) *Adv. Enzymol.* **38**, 413.
 T. C. Stadtman (1972) *The Enzymes*, 3rd ed., **6**, 539.

D-LYSINE 5,6-AMINOMUTASE

This cobalamin-dependent enzyme [EC 5.4.3.4], also called D- α -lysine mutase, catalyzes the interconversion of D-lysine and 2,5-diaminohexanoate.

- P. A. Frey & G. H. Reed (1993) *Adv. Enzymol.* **66**, 1.
 T. C. Stadtman (1973) *Adv. Enzymol.* **38**, 413.
 T. C. Stadtman (1972) *The Enzymes*, 3rd ed., **6**, 539.

LYSINE 6-AMINOTRANSFERASE

This enzyme [EC 2.6.1.36], also called lysine ϵ -aminotransferase, catalyzes the pyridoxal-phosphate-dependent reaction of L-lysine with α -ketoglutarate (or, 2-oxoglutarate) to produce 2-aminoadipate 6-semialdehyde

and L-glutamate. The aldehyde product, also known as allysine, then undergoes dehydration to form 1-piperidine 6-carboxylate.

- K. Tanizawa, S. Toyama & K. Soda (1985) *Meth. Enzymol.* **113**, 96.
 A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

LYSINE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 4.1.1.18] catalyzes the conversion of L-lysine to form cadaverine and carbon dioxide. 5-Hydroxy-L-lysine can also be decarboxylated by this enzyme.

- E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

**LYSINE DEHYDROGENASE
(α -DEAMINATING)**

This enzyme [EC 1.4.1.15] catalyzes the reaction of L-lysine with NAD^+ to produce 1,2-didehydropiperidine 2-carboxylate, ammonia, and NADH.

- N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

LYSINE 2-MONOOXYGENASE

This FAD-dependent enzyme [EC 1.13.12.2] catalyzes the reaction of L-lysine with dioxygen to produce 5-aminopentanamide, carbon dioxide, and water. Other diamino acids can serve as substrates as well.

- S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.
 V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

LYSOPHOSPHOLIPASE

This enzyme [EC 3.1.1.5] (also referred to as lecithinase B, lysolecithinase, and phospholipase B) catalyzes the hydrolysis of a 2-lysophosphatidylcholine to produce glycerophosphocholine and a fatty acid anion.

- E. A. Dennis, ed. (1991) *Meth. Enzymol.* **197**.
 J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.
 E. A. Dennis (1983) *The Enzymes*, 3rd ed., **16**, 307.

LYSOPHOSPHOLIPASE D

This enzyme [EC 3.1.4.39], also known as alkylglycerophosphoethanolamine phosphodiesterase, catalyzes the hydrolysis of 1-alkyl-*sn*-glycero-3-phosphoethanolamine to produce 1-alkyl-*sn*-glycerol 3-phosphate and ethanolamine. The enzyme will also act on the acyl and choline analogs of the lipid.

- R. L. Wykle & J. C. Strum (1991) *Meth. Enzymol.* **197**, 583.

LYSOPINE DEHYDROGENASE

This enzyme [EC 1.5.1.16], also referred to as D-lysopine synthase, catalyzes the reaction of N^2 -(D-1-carboxyethyl)-L-lysine with NADP^+ and water to produce L-lysine, pyruvate, and NADPH. In the reverse reaction, a number of L-amino acids can serve as substrates instead of L-lysine.

J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.

LYSOZYME

This enzyme [EC 3.2.1.17], also called muramidase, catalyzes the hydrolysis of the 1,4- β -linkages between *N*-acetyl-D-glucosamine and *N*-acetylmuramic acid in peptidoglycan heteropolymers of prokaryotic cell walls. Some chitinases [EC 3.2.1.14] also exhibit this activity.

See Kinetic Isotope Effect

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

H. A. McKenzie & F. H. White, Jr. (1991) *Adv. Protein Chem.* **41**, 174.

L. N. Johnson, J. Cheetham, P. J. McLaughlin, K. R. Acharya, D. Barford & D. C. Phillips (1988) *Curr. Top. Microbiol. Immunol.* **139**, 81.

A. J. Kirby (1987) *Crit. Rev. Biochem.* **22**, 283.

T. Imoto, L. N. Johnson, A. C. T. North, D. C. Phillips & J. A. Rupley (1972) *The Enzymes*, 3rd ed., **7**, 665.

A. Tsugita (1971) *The Enzymes*, 3rd ed., **5**, 343.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

N-Acetylglucosamine interaction, **64**, 275; antigenic sites, **70**, 46, 47; binding sites, **61**, 515-522; use in cell lysis, **65**, 119, 168, 186, 861; chemical modification, **63**, 213; clefts, **61**, 445-448 [^3C NMR, **61**, 469, 471-474, 477-479, 491-510, 531, 533; carbon count, **61**, 482, 488; chemical modification, **61**, 515, 526, 538-541; linewidths, **61**, 486; proton-coupling conditions, **61**, 492, 497; self-association, **61**, 542-544; specific assignments, **61**, 531]; conformation studies, **74**, 257; cryoenzymology, **63**, 338; calorimetry of denaturation, **61**, 266; difference spectral titration, **63**, 332, 333; effect of structure on antibody specificity, **70**, 42-46; frictional coefficient, **61**, 111; inhibitor, **63**, 401; interaction parameters, **61**, 48; low-frequency dynamics, **61**, 443-445; microcalorimetric studies of binding, **61**, 302; near-UV circular dichroism, **61**, 349, 350, 377; partial specific volumes, **61**, 48; pH study, **63**, 231-234; perturbed pK value, **63**, 210, 233; spin immunoassay, **74**, 151; structure, **70**, 43; subsite binding energy, **64**, 260, 275.

LYSYL ENDOPEPTIDASE

This enzyme [EC 3.4.21.50] catalyzes the hydrolysis of peptide bonds with a preference for Lys—Xaa (including Lys—Pro).

F. Sakiyama & T. Masaki (1994) *Meth. Enzymol.* **244**, 126.

LYSYL HYDROXYLASE

This enzyme [EC 1.14.11.4] catalyzes the reaction of a procollagen L-lysyl residue with α -ketoglutarate (or, 2-

oxoglutarate) and dioxygen to produce a procollagen 5-hydroxy-L-lysyl residue, succinate, and carbon dioxide. Iron and ascorbate are essential cofactors.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

LYSYL OXIDASE

This enzyme [EC 1.4.3.13], also referred to as protein-lysine 6-oxidase, catalyzes the reaction of a peptidyl-L-lysyl-peptide with dioxygen and water to produce a peptidyl-allysyl-peptide, ammonia, and hydrogen peroxide. The enzyme will also act on 5-hydroxylysyl residues in proteins.

C. Anthony (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 155.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.

C. Hartmann & W. S. McIntire (1997) *Meth. Enzymol.* **280**, 98.

J. P. Klinman (1996) *J. Biol. Chem.* **271**, 27189.

H. M. Kagan & P. Cai (1995) *Meth. Enzymol.* **258**, 122.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

G. J. Cardinale & S. Udenfriend (1974) *Adv. Enzymol.* **41**, 245.

LYSYLTRANSFERASE

This enzyme [EC 2.3.2.3], also referred to as lysyl-tRNA:phosphatidylglycerol transferase, catalyzes the reaction of L-lysyl-tRNA with phosphatidylglycerol to produce tRNA and 3-phosphatidyl-1'-(3'-*O*-L-lysyl) glycerol.

R. L. Soffer (1974) *Adv. Enzymol.* **40**, 91.

α -LYTIC PROTEINASE

This enzyme [EC 3.4.21.12] catalyzes the hydrolysis of peptide bonds in proteins, especially at peptide bonds adjacent to alanyl and valyl residues of bacterial cell walls, elastin, and other proteins. The enzyme is a member of the peptidase family S2A. **See also** *Chymotrypsin; Catalytic Triad*

K. Morihara (1974) *Adv. Enzymol.* **41**, 179.

LYXOSE ISOMERASE

This enzyme [EC 5.3.1.15], also referred to as lyxose ketol-isomerase, catalyzes the interconversion of D-lyxose and D-xylulose.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

M

MACROMOLECULAR SEQUENCE ANALYSIS (Computer Methods)

There are numerous computer software programs that permit one to investigate sequence alignment and phylogenetic relationships among (a) various proteins, domains, motifs, modules, *etc.*, and (b) nucleic acid sequences in DNA and RNA. In addition to those presented below, various Internet-based algorithms afford rapid and convenient analysis of macromolecular sequences.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Databases and Resources: Information services of European Bioinformatics Institute, **266**, 3; TDB: new databases for biological discovery, **266**, 27; PIR-international protein sequence database, **266**, 41; superfamily classification in PIR-international protein sequence database, **266**, 59; gene classification artificial neural system, **266**, 71; blocks database and its applications, **266**, 88; indexing and using sequence databases, **266**, 105; SRS: information retrieval system for molecular biology data banks, **266**, 114.

Searching through Databases: Applications of network BLAST server, **266**, 131; Entrez: molecular biology database and retrieval system, **266**, 141; applying motif and profile searches, **266**, 162; consensus approaches in detection of distant homologies, **266**, 184; identification of sequence patterns with profile analysis, **266**, 198; effective large-scale sequence similarity searches, **266**, 212; effective protein sequence comparison, **266**, 227; discovering and understanding genes in human DNA sequence using GRAIL, **266**, 259; linguistic analysis of nucleotide sequences: algorithms for pattern recognition and analysis of codon strategy, **266**, 281; protein sequence comparison at genome scale, **266**, 295; iterative template refinement: protein-fold prediction using iterative search and hybrid sequence/structure templates, **266**, 322.

Multiple Alignment and Phylogenetic Trees: Multiple protein sequence alignment: algorithms and gap insertion, **266**, 343; progressive alignment of amino acid sequences and construction of phylogenetic trees from them, **266**, 368; using CLUSTAL for multiple sequence alignments, **266**, 383; combined DNA and protein alignment, **266**, 402; inferring phylogenies from protein sequences by parsimony, distance, and likelihood methods, **266**, 418; reconstruction of gene trees from sequence data, **266**, 427; estimating evolutionary distances between DNA sequences, **266**, 449; local alignment statistics, **266**, 460; parametric and inverse-parametric sequence alignment with XPARAL, **266**, 481.

Secondary Structure Considerations: Identification of functional

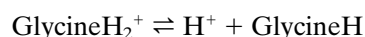
residues and secondary structure from protein multiple sequence alignment, **266**, 497; prediction and analysis of coiled-coil structures, **266**, 513; PHD: predicting one-dimensional protein structure by profile-based neural networks, **266**, 525; GOR method for predicting protein secondary structure from amino acid sequence, **266**, 540; analysis of compositionally biased regions in sequence databases, **266**, 554.

Three-Dimensional Considerations: Discrimination of common protein folds: application of protein structure to sequence/structure comparisons, **266**, 575; three-dimensional profiles for measuring compatibility of amino acid sequence with three-dimensional structure, **266**, 598; SSAP: sequential structure alignment program for protein structure comparison, **266**, 617; understanding protein structure: using scop for fold interpretation, **266**, 635; detecting structural similarities: a user's guide, **266**, 643; alignment of three-dimensional protein structures: network server for database searching, **266**, 653; converting sequence block alignments into structural insights, **266**, 662.

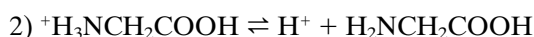
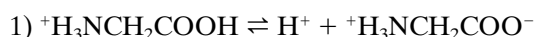
MACROSCOPIC CONSTANTS

Observable constants, usually thermodynamic dissociation constants, for the association of a particular ligand with two or more sites on a larger molecular entity (*e.g.*, a macromolecule). Macroscopic constants are composites of microscopic (*i.e.*, intrinsic) constants.

The larger molecular entity need not be a macromolecule. Consider the proton dissociation steps of the simplest amino acid, glycine. The dissociation of the first proton (*i.e.*, the most acidic proton) can be represented by



and has an observable, macroscopic dissociation constant ($K_1 = 4.45 \times 10^{-3}$, $\text{p}K_1 = 2.35$). This macroscopic constant is a composite of two different dissociations:



having the microscopic dissociation constants k_1 and k_2 , respectively.

As can be seen in the entry on zwitterions, $K_1 = k_1 + k_2$. The observable macroscopic constant is indeed a com-

posite of the intrinsic microscopic constants. **See** *Microscopic Constant; Zwitterion*

MACROSCOPIC DIFFUSION CONTROL

The limiting reaction rate achieved by diffusion occurring after mixing solutions containing two or more reactants. This form of control, also called mixing control, is typically more relevant for reactions involving heterogeneous systems (*e.g.*, solid/liquid or liquid/gas). **See also** *Microscopic Diffusion Control*

MAGNESIUM ION (Intracellular)

Beyond the effect of magnesium ion concentration on the equilibrium hydrolysis of adenosine triphosphate to adenosine diphosphate^{1,2}, there is ample evidence that MgATP^{2-} is generally the most widespread substrate in kinase-type phosphotransferase reactions as well as other ATP-dependent processes. The extent to which MgATP^{2-} is formed in solution depends on the free (or uncomplexed) magnesium ion concentration, as shown by the following equilibrium constant:

$$K_{\text{formation}} = [\text{MgATP}^{2-}] / [\text{Mg}_{\text{free}}^{2+}] [\text{ATP}^{4-}]$$

For example, Bachelard³ used $[\text{Mg}_{\text{total}}] / [\text{ATP}_{\text{total}}^{4-}] = 1$ in his rate studies, and he obtained a slightly sigmoidal plot of initial velocity versus “substrate ATP” concentration. This culminated in the erroneous proposal that brain hexokinase was allosterically activated by magnesium ions and by magnesium ion-adenosine triphosphate complex. Purich and Fromm⁴ demonstrated that failure to achieve adequate experimental control over the free magnesium ion concentration can wreak havoc on the examination of enzyme kinetic behavior. Indeed, these investigators were able to account fully for the effects obtained in the previous hexokinase study³.

Nonetheless, there remained some uncertainty regarding the actual intracellular concentration of uncomplexed magnesium ion, which controls the fraction of total ATP complexed with Mg^{2+} :

$$\text{Fraction}_{\text{MgATP}} = [\text{MgATP}^{2-}] / \{ [\text{ATP}^{4-}] + [\text{MgATP}^{2-}] \}$$

or

$$\text{Fraction}_{\text{MgATP}} = 1 / \{ [\text{ATP}^{4-}] / [\text{MgATP}^{2-}] + 1 \}$$

By substituting from the first equilibrium expression (*i.e.*, $[\text{ATP}^{4-}] / [\text{MgATP}^{2-}] = [\text{Mg}_{\text{free}}^{2+}] / K_{\text{formation}}$), this fraction becomes:

$$\text{Fraction}_{\text{MgATP}} = 1 / \{ 1 + [\text{Mg}_{\text{free}}^{2+}] / K_{\text{formation}} \}$$

This underscores the need to know two parameters, namely $[\text{Mg}_{\text{free}}^{2+}]$ and $K_{\text{formation}}$ to assess $\text{Fraction}_{\text{MgATP}}$.

Accurate determination of the free intracellular magnesium ion concentration has been challenging because the available methods, including dye injection, microelectrodes, and nuclear magnetic resonance measurements, are either invasive or indirect. On the basis of substantially identical ³¹P NMR measurements of the resonances in the spectrum of MgATP^{2-} , Wu *et al.*⁵ put $[\text{Mg}_{\text{free}}^{2+}]$ at 2.5 mM in guinea pig heart muscle, whereas other investigators^{6,7} arrived at a value of 0.6 mM. Later assessment⁸ of the experimental evidence regarding the free intracellular magnesium concentration (as well as the limitations of the analytical methods) suggested that intracellular concentration of free magnesium is low, probably around 0.4 mM, and varies with time and conditions. Clarke *et al.*⁹ recently introduced the β/α phosphoryl peak height ratio of ATP as a means for measuring free $[\text{Mg}^{2+}]$ using ³¹P NMR. **See** *Energy Charge; Metal Ions in Nucleotide-Dependent Reactions; Biochemical Thermodynamics*

¹R. A. Alberty (1968) *J. Biol. Chem.* **243**, 1337.

²R. C. Phillips, P. George & R. J. Rutman (1969) *J. Biol. Chem.* **244**, 3330.

³H. S. Bachelard (1971) *Biochem. J.* **125**, 249.

⁴D. L. Purich & H. J. Fromm (1972) *Biochem. J.* **130**, 63.

⁵S. T. Wu, G. M. Pieper, J. M. Salhany & R. S. Eliot (1981) *Biochemistry* **20**, 7399.

⁶R. K. Gupta & R. D. Moore (1980) *J. Biol. Chem.* **255**, 3987.

⁷R. K. Gupta, J. L. Benkovic & Z. B. Rose (1978) *J. Biol. Chem.* **253**, 6165.

⁸L. Garfinkel, R. A. Altschuld & D. Garfinkel (1986) *J. Mol. Cell. Cardiol.* **18**, 1003.

⁹K. Clarke, Y. Kashiwaya, M. T. King, D. Gates, C. A. Keon, H. R. Cross, G. K. Radda & R. L. Veech (1996) *J. Biol. Chem.* **271**, 21142.

MAGNESIUM-PROTOPORPHYRIN O-METHYLTRANSFERASE

This enzyme [EC 2.1.1.11] catalyzes the reaction of *S*-adenosylmethionine with magnesium protoporphyrin to produce *S*-adenosylhomocysteine and magnesium protoporphyrin monomethyl ester.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

MAGNETICALLY ANISOTROPIC GROUP

A functional moiety in a molecule that does not behave identically along all three axes when placed in a magnetic field. Examples are aromatic rings and triple bonds.

MAGNETIC SUSCEPTIBILITY

The ratio of the magnetization M to the magnetic field strength H , multiplied by $[(\mu/\mu_0) - 1]$, where the permeability μ is the magnetic counterpart of the electrical permittivity, and μ_0 is the permeability of a vacuum. Magnetic susceptibility is measured with a magnetic balance, in which a sample suspended from a weighing balance is situated between the poles of a strong electromagnet. A diamagnetic substance will be pushed out of the field, whereas ferromagnetic and paramagnetic substances will be drawn into the field. Using a magnifying viewer, the experimentalist measures the force necessary to restore the sample to the same position as when the field is turned off. Paramagnetism is related to orbital angular momenta and electron spins within a chemical substance, and measurements of magnetic susceptibility can provide valuable mechanistic information related to ligand binding interactions with certain metal ions. **See** *Hemoglobin; Diamagnetic; Paramagnetic*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theoretical aspects, **76**, 354-356; diamagnetic contribution, **76**, 358-359; experimental methods, **76**, 356-360; Faraday balance technique, **76**, 360-361; Gouy technique, **76**, 357, 360; instrumentation, **76**, 360-369; oxygen contribution, **76**, 360, 368; sources of experimental errors, **76**, 359-360; SQUID magnetometer use, **76**, 364-365; thermal equilibria, **76**, 358, 370; thermal expansion, **76**, 358.

MAGNETIZATION TRANSFER

An NMR method for determining the kinetics of chemical exchange by perturbing (*e.g.*, by saturation or inversion) the magnetization of nuclei in a particular site or sites and following the rate at which the magnetic equilibrium is restored.

MAGNETOTACTIC

Referring to the ability to become oriented with respect to a magnetic field. In addition to describing the properties of certain chemical substances, the term also applies to those organisms that can both sense the presence of, and orient relative to, the earth's magnetic field.

**MALAPRADE REACTION
(Periodate Oxidation)**

If a compound contains two hydroxyl groups, each attached to adjacent carbon atoms (*i.e.*, 1,2-glycols) or contains one hydroxyl group and one primary or secondary amino group (*i.e.*, β -amino alcohols), each attached to adjacent carbon atoms, then periodic acid (HIO_4) will cleave that carbon-carbon bond. If neither of those carbon atoms were terminal carbons, then each will be converted to the corresponding aldehyde or ketone, depending on what other groups are attached to those carbon atoms: *e.g.*, $\text{RCH}(\text{OH})\text{CH}(\text{OH})\text{R}' + \text{HIO}_4 \rightarrow \text{RCHO} + \text{R}'\text{CHO} + \text{H}_2\text{O} + \text{HIO}_3$ and $\text{RCH}(\text{OH})\text{CH}(\text{NH}_2)\text{R}' + \text{HIO}_4 \rightarrow \text{RCHO} + \text{R}'\text{CHO} + \text{NH}_3 + \text{HIO}_3$. If there are three or more hydroxyl groups on three or more adjacent carbon atoms, the middle carbon atoms are converted to formic acid.

Periodic acid oxidation will also act on α -hydroxy aldehydes and ketones, α -diketones, α -keto aldehydes, and glyoxal. If the two neighboring hydroxyl groups are on an aromatic ring, the carbon-carbon bond is not cleaved but the reactant is still oxidized. Thus, catechol is oxidized to the corresponding quinone.

Periodic acid reacts well in aqueous solution. Usually, if the reactant has to be run in organic solvents, lead tetraacetate is used as the reagent. Interestingly, periodic acid will not act on α -keto acids or α -hydroxy acids whereas lead tetraacetate will. The corresponding reactions are actually oxidative decarboxylations.

Periodic acid oxidation has proved to be a very useful tool in enzymology since a wide variety of biochemicals contain hydroxyl groups on adjacent carbon atoms. For example, periodate-oxidized ATP (also called adenosine 5'-triphosphate 2',3'-dialdehyde) has often been used as an alternative substrate or an irreversible inhibitor for a wide variety of ATP-utilizing enzymes. This compound, and many others, are now commercially available, even though they are readily synthesized: *e.g.*, periodic acid oxidized ADP, AMP, adenosine, P^1, P^5 -di(adenosine-5')pentaphosphate, P^1, P^4 -di(adenosine-5')tetraphosphate, GTP, GDP, GMP, guanosine, CTP, CDP, CMP, *etc.* In the case of the nucleosides, commercial sources also can supply the dialcohol form of the nucleoside: *i.e.*, the nucleoside has first been oxidized with periodic acid and then reduced to the dialcohol with borohydride.

- L. Malaprade (1928) *Bull. Soc. Chim. France* [4] **43**, 683.
 L. Malaprade (1928) *Compte Rend.* **186**, 382.
 K. W. Bentley (1973) in *Elucidation of Organic Structures by Physical and Chemical Methods*, pt. 2 (K. W. Bentley & G. W. Kirby, eds.), pp. 137-254 (vol. 4 of A. Weissberger, *Techniques in Chemistry*), Wiley, New York.
 H. O. House (1972) *Modern Synthetic Reactions*, 2nd ed., Benjamin, New York.
 B. Sklarz (1967) *Q. Rev., Chem. Soc.* **21**, 3.

MALATE DEHYDROGENASE

1. Malate dehydrogenase [EC 1.1.1.37], also called malic dehydrogenase, catalyzes the reaction of (*S*)-malate with NAD⁺ to produce oxaloacetate and NADH. Other 2-hydroxydicarboxylic acids can act as substrates as well. 2. Malate dehydrogenase (NADP⁺) [EC 1.1.1.82] catalyzes the reaction of (*S*)-malate with NADP⁺ to produce oxaloacetate and NADPH. This enzyme is activated by light. 3. Malate dehydrogenase (acceptor) [EC 1.1.99.16] catalyzes the FAD-dependent reaction of (*S*)-malate with an acceptor to produce oxaloacetate and the reduced acceptor. 4. Malate dehydrogenase (oxaloacetate-decarboxylating) [EC 1.1.1.38], also referred to as malic enzyme and pyruvic-malic carboxylase, catalyzes the reaction of (*S*)-malate with NAD⁺ to produce pyruvate, carbon dioxide, and NADH. The enzyme will also decarboxylate added oxaloacetate. 5. Malate dehydrogenase (decarboxylating) [EC 1.1.1.39], also referred to as the malic enzyme, catalyzes the reaction of (*S*)-malate with NAD⁺ to produce pyruvate, carbon dioxide, and NADH. This enzyme will not decarboxylate added oxaloacetate. 6. Malate dehydrogenase (oxaloacetate-decarboxylating, NADP⁺) [EC 1.1.1.40], also referred to as malic enzyme and pyruvic-malic carboxylase, catalyzes the reaction of (*S*)-malate with NADP⁺ to produce pyruvate, carbon dioxide, and NADPH. The enzyme will also decarboxylate added oxaloacetate. 7. D-Malate dehydrogenase (decarboxylating) [EC 1.1.1.83] catalyzes the reaction of (*R*)-malate with NAD⁺ to produce pyruvate, carbon dioxide, and NADH. **See Abortive Complexes**

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 H. Eisenberg, M. Mevarech & G. Zaccai (1992) *Adv. Protein Chem.* **43**, 1.
 W. W. Cleland (1980) *Meth. Enzymol.* **64**, 104.
 L. J. Banaszak & R. A. Bradshaw (1975) *The Enzymes*, 3rd ed., **11**, 369.
 K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.
 M. G. Rossman, A. Liljas, C.-I. Brändén & L. J. Banaszak (1975) *The Enzymes*, 3rd ed., **11**, 61.

MALATE SYNTHASE

This enzyme [EC 4.1.3.2], also known as malate condensing enzyme and glyoxylate transacetylase, catalyzes the reaction of L-malate with coenzyme A to produce acetyl-CoA, water, and glyoxylate.

- R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 M. J. P. Higgins, J. A. Kornblatt & H. Rudney (1972) *The Enzymes*, 3rd ed., **7**, 407.

MALEATE ISOMERASE

This enzyme [EC 5.2.1.1] catalyzes the interconversion of maleate and fumarate.

- S. Seltzer (1972) *The Enzymes*, 3rd ed., **6**, 381.

MALEYLACETOACETATE ISOMERASE

This enzyme [EC 5.2.1.2] catalyzes the interconversion of 4-maleylacetoacetate and 4-fumarylacetoacetate. Maleylpyruvate can also act as a substrate.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.
 S. Seltzer (1972) *The Enzymes*, 3rd ed., **6**, 381.

MALEYLPYRUVATE ISOMERASE

This enzyme [EC 5.2.1.4] catalyzes the interconversion of 3-maleylpyruvate and 3-fumarylpyruvate.

- K. T. Douglas (1987) *Adv. Enzymol.* **59**, 103.
 S. Seltzer (1972) *The Enzymes*, 3rd ed., **6**, 381.

MALONYL-CoA:[ACYL-CARRIER-PROTEIN] TRANSACYLASE

This enzyme [EC 2.3.1.39], also known as acyl-carrier protein malonyltransferase, catalyzes the reaction of malonyl-CoA with an acyl-carrier-protein to produce coenzyme A and a malonyl-[acyl-carrier-protein].

- S. J. Wakil & J. K. Stoops (1983) *The Enzymes*, 3rd ed., **16**, 3.
 P. R. Vagelos (1973) *The Enzymes*, 3rd ed., **8**, 155.

MALTASE

This enzyme [EC 3.2.1.20] catalyzes the hydrolysis of terminal, nonreducing 1,4-linked D-glucose residues with resulting release of D-glucose. This classification represents a set of enzymes with specificity directed mainly toward the exohydrolysis of 1,4- α -glucosidic linkages, as well as enzymes that hydrolyze oligosaccharides rapidly, relative to polysaccharides. The intestinal enzyme also hydrolyzes polysaccharides, thereby exhibiting a glucan

1,4- α -glucosidase [EC 3.2.1.3] activity. This enzyme will also catalyze the hydrolysis of 1,6- α -D-glucose linkages, albeit less effectively. **See also** *α -Glucosidase*

E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, 197.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

M. Vihinen & P. Mäntsäälä (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 329.

MALTOSE PHOSPHORYLASE

This enzyme [EC 2.4.1.8] catalyzes the reaction of maltose with orthophosphate to produce D-glucose and β -D-glucose 1-phosphate.

J. J. Mieyal & R. H. Abeles (1972) *The Enzymes*, 3rd ed., **7**, 515.

MALYL-CoA LYASE

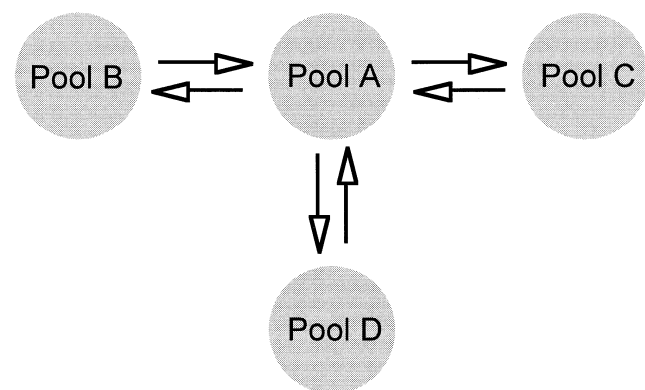
This enzyme [EC 4.1.3.24] catalyzes the conversion of (3S)-3-carboxy-3-hydroxypropanoyl-CoA to acetyl-CoA and glyoxylate.

P. M. Goodwin (1990) *Meth. Enzymol.* **188**, 361.

A. J. Hacking & J. R. Quayle (1990) *Meth. Enzymol.* **188**, 379.

P. J. Arps (1990) *Meth. Enzymol.* **188**, 386.

MAMILLARY MODEL



One of several general models for metabolite compartmentation in which a central compartment is directly linked to or feeds from (hence the name) other compartments that do not communicate with each other aside from their connection to the central pool. **See** *Catenary Model; Compartmental Analysis*

MANDELATE 4-MONOOXYGENASE

This iron-dependent enzyme [EC 1.14.16.6], also known as L-mandelate 4-hydroxylase, catalyzes the reaction of (S)-2-hydroxy-2-phenylacetate with tetrahydropteridine and dioxygen to produce (S)-4-hydroxymandelate, dihydropteridine, and water.

R. Y. Stanier (1955) *Meth. Enzymol.* **2**, 273.

MANDELATE RACEMASE

This enzyme [EC 5.1.2.2] catalyzes the interconversion of (S)-mandelate and (R)-mandelate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

G. L. Kenyon & G. D. Hegeman (1979) *Adv. Enzymol.* **50**, 325.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

MANDELONITRILE LYASE

This enzyme [EC 4.1.2.10], also known as hydroxynitrile lyase and (R)-oxynitrilase, catalyzes the conversion of mandelonitrile to cyanide and benzaldehyde.

B. A. Palfev & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

MANGANESE PEROXIDASE

This heme-dependent enzyme [EC 1.11.1.13] catalyzes the reaction of two Mn^{2+} with two H^+ and hydrogen peroxide to generate two Mn^{3+} and two water. The enzyme participates in the oxidative degradation of lignin in white rot basidiomycetes.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

J. E. Penner-Hahn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 439.

β -MANNANASE

This enzyme [EC 3.2.1.78], also known as mannan endo-1,4- β -mannosidase and endo-1,4-mannanase, catalyzes the random hydrolysis of 1,4- β -D-mannosidic linkages in mannans, galactomannans, glucomannans, and galactoglucomannans.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

MANNONATE DEHYDRATASE

This enzyme [EC 4.2.1.8], also known as D-mannonate hydro-lyase, catalyzes the conversion of D-mannonate to 2-dehydro-3-deoxy-D-gluconate and water.

J. Robert-Baudouy, J. Jimeno-Abendano & F. Stoeber (1982) *Meth. Enzymol.* **90**, 288.

W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

MANNOSE ISOMERASE

This enzyme [EC 5.3.1.7] catalyzes the interconversion of D-mannose and D-fructose. Both D-lyxose and D-rhamnose can act as substrates as well.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

MANNOSE-1-PHOSPHATE GUANYLYLTRANSFERASE

This enzyme [EC 2.7.7.13], also known as GTP-mannose-1-phosphate guanylyltransferase, catalyzes the reaction of GTP with α -D-mannose 1-phosphate to produce GDP-mannose and pyrophosphate (or, diphosphate). The bacterial enzyme can also use ITP and dGTP as the phosphoryl donors.

J. Preiss (1966) *Meth. Enzymol.* **8**, 271.

MANNOSE-6-PHOSPHATE ISOMERASE

This zinc-dependent enzyme [EC 5.3.1.8] (also known as phosphomannose isomerase, phosphohexoisomerase, and phosphohexomutase) catalyzes the interconversion of D-mannose 6-phosphate and D-fructose 6-phosphate.

K. U. Yüksel & R. W. Gracy (1991) *A Study of Enzymes* **2**, 457.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

I. A. Rose (1975) *Adv. Enzymol.* **43**, 491.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

MANNOSIDASES

α -Mannosidase [EC 3.2.1.24] catalyzes the hydrolysis of terminal, nonreducing α -D-mannose residues in α -D-mannosides. In addition, α -D-lyxosides and heptopyranosides (with the same configuration at C2, C3, and C4 as mannose) will serve as substrates as well.

β -Mannosidase [EC 3.2.1.25], also known as mannanase and mannanase, catalyzes the hydrolysis of terminal, nonreducing β -D-mannose residues in β -D-mannosides.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

S. C. Fry (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 497.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

K. A. Presper & E. C. Heath (1983) *The Enzymes*, 3rd ed., **16**, 449.

H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

MANOMETRIC ASSAY METHODS

For reactions in which one or more reactants or products is a gas, manometry (the measurement of pressure differences) can provide a convenient means for monitoring the course and kinetics of the reaction¹. Thus, enzymes that can be assayed with this method include oxidases, urease, carbonic anhydrase, hydrogenase, and decarboxylases. For example, bacterial glutamate decarboxylase is readily assayed by utilizing a Warburg flask and measuring the volume of gas evolved at different times using a constant-pressure respirometer².

Note that manometric methods can also be used with reactions that generate acids³. If bicarbonate is a component of the reaction mixture, any acid formed will result in the generation of CO₂. Thus, even peptidases, proteinases, and esterases can be assayed manometrically^{4,5}. Even some phosphotransferases have been assayed with this procedure⁶. **See also** *Eudiometer*

¹M. Dixon (1951) *Manometric Methods*, Cambridge Univ. Press, Cambridge.

²M. L. Fonda (1985) *Meth. Enzymol.* **113**, 11.

³D. E. Green, D. M. Needham & J. G. Dewan (1937) *Biochem. J.* **31**, 2327.

⁴H. A. Krebs (1930) *Biochem. Z.* **220**, 283.

⁵R. Ammon (1934) *Pflug. Arch. Ges. Physiol.* **233**, 486.

⁶S. P. Colowick & H. M. Kalcker (1941) *J. Biol. Chem.* **137**, 789.

MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA

Although X-ray crystallography provides the most definitive information on the stereochemical considerations governing substrate binding, atomic structures of enzyme complexes often fail to reveal the nature of ligand-induced conformational changes that strongly influence the kinetics and thermodynamics of enzyme-ligand interactions. Nonetheless, one can frequently infer stereochemical relationships by comparing observed binding constants, inhibition constants, and in some cases Michaelis constants, for interactions of inhibitors and substrates with enzymes.

In the case of sheep brain glutamine synthetase, the goal was to discover why both D- and L-glutamate serve as substrates, and Gass and Meister¹ studied these substrates along with ten of their monomethyl derivatives. Of the latter, only three were active as substrates, and by making the assumption that amino acid substrates bind to glutamine synthetase in their most extended conformations, these investigators found that those hydrogens, whose methyl substituents yielded enzymatic activity, lie on one side of the molecule (*i.e.*, the side that resides behind the plane for the structures of L-glutamate and D-glutamate as shown in Fig. 1). This led to the hypothesis that the enzyme approaches its amino acid substrate from above the plane of the paper, thereby permitting methyl groups of the three active methyl derivatives to project backward and away from the enzyme's surface. To test this proposal, these investigators prepared L-*cis*-1-amino-1,3-dicarboxycyclohexane as a conformationally restricted L-glutamate analogue

wherein the two backward-extending hydrogens of D- and L-glutamate are connected by the three additional methylene groups within the cyclohexane ring (see Fig. 1). This analogue proved to be an effective substrate for the sheep brain enzyme, thereby fixing the stereochemistry of enzyme-substrate complexation without any atomic-level information on the enzyme's structure. In retrospect, this conformationally restricted derivative would have also served as a valuable mechanistic probe of the mechanism of carboxyl group activation, because reaction of the analogue with ATP in an ammonia-free solution should form the corresponding acyl phosphate intermediate, and any intramolecular attack of the α -amino group to a pyrrolidone-like product (resembling a pyrrolidone carboxylate) should be far less likely. A similar argument applies to the use of β -glutamate which should form an acyl phosphate compound that resists intramolecular attack by the amino group.

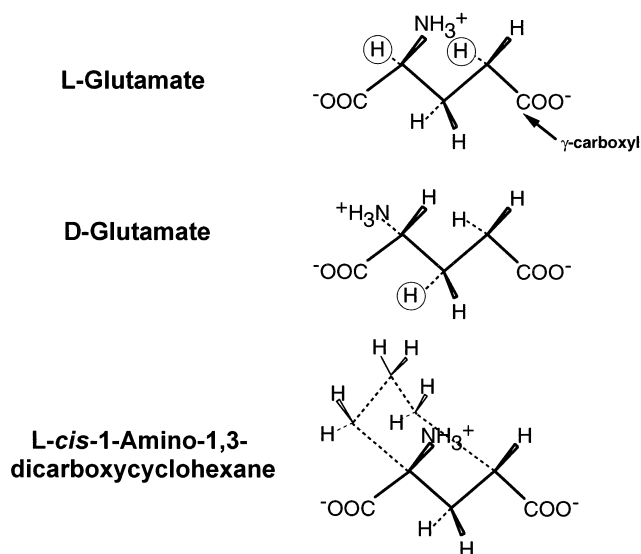


Figure 1. Stereochemical representation of L-glutamate, D-glutamate, and L-cis-1-amino-1,3-dicarboxycyclohexane. Both D- and L-glutamate as well as α -aminoadipic acid are substrates for the enzyme, but only three of ten monomethyl derivatives of D- and L-glutamate are substrates. Assuming that the substrates are completely extended when bound to the enzyme, active substrates were obtained when the circled hydrogens are substituted by a methyl group. These hydrogens lie behind the plane of the paper, and they most probably face outward from the enzyme-bound substrate toward the bulk solvent. L-cis-1-Amino-1,3-dicarboxycyclohexane is an alternative substrate for glutamate in the glutamine synthetase reaction. Note that the cyclohexane ring is connected by three successive methylenes that bridge the two replaceable (circled) hydrogen atoms in L-glutamate.

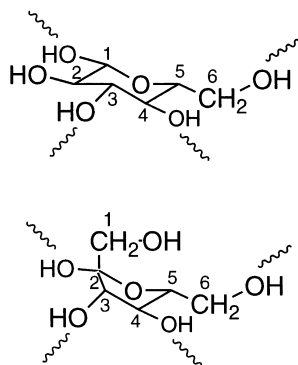
Hexokinase provides another example of the stereochemical inferences that can be drawn on the basis of kinetic studies with alternative substrates and competi-

tive inhibitors². Although the hexokinases show broad specificity with regard to their hexose substrates, they often differ with respect to the relative affinities and relative rates of phosphorylation of the sugar substrate. To rationalize the different affinities and relative rates of hexose phosphorylation, Robert Crane presented an elaborate model for the interaction of hexokinase with its sugar substrates. He proposed that (a) β -D-glucopyranose as the C1 conformer (chair equatorial) combines with hexokinase through the interaction of specific binding sites on the phosphotransferase with the hydroxyl groups located on carbon atoms 1, 3, and 6 of the sugar molecule; (b) this enzyme-substrate Michaelis complex then undergoes a contraction (or induced fit) to yield the C1 conformer (chair axial); and (c) only with this latter enzyme-substrate complex does phosphorylation occur. The chief reason for his postulating a substrate-induced conformational change was to rationalize the inability of hexokinase to act on compounds like *N*-acetylglucosamine which do act as potent competitive inhibitors with respect to the sugar substrate. The presence of such bulky substituents as the *N*-acetyl group, Crane reasoned, should stabilize the hexose in the C1 conformation, which is inactive.

While this model explained the action of the brain enzyme on a number of hexose substrates and nonsubstrate inhibitory analogs, the model had its weaknesses. It assumed that the other conformations of a hexose that are in equilibrium with the "active" conformer act as competitive inhibitors relative to this conformer. One cannot evaluate the effect of a competitive inhibitor which is present in a constant proportion relative to the active substrate by initial velocity measurements. Moreover, the use of apparent Michaelis constants may not provide accurate estimates of affinity, which is more directly related to a dissociation constant. The chief limitation of the model, however, is that an equally great number of experimental facts can be satisfactorily explained in terms of a simpler scheme involving the binding and phosphorylation of the C1 conformer. Furthermore, one can understand more directly how the enzyme can phosphorylate glucopyranose and fructofuranose equally well.

Purich *et al.*² offered the following scheme which illustrates the structural similarities between C1 β -D-gluco-

pyranose and the corresponding conformer of β -D-fructofuranose (Fig. 2):



Hexokinase Interactions with Glucose (top) and Fructose (bottom)

Figure 2. Hypothetical scheme for the selectivity-conferring interactions (indicated as wavy lines) of hexokinase with keto- and ald-hexoses. This model accounts for all known specificity and selectivity of yeast hexokinase toward hexose substrates and pentose ligands.

They noted that the hydroxyl substituents on carbon atoms 1, 3, 4, and 6 of glucose are oriented in approximately the same way as the hydroxyl groups located on carbon atoms 2, 3, 4, and 6 of fructose. By hypothesizing that these positions are the specificity-imparting groups on the hexose (as indicated by the wavy lines), they could rationalize the relative unimportance of the hydroxyl substituent at carbon atom 2 of glucose, mannose, and 2-deoxyglucose. The relative importance of the hydroxyl substituent at carbon 1 of glucose can be further appreciated by the fact that 1,5-anhydro-D-glucitol and 1,5-anhydro-D-mannitol are bound less tightly than their respective hexoses. That these anhydro sugar alcohols can be phosphorylated, however, indicates that the presence of a hydroxyl group in this position is not required. On the other hand, the 4 position of glucose appears essential in that galactose is bound very poorly and is not phosphorylated. It would be of interest to know whether 4-deoxyglucose can be bound and phosphorylated; this information would permit one to determine whether the inactivity of galactose is due to the axial orientation of the hydroxyl group or to the lack of a hydroxyl group in the equatorial orientation for binding. Finally, the inability of brain hexokinase to act on *N*-acetylglucosamine can be attributed to interaction of the bulky *N*-acetyl substituent with the groups on the enzyme responsible for recognizing the hydroxyl group at carbon atom 1 or 3 of glucose.

¹J. D. Gass & A. Meister (1970) *Biochemistry* **9**, 1380.

²D. L. Purich, F. B. Rudolph & H. J. Fromm (1973) *Advan. Enzymol.* **39**, 249.

MARCUS EQUATION

An equation relating the Gibbs energy of activation ($\Delta G^\ddagger$) with the driving force of the reaction ($\Delta_r G^\ominus$): $\Delta G^\ddagger = (\lambda/4)(1 + \Delta_r G^\ominus/\lambda)^2$ where λ is the reorganization energy, $\lambda/4$ is the intrinsic barrier of the reaction, and $\Delta_r G^\ominus$ is the standard Gibbs free energy for the reaction corrected for the electrostatic work required to bring the reactants together. This equation has been applied to electron transfer reactions as well as atom and group transfer reactions.

R. A. Marcus (1964) *Ann. Rev. Phys. Chem.* **15**, 155.

W. J. Albery (1980) *Ann. Rev. Phys. Chem.* **31**, 227.

MARCUS RATE THEORY

The Marcus rate theory^{1,2} provides a means for addressing issues of proton transfer reactions, particularly in the analysis of Brønsted plots. The slope β in a Brønsted plot is often used to assist in characterizing a reaction mechanism and to estimate the degree of charge transfer between reactant(s) and product(s) in the transition state. In the Marcus rate theory, this slope is $d\Delta G^\ddagger/d\Delta G_R^\ominus$ where $\Delta G^\ddagger$ is the Gibbs activation energy for the proton transfer and $\Delta G_R^\ominus$ is the standard Gibbs free energy of the reaction with the required active site conformation. The Marcus rate theory has been applied to carbonic anhydrase³⁻⁵.

¹R. A. Marcus (1968) *J. Phys. Chem.* **72**, 891.

²A. J. Kresge (1975) *Acc. Chem Res.* **8**, 354.

³R. S. Rowlett & D. N. Silverman (1982) *J. Amer. Chem. Soc.* **104**, 6737.

⁴D. N. Silverman, C. Tu, X. Chen, S. M. Tanhauser, A. J. Kresge & P. J. Laipis (1993) *Biochemistry* **32**, 10757.

⁵D. N. Silverman (1995) *Meth. Enzymol.* **249**, 479.

MARKOV CHAIN

Any process comprising a sequence of events (or states), such that the probability of any one state only depends on the immediately preceding event. Markov (or Markoff) chains are frequently used to describe stochastic processes, such as Brownian motion, the random opening/closing of ion channels, and population genetics. This treatment affords the opportunity to treat chemical events, such as self-assembly, as discrete events rather than that achieved using continuous functions which might result in the absurdity of loss or gain of a fraction

of one subunit in a depolymerization or polymerization event.

W. R. Gilks (1996) *Markov Chain Monte Carlo in Practice*, Chapman & Hall, New York.

A. Borodin & P. Salminen (1996) *Handbook of Brownian Motion — Facts and Formulae*, Springer, Berlin.

MARQUARDT ALGORITHM

An algorithm used in the statistical fitting of cooperative binding data via a nonlinear least-squares analysis. This algorithm is an efficient method for analyzing nonlinear binding or nonlinear kinetic functions. It is applicable for any type of saturation function¹⁻³. In this process, a parameter, λ , is defined that controls “the interpolation between the extremes of the analytical or the gradient search by increasing the diagonal of the curvature matrix³.” **See** *Computer Algorithms & Software*

¹P. R. Bevington (1969) *Data Reduction and Error Analysis for the Physical Sciences*, McGraw-Hill, New York.

²P. J. Kasvinsky, N. B. Madsen, R. J. Fletterick & J. Sygusch (1978) *J. Biol. Chem.* **253**, 1290.

³K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

MASS-ACTION RATIO

The ratio of the product of the concentrations (or activities) of all the products of a reaction to the product of the concentration (or activities) of all the reactants (or substrates). This ratio, often symbolized by Γ , will change as the reaction progresses until equilibrium is reached (*i.e.*, at $t = \infty$), at which point $\Gamma = K_{eq}$.

MASSIEU FUNCTION

A thermodynamic function, symbolized by J , equal to the negative of the Helmholtz energy divided by the absolute temperature: thus, $J = -A/T$. The SI units are joules per kelvin. **See also** *Planck Function*; *Helmholtz Energy*

MASS SPECTROMETRY

An analytical technique that exploits magnetic fields to analyze molecules on the basis of their mass and electrical properties to determine (a) qualitative separation of mixtures of inorganic and organic species, (b) quantitative determination of the amount of substance, (c) isotopic abundance of atoms in simple and complex molecules, and (d) structures of biological and other organic molecules by use of special fragmentation methods.

General Features of Mass Spectrometers. To separate molecules into discrete chemical species, mass spectrometers utilize a sample inlet device, an ion source for imparting charge by bombardment with electrons, chemical ions, molecules, or photons. These systems are maintained at high vacuum so that the input sample is volatilized to a gaseous state and ported into the mass analyzer and detector. Mass analyzers differentiate molecules on the basis of their mass-to-charge (m/z) ratio by using magnetic sector analyzers, double-focusing analyzers, quadrupole mass filters, ion trapped analyzers, or time-of-flight analyzers. The power of mass spectrometry to separate chemical species is expressed in terms of resolution, which can be defined as $R = m/\Delta m$, where m is the mass of the first peak and Δm is the difference in mass between two successive peaks. Typical commercial spectrometers have resolutions of up to 500,000.

Types of Ion Sources. The versatility of mass spectrometry relates to the ability to generate ionized species using a variety of ion sources. Most frequently one utilizes electron ionization (EI) in which molecules in the gas phase are ionized by energetic electrons typically emitted from a tungsten or rhenium and accelerated by an electric potential of around 70 volts. The second most popular method for generating ionized species is chemical ionization (CI), a technique that relies on collision of analyte molecules with ions that have been previously produced by bombarding a reagent gas with electrons. Here, a common reagent gas is methane which upon electron bombardment can generate the following reactive species: CH_4^+ , CH_3^+ , and CH_2^+ . High-molecular-weight analytes and polyelectrolytes cannot be readily introduced in the gaseous state, and in such cases, one can use fast atom bombardment (FAB) to produce positive and negative analyte ions that are desorbed from the surface of the source. In the case of proteins, matrix-assisted laser desorption ionization (MALDI) is utilized to introduce the macromolecule into a mass analyzer, typically of the time-of-flight type. Another ion source technique is thermospray (or electrospray) ionization (ES) which imparts positive charges to a nebularized analyte solution.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

General Techniques: Methods of ionization, **193**, 3; mass analyzer, **193**, 37; detectors, **193**, 61; selected-ion measurements, **193**, 86; liquid chromatography-mass spectrometry, **193**, 107; tandem MS: multisector magnetic instruments, **193**, 131; tandem MS:

quadrupole and hybrid instruments, **193**, 154; matrix selection for liquid secondary ion and fast atom bombardment MS, **193**, 201; continuous-flow fast atom bombardment MS, **193**, 214; collision-induced dissociation, **193**, 237; mass spectrometry of peptides and proteins by matrix-assisted ultraviolet laser desorption/ionization, **193**, 280; exact mass measurements, **193**, 295; mass reference standards, **193**, 305; introduction of deuterium by exchange, **193**, 329; preparation of labeled molecules by exchange with oxygen-18 water, **193**, 338; mass and abundance values, **193**, 869; reference ions for exact mass measurements, **193**, 870; mass spectra of matrix materials, **193**, 875; calculation of isotopic abundance distributions, **193**, 882; nomenclature for peptide fragment ions (positive ions), **193**, 886; electron ionization mass spectra of trimethylsilyl derivatives of nucleosides, **193**, 825.

Peptides and Proteins: Residue mass for peptides, **193**, 888; electron ionization mass spectra of trimethylsilyl derivatives of nucleosides, **193**, 825, 361; strategies for locating disulfide bonds in proteins, mass and abundance values, **193**, 869; reference ions for exact mass measurements, **193**, 870; mass spectra of matrix materials, **193**, 875; calculation of isotopic abundance distributions, **193**, 882; nomenclature for peptide fragment ions (positive ions), **193**, 886; residue mass for peptides, **193**, 888; recombinant proteins, **193**, 519; electron ionization mass spectra of trimethylsilyl derivatives of nucleosides, **193**, 825; reversed-phase HPLC fractionation of enzymatic digests and chemical cleavage products, **193**, 389; electrospray ionization, **193**, 412; sample preparation for plasma desorption MS, **193**, 432; protein molecular weight analysis, **193**, 441; peptide sequencing by tandem MS and high-energy collision-induced dissociation, **193**, 455; liquid secondary ion MS of phosphorylated and sulfated peptides and proteins, **193**, 480; attachment sites and structural classes of asparagine-linked carbohydrates in glycoproteins, **193**, 501.

MATRIX ISOLATION

The isolation of a reactive or unstable molecular entity by dilution in an inert matrix.

MAXIMUM LIKELIHOOD METHOD

Maximum (maximized) likelihood is a statistical term that refers to the probability of randomly drawing a particular sample from a population, maximized over the possible values of the population parameters.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **210**, 203; testing by simulations, **210**, 225; computer applications for, **210**, 233; fitting of sums of exponentials to dwell-time distributions, **207**, 772; fluorescence data analysis, **210**, 200; phylogenetic analysis of restriction site data, **224**, 446.

MAXWELL

A unit, symbolized by Mx, for magnetic flux. It is the flux through a square centimeter normal to a field of one gauss. One maxwell is equal to 10^{-8} weber.

MAXWELL'S THERMODYNAMIC RELATIONS

Fundamental relationships involving first partial differentials of absolute temperature (T), pressure (P), vol-

ume (V), and entropy (S) for a homogeneous system with a constant, given mass.

$$(\partial T/\partial V)_s = -(\partial P/\partial S)_v$$

$$(\partial T/\partial P)_s = (\partial V/\partial S)_p$$

$$(\partial P/\partial T)_v = (\partial S/\partial V)_T$$

$$(\partial V/\partial T)_p = -(\partial S/\partial P)_T$$

In this set of four fundamental thermodynamic expressions, one should note that only entropy cannot be directly determined experimentally, whereas P , V , and T are all measurable properties.

MEAN TRANSIT TIME

The average period of time during which a metabolite (or drug) remains in the volume of distribution; also referred to as the mean residence time. **See** *Compartmental Analysis*

MECHANISM-BASED INHIBITOR

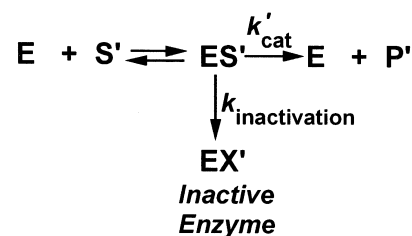
A molecular entity that (a) resembles the substrate (or product) of an enzymic reaction, (b) is chemically unreactive in the absence of the enzyme, (c) binds to the enzyme; (d) undergoes activation by the enzyme to yield a chemically reactive species, (e) can be converted to product, and (f) occasionally inactivates the enzyme prior to its release from that protein.

Suicide inhibition follows the general scheme^{1,2}:

Michaelis-Menten Scheme

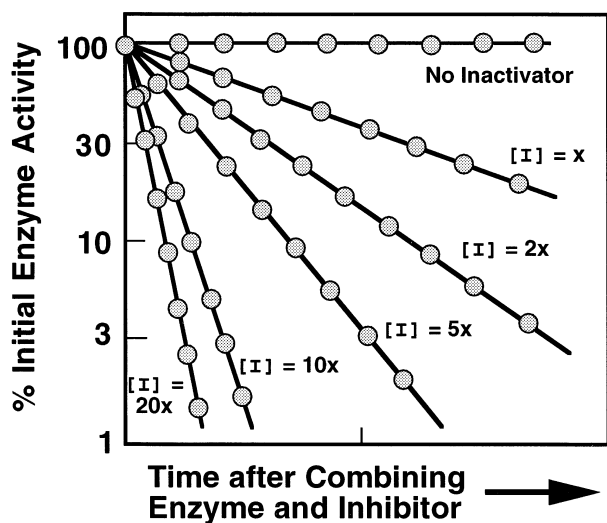


Suicide Inhibitor Scheme



where enzyme and inhibitor combine to form an EI complex which is interconverted to EX, the chemically activated species that either forms product P or covalently inactivates the enzyme. The kinetic behavior of a suicide

inhibitor is shown below. The amount of product formed before complete inactivation is a direct measure of the relative magnitudes of rate constants for the product-forming and enzyme-inactivating steps.



The synthesis of suicide inhibitors and the kinetics of enzyme inactivation have greatly advanced the study of enzyme mechanisms as well as the design of drugs and antimetabolites.

¹R. B. Silverman (1988) *Mechanism-Based Enzyme Activation: Chemistry and Enzymology*, vols. 1 and 2, CRC Press, Boca Raton.

²R. B. Silverman (1995) *Meth. Enzymol.* **249**, 240.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Applications, **249**, 263-283; basic kinetics, **249**, 243-245; biphasic kinetics, **249**, 253; competitive inhibitor protection, **249**, 247, 256; criteria for, **249**, 247-249; definition, **249**, 240-242; in drug design, **249**, 246-247; experimental protocols for, **249**, 249-263; inactivation prior to release of active species, **249**, 248-249, 259-260; involvement of catalytic step, **249**, 248, 259; irreversibility, **249**, 248, 256-257; kinetic constants, determination, **249**, 254-256; non-pseudo-first-order kinetics, **249**, 251-253; partition ratio, **249**, 251; partition ratio determination, **249**, 260-263; pseudo-first-order kinetics, **249**, 250-251; saturation, **249**, 247, 254-256; stoichiometry, **249**, 248, 257-259; in study of enzyme mechanisms, **249**, 245-246; substrate protection, **249**, 247, 256; time dependence of inactivation, **249**, 247, 249-253; uses, **249**, 245-247.

MEDIATED AMPEROMETRIC BIOSENSOR

Any electrochemical device using a low molecular weight redox couple to shuttle electrons from the redox center of an enzyme to the surface of an indicator electrode, thereby increasing the effectiveness of amperometry in the detection of a substrate for the particular enzyme. The internal cavities of six-, seven-, and eight-membered cyclodextrins are trapezoids of revolution with larger "open mouths" dimensions (*i.e.*, respective diameters of

0.49, 0.62, and 0.79 nm and depths typically of 0.8 nm) suitable for the binding of many redox-active substances as host-guest complexes. **See Biosensor**

J. H. T. Luong, R. S. Brown, K. B. Male, M. V. Cattaneo & S. Zhao (1995) *Trends in Biotechnology* **13**, 457.

MEDIUM

1. The phase or environment in which one or more molecular entities and/or their reactions are being studied.
2. The solvent holding one or more solutes in solution or suspension.
3. Culture medium; a substance or mixture used for cultivation, growth, storage, *etc.*, of microorganisms.

MEERWEIN-PONNDORFF-VERLEY REDUCTION

The reversible reduction of aldehydes or ketones to the corresponding alcohols using aluminum alkoxides such as aluminum isopropoxide ($\text{Al}[\text{OCH}(\text{CH}_3)_2]_3$), the reducing alcohol usually being isopropyl alcohol (*i.e.*, $\text{RCOR}' + \text{CH}_3\text{CH}(\text{OH})\text{CH}_3 \rightarrow \text{RCH}(\text{OH})\text{R}' + \text{CH}_3\text{COCH}_3$)¹⁻³. The reaction is facilitated by the removal of acetone by distillation⁴⁻⁶. This reaction has served as a model for metal-ion-dependent hydride transfer reactions catalyzed by certain enzymes. The reverse reaction is known as the Oppenauer oxidation^{7,8}.

¹H. Meerwein & R. Schmidt (1925) *Ann.* **444**, 221.

²W. Ponndorf (1926) *Angew. Chem.* **39**, 138.

³A. Verley (1925) *Bull. Soc. Chim. France* **37**, 537 and 871.

⁴J. Hutton (1979) *Synth. Commun.* **9**, 483.

⁵J. L. Namy, J. Soupe, J. Collin & H. B. Kagen (1984) *J. Org. Chem.* **49**, 2045.

⁶T. Okano, M. Matsuoka, H. Konishi & J. Kiji (1987) *Chem. Lett.* **1987**, 181.

⁷R. V. Oppenauer (1937) *Rec. Trav. Chim.* **56**, 137.

⁸C. Djerassi (1951) *Org. React.* **6**, 207.

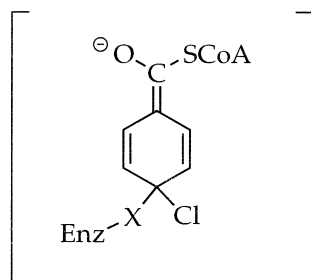
MEGA-

A prefix, symbolized by M, used in multiples of units and corresponding to a value of 10^6 .

MEISENHEIMER ADDUCT

A cyclohexadienyl Lewis adduct or salt formed by the reaction of a Lewis base with an aromatic compound¹. Such an adduct is apparently formed from the reaction of OH^- with 4-(*N*-2-aminoethyl-2'-pyridyl disulfide)-7-nitrobenzo-2-oxa-1,3-diazole (2PROD)². 2PROD is a two-protonic-state electrophile used as a probe for enzyme active site nucleophiles and as a fluorescent re-

porter-group delivery vehicle³. **See** *Two-Protonic State Electrophiles*



Yang *et al.*⁴ recently reported on the mechanism of 4-chlorobenzoyl coenzyme A dehalogenase, an enzyme that catalyzes the hydrolytic dehalogenation of 4-chlorobenzoyl coenzyme A (4-CBA-CoA) to form 4-hydroxybenzoyl coenzyme A (4-HBA-CoA). The mechanism involves attack of an active site carboxylate at C4 of the substrate benzoyl ring to form a Meisenheimer complex (shown above). Loss of chloride ion from this intermediate then forms an arylated enzyme intermediate that is hydrolyzed to free enzyme plus 4-HBA-CoA by the addition of water at the acyl carbon. In later work, Taylor *et al.*⁵ examined the activation of the 4-CBA-CoA toward nucleophilic attack by the active site carboxylate group.

¹J. Meisenheimer (1902) *Liebig's Ann. Chem.* **323**, 205.

²B. S. Baines, G. Allen & K. Brocklehurst (1977) *Biochem. J.* **163**, 189.

³K. Brocklehurst (1982) *Meth. Enzymol.* **87**, 427.

⁴G. Yang, P.-H. Liang & D. Dunaway-Mariano (1994) *Biochemistry* **33**, 8527.

⁵K. L. Taylor, R. Q. Liu, P. H. Liang, J. Price, D. Dunaway-Mariano, P. J. Tonge, J. Clarkson & P. R. Carey (1995) *Biochemistry* **34**, 13881.

MELILOTATE HYDROXYLASE

This FAD-dependent enzyme [EC 1.14.13.4], also known as melilotate 3-monooxygenase and 2-hydroxyphenylpropionate hydroxylase, catalyzes the reaction of 3-(2-hydroxyphenyl)propanoate with NADH and dioxygen to produce 3-(2,3-dihydroxyphenyl)propanoate, NAD⁺, and water.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

MELTING

1. The process by which a solid converts into a liquid, usually as a result of an increase in temperature. [The melting point is the temperature at which the solid and

liquid phases are in equilibrium with each other at a defined pressure (usually one atmosphere)]. While the melting point should have the same value as the freezing point of a substance, "freezing" is defined as the conversion of a liquid to a solid. 2. A term typically used in biochemistry to describe loss of structure attending thermal denaturation (most frequently applied to polynucleotides such as DNA). This denaturation is a cooperative process and the kinetics of the process depends on a number of factors: for example, the GC content, the ionic strength, the nature of the solvent, the pH, *etc.* The melting temperature, T_m , is the midpoint of a melting curve. Crothers *et al.*¹ have studied the kinetics and mechanism of the thermal melting of *E. coli* tRNA_f^{Met}.

¹D. M. Crothers, P. E. Cole, C. W. Hilbers & R. G. Shulman (1974) *J. Mol. Biol.* **87**, 63.

MEMBRANE POTENTIAL

The voltage difference, sometimes referred to as the transmembrane potential or the membrane electric potential, that is due to the difference of concentrations of ions on either side of the membrane. Typically, such potentials are measured in millivolts (mV) and usually have values between -30 and -70 mV. In most cells, the interior of a cell is negative with respect to the exterior.

For the simplified case of only a single permeant ion, the equilibrium or Nernst potential defines the condition at which there is no net change in ion concentration. Under this condition of equilibrium, the electrochemical potential μ_j for moving the ion j^{z+} across the membrane will be zero, and

$$\mu_j = RT \ln[j]_{\text{exterior}}/[j]_{\text{interior}} - z\mathcal{F}E$$

where R is the gas constant, T is the absolute temperature, $\mathcal{F}$ is Faraday's constant (equal to 96,500 coulombs/mol), z is the ionic charge, and E is the transmembrane potential. Thus,

$$\begin{aligned} E_j &= (RT/z\mathcal{F}) \ln[j]_{\text{exterior}}/[j]_{\text{interior}} \\ &= (58 \text{ mV}/z) \log_{10}[j]_{\text{exterior}}/[j]_{\text{interior}} \end{aligned}$$

Thus, a 10:1 transmembrane gradient of a single monovalent ion, say potassium, will generate a membrane potential of 58 mV. **See** *Resting Potential; Action Potential; Depolarization; Threshold Potential; Nernst Equation; Goldman Equation; Patch-Clamp Technique*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

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MEMBRANE TRANSPORT

Transporters, particularly those carrying nonlipophilic species across biomembranes or model membranes, can be regarded as vectorial catalysts¹ (and are also called carriers, translocators, permeases, pumps, and ports [e.g., symports and antiports]). Many specialized approaches and techniques have been developed to characterize such systems. This is reflected by the fact that there are currently twenty-three volumes in the *Methods in Enzymology* series (vols. 21, 22, 52-56, 81, 88, 96-98, 125-127, 156-157, 171-174, and 191-192) devoted to biomembranes and their constituent proteins. Chapters in each of these volumes will be of interest to those investigating transport kinetics. Other volumes are devoted to ion channels (207), membrane fusion techniques (220 and 221), lipids (14, 35, 71, and 72), plant cell membranes (148), and a volume on the reconstitution of intracellular transport (219). **See** *Ion Pumps*

BASIC TRANSPORTER KINETICS. In the kinetic identification and characterization of a transport system [including a demonstration of the ion dependency (e.g., Na⁺-dependency) or lack of dependency], it is essential to demonstrate that saturation of the transport of a particular substrate occurs²⁻⁴. Simple diffusion of a substrate

across a membrane is nonsaturable. (Note that for many substrates, a slow rate of diffusion across a membrane may take place simultaneously with the carrier-linked transport system, and appropriate corrections for basal rates of diffusion are necessary.) In addition, mediated-transport systems are typically very specific with respect to the nature and structure of the transporting molecular entity. An excellent probe of carrier-mediated transport is the stimulation observed of substrate uptake by the presence of a “countersubstrate” on the other side of the membrane. This acceleration by an internal substrate represents a facultative counterexchange and is a stringent criteria of carrier-mediated transport⁴.

The rate equation for a single class of transporters acting on a metabolite is analogous to the Michaelis-Menten equation, except that one must also account for the effects of passive diffusion:

$$v = (v_{\text{carrier}} + v_{\text{diffusion}}) = \left\{ \frac{V_{\text{max}}[S]}{[S] + K_m} + k_D[S] \right\}$$

The contribution of diffusion to the rate does not follow rate saturation. Therefore, as one proceeds toward carrier saturation with substrate S, the rate will not simply reach a plateau, as is most often the case for one-substrate enzymes operating on extremely slow uncatalyzed reactions. Instead, a plot of v versus $[S]$ will continue to rise with a constant slope, equal to k_D . It is usually a simple matter to extrapolate this noncarrier-mediated contribution back to each time-point for which a measurement was obtained and to obtain the corrected carrier-mediated rate by subtraction of the diffusional contribution.

Net carrier transport can be examined whenever a transportable substrate is present on both sides of a membrane. Again for the case of a single class of carriers, the inward transport rate and outward transport rates show rate-saturation behavior:

$$v_{\text{inward}} = \frac{V_{\text{max}}^{\text{out}} [S_{\text{outside}}]}{[S_{\text{outside}}] + K_m^{\text{out}}} \quad \text{and} \quad v_{\text{outward}} = \frac{V_{\text{max}}^{\text{in}} [S_{\text{inside}}]}{[S_{\text{inside}}] + K_m^{\text{in}}}$$

and the expression for net transfer, $v_{\text{net}} = (v_{\text{outside}} - v_{\text{inside}})_{\text{outward}}$, including the contribution from passive diffusion, can be stated as follows:

$$v_{\text{net}} = \left\{ \frac{V_{\text{max}}^{\text{out}}[S_{\text{outside}}]}{[S_{\text{outside}}] + K_{\text{m}}^{\text{out}}} \right\} - \left\{ \frac{V_{\text{max}}^{\text{in}}[S_{\text{inside}}]}{[S_{\text{inside}}] + K_{\text{m}}^{\text{in}}} \right\} + k_{\text{D}}([S_{\text{outside}}] - [S_{\text{inside}}])$$

Inhibition of carrier-mediated transport usually obeys the same rate laws describing competitive inhibition and noncompetitive inhibition for one-substrate enzymes.

If more than one class of carriers is mediating the transport of a common substrate S, then the rate law for transport becomes:

$$v = (v_1 + v_2) = \left\{ \frac{V_{\text{max},1}[S]}{[S] + K_{\text{m},1}} + \frac{V_{\text{max},2}[S]}{[S] + K_{\text{m},2}} \right\}$$

KINETIC CHARACTERIZATION OF TRANSPORTERS. Stein^{3,5} has provided a useful outline of kinetic approaches to characterize transport systems. In each of the approaches, the investigator must accurately determine the amount of substrate transported at a defined point of time after the substrate and the membrane system have contacted each other. The kinetic approaches include:

(1) Equilibrium Exchange Methods: The membrane-enclosed container (*e.g.*, a cell, an organelle, or a vesicle) is preloaded with substrate at a particular concentration and then placed in a medium containing the substrate at the same concentration. (Note: One of the two “pools” of substrate is isotopically labeled.) The investigator then measures the amount of radiolabel transferred from vesicle to medium or *vice versa* as a function of time. A plot of the logarithm of the fraction of label exchange will be linear provided the system is homogeneous. It should also be noted that the same rate constant for equilibrium exchange should be obtained for both efflux and influx experiments. This should be demonstrated for each transport system studied. Finally, the rate constant for equilibrium exchange should be determined over a wide range of substrate concentrations. The dependency of the rate constant (or lack of) on the substrate concentration will reflect the type(s) of the transport system(s) present.

(2) Zero-Trans Procedures: These methods involve measurements of substrate transport into or out of a container (*i.e.*, a cell, organelle, vesicle, *etc.*) in which one side of the membrane (the trans side) is free of the

presence of the substrate. The other side of the membrane (termed the cis side) is preloaded with different concentrations of the substrate and the rate of transport is measured. This technique has a number of experimental difficulties. For example: measurements must be made rapidly; otherwise, integrated equations are required to account for the time evolution of the process. Thus, measurements must commence as soon as the cells are loaded. Because transport begins immediately, efflux and influx studies often yield different results as a consequence of the inherent asymmetry of the transporter. Transport of the substrate to the trans side is also often accompanied by osmotic and/or pH changes. When such experiments are done and these technical difficulties are properly addressed, the velocity of zero-trans transport can be plotted as a function of substrate concentration to obtain valid kinetic parameters.

(3) Infinite Trans Procedures: In this experimental methodology, the membrane-enclosed container (*e.g.*, cells, organelles, vesicles, *etc.*) is preloaded with a high concentration of unlabeled substrate. They are then placed in a medium containing the labeled substrate and the rate of influx is measured. The experiment is then repeated at different concentrations of the substrate in the surrounding medium. Note that the concentration of the substrate within the cell must be very high (in effect, infinite). This requires some awareness of the approximate magnitude of the K_{m} value of the substrate, and preliminary experiments should be undertaken. An investigator may also wish to demonstrate that an increase in the concentration of the internalized unlabeled substrate does not result in changes in the value of the observed kinetic parameters. While one could also do an infinite trans experiment for efflux transport (in which the medium contains a high concentration of substrate), this is difficult on technical grounds.

(4) Infinite Cis Methods: With this approach, membrane-enclosed containers (once again, cells, organelles, vesicles, *etc.*) are preloaded with high concentrations of unlabeled substrate. The cells are then placed in a medium containing various concentrations of unlabeled substrate (including no substrate and a saturating level of substrate). The investigator then measures the net loss of substrate from the container.

(5) Counterflow Experiments: Cells are once again preloaded with a high concentration of unlabeled substrate. The cells are then placed in a medium containing a low concentration of labeled substrate or a low concentration of an alternative substrate which utilizes the same transport system. The investigator then measures the rate of entry of the labeled substrate or of the alternative substrate.

EFFECT OF ELECTRICAL POTENTIAL ON TRANSPORT. The Gibbs free energy associated with membrane transport can be expressed as $\Delta G = RT \ln(C_{\text{internal}}/C_{\text{external}})$. Nonetheless, the actual concentrations achieved depend on the generation of transmembrane electrical potential, itself written as $\Delta G = z\mathcal{F}E$, where z is the valence, $\mathcal{F}$ is Faraday constant (23.5 kcal/V·mol). E is the transmembrane potential difference, expressed in volts. Thus the minimal amount of free energy change associated with the movement of 1 mol solute can be written as:

$$\Delta G = RT \ln(C_{\text{internal}}/C_{\text{external}}) + z\mathcal{F}E$$

For a more detailed discussion of this topic, **See Membrane Potential**

¹G. Sachs & S. Fleischer (1989) *Meth. Enzymol.* **171**, 3.

²T. Rosenberg & W. Wilbrandt (1957) *J. Gen. Physiol.* **41**, 289.

³W. D. Stein (1986) *Transport and Diffusion Across Cell Membranes*, Academic Press, Orlando, Florida.

⁴M. Klingenberg (1989) *Meth. Enzymol.* **171**, 12.

⁵W. D. Stein (1989) *Meth. Enzymol.* **171**, 23.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

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MEPRIN

These zinc-dependent endopeptidases (meprin A [EC 3.4.24.18] and meprin B [EC 3.4.24.63]) are members of the peptidase family M12A. They catalyze the hydrolysis of peptide bonds in proteins and peptide substrates. Meprin A, a membrane-bound enzyme that has been isolated from mouse and rat kidney and intestinal brush borders as well as salivary ducts, acts preferentially on carboxyl side of hydrophobic amino acyl residues. Meprin A and B are insensitive to inhibition by phosphoramidon and thiorphan.

R. L. Wolz & J. S. Bond (1995) *Meth. Enzymol.* **248**, 325.
J. S. Bond & P. E. Butler (1987) *Ann. Rev. Biochem.* **56**, 333.

3-MERCAPTOPYRUVATE SULFURTRANSFERASE

This zinc-dependent enzyme [EC 2.8.1.2] catalyzes the reaction of 3-mercaptopyruvate with cyanide to produce pyruvate and thiocyanate. Other substrates include sulfite, sulfinates, mercaptoethanol, and mercaptopyruvate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

MERCURY(II) REDUCTASE

This enzyme [EC 1.16.1.1], also known as mercuric reductase, catalyzes the reaction of Hg with NADP⁺ and H⁺ to produce Hg²⁺ and NADPH.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

MESO FORM (or, MESO ISOMER)

A stereoisomer having more than one chiral center and having an internal plane of symmetry. Hence, *meso* compounds do not exhibit optical activity.

Meso isomers can prove to be quite useful at times, particularly in assessing the topology of the active site of a protein. γ -Glutamyl transpeptidase can catalyze the transfer of the γ -glutamyl moiety of glutathione to an amino acid, forming a γ -glutamyl amino acid isopeptide¹. The enzyme is stereospecific for L-amino acids and L-cystine serves as an excellent substrate. As one might expect, *meso*-cystine can also act as a substrate although there can only be one productive binding mode.

¹R. D. Allison (1985) *Meth. Enzymol.* **113**, 419.

MESOLYTIC CLEAVAGE

Cleavage of a bond in a radical ion such that a separate radical product and ion product are formed.

P. Maslak & J. N. Narvaez (1990) *Angew. Chem. Int. Ed. Engl.* **29**, 283.

METABOLIC CONTROL ANALYSIS

An approach for interpreting changes in metabolic flux (symbolized by *J*) for a pathway made up of multiple steps, each catalyzed by an enzyme. The basic tenets of this approach are that no single enzyme-catalyzed step controls flux through a pathway and that the relative importance of individual steps in determining the overall flux can be altered by changes in metabolic state or by externally applied conditions¹. The sensitivity of a pathway to changes in the activity of a particular enzyme-catalyzed step is typically evaluated by using inhibitors or molecular genetic approaches to alter the activity of the particular step.

The formalized application of metabolic control analysis deals with several parameters²: (a) The *flux control coefficient* is defined as the fractional change in pathway flux

that attends an infinitesimal fractional change in activity of one particular enzyme in the pathway. A value of zero indicates that the metabolic flux is completely insensitive to a change in the activity of a particular enzyme, and a value of one would mean that the pathway is completely dominated by the activity of one particular step. (b) The *summation theorem* requires that all flux control coefficients sum to unity. (c) *Elasticity coefficients* represent measures of the fraction change in metabolic flux caused by an infinitesimal change in the concentration of a positive or negative effector. (d) The *connectivity theorem* indicates that the control of a system can be solved by determining the flux control coefficients and elasticity coefficients for each component step in a pathway. **See Top-Down Control Analysis**

¹H. Kacser & J. A. Burns (1987) *Trends in Biochem. Sci.* **12**, 5.

²P. A. Quant (1993) *Trends in Biochem. Sci.* **18**, 26.

METABOLONS

Organized, multienzyme structures of particular biochemical pathways such that they provide efficient flux through the sequential reactions of those pathways¹. Examples of pathways that have been suggested to act through metabolons include the tricarboxylic acid cycle², β -oxidation^{3,4}, the biosynthesis of specific classes of isoprenoids^{5,6}, and nucleotide biosynthesis⁷⁻⁹. There are now excellent kinetic criteria for judging whether a system involves substrate channeling. **See Substrate Channeling**

¹P. A. Srere (1987) *Ann. Rev. Biochem.* **56**, 89.

²B. Sumegi, A. D. Sherry, C. R. Malloy, C. Evans & P. A. Srere (1991) *Biochem. Soc. Trans.* **4**, 1002.

³P. A. Srere & B. Sumegi (1994) *Biochem. Soc. Trans.* **22**, 446.

⁴M. A. Nada, W. J. Rhead, H. Sprecher, H. Schulz & C. R. Roe (1995) *J. Biol. Chem.* **270**, 530.

⁵J. Chappell (1995) *Plant Physiol.* **107**, 1.

⁶J. Chappell (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 521.

⁷R. J. Zeleznikar, R. A. Heyman, R. M. Graeff, T. F. Walseth, S. M. Davis, E. A. Butz & N. D. Goldberg (1990) *J. Biol. Chem.* **265**, 300.

⁸F. N. Gellerich, Z. A. Khuchua & A. V. Kuznetsov (1993) *Biochem. Biophys. Acta* **1140**, 327.

⁹F. Savabi (1994) *Mol. Cell Biochem.* **134**, 145.

METAL ION CATALYSIS

Acceleration of reaction rate as a result of the presence of one or more metal ions. In many cases, the general acid properties of the metal ion contribute to catalysis. Metal ions play organizing roles as templates for reactants. The metal ion may also stabilize the transition

state. Enzyme-bound metal ions (many enzymes are metalloproteins with tightly bound ions) can transiently undergo changes in their oxidation state to facilitate catalysis.

METAL ION COMPLEXES (Determining Stability Constants)

An essential aspect to understanding the influence of metal ions on enzyme-catalyzed reactions is the knowledge of how tight different metal ions bind to a wide variety of substrates (particularly nucleotides and other phosphoryl-containing molecular entities), products, and effectors and that binding phenomena are altered by the experimental conditions (*e.g.*, the effects of pH, temperature, ionic strength, *etc.*). This necessitates the experimental determination of the stability constant (an association constant) for the metal ion-ligand complex. O'Sullivan and Smithers¹ have reviewed the theory and the various techniques for such determinations and have provided values for many of the more common, biochemically relevant complexes.

¹W. J. O'Sullivan & G. W. Smithers (1979) *Meth. Enzymol.* **63**, 294.

METAL ION COORDINATION REACTIONS

Metal ions interact with many substances, and the term "ligand" (from the Latin verb *ligare*: to join together) refers to a substance that interacts ionically to form an electrovalent bond or donates electrons to form coordinate covalent bonds. In biological systems, metal ions are typically aquated (*i.e.*, complexed to water) as well as combined with other ligands. Because water is so plentiful, aqua complexes of metal ions must first lose one or more H₂O molecules before other ligands can take their place. Water is neutral, and its reaction with a complex, say [L₅MX] where X is an anionic or electron-rich substance, is usually relatively slow. **See Water; Water Exchange Rates of Aqua Metal Ions.** By contrast, the displacement of a neutral water molecule by a negatively charged ligand (*e.g.*, [L₅M(H₂O)]ⁿ⁺ + X⁻ → [L₅MX]ⁿ⁻¹ + H₂O) generally proceeds rapidly.

Acid hydrolysis of an octahedral metal ion complex is typically a dissociative or S_N1-type reaction. In the case of base hydrolysis, reactions tend to display S_N2-type reaction mechanisms, although others take place by what is termed an S_N1-conjugate base mechanism. The latter involves attack by an electrophile to abstract a proton

from a bound ligand to yield its conjugate base form, and this route is limited to ligands that contain protons that can dissociate.

Oxidation-reduction reactions can occur if two redox-active atoms or groups come sufficiently close to each other for electron transfer to occur. With metal ion complexes, a change in the oxidation state of the metal ion may lead to a rearrangement of ligand bonding geometry. The term “outer-sphere” mechanism describes the situation where the ligands within the metal ion complex are so firmly held that the electron from the oxidant must transfer through this fully occupied coordination shell. By contrast, “inner-sphere” electron transfer reactions occur if the oxidizing agent binds to or shares a ligand with the metal ion complex. **See also** *Water; Water Exchange Rates of Aqua Metal Ions*

F. A. Cotton & G. Wilkinson (1988) *Advanced Inorganic Chemistry*, 5th ed., Wiley, New York.

METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS

The actual substrate in most nucleotide-dependent reactions is not the free nucleotide (*i.e.*, NTP^{4-} [or NTPH^{3-}] or NDP^{3-} [or NDPH^{2-}]); rather, the actual phosphoryl donor species is the metal ion-nucleotide complex. Examples include the hexokinase reaction ($\text{MgATP}^{2-} + \text{glucose} \rightleftharpoons \text{MgADP}^{1-} + \text{glucose 6-phosphate}$) and the adenylate kinase reaction ($\text{MgADP}^{1-} + \text{ADP}^{3-} \rightleftharpoons \text{MgATP}^{2-} + \text{AMP}^{2-}$). In fact, the uncomplexed nucleotide may even act as a potent inhibitor. Such is the case with the brain hexokinase reaction in which the true phosphoryl donor is MgATP^{2-} complex, whereas uncomplexed ATP^{4-} is a competitive inhibitor¹. Interested readers should consult Morrison² for a detailed account of recommended procedures for studying enzymes that utilize metal-nucleotide substrates.

A surprisingly common, but ill-advised practice in studies of ATP-dependent kinases, synthetases, ATPases, and related enzymes is the use a single, high concentration of divalent metal ion throughout the entire kinetic investigation. The resultant problem is that elevated concentrations of metal ions often prove to be inhibitory; even more significant is the fact that the free metal ion concentration will not remain constant as the nucleotide concentration is increased. The other substrates in a multisub-

strate reaction may also nonspecifically bind the metal ions that are present at elevated concentrations. Another erroneous practice is to vary the absolute concentrations of metal ion and nucleotide simultaneously, say by maintaining both at constant ratio relative to each other. In such cases, the fraction of nucleotide present as the metal-nucleotide complex will change with absolute concentration, a phenomenon that can result in false cooperativity¹ observed as sigmoidal plots of v_o versus [substrate]. These effects are illustrated in Fig. 1 for a hypothetical ATP-dependent enzyme utilizing MgATP^{2-} complex as the substrate.

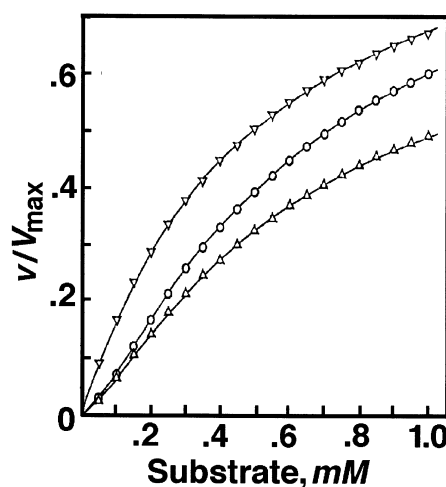


Figure 1. Plot of $v/V_{\max}$ versus the millimolar concentration of total substrate for a model enzyme displaying Michaelis-Menten kinetics with respect to its substrate MA (*i.e.*, metal ion M complexed to otherwise inactive ligand A). The concentrations of free A and MA were calculated assuming a stability constant of $10,000 \text{ M}^{-1}$. The Michaelis constant for MA and the inhibition constant for free A acting as a competitive inhibitor were both assumed to be 0.5 mM . The ratio $v/V_{\max}$ was calculated from the Michaelis-Menten equation, taking into account the action of a competitive inhibitor (when present). The upper curve represents the case where the substrate is both A and MA. The middle curve deals with the case where MA is the substrate and where A is not inhibitory. The bottom curve describes the case where MA is the substrate and where A is inhibitory. In this example, $[\text{M}]_{\text{total}} = [\text{A}]_{\text{total}}$ at each concentration of A plotted on the abscissa. Note that the bottom two curves are reminiscent of allosteric enzymes, but this false cooperativity arises from changes in the fraction of total “substrate A” that has metal ion bound. For a real example of how brain hexokinase cooperatively was debunked, consult D. L. Purich & H. J. Fromm (1972) *Biochem. J.* **130**, 63.

For kinetic experiments that are designed to determine the rate dependence of an enzyme-catalyzed reaction on nucleotide concentration, care must be exercised to maintain a constant free metal ion concentration under all experimental conditions. Too low a metal ion concentration allows uncomplexed nucleotides to inhibit an en-

zyme, and too high a metal ion concentration can lead to nonspecific binding and inhibition. In cases where the cosubstrate also binds metal ions, one may need to consider these binding interactions as well. Likewise, one may also want to use a buffer that does not bind metal ions and does not interfere with the strength of metal-nucleotide complexation².

Preliminary rate measurements should allow one to make a plot of initial velocity v_o versus [metal ion], and this should provide information on the optimal metal ion concentration. (For many MgATP^{2-} -dependent enzymes, the optimum is frequently 1–3 mM uncomplexed magnesium ion.) Then, by utilizing published values for formation constants (also known as stability constants) defining metal ion-nucleotide complexation, one can readily design experiments to keep free metal ion concentration at a fixed level. To compensate properly for metal ion complexation in ATP-dependent reactions, one must choose a buffer for which a stability constant is known. For example, in 25 mM Tris-HCl (pH 7.5), the stability constant for MgATP^{2-} is approximately $20,000 \text{ M}^{-1}$. Thus, one can write the following equation:

$$K = 20,000 \text{ M}^{-1} = [\text{MgATP}^{2-}]/[\text{Mg}^{2+}\text{free}][\text{ATP}^{4-}]$$

and if, for example, one wishes to work at 1 mM free (or uncomplexed) magnesium ion, the following holds:

$$(20,000 \text{ M}^{-1})[\text{Mg}^{2+}\text{free}] = 20 = [\text{MgATP}^{2-}]/[\text{ATP}^{4-}]$$

The ratio $[\text{MgATP}^{2-}]/[\text{ATP}^{4-}]$ can be treated as x/y , where x represents $[\text{MgATP}^{2-}]$, y represents $[\text{ATP}^{4-}]$, and $(x + y)$ equals $[\text{ATP}_{\text{total}}]$, the total nucleotide concentration³. For the above case, we see that $x = (20/21)[\text{ATP}_{\text{total}}]$ and $y = (1/21)[\text{ATP}_{\text{total}}]$, such that $(20/21)[\text{ATP}_{\text{total}}] + (1/21)[\text{ATP}_{\text{total}}] = [\text{ATP}_{\text{total}}]$. If one chooses to work with 25 mM HEPES (pH 7.5) as the buffer, the stability constant for MgATP^{2-} is $100,000 \text{ M}^{-1}$; in this case, at 1 mM free magnesium ion, $x = (100/101)[\text{ATP}_{\text{total}}]$ and $y = (1/101)[\text{ATP}_{\text{total}}]$.

To prepare a series of assay samples with different concentrations of phosphoryl acceptor substrate (designated here as “phosphoryl acceptor”) and ATP, while still maintaining the desired $[\text{Mg}^{2+}]_{\text{free}}$, the following procedure is recommended. First, one prepares four different component solutions:

0.30 mL Buffer- $\text{Mg}^{2+}_{\text{free}}$ solution

0.30 mL Phosphoryl-Acceptor solution (at highest concentration)

0.30 mL ATP- $\text{Mg}^{2+}_{\text{bound}}$ solution (at highest concentration)

0.10 mL Enzyme solution

1.00 mL Reaction sample when combined

(The total volume here is chosen for most UV/Vis spectrophotometric assays, but the total reaction volume can be whatever the experimenter desires.) Note also that the magnesium ion is present in *two* solutions, and only when they are combined in the overall 1.00 mL sample will the final free concentration be what one desires.

Let us, for example, prepare 10 mL of the component solution that is 0.025 M buffer, 1 mM magnesium ion, and 5 mM of a nonvaried substance Y from stock solutions of buffer (say 1 M), magnesium ion (say 0.1 M), and Y (say 20 mM). The following equation allows us to calculate the volume of each stock solution while accounting for the dilution factor (in this case, 3.33 or 1.0 divided by 0.30):

$$\frac{(1.00/0.30) \times 0.025 \text{ M} \times 10 \text{ mL}}{1 \text{ M Buffer Stock Solution}} = 0.83 \text{ mL Stock Buffer}$$

$$\frac{(1.00/0.30) \times 1 \text{ mM} \times 10 \text{ mL}}{100 \text{ mM Magnesium Stock Solution}} = 0.33 \text{ mL Stock Magnesium}$$

$$\frac{(1.00/0.30) \times 5 \text{ mM} \times 10 \text{ mL}}{25 \text{ mM “Y” Stock Solution}} = 6.67 \text{ mL Stock “Y”}$$

The solutions are combined and then brought to 10 mL final volume with the addition of 2.17 mL deionized water. [Note: The (1.00/0.30)-factor corrects for the final combination of all four component solutions into the final reaction sample.]

To prepare say 10 mL of the 2 mM Phosphoryl-Acceptor (or Acc) solution from a 10 mM stock solution, we use the same procedure:

$$\frac{(1.00/0.30) \times 2 \text{ mM} \times 10 \text{ mL}}{10 \text{ mM Phosphoryl-Acc. Stock Solution}} = 6.67 \text{ mL Stock Phosphoryl-Acc.}$$

The solutions are combined and then brought to 10 mL final volume with the addition of deionized water. Fur-

ther dilutions of this well-mixed solution can be used to generate working solutions, such as $[\text{Acc}]_1$, $[\text{Acc}]_3$, $[\text{Acc}]_5$, $[\text{Acc}]_7$, and $[\text{Acc}]_9$, by combining one volume of solution at $[\text{Acc}]$ with 0, 2, 4, 6, and 8 volumes of deionized water.

Next, we can prepare 10 mL $\text{ATP-Mg}^{2+}_{\text{bound}}$ solution in a similar manner:

$$\frac{(1.00/0.30) \times 10 \text{ mM} \times 10 \text{ mL}}{100 \text{ mM ATP Stock Solution}} = \frac{3.33 \text{ mL}}{\text{Stock ATP}}$$

$$\frac{(1.00/0.30) \times (20/21) \times 10 \text{ mM} \times 10 \text{ mL}}{1.00 \text{ M Magnesium Stock Solution}} = \frac{0.79 \text{ mL Stock}}{\text{Magnesium}}$$

These latter two aliquots are combined and brought to 10 mL final volume with the addition of 5.88 mL deionized water. [Note that the complexation factor has a value of 20/21 in the second entry. This value will depend on the stability constant and the desired free magnesium ion concentration. For example, in 25 mM Tris-HCl (pH 7.5), the stability constant is $20,000 \text{ M}^{-1}$; thus, the complexation factor will be (20/21) at 1 mM uncomplexed magnesium ion. With 25 mM HEPES (pH 7.5), the stability constant is $100,000 \text{ M}^{-1}$, and the complexation factor value would be (100/101) at 1 mM free magnesium ion. If another free magnesium ion concentration is desired or if another buffer is used, then one must recalculate the fraction of bound metal ion.] Further dilutions of this well mixed solution can be used to generate $[\text{ATP}]_1$, $[\text{ATP}]_3$, $[\text{ATP}]_5$, . . . , working solutions as described above. When finally combined with the three other component solutions to make up a reaction sample, the free metal ion will always be 1 mM for the example given.

Lastly, to prepare the enzyme solution, say 10 nM from a 200 nM stock enzyme solution, we use the same procedure, except that the dilution factor in this case is (1.00/0.10) in terms of the 0.1 mL enzyme solution volume cited above.

$$\frac{(1.00/0.10) \times 10 \text{ nM} \times 1 \text{ mL}}{200 \text{ nM Enzyme Stock Solution}} = 0.50 \text{ mL Stock Enzyme}$$

This volume is then brought to the specified 1 mL final volume with the addition of 0.5 mL deionized water (or buffer and any other factors needed to maintain enzyme stability).

In this example, one can construct the following matrix of samples for determining the kinetic parameters of a two-substrate enzymic reaction:

$[\text{Acc}]_1 + [\text{ATP}]_1$	$[\text{Acc}]_3 + [\text{ATP}]_1$	$[\text{Acc}]_5 + [\text{ATP}]_1$	$[\text{Acc}]_7 + [\text{ATP}]_1$	$[\text{Acc}]_9 + [\text{ATP}]_1$
$[\text{Acc}]_1 + [\text{ATP}]_3$	$[\text{Acc}]_3 + [\text{ATP}]_3$	$[\text{Acc}]_5 + [\text{ATP}]_3$	$[\text{Acc}]_7 + [\text{ATP}]_3$	$[\text{Acc}]_9 + [\text{ATP}]_3$
$[\text{Acc}]_1 + [\text{ATP}]_5$	$[\text{Acc}]_3 + [\text{ATP}]_5$	$[\text{Acc}]_5 + [\text{ATP}]_5$	$[\text{Acc}]_7 + [\text{ATP}]_5$	$[\text{Acc}]_9 + [\text{ATP}]_5$
$[\text{Acc}]_1 + [\text{ATP}]_7$	$[\text{Acc}]_3 + [\text{ATP}]_7$	$[\text{Acc}]_5 + [\text{ATP}]_7$	$[\text{Acc}]_7 + [\text{ATP}]_7$	$[\text{Acc}]_9 + [\text{ATP}]_7$
$[\text{Acc}]_1 + [\text{ATP}]_9$	$[\text{Acc}]_3 + [\text{ATP}]_9$	$[\text{Acc}]_5 + [\text{ATP}]_9$	$[\text{Acc}]_7 + [\text{ATP}]_9$	$[\text{Acc}]_9 + [\text{ATP}]_9$

The most important point is that this procedure provides a reliable and effective way for dealing with metal ion interactions with nucleotides. By maintaining the free magnesium ion at a single fixed concentration, the experimenter avoids any chance of uncontrolled variation in the $[\text{MgATP}^{2-}]/[\text{ATP}^{4-}]$ ratio.

The following table lists several frequently used stability constants for metal ion-nucleotide complexes. Because complexation is driven by neutralization of electric charge on the components, one should immediately appreciate that the values apply only for the specified solution conditions. A more complete list is provided elsewhere⁴⁻⁵.

Stability Constants for Complexes of ATP^a

Complex Formed	$K_{\text{stability}} (\text{M}^{-1})$
MgH_2ATP	20
MgHATP^{1-}	500
MgATP^{2-}	73,000
CaH_2ATP	20
CaHATP^{1-}	400
CaATP^{2-}	35,000
MnH_2ATP	~100
MgHATP^{1-}	~1000
MgATP^{2-}	100,000
KATP^{3-}	15
NaATP^{3-}	14

^aFrom reference 4.

Finally, because nucleotides can also act as allosteric effectors, an investigator seeking to characterize the kinetic properties of allosteric enzymes should consider using the methods described above to determine whether the true effector is the free nucleotide or the form complexed with metal ion.

¹D. L. Purich & H. J. Fromm (1972) *Biochem. J.* **130**, 63.

²J. F. Morrison (1979) *Meth. Enzymol.* **63**, 257. (This reference provides additional advice for those seeking to characterize metal ion-dependent enzymes.)

³Recall that if one knows the ratio, r , of any two quantities a and b , the relative amounts of those two quantities is also known, because $r = a/b = x/(1 - x)$.

⁴D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Wiley, New York. (This handy reference provides detailed procedure that allow one to compensate for pH and temperature effects on metal-ligand interactions as well as buffers.)

⁵W. J. O'Sullivan & G. W. Smithers (1979) *Meth. Enzymol.* **63**, 295. (This reference lists many stability constants and also provides a detailed account of the procedures for determining stability constants.)

METALLO-BIOCHEMICAL METHODS

Because metal ions bind to and modify the reactivity and structure of enzymes and substrates, a wide spectrum of techniques has been developed to examine the nature of metal ions which serve as templates, redox-active cofactors, Lewis acids/bases, ion-complexing agents, *etc.*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

General Methods: Preparation of metal-free water, **158**, 3; elimination of adventitious metals, **158**, 6; metal-free dialysis tubing, **158**, 13; metal-free chromatographic media, **158**, 15; preparation of metal-free enzymes, **158**, 21; metal-buffered systems, **158**, 33; standards for metal analysis, **158**, 56; methods for metal substitution, **158**, 71; preparation of metal-hybrid enzymes, **158**, 79; introduction of exchange-inert metal ions into enzymes, **158**, 95; use of chelating agents to inhibit enzymes, **158**, 110.

Analytical Techniques: Atomic absorption spectrometry, **158**, 117; multielement atomic absorption methods of analysis, **158**, 145; ion microscopy in biology and medicine, **158**, 157; flame atomic emission spectrometry, **158**, 180; inductively coupled plasma-emission spectrometry, **158**, 190; inductively coupled plasma-mass spectrometry, **158**, 205; atomic fluorescence spectrometry, **158**, 222; electrochemical methods of analysis, **158**, 243; neutron activation analysis, **158**, 267.

Analysis of Specific Metals: Aluminum, **158**, 289; total calcium, **158**, 302; ionized calcium, **158**, 320; chromium, **158**, 334; cobalt, **158**, 344; copper, **158**, 351; complexed iron, **158**, 357; magnesium, **158**, 365; molybdenum, **158**, 371; nickel, **158**, 382; selenium, **158**, 391; vanadium, **158**, 402; zinc, **158**, 422.

Probing Metalloproteins: Electronic absorption spectroscopy of copper proteins, **226**, 1; electronic absorption spectroscopy of nonheme iron proteins, **226**, 33; cobalt as probe and label of proteins, **226**, 52; biochemical and spectroscopic probes of mercury(II) coordination environments in proteins, **226**, 71; low-temperature optical spectroscopy: metalloprotein structure and dynamics, **226**, 97; nanosecond transient absorption spectroscopy, **226**, 119; nanosecond time-resolved absorption and polarization dichroism spectroscopies, **226**, 147; real-time spectroscopic techniques for probing conformational dynamics of heme proteins, **226**, 177; variable-temperature magnetic circular dichroism, **226**, 199; linear dichroism, **226**, 232; infrared spectroscopy, **226**, 259; Fourier transform infrared spectroscopy, **226**, 289; infrared circular dichroism, **226**, 306; Raman and resonance Raman spectroscopy, **226**, 319; protein structure from ultraviolet resonance Raman spectroscopy, **226**, 374; single-crystal micro-Raman spectroscopy, **226**, 397; nanosecond time-resolved resonance Raman spectroscopy, **226**, 409; techniques for obtaining resonance Raman spectra of metalloproteins, **226**, 431; Raman optical activity, **226**, 470; surface-enhanced resonance Raman scattering, **226**, 482; luminescence

spectroscopy, **226**, 495; circularly polarized luminescence, **226**, 539; low-temperature stopped-flow rapid-scanning spectroscopy: performance tests and use of aqueous salt cryosolvents, **226**, 553; transition-metal complexes as spectroscopic probes for selective covalent labeling and cross-linking of proteins, **226**, 565; ruthenium complexes as luminescent reporters of DNA, **226**, 576; detecting metal-metal interactions and measuring distances between metal centers in metalloproteins, **226**, 594; two-dimensional nuclear magnetic resonance of paramagnetic metalloproteins, **227**, 1; cadmium-113 nuclear magnetic resonance applied to metalloproteins, **227**, 16; lanthanide shift reagents, **227**, 43; alkali metal nuclear magnetic resonance, **227**, 78; calcium nuclear magnetic resonance, **227**, 107; pulsed electron nuclear multiple resonance spectroscopic methods for metalloproteins and metalloenzymes, **227**, 118; continuous wave electron nuclear double resonance spectroscopy, **227**, 190; vanadyl(IV) electron nuclear double resonance/electron spin echo envelope modulation spin probes, **227**, 232; multidimensional nuclear magnetic resonance methods to probe metal environments in proteins, **227**, 244; optically detected magnetic resonance of triplet states in proteins, **227**, 290; electron paramagnetic resonance, **227**, 330; electron paramagnetic resonance spectroscopy of iron complexes and iron-containing proteins, **227**, 353; intrinsic and extrinsic paramagnets as probes of metal clusters, **227**, 384; electron paramagnetic resonance spectroelectrochemical titration, **227**, 396; magnetic susceptibility, **227**, 412; multifield saturation magnetization of metalloproteins, **227**, 437; combining Mössbauer spectroscopy with integer spin electron paramagnetic resonance, **227**, 463; voltammetric studies of redox-active centers in metalloproteins adsorbed on electrodes, **227**, 479; direct and indirect electrochemical investigations of metalloenzymes, **227**, 501; pulse radiolysis, **227**, 522; physical methods to locate metal atoms in biological systems, **227**, 535.

METALLOTHIONEINS

Thioneins are apoproteins that are exceptionally sulfur-rich (composed of greater 30 mol% cysteine). These proteins are found in high abundance in liver and kidney cytoplasm where they form metallothioneins (the holoprotein forms) upon complexation with metal ions. Thionein synthesis is induced by the presence of metals, especially zinc, copper, mercury, and cadmium.

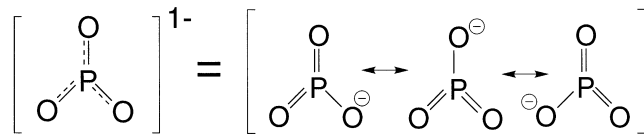
J. F. Riordan & B. L. Vallee, eds. (1991) *Meth. Enzymol.*, vol. **205**, Academic Press, San Diego.

META-MODEL

A computer program¹ for calculating steady-state fluxes and metabolite concentrations of metabolic systems on the IBM-PC and compatible computers. The software provides a simple means for calculating the control structure of a pathway.

¹A. Cornish-Bowden & J. H. Hofmeyr (1991) *Comput. Appl. Biosci.* **7**, 89.

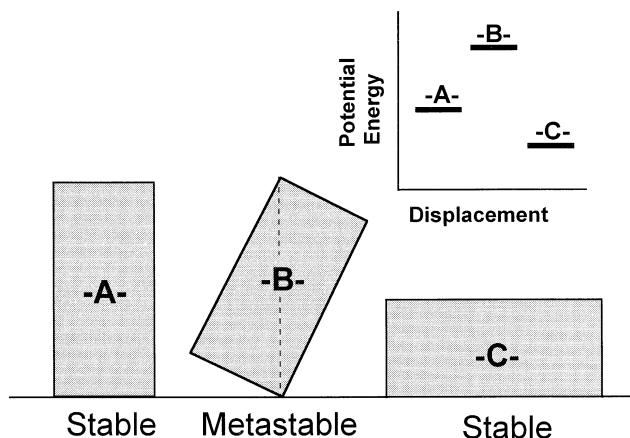
METAPHOSPHATE



The unstable PO_3^- ion proposed to be an intermediate in a number of enzyme-catalyzed phosphotransfer reactions. **See** *Acyl-Phosphate*

METASTABLE STATE

An unstable equilibrium state (or configuration) that is at maximal potential energy. A metastable state is at equilibrium, and its potential energy [written here as $U(x)$] is such that any displacement (dx) from $x_{\text{equilibrium}}$ will result in the loss of potential energy.



Although similar principles hold for transition states of chemical reactions, use of the term “metastable state” to refer to the transition state is actively discouraged by scientific convention. **See** *Transient Chemical Species; Jablonski Diagram*

METHANE MONOOXYGENASE

This enzyme [EC 1.14.13.25] catalyzes the reaction of methane with NAD(P)H and dioxygen to produce methanol, NAD(P)^+ , and water. This enzyme is reported to exhibit a broad specificity. Many alkanes can be hydroxylated and alkenes are converted into the corresponding epoxides. Carbon monoxide is oxidized to carbon dioxide, ammonia is oxidized to hydroxylamine, and some aromatic compounds and cyclic alkanes can also be hydroxylated, albeit not as efficiently.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.

J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.

S. J. Pilkington & H. Dalton (1990) *Meth. Enzymol.* **188**, 181.

P. G. Fox, W. A. Froland, D. R. Jollie & J. D. Lipscomb (1990) *Meth. Enzymol.* **188**, 191.

METHANOL DEHYDROGENASE

This enzyme [EC 1.1.99.8], also known as alcohol dehydrogenase (acceptor), catalyzes the reaction of a primary alcohol with an acceptor to produce an aldehyde and the reduced acceptor. A wide range of primary alcohols (e.g., methanol) can serve as substrates.

C. Anthony (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 155.

J. P. Klinman & D. Mu (1994) *Ann. Rev. Biochem.* **63**, 299.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

J. Frank & J. A. Duine (1990) *Meth. Enzymol.* **188**, 202.

D. J. Day & C. Anthony (1990) *Meth. Enzymol.* **188**, 210.

A. R. Long & C. Anthony (1990) *Meth. Enzymol.* **188**, 216.

N. Arfman & L. Dijkhuizen (1990) *Meth. Enzymol.* **188**, 223.

J. A. Duine, J. Frank, & J. A. Jongejan (1987) *Adv. Enzymol.* **59**, 169.

5,10-METHENYLTETRAHYDROFOLATE CYCLOHYDROLASE

This enzyme [EC 3.5.4.9] catalyzes the hydrolysis of 5,10-methenyltetrahydrofolate to produce 10-formyltetrahydrofolate.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

V. Schirch (1997) *Meth. Enzymol.* **281**, 146.

R. E. MacKenzie (1997) *Meth. Enzymol.* **281**, 171.

S. W. Ragsdale (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 261.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

5,10-METHENYLTETRAHYDROFOLATE SYNTHETASE

This enzyme [EC 6.3.3.2], also known as 5-formyltetrahydrofolate cyclo-ligase, catalyzes the reaction of ATP with 5-formyltetrahydrofolate to produce ADP, orthophosphate, and 5,10-methenyltetrahydrofolate.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

V. Schirch (1997) *Meth. Enzymol.* **281**, 146.

J. Jolivet (1997) *Meth. Enzymol.* **281**, 162.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

METHIONINE ADENOSYLTRANSFERASE

This enzyme [EC 2.5.1.6], also known as *S*-adenosylmethionine synthetase, catalyzes the reaction of ATP

with L-methionine and water to produce *S*-adenosyl-L-methionine, orthophosphate, and pyrophosphate (or, diphosphate).

- I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bârzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.
 M. H. Stipanuk (1986) *Ann. Rev. Nutr.* **6**, 179.
 C. W. Tabor & H. Tabor (1984) *Adv. Enzymol.* **56**, 251.
 S. H. Mudd (1973) *The Enzymes*, 3rd ed., **8**, 121.

METHIONINE γ -LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.11], also known as L-methioninase, catalyzes the conversion of L-methionine to methanethiol, ammonia, and α -ketobutyrate (or, 2-oxobutanoate).

- M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.
 N. Esaki & K. Soda (1987) *Meth. Enzymol.* **143**, 459.

METHIONINE SULFOXIDE REDUCTASE

This enzyme [EC 1.8.4.5], also known as methionine *S*-oxide reductase, catalyzes the reaction of L-methionine with oxidized thioredoxin to produce L-methionine *S*-oxide and reduced thioredoxin. Dithiothreitol can substitute for reduced thioredoxin in the reverse reaction. In addition, other methyl sulfoxides can replace methionine sulfoxide in the reverse reaction.

- S. Black (1962) *Meth. Enzymol.* **5**, 992.

METHIONYL-tRNA FORMYLTRANSFERASE

This enzyme [EC 2.1.2.9] catalyzes the reaction of 10-formyltetrahydrofolate with L-methionyl-tRNA and water to produce tetrahydrofolate and *N*-formylmethionyl-tRNA.

- V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.
 J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

METHIONYL-tRNA SYNTHETASE

This enzyme [EC 6.1.1.10], also known as methionine:tRNA ligase, catalyzes the reaction of methionine with tRNA^{Met} and ATP to produce methionyl-tRNA^{Met}, AMP, and pyrophosphate (or, diphosphate). **See also** *Aminoacyl-tRNA Synthetases*

- E. A. First (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 573.
 P. Schimmel (1987) *Ann. Rev. Biochem.* **56**, 125.

4-METHOXYBENZOATE MONOOXYGENASE (O-Demethylating)

This enzyme [EC 1.14.99.15] catalyzes the reaction of 4-methoxybenzoate with AH₂ and dioxygen to produce 4-hydroxybenzoate, formaldehyde, A, and water. The bacterial enzyme consists of a ferredoxin-type protein and an iron-sulfur flavoprotein (FMN). 4-Ethoxybenzoate, *N*-methyl-4-aminobenzoate, and toluate can serve as substrates as well. The fungal enzyme acts best on veratrate.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
 V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

3-METHYLADENINE-DNA GLYCOSYLASE

3-Methyladenine-DNA glycosylase I [EC 3.2.2.20], also known as DNA-3-methyladenine glycosylase I and DNA glycosidase I, catalyzes the hydrolysis of alkylated DNA, releasing 3-methyladenine. 3-Methyladenine-DNA glycosylase II [EC 3.2.2.21], also known as DNA-3-methyladenine glycosylase II and DNA glycosidase II, catalyzes the hydrolysis of alkylated DNA, releasing 3-methyladenine, 3-methylguanine, 7-methylguanine, and 7-methyladenine.

- A. B. Britt (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 75.
 A. Sancar & G. B. Sancar (1988) *Ann. Rev. Biochem.* **57**, 29.
 B. K. Duncan (1981) *The Enzymes*, 3rd ed., **14**, 565.

N-METHYL-L-ALANINE DEHYDROGENASE

This enzyme [EC 1.4.1.17] catalyzes the reaction of *N*-methyl-L-alanine with NADP⁺ and water to produce pyruvate, methylamine, and NADPH.

- N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

METHYLAMINE DEHYDROGENASE

This enzyme [EC 1.4.99.3], also known as amine dehydrogenase and primary-amine dehydrogenase, catalyzes the reaction of R-CH₂-NH₂ with water and an acceptor to produce R-CHO, ammonia, and the reduced acceptor. Tryptophan tryptophylquinone (TTQ) is the co-factor for this enzyme. **See** *Resonance Raman Spectroscopy; Topaquinone*

- C. Anthony (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 155.
 C. Hartmann & W. S. McIntire (1997) *Meth. Enzymol.* **280**, 98.
 W. S. McIntire (1995) *Meth. Enzymol.* **258**, 149.
 V. L. Davidson, H. B. Brooks, M. E. Graichen, L. H. Jones & Y.-L. Hyun (1995) *Meth. Enzymol.* **258**, 176.

- J. P. Klinman & D. Mu (1994) *Ann. Rev. Biochem.* **63**, 299.
V. L. Davidson (1990) *Meth. Enzymol.* **188**, 241.
J. A. Duine, J. Frank, & J. A. Jongejan (1987) *Adv. Enzymol.* **59**, 169.

3-METHYLASPARTATE AMMONIA-LYASE

This cobalamin-dependent enzyme [EC 4.3.1.2], also known as β -methylaspartase, catalyzes the conversion of L-threo-3-methylaspartate to mesaconate and ammonia.

- K. R. Hanson & E. A. Havir (1972) *The Enzymes*, 3rd ed., **7**, 75.

β -METHYLASPARTATE-GLUTAMATE MUTASE

This cobalamin-dependent enzyme [EC 5.4.99.1], also known as glutamate mutase or methylaspartate mutase, catalyzes the interconversion of L-threo-3-methylaspartate and L-glutamate.

- B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.
H. A. Barker (1985) *Meth. Enzymol.* **113**, 121.
B. M. Babior & J. S. Krouwer (1979) *Crit. Rev. Biochem.* **6**, 35.
H. A. Barker (1972) *The Enzymes*, 3rd ed., **6**, 509.

METHYLATION (As a Mechanistic Probe)

Certain active site residues and/or reaction intermediates can be detected by direct methylation using diazomethane or nitromethane. Among the numerous examples are the treatment of the N-1'-carboxybiotin in carboxylases with diazomethane to yield the corresponding stable dimethyl ester¹ and the reaction of nitromethane at the active-site dehydroalanyl residue of phenylalanine ammonia lyase². Another example is the use of diazomethane to identify the amino acid sequence of a peptide bearing the activator CO₂ in the ribulose-bisphosphate carboxylase reaction³. This enzyme was first activated by reaction of an activator CO₂, forming an ϵ -amino carbamate that was converted to the methoxycarbonyl derivative by diazomethane treatment, followed by trypsinization and sequencing of the resultant labeled peptides. The lysyl residue bearing the activator CO₂ is 26 residues removed from a lysyl residue identified by using N-bromoacetyethanolamine phosphate to react with the enzyme's active-site.

¹F. Lynen, J. Knappe, E. Lorch, G. Jutting & E. Ringelman (1959) *Angew. Chem. Int. Ed. Engl.* **71**, 481.

²E. A. Havir & K. R. Hanson (1975) *Biochemistry* **14**, 1620.

³G. H. Lorimer (1981) *Biochemistry* **20**, 1236.

2-METHYL-BRANCHED-CHAIN ACYL-CoA DEHYDROGENASE

The acyl-CoA dehydrogenases are a family of mitochondrial flavoenzymes involved in fatty acid and branched chain amino-acid metabolism. In addition to long chain acyl-CoA dehydrogenases (LCADs), there are short/branched chain acyl-CoA dehydrogenase (SBCAD) that act on 2-methyl branched chain acyl-CoA substrates of varying chain lengths.

- Y. Ikeda & K. Tanaka (1988) *Meth. Enzymol.* **166**, 360.

β -METHYLCROTONYL-CoA CARBOXYLASE

This biotin-dependent enzyme [EC 6.4.1.4] catalyzes the reaction of ATP with 3-methylcrotonyl-CoA and bicarbonate to produce ADP, orthophosphate, and 3-methylglutaconyl-CoA.

- J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.
F. Lynen (1979) *Crit. Rev. Biochem.* **7**, 103.
A. W. Alberts & P.R. Vagelos (1972) *The Enzymes*, 3rd ed., **6**, 37.

2-METHYLENEGLUTARATE MUTASE

This cobalamin-dependent enzyme [EC 5.4.99.4] catalyzes the interconversion of 2-methyleneglutarate and 2-methylene-3-methylsuccinate.

- B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.
B. M. Babior & J. S. Krouwer (1979) *Crit. Rev. Biochem.* **6**, 35.
H. A. Barker (1972) *The Enzymes*, 3rd ed., **6**, 509.

5,10-METHYLENETETRAHYDROFOLATE DEHYDROGENASE

5,10-Methylenetetrahydrofolate dehydrogenase (NADP⁺) [EC 1.5.1.5] catalyzes the reaction of 5,10-methylenetetrahydrofolate with NADP⁺ to produce 5,10-methenyltetrahydrofolate and NADPH. 5,10-Methylenetetrahydrofolate dehydrogenase (NAD⁺) [EC 1.5.1.15] catalyzes the reaction of 5,10-methylenetetrahydrofolate with NAD⁺ to produce 5,10-methenyltetrahydrofolate and NADH.

- V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.
V. Schirch (1997) *Meth. Enzymol.* **281**, 146.
R. E. MacKenzie (1997) *Meth. Enzymol.* **281**, 171.
D. R. Appling & M. G. West (1997) *Meth. Enzymol.* **281**, 178.
S. W. Ragsdale (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 261.

5,10-METHYLENETETRAHYDROFOLATE REDUCTASE

5,10-Methylenetetrahydrofolate reductase (NADPH) [EC 1.5.1.20] is an FAD-dependent enzyme that catalyzes the reaction of 5-methyltetrahydrofolate with NADP⁺ to produce 5,10-methylenetetrahydrofolate and NADPH. 5,10-Methylenetetrahydrofolate reductase (FADH₂) [EC 1.7.99.5] is an FAD-dependent enzyme that catalyzes the reaction of 5-methyltetrahydrofolate with an acceptor to produce 5,10-methylenetetrahydrofolate and the reduced acceptor.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

S. W. Ragsdale (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 261.

5,10-METHYLENETETRAHYDROFOLATE: tRNA (URACIL-5-)-METHYLTRANSFERASE (FADH₂ OXIDIZING)

This enzyme [EC 2.1.1.74], also known as folate-dependent ribothymidyl synthase, catalyzes the reaction of 5,10-methylenetetrahydrofolate with tRNA (with U ψ C) and FADH₂ to produce tetrahydrofolate, tRNA (with T ψ C) and FAD.

D. Söll & L. K. Kline (1982) *The Enzymes*, 3rd ed., **15**, 557.

3-METHYLGLUTACONYL-CoA HYDRATASE

This enzyme [EC 4.2.1.18] catalyzes the conversion of (*S*)-3-hydroxy-3-methylglutaryl-CoA to *trans*-3-methylglutaconyl-CoA and water.

K. M. Gibson (1988) *Meth. Enzymol.* **166**, 214.

N-METHYLGLUTAMATE SYNTHASE

This enzyme [EC 2.1.1.21], also known as methylamine-glutamate methyltransferase, catalyzes the reaction of methylamine and L-glutamate to produce ammonia and N-methyl-L-glutamate.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

L. B. Hersh (1985) *Meth. Enzymol.* **113**, 36.

 γ -METHYL- γ -HYDROXY- α -KETOGLUTARATE ALDOLASE

This enzyme [EC 4.1.3.17], also known as 4-hydroxy-4-methyl-2-oxoglutarate aldolase and 4-hydroxy-4-methyl-2-oxoglutarate pyruvate-lyase, catalyzes the reversible conversion of 4-hydroxy-4-methyl-2-oxoglutarate to pro-

duce two pyruvate. 4-Hydroxy-4-methyl-2-oxoadipate and 4-carboxy-4-hydroxy-2-oxohexadioate can serve as substrates as well.

W. A. Wood (1972) *The Enzymes*, 3rd ed., **7**, 281.

METHYLHYDROXYPYRIDINECARBOXYLATE DIOXYGENASE

This FAD-dependent enzyme [EC 1.14.12.4], also known as 3-hydroxy-2-methylpyridinecarboxylate dioxygenase, catalyzes the reaction of 3-hydroxy-2-methylpyridine-5-carboxylate with NAD(P)H and dioxygen to produce 2-(acetamidomethylene)succinate and NAD(P)⁺.

B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

METHYLITACONATE Δ -ISOMERASE

This enzyme [EC 5.3.3.6] catalyzes the interconversion of methylitaconate and dimethylmaleate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

(R)-2-METHYLMALATE DEHYDRATASE

This iron-dependent enzyme [EC 4.2.1.35], also known as citraconase and citraconate hydratase, catalyzes the conversion of (*R*)-2-methylmalate to 2-methylmaleate and water.

H. A. Lardy (1969) *Meth. Enzymol.* **13**, 314.

(S)-2-METHYLMALATE DEHYDRATASE

This enzyme [EC 4.2.1.34], also known as mesaconase and mesaconate hydratase, catalyzes the conversion of (*S*)-2-methylmalate to 2-methylfumarate and water. The enzyme will also catalyze the hydration of fumarate to (*S*)-malate.

H. A. Lardy (1969) *Meth. Enzymol.* **13**, 314.

C. C. Wang & H. A. Barker (1969) *Meth. Enzymol.* **13**, 331.

METHYLMALONATE-SEMIALDEHYDE DEHYDROGENASE (Acylyating)

This enzyme [EC 1.2.1.27] catalyzes the reaction of 2-methyl-3-oxopropanoate with coenzyme A and NAD⁺ to produce propanoyl-CoA, carbon dioxide, and NADH. The enzyme will also catalyze the conversion of propanal to propanoyl-CoA.

K. Hatter & J. R. Sokatch (1988) *Meth. Enzymol.* **166**, 389.

METHYLMALONYL-CoA DECARBOXYLASE

This enzyme [EC 4.1.1.41], also known as propionyl-CoA carboxylase, catalyzes the conversion of (*S*)-2-methyl-3-oxopropanoyl-CoA to propanoyl-CoA and carbon dioxide. The enzyme from *Veillonella alcalescens* is a biotinyl-protein, requires sodium ions, and acts as a sodium pump.

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.

METHYLMALONYL-CoA EPIMERASE

This enzyme [EC 5.1.99.1] interconverts (*R*)-2-methyl-3-oxopropanoyl-CoA (or, (*R*)-methylmalonyl-CoA) to (*S*)-2-methyl-3-oxopropanoyl-CoA (or, (*S*)-methylmalonyl-CoA).

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

S. P. Stabler & R. H. Allen (1988) *Meth. Enzymol.* **166**, 400.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

D-METHYLMALONYL-CoA HYDROLASE

This enzyme, officially known as (*S*)-methylmalonyl-CoA hydrolase [EC 3.1.2.17], catalyzes the hydrolysis of (*S*)-methylmalonyl-CoA to form methylmalonate and coenzyme A.

R. J. Kovachy, S. P. Stabler & R. H. Allen (1988) *Meth. Enzymol.* **166**, 393.

L-METHYLMALONYL-CoA MUTASE

This cobalamin-dependent enzyme [EC 5.4.99.2] catalyzes the conversion of (*R*)-2-methyl-3-oxopropanoyl-CoA (or, (*R*)-methylmalonyl-CoA) to succinyl-CoA.

B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.

M. L. Ludwig & R. G. Matthews (1997) *Ann. Rev. Biochem.* **66**, 269.

J. F. Kolhouse, S. P. Stabler & R. H. Allen (1988) *Meth. Enzymol.* **166**, 407.

H. A. Barker (1972) *The Enzymes*, 3rd ed., **6**, 509.

N⁵-METHYLTETRAHYDROFOLATE: HOMOCYSTEINE METHYLTRANSFERASE

This cobalamin-dependent enzyme [EC 2.1.1.13], also known as methionine synthase and tetrahydropteroyl-glutamate methyltransferase, catalyzes the reaction of 5-methyltetrahydrofolate with L-homocysteine to produce tetrahydrofolate and L-methionine. Interestingly, the bacterial enzyme is reported to require *S*-adenosyl-L-methionine and FADH₂. **See also** *Tetrahydropteroyl-triglutamate Methyltransferase*

M. L. Ludwig & R. G. Matthews (1997) *Ann. Rev. Biochem.* **66**, 269.

R. Banerjee, Z. Chen & S. Gulati (1997) *Meth. Enzymol.* **281**, 189.

F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.

J. T. Jarrett, C. W. Goulding, K. Fluhr, S. Huang & R. G. Matthews (1997) *Meth. Enzymol.* **281**, 196.

R. T. Taylor & H. Weissbach (1973) *The Enzymes*, 3rd ed., **9**, 121.

METHYLTETRAHYDROMETHANOPTERIN: COENZYME M METHYLTRANSFERASE

This cobalamin-dependent enzyme catalyzes the reaction of methyltetrahydromethanopterin with coenzyme M to produce methyl-coenzyme M and tetrahydromethanopterin.

M. L. Ludwig & R. G. Matthews (1997) *Ann. Rev. Biochem.* **66**, 269.

K. M. Noll (1995) *Meth. Enzymol.* **251**, 470.

METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE

This enzyme [EC 3.2.2.16], also known as methylthioadenosine nucleosidase, catalyzes the hydrolysis of methylthioadenosine to produce adenine and 5-methylthio-D-ribose. *S*-Adenosylhomocysteine is not a substrate for this enzyme. **See also** *S-Adenosylhomocysteine Nucleosidase*

F. Schlenk (1983) *Adv. Enzymol.* **54**, 195.

METHYL TRANSFER REACTIONS USING S-ADENOSYLMETHIONINE

Methyltransferases that utilize *S*-adenosyl-L-methionine as the methyl donor (and thus generating *S*-adenosyl-L-homocysteine) catalyze: (a) *N*-methylation (e.g., norepinephrine methyltransferase, histamine methyltransferase, glycine methyltransferase, and DNA-(adenine-N⁶) methyltransferase), (b) *O*-methylation (e.g., acetylserotonin methyltransferase, catechol methyltransferase, and tRNA-(guanosine-O²) methyltransferase), (c) *S*-methylation (e.g., thiopurine methyltransferase and methionine *S*-methyltransferase), (d) *C*-methylation (e.g., DNA-(cytosine-5) methyltransferase and indolepyruvate methyltransferase), and even (e) Co(II)-methylation during the course of the reaction catalyzed by methionine synthase¹⁻³.

Crystal structures of methyltransferases suggest that *S*-adenosyl-L-methionine binds in an extended conforma-

tion. The stereochemistry of many SAM-dependent methyltransferases has been determined by using chiral methyl groups containing all three isotopes of hydrogen. In the enzymes studied, the methyl group undergoes inversion of configuration, suggesting that the transfer of the methyl group is a direct S_N2 reaction with the methyl acceptor.

Either hydroxyl group of catechol can be methylated by catechol *O*-methyltransferase, albeit at different rates (*i.e.*, the enzyme does not exhibit absolute regioselectivity). The k_{cat} value for the 3-hydroxyl group is about 1 s^{-1} whereas that at the 4-position is about 0.1 or 0.2 s^{-1} . The mechanism has been reported to be ordered with SAM binding first, followed by magnesium ion, and then catechol. Interestingly, it appears that the rate-limiting step is the actual catalytic event.

Glycine *N*-methyltransferase is also reported to have an ordered binding mechanism with SAM binding first to the enzyme, there being no metal-ion dependency. Cooperative behavior is observed with SAM binding. The cooperative nature can be eliminated by the tryptic hydrolysis of the N-terminal eight amino acid residues.

With DNA-(cytosine- C^5) methyltransferases, the binding mechanism also appears to be ordered with an active-site cysteinyl residue required in the enzyme reaction path. However, in this case SAM is the second substrate. The binding mechanism appears to be random with DNA-(adenine- N^6) methyltransferase.

¹R. T. Borchardt, Y. S. Wu, J. A. Huber & A. F. Wycpalek (1976) *J. Med. Chem.* **19**, 1104.

²J. Olsen, Y. S. Wu, R. T. Borchardt & R. L. Schowen (1979) *Transmethylation* (R. T. Borchardt, C. R. Creveling & E. Esdin, eds.), pp. 127-133, Elsevier, New York.

³F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

MEVALDATE REDUCTASE

Mevaldate reductase [EC 1.1.1.32] catalyzes the reaction of (*R*)-mevalonate with NAD^+ to produce mevaldate and NADH. Mevaldate reductase (NADPH) [EC 1.1.1.33] catalyzes the reaction of (*R*)-mevalonate with $NADP^+$ to produce mevaldate and NADPH. There are reports that this enzyme may be identical with alcohol dehydrogenase ($NADP^+$) [EC 1.1.1.2].

G. Popják (1969) *Meth. Enzymol.* **15**, 393.

MEVALONATE KINASE

This enzyme [EC 2.7.1.36] catalyzes the reaction of ATP with (*R*)-mevalonate to produce ADP and (*R*)-5-phosphomevalonate. The nucleotide substrate can also be provided with CTP, GTP, or UTP.

R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

G. J. Schroepfer, Jr. (1981) *Ann. Rev. Biochem.* **50**, 585.

E. D. Beytía & J. W. Porter (1976) *Ann. Rev. Biochem.* **45**, 113.

MEYERHOF OXIDATION QUOTIENT

A quantitative index describing the effect of the absence or presence of molecular oxygen on the rate of glycolysis or fermentation:

$$Q_{\text{Meyerhof}} = \frac{\text{Rate}_{\text{Anaerobic Fermentation}} - \text{Rate}_{\text{Aerobic Fermentation}}}{\text{Rate}_{\text{Molecular Oxygen Uptake}}}$$

The quotient is usually about two for most cells, meaning that the consumption of one O_2 molecule usually diverts two molecules of either lactate or ethanol by forming their oxidative counterparts.

MICELLES

Organized supramolecular aggregates formed by surface active substances in aqueous solutions (Fig. 1). Micellar monolayers or bilayers are formed by burying apolar groups toward the interior while exposing hydrophilic head groups at the water-surfactant interface. Two types of surfactants form micelles: ionic surfactants (*i.e.*, those containing positively or negatively charged head groups) or nonionic surfactants (*i.e.*, those with nonionic hydrophilic head groups).

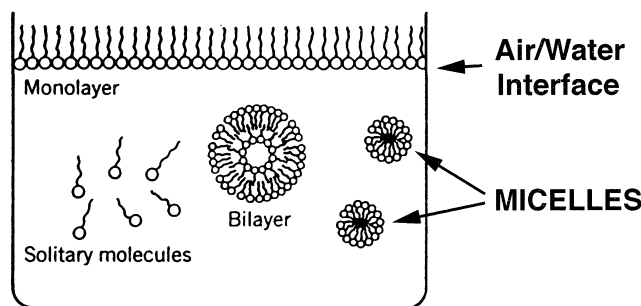


Figure 1. Various physical states of phospholipids in aqueous solution. Note the following features: (a) phospholipids residing at the air/water interface are arranged such that their polar head groups maximize contact with the aqueous environment, whereas apolar side chains extend outward toward the air; (b) solitary phospholipid molecules remain in equilibrium with various monolayer and bilayer structures; (c) bilayer vesicles and micelles remain in equilibrium with solitary phospholipid molecules, provided that the total lipid content exceeds the critical micelle concentration.

The term “mixed micelle” refers to those micelles composed of two or more surface active agents. The sizes of micelles in a solution obey a distribution function that is characteristic of their chemical composition and the ionic nature of the solution in which they reside.

FORMATION. Aqueous solutions of highly surface-active substances spontaneously tend to reduce interfacial energy of solute-solvent interactions by forming micelles. The critical micelle concentration (or, c.m.c.) is the threshold surfactant concentration, above which micelle formation (also known as micellization) is highly favorable. For sodium dodecyl sulfate, the c.m.c. is 5.6 mM at 0.01 M NaCl or about 3.1 mM at 0.03 M NaCl. The lower c.m.c. observed at higher salt concentration results from a reduction in repulsive forces among the ionic head groups on the surface of micelles made up of ionic surfactants. As would be expected for any entropy-driven process, micelle formation is less favorable as the temperature is lowered.

At their critical micelle concentrations, surface active agents (such as sodium dodecyl sulfate, Triton X-100, lysolecithin, and bile salts) self-associate into spherical or rod-shaped structures. Because dilution to below the c.m.c. results in rapid disassembly or dissolution of these detergent micelles, micelles are in dynamic equilibrium with other dissolved detergent molecules in the bulk solution.

ENERGETICS. The standard free energy change for micelle formation from 1.0 mole surfactant is given by the relationship,

$$\Delta G^\circ = -(RT/m)\ln K$$

where K is the equilibrium constant and m is number of monomeric units per micelle. If c is taken to be the stoichiometric concentration of the surfactant solution, and if x designates the fraction of monomers in the aggregated state, then:

$$K = (cx/m)/[c(1-x)]^m$$

and the standard state Gibbs free energy can also be restated as:

$$\Delta G^\circ = -(RT/m)\ln[cx/m] + RT\ln[c(1-x)]$$

At the critical micelle concentration (or c.m.c.), x must be zero because all the monomers are disaggregated; thus,

$$\Delta G^\circ = RT\ln[c.m.c.]$$

(Note the change of sign.) Likewise, we can define the standard entropy and enthalpy for micellization as follows:

$$\Delta S^\circ = d(\Delta G^\circ)/dT = -RT d\{\ln[c.m.c.]\}/dT - RT\ln[c.m.c.]$$

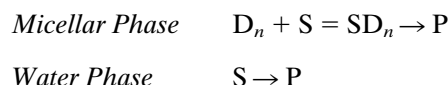
$$\Delta H^\circ = -RT^2 d\{\ln[c.m.c.]\}/dT$$

In general, the standard enthalpy of micellization is large and negative, and an increase in temperature results in an increase in the c.m.c.; the positive entropy of micellization relates to the increased mobility of hydrocarbon side chains deep within the micelle as well as the hydrophobic effect. Hoffmann and Ulbricht¹ have provided a detailed account of the thermodynamics of micellization, and the interested reader will find that their tabulated thermodynamic values and treatment of models for micellar aggregation processes are especially worthwhile.

MICELLAR CATALYSIS. Chemical reactions can be accelerated by concentrating reactants on a micelle surface or by creating a favorable interfacial electrostatic environment that increases reactivity. This phenomenon is generally referred to as micellar catalysis. As pointed out by Bunton², the term “micellar catalysis” is used loosely because enhancement of reactivity may actually result from a change in the equilibrium constant for a reversible reaction. Because catalysis is strictly viewed as an enhancement of rate without change in a reaction’s thermodynamic parameters, one must exercise special care to distinguish between kinetic and equilibrium effects. This is particularly warranted when there is evidence of differential interactions of substrate and product with the micelle. Micelles composed of optically active detergent molecules can also display stereochemical action on substrates³.

Catalysis can occur when micellized surfactants are themselves chemically inert, and this effect relies on electrostatic and hydrophobic interactions that alter the susceptibility of reactants to nucleophilic attack or electron

transfer. Catalysis can also take place with the micelle providing the acidic/basic as well as nucleophilic/electrophilic groups needed for reaction. In either case, the rate enhancement is designated as k_m/k_w where the subscripts indicate rate constants applying to the micellar and water phases:



where D_n is made up of n detergent molecules, and S and P are substrate and product, respectively. (Note the analogy between the micellar path and the Michaelis-Menten kinetic mechanism for enzyme action. Both display saturation kinetics, the former dependent on substrate binding at the interface of the micelle instead of at a discrete enzyme active site.) In Bunton's treatment², the concentration of micelles C_M is equal to $(C_D - c.m.c.)/N$ where C_D is the total concentration of surfactant (detergent) and N is the number of detergent molecules per micelle. The rate constant for the above mechanism can be obtained as,

$$k_\psi = (k_w + k_m KC_m)/(1 + KC_m)$$

This equation indicates that the observed rate will be equal to the water phase reaction rate until the added surfactant is present at sufficient concentrations to form micelles. Added complexity arises if the substrate selectively interacts with a fraction of the micelles having a particular size and/or shape.

MICELLAR SUBSTRATES. Phospholipids in micelles are frequently found to be more active substrates in lipolysis than those phospholipids residing in a lipid bilayer⁴. Dennis⁴ first described the use of Triton X-100 to manipulate the amount of phospholipid per unit surface area of a micelle in a systematic analysis of the interfacial interactions of lipases with lipid micelles. Verger⁵ and Jain *et al.*⁶ have presented cogent accounts of the kinetics of interfacial catalysis by phospholipases. The complexity of the problem is illustrated in the diagram shown in Fig. 2 showing how the enzyme in the aqueous phase can bind to the interface (designated by the asterisk) and then become activated. Once this is achieved, E^* catalyzes conversion of S^* to release P^* .

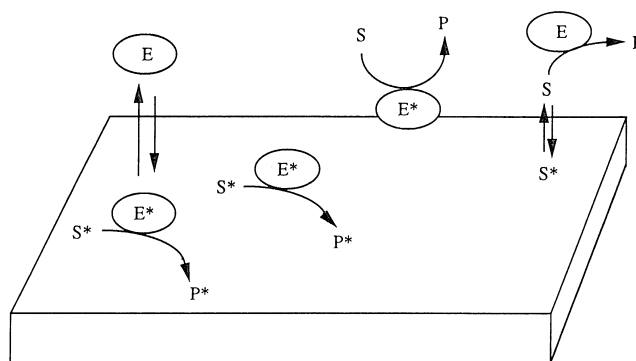


Figure 2. Behavior of membrane-associated lipases. From left to right: (a) catalytic action of an enzyme that first requires attachment to the substrate at the water-membrane interface; (b) action of an integral membrane enzyme that remains attached to the membrane where the enzyme finds its substrate; (c) action of a membrane-bound enzyme on substrates in the aqueous medium; and (d) action of an enzyme in the aqueous phase on a substrate that must first desorb from the membrane before it can interact with enzyme. From Jain *et al.*⁶ with permission of the authors.

¹H. Hoffmann & W. Ulbricht (1986) *Thermodynamic Data for Biochemistry and Biotechnology* (H.-J. Hinz, ed.), pp. 297-348, Springer-Verlag, Berlin.

²C. A. Bunton (1976) in *Techniques of Chemistry*, **10**, pp. 731-814, Wiley, New York.

³R. A. Moss, C. J. Talkowski, D. W. Reger & C. E. Powell (1973) *J. Amer. Chem. Soc.* **95**, 3215.

⁴E. A. Dennis (1973) *J. Lipid Res.* **14**, 152.

⁵R. Verger (1980) *Meth. Enzymol.* **64**, 340.

⁶M. K. Jain, M. H. Gelb, J. Rogers & O. G. Berg. (1995) *Meth. Enzymol.* **249**, 567.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Activation of lipolytic enzymes by interfaces, **64**, 341; model for lipase action on insoluble lipids, **64**, 345; interfacial enzyme inactivation, **64**, 347; reversibility of the adsorption step, **64**, 347; monolayer substrates, **64**, 349; kinetic models applicable to partly soluble amphiphilic lipids, **64**, 353; surface dilution model, **64**, 355 and 364; practical aspects, **64**, 357.

MICHAELIS COMPLEX (Distal Motions Affecting E·S Formation)

Two-dimensional heteronuclear (^1H - ^{15}N) nuclear magnetic relaxation studies indicate that the dihydrofolate reductase-folate complex exhibits a diverse range of backbone fluctuations on the time-scale of picoseconds to nanoseconds¹. To assess whether these dynamical features influence Michaelis complex formation, Miller *et al.*² used mutagenesis and kinetic measurements to assess the role of a strictly conserved residue, namely Gly-121, which displays large-amplitude backbone motions on the nanosecond time scale. Deletion of Gly-121 dramatically reduces the hydride transfer rate by 550 times; there is also a 20-times decrease in NADPH cofactor binding affinity and a 7-fold decrease for NADP⁺ relative to wild-type. Insertion mutations significantly decreased both

substrate and cofactor binding. Their results suggest that distant residues, such as Gly-121 in DHFR, can strongly influence the formation of liganded complexes as well as the proper orientation of substrate and cofactor during the catalytic cycle.

¹D. M. Epstein, S. J. Benkovic & P. E. Wright (1995) *Biochemistry* **34**, 11037.

²G. P. Miller & S. J. Benkovic (1998) *Biochemistry* **37**, 6327.

MICHAELIS CONSTANT

1. The quotient of rate constants obtained in steady-state treatments of enzyme behavior to define a substrate's interaction with an enzyme. While the Michaelis constant (with overall units of molarity) is a rate parameter, it is not itself a rate constant. Likewise, the Michaelis constant often is only a rough gauge of an enzyme's affinity for a substrate. 2. Historically, the term "Michaelis constant" referred to the true dissociation constant for the enzyme-substrate binary complex, and this parameter was obtained in the Michaelis-Menten rapid-equilibrium treatment of a one-substrate enzyme-catalyzed reaction. In this case, the Michaelis constant is usually symbolized by K_s . 3. The value equal to the concentration of substrate at which the initial rate, v , is one-half the maximum velocity ($V_{\max}$) of the enzyme-catalyzed reaction under steady state conditions.

In the Briggs-Haldane steady-state treatment of a one-substrate enzyme system, the Michaelis constant, usually symbolized by K_m , is $(k_2 + k_3)/k_1$. For more complex reactions (e.g., with several substrates and/or isomerization steps), the Michaelis constant for a given substrate is a more complex collection of rate constants. For a multisubstrate enzyme having substrates A and B, the Michaelis constants are usually symbolized by K_a and K_b , by K_{mA} and K_{mB} , or by K_m^A and K_m^B , respectively.

MICHAELIS CONSTANT (Apparent)

1. A parameter that depends on the concentration of another substrate in a multisubstrate reaction or on one or more cofactors or substances that influence reaction rate. It is an approximation of the true Michaelis constant. 2. A parameter obtained under conditions that do not rigorously satisfy the requirements of initial rate measurements.

MICHAELIS-MENTEN EQUATION

A model first advanced by Victor Henri¹ and later by Leonor Michaelis and Maud Menten² to account for the kinetic properties of a one-substrate, one-product enzyme-catalyzed reaction.

ASSUMPTIONS. Although one does not frequently state all of the requirements for deriving the Michaelis-Menten equation, the model assumes (a) that all experimental conditions (e.g., pH, temperature, ionic strength, etc.) remain constant over the course of the experiment; (b) that the enzyme, substrate, and product are all stable under the assay conditions over the time course of the experiment; (c) that at least one binary central complex does, in fact, form (e.g., ES); (d) that E and S react rapidly and reversibly to form this complex; (e) that the binding of S to E can be described by an equilibrium dissociation constant, $K_s = k_2/k_1$; (f) that the conversion of ES to E + P is slow and irreversible (or, if reversible, the concentration of [P] over the time course of the experiment is so low that the reverse reaction is negligible); (g) that the total enzyme concentration $[E_{\text{total}}]$ is the sum of uncomplexed enzyme concentration, [E], and the enzyme-substrate central complex (i.e., $[E_{\text{total}}] = [E] + [ES]$); (h) that the velocity, v , of the reaction, equal to $d[P]/dt$, equals [ES] multiplied by the unimolecular rate constant k_3 (thus, only considering the forward reaction [assumption (f) above]); (i) that the concentration of substrate is essentially unchanged during the initial velocity period (i.e., $\Delta[S] \ll [S_{\text{initial}}]$); (j) that the total substrate concentration, $[S_{\text{total}}]$, far exceeds the total enzyme concentration, $[E_{\text{total}}]$, such that $[S_{\text{total}}] \approx [S]$, where [S] is the uncomplexed substrate concentration.

DERIVATION OF RAPID EQUILIBRIUM EXPRESSION. We can begin with two equations:

$$K_s = [E][S]/[ES] \quad \text{and} \quad [E_{\text{total}}] = [E] + [ES]$$

The first equation can be rewritten as $[E] = K_s[ES]/[S]$ and substituted for [E] in the second equation to yield

$$[E_{\text{total}}] = (K_s[ES]/[S]) + [ES]$$

Dividing both sides by [ES], we get

$$[E_{\text{total}}]/[ES] = (K_s/[S]) + 1$$

Then, by multiplying the left side by k_3/k_3 (noting that $v = k_3[ES]$ and defining the maximum velocity, $V_{\max}$, as being $k_3[E_{\text{total}}]$), we get:

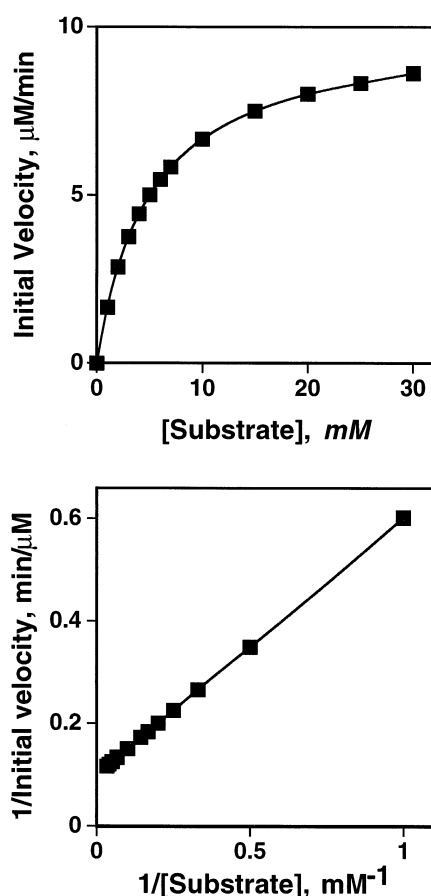
$$v = \frac{V_{\max}[S]}{K_s + [S]}$$

The derivation of the steady-state enzyme rate equation for the single substrate enzyme-catalyzed reaction is provided in the entry entitled *Enzyme Rate Equations* (I. The Basics).

GRAPHICAL REPRESENTATION. The above expression represents the equation of a hyperbola (*i.e.*, $f(x) = ax/(b + x)$ where a and b are constants) and a plot of the initial velocity as a function of $[S]$ will result in a rectangular hyperbola. Another way for representing the Michaelis-Menten equation is by using the double-reciprocal (or, Lineweaver-Burk³) transformation:

$$\frac{1}{v} = \frac{1}{V_{\max}} + \left(\frac{K_m}{V_{\max}}\right) \frac{1}{[S]}$$

This method is widely used because it is a linear transformation of the hyperbolic function describing the rate saturation process. Both plots are shown below for the same data set.



Double-reciprocal plots can be reasonably accurate if rate data can be obtained over a reasonable range of saturation, say from $0.3V_{\max}$ to $0.8V_{\max}$. Any indication of a curved double-reciprocal plot necessitates consideration of substrate activation (downward curvature as one proceeds to higher $(1/[S])$ values), substrate inhibition (usually evident by upward curvature as one approaches the $1/v$ -axis), multiple binding of S, failure to measure true initial rates, or cooperativity.

INTEGRATED MICHAELIS-MENTEN EQUATION.

Integrated rate equations for the treatment of enzyme kinetics can be traced back to the studies of Henri⁴. In principle, the integrated rate equation approach offers the advantage that a single reaction progress curve can provide all of the rate parameters typically obtained in initial rate studies carried out in the absence and presence of product (**See Progress Curve Analysis**). Furthermore, the method avoids mixing errors that can influence the outcome of a series of initial rate measurements made separately at different substrate and product concentrations. Stayton and Fromm⁵ have written an illuminating discussion regarding the validity of the integrated Michaelis-Menten equation, based on their detailed computer simulation studies.

¹V. Henri (1903) *Lois Generales de l'Action des Diastases* (translated: "General Laws of Diastase Action"), Hermann, Paris.

²L. Michaelis & M. L. Menten (1913) *Biochem. Zeit.* **49**, 333.

³H. Lineweaver & D. Burk (1934) *J. Amer. Chem. Soc.* **56**, 658.

⁴V. Henri (1904) *Arch. Fisiol.* **1**, 299.

⁵M. M. Stayton & H. J. Fromm (1979) *Meth. Enzymol.* **78**, 309.

MICHAELIS-MENTEN KINETICS

An enzyme is said to "obey" Michaelis-Menten kinetics, if a plot of the initial reaction rate (in which the substrate concentration is in great excess over the total enzyme concentration) *versus* substrate concentration(s) produces a hyperbolic curve. There should be no cooperativity apparent in the rate-saturation process, and the initial rate behavior should comply with the Michaelis-Menten equation, $v = V_{\max}[A]/(K_a + [A])$, where v is the initial velocity, $[A]$ is the initial substrate concentration, $V_{\max}$ is the maximum velocity, and K_a is the dissociation constant for the substrate, A, binding to the free enzyme. The original formulation of the Michaelis-Menten treatment assumed a rapid pre-equilibrium of E and S with the central complex EX. However, the steady-state or Briggs-Haldane derivation yields an equation that is iso-

morphic: $v = V_{\max}[A]/(K_m + [A])$ where K_m is a composite of a number of rate constants and is numerically equivalent to the substrate concentration at which the initial velocity is $0.5V_{\max}$. Thus, the general rubric of “Michaelis-Menten kinetics” extends to systems exhibiting hyperbolic kinetics, irrespective of whether one relies on the original rapid equilibrium condition or the later steady-state assumption.

Some investigators also unnecessarily apply the further restriction that Michaelis-Menten kinetics refers only to enzymes catalyzing the conversion of a single substrate to a single product. Were this taken to its extreme, only isomerases would qualify, because most one-substrate systems utilize water as a second substrate or product. **See** *Michaelis-Menten Equation; Uni Uni Mechanism; Enzyme Rate Equations (1. The Basics)*

MICRO-

A prefix, symbolized by μ , used in submultiples of units and corresponding to a value of 10^{-6} .

MICROENVIRONMENT

The ionic and solvent environment of a specific localized region (*e.g.*, the near environment surrounding a substrate or residue residing within an enzyme’s active site).

MICROSCOPIC DIFFUSION CONTROL

Any limitation on reaction rates in a homogeneous medium by the frequency of forming encounter complexes from reacting molecular entities (also called “encounter control”). For any instantaneously occurring bimolecular reaction in a homogeneous medium (*i.e.*, a reaction proceeding significantly more quickly than the encounter rate), the overall rate of the reaction is governed solely by the diffusion rates of the reactants. The alternative term is “total microscopic diffusion control”. The term “partial microscopic diffusion control” is often used to describe cases where diffusion and chemical conversion occur on comparable time-scales. **See also** *Macroscopic Diffusion Control*

MICROSCOPIC REVERSIBILITY

A principle requiring that, under equilibrium conditions, any molecular process (or reaction) and the reverse of that process (or reaction) will occur with the same frequency. This principle was verified through the use of

certain quantum mechanical expressions for transition probabilities.

The Principle of Microscopic Reversibility and its large-scale consequence, known as the Principle of Detailed Balancing enable investigators to understand the mechanism of the reverse reaction to the same level of accuracy as that achieved for the forward reaction.

This principle has only limited application to reactions that are not at equilibrium. Furthermore, the Principle of Microscopic Reversibility does not apply to reactions commencing with photochemical excitation. **See also** *Detailed Balancing, Principle of; Chemical Reaction; Chemical Kinetics*

MICROTUBULE ASSEMBLY KINETICS

The $\alpha\beta$ tubulin dimer contains all of the structural information required for the spontaneous formation of microtubules^{1,2}. Tubulin polymerization readily occurs *in vitro* when the concentration of the $\alpha\beta$ tubulin dimer is sufficient to support spontaneous nucleation and elongation (see below). The following sections describe *in vitro* microtubule (MT) self-assembly. Intracellular MT assembly and disassembly uses similar elongation, dynamic instability, and treadmilling processes. However, intracellular MTs are stabilized through the association of one MT end with microtubule-organizing centers (*e.g.*, centrioles and ciliary/flagellar axonemes). **See also** *Biochemical Self-Assembly; Actin Assembly Kinetics; Hemoglobin S Polymerization; Indefinite Polymerization; Isodesmic Polymerization; Microtubule Treadmilling*

1. NUCLEATION. The polymer self-assembly theory of Oosawa and Kasai³ treats nucleation as a highly cooperative and unfavorable event. Their kinetic theory for nucleation permits one to obtain information about the size of the polymerization nuclei, provided that two basic assumptions can be satisfied experimentally. First, the rate of nuclei formation is assumed to be proportional to the i_0 th power of the protomer concentration, with i_0 representing the number of protomers required to create the nucleus. Second, the treatment deals only with that period during which the polymerization rate greatly exceeds the rate of protomer loss from the polymers (*i.e.*, the initial stage of polymerization when the protomer concentration is the highest).

1a. Theoretical Considerations. Oosawa and Kasai³ also assumed nucleation involved formation of a higher order “closed” structure from an “open” structure with the same number of protomers. The nucleation step involved the conversion of an unstable linear trimer to a closed helical trimer; the greater stability of the closed structure arises from additional interprotomer bonding in the closed structure. Assuming that the rate of nuclei formation is proportional to the third power of the protomer concentration, c_1^3 , the following equation results:

$$\frac{dc_p}{dt} = (k_+c_1 - k_-) \int_0^t Ac_1^3 dt = -\frac{dc_1}{dt}$$

where c_p is the total concentration of protomer in the polymer form, k_+ and k_- are the assembly and disassembly rate constants, and A is yet another constant which incorporates the rate constants for nucleation. The integral in this equation calculates the polymer number concentration at time t , and the net rate of polymer formation, dc_p/dt , is simply the number concentration times the rate of polymer formation for a single polymer (*i.e.*, $k_+c_1 - k_-$). The last part of the equality reflects the definition that polymer formation parallels the rate of loss of protomer concentration.

The above equation can be solved only if one adds a further restriction that k_+c_1 greatly exceeds k_- . Under this condition, we arrive at the following expression:

$$\ln \left[\frac{1 + [1 - (c_1/c_0)^3]^{1/2}}{1 - [1 - (c_1/c_0)^3]^{1/2}} \right] = 3[(2/3)k_+A]^{1/2}c_0^{3/2}t$$

Here, c_0 is the total protein concentration, and a plot of polymer formed *versus* reaction time can be made by substituting decreasing values of c_1 (recalling that c_p equals c_0 minus c_1) into this second expression and solving for t . Oosawa and Asakura⁴ presented a more general solution for a nucleus composed of i_0 protomers, and the observed cooperativity depends on the number of protomers engaging in nucleation, as shown in the following expression:

$$\ln \left[\frac{1 + [1 - (c_1/c_0)^{i_0}]^{1/2}}{1 - [1 - (c_1/c_0)^{i_0}]^{1/2}} \right] = 2i_0^{1/2}[k_+k_+^*]^{1/2}c_0^{i_0/2}t$$

Note that k_+^* is the forward rate constant for nucleation. If one chooses a particular value of c_1/c_0 (*e.g.*, one-half), then the left-hand side of the equation will remain constant, and one may measure the time, t , required to

obtain this particular c_1/c_0 ratio for different values of c_0 . Under these conditions, this expression can be rearranged into the following forms:

$$t = (\text{constant})c_0^{-i_0/2}$$

$$\log t = -(i_0/2)\log c_0 + \log (\text{constant})$$

A plot of $\log t$ *versus* $\log c_0$, where t is the time required to achieve a final c_1/c_0 ratio, yields the number of protomers in the nucleus by determining the slope. Although the Oosawa treatment originally used a c_1/c_0 value equal to 0.5, detailed computer simulation work shows that the $k_+c_1 \gg k_-$ assumption introduces significant error in the estimation of i_0 by the time that the reaction proceeds to the point that c_1/c_0 is at the value of one-half¹. Thus, one must actually work under conditions where a smaller c_1/c_0 value is obtained. The above theory resembles the Hill Equation for cooperative ligand binding, because the nucleation step is assumed to be of high reaction order and no intermediate states of fewer than i_0 protomers accumulate. A difficulty with this theory is the need to establish that there is only protomer in solution. In the case of cold-depolymerized microtubule (MT) protein, for example, there are many oligomeric species that complicate the experimental outcome.

Wegner and Engel⁵ extended the Oosawa analysis by dropping the assumption of irreversibility and by applying a steady-state approximation. They focused on a dimeric nucleus, and there is no readily available way for extending their treatment to more cooperative systems to yield tractable rate expressions. Nonetheless, their steady-state solutions agreed well with those obtained by direct numerical integration of the rate laws for two different sets of dimerization rate constants, and their analysis provides a rather satisfying view for actin polymerization.

1b. Experimental Considerations. Weisenberg⁶ first showed MT reassembly *in vitro* is a spontaneous process, one that does not require addition of nucleating centers. Borisy and Olmsted⁷ reported that microtubule assembly from porcine brain homogenates was clearly stimulated by the presence of high-molecular-weight structures and not just tubulin monomers. Cell extracts obtained by high-speed centrifugation formed few tubules, even though they were rich in tubulin. By electron microscopy,

they observed characteristically disk-shaped structures in the low-speed supernatants that disappeared as MT reassembly progressed. That these ring structures disappear during tubule formation does not establish a direct causal relationship between ring structures and the true nucleating species for reassembly. In the strictest sense, one must reserve the designation “nucleation center” for the simplest “embryonic” oligomeric structures that permit elongation reactions. Nonetheless, their discovery of a means for achieving seeded assembly from the ring structures stimulated widespread interest in oligomeric rings as microtubule precursors.

There are several potential fates of ring structures during MT assembly¹. First, they might serve as direct intermediates in MT assembly in much the same way that tobacco mosaic virus (TMV) protein disks stack to form rodlike particles. Second, rings may act as templates controlling the geometrical arrangement of protomers during assembly. Third, rings could break open into linear structures resembling protofilaments or they might open slightly to form “lock-washer” structures with their own incipient helical configuration. Such protofilament-like structures could interact with each other laterally to form structures that close to form a MT-like arrangement. Fourth, the ring structure *per se* might not be a true nucleating center; yet, oligomeric structures issuing from rings could isomerize into nucleation centers. Finally, ring structures could merely be a cold-stable tubulin polymorph, lacking function in reassembly formation. In this case, the ring structures could still be a depot or reservoir of tubulin. Timasheff and Grisham⁸ also pointed out: (a) that the various mechanistic proposals need not be mutually exclusive; (b) that solvent composition could influence the pathway of reassembly; and (c) that ring structures, while interesting structural isomers of oligomeric species, may have no direct role in assembly.

Johnson and Borisy⁹ first showed that the lag phase in the plot of turbidity (*i.e.*, polymer weight concentration) *versus* time accounted for only 5–10% of the entire amplitude obtained upon completion of the polymerization process. By fitting the elongation phase to a single exponential process, these investigators arrived at the correct conclusion that microtubule number concentration becomes relatively stable within the first minutes

after warming of cold-depolymerized microtubule protein. The constancy of microtubule number concentration during elongation has also been directly observed with bovine brain microtubule assembly by electron microscopy using samples fixed by dilution into microtubule stabilizing buffer under conditions which disfavor further nucleation¹⁰. Thus, nucleation does not occur to any appreciable extent once the lag phase ends.

Consider the case where the protein consists of $\alpha\beta$ dimers exclusively at the very beginning of an assembly experiment. Suppose further that spontaneous nucleation is sufficiently infrequent as the polymerization reaction reaches 5–10% of its maximal amplitude achieved over the remaining course of elongation. In this case, a reduction of the protomer concentration from about 20 to 18 μM would reduce the apparent extent of nucleation by a factor of about 10–20, such that the polymer number concentration remains fixed throughout the ensuing elongation phase. If nucleation were viewed as a one-step cooperative event, then the rate of nucleation would be proportional to the i th power of the protomer concentration if I protomers cooperatively form the polymerization nucleus:

$$i(\text{Protomers}) = (\text{Nucleus})$$

The equilibrium constant for this process would be given as follows.

$$K = [\text{Protomer}]^i / [\text{Nucleus}]$$

One would conclude that I must approximately equal 28 for this process! Hofrichter *et al.*¹¹ found a similar behavior in nucleation of human hemoglobin S (HbS); the apparent reaction order for the nucleation of HbS aggregation was about 32 (**See Hemoglobin S Polymerization**). Of course, such analyses are not fully justifiable, because one cannot assume ideality in the solution properties at high protein concentrations (**See Molecular Crowding**).

Engelbroughs *et al.*¹² carried out a kinetic analysis with the porcine brain protein utilizing exposure of rings to 0.8 M without reformation of ring structures to any major extent, as inferred from the much slower rate of assembly of such diluted protein as compared with diluted protein not receiving the high-salt treatment. They concluded that nucleation starts by the association of rings that are probably broken open upon warming. Because salt

treatment obviates the need to fractionate the microtubule-associated proteins (MAPs) from the tubulin, their studies provided a means for analyzing the system at constant MAPs/tubulin ratios.

Karr and Purich¹³ concluded that ring structures are themselves not microtubule assembly intermediates. They estimated that a 1–2 mg/mL sample of whole microtubule protein would contain around 10^{-7} M ring structures, if nearly 40% of the total cold-depolymerized protein elutes with the ring fraction. Yet, the MT number concentration during self-assembly is much lower, especially because the free tubulin concentration would be in considerable excess over the critical MT protein concentration. Karr and Purich¹³ used 1 mM GDP to inhibit microtubule elongation and followed the time-course of ring disintegration after warming to 30°C. By assuming the change in turbidity at 350 nm is a linear measure of the protein weight concentration, the loss in ring structure concentration followed an exponential decay ($t_{0.5} = 73$ seconds). Ring dissociation was sufficiently long-lived to permit production of smaller oligomers that might serve as nucleation centers. Terry and Purich¹⁴ also carried out pulse-chase measurements that demonstrated nearly complete equilibration of radiolabeled tubulin (Tb) E-site nucleotide warmed in the presence of unlabeled nucleotide. The MT protein concentrations used in these experiments were in the range of 1–2 mg/mL; at higher concentrations, elongation may be sufficiently rapid so as to incorporate more tubulin oligomers into microtubule polymer.

Mandelkow *et al.*¹⁵ provided direct kinetic studies of nucleation by using 1 Å synchrotron radiation to obtain time-resolved scattering data during cycles of assembly and disassembly after temperature shifts between 4 and 36°C. Small-angle scattering theory requires independent scattering from all particles in the solution, and the theory relates the intensity of scattering to other parameters as follows:

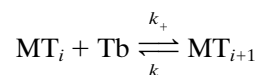
$$I(h, t) \propto \sum x_k(t) p_k(t) i_k(h)$$

where h is $(4\pi \sin \theta)/\lambda$ with θ the angle of scattering and λ the wavelength of X-radiation. The sum, $\sum x_k(t)$ is taken over all aggregates k ; $x_k(t)$ is the fraction of Tb promoters in an aggregate k ; $p_k(t)$ represents the degree of polymer-

ization (*i.e.*, the number of protomers in a given aggregate); and $i_k(h)$ is the scattering of function characteristic of an aggregate k normalized to unity at $h = 0$. Because $x_k(t)$ is the fraction of subunits in a particular aggregate, $\sum x_k(t)$ represents a conservation term. The $i_k(h)$ term contains structural information about each aggregate. Mandelkow *et al.*¹⁵ treated the rings as having apparent diameters of 375 Å and the MTs as having a mean diameter of about 350 Å. They recorded sufficient intensity data over time frames of 15 seconds during the assembly and disassembly events, and the angular dependence was evaluated from 0 to 30 mrad. They confirmed that ring structures breakdown to form small oligomers within 1 minute following warming to 36°C. Loss of rings leads to a condition where the main contribution to the scattering intensity can be interpreted in terms of a solution containing tubulin protomers and oligomers for the most part.

2. ELONGATION KINETICS. Elongation is the repetitive addition of protomers to the growing points at the ends of the polymer¹. The foregoing discussion indicates that the life-span of nucleation should be relatively short. In this respect, nucleation and elongation phases are temporally separate and can be treated as only slightly overlapping kinetic phases. Elongation can also be studied in the absence of spontaneous nucleation by using pre-existing microtubules or microtubule organizing centers (MTOCs).

2a. Theoretical Considerations. To treat elongation kinetics¹, one typically assumes: (1) that all of the growing points are identical and well described by a single bimolecular rate constant for addition and a second rate constant for protomer loss; (2) that the addition of a protomer to a growing point creates a new growing point of identical reactivity, such that the rate for addition or loss does not depend on polymer length; (3) that continuous nucleation does not occur and the polymer number concentration remains constant; and (4) that other pathways for elongation, including addition of oligomers or polymers, are negligible. If these assumptions hold, elongation reactions can be written simply as follows:



where MT_i is a microtubule containing i protomers, T_b is a tubulin protomer, and k_+ and k_- are the corresponding rate constants for protomer addition and loss, respectively. The rate equation for this process is

$$dc_1/dt = k_-m - k_+mc_1 = (k_- - k_+c_1)m$$

where c_1 is the concentration of free protomers, and m is the constant concentration of polymer ends. Assuming that only a small fraction of the initial protein concentration (c_0) is depleted to create the nuclei during nucleation, then c_1 at the onset of elongation equals c_0 . This also requires that any preexisting oligomers (*e.g.*, ring structures and protofilaments) comprise only a small part of the total protein at the start of the elongation phase. This rate equation describes a single exponential process, and the following integrated rate law describes the time evolution of the elongation process:

$$c_1(t) = k_-/k_+ + [c_0 - (k_-/k_+)]\exp(-k_+mt)$$

The equilibrium value of c_1 is k_-/k_+ or K_D , and the equilibrium constant should be independent of the total protein concentration in the polymerization reaction. Because all of the protein is considered to be in either the protomer or polymer forms, we may use the relation that $c_p = c_0 - c_1$, where c_p is the total concentration of protomers in the polymeric state. This relationship can be substituted into the above equation to yield a result for the time course of polymer formation:

$$c_p(t) = [c_0 - [k_-/k_+]][1 - \exp(-k_+mt)]$$

The quantity $c_p(t)$ can be measured experimentally by several techniques, but turbidimetric procedures are generally the preferred means. If the polymer is to form, c_0 must be greater than k_-/k_+ . Thus, Oosawa's model predicts that there is a critical protomer concentration of k_-/k_+ (or K_D) below which polymer cannot exist.

The above theory can be extended to deal with other more complex cases. For example, the two ends of a biopolymer need not behave identically, and, as noted earlier, MTs are helical polymers of asymmetric protomer units. Thus, two sets of on- and off-constants might be necessary. In other cases, such as in the polymerization of tubulin in the presence of tubulin-colchicine complex (Sternlicht *et al.*¹⁶), one may need to consider copolymerization. The kinetics of microtubule depolymerization are the reverse of elongation, and are gener-

ally unaffected by processes as complicated as nucleation. Thus, depolymerization can often provide elongation rate constants more directly.

2b. Experimental Considerations. There are several methods for obtaining elongation rate constants. The rate constant for polymerization, k_+ , can be determined from assembly kinetics data and a knowledge of the polymer number concentration, m at time t and equilibrium (eq), by plotting $\ln\{[c_p(t) - c_p(\text{eq})]/c_p(\text{eq})\}$ versus time. The slope of this plot is $(-k_+m)$, as can be readily determined using a value of m obtained by microtubule length distribution measurements from dark-field¹⁶ or electron¹⁷ microscopy. In both cases, the average polymer length is determined and subsequently divided into the total polymer length. This quantity is estimated from the amount of protein in the polymer form and the number of protomers per unit length of polymer. Once k_+ is evaluated, one may obtain k_- with a knowledge of the critical concentration, K_D . A variation on this theme is provided by utilizing needed assembly with a known number of microtubule seeds prepared by mechanical shearing and measuring the initial rate of assembly at various concentrations of protomer. Alternatively, one may resort to depolymerization experiments to get k_- under conditions where the polymers are induced to disassemble irreversibly. This may be accomplished by rapidly dropping the temperature of the polymer solution or by rapidly reducing the protomer concentration to a value far below the prevailing critical concentration. The latter has the advantage that the rate constants refer to the temperature at which other tubule elongation reactions are most typically examined¹. The kinetics of irreversible disassembly are presented elsewhere. **See Dilution-Induced Depolymerization**

2c. Role of GTP Hydrolysis. Enzymologists typically seek to identify those mechanistic features that explain catalytic rate enhancement; however, cell biologists focus their attention on regulation as the key feature of biological catalysis¹⁸. Tubulin may be considered to be a low- k_{cat} GTPase that acts as a highly regulated switch for controlling cell shape and motility. An even larger family of GTPases, commonly referred to as GTP- (or G-) regulatory systems, links the phosphorylation state of the bound guanine nucleotide to their inhibitor or activator potency. Each G-protein has two states, each character-

ized by the binding of GTP or GDP, with their interconversion accomplished directly by GTPase or indirectly by nucleotide exchange. In much the same way, tubulin exists in two principal states: the assembly-competent $Tb_E \cdot GTP$ complex and the assembly-incompetent $Tb_E \cdot GDP$ complex. Signal processing arises from the conformationally restricted properties of $Tb_E \cdot GTP$ and $Tb_E \cdot GDP$ in unpolymerized and polymerized forms, and although the microtubule literature is burgeoning, the scope of the present review is limited to this issue.

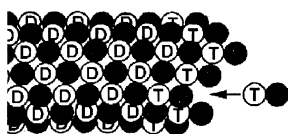
MacNeal and Purich¹⁹ first demonstrated that hydrolysis of exchangeable site GTP is tightly coupled to assembly. These observations, along with changes in the critical concentration in the presence of GDP, led Karr *et al.*²⁰ to propose their boundary stabilization model for microtubule growth and stabilization. Here, $Tb_E \cdot GTP$ participates in tight-binding interactions that result in tubule stabilization. In this model, $Tb_E \cdot GDP$ (itself arising from assembly-induced hydrolysis) occurs within the interior microtubule lattice. Each internal $Tb_E \cdot GDP$ is stabilized by its surrounding $Tb_E \cdot GDP$ neighbors or by $Tb_E \cdot GTP$ at the microtubule ends which are referred to as boundaries. Loss of the single terminal layer (or boundary) would raise the critical concentration as a consequence of the much weaker binding of $Tb_E \cdot GDP$ molecules at the tubule ends. Such a model is also fully consistent with GDP inhibition of assembly as well as with the ability of enzymatic conversion of GTP to GDP resulting in microtubule disassembly. Delayed GTP hydrolysis is a model where $Tb_E \cdot GTP$ molecules persist randomly throughout the microtubule lattice for a period following polymerization¹⁸. There is no evidence supporting such a model, and isotope exchange data obtained by Angelastro and Purich²¹ would suggest that this is an unlikely mechanism. Indeed, the microtubule lattice may exclude tubulin-bound GTP except at the terminal boundary. A third model involves formation of a multilayer “cap” comprised of many $Tb_E \cdot GTP$ molecules; this possibility suggested by the results of experiments conducted in the presence of the microtubule-stabilizing drug taxol²². Assembly under what may be regarded as “forcing conditions” probably has little or no relevance to microtubule stabilization *in vivo*, especially when one recognizes that $Tb_E \cdot GTP$, $Tb_E \cdot GDP$, and even E-site nucleotide-free Tb readily polymerize in the presence of taxol.

There is, in fact, no evidence supporting the existence of a “cap” containing many $Tb_E \cdot GTP$ molecules, even at very high tubulin concentrations where rapid polymerization is favored. Over an extraordinarily large range of tubulin concentration, O’Brien and Erickson²³ observed no accumulation of GTP beyond that anticipated on the basis of the boundary stabilization model of Karr *et al.*²⁰ This was confirmed independently by Jordan and Wilson²⁴.

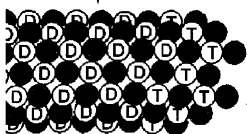
Based on the use of aluminum tetrafluoride anion (AlF_4^-) as a phosphate analogue, Carrier *et al.*²⁵ suggested that orthophosphate may bind within the γ -phosphoryl pocket of the exchangeable nucleotide site in a manner that confers at least some of the stabilization displayed by the bound-GTP conformation. In particular, they proposed that microtubule-bound tubulin-GDP- P_i is a stable intermediate in the GTPase reaction, and they suggested that the dynamic instability of microtubules could be governed by loss of such “caps” or by P_i release into the medium. Formation of such a stable intermediate creates a paradox; if release of P_i increases the instability of microtubule ends (*i.e.*, $\Delta G_{assembly}$ is more positive), this would require a correspondingly less favorable dissociation of P_i . Were $Tb_E \cdot GDP \cdot P_i$ and $Tb_E \cdot GTP$ isoenergetic, or even nearly so, the stable oxygen-18 exchange studies²¹ would have readily indicated this through the loss of oxygen-18 atoms during such reversals.

Microtubules contain tubulin protomers that are regularly arranged in a head-to-tail manner¹. Biased growth by addition of tubulin to one end is a consequence of the head-to-tail arrangement, and the transition states for tubulin addition reactions need not be identical for both polymer ends. Activation energies for the reactions at the two ends will in general be different, and the corresponding bimolecular rate constants will then be different. Microtubule polarity also accounts for the maintenance of different critical concentrations in the presence of GTP, and each end has different equilibrium constants for tubulin addition/release reactions. Such differential affinity arises from unequal retention of the Gibbs free energy of GTP hydrolysis; the less stable end (*i.e.*, the (–)-end) does not release as much free energy of hydrolysis during each step in microtubule elongation.

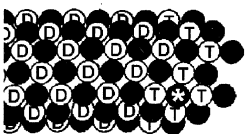
Delayed GTP Hydrolysis: Boundary Stabilization Model



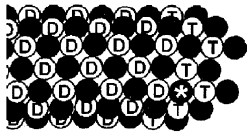
Insertion of Tb-GTP at Growth Site



Activation of Penultimate Tb-GTP
(marked with asterisk)

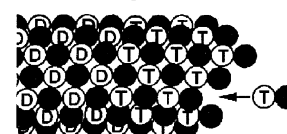


GTP Hydrolysis Without
Affecting MT Stability

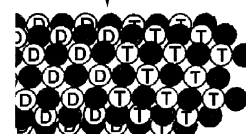


Comment:
Treadmilling Can
Occur in this Model

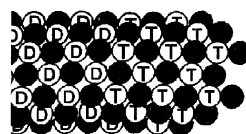
Delayed GTP Hydrolysis: GTP-Cap Model



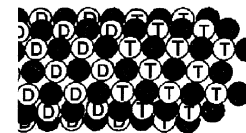
Insertion of Tb-GTP at Growth Site



Assembly & Nucleotide Hydrolysis
are NOT Coupled to Each Other

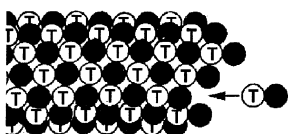


GTP-Cap Depends on Relative Rates
of GTP Hydrolysis & Elongation

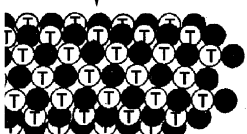


Comment:
Treadmilling Can
Occur in this Model

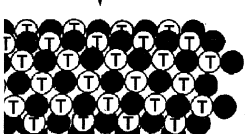
No GTP Hydrolysis: (Extreme Case)



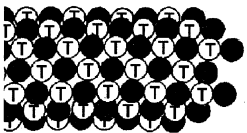
Insertion of Tb-GTP at Growth Site



No Hydrolysis

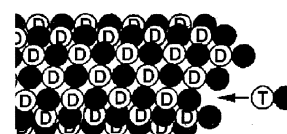


MT Lattice Contains
only Tb-GTP Dimers

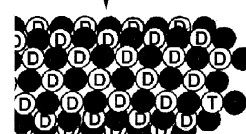


Comment:
Treadmilling Cannot
Occur in this Model

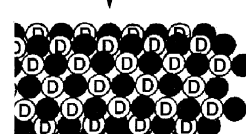
Instantaneous GTP Hydrolysis: (Extreme Case)



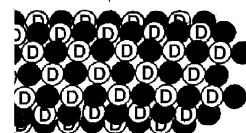
Insertion of Tb-GTP at Growth Site



Immediate GTP Hydrolysis



No Stabilization by Boundary or Cap



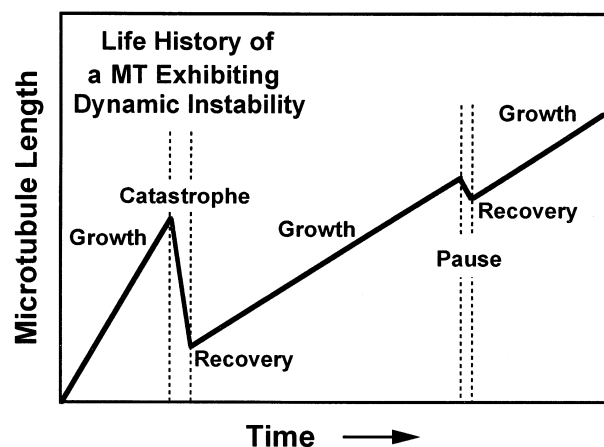
Comment:
Treadmilling Cannot
Occur in this Model

3. MICROTUBULE ASSEMBLY/DISASSEMBLY KINETICS. Cellular microtubules must undergo turnover, and nucleotide hydrolysis appears to play a central role in priming microtubules for their eventual disassembly. Two fundamentally different assembly/disassembly mechanisms persist during what has been termed steady-state polymerization; both rely on GTP hydrolysis to provide a source of Gibbs free energy to sustain the steady-state condition^{1,18}.

3a. Treadmilling. In the original formulation for actin dynamics²⁶, the concept of cytoskeletal polymer “treadmilling” was shown to arise from ATP hydrolysis-induced differences in the critical subunit concentrations in equilibrium with each filament end. Such a process implies that the macroscopic critical concentration [*i.e.*, K_{∞} equals the algebraic sum of the off-rate constants for both ends divided by the algebraic sum of the on-rate constants for both ends] lies between the microscopic critical concentrations for the more stable and less stable filament ends; at a steady state of assembly/disassembly, this will consequentially lead to loss of subunits from the less stable filament end and accumulation of subunits at the more stable end. In the context of microtubule dynamics, Margolis and Wilson²⁷ proposed a microtubule treadmilling model that operates *via* the exclusive addition and release of tubulin dimers from opposite ends of microtubules. Because rapid dilution of microtubule protein to below the critical concentration resulted in prompt depolymerization with a rate constant of around 0.1 min^{-1} for complete loss of polymer mass, Karr and Purich²⁸ noted that this disassembly rate exceeded the steady-state treadmilling rate by a factor of 1000. They therefore proposed a minimal model wherein the possibility of reversibility at the so-called primary disassembly end was also indicated by use of a broken arrow. Caplow²⁹ correctly indicated that *all* four rate constants must be considered to account for the assembly/disassembly kinetics of tubulin dimer interactions with microtubule ends. **See also** *Actin Assembly Kinetics*

3b. Dynamic Instability. Mitchison and Kirschner³⁰ presented another explanation for microtubule steady-state dynamics in terms of GTP hydrolysis and the stochastics of losing the stabilizing cap (or boundary) of tubulin molecules containing unhydrolyzed GTP at their E-sites. In the “dynamic instability” model, length changes in

microtubules at steady-state are thought to arise from the overall balance of two phases: the first involving slow growth of the majority of microtubule polymers; and the second arising from the rapid disassembly of a smaller fraction of polymers. They proposed that microtubules at steady-state contain $\text{Tb}_E \cdot \text{GTP}$ protomers at the polymer ends, forming a cap of stably bound protomers. In contrast, the microtubule interior lattice contains largely $\text{Tb}_E \cdot \text{GDP}$ protomers that are lost by endwise depolymerization whenever microtubules lose $\text{Tb}_E \cdot \text{GTP}$ protomers stabilizing their ends. Taken with the mounting evidence (*vide supra*) that there is no cap, the dynamic instability model most probably arises from the presence or loss of the stabilizing boundary formed by a few terminally bound $\text{Tb}_E \cdot \text{GTP}$ protomers.



Tubules grow and remain stable as a consequence of $\text{Tb}_E \cdot \text{GTP}$ bound at the growth sites. They undergo stochastic disassembly in the improbable case that all growth points lose their $\text{Tb}_E \cdot \text{GTP}$ molecules. Under the influence of as yet unidentified intracellular signal(s), terminal $\text{Tb}_E \cdot \text{GTP}$ molecules are probably released from growth points. Then, disassembly promptly takes place in what can be regarded as a catastrophe. Depending on the circumstance, disassembling tubules may completely depolymerize or may recover such that regrowth begins the cycle anew. Kristofferson *et al.*³¹ used biotinylated tubulin and antibody methods to analyze the time evolution of microtubule length redistribution which is a measure of steady-state microtubule dynamics. They were able to confirm the basic tenets of the dynamic instability model, and they clearly demonstrated that some tubules exhibit catastrophic depolymerization. Microtubules showed no evidence of treadmilling.

Video microscopy has permitted direct observation of microtubule assembly/disassembly dynamics *in vitro*. Horio and Hotani³² first used dark-field optics to observe the growth and shrinkage phases, but so-called Allen video-enhanced contrast microscopy has become most convenient.

4. INTRACELLULAR DYNAMICS. Evaluating microtubule assembly/disassembly dynamics directly in living cells has proven to be a challenging task requiring great technical ingenuity. No single method provides both temporal and spatial resolution for obtaining (a) accurate estimates of kinetic constants and (b) the size and intracellular location of stable and dynamic microtubule pools. Direct microscopic observation of microtubules, for example, offers an attractive means for analyzing net rates of tubulin gain or loss^{32–35}, but the resolving power of light and fluorescence microscopy cannot distinguish single tubules from bundled microtubules or even a pair of microtubules that run closely parallel to each other. Thus, even as image reconstruction techniques are improved electronically, the physics of light refraction still will limit the technique to cells or cell regions containing only a few microtubules. In the case of microinjection, proteins can be introduced into unfertilized oocytes³⁶, thereby allowing the investigator to observe details of cytoskeletal assembly and disassembly after the subsequently fertilized egg proceeds through embryogenesis. Such an approach works best with large oocytes, as in the case of *Xenopus*, so that the embryo's development does not lead to significant dilution of the original microinjected reporter protein. Recent studies in *Xenopus* exemplify how microinjection of rhodamine-labeled or bis-“caged” fluorescein-labeled tubulin during the first cleavage division can be used to examine microtubule movement. At the appropriate stage of embryonic development, the “caged” fluorophore can be photoactivated by brief exposure to a focused light beam, and intracellular tubule dynamics can be recorded with a silicon-intensified target video camera. In experiments of this sort one must contend with problems of photo-induced oxidation of the fluorescent tag itself as well as damage to light-sensitive subcellular components. Because dioxygen is essential for photooxidation, efforts to eliminate O₂ by redox scavengers or by physically excluding this gas helps to minimize photo-damage. However, oxygen is an essential nutrient for maintaining

the [ATP]/[ADP] and [GTP]/[GDP] concentration ratios *via* oxidative phosphorylation. Likewise, the [ATP]/[ADP] ratio is important in modulating the action of microtubule-based motors [*i.e.*, dynein and kinesin], and various regulatory protein kinases. While the metabolic labeling technique can provide estimates of rates and extents of tubulin-microtubule exchange, results obtained with this approach must be interpreted using other data to gain insight about the cellular locations of stable and dynamic tubules. For example, there is mounting evidence for the occurrence of stable microtubules located in neurite outgrowths of PC12 cells grown in the presence of nerve growth factor.

The transition from G to M stages of the cell cycle is accompanied by complete microtubule cytoskeleton disassembly. The interphase microtubule cytoskeleton must be fully dismantled to allow mitosis to commence. Thus, upon transition from a growing to a shrinking tubule in the dynamic instability model, totality of depolymerization occurs, and this may be a very desirable outcome. During microtubule disassembly, Tb_a·GDP molecules rapidly issue from microtubule ends and probably crowd the vicinity of each unwinding microtubule end, and any entry and binding of Tb_E·GTP molecules would probably be most unlikely. Moreover, restoration of a Tb_E·GTP boundary layer and restabilization of a microtubule end would be especially improbable if Tb_E·GTP molecules must coexist at several growth sites simultaneously before stabilization can be restored. There must be multiple hierarchies of regulatory interactions that trigger global microtubule disassembly. Modulation of microtubule dynamics may be elicited by microtubule-associated proteins (MAPs) and enzymes, metabolic signals, as well as other low-molecular-weight factors. Protein kinase-mediated MAP phosphorylation releases this constraint of the latter *in vitro*³⁷. Even more intriguing is the possibility of enhanced instability that may be achieved through the action of a *Xenopus* oocyte protein found to sever microtubules after mitotic activation³⁸. Microtubules have their (+)-ends near the cell margin and distal to the centrosomes which appear to bind and stabilize the (–)-ends; loss of subunits only from the (+)-ends may be insufficient to permit rapid microtubule disassembly during transitions in the cell cycle. If both ends were free to disassemble, tubulin dimer release could proceed with rate constants of 200–500 s^{–1}, correspond-

ing to length changes of about 6–12 $\mu\text{m}/\text{min}$. Severing long tubules into several shorter fragments could thus increase both the number and kind of disassembling ends, thereby allowing for rates upward of 50 $\mu\text{m}/\text{min}$. Vale's experiments on the severing of taxol-stabilized tubules are consistent with extensive, multiple fragmentation. In terms of control mechanisms, one should also not discount the opportunity for tubule disassembly in response to local surges of calcium ion, and any factor causing a drop in the cellular $[\text{GTP}]/[\text{GDP}]$ poise could likewise tip the balance to favor disassembly. Post-translational modification² of tubulin [*e.g.*, acetylation, tyrosination, glutamylation, glycylation, and ADP-ribosylation] also may control the stability and dynamics of microtubules.

- ¹D. L. Purich & D. Kristofferson (1984) *Adv. Prot. Chem.* **36**, 133.
- ²B. J. Terry & D. L. Purich (1982) *Adv. Enzymol.* **53**, 113.
- ³F. Oosawa & S. Asakura (1975) *Thermodynamics of the Polymerization of Protein*, Academic Press, New York.
- ⁴F. Oosawa & M. Kasai (1962) *J. Mol. Biol.* **4**, 10.
- ⁵A. Wegner & J. Engel (1975) *Biophys. Chem.* **3**, 215.
- ⁶R. Weisenberg (1972) *Science* **177**, 1104.
- ⁷G. G. Borisy & J. B. Olmsted (1972) *Science* **177**, 1196.
- ⁸S. N. Timasheff & L. M. Grisham (1980) *Annu. Rev. Biochem.* **49**, 565.
- ⁹K. A. Johnson & G. G. Borisy (1977) *J. Mol. Biol.* **117**, 1.
- ¹⁰D. Kristofferson & D. L. Purich (1981) *Arch. Biochem. Biophys.* **211**, 222.
- ¹¹J. Hofrichter, P. D. Ross & W. A. Eaton (1974) *Proc. Natl. Acad. Sci. U.S.A.* **71**, 4864.
- ¹²Y. Engelbroughs, L. C. M. DeMaeyer & N. Overbergh (1977) *FEBS Lett.* **80**, 81.
- ¹³T. L. Karr & D. L. Purich (1980) *Biochem. Biophys. Res. Commun.* **95**, 1885.
- ¹⁴B. J. Terry & D. L. Purich (1980) *J. Biol. Chem.* **255**, 10532.
- ¹⁵E.-M. Mandelkow, A. Harmsen, E. Mandelkow & J. Bordonas (1980) *Nature* **287**, 595.
- ¹⁶H. Sternlicht & I. Ringel (1979) *J. Biol. Chem.* **254**, 10540.
- ¹⁷T. L. Karr, D. Kristofferson & D. L. Purich (1980) *J. Biol. Chem.* **255**, 8560 and 8567.
- ¹⁸D. L. Purich & J. M. Angelastro (1994) *Adv. Enzymol.* **69**, 121.
- ¹⁹R. K. MacNeal & D. L. Purich (1978) *J. Biol. Chem.* **253**, 4683.
- ²⁰T. L. Karr, A. E. Podrasky & D. L. Purich (1979) *Proc. Natl. Acad. Sci. U.S.A.* **76**, 5475.
- ²¹J. M. Angelastro & D. L. Purich (1990) *Eur. J. Biochem.* **191**, 507.
- ²²M.-F. Carlier & D. Pantaloni (1981) *Biochemistry* **20**, 1918.
- ²³E. T. O'Brien & H. P. Erickson (1989) *Biochemistry* **28**, 1413.
- ²⁴M. A. Jordan & L. Wilson (1990) *Biochemistry* **29**, 2730.
- ²⁵M.-F. Carlier, D. Didry, R. Melki, M. Chabre & D. Pantaloni (1988) *Biochemistry* **27**, 3555.
- ²⁶A. Wegner (1976) *J. Mol. Biol.* **108**, 139.
- ²⁷R. L. Margolis & L. Wilson (1978) *Cell* **13**, 1.
- ²⁸T. L. Karr & D. L. Purich (1979) *J. Biol. Chem.* **254**, 10885.
- ²⁹M. Caplow (1992) *Curr. Opin. Cell Biol.* **4**, 58.
- ³⁰T. Mitchison & M. W. Kirschner (1984) *Nature* **312**, 237.
- ³¹D. Kristofferson, T. Mitchison & M. W. Kirschner (1986) *J. Cell. Biol.* **102**, 1007.
- ³²T. Horio & H. Hotani (1986) *Nature* **321**, 605.

- ³³M. Caplow, R. Ruhlen, J. Shanks, R. A. Walker, & E. D. Salmon (1989) *Biochemistry* **28**, 8136.
- ³⁴L. Cassimeris, N. K. Pryer & E. D. Salmon (1988) *J. Cell Biol.* **107**, 2223.
- ³⁵D. G. Drubin, S. C. Feinstein, E. M. Shooter & M. W. Kirschner (1985) *J. Cell Biol.* **101**, 1799.
- ³⁶E. M. Tanaka & M. W. Kirschner (1991) *J. Cell. Biol.* **115**, 345.
- ³⁷N. Raffaelli, P. S. Yamauchi & D. L. Purich (1992) *FEBS Letters* **296**, 21.
- ³⁸R. D. Vale (1991) *Cell* **64**, 827.

MICROTUBULE CYTOSKELETON AS A "SENSISOME"

The "sensisome" is a hypothetical intracellular sensory organelle that may exploit the internal channels of microtubules (MTs) to acquire and conduct regulatory signals from the cell's periphery to its deepest interior via the centrosome¹. Although the structural and mechanical properties of microtubules remain a central focus of modern cell biology, one inescapable feature of the microtubule—namely its lumen or internal channel—has attracted surprisingly little attention. When viewed in terms of MT cytoskeletal organization and assembly/disassembly dynamics, the MT lumen appears to be uniquely suited to serve as a sensory organelle.

Relative Dimensions of Microtubule Channels and Other Cell Components. The microtubule channel is surprisingly large. If filled only with water, the channel of a one micrometer-long microtubule would contain upwards of 1.2×10^6 water molecules. The channel's 15–16 nm diameter has an approximate cross-sectional area of about 180 nm², whereas the approximate cross-sectional areas of 100 kDa globular proteins fall in the range of 35–40 nm², suggesting that the microtubule channel is of sufficient dimension to conduct most biological molecules. Particles the size of the ribosome (molecular mass = 2.5×10^6 Da; long-axis = 25 nm) obviously cannot enter these channels, but even smaller molecules may experience limited access. It is well known that molecular diffusion through large pores depends on the pore's cross-sectional area as well as the size and shape of the diffusing molecule^{2–4}. By analogy, as the dimension of a molecule approaches that of the MT channel diameter, movement into and through the channel will become hindered. This is to be expected, because molecules must enter without hitting the rim of the channel. One can most readily deal with the case where (a) no electrostatic interactions occur, (b) bulk solvent flow is inconsequential, and (c) the motion of a rigid spherical protein

through the channel's orifice occurs by pure diffusion. In this limiting case, the channel's effective cross-sectional area for entry, $A_{\text{effective}}$, will be reduced by the radius of the entering molecule. For a sphere entering a cylindrical orifice, $A_{\text{effective}} = A_o(r - a)2/K_1r^2$, where A_o is the actual cross-sectional area, a is the sphere's radius, and K_1 is the drag coefficient; the latter of which is a complex function of a/r . These considerations indicate that molecules with molecular masses above 300 kDa or with one dimension exceeding 10 nm may be effectively excluded from entry into the channel. Another factor that can alter the rate of diffusion within narrow-bore tubes is hindered motion. This phenomenon relates to friction imparted by the inner MT wall, and the effect is likely to be significant for larger protein molecules (molecular masses > 200 kDa) that might be present in microtubule channel. Even 100 kDa molecules might experience a 2-3 times reduction in diffusional mobility as a consequence of hindered motion, but experiments are needed to settle this issue definitively. Even if hindered motion does in fact occur, this need not detract from the potential suitability of the MT channels acting collectively as a "sensisome."

Nature of One-Dimensional Diffusion. The 15-16 nm diameter of the microtubule channel is comparatively small relative to the 5-50 micrometer microtubule lengths observed in most living cells. This feature allows one to treat diffusion within a microtubule virtually as a one-dimensional process, and we may begin by considering a long channel, or tube, running parallel with respect to the x -axis. Suppose further that the system is initially configured, such that the concentration to the left of the origin is c_1 and that to the right of the origin is c_2 . For the case where $c_1 < c_2$, the mean square displacement ($\overline{x^2}$) of a particle can be described by the following function: $\overline{x^2} = (2k_B T/f)\tau$, where k_B is the Boltzmann constant, f is the frictional coefficient (Note: $f = -F/v$, for friction force F and particle velocity v), T is the absolute temperature, and τ is time over which a particle has a certain kinetic energy. During this small period τ , the diffusing molecules will possess the above-defined mean-square displacement; because the molecules can move in either direction, 50% of the molecules within a distance of $\overline{x^2}$ will pass the origin. Accordingly, leftward and rightward displacements will correspond to $(c_1 A/2)(\overline{x^2})^{1/2}$ and $(c_2 A/2)(\overline{x^2})^{1/2}$, respectively, and the net flow in the (+)- x direction is: $dn/dt = n/\tau = (c_1 -$

$c_2)A/2\tau$. The average change in the concentration gradient over time τ is: $dc/dn = (c_1 - c_2)A/2(\overline{x^2})^{1/2}$, and upon substitution into the prior equation, one arrives at: $dn/dt = -(A\overline{x^2}/2\tau)(\partial c/\partial x)$. [Note: This treatment describes motion along the x -axis; were one dealing with three-dimensional diffusion, the mean squared displacement $\overline{r^2}$ would be equal to $\overline{x^2} + \overline{y^2} + \overline{z^2}$, and $\overline{r^2} = (6kT/f)\tau$.] Based on the magnitude of diffusion constants for globular (*i.e.*, spherical) protein molecules, one can estimate that proteins like hemoglobin (mass = 68 kDa; diffusivity = $6.2 \text{ m}^2/\text{s} \times 10^{11}$) can readily diffuse 20–30 micrometers within 10–20 seconds. By contrast, low-molecular-weight metabolites, such as glycine (mass = 75; diffusivity = $95 \text{ m}^2/\text{s} \times 10^{11}$) and arginine (mass = 174; diffusivity = $58 \text{ m}^2/\text{s} \times 10^{11}$), will rapidly traverse these distances in a few seconds or less.

Advantages of Reducing the Dimensionality of Diffusion. Although molecular diffusion within bulk solutions is often best treated in a manner akin to three-dimensional Brownian motion, many biological processes are constrained geometrically, and the timing and efficiency of certain events can be strongly influenced by the ability of living systems to limit or reduce the dimensionality of diffusion. For example, two-dimensional surface diffusion increases the probability that a component will arrive at a particular locus within the cell. While transport along membrane surfaces represents an obvious example, movement within the microtubule channel represents a form of one-dimensional diffusion. Some of the virtues of one-dimensional diffusion have been indicated by Eigen who first proposed that the rapid action of repressors and other gene regulatory proteins might arise from a reduction in diffusion resulting in one-dimensional movement along DNA. The major advantages are (a) speed, (b) directionality, (c) processivity, and (d) the certainty of signal arrival at a desired destination.

Adam and Delbrück⁵ pointed out how the ability to manage the dimensionality of diffusion represents a selective advantage in evolution, suggesting that fully developed, internally compartmentalized cells enjoy a great range of options for controlling metabolic processes. Briefly, the mean time τ_i for diffusion to a target can be described by the following two-component equation:

$$\tau^{(i)} = \frac{b^2}{D^{(i)}} f^{(i)}(a/b)$$

The first component $b^2/D^{(1)}$ is the period of time required to traverse a distance b in any direction, whereas the second term $f^{(i)}(a/b)$ strongly depends on the dimensionality. Adam and Delbrück⁶ define appropriate boundary conditions and equations describing the concentration of molecules in the diffusion space in terms of space coordinates and time. They treated four cases: (1) one-dimensional diffusion in the linear interval $a \leq x \leq b$; (2) two-dimensional diffusion on the circular ring $a \leq r \leq b$; (3) three-dimensional diffusion in a spherical shell $a \leq r \leq b$; and (4) combined three-dimensional and surface diffusion. They provide a useful account of how reduced dimensionality of diffusion can (a) lower the time required for a metabolite or particle originating at point P to reach point Q , and (b) improve the likelihood for capture (or catch) of regulatory molecules by other molecules localized in the immediate vicinity of some target point Q .

Properties of Microtubule Cytoskeleton that Commend its Use as a “Sensisome.” The intriguing notion that the internal channels of cytoskeletal microtubules may act as a sensory organelle rests on several properties. First, microtubule channels should, on the basis of the above discussion, readily conduct small- and medium-sized molecules within seconds over cellular dimensions. Second, microtubules adopt and maintain a well defined cytoskeletal organization; during interphase, their (–)-ends are attached to the centrosome which resides in a perinuclear locale near the Golgi apparatus, and during mitosis they assemble into astral, pole-to-pole, and pole-to-kinetochore arrays. The (+)-ends of the microtubule in nondividing cells are localized in the cell’s periphery, nearest to the plasma membrane where metabolic transport and signal transduction are most active. Third, the (+)-ends of microtubules in living cells are known to undergo stochastic disassembly and regrowth as a consequence of dynamic instability. This property allows MTs to act as self-filling “syringes” that can quickly capture an aliquot of the peripheral cytoplasm, thereby allowing the centrosome to be informed about recent changes in the solute composition of the peripheral cytoplasm. A second outcome of dynamic instability is that microtubules should be self-clearing, and larger molecules that become accidentally lodged in the inner MT channels can be released during catastrophic depolymerization. Fourth, the continuity of the microtubule channels from

the cell periphery to the centrosome makes certain that metabolic signals will arrive at the centrosome. Finally, in some cells the total MT channel volume is likely to occupy a significant fraction of cellular space. In the case of the axon, microtubules appear as well spaced colinear arrays, and the MT-to-MT distances observed in EM cross-sections suggests that the inner channels may contain upwards of 2% of the axon’s total volume.

If one accepts the view that Nature is extraordinarily opportunistic—never wasting even a gesture, then the microtubule lumen presents the opportunity for use as an elegant sensory organelle that may be present in all eukaryotic cells.

¹One of the authors (D.L.P.) wishes to thank Ms. Dagny Ulrich for her continuing work on molecular diffusion within microtubule channels. Professor Robert J. Cohen also provided helpful insights about molecular diffusion as well as references to Smoluchowski’s early work.

²C. M. Feldherr, R. J. Cohen & J. A. Ogburn (1983) *J. Cell Biol.* **96**, 1486.

³W. L. Haberman & R. M. Sayre (1958) Motion of Rigid and Fluid Spheres in Stationary and Moving Liquids Inside Cylindrical Tubes, *David Taylor Model Basin Report No. 1143*, U. S. Navy, Washington, D. C.

⁴P. L. Paine & P. Scherr (1975) *Biophys. J.* **15**, 1087.

⁵G. Adam & M. Delbrück (1968) in *Structural Chemistry and Biology* (A. Rich & N. Davidson, eds.) pp. 198-215, Freeman, San Francisco.

MICROTUBULE TREADMILLING

The biased opposite-end assembly and disassembly behavior of tubulin dimer interactions with assembled microtubules resulting from the differentially greater stability of the (+)-end, a consequence of that end’s greater release of free energy attending GTP hydrolysis. The treadmilling phenomenon was originally observed *in vitro* with actin filaments^{1,2}. Margolis and Wilson³ first observed microtubule treadmilling and suggested that the microtubule assembly-disassembly “equilibrium” is the summation of two different steady-state reactions occurring at the opposite ends of the microtubule. They proposed that assembly and disassembly occur predominantly (if not exclusively) at the opposite ends under steady-state conditions *in vitro*. Terry and Purich⁴ demonstrated that GTP hydrolysis was absolutely essential for the treadmilling process and that nonhydrolyzable analogues such as Gpp(NH)p (or, 5′-guanylylimidodiphosphate) and Gpp(CH₂)p (or, β,γ-methyleneguanosine 5′-triphosphate) promoted microtubule assembly, but not treadmilling.

The significance of microtubule treadmilling in living cells has remained a matter of dispute, especially because cytoplasmic formation and growth of microtubules are thought to be template-driven, using so-called microtubule-organizing centers such as the centrosome. With the minus ends anchored at the centrosome, microtubule turnover is believed to be dominated by dynamic instability⁵ which reflects the disassembly/reassembly properties of the plus ends of microtubules. Nonetheless, Rodionov and Borisov⁶ recently observed that microtubules in cytoplasmic fragments of fish melanophores detach from their nucleation site and depolymerize from their minus ends. Free microtubules moved toward the periphery by treadmilling growth at one end and shortening from the opposite end. Rodionov and Borisov⁶ suggested that frequent release from nucleation sites may be a general centrosome property that permits minus-end release and microtubule turnover by treadmilling. **See** *Microtubule Assembly Kinetics*

¹A. Wegner (1975) *Biophys. Chem.* **3**, 215.

²A. Wegner (1976) *J. Mol. Biol.* **108**, 139.

³R. L. Margolis & L. Wilson (1978) *Cell* **13**, 1.

⁴B. J. Terry & D. L. Purich (1980) *J. Biol. Chem.* **255**, 10532.

⁵T. Mitchison & M. W. Kirschner (1984) *Nature* **312**, 237.

⁶V. I. Rodionov & G. G. Borisov (1997) *Science* **275**, 215.

MIGRATION

1. The movement, relocation, or transfer of an atom, moiety, or group in a rearrangement reaction. Such transfer is usually intramolecular, and the migrating group can move with its electron pair (often regarded as a nucleophile), without its electron pair (thus regarded as an electrophile), or with a single electron (as in free-radical rearrangements). 2. The movement of a bond (within a given molecular entity) to a new position; often referred to as bond migration. Such a bond migration occurs in the conversion of zymosterol to $\Delta^{7,24}$ -cholesta-dienol during cholesterol biosynthesis. 3. The movement of ions while under the influence of an electric field. Because cations and anions need not always move at the same velocity, they may transport different fractions of the current. **See also** *Molecular Rearrangement*

MINICHAPERONES

A polypeptide consisting of residues 191-345 of GroEL that, when immobilized on agarose, acts as a very efficient chaperone with proteins that resist renaturation by conventional refolding methods. Immobilized minichap-

erones may be useful in column chromatography procedures or batchwise to renature proteins in high yield and with biological activity.

M. M. Altamirano, R. Golbik, R. Zahn, A. M. Buckle & A. R. Fersht (1997) *Proc. Natl. Acad. Sci. U.S.A.* **94**, 3576.

A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

MINIMAL LETHAL DOSE

The minimum dose of a toxic substance that is lethal when assayed with various experimental animals. It is usually followed by a subscript number equal to the percentage of the animals that died under the assay conditions (*e.g.*, LD₅₀).

MINIMAL STRUCTURAL CHANGE, POSTULATE OF

A prediction that only the minimum number of bond-making and bond-breaking steps occurs in the conversion of reactant(s) to product(s). Thus, during the course of a substitution reaction, a new substituent enters the exact same position previously occupied by the displaced moiety. **See** *Molecular Rearrangement; Rearrangement Stage*

MITOCHONDRIAL INTERMEDIATE PEPTIDASE

This zinc-dependent enzyme [EC 3.4.24.59] of the peptidase M3 family catalyzes the hydrolysis of a peptide bond such that there is a release of an N-terminal octapeptide at the second stage of processing of some proteins imported into the mitochondrion. The natural substrates are precursor proteins that already have been processed by the mitochondrial processing peptidase.

G. Isaya & F. Kalousek (1995) *Meth. Enzymol.* **248**, 556.

MITOGEN-ACTIVATED PROTEIN KINASES

As integral elements of the signal transduction pathway connecting the binding of a protein growth factor on the extracellular surface to specific gene expression, these protein kinases migrate from cytosol to nucleus and phosphorylate various transcription factors, including Jun/AP-1, Fos, and Myc. The following are recent reviews on the molecular and physical properties of this phosphotransferase (or, MAP kinase).

- C. J. Marshall & S. J. Leever (1995) *Meth. Enzymol.* **255**, 273.
 D. R. Alessi, P. Cohen, A. Ashworth, S. Crowley, S. L. Leever & C. J. Marshall (1995) *Meth. Enzymol.* **255**, 279.
 C. A. Lange-Carter & G. L. Johnson (1995) *Meth. Enzymol.* **255**, 290.

MITOGEN-ACTIVATED PROTEIN KINASE KINASE

One of the downstream elements of the signal transduction pathway, MAPKK (or, MEK) is phosphorylated by a Ser/Thr protein kinase, known as Raf, and phosphorylated MAPKK then catalyzes phosphoryl transfer to MAP kinase. The following is a recent review on the molecular and physical properties of this phosphotransferase (or, MAP kinase kinase).

- D. R. Alessi, P. Cohen, A. Ashworth, S. Crowley, S. L. Leever & C. J. Marshall (1995) *Meth. Enzymol.* **255**, 279.

MIXED-FUNCTION OXIDASE

The following is review on the molecular and physical properties of this class of monooxygenases, which are also known as hydroxylases. A typical monooxygenase reaction is the hydroxylation of an alkane to an alcohol which involves a reduced cosubstrate that reduces a second atom within the O₂ molecule to form water. Flavin-containing monooxygenases include lysine oxygenase and 4-hydroxybenzoate hydroxylase. Reduced pteridines are involved in the phenylalanine hydroxylase and tryptophan hydroxylase reactions. **See also** *Cytochrome P-450*

- C. H. Williams, Jr. (1976) *The Enzymes*, 3rd ed., **13**, 89.

MIXED-TYPE INHIBITION

A limiting case of noncompetitive inhibition, characterized by *third*-quadrant convergence of double-reciprocal plots of $1/v$ versus $1/[S]$ in the absence and presence of several constant levels of the inhibitor.

Consider the standard Uni Uni mechanism, $E + S \rightleftharpoons EX \rightleftharpoons E + P$, in which an inhibitor can bind reversibly to either the free enzyme, E, or to the central complex, EX, albeit with different affinities. Using the Cleland nomenclature, let K_{is} represent the dissociation constant of the inhibitor with respect to the free enzyme (thus, $K_{is} = [E][I]/[EI]$) and K_{ii} is the corresponding dissociation constant with respect to the central complex ($K_{ii} = [EX][I]/[EXI]$). Thus, $1/v = \{(1/V_{\max})(1 + ([I]/K_{ii}))\} + \{(K_m/(V_{\max}[S]))(1 + ([I]/K_{is}))\}$. If $K_{is} < K_{ii}$, the intersec-

tion will occur in the second quadrant of a double-reciprocal plot and if $K_{is} > K_{ii}$, the intersection point will be in the third quadrant. Thus, note that mixed-type inhibition is really a subset of noncompetitive inhibition.

MIXED VALENCY

A condition where metal ions within a coordination complex or cluster are present in more than one oxidation state. In such systems, there is often complete delocalization of the valence electrons over the entire complex or cluster, and this is thought to facilitate electron-transfer reactions. Mixed valency has been observed in iron-sulfur proteins. Other terms for this behavior include mixed oxidation state and nonintegral oxidation state.

MIXING TIME

The period that elapses before two or more solutions are thoroughly mixed in a chemical kinetic experiment. In most manually controlled chemical kinetic studies, the mixing time is rarely a factor affecting accurate data acquisition; however, the mixing time can be significant in rapid kinetic processes studied by continuous and stopped-flow kinetic techniques.

MODELS

Schematic representations serving as tools for assessing current knowledge and for designing better experiments to reveal the integrated nature of a process.

In chemistry and biology, mechanistic inquiry requires models that offer the virtues of precision, simplicity, and generativity. Simplicity arises from the symbolic representation of the interactions involving a minimal, yet sufficient, number of components to account for all observed properties of the system under investigation. Precision emerges from the consideration of how rival models lead to nonisomorphic features (*i.e.*, testable differences) that distinguish each candidate model. Generativity results from the act of recombining the constituent elements from one or more models to arrive at a revised model and to achieve added clarity that stimulates a new round of experimental tests. In this recursive manner, effective use of models permits one to determine the fitness of one or more models in describing the system. In biochemistry and molecular biology, this includes efforts to account fully for the effects of concentrations of each component, the time evolution of bio-

chemical events, the accumulation of transient intermediates, changes in such positional properties as conformation, configuration, and/or location, as well as the thermodynamics or bioenergetics of all of the molecular interactions.

MODIFIER

A term applied to substances or agents that either depress or elevate the action of a particular catalyst. While often used synonymously with effector, the International Union of Biochemistry has suggested that the term modifier is more appropriate for those substances that are artificially added to *in vitro* systems. Other investigators restrict the use of the term modifier to those agents that covalently modify the structure of the catalyst, thereby altering the catalyst's activity. **See** *Effector; Activator; Inhibitor*

MODULATION

1. Any alteration in the kinetics or flux of an enzyme-catalyzed reaction or biochemical pathway. 2. The regulation of the rate of translation of mRNA by a modulating codon. 3. The fluctuation of cells, either functionally or morphologically or both, in response to one or more changes in the environment of the cells.

MOLAR ABSORPTION COEFFICIENT

A fundamental unit in spectroscopy (synonymous with molar extinction coefficient) and symbolized by ϵ . It is equal to the absorbance of light per unit path length and per unit concentration. **See** *Absorption Spectroscopy; Beer-Lambert Law*

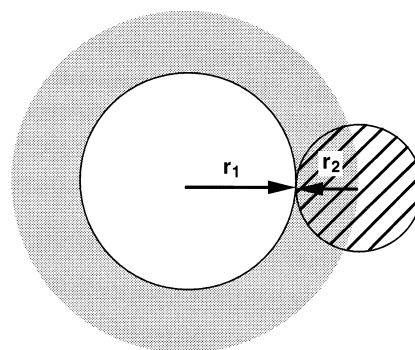
MOLAR GAS CONSTANT

A constant (symbolized by R) and used throughout the disciplines of kinetics and thermodynamics, equal to $8.314510 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ or $1.987216 \text{ cal}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. [The term was originally developed as the proportionality constant for the ideal gas law ($PV = nRT$ where P , V , n , and T are pressure, volume, number of moles, and absolute temperature of the ideal gas, respectively), hence the appearance of "gas" in its name.] **See** *Boltzmann Constant*

MOLECULAR CROWDING

A term referring to volume-exclusion effects of added solutes on the activity and reactivity of chemical species. Although most biochemical binding interactions take place in low reactant concentrations, the cumulative effect of a cell's numerous solutes is that nearly every cellular compartment is highly concentrated¹⁻³. In fact, one cannot escape the conclusion that cellular components are packed within cells to such an extent that certain nonideal behavior occurs. This is analogous to the behavior of a real gas whose molecules occupy a finite volume, rather than being infinitesimally small as in the case of ideal gas theory. Molecular crowding is probably of great significance in intracellular ligand binding and polymerization reactions.

Excluded Volume:



Volume from which solute molecules (of radii $r_1 + r_2$) are excluded by virtue of their mutual impenetrability.

$$V_{\text{excluded}}/\text{molecule} = (4/3) \pi [r_1 + r_2]^2$$

$$V_{\text{excluded}}/\text{mole} = (4/3) \pi N_o [r_1 + r_2]^2$$

Altered thermodynamic activity of proteins in solution arises when unreactive (or inert) macromolecules are added to a solution and occupy more than a few percent of total solution volume. Terms such as "unreactive", "background," or "inert" are used to emphasize that the added protein need not exhibit a direct binding interaction with the protein of interest. Instead, the consequences have more to do with molecular crowding, and approximate theoretical models show that this effect depends on the shapes and sizes of the macromolecules. Thus, biological fluids are anything but ideal or dilute solutions.

Muramatsu and Minton⁴ studied the diffusion of tracer proteins at low concentration in solutions containing “background” proteins at concentrations of up to 200 g/liter. The fractional reduction of the diffusion coefficient of tracer in the presence of a given weight per volume concentration of background species generally increases with increasing size of tracer species and with decreasing size of background species. The diffusion constants of three out of four tracer species depended upon the concentrations of four background species in a way that can be treated semiquantitatively by a simple hard particle model. They concluded that the diffusive transport of larger proteins and aggregates may be slower by several orders of magnitude than that occurring in dilute solution. These investigators applied scaled particle theory to evaluate the probability that the target volume for a step in a random walk is free of any macromolecules.

Minton *et al.*⁵ suggested that large excluded volume effects in cytoplasm (which they term “macromolecular crowding”) make otherwise soluble macromolecules associate with membrane proteins to a greater extent than might be expected on the basis of simple considerations of mass action. Han *et al.*⁶ extended this approach by choosing a different target volume which also allows the calculation to be extended to the diffusion of a hard sphere among hard spherocylinders. They concluded that, to the extent that proteins can be approximated as hard particles, the hindrance of globular proteins by other proteins is reduced when the background proteins aggregate. Greater aggregation has the effect of further decreasing particle surface area. Moreover, they indicate that the hindrance arising from the presence of rod-shaped background particles is reduced slightly when such rod-like particles are aligned.

Minsky *et al.*^{7,8} recently suggested that the emergence of eukaryotes led to a highly crowded environment that may have promoted DNA self-assembly, leading to extremely condensed and thermodynamically stable DNA aggregates such as nucleosomes and solenoid structures of chromatin.

¹A. P. Minton (1983) *Mol. Cell. Biochem.* **55**, 119.

²S. B. Zimmerman & A. P. Minton (1993) *Annu. Rev. Biophys. Biomol. Struct.* **22**, 27.

³A. P. Minton (1998) *Meth. Enzymol.* **295**, 127.

⁴N. Muramatsu & A. P. Minton (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 2984.

⁵A. P. Minton, G. C. Colclasure & J. C. Parker (1992) *Proc. Natl. Acad. Sci. U.S.A.* **89**, 10504.

⁶J. Han & J. Herzfeld (1993) *Biophys. J.* **65**, 1155.

⁷A. Minsky, R. Ghirlando & Z. Reich (1997) *J. Theor. Biol.* **188**, 379.

⁸A. P. Minton (1998) *Meth. Enzymol.* **295**, 127.

MOLECULAR DYNAMICS

Any of a series of computational methods for deducing molecular structure and reactivity by treating molecules or a system atoms comprising a segment of a macromolecule in terms of quantum and molecular mechanics. The basic goal is to explain molecular structure on the basis of the forces associated with and between bonded and nonbonded atoms. Quantum mechanical treatments become cumbersome when the number of atoms within the system is increased. For this reason one tends to use molecular mechanics, which is an empirical method, also called force-field calculations. These approximate methods allow one to estimate the relative energy differences among various molecular conformations by taking into account natural bond lengths and angles, influence of strain, torsional interactions, and attractive and/or repulsive electronic, dipolar, and van der Waals forces.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Application to [free energy calculations, **243**, 599-601; protein force field, **243**, 595-598; anti-hapten monoclonal antibodies, **203**, 31; and distance geometry, in NMR analysis of protein structure, **202**, 268; DNA structure modeling, **211**, 449; peptide modeling, **202**, 427; RATTLE algorithm, **243**, 597; restrained, **239**, 416, 429-433, 437-438; types of runs, **243**, 598; SHAKE algorithm, **243**, 597; simplification for computational speed, **243**, 596-598; simulations [applications, **241**, 178-182; in drug design, **241**, 382-384; in four dimensions, **239**, 645-646; historical background, **241**, 178-182; HIV-1 protease, **241**, 371-373, 376-381]; theoretical model, **241**, 187-189; methodology, **241**, 186-187; searching configurational space, **239**, 640; refinement [associated lowering of interval energy of distance geometry structures, **202**, 290; physical reliability, dependence on force field accuracy, **202**, 279; solvent inclusion during, **202**, 297]; simulations [in free energy evaluation, **202**, 499; metal force fields in proteins, **227**, 285; applications, **241**, 178-182; in drug design, **241**, 382-384; in four dimensions, **239**, 645-646; historical background, **241**, 178-182; HIV-1 protease, **241**, 371-373, 376-381 [Gln-88 mutant, **241**, 189-192; structure-function analysis, **241**, 178, 182-195; theoretical model, **241**, 187-189]; HIV-1 protease inhibitor design method, **241**, 370-384 [symmetric structures, **241**, 340]; HIV reverse transcriptase, **241**, 381; methodology, **241**, 186-187; modified, **239**, 641; multiple copy simultaneous search method, **241**, 383-384; searching configurational space, **239**, 640]; with time-averaged NOE restraints, **202**, 299.

MOLECULAR ENTITY

Any chemically or isotopically distinct atom, molecule, ion, radical, complex, *etc.*, capable of existing as a sepa-

rately distinguishable unit or entity. The term generally refers to single entities, irrespective of their nature. The term chemical species stands for a set or ensemble of molecular entities.

MOLECULAR "HITCH-HIKING"

Certain biological molecules exhibiting even relatively weak binding to cytoskeletal motors may "hitch-hike" for certain periods of time; when summed over time, this behavior can represent a physiologically significant form of vectoral conveyance. Depending on the persistence of their interactions with cytoskeletal motors, the effective diffusional dimensionality of these hitch-hiking molecules can be substantially reduced. During the course of their studies on the delivery of cytoplasmic proteins to the presynaptic terminals of CNS neurons, Garner *et al.*¹ found unique characteristics of one protein (designated p118) conveyed in slow component b (or SCb) of axonal transport, the large group of proteins representing the cytoplasmic matrix. Alone among the SCb group, p118 coisolated with the synaptic junctional complex, and sequencing of this protein revealed it is type I or brain hexokinase (HK, EC 2.7.1.1). Brain HK associates primarily with the mitochondrial outer membrane, but Garner *et al.*¹ found that most of type I HK is transported at a rate at least 10 times slower than mitochondria. Accordingly, they suggested that the enzyme's transient interactions with the more rapidly moving mitochondria may account for the kinetics of axonal transport of HK toward the presynaptic terminal.

¹J. A. Garner, K. D. Linse & R. K. Polk (1996) *J. Neurochem.* **67**, 845.

MOLECULARITY

An integer indicating the molecular stoichiometry of an elementary reaction. A fundamental assumption in chemical kinetics is that the kinetic form of a *one-step* reaction will be identical to its stoichiometric form. In terms of transition-state theory, molecularity equals the number of molecules (or entities) that are used to form the activated complex. For reactions in solution, solvent molecules are counted in the molecularity only if they enter into the overall process, not if they only exert an environmental or solvent effect¹⁻³.

The molecularity of a reaction is always an integer and only applies to elementary reactions. That is not always so for the order of a reaction, thus emphasizing the difference between molecularity and order. Molecularity

is the theoretical description of an elementary process, whereas order refers to the empirically derived rate expression. Usually, a bimolecular reaction is second order; however, the converse may not be true. Thus, unimolecular (sometimes called monomolecular), bimolecular, and ter- or trimolecular reactions refer to elementary reactions that involve one, two, or three entities, respectively, that combine to form an activated complex. The rate constant (or, specific reaction rate) for a reaction is commonly designated by k . The unit of time in the International System of units (SI), adopted in 1960 by the 11th General Conference on Weights and Measures, is the second (symbolized by s). Thus, unimolecular rate constants are typically expressed in s^{-1} , the value being concentration independent. A slight difficulty arises with regard to SI units and bi- and trimolecular rate constants. Concentrations in the SI system would be $mol \cdot m^{-3}$. Nevertheless, concentrations are typically expressed in $mol \cdot dm^{-3}$ (or, $mol \cdot L^{-1}$ or M^{-1}). Thus, a bimolecular rate constant typically has units of $M^{-1}s^{-1}$ whereas a trimolecular rate constant is expressed in units of $M^{-2}s^{-1}$. When gas-phase reactions are studied, (or, even with gases in solution), "concentrations" are often reported in terms of partial pressures rather than molarity. Nevertheless, it is usually preferable, when able, to convert such units to true concentration units.

Both unimolecular and bimolecular reactions are common throughout chemistry and biochemistry. Binding of a hormone to a receptor is a bimolecular process as is a substrate binding to an enzyme. Radioactive decay is often used as an example of a unimolecular reaction. However, this is a nuclear reaction rather than a chemical reaction. Examples of chemical unimolecular reactions would include isomerizations, decompositions, and disassociations. **See also** *Chemical Kinetics; Elementary Reaction; Unimolecular; Bimolecular; Transition-State Theory; Elementary Reaction*

¹Nomenclature Committee of the International Union of Biochemistry (1982) *Eur. J. Biochem.* **128**, 281.

²J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley-Interscience, New York.

³W. P. Jencks (1969) *Catalysts in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986) *Catalysis in Chemistry and Enzymology*, Dover Publications, New York.]

MOLECULAR ORBITALS

Geometrically defined regions of electron density surrounding two or more atomic nuclei. These orbitals are

considered a result of an overlap of atomic orbitals. Two types of molecular orbitals are generated: bonding and antibonding orbitals. Bonding orbitals have electron density positioned in a manner that favors bond formation, whereas the opposite is true for antibonding orbitals.

MOLECULAR REARRANGEMENT

A chemical process involving intramolecular repositioning or another alteration in the connectivity of atoms within a molecule, such that the product is isomeric with the reactant. Such reactions violate the principle of minimum structural change. Any reaction that includes migration of an atom, group, or bond falls under this class of molecular rearrangement reactions. If the entering group in a chemical reaction bonds to a different atom or position than was occupied by the leaving group, the reaction is also classified as a rearrangement.

In intermolecular rearrangement reactions (also referred to as “apparent rearrangement reactions”), the migrating group has been completely separated from the parent species and is then attached at a new position in a subsequent step. In intramolecular rearrangements the migrating group is never separated from the parent compound.

MOLECULAR SIMILARITY

1. A systematic approach^{1,2} for the rational design of extremely high-affinity transition-state analogues based on the use of detailed reaction mechanism information to deduce the most likely transition-state configuration, followed by molecular mechanical modeling of the electrostatic surface potential of that transition-state configuration as well as numerous candidate analogues to obtain the best matching analogue. Subsequent testing of the inhibitory action of the most promising analogues provides an additional route for refining the investigator's structural model of the transition state. 2. A numerical index for evaluating the similarity of geometric and electronic properties of an analogue and the most likely transition-state configuration for an enzyme-catalyzed reaction.

The “molecular similarity” strategy is based on Pauling's long-standing postulate that catalytic rate enhancement arises from the catalyst's ability to lower the energy input required to reach the transition state by selectively binding (and hence stabilizing) reactants in their transition-

state configuration. Vibrational modeling of a data set for experimentally determined kinetic isotope effects allows one to obtain a geometric model of the ground state and transition state for the nucleoside hydrolase reaction. From these data, one creates an electrostatic potential surface for free inosine, the above-mentioned transition state, as well as the reaction products.

¹B. A. Horenstein & V. L. Schramm (1993) *Biochemistry* **32**, 7089.

²B. A. Horenstein, D. W. Parkin, B. Estupinan & V. L. Schramm (1991) *Biochemistry* **30**, 10788.

MOLE FRACTION

A measure of the fractional composition of a solution containing one or more substances in amounts n_A , n_B , n_C , . . . (where n_i is the amount-of-substance i expressed in mol), such that for any component, say B:

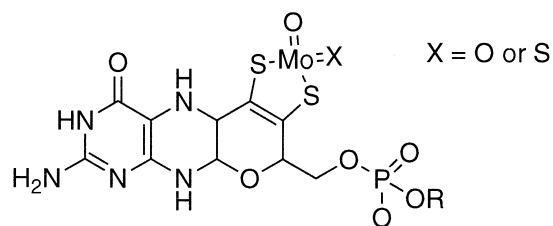
$$\text{mole fraction} = X_B = n_B/[n_A + n_B + n_C + \dots]$$

Technically, one includes solvent as one of the components when expressing mole fraction in chemical thermodynamics, but in describing dilute biological solutions, n_{solvent} is often omitted. With multicomponent solutions, one may choose to analyze the fractional composition of any two (or more) substances while not including others held constant in the experiment (*e.g.*, concentrations of buffer components, proton, supporting electrolyte(s), enzyme, *etc.*). For two components,

$$X_A = n_A/[n_A + n_B] \quad \text{and} \quad X_B = n_B/[n_A + n_B]$$

In this case, $X_A + X_B = 1$, and this formulation allows one to analyze binding interactions between two such components by the method of continuous variation. **See** *Concentration; Method of Continuous Variation*

MOLYBDENUM COFACTOR (MoCo)



The molybdenum complex of the molybdopterin prosthetic group (ligand). In the molybdenum cofactor the minimal coordination of the Mo atom is thought to be provided by the chelating dithiolene group of the molybdopterin and either two oxo or one oxo and one sulfido ligands.

MOLYBDOPTERIN

The prosthetic group associated with the molybdenum atom of the molybdenum cofactor found in most molybdenum-containing enzymes except nitrogenase (*See Molybdenum Cofactor*). Many of these enzymes catalyze two-electron redox reactions involving the net exchange of an oxygen atom between the substrate and water. In bacterial enzymes a nucleotide is linked to the phosphoryl group.

MONOD-WYMAN-CHANGEUX MODEL

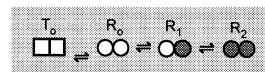
A ligand binding model¹ that explains the positive cooperativity of many allosteric proteins and enzymes. Contrary to what is often stated in general biochemistry textbooks, the MWC cooperativity does not increase the affinity of ligand binding as one progresses toward higher ligand concentration; indeed, the MWC model states that the intrinsic binding constants are the same for each site. Cooperativity in this model is actually the result of the allosteric transitions that increase sites available in a conformation permitting ligand binding.

The MWC model (also known as the all-or-nothing model, the concerted transition model, and symmetry-conserving model^{2,3}) relies on the following definitions and properties of allosteric systems: (1) Homotropic effects are interactions between binding sites for the same ligands. For example, the oxygenation of the four heme groups on hemoglobin are described as homotropic interactions. (2) Heterotropic interactions are interactions between binding sites for different ligands. For example, the effects of 2,3-bisphosphoglycerate on oxygenation of hemoglobin can be described as heterotropic interactions. (3) Most allosteric proteins are oligomeric, and changes in protein quaternary structure are factors in ligand binding cooperativity. (4) Homotropic interactions are most often cooperative for allosteric proteins, and few if any systems display only heterotropic interactions. (5) Cooperativity in the action of heterotropic ligands most likely stems from cooperative homotropic binding interactions. (6) Conditions, treatments, or mutations that alter heterotropic interactions also simultaneously alter the homotropic interactions.

The derivation of equilibrium ligand binding equations for this model involves the following assumptions: First, the protein is oligomeric and contains a finite number of identical subunits (called protomers). Second, the protein exists in two different symmetrical states (historically re-

ferred to as the R- and T-states), even in the absence of the binding ligand. Third, R- and T-states are in equilibrium with each other, even in the absence of bound ligand. Fourth, each state has a different affinity for the binding ligand or substrate (and the less abundant of the two states has the tighter affinity for the ligand); thus, the equilibrium that exists between the two forms of the protein can be altered by the presence of the ligand. Fifth, the symmetry of the oligomeric protein is preserved in the conformational transition between the two forms of the protein. (All binding sites on each state of the oligomeric protein are initially equivalent in the absence of bound ligand, and remain equivalent at all degrees of saturation.) And sixth, in the case of allosteric enzymes, binding of the substrate is treated as a rapid equilibrium phenomenon (that is, binding is rapid relative to any catalytic event(s)).

MONOD-WYMAN-CHANGEUX COOPERATIVITY MODEL



There is a pre-existing equilibrium between T_0 and R_0 , even in the absence of ligand F.

$$\mathcal{L} = [\square\square] / [\circ\circ]$$

One can use α to represent the so-called reduced (or normalized) concentration of ligand F..

$$\alpha = [F] / K_F$$

Ligand F can bind to either of the two open sites on R_0 , hence the "2" in the numerator.

$$K_F = 2 [\circ\circ][F] / [\circ\bullet] \quad \text{or} \quad [\circ\bullet] = 2[\circ\circ][F] / K_F \\ [\bullet\bullet] = 2\alpha[\circ\bullet]$$

Ligand F can bind to only one open site, but the "2" in the denominator indicates two ways for the ligand to be released from the fully bound dimer.

$$K_F = [\circ\bullet][F] / 2[\bullet\bullet] \quad \text{or} \quad [\bullet\bullet] = [\circ\bullet][F] / 2K_F \\ [\bullet\bullet] = \alpha[\circ\bullet] / 2 \\ [\bullet\bullet] = \alpha^2[\circ\circ]$$

The ligand saturation function accounts for the moles of bound ligand divided by the total moles of binding sites (hence the "n" in the denominator).

$$\bar{Y}_F = \{ [\circ\bullet] + 2[\bullet\bullet] \} / n \{ [\square\square] + [\circ\circ] + [\circ\bullet] + [\bullet\bullet] \}$$

The ligand saturation function can then be re-expressed as:

$$\bar{Y}_F = \{ 2\alpha[\circ\circ] + 2\alpha^2[\circ\circ] \} / 2 \{ \mathcal{L}[\circ\circ] + [\circ\circ] + 2\alpha[\circ\circ] + \alpha^2[\circ\circ] \}$$

$$\bar{Y}_F = \{ 2\alpha + 2\alpha^2 \} / 2 \{ \mathcal{L} + (1 + 2\alpha + \alpha^2) \}$$

$$= \alpha (1 + \alpha) / \{ \mathcal{L} + (1 + \alpha)^2 \}$$

Figure 1. Derivation of the MWC ligand saturation equation for a dimeric protein.

In Fig. 1, we show the full derivation of the saturation function for the case of ligand F binding to a dimer (open circles = unbound R-state; open squares = unbound T-state). Moreover, to demonstrate how the allosteric constant L influences the sigmoidicity of the ligand saturation curve, we present the dimer case again in Figure 2. At higher values of L (i.e., the greater the $[T_0]/[R_0]$ ratio), more of the ligand is present in the T-state, thereby requiring higher concentrations of ligand F (expressed here as the reduced concentration $\alpha = [F]/K_R$) to achieve significant saturation. The left-hand panel of Fig. 2 shows that the saturation curve becomes less S-shaped as the value of L approaches unity, a condition where the R_0 and T_0 states would be equally stable (i.e., $\Delta G_{R \rightarrow T\text{-transition}} \approx 0$). The right-hand panel shows how the concentrations T_0 , R_0 , R_1 , and R_2 change as a function of α for the case where $L = 1000$.

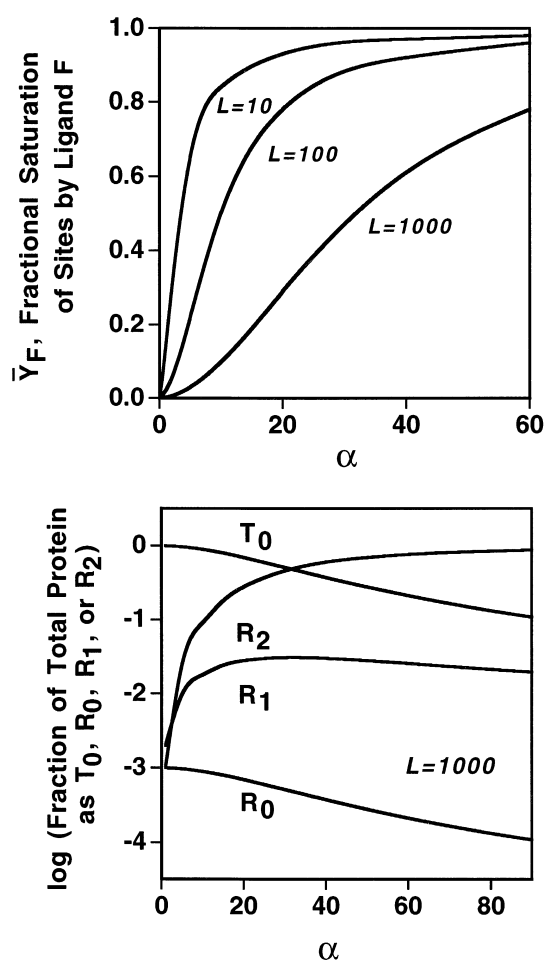


Figure 2. Top: Effect of the magnitude of the allosteric constant on ligand saturation behavior of a dimer obeying the MWC model. Bottom: Fraction of total dimeric protein present in various T- and R-species.

One can use free energy relationships to analyze the effect of ligand binding on the equilibrium between R and T states (see schematic diagrams shown in Fig. 3). Note that the energy difference between R_0 and R_1 is exactly the same as that for R_1 and R_2 . This is a required feature of the MWC model, because the strength of ligand F binding is exactly the same for each and every site in the R-state. Also note that the relative stability of T_0 with respect to R_0 will alter the fraction of total protein that is available for binding to ligand F.

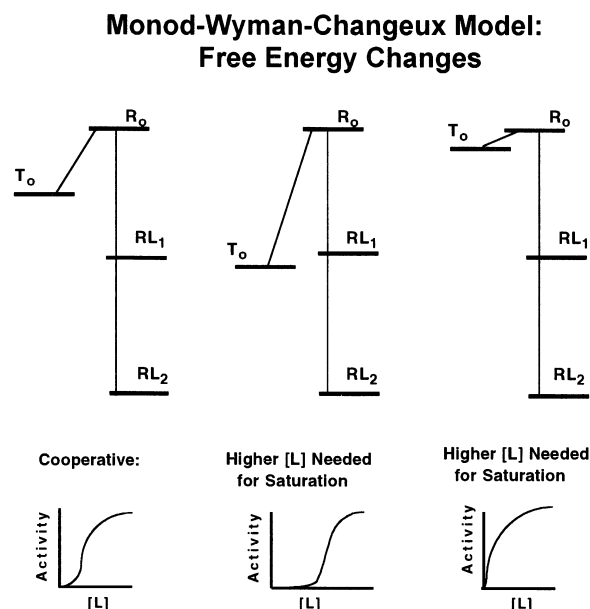


Figure 3. Free energy diagram for ligand binding interactions to an MWC dimer. Note that the relative stability of the T_0 to R_0 alters the ligand saturation curve as shown in each graph. The energy changes between each R-state are equivalent because each ligand binding interaction has the same equilibrium constant.

Shown in Fig. 4 are related energy diagrams showing (a) how inhibitor stabilizes the T-state and further reduces the ease of ligand F binding, and (b) how activator stabilizes the R state, increasing the fraction of total sites in the R state and thereby enhancing the binding of ligand F.

The energy diagrams presented in Fig. 5 allow one to compare the MWC model with the Koshland-Némethy-Filmer²⁻⁴ model for ligand-induced conformation changes in ligand affinity. While describing key aspects of positive cooperativity, the Monod-Wyman-Changeux model cannot account for negative cooperativity. In this respect, any experimental evidence demonstrating nega-

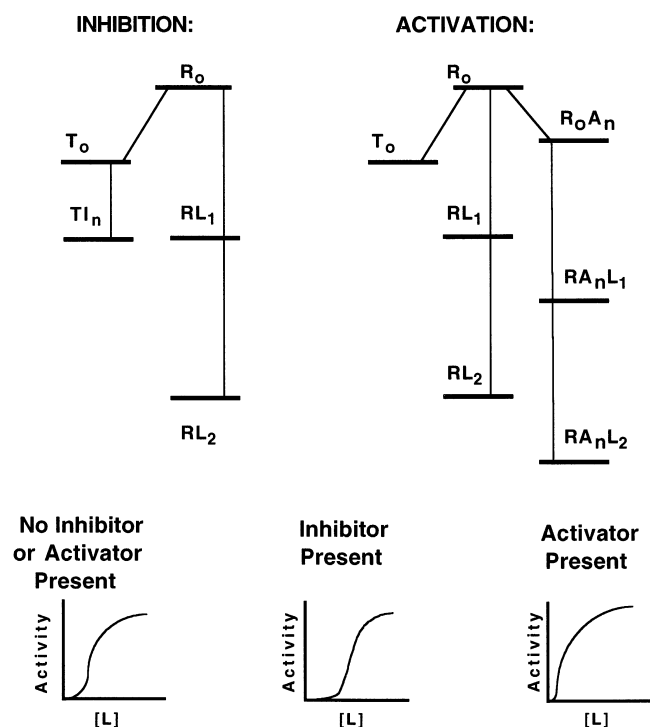


Figure 4. Free energy diagrams illustrating that binding of an inhibitor stabilizes the T-state and that binding of an activator stabilizes the R-state.

tive cooperativity immediately excludes the MWC model as a candidate mechanism.

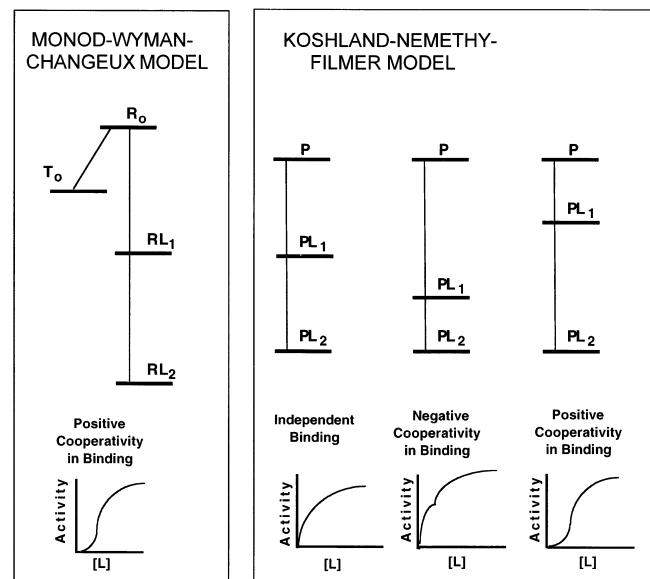


Figure 5. Free energy diagrams showing the salient differences between the Monod and Koshland models. The MWC model is a two-state model with equivalent ligand binding interactions (indicated here by the equal spacing between R₀ and RL₁ states and between RL₁ and RL₂ states). In the KNF model, the amount of energy released determines whether binding will be independent or show negative or positive cooperativity.

¹J. Monod, J. Wyman & J.-P. Changeux (1965) *J. Mol. Biol.* **12**, 88.

²K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

³K. E. Neet (1995) *Meth. Enzymol.* **249**, 519.

⁴D. E. Koshland, G. Némethy & D. Filmer (1966) *Biochemistry* **5**, 365.

MONOGLYCERIDE LIPASE

This enzyme [EC 3.1.1.23], also known as monoacylglycerol lipase, catalyzes the hydrolysis of glycerol monoesters of long-chain fatty acids. **See also** *Lipases*

J. C. Khoo & D. Steinberg (1983) *The Enzymes*, 3rd ed., **16**, 183.

P. Desnuelle (1972) *The Enzymes*, 3rd ed., **7**, 575.

MONO-ISO MECHANISMS

Enzyme reaction mechanisms that contain one ligand-independent enzyme isomerization step. **See** *Iso Mechanisms*.

MONO-ISO PING PONG BI BI REACTION SCHEME

A reaction scheme for a ping pong reaction in which there is a ligand-independent enzyme isomerization step following the second half-reaction. Thus, the first half-reaction of the scheme is represented by $E + A \rightleftharpoons EA \rightleftharpoons FP \rightleftharpoons F + P$, the second half-reaction by $F + B \rightleftharpoons FB \rightleftharpoons GQ \rightleftharpoons G + Q$, and the isomerization step by $G \rightleftharpoons E$. In the absence of product, the steady-state rate expression predicts parallel lines in the double-reciprocal plots, identical to the lines one would observe in a normal ping pong Bi Bi reaction scheme. However, the complete rate equation predicts different product inhibition patterns^{1,2}. This rate expression is identical to that of the straight-forward ping pong bi bi mechanism except that it includes additional terms in the denominator containing $[A][Q]$, $[A][B][Q]$, and $[A][P][Q]$.

¹F. B. Rudolph (1979) *Meth. Enzymol.* **63**, 411.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

MONOMER—POLYMER EXCHANGE KINETICS

Oosawa¹ originally described the phases of protein polymerization reactions as nucleation, growth, and attainment of monomer-polymer equilibrium. In the final phase, redistribution of polymer lengths causes the polymer length distribution to resemble a decaying exponential shape. Following the end of the rapid growth phase, no significant change occurs in the concentration of monomers. We use the term “steady state” to describe this period of constant monomer concentration. Wegner²

showed that head-to-tail polymerization may occur during steady state, but only if monomer addition to polymers is accompanied by nucleotide hydrolysis. He postulated this mechanism to account for the unusually rapid exchange of radioactive monomers into F-actin at steady state. Margolis & Wilson³ observed the same phenomenon for microtubules which use GTP. These investigators also recognized that radioactive label would be incorporated into polymers through the normal process of equilibrium exchange at the polymer ends. Polymers undergoing head-to-tail polymerization should incorporate label more quickly than by equilibrium exchange. The only theory describing equilibrium exchange rates was that of Kasai & Oosawa⁴ and was based on an approximation. Wegner used this approximation to compare the label uptake rates resulting from the two mechanisms.

Kristofferson & Purich⁵ discovered a simple solution that avoids Oosawa's approximation. Derivation of the equilibrium exchange rate of labeled monomer incorporation into biopolymers involves the following assumptions that are easily realized in practice. (1) The polymers are very long so there is no significant change in the number of polymers. (2) Exchange at one end will not interfere with the exchange dynamics at the opposite end of the polymer. (3) With equilibrium exchange, the probabilities of gaining or losing a monomer at a polymer end are the same. (4) All monomers free in solution are radiolabeled and dilution of label is negligible. One can also define the following quantities: N is the number of polymer ends; n is the number of "steps," where a "step" is either the addition or loss of a monomer at a polymer end. The average duration of a step is the reciprocal of the depolymerization rate constant at the polymer end. Karr, Kristofferson & Purich⁶ determined a value of 113 s^{-1} for the net microtubule rate constant, or an average of $56.5 \text{ s}^{-1}/\text{end}$. Thus, the average step duration is about 18 ms. One should note that the theory does not consider dilution of the specific radioactivity of label which would result in lower label uptake values after long periods of time.

¹F. Oosawa (1970) *J. Theor. Biol.* **27**, 69.

²A. Wegner (1976) *J. Mol. Biol.* **108**, 139.

³R. L. Margolis & L. Wilson (1978) *Cell* **13**, 1.

⁴M. Kasai & F. Oosawa (1969) *Biochim. Biophys. Acta.* **172**, 300.

⁵D. Kristofferson & D. L. Purich (1981) *J. Theor. Biol.* **92**, 85.

⁶T. L. Karr, D. Kristofferson & D. L. Purich (1980) *J. Biol. Chem.* **255**, 8560.

MONOMOLECULARITY

A synonym for "unimolecularity", referring to a reaction in which a single molecular entity is involved in the chemical event of an elementary reaction.

MONOPHENOL MONOOXYGENASE

This copper-dependent enzyme [EC 1.14.18.1] (also known as tyrosinase, phenolase, monophenol oxidase, and cresolase) catalyzes the reaction of L-tyrosine with L-dopa and dioxygen to produce L-dopa, dopaquinone, and water. This classification actually represents a set of copper proteins that also catalyze the reaction of catechol oxidase [EC 1.10.3.1] if only 1,2-benzenediols are available as substrates.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

V. J. Hearing, Jr. (1987) *Meth. Enzymol.* **142**, 154.

K. Lerch (1987) *Meth. Enzymol.* **142**, 165.

MONTÉ CARLO METHODS

A computer-based, computationally intensive technique for analyzing chemical dynamics by using simulations based on the integration of simultaneous equations describing the time evolution of motions, usually classical in nature, of colliding atoms or molecules. The method requires a mathematical algorithm for randomly assigning initial conditions, including positional coordinates, energies, velocities, and related impact parameters for each atom or molecule. The overall distribution of energies is typically restricted to those permitted on the basis of the Maxwell-Boltzmann equilibrium distribution. The randomness of assignment was initially likened to the haphazard behavior of roulette wheels and other games of chance practiced in Monte Carlo, hence the cognomen. The method has also been used extensively to obtain approximate solutions to otherwise mathematically intractable problems by defining a range of values, each having a calculable probability of being the problem's solution.

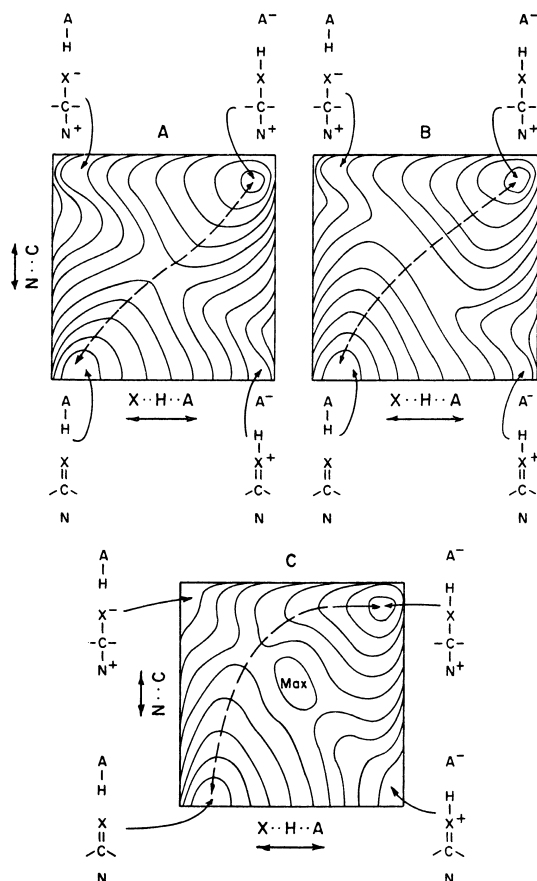
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Algorithms, **259**, 579-580; computational time, **240**, 430, 436-437; membrane modeling [attainment of equilibrium, **240**, 577-578, 591-592; biased sampling, **240**, 590; cluster statistics, **240**, 583-585; data collection, **240**, 579-580; decision making, **240**, 576-577, 591-592; defining Monte Carlo cycle, **240**, 575; end detection, **240**, 577; equilibrium fluctuation, **240**, 590-592; Gibbs energy change, **240**, 576; parameter fitting, **240**, 580; phase transition of small unilamellar dipalmitoylphosphatidylcholine vesicles, **240**, 569-570; random selection of lattice point, **240**, 575-

576; snapshot analysis, **240**, 578-579; trial conformation, **240**, 576, 590-591; umbrella sampling, **240**, 590, 592]; neighbor exclusion model analysis [nonlinear least-squares fitting, **240**, 602-603, 609; resolution of negative cooperativity, **240**, 606-612; simulation, **240**, 603-606]; nuclear Overhauser effect spectroscopy, **240**, 417, 419, 429-436; peptide conformation analysis, **259**, 580-584; photon migration [characteristics, **240**, 780; detection of re-emission, **240**, 765; frequency domain measurements, **240**, 769-770; interfering photon density measurements, **240**, 770-772; migration across boundaries, **240**, 763-764; photon loss by absorption, **240**, 763; photon packet migration, **240**, 762; theory, **240**, 760-762; time domain measurements, **240**, 765-769]; restrained refinement, **239**, 437-438; for searching configurational space, **239**, 640; simple neighbor exclusion, **240**, 603-606; structure refinement, 261, 94; validation of standard curve, **240**, 139, 141, 146, 148-149; in weighted nonlinear regression analysis [data synthesis, **232**, 586-588; nonlinear regression algorithm, **232**, 588-589; real data relationship to simulation, **232**, 593].

MORE O'FERRALL-JENCKS DIAGRAM

A depiction of a hypothetical potential energy surface for a reacting system as a function of two chosen coordinates (e.g., the lengths of two bonds being broken)¹⁻⁵. Such diagrams are useful in assessing structural effects on transition states for stepwise or concerted pathways. An example of More O'Ferrall-Jencks diagrams for β -elimination reactions is shown below.



Structural changes influencing vibrational modes of the transition states alter transition state geometry. Changes in the direction of the reaction coordinate (reactant or product stabilizing or destabilizing factors) cause changes in accordance with the Hammond principle. Structural changes perpendicular to the reaction coordinate (so-called “anti-Hammond” effects or perpendicular effects) cause changes that run counter to the Hammond behavior (i.e., the easier the process related to the structural change, the more advanced it will be at the transition state)¹.

¹IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

²R. A. More O'Ferrall (1970) *J. Chem. Soc. (B)*, 274.

³W. P. Jencks (1972) *Chem. Rev.* **72**, 705.

⁴W. P. Jencks (1980) *Acc. Chem. Res.* **13**, 161.

⁵D. A. Winey & E. R. Thornton (1975) *J. Amer. Chem. Soc.* **97**, 3102.

MÖSSBAUER SPECTROSCOPY

The Mössbauer effect refers to the absorption and re-radiation of a gamma ray by a nonrecoiling nucleus. Gamma rays of extremely narrowly defined wavelength exhibit this effect with certain nuclei (e.g., the ⁵⁷Fe nucleus), and Mössbauer spectroscopy can detect even very subtle changes in electron configuration, chemical bonding, and oxidation state during substrate binding and enzymic catalysis. A lucid description of the theory and application of Mössbauer spectroscopy to the characterization of an enzyme mechanism appears in vol. **4** of *Comprehensive Biological Catalysis: A Mechanistic Reference* (M. Sinnott, ed.).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Characteristic features of spectra, **76**, 334-335 [of Fe proteins, **69**, 775-777; of Mo-Fe proteins, **69**, 771-773]; cryogenic temperatures and external magnetic fields, **76**, 338-339; experimental running time, **76**, 342-344; experimental techniques, **76**, 340-347; of hemoglobins, **76**, 347-354 [by photodissociation, **76**, 349-351; by selective enrichment, **76**, 349; in strong magnetic fields, **76**, 351-354]; information sources, **76**, 345-347; Mössbauer effect as basis, **76**, 329-332; obtaining transmission spectra, **76**, 340-342; phase determination in X-ray structure analysis, **76**, 354; pulse processing equipment, **76**, 337-338; radiation shielding, **76**, 339-340; selective excitation double technique, **76**, 352-354; sources and detectors, **76**, 336-337; spectrometer components, **76**, 335-340; spectrum analysis, **76**, 344-345; superoperator approach, **76**, 345; transmission apparatus, **76**, 340; transmission geometry, **76**, 332-333.

mRNA (GUANINE-N⁷-)METHYLTRANSFERASE

This enzyme [EC 2.1.1.56] catalyzes the addition of the N⁷-methyl group to the cap of mRNA: thus, S-adenosyl-L-methionine reacts with G(5')pppR-RNA to produce

S-adenosyl-L-homocysteine and $m^7G(5')pppR$ -RNA (the R group may be guanosine or adenosine).

P. Narayan & F. M. Rottman (1992) *Adv. Enzymol.* **65**, 255.

MU (μ)

1. Symbol for micro-. 2. Symbol for micron. 3. Symbol for dynamic viscosity. 4. Symbol for magnetic or electric dipole moment of a molecule. 5. Symbol for chemical potential. 6. Symbol for permeability; μ_0 for permeability of vacuum; μ_r for relative permeability. 7. Symbol for reduced mass.

$\bar{\mu}$ Symbol for electrochemical potential.

m μ Symbol for millimicro- or millimicron.

MUCONATE CYCLOISOMERASE

This manganese-dependent enzyme [EC 5.5.1.1] catalyzes the interconversion of 2,5-dihydro-5-oxofuran-2-acetate and *cis,cis*-hexadienedioate. In the reverse reaction, 3-methyl-*cis,cis*-hexadienedioate can serve as a substrate and *cis,trans*-hexadienedioate can act as a weak substrate. The enzyme differs from chloromuconate cycloisomerase [EC 5.5.1.7] and dichloromuconate cycloisomerase [EC 5.5.1.11].

R. B. Meagher, K.-L. Ngai & L. N. Ornston (1990) *Meth. Enzymol.* **188**, 126.

MUCONOLACTONE Δ -ISOMERASE

This enzyme [EC 5.3.3.4] catalyzes the interconversion of 2,5-dihydro-5-oxofuran-2-acetate and 3,4-dihydro-5-oxofuran-2-acetate.

R. B. Meagher, K.-L. Ngai & L. N. Ornston (1990) *Meth. Enzymol.* **188**, 130.

MUCROLYSIN

This zinc-dependent endopeptidase [EC 3.4.24.54] from Chinese habu snake venom catalyzes the hydrolysis of the Ser⁹—His¹⁰, His¹⁰—Leu¹¹, Ala¹⁴—Leu¹⁵, Leu¹⁵—Tyr¹⁶, and the Tyr¹⁶—Leu¹⁷ peptide bonds in the insulin B chain.

J. B. Bjarnason & J. W. Fox (1995) *Meth. Enzymol.* **248**, 345.

MULTIENZYME

Any protein exhibiting more than one catalytic function as a result of distinct parts of a polypeptide chain ('domains') or distinct subunits, or both.

IUB Nomenclature Committee (1989) *Eur. J. Biochem.* **185**, 485.

MULTIPLE DEAD-END INHIBITION

Depressed catalytic activity occurring when an inhibitor binds more than once to a single enzyme form (or forms). While standard double-reciprocal plots are usually linear, secondary replots of the data (*i.e.*, plots of slopes and/or intercepts vs. [I], the concentration of the inhibitor) will be nonlinear depending on the relative magnitude of the [I]², [I]³, ..., and [I]^{*n*} terms in the rate expression.

Cleland¹ presented more complex examples of multiple dead-end inhibition in which the kinetic expression for the slope or intercept is a function containing polynomials of the inhibitor concentration in both the denominator and numerator. For example, if the slope is equal to a function having the form $(a + b[I] + c[I]^2)/(d + e[I])$ in which *a*, *b*, *c*, *d*, and *e* are constants or collections of constants, then the slope is said to be a 2/1 function (the numbers representing the highest power of [I] in the numerator and denominator, respectively). The nonlinearity of the slope replot, in this case, is dependent on the relative magnitudes of the constants in the expression.

¹W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 173.

MULTIPLE-TURNOVER CONDITIONS

Reaction conditions permitting a catalyst to pass through many catalytic rounds. Multiple-turnover conditions are usually obtained by maintaining the substrate concentration in excess over the concentration of active catalyst. This technique usually allows one the opportunity to evaluate the catalytic rate constant k_{cat} , which is the first-order decay rate constant for the rate-determining step for each cycle of catalysis, and one can evaluate the magnitude of other parameters such as the substrate's dissociation constant or Michaelis constant.

MULTIPLICATIVE MODEL

Any mechanism in which the observed effect of two or more modifiers equals the product of effects observed when those modifiers act alone. Two agents acting competitively (*i.e.*, by binding at the same site) cannot act multiplicatively.

MULTIPLICITY

For a given total electron spin quantum number (*S*), the multiplicity is the number of possible orientations of the spin angular momentum for the same spatial electronic wavefunction. Thus, the multiplicity equals $2S + 1$. For

a state of singlet multiplicity, $S = 0$ and $2S + 1 = 1$. If $S > L$, where L is the total electron orbital angular momentum quantum number, there are only $2L + 1$ possible orientations.

Comm. on Photochem. (1988), *Pure and Appl. Chem.* **60**, 1055.

MULTISITE PING PONG REACTION SCHEMES

Certain enzymes catalyze their reactions by way of a multisite mechanism in which the covalently linked intermediate is attached to a long arm that “swings” from one subsite to another subsite within the enzyme. In some cases, the covalently tethered intermediate can actually be transferred between subunits that form the active site. An example is *Propionibacterium shermanii* transcarboxylase¹, an enzyme that catalyzes the biotin-dependent conversion of methylmalonyl-CoA and pyruvate to propionyl-CoA and oxaloacetate. Carboxylated biotin allows the two catalytic subsites to operate on the same reaction intermediate.

While the initial rate and exchange studies indicated a standard ping pong mechanism for this enzyme, the product inhibition results were not consistent with the traditional single-site mechanism^{1,2}. The two-site ping-pong scheme accounts for the product inhibition results observed with the transcarboxylase. A more complex multisite ping pong mechanism has also been presented for pyruvate dehydrogenase³.

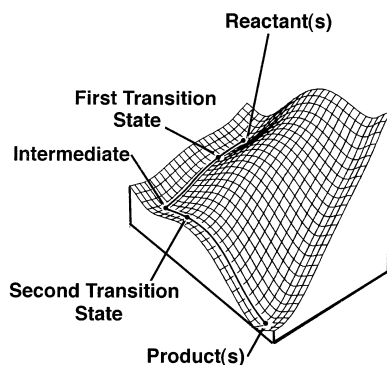
¹D. B. Northrop (1969) *J. Biol. Chem.* **244**, 5808.

²D. B. Northrop & H. G. Wood (1969) *J. Biol. Chem.* **244**, 5820.

³W. W. Cleland (1973) *J. Biol. Chem.* **248**, 8353.

MULTISTAGE REACTIONS

Chemical reactions with one or more intermediates separated by transition states (usually indicated by the superscript symbol ‡):



Potential Energy Surface for a Multistage Reaction

The overall rate of multistage reactions depends on the activation energy, the minimal energy needed by a reactant to reach the highest lying transition state. Any reactant(s) reaching this transition state can either reform reactant(s) or proceed to form products. In so doing, energy is released, usually by momentum exchanges with colliding molecules. If reactants can achieve equilibrium with products, then the Law of Microscopic Reversibility requires that the product-to-reactant path will be identical to the reactant-to-product path.

MULTISUBSTRATE ANALOGUE

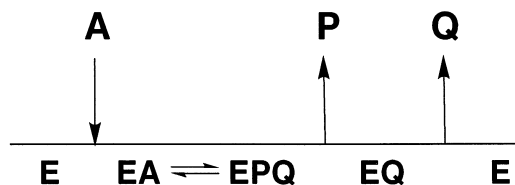
A molecule that possesses the capacity to span two or more substrate binding sites within the active site of an enzyme. These analogues are to be distinguished from transition-state analogues which typically mimic the geometry or bond order of the transition state. Early examples of multisubstrate analogues were P^1, P^4 -di(adenosine-5')tetraphosphate¹ and P^1, P^5 -di(adenosine-5')pentaphosphate² which were found to be potent inhibitors of adenylate kinase. **See Adenylate Kinase Inhibitors; Geometrical Analogue**

¹D. L. Purich & H. J. Fromm (1972) *Biochim. Biophys. Acta* **276**, 563.

²G. E. Lienhard & I. I. Secemski (1973) *J. Biol. Chem.* **248**, 1121.

MULTISUBSTRATE MECHANISMS

The following diagrams indicate the binding interactions for enzyme kinetic mechanisms. To conserve space, the notation used in this Handbook is a compact version of the diagrams first introduced by Cleland¹. His diagram for the Ordered Uni Bi Mechanism is as follows:



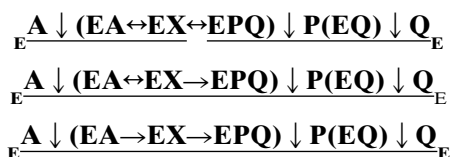
We use the following single-line compact form:



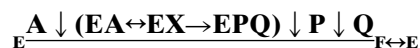
The enzyme surface is represented by the underline, in this case preceded by a subscript E to indicate the unbound (or free) enzyme before any substrate addition and after all products desorb. An arrow pointing to the line indicates binding, and the reader should understand that reversible binding is taken for granted. Moreover,

unlike the original Cleland diagram, product release always is written by a downward arrow, because this systematic usage emphasizes the symmetry of certain mechanisms. The symbol " $\underline{\mathbf{A}} \downarrow$ " indicates that substrate A adds; the symbol " $\underline{\mathbf{A}} \text{ or } \underline{\mathbf{B}} \downarrow \downarrow$ ", " $\underline{\mathbf{A}} \text{ or } \underline{\mathbf{B}} \text{ or } \underline{\mathbf{C}} \downarrow \downarrow \downarrow$ ", etc., indicates *random* addition of two or three substrates, respectively. Interconversions of enzyme-bound reactants can be reversible ($\leftrightarrow$) or irreversible ($\rightarrow$).

Taking the Ordered Uni Bi mechanism as an example, we can consider several additional possibilities:



For the case of so-called iso mechanisms, the compact diagrams are as follows:



where $\mathbf{F} \rightleftharpoons \mathbf{E}$ represents the isomerization step.

The following mechanisms are typically encountered in studies of enzymic catalysis:

Ordered Uni Bi Mechanism



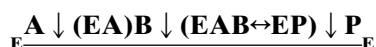
Random Uni Bi Mechanism



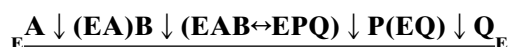
Random Bi Uni Mechanism



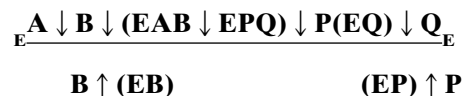
Ordered Bi Uni Mechanism



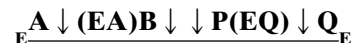
Ordered Bi Bi Mechanism



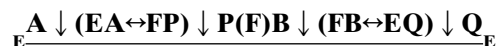
Ordered Bi Bi - Subsite Mechanism



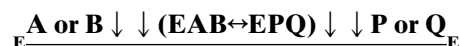
Ordered Bi Bi-Theorell Chance Mechanism



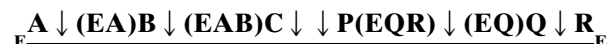
Ping Pong Bi Bi Mechanism



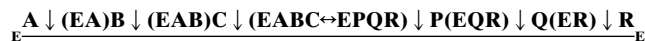
Random Bi Bi Mechanism



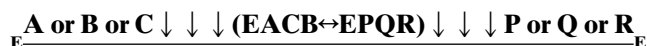
Ordered Ter Ter-Theorell Chance Mechanism



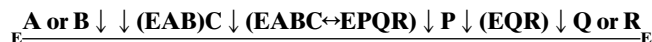
Ordered Ter Ter Mechanism



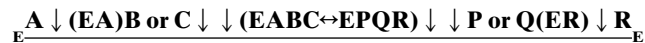
Random Ter Ter Mechanism



Random AB, Random QR Ter Ter Mechanism



Random BC, Random PQ Ter Ter Mechanism



MUSCLE MASS DETERMINATION BY ISOTOPIC TRACERS

The isotope tracer method¹ for estimating the total muscle mass of an organism by determining the amount of creatinine clearance. The method relies on the fact that muscle creatine phosphate is converted to creatine, and the latter is metabolized into creatinine.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, pp. 193, Wiley-Liss, New York.

MUTAROTASES

Enzymes that catalyze the interconversion of the anomeric carbon of saccharides¹⁻³. The name is derived from

the ability of these enzymes to change or “mutate” the optical rotatory properties of sugars.

These enzymes catalyze mutarotation via acid/base catalysis, and the open-chain form of the sugar is an enzyme-bound intermediate in the reaction pathway. In principle, any protein that binds the ring-open form of a sugar will be capable of catalyzing mutarotation, and such an activity may have little to do with the physiologic function of that protein. These enzymes catalyze epimerization reactions at the hemiacetal or hemiketal carbon. Aldose 1-epimerase [EC 5.1.3.3], also known as mutarotase and aldose mutarotase, catalyzes the interconversion of α -D-glucose and β -D-glucose. The enzyme is reported to also utilize L-arabinose, D-xylose, D-galactose, maltose, and lactose as substrates. **See also** *Anomers; Fischer Projection; Aldose 1-Epimerase*

¹J. M. Bailey, P. G. Pentchev, P. H. Fishman, S. A. Mulhern & J. W. Kusiak (1973) *Adv. Chem. Ser.* **117**, 264.

²E. Adams (1976) *Adv. Enzymol.* **44**, 69.

³M. E. Tanner (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 208.

MUTAROTATION KINETICS

Anomeric interconversion of sugars and their derivatives is typically relatively slow, but anomerization of phosphorylated sugars is usually faster. Values for the rate constants can be obtained by nuclear magnetic resonance techniques. For example, the conversion of β -fructose 1,6-bisphosphate to the straight-chain form, at 25°C and pH 7.2, has a rate constant of 35 s⁻¹, and the rate constant for the reverse reaction is 1450 s⁻¹. Conversion of the α -anomer to the straight-chain form has a rate constant of 8.5 s⁻¹ and a 70 s⁻¹ value for the reverse reaction².

¹S. J. Benkovic (1979) *Meth. Enzymol.* **63**, 370.

²C. F. Midelfort, R. K. Gupta & I. A. Rose (1976) *Biochemistry* **15**, 2178.

MYELOBLASTIN

This enzyme [EC 3.4.21.76] catalyzes the hydrolysis of peptide bonds in proteins, including elastin.

J. R. Hoidal, N. V. Rao & B. Gray (1994) *Meth. Enzymol.* **244**, 61.

MYELOPEROXIDASE

This enzyme [EC 1.11.1.7] catalyzes the reaction of a donor substrate with hydrogen peroxide to produce an oxidized donor and two water.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

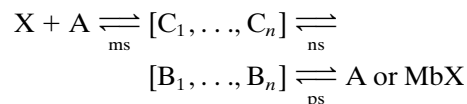
E. Cadenas (1989) *Ann. Rev. Biochem.* **58**, 79.

J. K. Hurst & W. C. Barrette, Jr. (1989) *Crit. Rev. Biochem. Mol. Biol.* **24**, 271.

A. Naqui, B. Chance & E. Cadenas (1986) *Ann. Rev. Biochem.* **55**, 137.

MYOGLOBIN OXYGENATION

Although myoglobin was the first protein for which the atomic structure was determined, the detailed molecular mechanism of reversible oxygenation remains an area of active research¹. The binding of oxygen is a multistep process, involving displacement of a noncoordinated water molecule at a distal pocket, iron atom movement into the plane of the porphyrin ring to create a hexacoordinate complex, and formation of a hydrogen bond between His-64's N^ε and the second bound oxygen atom². This can be described schematically as follows:



where X is the free ligand, [C₁, . . . , C_n] is the water molecule displacement step, [B₁, . . . , B_n] is the formation of a contact pair, and A or MbX is the ground state complex. The overall association rate constants are 0.5 $\mu\text{M}^{-1}\text{s}^{-1}$ for CO, 17 $\mu\text{M}^{-1}\text{s}^{-1}$ for O₂, and 22 $\mu\text{M}^{-1}\text{s}^{-1}$ for NO[•]. Because photodissociation of the heme-bound ligand can be achieved by using an intense excitation pulse, the kinetics of these events can be recorded on the time scales shown beneath each species in the above scheme.

¹Q. H. Gibson (1989) *J. Biol. Chem.* **264**, 20155.

²J. S. Olson & G. N. Phillips, Jr. (1996) *J. Biol. Chem.* **271**, 17593.

The following are other recent reviews on the molecular and physical properties of this oxygen binding protein.

H. B. Gray & J. R. Winkler (1996) *Ann. Rev. Biochem.* **65**, 537.

J. N. Onuchic, D. N. Beratan, J. R. Winkler & H. B. Gray (1992) *Ann. Rev. Biophys. Biomol. Struct.* **21**, 349.

J. B. Wittenberg & B. A. Wittenberg (1990) *Ann. Rev. Biophys. Biophys. Chem.* **19**, 217.

R. E. Dickerson & I. Geis (1983) *Hemoglobin*, Benjamin/Cummings, Menlo Park, CA.

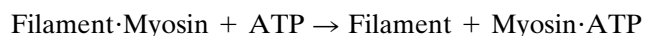
MYOSIN ATPase

This ATPase activity [EC 3.6.1.32] is directly responsible for muscle contraction. In the absence of actin filaments, myosin is a feeble ATPase with a k_{cat} of only 0.05 s⁻¹, because product release is much slower than the rapid release of a proton in the P—O—P bond-cleavage step forming ADP and P_i from bound ATP. Interaction with

actin filaments, however, strongly stimulates the ATPase activity, increasing k_{cat} by a factor of about 200. This acceleration occurs as a consequence of the expedited release of P_i and then ADP.

In the so-called Lymn-Taylor model, the actomyosin ATPase reaction proceeds in a stepwise manner:

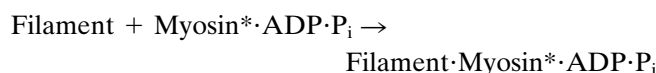
(1) Myosin binds ATP and releases the actin filament.



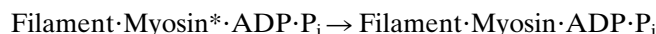
(2) Myosin·ATP complex is hydrolyzed rapidly to produce an conformationally energized state (often designated by an asterisk*).



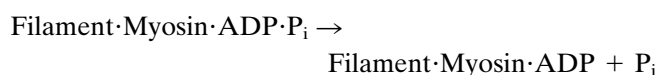
(3) Myosin*·ADP· P_i then binds to an actin filament, presumably at another position along the filament.



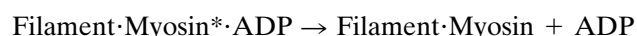
(4) The filament-bound complex relaxes through the release of stored conformational energy and returns to the resting state, myosin·ADP· P_i .



(5) This conformational relaxation is followed by release of P_i .



(6) Finally, ADP is released to regenerate filament-bound myosin.



When appropriately arranged in the context of the thick and thin filaments of the myofibril, repeated cycles of reaction steps 1-6 result in contraction. Other forms of myosin are responsible for contractile steps that are associated with the crawling of nonmuscle cells. **See also** *Force Effects on Molecular Motors*

S. Khan & M. Sheetz (1997) *Ann. Rev. Biochem.* **66**, 785.

I. Rayment (1996) *J. Biol. Chem.* **271**, 15850.

K. M. Ruppel & J. A. Spudich (1996) *Ann. Rev. Cell Dev. Biol.* **12**, 543.

M. S. Mooseker & R. E. Cheney (1995) *Ann. Rev. Cell Dev. Biol.* **11**, 633.

I. Rayment & H. M. Holden (1994) *Trends Biochem Sci.* **19**, 129.

K. A. Johnson (1992) *The Enzymes*, 3rd ed., **20**, 1.

J. R. Sellers & R. S. Adelstein (1987) *The Enzymes*, 3rd ed., **18**, 381.

Y. Tonomura & F. Oosawa (1972) *Ann. Rev. Biophys. Bioeng.* **1**, 159.

N

n→ π^* TRANSITION

An electronic transition in which an electron in a non-bonding (e.g., lone pair) orbital (called an n-orbital) is promoted to a π -antibonding orbital. The excited state arising from such a promotion is often referred to as an n- π^* state. An electron in an n-orbital typically interacts strongly with a polar solvent; this is less likely to be the case for an electron in a π^* orbital. Therefore, the energy difference between n and π^* orbital electrons will increase when a substance is placed in a more polar solvent; this is manifested as a shift to shorter wavelength (often called a blue shift) for light absorption. **See** *Absorption Spectroscopy*

n→ σ^* DELOCALIZATION

Delocalization of a free electron pair into an antibonding σ -orbital. **See** *Hyperconjugation; Resonance; Absorption Spectroscopy*

n→ σ^* TRANSITION

An electronic transition in which an electron in a non-bonding (e.g., lone-pair) orbital is promoted to an antibonding σ orbital. **See** *Absorption Spectroscopy*

NAD⁺ GLYCOHYDROLASE

This enzyme [EC 3.2.2.5] (also known as NAD⁺ nucleosidase, NADase, DPNase, DPN hydrolase, and ADP-ribosylcyclase) catalyzes the hydrolysis of NAD⁺ to nicotinamide and ADP-ribose. The enzyme from some mammalian sources also catalyzes the transfer of ADP-ribose residues.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.

NADH-CYTOCHROME *b*₅ REDUCTASE

This FAD-dependent enzyme [EC 1.6.2.2], also known as cytochrome *b*₅ reductase, catalyzes the reaction of NADH with two molecules of ferricytochrome *b*₅ to produce NAD⁺ and two ferrocytochrome *b*₅.

P. W. Holloway (1983) *The Enzymes*, 3rd ed., **16**, 63.

C. H. Williams, Jr. (1976) *The Enzymes*, 3rd ed., **13**, 89.

NADH DEHYDROGENASE

NADH dehydrogenase (ubiquinone) [EC 1.6.5.3] (also called ubiquinone reductase, type I dehydrogenase, and complex I dehydrogenase) catalyzes the reaction of NADH with ubiquinone to produce NAD⁺ and ubiquinol. The complex, which uses FAD and iron-sulfur proteins as cofactors, is found in mitochondrial membranes and can be degraded to form NADH dehydrogenase [EC 1.6.99.3]. NADH dehydrogenase [EC 1.6.99.3] catalyzes the reaction of NADH with an acceptor to produce NAD⁺ and the reduced acceptor. Iron-sulfur and flavoproteins are still being used as cofactors with this component of EC 1.6.5.3. Interestingly, after certain preparations have been followed, cytochrome *c* may serve as the acceptor substrate.

U. Schulte & H. Weiss (1995) *Meth. Enzymol.* **260**, 3.

J. E. Walker, J. M. Skehel & S. K. Buchanan (1995) *Meth. Enzymol.* **260**, 14.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

NADH DEHYDROGENASE (Other Than Complex I)

Other NADH dehydrogenases include NADH dehydrogenase (quinone) [EC 1.6.99.5] which catalyzes the reaction of NADH with an acceptor to produce NAD⁺ and the reduced acceptor. Menaquinone can serve as the acceptor substrate. This dehydrogenase is inhibited by AMP and 2,4-dinitrophenol but not by dicoumarol or folic acid derivatives.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

NADH/NADPH SURFACE FLUORESCENCE IN LIVING TISSUES

Surface fluorescence of NADH/NADPH can be recorded continuously with a DC fluorimeter and correlated with changes in experimental conditions. A mercury arc lamp (with a 340-375 nm filter in front) is used as a light source for fluorescence excitation. The fluorescence response of reduced NADH/NADPH was measured at 450-510 nm. The DC fluorimeter and the Hg arc lamp are connected to the kidney by a trifurcated fiber optics light guide. NADH/NADPH fluorescence emission can be corrected for changes in tissue opacity by a 1:1 subtraction of reflectance changes at 340-375 nm from the fluorescence. To determine NADH/NADPH redox state of the total surface area of kidney cortex and to evaluate whether certain areas were insufficiently perfused, fluorescence photographs of the total surface area were taken. The study demonstrated that the surface fluorescence method is simple and provides specific information about the mitochondrial oxidation-reduction state.

H. Franke, C. H. Barlow & B. Chance (1980) *Contrib. Nephrol.* **19**, 240.

NADH PEROXIDASE

This FAD-dependent enzyme [EC 1.11.1.1] catalyzes the reaction of NADH with hydrogen peroxide to produce NAD^+ and two water. Ferricyanide, quinones, and other compounds can replace hydrogen peroxide in this enzyme-catalyzed reaction.

B. A. Palfrey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.

NAD^+ KINASE

This enzyme [EC 2.7.1.23] catalyzes the reaction of ATP with NAD^+ to produce ADP and NADP^+ . This reaction has the potential for altering the intracellular $[\text{NADP}^+]/[\text{NADPH}]$ redox potential by converting NAD^+ to NADP^+ , but details on the regulation of this enzyme have not been worked out.

E. P. Anderson (1973) *The Enzymes*, 3rd ed., **9**, 49.
S. Pinder, J. B. Clark & A. L. Greenbaum (1971) *Meth. Enzymol.* **18B**, 20.

NADPH DEHYDROGENASE

This enzyme [EC 1.6.99.1] (also known as old yellow enzyme and NADPH diaphorase) catalyzes the reaction of NADPH with an acceptor to produce NADP^+ and

the reduced acceptor. The yeast enzyme uses FMN whereas the plant enzyme uses FAD.

B. A. Palfrey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
L. M. Schopfer & V. Massey (1991) *A Study of Enzymes* **2**, pp. 247.
H. J. Bright & D. J. T. Porter (1975) *The Enzymes*, 3rd ed., **12**, 421.

NAD(P)H DEHYDROGENASE (Quinone)

This enzyme [EC 1.6.99.2] (also known as NAD(P)H:quinone reductase, DT diaphorase, quinone reductase, azoreductase, phyloquinone reductase, and menadione reductase) catalyzes the reaction of NAD(P)H with an acceptor to produce NAD(P)^+ and the reduced acceptor. This FAD-dependent enzyme is inhibited by dicoumarol.

B. A. Palfrey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
E. Cadenas, P. Hochstein & L. Ernster (1992) *Adv. Enzymol.* **65**, 97.
C. Lind, E. Cadenas, P. Hochstein & L. Ernster (1990) *Meth. Enzymol.* **186**, 287.

NAD(P)^+ TRANSHYDROGENASE

This enzyme [EC 1.6.1.1] (also known as NAD(P)^+ transhydrogenase (B-specific), pyridine nucleotide transhydrogenase, and nicotinamide nucleotide transhydrogenase) catalyzes the reversible reaction of NADPH with NAD^+ to produce NADP^+ and NADH. This FAD-dependent enzyme is B-specific with respect to both pyridine coenzymes. In addition, deamino coenzymes will also serve as substrates.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
J. Rydström, J. B. Hoek & L. Ernster (1976) *The Enzymes*, 3rd ed., **13**, 51.

NAD^+ SYNTHETASE

NAD^+ synthetase (ammonia-utilizing) [EC 6.3.1.5] catalyzes the reaction of ATP with deamido- NAD^+ and ammonia to produce NAD^+ , AMP, and pyrophosphate (or, diphosphate). NAD^+ synthetase (glutamine-utilizing) [EC 6.3.5.1] catalyzes the reaction of ATP with deamido- NAD^+ , glutamine, and water to produce NAD^+ , glutamate, AMP, and pyrophosphate.

V. Micheli & S. Sestini (1997) *Meth. Enzymol.* **280**, 211.
H. Zalkin (1993) *Adv. Enzymol.* **66**, 203.
H. Zalkin (1985) *Meth. Enzymol.* **113**, 297.
J. M. Buchanan (1973) *Adv. Enzymol.* **39**, 91.

NANO-

A prefix, symbolized by n, used in submultiples of units and corresponding to a value of 10^{-9} .

NANOSECOND ABSORPTION SPECTROSCOPY

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Overview, **226**, 119, 147; absorption apparatus, **226**, 131; apparatus, **226**, 152; detectors, **226**, 126; detector systems, **226**, 125; excitation source, **226**, 121; global analysis, **226**, 146, 155; kinetic applications, **226**, 134; heme proteins, **226**, 142; multiphoton effects, **226**, 141; nanosecond time-resolved recombination, **226**, 141; quantum yields, **226**, 139; singular value decomposition, **226**, 146, 155; spectral dynamics, **226**, 136; time delay generators, **226**, 130.

NARCISSISTIC REACTION

A chemical reaction in which the reactant is converted into its mirror image, hence the name alluding to the self-admiring nature of the mythic Narcissus. In a narcissistic process, a chiral reactant can be converted into its enantiomer. Racemases catalyze the most obvious narcissistic biochemical reactions. Another example of a narcissistic reaction is degenerate rearrangement.

L. Salem (1971) *Acc. Chem. Res.* **4**, 322.

NATURAL ABUNDANCE

A quantitative measure of isotope composition relative to the abundance of all isotopic forms found in nature. Values for those stable isotopes most commonly employed in biological tracer experiments are: ^1H , 99.985%; ^2H , 0.015%; ^{12}C , 98.90%; ^{13}C , 1.10%; ^{14}N , 99.63%; ^{15}N , 0.37%; ^{16}O , 99.76%; ^{17}O , 0.04%; and ^{18}O , 0.20%. **See** *Isotope Exchange Kinetics; Compartmental Analysis*

NEGATIVE COOPERATIVITY

Any set of ligand interactions with oligomeric or polymeric macromolecules where binding of the the first (or preceding) ligand molecule decreases the likelihood for binding of the next (or subsequent) ligand molecule. In the Koshland-Némethy-Filmer model, negative cooperativity occurs when the dissociation constant for ligand binding to the $(i + 1)$ -site is greater than the dissociation constant for ligand binding to the i th site.

Note that negative cooperativity cannot occur in the Monod-Wyman-Changeux allosteric transition model, because the dissociation constant is equivalent for all sites. Thus, positive cooperativity can only result in this binding mechanism as a consequence of the “recruitment” of binding sites from the T-state in an all-or-none transition to the R-state. Any occurrence of negative cooperativity can be regarded as *prima facie* evidence

against the applicability of the Monod-Wyman-Changeux model to the system under investigation. **See** *Ligand Binding Cooperativity; Koshland-Némethy-Filmer Model; Scatchard Equation; Ligand-Induced Conformational Change*

NEIGHBORING GROUP MECHANISM

The condition observed with aliphatic nucleophilic substitution reactions displaying either a greater than expected rate of reaction or an unexpected retention of configuration at a chiral carbon atom. March¹ indicates that such effects are usually attributable to the presence of a group with an unshared electron pair located one or more carbon atoms away from the reaction center. A neighboring group mechanism involves two $\text{S}_{\text{N}}2$ reactions, with apparent overall retention of configuration occurring by two successive inversions. First, the neighboring group behaves as the nucleophile and facilitates expulsion of the leaving group; then the external nucleophile attacks from an angle of 180° with respect to the neighboring group, thereby expelling the neighboring group and resulting in a chirality that matches that of the original reactant. Because the neighboring group is contained as part of the reactant, its local concentration is high, thereby accounting for the higher observed reactivity. **See also** *Anchimeric Assistance*

¹J. March (1992) *Advanced Organic Chemistry*, 4th ed., Wiley, New York.

NEIGHBORING-GROUP PARTICIPATION

The direct interaction of the reaction center of a molecular entity with a lone pair of electrons of an atom within that same molecular entity that is not associated with the reaction center; or, interaction of the reaction center with the σ - or π -electrons of a bond that is neither a part of the reaction center nor conjugated with the reaction center. Rate acceleration by such a process is referred to as anchimeric assistance. **See** *Intramolecular Catalysis; Synergistic Acceleration*.

N-END RULE

A model¹ that accounts for the selective degradation of proteins based on the amino acid that is present on the amino- or N-end of nascent proteins. Intracellular processing of nascent, noncompartmentalized proteins generates the mature protein via the action of amino-terminal peptidases. In model studies using β -galactosidase

with different N-terminal aminoacyl residues, the time scale of degradation can be as short as 3 min or as long as a day. Commitment to degrade is encoded by classes of amino acids whose presence at the N-terminus places them into three general classes of turnover kinetics (*e.g.*, fast, intermediate, and slow), depending on whether they are strongly destabilizing, mildly destabilizing, or stabilizing with respect to turnover. The most stable contain N-terminal methionyl, seryl, alanyl, glycyl, threonyl, and valyl residues. Those of intermediate stability contain isoleucyl, glutamyl, glutamyl, and tyrosyl residues, whereas the fast degrading proteins are strongly destabilized by the presence of leucyl, phenylalanyl, aspartyl, lysyl, or arginyl residues at the amino terminus. **See Protein Turnover Kinetics**

¹A. Bachmair, D. Finley & A. Varshavsky (1986) *Science* **234**, 179.

NEPRILYSIN

This zinc-dependent endopeptidase [EC 3.4.24.11] catalyzes the hydrolysis of peptide bonds and exhibits preferential cleavage at the amino group of hydrophobic residues in proteins and polypeptides. Neprilysin is a membrane-bound glycoprotein that is inhibited by phosphoramidon and thiorphan.

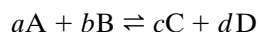
C. Li & L. B. Hersh (1995) *Meth. Enzymol.* **248**, 253.

B. P. Roques, F. Noble, P. Crine & M.-C. Fournié-Zaluski (1995) *Meth. Enzymol.* **248**, 263.

NERNST EQUATION

The thermodynamic equation that expresses the electromotive force E in terms of the standard potential E° and the concentrations of reactants and products in a redox reaction.

Basic Principles of the Nernst Equation. Consider the following generalized oxidation/reduction reaction



where the lower-case letters indicate the stoichiometry. In this case, the Gibbs free energy is given in the usual way as:

$$\Delta G = \Delta G^\circ + RT \ln\{([C]^c[D]^d)/([A]^a[B]^b)\}$$

in terms of activities where R is the universal gas constant and T is the absolute temperature or in terms of reactant and product concentrations:

$$\Delta G = \Delta G^\circ + RT \ln\{[C]^c[D]^d/([A]^a[B]^b)\}$$

The expressions $\{\Delta G = -n\mathcal{F}E$ and $\Delta G^\circ = -n\mathcal{F}E^\circ$ where $\mathcal{F}$ is the Faraday constant} allow us to rewrite this expression in the following form:

$$-n\mathcal{F}E = -n\mathcal{F}E^\circ + RT \ln\{[C]^c[D]^d/([A]^a[B]^b)\}$$

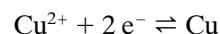
Dividing each side of the expression by $(-n\mathcal{F})$, we obtain the Nernst equation:

$$E = E^\circ - (RT/n\mathcal{F}) \ln\{[C]^c[D]^d/([A]^a[B]^b)\}$$

At a temperature of 298.15 K, with $R = 8.314510 \text{ J/mol}^\circ\text{K}^{-1}$, the faraday $\mathcal{F}$ equals 96485.309 coulombs/mol, and converting to decadic logarithms by multiplying by the reciprocal of $\log_{10}e$ (or, 2.302585), the Nernst equation has the following abbreviated form (*useful only at 25°C*):

$$E = E^\circ - (0.05916/n) \log\{[C]^c[D]^d/([A]^a[B]^b)\}$$

Using the Nernst Equation. One may wish to evaluate $E_{\text{Cu}^{2+},\text{Cu}}$ for the following half-reaction, written here as the reduction (*i.e.*, gain of electrons) by Cu^{2+} to form elemental copper:



Because elemental copper is insoluble, its activity or concentration is by convention set to a value of 1, and the number of equivalents of transferred electrons is two (from the equation immediately above). Therefore the Nernst equation is:

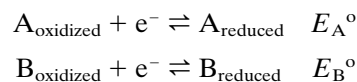
$$E_{\text{Cu}^{2+},\text{Cu}} = E_{\text{Cu}^{2+},\text{Cu}}^\circ - (0.05916/2) \log\{1/[\text{Cu}^{2+}]\}$$

or by inserting +0.3419 as the tabulated value for E° and changing the sign on the logarithm upon inverting the Cu^{2+} concentration, we get

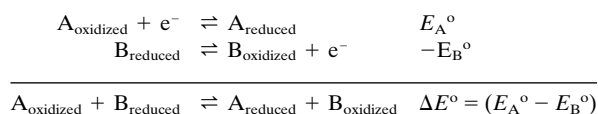
$$E_{\text{Cu}^{2+},\text{Cu}} = +0.3419 + 0.02958 \log [\text{Cu}^{2+}]$$

If one knows the concentration of cupric ions in solution, the value of $E_{\text{Cu}^{2+},\text{Cu}}$ can be evaluated quantitatively.

Calculation of Electromotive Force for an Overall Redox Reaction. In this case, there are two half-reactions:



And these must be written to obtain an electronically and stoichiometrically balanced equation:



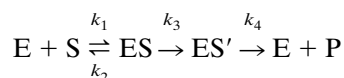
Rees & Farrelly¹ give an excellent account of how solution conditions affect the electromotive force of biological electron transfer reactions. Moreover, they describe how the protein can influence the value of E° : (a) by controlling the chemical nature of the ligands; (b) by imposing geometrical constraints on the redox center; (c) by altering the electrostatics by such noncovalent interactions as steric effects, solvent effects, placement of hydrophobic groups, and ionic effects (experienced directly as a result of ion binding to the redox protein or indirectly by altering the solution environment). **See also** *Electron Transfer Kinetics; Half-Reactions*

¹D. C. Rees & D. Farrelly (1990) *The Enzymes* **19**, 37.

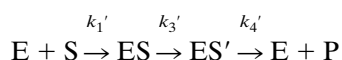
NET RATE CONSTANT METHOD FOR DERIVING ENZYME EQUATIONS

Cleland¹ introduced the net rate constant method to simplify the treatment of enzyme kinetic mechanisms that do *not* involve branched pathways. This method can be applied to obtain rate laws for isotope exchange, isotope partitioning, and positional isotope exchange. Since the net-rate constant method allows one to obtain V_{max}/K_m and K_m in terms of the individual rate constants, this method has greatest value for the characterization of isotope effects on V_{max}/K_m and K_m . Because only nonbranched kinetic schemes can be treated, most individuals will find Fromm's method^{2,3} or Huang's modification⁴ of Fromm's method to be much more generally useful, while still avoiding the inconvenience of the King-Altman method.

EXAMPLE 1. ONE-SUBSTRATE STEADY-STATE RATE EQUATION. We begin with the simplest two-intermediate example of a one-substrate mechanism:



One then converts the scheme to a series of unidirectional or net rate constants (indicated by primes). The steady-state flux through each step as well as the distribution of enzyme species remains the same, and we get



where each step has an associated net rate constant k_i' . Because the mechanism is a linear series, the rate for the overall reaction is the sum of the inverse of the net-rate constants through the individual steps. Thus, the rate equation must be:

$$(v/[E_{\text{total}}]) = 1/\{1/k_1' + 1/k_3' + 1/k_4'\}$$

where these net rate constants are expressed in terms of the actual rate constants as follows. One begins with any irreversible step in the original reaction scheme. In this case, there are two such steps, that converting ES into ES' and the other converting ES' to E + P. For an irreversible step, the net rate constant and the real rate constant will be identical:

$$k_3' = k_3 \quad \text{and} \quad k_2' = k_2$$

Moving leftward, we express k_1' as the real forward rate constant (in this case k_1) multiplied by a factor that relates the fraction of ES reacting in the forward direction as opposed to the fraction that returns to E. [Note: All of these constants are unimolecular rate constants, and all bimolecular rate constants must be multiplied by the concentration of substrate or product. This will become clearer in the later steps of this algorithm for deriving rate equations.] In this case, we get:

$$k_1' = k_1[S]\{k_3/(k_2 + k_3)\}$$

Now returning to the first equation and substituting the k_i' terms above

$$v = [E_{\text{total}}]/\{1/k_1' + 1/k_3' + 1/k_4'\}$$

$$v = k_1 k_3 k_4 [S][E_{\text{total}}]/\{k_4(k_2 + k_3) + k_1[A](k_3 + k_4)\}$$

$$v = \{k_3 k_4 / (k_3 + k_4)\} [E_{\text{total}}] / \{1 + \{k_4(k_2 + k_3) / k_1(k_3 + k_4)\} / [S]\}$$

Because this equation is now in the form

$$v = V_{\text{max}} / \{1 + K_m/[S]\}$$

then,

$$V_{\text{max}} = \{k_3 k_4 / (k_3 + k_4)\} [E_{\text{total}}]$$

$$K_m = k_4(k_2 + k_3) / \{k_1(k_3 + k_4)\}$$

When the goal is to obtain V_{max} or V_{max}/K_m , the following procedure avoids the need to derive the entire rate equation:

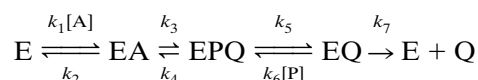
(a) V_m is obtained at saturating S concentrations, such that k_1' will be infinite and the $(1/k_1')$ term drops out, such that for the above mechanism

$$V_{\max} = [E_{\text{total}}]/\{1/k_3' + 1/k_4'\} = \{k_3k_4/(k_3 + k_4)\}[E_{\text{total}}]$$

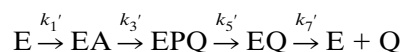
(b) $V_{\max}/K_m$ is obtained as [S] is extrapolated to near zero. In this case, the k_1' term becomes the smallest net rate constant, and

$$V_{\max}/K_m = k_1'[E_{\text{total}}] = k_1k_3[S][E_{\text{total}}]/(k_2 + k_3)$$

EXAMPLE 2. ORDERED UNI BI MECHANISM. We now repeat the process for a slightly more complicated case.



One converts the scheme to a series of unidirectional or net rate constants. The steady-state flux through each step as well the distribution of enzyme species remains the same, and we get



Because the mechanism is a linear series, the rate for the overall reaction is the sum of the inverse of the net rate constants through the individual steps:

$$(v/[E_{\text{total}}]) = 1/\{1/k_1' + 1/k_3' + 1/k_5' + 1/k_7'\}$$

where these net rate constants are expressed in terms of the actual rate constants below. [Recall again that these constants are unimolecular rate constants, and all bimolecular rate constants must be multiplied by the concentration of substrate or product.]

One begins with any irreversible step in the original reaction scheme. In this case, there is only one such irreversible step, that of converting EQ into E + Q. For an irreversible step, the net rate constant and the real rate constant are identical:

$$k_7' = k_7$$

Moving leftward, we express k_5' as the real forward rate constant multiplied by a factor that relates the fraction of EQ reacting in the forward direction as opposed to the fraction that returns to EPQ. In this case, because $k_7' = k_7$, we get:

$$k_5' = k_5\{k_7/(k_7 + k_6[P])\} = k_5k_7/(k_7 + k_6[P])$$

Again moving leftward, we get

$$k_3' = k_3\{k_5'(k_4 + k_5')\}$$

then replacing k_5' by $k_5k_7/(k_7 + k_6[P])$, one gets

$$k_3' = \{k_3k_5k_7/(k_7 + k_6[P])\}/\{k_4 + \{k_5k_7/(k_7 + k_6[P])\}\}$$

$$k_3' = k_3k_5k_7/\{k_5k_7 + k_4k_6[P] + k_4k_7\}$$

And then we deal with the leftwardmost term k_1' :

$$k_1' = k_1[A]k_3'/(k_2 + k_3')$$

which upon substituting k_3' by $k_3k_5k_7/\{k_5k_7 + k_4k_6[P] + k_4k_7\}$, gives

$$k_1' = k_1k_3k_5k_7[A]/\{k_2k_4k_6[P] + k_2k_4k_7 + k_2k_5k_7 + k_3k_5k_7\}$$

Now returning to the first equation and substituting the k_i' terms above

$$v = [E_{\text{total}}]/\{1/k_1' + 1/k_3' + 1/k_5' + 1/k_7'\}$$

$$\begin{aligned} v &= [E_{\text{total}}]/\{[(k_2k_4k_6[P] + k_2k_4k_7 + k_2k_5k_7) \\ &\quad + k_3k_5k_7]/\{k_1k_3k_5k_7[A]\} + \{(k_5k_7 + k_4k_6[P] \\ &\quad + k_4k_7)/(k_3k_5k_7)\} + \{(k_7 + k_6[P])/(k_5k_7)\} + (1/k_7)\} \\ &= k_1k_3k_5k_7[E_{\text{total}}][A]/\{k_2k_4k_7 + k_2k_5k_7 + k_3k_5k_7 \\ &\quad + k_2k_4k_6[P] + k_1(k_7(k_3 + k_4 + k_5) + k_3k_5)[A] \\ &\quad + k_1k_6(k_3 + k_4)[A][P]\} \end{aligned}$$

¹W. W. Cleland (1975) *Biochemistry* **14**, 3220.

²H. J. Fromm (1970) *Biochem. Biophys. Res. Commun.* **40**, 692.

³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁴C. Y. Huang (1979) *Meth. Enzymol.* **63**, 54.

NEUROLYSIN

This endopeptidase [EC 3.4.24.16] catalyzes the hydrolysis of peptide bonds, preferentially cleaving neurotensin between Pro-10 and Tyr-11. However, there is no absolute requirement for a prolyl bond.

A. J. Barrett, M. A. Brown, P. M. Dando, C. G. Knight, N. McKie, N. D. Rawlings & A. Serizawa (1995) *Meth. Enzymol.* **248**, 529.

F. Checcher, H. Barelli, P. Danch, V. Dive, B. Vincent & J. P. Vincent (1995) *Meth. Enzymol.* **248**, 593.

NEWTON

The SI unit (symbolized by N) for force; equal to kilogram-meter-second⁻². The name of this classical mechanical term stems from Isaac Newton's work showing that $\mathbf{F} = m\mathbf{a}$, where $\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$.

NEWTON-RAPHSON METHOD

An algorithm for solving integrated rate equations, thereby facilitating the systematic analysis of progress curves of enzyme-catalyzed reactions.

I. A. Nimmo & G. L. Atkins (1974) *Biochem. J.* **141**, 913.

R. G. Duggleby & J. F. Morrison (1977) *Biochim. Biophys. Acta* **481**, 297.

R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 61.

N-H EXCHANGE

Any technique^{1,2} designed to probe protein structure and/or stability in terms of the rate and extent of deuterium-protium (^2H - ^1H) or tritium-protium (^3H - ^1H) exchange with peptide N-H groups. These methods are based on the resistance to exchange by N-H groups participating in hydrogen bonding with O=C-groups within α -helices. These hydrogen bonds must be broken for exchange to occur, and this fact forms the basis of the technique which can be applied (a) grossly by using one-dimensional NMR to measure exchange of N-H groups in a protein or (b) specifically by using two-dimensional NMR spectroscopy and site-directed mutagenesis to assign each resonance unambiguously to a certain residue in the protein. The observed rate constant for exchange by a single N-H group of residue i is of the product of the intrinsic rate of exchange times the relative fraction of groups in the nonbonded state (i.e., $k_{\text{observed}} = k_{\text{intrinsic}}[1 - fB(i)]$).

¹C. A. Rohl & R. L. Baldwin (1998) *Meth. Enzymol.* **295**, 1.

²S. W. Englander & N. R. Kallenbach (1984) *Quart. Rev. Biophys.* **16**, 521.

NICKEL-DEPENDENT ENZYMES

Perhaps the best known nickel-dependent enzyme is urease (EC 3.5.1.5) which catalyzes the hydrolysis of urea to ammonium ion and carbamate; carbamate then undergoes nonenzymatic hydrolysis to bicarbonate and ammonia. Jack bean urease, the first ever enzyme crystallized, contains two Ni(II) atoms. Why urease utilizes Ni(II), perhaps the least Lewis-acidic of the transition metals, instead of another cation remains unclear¹. Other nickel-dependent enzymes include hydrogenase (EC 1.18.99.1), carbon monoxide dehydrogenase (EC 1.2.99.2), hydrogen dehydrogenase (EC 1.12.1.2), and coenzyme F420 hydrogenase (EC 1.12.99.1).

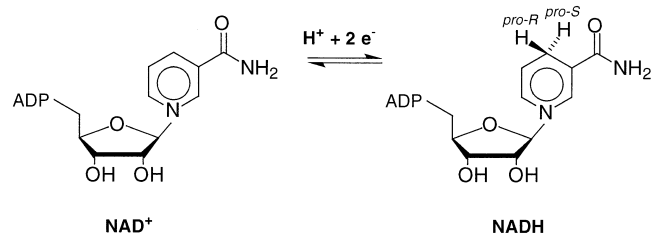
¹M. A. Halcrow (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 506.

NICOTINAMIDASE

This enzyme [EC 3.5.1.19] catalyzes the hydrolysis of nicotinamide to produce nicotinate and ammonia.

S. Pinder, J. B. Clark & A. L. Greenbaum (1971) *Meth. Enzymol.* **18B**, 20.

NICOTINAMIDE COENZYMES



Reduced forms of these coenzymes absorb ultraviolet light near 340 nm, whereas the oxidized forms do not. For NAD⁺ at neutral pH, the maximal absorbance band ($\epsilon = 18000 \text{ M}^{-1}\text{cm}^{-1}$) occurs at 260 nm; another absorbance band ($\epsilon = 8000 \text{ M}^{-1}\text{cm}^{-1}$) occurs at 230 nm. For NADH, the maximal absorbance band ($\epsilon = 16900 \text{ M}^{-1}\text{cm}^{-1}$) occurs at 259 nm; a second absorbance band ($\epsilon = 6220 \text{ M}^{-1}\text{cm}^{-1}$)¹ occurs at 339 nm; two weaker bands occur at 234 and 290 nm with respective ϵ values of 6600 $\text{M}^{-1}\text{cm}^{-1}$ and 1300 $\text{M}^{-1}\text{cm}^{-1}$. The same is true for NADP⁺ and NADPH at 339 nm. Occasionally, investigators have used thio-NADH which has a much stronger absorbance around 366 nm. **See Absorption Spectroscopy** (Fig. 4, pg. 5); *Nicotinic Acid: Analogs and Coenzymes*

¹The International Federation of Clinical Chemistry recommends a value of 6300 $\text{M}^{-1}\text{cm}^{-1}$ at 339 nm and 30°C.

NICOTINAMIDE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.1] catalyzes the reaction of *S*-adenosyl-L-methionine with nicotinamide to produce *S*-adenosyl-L-homocysteine and 1-methylnicotinamide.

G. L. Cantoni (1955) *Meth. Enzymol.* **2**, 257.

NICOTINAMIDE MONONUCLEOTIDE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.1], also known as NAD⁺ pyrophosphorylase, catalyzes the reaction of ATP with nicotinamide ribonucleotide to produce NAD⁺ and pyrophosphate (or, diphosphate). Nicotinate nucleotide is also a substrate. Nicotinate-nucleotide adenylyltransferase [EC 2.7.7.18] uses nicotinate ribonucleotide as the pyridine substrate, thereby producing deamido-NAD⁺ and pyrophosphate.

- G. Magni, N. Raffaelli, M. Emanuelli, A. Amici, P. Natalini & S. Ruggieri (1998) *Adv. Enzymol.* **73**, 240.
 V. Micheli & S. Sestini (1997) *Meth. Enzymol.* **280**, 211.
 G. Magni, M. Emanuelli, A. Amici, N. Raffaelli & S. Ruggieri (1997) *Meth. Enzymol.* **280**, 241 and 248.

NICOTINAMIDE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.12], also known as NMN pyrophosphorylase, catalyzes the reaction of nicotinamide with 5-phospho- α -D-ribose 1-diphosphate to produce nicotinamide D-ribonucleotide and pyrophosphate.

- V. Micheli & S. Sestini (1997) *Meth. Enzymol.* **280**, 211.
 W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

NICOTINATE DEHYDROGENASE

This enzyme [EC 1.5.1.13], also known as nicotinate hydroxylase, catalyzes the reaction of nicotinate with NADP⁺ and water to produce 6-hydroxynicotinate and NADPH. This iron-dependent flavoprotein will also oxidize NADPH.

- T. C. Stadtman (1980) *Ann. Rev. Biochem.* **49**, 93.
 L. Tsai & E. R. Stadtman (1971) *Meth. Enzymol.* **18B**, 233.

NICOTINATE MONONUCLEOTIDE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.18], also known as nicotinate-nucleotide adenylyltransferase and deamido-NAD⁺ pyrophosphorylase, catalyzes the reaction of ATP with nicotinate ribonucleotide to produce deamido-NAD⁺ and pyrophosphate.

- V. Micheli & S. Sestini (1997) *Meth. Enzymol.* **280**, 211.
 S. Pinder, J. B. Clark & A. L. Greenbaum (1971) *Meth. Enzymol.* **18B**, 20.

NICOTINATE-NUCLEOTIDE DIMETHYLBENZIMIDAZOLE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.21] catalyzes the reaction of β -nicotinate D-ribonucleotide with dimethylbenzimidazole to produce nicotinate and *N*¹-(5-phospho- α -D-ribosyl)-5,6-dimethylbenzimidazole. Benzimidazole also serves as a substrate. The clostridial enzyme acts on adenine to produce 7- α -D-ribosyladenine 5'-phosphate.

- J. A. Fyfe & H. C. Friedmann (1971) *Meth. Enzymol.* **18B**, 197.

NICOTINATE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.11] catalyzes the reaction of nicotinate with 5-phospho- α -D-ribose 1-diphosphate to produce nicotinate D-ribonucleotide and pyrophosphate.

- V. Micheli & S. Sestini (1997) *Meth. Enzymol.* **280**, 211.
 W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

NICOTINIC ACID: ANALOGUES AND COENZYMES

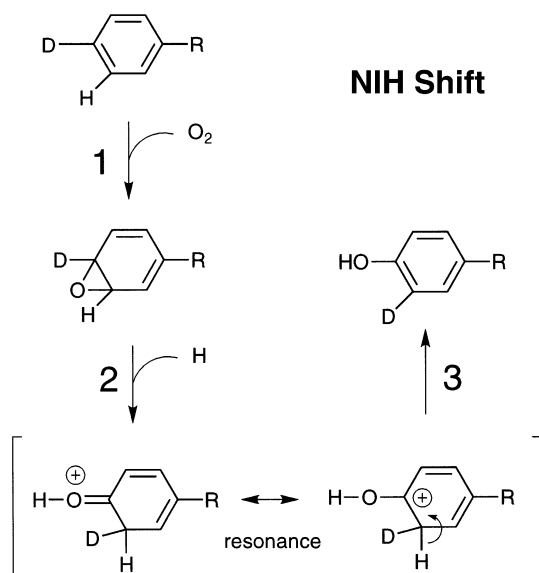
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Determination of nicotinamide, **66**, 3; fluorometric quantitation of picomole amounts of 1-methylnicotinamide and nicotinamide in serum, **66**, 5; temperature dependence of the spectroscopic properties of NADH, **66**, 8; purification of commercial NADH, **66**, 11; isolation and analysis of pyridine nucleotides and related compounds by liquid chromatography, **66**, 23; affinity chromatography of NAD⁺ on immobilized dehydrogenase columns, **66**, 39; preparation of stereospecific tritium-labeled reduced nicotinamide adenine dinucleotide phosphate, **66**, 51; kinetic methods for detecting inhibitors in NADH for NADH-dependent enzymes, **66**, 55; simple methods for preparing nicotinamide mononucleotide and related analogs, **66**, 62; preparation and purification of nicotinamide mononucleotide analogs, **66**, 71; preparation of 3-aminopyridine adenine dinucleotide and 3-aminopyridine adenine dinucleotide phosphate, **66**, 81; preparation of α -NADP⁺, **66**, 87; an improved method for measuring quinolinic acid in biological specimens, **66**, 91; NAD⁺ kinase from sea urchin eggs, **66**, 101; nicotinic acid adenine dinucleotide phosphate (NAADP⁺), **66**, 105; preparation of 2'-P-ADP, **66**, 112; convenient method for enzymic synthesis of [¹⁴C]nicotinamide riboside, **66**, 120; formation of nicotinamide ribose diphosphate ribose, a new metabolite of the NAD⁺ pathway, by *Aspergillus niger*, **66**, 123; NAD⁺ glycohydrolase from bovine seminal plasma, **66**, 144; NAD⁺ glycohydrolases from rat liver nuclei, **66**, 151; poly(ADP-ribose) synthetase from rat liver nuclei, **66**, 154; poly(ADP-ribose) synthetase from calf thymus, **66**, 159; extraction and quantitative determination of larger than tetrameric endogenous polyadenosine diphosphoribose from animal tissues, **66**, 165; covalent modification of proteins by metabolites of NAD⁺, **66**, 168; coenzyme activity of NAD⁺ bound to polymer supports through the adenine moiety, **66**, 176; use of differently immobilized nucleotides for binding NAD⁺-dependent dehydrogenases, **66**, 192.

NIH SHIFT

An intramolecular hydrogen migration observed in the hydroxylation of aromatic rings in certain enzyme-catalyzed reactions as well as some chemical reactions. The rearrangement was first observed¹⁻³ at the National Institutes of Health (hence the name "NIH") in studies of the synthesis of L-tyrosine from L-phenylalanine via phenylalanine hydroxylase. Observation of this shift requires appropriate deuteration of the aromatic reactant.

Other enzymes exhibiting this migration include cytochrome P450 and *p*-hydroxyphenylpyruvate dioxygenase



(which involves an alkyl group migration rather than a hydride ion). In general, the NIH shift is attributed to the formation of an arene oxide intermediate, although other pathways have been suggested. Nonenzymatic routes that appear to involve NIH shifts have been reported⁴.

¹G. Guroff, D. Jerina, J. Rensen, S. Udenfriend & B. Witkop (1967) *Science* **157**, 1524.

²D. Jerina (1973) *Chem. Technol.* **4**, 120.

³R. P. Hanzlik, K. Hogberg & C. M. Judson (1984) *Biochemistry* **23**, 3048.

⁴T. C. Bruice & P. Bruice (1976) *Acc. Chem. Res.* **9**, 378.

NITRATE REDUCTASE

(1) Nitrate reductase (NADH) [EC 1.6.6.1], also known as assimilatory nitrate reductase, catalyzes the reaction of NADH with nitrate to produce NAD⁺, nitrite, and water. This enzyme uses FAD or FMN, heme, and a molybdenum ion as cofactors. (2) Nitrate reductase (NAD(P)H) [EC 1.6.6.2], also known as assimilatory nitrate reductase, catalyzes the reaction of NAD(P)H with nitrate to produce NAD(P)⁺, nitrite, and water. This enzyme uses FAD or FMN, heme, and a molybdenum ion as cofactors. (3) Nitrate reductase (NADPH) [EC 1.6.6.3] catalyzes the reaction of NADPH with nitrate to produce NADP⁺, nitrite, and water. This enzyme uses FAD, heme, and a molybdenum ion as cofactors. (4) Nitrate reductase (cytochrome) [EC 1.9.6.1] catalyzes the reaction of nitrate with ferrocyanochrome to produce nitrite and ferrocyanochrome. (5) Nitrate reductase (acceptor) [EC 1.7.99.4], also known as respiratory nitrate

reductase, catalyzes the reaction of nitrate with a reduced acceptor to produce nitrite and the oxidized acceptor. The *Pseudomonas* enzyme is a cytochrome-dependent system, but the enzyme from *Micrococcus halodenitrificans* is an iron protein containing molybdenum.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.

H. C. Huppe & D. H. Turpin (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 577.

L. P. Solomonson & M. J. Barber (1990) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **41**, 225.

M. G. Guerrero, J. M. Vega & M. Losada (1981) *Ann. Rev. Plant Physiol.* **32**, 169.

R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

NITRENE

A molecule containing a neutral nitrogen atom with four nonbonding electrons (R-N:). While two of these nonbonding electrons are paired, the other two may have parallel spins (thus, triplet state) or antiparallel spins (singlet state). Nitrenes are the nitrogen analogues of carbenes.

W. Lwowski (1970) *Nitrenes*, Interscience, New York.

E. F. V. Scriven (1982) *React. Intermed.* **2**, 1.

NITRENIUM ION

A cationic species exhibiting significant positive charge residing on a dicoordinated nitrogen atom. Nitrenium ions are the nitrogen analogs of carbenium ions and may exist as intermediates in certain reactions. There are at least two types of nitrenium ion, R-N⁺-R' (note that if R is a proton, this species is simply a protonated nitrene) and RR'C=N⁺. These ions can exist in either triplet or singlet states.

P. G. Gassman (1970) *Acc. Chem. Res.* **3**, 26.

P. T. Lansbury (1970) in *Nitrenes* (W. Lwowski, ed.), pp. 405-419, Interscience, New York, pp. 405-419.

NITRIC OXIDE SYNTHASE

This enzyme [EC 1.14.13.39] catalyzes the reaction of L-arginine with 1.5 NADPH, 0.5 H⁺, and two dioxygen to produce L-citrulline, nitric oxide, 1.5 NADP⁺, and two water. The enzyme in brain, but not that induced in lung or liver by endotoxin, requires calcium ions.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

L. Packer, ed. (1996) *Meth. Enzymol.*, vols. **268** and **269**.

S. R. Jaffrey & S. H. Snyder (1995) *Ann. Rev. Cell Dev. Biol.* **11**, 417.

- D. S. Brecht & S. H. Snyder (1994) *Ann. Rev. Biochem.* **63**, 175.
 M. Marletta (1993) *J. Biol. Chem.* **268**, 12231.
 D. J. Stuehr & O. W. Griffith (1992) *Adv. Enzymol.* **65**, 287.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Assays, **233**, 250-258, 265 [citrulline method, **233**, 252, 255-256; hemoglobin method, **233**, 252-255; nitrite/nitrate method, **233**, 252, 256-258]; biopterin content determination, **233**, 262-264; co-factor requirements, **233**, 251, 259; distribution, **233**, 250, 258; flavin content determination, **233**, 262; isoforms [activity, **233**, 258-259; assay considerations, **233**, 251-252; cDNA, isolation and cloning, **233**, 266-269; properties, **233**, 258, 263; purification (from bovine aortic endothelial cells, **233**, 261-262; from induced RAW 264.7 macrophages, **233**, 260-261; from neuronal tissue, **233**, 259-260, 265-266); isolation, **233**, 250-251]; pH optimum, **233**, 251-252; properties, **233**, 250; reaction catalyzed by, **233**, 250-251, 258, 264-265; stability, **233**, 252; storage, **233**, 252; structure, **233**, 250.

NITRILE HYDRATASE

Nitrilase [EC 3.5.5.1], also known as nitrile aminohydrolase and nitrile hydratase, catalyzes the hydrolysis of a nitrile to produce a carboxylate and ammonia. The enzyme acts on a wide range of aromatic nitriles. Nitrile hydratase [EC 4.2.1.84], also known as nitrilase, catalyzes the hydrolysis of a nitrile to produce an aliphatic amide. The enzyme acts on short-chain aliphatic nitriles, converting them into the corresponding acid amides. However, this particular enzyme does not further hydrolyze these amide products; nor does the enzyme act on aromatic nitriles.

- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

NITRITE REDUCTASE

(1) Nitrite reductase (NAD(P)H) [EC 1.6.6.4] catalyzes the reaction of three NAD(P)H with nitrite to yield three NAD(P)⁺, NH₄OH, and water. Cofactors for this enzyme include FAD, non-heme iron, and siroheme. (2) Nitrite reductase (cytochrome) [EC 1.7.2.1] is a copper-dependent system that catalyzes the reaction of nitric oxide with two ferricytochrome *c* and water to produce nitrite and two ferrocyanochrome *c*. (3) Ferredoxin–nitrite reductase [EC 1.7.7.1], a heme- and iron-dependent enzyme, catalyzes the reaction of ammonia with three oxidized ferredoxin to produce nitrite and three reduced ferredoxin. (4) Nitrite reductase [EC 1.7.99.3] is a copper- and FAD-dependent enzyme that catalyzes the reaction of two nitric oxide with an acceptor substrate and two water to produce two nitrite and the reduced acceptor.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

- A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 401.

- M.-C. Liu, C. Costa & I. Moura (1994) *Meth. Enzymol.* **243**, 303.
 E. T. Adman (1991) *Adv. Protein Chem.* **42**, 145.
 T. E. Meyer & M. D. Kamen (1982) *Adv. Protein Chem.* **35**, 105.
 Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

NITROGENASE

(1) Nitrogenase (ferredoxin) [EC 1.18.6.1] catalyzes the reaction of three reduced ferredoxin molecules with protons, N₂, and *n* ATP molecules to produce three oxidized ferredoxin molecules, two ammonia molecules, *n* ADP molecules, and *n* orthophosphate molecules where *n* is between 12 and 18. This iron-sulfur system also uses either molybdenum or vanadium ions. (2) Nitrogenase (flavodoxin) [EC 1.19.6.1] catalyzes the reaction of six reduced flavodoxin molecules with protons, N₂, and *n* ATP molecules to produce six oxidized flavodoxin molecules, two ammonia molecules, *n* ADP molecules, and *n* orthophosphate molecules. This system uses iron-sulfur and molybdenum ions.

- G. J. Leigh (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 349.
 P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.
 J. B. Howard & D. C. Rees (1994) *Ann. Rev. Biochem.* **63**, 235.
 L. E. Mortenson, L. C. Seefeldt, T. V. Morgan & J. T. Bolin (1993) *Adv. Enzymol.* **67**, 299.
 H. Deng & R. Hoffman (1993) *Angew. Chem. Int. Ed. Engl.* **32**, 1062.
 J. T. Bolin, N. Campobasso, S. W. Muchmore, T. V. Morgan & L. E. Mortenson (1993) in *Molybdenum Enzymes, Cofactors & Model Systems* (E. I. Stiefel, D. Coucouvanis & W. E. Newton, eds.) pp. 186-195, Amer. Chem. Soc., Washington, D.C.
 S. Hunt & D. B. Layzell (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 483.
 R. H. Burris (1991) *J. Biol. Chem.* **266**, 9339.
 R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.
 G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

NITROGEN FIXATION

The process that assimilates molecular nitrogen, N₂, into ammonia and other nitrogenous metabolites. The first reaction is catalyzed by nitrogenase [EC 1.18.2.1] which is composed of two different protein components—one containing iron, molybdenum, and acid-labile sulfur (variously named as Mo-Fe protein, Component I, or molybdoferredoxin) and a second component containing iron and sulfur only (called Fe protein, Component II, or azoferredoxin). In bacteria, nitrogen fixing polypeptides are encoded in *nif* operon, and for *E. coli*, *Klebsiella*, and *Salmonella*, the *nif* operon is tightly coupled to the adenylylation/deadenylylation cascade that controls the

synthesis of glutamine. In fact, transcription of the *nif* operon is promoted by glutamine synthetase (unadenylylated), the enzyme form that persists when ammonia stores are low.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Nitrogen fixation-assay methods and techniques, **24**, 415; preparation and properties of clostridial ferredoxins, **24**, 431; purification of nitrogenase from *Clostridium pasteurianum*, **24**, 446; nitrogenase complex and its components, **24**, 456; preparation of nitrogenase from nodules and separation into components, **24**, 470; clostridial rubredoxin, **24**, 477; purification of nitrogenase and crystallization of its Mo-Fe protein, **24**, 480; N_2 -fixing plant-bacterial symbiosis in tissue culture, **24**, 497.

NOISE

Any unwanted, extraneous, and random disturbance that degrades accuracy and/or precision in the recording or measuring of a signal. The occurrence of noise is manifested in experimental science as a threshold that defines the sensitivity of measurement.

One usually evaluates the quality of an analytical measuring device on the basis of its signal-to-noise (or, S/N) ratio. The average signal strength or output of an instrument is proportional to the concentration of an analyte, whereas the amplitude (*i.e.*, strength) of noise is not. For this reason, the signal strength is proportional to some mean value $\bar{x}$, and the noise is best described by a standard deviation sd . The relative standard deviation $sd/\bar{x}$ is therefore the reciprocal of S/N. In most cases, accuracy and precision will be tolerant to S/N values that do not drop below 2-3. Noise can be categorized according to its source. Instrument noise involves thermal noise, shot noise, environmental noise, input power fluctuations, and a less well understood phenomenon known as flicker. Any impurities in chemical reagents or any inhomogeneously mixed sample can be a source of noise that is more related to the chemistry or physics of the sample whose properties are being measured.

Thermal noise (also known as Johnson noise) relates to the thermal motions or agitations of electrons in electrical devices. For example, a photomultiplier tube is set at a high voltage so that impinging photons can release electrons from the photocathode surface. Proper voltage adjustments must be made for optimal response to absorbed photons: (a) too low a voltage will result in no observable signal, and (b) too high a voltage will result in the spurious release of too many electrons, even in

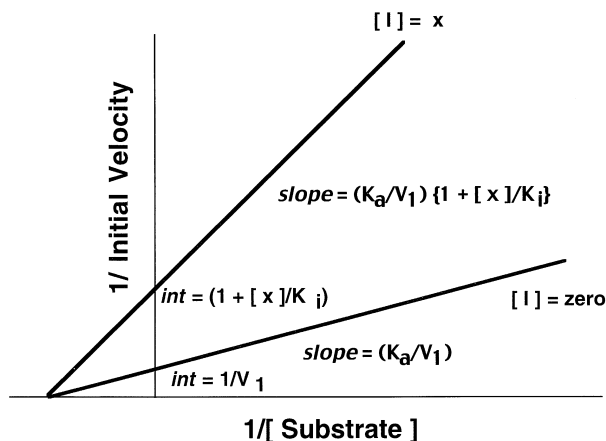
the absence of photoexcitation. Thus, thermal noise is an unavoidable or intrinsic property. In instruments relying on photomultiplier tubes, thermal noise is measured as a dark current: this simply means that there is a measurable electrical output even in the absence of light. At 298 K, the dark current in an RCA 928 photomultiplier tube is often around 400-500 counts/second; however, by reducing the temperature to around 240 K, the dark current drops into the 12-20 cpm range. This means that a system that is thermal noise-limited will experience a 25-40 times better signal-to-noise ratio at -33°C than at 25°C . For this reason, photomultiplier tubes in many spectrophotometers and fluorimeters are often housed within Peltier cooling devices.

Shot noise relates to randomly fluctuating electron movements across the anode-to-cathode junction. While this behavior does not alter the shape of a visible or UV spectrum, the presence of shot noise can reduce sensitivity.

Environmental noise refers to the effects of fluctuations in the surrounding electromagnetic fields generated by lightning, the presence of nearby high-voltage electrical power transmission lines, radiofrequency noise, and arcing from nearby high-current electrical motors or switches.

NONCOMPETITIVE INHIBITION

Reversible inhibition of an enzyme by an inhibitor binding to the enzyme whether or not the substrate is bound at the active site^{1,2}. A noncompetitive inhibitor binds at a site that is distinct from the active site, and saturating levels of substrate cannot completely remove the inhibitory influence of this agent.



Consider the standard Uni Uni mechanism ($E + A \rightleftharpoons EX \rightarrow E + P$). A noncompetitive inhibitor, I, can bind reversibly to either the free enzyme (E) to form an EI complex (having a dissociation constant K_{is}), or to the central complex (EX) to form the EXI ternary complex (having a dissociation constant K_{ii}). Both the slope and vertical intercept of the standard double-reciprocal plot ($1/v$ vs. $1/[A]$) are affected by the presence of the inhibitor. If the secondary replots of the slopes and the intercepts (thus, slopes or vertical intercepts vs. $[I]$) are linear (See *Nonlinear Inhibition*), then the values of those dissociation constants can be obtained from these replots. If $K_{is} = K_{ii}$, then a plot of $1/v$ vs. $1/[A]$ at different constant concentrations of the inhibitor will have a common intersection point on the horizontal axis (if not, See *Mixed-Type Inhibition*). Note that the above analysis assumes that the inhibitor binds in a rapid equilibrium fashion. If steady-state binding conditions are present, then nonlinearity may occur, depending on the magnitude of the $[I]^2$ and $[A]^2$ terms in the rate expression. See also *Mixed Type Inhibition*

¹J. A. Todhunter (1979) *Meth. Enzymol.* **63**, 383.

²H. J. Fromm (1995) *Meth. Enzymol.* **249**, 123.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Computer program, **63**, 115, 128, 129, 137; coupled enzyme assay, **63**, 40; experimental design, **63**, 394; graph, **63**, 389; metal-ion, **63**, 292; one-substrate, **63**, 470-472; partial, **63**, 385; pH effect, **63**, 214; product, **63**, 412-414; progress curve, **63**, 164, 170, 171; rate expression, **63**, 393; tight-binding, **63**, 470-472.

NONCOVALENT INTERACTIONS

Any interaction between ions, molecules, or functional groups that does not involve covalency (*i.e.*, the sharing of a pair of electrons within a well defined molecular orbital). While such interactions are considerably weaker than those involved in the making/breaking of covalent bonds, significant binding energy can nonetheless arise owing to the great number of such interactions within structurally organized biological macromolecules. The same can be said for ligand binding interactions with macromolecules. The most common examples of noncovalent interactions are: (a) ion-ion interactions¹ (typically involving interaction energies in the range of 10 to 70 kcal/mol (or, 40 to 335 kJ/mol); (b) dipole-dipole interactions² (frequently characterized by interaction energies of less than 1 kcal/mol); (c) ion-induced dipole interactions³ (with interaction energies also less than 1 kcal/mol); (d) dipole-induced dipole interactions⁴ (with

interaction energies of 1 to 10 kcal/mol); and (e) London dispersion interactions⁵ (also in the 1–10 kcal/mol range). Hydrophobic interactions may also be included along with this general group of noncovalent interactions; however, hydration involves weak coordinate covalent interactions, and hydrogen bonding is most often treated as a unique three-center interaction.

¹The energy for ion-ion interactions is proportional to the product of the charges, q_1 and q_2 , divided by the product of the dielectric constant D and the distance between the charges. If q_1 and q_2 are both of the same sign, the energy will be positive (*i.e.*, unfavorable), as would be expected for repulsion. If q_1 and q_2 are of opposite sign, the noncovalent interaction will be negative (*i.e.*, favorable).

²Dipole-dipole interaction energies are proportional to the product of the dipole moments, μ_1 and μ_2 , divided by the product of the dielectric constant D and the sixth power of the distance between the dipoles.

³Ion-induced dipole interaction energies are proportional to the product of the square of the charge and the polarizability α of the atom/group with which the ion interacts, divided by the product of the dielectric constant D and the fourth power of the distance between the dipoles.

⁴Dipole-induced dipole interaction energies are proportional to the product of the square of the dipole moment and the polarizability α of the atom/group with which the ion interacts, divided by the product of the square of dielectric constant D and the sixth power of the distance between the dipole and the polarizable group.

⁵Dispersion interactions have energies that are proportional to the product of polarizabilities α_1 and α_2 , divided by the sixth power of the distance between two polarizable atoms (or groups of atoms).

NONEQUILIBRIUM THERMODYNAMICS (A Primer)

Although many chemical and physical processes readily attain thermodynamic equilibrium, living systems characteristically maintain steady-state flows of energy and/or matter. Heat flow from a region of high temperature to one at low temperature and a gas expanding through a nozzle into a vacuum are examples of irreversible processes. So too are electric current flow and Fick's law of diffusion for neutral particles. In living systems, other examples include the steady-state rates of metabolism as well as the downward flow of protons along transmembrane gradients that drive ATP synthesis. Indeed, the cardinal feature of living systems is homeostasis—the compensatory adjustment from one steady state to another in response to an external stimulus or to changes in the availability of energy and/or matter. Any detailed thermodynamic description of nonequilibrium processes is necessarily more complicated, requiring additional parameters beyond those needed to specify the thermody-

dynamic properties of systems at equilibrium. Even for the simplest nonequilibrium processes (such as the unidirectional transfer of only a single component), one must account for spatial constraints as well as the rates of change in the entropy of the system¹⁻⁴.

Steady-state processes are unidirectional and irreversible. Their special thermodynamic behavior stems from a lack of balance in the system. In all irreversible processes, ΔS_{total} will be a positive quantity. In the above mentioned case of the transfer of a quantity q of heat from a hot body to a cold body, we can explicitly express the ΔS_{system} as follows:

$$\Delta S_{\text{system}} = -q/T_{\text{hot}} + q/T_{\text{cold}}$$

where T_{hot} and T_{cold} are the temperatures of the hot and cold bodies. ΔS_{total} is the sum of ΔS_{system} and $\Delta S_{\text{surroundings}}$, and in the absence of any exchange with the surrounding environment, the total entropy change will equal the system's entropy change:

$$\Delta S_{\text{total}} = -q/T_{\text{hot}} + q/T_{\text{cold}} > 0$$

such that $q/T_{\text{cold}} > q/T_{\text{hot}}$.

Any directional flow of energy and/or matter between two interconnected systems, say A and B, can be treated as occurring across a metastable zone joining A and B. One can assume that systems A and B are each at equilibrium internally (*i.e.*, there are no differences within A or B in terms of temperature, pressure, composition, thermodynamic activities of each component, or even chemical reactivity). Nonetheless, systems A and B are not at equilibrium with respect to each other, thereby allowing for the net transfer of energy and/or matter from the system of higher energy to the other. Within the zone of transfer, one can apply the so-called method of local isolation to examine the thermodynamic properties describing a point or region. This point or region will experience the flow of energy/matter from A to B, and if the rate into that point or region is exactly balanced by its rate of loss of energy/matter, then that point or region is said to satisfy the steady-state approximation. This virtually time-independent rate of transfer can be described by writing out expressions of JX_i , the nonequilibrium flux in terms of each component X_i :

$$JX_i = -dX_{iA}/dt = dX_{iB}/dt$$

where the subscripts indicate component X_i present in system A and B.

This flux through the transfer zone leads to a net increase in entropy. While it is certainly true that the earlier definition of entropy requires that we consider reversible phenomena (*i.e.*, $dS = dq_{\text{reversible}}/T$), this limitation can be overcome, in principle, by treating the transition or transfer zone as a series of layers running perpendicular to the direction of flux. This approach offers the virtue that one can assume that T and P are well defined and constant within each of the layers or lamina. Onsager succeeded in achieving a rigorous theoretical framework that allows one to specify the thermodynamic properties of irreversible systems, including those at steady state. While a detailed consideration of the Onsager method lies beyond the scope of this brief description, the following statements comprise its essence. One uses the Principle of Microscopic Reversibility which clearly states that under equilibrium conditions, any process and its reverse follow the same path and take place at the same rate on a time-averaged basis. The thermodynamic equations of motion for each transport process maybe written with the flux equal to the sum of terms accounting for the transfer to and from systems A and B in a manner that is proportional to the thermodynamic force. Note that the forces characterizing irreversible thermodynamics should not be confused with those of classical mechanics, and no accelerative effects are implied. Rather each flux has a conjugate force associated with it, and the multiplicative product of a flux and its conjugate force will always have units corresponding to the rate of entropy production. For a chemical reaction, the generalized flux J equals $(1/V)d\xi/dt$ (where ξ is the extent of reaction), and the conjugate force is the affinity $\mathcal{A}$, equal to $-\sum \nu_i \mu_i$ (where ν_i and μ_i are the stoichiometric coefficient and the chemical potential of the i -th component).

For a system in which a chemical reaction occurs at constant T and P , $d_e S$ is the flow of entropy into the system from the exterior, such that

$$Td_e S = dq = dU + PdV$$

The rate of change in $d_i S$ (defined as the time-dependent change in the entropy change within the system) can then be written as:

$$d_i S/dt = -(1/T)(d\xi/dt) \sum \nu_i \mu_i = (d\xi/dt) \mathcal{A}/V = \mathbf{v} \cdot \mathcal{A}$$

where the flux J is the reaction rate $\mathbf{v}$ (or, $(d\xi/dt)$ per unit volume V) and the generalized force $\mathcal{A}$ is as defined earlier. For systems undergoing steady state transfer, it can be further demonstrated that the rate of entropy production is at a minimum. Prigogine later demonstrated that the steady state can be quite stable, and small perturbations will not substantially alter steady state flux.

¹B. C. Eu (1994) *Kinetic Theory and Irreversible Thermodynamics*, Wiley, New York.

²D. K. Kondepudi & I. Prigogine (1998) *Modern Thermodynamics: From Heat Engines to Dissipative Structures*, Wiley, New York.

³D. Jou & J. E. Llebot (1990) *Introduction to the Thermodynamics Processes of Biological Processes*, Prentice Hall, New York.

⁴H. J. Kreuzer (1981) *Nonequilibrium Thermodynamics and Its Statistical Foundations*, Oxford University Press, London.

NONHYPERBOLIC KINETICS

Referring to an enzyme whose kinetic properties do not yield hyperbolic saturation curves in plots of the initial rate as a function of the substrate concentration.

While the term is most typically applied to cooperative enzymes, nonhyperbolic behavior may indicate: (1) that the steady-state condition is not a valid description of an enzyme's behavior; (2) that one or more of the experimental conditions (*e.g.*, pH, temperature, ionic strength, *etc.*) changes over the course of the experiment or varies with the substrate concentration; (3) that the total enzyme concentration is not appreciably lower than the substrate concentration or the concentration of an effector; (4) that nonlinear substrate inhibition occurs; (5) that more than one molecule of a substrate is consumed during one catalytic "round" (*e.g.*, MgATP in carbamoyl phosphate synthetase or glutathione as both the γ -glutamyl donor and acceptor in γ -glutamyltranspeptidase); (6) that the product(s) has (have) a relatively low K_i value(s), thereby resulting in significant product inhibition; and (7) that the enzyme obeys a steady-state random multisubstrate reaction mechanism, such that squared terms in substrate concentration are evident.

NONLINEAR INHIBITION

This term usually applies to reversible inhibition of an enzyme-catalyzed reaction in which nonlinearity is detected (a) in a double-reciprocal plot (*i.e.*, $1/v$ versus $1/[S]$) in the presence of different, constant concentrations of inhibitor or (b) in replots of slope or intercept values obtained from primary plots of $1/v$ versus $1/[S]$). Nonlinearity

replots can be attributable to a number of factors: cooperativity; multiple inhibition (for example, the formation of an EI_2 complex); partial inhibition; substrate inhibition; inhibitor-induced substrate inhibition; tight-binding enzyme inhibition; or in certain cases of product inhibition. Before proceeding to consider these more complicated explanations, one should first take measures to exclude trivial reasons for nonlinearity, such as lack of constant pH, temperature variability, instability of substrate and protein, *etc.* **See also** Cleland Rules for Dead-End Inhibition; Abortive Complex

NONLINEAR LEAST SQUARES ANALYSIS

A class of statistical methods frequently used to analyze kinetic and thermodynamic data. Most of these methods require a preliminary estimate of the constants followed by cycles of iterative calculations that converge on a final value(s). Cleland¹ has presented a protocol for the statistical estimation of kinetic data. A nonlinear analysis has also been applied to progress curves².

¹W. W. Cleland (1979) *Meth. Enzymol.* **63**, 103.

²R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 61.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Accuracy of computer programs, **240**, 21-22; accuracy of method, **240**, 6; algorithms, **240**, 5, 13; assumptions, **240**, 7-14, 55; curve fitting, **240**, 3-5, 10; data points [minimal number, **240**, 14; smoothing, **240**, 14]; determination of confidence intervals, **240**, 15-21, 28, 462; determination of statistical significance, **240**, 18-20; deviations, **240**, 3; effect of instrument delay, **240**, 13; fitting of heat capacity function, **240**, 510, 543; weighted sum, **240**, 3.

NONRADIATIVE DECAY

Disappearance of an excited molecular species arising from a radiationless transition. The energy of a nonradiative decay is dissipated vibrationally as thermal energy.

D-NOPALINE DEHYDROGENASE

This enzyme [EC 1.5.1.19], also known as D-nopaline synthase, catalyzes the reaction of N^2 -(D-1,3-dicarboxypropyl)-L-arginine with NADP⁺ and water to produce L-arginine, NADPH, and α -ketoglutarate (or, 2-oxoglutarate). In the reverse direction, the enzyme catalyzes the formation of D-nopaline from L-arginine as well as D-ornaline from L-ornithine.

J. Thompson & J. A. Donkersloot (1992) *Ann. Rev. Biochem.* **61**, 517.

J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.

NOREPINEPHRINE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.28], also known as phenylethanolamine N-methyltransferase, catalyzes the reaction of S-adenosyl-L-methionine with phenylethanolamine to produce S-adenosyl-L-homocysteine and N-methylphenylethanolamine. The enzyme will act on a number of phenylethanolamines and will catalyze the conversion of noradrenalin (or norepinephrine) into adrenalin (or epinephrine).

R. W. Fuller (1987) *Meth. Enzymol.* **142**, 655.

NORMAL DISTRIBUTION

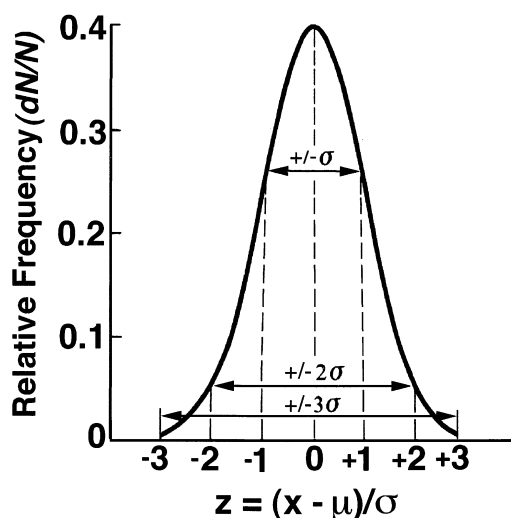
A symmetrical bell-shaped curve corresponding to a graph of the frequency $f(x)$ of a value x , when μ is the mean and σ^2 is the variance, such that

$$f(x) = \{\exp[-(x - \mu)^2/2\sigma^2]\}/\sigma(2\pi)^{1/2}$$

When this function appears to satisfy the observed distribution of quantitative measurements, we say that such measurements are normally distributed. **See** *Statistics (A Primer)*

NORMAL ERROR CURVE

A statistical method for plotting the relative frequency (dN/N) of a probable error in a single measured value x versus the deviation (z) from μ , the mean of the data, in units of standard deviation (σ), such that $z = (x - \mu)/\sigma$. The standard error curve (shown below) does not depend on either the magnitude of the mean or the standard deviation of the data set. The maximum of the normal error curve is poised at zero, indicating that the mean is the most frequently observed value.



The area beneath the standard error curve can be obtained by integrating the following equation:

$$dN/N = (1/2\pi) \exp[-z^2/2] dz$$

The area under the entire curve equals a probability of unity. Over the interval from -1σ to $+1\sigma$, the integrated area equals 68.3% of the total area under the curve; thus, the probability of finding a value within that interval is 0.683. For $\pm 2\sigma$, the probability rises to 0.955, and for $\pm 3\sigma$, the probability is 0.997. Note that for a set of N data values, the estimated mean will progressively deviate less and less from the true mean as the number of data points increases. This so-called standard error of the mean (s_m) is inversely related to $N^{1/2}$. **See** *Statistics (A Primer)*

NORMALITY (or, NORMAL CONCENTRATION)

A unit of concentration (symbolized by N) equal to the number of equivalents of a chemical substance (*i.e.*, the grams of the substance divided by the equivalent weight) per liter of solution. The term “normal solution” refers to a solution of 1.0 N. For example, the gram-molecular-weight and the gram-equivalent-weight for a monoprotic acid are the same; hence, a 1.0 M HNO_3 is also 1.0 N. For a diprotic acid, the gram-molecular-weight is exactly twice the gram-equivalent-weight, and a 1.0 M H_2SO_4 solution is therefore 2.0 N. **See also** *Weight Normality*

NORTHROP ISOTOPE EFFECT METHOD

A procedure using both deuterium and tritium isotope effects on $V_{\max}/K_m$ to obtain *intrinsic* isotope effects for enzyme-catalyzed reactions¹⁻³. **See** *Kinetic Isotope Effects*

¹D. B. Northrop (1975) *Biochemistry* **14**, 2644.

²W. W. Cleland, M. H. O'Leary & D. B. Northrop, eds. (1977) *Isotope Effects on Enzyme-Catalyzed Reactions*, University Park Press, Baltimore.

³W. W. Cleland (1995) *Meth. Enzymol.* **249**, 341.

NOTEBOOK ENTRIES

Although every experimentalist develops her/his own habits for recording research observations, maintenance of adequate and timely documentation is still insufficiently stressed. Exceptions to this statement may be cited, especially in industry where laboratory notebooks are read and endorsed on a regular schedule for proprietary motives. Next to the writing of monthly or quarterly progress reports, nothing seems at first glance more inef-

ficient than the humdrum recording of day-to-day observations. Nonetheless, laboratory notebooks and other supporting documentation are the stuff of science, and while there are no rules for keeping a laboratory notebook, the following are a few suggestions that can be used to good advantage.

(1) Chose a sturdy notebook, one that will survive your need to open and close it on innumerable occasions and one that will not burst when you have taped or pasted in chart paper tracings, scintillation counter output, chemical manufacturer's product information slips, graphs, *etc.* (Even superbly constructed laboratory notebooks can be damaged through normal use; so protect your notebook out of respect for its archival value.)

(2) Lab notebooks should be permanently bound, with all pages numbered. Reserve the first 3-4 pages for a fully annotated Table of Contents. This will speed up data retrieval when you're rushing to complete a required report or to beat the competition in submitting a manuscript for publication.

(3) Use a pen with permanent ink, and NOT a pencil, to make your entries. Date each entry, and be sure to put a date next to comments that you find necessary to add on a later occasion. (Such careful chronicling will allow you to recollect why you made certain changes in protocol or observations, especially those based on some later realization.)

(4) While you should keep your notebooks out of harm's way during an experiment, the lab notebook rightfully belongs at the bench, and not the desk. Record data directly into the book, rather than committing data to memory or temporarily using a sheet of paper (or worse, a paper towel) in place of the notebook itself.

(5) Always opt for recording primary data, because errors and assumptions can influence your reduction of primary data into their final report-ready form. In kinetic experiments, you should always try to avoid the use of unitless parameters, such as "relative activity", because the ratio of two activity measurements, if recorded in the absence of the originally measured rate values, obscures later analysis of your experiments.

(6) Describe your experiments in sufficient detail to ensure that others can replicate your findings. Avoid codes or your own jargon, unless you have clearly defined the term in writing. Many biochemical experiments involve many operations that, while combined in different ways, should be clearly described in their order of execution for each experiment.

(7) Get in the habit of using any hiatus, however brief, during your experiments to keep your notebook current. (Reward yourself by combining this activity with time-out for some refreshment or relaxing music.)

(8) Use flow charts and "time lines" as a convenient means for recording an experiment. You can frequently write out a better diagram (especially as your experiment proceeds long into the night) than you can record any paragraph of sentences. The following is an example of a time line describing the assay of coenzyme-A-linked acetaldehyde dehydrogenase (abbreviated AldDH). The horizontal axis is the time axis and vertical downward arrows are used to indicate the sequence of various operations and/or treatments.

MIX: (at 4°C in cuvette)

0.5 mL standard buffer

(for composition, see

p. 25 in Notebook)

including:

10 mM DTT

23 μ M CoASH

0.5 mM NADH

+

0.5 mL AldDH sample

(dialyzed overnight

versus 100 vols

standard buffer)

Start reaction with

1.0 mL 50 mM

acetaldehyde

in std. buffer at 28°

make duplicate

assay

15 min at 28°C
to activate enzyme

assay immediately
in SPEX Fluorolog
excite at 340 nm; read at 460 nm

(9) Note any unusual occurrences, even if they indicate some laziness or inattentiveness on your part. Despite all earnest attempts to keep an experiment from going awry, the complexity of most experiments conspires to introduce error or accidents into all observations. (It's far better to acknowledge an erroneous or suspicious measurement than to send later readers (including yourself) off on a series of experiments to follow up on a misleading observation that appears in your notebook without comment.)

(10) Record the source of the kinetic treatment (or some new derivation) directly into your notebook. Because kinetic measurements usually involve use of standard or newly derived rate equations, you should specify a complete list of assumptions and comments on the reasonableness of each assumption in your particular experiments.

(11) The source data for all published graphs and tables also should be marked in the notebook's Table of Contents. Likewise, after submitting the corrected version of your doctoral dissertation, the notebook's Table of Contents should be marked to indicate all figures and tables that appear in the dissertation. This will help your successors, should they wish to reanalyze a figure in your publications or dissertation.

(12) Finally, keep in mind that scientists must maintain high standards in conducting and reporting their findings. The following statements are excerpted from the 1989 report, entitled *On Being a Scientist*, published by the National Academy of Sciences' Committee on the Conduct of Science:

"Error caused by inherent limits on scientific theories can be discovered only through the gradual advancement of science, but error of a more human kind also occurs in science. Scientists are not infallible; nor do they have limitless working time or access to unlimited resources. Even the most responsible scientist can make an honest mistake. When such errors are discovered, they should be acknowledged, preferably in writing in the same journal in which the mistaken information was published. Scientists who make such acknowledgements promptly and graciously are not usually condemned by colleagues. Others can imagine making similar mistakes."

"Mistakes made while trying to do one's best are tolerated in science; mistakes made through negligent work are not. Haste, carelessness, inattention—any number of faults can lead to work that does not meet the standards demanded in science. In violating the methodological standards required by a discipline, a scientist damages not only his or her own work but the work of others. Furthermore, because the source of error may be hard to identify, sloppiness can cost years of effort, both for the scientist who makes the error and for others who try to build on that work."

The most recent version of *On Being a Scientist* is available over the internet at <http://www.nas.edu/nap/online/obas/>

NOYES EQUATION

An equation allowing an investigator to determine the chemical reaction order of a non-enzyme-catalyzed reaction and the rate expression for a non-first-order process by noting that half-lives for non-first-order reactions are dependent on the initial reactant concentration.

NU (ν)

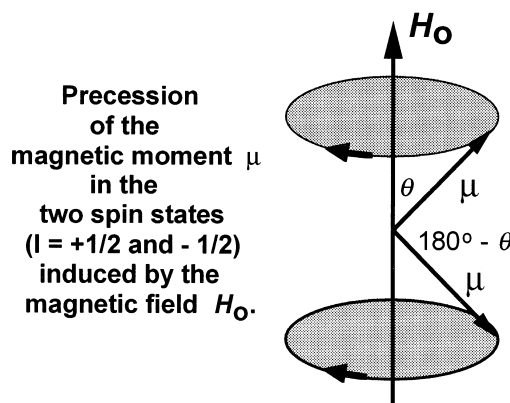
1. Symbol for frequency. 2. Symbol for stoichiometric number. 3. Symbol for kinematic viscosity. 4. Symbol (with overbar) for the number of moles of ligand bound per mole of macromolecule.

NUCLEARITY

The number of central atoms joined in a single coordination entity or cluster by bridging ligands or by metal-metal bonds. Such complexes are referred to as being dinuclear, trinuclear, tetranuclear, polynuclear, etc.

NUCLEAR MAGNETIC RESONANCE

Space constraints permit us only to consider the essence of the phenomenon known as nuclear magnetic resonance. An applied magnetic field H_0 interacts with the magnetic field generated by a nucleus (here, for simplicity, a proton with nuclear spin states of $+1/2$ and $-1/2$) having angular momentum as a consequence of its spinning motion. The spinning nucleus has a magnetic field strength, indicated by its magnetic moment μ which becomes directionally oriented with the applied magnetic field. Because the nucleus precesses, μ is oriented at angles θ and $(180^\circ - \theta)$ relative to the lines of force in an applied field (Fig. 1).



At angle θ , μ is aligned with the direction of the applied field, whereas at angle $(180^\circ - \theta)$, the opposite is true. For this reason, the respective potential energy of these two states is $-\mu H_o \sin \theta$ and $+\mu H_o \sin \theta$, resulting in a quantized energy difference ΔE given by the expression:

$$\Delta E = 2\mu H_o \sin \theta = h\nu$$

where h is Planck's constant. Thus, electromagnetic radiation with a frequency ν can be absorbed, thereby promoting the system to the higher energy state. Such energies correspond to photons in the radiofrequency range, and a radiofrequency generating coil is positioned around the sample within the cavity of the NMR spectrometer. NMR provides valuable information regarding molecular structure, and we can briefly consider two phenomena, namely chemical shift and spin-spin coupling.

A chemical shift is defined as a displacement in the magnetic resonance frequency of a nucleus as a consequence of the electronic environment in which the nucleus resides. Because moving electrons generate their own magnetic fields, a nucleus surrounded by these electrons experiences an effective field, H_{eff} , which is defined by $(1 - \alpha)H_o$, where α is the so-called screening constant and H_o is the applied magnetic field. A chemical shift is typically reported as a dimensionless displacement (units = parts per million, or simply ppm) from a reference standard. If the magnetic field is varied while the radiofrequency ν is held constant, then the chemical shift (ppm) equals $\{(H_{\text{sample}} - H_{\text{reference}})/H_{\text{reference}}\} \times 10^6$. If the radiofrequency is varied at constant magnetic field, the chemical shift (ppm) equals $\{(\nu_{\text{sample}} - \nu_{\text{reference}})/\nu_{\text{reference}}\} \times 10^6$. Tetramethylsilane is frequently used as a proton chemical shift reference compound in organic chemistry, because it yields a single sharp NMR signal in a region well removed from the resonance lines of most other protons.

One begins to sense the great versatility and power of NMR as a structural probe by considering so-called splitting patterns (or, more formally, absorption band multiplicities). These splitting patterns define the positions of nuclei within a molecule. Splitting is a consequence of reciprocal magnetic interactions between spinning nuclei, and the interaction likewise depends on electronic structure. Spin-spin coupling creates multiplets from

what would otherwise appear as sharp resonance lines, and the coupling strength J depends on the particular nuclei under examination. For example, in the case of n spin-coupled protons, the multiplet will have $n + 1$ lines. A single nearby proton will split the observed proton resonance into a doublet of equal or (1:1) intensity, two nearby electrons will split the observed proton resonance to a triplet with intensities of 1:2:1, and so on. The intensities are always coefficients of a binomial expansion. **See** *Pascal's Triangle*

So-called multidimensional NMR techniques can provide important information about macromolecular conformation. In these cases, the sequence of a protein is already known, and establishing covalent connectivity between atoms is not the goal. Rather, one seeks through-space information that can reveal the solution conformation of a protein or other macromolecule. Two- or three-dimensional techniques use pulses of radiation at different nuclear frequencies, and the response of the spin system is then recorded as a free-induction decay (FID). Techniques like COSY and NOESY allow one to deduce the structure of proteins with molecular weights less than 20,000–25,000.

Finally, as tabulated below, many elements having useful natural abundance and nuclear spin are naturally present in biomolecules. Still others can be substituted into biological molecules to provide a diverse range of opportunities. For example, fluorine can be substituted for hydrogen in many cases.

Table I
Several Biologically Important Nuclei for NMR Experiments

Nuclide	Natural Abundance	Spin
^1H	99.985%	$I = 1/2$
^2H	0.015%	$I = 1$
^{13}C	1.10%	$I = 1/2$
^{15}N	0.366%	$I = 1/2$
^{17}O	0.003%	$I = 5/2$
^{19}F	100%	$I = 1/2$
^{31}P	100%	$I = 1/2$

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Techniques: Proton nuclear magnetic resonance in aqueous solutions, **49**, 253; fluorine nuclear magnetic resonance studies of proteins, **49**, 270; stopped-flow nuclear magnetic resonance spectroscopy, **49**, 295; nuclear relaxation measurements of the geometry of enzyme-bound substrates and analogs, **49**, 322; nuclear magnetic resonance kinetics viewed as enzyme kinetics, **49**,

359; saturation-transfer spectroscopy, **49**, 480; one-dimensional nuclear Overhauser effects and peak intensity measurements, **176**, 3; two-dimensional nuclear magnetic resonance spectroscopy of proteins: an overview, **176**, 12; solvent suppression, **176**, 64; sample preparation, **176**, 78; heteronuclear nuclear magnetic resonance experiments for studies of protein conformation, **176**, 93; multiple-quantum nuclear magnetic resonance, **176**, 114; detection of insensitive nuclei, **176**, 134; homonuclear Hartmann-Hahn experiments, **176**, 151; two-dimensional nuclear Overhauser effect: complete relaxation matrix analysis, **176**, 169; selective relaxation techniques, **176**, 184; rotating-frame nuclear Overhauser effect, **176**, 199; modern spectrum analysis in nuclear magnetic resonance: alternatives to the Fourier transform, **176**, 216; solid-state nuclear magnetic resonance structural studies of proteins, **176**, 242; fluorine nuclear magnetic resonance of fluorinated ligands, **177**, 3; deuteration in protein proton magnetic resonance, **177**, 23; expression and nitrogen-15 labeling of proteins for proton and nitrogen-15 nuclear magnetic resonance, **177**, 44; isotopic labeling with hydrogen-2 and carbon-13 to compare conformations of proteins and mutants generated by site-directed mutagenesis, Part I, **177**, 74, Part II, **177**, 282; spin labeling of proteins, **177**, 86; practical introduction to theory and implementation of multinuclear, multidimensional nuclear magnetic resonance experiments, **239**, 3; measurement of homo- and heteronuclear J couplings from quantitative J correlation, **239**, 79; two-dimensional exchange spectroscopy of proteins, **239**, 106; pulsed-field gradients: theory and practice, **239**, 145; selective pulses and their applications to assignment and structure determination in nuclear magnetic resonance, **239**, 207; modifications of older model nuclear magnetic resonance console for collection of multinuclear, multidimensional spectral data, **239**, 247; using maximum likelihood spectral deconvolution in multidimensional nuclear magnetic resonance, **239**, 257; automated and semiautomated analysis of homo- and heteronuclear multidimensional nuclear magnetic resonance spectra of proteins: the program Pronto, **239**, 288; automated assignment of multidimensional nuclear magnetic resonance spectra, **239**, 308; quantitative information from complicated nuclear magnetic resonance spectra of biological macromolecules, **239**, 318

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Enzyme Mechanisms: NMR and EPR studies of chromium and cobalt nucleotides and their interactions with enzymes, **87**, 180; use of $^{31}\text{P}(\text{H}_2\text{O})$, $^{31}\text{P}(\text{D}_2\text{O})$, and ^{17}O NMR methods to study enzyme mechanisms involving phosphorus, **87**, 235; two-dimensional nuclear magnetic resonance studies of enzyme kinetics and metabolites *in vivo*, **176**, 330; isotope exchange, **176**, 342; phosphorus-31 nuclear magnetic resonance of enzyme complexes: bound ligand structure, dynamics, and environment, **177**, 295; ligand-protein interactions via nuclear magnetic resonance of quadrupolar nuclei, **177**, 317; ligand conformations and ligand-enzyme interactions as studied by the nuclear Overhauser effect, **177**, 333; determination of equilibrium constants of enzyme-bound reactants and products by nuclear magnetic resonance, **177**, 358; mechanistic studies utilizing oxygen-18 analyzed by carbon-13 and nitrogen-15 nuclear magnetic resonance spectroscopy, **177**, 376; positional isotope exchange using phosphorus-31 nuclear magnetic resonance, **177**, 390; paramagnetic probes of macromolecules, **177**, 403; enzyme regulation of metabolic flux, **177**, 417; monitoring intracellular metabolism by nuclear magnetic resonance, **177**, 435.

Protein-Ligand Interactions: Exchange processes and determination of ligand conformation and protein-ligand contacts, **238**, 657; nuclear magnetic studies of protein-peptide complexes, **238**, 700; nuclear magnetic resonance methods for studying protein-ligand complexes, **238**, 717; use of nuclear magnetic resonance in probing ligand-macromolecule interactions, **238**, 739.

DNA and RNA Structure: How to generate accurate solution structures of double-helical nucleic acid fragments using nuclear magnetic resonance and restrained molecular dynamics, **261**, 3; comparison of X-ray and NMR-determined nucleic acid structures, **261**, 45; distance geometry in NMR determination of solution conformation of nucleic acids: application of d-ACCGT-TAACGGT, **261**, 73; statistical analysis of DNA duplex structural features, **261**, 90; structure determination and analysis of local bending in an A-tract DNA duplex: comparison of results from crystallography, nuclear magnetic resonance, and molecular dynamics simulation on d(CGCAAAATGCG), **261**, 121; DNA mismatches and modified bases, **261**, 145; NMR studies of complex DNA structures: the Holliday junction intermediate in genetic recombination, **261**, 163; structural studies of DNA three-way junctions, **261**, 183; parallel-stranded duplex DNA: an NMR perspective, **261**, 207; ^1H NMR spectroscopy of DNA triplexes and quadruplexes, **261**, 225; solid-state NMR of DNA, **261**, 256; systems for the NMR study of modified nucleoside-dependent,

metal-ion induced conformational changes in nucleic acids, **261**, 270; preparation of isotopically enriched RNAs for heteronuclear NMR, **261**, 300; biochemical and NMR studies of RNA conformation with an emphasis on RNA pseudoknots, **261**, 323; multidimensional heteronuclear NMR experiments for structure determination of isotopically labeled RNA, **261**, 350

Nucleic Acid Dynamics: Studies of base pair kinetics by NMR measurement of proton exchange, **261**, 383; determination of fast dynamics of nucleic acids by NMR, **261**, 413; isotope labeling for ^{13}C relaxation measurements on RNA, **261**, 436; site-specific dynamics in DNA: theory and experiment, **261**, 451; NMR and nucleic acid-protein interactions: the lac repressor operator system, **261**, 513; induced structural changes in protein-DNA complexes, **261**, 524; uses of ^{13}C - and ^{15}N -labeled RNA in NMR of RNA-protein complexes, **261**, 542; studies of nucleic acids and their protein interactions by ^{19}F NMR, **261**, 560; NMR studies of drug-DNA complexes, **261**, 575.

NUCLEASE S1

This zinc-dependent endonuclease [EC 3.1.30.1] can act on either ribo- or deoxyribonucleic acids, with a preference for single-stranded substrates. It catalyzes the endonucleolytic cleavage of those nucleic acids to 5'-phosphomononucleotide and 5'-phosphooligonucleotide end-products.

I. R. Lehman (1981) *The Enzymes*, 3rd ed., **14**, 193.

NUCLEATION

Any highly cooperative chemical or physical process that serves to generate structures that then grow by linear accretion or elongation. Nucleation is an important phase in indefinite polymerization processes (e.g., actin filament and microtubule assembly). Nucleation is typically of a high kinetic order,

$$\text{Nucleation Rate} = k[\text{X}]^n$$

where n is the number of X subunits that must simultaneously collide to form the critical nucleus for further growth. In the case of actin assembly, n is around three. Once formed, these nuclei are frequently unstable and dissociate unless additional growth occurs. **See** *Protein Polymerization; Actin Assembly Kinetics; Microtubule Assembly Kinetics*

NUCLEOFUGE

A leaving group that departs with the bonding electron pair. The tendency of such a moiety to leave is referred to as nucleofugicity. **See also** *Electrofuge; Nucleophile*

NUCLEOPHILE

A molecular entity preferentially attracted to an electrophilic region or some other site of low electron density

in another molecule. A nucleophile typically brings a pair of electrons to the second molecular entity, thereby qualifying it as an electron-pair donor (or Lewis base).

See also *Electrophile*

NUCLEOPHILIC CATALYSIS

Catalysis by a Lewis base in which a Lewis adduct is a reaction intermediate.

NUCLEOPHILIC COMPETITION

A mechanistic strategy for detecting the presence of an electrophilic intermediate in a two-step enzyme-catalyzed hydrolysis reaction by introducing a second nucleophile to compete with water and thereby create a new reaction product. An illustrative case is catalysis of the *Escherichia coli* β -galactosidase reaction in the absence of the small subunit¹. Rate-determining hydrolysis of the glycosyl-enzyme intermediate was demonstrated by pre-steady-state measurements and nucleophilic competition, the latter accomplished by using methanol as the competitive nucleophile to form the corresponding methylglycoside.

¹S. V. Calugaru, B. G. Hall & M. L. Sinnott (1995) *Biochem. J.* **312**, 281.

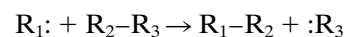
NUCLEOPHILICITY

1. The relative reactivity of a nucleophilic reagent. Nucleophilicity of a Lewis base is typically measured by relative rate constants of different reagents toward a common substrate. 2. The property of being nucleophilic.

See *Electrophilicity*

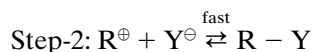
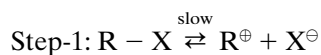
NUCLEOPHILIC SUBSTITUTION REACTION

A heterolytic reaction in which a nucleophilic reagent brings (i.e., attacks with) a free pair of electrons on a molecular entity.

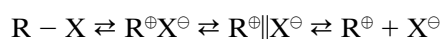


This pair of electrons forms a new bond, and the leaving group (the nucleofuge) exits with an electron pair.

Types of Nucleophilic Substitution Reactions. A reaction described as $\text{S}_{\text{N}}1$, short for substitution, nucleophilic (unimolecular), comprises two steps:



The first reaction is the slow, or rate-determining, ionization of the substrate to form a carbocation intermediate. The products of this first step will tend to be stabilized best in polar solvents. The S_N1 type of reaction can also lead to the formation of ion pair intermediates, as shown in the following reaction scheme:



where the first of the two intermediates is called an intimate ion pair, and the second is termed a loose ion pair. The intimate ion pair is apt to undergo recombination to form the original substrate, and this process is frequently termed internal return.

A reaction described as S_N2 , abbreviation for substitution, nucleophilic (bimolecular), is a one-step process, and no intermediate is formed. This reaction involves the so-called backside attack of a nucleophile Y on an electrophilic center RX , such that the reaction center (*i.e.*, the carbon or other atom attacked by the nucleophile) undergoes inversion of stereochemical configuration. In the transition-state nucleophile and exiphile (leaving group) reside at the reaction center. Aside from stereochemical issues, other evidence can be used to identify S_N2 reactions. First, because both nucleophile and substrate are involved in the rate-determining step, the reaction is second order overall: $\text{rate} = k[RX][Y]$. Moreover, one can use kinetic isotope effects to distinguish S_N1 and S_N2 cases (**See Kinetic Isotope Effects**).

A less frequent case is a reaction described as S_Ni , short for substitution, nucleophilic (internal). This reaction type involves a unimolecular process in which the substrate dissociates to form an intimate ion pair; however, no inversion of configuration occurs, presumably because steric hindrance forces the nucleophile to enter at the same side from which the leaving group departed.

NUCLEOSIDE DEOXYRIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.6] catalyzes the reaction of 2-deoxy-D-ribose-base¹ with base² to produce 2-deoxy-D-

ribose-base² and base¹ in which base¹ and base² represent various purines and pyrimidines.

W. S. McNutt (1955) *Meth. Enzymol.* **2**, 464.

NUCLEOSIDE DIPHOSPHATASE

This enzyme [EC 3.6.1.6] catalyzes the hydrolysis of a nucleoside diphosphate to produce a nucleotide (that is, a nucleoside monophosphate) and orthophosphate. NDP substrates include IDP, GDP, UDP, as well as D-ribose 5-diphosphate.

G. W. E. Plaut (1963) *Meth. Enzymol.* **6**, 231.

NUCLEOSIDE DIPHOSPHATE KINASE

This enzyme [EC 2.7.4.6], also known as nucleoside diphosphokinase and nucleoside 5'-diphosphate phosphotransferase, catalyzes the reaction of ATP with a nucleoside diphosphate to produce ADP and a nucleoside triphosphate. ATP can be substituted by a number of nucleoside triphosphate and deoxynucleoside triphosphate compounds.

N. B. Ray & C. K. Mathews (1992) *Curr. Top. Cell. Reg.* **33**, 343.

P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.

D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **64**, 3.

R. E. Parks, Jr., & R. P. Agarwal (1973) *The Enzymes*, 3rd ed., **8**, 307.

NUCLEOSIDE PHOSPHOTRANSFERASE

This enzyme [EC 2.7.1.77] catalyzes the reaction of a nucleotide with a 2'-deoxynucleoside to produce a nucleoside and a 2'-deoxynucleoside 5'-monophosphate. The nucleotide substrate can be substituted with phenyl phosphate and nucleoside 3'-phosphates, although they are not as effective.

P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

P. A. Frey (1989) *Adv. Enzymol.* **62**, 119.

NUCLEOSIDE TRIPHOSPHATE:HEXOSE-1-PHOSPHATE NUCLEOTIDYLTRANSFERASES

This enzyme [EC 2.7.7.28], also known as NDP-hexose pyrophosphorylase, catalyzes the reaction of a nucleoside triphosphate with a hexose 1-phosphate to produce a NDP-hexose and pyrophosphate (or, diphosphate). In the reverse reaction the NDP-hexose can be, in decreasing order of activity, guanosine, inosine, and adenosine diphosphate hexoses in which the sugar is either glucose or mannose.

K. A. Presper & E. C. Heath (1983) *The Enzymes*, 3rd ed., **16**, 449.

NUCLEOSIDE 5'-TRIPHOSPHATE REGENERATION

The free energy of hydrolysis ($\Delta G_{\text{hydrolysis}}$) of nucleoside 5'-triphosphate (NTP) is thermodynamically coupled to many biosynthetic reactions as well as polymerization processes, and nucleoside 5'-diphosphate (NDP) accumulates during the course of experiments. Proper design of kinetic studies on these systems requires maintenance of NTP concentration throughout the course of any measurements, and a NTP-regenerating reaction should be incorporated into all such reaction samples. The late Ephraim Racker first introduced the practice of supplementing a reaction sample with one or more auxiliary enzymes and metabolites to regenerate any NTP that had become converted to NDP.

Three enzymes are typically employed for this purpose: (a) creatine kinase, (b) acetate kinase, and (c) pyruvate kinase. Their NDP specificity, the nature of the co-substrate, and both the kinetic and thermodynamic properties of each of these enzyme-catalyzed reactions commends their use in NTP regeneration. Likewise, their commercial availability at high purity also factors into their widespread usage. These three enzyme systems are not interchangeable, and judicious choice of regenerating enzyme is to be strongly advised. In particular, the experimenter will wish to confirm that the regenerating enzyme and the cosubstrate do not possess any contaminating enzymatic activity or inhibitory/activating property that interferes with the primary reaction under investigation.

CREATINE KINASE. The creatine kinase reaction¹ can be written as follows:



The enzyme preferentially acts on adenine nucleotide, and the equilibrium constant for the reaction in the direction written above is approximately 40, depending on magnesium ion and proton concentrations. The enzyme is inhibited by 20 mM chloride ion. Typically, one uses 1–2 international units of the rabbit muscle enzyme along with 10 mM phosphocreatine.

ACETATE KINASE. MacNeal *et al.*² first proposed the use of *Escherichia coli* acetate kinase as a nearly ideal

system for ATP (or GTP) resynthesis from ADP (or GDP). The reaction catalyzed is:



or



Earlier work^{3,4} indicated that ADP and GDP were efficient phosphoryl-acceptor substrates. The equilibrium constant for these reactions is highly favorable ($K_{\text{eq}} = 3000$); thus, even when 90% of the initial acetyl phosphate concentration is converted to acetate, the $[\text{GTP}]/[\text{GDP}]$ or $[\text{ATP}]/[\text{ADP}]$ concentration ratio will still exceed 200–300, provided sufficient enzyme is present to maintain the equilibrium. Because acetate kinase and acetyl phosphate are not present in eukaryotes, one gains an added advantage over the above-mentioned regenerating systems: there are no contaminating levels of either even in crude animal tissue extracts. Use of acetate kinase and acetyl phosphate is highly recommended. The lithium salt of acetyl phosphate is stable for long-term storage in a desiccator at freezer temperatures.

For most reactions, one can use 1–2 mM acetyl phosphate and 1–2 international units of acetate kinase. The enzyme is usually supplied as a crystalline suspension in 3 M ammonium sulfate, and 10 mM ammonium ion is inhibitory. Therefore, a useful practice is to snip off 0.5 cm from a disposable Eppendorf micropipette tip to facilitate removal of 10–20 microliters of the crystalline suspension; then spin down the enzyme in a 1.5 ml disposable conical plastic centrifuge tube, and remove the ammonium sulfate solution with a wick of twisted Kimwipe. The enzyme precipitate can now be taken up directly into your working buffer. *Note:* Acetate kinase is inactivated by cold exposure, but incubation with 10^{-4} M ATP or GTP reactivates the enzyme if warmed to room temperature for 5–10 min.

PYRUVATE KINASE. This enzyme catalyzes phosphoryl transfer from the glycolytic intermediate phosphoenolpyruvate:



The pyruvate kinase reaction is virtually irreversible under typical intracellular conditions⁵, and the enzyme requires Mg^{2+} or Mn^{2+} in addition to the monovalent cation

K^+ . Pyruvate kinase is inhibited by 3-5 mM ATP, as well as by acetyl-S-CoA and long-chain fatty acids. Rabbit muscle pyruvate kinase is typically used in ATP regeneration.

REGENERATION OF NTP'S OTHER THAN ATP OR GTP. Although most experiments require regeneration of either ATP or GTP, occasionally one may wish to maintain the phosphorylation state of other nucleotides. This is readily accomplished by using acetate kinase in combination with nucleoside-5'-diphosphate kinase (NDPK) in the presence of ATP and acetyl phosphate. For the sake of exemplifying this method, let us consider the regeneration of UTP from UDP. The overall equilibrium constant for rephosphorylation of UDP to UTP is the same as the acetate kinase equilibrium constant, because the equilibrium constant for the NDPK reaction is approximately one. Typically, one can include 2-4 international units of commercially available yeast NDPK along with 0.5 mM ATP in the reaction mixture.

¹F. B. Rudolph, B. W. Baugher & R. S. Beissner (1979) *Meth. Enzymol.* **63**, 22.

²R. K. MacNeal, B. C. Webb & D. L. Purich (1977) *Biochem. Biophys. Res. Commun.* **74**, 440.

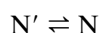
³R. S. Anthony & L. B. Spector (1972) *J. Biol. Chem.* **247**, 2120.

⁴B. C. Webb, J. A. Todhunter & D. L. Purich (1976) *Arch. Biochem. Biophys.* **173**, 282.

⁵D. E. Metzler (1976) *Biochemistry: The Chemical Reactions of Living Cells*, p. 401, Academic Press, New York.

NUCLEOSOME CORE PARTICLE SELF-ASSEMBLY

The self-assembly of this fundamental building block of chromatin is a topic of enduring interest. DNase I digestion experiments as well as spectroscopic studies indicate that nucleosome core particles can be reconstituted by salt-jump (*i.e.*, diluting NaCl concentration from 2.0 to 0.2 M) or by direct mixing of histones and DNA at the lower salt concentration. Daban and Cantor¹ used the increase in eximer fluorescence to investigate the reassembly process in terms of a two-state model:

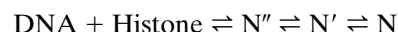


and

$$K_{\text{obs}} = [N]/[N'] = (F_e - F_{N'})/(F_N - F_e)$$

where K_{obs} is the equilibrium constant between the N and N' states, F_e is the fluorescence intensity of the equilibrium mixture, $F_{N'}$ is the fluorescence intensity of

N', and F_N is the fluorescence intensity of the N state. Their studies actually produced evidence favoring a slightly more complicated scheme:



where DNA and histone bind rapidly and a two-exponential rate equation gave values of 10 s^{-1} and $7 \times 10^{-3} \text{ s}^{-1}$ for the N''-to-N' and N'-to-N steps, respectively. From the concentration dependence of the equilibrium constant, Daban and Cantor¹ suggest that about 20 NaCl molecules are involved in the process.

In a companion study, Daban and Cantor² proposed a role for histone pairs H2A,H2B and H3,H4 in the self-assembly of nucleosome core particles. A modified mechanism for *in vitro* nucleosome core particle formation was offered later by Aragay *et al.*³ who probed the interaction of different histone oligomers with nucleosomes by using nondenaturing gel electrophoresis. Their experimental results indicate that H2A,H2B, H3,H4, and the four core histones can migrate spontaneously from the aggregated nucleosomes containing excess histones to free core DNA. Moreover, on the basis of electron microscopy, solubility, and supercoiling assays, transfer of excess histones from oligonucleosomes to free circular DNA can be demonstrated. Their results suggest that nucleosome core particles can be formed in 0.2 M NaCl by the following mechanisms: (1) by transfer of excess core histones from oligonucleosomes of free DNA; (2a) by transfer to excess H2A,H2B and H3,H4 associated separately with oligonucleosomes to free DNA or (2b) by transfer to excess H2A,H2B initially associated with oligonucleosomes to DNA, followed by the reaction of the resulting DNA-(H2A,H2B) complex with oligonucleosomes containing excess H3,H4, and (3) a two-step transfer reaction similar to that indicated in (2b), in which excess histones H3,H4 are transferred to DNA before the reaction with oligonucleosomes containing excess H2A,H2B. Samso and Danban⁴ recently described synchrotron X-ray scattering data on core DNA-(H2A,H2B) complexes, using different histone to DNA weight ratios and ionic conditions ranging from very low ionic strength to 0.2 M NaCl to show that histones H2A,H2B are unable to fold core DNA. Reconstituted complexes prepared at a physiological salt concentration either with the four core histones or with histones H3,H4 without H2A,H2B were found to be completely folded

particles with a radius of gyration that is similar to that of native nucleosome core. They found that the DNA of the extended complexes containing histones H2A,H2B becomes completely folded after the histone pair exchange reaction which occurs spontaneously between preformed DNA-(H2A,H2B) and DNA-(H3,H4) complexes.

¹J.-R. Daban & C. R. Cantor (1982) *J. Mol. Biol.* **156**, 749.

²J.-R. Daban & C. R. Cantor (1982) *J. Mol. Biol.* **156**, 771.

³A. M. Aragay, X. Fernandez-Busquets & J. R. Daban (1991) *Biochemistry* **30**, 5022.

⁴M. Samso & J. R. Daban (1993) *Biochemistry* **32**, 4609.

3'-NUCLEOTIDASE

This enzyme [EC 3.1.3.6] catalyzes the hydrolysis of a 3'-ribonucleotide to produce a ribonucleoside and orthophosphate. The enzyme exhibits a wide specificity for the 3'-nucleotides.

G. I. Drummond & Y. Yamamoto (1971) *The Enzymes*, 3rd ed., **4**, 337.

5'-NUCLEOTIDASE

This enzyme [EC 3.1.3.5] catalyzes the hydrolysis of a 5'-ribonucleotide to produce a ribonucleoside and orthophosphate. The enzyme exhibits a wide specificity for the 5'-nucleotides.

G. I. Drummond & Y. Yamamoto (1971) *The Enzymes*, 3rd ed., **4**, 337.

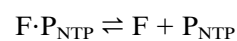
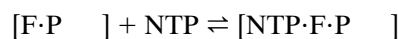
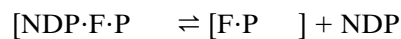
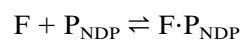
NUCLEOTIDE EXCHANGE FACTORS

Proteins that catalyze the exchange of protein-bound nucleotide with nucleotide free in solution. These exchange factors represent an unclassified type of isomerase, but there has been no Enzyme Commission of this class.

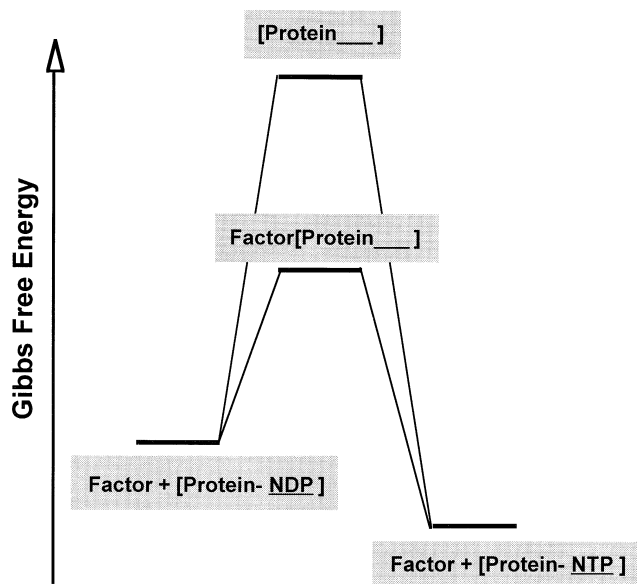
Several well-known regulatory¹⁻³ and cytoskeletal⁴ proteins carry out the metabolic functions as "two-state" proteins. Binding of a nucleoside 5'-triphosphate (NTP) induces one conformational state and binding the corresponding nucleoside 5'-diphosphate (NDP) produces a second state. In such models, only one form is active, and the free energy of NTP hydrolysis drives the formation of the second NDP-bound state. The rate of hydrolysis serves as a metabolic clock that sets the delay in the conversion from NTP-bound to NDP-bound forms. Moreover, rebinding of NTP is needed to reload the regulatory protein, and this requires nucleotide exchange which can occur spontaneously, albeit slowly, even in the absence of a nucleotide exchange factor. Cells achieve a

more robust range of responses through the biosynthesis of nucleotide exchange factors that accelerate exchange in response to other metabolic signals. The GTP-binding proteins in signal transduction cascades operate through the use of low-molecular-weight proteins that act as exchange factors. Likewise, the high rates of intracellular actin filament assembly/disassembly may lead to rate-limiting exchange of actin-bound ADP with cytosolic ATP to regenerate actin-ATP complex, the most active polymerizing form of this cytoskeletal protein. The 15 kDa protein profilin may fulfill this role in promoting actin-based motility. **See Actin-Based Motility**

How then might one represent the facilitated exchange process? The facilitation of nucleotide exchange is in all respects analogous to an enzyme-catalyzed reaction (*i.e.*, the rate of a reaction is accelerated without altering the equilibrium position of the reaction). In this way, one can represent the facilitated nucleotide exchange reaction as one would any enzymic process:



where F is the factor, and P is the protein in the various nucleotide-bound and nucleotide-free forms.



The diagram above also illustrates how the nucleotide-free form (indicated as [Protein___]) is stabilized by binding of the protein to the exchange factor. The nucleotide-free form is analogous to the transition state, and it lies at higher free energy (*i.e.*, it is less stable) than the bound NTP and NDP forms. Note also that the concentrations of NTP and/or NDP in the solution serve as a driving force favoring binding, as opposed to unbound forms, of the regulatory protein. Note also that the energy change from the P_{NDP} state to the P_{NTP} state is unaffected by the factor in this model.

¹R. D. Vale (1996) *J. Cell Biol.* **135**, 291.

²A. G. Gilman (1987) *Ann. Rev. Biochem.* **56**, 615.

³L. Stryer (1991) *J. Biol. Chem.* **266**, 10711.

⁴D. L. Purich & J. M. Angelastro (1994) *Adv. Enzymol.* **69**, 121.

NUMERICAL COMPUTER METHODS

Enzyme kinetic data has been routinely presented in the form of variously named plots (*e.g.*, Lineweaver-Burk plot, Eadie-Hofstee plot, *etc.*), and the same is true for ligand binding (*e.g.*, Scatchard plot, Hill plot, *etc.*). With the advent of inexpensive digital computers, one can make strong arguments that direct linear plots (such as Initial Rate *versus* [Substrate] or Degree of Saturation *versus* [Ligand]) are more reliable and easier to treat statistically. The entries presented below illustrate how numerical computer methods can provide better ways for analyzing rate and equilibrium binding data.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Parameter estimation by least-squares methods, **210**, 1; global analysis of biochemical and biophysical data, **210**, 37; use of weighting functions in data fitting, **210**, 68; analysis of residuals: criteria for determining goodness-of-fit, **210**, 87; analysis of ligand-binding data with experimental uncertainties in independent variables, **210**, 106; Monte Carlo method for determining complete confidence probability distributions of estimated model parameters, **210**, 117; method of moments and treatment of nonrandom error, **210**, 237; interpolation methods, **210**, 305; practical aspects of kinetic analysis, **210**, 374; compartmental analysis of enzyme-catalyzed reactions, **210**, 391; numerical analysis of binding data: advantages, practical aspects, and implications, **210**, 481; oxygen equilibrium curve of concentrated hemoglobin, **232**, 486; Adair fitting to oxygen equilibrium curves of hemoglobin, **232**, 559; weighted nonlinear regression analysis of highly cooperative oxygen equilibrium curves, **232**, 576; dimer-tetramer equilibrium in Adair fitting, **232**, 597; effects of wavelength on fitting Adair constants for binding of oxygen to human hemoglobin, **232**, 606; Adair equation: rederiving oxygenation parameters, **232**, 632; linkage thermodynamics, **232**, 655; simulation of hemoglobin kinetics using finite element numerical meth-

ods, **232**, 517; least-squares techniques in biochemistry, **240**, 1; parameter estimates from nonlinear models, **240**, 23; sequential versus simultaneous analysis of data: differences in reliability of derived quantitative conclusions, **240**, 89; model-independent quantification of measurement error: empirical estimation of discrete variance function profiles based on standard curves, **240**, 121; impact of variance function estimation in regression and calibration, **240**, 150; application of Kalman filter to computational problems in statistics, **240**, 171; modeling chemical reactions: Jacobian paradigm and related issues, **240**, 181; the mathematics of biological oscillators, **240**, 198; analysis of diffusion-modulated energy transfer and quenching by numerical integration of diffusion equation in Laplace space, **240**, 216; applications of KinSim and FitSim computer simulation and fitting programs, **240**, 311; determination of rate and equilibrium binding constants for macromolecular interactions by surface plasmon resonance, **240**, 323; characterization of enzyme-complex formation by analysis of nuclear magnetic resonance line shapes, **240**, 438; analysis of drug-DNA binding isotherms: a Monte Carlo approach, **240**, 593; estimating binding constants for site-specific interactions between monovalent ions and proteins, **240**, 645; applying bifurcation theory to enzyme kinetics, **240**, 781.

NUMERICAL INTEGRATION

Any mathematical transformation that maps a function ($t = 0, q_0 \rightarrow (t = x, q_x)$) and is then iterated to map ($t = x, q_x \rightarrow (t = x + y, q_x + y)$), and so forth to solve for the time evolution of an initial value problem.

Although one cannot obtain a general solution for solving the initial value problem

$$dy/dt = f(t, y) \quad \text{where} \quad y(t_0) = y_0$$

there are a number of numerical procedures. These methods allow one to obtain reaction progress curves for kinetic systems that are characterized by one or more first-order differential equations. While a detailed consideration of these methods lies beyond the scope of this Handbook, many engineering mathematics textbooks describe the subject. A most useful reference by Braun¹ describes some of the most widely used algorithms, including the Euler method, the Runge-Kutta procedure, the three-term Taylor series method, and an improved Euler method. Also presented are short-statement Fortran and Pascal programs for these techniques, along with some practical advice. **See** *Gear Algorithm; Numerical Computer Methods; Stiffness*

¹M. Braun (1991) *Differential Equations and Their Applications: An Introduction to Applied Mathematics*, pp. 96-126, Springer-Verlag, New York.



OCCAM'S RAZOR

The concept (known also as the Principle of Parsimony) grounded in scientific logic and stating that, in the absence of evidence to the contrary, one should always adopt the simplest explanation or model¹. The metaphor “razor” (probably introduced by the French logician Condillac) is used to indicate a sharp and swift device that one can use to cut away and dispose of unneeded complexity. First advanced by William of Occam (1285–1349), this philosophic strategy tacitly assumes that Nature is straightforward, rarely wasting even a gesture^{2,3}. This principle is of particular use in biochemical kinetics where the temptation to adopt complexity unnecessarily is to be resisted. **See Model**

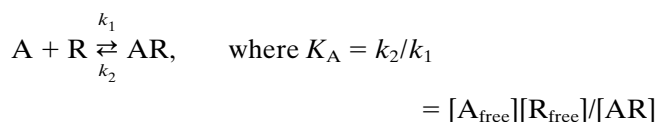
¹Stated in the Latin, “*Entia non sunt multiplicanda præter necessitate*,” or in English, “Entities should not be multiplied unnecessarily.”

²Such being evident from numerous examples, one should not assume that Nature has achieved perfection in the elimination of inefficiency. Evolution works through accumulated advantage, the product of which may only be sufficiently adapted, while still appearing to be perfectly suited for a particular purpose.

³E. A. Moody (1967) in *The Encyclopedia of Philosophy* (P. Edwards, ed.) vol. 8, pp. 306-317, Macmillan, New York.

OCCUPANCY THEORY OF DRUG ACTION

A mass action treatment of drug binding at a binding site on a receptor, such that the degree of biological response (either positive or negative) to drug binding is proportional to the drug's fractional occupancy of the receptor. Consider the binding of drug A to receptor R, with the formation of an AR complex:



in which k_1 and k_2 are the respective bimolecular and unimolecular rate constants for complex formation and

dissociation, respectively. At equilibrium, we can use K_A to define the right-hand expression, where $[A_{\text{free}}]$ is the concentration of uncomplexed drug, $[R_{\text{free}}]$ is the concentration of uncomplexed receptor, and $[AR]$ is the concentration of drug-receptor complex. If the total drug concentration, $[A_t]$, greatly exceeds the total receptor concentration, $[R_{\text{tot}}]$, then $[A_{\text{tot}}] \approx [A_{\text{free}}]$, whereas $[R_{\text{tot}}] = [R_{\text{free}}] + [AR]$. From these relationships, we get:

$$[AR]/[R_{\text{tot}}] = [A_{\text{free}}]/([A_{\text{free}}] + K_A)$$

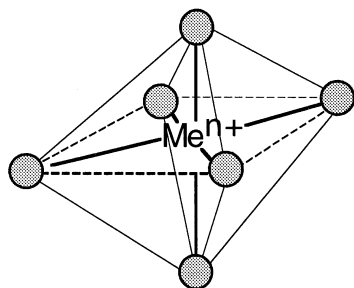
If the biological response elicited by drug binding is directly proportional to occupancy, then

$$E_a/E_m = [A_{\text{free}}]/([A_{\text{free}}] + K_A)$$

where E_a is the effect observed as AR is formed, and E_m is the maximal effect occurring at full occupancy of available sites R_t . Typical dose-response (or more correctly, concentration-response) curves show hyperbolic functionality. This behavior only applies if $[A_t] \gg [R_t]$ and the drug displays either rapid equilibrium or steady-state binding to the receptor. Very tight-binding or slow-binding drugs will not give the concentration-response curves shown above. Furthermore, drugs may not elicit just one effect, and side effects are common for almost all. If more than one receptor binds a particular drug, then the drug's concentration-response curve is likely to be more complicated than that shown above. Other confounding variables include: (a) nonuniform access of the drug to its receptor (**See Diffusion of Ligand to Receptor**); (b) cooperativity in drug binding, such that a single equilibrium cannot accurately account for the binding process; and (c) failure of the system to obey equilibrium or steady-state conditions required in deriving the mass action equilibrium.

OCTAHEDRAL COORDINATION

The eight-sided geometric solid corresponding to a hexacoordinate metal-ligand complex, as observed with ferricyanide $[\text{Fe(III)CN}_6]^{3-}$ or calcium ion complex with EGTA.



The six coordination sites are generated by hybridization of two $3d$, one $4s$, and three $4p$ orbitals to yield six equivalent d^2sp^3 orbitals, each situated at the corners of a regular octahedron.

OCTANOL DEHYDROGENASE

This enzyme [EC 1.1.1.73] will catalyze the reaction of 1-octanol with NAD^+ to produce 1-octanal and NADH. The enzyme can act on other long-chain alcohols, albeit not as effectively.

M. Ueda & A. Tanaka (1990) *Meth. Enzymol.* **188**, 171.

trans-OCTAPRENYLTRANSTRANSFERASE

This enzyme [EC 2.5.1.11], also known as nonaprenyl-diphosphate synthase and solanesyl-diphosphate synthase, catalyzes the reaction of all-*trans*-octaprenyl diphosphate with isopentenyl diphosphate to produce all-*trans*-nonaprenyl diphosphate and pyrophosphate. The enzyme will also utilize geranyl diphosphate and all-*trans*-prenyl diphosphates of intermediate size as substrates, but it will not use dimethylallyl diphosphate.

H. Kleinig (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 39.
E. D. Beytia & J. W. Porter (1976) *Ann. Rev. Biochem.* **45**, 113.

OCTOPINE DEHYDROGENASE

This enzyme [EC 1.5.1.11], also known as D-octopine synthase, catalyzes the reversible reaction of N^2 -(D-1-carboxyethyl)-L-arginine with NAD^+ and water to produce L-arginine, pyruvate, and NADH. The enzyme, in the reverse direction, will also act on L-ornithine, L-lysine, and L-histidine.

J. Thompson & J. A. Donkersloot (1992) *Ann. Rev. Biochem.* **61**, 517.
J. Thompson & S. P. F. Miller (1991) *Adv. Enzymol.* **64**, 317.
H. F. Fisher (1988) *Adv. Enzymol.* **61**, 1.

OFFORD PLOT

A plot of the logarithm of electrophoretic mobility of peptides as a function of molecular weight. Parallel lines are obtained with peptides having different net charges.

R. E. Offord (1977) *Meth. Enzymol.* **47**, 51.

OHM'S LAW

$I = E/R$ where I is the current, E is the electromotive force, and R is the resistance. The SI units for each of these is amperes, volts, and ohms, respectively. Ohm's law is also expressed as $I = \Delta F/R$ where ΔF is the difference in electric potential. The resistance is dependent upon the dimensions of the conductor.

2',5'-OLIGOADENYLATE SYNTHETASE

The following are recent reviews on the molecular and physical properties of this ligase (or, 2',5'-oligo(A) polymerase or oligo-isoadenylate synthetase E).

A. V. Itkes & E. S. Severin (1987) *Adv. Enzymol.* **59**, 213.
P. Lengyel (1982) *Ann. Rev. Biochem.* **51**, 251.
L. A. Ball (1982) *The Enzymes*, 3rd ed., **15**, 281.

OLIGOMERIZATION

The formation of multisubunit complexes (for example, protein complexes). Such complex formation is usually a cooperative phenomenon.

OMEGA (Ω or ω)

Ω Symbol for ohm, the SI unit of electrical resistance.
 ω 1. Symbol for solid angle. 2. Symbol for circular frequency. 3. Symbol for angular velocity.

OMPTIN

This serine-protease [EC 3.4.21.87] catalyzes the hydrolysis of peptide bonds at Xaa—Yaa in which there is a preference for arginyl or lysyl residues at Xaa and Yaa.

W. F. Mangel, D. L. Toleda, M. T. Brown, K. Worzalla, M. Lee & J. J. Dunn (1994) *Meth. Enzymol.* **244**, 384.

OPEN SYSTEM

Any chemical, biological, or mechanical system in which both heat and matter within the system can be transferred to and from its surrounding environment. Thus, a change in the immediate environment can result in a change in the system under consideration. **See also** *Closed System*; *Isolated System*

OPTICAL ACTIVITY

The property of a substance, molecular entity, or material to rotate the plane of polarized light as that light passes through a crystal, liquid, or solution containing that substance. The degree of rotation is dependent on a number of factors (*e.g.*, concentration, temperature, pressure, *etc.*) as well as on the wavelength of the light.

OPTICAL DENSITY

A now archaic synonym, abbreviated by O.D., for absorbance or turbidity. **See** *Absorbance*; *Turbidity*; *Absorption Spectroscopy*

OPTICAL ROTATION

The change, measured in terms of an angle α , of the plane of polarized light of a given wavelength passing through an optically active substance. The specific optical rotation, symbolized by $[\alpha]$, is the angle that the plane of polarized light has changed upon passing through a solution of one gram of an optically active substance per milliliter of solution having a path length of one decimeter, using light of wavelength equal to the D line of sodium. Optical rotation is a useful property in structure studies and has often proved valuable in kinetics as well.

OPTICAL TWEEZERS

A force-measuring technique that uses the photic energy of a tightly focused laser beam to “grab” hold of small particles through momentum transfers from impinging photons, the wavelength of which must be in nonabsorbing region of the spectrum. No mechanical contact with the object is required. In most biological applications, manipulated objects are large compared to the wavelength of light, and the resultant forces are explained in terms of principles of momentum conservation arising as a consequence of the wave-particle duality of light. The beam of light is a stream of photons that exerts what has been termed “radiation pressure” which can stop organelle movement when appropriately oriented relative to direction of force exerted by molecular motors.

Ashkin and Dziedzic¹ first reported the optical trapping and manipulation of viruses and bacteria by laser radiation pressure with single-beam gradient traps. Individual tobacco mosaic viruses and dense oriented arrays of viruses were trapped in aqueous solution with no apparent damage, even at ~120 milliwatts of argon laser power. Trapping and manipulation of individual motile bacteria were also demonstrated in a high-resolution microscope at powers of a few milliwatts. Ashkin *et al.*² also introduced the use of infrared (IR) light to greatly improve laser traps with much less optical consequential damage to living cells. Ashkin *et al.*³ later examined organelle transport along microtubules and estimated the force generated by a single molecular motor to be roughly

$2-3 \times 10^{-7}$ dynes. **See** *Force Effects on Molecular Motors*

¹A. Ashkin & J. M. Dziedzic (1987) *Science* **235**, 1517.

²A. Ashkin A, J. M. Dziedzic & T. Yamane (1987) *Nature* **330**, 769.

³A. Ashkin, K. Schutze, J. M. Dziedzic, U. Euteneuer & M. Schliwa (1990) *Nature* **348**, 346.

OPTICAL YIELD

The ratio of the optical purity of the product of a reaction (involving chiral reagents and products) to the optical purity of the product's precursor. The optical yield of a reaction is not related to the chemical yield.

OPTIMAL pH

The pH at which a particular reaction or process is most effective; often symbolized by pH_{opt} . In most instances, the optimum pH value for an enzyme-catalyzed reaction or biological process will fall within normal physiological pH ranges. However, when the pH_{opt} value is not within physiological ranges, then a number of possibilities are worth considering:

- (1) The specific assay conditions have altered or modified the protein(s) or system such that the experimentally observed pH_{opt} is not the physiological pH_{opt} . For example, some component may be missing from the reaction mixture, a component that is present in the reaction mixture (and not present in physiological contexts) has altered the system, a nonphysiological ionic strength or temperature has altered the pH_{opt} , *etc.* **See** *Anion Effects on pH-Rate Data*
- (2) The physiological pH value is not the pH value experienced by the *in vivo* system. The *in vivo* protein is optimally active in a pH range present in a localized volume (or, compartment) that is different than the pH range of the larger physiological environment.
- (3) The protein(s) is relatively unstable at its true pH_{opt} , and this lack of stability has not been corrected in the pH-activity plot. Thus, the observed pH_{opt} is a compromise of the effect of pH on both catalytic activity (under the assay conditions) and protein denaturation and/or conformation.
- (4) The pH_{opt} may not reflect the effect of pH on protein activity (or on biological processes) but may be a result of the effect of pH on the ionization(s) of the substrate(s). This may be particularly true when utilizing artificial substrates having pK_a value(s) significantly dif-

ferent that the naturally occurring substrate's pK_a value(s). Note that it is not uncommon that use of different substrates will result in different pH-activity curves.

(5) The observed pH_{opt} is identical to the physiological pH_{opt} even though it lies outside the normal pH range. If such a situation is true, then solution acidity becomes an important regulatory control of the protein's biological activity.

ORBITAL STEERING

A model attempting to account for the remarkable rate enhancements observed in enzyme catalysis in terms of an apparent reduction in reaction order and the optimal alignment of molecular orbitals to promote reactivity^{1,2}. This model was introduced at a time when Woodward & Hoffmann³ formalized their ideas for predicting the nature of organochemical reactions using molecular orbital calculations. Proponents of orbital steering argued that the angular orientation of each reactant could be restricted to sufficiently small angles, such that rate enhancements of eight orders of magnitude would be accessible. The model drew almost immediate criticism⁴ on the grounds that the proposed angular orientation would lead to unfavorable thermodynamics (*i.e.*, too much energy would be required to reduce the rotational degrees of freedom to give the needed angular orientation). Page and Jencks^{5,6} also estimated the magnitude of rate enhancements achieved by bringing two reactants from the freedom of the bulk solution to form what amounts to a unimolecular complex on the enzyme's surface. Hackney⁷ has presented a useful account of the orbital steering model in his review on the role of binding energy in promoting enzyme catalysis. We should note that while the orbital steering hypothesis remains out of favor, the merit of this proposal will only be known when it is possible to assess quantitatively how much binding energy plays a role in distorting the ground-state ES complex to reach its transition-state configuration. There is still much work needed to establish the energetics of complementarity of the enzyme and its transition state.

¹D. R. Storm & D. E. Koshland, Jr. (1970) *Proc. Natl. Acad. Sci. U.S.A.* **66**, 445; (1972) *J. Am. Chem. Soc.* **94**, 5805.

²A. Dafforn & D. E. Koshland, Jr. (1971) *Bioorg. Chem.* **1**, 129; *Proc. Natl. Acad. Sci. U.S.A.* **66**, 445.

³R. B. Woodward & R. Hoffmann (1969) *Angew. Chem. Int. Ed. Engl.* **8**, 781.

⁴T. C. Bruice, A. Brown & D. O. Harris (1971) *Proc. Natl. Acad. Sci. U.S.A.* **68**, 658.

⁵M. I. Page & W. P. Jencks (1971) *Proc. Natl. Acad. Sci. U.S.A.* **68**, 1678.

⁶M. Page (1973) *Chem. Soc. Revs.* **2**, 295.

⁷D. Hackney (1990) *The Enzymes*, 3rd ed., **19**, 1.

ORBITAL SYMMETRY CONSIDERATIONS

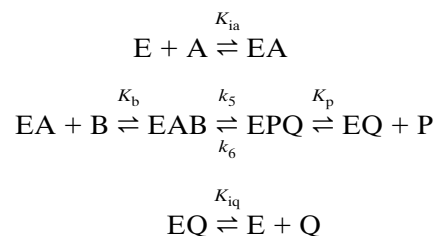
The interpretation of chemical reactivity in terms of molecular orbital symmetry. The central principle is that orbital symmetry is conserved in concerted reactions. An orbital must retain a certain symmetry element (for example, a reflection plane) during the course of a molecular reorganization in concerted reactions. It should be emphasized that orbital-symmetry rules (also referred to as Woodward-Hoffmann rules) apply only to concerted reactions. The rules are very useful in characterizing which types of reactions are likely to occur under thermal or photochemical conditions. Examples of reactions governed by orbital symmetry restrictions include cycloaddition reactions and pericyclic reactions.

R. B. Woodward & R. Hoffmann (1969) *Angew. Chem. Int. Ed. (Engl.)* **8**, 781.

ORDERED BI BI ENZYME MECHANISM

A sequential enzyme-catalyzed reaction mechanism in which two substrates react to form two products and in which there is a preferred order in the binding of substrates and release of products. Several enzymes have been reported to have this type of binding mechanism, including alcohol dehydrogenase¹, carbamate kinase², lactate dehydrogenase³⁻⁵, and ribitol dehydrogenase⁶.

The Rapid Equilibrium Mechanism: In this scheme, the rate-determining step is the interconversion of the central complex, such that all binding steps can be described via dissociation constants.



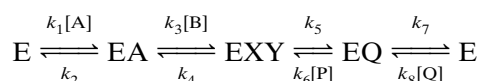
Under initial rate conditions (*i.e.*, $[P]$ and $[Q] \approx 0$) and no abortive complexes being formed, the initial-rate expression is

$$v = V_{max}[A][B]/\{K_{ia}K_b + K_b[A] + [A][B]\}$$

where $V_{max} = k_5[E_{total}]$, $K_{ia} = [E][A]/[EA]$, and $K_b = [EA][B]/[EAB]$. When $1/v$ is plotted as a function of

$1/[A]$ at different constant concentrations of B, a series of straight lines will be obtained having a common intersection point. The intersection point will always be in the second quadrant. A double-reciprocal plot of $1/v$ vs. $1/[B]$ at different constant concentrations of A will consist of a series of straight lines all intersecting on the vertical axis. This observation is characteristic of the rapid equilibrium ordered system but not of the steady-state scheme.

The Steady-State Mechanism: If the interconversion step is not the sole rate-determining step and binding steps are not in rapid equilibrium, then a steady-state description of the reaction is applicable. If there is a single ternary complex, EXY, (thus, $[EXY] = [EAB] + [EPQ]$), the scheme can be depicted as



The steady-state determinants are:

$$[E] = k_2k_7(k_4 + k_5) + k_3k_5k_7[B] + k_2k_4k_6[P]$$

$$[EA] = k_1k_7(k_4 + k_5)[A] + k_1k_4k_6[A][P] + k_4k_6k_8[P][Q]$$

$$[EXY] = k_1k_3k_7[A][B] + k_2k_6k_8[P][Q] + k_1k_3k_6[A][B][P] + k_3k_6k_8[B][P][Q]$$

$$[EQ] = k_2k_8(k_4 + k_5)[Q] + k_1k_3k_5[A][B] + k_3k_5k_8[B][Q]$$

A double-reciprocal plot of $1/v$ vs. $1/[A]$ at varying constant concentrations of B will be a series of straight lines having a common intersection point. A double-reciprocal plot of $1/v$ vs. $1/[B]$ at varying constant concentrations of A will also be a series of straight lines having a common intersection point. Secondary replots of the initial-rate data will yield values for the kinetic parameters. The overall steady-state rate expression is

$$v =$$

$$\frac{V_{\max,f}V_{\max,r}([A][B] - [P][Q]/K_{eq})}{\left\{ V_{\max,r}K_{ia}K_b + V_{\max,r}K_b[A] + V_{\max,r}K_a[B] + V_{\max,r}[A][B] + \frac{V_{\max,f}K_q[P]}{K_{eq}} + \frac{V_{\max,f}K_p[Q]}{K_{eq}} + \frac{V_{\max,f}[P][Q]}{K_{eq}} + \frac{V_{\max,f}K_q[A][P]}{K_{ia}K_{eq}} + \frac{V_{\max,r}K_a[B][Q]}{K_{iq}} + \frac{V_{\max,r}[A][B][P]}{K_{ip}} + \frac{V_{\max,r}[B][P][Q]}{K_{ip}K_{eq}} \right\}}$$

where

$$V_{\max,f} = k_5k_7[E_{\text{total}}]/(k_5 + k_7)$$

$$V_{\max,r} = k_2k_4[E_{\text{total}}]/(k_2 + k_4)$$

$$K_{ia} = k_2/k_1$$

$$K_{ib} = (k_2 + k_4)/k_3$$

$$K_{ip} = (k_5 + k_7)/k_6$$

$$K_{iq} = k_7/k_8$$

$$K_a = k_5k_7/[k_1(k_5 + k_7)]$$

$$K_b = k_7(k_4 + k_5)/[k_3(k_5 + k_7)]$$

$$K_p = k_2(k_4 + k_5)/[k_6(k_2 + k_4)]$$

$$K_q = k_2k_4/[k_8(k_2 + k_4)]$$

$$K_{eq} = k_1k_3k_5k_7/[k_2k_4k_6k_8] = V_{\max,f}K_pK_{iq}/[V_{\max,r}K_{ia}K_b]$$

See also Random Bi Bi Mechanism; Ordered Bi Bi Theorell-Chance Mechanism; Multisubstrate Mechanisms

¹C. C. Wratten & W. W. Cleland (1963) *Biochemistry* **2**, 935.

²M. Marshall & P. P. Cohen (1966) *J. Biol. Chem.* **241**, 4197.

³M. T. Hakala, A. J. Glaid & G. W. Schuert (1956) *J. Biol. Chem.* **221**, 191.

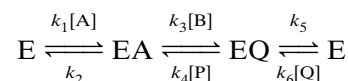
⁴V. Zewe & H. J. Fromm (1965) *Biochemistry* **4**, 782.

⁵S. R. Anderson, J. R. Florini & C. S. Vestling (1964) *J. Biol. Chem.* **239**, 2991.

⁶R. C. Nordlie & H. J. Fromm (1959) *J. Biol. Chem.* **234**, 2523.

ORDERED BI BI THEORELL-CHANCE MECHANISM

A sequential enzyme-catalyzed reaction mechanism in which two substrates react to form two products, in which the substrates bind and products are released in an ordered fashion and in which the ternary complex intermediate is not kinetically relevant under the reaction conditions:



The steady-state initial rate expression, in the absence of products and abortive complexes, is:

$$v = V_{\max}[A][B]/\{K_{ia}K_b + K_b[A] + K_a[B] + [A][B]\}$$

where $V_{\max} = k_5[E_{\text{total}}]$, $K_{ia} = k_2/k_1$, $K_a = k_5/k_1$, and $K_b = k_5/k_3$. The complete steady-state determinants for the individual enzyme forms are: $[E] = k_2k_5 + k_3k_5[B] +$

$k_2k_4[P]$; $[EA] = k_1k_5[A] + k_1k_4[A][P] + k_4k_6[P][Q]$; and $[EQ] = k_2k_6[Q] + k_1k_3[A][B] + k_3k_6[B][Q]$. A double-reciprocal plot of $1/v$ vs. $1/[A]$ at varying constant concentrations of $[B]$ will produce a series of straight lines having a common intersection point. Similarly, a double-reciprocal plot of $1/v$ vs. $1/[B]$ at varying constant concentrations of A will also yield a series of straight lines having a common intersection point. Replots of the slopes and intercepts as a function of the reciprocal of the concentration of the other substrate will yield values for the kinetic parameters.

ORDERED BI TER MECHANISM

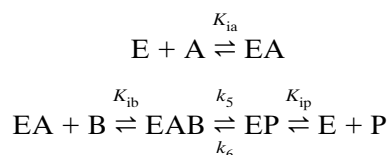
An enzyme reaction scheme in which there are two substrates (A and B) and three products (P , Q , and R) and in which the substrates bind and the products are released in an ordered fashion. This reaction scheme is exemplified by the malic enzyme¹. The initial rate expression, in the absence of abortive complexes and products, is identical to the corresponding equation for the ordered Bi Bi mechanism. **See** *Multisubstrate Mechanisms; Ordered Bi Bi Mechanism*

¹R. Y. Hsu, H. A. Lardy & W. W. Cleland (1967) *J. Biol. Chem.* **242**, 5315.

ORDERED BI UNI MECHANISM

A sequential enzyme-catalyzed reaction scheme in which two substrates (A and B) react and form a single product and in which the substrates bind to the enzyme in a distinct order (*i.e.*, only A and the product P can bind to the free enzyme). The reverse scheme of this mechanism is the ordered Uni Bi system. (**See also** *Ordered Uni Bi Mechanism*)

The Rapid Equilibrium Mechanism: The following scheme depicts this mechanism:



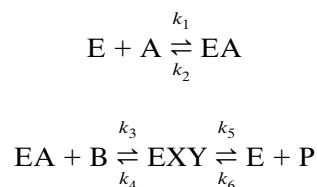
In this rapid equilibrium mechanism, with all binding steps expressed with dissociation constants, then under initial rate conditions (*i.e.*, $[P] \approx 0$) the initial rate expres-

sion for this scheme (in the absence of abortive complexes and isomerization steps) is

$$v = V_{\max}[A][B]/\{K_{ia}K_{ib} + K_{ib}[A] + [A][B]\}$$

where $V_{\max} = k_5[E_{\text{total}}]$, $K_{ia} = [E][A]/[EA]$, and $K_{ib} = [EA][B]/[EAB]$. A double-reciprocal plot of $1/v$ vs. $1/[A]$ at different constant concentrations of B will result in a series of straight lines. A replot of the slopes vs. $1/[B]$ yields a line having a slope of $K_{ia}K_{ib}/V_{\max}$. A replot of the vertical intercepts vs. $1/[B]$ will yield a horizontal intercept of $-1/K_{ib}$. Correspondingly, a double-reciprocal plot of $1/v$ vs. $1/[B]$ at different constant concentrations of A will produce a series of straight lines having a common intersection point on the vertical axis (at $1/v = 1/V_{\max}$).

The Steady-State Mechanism: If we consider the ordered Bi Uni mechanism with only one central complex,



then the steady-state enzyme determinants are: $[E] = k_2(k_4 + k_5) + k_3k_5[B]$; $[EA] = k_1(k_4 + k_5)[A] + k_4k_6[P]$; and $[EXY] = k_1k_3[A][B] + k_3k_6[B][P] + k_2k_6[P]$.

The initial rate expression (in which $[P] = 0$) is

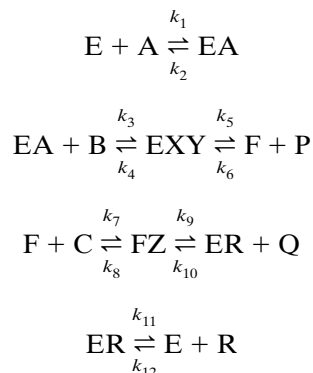
$$v = V_{\max}[A][B]/\{K_{ia}K_b + K_b[A] + K_a[B] + [A][B]\}$$

where $V_{\max} = k_5[E_{\text{total}}]$, $K_{ia} = k_2/k_1$, $K_b = (k_4 + k_5)/k_3$, and $K_a = k_5/k_1$. A double-reciprocal plot of $1/v$ vs. $1/[A]$ at different constant concentrations of B yields a set of straight lines. Correspondingly, a double-reciprocal plot of $1/v$ vs. $1/[B]$ at different constant concentrations of A will be a series of straight lines. Secondary replots of the initial rate data will yield values for the kinetic parameters.

ORDERED BI UNI UNI BI PING PONG MECHANISM

Melilotate hydrolase¹ and threonyl-tRNA synthetase² have been reported to use this reaction scheme. The

reaction scheme for this three-substrate (A, B, and C), three-product (P, Q, and R) mechanism can be depicted as



The initial rate expression for this scheme, in the absence of products or abortive complexes, is

$$v = \frac{V_{\max}[\text{A}][\text{B}][\text{C}]}{\{K_{ia}K_b[\text{C}] + K_c[\text{A}][\text{B}] + K_b[\text{A}][\text{C}] + K_a[\text{B}][\text{C}] + [\text{A}][\text{B}][\text{C}]\}}$$

where $V_{\max} = k_5k_9k_{11}[\text{E}_{\text{total}}]/\{k_5k_9 + k_5k_{11} + k_9k_{11}\}$, $K_{ia} = k_2/k_1$, $K_a = k_5k_9k_{11}/\{k_1(k_5k_9 + k_5k_{11} + k_9k_{11})\}$, $K_b = k_9k_{11}(k_4 + k_5)/\{k_3(k_5k_9 + k_5k_{11} + k_9k_{11})\}$, and $K_c = k_5k_{11}(k_8 + k_9)/\{k_7(k_5k_9 + k_5k_{11} + k_9k_{11})\}$.

See Multisubstrate Mechanisms

¹S. Strickland & V. Massey (1973) *J. Biol. Chem.* **248**, 2953.

²C. C. Allende, H. Chaimovich, M. Gatica & J. E. Allende (1970) *J. Biol. Chem.* **245**, 93.

ORDERED BI UNI UNI TER PING PONG MECHANISM

A three-substrate, four-product enzyme reaction scheme in which the first two substrates bind in an ordered fashion followed by the release of the first product. The third substrate then binds and the remaining three products are released in order (thus, $\text{E} + \text{A} \rightleftharpoons \text{EA}$ is followed by $\text{EA} + \text{B} \rightleftharpoons (\text{EAB} \rightleftharpoons \text{FP}) \rightleftharpoons \text{F} + \text{P}$ and then $\text{F} + \text{C} \rightleftharpoons (\text{FC} \rightleftharpoons \text{EQRS}) \rightleftharpoons \text{ERS} + \text{Q}$ followed by $\text{ERS} \rightleftharpoons \text{ES} + \text{R}$ and $\text{ES} \rightleftharpoons \text{E} + \text{S}$). There are indications that asparagine synthetase from certain sources may have this reaction scheme.

ORDERED ON-RANDOM OFF BI BI DUAL-THEORELL-CHANCE MECHANISM

A sequential enzyme-catalyzed binding mechanism for a two substrate-two product system in which substrates A and B have to bind in a certain order but either P or Q can be released in a Theorell-Chance step upon the binding of B. Following this step, the other substrate is

released (thus, $\text{E} + \text{A} \rightleftharpoons \text{EA}$ followed by either $\text{EA} + \text{B} \rightleftharpoons \text{EQ} + \text{P}$ or $\text{EA} + \text{B} \rightleftharpoons \text{EP} + \text{Q}$; then $\text{EQ} \rightleftharpoons \text{E} + \text{Q}$ and $\text{EP} \rightleftharpoons \text{E} + \text{P}$).

ORDERED ON-RANDOM OFF BI BI MECHANISM

An enzyme reaction mechanism involving A binding before B and followed with the random release of products. In the absence of products and abortive complexes, the steady-state rate expression is identical to the rate expression for the ordered Bi Bi mechanism^{1,2}. A random on-ordered off Bi Bi mechanism has been proposed for a mutant form of alcohol dehydrogenase³.

¹C. Y. Huang (1979) *Meth. Enzymol.* **63**, 54.

²S. Cha (1968) *J. Biol. Chem.* **243**, 820.

³A. J. Ganzhorn & B. V. Plapp (1988) *J. Biol. Chem.* **263**, 5446.

ORDERED TER BI MECHANISM

A three-substrate (A, B, and C), two-product (P and Q) enzyme reaction scheme in which all substrates and products bind and are released in an ordered fashion. Glyceraldehyde-3-phosphate dehydrogenase¹ has been reported to have this reaction scheme. The steady-state and rapid equilibrium expressions, in the absence of products and abortive complexes, are identical to the ordered Ter Ter mechanism. **See Ordered Ter Ter Mechanism**

¹B. A. Orsi & W. W. Cleland (1972) *Biochemistry* **11**, 102.

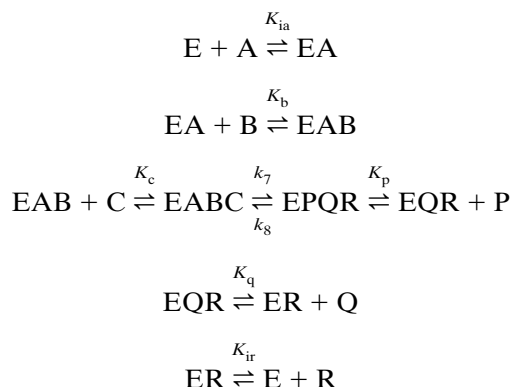
ORDERED TER BI THEORELL-CHANCE MECHANISM

A three-substrate, two-product enzyme reaction scheme in which the substrates bind and products are released in an ordered pattern and the third substrate on and first product off participate in a Theorell-Chance step. Thus, $\text{E} + \text{A} \rightleftharpoons \text{EA}$ is followed by $\text{EA} + \text{B} \rightleftharpoons \text{EAB}$ and $\text{EAB} + \text{C} \rightleftharpoons \text{EQ} + \text{P}$ after which comes $\text{EQ} \rightleftharpoons \text{E} + \text{Q}$.

ORDERED TER TER MECHANISM

A three-substrate, three-product enzyme reaction scheme in which all substrates bind and all products are released in an ordered sequence. Glutamate dehydrogenase¹ at a pH 8.8 has been reported to have this reaction scheme.

The Rapid Equilibrium Reaction: If the rate-determining step for the overall reaction is the interconversion step of the central complexes, then all of the binding steps can be described in terms of dissociation constants:

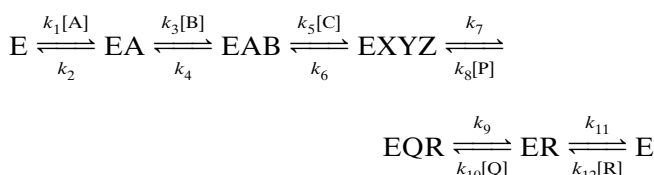


The rate expression, in the absence of products or abortive complexes, is:

$$v = \frac{V_{\max}[A][B][C]}{\{K_{ia}K_bK_c + K_bK_c[A] + K_c[A][B] + [A][B][C]\}}$$

where $V_{\max} = k_7[E_{\text{total}}]$, $K_{ia} = [E][A]/[EA]$, $K_b = [EA][B]/[EAB]$, and $K_c = [EAB][C]/[EABC]$.

The Steady-State Mechanism: If substrate binding and product release are not rapid, then a steady-state description of the reaction is applicable. If one considers the reaction scheme with a single central complex (that is, EXYZ), the reaction can be depicted as



The steady-state expression for this reaction scheme, in the absence of any products and abortive complexes, is

$$v = \frac{V_{\max}[A][B][C]}{\{K_{ia}K_{ib}K_c + K_{ib}K_c[A] + K_{ia}K_b[C] + K_c[A][B] + K_b[A][C] + K_a[B][C] + [A][B][C]\}}$$

where

$$\begin{aligned}
 V_{\max} &= k_7k_9k_{11}[E_{\text{total}}]/\{k_7k_9 + k_7k_{11} + k_9k_{11}\} \\
 K_{ia} &= k_2/k_1 \\
 K_{ib} &= k_4/k_3 \\
 K_a &= k_7k_9k_{11}/\{k_1(k_7k_9 + k_7k_{11} + k_9k_{11})\}
 \end{aligned}$$

$$K_b = k_7k_9k_{11}/\{k_3(k_7k_9 + k_7k_{11} + k_9k_{11})\}$$

$$K_c = k_9k_{11}(k_6 + k_7)/\{k_5(k_7k_9 + k_7k_{11} + k_9k_{11})\}$$

¹E. Silverstein (1974) *Biochemistry* **13**, 3750.

ORDERED UNI BI MECHANISM

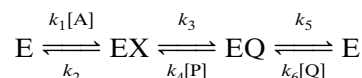
An enzyme reaction involving the conversion of a single substrate (A) to two products (P and Q) which are released in a preferred order ^{1,2}.

The Rapid Equilibrium Reaction: If the interconversion step between the central complexes is rate-determining (*i.e.*, rapid equilibrium), then for the reaction $E + A \xrightleftharpoons[k_2]{k_1} EA \xrightarrow{k_3} \text{products}$, in the absence of products and abortive complexes, the rate expression is identical to the rapid equilibrium Michaelis-Menten equation,

$$v = V_{\max}[A]/(K_{ia} + [A])$$

where $K_{ia} = k_2/k_1 = [E][A]/[EA]$ and $V_{\max} = k_3[E_{\text{total}}]$.

The Steady-State Reaction: If central complex interconversion is not the sole rate-determining step (or, some other step is rate-limiting) then,



The steady-state determinants are: $[E] = k_5(k_2 + k_3) + k_2k_4[P]$; $[EA + EPQ] = [EX] = k_1k_5[A] + k_1k_4[A][P] + k_4k_6[P][Q]$; and $[EQ] = k_6(k_2 + k_3)[Q] + k_1k_3[A]$. In the absence of products P and Q, with no abortive complexes and without isomerization steps,

$$v = V_{\max,f}[A]/(K_a + [A])$$

where $V_{\max,f} = k_3k_5[E_{\text{total}}]/(k_3 + k_5)$ and the Michaelis constant for A = $K_a = k_5(k_2 + k_3)/(k_1(k_3 + k_5))$. The complete rate expression^{3,4} is

$v =$

$$\frac{V_{\max,f} V_{\max,r} ([A] - [P][Q]/K_{eq})}{\left\{ V_{\max,r} K_a + V_{\max,r} [A] + \frac{V_{\max,f} K_q [P]}{K_{eq}} + \frac{V_{\max,f} K_p [Q]}{K_{eq}} + \frac{V_{\max,f} [P][Q]}{K_{eq}} + \frac{V_{\max,r} [A][P]}{K_{ip}} \right\}}$$

where $V_{\max,r} = k_2[E_{\text{total}}]$, $K_p = (k_2 + k_3)/k_4$, $K_q = k_2/k_6$, $K_{ip} = (k_3 + k_5)/k_4$, $K_{ia} = k_2/k_1$, $K_{iq} = k_5/k_6$, and $K_{eq} = k_1k_3k_5/\{k_2k_4k_6\} = V_{\max,f}K_{ip}K_q/\{V_{\max,r}K_{ia}\} = V_{\max,f}K_pK_{iq}/\{V_{\max,r}K_a\}$.

Adenylosuccinate lyase⁵, malyl coenzyme A lyase⁶, and NADase⁷ are reported to have this mechanism. It should also be recognized that the termed “ordered Uni Bi” is often used to refer to enzyme-catalyzed reactions which are actually “ping pong Bi Bi” mechanisms in which water is the second substrate. **See Multisubstrate Mechanisms**

¹W. W. Cleland (1963) *Biochem. Biophys. Acta* **67**, 104.

²H. J. Fromm (1979) *Meth. Enzymol.* **63**, 42.

³I. H. Segel (1975) *Enzyme Kinetics*, Wiley, New York.

⁴H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

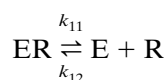
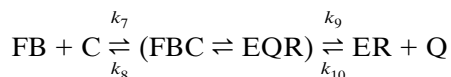
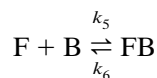
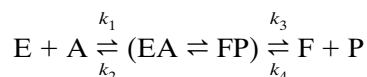
⁵W. A. Bridger & L.H. Cohen (1968) *J. Biol. Chem.* **243**, 644.

⁶L. B. Hersh (1974) *J. Biol. Chem.* **249**, 5208.

⁷L. J. Zatman, N. O. Kaplan & S. P. Colowick (1953) *J. Biol. Chem.* **200**, 197.

ORDERED UNI UNI BI BI PING PONG MECHANISM

A three-substrate, three-product enzyme reaction scheme in which the first substrate binds and the first product is release before the ordered binding of the second and third substrates:



The steady-state rate expression, in the absence of products, abortive complexes, and any isomerization steps is

$v =$

$$\frac{V_{\max}[A][B][C]}{\{K_c K_{ib} + K_c[A][B] + K_b[A][C] + K_a[B][C] + [A][B][C]\}}$$

where $V_{\max}$ is the maximum forward velocity equal to $k_3 k_9 k_{11} [E]_{\text{total}} / \{k_3 k_9 + k_3 k_{11} + k_9 k_{11}\}$, K_{ib} is equal to k_6 / k_5 , $K_a = k_9 k_{11} (k_2 + k_3) / \{k_1 (k_3 k_9 + k_3 k_{11} + k_9 k_{11})\}$, $K_b = k_3 k_9 k_{11} / \{k_5 (k_3 k_9 + k_3 k_{11} + k_9 k_{11})\}$, and $K_c = k_3 k_{11} (k_8 + k_9) / \{k_7 (k_3 k_9 + k_3 k_{11} + k_9 k_{11})\}$.

ORDERED UNI UNI, UNI BI PING PONG MECHANISM

A two-substrate, three-product enzyme reaction scheme in which the first substrate (A) binds to the enzyme,

producing the first product (P) in a ping pong half-reaction. The second substrate (B) and the remaining two products then follow an ordered Uni Bi scheme.

ORDER OF REACTION

The overall order of a chemical reaction is equal to the sum of the exponents of all the concentration terms in the differential rate expression for a reaction considered in one direction only. For example, if the empirical rate law for a particular chemical reaction was

$$v = k[A]^\alpha[B]^\beta[C]^\gamma$$

then the overall order of the reaction would be $\alpha + \beta + \gamma$. In addition to the overall order, there are also partial reaction orders with respect to individual reactants. For the rate expression above, the reaction is of partial order α with respect to reactant A, partial order β with respect to B, and partial order γ with respect to C. If the concentration of one or more of the reactants is held constant, or nearly constant, over the time frame of experiment, then the reaction will be pseudo- n th order in which n is the sum of the exponents of the concentration terms in the rate expression that are not held constant. This is precisely the case with an enzyme, since the catalyst concentration remains constant.

With chemical reactions, the exponents in a rate expression are usually integers. However, the exponents can be fractions or even negative depending on the complexity of the reaction. Reaction order should not be confused with molecularity. Order is an empirical concept whereas molecularity refers to the actual molecular process. However, for elementary reactions, the reaction order equals the molecularity. **See Chemical Kinetics; Molecularity; First-Order Reactions; Rate Constants**

D-ORNITHINE 4,5-AMINOMUTASE

This enzyme [EC 5.4.3.5] catalyzes the interconversion of D-ornithine and D-threo-2,4-diaminopentanoate. The enzyme requires cobalamin, dithiothreitol, and pyridoxal phosphate.

P. A. Frey & G. H. Reed (1993) *Adv. Enzymol.* **66**, 1.

T. C. Stadtman (1972) *The Enzymes*, 3rd ed., **6**, 539.

ORNITHINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.13] (also known as ornithine:oxo-acid aminotrans-

ferase and ornithine keto-acid aminotransferase) catalyzes the reversible reaction of L-ornithine with an α -keto acid (or, a 2-oxo acid) to produce L-glutamate 5-semialdehyde and an L-amino acid.

H. Hayashi, H. Wada, T. Yoshimura, N. Esaki & K. Soda (1990) *Ann. Rev. Biochem.* **59**, 87.

A. J. L. Cooper (1985) *Meth. Enzymol.* **113**, 76.

E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

ORNITHINE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 4.1.1.17] catalyzes the conversion of L-ornithine to putrescine and carbon dioxide.

R. A. John (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 173.

S.-I. Hayashi (1995) *Essays in Biochem.* **30**, 37.

T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.

E. A. Boeker & E. E. Snell (1972) *The Enzymes*, 3rd ed., **6**, 217.

ORNITHINE TRANSCARBAMOYLASE

This enzyme [EC 2.1.3.3], also known as ornithine carbamoyltransferase and citrulline phosphorylase, catalyzes the reaction of carbamoyl phosphate with L-ornithine to produce L-citrulline and orthophosphate.

P. P. Cohen (1981) *Curr. Top. Cell. Reg.* **18**, 1.

H. J. Vogel & R. H. Vogel (1974) *Adv. Enzymol.* **40**, 65.

S. Ratner (1973) *Adv. Enzymol.* **39**, 1.

OROTATE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.10], also known as orotidylic acid phosphorylase and orotidine-5'-phosphate pyrophosphorylase, catalyzes the reaction of orotate with 5-phospho- α -D-ribose 1-diphosphate to produce orotidine 5'-phosphate and pyrophosphate (or, diphosphate).

W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

M. E. Jones (1980) *Ann. Rev. Biochem.* **49**, 253.

OROTIDYLATE DECARBOXYLASE

This enzyme [EC 4.1.1.23], also known as orotidine-5'-phosphate decarboxylase, catalyzes the conversion of orotidine 5'-phosphate to UMP and carbon dioxide.

M. E. Jones (1992) *Curr. Top. Cell. Reg.* **33**, 331.

ORTHOPHOSPHATE CONTINUOUS ASSAY

Because ATP and GTP hydrolysis play such central roles in biochemical processes, kinetic assays of orthophos-

phate (or, P_i) formation are widely utilized in mechanistic investigations. Traditionally, this has involved stopped-time assays in which samples must be removed from a reaction mixture and quenched prior to colorimetric determination of phosphate content. The discovery of a phosphate-binding protein that becomes fluorescent upon phosphate binding now permits continuous fluorimetric assay¹ in the same way that one measures NADH or NADPH production. The technique utilizes the A197C mutant of *Escherichia coli* phosphate binding protein containing the fluorophore *N*-[2-(1-maleimidyl)ethyl]-7-(diethylamino)coumarin-3-carboxamide attached to the introduced thiol¹. The modified protein has an excitation maximum at 425 nm and emission maximum at 474 nm in the absence of P_i , which shifts to 464 nm with a 5.2-times enhancement of fluorescence upon complexation with P_i . The binding reaction occurs at pH 7.0, low ionic strength, 22°C, and similar spectral properties are observed at pH 8 or at ionic strength up to 1 M. The binding of orthophosphate is characterized by a K_d of about 0.1 μ M, a bimolecular rate constant of $1.36 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$, and a dissociation rate constant of 21 s^{-1} . ATP only weakly inhibits P_i -induced fluorescence enhancement. In view of the presence of phosphate in many biological samples, Brune *et al.*¹ depleted endogenous orthophosphate to concentrations below 0.1 μ M using 7-methylguanosine and purine nucleoside phosphorylase, with conversion of free P_i to ribose 1-phosphate. Using this probe to measure single turnover P_i release rates for actomyosin subfragment 1 hydrolysis of ATP, these investigators observed a small lag prior to rapid P_i release at both high and low [ATP].

White *et al.*² measured the rates of phosphate release during a single turnover of actomyosin nucleoside triphosphate (NTP) hydrolysis using a double-mixing stopped-flow spectrofluorometer, at very low ionic strength to increase the affinity of myosin-ATP and myosin-ADP- P_i to actin. Myosin subfragment 1 and a series of nucleoside triphosphates were mixed and incubated for approximately 1–10 s to allow NTP to bind to myosin and generate a steady-state mixture of myosin-NTP and myosin-NDP- P_i . The steady-state intermediates were then mixed with actin.

¹M. Brune, J. L. Hunter, J. E. Corrie & M. R. Webb (1994) *Biochemistry* **33**, 8262.

²H. D. White, B. Belknap & M. R. Webb (1997) *Biochemistry* **36**, 11828.

OSTWALD RIPENING

A process by which a two-phase crystalline suspension (experiencing a positive interfacial tension) proceeds to decrease in free energy by minimizing the interfacial surface area at constant pressure and temperature¹. This process of “ripening” reaches equilibrium only when the surface area reaches a minimum. As a result, a population of small crystals or a noncrystalline suspension will spontaneously dissolve, allowing its component subunits to add to other larger crystals. Some of these larger crystals then dissolve away and their subunits add to still fewer crystals, and the process ultimately culminates in the survival of a single crystal. **See** *Biom mineralization*

¹A. E. Neilsen (1964) *Kinetics of Precipitation*, pp. 108-117, Pergamon Press, Oxford.

OSTWALD SOLUBILITY COEFFICIENT

The volume in milliliters of gas dissolved per milliliter of liquid at one atmosphere of partial pressure of the gas at any given temperature. Thus, the Ostwald coefficient (Λ) differs from the Bunsen solubility coefficient (α) which is based on standard temperature and pressure. The two coefficients are related by the equation $\Lambda = \alpha(1 + 0.00367t)$ where t is the temperature in degrees Celsius.

OUTER SPHERE ELECTRON TRANSFER REACTION

A well-established electron transfer mechanism in inorganic chemistry, typically observed when two reacting species take longer to exchange ligands than undergo electron transfer. Each complex maintains its full complement of bound ligands during electron transfer, and the transferred electron must traverse both coordination shells. **See** *Metal Ion Coordination (Basic Chemical Behavior)*

OVERSATURATION

A phenomenon associated with noncompetitive product inhibition, wherein the half-times for approach to equilibrium divided by the initial substrate concentration are observed to increase with increasing substrate concentrations. As pointed out by Cleland¹, this would not be the case for such time courses (normalized with respect to substrate concentration) if the product inhibition were competitive. In the case of proline racemase², the observation of oversaturation suggests that the enzyme oper-

ates by an Iso mechanism (*i.e.*, the form of free enzyme reacting with the L-isomer is different from the enzyme form reacting with the D-isomer. Another case of oversaturation was described by Silverman *et al.*³ for carbonic anhydrase.

¹W. W. Cleland (1990) *The Enzymes* **19**, 99.

²L. M. Fisher, W. J. Albery & J. R. Knowles (1986) *Biochemistry* **25**, 2529.

³D. N. Silverman & S. H. Vincent (1983) *Crit. Rev. Biochem.* **14**, 207.

OXALOACETATE DECARBOXYLASE

This biotinylated enzyme [EC 4.1.1.3] catalyzes the decarboxylation of oxaloacetate to pyruvate and carbon dioxide.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

OXALYL-CoA DECARBOXYLASE

This thiamin pyrophosphate-dependent enzyme [EC 4.1.1.8] catalyzes the conversion of oxalyl-CoA to formyl-CoA and carbon dioxide.

J. R. Quayle (1969) *Meth. Enzymol.* **13**, 369.

OXIDOREDUCTASES

A major class of enzymes that catalyze oxidation-reduction reactions. This class includes dehydrogenases, reductases, oxygenases, peroxidases, and a few synthases. Examples include alcohol dehydrogenase (EC 1.1.1.1), aldehyde oxidase (EC 1.2.3.1), orotate reductase (EC 1.3.1.14), glutamate synthase (EC 1.4.1.14), NAD(P)⁺ transhydrogenase (EC 1.6.1.1), and glutathione peroxidase (EC 1.11.1.9).

5-OXOPROLINASE

This enzyme [EC 3.5.2.9], also known as pyroglutamase (ATP-hydrolyzing), catalyzes the reaction of ATP with 5-oxo-L-proline and two water to produce L-glutamate, ADP, and orthophosphate.

A. Meister, O. W. Griffith & J. M. Williamson (1985) *Meth. Enzymol.* **113**, 445.

A. P. Seddon, L. Li & A. Meister (1985) *Meth. Enzymol.* **113**, 451.

P. Van Der Werf & A. Meister (1975) *Adv. Enzymol.* **43**, 519.

OXYGEN ELECTRODE

An electrochemical device for detecting dissolved oxygen content. The well known Clark electrode¹ consists of a platinum wire tip surrounded by a thin film of electrolyte solution that is shrouded by a plastic membrane. The membrane is permeable to oxygen, but impermeable to

water and other solutes. When an appropriate potential is applied, oxygen is electrochemically destroyed on the metal's surface, and the flow of electrons from the tip to the infusing oxygen can be used to measure dissolved oxygen *in vitro* as well as in physiologic fluids (such as plasma). Although manometry was first applied to investigate mitochondrial oxidation, the O_2 electrode offers the advantage of rapid detection and continuous recording of changes in oxygen concentration². Briefly, when ADP is added to a suspension of tightly coupled mitochondria, one can measure the P/O ratio by measuring ATP formation and molecular oxygen uptake.

If one assumes that the reaction system is at thermodynamic equilibrium, the oxygen electrode method also allows one to infer the partial pressure of oxygen in equilibrium with the test solution.

¹L. C. Clark (1962) *Ann. N. Y. Acad. Sci.* **102**, 29.

²R. W. Estabrook (1967) *Meth. Enzymol.* **10**, 41.

OXYGEN ELECTRODE FOR MONITORING HEMOGLOBIN OXYGENATION

Polarography¹ can be used to measure the hemoglobin O_2 saturation in whole blood, employing up to 10 μ L sample in an anaerobic stainless-steel cuvette (1 mL). Three oxygen tension values are determined: (a) that of an air-equilibrated buffer before the addition of the sample; (b) that after the addition of the sample; and (c) that after the addition of an oxidant. The oxygen saturation is then calculated from the three oxygen tension values, the volume of the reagents, and the solubility coefficient of oxygen. This simple, inexpensive, and accurate method correlates well with other standard methods.

¹M. Samaja & E. Rovida (1983) *J. Biochem. Biophys. Methods* **7**, 143.

OXYGEN, OXIDES, & OXO ANIONS

DIOXYGEN or O_2 . The most common allotrope of this element, dioxygen is a colorless gas (mp = -218.4°C and bp = -182.692°C) that reacts with all elements, except halogens, noble gases, and a few metals. At saturation, O_2 is about 1.4 mM in water, and is readily soluble in organic solvents. Indeed, the act of pouring an organic solvent in the presence of air will often result in saturating the solvent with O_2 . This observation regarding or-

ganic solvents should always be a consideration in handling air-sensitive materials.

Elemental oxygen ($Z = 8$ and $A_r = 15.9994$) occurs as 13 isotopes, but only four are of particular importance. Oxygen-16 (^{16}O) is the most common of the oxygen isotopes, representing 99.76% of all naturally occurring oxygen. Oxygen-17 (^{17}O) is a stable isotope representing only 0.04% of naturally occurring oxygen. With a nuclear spin of 5/2, ^{17}O is utilized in resonance studies. The remaining 0.20% of natural oxygen is oxygen-18 (^{18}O), a stable isotope used extensively in mass spectroscopy studies and in NMR experiments on enzymic systems. Oxygen-15 is the longest half-life (122.2 seconds; positron-emitting) radioisotope. ^{15}O is utilized in respiration studies, with detection achieved by positron emission tomography.

The structure of dioxygen is predicted by valence bond theory to be $\text{O}=\text{O}$, having one σ and one π bond, with each oxygen having two pairs of unshared electrons and a bond length of 0.1208 nm. Such a structure does not account for the observation that O_2 is paramagnetic in all phases, and molecular orbital theory correctly predicts the two unpaired electrons in dioxygen both in π^* orbitals. The singlet state of oxygen (the $^1\Delta_g$ state), in which the two π electrons are paired, is actually an excited state and is about 92 $\text{kJ}\cdot\text{mol}^{-1}$ above the triplet ground state (the $^3\Sigma_g^-$ state). Singlet dioxygen can react with many unsaturated compounds and can be readily generated by numerous photochemical reactions. Singlet oxygen is generated and/or utilized in many biochemical processes, particularly in the presence of light.

OZONE or O_3 . The other molecular oxygen allotrope is ozone, a colorless gas (mp = -192.7°C and bp = -111.9°C) that is also a powerful oxidant. The molecule is symmetrical and bent, having a bond angle of 116.8° and an oxygen—oxygen bond length of 0.1278 nm (another report lists 117.47° and 0.12716 nm). There appears to be considerable double-bond character present which can be accounted by resonance: $\text{O}=\text{O}=\text{O} \leftrightarrow \text{O}=\text{O}-\text{O}$. The conversion of ozone to dioxygen ($\text{O}_3 \rightleftharpoons (3/2) \text{O}_2$) is substantially endothermic ($\Delta H = -142 \text{ kJ}\cdot\text{mol}^{-1}$). This reaction is slow in the absence of catalysts and of light. One centrally important biological role of ozone is its absorption of ultraviolet light. Approximately 27% of

this planet's atmosphere between the altitudes of 15 and 25 km (or, about 9 to 16 miles) is ozone, and this layer provides a UV-protective shield for many living organisms.

SUPEROXIDE or $\text{O}_2^{\cdot-}$. This paramagnetic radical anion is generated in a number of biochemical processes. For example, $\text{O}_2^{\cdot-}$ is the product of the reaction catalyzed by xanthine oxidase. Superoxide is a strong oxidizing agent and can have significant effects on a cell. Several superoxide dismutases (EC 1.15.1.1) catalyze the dismutation of superoxide: $2 \text{O}_2^{\cdot-} + 2 \text{H}^+ \rightarrow \text{O}_2 + \text{H}_2\text{O}_2$. In aqueous solutions, $\text{O}_2^{\cdot-}$ is highly hydrated. The oxygen–oxygen bond length is slightly longer (0.133 nm) in superoxide than in dioxygen. Stable ionic superoxides are formed by certain of the alkali metals. Potassium superoxide (KO_2) is the most common of these. It is a yellow crystal (mp = 380°C) that is soluble (and decomposes) in water. Transient levels of superoxide in aqueous solution can be generated a number of ways.

OXIDES. These are compounds consisting of oxygen and another element or radical. Nonmetal oxides are usually either acidic or neutral. Metal oxides are usually ionic and tend to generate a base upon reaction with water. When insoluble in water, metal oxides usually dissolve in dilute acid (e.g., $\text{MgO}_{(\text{s})} + 2\text{H}^+ \rightarrow \text{Mg}_{(\text{aq})}^{2+} + \text{H}_2\text{O}$). Amphoteric oxides act as a base with respect to a strong acid and as an acid with respect to a strong base. An example is zinc oxide ($\text{ZnO} + 2\text{H}_{(\text{aq})}^+ \rightarrow \text{Zn}^{2+} + \text{H}_2\text{O}$ and $\text{ZnO} + 2\text{OH}^- + \text{H}_2\text{O} \rightarrow \text{Zn}(\text{OH})_4^{2-}$). Such oxides can often be found with the metalloids (e.g., boron, arsenic, germanium). If a particular element forms several different oxides, then that with the element in its highest oxidation state is the most acidic oxide. Many oxides have important roles in biological processes:

(1) Water or H_2O . The O—H bond lengths in water are 0.0958 nm (or 0.958 Å) and the bond angle is 104.45° (another report has 0.09575 nm and 104.51°). **See** *Water*

(2) Hydroxide anion or OH^- . This anion is the conjugate base of water. For electropositive elements such as sodium, potassium, and barium, discrete OH^- ions exist. For such hydroxides, the substance will act as a strong base when dissolved in water: thus, $\text{M}^+\text{OH}_{(\text{s})}^- + n\text{H}_2\text{O}$

$\rightarrow \text{M}_{(\text{aq})}^+ + \text{OH}_{(\text{aq})}^-$. However, when there is a strong covalent bond between M and the oxygen, then the substance can act as an acid: that is, $\text{MOH} + n\text{H}_2\text{O} = \text{MO}_{(\text{aq})}^- + \text{H}_3\text{O}_{(\text{aq})}^+$. Amphoteric hydroxides will act as a base in the presence of a strong acid and as an acid in the presence of a strong base. Hydroxide ions can also form bridges between metal ions. **See** *Water*; pK_w ; Bases

(3) Hydronium ion or H_3O^+ . **See** *Water*; pK_w ; Acids

(4) Hydroxide radical or $\text{HO}\cdot$. The hydroxide radical is an extremely reactive species that can damage cells. It can be generated by activated macrophages via superoxide and nitric oxide (especially peroxynitrite).

(5) Hydrogen peroxide or HOOH . Hydrogen peroxide is a relatively unstable compound and readily breaks down to water and dioxygen, a reaction catalyzed by powdered metals as well as the enzyme catalase ($2\text{H}_2\text{O}_2 = \text{O}_2 + 2\text{H}_2\text{O}$). The O—O bond length is 0.1475 nm and the O—H length is 0.097 nm. The O—O—H bond angle is 96.9°; the two hydrogens do not lie in the same plane, and the dihedral angle is 93.9°. Many physical properties of hydrogen peroxide resemble those of water. There is extensive hydrogen bonding, and the density of neat hydrogen peroxide is greater than that of water. The pure, colorless liquid (melting point of -0.41°C ; boiling point of 150.2°C) has a high dielectric constant that is slightly larger than water (e.g., a 65% aqueous hydrogen peroxide solution has a dielectric constant of 120). H_2O_2 is also more acidic than water when in dilute aqueous solutions ($\text{H}_2\text{O}_2 = \text{H}^+ + \text{HO}_2^-$, $K_a = 1.5 \times 10^{-12}$ at 20°C). Dilute (e.g., 30%) solutions of H_2O_2 are commonly used as oxidants, oxidation proceeding more rapidly in basic conditions than in acid solutions (thus, excess H_2O_2 can be best destroyed by heating in a basic solution). Hydrogen peroxide can be generated naturally (e.g., via amino acid oxidases, amine oxidases, glycolate oxidase, glycerol-3-phosphate oxidase, *etc.*) or by the autoxidation of a number of pharmaceuticals. Elevated levels of H_2O_2 can be very damaging to cells. For example, H_2O_2 (a) can oxidize iron(II) present in a number of heme proteins to Fe(III), (b) can act on double bonds of unsaturated fatty acyl groups of phospholipids of cell membranes (which can ultimately lead to rupture of the membrane), (c) can participate in the nonenzymatic oxidative decarboxylation of α -keto acids, and (d) can oxi-

dize methionyl and cysteinyl residues in proteins. Hydrogen peroxide is also acted upon by several other enzymes (EC 1.11.1.x): for example, the seleno-protein glutathione peroxidase [$2 \text{ glutathione} + \text{H}_2\text{O}_2 = 2 \text{ H}_2\text{O} + \text{glutathione disulfide}$] has an important role in regulating cellular H_2O_2 levels.

(6) Hydroperoxy radical or $\text{HOO}\cdot$. This is a very reactive free radical that can be generated from hydrogen peroxide, particularly via metal ion-catalyzed decomposition. This radical can generate superoxide.

(7) Carbon monoxide or CO . Carbon monoxide is a colorless and practically odorless gas ($\text{mp} = -199^\circ\text{C}$ and $\text{bp} = -191.5^\circ\text{C}$). The C-to-O bond length is 0.11283 nm. The linear carbon monoxide is able to use vacant p orbitals to form end-on bonds with metal ions. Thus, CO can inhibit a wide range of metalloproteins, particularly heme proteins such as hemoglobin and cytochrome c oxidase (the fact that carbon monoxide complexed with heme proteins can dissociate by the action of light was an important tool in the early history of the characterization of heme proteins and electron transport). At least two enzymes catalyze reactions producing CO : the mammalian heme oxygenase (EC 1.14.99.3) and 3,4-dihydroxyquinoline 2,4-dioxygenase (EC 1.13.99.5). Other enzymes utilize CO as a substrate: *e.g.*, carbon monoxide oxygenase (or, cytochrome b -561) (EC 1.2.2.4), carbon monoxide oxidase (EC 1.2.3.10), and carbon monoxide dehydrogenase (EC 1.2.99.2). CO has been reported to be an alternative activator of guanylate cyclase. CO may play a role as a cellular messenger in mammals.

(8) Carbon dioxide or CO_2 . Carbon dioxide is a linear molecule, $\text{O}=\text{C}=\text{O}$ with a mean carbon-oxygen bond length of 0.116 nm. **See Carbon Dioxide**

(9) Nitric oxide or $\cdot\text{N}=\text{O}$. This oxide is a colorless gas ($\text{mp} = -163.6^\circ\text{C}$ and $\text{bp} = -151.8^\circ\text{C}$) that is very soluble in water. The N—O bond length is 0.11506 nm. The molecule is paramagnetic, and the electron configuration is $(\sigma_1)^2(\sigma_2)^2(\sigma_3)^2(\pi)^4(\pi^*)$. Note that the unpaired electron is in an antibonding orbital and is relatively easily lost. Hence, the nitrogen-oxygen bond in the nitrosonium ion, NO^+ , is actually stronger and shorter (0.1062 nm) than the corresponding bond in nitric oxide. The bond order in nitric oxide is 2.5. In physiological solu-

tions, NO has a short half-life ($\sim$ five seconds), reacting rapidly with dissolved dioxygen to form the nitrogen dioxide radical ($2\cdot\text{NO} + \text{O}_2 = 2\cdot\text{NO}_2$). The latter, a free radical, then forms dinitrogen tetroxide which, in aqueous solutions, forms the stable anions nitrite and nitrate or will react with nitric oxide to form nitrous anhydride.

(10) Phosphate (or, Orthophosphate) or PO_4^{3-} . The conjugate base of HPO_4^{2-} , one of the oxo anions of phosphoric acid (the pK_a values of H_3PO_4 are, at 25°C , 2.15, 7.20, and 12.33). The structure of the resonance-stabilized PO_4^{3-} is tetrahedral and can form complexes with metal ions and other cations.

(11) Pyrophosphate (also known as diphosphate) or $\text{P}_2\text{O}_7^{4-}$. The conjugate base of pyrophosphoric acid (or, diphosphoric acid), $\text{H}_4\text{P}_2\text{O}_7$ (having pK_a values, at 25°C , of 0.85, 1.96, 6.60, and 9.41). Crystals of tetrasodium pyrophosphate ($\text{mp} = 988^\circ\text{C}$) are very soluble in water. The pyrophosphate anion hydrolyzes to orthophosphate; however, the rate of hydrolysis is much slower than that of pyrophosphoric acid. The hydrated form of tetrasodium pyrophosphate exhibits a slight degree of efflorescence.

(12) Metaphosphate or PO_3^- . This ion, which is stable in the gas phase, is quickly converted to H_2PO_4^- in aqueous solutions (note that the rate of conversion is slower at low temperatures). Metaphosphate has been postulated to be an intermediate in a number of enzyme-catalyzed reactions. **See Metaphosphate**

(13) Vanadium oxides. There are several oxides of vanadium, primarily depending on the oxidation state of this transition metal: vanadium(V) oxide (V_2O_5), vanadium(IV) oxide (VO_2), vanadium(III) oxide (V_2O_3), and vanadium(II) oxide (VO). In VO , the V—O bond length is 0.15893 nm. The vanadates consist of VO_4^{3-} (the conjugate base of $\text{VO}_3(\text{OH})^{2-}$ which has a pK_a value of about 1.0) and a series of aggregated structures (*e.g.*, $\text{V}_{10}\text{O}_{26}^{6-}$). Sodium orthovanadate (Na_3VO_4) is commercially available. Orthovanadate is an analog of phosphate and has proved very useful in its action as an inhibitor of a wide variety of phosphate-dependent reactions. In the presence of orthovanadate, $(\text{Na}^+/\text{K}^+)\text{-ATPase}$ will form VO_5^{5-} , which has a trigonal bipyramidal structure.

Metavanadate (VO_3^-) can form indefinite chains as well as cyclic structures. Sodium metavanadate is also commercially available. Hence, the investigator should be aware of which vanadate is being utilized in the studies.

(14) Arsenate. The conjugate base (AsO_4^{3-}) of HAsO_4^{2-} , an oxo anion of arsenic acid, H_3AsO_4 (pK_a values, at 18°C , of 2.25, 6.77, and 11.60). **See Arsenolysis**

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Lipid Peroxidation: Chemistry, **105**, 273; overview of methods used for detecting lipid peroxidation, **105**, 283; chemical methods for the detection of lipid hydroperoxides, **105**, 293; comparative studies on different methods of malonaldehyde determination, **105**, 299; concentrating ethane from breath to monitor lipid peroxidation *in vivo*, **105**, 305.

Production, Detection, and Characterization of Oxygen Radicals and Other Species:

Pulse radiolysis in study of oxygen radicals, **186**, 89; free radical initiators as source of water- or lipid-soluble peroxy radicals, **186**, 100; generation of iron(IV) and iron(V) complexes in aqueous solutions, **186**, 108; sources of compilations of rate constants for oxygen radicals in solution, **186**, 113; use of polyaminocarboxylates as metal chelators, **186**, 116; preparation of metal-free solutions for studies of active oxygen species, **186**, 121; use of ascorbate as test for catalytic metals in simple buffers, **186**, 125; spin-trapping methods for detecting superoxide and hydroxyl free radicals *in vitro* and *in vivo*, **186**, 127; detection of singlet molecular oxygen during chloride peroxidase-catalyzed decomposition of ethyl hydroperoxide, **186**, 133; detection and quantitation of hydroxyl radical using dimethyl sulfoxide as molecular probe, **186**, 137; distinction between hydroxyl radical and ferryl species, **186**, 148; determination of hydroperoxides with fluorometric reagent diphenyl-1-pyrenylphosphine, **186**, 157; phycoerythrin fluorescence-based assay for reactive oxygen species, **186**, 161; sulfur-centered free radicals, **186**, 168; quinoid compounds: high-performance liquid chromatography with electrochemical detection, **186**, 180; preparation of tocopheroxyl radicals for detection by electron spin resonance, **186**, 197.

Enzymes Involved in Forming or Removing Oxygen Radicals and Derived Products:

Determination of superoxide dismutase activity by purely chemical system based on NAD(P)H oxidation, **186**, 209; assays for superoxide dismutase autooxidation of hematoxylin, **186**, 220; assay for superoxide dismutase based on chemiluminescence of luciferin analog, **186**, 227; automated assay of superoxide dismutase in blood, **186**, 232; oxidative reactions of hemoglobin, **186**, 265; assays for cytochrome P-450 peroxylase activity, **186**, 273; reductive cleavage of hydroperoxides by cytochrome P-450, **186**, 278; prostaglandin H synthase, **186**, 283; DT-diaphorase: purification, properties, and function, **186**, 287.

P

PACEMAKER REACTION

The slowest enzyme-catalyzed step in a metabolic pathway; sometimes referred to as the rate-limiting step (although that term is also used to describe a step in the reaction pathway of a single reaction). The pacemaker step may change, depending on availability of substrates and/or effectors as well as reaction conditions. For example, in red blood cells, hexokinase is thought to be the pacemaker for glycolysis, but hexokinase and phosphofructokinase both act as glycolytic pacemakers in many other cell types. **See also** Rate-Determining Step

H. A. Krebs (1957) *Endeavour* **16**, 125.

PACKING DENSITY

The fraction of total volume occupied by atoms in a molecule or solid. A van der Waals envelope for a molecule or segment of molecules can be readily determined from atomic models based on X-ray diffraction studies of crystals. The packing density of closest packed spheres is 0.74, that of closest packed infinite cylinders is 0.91, and that of a continuous solid is taken to be unity. The average packing density of ribonuclease S¹ is nearly 0.75, with several regions reaching 0.82–0.84. From the work of Monaco *et al.*², it is clear that the packing density at the subunit interfaces of aspartate transcarbamylase is also nearly 0.75. These values attest to the close-packed nature of polypeptide chains in globular proteins.

¹F. M. Richards (1974) *J. Mol. Biol.* **82**, 1.

²H. L. Monaco, J. L. Crawford & W. N. Lipscomb (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 5276.

PANTOTHENIC ACID, COENZYME A, AND DERIVATIVES

Metabolic derivatives of pantothenic acid are of fundamental importance in acyl transfer reactions and in condensation reactions requiring an acidic α -proton. The

discovery that coenzyme A was the so-called “acetylation coenzyme” resulted in the award of the 1953 Nobel Prize in Medicine and Physiology to Fritz Lipmann. Pantothenate is also found in acyl-carrier protein (ACP) which functions in a manner akin to CoA.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Gas chromatography of pantothenic acid, **18A**, 311; pantothenic acid and coenzyme A: determination of CoA by phosphotransacetylase from *Escherichia coli* B, **18A**, 314; preparation and assay of CoA-SS-glutathione, **18A**, 318; pantothenic acid and coenzyme A: preparation of CoA analogs, **18A**, 322; synthesis of coenzyme A analogues, **18A**, 338; pantothenic acid and coenzyme A: phosphopantothenoylecysteine synthetase from rat liver (pantothenate 4'-phosphate:L-cysteine ligase, EC 6.3.2.5), **18A**, 350; pantothenic acid and coenzyme A: phosphopantothenoylecysteine decarboxylase from rat liver [4'-phospho-N-(D-pantothenoyl)-L-cysteine carboxy-lyase], **18A**, 354; preparation of pantothenate-¹⁴C labeled pigeon liver fatty acid synthetase: release of the 4'-phosphopantetheine ¹⁴C-labeled prosthetic group, **18A**, 364; enzymatic synthesis of ¹⁴C-labeled coenzyme A, **18A**, 371; microbiological assay of pantothenic acid, **62**, 201; purification and properties of ketopantoate hydroxymethyl-transferase, **62**, 204; ketopantoyle lactone reductases, **62**, 209; pantothenate synthetase from *Escherichia coli* [D-pantoate: β -alanine ligase (AMP-forming)], **62**, 215; microbial synthesis of sugar derivatives of D-pantothenic acid, **62**, 220; syntheses of glycosyl derivatives of pantothenic acid and pantetheine, **62**, 227; synthesis of coenzyme A and its biosynthetic intermediates by microbial processes, **62**, 236; phosphopantothenoylecysteine decarboxylase from horse liver [4'-phospho-N-(D-pantothenoyl)-L-cysteine carboxy-lyase], **62**, 245; interconversion of apo- and holo fatty acid synthetases of rat and pigeon liver, **62**, 249; purification and properties of pantetheinase from horse kidney, **62**, 262; pantothenase from *Pseudomonas fluorescens*, **62**, 267.

PAPAIN

This endopeptidase [EC 3.4.22.2], a member of the C1 peptidase family hydrolyzes peptide bonds in proteins, exhibiting a broad specificity for those bonds. There is a preference for an amino acyl residue bearing a large hydrophobic side chain at the P₂ position and the enzyme does not accept a valyl residue at P₁.

A. C. Storer & R. Ménard (1994) *Meth. Enzymol.* **244**, 486.

D. J. Buttle (1994) *Meth. Enzymol.* **244**, 539.

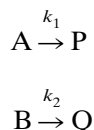
E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

J. Drenth, J. N. Jansonius, R. Koekoek & B. G. Wolthers (1971) *The Enzymes*, 3rd ed., **3**, 484.

A. N. Glazer & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 502.

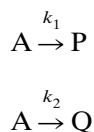
PARALLEL FIRST-ORDER REACTIONS

Two Different Reactants. A set of two or more first-order reactions, the reactants of which are independent of each other, generating at least one product that is common for each chemical reaction. For example, consider the following set of parallel reactions:



If the reactions are followed by the rates of disappearance of A and B (*i.e.*, $-d[A]/dt$ and $-d[B]/dt$), then the first-order rate constants k_1 and k_2 are directly obtained via standard analysis. However, if the rate of formation of the common product concentration or the rate of disappearance of the sum of A and B is measured, the evaluation of the individual rate constants is more complex.

One Common Reactant. A set of two or more chemical reactions for a given reactant, each of which is first-order.



(An example would be the reaction or decomposition of a substance by two pathways.) The complete rate law must include all of the alternative pathways. If $[A_0]$ represents the concentration of A at time $t = 0$, and $[A]$, $[P]$, and $[Q]$ represent the concentrations of the corresponding substances at $t = t$, then $-d[A]/dt = k_1[A] + k_2[A] = (k_1 + k_2)[A]$. Integration yields $\ln([A_0]/[A]) = (k_1 + k_2)t$ or $[A] = [A_0]e^{-(k_1+k_2)t}$. Note that the overall reaction is first order with respect to A.

PARALLEL REACTIONS

1. A set of reactions in which a specific reactant can proceed to react by two or more pathways, each independent of the others. 2. A set of reactions in which a common product is produced from different reactants, each reaction being independent of the others. 3. A less satisfactory usage of the term “parallel reactions” is as a

synonym for “side reactions.” **See** *Specific type of parallel reaction; Parallel First-Order Reactions*

PARAMAGNETIC

A physical property of substances having a positive, non-zero magnetic susceptibility (albeit small). All substances with unpaired electrons will exhibit paramagnetism, as will metals (as a consequence of the magnetic moments associated with the spins of conducting electrons). **See also** *Diamagnetic; Permeability; Magnetic Susceptibility; Electron Spin Resonance; Diamagnetic*

PARTIAL COMPETITIVE INHIBITION

This type of inhibition differs from that exhibited by classical competitive inhibitors, because the substrate can still bind to the EI complex and the EIS complex can go on to form product (albeit at a slower rate) without the inhibitor being released from the binding site. While standard double-reciprocal plots of partial competitive inhibitors will be linear (except for some steady-state, *i.e.*, non-rapid-equilibrium, cases), secondary slope replots will be nonlinear. **See** *Nonlinear Inhibition*

Consider the simplest case on a rapid equilibrium Uni Uni enzyme (*i.e.*, an enzyme in which all of the binding steps are fast relative to the catalytic step and all binding steps can be characterized by dissociation constants). In the absence of the inhibitor, $E + A \rightleftharpoons EX \rightarrow \text{product}$ where $K_{ia} = [E][A]/[EX]$ and the conversion step has the unimolecular rate constant k_3 . The inhibitor I can bind to the free enzyme to form the EI complex ($E + I \rightleftharpoons EI$) in which $K_i = [E][I]/[EI]$ and the inhibitor can bind to the EX complex to form EXI (where $K_{ii} = [EX][I]/[EXI]$). The EI complex can bind substrate and also form EXI (in which $K_{iii} = [EI][A]/[EXI]$). The ternary EXI complex can now proceed to form product, $EXI \rightarrow \text{product}$. In this simplest case, let the rate constant for the formation of product from EXI also be equal to k_3 . Hence, $d[\text{product}]/dt = v = k_3([EX] + [EXI])$. The derivation of the rate expression is fairly straightforward, yielding $v = V_{\max}[A](1 + ([I]/K_{ii})) / \{K_{ia}(1 + ([I]/K_i)) + [A](1 + ([I]/K_{ii}))\}$ where $V_{\max} = k_3[E_{\text{total}}]$. A double-reciprocal plot ($1/v$ vs. $1/[A]$ at different constant concentrations of I) will consist of a series of straight lines, all intersecting at a common point on the vertical axis (at $1/v = 1/V_{\max}$), as would be the case for a normal, that is, linear, competitive inhibitor. A secondary replot

of the slopes, however, will produce a curve that is either hyperbolic up or down, depending on the relative sizes of the dissociation constants for the inhibitor. If $K_i = K_{ii}$, no inhibition will be detected. Numerical analysis of the rate data can provide good estimates of the dissociation constants.

For non-rapid-equilibrium cases (*i.e.*, steady-state cases) the enzyme rate expression is much more complex, containing terms with $[A]^2$ and with $[I]^2$. Depending on the relative magnitude of those terms in the initial rate expression, there may be nonlinearity in the standard double-reciprocal plot. In such cases, computer-based numerical analysis may be the only means for obtaining estimates of the magnitude of the kinetic parameters involving the partial inhibition. **See** *Competitive Inhibition*

PARTIAL DERIVATIVES (Basic Properties)

Because partial derivatives are used so prominently in thermodynamics (**See** *Maxwell's Relationships*), we briefly consider the properties of partial derivatives for systems having three variables x , y , and z , of which two are independent. In this case, $z = z(x, y)$, where x and y are treated as independent variables. If one deals with infinitesimal changes in x and y , the corresponding changes in z are described by the partial derivatives:

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy$$

The subscripts indicate the variables that are held constant when taking the partial derivatives. Likewise, one can choose to treat y and z as independent variables, thereby writing x in terms of these variables, such that $x = x(y, z)$. In this case, we can obtain the following expression for infinitesimal changes in these independent variables:

$$dx = \left(\frac{\partial x}{\partial y} \right)_z dy + \left(\frac{\partial x}{\partial z} \right)_y dz$$

If z is kept constant, we can evaluate $(\partial x / \partial y)_z$ as follows, noting that $dz = 0$:

$$0 = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy$$

and

$$\frac{dx}{dy} = - \frac{(\partial z / \partial y)_x}{(\partial z / \partial x)_y}$$

PARTIAL INHIBITION

A form of reversible inhibition in which the inhibitor-substrate(s)-enzyme complex can still generate product(s), without the inhibitor dissociating (albeit at a slower rate when compared to the inhibitor-free system).

See *Partial Competitive Inhibition; Inhibition*

PARTIAL MOLAR VOLUMES

A thermodynamic parameter $(\partial V / \partial n_B)_{T, P, n_{j \neq B}}$ which describes how the volume of component B in a multicomponent system depends on the change in its amount expressed in mol. Høiland¹ recently summarized the partial molar volumes of numerous biochemical compounds in aqueous solution. **See** *Dalton's Law of Partial Pressures; Concentrations; Molecular Crowding*

¹H. Høiland (1986) in *Thermodynamic Data for Biochemistry and Biotechnology* (H.-J. Hinz, ed.), pp. 1-44, Springer-Verlag, Berlin.

PARTIAL PRESSURE

For a mixture of gases in a specified volume, the partial pressure of a component of the mixture is the pressure exerted by that compound if it were the sole occupant of the volume. For example, the partial pressure of oxygen is around 152 mm of mercury or about 0.21 atmospheres at sea level and 298 K. **See** *Dalton's Law of Partial Pressures; Concentration*

PARTIAL RATE FACTOR

The rate of substitution for a given group and a given reaction at a single position in an aromatic compound relative to the rate of substitution at one position in benzene¹.

Partial rate factors have proven quite useful in predicting isomer distributions from related aromatic reactants under the same reaction conditions. If two or more substituents are present on the aromatic ring, the effect of those substituents are usually considered to be additive. However, the procedure is less effective in the presence of steric effects, *ipso* attack, and resonance interactions between substituents.

The term has also been applied to Hammett equations (**See** *Selectivity Factor*) and for other substituted reactants undergoing parallel reactions at different sites with the same rate law.

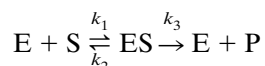
¹L. M. Stock & H. C. Brown (1963) *Adv. Phys. Org. Chem.* **1**, 35.

PARTITION FACTOR

A parameter (often symbolized as R) used in the analysis of kinetic isotope effects as a measure of the rate of a particular step relative to the rate of all preceding steps in the reaction sequence. This factor indicates the fate of the intermediate that immediately precedes the step in which the change in bonding to the isotopic atom occurs¹. If the key step is described by the constant k_i (and its isotopically altered value k_i^*), then

$$(k/k^*)_{\text{observed}} = (k_i/k_i^* + R)/(1 + R)$$

where the value of R depends on all rate constants up to and including k_{i-1} . For example, in the one-intermediate, one-substrate mechanism:



where k_3 is the isotopically sensitive step. $R = k_3/k_2$.

¹M. H. O'Leary (1976) in *Isotope Effects on Enzyme-Catalyzed Reactions* (W. W. Cleland, M. H. O'Leary & D. B. Northrop, eds) pp. 236-238, University Park Press, Baltimore.

PARVULIN

The smallest member of a new family of prolyl isomerases (unrelated to the cyclophilins or the FK-506 binding proteins) that catalyzes the proline-limited folding of a variant of ribonuclease T1 with a k_{cat}/K_m value of $30,000 \text{ M}^{-1} \text{ s}^{-1}$. With the tetrapeptide succinyl-Ala-Leu-Pro-Phe-4-nitroanilide as a substrate in parvulin-catalyzed prolyl isomerization, this parameter is $1.1 \times 10^7 \text{ M}^{-1} \text{ s}^{-1}$. Parvulin also accelerates its own refolding in an autocatalytic fashion.

C. Scholz, J. Rahfeld, G. Fischer & F. X. Schmid (1997) *J. Mol. Biol.* **273**, 752.

PASCAL

The SI unit (symbolized by Pa), equal to one newton per square meter, for pressure or stress. One pascal equals ten baryes, 10^{-5} bar , and $9.869233 \times 10^{-6} \text{ atm}$ -sphere.

PASCAL'S TRIANGLE

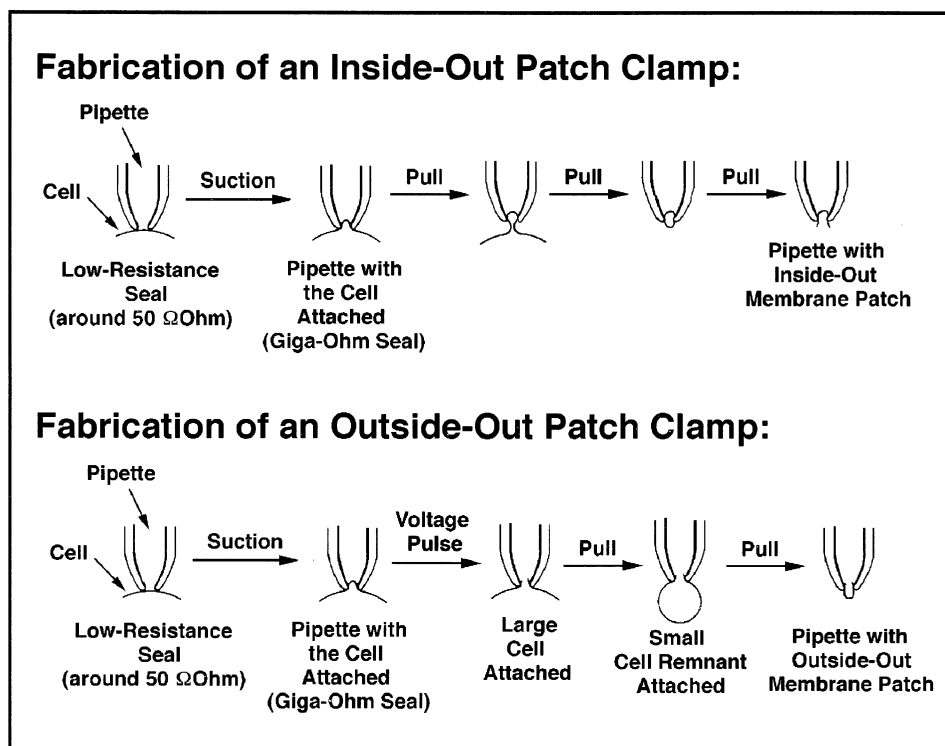
A systematic scheme for generating the coefficients of terms in a polynomial formed by raising a binomial $(a + b)$ to the exponent n , such that: $(a + b)^0 = 1$; $(a + b)^1 = 1a + 1b$; $(a + b)^2 = 1a + 2ab + 1b$; $(a + b)^3 = 1a + 3a^2b + 3ab^2 + 1b^3$; $(a + b)^4 = 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1a^4$; etc. Pascal's triangle offers a useful check on coefficients used in derivations of multiple equilibria models of ligand-receptor interactions.

1	1	1	1	1	1	1	1
1	2	3	4	5	6	7	
1	3	6	10	15	21		
1	4	10	20	35			
1	5	15	35				
1	6	21					
1	7						
1							

The triangle can be generated by remembering three properties: (a) the first row and first column are filled at every step with 1's; (b) the second row and second column are sets of counting numbers (*i.e.*, 1, 2, 3, . . .); and (3) each of the remaining numbers is the sum of numbers to its left and above it. Shown here is Pascal's triangle for the first seven steps.

PATCH CLAMP

An electrophysiologic ion current-measuring device¹ that comprises (a) an excised membrane patch that has become sealed to the orifice of a glass micropipette and (b) a voltage clamp electronic circuit for measuring ion channel currents. The patch clamp relies on formation of a gigaohm seal between the recording pipette and the cell membrane. This approach allows the investigator to prepare (a) inside-out patches with their cytoplasmic membrane face exposed to the bath solution, and (b) outside-out patches with their extracellular membrane face exposed to the bath solution (see diagram). The technique offers the advantage that solution composition on each side of the membrane patch can be readily manipulated. Electrophysiologic measurements are both accurate and precise.



Sakman and Neher² should be consulted regarding refinements in the extracellular patch clamp method that enable higher current resolution, direct membrane patch potential control, and physical isolation of membrane patches. They describe a convenient method for the fabrication of patch recording pipettes and procedures for achieving so-called “giga-seals”—namely pipette-membrane seals with resistances of 10^9 – 10^{11} Ω . The basic patch clamp recording circuit and properties of single acetylcholine-activated channels in muscle membrane illustrate the improved current and time-resolution that can be achieved with giga-seals. This monograph also considers the various ways for physically isolating patches of membrane from cells, thereby enabling one to record single channel currents with well-defined solutions on both sides of the membrane.

¹E. Neher & J. H. Steinbach (1978) *J. Physiol.* **277**, 153.

²B. Sakman & E. Neher, eds. (1983) *Single-Channel Recording*, Plenum, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Technique and sample preparation, **207**, 159; outside-out configuration, **207**, 10; amplifiers, **238**, 153, 310; scrape loading, **238**, 363–364; trituration method, **238**, 374; data acquisition with fluorescence measurements, **238**, 312–313, 318–320; exocytosis monitoring, **238**, 320; pipette, **238**, 314–315; recording, **238**, 142, 152–153; stimulator, **238**, 311; temperature control, **238**, 314; tip-dip method, **238**, 344–348; ion channels from [*Esche-*

richia coli, giant cells, and giant spheroplasts, **207**, 687; microves, **207**, 681; organelles, **207**, 673; yeast (mitochondria, **207**, 686; spheroplasts, **207**, 685; vacuoles, **207**, 686).

pCa

The negative decadic (*i.e.*, base 10) logarithm of the calcium ion concentration.

pD

The negative decadic (*i.e.*, base 10) logarithm of the D^+ (deuteron) concentration. If a pH electrode has been standardized with aqueous buffers, it can be used with D_2O systems with the following relationship: $pD = (\text{meter reading}) + 0.4$.

PECTINESTERASE

This enzyme [EC 3.1.1.11] also known as pectin methyl-esterase, pectin demethoxylase, and pectin methoxylase, catalyzes the reaction of pectin with n water to produce pectate and n methanol.

R. L. Fischer & A. B. Bennett (1991) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **42**, 675.

PENICILLOPEPSIN

This endopeptidase [EC 3.4.23.20], a member of the peptidase family A1, hydrolyzes proteins with a broad specificity similar to that of pepsin A.

D. R. Davies (1990) *Ann. Rev. Biophys. Biophys. Chem.* **19**, 189.

PEPSINS AND PEPSINOGENS

(1) Pepsin A [EC 3.4.23.1], the predominant endopeptidase in the gastric juice of vertebrates, is formed from pepsinogen A by limited proteolysis. Occurring in five molecular forms, this member of the A1 peptidase family preferentially cleaves peptide bonds in which hydrophobic, preferably aromatic, amino acyl residues are in the P₁ and P_{1'} positions. (2) Pepsin B [EC 3.4.23.2] is an endopeptidase that catalyzes the degradation of gelatin. It exhibits very little activity with hemoglobin. The enzyme is formed from pepsinogen B and is a member of the peptidase family A1. (3) Pepsin C [EC 3.4.23.3], better known as gastricsin, is formed from progastricsin. Gastricsin exhibits an even more restricted specificity than pepsin A, with preferential cleavage at Tyr—Xaa bonds.

T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.

J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Pepsin: Activation energy, **63**, 243-245; isotope exchange, **64**, 10; kinetic constants, **63**, 244; kinin-releasing enzyme, **80**, 174; porcine, homology to cathepsin D, **80**, 578; spectrokinetic probe, **61**, 323; tight-binding inhibition, **63**, 462, 466; active site, structure, **241**, 214-215; amino-terminal lobe, pep.pep homodimer, **241**, 198, 201-203; assay, **241**, 213-214; crystal structure, **241**, 214-216; kinetic parameters, **241**, 222; pK_a values, **241**, 222

Pepstatin: As aspartyl protease inhibitor, **241**, 160, 183; inhibition of aspartic peptidases, **244**, 10; pepstatin-protease complex, **241**, 170-171.

PEPTIDE METHIONINE SULFOXIDE REDUCTASE

This reductase, officially known as protein-methionine-S-oxide reductase [EC 1.8.4.6], catalyzes the reaction of protein L-methionine with oxidized thioredoxin to yield protein L-methionine S-oxide and reduced thioredoxin. Dithiothreitol can replace reduced thioredoxin in the reverse reaction. Free methionine is not a substrate; however free methionine is similarly acted upon by an analogous reductase [EC 1.8.4.5].

N. Brot, M. A. Rahman, J. Moskovitz & H. Weissbach (1995) *Meth. Enzymol.* **251**, 462.

PEPTIDOMIMETIC COMPOUND

Analogues in which one or more of the naturally occurring peptide bonds in a substrate, effector, or ligand has been replaced by some other chemical structure. Such compounds are useful in characterizing binding specificity (particularly conformational specificity), and

by virtue of omitting the peptide bond(s), they often exhibit much longer *in vivo* half-life. Peptidomimetics can also occur naturally. Although morphine's structure has been known from Knorr's work nearly a century ago, the peptidomimetic nature of this analgesic agent wasn't recognized until about seventy years later.

PEPTIDYL-ASPARTATE ENDOPEPTIDASE

This metalloendoproteinase [EC 3.4.24.33] catalyzes the hydrolysis of Xaa—Asp, Xaa—Glu, and Xaa—cysteic acid bonds. Because of its limited specificity, this enzyme is useful in protein sequencing applications.

M.-L. Hagmann, V. Geuss, S. Fischer & G.-B. Kresse (1995) *Meth. Enzymol.* **248**, 782.

PEPTIDYL GLYCINE α -AMIDATING MONOOXYGENASE

Peptidylglycine monooxygenase [EC 1.14.17.3], also known as peptidyl α -amidating enzyme and peptidylglycine 2-hydroxylase, catalyzes the reaction of a peptidylglycine with ascorbate and dioxygen to produce a peptidyl(2-hydroxyglycine), dehydroascorbate, and water.

A. S. Kolhekar, R. E. Mains & B. A. Eipper (1997) *Meth. Enzymol.* **279**, 35.

PEPTIDYLTRANSFERASE

This ribosomal enzyme [EC 2.3.2.12] catalyzes the acyl transfer from a peptidyl-tRNA¹ with an aminoacyl-tRNA² to produce tRNA¹ and a peptidylaminoacyl-tRNA².

H. F. Noller (1993) *J. Bacteriol.* **175**, 5297.

H. F. Noller, V. Hoffarth & L. Zimniak (1992) *Science* **256**, 1416.

R. L. Soffer (1974) *Adv. Enzymol.* **40**, 91.

PERIODATE CLEAVAGE

The oxidative C—C bond-breaking reaction that occurs when periodate ion (IO₄[−]) reacts with an organic compound having one or more vicinal hydroxyl groups. The specificity of the reaction results from the formation of an initial complex of the 1,2-diol with periodate prior to the oxidative cleavage step. This reaction is useful in determining the cyclic structures of carbohydrates as well as in the preparation of electrophilic substrate analogues that can be immobilized for affinity chromatography used in enzyme purification. **See Malaprade Reaction**

PERMEABILITY

1. The scalar, symbolized by μ , that relates the magnetic flux density, B , in a body or medium with the external

magnetic field strength, H , inducing it. Thus, $B = \mu H$ where μ has SI units of newtons per square ampere (which are equivalent to henrys per meter). The permeability of a vacuum, symbolized by μ_0 and synonymous with the term magnetic constant, is set equal to $4\pi \times 10^{-7} \text{ N A}^{-2}$ (or, $12.56637 \dots \times 10^{-7} \text{ N A}^{-2}$). The relative permeability, μ_r , is equal to μ/μ_0 . For diamagnetic substances, $\mu_r < 1$. If the substance is paramagnetic, $\mu_r > 1$. The μ_r values for ferromagnetic substances are large and vary with the field strength. **See also** *Magnetic Susceptibility*

2. The ability of a substance or ion to pass through a membrane; typically measured by permeability constants. **See** *Permeability Constant*

PERMEABILITY COEFFICIENT

A parameter (usually symbolized by P , and often containing a subscript to indicate the specific ion) that is a measure of the ease with which an ion can cross a unit area of membrane by simple (or passive) diffusion through a membrane experiencing a 1.0 M concentration gradient. For a particular biological membrane, the permeabilities are dependent on the concentration and activity of various channel or transporter proteins. In an electrically active cell (*e.g.*, a neuron), increasing the permeability of K^+ or Cl^- will usually result in hyperpolarization of the membrane. Increasing P_{Na} will cause depolarization.

The permeability coefficient (units = cm/s) can be written as follows:

$$P = -\frac{(c_2 - c_1)}{J} = \frac{KD}{l}$$

where $(c_2 - c_1)$ represents the concentration difference across a semipermeable membrane, J is the net transport rate (usually expressed as moles transferred per square centimeter per second), K is the lipid-water partition coefficient, D is the diffusion coefficient, and l is the thickness of the membrane. For the human erythrocyte membrane, the approximate permeability coefficients (cm/s) for K^+ , Na^+ , Cl^- , glucose, and water are 2×10^{-10} , 10^{-10} , 10^{-4} , 2×10^{-5} , and 5×10^{-3} . For the peripheral membrane of giant squid axon, the approximate permeability coefficients (cm/s) for K^+ , Na^+ , and Cl^- are: (un-

der resting conditions) 6×10^{-8} , 2×10^{-8} , and 10^{-8} ; and (upon excitation) 2×10^{-4} , 5×10^{-6} , and 10^{-8} .

M. K. Jain & R. C. Wagner (1980) *Introduction to Biological Membranes*, Wiley, New York.

PERMISSIVE ACTIVE SITE

A descriptor for an enzyme active site that permits binding of a family of related compounds (*e.g.*, mimics of the reaction intermediate) that can be derived from the initial binding and conformational changes in the substrate. This concept arose from the observation that a number of monoterpene cyclases were incapable of discriminating between enantiomers of the reaction intermediate, even though the enzyme catalyzes the synthesis of an enantiomerically pure product from an achiral substrate. An example is trichodiene synthase¹ which catalyzes the cyclization of farnesyl diphosphate to trichodiene.

¹D. E. Cane, G. Yang, R. M. Coates, H.-J. Pyun & T. M. Hohn (1992) *J. Org. Chem.* **57**, 3454.

PERMUTATIONS AND COMBINATIONS

A permutation is a change that is primarily based on a rearrangement of existing elements or things. A combination of elements or things is any selection of one or more, taken without concern regarding their order.

Permutations. There are several important statements about permutations and their properties. First, the number of permutations of n different things in a row, taken all at a time, is given by the expression:

$$n! = (1)(2)(3) \dots (n-1)(n)$$

where $n!$ is "n factorial". This rule is reasonable because the first row can have n possible things, the second row can have $(n-1)$ things, the third row $(n-2)$ things, and so forth until the n th row contains only a single thing. Second, if n things are divided into c classes, where the same class contains things that are alike, and where those within different classes are different, the number of permutations of these things, taken all at a time, is given by the expression:

$$\frac{n!}{(n_1!)(n_2!)(n_3!) \dots (n_c!)}$$

where the subscripts indicate all the classes from 1 to c , and $n = n_1 + n_2 + n_3 + \dots + n_c$. Third, the number of

different permutations of n different things taken k at a time without repetitions is given by:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}$$

Combinations. The number of different combinations of n different things, taken k at a time, without repetitions, is given by the expression:

$$\left(\frac{n}{k}\right) = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-k+1)}{k!(n-k)!}$$

Stirling's Approximation. When n gets large, evaluation of $n!$ becomes cumbersome, even with modern hand-held calculators. In such cases, one can use an approximation first recognized by the Scottish mathematician James Stirling:

$$n \simeq (2\pi n)^{1/2}(n/e)^n$$

See also *Statistics (A Primer); Linked Functions*

PEROXIDASE

This heme-dependent enzyme [EC 1.11.1.7] catalyzes the reaction of a donor with hydrogen peroxide to produce an oxidized donor and two water. **See also** *Horseradish Peroxidase; Oviperoxidase; Myeloperoxidase*

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

F. P. Guengerich (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 97.

PERSISTENCE TIME

1. The period over which a system sustains a particular chemical behavior or biological response. 2. The time of continual progress of a chemotactic microorganism locomoting in one direction. **See** *Twiddling*

PERSISTENT

Referring to a reaction intermediate or free radical that has a lifetime longer than that of a transient species, typically on the time-scale of at least several minutes in dilute solution in inert solvents. Persistence is therefore a kinetic property related to reactivity. The *stability* of an intermediate or free radical is a thermodynamic property, often expressed in terms of the appropriate bond strengths. **See** *Transient Chemical Species*

D. Griller and K. U. Ingold (1976) *Acc. Chem. Res.* **9**, 13.

PERTUSSIS TOXIN

An ADP-ribosyltransferase produced by the *Bordetella pertussis* that catalyzes the transfer of an ADPR group onto the G-protein $G_{i\alpha}$, thereby blocking exchange of bound GDP for GTP.

D. J. Carty (1994) *Meth. Enzymol.* **237**, 63.

L. Passador & W. Iglewski (1994) *Meth. Enzymol.* **235**, 617.

N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., **20**, 453.

C.-Y. Lai (1986) *Adv. Enzymol.* **58**, 99.

PHARMACOKINETICS

A special branch of drug metabolism/physiology dealing with the development of quantitative models for treating the time evolution of processes involved in drug absorption by the body, drug distribution throughout the various compartments (plasma, organs, tissues, populations of single cells, as well as defined regions within cells), drug conversion into various metabolized forms, and finally, drug excretion. These models are absolutely essential in developing an understanding of how various individuals dispose of a drug, and the information can be of paramount importance in efforts to improve the efficacy of a drug.

Many factors can influence drug response, including but not limited to the following: (a) physiologic factors such as gastrointestinal function, cardiovascular function, liver function, immune responsiveness, renal function, and stress (additionally for females: luteal cycle, pregnancy, lactation, and menopause); (b) physical factors such as season, time of day, duration of daylight, altitude, and ambient temperature; (c) individual factors such as genetics, age, sex, weight (obesity), state of health, exercise, cigarette smoking, occupational hazards, chemical exposure, alcohol consumption, and immunization; (d) site(s) and mode(s) of drug administration; (e) the various derivatized and/or formulated forms of the drug, including factors influencing drug stability and dissolution kinetics; (f) interactions with other drugs taken by the subject; and (g) biochemical and physiologic individuality (such as blood albumin levels, mutations affecting receptors, transporters, acid/base balance, *etc.*).

Detailed pharmacokinetic investigations are a necessary prelude to the formal approval protocols of the Federal Drug Administration and its counterparts worldwide. Although this topic lies beyond the scope of this Hand-

book, those interested will want to consult an excellent treatment, entitled “Basics Pharmacokinetics”. This internet textbook developed by M. C. Makoid, P. J. Vuchetich, and U. V. Banakar at Creighton University (<http://kiwi.creighton.edu>) includes a mathematical review (through Laplace transforms) as well as chapters on pharmacological response, kinetics of intravenous bolus and intravenous infusion, biopharmaceutic factors, oral dosing, bioavailability, clearance, dosage, multicompartment modeling, protein binding, and nonlinear kinetics behavior.

PHASE

1. Any homogeneous part of a system that itself is heterogeneous (*i.e.*, separated from the other parts of the system by some physical boundary). For example, there are three phases present in a glass half-full of water with ice floating in the water: ice, liquid water, and air (containing water vapor). 2. A particular state of some periodic function or operation. 3. A point in time or fraction of a period that has elapsed in which the item undergoing change is of harmonic motion, uniform circular motion, or any periodic change (such as rotation, oscillation, *etc.*) that follows simple harmonic laws. The phase of such a harmonic quantity usually refers to some standard position or direction such as at the instant of starting. Particles in periodic motion are in phase if they are moving in the same direction as the relative displacement. Waves having the same frequency and waveform are in phase if they reach the same values simultaneously.

PHASE MODULATION FLUORESCENCE

A kinetic technique for determining a fluorophore's excited state lifetime by using a light source whose intensity is modulated sinusoidally at a certain frequency, such that the intensity of the fluorescence emission likewise varies sinusoidally but with an added delay from the finite relaxation constant for fluorescence decay. The period of the sinusoidal modulation is chosen to be in the neighborhood of the magnitude of the fluorescence lifetime.

J. R. Lakowicz (1983) *Principles of Fluorescence Spectroscopy*, pp. 53-56, Plenum, New York.

PHASE-TRANSFER CATALYSIS

The rate enhancement observed in a reaction occurring between chemical species present in different phases (for

example, in two immiscible solvents) by the addition of a small amount of some agent (often referred to as the phase-transfer catalyst). In many cases in biological systems, this agent is a surfactant that extracts one of the reactants from one phase to the other phase, thus permitting the reaction to proceed.

pH EFFECTS ON ENZYMIC CATALYSIS

Because enzymes and/or substrates often contain one or more catalytically functional acid/base groups, it is not at all surprising that pH is a major factor influencing enzymic catalysis. That enzymes often display such interpretable and quantitatively analyzable behavior is quite remarkable, considering the great number of ionizable groups that they contain. This fact suggests (a) that the active center of an enzyme is reasonably well isolated from the effects of other ionizable groups outside the active site and (b) that an enzyme employs relatively few acid or base groups to achieve its great rate enhancement.

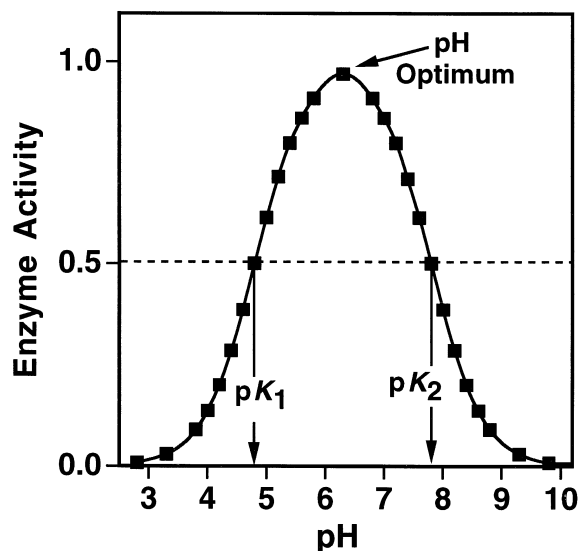
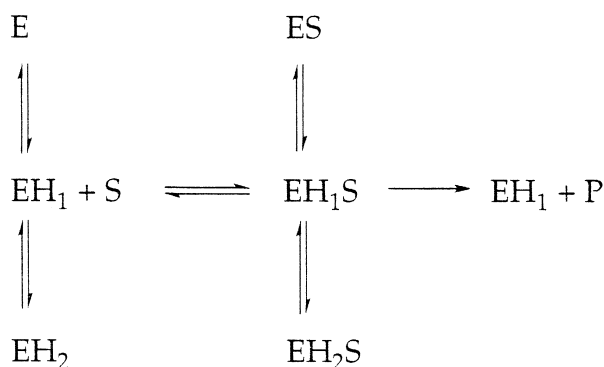


Figure 1. Bell-shaped pH-rate profile for an enzyme that has two ionizable groups affecting enzyme activity. The rate is directly proportional to the degree of ionization of the first group (determined by pK_1) and inversely proportional to the degree of ionization of the second group (governed by pK_2).

In characterizing the pH dependence of enzyme activity, one often observes (a) a bell-shaped curve in plots of activity *versus* pH (Fig. 1), or (b) “S”-shaped activity *versus* pH curves (either falling from optimal activity or rising to optimal activity) with an inflection point at some

pH value. Michaelis and Davidsohn¹ were among the first to offer a kinetic scheme for explaining the nature of pH-rate profiles for enzyme-catalyzed reactions. Von Euler² extended these ideas by writing the following generalized scheme involving three enzyme states:



where the pertinent equilibrium constants can be written for the acidic (designated by subscript “a”) and basic (designated by subscript “b”) equilibria.

$$\begin{aligned}
 K_S &= [\text{E}][\text{S}]/[\text{ES}] \\
 K_{\text{Eb}} &= [\text{E}][\text{H}^+]/[\text{EH}_1] \\
 K_{\text{ESb}} &= [\text{ES}][\text{H}^+]/[\text{EH}_1\text{S}] \\
 K_{\text{Ea}} &= [\text{EH}_1][\text{H}^+]/[\text{EH}_2] \\
 K_{\text{ESa}} &= [\text{EH}_1\text{S}][\text{H}^+]/[\text{EH}_2\text{S}]
 \end{aligned}$$

If EH_1 and S combine rapidly, and if the conversion of EH_1S to $\text{EH}_1 + \text{P}$ is the slow step, one can write a rate law similar to the Michaelis-Menten equation. In this case, however, the enzyme is distributed into additional forms (including four nonproductive forms E , ES , EH_2 and EH_2S):

$$\begin{aligned}
 [\text{E}_{\text{total}}] &= [\text{ES}] + [\text{EHS}] + [\text{EH}_2\text{S}] \\
 &\quad + [\text{E}] + [\text{EH}] + [\text{EH}_2]
 \end{aligned}$$

The resulting rate law can be written as:

$$v = V_{\text{max}} / \{ [1 + [\text{H}^+]/K_{\text{ESa}} + K_{\text{ESb}}/[\text{H}^+]] + (K_S/[\text{S}])[1 + [\text{H}^+]/K_{\text{ESa}} + K_{\text{ESb}}/[\text{H}^+]] \}$$

If $pK_{\text{Eb}} \approx pK_{\text{ESb}}$ and $pK_{\text{Ea}} \approx pK_{\text{ESa}}$, and if the two sets of pK 's are separated sufficiently on the pH scale, then one can demonstrate that a plot of velocity *versus* pH will rise to a maximum at $(pK_{\text{Eb}} + pK_{\text{Ea}})/2$. At a pH value

below this point, the increased hydrogen ion “drives” the enzyme into the inactive EH_2 and EH_2S forms; similarly, as one moves toward higher pH, the inactive E and ES forms accumulate (Fig. 2).

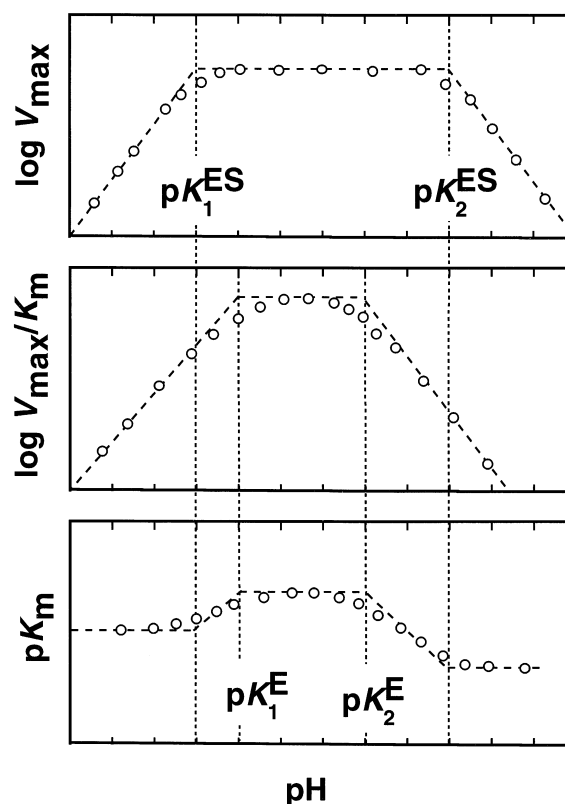
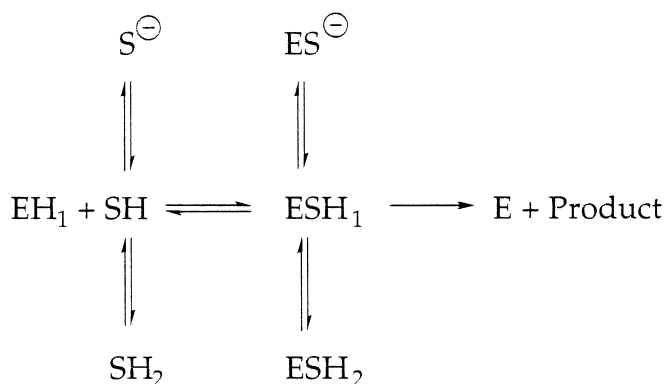


Figure 2. Analysis of the pH-dependence of the key parameters influencing the initial rate behavior of an enzyme containing two ionizable groups that determine catalytic activity.

The reader should note that ionizable groups on the substrate could also result in pH-rate profiles that pass through a maximum. For example, one can write the following scheme:



pH kinetic studies can help one to identify the type of amino acid side-chain group that is participating in catalysis. When combined with studies of temperature effects, solvent effects, isotope effects, *etc.*, pH kinetics can be a powerful tool. A thorough treatment of pH kinetics for enzymes lies beyond the scope of this Handbook, and the interested reader is encouraged to consult Fromm³, Tipton & Dixon⁴, and Cleland⁵ for more details on the effects of pH on enzymic activity.

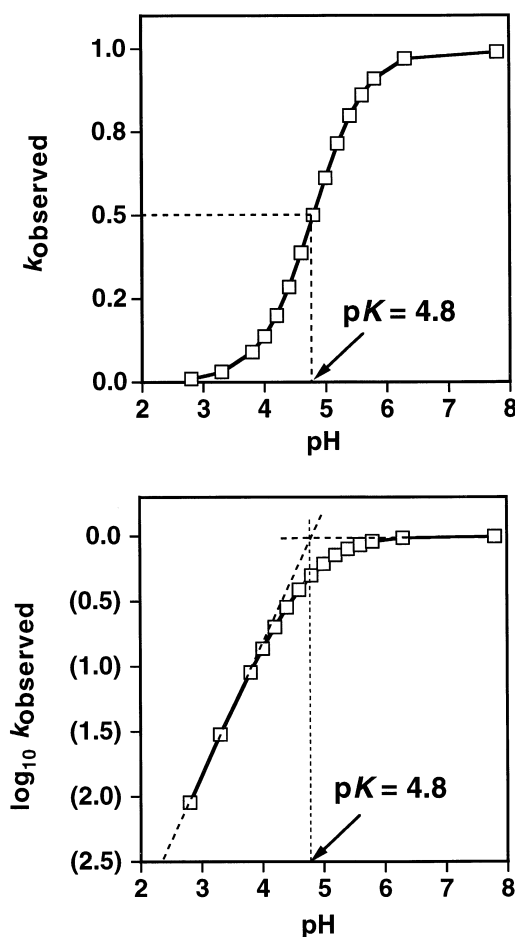


Figure 3. pH-dependence of an observed rate constant k_{obs} .

When it is feasible to identify and characterize a single rate process, the effect of pH is more straightforward. This is frequently the case for nonenzymatic reactions obeying simple rate equations, but fast reaction techniques sometimes allow one to examine a single elementary reaction in an enzymic process. In either case, the behavior shown in Fig. 3 typifies the manner in which an observed reaction rate constant depends on the pK_a of an ionizable group.

¹L. Michaelis & H. Davidsohn (1911) *Biochem Z.* **35**, 386.

²H. v. Euler, K. Josephson & K. Myrbäck (1924) *Hoppe-Seyler's Z. Physiol. Chem.* **134**, 39.

³H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, pp. 201-224, Springer-Verlag, New York.

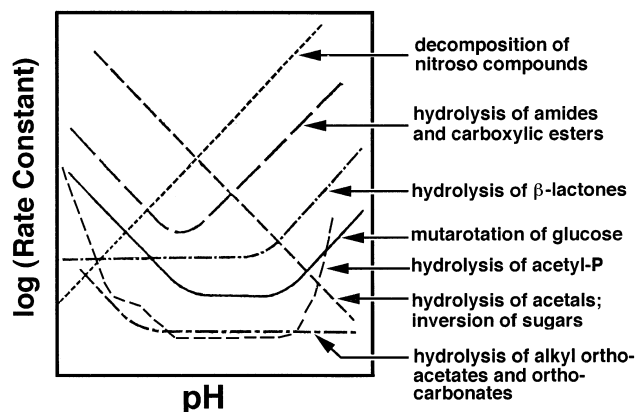
⁴K. F. Tipton & H. B. F. Dixon (1979) *Meth. Enzymol.* **63**, 183.

⁵W. W. Cleland (1982) *Meth. Enzymol.* **87**, 390.

pH EFFECTS ON NONENZYMATIC HYDROLYSIS

In his famous treatise on acid and base catalysis, Bell¹ consolidated modern views on acids and bases, particularly with respect to their effects on the rates of hydrolysis of various substances. Though not the originator of the concepts of general acid and general base catalysis, he promoted broader recognition that catalysis does not always occur simply through the action of hydrogen and hydroxyl ions. Bell also advanced the acidity and basicity concepts of Brønsted and Lewis, while investigating the mechanistic origin for the fact that hydrolysis of certain substances has a “signature” profile of $\log(\text{reaction rate})$ versus pH.

Typical Profiles for Rate versus pH



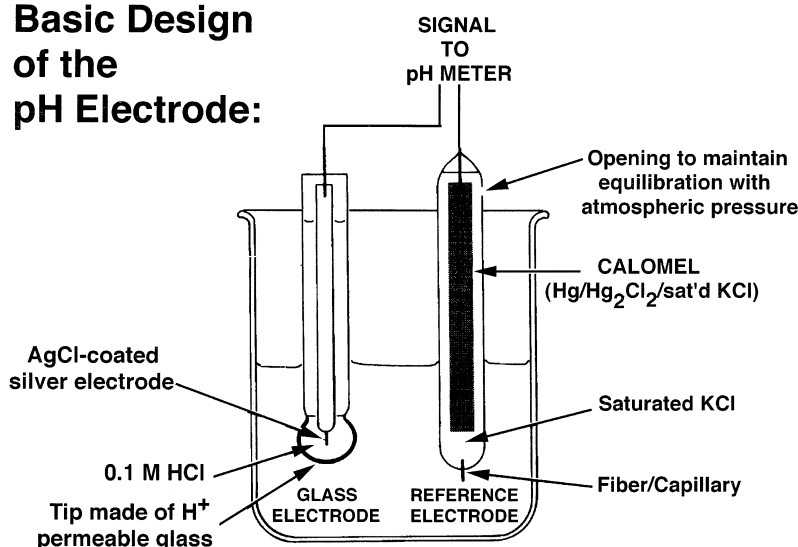
The diagram shown above is based, in part, on Bell's compilation of pH-rate profiles. The interested reader will wish to consult his classic presentation as well as specific entries on the properties and actions of acids and bases throughout this Handbook.

¹R. P. Bell (1941) *Acid-Base Catalysis*, Clarendon Press, Oxford.

pH ELECTRODE

A glass electrode which takes advantage of rapid and reproducible ion exchange across a glass membrane located as a thin-walled bulb at the tip of the electrode. This membrane has the following liquid-membrane junction:

Basic Design of the pH Electrode:



By its chemical nature, the very thin dry layer of glass permits ion exchange without any possibility of electrochemical redox reactions. If the membrane is a sufficiently thin layer, one can achieve millisecond response times to changes in pH, and these properties can be put to full advantage with rapid mixing devices to study chemical reactions with half-lives in the 5–10 msec range.

PHENOL HYDROXYLASE

This FAD-dependent enzyme [EC 1.14.13.7], also known as phenol 2-monooxygenase, catalyzes the reaction of phenol with NADPH and dioxygen to produce catechol, NADP⁺, and water. Other substrates include resorcinol and *o*-cresol.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
- B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
- S. Yamamoto & Y. Ishimura (1991) in *A Study of Enzymes* (S. A. Kuzy, ed.), vol. **2**, pp. 315-344, CRC Press, Boca Raton.
- J. H. Law, P. E. Dunn & K. J. Kramer (1977) *Adv. Enzymol.* **45**, 389.
- V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.
- V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

PHENYLALANINE AMINOTRANSFERASE

This enzyme [EC 2.6.1.58], also referred to as phenylalanine(histidine) aminotransferase, catalyzes the reversible reaction of L-phenylalanine with pyruvate to produce phenylpyruvate and L-alanine. This enzyme will also use L-histidine and L-tyrosine as substrates. In the reverse reaction, L-methionine, L-serine, and L-gluta-

mine can replace L-alanine. **See also** *Aromatic Amino Acid Aminotransferase*

M. Fujioka, Y. Morino & H. Wada (1970) *Meth. Enzymol.* **17A**, 585.

PHENYLALANINE AMMONIA-LYASE

This enzyme [EC 4.3.1.5] catalyzes the conversion of L-phenylalanine to *trans*-cinnamate and ammonia. The enzyme may also act on L-tyrosine. **See also** *Dehydroalanine; Borohydride Reduction*

- V. E. Anderson (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 115.
- C. W. Abell & R.-S. Shen (1987) *Meth. Enzymol.* **142**, 242.
- E. A. Havir (1987) *Meth. Enzymol.* **142**, 248.
- R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.
- K. R. Hanson & E. A. Havir (1972) *The Enzymes*, 3rd ed., **7**, 75.

PHENYLALANINE DECARBOXYLASE

This pyridoxal-phosphate-dependent enzyme [EC 4.1.1.53] catalyzes the conversion of L-phenylalanine to phenethylamine and carbon dioxide. Tyrosine and other aromatic amino acids can also serve as substrates.

M. Fujioka, Y. Morino & H. Wada (1970) *Meth. Enzymol.* **17A**, 585.

PHENYLALANINE DEHYDROGENASE

This enzyme [EC 1.4.1.20] catalyzes the reaction of L-phenylalanine with NAD⁺ and water to produce phenylpyruvate, ammonia, and NADH. The enzymes isolated from *Bacillus badius* and *Sporosarcina ureae* are highly specific for L-phenylalanine, whereas that isolated from *Bacillus sphaericus* also acts on L-tyrosine.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

PHENYLALANINE HYDROXYLASE (Phenylalanine 4-Monooxygenase)

This iron-dependent enzyme [EC 1.14.16.1], more widely known as phenylalanine hydroxylase and phenylalanine, catalyzes the reaction of L-phenylalanine with tetrahydrobiopterin and dioxygen to produce L-tyrosine, dihydrobiopterin, and water.

- P. F. Fitzpatrick (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 181.
 S. Kaufman (1993) *Adv. Enzymol.* **67**, 77.
 S. Kaufman (1987) *The Enzymes*, 3rd ed., **18**, 217.
 V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.
 V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

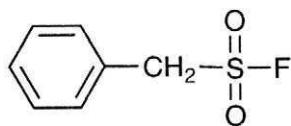
PHENYLALANINE RACEMASE

This enzyme [EC 5.1.1.11], also known as phenylalanine racemase (ATP-hydrolyzing), catalyzes the reaction of ATP with L-phenylalanine to produce D-phenylalanine, AMP, and pyrophosphate. In this unusual racemase reaction, a thiol group of an enzyme-bound pantotheine forms a thiolester from an initial aminoacyl-AMP intermediate; then, as is typical of acyl thioesters, the α -proton becomes labile, thereby permitting reversible inversion of configuration to produce an equilibrated mixture of thiolester-bound enantiomers. Hydrolysis of the thiolester yields the product.

- M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
 E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

PHENYLMETHYLSULFONYL FLUORIDE

A sulfonylating agent (abbreviated PMSF) that irreversibly inhibits many "serine esterases" and "serine proteases". Target enzymes usually react with PMSF and related alkylating agents through the activated imidazole group of a histidyl residue that is part of the catalytic triad.



Gold¹ suggests that the reaction can be written analogously to the Michaelis-Menten equation:



Because PMSF fails to inactivate acetylcholinesterase, this reagent is much less toxic than diisopropylfluorophosphate, and is also recommended as an alternative to the neurotoxic fluorophosphates and fluorophosphonates. PMSF is freshly prepared as a 1–3 mM solution in water (higher concentrations will precipitate spontaneously). A better procedure is to first prepare a 20 mM PMSF solution in 2-propanol or dioxane; this solution can then be added to the biological fluid with vortex mixing to achieve a 1–3 mM final concentration as a homogeneous solution. One should confirm that the alcohol or dioxane has little or no undesirable effect on enzymes or proteins of interest. **See Chymotrypsin; Protease Inhibitor "Cocktails"**

¹A. M. Gold (1967) *Meth. Enzymol.* **11**, 706.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Sulfonylation reaction, **11**, 706; reaction kinetics, **11**, 707; second-order rate constants for inactivation of chymotrypsin, trypsin, and acetylcholine esterase by PMSF and related sulfonylating agents, **11**, 707; reactivation of PMS-chymotrypsin, **11**, 710; as inhibitor [of calcium-activated factor, **80**, 674; of cathepsin G, **80**, 565; of crayfish trypsin, **80**, 639; of elastase, **80**, 587; of prolylcarboxypeptidase, **80**, 465; of protease Re, **80**, 691; of protease So, **80**, 695; of protein C_s, **80**, 329]; proteolysis, **76**, 7.

PHENILPYRUVATE SYNTHASE

This enzyme [EC 1.3.1.12], also known as hydroxyphenylpyruvate synthase or prephenate dehydrogenase, catalyzes the NAD⁺-dependent oxidation of prephenate to form 4-hydroxyphenylpyruvate, CO₂ and NADH. The following is a recent review on the molecular and physical properties of this synthase.

B. B. Buchanan (1972) *The Enzymes*, 3rd ed., **6**, 193.

PHI (Φ or ϕ)

Φ 1. Symbol for potential energy. 2. Symbol for magnetic flux. 3. Symbol for radiant energy per time (radiant power).

ϕ 1. Symbol for quantum yield. 2. Symbol for one of the space coordinates in the three-dimensional, spherical polar coordination system. 3. Symbol for electric potential. 4. Symbol for volume fraction. 5. With a subscript designation, symbol for a Dalziel coefficient. 6. Symbol for fugacity coefficient. 7. Symbol for osmotic coefficient. 8. Symbol for heat flow rate.

PHI VALUES IN BISUBSTRATE MECHANISMS

Initial rate enzyme kinetics are useful in defining the order of substrate binding interactions in multisubstrate

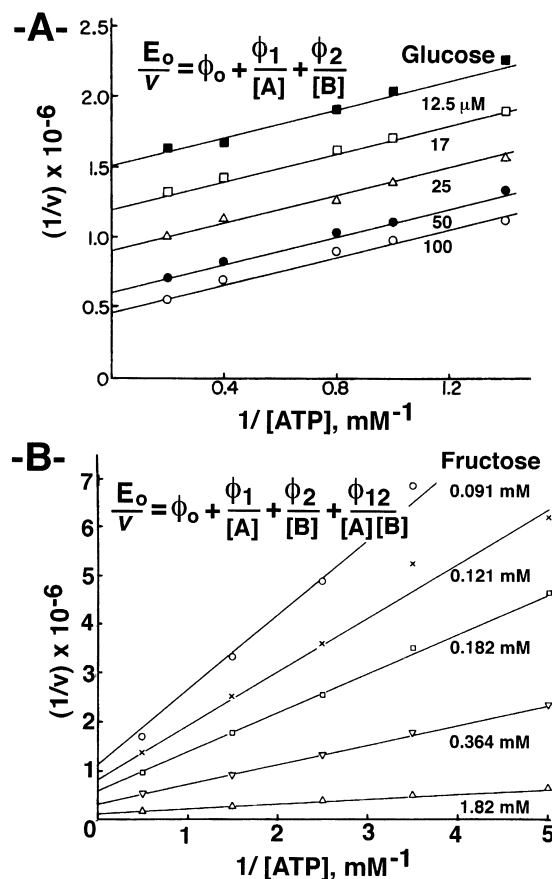
kinetic mechanisms. Even in the absence of corroborating evidence (from competitive inhibition studies, product inhibition, isotope exchange at equilibrium, *etc.*), sequential mechanisms should be discernible from the ping pong case. Indeed, as shown below for bisubstrate cases, sequential and ping pong mechanisms have two different functional dependencies on changes in the concentrations of substrates:

$$\text{Sequential Mechanisms: } E_o/v = \phi_o + \phi_1/[A] + \phi_2/[B] + \phi_{12}/[A][B]$$

$$\text{Ping-Pong Mechanism: } E_o/v = \phi_o + \phi_1/[A] + \phi_2/[B]$$

where the phi values represent collections of rate constants. Concentration ranges for substrates A and B can be chosen to achieve adequate saturation of the enzyme, thereby yielding a suitable range of experimentally determined reaction velocities (**See Initial Rate Enzyme Assays**). Associated with each A_xB_y pair will be an experimentally determined initial reaction velocity v_{xy} , and these can be plotted (1) as $1/v$ versus $1/[A]$, yielding a family of five straight-line plots corresponding to the five constant levels of B, and (2) as $1/v$ versus $1/[B]$, yielding a family of five straight-line plots corresponding to the five constant levels of A. Because sequential mechanisms have a $\phi_{12}/[A][B]$ term, each family of curves will converge to the left of the $(1/v)$ -axis. This was observed to be the case in the yeast hexokinase reaction¹. The absence of the $\phi_{12}/[A][B]$ term in ping pong mechanisms means that one should obtain two families of parallel line data, as shown for yeast nucleoside-5'-diphosphate kinase².

While such considerations suggest that initial rate studies alone can be a useful diagnostic of mechanism type, parallel-line data may result whenever the $\phi_{12}/[A][B]$ is sufficiently small so as to prevent the experimenter from observing convergent-line data characteristic of sequential mechanisms. This was recognized as a possibility in the case of the brain hexokinases³⁻⁶ which yield parallel-line (or nearly so) data in initial rate studies with D-glucose and MgATP²⁻. Because this class of enzymes exhibits activity with other hexoses, such as D-fructose or D-mannose, the initial rate results conducted with the former monosaccharide were informative regarding the kinetic mechanism of the brain hexokinases. The double-reciprocal plots below indicate the initial rate behavior of rat brain hexokinase with D-glucose and D-fructose and MgATP²⁻.



Initial rate kinetics of bovine brain hexokinase (A) with glucose and ATP as substrates, and (B) with fructose and ATP. Note that the apparent parallel-line kinetics observed with glucose conform to a rate equation lacking a ϕ_{12} term.

The results indicate that brain hexokinase does indeed operate by way of a sequential kinetic mechanism, and subsequent kinetic studies with reversible inhibitors support this conclusion. These comments reinforce the wisdom of remaining wary of mechanistic inferences drawn solely on the basis of initial rate studies alone. **See also Initial Rate Enzyme Assays**

¹H. J. Fromm & V. Zewe (1962) *J. Biol. Chem.* **237**, 3027.

²E. Garces & W. W. Cleland (1967) *Biochemistry* **8**, 633.

³M. Copley & H. J. Fromm (1967) *Biochemistry* **6**, 3503.

⁴H. J. Fromm & J. Ning (1968) *Biochem. Biophys. Res. Commun.* **32**, 672.

⁵J. Ning, D. L. Purich & D. L. Fromm (1969) *J. Biol. Chem.* **244**, 3840.

⁶D. L. Purich & H. J. Fromm (1971) *J. Biol. Chem.* **246**, 3456.

PHOSPHATE ACETYLTRANSFERASE (Phosphotransacetylase)

This enzyme [EC 2.3.1.8], perhaps more widely known as phosphotransacetylase and less so as phosphoacylase, catalyzes the reaction of acetyl-CoA with orthophosphate to produce coenzyme A and acetyl phosphate.

Other short-chain acyl-CoA derivatives can act as substrates.

L. L. Barton (1994) *Meth. Enzymol.* **243**, 94.

PHOSPHATE CONTINUOUS ASSAY

See *Orthophosphate Continuous Assay*

PHOSPHATE DIESTER HYDROLYSIS

Phosphate diesters resist acid hydrolysis, and they are only inefficiently hydrolyzed in alkaline solutions. The exception is any five-membered ring containing a phosphodiester, such as that formed upon alkaline treatment of RNA. Such five-membered cyclic phosphodiesters are rapidly hydrolyzed in alkaline solutions.

PHOSPHATE MONOESTER HYDROLYSIS

Phosphate monoesters $[R-OPO(OH)_2]$ resist alkaline hydrolysis by converting to unreactive dianions $[R-OPO_3^{2-}]$. Phosphate monoester monoanions $[R-OPO(OH)O^{1-}]$ are, however, susceptible to facile hydrolysis under slightly acidic conditions.

PHOSPHATIDATE CYTIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.41], also known as CTP:phosphatidate cytidylyltransferase and CDP-diacylglycerol synthase, catalyzes the reaction of CTP with phosphatidate to produce CDP-diacylglycerol and pyrophosphate.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

C. P. Sparrow (1992) *Meth. Enzymol.* **209**, 237.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

PHOSPHATIDATE PHOSPHATASE

This enzyme [EC 3.1.3.4] catalyzes the hydrolysis of a 3-*sn*-phosphatidate to generate a 1,2-diacyl-*sn*-glycerol and orthophosphate.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

G. M. Carman & J. J. Quinlan (1992) *Meth. Enzymol.* **209**, 219.

G. M. Carman & Y.-P. Lin (1991) *Meth. Enzymol.* **197**, 548.

R. M. Bell & R. A. Coleman (1983) *The Enzymes*, 3rd ed., **16**, 87.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Phosphatidate phosphatase from yeast, **197**, 548; characterization and assay of phosphatidate phosphatase, **197**, 553.

PHOSPHATIDYLCHOLINE:SERINE O-PHOSPHATIDYLTRANSFERASE

This transferase, officially known as CDP-diacylglycerol—serine *O*-phosphatidyltransferase [EC 2.7.8.8],

catalyzes the reaction of CDP-diacylglycerol with L-serine to form CMP and *O*-*sn*-phosphatidyl-L-serine.

W. R. Bishop & R. M. Bell (1988) *Ann. Rev. Cell Biol.* **4**, 579.

PHOSPHATIDYLETHANOLAMINE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.17] catalyzes the reaction of *S*-adenosyl-L-methionine with phosphatidylethanolamine to produce *S*-adenosyl-L-homocysteine and phosphatidyl-*N*-methylethanolamine.

D. Rhodes & A. D. Hanson (1993) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **44**, 357.

N. D. Ridgway & D. E. Vance (1992) *Meth. Enzymol.* **209**, 366.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

PHOSPHATIDYLGLYCEROL PHOSPHATE PHOSPHOHYDROLASE

This enzyme [EC 3.1.3.27], also known as phosphatidylglycerophosphatase, catalyzes the hydrolysis of phosphatidylglycerophosphate to produce phosphatidylglycerol and orthophosphate.

W. Dowhan & C. R. Funk (1992) *Meth. Enzymol.* **209**, 224.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

1-PHOSPHATIDYLINOSITOL 4-KINASE

This enzyme [EC 2.7.1.67] catalyzes the reaction of ATP with 1-phosphatidyl-1D-*myo*-inositol to produce ADP and 1-phosphatidyl-1D-*myo*-inositol 4-phosphate.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

G. M. Carman, C. J. Belunis & J. T. Nickels, Jr. (1992) *Meth. Enzymol.* **209**, 183.

A. Moritz, J. Westerman, P. N. E. de Graan & K. W. A. Wirtz (1992) *Meth. Enzymol.* **209**, 202.

PHOSPHATIDYLINOSITOL-4-PHOSPHATE 5-KINASE

This enzyme [EC 2.7.1.68], also known as diphosphoinositide kinase, catalyzes the reaction of ATP with 1-phosphatidyl-1D-*myo*-inositol 4-monophosphate to produce ADP and 1-phosphatidyl-1D-*myo*-inositol 4,5-bisphosphate.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

C. E. Bazenet & R. A. Anderson (1992) *Meth. Enzymol.* **209**, 189.

A. Moritz, J. Westerman, P. N. E. de Graan & K. W. A. Wirtz (1992) *Meth. Enzymol.* **209**, 202.

1-PHOSPHATIDYLINOSITOL PHOSPHODIESTERASE

This enzyme [EC 3.1.4.10], also known as monophosphatidylinositol phosphodiesterase and phosphatidylinositol

phospholipase C, catalyzes the conversion of 1-phosphatidyl-1D-*myo*-inositol to D-*myo*-inositol 1,2-cyclic phosphate and a diacylglycerol. The animal enzyme, but not that from bacteria, also hydrolyzes the cyclic phosphate, forming inositol 1-phosphate as the final product.

J. E. Bleasdale, J. C. McGuire & G. A. Bala (1990) *Meth. Enzymol.* **187**, 226.

PHOSPHATIDYLINOSITOL SYNTHASE

This enzyme [EC 2.7.8.11], also known as CDP-diacylglycerol—inositol 3-phosphatidyltransferase, catalyzes the reaction of CDP-diacylglycerol with *myo*-inositol to produce CMP and phosphatidyl-1D-*myo*-inositol.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

G. M. Carman & A. S. Fischl (1992) *Meth. Enzymol.* **209**, 305.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

PHOSPHATIDYL-N-METHYLETHANOLAMINE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.71] catalyzes the reaction of *S*-adenosyl-L-methionine with phosphatidyl-N-methylethanolamine to produce *S*-adenosyl-L-homocysteine and phosphatidyl-N-dimethylethanolamine. The enzyme will catalyze the transfer of an additional methyl group, thus producing phosphatidylcholine.

J. Bremer (1969) *Meth. Enzymol.* **14**, 125.

PHOSPHATIDYLSERINE DECARBOXYLASE

These pyridoxal-phosphate-dependent (or pyruvate-dependent) enzymes [EC 4.1.1.65] catalyze the decarboxylation of phosphatidyl-L-serine to produce phosphatidylethanolamine and carbon dioxide.

M. L. Hackert & A. E. Pegg (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 201.

W. Dowhan (1997) *Meth. Enzymol.* **280**, 81.

W. Dowhan & Q.-X. Li (1992) *Meth. Enzymol.* **209**, 348.

D. R. Voelker & E. B. Gordon (1992) *Meth. Enzymol.* **209**, 360.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

PHOSPHATIDYLSERINE SYNTHASE

This enzyme [EC 2.7.8.8], also known as CDP-diacylglycerol:L-serine *O*-phosphatidyltransferase, catalyzes the reaction of CDP-diacylglycerol with L-serine to produce CMP and *O*-*sn*-phosphatidyl-L-serine.

C. Kent (1995) *Ann. Rev. Biochem.* **64**, 315.

W. Dowhan (1992) *Meth. Enzymol.* **209**, 287.

G. M. Carman & M. Bae-Lee (1992) *Meth. Enzymol.* **209**, 298.

R. A. Pieringer (1983) *The Enzymes*, 3rd ed., **16**, 255.

PHOSPHOACETYLGLUCOSAMINE MUTASE

This enzyme [EC 5.4.2.3], also known as acetylglucosamine phosphomutase and *N*-acetylglucosamine-phosphate mutase, catalyzes the interconversion of *N*-acetyl-D-glucosamine 1-phosphate and *N*-acetyl-D-glucosamine 6-phosphate. The enzyme is activated by *N*-acetyl-D-glucosamine 1,6-bisphosphate.

J. L. Reissig & L. F. Leloir (1966) *Meth. Enzymol.* **8**, 175.

D. M. Carlson (1966) *Meth. Enzymol.* **8**, 179.

3'-PHOSPHOADENYLYL SULFATE SULFOHYDROLASE

This manganese-dependent enzyme [EC 3.6.2.2], also known as phosphoadenylylsulfatase, catalyzes the hydrolysis of 3'-phosphoadenylyl sulfate to yield adenosine 3',5'-bisphosphate and sulfate.

A. B. Roy (1971) *The Enzymes*, 3rd ed., **5**, 1.

PHOSPHODIESTERASES

Venom exonuclease [EC 3.1.15.1], also known as venom phosphodiesterase, catalyzes the exonucleolytic cleavage of RNA or DNA (preferring single-stranded substrates) in the 3' to 5' direction to yield 5'-phosphomononucleotides. Similar enzymes include hog kidney phosphodiesterase and the *Lactobacillus* exonuclease. **See also** *specific phosphodiesterase*

J. A. Gerlt (1992) *The Enzymes*, 3rd ed., **20**, 95.

PHOSPHOENOLPYRUVATE CARBOXYKINASE

Phosphoenolpyruvate carboxykinase (GTP) [EC 4.1.1.32], also known as phosphoenolpyruvate carboxylase and phosphopyruvate carboxylase, catalyzes the reaction of GTP with oxaloacetate to produce GDP, phosphoenolpyruvate, and carbon dioxide. ITP can replace GTP as the phosphorylating substrate.

Phosphoenolpyruvate carboxykinase (ATP) [EC 4.1.1.49], also known as phosphopyruvate carboxylase (ATP) and phosphoenolpyruvate carboxylase, catalyzes the reaction of ATP with oxaloacetate to produce ADP, phosphoenolpyruvate, and carbon dioxide.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

A. Matte, L. W. Tari, H. Goldir & L. T. J. Delbaere (1997) *J. Biol. Chem.* **272**, 18105.

- R. W. Hanson & Y. M. Patel (1994) *Adv. Enzymol.* **69**, 203.
 J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.
 M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.
 M. F. Utter & H. M. Kolenbrander (1972) *The Enzymes*, 3rd ed., **6**, 117.

PHOSPHOENOLPYRUVATE CARBOXYKINASE (Pyrophosphate)

This enzyme [EC 4.1.1.38] (also known as phosphoenolpyruvate carboxytransphosphorylase, phosphopyruvate carboxylase, and phosphoenolpyruvate carboxylase) catalyzes the reaction of phosphoenolpyruvate with orthophosphate and carbon dioxide to produce oxaloacetate and pyrophosphate (or diphosphate). The enzyme also catalyzes the reaction of phosphoenolpyruvate with orthophosphate to produce pyruvate and pyrophosphate.

- H. G. Wood, W. E. O'Brien & G. Michaels (1977) *Adv. Enzymol.* **45**, 85.
 M. F. Utter & H. M. Kolenbrander (1972) *The Enzymes*, 3rd ed., **6**, 117.

PHOSPHOENOLPYRUVATE CARBOXYLASE

This enzyme [EC 4.1.1.31] catalyzes the reaction of orthophosphate with oxaloacetate to produce phosphoenolpyruvate, carbon dioxide, and water.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 R. Chollet, J. Vidal & M. H. O'Leary (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 273.
 M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.
 P. M. Goodwin (1990) *Meth. Enzymol.* **188**, 361.
 M. F. Utter & H. M. Kolenbrander (1972) *The Enzymes*, 3rd ed., **6**, 117.

PHOSPHOENOLPYRUVATE MUTASE

This enzyme [EC 5.4.2.9], also known as phosphoenolpyruvate phosphomutase, catalyzes the interconversion of phosphoenolpyruvate and 3-phosphonopyruvate. This enzyme participates in the biosynthesis of the C—P bond. However, the equilibrium greatly favors phosphoenolpyruvate.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 A. C. Hengge (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 517.

PHOSPHOENOLPYRUVATE PHOSPHATASE

This phosphatase [EC 3.1.3.60] catalyzes the hydrolysis of phosphoenolpyruvate to form pyruvate and phos-

phate. The enzyme also acts less efficiently on a wide range of other monophosphates.

- W. C. Plaxton (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 185.

PHOSPHOFRUCTOKINASE

This enzyme [EC 2.7.1.11], also known as phosphohexokinase and phosphofructokinase I, catalyzes the reaction of ATP with D-fructose 6-phosphate to produce ADP and D-fructose 1,6-bisphosphate. Both D-tagatose 6-phosphate and sedoheptulose 7-phosphate can act as the sugar substrate. UTP, CTP, GTP, and ITP all can act as the nucleotide substrate. This enzyme is distinct from that of 6-phosphofructo-2-kinase. **See also** *ATP & GTP Depletion*

- H. C. Huppe & D. H. Turpin (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 577.
 T. Schirmer & P. R. Evans (1990) *Nature* **343**, 140.
 S. J. Pilkis, T. H. Claus, P. D. Kountz & M. R. El-Maghrabi (1987) *The Enzymes*, 3rd ed., **18**, 3.
 E. Van Schaffingen (1987) *Adv. Enzymol.* **59**, 315.
 K. Uyeda (1979) *Adv. Enzymol.* **48**, 193.
 D. P. Bloxham & H. A. Lardy (1973) *The Enzymes*, 3rd ed., **8**, 239.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Activation, **64**, 221; anomeric specificity, **63**, 373, 374; assay, **63**, 376; burst analysis, **64**, 197; cascade control, **64**, 325; contamination, **63**, 7; equilibrium shift, **64**, 220; hysteresis, **64**, 195; inactivation, **64**, 221; mechanism, **63**, 51; physiological damping, **64**, 217, 220, 221; product inhibition, **63**, 433; rate properties, initial, **63**, 156; substrate analog, **63**, 373, 374; substrate inhibition, **63**, 501.

6-PHOSPHOFRUCTO-2-KINASE

This enzyme [EC 2.7.1.105], also known as phosphofructokinase 2, catalyzes the reaction of ATP with D-fructose 6-phosphate to produce ADP and D-fructose 2,6-bisphosphate. This enzyme is distinct from that of 6-phosphofructokinase. It copurifies with fructose-2,6-bisphosphatase.

- S. J. Pilkis, T. H. Claus, I. J. Kurland & A. J. Lange (1995) *Ann. Rev. Biochem.* **64**, 799.
 S. J. Pilkis, T. H. Claus, P. D. Kountz & M. R. El-Maghrabi (1987) *The Enzymes*, 3rd ed., **18**, 3.
 E. Van Schaffingen (1987) *Adv. Enzymol.* **59**, 315.

6-PHOSPHO- β -GALACTOSIDASE

This enzyme [EC 3.2.1.85], also known as β -D-phosphogalactoside galactohydrolase, catalyzes the hydrolysis of a 6-phospho- β -D-galactoside to produce an alcohol and 6-phospho-D-galactose.

- M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

PHOSPHOGLUCOMUTASE

This enzyme [EC 5.4.2.2], also known as glucose phosphomutase, catalyzes the interconversion of α -D-glucose 1-phosphate and α -D-glucose 6-phosphate. For this reaction to occur, α -D-glucose 1,6-bisphosphate must be present. This bisphosphate is an intermediate in the reaction, being formed by the transfer of a phosphate residue from the enzyme to the substrate. However, the dissociation of bisphosphate from the enzyme complex is much slower than the overall isomerization.

- S. Shankar, R. W. Ye, D. Schlichtman & A. M. Chakrabarty (1995) *Adv. Enzymol.* **70**, 221.
 J.-B. Dai, Y. Liu, W. J. Ray, Jr. & M. Konno (1992) *J. Biol. Chem.* **267**, 6322.
 I. A. Rose (1975) *Adv. Enzymol.* **43**, 491.
 W. J. Ray, Jr., & E. J. Peck, Jr. (1972) *The Enzymes*, 3rd ed., **6**, 407.

6-PHOSPHOGLUCONATE DEHYDROGENASE

This enzyme [EC 1.1.1.43] catalyzes the reaction of 6-phospho-D-gluconate with NAD(P)⁺ to produce 6-phospho-2-dehydro-D-gluconate and NAD(P)H.

- M. J. Adams, G. H. Ellis, S. Gover, C. E. Naylor & C. Phillips (1994) *Structure* **2**, 651.
 M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.
 R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 L. V. Kletsova, M. Y. Kiriukhin, A. Y. Chistoserdov & Y. D. Tsyganov (1990) *Meth. Enzymol.* **188**, 335.
 A. P. Sokolov & Y. A. Trotsenko (1990) *Meth. Enzymol.* **188**, 339.
 K. Dalziel (1975) *The Enzymes*, 3rd ed., **11**, 1.

PHOSPHOGLUCOSE ISOMERASE

This enzyme [EC 5.3.1.9] (also known as glucose-6-phosphate isomerase, phosphohexose isomerase, phosphohexomutase, oxoisomerase, hexosephosphate isomerase, phosphosaccharomutase, phosphoglucoisomerase, and phosphohexoisomerase) catalyzes the interconversion of D-glucose 6-phosphate and D-fructose 6-phosphate. The enzyme also catalyzes the anomerization (or, mutarotation) of D-glucose 6-phosphate.

- S. H. Seeholzer (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 1237.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **9**, 323.
 E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

6-PHOSPHO- β -GLUCOSIDASE

This enzyme [EC 3.2.1.86] catalyzes the hydrolysis of 6-phospho- β -D-glucoside-(1,4)-D-glucose to produce D-glucose 6-phosphate and glucose. The enzyme will also

catalyze the hydrolysis of several other phospho- β -D-glucosides; but not their nonphosphorylated forms.

- M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

3-PHOSPHOGLYCERATE DEHYDROGENASE

This enzyme [EC 1.1.1.95] catalyzes the reaction of 3-phosphoglycerate with NAD⁺ to produce 3-phosphohydroxypyruvate and NADH.

- A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.
 L. I. Pizer & E. Sugimoto (1971) *Meth. Enzymol.* **17B**, 325.
 H. J. Sallach (1966) *Meth. Enzymol.* **9**, 216.

3-PHOSPHOGLYCERATE KINASE

This enzyme [EC 2.7.2.3] catalyzes the reaction of ATP with 3-phospho-D-glycerate to produce ADP and 3-phospho-D-glyceroyl phosphate.

- R. Jaenicke, H. Schurig, N. Beaucamp & R. Ostendorp (1996) *Adv. Protein Chem.* **48**, 181.
 P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

PHOSPHOGLYCERATE MUTASE

This enzyme [EC 5.4.2.1] (also known as phosphoglycerate phosphomutase and phosphoglyceromutase) catalyzes the reaction of 2-phospho-D-glycerate with 2,3-diphosphoglycerate to produce 3-phospho-D-glycerate and 2,3-diphosphoglycerate. The enzymes isolated from mammals and from yeast are phosphorylated by (2R)-2,3-bisphosphoglycerate, which is also an intermediate. With the rabbit muscle enzyme, the dissociation of bisphosphate from the enzyme is much slower than the overall isomerization.

- A. C. Hengge (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 517.
 L. A. Fothergill-Gilmore & H. C. Watson (1989) *Adv. Enzymol.* **62**, 227.
 Z. B. Rose (1982) *Meth. Enzymol.* **87**, 42.
 Z. B. Rose (1980) *Adv. Enzymol.* **51**, 211.
 W. J. Ray, Jr., & E. J. Peck, Jr. (1972) *The Enzymes*, 3rd ed., **6**, 407.

PHOSPHOGLYCERIDE: LYSOPHOSPHATIDYLGLYCEROL ACYLTRANSFERASE

The following is a recent review on the molecular and physical properties of this transferase.

- K. Y. Hostetler, J. Huterer & J. R. Wherrett (1992) *Meth. Enzymol.* **209**, 104.

PHOSPHOKETOLASE

This thiamin pyrophosphate-dependent enzyme [EC 4.1.2.9] catalyzes the reaction of D-xylulose 5-phosphate with orthophosphate to produce acetyl phosphate, D-glyceraldehyde 3-phosphate, and water.

M. Goldberg, J. M. Fessenden & E. Racker (1966) *Meth. Enzymol.* **9**, 515.

B. L. Horecker (1962) *Meth. Enzymol.* **5**, 261.

PHOSPHOLIPASE

This broad class of hydrolases constitutes a special category of enzymes which bind to and conduct their catalytic functions at the interface between the aqueous solution and the surface of membranes, vesicles, or emulsions. In order to explain the kinetics of lipolysis, one must determine the rates and affinities that govern enzyme adsorption to the interface of insoluble lipid structures^{1,2}. One must also account for the special properties of the lipid surface as well as for the ability of enzymes to scoot² along the lipid surface. **See** *specific enzyme*; *Micelle*; *Interfacial Catalysis*

¹R. Verger (1980) *Meth. Enzymol.* **64**, 340.

²M. K. Jain, M. H. Gelb, J. Rogers & O. G. Berg (1995) *Meth. Enzymol.* **249**, 567.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Assays & Kinetics: Assay strategies and methods for phospholipases, **197**, 3; utilization of labeled *Escherichia coli* as phospholipase substrate, **197**, 24; nuclear magnetic resonance spectroscopy to follow phospholipase kinetics and products, **197**, 31; monolayer techniques for studying phospholipase kinetics, **197**, 49; thio-based phospholipase assay, **197**, 65; chromogenic substrates and assay of phospholipases A₂, **197**, 75; coupled enzyme assays for phospholipase activities with plasmalogen substrates, **197**, 79; phospholipase A₂ assays with fluorophore-labeled lipid substrates, **197**, 90; assays of phospholipases on short-chain phospholipids, **197**, 95; phospholipase A₂-catalyzed hydrolysis of vesicles: uses of interfacial catalysis in the scooting mode, **197**, 112; assay of phospholipases C and D in presence of other lipid hydrolases, **197**, 125; preparation of alkyl ether and vinyl ether substrates for phospholipases, **197**, 134; inositol phospholipids and phosphates for investigation of intact cell phospholipase C substrates and products, **197**, 149; chromatographic analysis of phospholipase reaction products, **197**, 158; assays for measuring arachidonic acid release from phospholipids, **197**, 166; phosphoinositide-specific phospholipase C activation in brain cortical membranes, **197**, 183; quantitative analysis of water-soluble products of cell-associated phospholipase C- and phospholipase D-catalyzed hydrolysis of phosphatidylcholine, **197**, 191.

Structure-Function: Dissection and sequence analysis of phospholipases A₂, **197**, 201; cloning, expression, and purification of porcine pancreatic phospholipase A₂ and mutants, **197**, 214; preparation of antibodies to phospholipases A₂, **197**, 223; thermodynamics of phospholipase A₂-ligand interactions, **197**, 234; activation of phospholipase A₂ on lipid bilayers, **197**, 249;

phospholipase stereospecificity at phosphorus, **197**, 258; phospholipase A₂: microinjection and cell localization techniques, **197**, 269; analysis of lipases by radiation inactivation, **197**, 280; phosphorylation of phospholipase C *in vivo* and *in vitro*, **197**, 288.

PHOSPHOLIPASE A₁

This calcium-dependent enzyme [EC 3.1.1.32] catalyzes the hydrolysis of a phosphatidylcholine to generate 2-acylglycerophosphocholine and a fatty acid anion. This enzyme has a much broader specificity than phospholipase A₂. **See** *Micelle*

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

E. A. Dennis (1983) *The Enzymes*, 3rd ed., **16**, 307.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Detergent-resistant phospholipase A₁ from *Escherichia coli* membranes, **197**, 309; phospholipase A₁ activity of guinea pig pancreatic lipase, **197**, 316; purification of rat kidney lysosomal phospholipase A₁, **197**, 325; purification and substrate specificity of rat hepatic lipase, **197**, 331; human postheparin plasma lipoprotein lipase and hepatic triglyceride lipase, **197**, 339; phospholipase activity of milk lipoprotein lipase, **197**, 345.

PHOSPHOLIPASE A₂

This calcium-dependent enzyme [EC 3.1.1.4] (also known as phosphatidylcholine 2-acylhydrolase, lecithinase A, phosphatidase, and phosphatidolipase) catalyzes the hydrolysis of phosphatidylcholine to produce 1-acylglycerophosphocholine and a fatty acid anion. The enzyme can also hydrolyze phosphatidylethanolamine, choline plasmalogen, and phosphatides, removing the fatty acyl group attached to the C2 position of the glycerol moiety. **See** *Micelle*

D. M. Quinn & S. R. Feaster (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 455.

M. H. Gelb, M. K. Jain, A. M. Hanel & O. G. Berg (1995) *Ann. Rev. Biochem.* **64**, 653.

M. K. Jain, M. H. Gelb, J. Rogers & O. G. Berg (1995) *Meth. Enzymol.* **249**, 567.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

E. A. Dennis (1983) *The Enzymes*, 3rd ed., **16**, 307.

R. Verger (1980) *Meth. Enzymol.* **64**, 340.

D. J. Hanahan (1971) *The Enzymes*, 3rd ed., **5**, 71.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Cobra venom phospholipase A₂: *Naja naja naja*, **197**, 359; phospholipase A₂ from rat liver mitochondria, **197**, 365; assay and purification of phospholipase A₂ from human synovial fluid in rheumatoid arthritis, **197**, 373; purification of mammalian nonpancreatic extracellular phospholipases A₂, **197**, 381; spleen phospholipases A₂, **197**, 390; purification and characterization of cytosolic phospholipase A₂ activities from canine myocardium and sheep platelets, **197**, 400.

PHOSPHOLIPASE B

This enzyme, known officially as lysophospholipase [EC 3.1.1.5] as well as by other common names: lecithinase B, lysolecithinase, and phospholipase B, catalyzes the hydrolysis of 2-lysophosphatidylcholine to yield glycerophosphocholine and a fatty acid. **See** *Micelle*

K. Saito, J. Sugatani & T. Okumura (1991) *Meth. Enzymol.* **197**, 446.

PHOSPHOLIPASE C

This zinc-dependent enzyme [EC 3.1.4.3] (also known as lipophosphodiesterase I, lecithinase C, *Clostridium welchii* α -toxin, and *Clostridium oedematiens* β - and γ -toxins) catalyzes the hydrolysis of a phosphatidylcholine to produce 1,2-diacylglycerol and choline phosphate. The enzyme isolated from bacterial sources also acts on sphingomyelin and phosphatidylinositol; however, the enzyme isolated from seminal plasma does not act on phosphatidylinositol. **See** *Micelle*

N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.

W. D. Singer, H. A. Brown & P. C. Sternweis (1997) *Ann. Rev. Biochem.* **66**, 475.

J. D. Esko & C. R. H. Raetz (1983) *The Enzymes*, 3rd ed., **16**, 207.

E. A. Dennis (1983) *The Enzymes*, 3rd ed., **16**, 307.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Phosphatidylinositol-specific phospholipases C from *Bacillus cereus* and *Bacillus thuringiensis*, **197**, 493; assays for phosphoinositide-specific phospholipase C and purification of isozymes from bovine brain, **197**, 502; properties of phospholipase C isozymes, **197**, 511; phosphatidylinositol-specific phospholipase C from human platelets, **197**, 518; purification of guinea pig uterus phosphoinositide-specific phospholipase C, **197**, 526.

PHOSPHOLIPASE D

This enzyme [EC 3.1.4.4] (also known as lipophosphodiesterase II, lecithinase D, and choline phosphatase) catalyzes the hydrolysis of a phosphatidylcholine to produce choline and a phosphatidate. The enzyme will also act on other phosphatidyl esters. **See** *Micelle*

J. H. Exton (1997) *J. Biol. Chem.* **272**, 15579.

H. A. Brown & P. C. Sternweis (1995) *Meth. Enzymol.* **257**, 313.

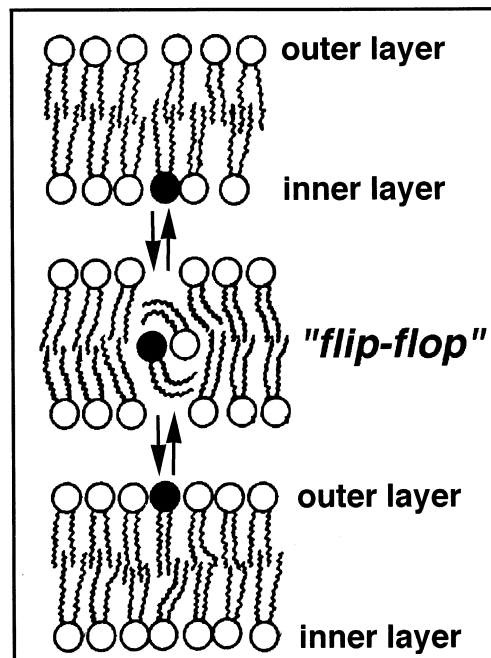
E. P. Bowman, D. J. Uhlinger & J. D. Lambeth (1995) *Meth. Enzymol.* **256**, 246.

E. A. Dennis (1983) *The Enzymes*, 3rd ed., **16**, 307.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Glycosylphosphatidylinositol-specific phospholipase D, **197**, 567; solubilization and purification of rat tissue phospholipase D, **197**, 575; lysophospholipase D, **197**, 583.

PHOSPHOLIPID FLIP-FLOP



The pair-wise exchange of phospholipid molecules across a membrane bilayer. The polar head groups of phospholipids resist entry into the apolar core of the membrane bilayer, and phospholipid flip-flop is both improbable and sluggish (*i.e.*, $k_{\text{exchange}} \approx 10^{-5} \text{ s}^{-1}$). Kornberg and McConnell¹ used nitroxide spin-labeled phospholipid vesicles to follow the time course of flip-flop. In their clever experiments, ascorbic acid was added to the bulk solvent to reduce any nitroxide spin label on the exterior surface of the lipid. Then, the time course of flip-flop could be determined by the time course of the diminution in the amplitude of the ESR spectrum that measured the amount of spin-label remaining on the inner, and thus solvent-inaccessible membrane surface.

¹R. Kornberg & H. M. McConnell (1971) *Biochemistry* **10**, 1111.

PHOSPHOMANNOMUTASE

This enzyme [EC 5.4.2.8] catalyzes the interconversion of D-mannose 1-phosphate and D-mannose 6-phosphate. Either D-mannose 1,6-bisphosphate or D-glucose 1,6-bisphosphate can act as the cofactor.

S. Shankar, R. W. Ye, D. Schlichtman & A. M. Chakrabarty (1995) *Adv. Enzymol.* **70**, 221.

5-PHOSPHOMEVALONATE KINASE

This enzyme [EC 2.7.4.2] catalyzes the reaction of ATP with (R)-5-phosphomevalonate to produce ADP and (R)-5-diphosphomevalonate.

PHOSPHOPANTOTHENOYL-CYSTEINE SYNTHETASE

This enzyme [EC 6.3.2.5], also known as phosphopantothenoylcysteine ligase, catalyzes the reaction of CTP with (*R*)-4'-phosphopantothenate and L-cysteine to produce as yet unidentified products of CTP breakdown and (*R*)-4'-phosphopantothenoyl-L-cysteine. Cysteine can be replaced in this reaction by some of its derivatives.

P. D. van Poelje & E. E. Snell (1990) *Ann. Rev. Biochem.* **59**, 29.

PHOSPHOPENTOMUTASE

This enzyme [EC 5.4.2.7], also known as phosphodeoxyribomutase, catalyzes the interconversion of D-ribose 1-phosphate and D-ribose 5-phosphate. This enzyme will also catalyze the interconversion of 2-deoxy-D-ribose 1-phosphate and 2-deoxy-D-ribose 5-phosphate. The bisphosphate compound is the cofactor: thus, D-ribose 1,5-bisphosphate, 2-deoxy-D-ribose 1,5-bisphosphate, or even α -D-glucose 1,6-bisphosphate.

W. J. Ray, Jr., & E. J. Peck, Jr. (1972) *The Enzymes*, 3rd ed., **6**, 407.

PHOSPHOPROTEIN PHOSPHATASES

This set of enzymes [EC 3.1.3.16], also known as serine/threonine-specific protein phosphatases, catalyzes the hydrolysis of a seryl- or threonyl-bound phosphate group from a wide range of phosphoproteins, including a number of enzymes which have been phosphorylated under the action of a kinase. Thus, phosphoprotein + water yields protein + orthophosphate.

A. C. Hengge (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 517.

R. D. Smith & J. C. Walker (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 101.

S. I. Walaas & P. Greengard (1987) *The Enzymes*, 3rd ed., **18**, 285.

L. M. Ballou & E. H. Fischer (1986) *The Enzymes*, 3rd ed., **17**, 311.

PHOSPHORESCENCE

1. Long-lived luminescence. 2. Luminescence involving a change in spin multiplicity (*e.g.*, triplet-to-singlet, singlet-to-triplet, quartet state-to-doublet state, *etc.*). **See** *Fluorescence*; *Jablonski Diagram*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

PHOSPHORIBOSYLAMINOIMIDAZOLE-CARBOXAMIDE FORMYLTRANSFERASE

This enzyme [EC 2.1.2.3], also known as AICAR transformylase, catalyzes the reaction of 10-formyltetrahydrofolate with 5'-phosphoribosyl-5-amino-4-imidazole-

carboxamide to produce tetrahydrofolate and 5'-phosphoribosyl-5-formamido-4-imidazolecarboxamide.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

PHOSPHORIBOSYL-AMP CYCLOHYDROLASE

This enzyme [EC 3.5.4.19] catalyzes the hydrolysis of 5-phosphoribosyl-AMP to produce 5-(5-phospho-D-ribosylaminoformimino)-1-(5-phosphoribosyl)imidazole-4-carboxamide. The enzyme isolated from *Neurospora crassa* can also catalyze the reactions of histidinol dehydrogenase and phosphoribosyl-ATP pyrophosphatase.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

PHOSPHORIBOSYLANTHRANILATE ISOMERASE

This enzyme [EC 5.3.1.24], also known as *N*-(5'-phosphoribosyl)anthranilate isomerase, catalyzes the interconversion of *N*-(5-phospho- β -D-ribosyl)anthranilate and 1-(2-carboxyphenylamino)-1-deoxy-D-ribulose 5-phosphate. In some organisms, this enzyme is part of a multifunctional protein, together with one or more components of the system for the biosynthesis of tryptophan (anthranilate phosphoribosyltransferase, indole-3-glycerol-phosphate synthase, anthranilate synthase, and tryptophan synthase).

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

K. Kirschner, H. Szadkowski, T. S. Jardetzky & V. Hager (1987) *Meth. Enzymol.* **142**, 386.

PHOSPHORIBOSYL-ATP PYROPHOSPHATASE

This enzyme [EC 3.6.1.31] catalyzes the hydrolysis of 5-phosphoribosyl-ATP to produce 5-phosphoribosyl-AMP and pyrophosphate (or, diphosphate). The *Neurospora crassa* enzyme also catalyzes the reactions of histidinol dehydrogenase and phosphoribosyl-AMP cyclohydrolase.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

N-(5'-PHOSPHORIBOSYLFORMIMINO)-5-AMINO-1-(5"-PHOSPHORIBOSYL)-4-IMIDAZOLECARBOXAMIDE ISOMERASE

This enzyme [EC 5.3.1.16] catalyzes the interconversion of *N*-(5'-phospho-D-ribosylformimino)-5-amino-1-(5"-

phosphoribosyl)-4-imidazolecarboxamide and *N*-(5'-phospho-D-1'-ribulosylformimino)-5-amino-1-(5''-phosphoribosyl)-4-imidazolecarboxamide.

R. G. Martin, M. A. Berberich, B. N. Ames, W. W. Davis, R. F. Goldberger & J. D. Yourno (1971) *Meth. Enzymol.* **17B**, 3.

PHOSPHORIBOSYLGLYCINAMIDE FORMYLTRANSFERASE

This enzyme [EC 2.1.2.2], also known as 5'-phosphoribosylglycinamide transformylase, catalyzes the reaction of 10-formyltetrahydrofolate with 5'-phosphoribosylglycinamide to produce tetrahydrofolate and 5'-phosphoribosyl-*N*-formylglycinamide.

V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

PHOSPHORUS-32 (³²P)

A radioactive β -emitting phosphorus nuclide with a half-life of 14.28 days, and a beta energy of 1.71 MeV. The following decay data indicate the time (in days) followed by the fraction of original amount at the specified time: 0.0, 1.000; 0.5, 0.976; 1.0, 0.953; 2.0, 0.908; 3.0, 0.865; 4.0, 0.824; 5.0, 0.785; 10.0, 0.616; 15, 0.483; 20, 0.379; 25, 0.297; 30, 0.233; 35, 0.183; 40, 0.144; 45, 0.113; 50, 0.088; 55, 0.069; 60, 0.054.

Special precautions: The maximum range in air is 6 meters, and the maximum range in water is 8 mm. Because bone concentrates phosphate, ³²P is the critical organ in terms of dose. Store this radioisotope behind 1–3 cm Lucite™ shielding. Avoid working over open containers, and exercise institutional guidelines on the type and location of dosimeters.

PHOSPHORYLASE KINASE

This enzyme [EC 2.7.1.38], also known dephosphophosphorylase kinase, catalyzes the reaction of four ATP with two phosphorylase *b* to produce four ADP and phosphorylase *a*.

E. G. Krebs (1986) *The Enzymes*, 3rd ed., **17**, 3.

G. M. Carlson, P. J. Bechtel & D. J. Graves (1979) *Adv. Enzymol.* **50**, 41.

D. A. Walsh & E. G. Krebs (1973) *The Enzymes*, 3rd ed., **8**, 555.

PHOSPHORYLASE PHOSPHATASE

This enzyme [EC 3.1.3.17] catalyzes the hydrolysis of phosphorylase *a* to produce two phosphorylase *b* and four orthophosphate.

L. M. Ballou & E. H. Fischer (1986) *The Enzymes*, 3rd ed., **17**, 311.

N. D. Madsen (1986) *The Enzymes*, 3rd ed., **17**, 365.

D. J. Graves & T. M. Martensen (1980) *Meth. Enzymol.* **64**, 325.

PHOSPHORYLATION STATE RATIO

A metabolic status parameter (abbreviated R_p) that indicates the phosphorylating power of ATP in cells as well as the availability of ADP and orthophosphate for oxidative phosphorylation: $R_p = [ATP]/([ADP][Orthophosphate])$. This ratio is buffered on the immediate time scale by substrate-level phosphorylation of ADP and on the intermediate time scale by mitochondrial phosphorylation of ATP which is then translocated to the cytoplasm and other compartments.

PHOSPHOSERINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.52] catalyzes the reaction of *O*-phospho-L-serine with α -ketoglutarate (or, 2-oxoglutarate) to produce 3-phosphonooxypyruvate and L-glutamate.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

3-PHOSPHOSHIKIMATE 1-CARBOXYVINYLTRANSFERASE

This enzyme [EC 2.5.1.19], also known as 5-enolpyruvylshikimate-3-phosphate synthase and 3-enolpyruvylshikimate-5-phosphate synthase, catalyzes the reaction of phosphoenolpyruvate with 3-phosphoshikimate to produce orthophosphate and 5-*O*-(1-carboxyvinyl)-3-phosphoshikimate.

K. J. Gruys & J. A. Sikorski (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 273.

A. Lewendon & J. R. Coggins (1987) *Meth. Enzymol.* **142**, 342.

D. M. Mousdale & J. R. Coggins (1987) *Meth. Enzymol.* **142**, 348.

PHOSPHOTRANSFERASE SYSTEM

The following are recent reviews on the molecular and physical properties of this transferase involved in group translocation.

M. Feese, D. W. Pettigrew, N. D. Meadow, S. Roseman & S. J.

Remington (1994) *Proc. Natl. Acad. Sci. U.S.A.* **91**, 3544.

B. Erni (1992) *Int. Rev. Cytology* **137A**, 127.

N. D. Meadow, D. K. Fox & S. Roseman (1990) *Ann. Rev. Biochem.* **59**, 497.

PHOSPHOTYROSINE PROTEIN PHOSPHATASE

This enzyme [EC 3.1.3.48], also known as protein-tyrosine-phosphatase, catalyzes the hydrolysis of a phosphory-

lated tyrosyl residue in a phosphoprotein (the result of the action of a protein-tyrosine kinase) to produce a dephosphorylated tyrosyl residue and orthophosphate.

See also *Phosphoprotein Phosphatases*

A. C. Hengge (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 517.

Z.-Y. Zhang & J. E. Dixon (1994) *Adv. Enzymol.* **68**, 1.

R. L. Stone & J. E. Dixon (1994) *J. Biol. Chem.* **269**, 31323.

L. M. Ballou & E. H. Fischer (1986) *The Enzymes*, 3rd ed., **17**, 311.

PHOTOACOUSTIC SPECTROSCOPY

A spectroscopic technique based on the photoacoustic effect—a phenomenon occurring when a sample is irradiated with monochromatic light passing through a beam chopper at a frequency in the acoustical frequency range, such that light absorption causes periodic pressure fluctuations that are detected by a sensitive microphone. This technique allows one to investigate the electronic and infrared spectral properties of molecules in solid and semi-solid samples. For example, one can detect the oxyhemoglobin spectrum in whole blood or red blood cells directly. Furthermore, spectra can be obtained from samples present as spots on thin-layer chromatograms.

See *Fourier Transform IR/Photoacoustic Spectroscopy to Assess Secondary Structure*

J. W. Lin & L. P. Dubek (1979) *Anal. Chem.* **51**, 1627.

A. Rosencwaig (1980) *Photoacoustics and Photoacoustic Spectroscopy*, Wiley, New York.

PHOTOAFFINITY LABELING

A technique and mechanistic tool, pioneered by American chemist Frank Westheimer, in which a photochemically reactive substance is photoexcited in order to covalently attach a molecular entity to another chemical species, often a biomolecule. The advantage of this technique is that photochemically generated species are highly reactive and tend to be promptly captured by electrophiles in the near vicinity of the ligand, such that ligand release typically does not compete with the photocross-linking event. A disadvantage is that unbound photoaffinity ligands also absorb light and tend to react with solutes and/or solvent, thereby reducing the amount of useful reagent.

Chowdhry & Westheimer¹ described many of the strategies and applications of photoaffinity labeling to the investigation of biological systems. (**See** *Azido Photoaffinity Reagents*) To provide an illustrative case, we describe

the recent studies of Bukhtiyarov *et al.*² who used photo-reactive analogues of prenyl diphosphates with different carbon chain lengths as probes of recombinant human protein prenyltransferases. A geranylgeranyl diphosphate analogue, 2-diazo-3,3,3-trifluoropropionyloxy-farnesyl diphosphate (DATFP-FPP), was found to be the most effective inhibitor of both protein farnesyltransferase (PFT) and protein geranylgeranyltransferase-I (PGGT-I). Shorter photoreactive isoprenyl diphosphate analogues with geranyl and dimethylallyl moieties and the DATFP derivative of farnesyl monophosphate were much poorer inhibitors. DATFP-FPP was a competitive inhibitor of both PFT and PGGT-I with K_i values of 100 and 18 nM, respectively. [³²P]DATFP-FPP specifically photoradiolabeled the β -subunits of both PFT and PGGT-I. Photoradiolabeling of PGGT-I was inhibited more effectively by geranylgeranyl diphosphate than farnesyl diphosphate, whereas photoradiolabeling of PFT was inhibited better by farnesyl diphosphate than geranylgeranyl diphosphate. These findings led the authors to conclude that the β -subunits of protein prenyltransferases must contribute significantly to the recognition and binding of the isoprenoid substrate. **See also** *Azido Photoaffinity Reagents; Affinity Labeling*

¹V. Chowdhry & F. H. Westheimer (1979) *Annu. Rev. Biochem.* **48**, 293.

²Y. E. Bukhtiyarov, C. A. Omer & C. M. Allen (1995) *J. Biol. Chem.* **270**, 19035.

PHOTOCHEMICAL REACTION

A chemical reaction induced by the absorption of a photon having energy corresponding to electromagnetic radiation in the ultraviolet or visible range.

PHOTOCHROMISM

A photo-induced transformation of a molecular entity in which there is a spectral change (usually, but not always, in the visible region of the spectrum). Such processes are photochemically or thermally reversible.

PHOTO-CROSS-LINKING

The photochemically induced formation of a covalent linkage between two different molecules (usually biomacromolecules) or between different moieties in the same molecule. **See** *Photoaffinity Labeling*

PHOTODIMERIZATION

A photochemical reaction in which the generation of free radicals results in dimerization of the starting mole-

cule. The cross-linking of pyrimidine bases in DNA represents a fundamentally important example of photodimerization in biology.

PHOTOISOMERIZATION

A photochemical reaction in which the reactant is converted into an isomer. The most common example of such a reaction is photochemical *cis-trans* isomerization. Another example of photoisomerization is the effect of light on cholesta-3,5-diene.

PHOTOLYSIS

A bond-cleavage reaction that has been light-induced. The term has also been used to describe irradiation of a sample by light; however, such usage is discouraged. **See** *Flash Photolysis; Photoreactive Caged Compounds*

PHOTON COUNTING

The recording of sequential photon pulses for measurements of low levels of electromagnetic radiation as well as the recording of emission decays. The pulses are recorded from electron emission events from some photosensitive layer in conjunction with a photomultiplier system. **See also** *Time-Correlated Single Photon Counting; Fluorescence*

PHOTOOXIDATION

An oxidation reaction induced by light. For example, the loss of an electron as a result of photoexcitation or the reaction with O₂ due to the influence of light. **See** *Photooxygenation; Photoreduction*

PHOTOXYGENATION

A photooxidative reaction in which molecular oxygen is incorporated into the reaction products(s). Three mechanisms appear to be common for such processes: (a) reaction of triplet O₂ with free radicals that have been generated photochemically; (b) reaction of photochemically produced singlet oxygen with a molecular species; and (c) the production of superoxide anion which then acts as the reactive species. **See also** *Photooxidation*

PHOTOPHYSICAL PROCESS

A photoexcited process that, while exhibiting no chemical change, leads to different states of a molecular entity by way of radiation and radiationless transitions. Light absorption is a photophysical process, as is fluorescence.

PHOTOREACTIVATION

The repair of macromolecules and biological systems by electromagnetic irradiation. This process is wavelength dependent. An example is DNA repair by certain enzymes.

PHOTOREACTIVE CAGED COMPOUNDS

Photosensitive precursors that generate chemically reactive species upon irradiation with a laser or flash-lamp. Suitably caged derivatives of ATP and other nucleotides are obtained by reaction of the metabolite with diazo compounds generated from 2-nitrobenzyl diazomethane. As pointed out in the excellent review of this topic by McCray & Trentham¹, 20 mJ from a near-UV light source (wavelength = 347 nm) can promptly liberate 2 mM ATP from a 5 mM initial concentration of caged P³-1-(2-nitrophenyl)ethyl ester of ATP over an area of 10 mm² and at a depth of 0.1 mm. If the photolysis is sufficiently rapid relative to the later utilization of the liberated metabolite, one can synchronize the kinetics of the subsequent processes. Caged compounds have been used to investigate the kinetics of such diverse processes as muscle contraction, ion-pump ATPases, ion channels, receptors, bacterial flagella motility. **See** *Photolysis*

¹J. A. McCray & D. R. Trentham (1989) *Ann. Rev. Biophys. Biophys. Chem.* **18**, 239.

PHOTOREDUCTION

A reduction reaction induced by the absorption of a photon in the UV or visible wavelength range of light; for example, the addition of one or more electrons to a photoexcited species and the photochemical hydrogenation of a substance.

PHOTOSYNTHESIS

The photoreductive synthetic process that promotes the assimilation of carbon dioxide into carbohydrates, other reduced metabolites, as well as ATP (synthesis of the latter is termed photophosphorylation). Photosynthesis is the primary mechanism for transducing solar energy into biomass, and green plants utilize chlorophyll *a* to capture a broad spectrum of solar radiant energy reaching the Earth's surface. Photosynthetic bacteria typically produce NADPH, the reductive energy of which is converted to ATP.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Absorption spectroscopy, **24**, 3; flash kinetic spectrophotometry, **24**, 25; ion transport (H^+ , K^+ , Mg^{2+} exchange phenomena), **24**, 68; light-induced paramagnetism in photosynthetic materials by means of electron paramagnetic resonance spectroscopy, **24**, 74; ATP formation, **24**, 88; light-triggered and light-dependent ATPase activities in chloroplasts, **24**, 92; light-influenced ATPase activity: bacterial, **24**, 96; two-stage phosphorylation techniques: light-to-dark and acid-to-base procedures, **24**, 103; oxygen electrode, **24**, 113; modulated light source use with the oxygen electrode, **24**, 123; measuring the delayed fluorescence and chemiluminescence of chlorophyll *in vivo*, **24**, 135; quantum yield determinations of photosynthetic reactions, **24**, 139; measurement of Hill reactions and photoreduction, **24**, 146; simultaneous measurement of photosynthesis and respiration using isotopes, **24**, 165; measurement of size distribution with the Coulter counter, **24**, 171; methods for the measurement of chloroplast volume and structure *in vitro* and *in vivo*, **24**, 181; circular dichroism and optical rotatory dispersion of photosynthetic organelles and their component pigments, **24**, 206; estimation of pool sizes and kinetic constants, **24**, 218; photosensitization reactions of chlorophyll, **24**, 238; measurement of photorespiration, **24**, 246; measurement of the intermediates of the photosynthetic carbon reduction cycle, using enzymatic methods, **24**, 261; light intensity measurement; characteristics of light sources; filters and monochromators, **24**, 268; enhancement, **24**, 293; steady-state relaxation spectrophotometry, **24**, 297; contact shifts and magnetic susceptibilities in iron-sulfur proteins as determined from nuclear magnetic resonance spectra, **24**, 304; exchange reactions (plant) $^{32}P_i \leftrightarrow ATP$, $ATP \leftrightarrow H^{18}OH$, and $P_i \leftrightarrow H^{18}OH$, **24**, 318; a green safe-light for the study of chloroplast development and other photomorphogenetic phenomena, **24**, 321; principles of differential spectrophotometry with special reference to the dual wavelength method, **24**, 322; acid-base indicator for the measurement of rapid changes in hydrogen ion concentration, **24**, 336; spectrophotometric measurement of calcium by murexide, **24**, 343; amperometric hydrogen electrode, **69**, 409; acceptors and donors for chloroplast electron transport, **69**, 413; electrophoretic analysis of chloroplast proteins, **69**, 434; isolation of thylakoid proteins, **69**, 446; measurement of photorespiratory activity and the effect of inhibitors, **69**, 453; cycling assay for nicotinamide adenine dinucleotides, **69**, 465; enhancement, **69**, 470; measurement of spillover, **69**, 474; binding of fluorescent nucleotide analogs to chloroplast coupling factor, **69**, 482; antibody approach to membrane architecture, **69**, 492; analysis of membrane structure-function relationships: a chemical approach, **69**, 502; analysis of membrane architecture: fluorimetric approach, **69**, 510; analysis of dynamic changes in membrane architecture: electron microscopic approach, **69**, 520; measurement of membrane ΔpH , **69**, 538; measurement of the oxygen cycle: the mass spectrometric analysis of gases dissolved in a liquid phase, **69**, 547; measurement of translocation, **69**, 561; system for long-term exposure of aerial part of plants to altered oxygen and/or carbon dioxide concentration, **69**, 571; measurement of photophosphorylation associated with photosystem II, **69**, 576; techniques for studying ionic regulations of chloroplasts, **69**, 585; measurement of metabolite movement across the envelope and of the pH in the stroma and the thylakoid space in intact chloroplasts, **69**, 604; stabilization of chloroplasts and subchloroplast particles, **69**, 613; H_2 production by an *in vitro* chloroplast, ferredoxin, hydrogenase reconstituted system, **69**, 625; energy-dependent reverse electron transport in chloroplasts, **69**, 630; single cell photosynthesis, **69**, 642; measurement of phosphorylation associated with photosystem I, **69**, 648; quantum requirements of photosynthetic reactions, **69**, 659; light-emitting diodes as a

light source in photochemical and photobiological work, **69**, 667.

PHOTOSYNTHETIC REACTION CENTER

The following are recent reviews on the molecular and physical properties of this plant redox system.

- H. B. Gray & J. R. Winkler (1996) *Ann. Rev. Biochem.* **65**, 537.
 J. Deisenhofer & J. R. Norris, eds. (1993) *The Photosynthetic Reaction Center*, vols. I & II, Academic Press, Orlando.
 M. Y. Okamura & G. Feher (1992) *Ann. Rev. Biochem.* **61**, 861.
 J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.
 D. C. Rees & D. Farrelly (1990) *The Enzymes*, 3rd ed., **19**, 37.

PHOTOSYNTHETIC WATER OXIDATION

The oxidation of water to dioxygen occurs as the consequence of Photosystem II-dependent generation of a very strong oxidant. Protons liberated by the water-oxidation reaction then contribute to the thylakoid transmembrane electrochemical gradient that drives ATP synthesis. Brudvig *et al.*¹ describe how flash-induced proton-release measurements have resolved key steps that provide insights on how the O_2 -evolving center of PSII mediates this four-electron oxidation of water.

- ¹G. W. Brudvig, W. F. Beck & J. C. de Paula (1989) *Ann. Rev. Biophys. Chem.* **18**, 25.

PHOTOSYSTEM I

A macromolecular complex that allows plants to harvest the sun's photic energy by absorbing photons and using their energy to catalyze photooxidation of plastocyanin, the copper protein situated in the lumen of thylakoid membranes, which undergoes subsequent electron transfer reactions. These reactions are illustrated in Fig. 1.

The following are recent reviews on the molecular and physical properties of Photosystems I and II.

- H. Levanan & K. Möbius (1997) *Ann. Rev. Biophys. & Biomol. Struct.* **26**, 495.
 B. Hankamer, J. Barber & E. J. Boekema (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 641.
 P. Horton, A. V. Ruban & R. G. Walters (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 655.
 B. R. Green & D. G. Durnford (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 685.
 S. Krömer (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 45.
 B. A. Barry (1995) *Meth. Enzymol.* **258**, 303.
 J. Barber, ed. (1992) *The Photosystems: Structure, Function & Molecular Biology*, Elsevier, New York.
 G. W. Brudvig (1991) *Acc. Chem. Res.* **24**, 311.
 L. Bogorad & I. K. Vasil (eds.) (1991) *The Photosynthetic Apparatus*, Academic Press, Orlando.

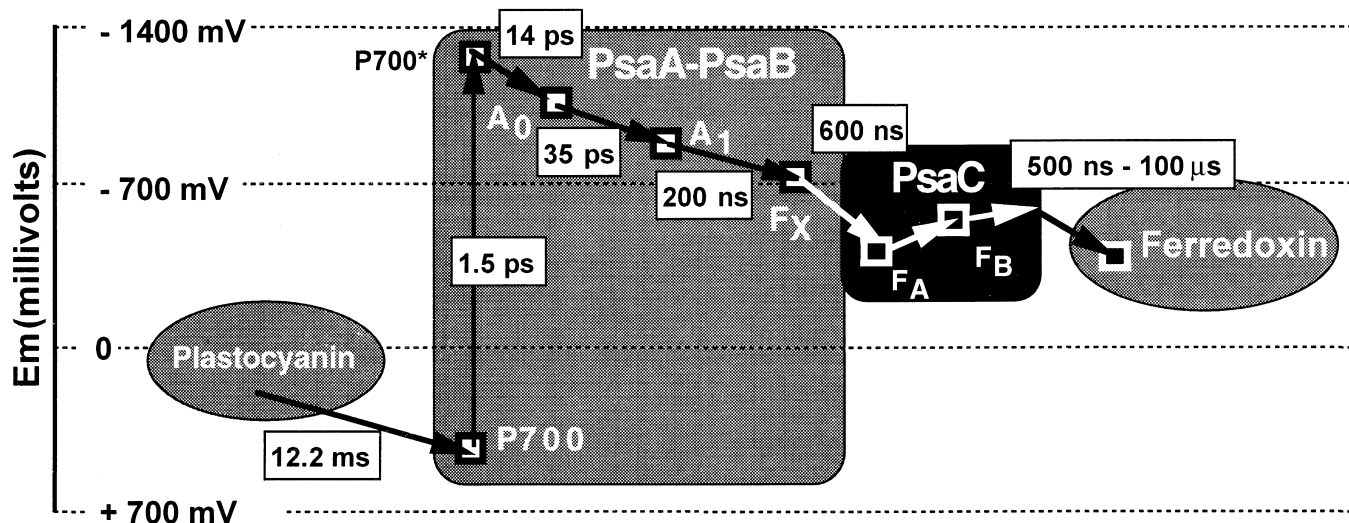


Figure 1. Electron transfer in Photosystem I. The values plotted in the vertical direction are the mid-point potentials, and the numbers next to each arrow are the half-lives for electron transfer. P700 is the primary reaction center; PsaA and PsaB are transmembrane proteins; and PsaC is a peripheral cytoplasmic component. Courtesy of Professor Parag Chitnis, Iowa State University.

pH-RATE PROFILE

A plot of the rate constant, the initial velocity, the maximum velocity of a catalyzed reaction, or the $V_{\max}/K_m$ ratio of an enzyme-catalyzed reaction (or the decadic logarithm of any of these quantities) as a function of the pH value of the solution, all other variables being held constant. **See also** pH Effects

pH STAT

Any electromechanical device that utilizes an automated feedback servomotor to regulate the addition of titrant (a standardized solution of acid or base within a syringe) into a reaction vessel or sample to maintain pH. The rate at which the syringe expels its contents allows one to determine the rate of a chemical reaction producing or consuming protons. There are many such enzyme-catalyzed reactions whose kinetics can be examined with a pH Stat. For maximal sensitivity, one must use weakly buffered solutions. In his classical kinetic investigation of DNA bond scission by DNase, Thomas¹ measured the rate of base addition in a pH Stat. The number of bonds cleaved was linear with time, and this was indicative of random scission.

Another means for following the kinetics of reactions that produce or consume protons is through the use indicator dyes and spectrophotometry, and an excellent example is the examination of hexokinase kinetics². In

such cases, the pK of the indicator dye must be carefully chosen to assure maximal sensitivity.

Although the glass electrode displays sufficient sensitivity and time response for many chemical reactions, special circumstances may require an ultrathin glass electrode to obtain millisecond response times. Less sensitive types of the pH Stat are also used in fermentation chambers to determine the rate/extent of fermentation or to maintain the viability and efficiency of microbes during fermentation.

¹C. A. Thomas, Jr. (1956) *J. Am. Chem. Soc.* **78**, 1861.

²G. G. Hammes & D. Kochavi (1962) *J. Am. Chem. Soc.* **84**, 2069, 2073, and 2076.

pH VALUE

The symbol pH for $-\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration (or, alternatively, $pH = -\log a_{H^+}$ where a_{H^+} is the hydrogen ion activity.) The two letters “p” and “H” stand for: “potency of H”. Typically, pH values are measured with glass electrodes in which the electromotive force (e.m.f.) of an unknown solution is converted to a pH measurement. **See** pH Measurement

S. P. L. Sørensen (1909) *Biochem. Zeit.* **21**, 131.

A. Albert & E. P. Serjeant (1984) *The Determination of Ionization Constants*, 3rd ed., Chapman and Hall, London.

pH* VALUE

The apparent pH value of an aqueous-organic solvent system; an important factor in cryoenzymology experi-

ments¹. In such cases, $\text{pH}^* = -\log a_{\text{H}^{+*}}$ where $a_{\text{H}^{+*}}$ is the proton activity in the mixed solvent.

¹A. L. Fink & M. A. Geeves (1979) *Meth. Enzymol.* **63**, 336.

PHYSICAL ORGANIC CHEMISTRY TERMINOLOGY (IUPAC Recommendations)

If the word you are searching for in the Word Finder directed you here, then it is a term that lies beyond the scope of this Handbook. The listed word will be found in an authoritative internet-accessible glossary dealing with physical organic chemistry: <http://www.chem.qmw.ac.uk/iupac/gtpoc/>. The glossary was developed under the auspices of the IUPAC Organic Chemistry Division's Commission on Nomenclature of Organic Chemistry and Commission on Physical Organic Chemistry. All definitions used in the internet glossary are identical to those in the published document¹, which the reader should cite if the definition is used verbatim or nearly so.

¹P. Müller (1994) *Pure and Appl. Chem.* **66**, 1077.

6-PHYTASE

This enzyme [EC 3.1.3.26], also known as phytase, phytate 6-phosphatase, and *myo*-inositol-hexaphosphate 6-phosphohydrolase, catalyzes the hydrolysis of *myo*-inositol hexakisphosphate to produce 1-*myo*-inositol 1,2,3,4,5-pentakisphosphate and orthophosphate.

F. A. Loewus & M. W. Loewus (1983) *Ann. Rev. Plant Physiol.* **34**, 137.

PHYTOENE SYNTHASE COMPLEX

Phytoene synthase [EC 2.5.1.32] (also known as geranylgeranyl-diphosphate geranylgeranyltransferase and prephytoene-diphosphate synthase) catalyzes the reaction of two geranylgeranyl diphosphate to produce pyrophosphate (or, diphosphate) and prephytoene diphosphate. Isopentenyl pyrophosphate isomerase [EC 5.3.3.2] catalyzes the interconversion of isopentenyl diphosphate and dimethylallyl diphosphate. **See also** *Geranylgeranyl Diphosphate Geranylgeranyltransferase*

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

J. Chappel (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 521.

B. Camara (1993) *Meth. Enzymol.* **214**, 352.

O. Dogbo, A. Laferrière, A. d'Harlingue & B. Camara (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 7054.

PI (Π, π)

Π 1. Symbol for osmotic pressure. 2. Symbol for the mathematical product of a series. π 1. Symbol for surface pressure. 2. Symbol for the ratio of the circumference of a circle to its diameter (3.14159...).

π-BOND (PI-Bond)

A bond (a π molecular orbital) between adjacent atoms that has a node plane which includes the internuclear bond axis.

π → π* TRANSITION

An electronic transition in which a π orbital electron is promoted to an antibonding orbital. The excited state arising from such a promotion is often referred to as a π-π* state. **See** *Absorption Spectroscopy*

π → σ* TRANSITION

An electronic transition in which an π orbital electron is promoted to a σ antibonding orbital. **See** *Absorption Spectroscopy*

PICO-

A prefix, symbolized by p, used in submultiples of units and corresponding to a value of 10⁻¹². **See also** *Nano-*; *Femto-*

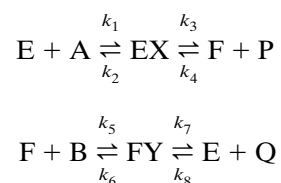
PICORNAIN

Picornain 2A [EC 3.4.22.29] catalyzes the hydrolysis of peptide bonds including the selective cleavage of a particular peptide bond in the picornavirus polyprotein. Picornain 3C [EC 3.4.22.28] catalyzes the selective hydrolysis of the Gln—Gly bond in the poliovirus polyprotein.

T. Skern & H.-D. Liebig (1994) *Meth. Enzymol.* **244**, 583.

PING PONG BI BI MECHANISM

An enzyme-catalyzed “double-displacement” reaction mechanism in which two substrates react to form two products, but one product must be released before the second substrate binds to the enzyme.



In the scheme above, E represents free enzyme, A and B are the substrates, P and Q are the products, EX represents the binary complexes for the first substrate-product pair ($EA \rightleftharpoons \dots \rightleftharpoons FP$), F is a modified form of the enzyme, and FY represents the binary complexes of the second substrate-product pair ($FB \rightleftharpoons \dots \rightleftharpoons EQ$). A large number of enzymes have been reported to have this type of binding mechanism, including the amino-transferases, adenine phosphoribosyltransferase, glucose oxidase, coenzyme A transferases, *etc.*

The steady-state derivation of the ping pong Bi Bi mechanism provides the following determinants for each of the enzyme forms:

$$[E] = k_5k_7(k_2 + k_3)[B] + k_2k_4(k_6 + k_7)[P]$$

$$[EX] = k_1k_5k_7[A][B] + k_1k_4(k_6 + k_7)[A][P] + k_4k_6k_8[P][Q]$$

$$[F] = k_1k_3(k_6 + k_7)[A] + k_6k_8(k_2 + k_3)[Q]$$

$$[FY] = k_1k_3k_5[A][B] + k_5k_8(k_2 + k_3)[B][Q] + k_2k_4k_8[P][Q]$$

Under initial rate conditions where [P] and [Q] are effectively zero, the steady-state expression for this reaction scheme is $v = V_{\max}[A][B]/\{K_b[A] + K_a[B] + [A][B]\}$ where $V_{\max} = k_3k_7[E_{\text{total}}]/(k_3 + k_7)$, $K_a = k_7(k_2 + k_3)/\{k_1(k_3 + k_7)\}$, and $K_b = k_3(k_6 + k_7)/\{k_5(k_3 + k_7)\}$. Note that the denominator does not contain a substrate-independent term (*i.e.*, a term not multiplied by either [A] or [B] or both). This characteristic is distinctive from ordered or random mechanisms.

A double-reciprocal plot of $1/v$ vs. $1/[A]$ at varying levels of [B] will yield a series of parallel lines for this reaction scheme (having no abortives or isomerization steps), the slope equal to $K_a/V_{\max}$, the vertical intercepts equal to $(1/V_{\max})(1 + K_b/[B])$, and the horizontal intercepts equal to $1/[A] = -(1/K_a)(1 + K_b/[B])$. Thus, a replot of the vertical intercepts vs. $1/[B]$ will produce a straight line having a slope of $K_b/V_{\max}$, a vertical intercept of $1/V_{\max}$, and a horizontal intercept of $1/[B] = -1/K_b$. Similarly, a secondary replot of $1/v$ vs. $1/[B]$ data will provide a value for K_a .

The complete steady-state rate expression for the ping pong Bi Bi reaction is

$v =$

$$\frac{V_{\max,f}V_{\max,r}([A][B] - [P][Q]/K_{eq})}{\left\{ \begin{aligned} &V_{\max,r}K_b[A] + V_{\max,r}K_a[B] + V_{\max,r}[A][B] + \frac{V_{\max,f}K_q[P]}{K_{eq}} \\ &+ \frac{V_{\max,f}K_p[Q]}{K_{eq}} + \frac{V_{\max,f}[P][Q]}{K_{eq}} + \frac{V_{\max,f}K_q[A][P]}{K_{ia}K_{eq}} + \frac{V_{\max,r}K_a[B][Q]}{K_{iq}} \end{aligned} \right\}}$$

where $V_{\max,f}$ is identical to $V_{\max}$ in the previous expression, $V_{\max,r} = k_2k_6[E_{\text{total}}]/(k_2 + k_6)$, $K_q = k_2(k_6 + k_7)/\{k_8(k_2 + k_6)\}$, $K_p = k_8(k_2 + k_3)/\{k_4(k_2 + k_6)\}$, $K_{ia} = k_2/k_1$, and $K_{iq} = k_7/k_8$. Then the Haldane relationships are $K_{eq} = k_1k_3k_5k_7/\{k_2k_4k_6k_8\} = K_{ip}K_{iq}/(K_{ia}K_{ib}) = V_{\max,f}K_{ip}K_q/(V_{\max,r}K_{ia}K_b) = V_{\max,f}K_pK_{iq}/(V_{\max,r}K_aK_{ib}) = V_{\max,f}^2K_pK_q/(V_{\max,r}^2K_aK_b)$.

Another important characteristic of ping pong reaction is the presence of exchange reactions in the absence of other substrates. **See** *Isotope Exchange at Equilibrium; Multisite Ping Pong Bi Bi Mechanism; Half-Reaction; Substrate Synergism*

PLANCK FUNCTION

A thermodynamic function, symbolized by Y , equal to the negative of the Gibbs free energy divided by the absolute temperature: $Y = -G/T$. The SI units are joules per kelvin. **See also** *Gibbs Free Energy; Massieu Function*

PLANCK'S CONSTANT

A proportionality constant, symbolized by h , that relates the frequency of electromagnetic radiation to its quanta of energy (*i.e.*, $E = h\nu$): $h = 6.6260755 \times 10^{-34}$ J·s (or $6.6260755 \times 10^{-27}$ erg·seconds). The quantity $\hbar$ equals $h/2\pi$.

PLASMALOGEN SYNTHASE

This enzyme [EC 2.3.1.25], also known as 1-alkenyl-glycerophosphocholine acyltransferase, catalyzes the reaction of an acyl-CoA with 1-*O*-alk-1-enylglycero-3-phosphocholine to produce coenzyme A and plasmenylcholine.

P. C. Choy & C. R. McMaster (1992) *Meth. Enzymol.* **209**, 86.

PLASMANYLETHANOLAMINE Δ^1 -DESATURASE

This magnesium-dependent enzyme [EC 1.14.99.19], also known as alkylacylglycerophosphoethanolamine

desaturase, catalyzes the reaction of an *O*-1-alkyl-2-acyl-*sn*-glycero-3-phosphoethanolamine with AH_2 and dioxygen to produce *O*-1-alk-1-enyl-2-acyl-*sn*-glycero-3-phosphoethanolamine, A, and two water. Either NADPH or NADH can serve as the AH_2 substrate. The reaction may require the participation of cytochrome b_5 and it has been reported that the enzyme requires ATP.

M. L. Blank & F. Snyder (1992) *Meth. Enzymol.* **209**, 390.

PLASMIN

Plasmin [EC 3.4.21.7], also known as fibrinase and fibrinolysin, is a peptidase (a member of the peptidase family S1) that exhibits preferential cleavage at Lys—Xaa > Arg—Xaa (there is actually greater selectivity than displayed by trypsin). Plasmin converts fibrin into soluble products. It is formed from plasminogen by proteolysis, resulting in multiple forms of the active plasmin.

F. B. Ablondi & J. J. Hagan (1960) *The Enzymes*, 2nd ed., **4**, 176.

^{31}P MAGNETIC RESONANCE OF CONTRACTILE SYSTEMS

Because muscle cells are especially rich in terms of phosphorus-containing metabolites (*e.g.*, ATP, ADP, phosphocreatine, and orthophosphate), nuclear magnetic resonance^{1–3} has proved to be a valuable noninvasive probe of metabolic changes attending muscle activity. The spectral sensitivity of ^{31}P is especially high relative to other nuclei, and one can detect cellular concentrations as low as 0.5 mM as well as utilize chemical shift data to define intracellular pH and free magnesium ion concentrations.

See also *Nuclear Magnetic Resonance; Chemical Shift*

¹D. I. Hoult, S. J. W. Busby, D. G. Gadian, G. K. Radda, R. E. Richards & P. J. Seeley (1974) *Nature* **252**, 285.

²M. Barany & T. Glondek (1982) *Meth. Enzymol.* **85**, 624.

³M. J. Dawson, D. G. Gadian & D. R. Wilkie (1977) *J. Physiol. (London)* **267**, 703.

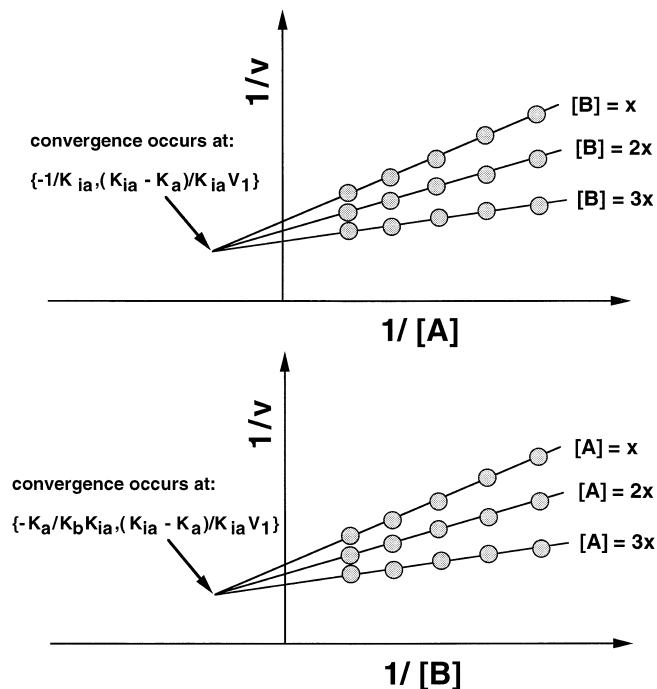
pMg

The negative base₁₀ logarithm of the magnesium ion concentration.

POINT-OF-CONVERGENCE METHOD

A procedure that assists in the characterization of binding mechanisms for sequential (*i.e.*, non-ping pong) reactions^{1,2}. The same general initial rate expression applies to the steady-state ordered Bi Bi reaction, the rapid-equilibrium random Bi Bi reaction, and the Theorell-

Chance Bi Bi mechanism. (This expression also applies to isomerization mechanisms for the ordered and Theorell-Chance schemes in which a stable enzyme form undergoes isomerization.) The points of intersection for each different mechanism can be used to exclude certain possibilities, depending on the quality of the rate data.



Another growing application for the technique is in evaluating mechanistic differences between wild-type enzymes and enzymes obtained from site-directed mutagenesis, as well as between different isozymes, between enzymes from different sources, or between the same enzyme acting on alternative substrates.

¹J. D. Lueck, W. R. Ellison & H. J. Fromm (1973) *FEBS Lett.* **30**, 321.

²F. B. Rudolph & H. J. Fromm (1979) *Meth. Enzymol.* **63**, 138.

POINT OF INFLECTION

A location on a curve identifiable when progress from a position prior to that point to a position beyond that point requires crossing of a tangent line to that point. In calculus, the point of inflection occurs wherever the second derivative is zero and changes sign of $f''(x)$ on either side of that point.

POISE

A unit (symbolized by P) of dynamic viscosity. One poise is equivalent to one dyne-second per square centimeter, to one gram per centimeter-second, and to 0.1 pascal-

second. This is the tangential force in dynes needed per unit area (in cm^2) to maintain a difference in velocity of one cm per second between two parallel plates separated by one cm of fluid.

POISSON DISTRIBUTION

The probability distribution for the number of events (occurring randomly at a rate μ) corresponding to the function

$$f(x) = (\mu^x/x!) \exp[-\mu]$$

where x equals 0, 1, 2, This probability density function is actually a limiting case of the binomial distribution. **See** *Statistics (A Primer)*

POLAR EFFECT

1. The effect of the relative electronegativity of a substituent (R) and/or the delocalization of electrons on a chemical property of a substance. Thus, for a molecule $R-R'$, the polar effect refers to all nonsteric influences and modifications of electrostatic forces operating at the reaction center (R'), relative to some standard molecule, R_o-R' . Hence, the term is synonymous with the electronic effect.

2. The nonsteric influence exerted by nonconjugated substituents on reaction rates. In this case, electron delocalization between the substituent and the rest of the molecule is not considered, and the polar effect is not synonymous with the electronic effect. **See** *Inductive Effect; Field Effect*

POLARITY

1. With respect to a solvent, the overall solvation capability for solutes. 2. A property of bodies or systems that have a distinct direction; *i.e.*, that have different or opposing physical properties or characteristics at different points. For example, an amino acid sequence in a polypeptide has polarity in that there is an amino end and a carboxyl end of the sequence. Similarly, microtubules and actin filaments have plus (+)-ends and minus (-)-ends that establish directionality for cellular and intracellular locomotion. 3. The state in which there is either a positive or negative aspect relative to the two poles of a magnet or to electrification. 4. Attraction toward an object or attraction in a specific direction. 5. In mathematics, the positive or negative sign of numbers.

POLARIZABILITY

A proportionality factor measuring the ease of distortion of the electron cloud of a chemical species as that species is subjected to an electric field (such as that artificially applied or due to the proximity of a charged reagent). The term “electric polarizability” has also been used. Polarizability (symbolized by the Greek letter “ α ”) is determined by measuring the magnitude of the induced electric dipole moment of the molecule, μ_{induced} , and the size of the local electric field strength, E , that is the source of the induced effect. Thus, $\mu_{\text{induced}} = \alpha E$. (In the more general case in which the direction of the induced dipole moment is not the same as the direction of the field, $\mu_{\text{induced}} = \tilde{\alpha} E$ where $\tilde{\alpha}$ is a tensor). The polarizability is the induced moment per unit field strength. In most cases, α is actually a mean polarizability, being an average over the three Cartesian coordinates of the molecule, and can be related to the index of refraction. With some molecules (*e.g.*, diatomic species such as O_2) distinctions can be made between longitudinal polarizability (*i.e.*, polarizability along the bond) and transverse polarizability (*i.e.*, perpendicular to the bond).

C. K. Ingold (1953) *Structure and Mechanism in Organic Chemistry*, Cornell Univ. Press, New York.

R. W. Taft, Jr., & R. D. Topson (1987) *Progr. Phys. Org. Chem.* **16**, 1.

POLARIZATION

1. Any action or phenomenon leading to a separation of electrical charge and/or magnetization, including: the linear and circular polarization of light, the unequal electron transfer across an electrode or battery pole (such that $i_{\rightarrow} \neq i_{\leftarrow}$), and separation of ionic or electric charge within molecules. 2. Any positional separation of biological components into morphologically differentiated areas at opposite ends of a living cell. Examples of the latter include: polarized distribution of membrane molecules that depend on tight junctions; antigen presentation on cell surfaces; localization of glucose transporters; epithelial polarity that is essential for vectorial reabsorption and secretion of ions; as well as the differential insertion of channels, transporters, and related proteins into apical or basolateral membranes. 3. Any separation of ions across a selectively permeable membrane, including those achieved biologically through the action of ion pumps as well as the separation of electrolyte solutions by reverse osmosis.

POLY(ADP-RIBOSE) SYNTHETASE

This enzyme [EC 2.4.2.30] (also referred to as NAD⁺ ADP-ribosyltransferase, poly(ADP) polymerase, poly(adenosine diphosphate ribose) polymerase, and ADP-ribosyltransferase (polymerizing)) catalyzes the reaction of NAD⁺ with [ADP-D-ribosyl]_n to produce nicotinamide and [ADP-D-ribosyl]_(n+1). The ADP-D-ribosyl group of NAD⁺ is transferred to an acceptor carboxyl group on a histone or on the enzyme itself, and further ADP-ribosyl groups are transferred to the 2'-position of the terminal adenosine moiety, building up a polymer with an average chain length of twenty to thirty units.

N. J. Oppenheimer & A. L. Handlon (1992) *The Enzymes*, 3rd ed., 20, 453.

POLYGALACTURONASE

This enzyme [EC 3.2.1.15], also known as pectin depolymerase and pectinase, catalyzes the random hydrolysis of 1,4- α -D-galactosiduronic linkages in pectate and other galacturonans.

H. J. Phaff (1966) *Meth. Enzymol.* 8, 636.

POLYMER FLEXING AND EXTENDING

Polymers spontaneously flex (or bend), and they can also extend. As pointed out by Oosawa and Asakura¹, any polymer behaving like a uniform rod will have an elastic (*i.e.*, reversible) modulus for flexing that is given by the following relationship:

$$\varepsilon = YI$$

where Y is the elastic modulus for stretching (often called the Young's modulus) and I is the moment of inertia ($I = \pi a^4/4$ for a cylinder of uniform radius a). When an assembled polymer composed of protein subunits flexes, there develops a restoring potential energy that is related to the energy required to stretch such bonds from their equilibrium bond length to their extended length. Oosawa and Asakura¹ demonstrate how one can obtain the change in the free energy of intersubunit bonding within the polymer's lattice when the degree of stretching is relatively small. They also show that deformation by bending is far more pronounced than any deformation attributable to stretching.

¹F. Oosawa & S. Asakura (1975) *Thermodynamics of Protein Polymerization*, pp. 162-168, Academic Press, New York.

POLYMERIZATION

1. The spontaneous self-assembly or template-directed assembly of component monomeric units into polymeric biological macromolecules. 2. The enzyme-catalyzed joining of monomeric units (such as amino acids, sugars, nucleotides) into covalently linked oligomeric or polymeric forms.

Biopolymers are typically formed from subunit components through covalent and/or noncovalent bonds. Examples of the covalently linked biopolymers include proteins, nucleic acids, and polysaccharides. Examples of biopolymers formed by noncovalent forces include hemoglobin ($\alpha_2\beta_2$), tubulin ($\alpha\beta$), and lactate dehydrogenase (M_4 , M_3H_1 , M_2H_2 , M_1H_3 , and H_4 tetramers formed from muscle M and heart H polypeptide chains) or polymeric structures (such as microtubules, filamentous actin, and intermediate filaments). These polymerize spontaneously in response to changes in pH, temperature, or in the presence/absence of other solutes (metal ions, metabolites, or other inorganic anions). **See also** Actin Assembly Kinetics; Microtubule Assembly Kinetics; Quasi-equivalence; Actin-Based Motility; Critical Concentration; Self-Assembly; Processivity

POLYMERIZATION KINETICS (Irreversible)

Consider a series of irreversible polymerization reactions for which the same rate constant k : applies to the initial and all subsequent steps.



etc.

The polymerization kinetics will be first-order with respect to $[A_{t=0}]$, and the polymerization reaction will exhibit zero-order kinetics for any set of rate measurements conducted at one fixed concentration of monomer.

If only monomer A_1 is initially present, the integrated rate law for this process can be written as:

$$[A_{\text{poly, total}}] = ([A_1] + [A_2] + [A_3] + \dots + [A_n])$$

$$= [A_{t=0}]kt[(4 + kt)/(2 + kt)^2]$$

where $[A_{\text{poly, total}}]$ is the total amount of polymeric species, and $[A_{t=0}]$ is the initial monomer concentration. The polymerization rate, $d[A_{\text{poly, total}}]/dt$ is obtained by taking the derivative of the quotient¹, yielding the result:

$$d[A_{\text{poly, total}}]/dt = 8[A_{t=0}]k/(2 + kt)^3$$

¹Recall that $d(u/v)/dx = \{(vdu/dx) - (udv/dx)\}/v^2 = \{(1/v)(du/dx) - \{(u/v^2)(dv/dx)\}$.

POLYNUCLEOTIDE 5'-HYDROXYL-KINASE

This enzyme [EC 2.7.1.78] catalyzes the reaction of ATP with 5'-dephospho-DNA to produce ADP and 5'-phospho-DNA. The enzyme can also act on 5'-dephospho-RNA 3'-mononucleotides. **See also** DNA Kinases

C. C. Richardson (1981) *The Enzymes*, 3rd ed., **14**, 299.

K. Kleppe & J. R. Lillehaug (1979) *Adv. Enzymol.* **48**, 245.

POLYNUCLEOTIDE PHOSPHORYLASE

This enzyme [EC 2.7.7.8], also known as polyribonucleotide nucleotidyltransferase, catalyzes the reaction of $[RNA]_{(n+1)}$ with orthophosphate to produce $[RNA]_n$ and a nucleoside diphosphate. In the reverse reaction, ADP, IDP, GDP, UDP, and CDP can act as substrates.

U. Z. Littauer & H. Soreq (1982) *The Enzymes*, 3rd ed., **15**, 517.

T. Godefroy-Colburn & M. Grunberg-Manago (1972) *The Enzymes*, 3rd ed., **7**, 533.

POLYPHOSPHATE GLUCOKINASE

This enzyme [EC 2.7.1.63], also known as polyphosphate-glucose phosphotransferase, catalyzes the reaction of $[\text{phosphate}]_n$ with D-glucose to yield $[\text{phosphate}]_{(n-1)}$ and D-glucose 6-phosphate. The enzyme requires the presence of a neutral salt (such as potassium chloride) for maximum activity. Glucosamine can also serve as a substrate.

H. G. Wood & J. E. Clark (1988) *Ann. Rev. Biochem.* **57**, 235.

POLYPHOSPHATE KINASE

This enzyme [EC 2.7.4.1] catalyzes the reaction of ATP with $[\text{phosphate}]_n$ to produce $[\text{phosphate}]_{(n+1)}$ and ADP.

H. G. Wood & J. E. Clark (1988) *Ann. Rev. Biochem.* **57**, 235.

POLYSACCHARIDE DEPOLYMERASE

This enzyme [EC 3.2.1.87], also called capsular-polysaccharide endo-1,3- α -galactosidase, catalyzes the random hydrolysis of 1,3- α -D-galactosidic linkages in the *Aerobacter aerogenes* capsular polysaccharide. It hydrolyzes the galactosyl- α -1,3-D-galactose linkages only in the complex substrate, bringing about depolymerization.

See also Amylases; Lysozyme

A. Tsugita (1971) *The Enzymes*, 3rd ed., **5**, 343.

POOL SIZE

The total amount or mass of a tracee in a pool, corresponding to the volume of distribution (V) multiplied by the tracee concentration in the same pool; frequently abbreviated Q_T . **See** Compartmental Analysis

P/O RATIO

A measure of oxidative phosphorylation, equal to the ratio of the number phosphate groups esterified (*i.e.*, ATP formation from ADP and phosphate) relative to the atoms of oxygen consumed by the mitochondria.

PORPHOBILINOGEN DEAMINASE

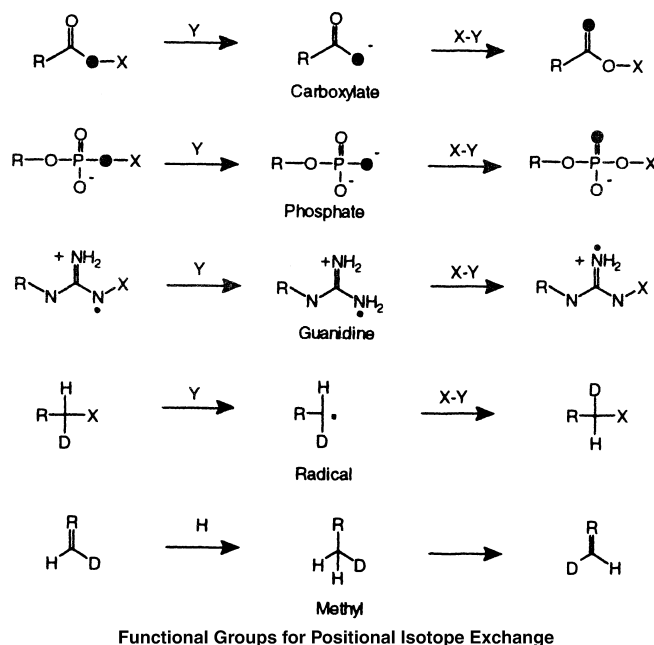
This enzyme [EC 4.3.1.8], also known as hydroxymethylbilane synthase, catalyzes the dipyrromethane-dependent reaction of four porphobilinogen molecules with water to produce hydroxymethylbilane and four molecules of ammonia. In the presence of a second enzyme, uroporphyrinogen-III synthase [EC 4.2.1.75], the product is cyclized to form uroporphyrinogen-III.

G. V. Louie, P. D. Brownlie, R. Lambert, J. B. Cooper, T. L. Blundell, S. P. Wood, M. J. Warren, S. C. Woodstock & P. M. Jordan (1992) *Nature* **359**, 33.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

POSITIONAL ISOTOPE EXCHANGE ("PIX")

Positional isotope exchange ("PIX") is a very valuable technique in determining enzyme mechanisms, particularly those utilizing ATP, GTP, or another NTP substrate^{1,2}. For example, the nucleotide substrate can be labeled with ¹⁸O in bridging (*i.e.*, P—O—P or phosphoanhydride oxygen) and/or its nonbridging positions.



By using ^{31}P NMR in the presence of the enzyme, an investigator can identify bridged-to-nonbridge (and the reverse) isotope exchanges and thereby identify probable intermediates on the reaction pathway. The procedure² is useful for any enzyme-catalyzed reaction in which the individual atoms of a functional group within a substrate, intermediate, or product become torsionally equivalent during the course of a reaction.

¹I. A. Rose (1979) *Adv. Enzymol.* **50**, 361.

²L. S. Mullins & F. M. Raushel (1995) *Meth. Enzymol.* **249**, 398.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Adenylosuccinate synthetase, **249**, 423; D-alanine-D-alanine ligase, **249**, 417-418; aminoacyl-tRNA synthetases, **249**, 423; argininosuccinate lyase, **249**, 413-414; carbamoyl-phosphate synthetase, **249**, 418-423; CTP synthase, **249**, 423-424; functional groups for, **249**, 400-401; galactose-1-phosphate uridylyltransferase, **249**, 416-417; glutamine synthetase, **249**, 453; as mechanistic probe, **249**, 398; phosphoenolpyruvate carboxykinase, **249**, 424; Ping Pong mechanism analysis, **249**, 409-413; pyruvate-phosphate dikinase, **249**, 424; qualitative and quantitative approaches, **249**, 401-404; sequential mechanism analysis, **249**, 404-409 [enhancement, **249**, 404-407; inhibition, **249**, 407-409]; UDPglucose pyrophosphorylase, **249**, 414-416; variation of nonlabeled substrates and products, **249**, 404-413.

POSITIVE COOPERATIVITY

Any set of ligand interactions with oligomeric or polymeric macromolecules, such that binding of the first (or preceding) ligand molecule increases the likelihood for binding of the next (or subsequent) ligand molecule. In

the Monod-Wyman-Changeux allosteric transition model, the dissociation constant for ligand interactions is equivalent for all sites, but cooperativity results from the disproportionate "recruitment" of binding sites from the so-called T-state in an all-or-none transition to the R-state. In the Koshland-Némethy-Filmer model, the dissociation constant for ligand binding to the $\text{PL}_{(i)}$ state is lower than the dissociation constant for ligand binding to the $\text{PL}_{(i-1)}$ state. **See** *Ligand Binding Cooperativity; Monod-Wyman-Changeux Model; Koshland-Némethy-Filmer Model; Hill Equation & Plot; Adair Equation; Allostereism; Linked Function; Hemoglobin; Aspartate Carbamoyltransferase; Scatchard Plot; Cooperativity; $\Delta\Delta G$ as Index of Cooperativity; Negative Cooperativity*

POSTTRANSLATIONAL MODIFICATION OF PROTEINS

Although allostery and protein turnover are effective regulatory mechanisms for enzymes, posttranslational modification represents another important means for achieving metabolic regulation. The range of enzyme-catalyzed posttranslational modification reactions is immense (e.g., phosphorylation, adenylation, ADPriboylation, hydroxylation, glycosylation, prenylation, fatty acylation, sulfation, proteolysis, acetylation, arginylation, glycylation, glutaminylation, glutamylation, etc.). Graves *et al.*¹ have provided a lucid account of the enzymology and kinetics of these modification reactions, including considerations of the structural consequences of such covalent modifications. Those interested in the stability and analysis of modified amino acids in proteins will wish to consult Table 4.2 in Graves *et al.*¹

¹D. J. Graves, B. L. Martin & J. H. Wang (1994) *Co- and Post-Translational Modification of Proteins: Chemical Principles and Biological Effects*, pp. 348ff, Oxford University Press, Oxford.

POTENTIAL ENERGY

The energy associated with position of a body or entity with respect to another body or entity or with respect to relative positions of parts within that same body. For an entity or body at position $\mathbf{r}$ associated with a force $\mathbf{F}$, the potential energy, E_p or U , is $-\int \mathbf{F} \cdot d\mathbf{r}$. The strictly classical mechanical definition of potential energy typically leads one to consider a system in terms of stored energy and position, and in that case, one deals with the magnitude of force and distance between interacting

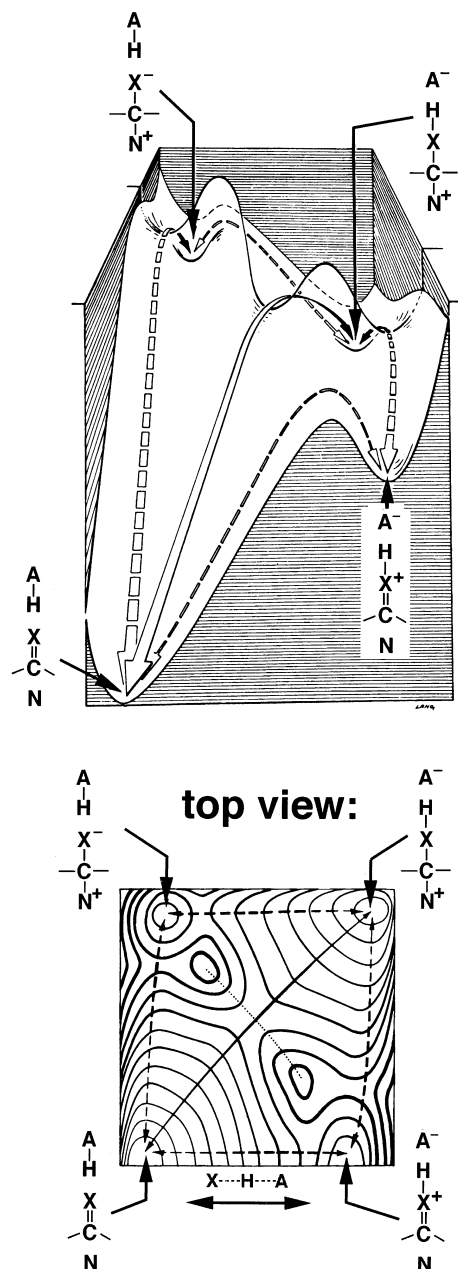
bodies or particles. In this case, a chemical bond is treated by the so-called Hooke's Law model: a pair of atoms or substituents of mass m_A and m_B are attached at opposite ends of a vibrating spring, and the potential energy of the bond at any instant is determined by the reduced mass, $m_A m_B / (m_A + m_B)$, the force constant k of the spring, and a distance indicating whether the spring is compressed, extended, or exactly at its equilibrium position. However, as was clearly demonstrated a century ago by the physicist James Clerk Maxwell, the concept of potential energy also embodies the contributions of electric field effects and magnetic field effects, integrated over all space. Quantum mechanical treatments can rigorously specify the potential energy, but this is most frequently accomplished for very simple systems having only a few atoms. **See** *Energy; Kinetic Energy*

POTENTIAL ENERGY PROFILE

A depiction of the variation of the potential energy of the system of atoms or molecules (*e.g.*, the reactants, intermediates, transition states, and products of a particular reaction) as a function of a reaction coordinate. For an elementary reaction, this horizontal reaction coordinate represents the path of the energetically easiest passage of reactant(s) to product(s). For a series of elementary reactions (*e.g.*, a set of successive reaction steps), the horizontal coordinate is equivalent to a set of successive reaction coordinates. **See also** *Reaction Coordinate Diagram; Potential Energy Surface; Gibbs Free Energy Diagram*

POTENTIAL ENERGY SURFACE

A diagram depicting the variation of the potential energy associated with the reactants and products of an elementary reaction in which the energetically easiest progress (for movement of a representative point of the reaction system) is plotted as a function of two or three coordinates (usually representing molecular geometries) of a multidimensional, potential energy hypersurface. Typically, the diagram is a depiction having isoenergetic contour lines and resembles a topographic map. In most two-coordinate systems, for elementary reactions, the coordinates chosen are related to two significant variables that change during the course of a reaction (*e.g.*, two distinct interatomic distances).



For more complicated reactions involving a number of elementary reactions, care should be exercised in selecting the coordinates in such systems. In some publications a third coordinate represents the standard Gibbs free energy instead of potential energy. The energetically easiest route on the contour map is referred to as the potential energy profile. **See** *Reaction Coordinate; Gibbs Free Energy Diagram; Potential Energy Profile; More-O'Ferrell-Jencks Diagram; Enzyme Energetics; Kinetic Isotope Effects*

J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, 3rd ed., Wiley, New York.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

PRECISION

The variability achieved in replicate measurements of the same quantity. The term high precision implies low variation within a set of determinations. **See also** *Accuracy; Statistics (A Primer)*

PRECURSOR

A metabolite, molecular entity, or some other event/process that precedes another component in a longer sequence of events or conversions. For example, the isoprenoid metabolite squalene is a precursor of cholesterol and glucose 6-phosphate is a precursor of glycogen, ribose, and pyruvate. **See** *Series First Order Reaction; Pulse-Chase Experiments*

PREDISSOCIATION

1. Dissociation occurring by tunneling from a bound to an unbound rovibronic state (*i.e.*, a state corresponding to a particular rotational energy level of a vibrational level of an electronic state). 2. The appearance of a diffuse band region within a series of sharp bands of an absorption spectrum.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

PREEQUILIBRIUM

A term (also referred to as “prior equilibrium”) denoting any reversible step that precedes an irreversible step or the rate-limiting step in a multistage reaction mechanism. The so-described reaction step must rapidly establish an equilibrium between its reactants and products. The first association/dissociation equilibrium leading to the formation of EX complex from E and S in the Michaelis-Menten treatment is an example of a preequilibrium. **See** *Michaelis-Menten Kinetics*

PREEXPONENTIAL FACTOR

A constant in the Arrhenius equation and commonly symbolized by *A*: $k = Ae^{-E_a/RT}$. It is also referred to as the *A*-factor. **See** *Arrhenius Equation; Collision Theory; Transition-State Theory; Entropy of Activation*

PRENYL-PROTEIN-SPECIFIC ENDOPEPTIDASE

A rat liver microsomal protease that cleaves an intact tripeptide, VIS, from S-farnesylated-CVIS tetrapeptide.

L. Liu, G.-F. Jang, C. C. Farnsworth, K. Yokoyama, J. A. Glomset & M. H. Gelb (1995) *Meth. Enzymol.* **250**, 189.
B. A. Gilbert, Y.-T. Ma & R. R. Rando (1995) *Meth. Enzymol.* **250**, 206.

PREORGANIZATION

1. The initial step(s) in a chemical reaction mechanism, as treated by Marcus Theory. (**See** *Marcus Theory*) 2. A term also applied to hosts, such as spherands, that have binding sites for ligands fully organized prior to any complexation. Studies of spherands and podands demonstrate that preorganization is a central determinant of binding strength^{1,2}. The principle of preorganization states that the more highly organized and desolvated are hosts and guests prior to their complexation, the more stable their complexes will be. Preorganization is governed by both enthalpic and entropic terms. **See also** *Podand; Spherand; Host; Guest; Crown Ether*

¹D. J. Cram (1988) *Angew. Chem. Int. Ed. Engl.* **27**, 1009.

²D. J. Cram, M. P. de Grandpro, C. B. Knobler & K. N. Trueblood (1984) *J. Amer. Chem. Soc.* **106**, 3286.

PREPHENATE DEHYDRATASE

This enzyme [EC 4.2.1.51] catalyzes the conversion of prephenate to phenylpyruvate, water, and carbon dioxide. This enzyme in enteric bacteria also possesses a chorismate mutase activity and converts chorismate into prephenate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

B. E. Davidson (1987) *Meth. Enzymol.* **142**, 432.

B. E. Davidson & G. S. Hudson (1987) *Meth. Enzymol.* **142**, 440.

R. Fischer & R. Jensen (1987) *Meth. Enzymol.* **142**, 507.

PREPHENATE DEHYDROGENASE

Prephenate dehydrogenase [EC 1.3.1.12] catalyzes the reaction of prephenate with NAD⁺ to produce 4-hydroxyphenylpyruvate, carbon dioxide, and NADH. This enzyme in enteric bacteria also possesses chorismate mutase activity and converts chorismate into prephenate. Prephenate dehydrogenase (NADP⁺) [EC 1.3.1.13] catalyzes the reaction of prephenate with NADP⁺ to produce 4-hydroxyphenylpyruvate, carbon dioxide, and NADPH.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

M. H. O'Leary (1989) *Ann. Rev. Biochem.* **58**, 377.

R. Fischer & R. Jensen (1987) *Meth. Enzymol.* **142**, 503.

PRESSURE-JUMP METHOD

A fast-reaction kinetic technique used to achieve a rapid change in external pressure that results in a sudden change in the equilibrium constant for a particular system. The investigator then analyzes the rate of approach of the system to the new equilibrium position. **See** *Chemical Kinetics*

PRE-STEADY-STATE PHASE

Refers to that initial period of nonlinear product formation, commencing with the initiation of the reaction and ending when the system is at steady state. Typically, the pre-steady-state phase lasts from milliseconds to a few seconds after mixing reactants. The time course of pre-steady-state rate processes often can be evaluated using stopped-flow, temperature-jump, and mix-quench methods.

PRIMARY PHOTOCHEMICAL PROCESS

Any elementary chemical process (also referred to as a primary photoreaction) in which an electronically excited molecular entity yields a primary photoproduct. **See** *Primary Photoproduct*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

PRIMARY PHOTOPRODUCT

The first observable entity, with distinctly different chemical properties compared to those of the reactant, produced in a primary photochemical process.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

PRIMITIVE CHANGE

A simple molecular change into which an elementary reaction can conceptually be further dissected. Primitive changes would include bond rupture, bond rotation, change in bond length, a redistribution of charge, *etc.* The concept of primitive change is useful in describing elementary reactions, but a primitive change does not itself represent a discrete chemical process. **See** *Elementary Reaction*

PRION PLAQUE FORMATION

Protein aggregation is a hallmark of scrapie, and scrapie protein can be induced to aggregate *in vitro* into forms that are indistinguishable from pathological brain-derived fibrils. Prusiner proposed that prion disease involves a mechanism for autocatalytic conversion of a host

protein into a genetically identical, but conformationally different, prion state. Lundberg *et al.*¹ examined the kinetics and mechanism of amyloid formation by the prion protein H1 peptide as determined by time-dependent ESR. Their analysis suggests a model whereby the amorphous aggregate has a previously unsuspected dual role of releasing monomer into solution and also providing initiation sites for fibril growth. These findings suggest that the β -sheet-rich scrapie prion protein may be stabilized by aggregation. Harper and Lansbury² have also discussed simple mechanistic models involving nucleation-dependent polymerization, suggesting (a) that with such a mechanism, aggregation would be dependent on protein concentration and time and (b) that amyloid formation can be seeded by preformed fibrils. Eigen³ has compared kinetic aspects of mechanisms of prion diseases based on the "protein only" hypothesis. He found that Prusiner's model almost surely also involves cooperativity, given realistic values of rate parameters. Eigen further states that were cooperativity occurring, then the model becomes phenomenologically indistinguishable from the Lansbury mechanism of plaque formation which is also a form of (passive) autocatalysis. Although both mechanisms differ with respect to relative stability of the two monomeric protein conformations, both require a thermodynamically favored aggregated state. While these considerations allow for a critical comparison of the mechanisms, they do not reveal the actual mechanism of infection. Nonetheless, Eigen argues that aggregation is likely to be a necessary, but possibly not sufficient, prerequisite of infection.

¹K. M. Lundberg, C. J. Stenland, F. E. Cohen, S. B. Prusiner & G. L. Millhauser (1997) *Chem. Biol.* **4**, 345.

²J. D. Harper & P. T. Lansbury, Jr. (1997) *Annu. Rev. Biochem.* **66**, 385.

³M. Eigen (1996) *Biophys. Chem.* **63**, A1.

PROBABILITY

A statistical measure of the frequency or regularity of occurrence of a measured outcome (or measurement) in an experiment subject to random fluctuations in the measurement. The fundamental axioms of mathematical probability are: (a) if E is any event in the sample space S , then the probability of E 's occurrence [written $P(E)$] ranges over the interval from zero to one; (b) over the entire sample space, $P(E)$ equals one; and (c) if events A and B are mutually exclusive, then the probability of their combined occurrence is the sum of the probabilities

of their occurring individually. **See** *Statistics (A Primer); Permutations and Combinations*

PROBABILITY DENSITY FUNCTION

A function applied in statistics to predict the relative distribution if the frequency of occurrence of a continuous random variable (*i.e.*, a quantity that may have a range of values which cannot be individually predicted with certainty but can be described probabilistically) from which the mean and variance can be estimated.

PROBABLE ERROR

A statistical term for the deviation from the true value within which lies an experimentally measured value with a probability of 0.50. This corresponds to 0.674σ (*i.e.*, 0.674 times the standard deviation). **See** *Statistics (A Primer); Normal Error Curve*

PROCESSIVITY

A measure of the tendency of an enzyme catalyzing a polymerization (or depolymerization) reaction to proceed through multiple catalytic rounds of bond making/breaking without dissociating from its polymeric substrate or template. The reader should note that processivity is technically a partition factor indicating the propensity of an enzyme to choose among several competing kinetic pathways. This partition constant has the general form: $k_{\text{catalysis}} / \{k_{\text{catalysis}} + k_{\text{dissociation}}\}$, and the greater its magnitude, the greater is the persistence of the enzyme on the polymer. One may also consider the processivity of molecular motors, such as dynein and myosin, in terms of their ability to proceed in the translocating/locomotory function by remaining attached to their cytoskeletal scaffold. Because polymerases, depolymerases, and cytoskeletal motors may be regulated by the concentration(s) of cosubstrate(s) and/or regulatory effector(s). Their processivity may well depend on the presence of these agents.

McClure and Chow¹ presented a useful protocol for determining the processivity of DNA polymerases. That approach relies on the ability to resolve discrete product lengths on acrylamide gels in 7 M urea and to analyze the resulting length distribution data using the Kuhn distribution law. The template challenge method² provides yet another means for observing the frequency of enzyme dissociation from its polymeric substrate,

thereby allowing its binding to a second competing substrate/inhibitor. One measures cycling-time perturbation², wherein steady-state velocity measurements allow one to estimate the average time that an enzyme spends on its template. In the case of DNA polymerases, processivity may be linked to fidelity of nucleotide incorporation, because the binding energy of enzyme-polymer interactions are apt to factor into so-called editing steps where incorrectly inserted nucleotides are hydrolyzed out of the polymer before the next round of phosphodiester synthesis. **See also** *Random Scission Kinetics*

¹W. R. McClure & Y. Chow (1980) *Meth. Enzymol.* **64**, 277.

²R. A. Bambara, D. Ujemura & T. Choi (1978) *J. Biol. Chem.* **253**, 413.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

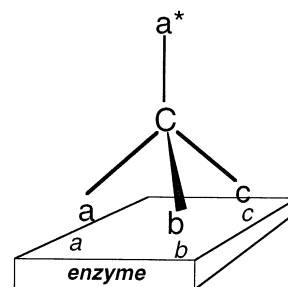
Assay of DNA polymerases [autoradiogram interpretation, **262**, 277; DNA traps, **262**, 272-273; enzyme concentration, **262**, 272-273; gel electrophoresis, **262**, 271, 276-277; primer-template concentration, **262**, 273-274; radiolabeling of substrate [end labeling of primer, **262**, 274-275; internal labeling, **262**, 274, 277-278]; three nucleotide assay, **262**, 270-271; assay of nucleases, **262**, 279-280; bacteriophage 29 DNA polymerase, **262**, 44, 49; DNA polymerase β , **262**, 117; DNA polymerase ϵ , **262**, 98; DNA polymerase II, **262**, 14; DNA polymerase III, **262**, 22; HIV-1 reverse transcriptase [ribonuclease H activity, **262**, 279-280; transcriptase activity, **262**, 143-144]; Klenow fragment, **262**, 264-265.

PROCESSIVITY OF KINESIN ATPase

Kinesin and dynein are two microtubule-associated ATP-dependent motors responsible for intracellular motility. Gilbert *et al.*¹ made direct measurements of the dissociation kinetics of kinesin from microtubules (MTs), the release of orthophosphate and ADP, and the rebinding of this motor to MTs. They observed processivity in ATP hydrolysis amounting to 10 molecules ATP per site at low salt concentration and 1 molecule of ATP per site at a higher concentration of salt. After hydrolysis, the dissociation of kinesin from the MT is rate-limiting, and rebinding of kinesin·ADP to MTs is fast. The authors discuss how this behavior differs from that of skeletal myosin.

¹S. P. Gilbert, M. R. Webb, M. Brune & K. A. Johnson (1995) *Nature* **373**, 671.

PROCHIRALITY



A stereochemical property of compounds arising from the ability of an enzyme's active site to distinguish between two chemically identical substituents covalently bound to a tetrahedral center (usually carbon and, in some cases, phosphorus). Prochirality is also termed prostereoisomerism. The classical example is citrate with its two carboxymethyl group substituents. Likewise, the C1 carbon atom of ethanol has two prochiral hydrogens. **See** *Chirality; Chirality Probes*

PROCOLLAGEN N-ENDOPEPTIDASE

This enzyme [EC 3.4.24.14], also known as procollagen N-proteinase, catalyzes the hydrolysis of the N-propeptide of the collagen chain α -1(I) at Pro—Gln and of α -2(II) chain at Ala—Gln. As a result, N-terminal propeptides of type I and II collagens are released prior to fibril assembly. However, it does not act on type III procollagen.

K. E. Kadler, S. J. Lightfoot & R. B. Watson (1995) *Meth. Enzymol.* **248**, 756.

PRODUCT

The substance or unstable species formed or generated by any chemical or photochemical reaction. The reaction product is usually symbolized by P; in reactions resulting in more than one product, P₁, P₂, . . . , or P, Q, R, . . . are commonly used.

PRODUCT-DETERMINING STEP

The step within a multistep reaction or process that determines the final product distribution. This step may be identical to or may even occur after the rate-determining step in the process.

PRODUCT DEVELOPMENT CONTROL

Referring to a kinetically controlled reaction in which selectivity parallels the relative thermodynamic stabilities of the products. Often in such reactions, the transition state occurs late along the reaction coordinate.

PRODUCT INHIBITION STUDIES

Experiments designed to reach conclusions about an enzyme-catalyzed reaction by examining how one or more products of the reaction alter the kinetic behavior of the enzyme. The diagnostic value of these approaches can be limited by formation of E·substrate·product abortive complexes in multisubstrate mechanisms.

(1) Evaluating the magnitude of dissociation constants of products can provide useful information on enzyme interactions of products. In most cases, these values are true dissociation constants, not Michaelis constants. Initial rate studies of the reverse reaction will provide K_m values and enable one to compare K_d and K_m values. This can aid in identifying key rate constants or steps in the reaction scheme.

(2) Under favorable circumstances, product inhibition studies can offer insights for distinguishing between rival kinetic schemes¹⁻³. For example, consider a two-substrate two-product enzyme. A series of initial rate studies are undertaken in which double-reciprocal plots are made with each substrate (*i.e.*, $1/v$ vs. $1/[A]$ and $1/v$ vs. $1/[B]$ in which the nonvaried cosubstrate is held at a high, albeit nonsaturating, level) at different concentrations of the inhibitor. Identification of what type of inhibitor the product is with respect to each substrate (*i.e.*, competitive, noncompetitive, uncompetitive) and which, if any, of the slope or intercept replots are nonlinear can be of great use in identifying possible mechanisms. Further information can be obtained by repeating the series of experiments at saturating levels of the nonvaried cosubstrate. Cleland⁴ has presented a set of rules to predict the type of inhibition one might expect for certain enzyme schemes.

Fromm¹ and Rudolph² have discussed the practical limitations on interpreting product inhibition experiments. The table below illustrates the distinctive kinetic patterns observed with bisubstrate enzymes in the absence or presence of abortive complex formation. It should also be noted that the random mechanisms in this table (and in similar tables in other texts) are usually for rapid equilibrium random mechanism schemes. Steady-state random mechanisms will contain squared terms in the product concentrations in the overall rate expression. The presence of these terms would predict nonlinearity in product inhibition studies. This nonlinearity might not be obvious under standard initial rate protocols, but products that would be competitive in rapid equilibrium systems might appear to be noncompetitive in steady-state random schemes^{1,4,5}, depending on the relative magnitude of those squared terms. **See** *Abortive Complex*

(3) Product inhibition studies are also very useful in identifying and characterizing the mechanisms of enzymes involving isomerization steps^{1,3,4}.

Use of Product Inhibitors to Distinguish Certain Bireactant Enzyme Systems^a

Mechanism	Product	1/v versus 1/[A]	1/v versus 1/[A]
Theorell-Chance	P	N	C
	Q	C	N
Theorell-Chance (with abortives)	P	N	C
	Q	C	N
Ordered Bi Bi	P	N	N
	Q	C	C
Ordered Bi Bi (with abortives)	P	N ^b	N ^c
	Q	C	N
Random Bi Bi	P	C	C
	Q	C	C
Random Bi Bi (with abortives)	P	N	C
	Q	C	N
Ping Pong Bi Bi	P	N	C
	Q	C	N
Ping Pong Bi Bi (with abortives)	P	N ^c	NL ^d
	Q	C	NL ^d

^aC = Competitive inhibition pattern; N = Noncompetitive inhibition pattern, and U = Uncompetitive inhibition pattern.

^bIntercept replots versus concentration will be parabolic (concave up).

^cSlope replots versus inhibitor concentration will be parabolic (concave up).

^dHyperbolic (concave up).

(5) With respect to enzymes having three or more products in the enzyme-catalyzed reaction, it is also possible to vary the concentration of two products (while keeping their ratio at a constant value) at the same time. If the rate expression for one reaction scheme has terms containing both those products, potentially that scheme can be distinguished from alternative reaction mechanisms that don't have those same terms in the rate expression.

¹H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

²F. B. Rudolph (1979) *Meth. Enzymol.* **63**, 411.

³K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

⁴W. W. Cleland (1963) *Biochim. Biophys. Acta* **67**, 188.

⁵F. B. Rudolph & H. J. Fromm (1971) *J. Biol. Chem.* **246**, 6611.

PRODUCTION

A term used in compartmental analysis to designate the appearance of a tracee in its pool (or, more properly, its volume of distribution), the result from tracee biosynthesis and/or tracee transit from another pool. **See** *Compartmental Analysis*

PRODUCT RULE

A procedure in calculus for obtaining the derivative of the product of two functions, $f(x)$ times $g(x)$,

$$d[f(x)g(x)]/dx = f(x)[dg(x)/dx] + g(x)[df(x)/dx]$$

where each function must be differentiable.

PROGRESS CURVE ANALYSIS

While traditional steady-state enzyme kinetic experiments have focused on initial velocity treatments^{1,2}, the complete time course or reaction progress curve is actually a more robust source of kinetic information. Initial rates are estimated from the slope of a tangent line drawn to the earliest portion of a reaction progress curve, allowing one to obtain a velocity measurement as the change in product (or substrate) concentration per unit time. Initial velocity measurements offer the advantage that the instantaneous substrate concentrations, as well as those of products and/or other effector molecules, can be directly manipulated independently of stoichiometric relationships that hold during the course of the complete reaction process. Moreover, the measurements are usually accomplished under conditions where the state of the enzyme is generally well characterized and is unlikely to change by any irreversible process during the few moments over which a measurement is made. On the other hand, the complete reaction progress curve offers the opportunity to characterize the rate behavior at various extents of substrate depletion and product accumulation. Duggleby³ has recently presented a useful detailed description of the progress curve analysis method, including nonlinear regression methods for evaluating the experimental data. He also discusses general approaches for deriving the integrated rate equations that describe the progress of one- and two-substrate enzyme systems.

See *KINSIM/FITSIM; Chemical Kinetics*

¹H. J. Fromm (1975) *Initial Velocity Enzyme Kinetics*, Springer-Verlag, New York.

²W. W. Cleland (1990) *The Enzymes* **19**, 99.

³R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 61.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Theory, **63**, 159-162; activation effect, **63**, 174, 175; analysis, **63**, 140, 159-183; burst, **64**, 20, 203, 215; enzyme concentration, **63**, 175-177; hysteresis, **64**, 197, 200-204; limitations, **63**, 181-183; plotting, **63**, 177-180; practical methods, **63**, 175-177; reversible inhibitor action, **63**, 163-175; reversible reaction, **63**, 171-175; simulation of, **63**, 180; advantages and disadvantages, **249**, 61-62; analysis, in kinetic models of inhibition, **249**, 168-169; concave-down, **249**, 156; concave-up, **249**, 156; with enzyme-product complex instability, **249**, 88; with enzyme-substrate instability,

ity, **249**, 88; equations [for classical inhibition, **249**, 176; for tight-binding inhibition, **249**, 174-178, 180]; inhibition rate constant determination, **244**, 448-449; in kinetic models of inhibition, **249**, 168; with noncompetitive product inhibition [Foster-Niemann plots, **249**, 226-228; nonlinear regression analysis, **249**, 225-226; oversaturation plots, **249**, 228]; nonlinear regression analysis, **249**, 61-90 [AGIRE computer program for, **249**, 79-81, 225-226; comparison to analysis based on rates, **249**, 61-63; complex reactions, **249**, 75-78; experimental design, **249**, 84-85; inhibitor effects, **249**, 71-75; potato acid phosphatase product inhibition, **249**, 73-74; preliminary fitting, **249**, 82-84; prephenate dehydratase product inhibition, **249**, 72-73; product inhibition effects, **249**, 72-73; prostate acid phosphatase phenyl phosphate hydrolysis, **249**, 70; reactions with two substrates, **249**, 75-77; reversible reactions, **249**, 77-78; with simple Michaelian enzyme, **249**, 63-71 [fitting equations, **249**, 63]; with slow-binding inhibitors, **249**, 88; with unstable enzymes, for kinetic characterization, **249**, 85-89.

PROLINE DEHYDROGENASE

This FAD-dependent enzyme [EC 1.5.99.8] catalyzes the reaction of L-proline with an acceptor and water to produce (S)-1-pyrroline 5-carboxylate and the reduced acceptor.

R. C. Scarpulla & R. L. Soffer (1978) *J. Biol. Chem.* **253**, 5997.
E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

PROLINE DIPEPTIDASE

This manganese-dependent enzyme [EC 3.4.13.9] (also known as Xaa-Pro dipeptidase, X-Pro dipeptidase, imidodipeptidase, prolidase, peptidase D, and γ -peptidase) catalyzes the hydrolysis of Xaa-Pro dipeptides (except for prolylproline). The dipeptidase also acts on aminoacylhydroxyproline derivatives. This cytosolic enzyme, a member of the peptidase family M24A, is found in most animal tissues.

N. C. Davies & E. L. Smith (1957) *J. Biol. Chem.* **224**, 261.
E. L. Smith (1955) *Meth. Enzymol.* **2**, 93.

PROLINE IMINOPEPTIDASE

This manganese-dependent enzyme [EC 3.4.11.5] catalyzes the release of an N-terminal prolyl residue from a peptide. The mammalian enzyme, which is not specific for prolyl bonds, is possibly identical with cytosol aminopeptidase [EC 3.4.11.1].

R. J. DeLange & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 81.
S. Sarid, A. Berger & E. Katchalski (1962) *J. Biol. Chem.* **237**, 2207.

PROLINE RACEMASE

This enzyme [EC 5.1.1.4] catalyzes the interconversion of L-proline and D-proline. **See Enzyme Energetics (The Case of Proline Racemase)**

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.
E. Adams (1976) *Adv. Enzymol.* **44**, 69.
E. Adams (1972) *The Enzymes*, 3rd ed., **6**, 479.

PROLINE REDUCTASE

This pyruvate-dependent enzyme [EC 1.4.4.1], also referred to as D-proline reductase (dithiol), catalyzes the reaction of 5-aminopentanoate with lipoate to produce D-proline and dihydrolipoate. Other dithiols can function as reducing agents.

P. D. van Poelje & E. E. Snell (1990) *Ann. Rev. Biochem.* **59**, 29.
P. A. Recsei & E. E. Snell (1984) *Ann. Rev. Biochem.* **53**, 357.

PROLIPOPROTEIN SIGNAL PEPTIDASE

This enzyme [EC 3.4.99.35] (better known as signal peptidase II and also known as bacterial leader peptidase I) catalyzes the cleavage of N-terminal leader sequences from membrane prolipoproteins.

K. Sankaran & H. C. Wu (1995) *Meth. Enzymol.* **248**, 169.

PROLYL 3-HYDROXYLASE

This enzyme [EC 1.14.11.7], also known as procollagen:proline, 2-oxoglutarate 3-dioxygenase, catalyzes the reaction of a prolyl residue in procollagen with dioxygen and α -ketoglutarate (or, 2-oxoglutarate) to produce a *trans*-3-hydroxyprolyl residue in procollagen, succinate, and carbon dioxide. This reaction also requires iron ions and ascorbate.

K. I. Kivirikko & T. Pihlajaniemi (1998) *Adv. Enzymol.* **72**, 325.
S. Englard & S. Seifter (1986) *Ann. Rev. Nutr.* **6**, 365.
E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

PROLYL 4-HYDROXYLASE

This enzyme [EC 1.14.11.2], also known as procollagen:proline, 2-oxoglutarate 4-dioxygenase, catalyzes the reaction of an L-prolyl residue in procollagen with dioxygen and α -ketoglutarate (or, 2-oxoglutarate) to produce a *trans*-4-hydroxyprolyl residue in procollagen, succinate, and carbon dioxide. This reaction also requires iron ions and ascorbate.

K. I. Kivirikko & T. Pihlajaniemi (1998) *Adv. Enzymol.* **72**, 325.
B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
A. G. Prescott & P. John (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 245.
O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

PROLYL ISOMERASE

This enzyme [EC 5.2.1.8] (also known as peptidylprolyl *cis-trans* isomerase, peptidylprolyl isomerase, and cyclophilin) catalyzes the interconversion of a peptidylprolyl derivative (in which $\omega = 180^\circ$) and a peptidylprolyl derivative (in which $\omega = 0^\circ$).

- F. X. Schmid (1993) *Ann. Rev. Biophys. Biomol. Chem.* **22**, 123.
 R. L. Stein (1993) *Adv. Protein Chem.* **44**, 1.
 F. X. Schmid, L. M. Mayr, M. Mücke & E. R. Schönbrunner (1993) *Adv. Protein Chem.* **44**, 25.

PROPAGATION

Those steps or reactions within chain reactions that generate one or more reactive intermediates (often free radicals). In such propagation reactions, the newly formed reactive intermediate (*e.g.*, free radical) can then react with another molecule to produce another reactive intermediate. **See** Chain Reaction; Chain Transfer; Initiation; Termination

PROPANEDIOL DEHYDRATASE

This cobalamin-dependent enzyme [EC 4.2.1.28] catalyzes the conversion of propane-1,2-diol to propanal and water. The enzyme also dehydrates ethylene glycol to acetaldehyde.

- B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.
 J. C. Murrell & W. Ashraf (1990) *Meth. Enzymol.* **188**, 26.

PROPIONYL-CoA CARBOXYLASE

This biotin-dependent enzyme [EC 6.4.1.3] catalyzes the reaction of ATP with propanoyl-CoA and HCO_3^- to produce ADP, orthophosphate, and (*S*)-methylmalonyl-CoA. Butanoyl-CoA will also serve as a substrate and the enzyme will also catalyze transcarboxylations.

- J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 H. G. Wood & R. E. Barden (1977) *Ann. Rev. Biochem.* **46**, 385.
 A. W. Alberts & P. R. Vagelos (1972) *The Enzymes*, 3rd ed., **6**, 37.

PROPIONYL-CoA SYNTHETASE

This enzyme [EC 6.2.1.17], also known as propionate:CoA ligase, catalyzes the reaction of ATP with propanoate and coenzyme A to produce propanoyl-CoA,

AMP, and pyrophosphate (or, diphosphate). Propenoate can also act as the substrate. This enzyme is not identical with acetyl-CoA synthetase or with butyryl-CoA synthetase.

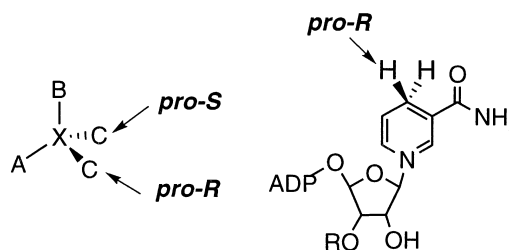
- J. C. Murrell & W. Ashraf (1990) *Meth. Enzymol.* **188**, 26.

PRO-PROCHIRAL

Referring to a compound containing an atom such that, replacement of an atom or group bonded to that particular atom with a different atom or group, would generate a prochiral compound. A simple example of a pro-prochiral molecule is provided by acetic acid. Replacement of one of the methyl hydrogens with another substituent (*e.g.*, Cl) will generate a prochiral compound (in this case, chloroacetic acid). The remaining two hydrogens are enantiotopic since compounds such as BrClCHCOOH are chiral. **See** Prochirality; Chirality

pro-R Group

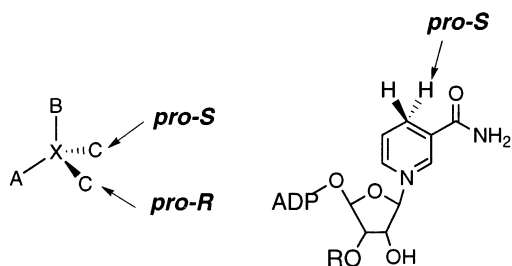
A stereochemical condition that describes a stereoheterotopic group C in the compound XABC_2 containing a prochiral tetrahedral atom X having substituents A, B, C, and C.



To make the assignment, *arbitrarily* assign a higher Cahn-Ingold-Prelog priority over the other stereoheterotopic group C, as though one had mentally substituted the first with a heavier isotope atom. Now, the “higher priority” group C is said to be *pro-R* if the configuration of the mentally generated chiral center is now *R*.

pro-S Group

A stereochemical condition that describes a stereoheterotopic group C in the compound XABC_2 containing a prochiral tetrahedral atom X having substituents A, B, C, and C.



To make the assignment, *arbitrarily* assign a higher Cahn-Ingold-Prelog priority over the other stereoheterotopic group C, as though one had mentally substituted the first with a heavier isotope atom. Now, the “higher priority” group C is said to be *pro-S* if the configuration of the mentally generated chiral center is now *S*.

PROSTACYCLIN SYNTHASE

This heme thiolate-dependent enzyme [EC 5.3.99.4] catalyzes the interconversion of prostaglandin H_2 and prostaglandin I_2 (prostacyclin).

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

PROSTAGLANDIN E: 9-KETO (α)-REDUCTASE

This enzyme [EC 1.1.1.189], also known as prostaglandin E_2 9-reductase, catalyzes the NADPH-dependent reduction of prostaglandin E_2 to prostaglandin $F_{2\alpha}$ and $NADP^+$.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

PROSTAGLANDIN H, PROSTAGLANDIN D ISOMERASE

This glutathione-dependent enzyme [EC 5.3.99.2], also known as prostaglandin D synthase and prostaglandin H_2 :D isomerase, catalyzes the opening of the epidioxy bridge in (5*Z*,13*E*)-(15*S*)-9 α ,11 α -epidioxy-15-hydroxyprosta-5,13-dienoate to yield (5*Z*,13*E*)-(15*S*)-9 α ,15-dihydroxy-11-oxoprost-5,13-dienoate.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

PROSTAGLANDIN H, PROSTAGLANDIN E ISOMERASE

This glutathione-dependent enzyme [EC 5.3.99.3], also known as prostaglandin E synthase, prostaglandin H_2 E isomerase, and endoperoxide isomerase, catalyzes the

opening of the epidioxy bridge, thereby converting (5*Z*,13*E*)-(15*S*)-9 α ,11 α -epidioxy-15-hydroxyprosta-5,13-dienoate into (5*Z*,13*E*)-(15*S*)-11 α ,15-dihydroxy-9-oxoprost-5,13-dienoate.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

PROSTAGLANDIN H_2 SYNTHASE

By acting both as a dioxygenase and as a peroxidase, this enzyme [EC 1.14.99.1] catalyzes the reaction of arachidonate with reduced acceptor AH_2 and two dioxygen molecules to produce prostaglandin H_2 , oxidized acceptor A, and water.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

W. L. Smith, R. M. Garavito & D. L. DeWitt (1996) *J. Biol. Chem.* **271**, 33157.

D. Picot, P. J. Loll & M. Garavito (1994) *Nature* **367**, 243.

J. A. Boyd (1990) *Meth. Enzymol.* **186**, 283.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

PROSTHETIC GROUP

A very tightly bound coenzyme. Examples of prosthetic groups include the hemes, cobalamin coenzymes, pyridoxal 5-phosphate, lipoic acid, and biotin. FAD and FMN are also very tightly held by most flavin-dependent oxidoreductases. In many instances, distinctions are made purely on the basis of affinity and do not apply consistently. For example, while NAD^+ and $NADP^+$ binding affinities are relatively high, dialysis is frequently sufficient for resolving (removing) these coenzymes from their binding sites on $NAD(P)^+$ -dependent oxidoreductases.

PROTEASE ACCESSIBILITY (as Probe of Protein Structure)

A method for analyzing protein structure based on limited proteolysis. This method is especially useful in investigations of membrane proteins whose membrane association limits the repertoire of techniques that can be gainfully applied to infer structural features. For example, Davis *et al.*¹ used four proteases to assess the topology of yeast H^+ -ATPase reconstituted into phosphatidylserine vesicles. Limited proteolysis by trypsin and α -chymotrypsin inactivates the enzyme and produces stable, membrane-bound fragments. Sequence analyses of

these peptides localized these cleavage sites to opposite sides of a central hydrophilic domain containing consensus sequences for the site of phosphorylation and fluorescein isothiocyanate binding of several related ATPases. Limited proteolysis of the ATPase by elastase cuts approximately fifty amino acids from the C terminus, and the remaining membrane-bound fragment is active. Proteolysis by carboxypeptidase Y in the presence and absence of detergent suggests that the C terminus is on the inside of the vesicle in this reconstitution. The authors presented a model for the transmembrane arrangement of the polypeptide, with the C terminus on the inside of the vesicle and the N terminus on the outside. They suggested that the ATP binding region is likewise on the outside, and that the polypeptide passes through the membrane a minimum of five times.

¹C. B. Davis & G. G. Hammes (1989) *J. Biol. Chem.* **264**, 370.

PROTEASE INHIBITOR "COCKTAILS"

Proteolysis can be arrested or minimized through the use of so-called cocktails of inhibitors of serine, cysteine, aspartic proteases and metalloproteases. Among the commonly used inhibitors are the sulfonylating agents acting on serine proteases, phenylmethanesulfonyl chloride (PMSF) and 4-(2-aminoethyl)benzenesulfonyl fluoride (AEBSF). Antipain is [(S)-carboxy-2-phenylethyl] carbamoyl-L-arginyl-L-valyl-L-arginal which has an I_{50} (i.e., inhibitor concentration that yields 50% inhibition) of 0.15 $\mu\text{g/mL}$ for papain, 0.25 $\mu\text{g/mL}$ for trypsin, 1.2 $\mu\text{g/mL}$ for cathepsin A, and is relatively ineffective with chymotrypsin. Leupeptin is *N*-propionyl-L-leucyl-L-leucyl-L-arginal which has a I_{50} of 2 $\mu\text{g/mL}$ with trypsin, 8 $\mu\text{g/mL}$ with plasmin, 0.5 $\mu\text{g/mL}$ with papain, 0.5 $\mu\text{g/mL}$ with cathepsin B and is relatively ineffective with chymotrypsin. Chymostatin (or, *N*-[[(S)-carboxy-2-phenylethyl]carbamoyl]- α -[2-imidohexahydro-4(*S*)-pyrimidyl]-L-glycyl-L-leucyl-L-phenylalaninal) is a potent chymotrypsin inhibitor (with a I_{50} of 0.1–0.2 $\mu\text{g/mL}$). Many of these cocktails are commercially available.

General Protease Cocktails. These should contain the following inhibitors: AEBSF, *trans*-epoxysuccinyl-L-leucylamido(4-guanidino)butane (E-64), bestatin, leupeptin, aprotinin (and sodium EDTA if the protein or enzyme being purified does not require a divalent metal cofactor for stability or activity).

Fungal and Yeast Protease Cocktail. These should contain the following inhibitors: AEBSF, pepstatin A, E-64, and 1,10-phenanthroline.

Mammalian Cell Protease Inhibitor Cocktail. These should contain AEBSF, pepstatin A, E-64, bestatin, leupeptin, and aprotinin. (Metal chelators can be added to suppress the activity of calcium ion-dependent proteases such as calpain. Again, one must determine whether the protein or enzyme being purified does not require a divalent metal cofactor for stability or activity.)

Bacterial Extract Protease Cocktail. Inhibitors present are pepstatin A, 4-(2-amino-ethyl)benzenesulfonyl fluoride (AEBSF), *trans*-epoxysuccinyl-L-leucylamido(4-guanidino)butane (E-64), bestatin, and sodium EDTA.

PROTEASE La

This enzyme [EC 3.4.21.53], also known as endopeptidase La, ATP-dependent serine proteinase, and ATP-dependent protease La, catalyzes the hydrolysis of peptide bonds in large proteins (for example, globin, casein, and denaturated serum albumin) in the presence of ATP (which is hydrolyzed to ADP and orthophosphate). Vanadate ion inhibits both reactions. A similar enzyme occurs in animal mitochondria. Protease La belongs to the peptidase family S16.

A. L. Goldberg, R. P. Moerschell, C. H. Chung & M. R. Maurizi (1994) *Meth. Enzymol.* **244**, 350.

S. Kuzela & A. L. Goldberg (1994) *Meth. Enzymol.* **244**, 376.

PROTEINASE A

This enzyme [EC 3.4.23.19], also known as aspergillopepsin II, proctase A, and *Aspergillus niger* var. *macrosporus* aspartic proteinase, catalyzes the hydrolysis of peptide bonds in proteins. It has been isolated from *Aspergillus niger* var. *macrosporus* and is distinct from aspergillopepsin I in specificity and in insensitivity to pepstatin.

K. Takahashi (1995) *Meth. Enzymol.* **248**, 146.

PROTEIN COMPLEMENTATION

A process that partially or fully restores the biological activity of two proteins, each lacking a functional domain or motif, through the formation of an oligomeric species.

See Domain Swapping

I. Zabin & M. R. Villarejo (1975) *Annu. Rev. Biochem.* **44**, 295.

PROTEIN DISULFIDE ISOMERASE

This enzyme [EC 5.3.4.1], also known as “S—S rearran-gase,” catalyzes the rearrangement of intrachain or interchain disulfide bonds within proteins to form their native structures. The enzyme requires reducing agents or the partly reduced enzyme. The reaction operates by sulfhydryl-disulfide interchange.

- H. F. Gilbert (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 609.
 R. B. Freedman, H. C. Hawkins & S. H. McLaughlin (1995) *Meth. Enzymol.* **251**, 397.
 W. W. Wells, Y. Yang & T. L. Deits (1993) *Adv. Enzymol.* **66**, 149.
 N. J. Bulleid (1993) *Adv. Protein Chem.* **44**, 125.
 R. C. Fahey & A. R. Sundquist (1991) *Adv. Enzymol.* **64**, 1.

PROTEIN-DISULFIDE REDUCTASE (Glutathione)

This enzyme [EC 1.8.4.2], also known as glutathione: insulin transhydrogenase and insulin reductase, catalyzes the reaction of two glutathione with a disulfide bond in a protein to produce glutathione disulfide and a protein with two new thiol groups. The enzyme can reduce insulin and a number of other proteins.

- H. H. Tomizawa (1971) *Meth. Enzymol.* **17B**, 515.

PROTEIN FARNESYLTRANSFERASES

This class of farnesyltransferases catalyzes the posttranslational modification of proteins by the cholesterol precursor, farnesylpyrophosphate. One of the substrates of this enzyme is Ras.

- R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.
 P. J. Casey & M. G. Seabra (1996) *J. Biol. Chem.* **271**, 5289.
 F. L. Zhang & P. J. Casey (1996) *Ann. Rev. Biochem.* **65**, 241.
 S. Clarke (1992) *Ann. Rev. Biochem.* **61**, 355.

PROTEIN GERANYLGERANYL-TRANSFERASES

The following are recent reviews on the molecular and physical properties of these transferases which transfer geranylgeranyl groups to defined sites on protein substrates.

- R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.
 P. J. Casey & M. G. Seabra (1996) *J. Biol. Chem.* **271**, 5289.
 F. L. Zhang & P. J. Casey (1996) *Ann. Rev. Biochem.* **65**, 241.
 Y. Jiang, G. Rossi & S. Ferro-Novick (1995) *Meth. Enzymol.* **257**, 21.
 S. A. Armstrong, M. S. Brown, J. L. Goldstein & M. C. Seabra (1995) *Meth. Enzymol.* **257**, 30.
 S. Clarke (1992) *Ann. Rev. Biochem.* **61**, 355.

PROTEIN KINASE

Any of a broad class of phosphoryl-transfer enzymes [EC 2.7.1.x] that catalyze the ATP-dependent phosphorylation of proteins, most often occurring at seryl, threonyl, and tyrosyl residues. These enzymes are central participants in cellular signal transduction pathways, and their discovery and recognition as primary control components of the cell culminated in the award of the 1992 Nobel Prize in Medicine and Physiology to American enzymologists Edwin Krebs and Edward Fischer. There is reason to believe that approximately 2% of the coding sequences in the human genome specify some 2000 different kinases that phosphorylate protein substrates. The prototypical enzyme is known as 3',5'-cAMP-stimulated protein kinase (or, protein kinase A). **See specific protein kinase**

- S. S. Taylor, D. R. Knighton, J. Zheng, L. F. Ten Eyck & J. M. Sowadski (1992) *Ann. Rev. Cell Biol.* **8**, 429.
 J. Larner (1990) *Adv. Enzymol.* **63**, 173.
 A. M. Edelman, D. K. Blumenthal & E. G. Krebs (1987) *Ann. Rev. Biochem.* **56**, 567.
 S. I. Walaas & P. Greengard (1987) *The Enzymes*, 3rd ed., **18**, 285.
 H. V. Rickenberg & B. H. Leichtling (1987) *The Enzymes*, 3rd ed., **18**, 419.
 E. G. Krebs (1986) *The Enzymes*, 3rd ed., **17**, 3.
 D. A. Walsh & E. G. Krebs (1973) *The Enzymes*, 3rd ed., **8**, 555.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

General: Protein kinase classification, **200**, 3; protein kinase catalytic domain sequence database: identification of conserved features of primary structure and classification of family members, **200**, 38; protein kinase phosphorylation site sequences and consensus specificity motifs: tabulations, **200**, 62.

Assay Methods: Enzyme activity dot blots for assaying protein kinases, **200**, 85; solid-phase protein-tyrosine kinase assays, **200**, 90; nonradioactive assays of protein-tyrosine kinase activity using anti-phosphotyrosine antibodies, **200**, 98; use of synthetic amino acid polymers for assay of protein-tyrosine and protein-serine kinases, **200**, 107; assay of phosphorylation of small substrates and of synthetic random polymers that interact chemically with adenosine 5'-triphosphate, **200**, 112; assay of protein kinases using peptides with basic residues for phosphocellulose binding, **200**, 115; design and use of peptide substrates for protein kinases, **200**, 121; synthetic peptide substrates for casein kinase II, **200**, 134; nucleotide photoaffinity labeling of protein kinase subunits, **200**, 477; separation of phosphotyrosine, phosphoserine, and phosphothreonine by high-performance liquid chromatography, **201**, 3; chemical properties and separation of phosphoamino acids by thin-layer chromatography and/or electrophoresis, **201**, 10; determination of phosphoamino acid composition by acid hydrolysis of protein blotted to Immobilon, **201**, 21; comparison of three methods for detecting tyrosine-phosphorylated proteins, **201**, 28; generation and use of antibodies to phosphothreonine, **201**, 44; generation and use of anti-phosphotyrosine antibodies raised against bacterially expressed Abl protein, **201**, 53; preparation and use of anti-phosphotyrosine antibodies to study structure and function of insulin receptor, **201**, 65; generation of monoclonal antibodies against phosphotyrosine and their use for affinity purification of phosphotyrosine-containing proteins, **201**, 79; isolation of tyro-

sine-phosphorylated proteins and generation of monoclonal antibodies, **201**, 92; generation and use of anti-phosphotyrosine antibodies for immunoblotting, **201**, 101.

Determination of Phosphorylation Stoichiometry: By biosynthetic labeling, **201**, 245; by separation of phosphorylated isoforms, **201**, 251; by microchemical determination of phosphate in proteins isolated from polyacrylamide gels, **201**, 261; production of phosphorylation state-specific antibodies, **201**, 264.

Protein Kinase Inhibitors: Pseudosubstrate-based peptide inhibitors, **201**, 287; utilization of the inhibitor protein of adenosine cyclic monophosphate-dependent protein kinase, and peptides derived from it, as tools to study adenosine cyclic monophosphate-mediated cellular processes, **201**, 304; use of sphingosine as inhibitor of protein kinase C, **201**, 316; properties and use of H-series compounds as protein kinase inhibitors, **201**, 328; use and specificity of staurosporine, UCN-01, and calphostin C as protein kinase inhibitors, **201**, 340; inhibition of protein-tyrosine kinases by tyrphostins, **201**, 347; use and specificity of genistein as inhibitor of protein-tyrosine kinases, **201**, 362; use and selectivity of herbimycin A as inhibitor of protein-tyrosine kinases, **201**, 370; use of erbstatin as protein-tyrosine kinase inhibitor, **201**, 379.

PROTEIN KINASE C

Members of the protein kinase C family promote signal transduction by catalyzing ATP-dependent protein phosphorylation [general EC number 2.7.1.37] in response to various signals that promote lipid hydrolysis. The primary activator is diacylglycerol. **See also** *Calcium/Calmodulin-Dependent Protein Kinase*

- A. C. Newton (1993) *Ann. Rev. Biophys. Biomol. Struct.* **22**, 1.
 S. S. Taylor, D. R. Knighton, J. Zheng, L. F. Ten Eyck & J. M. Sowadski (1992) *Ann. Rev. Cell Biol.* **8**, 429.
 J. Larner (1990) *Adv. Enzymol.* **63**, 173.
 U. Kikkawa, A. Kishimoto & Y. Nishizuka (1989) *Ann. Rev. Biochem.* **58**, 31.

PROTEIN OXIDATION (In Aging & Disease)

Berlett and Stadtman¹ recently discussed the pathways by which oxidatively modified forms of proteins accumulate during aging, oxidative stress, and in some pathological conditions. Reactive oxygen species (ROS), generated by physiologic as well as nonphysiologic means, result in amino acid residue side-chain oxidation, protein-protein cross-linkages, and protein backbone oxidation. Henle and Linn² have reviewed related issues regarding the formation, prevention, and repair of DNA damage by iron/hydrogen peroxide (so-called Fenton oxidants).

¹B. S. Berlett & E. R. Stadtman (1997) *J. Biol. Chem.* **272**, 20313.

²E. S. Henle & S. Linn (1997) *J. Biol. Chem.* **272**, 19095.

PROTEIN PRENYLTRANSFERASES

The following is a recent review on the molecular and physical properties of these transferases which prenylate suitable protein substrates.

Entire Volume on this Topic: P. J. Casey & J. E. Buss, eds. (1995) *Meth. Enzymol.*, vol. **250**.

PROTEIN-PROTEIN RECOGNITION KINETICS

Janin¹ considered the kinetics of protein-protein association based on a simple kinetic model for association that assumes proteins are relatively rigid bodies (*i.e.*, no large conformation change occurs) in the reaction. Association commences with the random collision at the rate $k_{\text{collision}}$ predicted by the Einstein-Smoluchowski equation, thereby creating an encounter pair whose evolution into a stable complex occurs provided that the two molecules are correctly oriented and positioned. This is treated by assigning a probability $p(r)$. In the absence of long-range interactions, the bimolecular association rate would be $p(r)k_{\text{collision}}$. He also noted that long-range electrostatic interactions should affect both $k_{\text{collision}}$ and $p(r)$. The collision rate is multiplied by $q(t)$, a factor larger than 1 when the molecules are oppositely charged; thus, coulombic attraction increases the frequency of collision. The parameter $q(t)$ is less than 1 for repulsive interactions. The product $p(r)q(t)$ represents the "steering effect" of electric dipoles acting to preorient the molecules even before collision. Janin¹ first applied this model on the kinetics of barnase-barstar association². He found that, when long-range electrostatic interactions are fully screened or abolished by site-directed mutagenesis (such that $p(r)q(t) \sim 1$), the rate of productive collision is approximately $10^5 \text{ M}^{-1} \text{ s}^{-1}$. Under such conditions, $p(r)$ is approximately 1.5×10^{-5} and is determined by the loss of rotational freedom. This value is in agreement with computer simulations of protein-protein docking, suggesting that a rotational entropy loss ΔS_{rot} is about 22 entropy units in the transition state for protein-protein recognition. As might be expected, low ionic strength enhances long-range electrostatic interactions and accelerates barnase-barstar association by a factor $q(t)q(r)$ of up to 10^5 , suggesting further that favorable charge-charge interactions work together with charge-dipole interactions to make the association rate much faster than otherwise expected on the basis of free diffusion.

Schreiber and Fersht² noted earlier that the association of oppositely charged proteins proceeds through the rate-determining formation of an early, weakly specific complex. The formation of such a complex is thought to be dominated by long-range electrostatic interactions³, and by precise docking to form the high affinity complex occurring afterward. They suggest that this mode of binding may be widespread in nature.

¹J. Janin (1997) *Proteins* **28**, 153.

²G. Schreiber & A. Fersht (1996) *Nature Struct. Biol.* **3**, 427.

³A. R. Fersht (1999) *Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysis and Protein Folding*, pp. 650, W. H. Freeman, New York.

PROTEIN SYNTHESIS INHIBITORS

Agents that retard or arrest protein synthesis by interacting with essential reactions in the multistep process. Cycloheximide effectively inhibits eukaryotic protein synthesis on cytoplasmic ribosomes, and 1 $\mu\text{g/mL}$ results in 70–75% inhibition for cell-free translation, whereas 30 $\mu\text{g/mL}$ almost totally blocks intracellular protein synthesis. Chloramphenicol blocks the peptidyltransferase reaction, but DNA synthesis can also be reduced by this inhibitor. Other protein synthesis inhibitors include: α -amino- β -butyric acid (an analogue of valine); aurintricarboxylic acid (an inhibitor of initiation at 50 μM); 7-azatryptophan (a tryptophan analogue that blocks bacteriophage synthesis); azetidine-2-carboxylic acid (a four-membered ring analogue of proline that can be incorporated into bacterial proteins); canavanine (an arginine analogue that inhibits bacterial protein synthesis) emetine (binds to the 40S eukaryotic ribosome and inhibits translocation); erythromycin (bacterial protein synthesis inhibitor); ethionine (methionine analogue that inhibits bacterial protein synthesis); 5-fluorotryptophan; 6-fluorotryptophan; methyltryptophan; fusidic acid; kasugamycin; 7-methylguanosine 5'-phosphate; ω -methyllysine; *O*-methylthreonine; norleucine; pactamycin; puromycin (potent inhibitor of prokaryotic and eukaryotic protein synthesis); selenomethionine; sparsomycin; streptomycin; tetracycline antibiotics; and tryptazan.

PROTEIN TURNOVER

Cyclical synthesis/degradation of proteins in cells. Schimke¹ treated the rate of protein synthesis as a zero-order process and the rate of protein degradation as a first-order reaction:

$$d[\text{P}_{\text{total}}]/dt = k_s - k_d[\text{P}_{\text{total}}]$$

with $[\text{P}_{\text{total}}]$ the concentration of a particular protein with k_s and k_d as the rate constants for synthesis and degradation. The zero-rate rate constant k_s (the units being molarity·time⁻¹) depends on many factors, such as a cell's mRNA content, ribosome content, pool size(s) for any potentially limiting amino acid(s) or aminoacyl-tRNA(s), and other factors affecting the efficiency of translation. In principle, the rate, extent, and location of mRNA synthesis/degradation can also strongly influence the magnitude of k_s . Although degradation rate is treated here as a first-order process, the suitability of the first-order approximation should always be verified for each individual protein. At a steady state,

$$k_s = k_d[\text{P}_{\text{total}}]_{\text{Steady-State}}$$

Moreover, with changes in cellular metabolism in response to changes in nutrients and metabolic effectors, the concentration of a specific protein can undergo a transition from one steady-state to another.

Experiments² on the incorporation of [¹⁴C]leucine into rat liver tryptophan oxygenase in response to treatment with tryptophan and/or hydrocortisone illustrate how the judicious application of enzyme activity and specific radioactivity measurements can provide powerful insights regarding the mechanisms controlling enzyme synthesis and degradation. (**See also** *Leucine Kinetics*) Yamamoto³ has discussed the related topic of steroid receptor-regulated transcription of specific genes and gene networks.

As shown in the table below, the half-lives of protein degradation are of widely disparate magnitudes.

Approximate Biological Half-Lives for Selected Proteins

Protein	Tissue Source	Half-life
Ornithine decarboxylase	Liver	12 min
Tyrosine aminotransferase	Liver	120 min
Glucokinase	Liver	12 h
Hexokinase	Liver	48–72 h
Fructose 1,6-bis-P aldolase	Muscle	>100 h
Lactate dehydrogenase	Muscle	>140 h
Cytochrome c	Muscle	>150 h

There are indications that the rates of degradation depend on the thermostability and oxidation of proteins, findings that suggest that the integrity of protein tertiary structure may in part determine the overall intracellular

lifetime of a protein. Dice and Goldberg⁴ have carried out a statistical analysis of the relationship between protein degradative rate and protein molecular weight. Likewise, the amino acid at the N terminus of a protein appears to influence the protein's lifetime in cells (**See N-End Rule**). Goldberg and St. John⁵ presented a very readable and informative account of investigations of intracellular protein degradation in mammalian and bacterial cells.

¹R. T. Schimke (1969) *Current Topics in Cellular Regulation* **1**, 77.

²R. T. Schimke, E. W. Sweeney & C. M. Berlin (1965) *J. Biol. Chem.* **240**, 322 and 4609.

³K. R. Yamamoto (1985) *Ann. Rev. Genet.* **19**, 209.

⁴J. F. Dice & A. L. Goldberg (1975) *Arch. Biochem. Biophys.* **170**, 213.

⁵A. L. Goldberg & A. C. St. John (1976) *Ann. Rev. Biochem.* **55**, 1091.

PROTocatechuate 3,4-DIOXYGENASE

This iron-dependent enzyme [EC 1.13.11.3], also known as protocatechuate oxygenase, catalyzes the reaction of 3,4-dihydroxybenzoate with dioxygen to produce 3-carboxy-*cis,cis*-muconate.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.

J. W. Whittaker, A. M. Orville & J. D. Lipscomb (1990) *Meth. Enzymol.* **188**, 82.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

PROTocatechuate 4,5-DIOXYGENASE

This iron-dependent enzyme [EC 1.13.11.8], also known as protocatechuate 4,5-oxygenase, catalyzes the reaction of protocatechuate with dioxygen to produce 4-carboxy-2-hydroxymuconate semialdehyde.

B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

D. M. Arciero, A. M. Orville & J. D. Lipscomb (1990) *Meth. Enzymol.* **188**, 89.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

PROTOchlorophyllide REDUCTASE

This enzyme [EC 1.3.1.33], also known as NADPH-protochlorophyllide oxidoreductase, catalyzes the reaction of chlorophyllide a with NADP⁺ to produce protochlorophyllide, having the (7*S*,8*S*)-configuration, and NADPH. It catalyzes a light-dependent *trans*-reduction of the D-ring of protochlorophyllide.

PROTOGENIC

A term applied to a solvent that can act as a proton donor. Water and ethanol are both protogenic and protophilic. **See** *Protophilic*; *Amphiprotic*

PROTON AFFINITY

The energy released when a proton (or, hydron) reacts with a molecular entity (in a gas-phase reaction, either real or hypothetical) to produce the corresponding conjugate acid; *i.e.*, the negative of the enthalpy change.

S. G. Lias, R. P. Liebman & R. P. Levin (1984) *J. Phys. Chem. Ref. Data* **13**, 695.

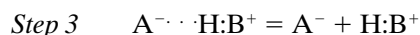
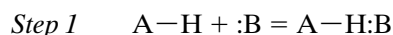
PROTON INVENTORY TECHNIQUE

An isotope effect technique applied in enzyme-catalyzed isotope exchange experiments measuring reaction rate as a function of the [D₂O]/[H₂O] ratio. Such studies often provide information concerning the effect of the environment of the transition state on exchangeable protons.

K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

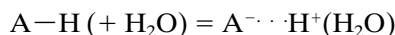
PROTON TRANSFER IN AQUEOUS SOLUTION

The transfer of protons in the vapor phase is strongly influenced by proton affinities as well as factors affecting the stability of hydrogen-bonded molecular complexes (among them are the linearity of hydrogen bonds, the distance between the two atoms sharing the bonding hydrogen, and the electrostatic potentials of those two atoms). Eigen and co-workers¹⁻³ treated proton transfer as the summation of three discernibly distinct steps:

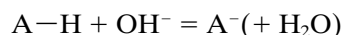


The first step is a bimolecular reaction leading to the formation of a hydrogen bond; the second step is the breaking of the hydrogen bond such that the protonated species H:B⁺ is formed; the third step is the dissociation reaction to form the products. In aqueous solutions, the bimolecular reaction proceeds much faster than would be predicted from gas phase kinetic studies, and this underscores the complexity of proton transfer in solvents with extensive hydrogen-bonding networks capable of creating parallel pathways for the first step. In their au-

thoritative consideration of proton transfer in aqueous solutions, Schuster *et al.*⁴ presented several summary statements about the nature of this process. First, transfer of a proton to solvent can occur equally well by protolysis:



and by hydrolysis



Second, rate constants for intermolecular proton transfer obey the Brønsted relationship, and the rates can be strongly influenced by electrostatics as well as the difference in the $\text{p}K_{\text{a}}$'s for the two acids. Third, proton transfer can proceed at slower than diffusion-limited values with compounds forming weak hydrogen bonds in water.

Carbonic anhydrase presents an instructive case where the catalytic efficiency is so great ($k_{\text{cat}} > 10^6 \text{ s}^{-1}$) that proton transfer becomes rate-limiting. The rate was found to depend on the concentration of the protonated form of buffers in the solution. Indeed, Silverman and Tu⁵ adduced the first convincing evidence for the role of buffer in carbonic anhydrase catalysis through their observation of an imidazole buffer-dependent enhancement in equilibrium exchanges of oxygen isotope between carbon dioxide and water. The effect is strictly on k_{cat} , and $k_{\text{cat}}/V_{\text{max}}$ is unaffected because the latter is described by a rate law that does not include proton transfer steps.

Other reviews on proton transfer deal with the following topics: hydrogen bonding and H^+ transfer⁶⁻⁸, general acid/base catalysis⁹⁻¹¹, and proton transfers in biological systems involving proteins^{12,13}.

¹M. Eigen, G. G. Hammes & K. Kustin (1960) *J. Am. Chem. Soc.* **82**, 3482.

²M. Eigen, W. Kruse, G. Maass & L. C. M. DeMaeyer (1964) *Prog. React. Kin.* **2**, 287.

³M. Eigen & G. G. Hammes (1963) *Adv. Enzymol.* **25**, 1.

⁴P. Schuster, W. Wolschann & K. Tortschanoff (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.) pp. 107-190, Springer-Verlag, Berlin.

⁵D. N. Silverman & C. K. Tu (1975) *J. Amer. Chem. Soc.* **97**, 2263.

⁶R. P. Bell (1973) *The Proton in Chemistry*, 2nd ed., Cornell Univ. Press, Ithaca.

⁷A. J. Kresge (1975) *Accts. Chem. Res.* **8**, 354.

⁸J. Crossley (1970) *Advan. Mol. Relax. Proc.* **2**, 69.

⁹E. F. Caldin & V. Gold (1975) *Proton Transfer Reactions*, Chapman-Hall, London.

¹⁰W. J. Albery (1967) *Progr. React. Kin.* **4**, 353.

¹¹E. Grunwald & E. K. Ralph (1971) *Accts. Chem. Res.* **4**, 107.

¹²G. G. Hammes & P. R. Schimmel (1970) *The Enzymes* **2**, 67.

¹³H. Gutfreund (1971) *Ann. Rev. Biochem.* **40**, 315.

PROTOPHILIC

A term applied to a solvent exhibiting affinity for protons and therefore acting as a proton acceptor. Water and ethanol are both protophilic; moreover, both are protogenic (*i.e.*, they can release protons in ionization reactions). **See** *Protogenic; Amphiprotic*

PROTOPORPHYRINOGEN OXIDASE

This FMN-dependent enzyme [EC 1.3.3.4] catalyzes the reaction of protoporphyrinogen-IX with dioxygen to produce protoporphyrin-IX and water. The enzyme will also use mesoporphyrinogen-IX as a substrate, albeit not as effectively.

T. A. Dailey & H. A. Dailey (1997) *Meth. Enzymol.* **281**, 340.

S. Granick & S. I. Beale (1978) *Adv. Enzymol.* **46**, 33.

PSEUDO BUFFERS

Solutions in which the buffering action is due to the solvent rather than any added solute¹. Strongly acidic or basic aqueous solutions will show little change in pH when additional increments of acid or base are added (recall that the $\text{p}K_{\text{a}}$ value for H_3O^+ is -1.74 , and that for H_2O is 15.74)^{1,2}. Because the solvent is in such high concentration, the buffering capacity for pseudo buffers is larger than for conventional buffers. **See** *Buffer Capacity*

¹D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

²Z. H. S. Harned & R. A. Robinson (1940) *Trans. Faraday Soc.* **36**, 973.

PSEUDO CATALYSIS

The acceleration of a reaction by a substance that is also *consumed* during the process. An example of such a phenomenon is the acceleration of a reaction by a Brønsted acid present in large excess or maintained by a nearly constant concentration by a buffer. The acceleration of the hydrolysis of an amide by a certain Brønsted acid is actually general acid promotion rather than general acid catalysis. The term "promotion" has been used as a synonym for pseudo catalysis.

PSEUDOHALIDES

Anions such as thiocyanate, azide, and cyanide that can act in place of halides as alternative substrates or inhibi-

tors in certain enzymatic reactions. The standard reduction potentials of Cl^- , Br^- , and I^- are 1.36 V, 1.05 V, and 0.53 V, respectively. The corresponding values for NCS^- and CN^- are 0.77 V and 0.375 V. Thus, haloperoxidase action on these anions is not at all surprising.

PSEUDOROTATION

A process observed in some pentacovalent phosphorane compounds (or intermediates) in which there is an interchange of apical and equatorial ligands.

In many nucleophilic substitution reactions on phosphorylated compounds, in the initial attack of the nucleophile to form a pentavalent intermediate, the entering nucleophile is in a so-called apical position, and the leaving group likewise resides in an apical position. However, in the case of an adjacent mechanism, the nucleophile enters on the same side that has the group destined to be the leaving group. Accordingly, Westheimer¹ described how the pentavalent phosphorus atom must undergo pseudorotation to place the eventual leaving group in an apical position. Only then will the leaving group depart from the reaction center. This mechanism was unambiguously demonstrated through the use of chiral thiophosphorus-containing substrates for the ribonuclease reaction in a classical set of experiments^{2,3} and heralded the widespread and systematic application⁴⁻⁸ of chiral [^{16}O , ^{17}O , ^{18}O]thiophosphate esters to define the stereochemical course of ATP/GTP-dependent reactions catalyzed by phosphatases, AMP- and ADP-forming synthetases, and phosphotransferases.

¹F. H. Westheimer (1968) *Acc. Chem. Res.* **1**, 70.

²D. A. Usher, D. I. Richardson & F. Eckstein (1970) *Nature (London)* **228**, 663.

³D. A. Usher, E. S. Erenrich & F. Eckstein (1972) *Proc. Natl. Acad. Sci. U.S.A.* **69**, 115.

⁴F. Eckstein, P. J. Romaniuk & B. A. Connolly (1982) *Meth. Enzymol.* **87**, 197.

⁵P. A. Frey, J. P. Richard, H. T. Ho, R. S. Brody, R. D. Sammons & K.-F. Sheu (1982) *Meth. Enzymol.* **87**, 213.

⁶M.-D. Tsai (1982) *Meth. Enzymol.* **87**, 235.

⁷S. L. Buchwald, D. E. Hansen, A. Hassett & J. R. Knowles (1982) *Meth. Enzymol.* **87**, 279.

⁸M. R. Webb (1982) *Meth. Enzymol.* **87**, 301.

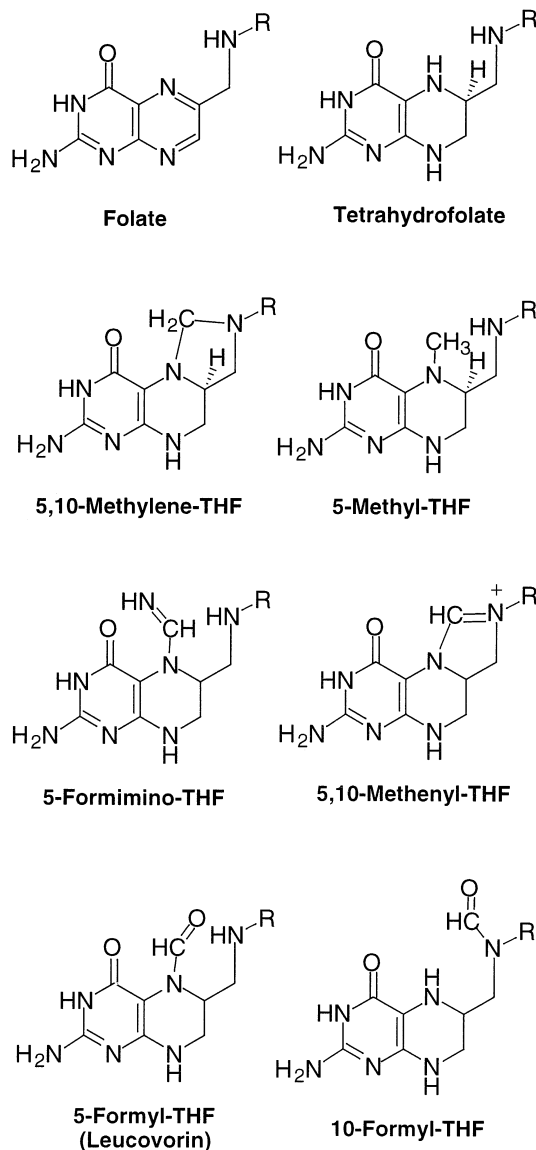
PSI (Ψ)

1. Symbol for electric flux. 2. Symbol for wave function.

PTERIDINES, ANALOGUES & FOLATE COENZYMES

Metabolites that contain a ring system formally defined by IUPAC as pyrimidino[4,5-*b*]pyrazine or pyrazine

[2,3-*d*]pyrimidine. These metabolites participate in various redox reactions, and the vitamin folic acid is the principal metabolic form from which these derivatives are synthesized. Folic acid is biosynthesized in microorganisms through the cleavage of the imidazole ring of GMP with loss of carbon-8 and subsequent ring closure to form the pteridine ring system.



Selected entries from *Methods in Enzymology* [vol, page(s)]:

Chromatographic analysis of pteridines, **66**, 429; thin-layer chromatography of pteroylmonoglutamates and related compounds, **66**, 437; chromatography of folates on Sephadex G-10, **66**, 443; separation of folic acid derivatives and pterins by high-performance liquid chromatography, **66**, 452; determination of folate by use of radioactive folate and binding proteins, **66**, 468; PMR characteristics of folic acid and analogs, **66**, 483; identification of the intracellular folate coenzymes of different cell types, **66**, 501;

isolation and preparation of pterins from biological materials, **66**, 508; methods for reduction, stabilization, and analyses of folates, **66**, 517; solid-phase synthesis of pteroylpolyglutamates, **66**, 523; preparation of tritiated dihydrofolic acid of high specific activity, **66**, 529; preparation of folic acid specifically labeled with carbon-13 in the benzoyl carbonyl, **66**, 533; preparation of aminopterin and *p*-aminobenzoylglutamic acid specifically labeled with carbon-13 in the benzoyl carbonyl, **66**, 536; folate and its reduced forms labeled with deuterium at carbon-7, **66**, 538; *N*-5-formyltetrahydrofolate (citrovorum factor) labeled stereospecifically with deuterium or tritium at carbon-6, **66**, 541; enzymic synthesis of (*d*)-L-5-methyltetrahydropteroylglutamate of high specific radioactivity, **66**, 545; synthesis of 2-amino-4-hydroxy-6-hydroxymethylpteridine pyrophosphate, **66**, 553; dihydro-neopterin aldolase from *Escherichia coli*, **66**, 556; radioassay for dihydropterotate-synthesizing enzyme activity, **66**, 560; a simple radioassay for dihydrofolate synthetase activity in *Escherichia coli* and its application to an inhibition study of new pterate analogs, **66**, 576; intracellular distribution, purification, and properties of dihydrofolate synthetase from pea seedlings, **66**, 581; synthetase, **66**, 585; formiminotransferase-cyclodeaminase: a bifunctional enzyme from porcine liver, **66**, 626; pteroylpolyglutamate synthase assay, **66**, 630; pteroylmonoglutamate conversion of into pteroylpolyglutamates, **66**, 638; conversion of pteroylmonoglutamates to pteroylpolyglutamates, **66**, 642; enzymic synthesis of 10-formyl-H₄-pteroyl- γ -glutamylglutamic acid from glutamic acid and 10-formyl-H₄-pteroylglutamic acid, **66**, 648; bacterial degradation of folic acid, **66**, 652; preparation and purification of pteric acid from pteroylglutamic acid (folic acid), **66**, 657; a radioactive assay of pteroylpolyglutamate hydrolases (conjugases), **66**, 660; assay of folypolyglutamate hydrolase using pteroyl-labeled substrates and selective short-term bacterial uptake for product determination, **66**, 663; purification of folate binding factors, **66**, 678; preparation and use of affinity columns with bovine milk folate-binding protein (FBP) covalently linked to Sepharose 4B, **66**, 686; purification and properties of the folate-binding protein, **66**, 690; kinetics of folate-protein binding, **66**, 694; purification of thymidylate synthetase from enzyme-poor sources by affinity chromatography, **66**, 709; preparation of an antiserum to sheep liver dihydropteridine reductase, **66**, 723.

PULLULANASE

This enzyme [EC 3.2.1.41] (also known as α -dextrin endo-1,6- α -glucosidase, pullulan 6-glucanohydrolase,

limit dextrinase, debranching enzyme, and amylopectin 6-glucanohydrolase) catalyzes the hydrolysis of (1,6)- α -glucosidic linkages in pullulan and starch to form maltotriose. It is the starch debranching enzyme.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

I. S. Pretorius, M. G. Lambrechts & J. Marmur (1991) *Crit. Rev. Biochem. Mol. Biol.* **26**, 53.

E. Y. C. Lee & W. J. Whelan (1971) *The Enzymes*, 3rd ed., **5**, 191.

PULSE-CHASE EXPERIMENTS

A radiometric technique for estimating the rates of intracellular synthesis and/or turnover of a metabolite. To measure the rates of synthesis, cells are exposed to a radioisotopically labeled precursor (such as [³H]-, [¹⁴C]-, [¹⁵N]-, [¹³C]-, [²H]leucine or [³⁵S]methionine in the case of proteins) for various periods, after which the cells are lysed and the protein of interest is purified (often by immunoprecipitation with specific antibodies). The time course of isotope incorporation gives information about the rate (slope of curve) and extent (amplitude of curve) of the proteins' synthesis. To measure degradation, cells are first pulse-labeled (*i.e.*, exposed to radiolabeled precursor for a fixed period, after which sufficient nonlabeled precursor is added to reduce the radiospecific activity of the precursor). Then, the cells are further incubated, and the radiospecific activity of a particular protein of interest is determined (again usually after immunoprecipitation or some other means for achieving its isolation from other cellular proteins). The key point is that the chase allows one to stop radiolabel uptake almost instantaneously, thereby permitting the kinetic

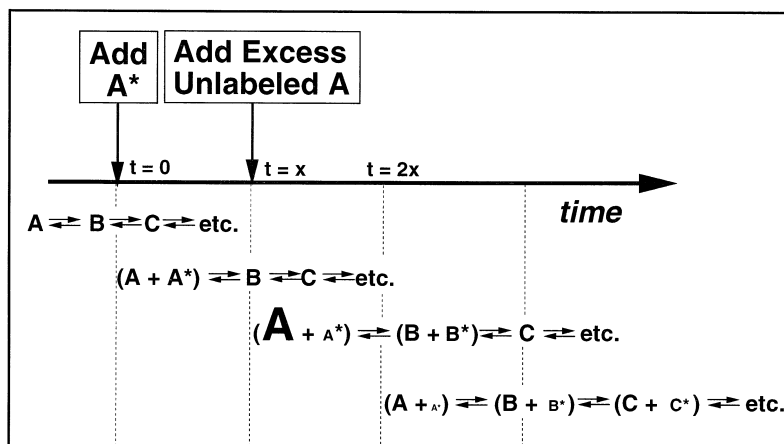


Diagram showing the addition of radiotracer A* at $t = 0$ (termed the "pulse") and the subsequent addition of excess unlabeled A (termed the "chase"). The relative size of the lettering serves to indicate the time-course of radiolabeled entry and exit from each metabolic pool.

analysis of subsequent steps in a reaction or series of reactions (see diagram above).

BASIC PULSE-CHASE KINETICS. For the case of a single pool of A that becomes promptly labeled with tracer A^* , bolus administration of tracer will initially increase the tracer/tracee ratio (indicated by the initial isotopic enrichment E_o), which will subsequently suffer exponential decay¹. As is the case for all first-order processes, the rate law is:

$$E_t = E_o e^{-kt}$$

where E_t is the isotopic enrichment at time t , and k is the first-order rate constant. Accordingly, a plot of $\ln(E_t/E_o)$ versus t should be linear, with a slope of $-k$. [In studies with whole organs or organisms, the kinetics of tracer incorporation into a large tracee pool may be sufficiently long so as to require the use of what is called a priming tracer dose².] When a tracee enters two different compartments, each with its own rate constant, the kinetics will be described as a sum of exponentials. If such a process exhibits a multiexponential form, the semilog plot will be nonlinear, often showing discrete linear segments if each successive exponential process is well resolved. Nonlinear least-squares fitting methods or software programs such as KINSIM and FITSIM can help one to simulate the time-evolution of E_t/E_o .

APPLICATIONS OF THE PULSE-CHASE TECHNIQUE. Consideration of a few illustrative cases should serve to underscore the utility of pulse-chase experiments in the characterization of biological systems.

Alice *et al.*³ studied the turnover kinetics of *Listeria monocytogenes*-secreted p60 protein (a murein hydrolase) by host cell cytosolic proteasomes. J774 cells, seeded in flasks and incubated overnight in culture medium, were infected with log-phase cultures of *L. monocytogenes* for 30 min, washed, and incubated in culture medium for 3 h, with gentamicin (50 $\mu\text{g/ml}$) added after the first 30 min to inhibit extracellular bacterial growth. Cells then were washed and placed in methionine-free medium with spectinomycin, gentamicin, the eukaryotic protein synthesis inhibitors [cycloheximide (50 $\mu\text{g/mL}$) and anisomycin (30 $\mu\text{g/ml}$),] and 25 μM calpain inhibitor I. After 30 min, [³⁵S]methionine was added, and the cells were pulse-labeled for periods of 20 to 60 min. Cells

were washed and chased for the time intervals in culture medium with 20 $\mu\text{g/mL}$ tetracycline to inhibit any further intracellular bacterial growth. Cells were harvested in 1% Triton X-100 lysis buffer containing protease inhibitors, and the detergent-containing lysates were clarified by centrifugation at 4°C. Then, p60 was immunoprecipitated with 25 μL protein A-Sepharose (50% slurry) and 5 μL anti-p60 antiserum for 1 h at 4°C. Beads were washed four times and resuspended in sample buffer, and the samples were electrophoresed on SDS-10% polyacrylamide gels under reducing conditions. Gels were enhanced with 0.5 M salicylate, 3% glycerol, dried, and exposed for autoradiography. Radioactive signals were quantified using a Bio-Rad GS-250 Molecular Imager. Half-lives were determined by plotting percentages of remaining p60 against the chase time intervals. The authors demonstrated that p60 is an N-end rule substrate. Because a protein's N-terminal amino acid can determine its rate of degradation, they mutagenized this residue in p60 into known stabilizing and destabilizing residues. N-terminal valine dramatically stabilized cytosolic p60, whereas aspartic acid substitution resulted in rapid degradation. **See N-End Rule**

Bai *et al.*⁴ recently studied how the pattern of sulfation of heparan affects its turnover by examining heparan sulfate catabolism in wild-type Chinese hamster ovary (CHO) cells and mutant cells that are defective in 2-O-sulfation of uronic acid residues. Pulse-chase experiments were used to demonstrate that both mutant and wild-type cells transport newly synthesized heparan sulfate proteoglycans to the plasma membrane, where they shed into the medium or move into the cell through endocytosis. Internalization of the cell-associated molecules leads to sequential endoglycosidase (heparanase) fragmentation of the chains and eventual lysosomal degradation. Both wild-type CHO and the mutant cells were grown to near confluence and pulse-labeled for 1 h with 100 $\mu\text{Ci/mL}$ $\text{Na}_2^{35}\text{SO}_4$ (25–40 Ci/mg) in sulfate-free culture medium. After removing the labeled medium, the cell layers were washed three times, and fresh F-12 medium supplemented with 1 mM Na_2SO_4 was added as a chase. At various times, the medium was collected, and the cells were washed three times with phosphate-buffered saline. Cell surface proteoglycans were released by treating the cells with 0.125% (w/v) trypsin for 5–10 min at 37°C. The cells were centrifuged at 800g for 3

min, and the released proteoglycans were recovered in the supernatant. The pellets were designated as the intracellular pool. [For steady-state labeling of heparan sulfate, wild-type CHO and mutant pgsF-17 cells were labeled to constant radiospecific activity after 24 h with $\text{Na}_2^{35}\text{SO}_4$ (50 $\mu\text{Ci/mL}$) in sulfate-free medium. The lack of sulfate in the medium did not cause undersulfation of the chains, because CHO cells derive adequate sulfate from the catabolism of cysteine and methionine.] Radio-labeled glycosaminoglycan chains were isolated from various fractions, and specific activities were determined. Their findings suggest that 2-*O*-sulfated iduronic acid residues in heparan sulfate are important for cleavage by endogenous heparanases but not for the overall catabolism of the chains.

To characterize further the cellular function of sterol carrier protein-2 (SCP-2), Baum *et al.*⁵ transfected McA-RH7777 rat hepatoma cells with a pre-SCP-2 cDNA expression construct. In stable transfectants, pre-SCP-2 processing resulted in an 8-fold increase in peroxisomal levels of SCP-2. SCP-2 overexpression increased the rates of newly synthesized cholesterol transfer to the plasma membrane and plasma membrane cholesterol internalization by 4-fold. To make these measurements, pulse-chase analysis of cholesterol biosynthesis was accomplished using cells (5×10^6) incubated in 10 mL medium containing 5% lipoprotein-deficient serum (LPDS) and [^3H]acetate (100 μCi) at 37°C. After 1 h, the medium was removed and fresh chase medium containing 10 mM sodium acetate (unlabeled) was added. At the indicated times, cells were extensively rinsed with phosphate-buffered saline, dissociated with trypsin, extracted, and analyzed by HPLC for the radioactivity in cholesterol and sterol intermediates. Radiolabeled low-density lipoprotein was used in pulse-chase studies to estimate the transfer rate of LDL-derived cholesterol to the plasma membrane as described previously with minor modifications. Cells were seeded into 60-mm dishes at 10^6 cells/dish. After 48 h, the medium was changed to culture medium with 2% LPDS, and cells were incubated for 24 h. Cells were then cooled by rinsing with PBS at 20°C, followed by the addition of fresh DMEM medium containing 2% LPDS, radiolabeled LDL (100 $\mu\text{g/mL}$), and compound 58-035 (2 $\mu\text{g/mL}$). After 2 h at 20°C, monolayers were extensively rinsed with PBS at 4°C followed by addition of 5 mL of DMEM medium with

2% LPDS, LDL (100 $\mu\text{g/mL}$), and compound 58-035 (2 $\mu\text{g/mL}$) at 37°C. At various times, cells were harvested, treated with cholesterol oxidase, and lipid extracts analyzed by HPLC.

Pulse-chase experiments are also powerful tools in enzyme kinetics. For example, Esteban *et al.*⁶ investigated the relationship between RNA folding and ribozyme catalysis through a detailed kinetic analysis of structural derivatives of the hairpin ribozyme. They used pulse-chase experiments to evaluate substrate dissociation. A small amount of 5- ^{32}P -labeled substrate (less than 1 nM) was incubated with a saturating excess of ribozyme (200 nM) in the reaction buffer for 2 min (wild type (wt) ribozymes) or 30 s (modified SV5 ribozymes) at 25°C. These incubation times allow essentially complete formation of the ribozyme-substrate complex since the binding half-times under these conditions are about 30 and 1 s, for the wt and the modified ribozymes, respectively. The chase step was initiated by adding a large excess (5 μM final concentration) of either a DNA oligonucleotide that was fully complementary to the wt substrate or, alternatively, a nonradiolabeled SV5 substrate for reactions carried out with wt or SV5 ribozymes, respectively. A complementary DNA oligonucleotide was used instead of the unlabeled wt RNA substrate because this molecule forms stable dimers at high concentrations. During the chase period, aliquots were removed and quenched with an equal volume of loading buffer (15 mM EDTA, 97% formamide). Samples were analyzed and quantified as described above. Parallel control reactions were carried out in the absence of the chase molecule. The efficiency of the chase step was evaluated by mixing the labeled substrate with the chase molecule (either the complementary DNA oligonucleotide or the unlabeled RNA substrate) prior to the addition of ribozyme. No significant cleavage of the labeled substrate was observed under these conditions, indicating that there is no rebinding of the labeled substrate during the chase step. Typically, time courses carried out in the presence of the chase molecule displayed monophasic behavior and, therefore, were fitted to single-exponential equations. Control reactions in the absence of chase showed biphasic kinetics, and hence double-exponential equations were used. Estimation of the kinetic parameters (amplitudes and rates) was carried out as described above.

These investigators⁶ also used pulse-chase experiments to measure the kinetics of substrate association. Several ribozyme concentrations, ranging from 12.5 nM to 200 nM for wt ribozymes or from 1 nM to 10 nM SV5 ribozymes, were combined with a trace amount (less than 0.1 nM) of the corresponding 5-³²P-labeled substrate in reaction buffer at 25°C. For each ribozyme concentration, several chase reactions were initiated at different times, ranging from 10 s to 4 min. The chase molecule was a complementary DNA oligonucleotide, in the case of the wt substrate, or unlabeled RNA substrate, in the case of the SV5 substrate. The final concentration of the chase molecule was 5 μ M or 1 μ M for reactions carried out with wt or SV5 substrates, respectively. Reactions were incubated for 1 h at 25°C after addition of the chase molecule. This time is sufficient to ensure a quantitative cleavage of the 5-³²P-labeled substrate-ribozyme complexes (the half-time for the cleavage reaction was about 5 min). Time courses of the cleavage reaction were fitted to single-exponential equations, and the observed rates were plotted versus ribozyme concentration to obtain k_{on} and k_{off} .

¹R. R. Wolfe (1992) *Radiative and Stable Isotope Tracers in Biomedicine: Principles and Practice of Kinetic Analysis*, pp. 471, Wiley-Liss, New York.

²H. L. Searle, E. H. Strisower & I. L. Chaikoff (1954) *Am. J. Physiol.* **176**, 190.

³A. J. A. M. Sijts, I. Pilip & E. G. Pamer (1997) *J. Biol. Chem.* **272**, 19261.

⁴X. Bai, K. J. Bame, H. Habuchi, K. Kimata & J. D. Esko (1997) *J. Biol. Chem.* **272**, 23172.

⁵C. L. Baum, E. J. Reschly, A. K. Gayen, M. E. Groh & K. Schadick (1997) *J. Biol. Chem.* **272**, 6490.

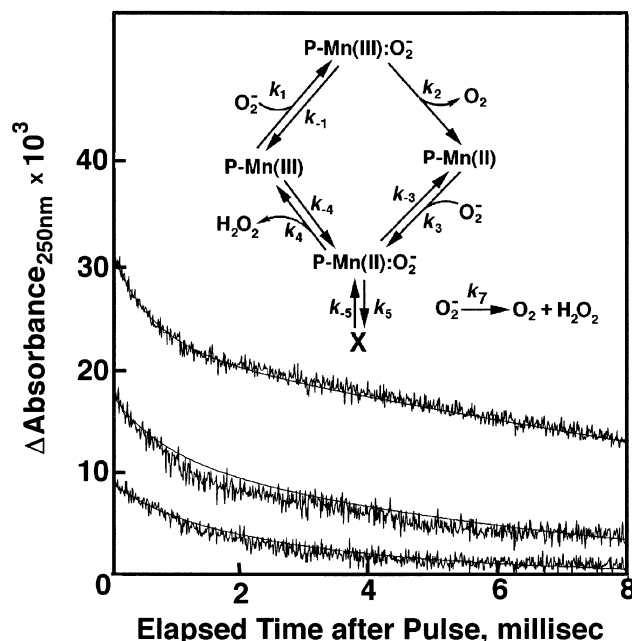
⁶J. A. Esteban, A. R. Banerjee & J. M. Burke (1997) *J. Biol. Chem.* **272**, 13629.

PULSE RADIOLYSIS

A rapid reaction kinetic technique (time scale = 10–1000 ps) that typically uses a Van de Graff accelerator or a microwave linear electron accelerator to promptly generate a pulse of electrons at sufficient power levels for excitation and ionization of target substances by electron impact. The technique is the direct radiation chemical analog of flash photolysis¹, and the ensuing kinetic measurements are accomplished optically by IR/visible/UV adsorption spectroscopy or by fluorescence spectroscopy.

A good example is the examination² of the time course of superoxide depletion catalyzed by human manganese

superoxide dismutase (MnSOD). The reaction was observed spectrophotometrically by measuring the absorbance of superoxide at 250–280 nm following pulse radiolysis to rapidly generate 6 μ M superoxide anion. Catalysis showed an initial burst of activity lasting approximately 1 ms, followed by the rapid emergence of a greatly inhibited catalysis of zero-order rate. These catalytic properties of human MnSOD are qualitatively similar to those reported³ for MnSOD from *Thermus thermophilus*. However, generation of the inhibited form is approximately 30-fold more rapid for human MnSOD. The turnover number for human MnSOD at pH 9.4 and 20°C was $k_{\text{cat}} = 4 \times 10^4 \text{ s}^{-1}$ and $k_{\text{cat}}/K_{\text{m}} = 8 \times 10^8 \text{ M}^{-1} \text{ s}^{-1}$, determined by KINSIM simulation and fitting pulse radiolysis data to the model of Bull *et al.*³



Time-course of human manganese superoxide dismutase reaction after generation of superoxide by pulse radiolysis². The experiment demonstrates decrease in absorbance at 250 nm ($\epsilon = 2000 \text{ M}^{-1} \text{ cm}^{-1}$) for a solution containing 0.5 μ M enzyme, 50 μ M EDTA, 10 mM sodium formate, and 2 mM sodium pyrophosphate at pH 9.4 and 20°C. The starting concentrations of superoxide were 11.6 μ M (upper curve), 6.5 μ M (middle curve), and 3.4 μ M (lower curve). The calculated progress curves shown as solid lines were obtained by using the KINSIM software and the model of Bull *et al.*³ Reproduced here with the permission of the authors and the American Chemical Society.

Other examples using pulse radiolysis include: (a) studies⁴ on cytochrome *c-d*₁ nitrate reductase from *Thiosphaera pantotropha* to provide evidence for a fast intramolecular electron transfer from *c*-heme to *d*₁-heme; (b) determination⁵ of the rate constant for the reaction of

hydroxyl and oxide radicals with cysteine in aqueous solution; (c) direct demonstration⁶ of the catalytic action of monodehydroascorbate reductase; (d) investigation⁷ of the generation and reactions of the disulfide radical anion derived from metallothionein; (e) the kinetics⁸ of NO $\cdot$ reactions with O $_2^{\cdot-}$ and HO $_2^{\cdot-}$; and the enhanced intramolecular electron transfer⁹ in the ascorbate oxidase reaction in the presence of oxygen.

¹L. M. Dorfman (1974) in *Investigation of Rates and Mechanisms of Reactions*, Part II, 3rd ed. (G. G. Hammes, ed.) vol. VI, pp. 463ff, Wiley-Interscience, New York.

²J. L. Hsu, Y. Hsieh, C. Tu, D. O'Connor, H. S. Nick & D. N. Silberman (1996) *J. Biol. Chem.* **271**, 17687.

³C. Bull, E. C. Niederhoffer, T. Yoshida & J. A. Fee (1991) *J. Am. Chem. Soc.* **113**, 4069.

⁴K. Kobayashi, A. Koppenhofer, S. J. Ferguson & S. Tagawa (1997) *Biochem.* **36**, 13611.

⁵S. P. Mezyk (1996) *Radiat. Res.* **145**, 102.

⁶K. Kobayashi, S. Tagawa, S. Sano & K. Asada (1995) *J. Biol. Chem.* **270**, 27551.

⁷X. Fang, J. Wu, G. Wei, H. P. Schuchmann & C. von Sonntag (1995) *Int. J. Radiat. Biol.* **68**, 459.

⁸S. Goldstein & G. Czapski (1995) *Free Radic. Biol. Med.* **19**, 505.

⁹O. Farver, S. Wherland & I. Pecht (1994) *J. Biol. Chem.* **269**, 22933.

PURINE NUCLEOSIDASE

This enzyme [EC 3.2.2.1], also known as inosine-uridine preferring nucleoside hydrolase and IU-nucleoside hydrolase, catalyzes the hydrolysis of an *N*-D-ribosylpurine to produce a purine and D-ribose.

T. P. Wang (1955) *Meth. Enzymol.* **2**, 456.

PURINE NUCLEOSIDE PHOSPHORYLASE

The enzyme [EC 2.4.2.1], also known as inosine phosphorylase, catalyzes the reaction of a purine nucleoside with orthophosphate to produce a purine and α -D-ribose 1-phosphate. The enzyme will also catalyze the activity of nucleoside ribosyltransferase.

G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.

R. E. Parks, Jr., & R. P. Agarwal (1972) *The Enzymes*, 3rd ed., **7**, 483.

PUTRESCINE CARBAMOYLTRANSFERASE

This enzyme [EC 2.1.3.6] catalyzes the reaction of carbamoyl phosphate with putrescine to produce orthophosphate and *N*-carbamoylputrescine. The plant enzyme will also catalyze the reactions of ornithine carbamoyltransferase, carbamate kinase, and agmatine deiminase.

C. W. Tabor & H. Tabor (1984) *Ann. Rev. Biochem.* **53**, 749.

PUTRESCINE N-METHYLTRANSFERASE

This enzyme [EC 2.1.1.53] catalyzes the reaction of *S*-adenosyl-L-methionine with putrescine to produce *S*-adenosyl-L-homocysteine and *N*-methylputrescine.

T. Hashimoto & Y. Yamada (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 257.

PUTRESCINE OXIDASE

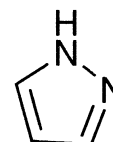
This FAD-dependent enzyme [EC 1.4.3.10] catalyzes the reaction of putrescine with dioxygen and water to produce 4-aminobutanal, ammonia, and hydrogen peroxide. 4-Aminobutanal then condenses nonenzymically to 1-pyrroline.

H. Yamada (1971) *Meth. Enzymol.* **17B**, 726.

P-VALUE

1. The lowest level of significance for which an obtained test statistic is significant. 2. The probability that a variate (often called the random variable would strictly by chance assume a value greater than or equal to the observed value.

PYRAZOLE



A potent five-membered heterocyclic inhibitor (the chemical systematic name is 1,2-diazole) that strongly inhibits liver alcohol dehydrogenase. **See Alcohol Dehydrogenase**

PYRIDOXAL 4-DEHYDROGENASE

This enzyme [EC 1.1.1.107] catalyzes the reaction of pyridoxal with NAD $^{+}$ to produce 4-pyridoxolactone and NADH. The enzyme acts on the hemiacetal form of the substrate.

R. W. Burg (1970) *Meth. Enzymol.* **18A**, 634.

PYRIDOXAL KINASE

This enzyme [EC 2.7.1.35] (also known as pyridoxine kinase, pyridoxamine kinase, and vitamin B $_6$ kinase) catalyzes the reaction of ATP with pyridoxal to produce ADP and pyridoxal 5'-phosphate. Pyridoxine, pyridoxamine, and various other derivatives can also act as substrates.

S. L. Ink & L. M. Henderson (1984) *Ann. Rev. Nutr.* **4**, 455.

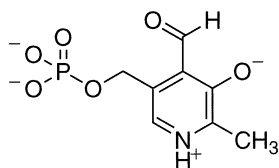
PYRIDOXAMINE:PYRUVATE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.30] catalyzes the reaction of pyridoxamine with pyruvate to produce pyridoxal and L-alanine.

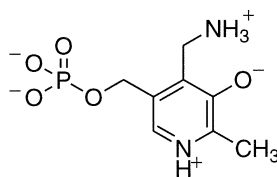
A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

PYRIDOXINE & PYRIDOXAL (Analogues And Derivatives)

Metabolites that contain pyridoxol (2-methyl-3-hydroxy-4,5-di[hydroxymethyl]pyridine), the water-soluble vitamin B₆. Phosphorylated forms of pyridoxal (the aldehyde form) and pyridoxamine (the amino form), known respectively as pyridoxal 5'-phosphate (PLP) and pyridoxamine 5'-phosphate (PMP), are tightly-bound coenzymes.



Pyridoxal 5'-phosphate



Pyridoxamine 5'-phosphate

PLP-dependent enzymes catalyze the following types of reactions: (1) loss of the α -hydrogen as a proton, resulting in racemization (example: alanine racemase), cyclization (example: aminocyclopropane carboxylate synthase), or β -elimination/replacement (example: serine dehydratase); (2) loss of the α -carboxylate as carbon dioxide (example: glutamate decarboxylase); (3) removal/replacement of a group by aldol cleavage (example: threonine aldolase; and (4) action via ketimine intermediates (example: selenocysteine lyase).

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Analyzing spectra of vitamin B₆ derivatives, **18A**, 433; fluorometric determination of pyridoxal and pyridoxal 5'-phosphate in biological materials by the reaction with cyanide, **18A**, 471; nuclear magnetic resonance spectroscopy of vitamin B₆, **18A**, 475; mass spectrometry of vitamin B₆, **18A**, 483; fluorometric assay of pyridoxal, **62**, 405; fluorometric determination of pyridoxal phosphate in enzymes, **62**, 407; high-voltage electrophoresis and thin-layer chromatographic separation of vitamin B₆ compounds, **62**, 410; high-performance chromatography of vitamin B₆, **62**, 415; carbon-13 nuclear magnetic resonance spectroscopy of the vitamin B₆ group, **62**, 422; ¹³C NMR spectroscopy of the vitamin B₆ group, **62**, 436; synthesis and biological activity of vitamin B₆ analogues, **62**, 454; vitamin B₆ antagonists of natural origin, **62**, 483; spin-labeled vitamin B₆ derivatives: synthesis and interaction with aspartate aminotransferase, **62**, 495; the use of 4'-N-(2,4-dinitro-5-fluorophenyl)pyridoxamine 5'-phosphate, **62**, 510; immobilization of pyridoxal 5'-phosphate and pyridoxal 5'-phosphate-dependent enzymes on Sepharose, **62**, 517; pyridoxal 5'-phosphate and analogs as probes of coenzyme-protein interac-

tion, **62**, 528; crystalline enzyme-substrate complexes of pyridoxal phosphate-dependent enzymes, **62**, 551; dithionite-induced changes in the spectra of free and enzyme-bound pyridoxal phosphate, **62**, 558; assay of pyridoxal phosphate and pyridoxamine phosphate, employing S-o-nitrophenyl-L-cysteine, a chromogenic substrate of tryptophanase, **62**, 561; pyridoxamine (pyridoxine) 5'-phosphate oxidase from rabbit liver, **62**, 568; pyridoxal (amine) 5'-phosphate hydrolase from rat liver, **62**, 574; plasma binding of B₆ compounds, **62**, 582.

PYRIMIDINE DIMER DNA GLYCOSYLASE

This enzyme [EC 3.2.2.17], also known as deoxyribodipyrimidine endonucleosidase, catalyzes the hydrolysis of the N-glycosidic bond between the 5'-pyrimidine residue in cyclobutadipyrimidine (in the DNA) and the corresponding deoxy-D-ribose residue.

B. K. Duncan (1981) *The Enzymes*, 3rd ed., **14**, 565.

PYROGLUTAMYL PEPTIDE HYDROLASE

This enzyme [EC 3.4.19.3], a member of the C15 peptidase family, is also known as pyroglutamyl-peptidase I, 5-oxoprolyl-peptidase, pyrrolidone-carboxylate peptidase, and pyroglutamyl aminopeptidase. This hydrolase catalyzes the conversion of a 5-oxoprolyl-peptide to produce 5-oxoproline and a peptide. The enzyme will not act on the 5-oxoprolyl peptide if the adjacent amino acid is L-proline. Enzyme activity is inhibited by thiol-blocking reagents.

E. Shaw (1990) *Adv. Enzymol.* **63**, 271.

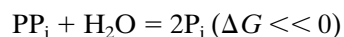
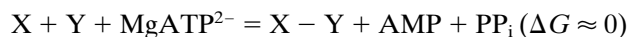
R. J. DeLange & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 81.

PYROLYSIS

Decomposition, involving the uncatalyzed cleavage of one or more covalent bonds, due to exposure to elevated temperatures; sometimes referred to as thermolysis.

PYROPHOSPHATASE, INORGANIC

Inorganic pyrophosphatase [EC 3.6.1.1] plays a central role in phosphorus metabolism by catalyzing the hydrolysis of the phosphoanhydride bond of inorganic pyrophosphate (or, diphosphate; PP_i). This cleavage reaction acts in conjunction with pyrophosphate-forming ligases to provide an additional thermodynamic impetus for certain biosynthetic reactions. For example:



Catalysis requires divalent metal ions which affect the apparent pK_a values of the essential general acid and

base on the enzyme, and the pK_a of the substrate¹⁻³. Three to five metal ions are required for maximal activity, depending on pH and enzyme source.

¹B. S. Cooperman (1982) *Meth. Enzymol.* **87**, 526.

²P. Heikinheimo, J. Lehtonen, A. Baykov, R. Lahti, B. S. Cooperman & A. Goldman (1996) *Structure* **4**, 1491.

³A. A. Baykov, T. Hyttia, S. E. Volk, V. N. Kasho, A. V. Vener, A. Goldman, R. Lahti & B. S. Cooperman (1996) *Biochemistry* **35**, 4655.

PYROPHOSPHATE FRUCTOSE-6-PHOSPHATE 1-PHOSPHOTRANSFERASE

This enzyme [EC 2.7.1.90] (also known as diphosphate:fructose-6-phosphate 1-phosphotransferase, 6-phosphofructokinase (pyrophosphate), and pyrophosphate-dependent 6-phosphofructose-1-kinase) catalyzes the reaction of pyrophosphate (or, diphosphate) with D-fructose 6-phosphate to produce orthophosphate and D-fructose 1,6-bisphosphate.

H. C. Huppe & D. H. Turpin (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 577.

E. Van Schaffingen (1987) *Adv. Enzymol.* **59**, 315.

H. G. Wood, W. E. O'Brien & G. Michaels (1977) *Adv. Enzymol.* **45**, 85.

PYROPHOSPHOMEVALONATE DECARBOXYLASE

This enzyme [EC 4.1.1.33] (also known as mevalonic acid 5-pyrophosphate anhydrodecarboxylase, mevalonate-5-diphosphate decarboxylase, and diphosphomevalonate decarboxylase) catalyzes the reaction of ATP with (R)-5-diphosphomevalonate to produce ADP, orthophosphate, isopentenyl diphosphate, and carbon dioxide.

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

R. A. Dixon, P. M. Dey & C. J. Lamb (1983) *Adv. Enzymol.* **55**, 1.

Δ^1 -PYRROLINE-5-CARBOXYLATE DEHYDROGENASE

This enzyme [EC 1.5.1.12] catalyzes the reaction of 1-pyrroline-5-carboxylate with NAD^+ and water to produce L-glutamate and NADH. The enzyme will also act on 3-hydroxy-1-pyrroline-5-carboxylate to generate 4-hydroxyglutamate.

E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

Δ^1 -PYRROLINE-5-CARBOXYLATE REDUCTASE

This enzyme [EC 1.5.1.2] catalyzes the reversible reaction of 1-pyrroline-5-carboxylate with $NAD(P)H$ to pro-

duce L-proline and $NAD(P)^+$. In addition, the enzyme will also act on 1-pyrroline-3-hydroxy-5-carboxylate, reducing it to L-hydroxyproline.

E. Adams & L. Frank (1980) *Ann. Rev. Biochem.* **49**, 1005.

D. M. Greenberg (1962) *Meth. Enzymol.* **5**, 959.

1-PYRROLINE-4-HYDROXY-2-CARBOXYLATE DEAMINASE

This enzyme [EC 3.5.4.22] catalyzes the hydrolysis of 1-pyrroline-4-hydroxy-2-carboxylate to generate 2,5-dioxopentanoate and ammonia.

E. Adams (1971) *Meth. Enzymol.* **17B**, 266.

PYRUVATE CARBOXYLASE

This biotin-dependent enzyme [EC 6.4.1.1] catalyzes the reaction of ATP with pyruvate and HCO_3^- to produce ADP, orthophosphate, and oxaloacetate. The enzyme from yeast requires zinc whereas the enzyme isolated from animal tissues needs manganese as well as acetyl-CoA.

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.

P. V. Attwood (1995) *Int. J. Biochem. Cell Biol.* **27**, 231.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

M. F. Utter, R. E. Barden & B. L. Taylor (1975) *Adv. Enzymol.* **42**, 1.

M. C. Strutton & M. R. Young (1972) *The Enzymes*, 3rd ed., **6**, 1.

PYRUVATE DECARBOXYLASE

This thiamin pyrophosphate-dependent enzyme [EC 4.1.1.1] catalyzes the conversion of an α -keto acid (or, a 2-oxo acid) to an aldehyde and carbon dioxide. This enzyme will also catalyze acyloin formation.

R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 217.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

A. Schellenberger, G. Hübner & H. Neef (1997) *Meth. Enzymol.* **279**, 131.

F. Dyda, W. Furey, S. Swaminathan, M. Sax, B. Farrenkopf & F. Jordan (1993) *Biochemistry* **32**, 6165.

Y. Lindqvist & G. Schneider (1993) *Curr. Opin. Struct. Biol.* **3**, 896.

M. H. O'Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

PYRUVATE DEHYDROGENASE

Pyruvate dehydrogenase (lipoamide) [EC 1.2.4.1], which requires thiamin pyrophosphate, catalyzes the reaction of pyruvate with lipoamide to produce S-acetyldihydrolipoamide and carbon dioxide. It is a component of the pyruvate dehydrogenase complex (which also includes dihydrolipoamide dehydrogenase [EC 1.8.1.4] and dihydrolipoamide acetyltransferase [EC 2.3.1.12]). Pyruvate dehydrogenase (cytochrome) [EC 1.2.2.2] catalyzes the

reaction of pyruvate with ferricytochrome b_1 and water to produce acetate, carbon dioxide, and ferrocycytochrome b_1 . It also requires thiamin pyrophosphate.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 A. Schellenberger, G. Hübner & H. Neef (1997) *Meth. Enzymol.* **279**, 131.
 R. N. Perham (1995) *Meth. Enzymol.* **251**, 436.
 R. N. Perham & P. N. Lowe (1988) *Meth. Enzymol.* **166**, 330.
 L. J. Reed & S. J. Yeaman (1987) *The Enzymes*, 3rd ed., **18**, 77.
 G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

PYRUVATE DEHYDROGENASE KINASE

This enzyme [EC 2.7.1.99] catalyzes the reaction of ATP with [pyruvate dehydrogenase (lipoamide)] to produce ADP and [pyruvate dehydrogenase (lipoamide)] phosphate. This is an enzyme that is associated with the pyruvate dehydrogenase complex. Phosphorylation of pyruvate dehydrogenase (lipoamide) [EC 1.2.4.1] inactivates that enzyme.

- L. J. Reed & S. J. Yeaman (1987) *The Enzymes*, 3rd ed., **18**, 77.
 D. A. Flockhart & J. D. Corbin (1982) *Crit. Rev. Biochem.* **12**, 133.
 D. A. Walsh & E. G. Krebs (1973) *The Enzymes*, 3rd ed., **8**, 555.

PYRUVATE DEHYDROGENASE PHOSPHATASE

This enzyme [EC 3.1.3.43] catalyzes the hydrolysis of the phosphorylated form of pyruvate dehydrogenase (lipoamide) to generate the dephosphorylated dehydrogenase and orthophosphate. The enzyme is associated with the pyruvate dehydrogenase complex.

- L. J. Reed & S. J. Yeaman (1987) *The Enzymes*, 3rd ed., **18**, 77.

PYRUVATE FORMATE LYASE

This enzyme [EC 2.3.1.54], also known as formate C-acetyltransferase, catalyzes the reaction of acetyl-CoA with formate to produce coenzyme A and pyruvate.

- J. Knappe & A. F. V. Wagner (1995) *Meth. Enzymol.* **258**, 343.
 E. J. Bruch & J. W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.
 J. A. Stubbe (1989) *Ann. Rev. Biochem.* **58**, 257.

PYRUVATE KINASE

This enzyme [EC 2.7.1.40] catalyzes the reaction of ADP with phosphoenolpyruvate to produce ATP and pyruvate. Other nucleotides that can be used as substrates include UDP, GDP, CDP, IDP, and dADP. The enzyme will also phosphorylate hydroxylamine and fluoride in the presence of carbon dioxide. **See** *Nucleoside 5'-Triphosphate Regeneration*

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

- H. C. Huppe & D. H. Turpin (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 577.

- J. J. Villafranca & T. Nowak (1992) *The Enzymes*, 3rd ed., **20**, 63.

- R. F. Colman (1990) *The Enzymes*, 3rd ed., **19**, 283.

- L. Engström, P. Ekman, E. Humble & Ö. Zetterqvist (1987) *The Enzymes*, 3rd ed., **18**, 47.

PYRUVATE, ORTHOPHOSPHATE DIKINASE

This enzyme [EC 2.7.9.1], also referred to as pyruvate, phosphate dikinase, catalyzes the reaction of ATP with pyruvate and orthophosphate to produce AMP, phosphoenolpyruvate, and pyrophosphate (or, diphosphate).

- P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
 F. M. Raushel & J. J. Villafranca (1988) *Crit. Rev. Biochem.* **23**, 1.
 N. H. Goss & H. G. Wood (1982) *Meth. Enzymol.* **87**, 51.
 H. G. Wood, W. E. O'Brien & G. Michaels (1977) *Adv. Enzymol.* **45**, 85.
 R. A. Cooper & H. L. Kornberg (1974) *The Enzymes*, 3rd ed., **10**, 631.

PYRUVATE OXIDASE

This enzyme [EC 1.2.3.3], which requires thiamin pyrophosphate and FAD, catalyzes the reaction of pyruvate with orthophosphate, dioxygen, and water to produce acetyl phosphate, carbon dioxide, and hydrogen peroxide.

- R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 217.
 A. Schellenberger, G. Hübner & H. Neef (1997) *Meth. Enzymol.* **279**, 131.
 Y. A. Muller, Y. Lindqvist, W. Furey, G. E. Schulz, F. Jordan & G. Schneider (1993) *Structure* **1**, 95.

PYRUVATE SYNTHASE

This enzyme [EC 1.2.7.1], also known as pyruvate:ferredoxin 2-oxido-reductase, catalyzes the reaction of pyruvate with coenzyme A and oxidized ferredoxin to produce acetyl-CoA, carbon dioxide, and reduced ferredoxin.

- L. L. Barton (1994) *Meth. Enzymol.* **243**, 94.
 E. J. Brush & J. W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.
 B. B. Buchanan (1972) *The Enzymes*, 3rd ed., **6**, 193.

PYRUVATE, WATER DIKINASE

This manganese-dependent enzyme [EC 2.7.9.2], also known as phosphoenolpyruvate synthase, catalyzes the reaction of ATP with pyruvate and water to produce AMP, phosphoenolpyruvate, and orthophosphate.

- P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.
 R. A. Cooper & H. L. Kornberg (1974) *The Enzymes*, 3rd ed., **10**, 631.

Q

Q_{10}

Symbol for the temperature coefficient, a quotient equal to v_{T+10}/v_T , where v_{T+10} and v_T are the rates of a process (*e.g.*, an enzyme-catalyzed reaction) at two temperatures differing by 10°C. This parameter is usually evaluated at saturating concentrations of substrate(s), so that temperature-dependent changes in Michaelis constant(s) are inconsequential. The Q_{10} value is a characteristic property of a particular enzyme from a specific organism and cell type. For example, one cannot use the Q_{10} value for one hexokinase from yeast to infer the temperature dependence of another hexokinase, say from rat brain. Likewise, the Q_{10} value need not remain the same for a mutant form and a wild-type enzyme.

Because the energy of activation, E_a , for a reaction having a rate constant k , equals $RT^2(d \ln k)/dT$ (where R is the universal gas constant and T is the absolute temperature), then, for a temperature increase of 10°, $E_a = \{RT(T + 10) \ln Q_{10}\}/10$. Since 10 is significantly less than T , this relationship reduces to $E_a = \{RT^2 \ln Q_{10}\}/10$. Q_{10} is temperature-dependent, and the above analysis assumes that all other reaction conditions are constant and that the enzyme is stable over this temperature range.

Fumarase¹ is reported to have a Q_{10} of 1.994 at 30°C and, at temperatures below 22°C, rat liver mitochondrial ATPase has a value of 29².

In addition, recall that $E_a = \Delta H^\ddagger + RT$. Thus, there is a minimal value for Q_{10} at a given temperature. This value is given by $Q_{10,\min} = e^{10/(T + 10)}$. Hence, at 25°C, $Q_{10,\min}$ is 1.034.

¹V. Massey (1953) *Biochem. J.* **53**, 72.

²J. K. Raison (1973) *Symposia of the Soc. for Exper. Biol.* **27**, 485.

Q_{CO_2}

Symbol for the amount (in microliters) of CO₂ given off (under standard conditions of pressure and temperature) per milligram of tissue per hour.

Q TEST

A means for deciding whether to keep or reject a particular measurement within a set of measurements.¹ For example, suppose an investigator weighed the same sample of a particular substance five separate times and obtained weights of 10.18 g, 9.98 g, 10.07 g, 10.20 g, and 10.38 g. The experimental Q value, Q_{exp} , is equal to the difference between the questionable measurement and its nearest neighbor [in the case of the 10.38 value, this is $(10.38 - 10.20) = 0.18$] divided by the overall spread of the entire set of measurements (thus, $10.38 - 9.98 = 0.40$). Hence, $Q_{\text{exp}} = 0.18/0.40 = 0.45$. If this value exceeds a certain critical value, termed Q_{crit} , then the measurement is rejected. For a 90% confidence level with five measurements, $Q_{\text{crit}} = 0.64$. Thus, the measurement would be retained in the example above. Tables of Q_{crit} are available¹ [for 90% confidence levels: for 3 samples or observations ($Q_{\text{crit}} = 0.94$), for 4 (0.76), for 5 (0.64), for 6 (0.56), for 7 (0.51), for 8 (0.47), for 9 (0.44), and for 10 (0.41)]. However, one should always be wary of rejecting certain data, particularly if the set of measurements is small. In addition, if the investigator is aware that there were errors or difficulties with a particular measurement, then he/she has a basis for rejecting that value, even if the value satisfies the Q test. One should always exercise good judgment. It is often a good practice to report median values rather than means. Finally, Q tests should not be applied to derived values (for example, K_m values). There are other statistical means for presenting such data.

¹R. B. Dean & W. J. Dixon (1951) *Anal. Chem.* **23**, 636.

QUADREACTANT SYSTEMS

Enzymes utilizing four substrates in a single reaction. An example is carbamoyl phosphate synthetase which

has four substrates (L-glutamine, CO₂, 2 ATP) in addition to the nonvaried substrate H₂O. Note that multi-enzyme complexes (e.g., pyruvate dehydrogenase complex) frequently use four substrates; however, these complexes generally involve a number of enzyme-catalyzed steps with fewer than four substrates in a single step. These would not be considered as quadreactant systems.

QUADRUPOLE

Two magnetic dipoles associated together as a single unit or four electric charges organized as a single unit. One of the isotopes of oxygen (¹⁷O) has a large quadrupole moment, and because quadrupolar nuclei have significant magnetic resonance properties, investigators have been able to use these properties in probes of enzyme mechanisms: for example, with ³¹P NMR^{1,2}.

¹M.-D. Tsai (1979) *Biochemistry* **18**, 1468

²M.-D. Tsai (1982) *Meth. Enzymol.* **87**, 235.

QUALITY INDEX

Symbolized by *Q*, the quality index is a parameter used in studying the stereochemical nature of a particular enzyme-catalyzed reaction using chiral [¹⁶O,¹⁷O,¹⁸O] phosphoric monoesters¹. This parameter, which is dependent on the fractional ¹⁷O content, allows the investigator a means to assess the reliability of the study(ies).

¹S. L. Buchwald, D. E. Hansen, A. Hassett & J. R. Knowles (1982) *Meth. Enzymol.* **87**, 279.

QUANTASOME

Small granular structural unit that serves as the basic building block of the thylakoid membrane and contains multiple chlorophyll molecules as well as cytochromes.

QUANTUM

1. A discrete unit or parcel of energy (electric, photic, vibrational, rotatory, etc.) absorbed by or released from a quantized state of matter, usually at the molecular, atomic, or subatomic level. Within the context of the wave-particle duality of electromagnetic radiation, the quantum and photon are the basic units of energy change. 2. The minimum amount by which a certain property (for example, angular momentum, energy, etc.) can change.

QUANTUM MECHANICAL TUNNELING

A kinetic condition wherein a small particle, such as a proton, electron, or a hydride ion may, as a consequence

of its de Broglie wavelength ($\lambda = h/mv$), “pass” or “tunnel” through a potential energy barrier rather than be required to be of adequate energy to surmount the activation energy barrier. This is a requirement in quantum mechanics: the Heisenberg Uncertainty Principle allows a small particle having a large enough de Broglie wavelength to exist on either side of a sufficiently narrow energy barrier. Protons have a de Broglie wavelength of 1-2 Å, just about the expected width for some proton transfer reactions. More massive particles will have correspondingly smaller de Broglie wavelengths, and quantum mechanical tunneling is less likely, even for deuterium. The net effect would be that tunneling should increase the magnitude of k_H/k_D . Tunneling is most likely in reactions possessing symmetrical transition states, a condition where the proton’s motion makes the greatest contribution to the reaction coordinate. Interestingly, steric hindrance is also believed to favor tunneling. The remaining mystery is how an enzyme might finely tune the configuration of catalytic acid/base groups near the substrate in order to take maximal advantage of quantum mechanical tunneling. **See** *Chemical Kinetics; Hydrogen Tunneling; Kinetic Isotope Effects*

QUANTUM REQUIREMENT

A stoichiometric number indicating the number of photons required per molecule of O₂ produced in biosynthesis. **See** *Quantum Yield*

QUANTUM YIELD

A value (symbolized by ϕ) equal to the number of molecules transformed via a reaction per quantum of light absorbed. It is synonymous to quantum efficiency and is the reciprocal of the quantum requirement.

A primary quantum yield is the fraction of light-absorbing molecules that are converted in a particular process. Unfortunately, such values are often very difficult to measure. A product quantum yield is the ratio of the number of molecules of product formed per number of quanta absorbed by the reactant. That product formation is the rate-determining step is a strong possibility if the observed ϕ value does not vary with experimental conditions.

The differential quantum yield is given by $\phi = \{d[x]/dt\}/n$ where $d[x]/dt$ is the rate of change of the concentra-

tion or of the amount of some measurable quantity and n is the amount (*e.g.*, moles) of photons absorbed per unit time. The integral quantum yield is simply defined as the number of events per number of photons absorbed¹. **See** *Actinometer; Fluorescence*

¹IUPAC (1988) *Pure and Appl. Chem.* **60**, 1055.

QUASI-EQUILIBRIUM ASSUMPTION

Another term used to indicate a rapid equilibrium assumption in a kinetic process. The most prominent example in biochemistry is the assumption that enzyme and substrate rapidly form a preequilibrium enzyme-substrate complex in the Michaelis-Menten treatment.

QUASI-EQUIVALENCE

The concept¹ used in biological self-assembly to account for the ability of otherwise chemically identical subunits to adopt different conformational states, such that alternative geometrical or stereochemical bonding interactions (or valences) within the final assembled structure can be satisfied by the same polymerization subunit. The classical case is the assembly of the coat protein of icosahedral viruses, wherein pentavalent and hexavalent interactions of the identical protein subunit generate a structure that has an overall minimum in free energy. Such behavior requires (a) that there be only small changes in the free energies of conformations capable of pentavalent and hexavalent bonding interactions, and/or (b) that the ability to form closed geometrical structures overcomes any difference in the free energy of these two conformational states. Because the transition states are likely to be different in forming pentavalent or hexavalent structures, the kinetics of individual steps during self-assembly may proceed at different rates.

¹D. L. D. Caspar (1980) *Biophys. J.* **32**, 103.

QUATERNARY COMPLEXES

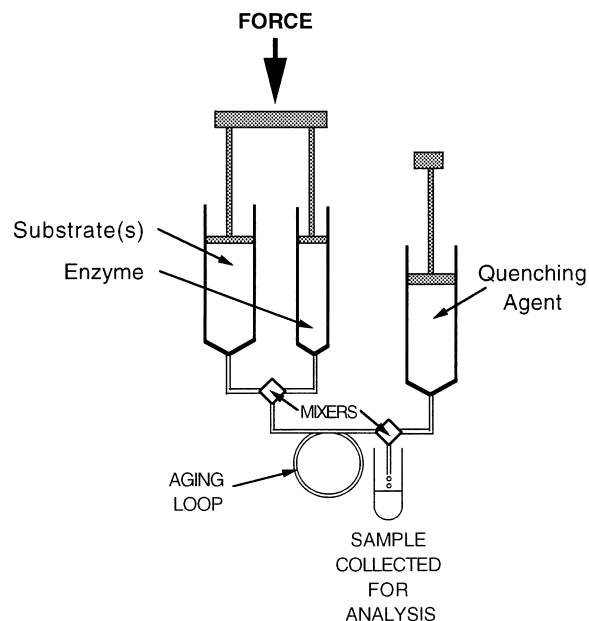
Complexes formed from the ordered or random combination of four distinct entities. Many three-substrate enzymes proceed via the formation of a reversible $E \cdot A \cdot B \cdot C$ quaternary complex.

QUENCHER

1. In a photochemical process, a molecular entity that deactivates an excited state of another molecular entity.
2. A substance, complex, body, system, or parameter that

stops, or slows down, a chemical or enzyme-catalyzed reaction. **See** *Quenching*

QUENCH-FLOW EXPERIMENTS



A method often used to characterize the transient kinetics of chemical and enzymic processes. The reaction is initiated by pushing two or more separate solutions through a single mixing chamber. The reaction mixture then flows through a tube for a known period of time (determined by the flow rate and the length of the tube) until a quenching solution is rapidly combined with the reacting mixture. When the substrate concentration is larger than that of the enzyme, one can often detect a burst of product formation that transiently exceeds the steady-state rate. When the concentration of substrate is the limiting factor, so-called single-turnover kinetics can be measured in rapid quench experiments.

K. A. Johnson (1995) *Meth. Enzymol.* **249**, 38.

QUENCHING

1. The process or procedure for arresting the progress of a chemical or enzyme-catalyzed reaction. The kineticist must ensure that the quenching procedure does, in fact, completely stop the reaction at the time of sampling (when measuring product formed or substrate remaining). In addition, it is necessary to demonstrate that the substrate and product (particularly labeled substrate and product) are stable throughout the quenching procedure. Another goal is to choose conditions for sufficiently

rapid quenching. An excellent test of quenching efficiency is to subject a typical reaction mixture (which includes all reaction components including the enzyme(s)) to the quenching protocol, *after which* a radiolabeled substrate is added. If labeled product is formed, then quenching was most probably incomplete. Methods for quenching include: (a) addition of specific agents (*e.g.*, organic solvents, EDTA, EGTA, thiol agents, acids, bases, *etc.*), (b) rapid changes in temperature (both heating and cooling have been used), and (c) dilution of the reaction mix to halt intermolecular processes.

2. A term often used in liquid scintillation counting in which the energy spectrum has been shifted to a lower energy. This often results from the presence of some interfering substance in the solution (*e.g.*, a colored agent or other foreign chemical). This phenomenon is probably responsible for many errors in rate studies, particularly when biomacromolecules or high salt concentrations cause phase changes in the scintillation cocktail. Three methods can be used to correct for the effects of quenching¹. First, most modern liquid scintillation counters are equipped with external sources that serve to detect quenching and to correct for its effects. After the radioactivity of a sample is determined, the external source is moved into place, and a standard dose (X_{standard}) from this source should result in a certain number of disintegrations per unit time. The observed value (X_{observed}) is a direct measure of the counting efficiency (*i.e.*, $\text{efficiency} = X_{\text{observed}}/X_{\text{standard}}$). Second, a known amount of a reference standard of the same radionuclide as is utilized in the tracer studies can be added to an experimental sample, and the counting efficiency can be determined by the ratio of the observed increase in radioactivity divided by the expected increase in radioactivity. The third method takes advantage of the shift in a radionuclide's energy spectrum to lower energy in the presence of a quenching agent. The "channel ratios" method allows one to use the pulse-height spectrum to divulge any evidence of quenching.

3. In Geiger counters, the process of inhibiting a continuous discharge. This process is usually accomplished by the presence of a gas such as methane mixed with argon or neon.

¹Y. Kobayashi & D. V. Marcidsley (1974) *Biological Applications of Liquid Scintillation Counting*, Academic Press, New York.

QUERCETIN 2,3-DIOXYGENASE

This enzyme [EC 1.13.11.24] catalyzes the reaction of quercetin, a flavonol, with dioxygen to produce 2-protocatechoylphloroglucinolcarboxylate and carbon monoxide. Copper ions are needed as a cofactor.

A. Messerschmidt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference*, **3**, 401.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

QUINOLINATE PHOSPHORIBOSYLTRANSFERASE

This enzyme [EC 2.4.2.19], also referred to as nicotinate mononucleotide pyrophosphorylase (carboxylating), catalyzes the reversible reaction of pyridine 2,3-dicarboxylate and 5-phospho- α -D-ribose 1-diphosphate to produce nicotinate D-ribonucleotide, carbon dioxide, and pyrophosphate.

L. M. Henderson (1983) *Ann. Rev. Nutr.* **3**, 289.

W. D. L. Musick (1981) *Crit. Rev. Biochem.* **11**, 1.

QUIN-2 TETRAACETOXY ESTER

A membrane-permeant form of Quin-2 (systematic name: 2-[(2-amino-5-methylphenoxy)methyl]-6-methoxy-8-aminoquinoline-*N,N,N',N'*-tetraacetate) which upon binding of calcium ion results in fluorescence. **See** *Fura-2*

QUOTIENT FUNCTION

Any ratio of parameters, such as V_{max}/K_m or the fraction of the protein in the R state to that fraction in the T state in the Monod-Wyman-Changeux allosterism model.

QUOTIENT, RESPIRATORY

A ratio of the amount of CO₂ produced by tissue metabolism to the amount of oxygen consumed.

QUOTIENT, SYNERGISM (Q_{syn})

A parameter used in isotope exchange experiments of enzyme-catalyzed reactions purported to have a ping pong mechanism. This parameter assesses whether substrate synergism is present and to what degree. For a ping pong Bi Bi mechanism, $Q_{\text{syn}} = \{(R_{\text{max, A-P}})^{-1} +$

$(R_{\max, \text{B-Q}})^{-1} / (V_{\max, \text{f}}^{-1} + V_{\max, \text{r}}^{-1})$ where $V_{\max, \text{f}}$ and $V_{\max, \text{r}}$ are the maximum initial-rate velocities for the forward and reverse reaction, respectively, and $R_{\max, \text{A-P}}$ and $R_{\max, \text{B-Q}}$ are the maximum rates of exchange seen in the half-reactions. A synergism quotient of unity indicates

a ping pong mechanism with no substrate synergism. For values significantly larger than one, either the mechanism is not a simple ping pong scheme or substrate synergism is present¹.

¹D. L. Purich & R. D. Allison (1980) *Meth. Enzymol.* **64**, 3.

R

RACEMASES

A class of enzymes that catalyze the interconversion of one enantiomer with its mirror image. Care must be exercised in applying this term. For example, the enzyme that interconverts D-methylmalonyl-CoA to L-methylmalonyl-CoA is not a racemase, but is instead an epimerase; the two coenzyme A derivatives are diastereoisomeric, and not enantiomeric, with respect to each other.

RACEMIC MIXTURE (or Racemate)

A mixture of equal amounts of two enantiomers, mirror images of each other. Such a mixture will not rotate the plane of polarized light. These mixtures can be designated by the prefix *RS*-, ($\pm$)-, or *DL*-. The prefix *dl*- should be avoided.

RACEMIZATION

Any reaction, either chemical or biochemical, which results in the conversion of an optically active molecular entity to a racemic mixture.

RAD

1. Unit for the dose absorbed from ionizing radiation. One rad equals 100 ergs of radiation absorbed per gram of tissue. One gray equals 100 rad. 2. Abbreviation for radian.

RADIAN

A supplementary SI unit (symbolized by rad) for plane angle: 2π radians equals 360° . Thus, one radian is equivalent to $57.2957795 \dots^\circ$ (or, $57^\circ 17' 44.806 \dots''$).

RADIANCE

A measure of the radiant power (P) for a parallel beam of electromagnetic radiation leaving or passing through

an infinitesimal element of surface in a given direction from the source divided by the orthogonally projected area of the element in a plane normal to the given direction of the beam (θ). It is symbolized by L and has SI units of watts per square meter. Thus, $L = (dP/dS)/\cos \theta$. If the radiant power is constant over the surface area, then $L = P/(S \cos \theta)$.

If the beam is divergent from an elementary cone of the solid angle ($d\Omega$) containing the given direction (θ), then $L = d^2P/(d\Omega dS \cos \theta)$. In this instance, the radiance has units of watts per square meter per steradian.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

RADIANT ENERGY

The radiant energy (Q) is equal to the total energy emitted, transferred, or received as radiation in a given period of time. It has SI units of joules. If the radiant power (P) is constant over the time period considered, then $Q = Pt$.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

RADIANT (or, ENERGY) FLUX

Symbolized by ϕ , the radiant flux is equivalent to the radiant power P . When the radiant energy Q is constant over the time frame considered, $\phi = Q/t$. Care should be exercised when using this term and its symbol in photochemical and related studies since ϕ is more commonly reserved for quantum yield.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

RADIATIONLESS TRANSITION

The transition that takes place without photon emission or absorption, between two states of a system. **See** *Fluorescence*

RADIATIVE LIFETIME

The lifetime of an excited state of a molecular entity (in the absence of any radiationless transition). Radiative

lifetime is symbolized by τ_0 and equals the reciprocal of the first-order rate constant (or, sum of first-order rate constants).

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

RADICAL (or FREE RADICAL)

1. Any molecular entity possessing an unpaired electron. The modifier “unpaired” is preferred over “free” in this context. The term “free radical” is to be restricted to those radicals which do not form parts of radical pairs. Further distinctions are often made, either by the nature of the central atom having the unpaired electron (or atom of highest electron spin density) such as a carbon radical (*e.g.*, $\cdot\text{CH}_3$) or whether the unpaired electron is in an orbital having more *s* character (thus, σ radicals) or more *p* character (hence, π radicals). Whenever presenting the radical molecular entity in a manuscript, the structure should always be written with a superscript dot or, preferably, a center-spaced bullet (*e.g.*, $\cdot\text{OH}$, $\cdot\text{CH}_3$, $\text{Cl}\cdot$).

2. Any substituent or moiety bound to a molecular entity. IUPAC suggests that this older term should be abandoned, preferring instead usage of “groups,” “moieties,” or “substituents.”

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

RADICAL CENTER

The atom within a molecule or ion on which the unpaired electron is mainly localized. This center could actually be a group of atoms of a moiety of the molecule.

RADICAL ION

An electrically charged radical. Hence, a radical cation carries a positive charge and a radical anion carries a negative charge. The charge and the odd electron are often localized with the same atoms of the molecular entity. According to IUPAC recommendations, if the unpaired electron and the charge cannot be associated with specific atoms of the molecular entity (or they are both localized to the same atom), then a superscript (or center space) dot or bullet should precede the superscript charge (+ or −). Thus, a benzene radical cation would be $\text{C}_6\text{H}_6^{+\cdot}$. This is opposite of the convention used in mass spectroscopy.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

RADIOACTIVE SUBSTANCES (General Precautions)

The following are general recommendations, which may be superseded by institutional requirements¹.

(1) Make certain that areas for radionuclide use are well marked, so that infrequent visitors to the laboratory will be immediately aware of areas to avoid. Ideally, these rooms should be locked and inaccessible to all but authorized personnel. A wise practice is to assume that these rooms will be entered by individuals lacking training in safe practices, and this mandates in favor of safe and secure storage procedures.

(2) Prohibit eating, drinking, and all mouth-pipetting in rooms containing designated radioactivity areas.

(3) Be properly dressed; *e.g.*, safety eyeglasses, long pants, labcoat, *closed-toe* shoes, gloves (these must be appropriate for the chemical properties of radiolabeled substances), lead apron (if recommended), and wrist guards. Dosimeters must be worn in recommended locations (wrist, finger, lapel, *etc.*).

(4) All radioactive material in storage and refuse containers should be clearly marked, preferably indicating the type of nuclide in use.

(5) Work with absorbent (preferably plastic-backed) benchtop coverings. Always use a spill tray containing an absorbent liner.

(6) Know the physical properties of the substances with which you are working. Keep in mind that some compounds (such as acetaldehyde and tritiated water) have low boiling points. Again, keep in mind that some gloves do not offer an adequate barrier to certain chemicals. Some compounds enter the body with such facility that special care must be exercised when they are in use. One example is dimethyl sulfoxide, which as a solvent facilitates the entry of many solutes into the body. There are many known cases where radiolabeled compounds contaminated individuals who failed to consider this power of DMSO as a solute vehicle.

(7) Handle any volatile substances in an approved and suitably marked fume hood or glove box. Chemical traps

may prove useful in preventing loss of certain volatile substances.

(8) Use a Geiger-Müller counter (and periodically use paper discs to swipe surfaces for scintillation counting) to determine the radioisotope as well as the sites and amounts of contaminating radioactivity.

(9) Keep accurate records on each compound, indicating radionuclide, amount remaining, location, date, and name of responsible user (usually the principal investigator's name).

(10) Maintain good housekeeping practices (*e.g.*, on ending an experiment, be sure to return all radionuclides to their safe storage locations, dispose of all contaminated items, and decontaminate surfaces). Remember that any unsuitably shielded waste can be hazardous to lab workers and custodial personnel.

(11) Again, one should always remember to WASH hands.

(12) Use of certain isotopes requires local or full-body scintillation counting to confirm the extent and location of any contaminant. Regulations concerning such monitoring vary according to institutional policy.

¹For certification by the U. S. Nuclear Regulatory Commission, each institution establishes its own standards of safe practice that must meet or exceed N. R. C. Standards. Practices of individual investigators must adhere to their own institutional policies and regulations.

RADIOCARBON DATING

A cosmogenic method for dating carbonaceous objects based on the generation of carbon-14 from atmospheric nitrogen by cosmic radiation in the form of thermal neutrons; newly formed ^{14}C is then oxidized to $^{14}\text{CO}_2$ which enters the biosphere/geosphere where active exchange occurs between living objects and nutrient carbon dioxide. When such objects die, they end active exchange with the biosphere, and their radiocarbon content decreases by radioactive decay. This technique was pioneered by Professor Willard F. Libby (University of California, Los Angeles) who calibrated the ^{14}C content using carbonaceous objects from Egyptian tombs well-dated by dynastic hieroglyphic inscriptions and/or other archeologic documentation. Because the half-life for carbon-

14 is 5715 years, and because the precision of measuring carbon-14 content ranges from 4–6 percent, the technique is limited to dating objects less than 50,000 years. One should recognize that the intensity of cosmic radiation impinging on earth's atmosphere is not constant. Volcanic eruptions can likewise release carbon dioxide in amounts sufficient to dilute the ^{14}C content in the atmosphere. Thus, radiocarbon dating can be inaccurate.

See *Cosmogenic Dating*

RADIOLYSIS

1. The cleavage or scission of a bond as a result of high-energy radiation. 2. The generation of any chemical process induced by high-energy radiation. **See** *Pulsed Radiolysis*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

RAMAN SCATTERING

A dispersive phenomenon occurring when the wavelength of scattered electromagnetic radiation in the mid-infrared spectral region is shifted relative to that of the incident beam of exciting radiation. Spectral excitation is typically measured at a nonabsorbing wavelength, and the Raman effect occurs when the polarizability of a bond varies with the internuclear distance, as specified by the equation:

$$\alpha = \alpha_o + (r - r_{eq})(\partial\alpha/\partial r)$$

where α_o is the bond's polarizability at an equilibrium distance r_{eq} , and r is the instantaneous internuclear distance. The internuclear distance depends on the vibrational frequency ν_{vib} , such that $(r - r_{eq}) = r_{max} \cos(2\pi\nu_{vib}t)$, and substitution into the first expression yields:

$$\alpha = \alpha_o + r_{maximal} \cos(2\pi\nu_{vibrational}t)(\partial\alpha/\partial r)$$

For the dipole moment μ , one can write a standard expression (*i.e.*, $\mu = \alpha E = \alpha E_o \cos(2\pi\nu_{ex}t)$, where ν_{ex} is the frequency of excitation). Upon substitution for α , we get

$$\begin{aligned} \mu &= \alpha_o E_o \cos(2\pi\nu_{ex}t) \\ &+ E_o r_{maximal} (\partial\alpha/\partial r) \cos(2\pi\nu_{vib}t) \cos(2\pi\nu_{ex}t) \end{aligned}$$

The last term contains the product of the two cosine functions¹ and can be re-expressed as:

$$\begin{aligned}\mu = & \alpha_o E_o \cos(2\pi\nu_{\text{ext}}t) \\ & + 0.5E_or_{\text{max}}(\partial\alpha/\partial r)\{\cos[2\pi(\nu_{\text{ex}} - \nu_{\text{vib}})t] \\ & + \cos[2\pi(\nu_{\text{ex}} + \nu_{\text{vib}})t]\}\end{aligned}$$

If only the $\alpha_o E_o \cos(2\pi\nu_{\text{ext}}t)$ term is present, there is no energy loss by the elastic collision of a photon and the molecule; in this case, we have the classical Rayleigh light scattering equation [i.e., $\mu = \alpha_o E_o \cos(2\pi\nu_{\text{ext}}t)$]. The term $0.5E_or_{\text{max}}(\partial\alpha/\partial r)\cos[2\pi(\nu_{\text{ex}} - \nu_{\text{vib}})t]$ represents the so-called Stokes frequency; and the term $0.5E_or_{\text{max}}(\partial\alpha/\partial r)\cos[2\pi(\nu_{\text{ex}} + \nu_{\text{vib}})t]$ is the so-called anti-Stokes frequency. One can say that the excitation wavelength has been changed or modulated when it experiences Raman scattering. We must further note that the energies represented by these inelastic collisions are too small to promote an electron to even the next electronic state; as such, they are not quantized, and they are treated as virtual states lying on a continuum between the ground and excited states. In resonance Raman spectroscopy, one takes advantage of the fact that Raman line intensities are enhanced greatly (often by a factor of 10^2 to 10^6) by excitation wavelengths approaching that of a electronic absorption in a molecule of interest. This increases the sensitivity of the technique, thereby lowering the concentration needed for spectral characterization. The availability of tunable dye lasers has stimulated the development of resonance Raman spectroscopy, and a fuller description of the technique and applications to enzyme studies are described elsewhere².

The Raman technique is particularly useful for investigating the nature of metal-ligand bonds, for which the vibrational energies typically range from 100–700 cm^{-1} . Spectra can be directly observed in aqueous solutions, offering a major advantage over infrared spectroscopy for the study of biological molecules owing to the opacity of water with respect to IR radiation.

¹Note: The product of $\cos a$ and $\cos b$ can be rewritten as $0.5[\cos(a + b) + \cos(a - b)]$

²M. Sinnott, C. D. Garner, E. First & G. Davies (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference*, vol. 4, p. 86, Academic Press, San Diego.

RANDOM AB/QR, ORDERED C/P TER TER MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which two substrates (A and B) can bind in any order but the third substrate (C) can only bind

after the other two have already bound. Then, following the catalytic event, a particular product (P) is released followed by the random release of the remaining two products (Q and R). Guanylate cyclase has been reported¹ to have this mechanism.

¹D. L. Garbers, J. G. Hardman & F. B. Rudolph (1974) *J. Biol. Chem.* **13**, 4166.

RANDOM AC/PR, ORDERED B/Q TER TER MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which a particular substrate (B) has to bind second, but the other two substrates (A and C) can either bind first or third. Then, following the catalytic event, a particular product (Q) has to be the second product released, but the other two products (P and R) can be either the first or third product released. **See Multisubstrate Mechanisms**

RANDOM BC/PQ, ORDERED A/R TER TER MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which one particular substrate (A) is the only substrate that can bind to the free enzyme. After the EA binary complex has formed, the other two substrates (B and C) can bind in any order. Following the catalytic event, two of the products (P and Q) can be released in a random sequence, but the third product (R) has to be the last product released. Citrate cleavage enzyme¹ and γ -glutamylcysteine synthetase² are reported to operate by this mechanism. **See Multisubstrate Mechanisms**

¹K. M. Plowman & W. W. Cleland (1967) *J. Biol. Chem.* **242**, 4239.

²B. P. Yip & F. B. Rudolph (1976) *J. Biol. Chem.* **251**, 3563.

RANDOM BI BI MECHANISM

A two-substrate, two-product enzyme-catalyzed reaction scheme in which both the substrates (A and B) and the products (P and Q) bind and are released in any order. Note that this definition does not imply that there is an equal preference for each order (that is, it is not a requirement that the flux of the reaction sequence in which A binds first has to equal the flux of the reaction sequence in which B binds first). In fact, except for rapid equilibrium schemes, this is rarely true. There usually is a distinct preference for a particular pathway in a random mechanism. A number of kinetic tools and protocols

(e.g., isotope exchange at equilibrium experiments) can assist the investigator in identifying those reaction pathways.

Quite a few enzymes have been reported to have this reaction scheme: for example, hexokinase¹, adenylate kinase², and phosphofructokinase³.

The rate expression for the rapid equilibrium random Bi Bi mechanism, in the absence of products, is $v = V_{\max}[A][B]/\{K_{ia}K_b + K_b[A] + K_a[B] + [A][B]\}$ where $V_{\max}$ is the maximum velocity in the forward direction (equal to $k_9[E_{\text{total}}]$ where k_9 is the rate constant for the $EAB \rightarrow EPQ$ conversion), $K_{ia} = [E][A]/[EA]$, $K_a = [EB][A]/[EAB]$, and $K_b = [EA][B]/[EAB]$. The steady-state random Bi Bi rate expression is a more complex equation containing additional terms of $[A]^2[B]$ and $[A][B]^2$ in the numerator and $[A]^2$, $[B]^2$, $[A]^2[B]$, and $[A][B]^2$ in the denominator. Rudolph and Fromm⁴ have looked at the effect of the magnitude of these other terms on initial rate and product inhibition studies. **See Multisubstrate Mechanisms**

¹J. Ning, D. L. Purich & H. J. Fromm (1969) *J. Biol. Chem.* **244**, 3840.

²D. G. Rhoads & J. M. Lowenstein (1968) *J. Biol. Chem.* **243**, 3963.

³R. L. Hanson, F. B. Rudolph & H. A. Lardy (1973) *J. Biol. Chem.* **248**, 7852.

⁴F. B. Rudolph & H. J. Fromm (1971) *J. Biol. Chem.* **246**, 6611.

RANDOM BI BI DUAL THEORELL-CHANCE MECHANISM

A two-substrate, two-product enzyme-catalyzed reaction scheme in which either substrate, A or B, can bind first. However, the second substrate participates in a Theorell-Chance step in which either one of the two products is released. **See Multisubstrate Mechanisms**

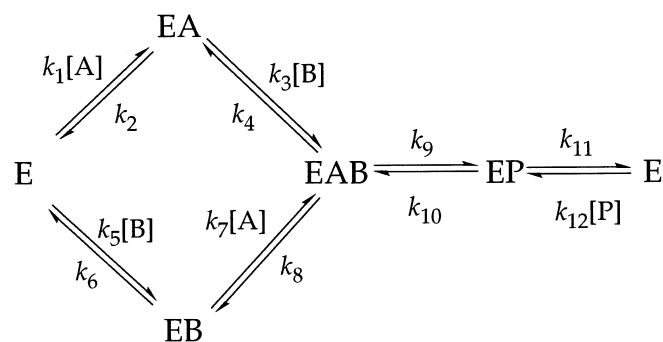
RANDOM BI UNI MECHANISM

An enzyme-catalyzed reaction scheme in which the two substrates (A and B) can bind in any order, resulting in the formation of a single product of the enzyme-catalyzed reaction (hence, this reaction is the reverse of the random Uni Bi mechanism). Usually the mechanism is distinguished as to being rapid equilibrium (i.e., the rate-determining step is the central complex interconversion, $EAB \leftrightarrow EP$) or steady-state (in which the substrate addition and/or product release steps are rate-contributing).

See Multisubstrate Mechanisms

Rapid Equilibrium Case. In the absence of significant amounts of product (i.e., initial rate conditions; thus, $[P] \approx 0$), the rate expression for the rapid equilibrium random Bi Uni mechanism is $v = V_{\max}[A][B]/(K_{ia}K_b + K_b[A] + K_a[B] + [A][B])$ where K_{ia} is the dissociation constant for the EA complex, K_a and K_b are the dissociation constants for the EAB complex with regard to ligands A and B, respectively, and $V_{\max} = k_9[E_{\text{total}}]$ where k_9 is the forward unimolecular rate constant for the conversion of EAB to EP. Double-reciprocal plots ($1/v$ vs. $1/[A]$ at different constant concentrations of B and $1/v$ vs. $1/[B]$ at different constant concentrations of A) will be intersecting lines. Slope and intercept replots will provide values for the kinetic parameters.

Steady-State Expression. In the absence of significant amounts of product, P (thus, initial rate conditions in which $[P] \approx 0$), the steady-state expression for the random Bi Uni mechanism having two central complexes



is $v = \{\Phi_{NAB}[A][B] + \Phi_{NA^2B}[A]^2[B] + \Phi_{NAB^2}[A][B]^2\} / \{\Phi_{D0} + \Phi_{DA}[A] + \Phi_{DB}[B] + \Phi_{DAB}[A][B] + \Phi_{DA^2}[A]^2 + \Phi_{DB^2}[B]^2 + \Phi_{DA^2B}[A]^2[B] + \Phi_{DAB^2}[A][B]^2\}$ where the Φ values are collections of rate constants (not to be confused with Dalziel Φ coefficients) and are designated in subscripts by N or D (representing numerator or denominator, respectively) and the corresponding letter(s) for substrate(s) (with exponent when needed) to which that Φ value is a coefficient. For the above scheme with two central complexes,

$$\Phi_{NAB} = k_9k_{11}(k_1k_3k_6 + k_2k_5k_7)[E_{\text{total}}]$$

$$\Phi_{NA^2B} = k_1k_3k_7k_9k_{11}[E_{\text{total}}]$$

$$\Phi_{NAB^2} = k_3k_5k_7k_9k_{11}[E_{\text{total}}]$$

$$\Phi_{D0} = k_2k_6((k_4 + k_8)(k_{10} + k_{11}) + k_9k_{11})$$

$$\Phi_{DA} = k_2k_7(k_4(k_{10} + k_{11}) + k_9k_{11}) + k_1k_6((k_4 + k_8)(k_{10} + k_{11}) + k_9k_{11})$$

$$\Phi_{DB} = k_3k_6(k_8(k_{10} + k_{11}) + k_9k_{11}) + k_2k_5((k_4 + k_8)(k_{10} + k_{11}) + k_9k_{11})$$

$$\Phi_{DAB} = (k_1k_3(k_6 + k_8) + k_5k_7(k_2 + k_4))(k_{10} + k_{11}) + k_9(k_3(k_1k_6 + k_7k_{11}) + k_2k_5k_7)$$

$$\Phi_{DA^2} = k_1k_7(k_4(k_{10} + k_{11}) + k_9k_{11})$$

$$\Phi_{DB^2} = k_3k_5(k_8(k_{10} + k_{11}) + k_9k_{11})$$

$$\Phi_{DA^2B} = k_1k_3k_7(k_9 + k_{10} + k_{11})$$

$$\Phi_{DAB^2} = k_3k_5k_7(k_9 + k_{10} + k_{11})$$

If the terms that contain $[A]^2$ or $[B]^2$ are negligible relative to the other terms, then one will still observe straight lines in double-reciprocal plots, $V_{\max} = \Phi_{NAB}/\Phi_{DAB}$, $K_a = \Phi_{DB}/\Phi_{DAB}$, and $K_b = \Phi_{DA}/\Phi_{DAB}$.

If one considers the simpler random Bi Uni scheme in which the two central complexes are grouped together as EXY (thus, $[EAB] + [EP] = [EXY]$ and k_{11} in the above scheme would become k_9 while k_{12} becomes k_{10}), the individual enzyme determinants are:

$$[E] = k_2k_6(k_4 + k_8 + k_9) + k_2k_7(k_4 + k_9)[A] + k_3k_6(k_8 + k_9)[B] + k_3k_7k_9[A][B]$$

$$[EA] = k_1k_6(k_4 + k_8 + k_9)[A] + k_1k_7(k_4 + k_9)[A]^2 + k_4k_5k_7[A][B] + k_4k_6k_{10}[P] + k_4k_7k_{10}[A][P]$$

$$[EB] = k_2k_5(k_4 + k_8 + k_9)[B] + k_3k_5(k_8 + k_9)[B]^2 + k_1k_3k_8[A][B] + k_2k_8k_{10}[P] + k_3k_8k_{10}[B][P]$$

$$[EXY] = (k_1k_3k_6 + k_2k_5k_7)[A][B] + k_1k_3k_7[A]^2[B] + k_3k_5k_7[A][B]^2 + k_3k_6k_{10}[B][P] + k_2k_7k_{10}[A][P] + k_3k_7k_{10}[A][B][P] + k_2k_6k_{10}[P]$$

RANDOM BI UNI UNI BI PING PONG MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which two substrates (A and B) can bind to the enzyme in any order. A catalytic event then occurs and a product (P) is released. After that desorption, the third substrate (C) binds, another catalytic event occurs, and the remaining two products (Q and R) are released in any order. Leucyl-tRNA synthetase has been re-

ported¹ to operate by this mechanism. **See** *Multisubstrate Mechanisms*

¹C.-S. Linn, R. Irwin & J. G. Chirikjian (1975) *J. Biol. Chem.* **250**, 9299.

RANDOM ERROR

Deviations that arise probabilistically and have two characteristics: (a) the magnitude of these errors is more typically small, and (b) positive and negative deviations of the same magnitude tend to occur with the same frequency. Random error is normally distributed, and the bell-shaped curve for frequency of occurrence *versus* magnitude of error is centered at the true value of the measured parameter. **See** *Statistics (A Primer)*

RANDOM EXPERIMENT

An experiment that can result in different outcomes, even though the experiment is repeated by exactly the same procedure every time. A random variable is a numerical outcome of a random experiment. **See** *Statistics (A Primer)*

RANDOMIZATION

A strategy that attempts to obtain a statistically unbiased sample or data set for a series of experimental measurements by simulating a chance distribution or chance sequence. **See** *Statistics (A Primer)*

RANDOM MECHANISMS

Multisubstrate or multiproduct enzyme-catalyzed reaction mechanisms in which one or more substrates and/or products bind and/or are released in a random fashion. Note that this definition does not imply that there has to be an equal preference for any particular binding sequence. The flux through the different binding sequences could very easily be different. However, in rapid equilibrium random mechanisms, the flux rates are equivalent. **See** *Multisubstrate Mechanisms*

RANDOM-ON/ORDERED-OFF BI BI MECHANISM

A two-substrate, two-product enzyme-catalyzed reaction scheme in which the two substrates (A and B) can bind to the enzyme in any order but the two products are released in a distinct order (P being released before Q). **See** *Multisubstrate Mechanisms*

RANDOM SCISSION KINETICS

Although some depolymerases act processively in cleaving their polymeric substrates, others act by what can be described as multiple attack which results in nonselective scission or random scission. The analysis of cleavage products during the course of enzyme-catalyzed depolymerization can provide important clues about the nature of the reaction. With random scission, the rate of bond scission must be proportional to the total number of unbroken bonds present in the solution. Thomas¹ measured the rate of base addition in a pH-Stat (a device with an automated feedback servomotor that expels titrant from a syringe to maintain pH) to follow the kinetics of DNA bond scission by DNase. The number of bonds cleaved was linear with time, and this was indicative of random scission. In other cases, one may apply the template challenge method to assess the processivity of nucleic acid polymerases². **See Processivity**

Tanford³ presented a cogent kinetic treatment of random scission of a polymer, and the complete analysis is beyond the scope of this Handbook. The basic idea is that a molecule M_y can yield a molecule M_x (where x and y indicate the degree of polymerization, and where $y > x$) in two different ways. The x -mer formation rate from y -mers is twice the rate of bond scission at the concentration of the y -mer; thus,

$$\frac{d[M_x]}{dt} = 2k \sum_{y=x+1}^{\infty} [M_y]$$

from y equal to $(x + 1)$ to infinity. The rate of conversion of x -mers into shorter polymer chains occurs by rupture of any of their $(x - 1)$ bonds, and this effectively amplifies the magnitude of the rate constant k :

$$-\frac{d[M_x]}{dt} = (x - 1)k[M_x]$$

The rate of change of the x -mer concentration is

$$-\frac{d[M_x]}{dt} = (x - 1)k[M_x] + 2k \sum_{y=x+1}^{\infty} [M_y]$$

again where from y equal to $(x + 1)$ to infinity. If $x = n$, then upon integration, this expression takes on the following first-order rate form:

$$[M_n] = [M_n]_0 e^{-k(n-1)t}$$

Likewise, we can write corresponding equations for $[M_{n-1}]$, $[M_{n-2}]$, $[M_{n-3}]$, etc.:

$$[M_{n-1}] = [M_n]_0 e^{-k(n-2)t} (1 - e^{-kt}) (2)$$

$$[M_{n-2}] = [M_n]_0 e^{-k(n-3)t} (1 - e^{-kt}) (3 - e^{-kt})$$

$$[M_{n-3}] = [M_n]_0 e^{-k(n-4)t} (1 - e^{-kt}) (4 - 2e^{-kt})$$

except for when it is undefined (*i.e.*, $x = n$). Then, by using N_{total} as the total number of subunits per volume, and by expressing $[M_x]$ as the number concentration of x -mers, this treatment can account for the time evolution of random polymer scission.

¹C. A. Thomas, Jr. (1956) *J. Amer. Chem. Soc.* **78**, 1861.

²W. R. McClure & Y. Chow (1980) *Meth. Enzymol.* **64**, 277.

³C. Tanford (1961) *Physical Chemistry of Macromolecules*, p. 615, Wiley, New York.

RANDOM TER TER MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which the three substrates (A, B, and C) and three products (P, Q, and R) can bind to and be released in any order. A number of enzymes have been reported to have this mechanism: for example, adenylosuccinate synthetase^{1,2}, glutamate dehydrogenase³, glutamine synthetase^{4,5}, formyltetrahydrofolate synthetase⁶, and tubulin:tyrosine ligase⁷. **See Multisubstrate Mechanisms**

¹F. B. Rudolph & H. J. Fromm (1969) *J. Biol. Chem.* **244**, 3832.

²G. D. Markham & G. H. Reed (1977) *Arch. Biochem. Biophys.* **184**, 24.

³P. C. Engel & K. Dalziel (1970) *Biochem. J.* **118**, 409.

⁴F. C. Wedler & P. D. Boyer (1972) *J. Biol. Chem.* **247**, 984.

⁵R. D. Allison, J. A. Todhunter & D. L. Purich (1977) *J. Biol. Chem.* **252**, 6046.

⁶B. K. Joyce & R. H. Hines (1966) *J. Biol. Chem.* **241**, 5716.

⁷N. L. Deans, R. D. Allison & D. L. Purich (1992) *Biochem. J.* **286**, 243.

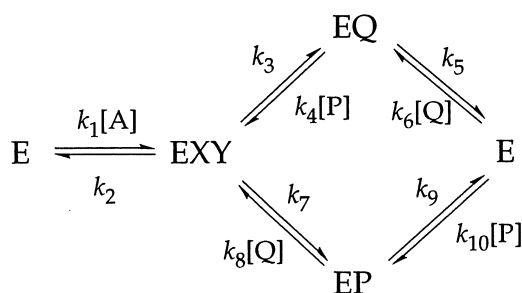
RANDOM UNI BI MECHANISM

A single-substrate, two-product enzyme-catalyzed reaction scheme in which a substrate (A) binds to the free enzyme and is converted to two products (P and Q) which can be released in any order. Many enzymes described with this mechanism are actually random pseudo-Uni Bi schemes in which the second substrate is water.

Rapid Equilibrium Mechanism. If the rate-determining step is the catalytic step and all binding steps can be described by dissociation constants (*e.g.*, $K_{ia} = [E][A]/[EA]$), then, in the absence of products (*i.e.*, $[P]$ and $[Q] \approx 0$), the initial rate equation for the rapid equilibrium Uni Bi mechanism is identical to that of the Uni Uni

mechanism: $v = V_{\max,f}[A]/\{K_{ia} + [A]\}$ where $V_{\max,f}$ is the maximum velocity in the forward direction and is equal to $k_3[E_{\text{total}}]$ where k_3 is the unimolecular forward rate constant for the $EA \leftrightarrow EPQ$ conversion. If product P (but not Q) is present, the rate expression (in the absence of abortive complexes and ligand-independent isomerization steps) becomes $v = V_{\max,f}[A]/\{K_{ia} + [A] + K_{ia}[P]/K_{ip}\}$ where $K_{ip} = [E][P]/[EP]$. As can readily be seen from the double-reciprocal form of this expression, P will be a competitive inhibitor with respect to A. A similar conclusion can be obtained for Q.

Steady-State Mechanism. Consider the reaction scheme with only one central complex (thus, $EA + EPQ = EXY$):



the individual steady-state enzyme determinants are:

$$[E] = k_5k_9(k_2 + k_3 + k_7) + k_4k_9(k_2 + k_7)[P] + k_5k_8(k_2 + k_3)[Q] + k_2k_4k_8[P][Q]$$

$$[EXY] = k_1k_5k_9[A] + k_1k_4k_9[A][P] + k_1k_5k_8[A][Q] + k_1k_4k_8[A][P][Q] + k_4k_6k_8[P][Q]^2 + k_4k_8k_{10}[P]^2[Q] + (k_4k_6k_9 + k_5k_8k_{10})[P][Q]$$

$$[EQ] = k_1k_3k_9[A] + k_6k_9(k_2 + k_3 + k_7)[Q] + k_6k_8(k_2 + k_3)[Q]^2 + k_1k_3k_8[A][Q] + k_3k_8k_{10}[P][Q]$$

$$[EP] = k_1k_5k_7[A] + k_5k_{10}(k_2 + k_3 + k_7)[P] + k_4k_{10}(k_2 + k_7)[P]^2 + k_1k_4k_7[A][P] + k_4k_6k_7[P][Q]$$

In the absence of products, the initial rate equation is

$$v = V_{\max,f}[A]/\{K_a + [A]\}$$

where

$$V_{\max,f} = k_5k_9(k_3 + k_7)[E_{\text{total}}]/\{k_9(k_3 + k_5) + k_5k_7\}$$

$$K_a = k_5k_9(k_2 + k_3 + k_7)/\{k_1(k_9(k_3 + k_5) + k_5k_7)\}$$

If the $[P]^2$ and $[Q]^2$ terms are negligible relative to the

other terms in the rate expression, the overall rate expression suggests that both P and Q may act like noncompetitive inhibitors (as opposed to the case in the rapid equilibrium scheme).

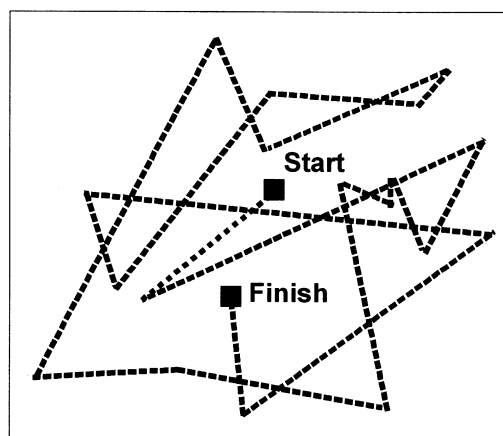
RANDOM UNI UNI BI BI PING PONG MECHANISM

A three-substrate, three-product enzyme-catalyzed reaction scheme in which a particular substrate (A) is the only substrate that can bind to the free, unmodified form of the enzyme. A chemical event then occurs, the first product (P) is released, and the remaining two substrates (B and C) can then bind in any order. Another catalytic event occurs, and the last two products (Q and R) are released in any order. **See** *Multisubstrate Mechanisms*

RANDOM VARIABLES

In statistics, a variable, often denoted by a capital letter, that associates a number with the outcome of a random experiment. More formally, a random variable is a function that assigns a real number to each outcome in the sample space of a random experiment. **See** *Statistics (A Primer)*

RANDOM WALK



An aimless path resulting from a set of successive steps (or excursions) whose individual lengths and directions are only randomly related to each other. As with any stochastic process, the preceding step has no influence at all on any of the subsequent steps. **See** *Diffusion; Persistence Time; Twiddling*

RAPID BUFFER EXCHANGE

Many biological molecules are stored in a stabilizing buffer containing solutes that interfere with kinetic experiments. This is especially true of commercial enzyme products, and the “cocktail” of stabilizing agents for maintaining a long shelf-life is sometimes a trade secret that a company may refuse to divulge. (This is particularly true for enzymes used for special purposes, such as in clinical autoanalyzers. For this reason, even though the same company offers two or more preparations of the same enzyme at the same specific activity, one cannot be absolutely certain that both are stabilized in an identical manner.) Commercial enzymes are often kept active by the presence of 1–3 M ammonium sulfate or 1–2 M glycerol. Other stabilizing substances include metal ion chelating agents (*e.g.*, EDTA, EGTA, or *o*-phenanthroline), antifungals and antibacterials, Bentonite clay, talcum powder, boric acid, and various protease inhibitors. In some cases, a reversibly bound ligand (including one or more of the substrates, an activator, or an inhibitor) is added to maintain the enzyme in its catalytically active conformation. Such agents must be removed before conducting kinetic experiments, especially detailed mechanistic investigations, and membrane dialysis or gel filtration on Sephadex G-25 is most frequently employed for this purpose. (Dialysis is known in some instances to introduce trace elements, plasticizers, and even minor levels of enzymatic activities.) The problem with dialysis and gel filtration is that large samples (> 1 mL) are required, and samples suffer dilution of up to a factor of three. Use of much smaller protein and enzyme aliquots (in the 25–150 μL range) are not manageable with dialysis or column gel filtration.

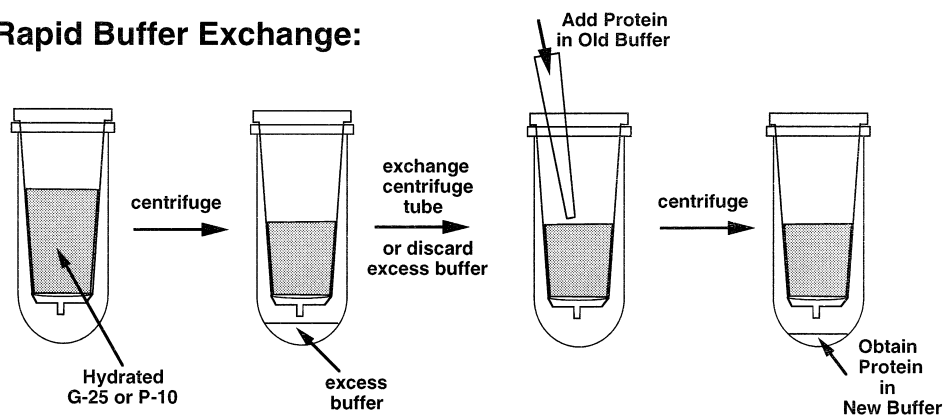
Penefsky¹ introduced the centrifuged-column procedure (see diagram below) that can be used for small aliquots of enzymes and other proteins (a) to exchange buffers and/or other solutes, (b) to measure ligand binding, and (c) to concentrate protein by as much as 10 times. His two-step protocol for rapid buffer exchange can be summarized as follows:

Step 1. Load a disposable mini-chromatography column^{2,3} with Sephadex G-50 (fine) dextran that was previously permitted to equilibrate with 5 volumes of the desired “new” buffer. Allow all the liquid to drain under the earth’s gravitation to leave a 1 mL bed volume. Transfer this spin-column to an appropriately sized centrifuge tube and spin at $100 \times g$ (calculated for the radial distance to the tip of the column itself, and *not* the bottom of the centrifuge tube) for 2 min in a swinging bucket rotor⁴. The volume of the now semi-dry gel bed will be reduced by 25–40%, and the media will appear as a gel mass that may even break away from the walls of the chromatography tube. (This is normal and will not affect performance.)

Step 2. Transfer the disposable spin-column to a clean centrifuge tube and then add 25–150 mL protein in “old” buffer. Centrifuge the tube exactly as before, and the filtrate (volume usually 90–110% the volume applied to the semi-dry gel) will contain 95–100% of the protein applied to the column.

By reducing the duration or speed of the second centrifugation step, Penefsky¹ found that the entire protein sample will become concentrated into a smaller volume;

Rapid Buffer Exchange:



hence, one can achieve up to 10x concentration of the protein with only a modest loss of enzyme.

¹H. H. Penefsky (1979) *Meth. Enzymol.* **56**, 527.

²BioRad produces Bio-Spin disposable polypropylene chromatography columns (Catalog No. 732-6008) that can be packed with up to 1.2 mL gel filtration media. Their micro Bio-Spin columns hold 0.8 mL media. Both are autoclavable columns that have snap-off column tips, polyethylene media bed supports, and fit standard centrifuge tubes.

³In his work, Penefsky¹ simply used 1 mL Plastipak tuberculin syringes fitted with porous disks cut with a cork borer (0.193 inch inside diameter) from polyethylene sheets (1.5 mm thick, 70 micrometer pore size) available from Bolab, Inc. (Deery, NH). Note: The rubberized plungers of most syringes frequently contain lubricants, and these agents may bind to enzymes or they may dissolve and contaminate some enzyme preparations.

⁴Because various swinging bucket centrifuges behave differently with respect to acceleration and braking, some optimization of centrifugation conditions may be advisable. This can be readily accomplished by UV and visible spectrophotometry if one uses BSA (or some other inexpensive protein sample) along with a small amount of dichromate as a yellow colored metal ion species.

RAPID EQUILIBRIUM MECHANISMS

Reaction mechanisms (and binding mechanisms for macromolecules not catalyzing reactions) in which the binding of substrates and products is rapid relative to the interconversion and isomerization steps. Hence, those binding steps can be easily characterized by dissociation constants. The derivations of enzyme rate expressions using the rapid equilibrium assumption are very easy and it is common to find such equations in publications addressing the kinetics of a particular enzyme. Nevertheless, it is relatively rare to find papers in which the authors have actually demonstrated that the enzyme under study is truly a rapid equilibrium case. The rapid equilibrium rate equation may not be the appropriate expression to describe that specific protein. Rapid equilibrium enzymes are not as common as one may have assumed. Product desorption and/or substrate adsorption are often rate-contributing or rate-limiting. In those cases, the steady-state rate expression may describe the initial rate results better.

It should also be noted that the common models of allosterism (e.g., the Monod-Wyman-Changeux model and the Koshland-Némethy-Filmer model) assume rapid equilibrium binding.

RAPID MIXING

As mentioned in an earlier entry (*See Mixing Time*), the time needed to mix all the components of a reaction

together should be as short as possible. In transient kinetic studies (i.e., rapid reaction kinetics or pre-steady-state kinetics) this factor is of even greater importance. In stopped-flow experiments, the mixing time should not be longer than 2 msec. *See Quench-Flow Experiments*

C. A. Fierke & G. G. Hammes (1995) *Meth. Enzymol.* **249**, 3.

RAPID SCAN SPECTROSCOPY OF FAST REACTIONS

Any advanced absorbance/fluorescence spectrophotometer designed for routine acquisition of absorption or emission on the subsecond time scale. The basic goal is to obtain a series of complete UV/visible or fluorescence spectra as a function of time, usually after samples are mixed in a stopped-flow device. Such data help the investigator to infer the most likely structures of transient intermediates whose electronic spectra or fluorescence spectra can be determined by deconvoluting the spectra with appropriate reaction kinetic simulation software or by some other global analysis method (Fig. 1).

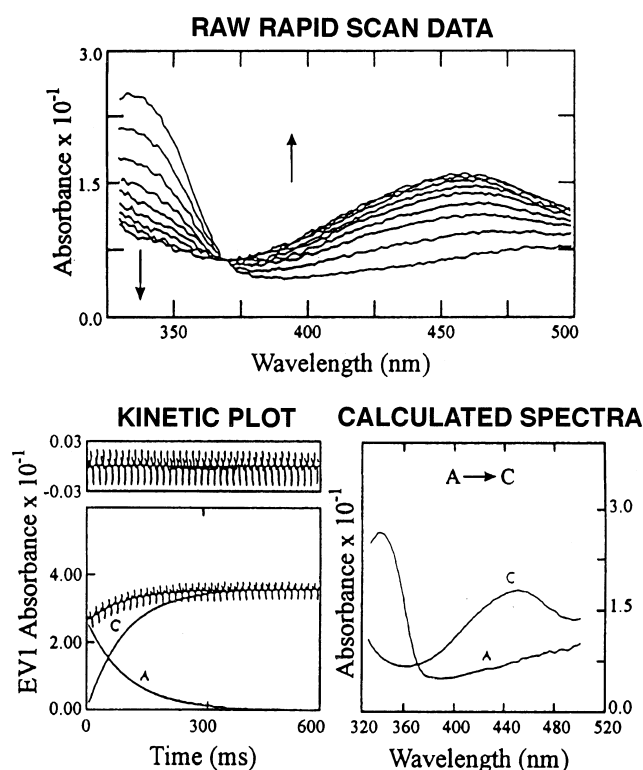


Figure 1. A series of rapid scan spectra obtained by the OLIS RSM-1000 Rapid-Scanning Monochromator (Courtesy of Olis Instruments, Inc., www.olisweb.com).

Rapid scan devices may employ a spinning prism, a diode array detector, or the newest spinning-disk technology. The OLIS spectrophotometers rely on the DeSa monochromator, a 0.25-meter double-grating monochromator. In place of a stationary central slit found in other monochromators is the “ScanDisk Module”, a spinning disk with 16 slits driven by a synchronous motor at a rotational speed of 62.5 Hz to achieve millisecond spectral scans (or “snap shots”) of a reaction. Because the DeSa monochromator is “subtractive”, the entire output beam is spectrally homogeneous, regardless of the chosen resolution. Photometric accuracy exceeds that achievable with CCD diode array detectors, and the spinning disk is mechanically rugged. The instrument produces spectra (*i.e.*, signal strength information *versus* wavelength) as a function of time, and data analysis proceeds in three steps: (a) by determining the number of species by factor analysis; (b) by determining the rate constants using kinetic modeling software; and (c) by reconstructing time-invariant spectra of intermediates.

RAPID-START COMPLEX

The complex that RNA polymerase forms at the promoter site just prior to initiation. Some bacterial promoters require high NTP concentrations to initiate efficient transcription, because this represents a “status report” on the stores of ATP, UTP, GTP, and CTP needed for RNA synthesis. Nature has evolved a kinetic control device: high initiating ATP and GTP concentrations must be present to stabilize an otherwise short-lived polymerase-promoter complex. The reader may also recall that bacterial translation is also tightly controlled, and amino acid starvation leads to ppGpp synthesis, the so-called stringent-response agent that also potently inhibits RNA polymerase. Such kinetic control ensures that NTP and amino acid concentrations are adequate before transcription and translation occur.

RATE CONSTANT

The proportionality constant allowing one to equate reaction velocity (having units of $M \cdot s^{-1}$) to the mo-

larity of reactant(s) involved in the reaction. For unimolecular processes, $v = k[X]$, where k has units of s^{-1} ; for bimolecular processes, $v = k[X]^2$ or $k[X][Y]$, where k has units of $M^{-1}s^{-1}$; and for trimolecular processes, $v = k[X]^3$ or $k[X]^2[Y]$ or $k[X][Y]^2$ or $k[X][Y][Z]$, where k has units of $M^{-2}s^{-1}$. **See** *Chemical Kinetics*

RATE-CONTRIBUTING STEP

Any reaction step having a rate constant whose magnitude is nearly that of the rate constant in the slowest step in a reaction mechanism. Such a step is said to be a contributing factor in slowing down the reaction rate.

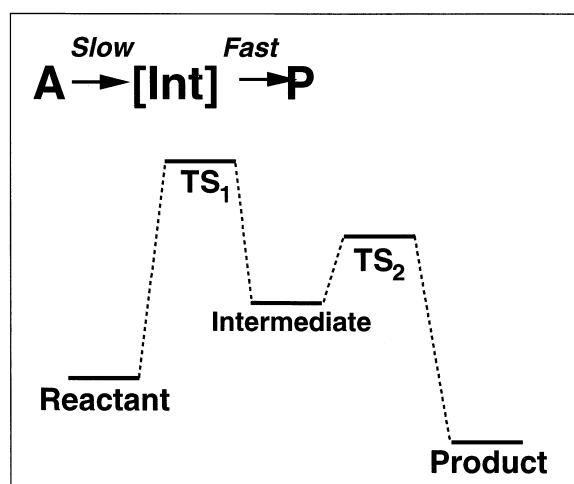
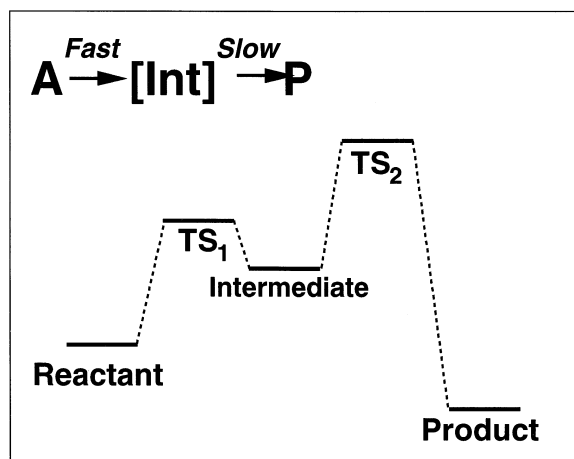
RATE-CONTROLLING STEP

Synonym for rate-determining step and for rate-limiting step. If a rate constant for an elementary reaction has a stronger influence on the overall rate than any other rate constant in the mechanism, then the step associated with that rate constant is referred to as the rate-controlling step. Note that the rate-controlling step may change with a corresponding change in reaction conditions.

The rate-controlling step is the elementary reaction that has the largest control factor (CF) of all the steps. The control factor for any rate constant in a sequence of reactions is the partial derivative of $\ln v$ (where v is the overall velocity) with respect to $\ln k_i$ in which all other rate constants (k_j) and equilibrium constants (K_j) are held constant. Thus, $CF = (\partial \ln v / \partial \ln k_i)_{K_j, k_j}$. This definition is useful in interpreting kinetic isotope effects. **See** *Rate-Determining Step; Kinetic Isotope Effects*

RATE-DETERMINING STEP OR RATE-LIMITING STEP

Designations of the slowest step(s) in a rate process. These terms should be regarded as synonymous to the preferred term “rate-controlling” step. However, all three are frequently used to describe a slow step within a series of otherwise more rapid steps. **See** *Rate-Controlling Step*



Energetics of Rate-Determining Steps. Although the initial and final states are the same in this reaction coordinate diagram, the rate of intermediate formation will be faster for the first example, where the barrier to intermediate formation is lower. Conversion of intermediate back to reactant or onward to product will also depend on the relative heights of the two barriers (marked here as TS₁ and TS₂). If the free energy barrier for conversion of a reaction intermediate in one direction is significantly lower than for conversion in the opposite direction, then the reaction will proceed many times faster by the lower energy route.

RATE LAW

A mathematical expression (usually in the form of a differential rate equation) showing how the rate of the reaction depends on the concentrations of chemical entities and rate constants (and/or equilibrium constants) and any partial orders of reactions. **See** *Chemical Kinetics*

RATE OF APPEARANCE

The rate of a reaction in which an increase in the concentration of one (or more) product(s) is actually what is

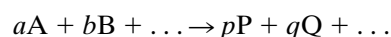
being measured. It is symbolized by v and sometimes has a subscript for the species being measured (e.g., v_P or v_{pyruvate}). **See** *Chemical Kinetics*

RATE OF DISAPPEARANCE

The rate of a reaction in which a decrease in the concentration of substrate or reactant (or several reactants) is actually what is being measured. It is symbolized by v and may have a subscript for the species being measured (e.g., v_A or v_{GTP}). **See** *Chemical Kinetics*

RATE OF REACTION

For a chemical reaction,



where A, B, ... , are the reactants, P, Q, ... , are the products, and the corresponding lower-case letters are the stoichiometry numbers. The rate of the reaction, v , is defined as

$$v = -(1/a)(d[A]/dt) = -(1/b)(d[B]/dt) = \dots \\ = (1/p)(d[P]/dt) = (1/q)(d[Q]/dt) = \dots$$

in which t is time (usually seconds) and the entities in brackets are concentrations.

The above definition applies only if there is no accumulation of intermediates or formation of side products. IUPAC strongly suggests that this term be applied only when those conditions have been experimentally established. IUPAC also recommends that the terms "rate of appearance" or "rate of disappearance" (or "rate of consumption") be used if these conditions are not met.

It should also be recalled that the term "rate of conversion" (or, rate of extent of reaction), symbolized by dx/dt and equal to $-(1/a)(dn_A/dt) = -(1/b)(dn_B/dt) = \dots = (1/p)(dn_P/dt) = (1/q)(dn_Q/dt) = \dots$ where n , with the appropriate subscript, designates the amount of substance (e.g., moles or multiple thereof) also has this restriction. **See** *Chemical Kinetics*

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

RATE SATURATION BEHAVIOR

A term used to refer to any chemical process whose rate depends upon saturation of a binding site. Rate saturation is observed in enzyme kinetics, metabolic

transport, host-guest reactions, and even heterogeneous catalysis such as hydrogenation on metallic surfaces of platinum and nickel.

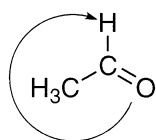
RAY-ROSCELLI TREATMENT (Isomerization)

A method¹ used to assess enzyme mechanisms which may have one or more enzyme isomerization steps². In this procedure, $[P_t]/[P]$ is plotted as a function of time at different initial concentrations of substrate, where $[P_t]$ is the concentration of the product at time t and $[P_\infty]$ is the concentration of P once the system has reached equilibrium. These data can be used to calculate the K_{iip} term, associated with the isomerization step, in the rate expression.

¹W. J. Ray, Jr., & G. A. Roscelli (1964) *J. Biol. Chem.* **239**, 3935.

²K. L. Rebholz & D. B. Northrop (1995) *Meth. Enzymol.* **249**, 211.

RE-



The *re* Face of Acetaldehyde

This stereochemical term (short for the Latin word *rectus*) is the designation for a stereoheterotopic face of a trigonal atom whose ligands appear in a clockwise sense in the order of the Cahn-Ingold-Prelog scheme when viewed from that side of the face.

G. P. Moss (1996) *Pure Appl. Chem.* **68**, 2193.

REACTANCY

The number of reactants partaking in an enzyme-catalyzed reaction. Because most enzyme reactions have an unequal number of substrates and products, one must specify the reactancy for a specified direction of the reaction. As an example, a multisubstrate reaction having two substrates and three products has a reactancy of two in the forward direction and three in the reverse direction. Cleland introduced the prefixes “Uni”, “Bi”, “Ter”, and “Quad” to indicate reactancies of one, two, three, and four, respectively. Thus, the example given above can be called a “Bi Ter” reaction. Water molecules and protons are not usually considered when specifying reactancy.

REACTING BOND RULES

A set of rules describing how motions can affect a molecule's transition-state structure and reactivity.

1. When considering a molecule's internal motion corresponding to progress along the reaction coordinate and over a transition state, any change making that motion more difficult will result in a new geometry at the energy maximum, in which the motion has proceeded further.

2. For an internal motion corresponding to a vibration, any change altering the equilibrium point of the vibration in a particular direction will serve to shift the equilibrium in that direction.

3. During the course of reaction, effects on bonds that are either made or broken are the most significant. Therefore, bonds nearest sites of structural change are apt to be most strongly affected.

IUPAC (1994) *Pure Appl. Chem.* **66**, 1077.

E. R. Thornton (1967) *J. Amer. Chem. Soc.* **89**, 2915.

REACTING ENZYME CENTRIFUGATION

A centrifugation technique used to determine the oligomeric or monomeric status of an enzyme during catalysis. For example, this method can address such questions as: Is the catalytically active form of the enzyme a dimer or monomer? Is there cooperativity in the interconversion between the forms?

The reacting enzyme centrifugation technique has been used with yeast hexokinase¹ and with phosphoenolpyruvate carboxykinase².

¹J. P. Shill & K. E. Neet (1975) *J. Biol. Chem.* **250**, 2259.

²Y.-B. Chaio (1975) Doctoral Dissertation, Case-Western Res. Univ., Cleveland, Ohio.

REACTION COORDINATE DIAGRAM

A graphical representation of changes in reactant structure and/or free energy along the reaction coordinate, which indicates the reaction's progress (*i.e.*, as the reactant(s) is(are) converted to product(s) in a process). For elementary reactions, a geometric parameter (such as bond length and/or angle) can represent a measure of the progress of the reaction. (Occasionally, the bond order is used to indicate progress along a reaction coordi-

nate.) In any case, the horizontal axis must not be confused with time; rather, the reaction coordinate only specifies that a certain stage of the reaction has been achieved.

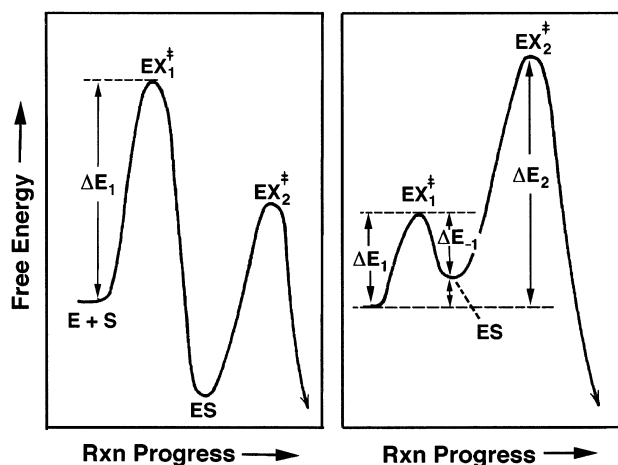


Figure 1. A hypothetical reaction coordinate diagram for an enzyme-catalyzed chemical reaction.

Reaction coordinate diagrams (Fig. 1) are important tools for analyzing the structure and energy relationships that describe a reaction mechanism. The following statements may prove helpful for constructing or interpreting reaction coordinate diagrams.

(a) The preferred unit for the vertical coordinate is ΔG° /mol. In the absence of such information, one may use a more arbitrary representation (such as the change in a system's potential energy) to construct a hypothetical reaction coordinate diagram.

(b) The baseline in reaction coordinate diagrams is arbitrary, and two different reactants (each with their own reaction coordinate diagram) may be placed at the same vertical position at the start of the reaction. This is true even if the reactants have different stability, because reactivity is only a function of the differences in energy of a reactant's initial, transition-state, and final positions.

(c) The vertical positions of each chemical species (e.g., reactants, products, intermediates, and enzyme-bound species) are defined by the ΔG° values for equilibria existing between them. Using the Additivity Principle, the differences in energy between successive steps are additive sums of their free energy changes, taking care to keep in mind that the signs of each energy change

must be written to agree with the direction of the reaction coordinate.

(d) Changes in free energy are estimated from experimentally determined equilibrium constants using the Gibbs equation ($\Delta G^\circ = -nRT \ln K_{eq}$). (Although one may occasionally rely on equilibrium data dealing with cognate nonenzymatic reactions, the internal equilibria between enzyme-bound species are frequently strongly influenced by the enzyme. This must be an anticipated feature of all catalyzed processes, because transition state stabilization is the dominant factor leading to catalytic rate enhancement.

(e) The heights of barriers between successive chemical species can, under favorable conditions, be determined from information obtained from studies of temperature dependence or through the use of kinetic isotope effects.

(f) The rate-determining step is the highest point on the diagram. Any change in reaction conditions that lowers the energy difference from reactants to transition state will result in faster reaction rate (*i.e.*, greater reactivity).

See Saddle-Point

(g) If the substrate (or reactant) is less stable than the product, the transition state is apt to appear early in the process and should resemble the starting material. And if the product is less stable, then the transition state will appear later and be more product-like in its structure.

(h) One can also infer that an unstable intermediate in a reaction should resemble the structure of the transition-state structure more closely than either substrate or product.

(i) Note that reversible reactions must pass through the same transition state, irrespective of whether the reaction is proceeding from reactant(s)-to-product(s) or *vice versa*.

(j) If a reactant has two pathways to product, and if the rates through paths A and B toward product are v_A and v_B , respectively, then the rates in the opposite directions along each pathway (and through their respective transition states) must have the same fractional contribution. This is a consequence of microscopic reversibility or the

law of detailed balance, and $v_A/(v_A + v_B)$ will be the same in both directions; so also will this hold true for $v_B/(v_A + v_B)$.

In what is now a classical study in enzyme kinetics, W. J. Albery and J. R. Knowles¹ developed a strategy for establishing a reaction coordinate diagram (shown in Fig. 2) for triose-phosphate isomerase catalysis using solvent exchange and kinetic isotope effect data.

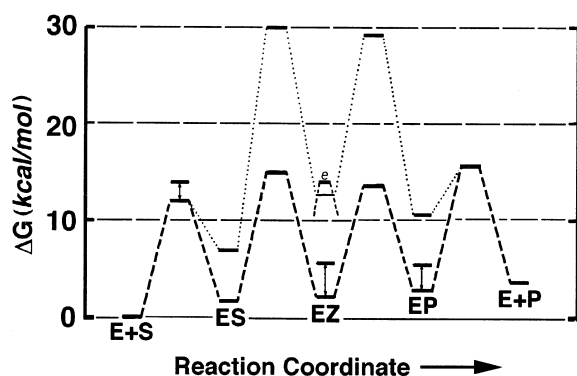


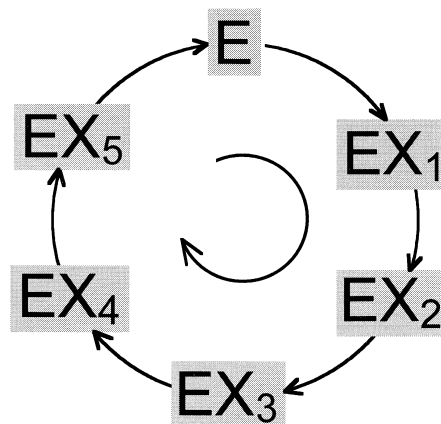
Figure 2. Free energy profile for converting dihydroxyacetone phosphate, the substrate (abbreviated S) and glyceraldehyde 3-phosphate, the product (abbreviated P), with intermediate formation of the enediolate (abbreviated Z). Catalysis occurs either by a free carboxyl group (levels connected by dotted lines) or by triose-phosphate isomerase (levels connected by dashed lines). The vertical arrows show the limits of those states that are less well defined as a result of uncertainty in the experimental data. The transition state marked "e" refers to the exchange of protons between the solvent and the enzyme-bound enediol intermediate (EZ). Reproduced with permission of the authors and the American Chemical Society.

In Fig. 2, the starting and ending positions for both reactions are the same, but the transition state lies a lower energy for the enzyme-catalyzed process. Their analysis led to the efficiency function, a quantitative expression that describes the effectiveness of a catalyst in accelerating a chemical reaction. The function, which depends on magnitude of the rate constants describing individual steps in the reaction, reaches a limiting value of unity when the reaction rate is controlled by diffusion. For the interconversion of dihydroxyacetone phosphate and glyceraldehyde 3-phosphate, the efficiency function equals 2.5×10^{-11} for a simple carboxylate catalyst and 0.6 for the enzyme-catalyzed process. Albery and Knowles¹ suggest that evolution has produced a nearly perfect catalyst. **See** *Enzyme Energetics (The Case of Proline Racemase); Potential-Energy Profile*

¹W. J. Albery & J. R. Knowles (1977) *Biochemistry* **15**, 5631.

REACTION CYCLE

A cyclical depiction of an enzyme-mediated reaction written to account for the regenerative nature of catalytic processes. The reaction cycle for a typical one-substrate one-product enzyme mechanism may be written as follows:



This reaction cycle has more steps than the simple Michaelis-Menten scheme. Nonetheless, the steady-state rate equations describing these reaction cycles have indistinguishable functions, and one cannot determine the number of intermediary steps by steady-state kinetics alone.

REACTION MECHANISM

A representation of all of the elementary reactions that lead to the overall chemical change being investigated. This representation would include a detailed analysis of the kinetics, thermodynamics, stereochemistry, solvent and electrostatic effects, and, when possible, the quantum mechanical considerations of the system under study. Among many items, this representation should be consistent with the reaction rate's dependence on concentration, the overall stoichiometry, the stereochemical course, presence and structure of intermediate, the structure of the transition state, effect of temperature and other variables, *etc.* **See** *Chemical Kinetics*

REACTION PROGRESS CURVES FOR UNSTABLE ENZYMES

Duggleby¹ provides a lucid account of how one can extract useful kinetic information from reaction progress curves. The nonlinear regression methods allow one to treat many cases, and they account for the fact that, after an enzyme is mixed with its substrate, the catalyzed rate

declines over time as a consequence of substrate utilization and product accumulation. In an earlier report², he also considered the effects of enzyme instability, occurring either spontaneously or as a result of some added agent. Under most circumstances, catalysis practically ceases before the substrate is totally exhausted, such that the amount of substrate remaining is related to the inactivation rate constants for various intermediates on the catalytic pathway. A graphical method for estimating these inactivation rate constants is presented, and expressions for the reaction half-time are given for some certain cases. **See** *Progress Curve Analysis*

¹R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 61.

²R. G. Duggleby (1995) *J. Theoret. Biol.* **123**, 67.

REACTION RATE/VELOCITY

The rate of a chemical reaction, typically expressed as: $v = \Delta[X]/\Delta t$, the change in the molarity of a reactant per unit time. **See** *Chemical Kinetics*

Reaction velocity is related to the molar concentrations of reactant(s) by the rate constant:

(A) For zero-order processes (where k has units of $M s^{-1}$):

$$v = k$$

(B) For first-order processes (where k has units of s^{-1}):

$$v = k[X]$$

(C) For second-order processes (where k has units of $M^{-1}s^{-1}$):

$$v = k[X]^2 \text{ or } k[X][Y]$$

(D) For third-order processes (where k has units of $M^{-2}s^{-1}$):

$$v = k[X]^3 \text{ or } k[X]^2[Y] \text{ or } k[X][Y]^2 \text{ or } k[X][Y][Z]$$

REACTIVATION KINETICS (Temperature Effects)

The study of temperature effects on the reactivation of an enzyme that has been completely unfolded allows one to distinguish between “reactivation” (referring to kinetic analysis exclusively) and “renaturation,” the latter of which would reflect both the refolding transition and the formation of misfolded or aggregated by-products.

For example, the temperature effects on the reconstitution of D-glyceraldehyde-3-phosphate dehydrogenase from the hyperthermophilic eubacterium *Thermotoga maritima* were recently investigated¹. The rate and yield of reactivation exhibited a complex behavior. At low temperatures, no significant regain of activity is seen. Between 35°C and 80°C reactivation exceeds 80%. At higher temperatures, thermal denaturation competes with refolding. Shifting the temperature from 0°C to a temperature equal to or greater than 30°C appears to release a trapped intermediate. High speed sedimentation equilibria of this trapped intermediate at 0°C reveals a nonlinear plot of $\ln c$ versus r^2 . The concentration dependency of reactivation shows that renaturation of the enzyme involves consecutive folding and association steps. At high enzyme concentrations, association reaches its diffusion-controlled limit so that the overall reaction is fully determined by the structure formation at the monomer level. Assuming that the tetrameric enzyme is the only active species, the authors were able to determine the corresponding first-order and second-order rate constants ($k_1 = 2.3 \times 10^{-4}s^{-1}$ for folding and $k_2 = 8.3 \times 10^3 M^{-1}s^{-1}$ for association).

¹V. Rehder & R. Jaenicke (1992) *J. Biol. Chem.* **267**, 10999.

REAGENT PURITY

Because enzymes are often inhibited/activated by low concentrations of metal ions, organic impurities, substrate or product analogs, and other effectors, it is incumbent that the investigator always be concerned with the purity of all reagents used, including water. For most of the common reagents used in a laboratory, purity standards based on the latest ACS specifications are normally sufficient. Researchers should always be aware of particular commercial lot specifications. Even at low concentrations, heavy metals are known to significantly inhibit some enzymes, and for some systems, further purification steps are mandatory. The investigator should always exercise caution with regards to the specification claims provided by the commercial supplier. It might prove useful to incubate the enzyme with a mixture containing all reaction components (minus the substrate(s)) for different times, followed by a standard assay protocol, to detect any significant time-dependent change in enzyme activity. **See also** *Substrate Purity*; *Water Purity*; *Enzyme Purity*

REDOX-ACTIVE AMINO ACIDS

Hydroxylated amino acid residues that participate in redox-active enzyme reactions. **See** *Topaquinone*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Precursors of quinone-tanning: Dopa-containing proteins, **258**, 1; isolation of 2,4,5-trihydroxyphenylalanine quinone (Topa quinone) from copper amine oxidases, **258**, 20; spectrophotometric detection of Topa quinone, **258**, 34; model studies of Topa quinone: synthesis and characterization of Topa quinone derivatives, **258**, 39; catalytic aerobic deamination of activated primary amines by a model for the quinone cofactor of mammalian copper amine oxidases, **258**, 53; detection of reaction intermediates in Topa quinone enzymes, **258**, 69; mass spectrometric studies of the primary sequence and structure of bovine liver and serum amine oxidase, **258**, 90; cloning of mammalian Topa quinone-containing enzymes, **258**, 114; isolation of active site peptides of lysyl oxidase, **258**, 122; resonance Raman spectroscopy of quinoproteins, **258**, 132; redox-cycling detection of dialyzable pyrroloquinoline quinone (PQQ) and quinoproteins, **258**, 140; tryptophan tryptophylquinone (TTQ) in bacterial amine dehydrogenases, **258**, 149; model studies of cofactor tryptophan tryptophylquinone (TTQ), **258**, 164; detection of intermediates in tryptophan tryptophylquinone (TTQ) enzymes, **258**, 176; biogenesis of pyrroloquinone quinone from ^{13}C -labeled tyrosine, **258**, 227; X-ray crystallographic studies of cofactors in galactose oxidase, **258**, 235; spectroscopic studies of galactose oxidase, **258**, 262; use of rapid kinetics methods to study the assembly of the diferri-tyrosyl radical cofactor of *E. coli* ribonucleotide reductase, **258**, 278; tyrosyl radicals in photosystem II, **258**, 303; role of tryptophans in substrate binding and catalysis by DNA photolyase, **258**, 319; glycyl free radical in pyruvate formate lyase: synthesis, structure characteristics, and involvement in catalysis, **258**, 343; characterization of a radical intermediate in the lysine 2,3-aminomutase reaction, **258**, 362; role of oxidized amino acids in protein breakdown and stability, **258**, 379.

REDOX POTENTIAL

The reduction-oxidation potential (typically expressed in volts) of a compound or molecular entity measured with an inert metallic electrode under standard conditions against a standard reference half-cell. Any oxidation-reduction reaction, or redox reaction, can be divided into two half-reactions, one in which a chemical species undergoes oxidation and one in which another chemical species undergoes reduction. In biological systems the standard redox potential is defined at pH 7.0 versus the hydrogen electrode and partial pressure of dihydrogen of 1 bar.

REDUCED CONCENTRATION

A normalization parameter used in treating ligand binding equilibria to convert two extensive variables, $K_{\text{dissociation}}$ and substrate concentration, into a parameter whose value is related to the fractional saturation of ligand binding sites. For the simple Michaelis-Menten treatment, $v = V_{\text{max}}/[1 + K_{\text{m}}/[S]]$, if R is the reduced

concentration parameter ($R = [S]/K_{\text{m}}$), then $v = V_{\text{max}}/[1 + R^{-1}]$. The extent of substrate saturation of two Michaelis-Menten enzymes with different K_{m} values will be identical if their R values are the same. For example, if $R = 1$, then $v/V_{\text{max}} = 0.5$ for both enzymes. Likewise, R values of 5 and 10 yield respective velocities of $0.833V_{\text{max}}$ and $0.909V_{\text{max}}$.

Reduced concentrations are also useful in the Monod-Wyman-Changeux cooperativity model, where $\alpha = [F]/K_{\text{r}}$ and $c\alpha = [F]/K_{\text{t}}$. This makes polynomial functions simpler to handle. For example, if ligand F binds exclusively to the R-state, then the ligand F saturation function, Y_{F} , for an n -site protein equals $(1 + \alpha)^{n-1}/[\mathcal{L} + (1 + \alpha)^n]$, where $\mathcal{L}$ is the allosteric constant. Similarly, $\beta = [I]/K_{\text{i}}$ and $\gamma = [A]/K_{\text{a}}$, where $[I]$ and $[A]$ are the inhibitor and activator concentrations, respectively. In this case, the effects of I and/or A on the ligand F saturation function Y_{F} are also easily expressed as: Y_{F} equals $(1 + \alpha)^{n-1}/[\mathcal{L}(1 + \gamma)^n/(1 + \beta)^n + (1 + \alpha)^n]$. Reduced concentrations have less utility for the Koshland-Némethy-Filmer ligand-induced conformational change model, because all the equilibrium constants for ligand binding need not be identical.

REDUCED MASS

A quantity, μ , used in collision theory for the collision of two molecules having masses m_{A} and m_{B} . It is equal to $m_{\text{A}}m_{\text{B}}/(m_{\text{A}} + m_{\text{B}})$. **See** *Hooke's Law Spring*; *Kinetic Isotope Effects*

REDUCING AGENT (or Reductant)

An agent that brings about a reduction of some other molecular entity. Hence, the reductant becomes oxidized during the course of the reaction. **See** *Reduction*

REDUCTIO AD ABSURDUM

A systematic method in logic for disproving a proposition by demonstrating how the application of the proposition results in absurd, self-contradictory conclusions (hence the Latin expression "reduction to absurdity"). It is also referred to as the rule of indirect proof.

REDUCTION

1. The transfer of one or more electrons to a molecular entity. The term is synonymous to electronation. 2. A less general (and earlier, historically) definition was the

loss of one or more oxygen atoms from a molecular entity. **See** *Oxidation; Electrode Kinetics*

REFERENCE REACTION

A nonenzymic reaction used to evaluate the magnitude of the catalytic rate enhancement achieved by a corresponding enzyme-catalyzed reaction. For example, Radzicka and Wolfenden¹ reported that orotic acid is decarboxylated with a $t_{1/2}$ of 78 million years (or, about 2.5×10^{15} seconds) at room temperature in neutral aqueous solution, as suggested by the kinetics determined in sealed quartz tubes maintained at elevated temperatures. Thus, based on the maximal rate of the extremely proficient enzyme orotidine-5'-phosphate decarboxylase, the rate enhancement is estimated to be somewhere around 10^{17} . Based on a transition-state binding model, Radzicka and Wolfenden¹ estimated the intrinsic binding energy of the altered substrate in its transition state corresponds to a dissociation constant lower than 5×10^{-24} M! There are limitations in the use of reference reactions for this purpose², and the chief concern relates to the possibility that two related reactions (or even the same reaction) may proceed by different mechanisms. Moreover, there can be changes in the molecularity of the nonenzymic and enzymic processes. If a covalent intermediate forms in the enzymic reaction, there may simply be no appropriate cognate nonenzymic reaction. Finally, the order of a multistep process may not be the same for both reactions, even though the reactions are otherwise fundamentally similar. **See** *Catalytic Proficiency*

¹A. Radzicka & R. Wolfenden (1995) *Science* **267**, 90.

²W. P. Jencks (1975) *Adv. Enzymol.* **43**, 373.

REGIOSELECTIVITY

1. Any reaction in which a bond-making or bond-breaking step exhibits a preference in a direction over all other directions. Regioselectivity^{1,2} is described as being complete or partial if the product exhibits 100% or $x\%$, respectively, preference of one site *versus* another site.

2. The ability of an enzyme to direct its catalysis toward a specific functional group in a substrate containing other identical or nearly identical group(s)^{3,4}. For example, catechol *O*-methyltransferase will catalyze the methylation of both hydroxyl groups of catechol, albeit at different rates. The effect of pH on the isotope effects of 3-methylation *vs.* 4-methylation suggests that the rate-limiting step for the more rapid 3-methylation changes

with more alkaline pH values whereas methyl transfer is still rate-determining for 4-methylation.

Regioselectivity can also have stereochemical issues associated with the catalytic event. For example, glutamine synthetase will catalyze amide bond formation using β -glutamate as an alternative substrate to produce each of the stereoisomeric β -glutamine products, albeit at different rates.

¹A. Hassner (1968) *J. Org. Chem.* **33**, 2684.

²D. L. Adams (1992) *J. Chem. Ed.* **69**, 451.

³S. E. Wu, W. P. Huskey, R. T. Borchardt & R. L. Schowen (1984) *J. Amer. Chem. Soc.* **106**, 5762.

⁴F. Takusagawa, M. Fujioka, A. Spies & R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 1.

REGULATION OF ENZYME ACTIVITY

The control via activation or inhibition of the rate(s) of an enzyme-catalyzed reaction(s). This control includes the increase or decrease in the stability or half-life of the enzyme(s). There are many different means by which control can be achieved. These include: 1. Substrate availability and reaction conditions (*e.g.*, pH, temperature, ionic strength, lipid interface activation); 2. Magnitude of $V_{\max}$ and K_m values; 3. Activation (particularly, feedforward activation); 4. Isozyme formation; 5. Compartmentalization and channeling; 6. Oligomerization/polymerization; 7. Feedback inhibition and cooperativity (particularly, allosterism and/or hysteresis); 8. Covalent modification; and 9. Gene regulation (induction & repression)

RELATIVE BIOLOGICAL EFFECTIVENESS

A ratio that assesses the biological effectiveness of absorbed radiation doses with respect to different types and energies of ionizing radiation. It is equal to the absorbed dose of a particular radiation divided by the absorbed dose of a standard radiation required to produce identical biological effects in a given organ, tissue, or organism.

RELATIVE DENSITY

The relative density, symbolized by d , is a unitless parameter equal to the density, r , of a substance at a given temperature, divided by the density of a standard, $\rho^\ominus$. Thus, $d = r/\rho^\ominus$. The standard density, for liquids and solids, is usually the density of pure water at the temperature of water's maximum density.

RELATIVE SUPERSATURATION (RS or RSS)

The ratio of the instantaneous solute concentration c to the solute's solubility s , where the latter is the solute concentration in equilibrium with its crystalline or precipitated phase. Hence, $RS = c/s$, and a supersaturated solution experiences a thermodynamic driving force ($\Delta G = -RT \ln[RS]$). A supersaturated solution will remain as a metastable state, because crystallization or precipitation requires a mechanism for relieving the supersaturated condition (e.g., nucleation or addition of crystallite/precipitate). **See** *Biom mineralization*

RELAXATION

The adjustment of a system of linked chemical reactions to a new state of equilibrium in response to the abrupt addition/removal of energy to/from the system. In temperature-jump relaxation methods, energy is added by electrical discharge of a capacitor which results in ohmic heating of ions suddenly moving toward their respective electrodes. In pressure-jump methods, the sudden rupture of a foil diaphragm causes depressurization, and the system responds to a loss of energy. In concentration correlation analysis, the chaotic movement of molecules leads to an increase or decrease in the statistically average number of molecules in a volume element, and the system relaxes in response to the perturbation from thermodynamic equilibrium. **See** *Chemical Kinetics*; *Fast Reaction Techniques*

RELAXATION PERIOD FOR PRECIPITATION

The time t that must elapse before a supersaturated solution reaches a quasi-stationary concentration¹. In dilute aqueous solutions, t is typically around 10^{-8} seconds, but for some solutes this period has proven to be much greater. For example, Dunning² found that the relaxation period for concentrated sucrose solutions is about 100 hours. **See** *Biom mineralization*

¹A. E. Nielsen (1964) *Kinetics of Precipitation*, pp. 20-23, Pergamon Press, Oxford.

²W. J. Dunning (1949) *Disc. Faraday Soc.* **5**, 79.

RELAXATION TIME

The period of time that must elapse for an exponentially decreasing variable to fall from its original value X_{initial} to X_{initial}/e (or, approximately $0.3679X_{\text{initial}}$). In 1866, J. C. Maxwell was the first to use the term "time of

relaxation" to represent the period needed for the elastic force of a fluid to decay to X/e , where X is the initial value of the imposed force. **See** *Fast Reaction Techniques*

RENATURATION

The return of a biomacromolecule back to its natural and active conformation. The kinetics of such processes often exhibit significant aspects of cooperativity, as well as being aided, physiologically, by other macromolecules.

RENIN

This enzyme [EC 3.4.23.15], also known as angiotensin-forming enzyme and angiotensinogenase, catalyzes the hydrolysis of the Leu—Leu bond in angiotensinogen to generate angiotensin I. It belongs to the peptidase family A1.

T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.

M. A. Ondetti & D. W. Cushman (1982) *Ann. Rev. Biochem.* **51**, 283.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Angiotensin I radioimmunoassay, **80**, 436-438; assay, **80**, 435-439; distribution, **80**, 427-429; fluorometric assay, **80**, 436, 438, 439; hog [amino acid composition, **80**, 441; isoelectric point, **80**, 440; kinetics, **80**, 439; molecular weight, **80**, 440; optimum pH, **80**, 440; properties, **80**, 440; purification, **80**, 431-433]; human [amino acid composition, **80**, 441; isoelectric point, **80**, 440; kinetics of activation, **80**, 439; molecular weight, **80**, 440; optimum pH, **80**, 440; properties, **80**, 440; purification, **80**, 433-435; stability, **80**, 435]; physiological function, **80**, 427; substrate specificity, **80**, 439-441; active site, **241**, 255-256; amino acid sequence, **241**, 216; pH of optimum human renin, **241**, 214-215; inhibitors, screening for HIV-1 protease inhibitors, **241**, 318-321; substrate specificity, **241**, 279.

REORGANIZATION ENERGY

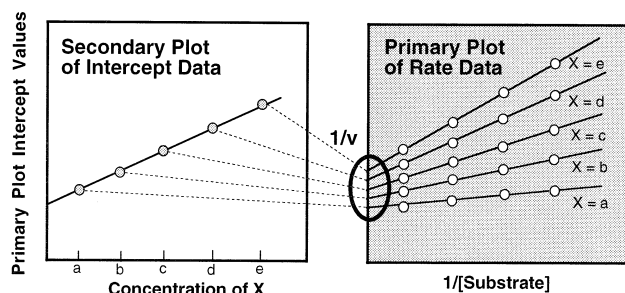
Symbolized by λ , the reorganization energy of a one-electron transfer reaction is that energy needed for all structural adjustments, not only in the two reactants but in the neighboring solvent molecules as well, required for the two reactants to assume the correct configuration needed to transfer the sole electron. **See** *Intrinsic Barrier*; *Marcus Equation*

REPLOTS

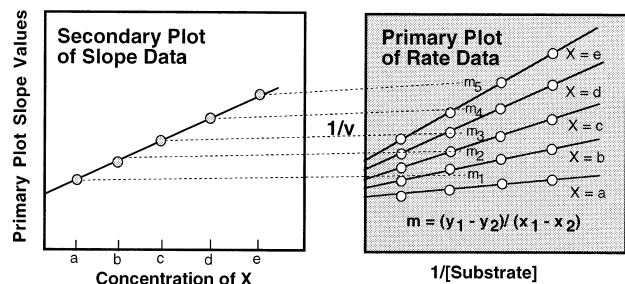
Secondary plots of kinetic data are used to obtain various rate constants and other kinetic parameters such as K_m and V_{max} . To simplify the analysis, one chooses an algebraic transform of the rate equation that allows the observed data to be graphed in a linear format.

For example, consider the double-reciprocal plot of a two-substrate enzyme ($1/v$ vs. $1/[A]$ at different, constant concentrations of the other substrate B). A replot of the primary plot would be either a plot of the slope or the vertical intercept as a function of $1/[B]$.

Intercept Replot:



Slope Replot:



In the examples above, the secondary plots utilized the slopes or intercepts of the original plot. However, replots are secondary plots for any functional dependency using data obtained from a primary graphing procedure. Secondary replots can also be used with inhibition studies. In these cases, the slope or intercept of a double-reciprocal plot is graphed as a function of the inhibitor concentration.

RESOLUTION

The separation (as well as the corresponding procedure) of a racemic mixture into its two enantiomeric components^{1,2}. There are several methods available, including:

1. Mechanical Separation of Crystals. The first instance of resolution was by L. Pasteur who was able to resolve crystals of sodium ammonium tartrate (which recrystallizes in two distinct, nonsuperimposable forms below 27°C). Although this procedure is rarely used, one might be able to seed a racemic solution resulting in only one

enantiomer crystallizing³. In a related fashion, the addition of a chiral reagent can result in enantioselective recrystallization⁴.

2. Formation of Diastereoisomers. The most common method of resolving a racemic mixture, also discovered by Pasteur, is the generation of diastereoisomers by reacting the mixture with an enantiomerically-pure reagent. The resulting two diastereoisomers will have different physical properties that permit separation (e.g., via solubility, fractional recrystallization, gas chromatography, HPLC, etc.). The purified diastereoisomer will now be used to regenerate the optically pure enantiomer (hence, the first step of reacting with the chiral reagent has to be reversible). If the racemic mixture has one or more carboxyl groups, it is a common procedure to react the mixture with an optically pure base (for example, *S*-brucine). Other reagents are available for other functional moieties. If the structure of the racemic mixture does not contain an available functional group, it may be possible to generate such a group and then reverse that process after the separation.

3. Chromatographic Techniques. If a chromatographic column consists of a chiral substance, it may be possible to resolve a racemic mixture. This is a very useful protocol and several chiral materials are now commercially available.

4. Enzymatic Reactions. Most enzymes are quite enantiomerically selective. Hence, reaction of a racemic mixture with the appropriate enzyme will destroy one enantiomer, leaving the other optically pure.

5. Reactivity Techniques. Enantiomers will react with chiral substances at different rates. Hence, it is possible to achieve at least partial resolution of the mixture when using chiral reagents⁵.

¹J. Jacques, A. Collett & S. H. Wilen (1981) *Enantiomers, Racemates, and Resolutions*, Wiley, New York.

²S. H. Wilen, A. Collett & J. Jacques (1977) *Tetrahedron* **33**, 2725.

³R. M. Secor (1963) *Chem. Rev.* **63**, 297.

⁴I. Weissbuch, L. Addadi, Z. Berkovitch-Yellin, E. Gati, S. Weinstein, M. Lahav & L. Leiserowitz (1983) *J. Amer. Chem. Soc.* **105**, 6615.

⁵L. Addadi, Z. Berkovitch-Yellin, I. Weissbuch, M. Lahav, & L. Leiserowitz (1986) *Top. Stereochem.* **16**, 1.

RESONANCE RAMAN SPECTROSCOPY

A spectroscopic technique that is associated with the enhancement of Raman line intensities upon photon absorption in the electronic spectral range corresponding to an absorption peak. **See** *Raman Spectroscopy*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Charge transfer transitions, **246**, 433, 438; data collection, **258**, 137-138; detectors for, **232**, 207; ferredoxin, **246**, 450-454; hemerythrin, **246**, 456-457; hemocyanin, **246**, 457-460; hemoglobin, **246**, 455-456; hemoglobin quaternary conformational changes, **232**, 58; high-potential iron protein, **246**, 453-454; instrumentation [excitation sources, **246**, 426-429; Fourier transform spectrometers, **246**, 425-426; grating spectrometers, **246**, 424-425; multichannel spectrometers, **246**, 424-425; sampling devices (flow cell, **246**, 430; low temperature, **246**, 430-432; NMR tube, **246**, 429-430; wavelength selection, **246**, 424-426)]; intensity of bands, **246**, 423, 438; intensity of scattered radiation, **246**, 420; isotopic substitution and band assignment, **258**, 139; laser-induced changes in sample monitoring, **258**, 137; laser selection, **258**, 136; low-temperature spectroscopy, **246**, 432; methylamine dehydrogenase, **258**, 138-139; multiple chromophore excitation, **246**, 439; picosecond, kinetic hole-burning experiment, **232**, 221-225; polarizability tensor component, **246**, 421-422; porphyrins, **246**, 435-437; as probe of geminate recombination, **232**, 212-228; protein concentration requirements, **246**, 387, 389; quinoprotein sample preparation [concentration, **258**, 135-136; derivatization, **258**, 133-135]; resonance enhancement, **246**, 382-383, 391, 417, 423; rubredoxin, **246**, 447-449; sample requirements, **246**, 391, 417; scattering mechanism, **246**, 418-423; sensitivity, **258**, 132, 137; Topa quinone peptides, **258**, 31-32, 34, 139; ultra-violet spectra of proteins, **246**, 386-387.

RESONANCE STABILIZATION ENERGY

The energy released by a molecular entity as a consequence of resonance, the magnitude of which equals the difference between the actual molecule's energy and the energy of the least energetic canonical form. Because the isolated canonical form does not exist factually, the resonance stabilization energy can only be estimated.

The following constitutes a brief description of how one makes effective use of the resonance method to assign the most likely resonance-stabilized structure(s): (1) Resonance structures must have the same number of paired electrons, and those that typically have identical locations for each pair are said to be the strongest contributors to the overall hybrid. (2) Ionic structures can be significant contributors if they contain elements of significantly different electronegativity. (3) The relative energies of each resonance structure can be estimated from bond energies, the presence of steric distortion,

and the presence of electron-attracting ionic structures. (4) When there are two or more contributing structures of identical or nearly equal energy, then resonance stabilization is greatest, and the overall hybrid will resemble these major contributors.

RESPIRATORY BURST OXIDASE

This FAD- and heme-containing oxidase catalyzes the NADPH-dependent conversion of two molecules of molecular oxygen to produce two molecules of superoxide. The enzyme plays a major role in destroying bacteria that are phagocytosed by neutrophils, eosinophils, and monocytes. **See also** *NADPH Oxidase*

C. Lamb & R. A. Dixon (1997) *Ann. Rev. Plant Physiol. & Mol. Biol.* **48**, 251.

S. J. Chanock, J. E. Benna, R. M. Smith & B. M. Babior (1994) *J. Biol. Chem.* **269**, 24519.

B. M. Babior (1992) *Adv. Enzymol.* **65**, 49.

B. M. Shapiro & P. B. Hopkins (1991) *Adv. Enzymol.* **64**, 291.

RESPIRATORY EXCHANGE RATIO

A ratio of the net output (in moles or volume at standard temperature and pressure per unit time) of CO₂ divided by the net uptake (in identical units) of O₂ at a given site.

RESTRICTION ENZYMES

Type I site-specific deoxyribonucleases [EC 3.1.21.3], also known as type I restriction enzymes, catalyze the ATP-dependent endonucleolytic cleavage of DNA to give random double-stranded fragments with terminal 5'-phosphates, also producing ADP and orthophosphate. These enzymes have an absolute requirement for ATP (or dATP) and S-adenosyl-L-methionine and recognize specific short DNA sequences, cleaving at sites remote from the recognition sequence. Type II site-specific deoxyribonucleases [EC 3.1.21.4], also known as type II restriction enzymes, catalyze the magnesium ion-dependent endonucleolytic cleavage of DNA to give specific double-stranded fragments with terminal 5'-phosphates. These enzymes recognize specific short DNA sequences and cleave either within, or at a short specific distance from, the recognition site. Type III site-specific deoxyribonucleases [EC 3.1.21.5], also known as Type III restriction enzymes, catalyze the endonucleolytic cleavage of DNA to give specific double-stranded fragments with terminal 5'-phosphates. There is an absolute requirement for ATP; however, the ATP is not hydrolyzed. S-Adenosyl-L-methionine stimulates the reac-

tion, but is not absolutely required. These enzymes recognize specific short DNA sequences and cleave a short distance away from the recognition sequence. **See also** *Deoxyribonucleases*

- N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.
 A. S. Bhagwat (1992) *Meth. Enzymol.* **216**, 199.
 P. Modrich (1982) *Crit. Rev. Biochem.* **13**, 287.
 B. Endlich & S. Linn (1981) *The Enzymes*, 3rd ed., **14**, 137.
 R. D. Wells, R. D. Klein & C. K. Singleton (1981) *The Enzymes*, 3rd ed., **14**, 157.
 R. J. Roberts (1976) *Crit. Rev. Biochem.* **4**, 123.

REVERSE TRANSCRIPTASE

This enzyme [EC 2.7.7.49], also known as RNA-directed DNA polymerase, DNA nucleotidyltransferase (RNA-directed), and revertase, catalyzes the RNA-template-directed extension of the 3'-end of a DNA strand by one deoxynucleotide at a time: n deoxynucleoside triphosphate to produce n pyrophosphate (or, diphosphate) and DNA $_n$. The enzyme cannot initiate a DNA chain *de novo* and requires a DNA or RNA primer. **See also** *Viral Polymerases*

- R. A. Katz & A. M. Skalka (1994) *Ann. Rev. Biochem.* **63**, 133.
 C. M. Joyce & T. A. Steitz (1994) *Ann. Rev. Biochem.* **63**, 777.
 A. M. Skalka & S. P. Goff (eds.) (1993) *Reverse Transcriptase*, Cold Spring Harbor Press, Cold Spring Harbor.
 J. A. Peliska & S. J. Benkovic (1992) *Science* **258**, 1112.
 I. M. Verma (1981) *The Enzymes*, 3rd ed., **14**, 87.

RHAMNOSE ISOMERASE

This enzyme [EC 5.3.1.14] catalyzes the interconversion of L-rhamnose and L-rhamnulose.

- E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

α -L-RHAMNOSIDASE

This enzyme [EC 3.2.1.40] catalyzes the hydrolysis of terminal nonreducing α -L-rhamnose residues in α -L-rhamnosides.

- P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

RHAMNULOSE-1-PHOSPHATE ALDOLASE

This enzyme [EC 4.1.2.19] catalyzes the reversible conversion of L-rhamnulose 1-phosphate to dihydroxyacetone phosphate (or, glyceraldehyde phosphate) and (S)-lactaldehyde.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.
 D. S. Feingold & P. A. Hoffee (1972) *The Enzymes*, 3rd ed., **7**, 303.

RHIZOPUSPEPSIN

This enzyme [EC 3.4.23.21], also known as *Rhizopus* aspartic proteinase, catalyzes the hydrolysis of peptide bonds in proteins with a broad specificity similar to that of pepsin A, preferring hydrophobic residues at P₁ and P_{1'}. It clots milk and activates trypsinogen. It does not hydrolyze the Gln4—His5 bond, but it does hydrolyze the His10—Leu11 and the Val12—Glu13 peptide bonds in the B chain of insulin. The enzyme, a member of the peptidase family A1, is isolated from the zygomycete fungus *Rhizopus chinensis*. A similar endopeptidase is found in *R. niveus*. **See** *Pepsin & Pepsinogen*

- T. D. Meek (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 327.
 D. R. Davies (1990) *Ann. Rev. Biophys. Biophys. Chem.* **19**, 189.

RHO (ρ)

1. Symbol for density. (*i.e.*, mass density). 2. Symbol for charge density. 3. Symbol for radiant energy density.

RHODANESE

This enzyme [EC 2.8.1.1] catalyzes the reaction of thio-sulfate with cyanide to produce sulfite and thiocyanate. A number of other sulfur-containing compounds can act as substrates.

- D. P. Kelly & A. P. Wood (1994) *Meth. Enzymol.* **243**, 501.
 J. Westley (1973) *Adv. Enzymol.* **39**, 327.

RHODOPSIN

The following are recent reviews on the molecular and physical properties of this centrally important visual protein. A light-sensitive chromoprotein that serves as the photon acceptor for rod cells, consisting of a protein (opsin) and an aldehyde derivative of vitamin A (11-*cis*-retinal). This cofactor forms a imine with the ϵ -amino group of a specific opsin lysyl residue. The primary photo event involves conversion of 11-*cis*-retinal to all-*trans*-retinal. Conformational changes in opsin upon *cis*-to-*trans* isomerization trigger signal transduction events culminating in a nerve impulse.

- W. D. Hoff, K.-H. Jung & J. L. Spudich (1997) *Ann. Rev. Biophys. Biomol. Struct.* **26**, 223.
 V. R. Rao & D. D. Oprea (1996) *Ann. Rev. Biophys. Biomol. Struct.* **25**, 287.
 B. J. Litman & D. C. Mitchell (1996) in *Rhodopsin and G-Protein Linked Receptors* (A. G. Lee, ed.), p. 1, JAI Press, Stamford, CT.
 H. G. Khorana (1992) *J. Biol. Chem.* **267**, 1.
 J. Nathans (1992) *Biochemistry* **31**, 4923.
 P. A. Hargrave & J. H. McDowell (1992) *FASEB J.* **6**, 2323.
 R. R. Birge (1990) *Ann. Rev. Phys. Chem.* **41**, 683.

RHODOPSIN KINASE

This enzyme [EC 2.7.1.125] catalyzes the reaction of ATP with rhodopsin to produce ADP and phosphorylated rhodopsin. The kinase acts on the bleached or activated form of rhodopsin.

X. Y. Zhao, K. Palczewski & H. Ohguro (1995) *Biophys. Chem.* **56**, 183.

A. M. Edelman, D. K. Blumenthal & E. G. Krebs (1987) *Ann. Rev. Biochem.* **56**, 567.

RHO-VALUE (ρ -Value)

A measure of the susceptibility to any substitution effect of reaction series for *families* of organic compounds as modified by σ -constants in an empirical $\rho\sigma$ -equation. The ρ -value, usually with subscripts and/or superscripts designating particular systems. For example, for comparing the effect of substituents at the *meta*- and *para*-positions of certain reactions of aromatic compounds, $\log(k_x/k_H) = \rho\sigma_x$ where σ_x is a constant characteristic of substituent X and its position.

Reactions with positive ρ -values are accelerated, provided that the σ -constant is also positive. **See also** *Hammett Equation*

RIBITOL-5-PHOSPHATE CYTIDYLYLTRANSFERASE

This enzyme [EC 2.7.7.40], also known as CDP-ribitol pyrophosphorylase, catalyzes the reaction of CTP with D-ribitol 5-phosphate to produce CDP-ribitol and pyrophosphate (or, diphosphate).

D. R. D. Shaw (1966) *Meth. Enzymol.* **8**, 244.

RIBITOL-5-PHOSPHATE 2-DEHYDROGENASE

This enzyme [EC 1.1.1.137] catalyzes the reaction of D-ribitol 5-phosphate with NAD(P)⁺ to produce D-ribulose 5-phosphate and NAD(P)H.

L. Glaser (1966) *Meth. Enzymol.* **8**, 240.

RIBOFLAVINASE

This enzyme [EC 3.5.99.1] catalyzes the hydrolysis of riboflavin to produce ribitol and lumichrome.

C. S. Yang & D. B. McCormick (1971) *Meth. Enzymol.* **18(B)**, 571.

RIBOFLAVIN KINASE

This enzyme [EC 2.7.1.26], also known as flavokinase, catalyzes the reaction of ATP with riboflavin to produce ADP and FMN.

E. P. Anderson (1973) *The Enzymes*, 3rd ed. **9**, 49.

D. B. McCormick (1971) *Meth. Enzymol.* **18(B)**, 544.

RIBOFLAVIN SYNTHASE

This enzyme [EC 2.5.1.9] catalyzes the conversion of two 6,7-dimethyl-8-(1-D-ribityl)lumazine to produce riboflavin and 4-(1-D-ribitylamino)-5-amino-2,6-dihydroxypyrimidine.

A. Bacher, S. Eberhardt, M. Fischer, S. Mörtl, K. Kis, K. Kugelbrey, J. Scheuring & K. Schott (1997) *Meth. Enzymol.* **280**, 389.

RIBONUCLEASE

This enzyme [often abbreviated by RNase; EC 3.1.27.5], also known as pancreatic ribonuclease, ribonuclease A, and ribonuclease I, catalyzes the endonucleolytic cleavage to 3'-phosphomononucleotides and 3'-phosphooligonucleotides ending in Cp or Up with 2',3'-cyclic phosphate intermediates. RNA hydrolysis occurs by general acid-base catalysis¹. During the enzyme-catalyzed reaction, nucleoside-2',3'-cyclic phosphodiester accumulate, indicating a two-step reaction: first, facilitated attack of the 2'-hydroxyl of RNA which leads to the formation of the cyclic intermediate; second, hydrolysis of the cyclic phosphodiester to yield the 3'-monoester. Now classical studies, chiefly from the laboratory of Moore & Stein, established that His-12 and His-119 play critically important roles, the former acting in the first step as a general base, and the latter serving as a general acid to protonate the leaving group. The second step is the reverse of the first, and the enzyme returns to its original protonation state upon completing the reaction cycle. *Aspergillus oryzae* RNase T1 partially digests RNA into short fragments by preferential cleavage after guanine residues, and pancreatic RNase catalyzes bond scission after pyrimidine residues. **See** *Pseudorotation*; *Diethylpyrocarbonate*

N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.

X. Parés, M. V. Nogués, R. de Llorens & C. M. Cuchillo (1991) *Essays in Biochem.* **26**, 89.

V. Shen & D. Schlessinger (1982) *The Enzymes*, 3rd ed. **15**, 501.

W. W. Cleland (1977) *Adv. Enzymol.* **45**, 373.

F. M. Richards & H. W. Wyckoff (1971) *The Enzymes*, 3rd ed., **4**, 647.

¹In general acid catalysis, the reaction rate increases because the transition state for the reaction is lowered by proton transfer from a Brønsted acid; in general base catalysis, the reaction rate increases by virtue of proton abstraction by a Brønsted base.

RIBONUCLEASE F

This enzyme [EC 3.1.27.7] catalyzes the endonucleolytic cleavage of an RNA precursor into two. The cleavage takes place between a cytosine and an adenine residue, leaving 5'-hydroxyl and 3'-phosphate groups.

- N. H. Williams (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 543.
 R. A. Katz & A. M. Skalka (1994) *Ann. Rev. Biochem.* **63**, 133.
 C. M. Joyce & T. A. Steitz (1994) *Ann. Rev. Biochem.* **63**, 777.
 J. M. Whitcomb & S. H. Hughes (1992) *Ann. Rev. Cell Biol.* **8**, 275.

RIBONUCLEASE II

Ribonuclease II [EC 3.1.13.1], also called exoribonuclease II, catalyzes the exonucleolytic cleavage of the polynucleic acid, preferring single-stranded RNA, in the 3'- to 5'-direction to yield 5'-phosphomononucleotides. The enzyme processes 3'-terminal extra-nucleotides of monomeric tRNA precursors, following the action of ribonuclease P. Similar enzymes include RNase Q, RNase BN, RNase PIII, and RNase Y. Ribonuclease T₂ [EC 3.1.27.1] is also known as ribonuclease II.

- V. Shen & D. Schlessinger (1982) *The Enzymes*, 3rd ed., **15**, 501.

RIBONUCLEASE III

This enzyme [EC 3.1.26.3], also known as RNase O and RNase D, catalyzes the endonucleolytic cleavage of RNA to 5'-phosphomonoesters. The enzyme cleaves multimeric tRNA precursors at the spacer region and is also involved in the processing of precursor rRNA, hnRNA, and early T7-mRNA. This enzyme can also act on double-stranded DNA.

- R. Kole & S. Altman (1982) *The Enzymes*, 3rd ed., **15**, 469.
 J. J. Dunn (1982) *The Enzymes*, 3rd ed., **15**, 485.

RIBONUCLEASE IV

This enzyme [EC 3.1.26.6], also known as endoribonuclease IV and poly(A)-specific ribonuclease, catalyzes the endonucleolytic cleavage of poly(A) to fragments terminated by 3'-hydroxyl and 5'-phosphate groups. Oligonucleotides are formed with an average chain length of ten.

- V. Shen & D. Schlessinger (1982) *The Enzymes*, 3rd ed., **15**, 501.

RIBONUCLEASE P

This system [EC 3.1.26.5] catalyzes the endonucleolytic cleavage of RNA, removing the 5'-extra-nucleotide from the tRNA precursor. This step is essential for tRNA processing. It generates the 5'-termini of mature tRNA

molecules. A similar activity is observed with RNase P3.
See Catalytic RNA

- J. A. Grasby (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 563.
 K. R. Groom, Y. L. Dang, G.-J. Gao, Y. C. Lou, N. C. Martin, C. A. Wise & M. J. Morales (1996) *Meth. Enzymol.* **264**, 86.
 S. Altman (1989) *Adv. Enzymol.* **62**, 1.
 R. Kole & S. Altman (1982) *The Enzymes*, 3rd ed., **15**, 469.

RIBONUCLEOTIDE REDUCTASE

Ribonucleotide reductase (diphosphate) [EC 1.17.4.1], also known as ribonucleoside-diphosphate reductase, catalyzes the reaction of a 2'-deoxyribonucleoside diphosphate with oxidized thioredoxin and water to produce a ribonucleoside diphosphate and reduced thioredoxin. This system requires the presence of iron ions and ATP. Ribonucleotide reductase (triphosphate) [EC 1.17.4.2], also known as ribonucleoside-triphosphate reductase, catalyzes the reaction of a 2'-deoxyribonucleoside triphosphate with oxidized thioredoxin and water to produce a ribonucleoside triphosphate and reduced thioredoxin. In this case, cobalt and ATP are cofactors.

- B. T. Golding & W. Buckel (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 239.
 B. G. Fox (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 261.
 J. E. Penner-Hahn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 439.
 J. M. Bollinger, Jr., W. H. Tong, N. Ravi, B. H. Huynh, D. E. Edmondson & J. Stubbe (1995) *Meth. Enzymol.* **258**, 278.
 P. Reichard (1993) *J. Biol. Chem.* **268**, 8383.
 W. W. Wells, Y. Yang & T. L. Deits (1993) *Adv. Enzymol.* **66**, 149.
 M. Fontecave, P. Nordlund, H. Eklund & P. Reichard (1992) *Adv. Enzymol.* **65**, 147.
 E. J. Brush & J. W. Kozarich (1992) *The Enzymes*, 3rd ed., **20**, 317.
 J. B. Howard & D. C. Rees (1991) *Adv. Protein Chem.* **42**, 199.
 J. Stubbe (1990) *Adv. Enzymol.* **63**, 349.
 M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

RIBOSE-5-PHOSPHATE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.35], also known as ADP-ribose phosphorylase, catalyzes the reaction of ADP with D-ribose 5-phosphate to produce orthophosphate and ADP-ribose.

- W. R. Evans (1971) *Meth. Enzymol.* **23**, 566.

RIBOSE-5-PHOSPHATE EPIMERASE

This enzyme [EC 5.3.1.6] (also known as phosphopentose isomerase, phosphoriboisomerase, and ribose 5-phos-

phate isomerase) catalyzes the interconversion of D-ribose 5-phosphate and D-ribulose 5-phosphate. Both D-ribose 5-diphosphate and D-ribose 5-triphosphate can also function as substrates.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

RIBOSEPHOSPHATE PYROPHOSPHOKINASE

This enzyme [EC 2.7.6.1], also known as phosphoribosyl pyrophosphate synthetase, catalyzes the reaction of ATP with D-ribose 5-phosphate to produce AMP and 5-phospho- α -D-ribose 1-diphosphate. dATP can also function as a substrate.

J. J. Villafranca & F. M. Raushel (1980) *Ann. Rev. Biophys. Bioengin.* **9**, 363.

M. A. Becker, K. O. Raivio & J. E. Seegmiller (1979) *Adv. Enzymol.* **49**, 281.

A. S. Mildvan (1979) *Adv. Enzymol.* **49**, 103.

W. N. Keley & J. B. Wyngaarden (1974) *Adv. Enzymol.* **41**, 1.

R. L. Switzer (1974) *The Enzymes*, 3rd ed., **10**, 607.

D-RIBULOSE-1,5-BISPHOSPHATE CARBOXYLASE/OXYGENASE

Probably the most abundant naturally occurring catalyst, this enzyme [EC 4.1.1.39], also known as “rubisco”, catalyzes the reaction of D-ribulose 1,5-bisphosphate with carbon dioxide to produce two 3-phospho-D-glycerate. The enzyme can also use dioxygen as a substrate instead of carbon dioxide, producing 3-phospho-D-glycerate and 2-phosphoglycolate.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

M. Stitt & U. Sonnewald (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 341.

S. Krömer (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 45.

F. C. Hartman & M. R. Harpel (1994) *Ann. Rev. Biochem.* **63**, 197.

D. R. Geiger & J. C. Servaites (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 235.

L. A. Kleczkowski (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 339.

F. C. Hartman & M. R. Harpel (1993) *Adv. Enzymol.* **67**, 1.

M. H. O’Leary (1992) *The Enzymes*, 3rd ed., **20**, 235.

M. I. Siegel, M. Wishnick & M. D. Lane (1972) *The Enzymes*, 3rd ed., **6**, 169.

RIBULOSE-5-PHOSPHATE 3-EPIMERASE

This enzyme [EC 5.1.3.1] (also known as phosphoribulose epimerase, erythrose-4-phosphate epimerase, and pentose-5-phosphate 3-epimerase) catalyzes the interconversion of D-ribulose 5-phosphate and D-xylulose 5-phosphate. The enzyme can also act on D-erythrose 4-phosphate.

M. E. Tanner & G. L. Kenyon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 7.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

RIBULOSE-5-PHOSPHATE 4-EPIMERASE

This enzyme [EC 5.1.3.4] catalyzes the interconversion of L-ribulose 5-phosphate and D-xylulose 5-phosphate.

E. Adams (1976) *Adv. Enzymol.* **44**, 69.

L. Glaser (1972) *The Enzymes*, 3rd ed., **6**, 355.

RIESKE IRON-SULFUR PROTEIN

A mitochondrial respiratory chain iron-sulfur protein containing a [2Fe–2S] cluster with two coordinated cysteinyl sulfur ligands and two coordinated histidyl imidazole ligands. The term is also applied to similar proteins isolated from photosynthetic organisms and microorganisms that contain similarly coordinated [2Fe–2S] clusters.

RITCHIE EQUATION

A linear free energy equation^{1,2} for reaction between nucleophiles and certain large organic cations (*e.g.*, triarylmethyl cations, aryltropylium cations, *etc.*). Thus, $\log k_N = \log k_o + N_+$ where k_N is the rate constant for the reaction of a particular cation with a given nucleophile in a given solvent, k_o is the corresponding rate constant for the same cation with water in water (*i.e.*, the solvent is water), and N_+ is the cation-independent parameter that is characteristic for that nucleophile system.

¹C. D. Ritchie (1972) *Acc. Chem. Res.* **5**, 348.

²C. D. Ritchie (1986) *Can. J. Chem.* **64**, 2239.

RNA LIGASE

This enzyme [EC 6.5.1.3], also known as polyribonucleotide synthase (ATP), catalyzes the reaction of ATP with [ribonucleotide]_n and [ribonucleotide]_m to produce [ribonucleotide]_(n+m), AMP, and pyrophosphate (or, diphosphate). This enzyme converts linear RNA to a circular form by the transfer of the 5′-phosphate to the 3′-hydroxyl terminus.

P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

O. C. Uhlenbeck & R. I. Gumport (1982) *The Enzymes*, 3rd ed., **15**, 31.

RNA POLYMERASES

RNA-directed RNA polymerase [EC 2.7.7.48] catalyzes the RNA-template-directed extension of the 3′-end of an RNA strand by one nucleotide at a time: thus, n

nucleoside triphosphate produces n pyrophosphate and RNA_n . This enzyme can initiate a chain *de novo*.

DNA-directed RNA polymerase [EC 2.7.7.6] catalyzes the DNA-template-directed extension of the 3'-end of an RNA strand by one nucleotide at a time: thus, n nucleoside triphosphate generate RNA_n and n pyrophosphate. The enzyme can initiate a chain *de novo*. Three forms of the enzyme have been distinguished in eukaryotes on the basis of sensitivity of α -amanitin and the type of RNA synthesized. **See also** *Replicase*

S. M. Uptain, C. M. Kane & M. J. Chamberlain (1997) *Ann. Rev. Biochem.* **66**, 117.

S. Adhya, ed. (1996) *Meth. Enzymol.*, vols. **273** and **274**.

M. E. Kang & M. E. Dahmus (1995) *Adv. Enzymol.* **71**, 41.

S. S. Sastry, H. P. Spielmann & J. E. Hearst (1993) *Adv. Enzymol.* **66**, 85.

J. E. Coleman (1992) *Ann. Rev. Biochem.* **61**, 897.

D. A. Erie, T. D. Yager & P. H. von Hippel (1992) *Ann. Rev. Biophys. Biomol. Struct.* **21**, 379.

P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

M. J. Chamberlin (1982) *The Enzymes*, 3rd ed., **15**, 61.

M. Chamberlin & T. Ryan (1982) *The Enzymes*, 3rd ed., **15**, 87.

M. K. Lewis & R. R. Burgess (1982) *The Enzymes*, 3rd ed., **15**, 110.

R. Losick & J. Pero (1976) *Adv. Enzymol.* **44**, 165.

P. Chambon (1974) *The Enzymes*, 3rd ed., **10**, 261.

M. J. Chamberlin (1974) *The Enzymes*, 3rd ed., **10**, 333.

ROENTGEN

Symbolized by R, the roentgen is the international unit of exposure dose. It is equal to that quantity of radiation that will produce a charge of 2.58×10^{-4} coulombs on all the ions of one sign, when all the electrons released in a volume of air (at STP) of a mass of one kilogram are completely stopped.

ROENTGEN-EQUIVALENT-MAN (rem)

A unit of dose equal to the amount of ionizing radiation that produces in humans the same biological effect as one rad of X-rays or gamma rays. One rem is equal to 0.01 sievert.

ROTATIONAL CORRELATION TIME

The period required for reduction of the fraction of some rotationally correlated property to $1/e$ (or 0.367) of its initial value. **See** *Correlation Function*

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Electron paramagnetic resonance [effect on line width, **246**, 596-598; motional narrowing spin label spectra, **246**, 595-598; slow motion spin label spectra, **246**, 598-601]; helix-forming peptides, **246**, 602-605; proteins, **246**, 595; Stokes-Einstein relationship, **246**, 594-595; temperature dependence, **246**, 602, 604.

ROTATIONAL DIFFUSION

The haphazard rotational motions of molecules or one or more segments of a molecule. This diffusional process strongly influences the mutual orientation of molecules (particularly large ones) as they encounter each other and proceed to form complexes. Rotational diffusion can be characterized by one or more relaxation times, τ_i , describing the motion of a molecule or segment of volume, V , in a medium of viscosity, η , as shown in the following equation:

$$\tau_i = \eta V / (RT)$$

where R and T are the universal gas constant and the absolute temperature, respectively. Because the rotating unit is rarely spherically symmetric in size and electronic charge, its rotational diffusional behavior is likely to be a composite of several tau values. Rod-like structures are especially likely to exhibit such behavior. The same is true for molecules having regions or segments that rotate relatively independently of each other. Under favorable conditions where the rotating unit is characterized by a fluorescence excited state lifetime of suitable magnitude, one can use time-resolved fluorescence anisotropy measurements to infer the time scale of rotational motions¹⁻³. For example, the rotational dynamics of the purified dicyclohexylcarbodiimide-sensitive H^+ -ATPase reconstituted into phospholipid vesicles and of the ATPase coreconstituted with the proton pump bacteriorhodopsin have been probed by time-resolved phosphorescence emission anisotropy⁴. These investigators used the phosphorescent probe erythrosin isothiocyanate to covalently label the γ -polypeptide of the ATPase before reconstitution. The rotational correlation time was independent of the viscosity of the external medium, but increased significantly as the microviscosity of the membrane increased. Thus, the rotational correlation times are a measure of the enzyme's motion in the membrane. At 4°C, the 100–180 μs correlation time was unaffected by the addition of substrates and the presence of a transmembrane pH gradient, suggesting further that molecular rotation does not appear to play an important role either in enzyme catalysis or ion pumping. **See** *Fluorescence*

¹G. Weber (1971) *J. Chem. Phys.* **55**, 2399.

²J. Ygueribide, H. Epstein & L. Stryer (1970) *J. Mol. Biol.* **51**, 573.

³J. R. Lakowicz & G. Weber (1980) *Biophys. J.* **32**, 591.

⁴K. M. Musier-Forsyth & G. G. Hammes (1990) *Biochemistry* **29**, 3236.

**rRNA (ADENINE-6-)-
METHYLTRANSFERASE**

This enzyme [EC 2.1.1.48] catalyzes the reaction of *S*-adenosyl-L-methionine with rRNA to produce *S*-adenosyl-L-homocysteine and rRNA containing *N*⁶-methyladenine. The enzyme will also methylate 2-aminoadenosine to 2-methylaminoadenosine.

S. J. Kerr & E. Borek (1973) *The Enzymes*, 3rd ed., **9**, 167.

**rRNA (GUANINE-N¹)-
METHYLTRANSFERASE**

This enzyme [EC 2.1.1.51] catalyzes the reaction of *S*-adenosyl-L-methionine with rRNA to produce *S*-adenosyl-L-homocysteine and rRNA containing *N*¹-methylguanine.

S. J. Kerr & E. Borek (1973) *The Enzymes*, 3rd ed., **9**, 167.

 R_s

A parameter used to assess the degree of cooperativity exhibited by an enzyme¹⁻³. R_s equals the ratio of $[S]_{0.9}/[S]_{0.09}$; that is, the ratio of the substrate concentration needed for 90% saturation divided by the substrate concentration needed for 10% saturation. For a normal, hyperbolic, noncooperative curve, R_s equals 81. Thus, positively cooperative systems will have an R_s ratio less than 81, whereas negatively cooperative systems will have values larger than 81. The ratio is insensitive to the shape of the curve and does not address any questions concerning the substrate concentration range between the 10% and 90% points.

¹D. E. Koshland, Jr., G. Némethy & D. Filmer (1966) *Biochemistry* **5**, 365.

²M. E. Kirtley & D. E. Koshland, Jr. (1967) *J. Biol. Chem.* **242**, 4192.

³K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

**RUBBER
cis-POLYPRENYLC/TRANSFERASE**

This enzyme [EC 2.5.1.20], also known as rubber allyltransferase and rubber transferase, catalyzes the reaction of poly-*cis*-polyprenyl diphosphate (or, rubber particles) with isopentenyl diphosphate to produce a poly-*cis*-

polyprenyl diphosphate longer by one C₅ unit and pyrophosphate (or, diphosphate).

R. A. Gibbs (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 31.

E. D. Beytía & J. W. Porter (1976) *Ann. Rev. Biochem.* **45**, 113.

B. L. Archer & E. G. Cockbain (1969) *Meth. Enzymol.* **15**, 476.

RUBREDOXIN

A class of iron-sulfur proteins lacking acid-labile sulfur, but similar in function to the ferredoxins. The iron center is coordinated by four sulfur-containing ligands, usually from cysteinyl residues. Where known, these proteins function as electron carriers.

RUBREDOXIN:NAD⁺ REDUCTASE

This iron-dependent enzyme [EC 1.18.1.1], also known as rubredoxin reductase, catalyzes the reaction of reduced rubredoxin and NAD⁺ to produce oxidized rubredoxin and NADH.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

RUNGE-KUTTA ALGORITHM

A systematic stepwise method for numerical integration of a rate expression [indeed, of any differential equation $y' = f(x,y)$ with an initial value $y(x_0) = y_0$] to determine the time evolution of the rate process. **See also** *Numerical Computer Methods; Numerical Integration; Stiffness; Gear Algorithm*

 R_v

A ratio used to assess the degree of cooperativity exhibited by an enzyme. It is equal to the true $V_{\max}$ value (typically extrapolated from the high-substrate-concentration end of a double-reciprocal plot) divided by the apparent $V_{\max}$ value obtained from extrapolating the asymptote in the low-substrate-concentration portion of the double-reciprocal plot. For a noncooperative system, R_v will equal one; positively cooperative systems will have values greater than one; and negatively cooperative systems will have values less than one^{1,2}. This method requires good estimates of the asymptotes.

¹G. R. Ainslie, Jr., J. P. Shill & K. E. Neet (1972) *J. Biol. Chem.* **247**, 7088.

²K. E. Neet (1980) *Meth. Enzymol.* **64**, 139.

S

$S_{0.5}$

A useful parameter in the investigation of cooperative enzymes. It is the concentration of substrate needed for one-half of the maximal saturation of the cooperative enzyme.

$S_{0.9}/S_{0.1}$ (or, R_S)

A kinetic parameter, introduced by Koshland¹, to indicate the ratio of substrate concentrations needed to achieve reaction velocities equal to $0.1V_{\max}$ and $0.9V_{\max}$. For an enzyme obeying the Michaelis-Menten equation, $S_{0.9}/S_{0.1}$ equals 81, indicating that such enzymes exhibit modest sensitivity of reaction rate relative to changes in the substrate concentration. Many positively cooperative enzymes have $S_{0.9}/S_{0.1}$ values between five and ten, indicating that they can be turned on or off over a relatively narrow substrate concentration range.

¹D. E. Koshland, Jr. (1969) *Curr. Top. Cell. Reg.* **1**, 1.

SAAM

An abbreviation commonly used for “simulation, analysis, and modeling” program, a set of digital computer algorithms based on the compartmental analysis approach introduced by Berman¹⁻³. Kinetic data are initially analyzed to determine the temporal and mass transport relationships between compartments by employing an interactive computer program for simulating kinetic models along with fitting procedures to evaluate the goodness of fit of the model to the experimental data. Each compartment can be related to one or more reacting chemical species. Hensley *et al.*⁴ provides an excellent example of the application of compartmental analysis in the characterization of *Bam*HI endonuclease action.

¹M. Berman (1971) in *Advances in Medical Physics* (J. S. Laughlin &

E. W. Webster, eds.), p. 279, Second International Conference on Medical Physics, Inc., Boston.

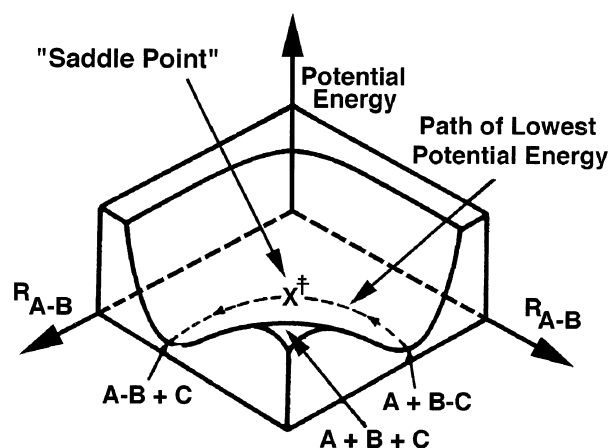
²M. Berman & M. F. Weiss (1978) *SAAM-27 Manual*, DHEW Publ. No. (NIH) 78-180.

³SAAM/CONSAAM software and manuals are available free of charge from the Laboratory of Mathematical Biology, Room 4B56/Building 10, National Cancer Institute, National Institutes of Health, Bethesda, MD 20892. For a listing of computer requirements as well as the down-loadable SAAM/CONSAAM software, the internet address is: <http://www.saam.nci.nih.gov/>

⁴P. Hensley, G. Nardone & M. E. Wastney (1992) *Meth. Enzymol.* **210**, 391.

SADDLE POINT

The point of minimal potential energy in the trajectory of reactants to products in a chemical reaction. A reaction's saddle point (or col¹) indicates the geometry and energy of reactants as they approach and pass the transition state of a reaction.



Graphical representation of the saddle point (here marked with an X) for the transfer of atom B as the substance A-B reacts with another species, C. Potential energy is plotted in the vertical direction. Note also that the surface resembles a horse saddle, with the horn of the saddle closest to the observer. As drawn here, the dissociation to form three discrete species ($A + B + C$) requires much more energy than that needed to surmount the path that includes the saddle point. A two-dimensional “slice” through a saddle point diagram is typically called a reaction-coordinate diagram or potential-energy profile.

The saddle point best represents the behavior of so-called elementary reactions occurring directly between small molecules; an example is the collision of gases and

ions in crossed-beam, or translational kinetic experiments. Computer programs allow investigators to determine reaction path and transition state properties, to perform a normal mode analysis at a specified geometry, to calculate a trajectory based on a data set of reactant coordinates and momenta, to ascertain the minimum energy geometry for a specified potential energy surface, to calculate a trajectory beginning with one or two reactants, and to calculate a trajectory starting at a saddle point and proceeding toward reactants or products.

Multistep or multistage reactions in aqueous solutions are far more complicated than the process depicted above. Nonetheless, the conceptual picture of a saddle point allows one to comprehend the spatial organization and energy constraints influencing chemical reactivity.

¹A col is a high pass through a mountain range or a saddle-like depression in the crest of a ridge.

SALICYLATE 1-MONOOXYGENASE

This FAD-dependent enzyme [EC 1.14.13.1], also known as salicylate hydroxylase, catalyzes the reaction of salicylate with NADH and dioxygen to produce catechol, NAD⁺, carbon dioxide, and water.

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 S. Yamamoto & Y. Ishimura (1991) *A Study of Enzymes* **2**, 315.
 B. Entsch (1990) *Meth. Enzymol.* **188**, 138.
 V. Massey & P. Hemmerich (1975) *The Enzymes*, 3rd ed., **12**, 191.

SAMPLE SPACE

A statistical term for the list or set (denoted as S) of all possible outcomes of a random experiment. In the case of a perfectly balanced, infinitely thin coin, there are only two outcomes, Heads and Tails, in the sample space. **See** *Random Experiment; Statistics (A Primer)*

SAPONIFICATION

The reaction of an ester (most often a fatty acid ester) with a base to produce a carboxylate anion and an alcohol.

SARCOSINE DEHYDROGENASE

This FMN-dependent enzyme [EC 1.5.99.1] catalyzes the reaction of sarcosine with an acceptor and water to produce glycine, formaldehyde, and the reduced acceptor.

- V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.
 M. H. Stipanuk (1986) *Ann. Rev. Nutr.* **6**, 179.

SARCOSINE OXIDASE

This FAD-dependent enzyme [EC 1.5.3.1] catalyzes the reaction of sarcosine with dioxygen and water to produce glycine, formaldehyde, and hydrogen peroxide.

- V. Schirch (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 211.

SATURATION

1. The condition wherein the concentration of a reactant, ligand, product, or effector is significantly greater than the dissociation constant of that same entity. 2. A solution becomes saturated with a particular substance when that substance has been dissolved to its maximum extent. 3. The process (or degree of) by which the number of π -bonds in a molecular entity decrease (that is, the removal of double and triple bonds and increase of σ -bonds).

SATURATION KINETICS

A term used to refer to any chemical process whose rate depends upon saturation of a binding site. Rate saturation is observed in enzyme kinetics, metabolic transport, host-guest reactions, and even heterogenous catalysis such as hydrogenation on metallic surfaces of platinum and nickel.

SCALAR COUPLING RELAXATION

A relaxation (and relaxation time) that contributes to the observed relaxation times (T_1 and T_2) in NMR experiments. Scalar coupling relaxation is the dominant relaxation process in ³¹P(¹⁷O) NMR.

SCATCHARD PLOT

A graphical procedure first used by Scatchard¹ to determine the stoichiometry and dissociation constant(s) for ligand binding to a macromolecule. Winzor and Sawyer² have pointed out that the Scatchard method^{1,3} can be misleading with respect to the derived parameters. In fact, there are many computer programs^{4,5} that permit one to analyze binding data without using graphical transformations like the Scatchard treatment.

Because the Scatchard equation is so widely used in the molecular life sciences, we show two different ways to

derive the equation and we shall consider several special types of binding interactions.

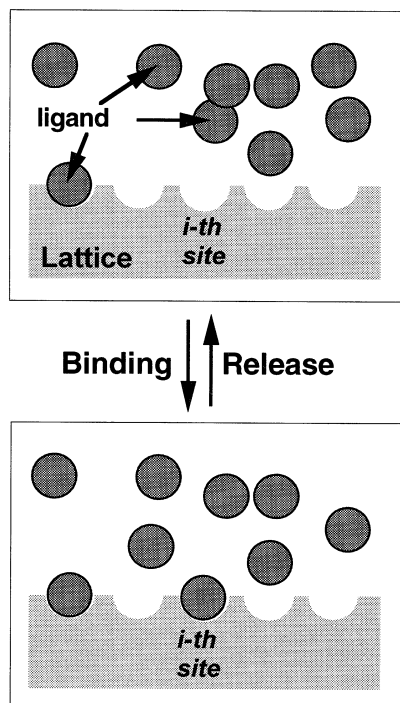
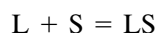


Figure 1. Ligand binding to a lattice of noninteracting binding sites.

Lattice Binding Model. Consider the diagram shown in Fig. 1 for a hypothetical array of binding sites on a protein, nucleic acid, organelle, or cell surface. Binding of small molecules (or ligands) can take place at any of the sites on the lattice. We assume: (a) that all sites are independent, such that occupancy at one site (marked here as the *i*-th site) in no way affects occupancy of other sites, even those immediately adjacent to the *i*-th site; (b) that all sites are capable of undergoing saturation by the ligand; (c) that the concentration of occupied sites can be experimentally determined using chemical analysis, but frequently relying on the use of radioisotopically labeled ligands or an appropriate spectroscopic technique that yields a change in signal to ligand occupancy. We can now define an equilibrium relationship:



where L signifies the ligand and S is the site on the lattice. Because all sites are equivalent and noninteracting, the total concentration of sites can be denoted as C_S , the total concentration of sites with ligand bound can be written as C_B , and the concentration of free ligand as

C_F . The equilibrium dissociation constant can then be expressed as follows:

$$K = C_F(C_S - C_B)/C_B$$

which can be rearranged to yield:

$$C_B/C_F = (C_S - C_B)/K = (C_S/K) - (C_B/K)$$

and a plot of C_B/C_F versus C_B will be linear with the following parameters: (a) the slope equals $-(1/K)$; (b) the vertical-axis intercept equals C_S/K ; and (c) the horizontal-axis intercept is C_S . The above expression is called the Scatchard equation, and the corresponding graphical representation is called a Scatchard plot (Fig. 2).

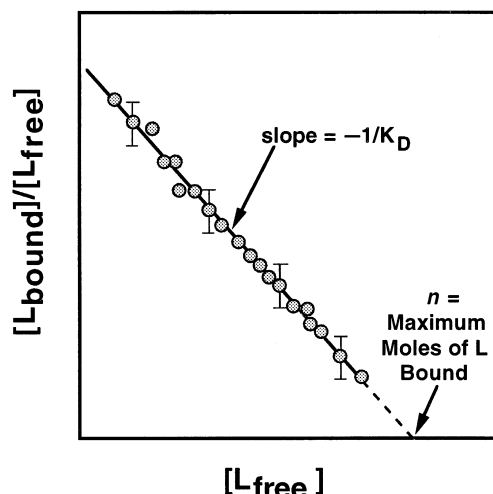


Figure 2. Scatchard plot for independent ligand binding.

How can the experimenter compensate for nonspecific binding of ligand? In addition to these high-affinity ligand binding sites, many macromolecules also bind ligands nonspecifically at one or more other sites, albeit with low affinity. This is often observed for drug binding at membrane receptor sites, but other cases, such as the binding of divalent cations to proteins, are also well documented. What is important to recognize about the Scatchard treatment is that one only considers high-affinity or so-called biospecific ligand binding. If there is an indication of weak secondary site interactions, one can typically correct for this by conducting binding studies with a nonphysiologic analogue. For drugs that act as antagonists or agonists of some physiologic process, this is often accomplished by using a pharmacologically ineffective enantiomer of an optically active drug. Such an analogue would not be expected to occupy the primary ligand sites on a receptor; otherwise, some pharma-

cologic effect would be manifested. Weak binding of ligand frequently shows a linear relation between amount bound and ligand concentration (see lower line in the upper panel of Fig. 3).

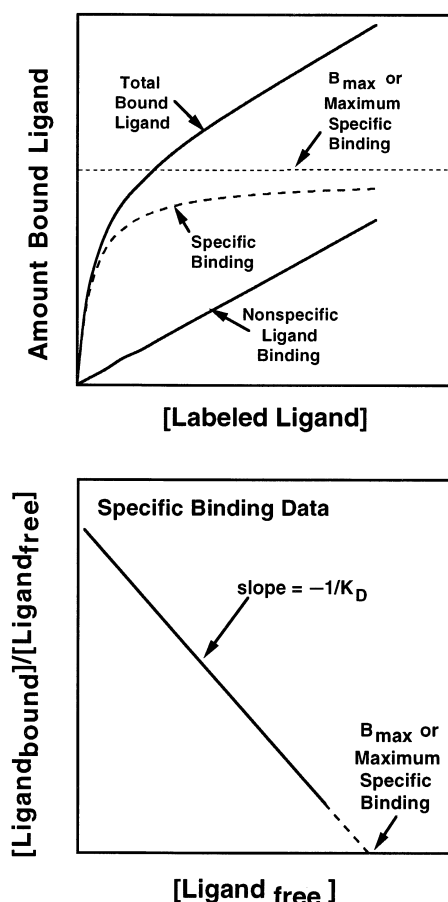


Figure 3. Treatment of specific and nonspecific binding of a ligand in equilibrium with a finite concentration of receptor sites.

This can be subtracted from the total amount of bound ligand (top curve) to arrive at the specific binding value for each ligand concentration. (Alternatively, in the absence of a pharmacologically ineffective ligand, one can extrapolate the slope of the total bound ligand curve to zero concentration, and then use this curve as the nonspecific binding curve. Of course, this requires one to first construct a linear plot starting at coordinates (0,0) with a slope matching that of the extrapolated line). Only then may one use the specific binding data to construct a Scatchard plot (shown in the lower panel of Fig. 3).

How may one extend the Scatchard treatment to deal with the case of steric hindrance among neighboring sites? This realistic situation occurs often, especially when the

distances between binding sites is not much greater than the molecular size of the ligand or when ligand binding disturbs the structure of its macromolecular binding partner. One such case is the binding of acridine dyes by intercalation between DNA base pairs. Another might be the binding of the dynein motor or a bulky microtubule-associated protein onto the tubulin lattice within a microtubule. In such a case, we can define the concentration of potential sites at theoretical saturation to be C_o ; we can define the fraction of sites actually bound at saturation as b_{ap} ; we define the concentration of free ligand as C_F ; and we can define $b_{ap}C_o$ as the concentration of total sites bound, or C_S . We then write the equilibrium expression:

$$K_{ap} = C_F(C_S - C_B)/C_B = C_F(b_{ap}C_o - C_B)/C_B$$

If we divide the right-hand numerator and denominator by C_o , and if we let r equal C_B/C_o , the above equation becomes

$$K_{ap} = C_F(b_{ap} - r)/r$$

Rearranging this expression yields a modified version of the Scatchard equation:

$$r/C_F = (b_{ap}/K_{ap}) - (r/K_{ap})$$

and a plot of r/C_F versus r will have a slope of $-(1/K_{ap})$ and a horizontal-axis intercept of b_{ap} . If such a plot is carried out for acridine binding to DNA, the value of b_{ap} is about 0.25, indicating that only about one in every four base pairs can accommodate a bound acridine molecule at saturation.

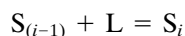
Standard Binding Model. The more commonly recognized form of the Scatchard equation can be written as follows:

$$\nu/[L] = K(n - \nu)$$

where ν is the fractional saturation of n binding sites by ligand L , and K is the dissociation constant. Consider the following sequence of ligand binding interactions for a macromolecule, where M_i represents the i th site on the macromolecule: $M_0 + L \rightleftharpoons M_1$; $M_1 + L \rightleftharpoons M_2$; ...; $M_{n-1} + L \rightleftharpoons M_n$. The binding expression described above can be derived by defining ν as follows:

$$\nu = \frac{\sum_{i=0}^n i[M_i]}{\sum_{i=0}^n [M_i]}$$

where the concentration of species are shown in brackets. We can write the equilibrium for L binding to the i -th site (S_i) as follows:



Now let ν_i be defined as the saturation function for the i -th site:

$$\nu_i = \frac{[\text{Ligand bound at Site}_i]}{[\text{Site}_{(i-1)}] + [\text{Ligand bound at Site}_i]}$$

$$\nu_i = [\text{Site}_i] / \{[\text{Site}_{(i-1)}] + [\text{Site}_i]\}$$

and (a) dividing the right-hand numerator and denominator by $[\text{Site}_{(i-1)}]$, and (b) using the expression $([S_i]/[S_{(i-1)}] = [L]/K_i)$ from the dissociation constant expression (*i.e.*, $K_i = [S_{(i-1)}][L]/[S_i]$), we get

$$\nu_i = ([L]/K_i) / \{1 + ([L]/K_i)\}$$

To convert the binding at the i -th site to the binding at all n sites, we use the expression:

$$\nu = \sum n\nu_i = n([L]/K_i) / \{1 + ([L]/K_i)\}$$

which directly rearranges to the familiar form of the Scatchard equation:

$$\nu = \frac{n[L]/K_i}{1 + ([L]/K_i)}$$

Upon rearrangement, we get the familiar Scatchard equation:

$$\nu/[L] = (n/K_i) - (\nu/K_i)$$

where a plot of $(\nu/[L])$ versus ν has a slope of $-(1/K_i)$ and a horizontal-axis intercept is the number of saturable sites n .

What about the likely case that there are two classes of binding interactions—one strong and one weak? To treat this case, we write two separate saturation functions, one for each class of sites:

$$\text{Class 1: } r_1/[L] = (b_1 - r_1)/K_1$$

$$\text{Class 2: } r_2/[L] = (b_2 - r_2)/K_2$$

where r_1 and r_2 are defined as the fraction of ligand bound to sites 1 and 2, respectively, and b_1 and b_2 are the fractions of bound sites that are type-1 and type-2. We can define r_T to be the sum of r_1 and r_2 , and the analytical technique generally allows us to measure total ligand bound, or $r_T[L_{\text{total}}]$. Thus, we can write $r_T/[L]$ as follows:

$$r_T/[L] = \{(b_1 - r_1)/K_1\} + \{(b_2 - r_2)/K_2\}$$

or, if we substitute $(r_T - r_2)$ for r_1 , we obtain

$$r_T/[L] = \{[b_1 - (r_T - r_2)]/K_1\} + \{(b_2 - r_2)/K_2\}$$

A plot of $(r_T/[L])$ versus r_T begins on the vertical axis at the intercept (equal to $b_1/K_1 + b_2/K_2$) and has an initial slope (*i.e.*, at r_T values ≈ 0) equal to $-(1/K_1)$; this initial linear region extrapolates to a horizontal-axis value of b_1 . As one progresses to even higher ligand concentrations, the curve bows to the right and eventually reaches the horizontal intercept r_T . Thus, on the basis of these experimentally derived values of K_1 and b_1 , one can iteratively fit the entire experimental binding curve.

This behavior is illustrated in Fig. 4 for the case where the first ligand binds with a dissociation constant that is 30 times lower than the equilibrium constant for dissociation of ligand from the second site. The dotted line shows the behavior of a Scatchard plot in the absence of second-site binding; in this case, the dotted line extrapolates to a value of 1.0 on the horizontal axis.

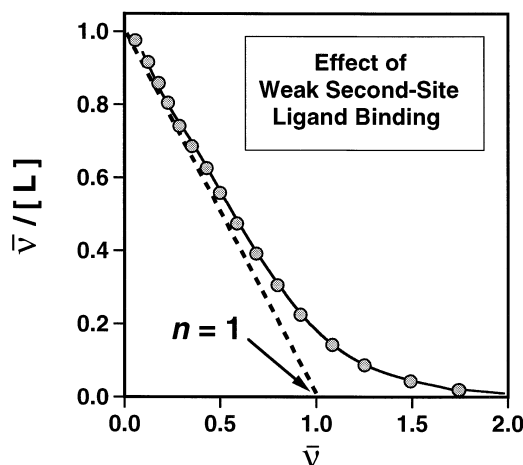


Figure 4. Scatchard plot for a ligand interacting at two different sites on the same protein. See text for details.

Scatchard *et al.*³ found that thiocyanate binding to human serum albumin gave 10 sites with a 1–2 mM dissociation constant, and another 30 with a dissociation constant that was nearly 40 mM.

Finally, with homodimeric proteins, both sites are identical in the absence of any ligand. If a homodimeric protein exhibits a Scatchard plot shaped like that shown in Fig. 4, such an observation is diagnostic for negative cooperativity in ligand binding. The Koshland-Neméthy-Filmer model for cooperativity treats cases where ligand binding at the first site on a homodimer induces a conformational change that reduces the affinity of ligand binding at a second site. Negative cooperativity is not a possibility in the Monod-Wyman-Changeux model for cooperative ligand binding.

¹G. Scatchard (1949) *New York Acad. Sci.* **51**, 660.

²D. J. Winzor & W. H. Sawyer (1995) *Quantitative Characterization of Ligand Binding*, p. 45, Wiley-Liss, New York.

³G. Scatchard, J. S. Coleman & A. L. Shen (1957) *J. Am. Chem. Soc.* **79**, 12.

⁴L. Brand & M. L. Johnson, eds., (1992) *Meth. Enzymol.*, vol. **210**, Academic Press, San Diego.

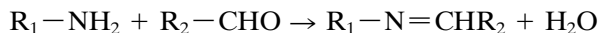
⁵M. L. Johnson & L. Brand, eds., (1994) *Meth. Enzymol.*, vol. **240**, Academic Press, San Diego.

SCAVENGER

Any compound, substance, or material that reacts with (or binds, chelates, or removes from a system) a trace component or reaction intermediate.

SCHIFF BASE (or Schiff's Base)

The product formed by the reaction of an aldehyde with an amine.



Historically, the amine was an aromatic amine but is now generalized to any amine. A Schiff base, also called an aldimine, is formed in the pyridoxal 5-phosphate-dependent aminotransferase reactions.

SCHRÖDINGER EQUATION

A class of partial differential equations first proposed by Erwin Schrödinger in 1926 to account for the so-called quantized wave behavior of molecules, atoms, nuclei, and electrons. Solutions to the Schrödinger equation are wave functions based on Louis de Broglie's proposal in 1924 that all matter has a dual nature, having properties of both particles and waves. These solutions are

of a probabilistic character, and they describe how the structure of atoms, molecules, and closely packed assemblies is altered discontinuously in discrete increments of energy change (called a quantum). While Schrödinger equations offer the opportunity to attain remarkable accuracy, the number of atoms within an enzyme (or even its active site complexes with substrates) greatly limits the accuracy needed to predict changes occurring during catalysis. For this reason, transition state theory of enzymic reactions is typically analyzed by classical mechanics. Nonetheless, the Schrödinger equation predicts that electron hydride as well as proton transfer can occur without surmounting an energy barrier. **See** *Quantum Mechanical Tunneling*

The Schrödinger equation for a one-electron system is:

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

where m is the mass of the electron, E and V are the electron's total and potential energy respectively, h is Planck's constant, and Ψ is the wave function for the electron. Chemical properties are determined by the energies of electrons as well as the relative positions of electrons with respect to the nucleus and other electrons within a molecule, ion, radical, or atom. Solutions to the Schrödinger equation can only be obtained for certain energy values, known as eigenvalues. They represent the allowable quantized states of the system, and for each eigenvalue, there is at least one eigenfunction Ψ .

P. S. C. Matthews (1986) *Quantum Chemistry of Atoms and Molecules*, Cambridge Univ. Press, Cambridge.

SCINTILLATORS

Compounds and materials, including mixtures, which can undergo radioluminescence and are used in the measurement of radioactivity. Characteristics of these substances include a high fluorescence quantum efficiency and a short fluorescence lifetime.

SCISSILE BOND

A covalent bond that is broken during the course of a chemical reaction.

SCREW SENSE OF METAL-NUCLEOTIDE COMPLEXES

The conformation and helical sense of the structure interconnecting the metal-coordinated oxygens and the phosphorus atoms of the nucleotide.

In the Δ series of CrATP, there are two screw-sense isomers, δ (pseudoequatorial) and λ (pseudoaxial). A number of nucleotide-dependent enzymes have been shown to be specific for one of the screw-sense isomers. **See** *Exchange-Inert Metal-Nucleotide Complexes*

D. Dunaway-Mariano & W. W. Cleland (1980) *Biochemistry* **19**, 1496.

D. Dunaway-Mariano & W. W. Cleland (1980) *Biochemistry* **19**, 1506.

W. W. Cleland (1982) *Meth. Enzymol.* **87**, 159.

SECOND

A unit of time (symbolized by s) that serves as one of the seven base SI units, equal to the duration corresponding to 9,192,631,770 times the period for transition between the two hyperfine levels of the ground state of an atom of ^{133}Cs .

SECONDARY ISOTOPE EFFECT

A change in reaction rate that results from isotopic substitution adjacent to the site of bond breaking/making. The classical example in enzymology is the hydrolysis of phenylglucosides by lysozyme^{1,2}, for which the so-called α -deuterium isotope effect (symbolized as $k_{\text{H}}/k_{\text{D}}$) was found to be 1.14. This result was consistent with an expected change in electronic hybridization at positions C-1 and oxygen (within the glucopyranose ring) if a carbonium ion (or carboxonium) intermediate formed during the reaction cycle. **See** *Kinetic Isotope Effect*

¹F. W. Dahlquist, T. Rand-Meir & M. A. Raftery (1968) *Proc. Natl. Acad. Sci. U.S.A.* **61**, 1194.

²L. E. H. Smith, L. H. Mohr & M. A. Raftery (1973) *J. Am. Chem. Soc.* **95**, 7497.

SECOND-ORDER REACTION

With a Single Reaction Component. A reaction of a substance A is said to be second order if the following rate expression applies:

$$\text{Reaction Rate} = v = -d[A]/dt = k[A]^2$$

where k is the second-order rate constant (having units of reciprocal concentration multiplied by reciprocal time) and t is time. Integration of this expression yields $[A]^{-1} = kt + [A_0]^{-1}$ where $[A_0]$ and $[A]$ are the concentrations of the reactant at $t = 0$ and $t = t$, respectively. Thus, a plot of $[A]^{-1}$ vs. t will result in a straight line having a slope¹ of k . If the reaction indeed follows the

above rate expression with only a single reaction component, the second-order rate constant can also be obtained by determining the reaction half-life, $t_{1/2}$. For a second-order reaction, $t_{1/2} = (k[A_0])^{-1}$. Notice that the half-life for a second-order process varies with the initial reactant concentration.

With Two Different Reaction Components. A reaction occurring between two substances, A and B, is said to be second order if the reaction rate agrees with the rate expression¹:

$$\text{Reaction Rate} = v = -d[A]/dt = k[A][B]$$

where t is time and k is the second-order rate constant. As mentioned in the entries on order and molecularity, this second-order rate constant is the bimolecular rate constant for an elementary reaction. Letting $[A_0]$ and $[B_0]$ represent the initial concentrations of the two reactants at $t = 0$, letting $[A]$ and $[B]$ represent the concentrations at $t = t$, and recalling that $[A_0] - [A] = [B_0] - [B]$, integration yields

$$\frac{1}{[B_0] - [A_0]} \ln \left(\frac{[A_0][B]}{[B_0][A]} \right) = kt$$

Plotting $([B_0] - [A_0])^{-1} \ln [(A_0][B]/([B_0][A]))$ vs. t or plotting $\ln([B]/[A])$ vs. t will yield a straight line having slope k . A value for k can also be obtained by substituting successive known values of $[A]$, $[B]$, and t into the above equation, calculate several values for k , and obtain a mean (assuming these values display some constancy). If x represents the decrease in reactant concentration at time t , the equation above becomes

$$\frac{1}{[B_0] - [A_0]} \ln \left(\frac{[A_0]([B_0] - x)}{[B_0]([A_0] - x)} \right) = kt$$

As before, values of k can be calculated using the equation or by graphical analysis (e.g., plotting $\ln([B_0] - x)/([A_0] - x)$ vs. t). If the initial concentrations of A and B are equal (i.e., $[A_0] = [B_0]$), the above equations have zero in the denominator, and are undefined.

If the concentration of the two reacting molecules remain identical throughout the course of reaction, then the rate expression becomes analogous to a second-order reaction with one component. Let $[D_0] = [A_0] + [B_0]$ and let $[D]$ represent the concentrations of A and B

at $t = t$. Then, the integrated rate expression becomes $[D]^{-1} = kt + [D_0]^{-1}$. Once again, plotting $[D]^{-1}$ vs. t will yield a straight line having a slope of k .

There are many examples of second-order reactions: for example, associations, addition reactions, *etc.* The decomposition of gaseous nitrogen dioxide ($2\text{NO}_2 \rightarrow 2\text{NO} + \text{O}_2$) follows the rate law, $-d[\text{NO}_2]/dt = k[\text{NO}_2]^2$. The binding of a hormone to a receptor protein is also second-order.

If the concentration of one of the reactants is much larger than that of the other (for example $[A_0] \gg [B_0]$), then that higher concentration can be treated as a constant as its value will not change much over the time course of the reaction. The rate expression then becomes

$$\ln \left(\frac{[B_0]}{[B_0] - x} \right) = [A_0]kt = k_{\text{obs}}t$$

where k_{obs} is the pseudo-first-order rate constant equal to $k[A_0]$. The second-order rate constant can be obtained by determining k_{obs} at different concentrations of A_0 and plotting k_{obs} vs. $[A_0]$. As Jencks has pointed out, pseudo-rate constants are very sensitive to impurities². It may be necessary to repurify the reactant. Nevertheless, pseudo-rate constants are of value as they provide the investigator with a means of separating variables that affect a chemical reaction mechanism². An example of a pseudo-first-order reaction is the hydrolysis (or, inversion) of

sucrose catalyzed by strong acids ($\text{H}_2\text{O} + \text{sucrose} \xrightarrow{\text{H}^+} \text{D-glucose} + \text{D-fructose}$). This reaction is of historical interest as it is the first reaction studied kinetically. Wilhelmy (1812–1864) used a polarimeter to follow the course of this reaction. The rate expression he reported, in his terminology, was $-dZ/dt = MZS$ where t was time, Z was the sucrose concentration, S was the acid concentration, and M was a constant³. This equation, in modern terminology, becomes $-d[\text{sucrose}]/dt = k[\text{sucrose}][\text{H}_2\text{O}][\text{H}^+]$. Wilhelmy is justly regarded by many as the father of chemical kinetics. **See** *Chemical Kinetics; Order of Reaction; Noyes Equation; Molecularity; Autocatalysis; First-Order Reaction*.

¹J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

²W. P. Jencks (1969) *Catalysis in Chemistry and Enzymology*, McGraw-Hill, New York. [Also available as Jencks, W. P. (1986)

Catalysis in Chemistry and Enzymology, Dover Publications, New York]

³L. Wilhelmy (1850) *Ann. der Physik und Chemie* **81**, 413 and 526.

SEDIMENTATION EQUILIBRIUM ULTRACENTRIFUGATION

An analytical ultracentrifugation method for determining the molecular mass, diffusion coefficient, and/or state of oligomerization of a macromolecule by conducting sedimentation conditions to establish an equilibrium distribution of the macromolecule from the meniscus to the bottom of the observation cell.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Association constant determination, **259**, 444-445; buoyant mass determination, **259**, 432-433, 438, 441, 443, 444; cell handling, **259**, 436-437; centerpiece selection, **259**, 433-434, 436; centrifuge operation, **259**, 437-438; concentration distribution, **259**, 431; equilibration time, estimation, **259**, 438-439; molecular weight calculation, **259**, 431-432, 444; nonlinear least-squares analysis of primary data, **259**, 449-451; oligomerization state of proteins [determination, **259**, 439-441, 443; heterogeneous association, **259**, 447-448; reversibility of association, **259**, 445-447]; optical systems, **259**, 434-435; protein denaturants, **259**, 439-440; retroviral protease, analysis, **241**, 123-124; sample preparation, **259**, 435-436; second virial coefficient [determination, **259**, 443, 448-449; nonideality contribution, **259**, 448-449]; sensitivity, **259**, 427; stoichiometry of reaction, determination, **259**, 444-445; terms and symbols, **259**, 429-431; thermodynamic parameter determination, **259**, 427, 443-444, 449-451.

SEDIMENTATION VELOCITY ULTRACENTRIFUGATION

An analytical ultracentrifugation method for determining the molecular mass, diffusion coefficient, and/or state of oligomerization of a macromolecule by conducting sedimentation conditions that allow the macromolecule to migrate at constant velocity. The sedimentation coefficient s equals the sedimentation rate divided by the relative centrifugal field: $s = dr/dt/\omega^2r$, where s is the sedimentation coefficient (expressed in svedberg units) coefficient, ω is the angular velocity in radians per second, and dr/dt is the sedimentation rate, and r is the radial displacement.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Boundary analysis [baseline correction, **240**, 479, 485-486, 492, 501; second moment, **240**, 482-483; time derivative, **240**, 479, 485-486, 492, 501; transport method, **240**, 483-486]; computation of sedimentation coefficient distribution functions, **240**, 492-497; diffusion effects, correction [differential distribution functions, **240**, 500-501; integral distribution functions, **240**, 501]; weight average sedimentation coefficient estimation, **240**, 497, 499-500.

SELECTION RULE

A rule in quantum chemistry which states when a particular transition is allowed or forbidden.

L-SELENOCYSTEINE

A naturally occurring amino acid that is identical in structure to cysteine, but contains the micronutrient selenium in place of sulfur. The pK_a of the R—SeH moiety is around 5.2, contrasting markedly from the corresponding 8.3 value for cysteine. Hence, at neutral pH selenocysteine is largely R—Se[−], whereas cysteine is largely R—SH. Selenocysteine is an essential component of glutathione peroxidase, an enzyme which protects cellular membranes from peroxidative damage. This amino acid is also present in bacterial formate dehydrogenase and glycine reductase. **See also** *L-Seryl-tRNA(SeC) Selenium Transferase*

T. C. Stadtman (1979) *Adv. Enzymol.* **48**, 1.

L-SELENOCYSTEINE β -LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.4.1.16], also known as selenocysteine reductase, catalyzes the reaction of L-selenocysteine with a reduced acceptor to produce hydrogen selenide, L-alanine, and the acceptor. The enzyme can use dithiothreitol or 2-mercaptoethanol as the reducing agent. Cysteine, serine, or chloroalanine are not alternative substrates for this enzyme.

T. C. Stadtman (1990) *Ann. Rev. Biochem.* **59**, 111.

N. Esaki & K. Soda (1987) *Meth. Enzymol.* **143**, 415.

N. Esaki & K. Soda (1987) *Meth. Enzymol.* **143**, 493.

SELENOPHOSPHATE SYNTHETASE

This magnesium-dependent enzyme [EC 2.7.9.3], also known as selenide,water dikinase and selenium donor protein, catalyzes the reaction of ATP with selenide and water to produce AMP, selenophosphate, and orthophosphate.

T. C. Stadtman (1996) *Ann. Rev. Biochem.* **65**, 83.

SELF-ABSORPTION

Self-absorption occurs when the fluorescence due to some excited molecular entity(ies) is absorbed by other molecular entities of the same species which are in the ground state.

SELF-PROTECTION MECHANISM

A mechanism seen on occasion in affinity labeling experiments when an active site-directed reagent binds to the active site of an enzyme and protects that active site against a bimolecular reaction with a second molecule

of the reagent. Appropriate design of the affinity labeling experiment will curtail this mechanism from occurring¹.

¹B. V. Plapp (1982) *Meth. Enzymol.* **87**, 469.

SELF-QUENCHING

The event which occurs when an excited atom or molecular entity is quenched by an interaction with another atom or molecular entity of the species, which is in the ground state.

SENSITIVITY

1. The capacity to respond to a change in the environment or the influence of some signal.

2. A relationship used in the analysis of enzyme cascade systems. It is symbolized by S (and also referred to as the sensitivity index) and is equal to $8.89[e_{0.5}]/([e_{0.9}] - [e_{0.1}])$ where $[e_{0.5}]$ is the concentration of effector required to attain 50% maximal amplitude, $[e_{0.9}]$ is the concentration of effector required for 90% maximal amplitude, $[e_{0.1}]$ is the concentration of effector required for 10% maximal amplitude, and 8.89 is a normalizing constant used to yield a reference value of 1.00 for a pure hyperbolic binding isotherm¹.

¹P. B. Chock & E. R. Stadtman (1980) *Meth. Enzymol.* **64**, 297.

SEPARATION OF VARIABLES

The process of solving for the integrated form of a differential equation, achieved by rearranging the equation into a form where each side of the resulting expression can be integrated with respect to one of the variables.

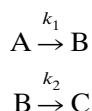
SEQUENTIAL MECHANISMS

Enzyme-catalyzed reaction schemes involving two or more substrates and/or two or more products in which there are no ping pong half-reaction steps. **See** *Multisubstrate Mechanisms*

SERIES REACTIONS

A set of reactions in which the product(s) of at least one reaction serves as a reactant for a subsequent reaction; also referred to as consecutive reactions.

The simplest set of series reactions involves two first-order reactions:



Harcourt and Esson¹ were the first to study such reactions (in their case, the reaction of potassium permanganate with oxalic acid). Esson² was the first to integrate the differential equation. This scheme can be treated exactly by considering the differential equations:

$$d[A]/dt = -k_1[A]$$

$$d[B]/dt = k_1[A] - k_2[B]$$

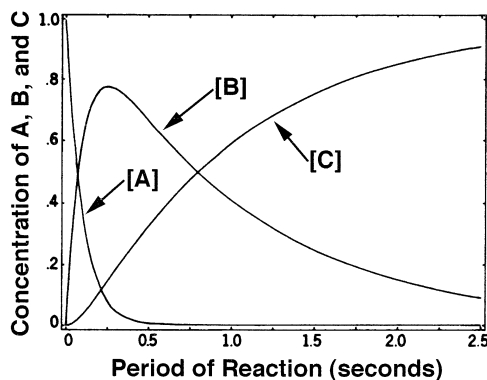
$$d[C]/dt = k_2[B]$$

If $[A_o]$ and $[A]$ are the concentrations of A at $t = 0$ and $t = t$, respectively, the integration of the first equation yields $[A] = [A_o]e^{-k_1t}$. Note that the concentration of A decreases with time exponentially, as would be expected for any first-order process. Substituting this expression for $[A]$ into the second differential equation and assuming $[B_o] = 0$ at $t = 0$, integration yields

$$[B] = [[A_o](k_1/k_2 - k_1)](e^{-k_1t} - e^{-k_2t})$$

To derive an integrated expression for C, two relationships must be used. The three differential equations above tell us that $d[A]/dt + d[B]/dt + d[C]/dt = 0$. And, the conservation relationship provides $[A] + [B] + [C] = [A_o]$ at any time t . Assuming $[B_o]$ and $[C_o]$ are zero at $t = 0$, the $[C] = \{[A_o] - [A] - [B]\}$ or

$$[C] = [A_o]\{1 + (k_1 - k_2)^{-1}(k_2e^{-k_1t} - k_1e^{-k_2t})\}$$

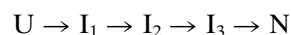


Time evolution of a series first-order process wherein $k_1 = 0.1 \text{ min}^{-1}$ and $k_2 = 0.05 \text{ min}^{-1}$.

As can be seen in the plot, the concentration of A decreases exponentially with time. The concentration of B increases to a maximum value and then decreases. The concentration of C increases gradually until, at $t = \infty$,

$[C_\infty] = [A_o]$. Formation of $[C]$ is slowest in the early stage of the progress curve (i.e. the induction period) which persists from $t = 0$ to the time at which the inflection point of the progress curve for $[C]$ is reached. This inflection point occurs at the same time that $[B]$ reaches its maximum concentration (since $d^2[C]/dt^2 = 0$ when $d[B]/dt = 0$). This is a characteristic of any series first-order sequence, however many steps are included, provided that only A is initially present. If $k_2 \ll k_1$ in the above scheme, the expression for $[A]$ remains the same, the expression for $[B]$ becomes $[B] = [A_o]e^{-k_2t}$, and $[C]$ becomes $[A_o](1 - e^{-k_2t})$. In this case, notice the concentration of the final product at any time t depends only on the smallest rate constant and the initial concentration of the first reactant. Note that when $k_2 \gg k_1$, $[C]$ will equal $[A_o](1 - e^{-k_1t})$.

Series first-order processes are especially important in understanding metabolic pathway kinetics. Likewise, one should recognize that unassisted protein folding^{3,4} is unlikely to be a single-step reaction (say $U \rightarrow N$), where U represents unfolded protein, and N is the native conformation. At a minimum, realistic schemes for protein folding are apt to occur as a series of isomerizations:



where I_1 , I_2 , and I_3 are intermediate states of folding. The greatest challenge is to develop spectroscopic tools that detect the intermediates with sufficient accuracy and precision so as to allow quantifiable kinetic measurements. If the number of steps exceeds two, one may utilize various software programs for numerical integration.

¹A. V. Harcourt & W. Esson (1865) *Proc. Roy. Soc.* **14**, 470.

²W. Esson (1866) *Phil. Trans. Roy. Soc. London Ser. A.* **156**, 220.

³P. S. Kim & R. L. Baldwin (1990) *Ann. Rev. Biochem.* **59**, 631.

⁴T. E. Creighton (1990) *Biochem. J.* **270**, 1.

SERINE DEHYDRATASE

L-Serine dehydratase [EC 4.2.1.13], also known as serine deaminase and L-hydroxyaminoacid dehydratase, catalyzes the pyridoxal-phosphate-dependent hydrolysis of L-serine to produce pyruvate, ammonia, and water. In a number of organisms, this reaction is also catalyzed by threonine dehydratase.

D-Serine dehydratase [EC 4.2.1.14], also known as D-hydroxyaminoacid dehydratase, catalyzes the pyridoxal-

phosphate-dependent hydrolysis of D-serine to produce pyruvate, ammonia, and water. The enzyme will also act on D-threonine, albeit not as effectively.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.
L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

SERINE DEHYDROGENASE

This enzyme [EC 1.4.1.7] catalyzes the reaction of L-serine with NAD⁺ and water to produce 3-hydroxypyruvate, ammonia, and NADH.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

SERINE HYDROXYMETHYLTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.1.2.1], which has a recommended EC name of glycine hydroxymethyltransferase, catalyzes the reversible reaction of 5,10-methylenetetrahydrofolate with glycine and water to produce tetrahydrofolate and L-serine. The enzyme will also catalyze the reaction of glycine with acetaldehyde to form L-threonine as well as with 4-trimethylammoniumbutanal to form 3-hydroxy-*N*⁶,*N*⁶,*N*⁶-trimethyl-L-lysine.

V. Schirch (1997) *Meth. Enzymol.* **281**, 146.
D. J. Oliver (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 323.
M. E. Lidstrom (1990) *Meth. Enzymol.* **188**, 365.
L. Schirch (1982) *Adv. Enzymol.* **53**, 83.
J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

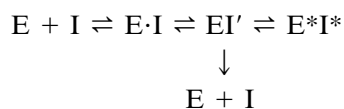
SERINE PALMITOYLTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.3.1.50] catalyzes the reaction of palmitoyl-CoA with L-serine to produce coenzyme A, 3-dehydro-D-sphinganine, and carbon dioxide.

A. H. Merrill, Jr., & E. Wang (1992) *Meth. Enzymol.* **209**, 427.

SERPINS (Inhibitory Mechanism)

A serpin is a *serine proteinase inhibitor* that forms catalytically inactive complexes which, after cleavage of the P¹–P^{1'} linkage, releases the inhibitor very slowly. O'Malley *et al.*¹ recently demonstrated that antichymotrypsin (I) binds to chymotrypsin (E) to form an E*I* complex *via* a three-step mechanism:



In this scheme, EI' retains the P¹–P^{1'} linkage and is formed in a partly, or largely, rate-determining step.

Using a rapid quench-flow kinetic assay for post-complex fragment formation, Nair and Cooperman² showed that the E·I encounter complex of serpin and enzyme forms both E*I* and the post-complex fragment with the same rate constant, indicating that both species arise from EI' conversion to E*I*. These results support the conclusions (a) that the peptide bond remains intact within the EI' complex, and (b) that E*I* is likely to be either the acyl-enzyme or the tetrahedral intermediate formed after water attack on acyl-enzyme.

¹K. H. O'Malley, S. A. Nair, H. Rubin & B. S. Cooperman (1997) *J. Biol. Chem.* **272**, 5354.

²S. A. Nair & B. S. Cooperman (1998) *J. Biol. Chem.* **273**, 17459.

L-SERYL-tRNA(SeC) SELENIUM TRANSFERASE

This enzyme [EC 2.9.1.1], also known as selenocysteine (or SeC) synthase, L-selenocysteinyl-tRNA(SeC) synthase, L-selenocysteinyl-tRNA(Sel) synthase, cysteinyl-tRNA(SeC)-selenium transferase, and cysteinyl-tRNA-(Sel)-selenium transferase, catalyzes the reaction of L-seryl-tRNA and selenophosphate to form L-selenocysteinyl-tRNA(SeC), orthophosphate, and water. The enzyme is responsible for the insertion of selenocysteine into selenoproteins. **See** *L-Selenocysteine*

SHIELDING

Interelectronic interactions that alter how any particular electron in a multi-electronic atom interacts with the nucleus and *vice versa*. These effects lead to so-called chemical shifts in NMR experiments, thereby providing valuable structural information concerning a molecule's bonding and conformation.

Atoms have a nuclear charge equal to the number of protons (*i.e.*, the atomic number), and an equivalent number of electrons are arranged in electronic orbitals, designated *s, p, d, f, g, . . .*. These orbitals are not equienergetic, and their shape and directionality creates a noncentrosymmetric electric field surrounding the nucleus. This results in shielding: the different degrees of electron-neutron attraction created by the electron occupancy of different orbitals. In an applied magnetic field (*H*_o), electrons move in a way that induces their own magnetic field (*H'*). The magnetic moment of this induced field opposes that of the applied field, making the net magnetic field (*H*) experienced by the nucleus (say a proton in an organic compound) less than that of the

applied field: $H = H_0 - H'$. Any nucleus experiencing a field that is smaller than the applied magnetic field is said to be shielded. Electromagnetic radiation at radiofrequency wavelengths is used to excite nuclei residing in the applied magnetic field of an NMR spectrometer, causing them to “spin flip” (*i.e.*, transit to a higher energy state), and *diamagnetic shielding* alters the radiofrequency at which this transition occurs. **See** *Chemical Shift*

SHIFTED BINDING

A phenomenon that has been observed with enzymes acting on a polymeric substrate. The active site of these enzymes consists of a number of subsites. Shifted binding is “the binding of a second substrate molecule in a non-productive mode within the subsites so as to sterically shift the otherwise more energetically favorable binding of a productively bound substrate.”¹ In such cases, it may be necessary to keep the concentration of the polymeric substrate low enough to avoid such problems. Nevertheless, the investigator should always examine the bond cleavage frequencies over a wide range of substrate concentrations. If shifted binding is present, the frequencies should change with increasing substrate concentrations.

¹J. D. Allen (1980) *Meth. Enzymol.* **64**, 248.

SHIFTING SPECIFICITY MODEL FOR ENZYME CATALYSIS

A new model^{1,2} for enzyme catalysis that challenges the long-standing concept of transition state complementarity as the sole source of enzymatic catalytic efficacy. This shifting model states that: (a) enzymes evolved to bind substrates; (b) enzyme·substrate complexes have evolved to bind transition states; and (c) stronger interactions of substrate with the enzyme facilitate rapid conversion to product. This model questions the concept that strong interactions of enzyme and substrate reduce catalytic efficiency.

¹B. M. Britt (1993) *J. Theor. Biol.* **164**, 181.

²B. M. Britt (1997) *Biophys. Chem.* **69**, 63.

SHIKIMATE DEHYDROGENASE

This enzyme [EC 1.1.1.25] catalyzes the reaction of shikimate with NADP⁺ to produce 5-dehydroshikimate and NADPH.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

S. Chaudhuri, I. A. Anton & J. R. Coggins (1987) *Meth. Enzymol.* **142**, 315.

SHIKIMATE KINASE

This enzyme [EC 2.7.1.71] catalyzes the reaction of ATP with shikimate to produce ADP and shikimate 3-phosphate.

R. Bentley (1990) *Crit. Rev. Biochem. Mol. Biol.* **25**, 307.

R. De Feyter (1987) *Meth. Enzymol.* **142**, 355.

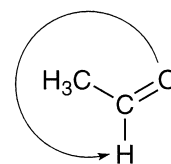
SHUTTLE GROUP

Any residue that facilitates entry or departure of a proton from an enzyme active site or to another region within an active site. Occasionally, proton shuttle groups can be removed by site-directed mutagenesis, and chemical rescue can be demonstrated by providing buffer salts to replace the shuttle group. **See** *Chemical Rescue*

An example is the hydration of CO₂, as catalyzed by carbonic anhydrase¹. The catalytic reaction requires proton transfer from the zinc-bound water at the active site to solution to regenerate Zn·OH[−] in each catalytic cycle. The most efficient isozyme forms use His-64 as a nearby proton shuttle group; other forms contain residues that are less effective in proton transfer and limit overall catalytic efficiency.

¹J. N. Earnhardt, M. Qian, C.-K. Tu, P. J. Laipis & D. N. Silverman (1998) *Biochemistry* **37**, 7649.

SI-



The *si* Face of Acetaldehyde

This stereochemical term (short for the Latin word *sine-trus*) is the designation for a stereoheterotopic face of a trigonal atom whose ligands appear in a counterclockwise sense in the order of the Cahn-Ingold-Prelog scheme when viewed from that side of the face.

G. P. Moss (1996) *Pure Appl. Chem.* **68**, 2193.

SIALIDASES (or Neuraminidases)

These enzymes [EC 3.2.1.18], better known as *exo-α*-sialidases and neuraminidases, catalyze the hydrolysis of 2,3-, 2,6-, and 2,8-glycosidic linkages joining terminal nonreducing *N*- or *O*-acylneuraminyl residues to galactose, *N*-acetylhexosamine, or *N*- or *O*-acylated neuraminyl residues in oligosaccharides, glycoproteins, glycolip-

ids, or colominic acid. The enzymes do not act on 4-*O*-acetylated sialic acids. Endo- α -sialidase [EC 3.2.1.129] catalyzes the endohydrolysis of α -2,8-sialosyl linkages in oligo- or poly(sialic) acids. Although the name endo-*N*-acetylneuraminidase has also been used for this enzyme, this is misleading since its activity is not restricted to acetyl-substituted substrates. Anhydrosialidase [EC 3.2.1.138] catalyzes the hydrolysis of α -sialosyl linkages in *N*-acetylneuraminic acid glycosides, releasing 2,7-anhydro- α -*N*-acetylneuraminic acid.

E. Conzelmann & K. Sandhoff (1987) *Adv. Enzymol.* **60**, 89.

R. O. Brady (1983) *The Enzymes*, 3rd ed., **16**, 409.

H. M. Flowers & N. Sharon (1979) *Adv. Enzymol.* **48**, 29.

A. Gottschalk & A. S. Bhargava (1971) *The Enzymes*, 3rd ed., **5**, 321.

O-SIALOGLYCOPROTEASE

This enzyme [EC 3.4.24.57], also known as *O*-sialoglycoprotein endopeptidase and glycoprotease, catalyzes the hydrolysis of *O*-sialoglycoproteins; for example, it cleaves the Arg³¹—Asp³² peptide bond in glycophorin A.

A. Mellors & R. Y. C. Lo (1995) *Meth. Enzymol.* **248**, 728.

SIALYLTRANSFERASES

This set of enzymes consists of those proteins that catalyze the transfer of sialyl moieties. (1) CMP-*N*-acetylneuraminate- β -galactoside α -2,6-sialyltransferase [EC

2.4.99.1] catalyzes the reaction of CMP-*N*-acetylneuraminate with β -D-galactosyl-1,4-acetyl- β -D-glucosamine to produce CMP and α -*N*-acetylneuraminyl-2,6- β -D-galactosyl-1,4-*N*-acetyl- β -D-glucosamine. (2) Monosialoganglioside sialyltransferase [EC 2.4.99.2] catalyzes the reaction of CMP-*N*-acetylneuraminate with D-galactosyl-*N*-acetyl-D-galactosaminyl-(*N*-acetylneuraminyl)-D-galactosyl-D-glucosylceramide to produce CMP and *N*-acetylneuraminyl-D-galactosyl-*N*-acetyl-D-galactosaminyl-(*N*-acetylneuraminyl)-D-galactosyl-D-glucosylceramide. This enzyme may be identical with β -galactoside α -2,3-sialyltransferase [EC 2.4.99.4]. (3) Lactosylceramide α -2,6-*N*-sialyltransferase [EC 2.4.99.11] catalyzes the reaction of CMP-*N*-acetylneuraminate with β -D-galactosyl-1,4- β -D-glucosylceramide to produce CMP and α -*N*-acetylneuraminyl-2,6- β -galactosyl-1,4- β -D-glucosylceramide.

Y. Kishimoto (1983) *The Enzymes*, 3rd ed., **16**, 357.

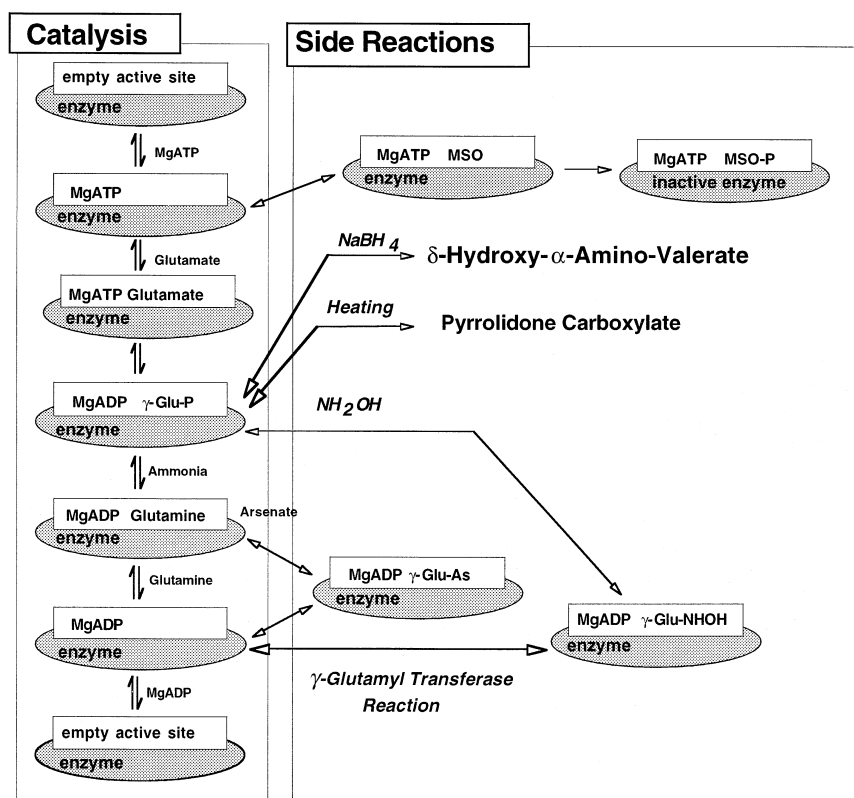
K. A. Presper & E. C. Heath (1983) *The Enzymes*, 3rd ed., **16**, 449.

R. J. Staneloni & L. F. Leloir (1982) *Crit. Rev. Biochem.* **12**, 289.

T. A. Beyer, J. E. Sadler, J. I. Rearick, J. C. Paulson & R. L. Hill (1981) *Adv. Enzymol.* **52**, 23.

SIDE REACTIONS

A reaction other than the physiologically important reaction catalyzed by an enzyme. There are numerous exam-



ples of side reactions in enzymic catalysis: (1) hexokinase displays an ATPase activity that is markedly enhanced when lyxose, a pentose that binds within the glucose binding site; (2) carbonic anhydrase can be an effective esterase, depending on reaction conditions; and (3) glutamine synthetase undergoes ATP-dependent inactivation by the metabolic inhibitor methionine sulfoximine.

Enzymes can also undergo other side reactions under conditions that divert a chemically reactive intermediate from its usual catalytic function. Again, glutamine synthetase is an excellent example (see figure above), because its side reactions include acyl-phosphate reduction by borohydride, pyroglutamate formation, and the formation of γ -glutamyl hydroxamate in the presence of hydroxylamine and arsenate.

SIGMA (σ)

1. The symbol for a symmetry designation of a molecular orbital. Such an orbital is without a nodal plane including the internuclear axis (as is the case with a π orbital). 2. The localized two-center bond associated with an s orbital. 3. The symbol for standard deviation. 4. The symbol for wavenumber. 5. The symbol for surface tension.

σ -ADDUCT (Sigma Adduct)

1. The product formed by the covalent attachment of some "entering" entity (an electrophile, nucleophile, or radical) to a ring carbon of an aromatic molecular entity, thus forming a new σ -bond and disrupting the original conjugation. 2. The product formed (as in definition 1) by the reaction of an "entering" entity to any unsaturated and conjugated system.

σ -CONSTANT (Sigma Constant)

1. The substituent constant for *meta*- and *para*-substituents in benzene derivatives, defined by Hammett on the basis of the K_a value of a substituted benzoic acid derivative in water at 25°C. 2. Referring to a set of related electronic substituent constants based on other standard reaction series (e.g., σ^+ , σ^- , σ°) or dissected on the basis of electronic effects (e.g., σ_I , and σ_R). A large negative value for a sigma constant suggests that the substituent has a high electron-releasing power relative to hydrogen (as the substituent). Correspondingly, a large positive value for a sigma constant suggests a high electron-with-

drawing power relative to hydrogen, either by inductive and/or resonance effects.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

N. B. Chapman & J. Shorter (eds.) (1972) *Advances in Linear Free Energy Relationships*, Plenum, New York.

C. D. Johnson (1973) *The Hammett Equation*, Cambridge Univ. Press, Cambridge.

J. Shorter (1973) *Correlation Analysis in Organic Chemistry. An Introduction to Linear Free Energy Relationships*, Oxford Univ. Press, Oxford.

N. B. Chapman & J. Shorter (eds.) (1978) *Correlation Analysis in Chemistry: "Recent Advances"*, Plenum, New York.

$\sigma \rightarrow \sigma^*$ TRANSITION

An electronic transition in which an electron is promoted from a bonding σ orbital to an antibonding σ^* orbital. The transition energies are often close to those of Rydberg transitions. **See** *Absorption Spectroscopy*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

SIGMATROPIC REARRANGEMENT

A molecular rearrangement in which a σ -bond is formed between two atoms of the chemical species not previously linked. There may also be a relocation of π -bonds. Nevertheless, the total number of σ - and π -bonds will remain unchanged. Some authors restrict this term to intramolecular pericyclic reactions.

SIGNAL AMPLIFICATION

A parameter used to assist in the characterization of enzyme cascade systems. Symbolized by SA, it is equal to $[e_{0.5E}]/[e_{0.5I}]$ where $[e_{0.5E}]$ is the concentration of effector required for 0.5 activation of the converter enzyme E and $[e_{0.5I}]$ is the concentration of effector at which 50% of the interconvertible enzyme (I) has been modified¹. **See** *Enzyme Cascade Kinetics*

¹P. B. Chock & E. R. Stadtman (1980) *Meth. Enzymol.* **64**, 297.

SIGNAL PEPTIDASE, EUKARYOTIC MICROSOMAL

This proteolytic enzyme catalyzes the removal of N-terminal signal peptide extensions of secretory proteins and certain intracellular proteins.

M. O. Lively, A. L. Newsome & M. Nusier (1994) *Meth. Enzymol.* **244**, 301.

SIGNAL PROCESSOR

Any device that modifies the transduced signal from a detector in a manner that increases the signal strength

(a process known as amplification) or converts an alternating current signal to a direct current output, or *vice versa*.

SIGNIFICANT FIGURES

The digits required to express the numerical results of a measurement to the precision with which that measurement is made. Suppose, for example, that the results of rate measurements, each made by two different methods (*e.g.*, absorbance and mass spectrometry in this case) gave values of 10.6 and 10.715 mM/min. Then the first measurement requires no fewer than three digits to convey the precision of the absorbance measurement, and the latter would not be fairly represented by any fewer than five in the case of the mass spectrometric determination. At the same time, any more than 3 or 5 digits for these respective cases would not be significant relative to the precision of the determinations. Note that these values could be equally well expressed as 1.06×10^{-2} and 1.0715×10^{-2} M/min, with the value to the left of the decimal point typically chosen to fall in the interval greater than one but less than 10. In this use of scientific notation, the exponent is utilized to locate the decimal place and does not reflect the number of significant figures. **See also** Precision

SINGLE-POOL MODEL FOR METABOLITE COMPARTMENTATION

A model based on the assumption that a metabolite is present within a single compartment with defined rate constants for absorption and elimination of the metabolite. The rate of appearance of a tracee and the infusion of tracer are assumed to take place in a single pool that is instantly well-mixed. Wolfe¹ has described in detail how the constant tracer infusion method allows one to calculate half-life, pool size, turnover time, mean residence time, and clearance time.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, Wiley-Liss, New York.

SINGLE-STEP REACTION

1. A reaction that cannot be reduced further, as in the case of *elementary reactions*. 2. Any reaction characterized by a single transition state.

SINGLET MOLECULAR OXYGEN

The excited singlet state of dioxygen (O₂). The term "singlet oxygen" should not be used since it is unclear

whether the writer is referring to dioxygen or the excited singlet oxygen atom (in a ¹S or ¹D state).

SINGLET STATE

Any state which has a total electron spin quantum number of zero.

SINGLET-TRIPLET ENERGY TRANSFER

The transfer of excitation energy present in an atom or molecular species in an excited singlet state to another atom or molecular species, generating an electronically excited species in a triplet state.

SINGLE-TURNOVER CONDITIONS

Reaction conditions that only permit a catalyst to pass through a single round of catalysis. Single-turnover conditions are usually obtained by limiting the substrate concentration relative to the concentration of active catalyst. Occasionally, single-turnover conditions can also be achieved by limiting the period of reaction.

SITE-SPECIFIC ISOTOPE FRACTIONATION IN BIOCHEMICAL MECHANISMS

Zhang-Yunianta and Martin¹ recently discussed how studies of site-specific natural isotope fractionation by nuclear magnetic resonance can provide information and insight concerning chemical or biochemical mechanisms. The natural abundance of isotopic distribution in fermentation ethanol has been shown to reflect the physiological and environmental conditions of its carbohydrate precursors. Redistribution coefficients relating the natural site-specific isotope contents in end products to those of their precursors characterize the traceability of specific hydrogens in end products to their parent hydrogens in the starting materials. Average hydrogen isotope effects associated with the transfer of hydrogen from the water pool to the methyl or methylene site of product can also be estimated.

¹B. L. Zhang-Yunianta & M. L. Martin (1995) *J. Biol. Chem.* **270**, 16023.

SI UNITS

The International System of Units (SI) built on seven base units: the unit of mass is the kilogram, the unit of time is the second, the unit of length is the meter, the unit of electric current is the ampere, the unit of tempera-

ture is kelvin, the unit of amount-of-substance is the mole, and the unit of luminous intensity is the candela.

SLOPE

1. The ratio of the rise to the run of any two points on a straight line or line segment. On a line drawn in a Cartesian coordinate system, it is also equal to the tangent of the angle that line makes with the horizontal axis (*i.e.*, the x -axis). 2. The first derivative of the equation of a curve at a given point on that curve.

SLOPE-INTERCEPT EQUATION

The equation of a line in the Cartesian coordinate system: $y = mx + b$ where x and y are the two coordinates of a point on the line, m is the slope of the line, and b is the vertical intercept.

SLOPE REPLOTS (or, Slope Secondary Replots)

Except for very simple systems, initial rate experiments of enzyme-catalyzed reactions are typically run in which the initial velocity is measured at a number of substrate concentrations while keeping all of the other components of the reaction mixture constant. The set of experiments is run again a number of times (typically, at least five) in which the concentration of one of those other components of the reaction mixture has been changed. When the initial rate data is plotted in a linear format (for example, in a double-reciprocal plot, $1/v$ vs. $1/[S]$), a series of lines are obtained, each associated with a different concentration of the other component (for example, another substrate in a multisubstrate reaction, one of the products, an inhibitor or other effector, *etc.*). The slopes of each of these lines are replotted as a function of the concentration of the other component (*e.g.*, slope vs. [other substrate] in a multisubstrate reaction; slope vs. $1/[\text{inhibitor}]$ in an inhibition study; *etc.*). Similar replots may be made with the vertical intercepts of the primary plots. The new slopes, vertical intercepts, and horizontal intercepts of these replots can provide estimates of the kinetic parameters for the system under study. In addition, linearity (or lack of) is a good check on whether the experimental protocols have valid steady-state conditions. Nonlinearity in replot data can often indicate cooperative events, slow binding steps, multiple binding, *etc.*

It is always good procedure to include the slope and intercept replots with the primary plots when reporting the results (perhaps best accomplished by having the replots as an inset of the primary graphs). They enable a reader to better evaluate the data.

SLOW-BINDING INHIBITOR

If there is a relatively long period of time before the interaction between a particular inhibitor and an enzyme has reached equilibrium (typically, on the order of 0.5–2 minutes, or longer), the inhibitor is considered to be a slow-binding inhibitor, and special procedures are used to characterize the kinetics^{1–3}. Note that slow-binding inhibition does not necessarily imply tight-binding inhibition. An important characteristic of slow-binding inhibitors is that the progress curve will change depending on whether the reaction is initiated with addition of enzyme or with addition of substrate.

There are two basic kinetic schemes for slow-binding enzyme inhibition: Mechanism A involves the slow interaction of an inhibitor with enzyme and Mechanism B involves the rapid formation of an E·I complex that subsequently undergoes a slow isomerization to E·I'. The initial interaction of enzyme and inhibitor may not be sufficiently fast to allow free enzyme, E, and E·I and E·I' to reach steady-state equilibrium. This would generate a more general mechanism. Sculley *et al.*⁴ attempted to ascertain whether it might be possible to distinguish between the three possible inhibition mechanisms by steady-state kinetic techniques. By using three different ratios for the two rate constants that determine which mechanism applies, these investigators modeled the kinetic behavior by creating theoretical data sets for the most general mechanism. These reaction progress curve data were then “fitted” to the rate equations for the other two mechanisms. Their findings show how the values for the kinetic parameters, as determined from fits of the data to the equations for Mechanisms A and B, can be considerably in error.

¹J. W. Williams & J. F. Morrison (1979) *Meth. Enzymol.* **63**, 437.

²J. F. Morrison & C. T. Walsh (1988) *Adv. Enzymol.* **59**, 201.

³S. E. Szedlacsek & R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 144.

⁴M. J. Sculley, J. F. Morrison & W. W. Cleland (1996) *Biochim. Biophys. Acta* **1298**, 78.

SLOW TEMPERATURE-JUMP METHODS

Biochemical processes such as protein unfolding/refolding and supramolecular assembly/disassembly take place on a time scale of seconds to minutes after readjusting the temperature of a system. Most commercially available glass-jacketed cuvettes are not suitable for temperature jumps on this time scale, as a result of the slow kinetics of heat transfer across substances with characteristically high dielectric constants, and their use can convolute the time scale of the temperature change onto the time scale

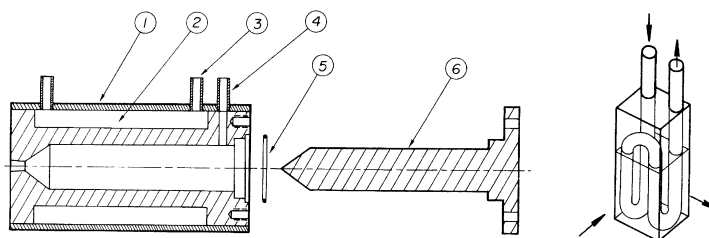
In all cases, degassing of solutions by stirring *in vacuo* or by centrifugation (*i.e.*, pressurized degassing) is advised, because gas bubbles can accumulate on cuvette optical faces. Experience indicates that bubble clearance frequently is first-order and surprisingly slow!

¹K. A. Johnson & G. G. Borisy (1977) *J. Mol. Biol.* **117**, 1.

²T. L. Karr, D. Kristofferson & D. L. Purich (1980) *J. Biol. Chem.* **255**, 8560 and 8567.

³F. M. Pohl (1968) *Eur. J. Biochem.* **4**, 373.

⁴F. M. Pohl (1977) in *Chemical Relaxation in Molecular Biology* (I. Pecht & R. Rigler, eds.) p. 289, Springer-Verlag, Berlin.



Rapid Heat Exchanger (left) and Heating/Cooling Cuvette (right)

Left: Diagram of a rapid heat exchanger. The overall block length is 10.2 cm with other dimensions drawn to scale. The heat exchanger block (1) was fabricated to accommodate a syringe with a Luer taper. Other features include the water jacket (2), water inlet and outlet ports (3), sample exit port (4), O-ring seal (5), and solid copper cold finger insert (6). This particular exchanger was built to minimize shear forces that might alter microtubule polymer length distributions, but one can also fabricate exchangers for dealing with smaller volumes to conserve sample. All copper parts are gold-plated to reduce chemical reactivity with biological molecules and buffer salts. Right: Temperature-regulated cuvette for examining temperature-dependent properties of biochemical reactions.

of the biochemical process *per se*. For example, early work on the kinetics of cold-induced microtubule depolymerization suggested that the process was essentially a zero-order process for over half of the disassembly process¹. Nonetheless, later studies² utilizing a gold-plated, copper-block flow device for rapid heat exchange demonstrated that the depolymerization process was clearly not zero-order. **See** *End-wise Depolymerization* Pohl³⁻⁴ has described a number of useful methods for achieving reasonably prompt changes in temperature. One approach (shown on the right in the above diagram) uses a fluorescence microcell fitted with an in-dwelling heating/cooling coil that does not obstruct the right-angle light path. The coil can be made of very thin-walled glass or, better still, of copper or aluminum with an appropriately inert, yet heat-conducting, coating. Pohl³ also describes the modification of a commercial flow cell from Zeiss, Inc., to achieve fast heat exchange (half-time of 0.3 seconds) with a 0.05 mL sample volume. He also suggests that one can use a stopped-flow mixer to combine two solutions, each held at different temperatures, to achieve faster temperature changes.

SLOW, TIGHT-BINDING INHIBITOR

Inhibition of enzyme activity by a chemical species that binds slowly and is tight-binding as well (*i.e.*, has a low dissociation constant). Such inhibitors require special kinetic analysis^{1,2}. The most common method of obtaining the inhibition parameters is by nonlinear regression analysis of the progress curves.

¹J. W. Williams & J. F. Morrison (1979) *Meth. Enzymol.* **63**, 437.

²S. E. Szedlacsek & R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 144.

SMOLUCHOWSKI TREATMENT FOR AGGREGATION

A probabilistic kinetic model describing the rapid coagulation or aggregation of small spheres that make contact with each other as a consequence of Brownian motion. Smoluchowski¹ recognized that the likelihood of a particle (radius = r_1) hitting another particle (radius = r_2 ; concentration = c_2) within a time interval (dt) equals the diffusional flux $(\partial c_2 / \partial \rho)_{\rho=R_{12}}$ into a sphere of radius R_{12} , equal to $(r_1 + r_2)$. The effective diffusion coefficient D_{12} was taken to be the sum of the diffusion coefficients

for the two particles, and the probability of a collision over the time interval may be stated as follows:

$$Pdt = 4\pi R_{12}^2 D_{12} (\partial c_2 / \partial \rho)_{\rho=R_{12}} dt = 4\pi R_{12}^2 D_{12} c_2 dt$$

This result implicitly requires: (a) that no strongly attractive/repulsive electrostatic interaction occurs as the particles approach each other, and (b) that a stationary diffusional concentration field surrounds the particle. [Note: Einstein's result² for spherical particle diffusivity (*i.e.*, $D = k_B T / 6\pi\eta r$, where k_B is the Boltzmann constant, T is temperature in kelvin, η is the viscosity of the medium, and r is the radius) indicates that $R_{12}D$ will be approximately $4k_B T / 6\pi\eta$.]

¹M. von Smoluchowski (1918) *Z. Physik Chem.* **92**, 129.

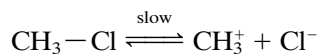
²A. Einstein (1906) *Annalen der Physik* **19**, 371.

S_N DISPLACEMENT REACTIONS

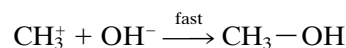
Substitution (type: nucleophilic) reactions in which an electron-rich agent (or nucleophile) attacks and forms a transient or stable bond with an electron-poor center (often carbon or phosphorus in biological systems) which has undergone prior or simultaneous expulsion of the leaving group (or exiphile).

The two limiting cases of nucleophilic displacement reactions are designated as S_N1 or S_N2 to indicate those mechanisms that respectively display overall first-order or second-order kinetics. These mechanisms are illustrated by the classical case of the reaction of hydroxide ion with chloromethane, and they differ with respect to the timing of the bond-breaking step relative to the bond-making step.

S_N1 REACTIONS. This mechanism can be represented as an initial rate-determining step, followed by the rapid reaction between the intermediate carbocation and the nucleophile (in this case, hydroxide ion):



In this first reaction, unless the process is occurring at very high temperatures, solvation of the ionic products is needed to provide the energy needed in the bond-breaking step. Because the carbocation has planar sp^2 hybridization, the second reaction can occur on either face. This second step may just as well be written in either of the following ways:



where the hydroxide ion attacks directly, or as the deprotonation of a hydrated carbocation whose positive charge is delocalized onto the oxygen atom of one solvent molecule:



The reaction rate (v) for an S_N1 reaction can be written as:

$$v = k[\text{CH}_3-\text{Cl}]^1[\text{OH}^-]^0 = k[\text{CH}_3-\text{Cl}]$$

In the above expression, the quantity to the right of the first equal sign indicates that the order of the reaction with respect to the specific reactants is one and zero, respectively; and because the order with respect to hydroxide is zero, then $[\text{OH}^-]^0 = 1$, and the overall reaction order is the sum of the orders of the respective reactants. Hence, a S_N1 reaction is first-order overall.

S_N2 REACTIONS. This bimolecular reaction mechanism can be represented as a backside attack, in which the nucleophile approaches at an angle of 180° with respect to the leaving group (hence, the frequently used descriptor: “in-line” mechanism). This geometrical constraint results in an inversion of configuration about the central atom. (Because such inversions were first described in the 1890's by the carbohydrate chemist Walden, they are often called Walden inversions.) Any energy for bond scission is provided by the simultaneous making of an incipient bond between the nucleophile and the reaction center. As the reaction reaches its transition state, the central atom is pseudo-pentavalent, and the central atom has progressed from its sp^3 hybridization in the ground state to an sp^2 hybridization in the transition state. (The bond order remains four, because the central atom has three nonreacting substituents as well as another bond that is the sum of the entering and leaving groups.) The reaction occurs in one step, and no intermediate is formed:



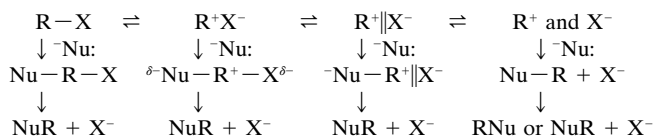
where the curved arrows indicate the pairwise movement of electrons.

The reaction rate (v) for an S_N2 reaction can then be written as:

$$v = k[\text{CH}_3\text{—Cl}]^1[\text{OH}^-]^1$$

For this expression, the quantity to the right of the first equal sign indicates that the order of the reaction with respect to either reactant is one, and the overall reaction order is the sum of the orders of the respective reactants. Hence, a S_N2 reaction is second-order overall. In certain cases, such as solvolysis, the rate expression takes on pseudo-first-order kinetic behavior.

Technically, for solution-phase nucleophilic substitution reactions, S_N1 and S_N2 mechanisms are limiting cases on a continuum of possible interactions:



where R—X is the covalently bonded form, R^+X^- is the tight (or intimate ion) ion pair, $\text{R}^+\|\text{X}^-$ is the loosely associated or solvent-separated ion pair, R^+ and X^- represent the fully dissociated ions, Nu: is the nucleophile, X^- is the counterion (or, gegenion), and RNu and NuR are the two optical isomers formed by attack on either face of the electrophilic center. **See** *Solvolysis*

SODIUM-22 (^{22}Na)

A radioactive sodium nuclide that has three principal radiation emissions β^+ (0.546 MeV), γ (1.275 MeV), and the typical annihilation peak at 0.512 MeV. The maximum range in air is 1.4 meters. The decay half-life is 2.605 years. The following decay data indicate the time (in days) followed by the fraction of the original amount at the specified time: 0.0, 1.000; 30, 0.978; 60, 0.957; 90, 0.936; 120, 0.916; 150, 0.896; 180, 0.877; 210, 0.858; 240, 0.839; 270, 0.821; 300, 0.804; 600, 0.646; 900, 0.519; 1200, 0.417; 1500, 0.335; 1800, 0.269; 2100, 0.216; 2400, 0.174; 2700, 0.140; 3000, 0.112.

Special precautions: Store behind lead shielding. Follow institutional regulations regarding type and location of dosimeter. Avoid working immediately above open containers. Because no organ selectively concentrates sodium metabolites, the entire body is considered to be

the critical organ in terms of dose. The effective *biological* half-life in humans is around 11 days.

SOLUBILITY PRODUCT

The product of the concentration of the ions in a saturated solution of a chemical species, typically symbolized by K_{sp} . This expression is valid for sparingly soluble salts.

If the product of the concentrations of the ionic species exceeds the K_{sp} value then either precipitation occurs or a metastable condition of supersaturation develops. When reporting solubility products, the temperature should always be provided. Examples of a few relevant K_{sp} values (at 25°C) are given below.

Barium carbonate (BaCO_3)	2.58×10^{-9}
Barium hydroxide octahydrate ($\text{Ba(OH)}_2 \cdot 8\text{H}_2\text{O}$)	2.55×10^{-4}
Calcium carbonate (CaCO_3)	4.96×10^{-9}
Calcium hydroxide (Ca(OH)_2)	4.68×10^{-6}
Calcium oxalate monohydrate ($\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$)	2.34×10^{-9}
Calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$)	2.07×10^{-33}
Iron(II) carbonate (FeCO_3)	3.07×10^{-11}
Iron(II) hydroxide (Fe(OH)_2)	4.87×10^{-17}
Iron(III) hydroxide (Fe(OH)_3)	2.64×10^{-39}
Magnesium carbonate (MgCO_3)	6.82×10^{-6}
Magnesium hydroxide (Mg(OH)_2)	5.61×10^{-12}
Magnesium oxalate dihydrate ($\text{MgC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)	4.83×10^{-6}
Magnesium phosphate ($\text{Mg}_3(\text{PO}_4)_2$)	9.86×10^{-25}
Manganese(II) carbonate (MnCO_3)	2.24×10^{-11}
Manganese(II) hydroxide (Mn(OH)_2)	2.06×10^{-13}
Manganese(II) oxalate dihydrate ($\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)	1.70×10^{-7}
Zinc carbonate (ZnCO_3)	1.19×10^{-10}
Zinc carbonate monohydrate ($\text{ZnCO}_3 \cdot \text{H}_2\text{O}$)	5.41×10^{-11}

SOLUTION

A homogeneous mixture of a liquid, known as the solvent, with one or more solids or gases (the solute) dissolved in that liquid. Solutions can also be formed by two or more miscible liquids. In such cases, the solvent is that liquid with the largest mole fraction.

SOLVATION

A stabilizing interaction between a solute and the solvent, or between groups on an insoluble substance and the surrounding solvent. **See** *Hydration Atmosphere*

SOLVENT-ACCESSIBLE SURFACE AREA

The surface¹ traced out by the center of a solvent probe molecule as it is rolled over the surface of a protein whose three-dimensional structure has been determined at the atomic level. Solvent-accessible surface areas can be calculated by various computer algorithms², and differences in solvent-accessible areas can be used to characterize the energetics of surface hydration as a function of changes in protein conformation, oligomerization, and complexation.

¹B. Lee & F. M. Richards (1971) *J. Mol. Biol.* **55**, 379.

²B. M. Baker & K. P. Murphy (1998) *Meth. Enzymol.* **295**, 294.

SOLVENT ISOTOPE EFFECT

A change in chemical reactivity occurring as a consequence of changes in the isotopic composition of the solvent. **See** *Proton Inventory; Kinetic Isotope Effects*

SOLVENT TRAPPING OF ENZYME INTERMEDIATES

An X-ray crystallographic technique^{1,2} resulting in the accumulation of transient enzyme-bound intermediates (in reactions using water as a substrate) by appropriately manipulating polarity through the addition of organic solvents. After enzyme crystals are grown normally to an acceptable size, they are placed in a capillary that permits flow of solvent. One can use protein cross-linking agents to stabilize a crystalline enzyme prior to solvent transfer¹. The crystals are dried by capillarity using filter paper wicks, and organic solvent is then introduced. (Included with the organic solvent are activated molecular sieves, and these zeolites contain pores that trap and remove all water from the sample.) This is followed by introduction of substrate(s) to allow enzyme-bound intermediates to accumulate. Under favorable conditions, crystals containing such solvent-trapped intermediates are suitable for structure determination, and the electron density map so obtained is a weighted average favoring the most abundant enzyme form.

¹P. A. Fitzpatrick, A. C. Steinmetz, D. Ringe & A. M. Klibanov (1993) *Proc. Natl. Acad. Sci. U.S.A.* **90**, 8653.

²N. H. Yennawar, H. P. Yennawar & G. K. Farber (1994) *Biochemistry* **33**, 7326.

SOLVOLYSIS

Whenever a nucleophilic substitution reaction uses the solvent as the nucleophile, the reaction can be referred to as a solvolysis reaction (or simply as solvolysis). Examples of solvolysis include hydrolysis (reaction with water), ethanolysis (reaction with ethanol), acetolysis (reaction with acetic acid), and formolysis (reaction with formic acid). All solvolysis reactions exhibit first-order kinetics, because the concentration of the solvent does not change to any appreciable extent during the solvolytic reaction. This fact is often misinterpreted as an indication that the reaction type is S_N1. To be sure, polar solvents such as water, alcohols, and amines should be regarded as good nucleophilic agents. A traditional method for distinguishing S_N1 and S_N2 reactions in physical organic chemistry is to add a low concentration of a substance that is a better nucleophile than the solvent. If the rate of substitution remains unchanged, then one can infer that the reaction is of the S_N1 type; however, should the nucleophilic substitution rate increase, an S_N2 type reaction is likely. Of course, the most direct means for distinguishing S_N1 and S_N2 reactions is by stereochemistry, when practicable.

SOMO

Abbreviation for singly occupied molecular orbital, a term that applies to, among other cases, the unpaired electrons of radicals.

SONICATION

The irradiation of a system with sound waves (usually ultrasound). Often used to disrupt cell membranes and in early steps in protein purification, it should also be noted that sonication can increase rates of reaction as well as assist in the preparation of vesicles.

SORBITOL DEHYDROGENASE

This enzyme [EC 1.1.1.14], also known as L-idoitol 2-dehydrogenase and polyol dehydrogenase, catalyzes the reaction of L-idoitol with NAD⁺ to produce L-sorbose and NADH. The enzyme will also convert D-glucitol to D-fructose.

J. Jeffery & H. Jornvall (1988) *Adv. Enzymol.* **61**, 47.

SPECIFIC ACTIVITY

1. The intrinsic variable expressed usually as international units (U) of enzyme activity per milligram protein

(or nucleic acid in the case of ribozymes). One U corresponds to the conversion of 1 micromole substrate into product per minute. A katal corresponds to the conversion of one mole substrate per second. Hence, 1 U corresponds to 16.67 nkat. As an experimental parameter, specific activity is especially useful during enzyme purification when one must develop strategies for maximizing specific activity while ensuring that adequate yield is maintained. Likewise, one may use specific activity as a measure of purity of any protein or other biomolecule that possesses some assayable response (such as drug binding or metabolite transport).

2. The intrinsic variable expressed as units of radioactivity (in becquerels or, more traditionally, curies) per mole of a substance. One Bq corresponds to 1 disintegration per second (dps) and one Ci to 3.70×10^{10} Bq. This parameter is especially useful in quantifying the amount of substance in biological samples. For example, if SA_s is the standard specific radioactivity (say, x dps/y mol) of a standard, and if SA_e is the experimental specific activity (say, x dps/(y + z) mol), then the content z in a sample can be determined from the expression $(SA_s/SA_e) = (y + z)/y$ or $z = y ([SA_s/SA_e] - 1)$. This intrinsic variable can also be expressed as the gram-atom excess of a stable isotope per mole of a substance. The numerator is typically determined using a ratio mass spectrometer, and the denominator can be estimated by chemical and/or spectroscopic techniques.

SPECIFIC CATALYSIS

The acceleration of a reaction by a unique catalyst rather than a family of related substances or materials. The term is commonly used with respect to H^+ or OH^- catalysis (thus, specific acid and specific base catalysis). **See** *General Acid Catalysis*; *General Base Catalysis*

SPECIFICITY

A measure and/or description of how specific an enzyme is toward a substrate or class of substrates or toward an effector or class of effectors. For effectors (or for ligands binding to macromolecules that are not enzymes), this specificity is readily measured by dissociation (or, association) constants. For enzymes, specificity is best quantitated by the V_{max}/K_m ratio. **See** *Specificity Constant*. It is crucial, in the complete characterization of an enzyme, that the specificity of the enzyme be known in detail.

This includes stereospecificity and anomeric specificity when appropriate. It should also be recalled that biomolecules often exist in different stable conformations and in different states of ionization and/or metal chelation. For example, for ATP-dependent enzymes, it may be necessary to know the relative specificity of $MgATP^{2-}$ and $MgHATP^-$ or with $MnATP^{2-}$ and $MnHATP^-$. If accurate values are known for the V_{max}/K_m ratios of a large set of substrates and alternative substrates, it may be possible to determine the topology of the enzyme's active site and to design specific active site inhibitors. **See** *Mapping Substrate Interactions Using Kinetic Data*

SPECIFICITY CONSTANT

The ratio of maximum velocity, V_{max} , or k_{cat} value to the true K_m value for a particular substrate, with units of a bimolecular rate constant $M^{-1}s^{-1}$. In comparing substances that act as substrates for a single enzyme, a higher V_{max}/K_m or k_{cat}/K_m ratio reflects a higher apparent rate of enzyme-substrate complexation.

SPIN LABEL

A stable paramagnetic group that has been attached to a molecular entity. Experiments with ESR measurements can reveal aspects of the microenvironment of the spin label. If the paramagnetic group is not covalently attached to the molecular entity of interest, the term "spin probe" is usually applied. **See also** *Electron Spin Resonance*

SPONTANEOUS PROCESS

Any reaction or change that is energetically poised to progress without the intervention of any external force or treatment. Although a spontaneous chemical process may be thermodynamically favorable, kinetic constraints may prevent the system from progressing to its equilibrium position. If the energy of activation is appreciable, then attainment of the transition state is less likely, and an otherwise spontaneous process may fail to progress. The descriptor "spontaneous" should not be confused with "instantaneous"; the latter describes a process that is immediate and occurs at the present, without any perceptible delay. In fact, the time-evolution or progress of many apparently instantaneous processes can be determined by using ultrafast methods for detecting and/or quantifying a change in the system.

SQUALENE SYNTHASE

This enzyme [EC 2.5.1.21], officially known as farnesyl-diphosphate farnesyltransferase (and also referred to as farnesyltransferase and presqualene-diphosphate synthase), catalyzes the conversion of two molecules of farnesyl diphosphate to yield presqualene diphosphate and pyrophosphate (or, diphosphate). In its polymeric form, the enzyme then catalyzes the NADPH-dependent reduction of presqualene diphosphate to form squalene. **See also** *Farnesyl Diphosphate Farnesyltransferase*

J. Chappel (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 521.

R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.

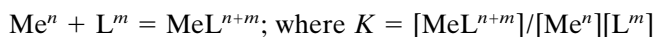
G. J. Schroepfer, Jr. (1982) *Ann. Rev. Biochem.* **51**, 555.

G. J. Schroepfer, Jr. (1981) *Ann. Rev. Biochem.* **50**, 585.

E. D. Beytia & J. W. Porter (1976) *Ann. Rev. Biochem.* **45**, 113.

STABILITY CONSTANT

An equilibrium association or formation constant (units expressed as molarity⁻¹ or M⁻¹) for ligand (L) binding to a metal (Me) ion:



where m and n are integers. **See** *Metal Ion Complexation*

STABILITY CONSTANT DETERMINATION BY A LINEAR PLOT METHOD

Li and Wang¹ described a convenient linear plotting protocol that permits one to evaluate stability constants for the formation of reversible metal-ligand complexes. Added accuracy is achieved over classical approaches, because their generalized equations quantitatively account for side-reactions of metal ion and/or ligand as well as the difference between the total and free ligand concentration.

¹H. Li & J. Wang (1989) *Analyt. Chim. Acta.* **226**, 323.

STABILITY, PRODUCT

A product of an enzyme-catalyzed reaction should be stable enough over the time course of the experiment, including isolation, identification, and measurement of that product. If not, it may be necessary to alter the reaction conditions to increase the stability of a product. A number of enzymes have been studied at lower temperatures for this reason. The comments made in the substrate stability entry are also pertinent here. **See** *Substrate Stability*

Identification of an alternative substrate may likewise allow one to reinvestigate a particular protein, especially if the resulting product is more stable than the earlier studied product.

STANDARD CURVE

Any calibrated curve showing the dependent variable (frequently Δ Absorbance) on the vertical axis and the independent variable (frequently concentration or time) on the horizontal axis. Error bars should be included to indicate the precision of experimental determinations.

See *Beer-Lambert Law; Absorption Spectroscopy*

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Generation, **240**, 122-123; confidence limits, **240**, 129-130; discrete variance profile, **240**, 124-126, 128-129, 131-133, 146, 149; error response, **240**, 125-126, 149-150; Monte Carlo validation, **240**, 139, 141, 146, 148-149; parameter estimation, **240**, 126-129; radioimmunoassay, **240**, 122-123, 125-127, 131-139; standard errors of mean, **240**, 135; unknown sample evaluation, **240**, 130-131; zero concentration response, **240**, 138, 150.

STANDARD DEVIATION

1. A measure, symbolized by σ , of the dispersion of a distribution and equal to the square root of the variance. Hence, it is used to describe the distribution about a mean value. For a normal distribution curve centered on some mean value, μ , multiples of the standard deviation provides information on what percentage of the values lie within $\pm n\sigma$ of that mean. Thus, 68.3% of the values lie within $\pm$ one standard deviation of the mean, 95.5% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$. 2. The corresponding statistic, s , used to estimate the true standard deviation: $\sigma^2 = (\sum(x_i - \bar{x})^2)/(n - 1)$. **See** *Statistics (A Primer)*

STAPHYLOCOCCAL NUCLEASE

This hydrolase, officially known as micrococcal nuclease [EC 3.1.31.1 (formerly 3.1.4.7)], catalyzes endonucleolytic cleavage of single-/double-stranded DNA (as well as RNA) to form 3'-phospho-mononucleotide and 3'-phospho-oligonucleotide end-products. The enzyme also displays exonucleolytic activity.

J. A. Gerlt (1992) *The Enzymes*, 3rd ed., **20**, 95.

A. S. Mildvan & D. C. Fry (1987) *Adv. Enzymol.* **59**, 241.

F. A. Cotton & E. E. Hazen, Jr. (1971) *The Enzymes*, 3rd ed., **4**, 153.

C. B. Anfinsen, P. Cuatrecasas & H. Taniuchi (1971) *The Enzymes*, 3rd ed., **4**, 177.

STARCH PHOSPHORYLASE

This enzyme [EC 2.4.1.1] catalyzes the reaction of starch (that is, [(1,4)- α -D-glucosyl]_{*n*}) with orthophosphate to

produce [(1,4)- α -D-glucosyl]_(n-1) and α -D-glucose 1-phosphate. **See** *Glycogen Phosphorylase*

STARCH SYNTHASE

Starch synthase [EC 2.4.1.11] catalyzes the reaction of UDP-glucose with starch (that is, [(1,4)- α -D-glucosyl]_n) to generate UDP and [(1,4)- α -D-glucosyl]_(n+1). Starch (bacterial glycogen) synthase [EC 2.4.1.21], also referred to as glycogen synthase and ADP-glucose-starch glucosyltransferase, catalyzes the reaction of ADP-glucose and [(1,4)- α -D-glucosyl]_n to produce ADP and [(1,4)- α -D-glucosyl]_(n+1). This enzyme is similar to glycogen synthase and starch synthase except that it utilizes ADP-glucose. The recommended name varies according to the source of the enzyme and the nature of its synthetic product; for example, starch synthase, bacterial glycogen synthase, *etc.*

A. M. Smith, K. Denyer & C. Martin (1997) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 67.

O. Nelson & D. Pan (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 475.

STATE FUNCTION

A function (such as energy) that is dependent only on the state of the system and not on the pathway as to how that state was reached. Examples of state functions include internal energy, enthalpy, entropy, Gibbs free

energy, Helmholtz free energy, pressure, volume, temperature, *etc.* Work and heat are not state functions.

STATIONARY STATE

1. A synonym for steady state. 2. In quantum mechanics, a state which does not change or evolve with time. **See** *Steady State*

STATISTICAL ANALYSIS OF ENZYME RATE DATA

The proper evaluation and assessment of the calculated or graphically determined values of the kinetic parameters requires the application of statistical analysis¹⁻⁵. This is also true when looking for possible patterns in the various plots (*e.g.*, parallel lines *vs.* intersecting lines). When reporting kinetic values, the error limits should always be provided. Programs are available that statistically evaluate kinetic data³. **See** *Statistics (A Primer)*

¹G. N. Wilkinson (1961) *Biochem. J.* **80**, 324.

²W. W. Cleland (1967) *Adv. Enzymol.* **29**, 1.

³W. W. Cleland (1979) *Meth. Enzymol.* **63**, 103.

⁴B. Mannervik (1982) *Meth. Enzymol.* **87**, 370.

⁵R. G. Duggleby (1995) *Meth. Enzymol.* **249**, 61.

STATISTICAL FACTORS

Terms appearing in binding equations describing the linked equilibria for the binding of a ligand (or proton)

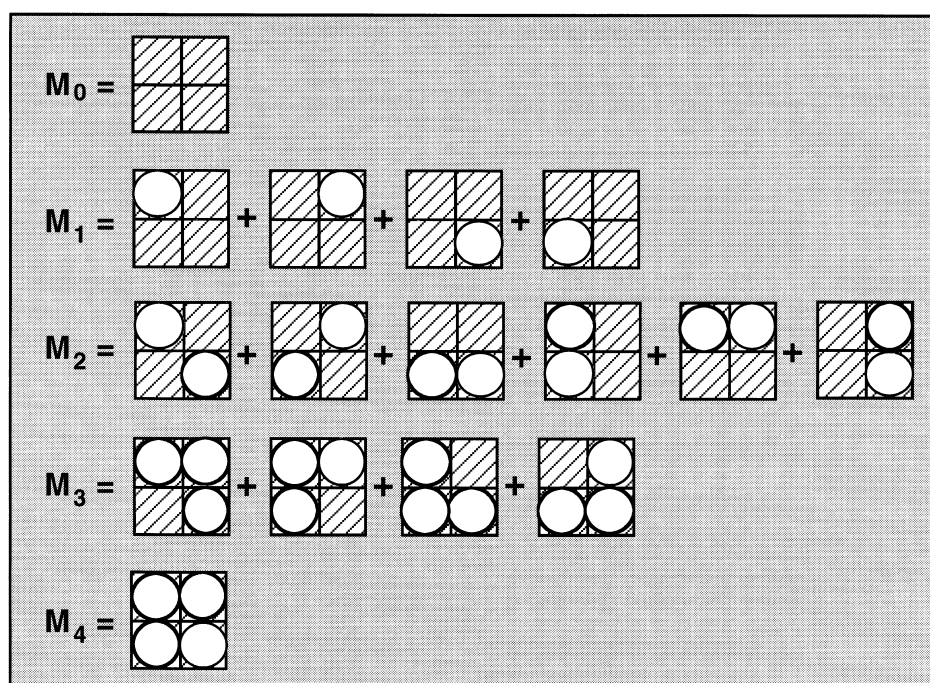


Figure 1. A schematic diagram indicating the multiple path by which a ligand (O) binds to an available site (■).

to polyvalent receptors. Statistical factors permit one to account for the multiple microscopically distinct species formed in the course of filling multiple sites on a polyvalent acid, a multisite protein, or any lattice of sites available for ligand binding. As discussed in detail by Edsall and Wyman¹ and Cantor and Schimmel², these terms are required to account for the various ways for adding a ligand from each state of ligand occupancy. Failure to include statistical factors, will result in an inability to obtain the polynomial function known as the ligand saturation function.

Taking ligand binding to a tetramer P_4 as an example (see Fig. 1), there are four ways that such a tetramer can bind the first molecule of ligand L: one for each of the four equivalent sites. In the general equation for binding of a ligand to a protein containing n sites that are equivalent (and therefore independent), the intrinsic microscopic dissociation constant k is the same for each ligand-receptor interaction. For the second ligand molecule, there are six different ways to convert a singly occupied tetramer P_4L_1 to generate a doubly occupied tetramer P_4L_2 . Generally, to bind i ligands on n -sites, there are $\Omega_{n,i} = n!/(n-i)!$ ways. Thus, for i equal to 1, $\Omega_{4,1} = 4$, and for i equal to 2, $\Omega_{4,2} = 6$.

¹J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, pp. 623-626, Academic Press, New York.

²C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, Part III, pp. 852-853, Freeman, San Francisco.

STATISTICS (A Primer)

The scientific method has several elemental stages: study, hypothesis, and experimentation. We rely on knowledge and insight to advance new conjectures about the nature of a process being examined, and one cannot minimize the importance of a good command of the associated literature. (Library research is time well spent.) A well considered hypothesis is the predicate for effective experimentation. A sound understanding of the particular experimental technique and all relevant theory is indispensable. Successful investigators almost invariably adopt research plans that allow a larger problem to be broken down into a series of easily tested parts. One should also heed the engineer's exhortation—"Keep it simple, stupid!"—when designing experiments. This adage also reminds us that straightforward experimental design often minimizes the chance introduction of unforeseen variables.

We usually seek to distinguish between two possibilities: (a) the *null hypothesis*—a conjecture that the observed set of results arises simply from the random effects of uncontrolled variables; and (b) the *alternative hypothesis* (or *research hypothesis*)—a trial idea about how certain factors determine the outcome of an experiment. We often begin by considering theoretical arguments that can help us decide how two rival models yield nonisomorphic (*i.e.*, characteristically different) features that may be observable under a certain set of imposed experimental conditions. In the latter case, the null hypothesis is that the observed differences are again haphazard outcomes of random behavior, and the alternative hypothesis is that the nonisomorphic feature(s) is (are) useful in discriminating between the two models.

The logic and rigor of statistics also reside at the heart of modern science. Statistical evaluation is the first step in considering whether a body of measurements allows one to discriminate among rival models for a biochemical process. Beyond their value in experimental sciences, probabilistic considerations also help us to formulate theories about the behavior of molecules and particles and to conceptualize stochastic and chaotic events.

We cannot possibly provide a detailed account of the full range of statistical methods and strategies that constitute the armamentarium of modern science. Yet, there is value in considering some basic statistical principles.

RANDOM SAMPLING AND RANDOM VARIABLES. A sample n is a representative subset of the larger population N . The sampling space, designated S , that is associated with an experiment is the set whose elements correspond to all possible sampling points. Experiments rarely yield the identical numerical value upon repeated sampling, and the value of the numerical event is called a random variable. In fact, the random variable is a real-valued function that is defined over the sample space. Sampling is said to be random if each of possible samples n_i , taken from N has an equal probability of occurring.

A second kind of error, namely systematic error, introduces bias in the data set, because systematic errors are most often the same for each observation made with a given apparatus or measuring device. This will tend to

displace the frequency distribution curve describing a random variable in one direction. Moreover, certain threshold phenomenon, such a quenching in liquid scintillation counting, may favor detection of higher energy β particles. Behavior of this sort may also have a nonrandom distribution of data about a population mean. Other problems arise from personal (or operator) error, and the occurrence of mistakes (honest errors) is unavoidable in most circumstances. Individual bias can also wreak havoc on experiments. While the effects of this type of error can be mitigated somewhat by double-blinded strategies, within the molecular life sciences, the detailed plan for most experiments is almost always known by the researcher during the course of the experiment. Two

related matters deserve consideration: first, a contemporaneous laboratory notebook record can prove invaluable as one justifiably discounts data points known or suspected to be erroneous; second, keeping short the time elapsed between data acquisition and analysis is an imperative. Except for Sherlock Holmes, most sleuths abhor a cold stiff body.

Randomization is a strategy that helps the experimenter to obtain a statistically unbiased sample or data set for a series of experimental measurements by simulating a chance distribution or chance sequence. Randomization often eliminates or minimizes interference by otherwise irrelevant variables. There are, in fact, many instances

Table I

A collection of "random" numbers, based on the first 1392 integers found in the numerical value of π

31	41	59	26	53	58	97	93	23	84	62	64	33	83	27	95	02	88	41	97	16	93	99	75	10	58	20	97	49
44	59	23	07	81	64	06	28	62	08	99	86	28	03	48	25	34	21	17	06	79	86	20	89	98	62	80	34	82
53	42	11	70	67	98	21	48	08	65	13	28	23	06	64	70	93	84	46	09	55	05	82	23	17	25	35	94	08
12	84	81	11	74	50	28	41	02	70	19	38	52	11	05	55	96	44	62	29	48	95	49	30	38	19	64	42	88
10	97	56	65	93	34	46	12	84	75	64	82	33	78	67	83	16	52	71	20	19	09	14	56	48	56	69	23	46
03	48	61	04	54	32	66	48	21	33	93	60	72	60	24	91	41	27	37	24	58	70	06	60	63	15	58	81	74
88	15	20	92	09	62	82	92	54	09	17	15	36	43	67	89	25	90	36	00	11	33	05	30	54	88	20	46	65
21	38	41	46	95	19	41	51	16	09	43	30	57	27	03	65	75	95	91	95	30	92	18	61	17	38	19	32	61
17	93	10	51	18	54	80	74	46	23	79	96	27	49	56	73	51	88	57	52	72	48	91	22	79	38	18	30	11
94	91	29	83	36	73	36	24	40	65	66	43	08	60	21	39	49	46	39	52	24	73	71	90	70	21	79	86	09
43	70	27	70	53	92	17	17	62	93	17	67	52	38	46	74	81	84	67	66	94	05	13	20	00	56	81	27	14
52	63	56	08	27	78	57	71	34	27	57	78	96	09	17	36	37	17	87	21	46	84	40	90	12	24	95	34	30
14	65	49	58	53	71	05	07	92	27	96	89	25	89	23	54	20	19	95	61	12	12	90	21	96	08	64	03	44
18	15	98	13	62	97	74	77	13	09	96	05	18	70	72	11	34	99	99	99	83	72	97	80	49	95	10	59	73
17	32	81	60	96	31	85	95	02	44	59	45	53	46	90	83	02	64	25	22	30	82	53	34	46	85	03	52	61
93	11	88	17	10	10	00	31	37	83	87	52	88	65	87	53	32	08	38	14	20	61	71	77	66	91	47	30	35
98	25	34	90	42	87	55	46	87	31	15	95	62	86	38	82	35	37	87	59	37	51	95	77	81	85	77	80	53
21	71	22	68	06	61	30	01	92	78	76	61	11	95	90	92	16	42	01	98	93	80	95	25	72	01	06	54	85
86	32	78	86	59	36	15	33	81	82	79	68	23	03	01	95	20	35	30	18	52	96	89	95	77	36	22	59	94
13	89	12	49	72	17	75	28	34	79	13	15	15	57	48	57	24	24	54	15	06	95	95	08	29	53	31	16	86
17	27	85	58	89	07	50	98	38	17	54	63	74	64	93	93	19	25	50	60	40	09	27	70	16	71	13	90	09
84	88	24	01	28	58	36	16	03	56	37	07	66	01	04	71	01	81	94	29	55	59	61	98	94	67	67	83	74
49	44	82	55	37	97	74	72	68	47	10	40	47	53	46	46	20	80	46	68	42	59	06	94	91	29	33	13	67
70	28	98	91	52	10	47	52	16	20	56	96	02	40	58	03	81	50	19	35	11	25	33	82	43	00	35	58	74

when randomization can prevent the introduction of human bias in the choice of sampling schemes. The mere anticipation of a certain result has been known to influence the outcome of a measurement, particularly when a measurement is of a qualitative nature. With enzyme rate assays, a decision to order all of the measurements from low to high substrate/effector concentration, or vice versa, may lead to an incorrect data set if the stock solution of enzyme or substrate undergoes some time-dependent change that influences chemical reactivity. In fact, time is a variable that is difficult to hold constant, because one can rarely carry out all measurements simultaneously. Likewise, various measuring instruments can introduce bias when the data collected are uncorrected for time-dependent drift in the signal strength, line voltage, and thermal noise.

Perhaps the simplest algorithm for obtaining a randomized pairing of variables is the toss of a coin. However, to randomize the order of executing a larger series of measurements, one typically employs a random number generator or a random number table to assign random numbers to each of the measurements.

With a random number table (such as the one shown above), one can arrange a small number of experimental conditions into a random order. If one were to have ten samples, initially arranged from lowest to highest substrate concentration, then the experimenter can use the table shown above to randomize these sample with respect to the order of performing the measurements. To assign a random number to the first sample, an eraser on the back of a pencil can be allowed to contact some number in the table, while the experimenter keeps his/her eyes closed. By proceeding an upward or downward (or leftward or rightward) direction according to some predetermined sequence, say every other number in the table, each of the subsequent samples can be assigned a number from the table. The measurements are then executed in the order that corresponds to the increasing (or decreasing) magnitude of the numbers chosen from the random number table. Should two identical numbers be assigned from the table, a coin flip can readily determine the order (e.g., “heads” before “tails”) of execution for these samples.

GENERAL DEFINITIONS. Effective use of statistical principles requires a clear understanding of certain fundamental terms.

The mean $\bar{x}$ (also known as the arithmetic mean or average) is given as:

$$\bar{x} = \frac{\sum_{i=1}^n X_i}{n}$$

The variance V is estimated by the mean value of the square of an observation's deviation from the arithmetic mean:

$$V = \frac{(X_1 - \bar{x})^2 + (X_2 - \bar{x})^2 + \cdots + (X_n - \bar{x})^2}{n - 1}$$

$$V = \frac{1}{n - 1} \sum_{i=1}^n n_i (X_i - \bar{x})^2$$

The standard deviation equals the square root of the variance V . This parameter, symbolized variously as S , s , or σ , is easier to visualize than the variance. Statisticians prefer to analyze a system in terms of variance. However, chemists and biochemists favor standard deviation which allows one to present the value for a particular set of measurements and its standard deviation, all in the same units.

The coefficient of variance (CV) is 100 times the ratio of the standard deviation divided to the arithmetic mean.

The frequency f_i equals n_i/n , where n_i is the number of things having the same value or property X_i divided by the total population of things. It follows that the mean can also be calculated from knowledge of the frequency, such that

$$\bar{x} = \sum_{i=1}^n (f_i X_i)$$

FREQUENCY DISTRIBUTIONS. All of the members of some class of things is referred to as a population, and the properties of such a population are inferred from the behavior of a smaller group of things called a sample. Any single property of an individual within a population may differ from the same property for another individual, and we usually measure a sample mean to obtain a mean for the entire population. One often wishes to

observe how a given property is distributed for a population of individuals, and the frequency distribution of a variable is estimated on the basis of a frequency distribution within the sample.

1. Binomial (or Bernoulli) Distribution. This distribution applies when we are concerned with the number of times an event A occurs in n independent trials of an experiment, subject to two mutually exclusive outcomes A or B . (Note: The descriptor “independent” indicates that the outcome of one trial has no effect on the outcome of any other trial.) In each trial, we assume that outcome A has a probability $P(A) = p$, such that q , the probability of outcome A not occurring, equals $(1 - q)$. Assuming that the experiment is carried out n times, we can consider the random variable X as the number of times that outcome A takes place. X takes on values $1, 2, 3, \dots, n$. Considering the event $X = x$ (meaning that A occurs in x of the n performances of the experiment), all of the outcomes A occur x times, whereas all the outcomes B occur $(n - x)$ times. The probability $P(X = x)$ of the event $X = x$ can be written as:

$$f(x) = \binom{n}{x} p^x q^{n-x}$$

where $x = 1, 2, 3, \dots, n$. The frequency distribution determined by this probability function is the Binomial distribution.

2. Poisson Distribution. This distribution is a limiting case of the binomial distribution and is described by the probability function:

$$f(x) = \frac{\mu^x}{x!} e^{-\mu}$$

This result is obtained from the binomial distribution if we let p approach 0 and n approach infinity. In this case, the mean $\mu = np$ approaches a finite value. The variance of a Poisson distribution is given as: $\sigma^2 = \mu$.

For a λ of fixed value and an experiment with outcome y equal to $1, 2, 3, \dots$, the Poisson distribution can also be expressed as:

$$p(y) = \frac{\lambda^y}{y!} e^{-\lambda}, \quad \text{where } y = 0, 1, 2, \dots$$

As an example, one may consider that a random process occurs where a protein may bind at a given location within the cell as often as $y = 1, 2, 3, \dots$, times per second, but on average the protein binds once per second. If y is described by a Poisson probability function, estimate the probability that the protein will not bind at the site during a one-second interval. In this case, the time period is one second and the mean binding steps is λ equal to 1. Therefore,

$$p(y) = \frac{(1)^y e^{-1}}{y!} = \frac{e^{-1}}{y!}$$

The binding event that is missed in a one second period corresponds to $y \geq 0$, such that

$$p(0) = \frac{e^{-1}}{0!} = \frac{e^{-1}}{1} = 0.368$$

The probability that the protein will bind at least once is the event that $y \geq 1$; therefore:

$$P(y \geq 1) = \sum_{y=1}^{\infty} p(y) = 1 - p(0) = 1 - e^{-1} = 0.632$$

3. Gaussian (or Bell-Shaped) Distribution. One of the most frequently encountered frequency distributions is the Gaussian, or normal, distribution. This function has the density:

$$f(x) = (1/\sigma\sqrt{2\pi})e^{-1/2((x-\mu)/\sigma)^2}$$

where μ is the mean and σ is the standard deviation of the distribution. The corresponding distribution curve is bell-shaped. A random variable having this distribution is said to be normally distributed, or just “normal.” If the area under this curve is Q , then (a) over the interval $\mu - \sigma$ to $\mu + \sigma$, the area is about $0.68Q$; (b) over the interval $\mu - 2\sigma$ to $\mu + 2\sigma$, the area is about $0.955Q$; and (c) over the interval $\mu - 3\sigma$ to $\mu + 3\sigma$, the area is about $0.997Q$.

For data that do not conform to a Gaussian (or normal) distribution, one often can use a mathematical procedure that transforms the observed values to create a corresponding Gaussian distribution. When one knows population distribution, transforming data values to create a Gaussian distribution becomes a useful means for adapting the data set for analysis with statistical tests requiring a Gaussian distribution. If the data are Counts C (from

a Poisson distribution), use the square root of C as the normalizing transformation; for Proportion P (from a binomial distribution), use the arc-sine of square root of P as the normalizing transformation; for a Measurement M (from a log-normal distribution), use the logarithm of M as the normalizing transformation; and for duration (D), use $1/D$ as the normalizing transformation.

CONFIDENCE LEVELS. The normal distribution curve can also be used to assign confidence limits. Recall that the population mean (μ) of a measurement is a constant that remains unknown unless one samples every individual within a population. In the absence of systematic errors, limits may be set within which the population mean is expected to lie with some given probability; these limits are called confidence limits. The population mean can be said to lie within $\mu \pm z\sigma$. For a confidence level of 50%, $z = 0.67$; for 68%, $z = 1.00$; for 80%, $z = 1.29$; for 90%, $z = 1.64$; for 95%, $z = 1.96$; for 96%, $z = 2.00$; for 99%, $z = 2.58$; for 99.7%, $z = 3.00$; and for 99.9%, $z = 3.29$. The equation for calculating a desired confidence level is:

$$\text{Confidence Level for } m = \bar{x} \pm \frac{z\sigma}{\sqrt{N}}$$

As an example, let's say that for a set of three determinations (*i.e.*, $N = 3$), the arithmetic mean for the rate measurements and its standard deviation are $1.63 \mu\text{M/s}$ and $0.1 \mu\text{M/s}$. If we wanted to know the range of values over which there is a 95% confidence of obtaining the true mean, we would carry out the following short computation:

$$\begin{aligned} 95\% \text{ Confidence Level} &= 1.63 \pm \frac{1.96 \times 0.1}{\sqrt{3}} \\ &= 1.63 \pm 0.11 \mu\text{M/s} \end{aligned}$$

Note: We used the value of 1.96 for the value of z given above for the 95% confidence level.

One can demonstrate that the effect of random error may be reduced without limit by using an ever larger sample size. In fact, the standard deviation of the mean (σ_m) can be written as:

$$\sigma_m = \frac{\sigma}{\sqrt{N}}$$

Therefore, one would need to make 100 times as many observations to increase the accuracy of a measurement by 10, and 10,000 times as many to improve accuracy by a factor of 100. This indicates that one quickly obtains diminishing returns in accuracy for increasing numbers of observations.

Analysis of Variance (ANOVA). Keeping in mind that the total variance is the sum of squares of deviations from the grand mean, this mathematical operation allows one to partition variance. ANOVA is therefore a statistical procedure that helps one to learn whether sample means of various factors vary significantly from one another and whether they interact significantly with each other. One-way analysis of variance is used to test the null hypothesis that multiple population means are all equal.

One uses ANOVA when comparing differences between three or more means. For two samples, the one-way ANOVA is the equivalent of the two-sample (unpaired) t test. The basic assumptions are: (a) within each sample, the values are independent and identically normally distributed (*i.e.*, they have the same mean and variance); (b) samples are independent of each other; (c) the different samples are all assumed to come from populations having the same variance, thereby allowing for a pooled estimate of the variance; and (d) for a multiple comparisons test of the sample means to be meaningful, the populations are viewed as fixed, meaning that the populations in the experiment include all those of interest.

Several or many factors may contribute simultaneously to the variance observed in an experiment. Thus the total variance can be expressed as:

$$\sigma_{total}^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \cdots + \sigma_n^2$$

This fact leads to the concept of total variance, a parameter corresponding to the sum of variances arising from different components of the variation. Analysis of variance represents a statistical means for partitioning the total variance into individual components. This method attempts to analyze variation by assigning portions of the variance to each set of independent variables. In general, the experimenter will not or cannot include all the variables influencing the response in an experiment, and random variation in the response is observed even

if all independent variables are kept constant. The variability of a set of n trial measurements is proportional to the sum of the deviations, $\sum_{i=1}^n (y_i - \bar{y})^2$. ANOVA separates the sum of the squares of deviations into parts that are individually attributable to one of the experiment's independent variables, such that

$$\begin{aligned} SS_{total} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= SS V_1 + SS V_2 + \cdots + SS V_n + SS E \end{aligned}$$

where $SS V_1$, $SS V_2$, and $SS V_n$ are the sums of squares for variables 1, 2 and n . The last term is the sum of the squares assigned to error. For independent random samples from two normally distributed populations having means of μ_1 and μ_2 and equal variances, $\sigma_1^2 = \sigma_2^2 = \sigma^2$, and the total variance of measurements around their respective means for the two samples can be written as:

$$\begin{aligned} SS_{total} &= \sum_{i=1}^2 \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2 \\ &= n_1 \sum_{i=1}^2 (\bar{y}_i - \bar{y})^2 + \sum_{i=1}^2 \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 \end{aligned}$$

where $\bar{y}_i$ is the average of the measurements in the i th sample. The first sum (on the rightmost side) is often called the sum-of-the-squares for treatments, and the second is the sum of squares of deviations that one uses to evaluate s^2 . Without presenting a detailed description, suffice it to say that analysis of variance allows one to evaluate when the sum of the squares for treatments is sufficiently large to detect a statistically significant difference between means of μ_1 and μ_2 .

LEAST SQUARES TREATMENT. Here, we deal briefly with the statistical algorithm, also known as linear regression analysis, for systems where Y (the random variable) is linearly dependent on another quantity X (the ordinary or controlled variable). The procedure allows one to fit a straight line through points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , $\cdots$, (x_n, y_n) where the values x_i are defined before the experiment and y_i values are obtained experimentally and are subject to random error. The best fit line through such a series of points is called a "least squares fit," and the protocol provides measures of the reliability of the data and quality of the fit.

For such a series of ordered pairs, the vertical displacement of a sample value, say (x_j, y_j) , from the straight line $y = a + bx$ is $|y_j - a - bx_j|$, and the sum of these squares is given as:

$$Q = \sum_{j=1}^n (y_j - a - bx_j)^2$$

Values of a and b are chosen to minimize the parameter Q which is referred to "the sum of the squares of $(y_j - a - bx_j)$ ", hence the name "Least Squares". Accordingly, the equation for the best fit least squares line is:

$$[y = a + bx]_{Q \min}$$

This line is frequently referred to as a regression line, and its slope b is the regression coefficient. This procedure can be extended to polynomial functions wherein segments can be treated as linear or nearly linear functions of Y on X .

COMPOUNDING OF ERRORS. Data collected in an experiment seldom involves a single operation, a single adjustment, or a single experimental determination. For example, in studies of an enzyme-catalyzed reaction, one must separately prepare stock solutions of enzyme and substrate, one must then mix these and other components to arrive at desired assay concentrations, followed by spectrophotometric determinations of reaction rates. A Lowry determination of protein or enzyme concentration has its own error, as does the spectrophotometric determination of ATP that is based on a known molar absorptivity. All operations are subject to error, and the error for the entire set of operations performed in the course of an experiment is said to involve the compounding of errors. In some circumstances, the experimenter may want to conduct an error analysis to assess the contributions of statistical uncertainties arising in component operations to the error of the entire set of operations. Knowledge of standard deviations from component operations can also be utilized to estimate the overall experimental error.

One begins by assuming that a final result, call it r , is related to a series of components x_i in some functional manner:

$$r = F(x_1, x_2, x_3, \dots, x_n)$$

Small variations in x_i can be written as dx_i , and the incremental effect on result r is dr , such that:

$$\begin{aligned} dr &= \frac{\partial F}{\partial x_1} dx_1 + \frac{\partial F}{\partial x_2} dx_2 + \frac{\partial F}{\partial x_3} dx_3 + \dots + \frac{\partial F}{\partial x_i} dx_i \\ &= \sum_{i=1}^n \frac{\partial F}{\partial x_i} dx_i \end{aligned}$$

As noted earlier, the variance V is the square of the standard deviation s ; thus, if each error is independent of the other, the overall variance for the above expression can be written as:

$$\sigma^2 = (dr)^2 = \sum_{i=1}^n \left(\frac{\partial F}{\partial x_i} \right)^2 dx_i^2$$

This confirms an earlier stated property of total variance:

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots + \sigma_i^2$$

Expressed in terms of the standard deviation,

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots + \sigma_i^2}$$

For example, compounding the error of three operations, each with standard deviations of in concentration, say ± 2.0 mM, ± 1.1 mM, and ± 1.4 mM, would yield an overall standard deviation:

$$\sigma_{total} = \sqrt{(2.0)^2 + (1.1)^2 + (1.4)^2} \approx 2.7$$

This exercise reminds us that the compounded error will always be larger than the single standard deviation for any single operation. If the standard deviations for component operations do not have the same units, they cannot be treated as described here. In many cases, the investigator can convert the uncertainty in one type of operation into units from another. For example, one might have a standard deviation of 0.01 mM in substrate concentration and a standard deviation of 0.6 absorbance units. If conversion of 0.01 mM substrate to product resulted in an absorbancy of 0.4 (*i.e.*, $A = \epsilon[S] = 0.4$), then the overall standard deviation would be $[(0.4)^2 + (0.6)^2]^{1/2}$ or 0.72 absorbance units.

One may also consider the compound effect of error when the result r is related to a ratio of components x_1/x_2 . In this case,

$$\left(\frac{\sigma}{r} \right) = \sqrt{\left(\frac{\sigma_1}{x_1} \right)^2 + \left(\frac{\sigma_2}{x_2} \right)^2}$$

CRITICAL t TEST FOR COMPARING AVERAGES.

Whenever two sets of analytical determinations have different arithmetic averages, but comparable standard deviations, one may apply the t test to assess the statistical significance of the difference in these averages.

For two averages, say $\bar{x}$ and $\bar{y}$, the parameter t is given by the following equation:

$$t = \frac{|\bar{x} - \bar{y}|}{s} \sqrt{nm(n+m)}$$

where the vertical bars indicate absolute value in order to make t greater than zero for two nonidentical averages. The coefficients m and n correspond to the number of values that were averaged to obtain $\bar{x}$ and $\bar{y}$.

Table II

Critical t Values for Comparing Averages

Degrees of Freedom ($n + m$) - 2	Confidence Level	
	95%	99%
1	12.7	63.7
2	4.3	9.9
3	3.2	5.8
4	2.8	4.6
5	2.6	4.0
6	2.5	3.7
8	2.3	3.4
10	2.3	3.2
15	2.1	2.9
20	2.1	2.8

If the calculated t value exceeds the tabulated value for the indicated degrees of freedom, then chances are 95-out-of-100 or 99-out-of-100 that the two averages are significantly different.

Before interpreting the results of a t test, one would do well to think carefully about whether an appropriate test has been chosen. First, one should not confuse t tests with other statistical concepts such as correlation and regression; the t test compares one variable between two groups, and one should use correlation and regression to learn how two different variables vary together. Second, do not confuse t tests with ANOVA; while t tests and related nonparametric tests exactly compare two groups, ANOVA compares three or more groups. Third, remember that t tests also must not be confused with analyses that use so-called contingency tables (*e.g.*, Fisher's or Chi Square (χ^2) tests); the t test is employed when one wishes to compare continuous variables,

while a contingency table is used to analyze categorical variables.

- W. Mendenhall, R. L. Schaeffer & D. D. Wackerly (1981) *Mathematical Statistics with Applications*, 2nd ed., Duxbury Press, Boston.
- A. Papoulis (1991) *Probability, Random Variables, and Stochastic Processes*, 3rd ed., McGraw Hill, New York.
- W. W. Daniel (1995) *Biostatistics*, 6th ed., Wiley, New York.
- H. R. Lindman (1992) *Analysis of Variance in Experimental Design*. Springer-Verlag, New York.
- R. G. Miller, Jr. (1996) *Beyond ANOVA, Basics of Applied Statistics*, 2nd ed., Chapman & Hall, London.
- J. Neter, W. Wasserman, & M. H. Kutner (1990) *Applied Linear Statistical Models*, 3rd ed., Irwin Press, Homewood, IL.
- B. Rosner (1995) *Fundamentals of Biostatistics*, 4th ed., Duxbury Press, Belmont, California.
- B. J. Winer, D. R. Brown & K. M. Michels (1991) *Statistical Principles in Experimental Design*, 3rd ed., McGraw-Hill, New York.
- J. H. Zar (1996) *Biostatistical Analysis*, 3rd ed., Prentice-Hall, Upper Saddle River, New Jersey.

STEADY STATE

1. A condition achieved in a multistep process when the rate of formation of unstable intermediates is time-invariant, or nearly so. 2. In stirred flow systems, the steady-state condition refers to the procedure in which the concentration of all components is kept constant, via additions to and withdrawals from the system.

STEADY-STATE ASSUMPTION

A mathematical simplification of rate behavior of a multistep chemical process assuming that over a period of time a system displays little or no change in the concentration(s) of intermediate species (*i.e.*, $d[\text{intermediate}]/dt \approx 0$). In enzyme kinetics, the steady-state assumption allows one to write and solve the differential equations defining the rates of interconversion of various enzyme species. This is especially useful in initial rate studies.

If the concentration of substrate is not at least 100 times the concentration of enzyme, the steady state will not persist over the time course of most experiments. In such cases, the resulting initial rate data cannot be analyzed by standard initial rate kinetic procedures. **See also** *Enzyme Kinetics; Numerical Integration*

STEPWISE REACTION

A sequence of at least two elementary reactions and having at least one reaction intermediate. **See** *Multistage Reaction*

STEREOCHEMICAL FIDELITY

Enzymatic catalysis that is stereochemically specific or highly selective, arising from the ability of the enzyme to orient the chiral center within the active site in a manner that leads to highly efficient stereochemical discrimination. An example is the hydride transfer from lactate's (*S*)-enantiomer to the *re*- or *A*-side of bound NAD^+ . This reaction proceeds with stereochemical fidelity of over 99.999998%. The occurrence of such stereochemical fidelity has at least three explanations¹: (a) that greater catalytic efficiency attends stereochemical fidelity; (b) that by chance evolution has inherited one type of transfer reaction; or (c) that its absence in some potentially stereoselective cases may reflect that the natural selective advantage for fidelity did not outweigh other advantages gained by its absence. In the case of NAD^+ -dependent enzymes such as LDH, the NADH product is equally active metabolically, suggesting that some catalytic advantage is gained².

¹D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 325.

²V. E. Anderson & R. D. LaReau (1988) *J. Am. Chem. Soc.* **110**, 3695.

STEREOCHEMICAL TERMINOLOGY (IUPAC Recommendations)

If the word you are searching for in the Wordfinder directed you here, then it is a term that lies beyond the scope of this Handbook. The listed word will be found in an authoritative internet-accessible glossary dealing with stereochemistry [<http://www.chem.qmw.ac.uk/iupac/stereo/>]. The glossary was developed under the auspices of the IUPAC Organic Chemistry Division's Commission on Nomenclature of Organic Chemistry and Commission on Physical Organic Chemistry. All definitions used in the internet glossary are identical to those in the published document¹, which the reader should cite if the definition is used verbatim or nearly so.

¹G. P. Moss (1996) *Pure and Appl. Chem.* **68**, 2193.

STEREOCHEMISTRY OF HYDRIDE TRANSFER TO/FROM NADH

NADH has two faces, designated the *A* (or *re*) face and the *B* (or *si*) face. Nearly fifty years ago, Vennesland, Westheimer, and their associates first developed a durable strategy for determining the stereochemical course of hydride transfer to and from the *A* or *B* face of

NADH. These experiments were pioneering with respect to contemporary enzymology, especially with regard to early recognition that coenzymes are held within enzyme active sites in stereochemically preferred ways. One typically utilizes NADH that contains a tritium or deuterium atom in the 4*R* or 4*S* position, and the success or failure of substrate deuteration/tritiation indicates the stereochemistry. Westheimer¹ has tabulated the known examples of dehydrogenases that exhibit specificity for a particular face of NADH. Creighton and Murthy² have reproduced this tabulation in their comprehensive review on the stereochemistry of enzyme-catalyzed reactions at carbon.

¹F. H. Westheimer (1987) in *Pyridine Nucleotide Coenzymes*, (D. Dolphin, R. Poulson, & O. Avramovic, eds.) Part A, p. 253, Wiley, New York.

²D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 325.

STEREOCHEMISTRY OF HYDROGEN TRANSFER TO/FROM FADH₂

Flavoenzymes use their tightly bound FAD or FMN coenzyme to promote oxidation-reduction reactions involving hydrogen transfer to the *re* or *si* face of their coenzyme. Fisher & Walsh¹ showed that the glutathione reductase reaction involves direct hydrogen transfer, and Pai and Schulz² advanced a general strategy for determining the stereochemical course of flavoenzyme-catalyzed reactions based on the stereochemistry of glutathione reductase. Creighton and Murthy³ present the stereochemistry of sixteen flavoenzyme reactions in their comprehensive review on the stereochemistry of enzyme-catalyzed reactions at carbon.

¹J. Fisher & C. T. Walsh (1974) *J. Am. Chem. Soc.* **96**, 4345.

²E. F. Pai & P. E. Schulz (1983) *J. Biol. Chem.* **258**, 1752.

³D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 324.

STEREOCHEMISTRY OF ISOMERASES INVOLVING PROTON TRANSFER

Creighton and Murthy¹ recently reviewed the stereochemistry and related mechanistic issues associated with enzyme-catalyzed isomerizations that proceed by 1,2-hydrogen transfer or by 1,3-allylic hydrogen transfer. In the first case, the prototypical aldose-ketose isomerase is triose-phosphate isomerase (or TPI), an enzyme that uses the carboxylate of Glu-165 as a base for abstracting a proton from the substrate during catalysis. Δ^5 -3-Keto-

steroid isomerase is likewise a leading example of a reaction involving 1,3-allylic proton transfer.

¹D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 344.

STEREOELECTRONIC

Referring to or pertaining to a dependence of the properties of a molecular entity or a transition-state species in a particular electronic state on the relative nuclear geometry. Such effects are usually due to different geometrical arrangements.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

STEREOSELECTIVITY

The preferential formation of one stereoisomer over another in a chemical reaction. If the stereoisomers of concern are enantiomers then the term enantioselectivity is often used. The degree of enantioselectivity is usually expressed by the enantiomeric excess. If the stereoisomers of concern are diastereoisomers, then the term used is diastereoselectivity. It is quantitatively measured by the diastereoisomer excess.

E. L. Eliel (1962) *Stereochemistry of Carbon Compounds*, McGraw-Hill, New York.

STEREOSPECIFICITY

1. A stereospecific chemical reaction is one in which starting substrates or reactants, differing only in their configuration, are converted into stereoisomeric products. Note, with this definition a stereospecific reaction has to be stereoselective whereas the inverse statement (that is, with respect to a stereoselective reaction or process) is not necessarily true. 2. Referring to reactions that act on only one stereoisomer (or, have a preference for one stereoisomer). Thus, many enzyme-catalyzed reactions are stereospecific, and characterization of that stereospecificity is always an issue to be addressed for a particular enzyme.

E. L. Eliel (1962) *Stereochemistry of Carbon Compounds*, McGraw-Hill, New York.

STERIC-APPROACH CONTROL

The means of achieving stereoselectivity due to steric hindrance toward the attack of a reagent or cosubstrate. The less hindered face will be the more susceptible side. Many enzymes utilize this method of product control.

STERIC HINDRANCE

A steric effect on chemical reactivity resulting from a crowding of substituents around an otherwise reactive center.

STEROID 11 β -HYDROXYLASE

This heme-thiolate-dependent enzyme [EC 1.14.15.4], also known as steroid 11 β -monooxygenase, catalyzes the reaction of a steroid with reduced adrenal ferredoxin and dioxygen to produce an 11 β -hydroxysteroid, oxidized adrenal ferredoxin, and water. The enzyme also catalyzes the hydroxylation of steroids at the 18-position and can catalyze the conversion of 18-hydroxycorticosterone into aldosterone.

M. J. Coon & D. R. Koop (1983) *The Enzymes*, 3rd ed., **16**, 645.
V. Ullrich & W. Duppel (1975) *The Enzymes*, 3rd ed., **12**, 253.

STEROID 17 α -HYDROXYLASE

This heme-thiolate-dependent enzyme [EC 1.14.99.9], also known as steroid 17 α -monooxygenase, catalyzes the reaction of a steroid with AH₂ and dioxygen to produce a 17 α -hydroxysteroid, A, and water.

M. J. Coon & D. R. Koop (1983) *The Enzymes*, 3rd ed., **16**, 645.

STEROID 21-HYDROXYLASE

This heme-thiolate-dependent enzyme [EC 1.14.99.10], also known as steroid 21-monooxygenase, catalyzes the reaction of a steroid with AH₂ and dioxygen to produce a 21-hydroxysteroid, A, and water.

M. J. Coon & D. R. Koop (1983) *The Enzymes*, 3rd ed., **16**, 645.

STEROID SULFOTRANSFERASE

This enzyme [EC 2.8.2.15] catalyzes the reaction of 3'-phosphoadenylylsulfate with a phenolic steroid to produce adenosine 3',5'-bisphosphate and a steroid O-sulfate. The enzyme is very similar in its activity to alcohol sulfotransferase. However, steroid sulfotransferase can utilize estrone as a substrate.

S. G. Ramaswamy & W. B. Jakoby (1987) *Meth. Enzymol.* **143**, 201.

STICKY SUBSTRATES

A designation given to a substrate (or ligand) that displays a strong tendency to remain bound to its binding site, and, in the case of enzymes, to undergo enzymatic catalysis with greater ease than to dissociate. Cleland¹ states that the stickiness of a substrate is a measure of the ratio of the net constant for reaction of the first

collision complex through the first irreversible step to the rate constant for dissociation of that collision complex. The strength of enzyme-substrate complexation can be altered by solution variables (*e.g.*, temperature, ionic strength, pH, solvent polarity, *etc.*) or factors such as proteolytic alteration of enzyme structure as in the case of yeast hexokinase's interaction with D-glucose². **See Isotope Trapping**

¹W. W. Cleland (1982) *Meth. Enzymol.* **87**, 631.

²I. A. Rose, E. L. O'Connell, S. Litwin & J. Bar-Tana (1974) *J. Biol. Chem.* **249**, 5163.

STIFFNESS INSTABILITY

A term used in numerical integration of rate processes to describe computational difficulties arising whenever reaction steps within a mechanism are each characterized by widely disparate time constants. In such a case, the most rapid step must be integrated repetitively over such small time intervals that these intervals are far too small for efficient integration of the slower reaction process. As a result, excessively long periods for computation are required for accurate integration of both steps in a mechanism. This problem was first noted in 1952¹, and Gear² has advanced a predictor-corrector algorithm to vary the size of the time interval to speed up numerical integration. **See Gear Algorithm**

¹C. F. Curtiss & J. O. Hirschfelder (1952) *Proc. Natl. Acad. Sci. U.S.A.* **38**, 235.

²C. W. Gear (1971) *Numerical Initial-Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, N. J.

STOICHIOMETRIC NUMBER

A number (usually symbolized by *n* followed by a subscript denoting the species) equal to the coefficient of that species in a particular reaction. For example, for the reaction A + 2B = 3P, the stoichiometric number for P is 3. By convention, stoichiometric numbers are positive for products and negative for reactants.

STOKES SHIFT

If an electronic transition results in both absorption and luminescence, then the Stokes shift is the difference (in either wavelength or frequency units) between the band maxima. If the luminescence occurs at a shorter wavelength, the difference is often referred to as an anti-Stokes shift.

STOPPED-FLOW MASS SPECTROMETRY IN ENZYME KINETICS

Northrop and Simpson¹ recently described how the interfacing of mass spectrometry with a stopped-flow mixing device promises to become a powerful approach for examining kinetic mechanisms as well as details of the chemical transformations attending catalysis. For example, one might apply this technique to an enzyme system suspected of forming a transient enzyme-substrate covalent intermediate. In this case, the covalently modified enzyme might be intercepted before it reacts with a second substrate or before it is transformed into a reaction product. Grieg *et al.*² already succeeded in determining dissociation constants for reversible protein·ligand complexes by using an electrospray injection port, and the technique may be especially useful for analyzing the properties of tight-binding inhibitors¹.

¹D. B. Northrop & F. B. Simpson (1997) *Bioorg. & Med. Chem.* **5**, 641.

²M. J. Grieg, H. Gaus, L. L. Cummins, H. Sasmor & R. H. Griffey (1995) *J. Am. Chem. Soc.* **117**, 10765.

STOPPED-FLOW TECHNIQUES

Any of the rapid kinetic methods^{1,2} relying on two syringes for passage of two solutions through a mixer and an observation cuvette into a third, so-called stopping-syringe. Filling of the latter mechanically trips a switch that starts the electronic acquisition of spectral data (usually accomplished by absorption or fluorescence spectroscopy). Such data provides kinetic information concerning the changes in reactant concentrations almost immediately after the chemical reaction is initiated by

the mixing of the reactant-containing (or reaction-initiating) solutions.

The dead-time of stopped-flow devices is the sum of two parameters, namely the mixing time and the delay time. The mixing time is controlled by the efficiency of turbulent flow, usually achieved in a multi-jet arrangement that promotes intimate mixing of the fluids. Mixing time is strongly influenced by the need to choose mixing conditions that avoid or minimize the effect of cavitation, the mechanically induced formation of a vapor-filled cavity within a rapid-flow device arising as a consequence of fluid viscosity and high shear forces. Cavitation within the cuvette compartment creates an inhomogeneous medium that scatters the incident light beam used in the photo-electronic detection of absorbance or fluorescence signals. Delay time, the second contributor to the device's dead-time, is the elapsed time required for all electromechanical events associated with signal detection.

A derivative technique, known as stopped-flow/temperature-jump³, allows one to take advantage of the stopped-flow strategy to establish a steady-state rate condition which can then be perturbed and analyzed by using the elements of a temperature-jump device.

¹B. Chance (1964) in *Rapid Mixing and Sampling Techniques in Biochemistry* (B. Chance, R. Eisenhardt, Q. H. Gibson & K. Lonberg-Holm, eds) p. 125, Academic Press, New York.

²Q. H. Gibson (1969) *Meth. Enzymol.* **16**, 187.

³G. G. Hammes (1974) in "Investigations of Rates and Mechanisms of Reactions: Part II, Investigation of Elementary Steps in Solu-

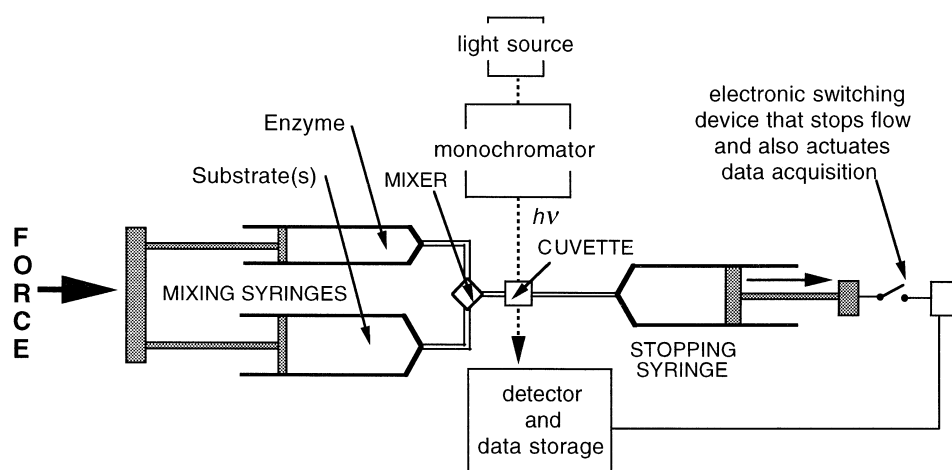


Figure 1. Construction and flow path of a stopped-flow device.

tion and Very Fast Reactions," *Techniques in Chemistry*, vol. VI (G. G. Hammes, ed.) p. 165, Wiley-Interscience, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Construction details, **16**, 189-194; Gibson-Milnes mixer, **16**, 192; temperature control, **16**, 194-197; light source, **16**, 197; monochromators, **16**, 198; photomultiplier and power supply, **16**, 198-200; noise levels, **16**, 205-207; problems (including air bubbles, leakage, sample contamination by diffusion from one working syringe to another, syringe breakage, anaerobic operations, sample poisoning by contact with stainless steel components), **16**, 207-210; Strittmatter microstopped-flow apparatus, **16**, 210-214; observation of rapid fluorescence changes, **16**, 214-218; pH detector for stopped flow, **16**, 218; polarographic recording, **16**, 218; dual-wavelength recording, **16**, 220-222; factors affecting dead time, **249**, 18; rapid scanning diode array, **249**, 19; pre-steady-state kinetics, **249**, 18; single turnover kinetics, **249**, 18; alcohol dehydrogenase, elucidation of mechanism, **246**, 16; circular dichroism, **246**, 61; copper amine oxidase [anaerobic conditions, **258**, 77; data analysis, **258**, 77-78, 81; deuterium isotope effects, **258**, 82-83; rapid-scanning analysis, **258**, 80-83; relaxation characteristics, **258**, 81-82]; dead time, **246**, 62, 172, 470, 472; high-pressure experiments, **259**, 373, 397; limitations, **249**, 18-19; methylamine dehydrogenase [amicyanin oxidation, **258**, 190; apparatus, **258**, 187, 189; methylamine reduction, **258**, 189-190]; ribonucleotide reductase diferric-tyrosyl radical cofactor assembly [absorbance changes, **258**, 282-283; apparatus, **258**, 281; mixing reactions, **258**, 281-282; rate constants, **258**, 283-285, 293; requirements, **258**, 280-281]; in thyl radical reaction analysis, **251**, 50-51; time resolution, **246**, 16, 62; **251**, 50.

STRAIN

1. A property present in a molecular entity (or a transition state) if the energy of that entity or state is enhanced due to unfavorable bond lengths, bond angles, or dihedral angles relative to some appropriate standard. It is the standard enthalpy of a structure relative to a "strainless" structure (real or hypothetical). 2. The change of volume or shape of a body, or portion of a body, due to the influence of one or more applied forces.

STREPTOKINASE

A metallopeptidase isolated from hemolytic streptococci. It hydrolyzes peptide bonds in plasminogen, producing plasmin.

J. T. Radek, D. J. Davidson & F. J. Castellino (1993) *Meth. Enzymol.* **223**, 145.

STURGILL-BILTONEN METHOD

A procedure used in the characterization of cooperative proteins in which the second moment of the derivative of the saturation function is measured.

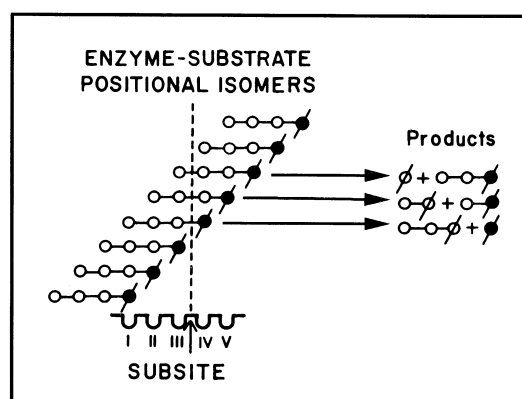
T. Sturgill & R. L. Biltonen (1976) *Biopolymers* **15**, 337.

SUBSITE MAPPING

A systematic kinetic analysis¹ of the binding site of an enzyme with the subunits of a polymeric substrate, in-

cluding specificity-imparting interactions at and surrounding the constellation of catalytically active groups participating in bond-making/breaking steps. The most rigorous approach relies on the additivity of subsite binding energies, based on the individual free energies of enzyme binding to the monomeric residues of the substrate. This approach requires thorough investigation of an enzyme's ability to bind polymeric substrates in various registrations, and this often can be inferred by studying the inhibitory properties of oligomeric competitive inhibitors.

In principle, the method affords the opportunity to predict the action pattern of an enzyme that is capable of binding a polymeric substrate in alternative ways, thereby leading to a preferred distribution of products. These methods are especially well suited for the investigation of polysaccharide hydrolyzing enzymes as well as proteinases. The binding of a maltotetraose to an enzyme possessing five glucosyl residue binding subsites illustrates the potential positional isomer interactions of the substrate: (a) only those substrate molecules spanning the cleavage site will undergo hydrolysis; (b) the action pattern of the hydrolase must be assessed by analysis of the limit dextran products; and (c) the other binding interactions, while not productive in terms of hydrolase activity, actually serve as inhibitors.



Hypothetical interactions of a five-subsite enzyme with a maltotetraose substrate shown initially with a terminally radiolabeled (filled circles) nonreducing end. The vertical arrow indicates the glycosyl bond cleavage region; the roman numerals indicate the subsites, and the 4 indicates the chain length. Only three binding interactions lead to hydrolysis: IV,4; V,4; and VI,4, and their different associated rate constants emphasize that the rates of hydrolysis need not be identical. From Allen and Thoma² with permission of the Biochemical Society (London).

¹J. D. Allen (1980) *Meth. Enzymol.* **64**, 248.

²J. D. Allen & J. A. Thoma (1976) *Biochem. J.* **159**, 105.

SUBSTITUENT

An atom, group of atoms bonded together, or moiety that can be considered to replace a hydrogen atom in a parent molecular entity (real or hypothetical).

SUBSTITUTION REACTION

A reaction in which an atom, group, or moiety in a molecular entity is replaced by another. **See** *S_N Reactions*

SUBSTRATE

1. The substance or reactant being acted upon by a catalyst. The substrate is often symbolized by S in one-substrate reactions. In multisubstrate reactions, the substrates are commonly symbolized by A, B, C, *etc.* 2. The base or foundation upon which an organism lives or grows. 3. The substance or compound of particular interest, with which a reaction with some other chemical reagent is under study.

SUBSTRATE-ASSISTED CATALYSIS

The facilitation of enzymic catalysis achieved by the presence of a substituent located on the substrate at a site away from the position of bond making/breaking. Substrate-assisted catalysis by serine proteases is well documented¹. Protease substrates are usually described as having sequences . . . P3–P2–P1–P1'–P2'–P3' . . . , where P1–P1' lie on each side of the scissile bond. Corey *et al.*² recently described how trypsin specificity is increased through substrate-assisted catalysis. They found that a strategically placed histidyl residue at position P2 or P1' in the peptide substrate was able to increase trypsin-catalyzed cleavage by 70–300 times. These investigators also used site-directed mutagenesis to explore this substrate-assisted catalysis.

¹P. Carter & J. A. Wells (1987) *Science* **237**, 394.

²D. R. Corey, W. S. Willet, G. S. Coombs & C. S. Craik (1995) *Biochemistry* **34**, 11521.

SUBSTRATE CHANNELING

Any process by which there is direct transfer of noncovalently bound substrate(s) or coenzyme from the active site of the enzyme producing the metabolite to the active site of an enzyme catalyzing a succeeding step in a metabolic pathway or to another enzyme utilizing the coenzyme product.

An implied feature of substrate channeling is that there must be direct contact or complexation between the two enzymes participating in the direct transfer. The following account dealing with conflicting interpretations regarding substrate channeling underscores the need for caution in proposing a kinetic model based on limited experimental findings and in the absence of an adequate kinetic theory. Srivastava and Bernhard¹ initially reported that reduction of benzaldehyde and *p*-nitrobenzaldehyde by NADH using horse liver alcohol dehydrogenase (LADH) was faster when NADH was bound to glyceraldehyde-3-phosphate dehydrogenase (GPDH) than with free NADH. The rate of reduction of aldehyde substrate with GPDH-NADH followed a Michaelis-type concentration dependence on GPDH-NADH. The reaction velocity was independent of GPDH concentration when [GPDH] was greater than [NADH]_{total}. The *K_m* for GPDH-NADH was higher than that for free NADH. They suggested that transfer of NADH from GPDH to LADH proceeds through the initial formation of a GPDH-NADH-LADH complex! These investigators² also proposed that glycolytic enzymes form multienzyme complexes for the direct transfer of metabolites from the producing enzyme to the utilizing one.

Chock and Gutfreund³ reexamined the kinetics of NADH transfer between its complexes with glycerol-3-phosphate dehydrogenase and with lactate dehydrogenase. Their work led them to conclude that coenzyme transfer between GPDH and LDH proceeds by a free-diffusion mechanism, *not* by direct transfer through a ternary complex. While their conclusions were disputed by Srivastava *et al.*⁴, Wu *et al.*⁵ carried out further studies on NADH binding to GPDH and LDH and on the displacement of enzyme-bound NADH by LDH or GPDH, adducing self-consistent evidence for a dissociative mechanism, as opposed to any direct transfer mechanism. Furthermore, while Brooks and Storey⁶ reported that they could reproduce the results of Srivastava *et al.*⁴, they also found that a mathematical solution of the direct-transfer-mechanism equations of Srivastava *et al.*⁶ did not adequately describe the experimental properties of the reaction rate at increasing LDH concentrations.

Those believing they have obtained evidence for direct nondissociative substrate channeling are strongly encouraged to consult the illuminating report of Wu *et al.*⁷,

who developed a rigorous set of criteria for differentiating mechanisms for ligand exchange reactions. They provide analytical expressions for the kinetics of (a) the dissociative mechanism (*i.e.*, where the leaving ligand must first dissociate prior to the binding of the incoming ligand) and (b) the associative mechanism (where a ternary complex is formed between the incoming ligand and the complex containing the leaving ligand). Analysis of these equations showed that an associative mechanism only can yield an increasing kinetic pattern for the observed pseudo-first-order rate constants as a function of increasing concentration of the incoming ligand and plateaus, in most cases, at a value higher than the off-rate constant of the leaving ligand. By contrast, the dissociative mechanism can produce increasing or decreasing conditions, depending on the magnitudes of the individual rate constants.

¹D. K. Srivastava & S. A. Bernhard (1984) *Biochemistry* **23**, 4538.

²D. K. Srivastava & S. A. Bernhard (1986) *Science* **234**, 1081.

³P. B. Chock & H. Gutfreund (1988) *Proc. Natl. Acad. Sci. U.S.A.* **85**, 8870.

⁴D. K. Srivastava, P. Smolen, G. F. Betts, T. Fukushima, H. O. Spivey & S. A. Bernhard (1989) *Proc. Natl. Acad. Sci. U.S.A.* **86**, 6464.

⁵X. M. Wu, H. Gutfreund, S. Lakatos & P. B. Chock (1991) *Proc. Natl. Acad. Sci. U.S.A.* **88**, 497.

⁶S. P. Brooks & K. B. Storey (1991) *Biochem. J.* **278**, 875.

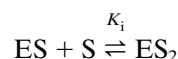
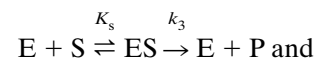
⁷X. M. Wu, H. Gutfreund & P. B. Chock (1992) *Biochemistry* **31**, 2123.

SUBSTRATE INHIBITION

The reduction in enzymatic activity that results from the formation of nonproductive enzyme complexes at high substrate concentration. The most straightforward explanation for substrate inhibition is that a second set of lower affinity binding sites exists for a substrate¹, and occupancy of these sites ties up the enzyme in nonproductive or catalytically inefficient forms. Other explanations include (a) the removal of an essential active site metal ion or other cofactor from the enzyme by high concentrations of substrate, (b) an excess of unchelated substrate (such as ATP⁴⁻, relative to the metal ion-substrate complex (such as CaATP²⁻ or MgATP²⁻) which is the true substrate^{2,3}; and (c) the binding of a second molecule of substrate at a subsite of the normally occupied substrate binding pocket, such that neither substrate molecule can attain the catalytically active conformation⁴. For multisubstrate enzymes, nonproductive dead-end complexes can also result in substrate inhibition in the presence of one of the reaction

products. In these cases, substrate inhibition can inform the experimentalist about the kinetic mechanism for substrate binding and/or product release⁴⁻⁶. **See Abortive Complex Formation**

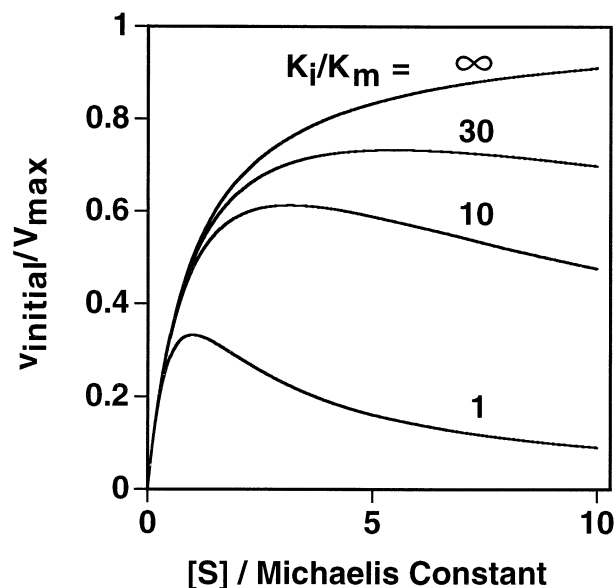
Haldane¹ was probably the first to treat the case of substrate inhibition arising as a consequence of second site binding. His kinetic scheme was as follows:



The rate equation for this mechanism is:

$$V_{\max}/v = 1 + K_s/[S] + [S]/K_i$$

where K_s and K_i are dissociation constants for the equilibria in the scheme above. The substrate inhibition behavior of an enzyme will depend on the relative magnitudes of K_s and K_i , and the figure presented below illustrates the general features of substrate inhibition.



Plot of the fractional saturation of enzyme active sites ($v/V_{\max}$) versus the reduced substrate concentration $[S]/K_s$ where $K_i/K_s = 1, 10, 30$, and infinity. Notice that there is an optimum in the rate dependence on substrate concentration.

¹J. B. S. Haldane (1930) *Enzymes*, p. 84, Longmans, Green, & Co., London.

²D. L. Purich & H. J. Fromm (1972) *Biochem. J.* **130**, 63.

³R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 1.

⁴W. W. Cleland (1971) *The Enzymes*, 3rd ed., **2**, 1.

⁵H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

⁶W. W. Cleland (1979) *Meth. Enzymol.* **63**, 500.

SUBSTRATE OXIDATION KINETICS

Wolfe¹ has presented an excellent description of the systematic application of stable and radioactive isotope tracers in determining the kinetics of substrate oxidation, carbon dioxide formation (including ¹³CO₂ breath tests), glucose oxidation, and fat oxidation in normal and diseased states. Quantification of the rate and extent of substrate oxidation can be achieved by using a specific ¹³C- or ¹⁴C-substrate which upon oxidation releases radioactive carbon dioxide.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, p. 235, Wiley-Liss, New York.

SUBSTRATE PURITY

Substrate purity, standardization, and handling are issues of great significance with respect to enzyme kinetics. The presence of impurities (or alternative substrates) is not only a source of experimental error; they can easily mislead an investigator regarding the mechanism of enzymatic catalysis and/or control. Incorrect values for the true substrate concentrations can lead to substantial errors in the reported values for the kinetic parameters. Thus, efforts should be made to confirm substrate purity, including the stereochemical purity, and this should be accomplished enzymatically, chemically, chromatographically, or spectrally prior to any kinetic experiment. If pure stereoisomers are not available, then a statement with regard to the stereochemical purity must be specified in any report of experimental findings. Biochemical substances tend to be relatively unstable, and impurities can be readily generated. All reagent bottles should be marked with the date on which the item was received as well as the date when the container was first opened. Such information is often quite valuable to co-workers in the same laboratory. The information provided by the commercial vendor regarding purity should not be accepted as definite, and some further purification may be required. For example, some commercial sources of amino acids contain heavy metal ions which may inhibit enzyme activity. Both L-glutamate and L-glutamine are usually recrystallized from solutions containing EDTA by investigators before kinetic studies are carried out. Substrate purity is of prime importance in investigations of partial exchanges in ping pong mechanisms. For example, the ATP ↔ AMP exchange observed with PRPP synthetase was reported to be due to impurities¹.

A good illustration of the influence of an impurity can be provided by noting the effect on initial-rate kinetics by the presence of a particular type of inhibitor in the substrate stock solution². Substrates and competitive inhibitors are often structural analogs of each other. It would not be surprising if an unknown competitive inhibitor was present with the substrate. Hence, upon serial dilutions of the stock substrate solution, the inhibitor concentration would be in constant ratio relative to the substrate. Thus, the initial rate expression in the presence of a competitive inhibitor,

$$1/v = (1/V_{\max}) + (K_m/V_{\max})(1/[S])(1 + [I]/K_i)$$

is transformed to

$$1/v = (1/V_{\max}) + (K_m/V_{\max})(1/[S])(1 + \alpha[S]/K_i)$$

where $\alpha = [I]/[S]$ and K_i is the dissociation constant for the EI complex. This form of the equation is indistinguishable from the standard expression for a one-substrate, enzyme-catalyzed reaction. Graphical analysis of initial-rate data will thus produce wrong estimates of the kinetic parameters; for example, the observed Michaelis constant from the above expression would actually be $K_m K_i / (K_i + \alpha K_m)$.

If a noncompetitive or an uncompetitive inhibitor were present with the substrate at constant ratio, then graphical analysis would suggest that the phenomenon of substrate inhibition is present. If an investigator analyzed the apparent substrate inhibition via a Marmasse plot, wrong estimates of both the K_m and K_{is} values would be reported and the investigator would be misled with respect to the kinetic mechanism. If partial inhibitors or alternative substrates are present in constant ratio, depending on the relative sizes of the $V_{\max}$ and K_m values, a variety of nonlinear graphical depictions can occur.

Kuby³ also discusses the possibility of the presence of an impurity which forms a complex with the substrate. If the impure ligand is in constant ratio with the substrate (*i.e.*, $[L_{\text{total}}] = a[A_{\text{total}}]$), then a kinetic equation of the form $\alpha[A] + \beta[A]^2/(\gamma + \delta[A] + \epsilon[A]^2)$ can be derived. Kuby points out that, in order for sigmoidal kinetics to be exhibited, $\beta\gamma > \alpha\delta$. However, this will not be true with this mechanism. Nevertheless, nonlinearity in double-reciprocal plots can still occur.

If an investigator remains concerned about the substrate purity following repurification procedures and subsequent analysis, then a number of other approaches may be considered. For example, different lots of substrate could be analyzed with the enzyme to learn if identical kinetic parameters are obtained. If one has some idea as to the identity of the possible contamination, then the impurity can be added to the stock substrate solution at a known amount and the accuracy of the kinetic parameters in the presence of the adduct can be assessed. Since the researcher already knows the degree of sensitivity of the various chemical, enzymatic, and/or spectral methods used to assess substrate purity, this known addition provides a means for determining the maximum amount of impurity present. Combined with the observations seen with the known addition of the impurity, such information will provide an idea on the level of accuracy of the kinetic parameters.

Closely related to issues of substrate purity are concerns related to the purity of the other components of the reaction mixture, especially water. For example, many enzymes that act on amine-containing molecules are inhibited by ammonium ions. As ammonia can be a common contaminant in many laboratories, doubly-distilled water can still contain enough ammonia to perturb certain enzyme activities. In such cases, it is important to test for ammonia and/or to determine the dissociation constant for ammonia with the enzyme under study.

Similarly, CO₂ and bicarbonate levels should be kept to a minimum by freshly preparing any alkaline buffer solutions. For similar reasons, stock solutions of NaOH, KOH, or other bases used to adjust pH values must always be freshly prepared.

When utilizing radiolabeled substrates, it is especially important to search for impurities resulting from radiolysis. This is particularly true if the stabilizing agent, often supplied by the vendor, has been removed during any laboratory procedure. It should be recalled that a radioactive compound can decompose chemically as well as via radioactive decay. In many cases, the rate of chemical decomposition for a radioactive compound is greater than the nonradioactive counterpart. This increased decomposition, which can result in a progressive increase in the apparent specific activity, may be a result of the

formation of free radicals by the action of the radiation on the solvent. Thus, it is prudent to analyze for chemical and radiopurity prior to use. It may be necessary to develop procedures to stabilize the radiolabel following purification.

Selection of the location of the radiolabel is also of importance. The position of the label within the compound should be such that no primary or secondary isotope effects perturb the kinetics of the system (unless, of course, the investigator is studying isotope effects).

¹R. L. Switzer & P. D. Simcox (1974) *J. Biol. Chem.* **249**, 5304.

²R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

³S. A. Kuby (1990) *A Study of Enzymes*, vol. 1, p. 285, CRC Press, Boca Raton.

⁴H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

SUBSTRATE STABILITY & HANDLING

The inherent instability of many biological compounds (particularly substances that can serve as substrates, products, inhibitors, and effectors of enzyme-catalyzed reactions) can be a major factor in kinetic studies if decay occurs within the time course of the experiment. For example, equilibrium exchange studies of malate dehydrogenase were done at 1°C to avoid problems with oxaloacetate instability¹. The degree of stability of each substrate and component of the assay mixture should be known under the specific assay conditions. For multisubstrate enzymes, it is a fairly straightforward procedure to examine each substrate for instability problems. For example, by using identical concentrations of two substrates containing different isotopic labels, one can verify the stoichiometry of multisubstrate reactions by observing the maintenance of the ratio of the two forms of radioactivity². For example, when using L-[³H]glutamate and [¹⁴C]ATP with glutamine synthetase, an alteration in the ³H:¹⁴C ratio as the reaction progresses would indicate that either one or both of the substrates are inherently unstable under the assay conditions or that a contaminating enzyme is present and preferentially acting on one of the substrates. Another common method for demonstrating the stability of a substrate is to have the substrate present in a reaction mixture containing all components except one of the other substrates. If a side reaction is present for the substrate under consideration, then the assay protocol³ for that substrate should demonstrate its existence.

Proper storage of substrates, stock solutions, and reagents is also an issue of importance in maintaining substrate, effector, and reagent stability as well as an issue of purity⁴. If stock solutions or reagents are to be kept frozen when not being used then, when needed, they should be thawed in a water bath and thoroughly mixed before use. If reagents are kept in a desiccator in the cold, the desiccator should be warmed to room temperature before opening. Due to the high instability of many biomolecules, efforts should also be made to keep the desiccant fresh. **See** *Substrate Purity; Enzyme Purity; Water Purity*

¹E. Silverstein & G. Sulebele (1969) *Biochemistry* **8**, 2543.

²R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

³H. U. Bergmeyer (1983) *Methods of Enzymatic Analysis*, 3rd ed., Springer-Verlag, Berlin.

⁴H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

SUBSTRATE SYNERGISM

The cardinal feature of Ping Pong mechanisms is the ability of the enzyme to catalyze partial exchange reactions as a result of the independence of the substrate's interactions with the enzyme. The second substrate is obliged to await the dissociation of the first product before it may bind to the enzyme. Multisubstrate enzymes frequently mediate such partial reactions which may be related to important steps in catalysis. One enzyme proposed to be of this sort is succinyl-CoA synthetase, but the partial reactions are relatively slow, and the participation of such reactions in catalysis becomes difficult to assess. Indeed, slow partial exchanges have been interpreted as proof of contamination of a particular enzyme with another enzyme or a small amount of the second substrate, an indication that the mechanism is not Ping Pong, or that the presence of other substrates may markedly increase the rates of elementary reactions giving rise to the partial exchange. Bridger *et al.*¹ proposed that the latter phenomenon be termed substrate synergism, and they examined this enzyme-substrate interaction by deriving appropriate rate laws for various exchanges. Their conclusion was that the rate of a partial exchange reaction must exceed the rate of the same exchange reaction in the presence of all the other substrates if the same catalytic steps and efficiencies are involved. If the opposite relation is observed, one must consider the possibility that synergism exists. Lueck and Fromm² examined the significance of partial exchange rate compari-

sons and focused on often misleading comparisons of exchange rates made with respect to initial velocity data.

¹W. A. Bridger, W. A. Millen & P. D. Boyer (1968) *Biochemistry* **7**, 3608.

²J. D. Lueck & H. J. Fromm (1973) *FEBS Lett.* **32**, 184.

SUBTILISIN

This enzyme [EC 3.4.21.62], a serine endopeptidase that evolved independently of chymotrypsin, contains no cysteinyl residues. This enzyme catalyzes the hydrolysis of peptide bonds in proteins and has a broad specificity, with a preference for a large uncharged aminoacyl residue in the P1 subsite.

E. Lolis & G. A. Petsko (1990) *Ann. Rev. Biochem.* **59**, 597.

R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

J. Kraut (1971) *The Enzymes*, 3rd ed., **3**, 547.

F. S. Markland, Jr., & E. L. Smith (1971) *The Enzymes*, 3rd ed., **3**, 562.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

Cleavage site specificity, **244**, 114; catalytic triad, **244**, 37; subtilisin family members, **244**, 37-39; oxyanion hole, **244**, 37; sequence homology, **244**, 37, 39; thiol dependence, **244**, 37, 39; unit activity, **235**, 566; zymography in nondissociating gels with copolymerized substrates, **235**, 588-590.

SUCCESSOR COMPLEX

A radical ion pair (*e.g.*, A^{•-} B^{•+}) produced immediately upon transfer of an electron from one molecular entity, designated here as B, to another molecular entity, in this case, A.

SUCCINATE DEHYDROGENASE

Succinate dehydrogenase (ubiquinone) [EC 1.3.5.1], a multiprotein complex found in the mitochondria, catalyzes the reaction of succinate with ubiquinone to produce fumarate and ubiquinol. The enzyme requires FAD and iron-sulfur groups. It can be degraded to form succinate dehydrogenase [EC 1.3.99.1], a FAD-dependent system that catalyzes the reaction of succinate with an acceptor to produce fumarate and the reduced acceptor, but no longer reacts with ubiquinone.

K. M. Robinson & B. D. Lemire (1995) *Meth. Enzymol.* **260**, 34.

Y. Anraku (1988) *Ann. Rev. Biochem.* **57**, 101.

G. von Jagow & W. Sebald (1980) *Ann. Rev. Biochem.* **49**, 281.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

T. P. Singer, E. B. Kearney & W. C. Kenney (1973) *Adv. Enzymol.* **37**, 189.

SUCCINYL-CoA SYNTHETASE

Succinyl-CoA synthetase (GDP) [EC 6.2.1.4], also known as succinate:CoA ligase, catalyzes the reversible reaction of GTP with succinate and coenzyme A to produce GDP, succinyl-CoA, and orthophosphate. The nucleotide substrate can be replaced with ITP and itaconate can substitute for succinate.

Succinyl-CoA synthetase (ADP) [EC 6.2.1.5] catalyzes the reversible reaction of ATP with succinate and coenzyme A to produce ADP, succinyl-CoA, and orthophosphate.

W. T. Wolodk, M. E. Fraser, M. N. G. James & W. A. Bridger (1994) *J. Biol. Chem.* **269**, 10883.

P. A. Frey (1992) *The Enzymes*, 3rd ed., **20**, 141.

J. S. Nishimura (1986) *Adv. Enzymol.* **58**, 141.

W. A. Bridger (1974) *The Enzymes*, 3rd ed., **10**, 581.

O-SUCCINYLMOMOSERINE (THIOL)-LYASE

This pyridoxal-phosphate-dependent enzyme [EC 4.2.99.9], also known as cystathionine γ -synthase, catalyzes the reaction of *O*-succinyl-L-homoserine with L-cysteine to produce cystathionine and succinate. The enzyme can also use hydrogen sulfide and methanethiol as substrates, producing homocysteine and methionine, respectively. In the absence of a thiol, the enzyme can also catalyze a β,γ -elimination reaction to form 2-oxobutanoate, succinate, and ammonia.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed., **19**, 323.

I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bärzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.

A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

SUCRASE

Sucrose α -D-glucosylhydrolase [EC 3.2.1.48], also known as sucrase and sucrase-isomaltase, catalyzes the hydrolysis of sucrose and maltose by an α -D-glucosidase-type action. This enzyme is isolated from intestinal mucosa as a single polypeptide chain that also exhibits activity toward isomaltose (an oligo-1,6-glucosidase activity, EC 3.2.1.10) by hydrolyzing 1,6- α -D-glucosidic linkages in isomaltose and dextrans. **See also** *Invertase; Isomaltase*

E. H. Van Beers, H. A. Büller, R. J. Grand, A. W. C. Einerhand & J. Dekker (1995) *Crit. Rev. Biochem. Mol. Biol.* **30**, 197.

H. Hauser & G. Semenza (1983) *Crit. Rev. Biochem.* **14**, 319.

SUCROSE PHOSPHORYLASE

This enzyme [EC 2.4.1.7], also known as sucrose glucosyltransferase, catalyzes the reaction of sucrose with orthophosphate to produce D-fructose and α -D-glucose 1-phosphate. In the forward reaction, arsenate may replace phosphate as the substrate. However, the resulting product is unstable in aqueous solutions. In the reverse reaction, various ketoses and L-arabinose may replace D-fructose. **See** *Arsenolysis*

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

J. J. Mieyal & R. H. Abeles (1972) *The Enzymes*, 3rd ed., **7**, 515.

SULFATE ADENYLYLTRANSFERASE

This enzyme [EC 2.7.7.4], also known as ATP sulfurylase, catalyzes the reaction of ATP with sulfate to produce adenylylsulfate and pyrophosphate.

C. Dahl, N. Speich & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 331.

C. Dahl & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 400.

J. A. Schiff & T. Saidha (1987) *Meth. Enzymol.* **143**, 329.

I. H. Segel, F. Renosto & P. A. Seubert (1987) *Meth. Enzymol.* **143**, 334.

H. D. Peck, Jr. (1974) *The Enzymes*, 3rd ed., **10**, 651.

SULFATE ADENYLYLTRANSFERASE (ADP)

This enzyme [EC 2.7.7.5], also known as ADP sulfurylase, catalyzes the reaction of ADP with sulfate to produce orthophosphate and adenylylsulfate.

A. Schmidt & K. Jäger (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 325.

H. D. Peck, Jr. (1974) *The Enzymes*, 3rd ed., **10**, 651.

SULFIDES

1. Inorganic sulfur compounds containing another (usually more electropositive) element. When the other element is an alkali or alkaline earth, the sulfide is ionic in character. Metal sulfides often have unusual stoichiometries. Examples of sulfides include H_2S , Na_2S , FeS , and HgS . 2. Organic sulfides are also referred to as thioethers and have the general structure $\text{R}-\text{S}-\text{R}'$. Biochemical examples of sulfides include methionine, cystathionine, and djenkolic acid. If the two R groups are identical, the substance can be referred to as a symmetrical sulfide (biological examples of which are lanthionine and homolanthionine).

SULFITE DEHYDROGENASE

This enzyme [EC 1.8.2.1] catalyzes the reaction of sulfite with two ferricytochrome *c* and water to produce sulfate

and two ferrocycytochrome *c*. The enzyme is associated with cytochrome *c*-551.

D. P. Kelly & A. P. Wood (1994) *Meth. Enzymol.* **243**, 501.

SULFITE OXIDASE

This heme- and molybdenum-dependent enzyme [EC 1.8.3.1] catalyzes the reaction of sulfite with dioxygen and water to produce sulfate and hydrogen peroxide.

C. Kisker, H. Schindelin & D. C. Rees (1997) *Ann. Rev. Biochem.* **66**, 233.

I. Suzuki (1994) *Meth. Enzymol.* **243**, 447.

R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.

R. C. Bray (1975) *The Enzymes*, 3rd ed., **12**, 299.

SULFITE REDUCTASE

Sulfite reductase (NADPH) [EC 1.8.1.2] catalyzes the reaction of H₂S with three NADP⁺ and three water molecules to produce sulfite and three NADPH. The enzyme requires FAD, FMN, and heme. Sulfite reductase (ferredoxin) [EC 1.8.7.1] catalyzes the iron-dependent reaction of H₂S with three oxidized ferredoxin and three water to produce sulfite and three reduced ferredoxin. Sulfite reductase (acceptor) [EC 1.8.99.1] catalyzes the iron-dependent reaction of H₂S with an acceptor and three water to produce sulfite and the reduced acceptor. A stoichiometry of six molecules of reduced methyl viologen per molecule of sulfide formed was reported. **See also** *Desulfofuscinidin*; *Desulforubidin*

I. Moura & A. R. Lino (1994) *Meth. Enzymol.* **243**, 296.

C. Dahl, N. Speich & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 331.

C. Dahl & H. G. Trüper (1994) *Meth. Enzymol.* **243**, 400.

H. G. Trüper (1994) *Meth. Enzymol.* **243**, 422.

Y. Hatefi & D. L. Stiggall (1976) *The Enzymes*, 3rd ed., **13**, 175.

G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

SULFONIUM SALTS

Compounds containing a pyramidally arranged (hence, chiral) sulfur to which are linked three alkyl or aryl groups, resulting in a net positive charge on the sulfur. A biologically important example is *S*-adenosyl-L-methionine chloride. Sulfonium salts can also be utilized as analogs or mimics of carbocation intermediates in enzyme-catalyzed reactions. For example, methyl-(4-methylpent-3-en-1-yl)vinylsulfonium perchlorate proved to be an excellent inhibitor ($K_i = 2.5 \mu\text{M}$) of the enzyme that catalyzes the formation of the bicyclic (+)- α -pinene¹.

¹R. Croteau, C. J. Wheeler, R. Aksela & A. C. Oehlschlager (1986) *J. Biol. Chem.* **261**, 7257.

SULFOTRANSFERASES

This class constitutes a large set of enzymes that catalyze the transfer of sulfate groups [EC 2.8.2.x]. **See also** *specific enzyme*

G. Lowe (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 627.

S. G. Ramaswamy & W. B. Jakoby (1987) *Meth. Enzymol.* **143**, 201.

SULFUR-35 (³⁵S)

A radioactive β -emitting sulfur nuclide with a decay half-life of 87.2 days and a beta energy of 0.167 MeV. The following decay data indicate the time (in days) followed by the fraction of original amount at the specified time: 0.0, 1.000; 3, 0.976; 6, 0.954; 9, 0.931; 12, 0.909; 15, 0.888; 18, 0.867; 21, 0.847; 24, 0.827; 27, 0.807; 30, 0.788; 60, 0.621; 90, 0.490; 120, 0.386; 150, 0.304; 180, 0.240; 210, 0.189; 240, 0.149; 270, 0.118; 300, 0.093.

Special precautions: Avoid sulfurous vapors by working in a chemical fume hood. Because no organ selectively concentrates sulfur metabolites, the entire body is the critical organ in terms of dose. The effective *biological* half-life in humans is around 90 days. The maximum range in air is 0.24 meters.

SULFUR DIOXYGENASE

This enzyme [EC 1.13.11.18] catalyzes the iron-dependent reaction of sulfur with dioxygen and water to produce sulfite.

D. P. Kelly & A. P. Wood (1994) *Meth. Enzymol.* **243**, 501.

SUPEROXIDE DETECTION/MEASUREMENT

Fridovich¹ recently summarized important aspects concerning the accurate detection and measurement of superoxide. He indicates that univalent reduction of O₂ to superoxide is a facile process, but the instability of superoxide in aqueous solutions hinders its detection and measurement. To measure intracellular superoxide, he favors use of the rapid inactivation of [4Fe-4S]-containing dehydratases (such as aconitase) by oxidation of their iron-sulfur clusters. **See** *Oxygen, Oxides & Oxygen Radicals*

¹I. Fridovich (1997) *J. Biol. Chem.* **272**, 18515.

SUPEROXIDE DISMUTASES

This enzyme [EC 1.15.1.1] catalyzes the reaction of two superoxide (2 O₂⁻) and two H⁺ to produce dioxygen and

hydrogen peroxide. The enzyme uses copper and zinc ions; or iron ions; or manganese ions.

- I. Fridovich (1995) *Ann. Rev. Biochem.* **64**, 97.
 E. Cadenas, P. Hochstein & L. Ernster (1992) *Adv. Enzymol.* **65**, 97.
 C. Bowler, M. Van Montagu & D. Inzé (1992) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **43**, 83.
 E. T. Adman (1991) *Adv. Protein Chem.* **42**, 145.
 I. Fridovich (1986) *Adv. Enzymol.* **58**, 61.
 B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.
 B. G. Malmström, L.-E. Andréasson & B. Reinhammer (1975) *The Enzymes*, 3rd ed., **12**, 507.
 I. Fridovich (1974) *Adv. Enzymol.* **41**, 35.

SUPERACID

A substance, mixture, or molecular entity having an unusually high degree of acidity, generally significantly higher than 100% sulfuric acid. An example is magic acid, an equimolar mixture of HSO_3F and SbF_5 .

- P. J. Gillespie (1968) *Acc. Chem. Res.* **1**, 202.
 G. A. Olah (1985) *Superacids*, Wiley, New York.

SUPERACID CATALYSIS

A term used occasionally in enzymology with respect to catalysis by metal ions.

SUPERBASE

A substance, mixture, or molecular entity having a very high degree of basicity. An example is lithium diisopropylamide.

SUPERIMPOSABILITY (or Superposability)

The ability to use translation and rotation to bring two stereochemical structures into three-dimensional coincidence without any breaking and remaking of bonds that alter the stereochemical configuration around each atom in the two stereochemical structures.

SUPERRADIANCE

Spontaneous emission that has been amplified by a passage through a population-inverted medium.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

SUPERSATURATION

A metastable condition in which the amount of a dissolved substance is greater than the condition in which the dissolved molecular entity is in equilibrium with the undissolved solid. **See** *Biomineralization*

SURFACE CROSSING

The crossing of two potential energy surfaces of the electron energies of two states having different symmetry.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

SURFACE TENSION

The work required to increase the area of a surface reversibly and isothermally by a unit area. The surface tension can also be described as the force exerted perpendicularly to the liquid surface. Surface tension is typically expressed in units of ergs/cm^2 or as dynes/cm .

SWAIN-LUPTON RELATION

An approach to the correlation analysis of substituent effects in chemical reactions and processes. In this treatment, there are two parameters, a field constant (F) and a resonance constant (R)¹⁻³. The approach has received some criticism⁴⁻⁸.

- ¹C. G. Swain & E. C. Lupton (1968) *J. Amer. Chem. Soc.* **90**, 4328.
²C. G. Swain, S. H. Unger, N. R. Rosenquist & M. S. Swain (1983) *J. Amer. Chem. Soc.* **105**, 492.
³IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.
⁴W. F. Reynolds & R. D. Topson (1984) *J. Org. Chem.* **49**, 1989.
⁵A. J. Hoefnagel, W. Oosterbeek & B. M. Wepster (1984) *J. Org. Chem.* **49**, 1993.
⁶M. Charton (1984) *J. Org. Chem.* **49**, 1997.
⁷C. G. Swain (1984) *J. Org. Chem.* **49**, 2005.
⁸C. Hanson, A. J. Leo & R. W. Taft (1991) *Chem. Rev.* **91**, 165.

SWAIN RELATIONSHIPS

Quantitative relationships¹⁻³ between magnitudes of deuterium and tritium primary kinetic isotope effects on chemical reactivity.

$$k_{\text{H}}/k_{\text{T}} = (k_{\text{H}}/k_{\text{D}})^{1.442}$$

$$k_{\text{H}}/k_{\text{T}} = (k_{\text{D}}/k_{\text{T}})^{3.26-3.34}$$

See *Kinetic Isotope Effects*

- ¹C. G. Swain, E. C. Stivers, J. F. Reuwer, Jr., & L. J. Schaad (1958) *J. Amer. Chem. Soc.* **80**, 5885.
²D. B. Northrop (1982) *Meth. Enzymol.* **87**, 607.
³B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

SYMBOLIC COMPUTING

The systematic application of artificial intelligence methods for the programmed manipulation of symbols in accordance with mathematical rules¹. Symbolic computing allows one to obtain symbolic solutions to equilibrium

binding expressions which otherwise require manual solution of simultaneous equations.

¹H. R. Halvorson (1992) *Meth. Enzymol.* **210**, 601.

SYMPORT

A transport system that couples the transport of two or more molecular entities and/or ions across a biomembrane in the same direction.

SYNCHROTRON X-RAY FOOTPRINTING

All footprinting techniques are designed to characterize binding interactions by determining the accessibility of the backbone of macromolecules to chemical/enzymatic cleavage or modification reactions. Synchrotron X-ray footprinting uses synchrotron radiation to generate reactive chemical species (such as hydroxide radical $\cdot\text{OH}$) to detect and quantitate backbone accessibility.

B. Sclavi, S. Woodson, M. Sullivan, M. Chance & M. Brenowitz (1998) *Meth. Enzymol.* **295**, 379.

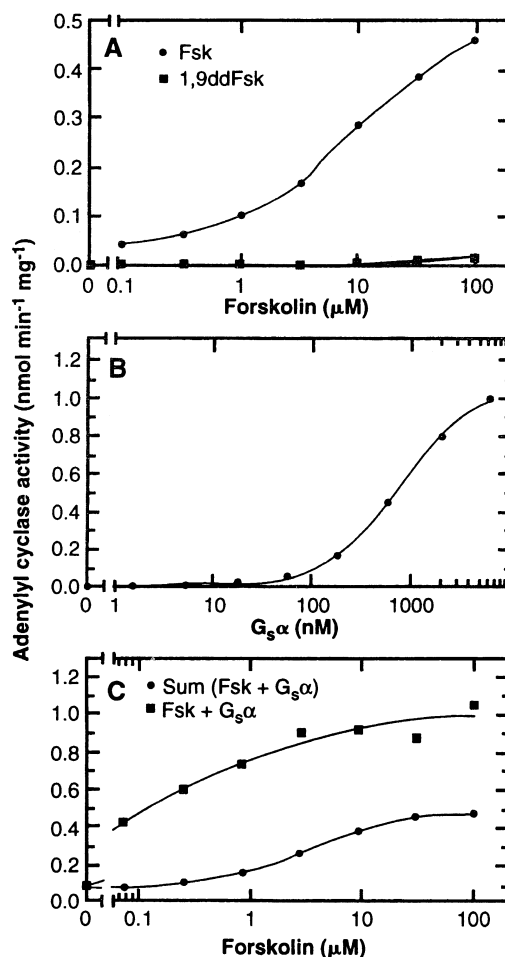
SYNERGISM

Any condition of enhanced potency achieved by two agents acting in concert, thereby exceeding the sum of each agent's separate effect. Synergism implies the cooperative or coordinated action of agents (frequently called agonists and effectors), and the molecular life sciences are replete with examples of synergism.

A classical case in pharmacology is the effect of cholinergic agents on the behavior of ion channels of the electric eel, *Electrophorus*. Measurements of membrane potentials using microelectrodes permitted Changeux and Podleski¹ to analyze the extent of depolarization as a function of the concentration of various agonists, either separately or in combination with each other. With phenyltrimethylammonium ion (or PTA), cooperativity was evident as a sigmoidal dose-response curve (Hill coefficient = 1.7). While antagonists had no effect on the cooperativity of PTA action, the presence of a second cholinergic agonist (decamethonium) completely abolished any evidence of PTA cooperativity. Thus, the two agonists acted synergistically, and their combined action was not simply the additive effect of their separate actions.

Another excellent example of synergism is the activation of a soluble adenylyl cyclase by forskolin and the α -

subunit of the heterotrimeric guanine nucleotide-binding protein (abbreviated: $G_s\alpha$)¹. Shown in the figure below are plots of cyclase activity *versus* the concentration of forskolin (Panel A) or $G_s\alpha$ (Panel B). The additive sum of these agents acting separately as well as their concerted action are plotted in Panel C.



Plot of the enzymatic activity² of recombinant adenylyl cyclase (IC₁₁IC₂-L₃). A, Activation by forskolin (Fsk), but not the inactive analogue 1,9-dideoxyforskolin (1,9-ddFsk). B, Activation by $G_s\alpha$ in the presence of GTP γ S. C, Synergistic activation by Fsk and $G_s\alpha$ (again in the presence of GTP γ S). (Reproduced with permission of the authors and the American Association for the Advancement of Science.)

Cooperative binding interactions of allosteric systems are also well-known cases of synergism. The archetype is hemoglobin which interacts with molecular oxygen, 2,3-bisphosphoglycerate, protons, and poisons such as carbon monoxide. In the Monod-Wyman-Changeux symmetry-conserving model for cooperativity, partial oxygenation displaces the so-called $R_0 \rightleftharpoons T_0$ equilibrium to provide a high concentration of sites for ligand binding with *no* effect on the affinity of each site for O₂. In the

Koshland-Némethy-Filmer ligand-induced conformational change model, binding of oxygen at the first site results in a conformational change that alters the affinity of neighboring sites for oxygen. The MWC model affects the concentration of available binding sites, and the KNF model achieves synergism by altering the binding constants for ligands. **See** *Allosterism; Cooperativity; Monod-Wyman-Changeux Model; Koshland-Némethy-Filmer Model*

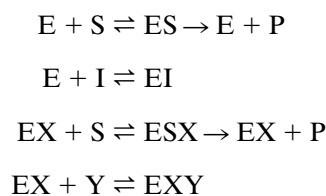
¹J. P. Changeux & T. R. Poleski (1968) *Proc. Natl. Acad. Sci. U.S.A.* **59**, 944.

²W. J. Tang & A. G. Gilman (1995) *Science* **268**, 1769.

SYNERGISTIC INHIBITION

A term used to describe inhibition induced by a second inhibitor. A particular inhibitor (I_1) may be a weak inhibitor for a certain enzyme; however, in the presence of a second inhibitor (I_2), inhibition is greatly enhanced. An example of such a system is the inhibitory effect of inorganic pyrophosphate on trichodiene synthase by certain aza analogues¹. Both enantiomers of serine are weak inhibitors of γ -glutamyl transpeptidase (L-serine has a K_i value of 10.7 mM), but in the presence of borate ion, they are potent inhibitors²⁻⁵.

Another example is the concerted inhibition model illustrated in the following scheme:



where X and Y are feedback inhibitors. Ligand X will have no effect on the reaction rate if the K_m and V_{max} remain the same as those obtained in the absence of X. Ligand Y is likewise ineffective alone, because the binding of ligand Y requires the presence of EX complex. However, in the presence of both X and Y, the enzyme is drawn into the catalytically inactive EXY complex.

¹D. E. Cane, G. Yang, R. M. Coates, H.-J. Pyun & T. M. Hohn (1992) *J. Org. Chem.* **57**, 3454.

²J. P. Revel & E. G. Ball (1959) *J. Biol. Chem.* **234**, 577.

³S. S. Tate & A. Meister (1978) *Proc. Natl. Acad. Sci. U.S.A.* **75**, 4806.

⁴G. A. Thompson & A. Meister (1977) *J. Biol. Chem.* **252**, 6792.

⁵R. D. Allison (1985) *Meth. Enzymol.* **113**, 419.

SYSTEM

A term used in thermodynamics to designate a region separated from the rest of the universe by definite boundaries. The system is considered to be isolated if any change in the surroundings (*i.e.*, the portion of the universe outside of the boundaries of the system) does not cause any changes within the system. **See** *Closed System; Isolated System; Open System*

SYSTEMATIC ERRORS

Those errors arising either from a specific step in the experimental design or as a result of error introduced by an apparatus or instrument (*e.g.*, calibrated micropipette, pH meter or electrode, analytical balance, scintillation counter, *etc.*). Systematic errors are frequently present when one first attempts to institute a new experimental protocol, and thoughtful consideration of each step in a procedure helps to eliminate or minimize systematic errors. There are three types of error: (a) systematic error; (b) random error, itself related to random variation in sample quality, handling, or measurement; and (c) consistent error which is constant and independent of experimental conditions. As an example of systematic error, the failure to remove a yellow colored substance introduced in a specific step in an experiment may lead to quenching of radioactive samples during liquid scintillation counting. This would introduce a one-sided error by yielding lower than actual values for the observed counts per minute (cpm) for all radioactive samples measured in the experiment. Impurity can often prove to be the source of many systematic errors. Likewise, some pH electrodes do not give the correct pH in the presence of Tris buffer; failure to recognize this problem would introduce systematic error. **Also see** *Random Error; Statistics (A Primer)*

T

TANDEM MASS SPECTROMETRY

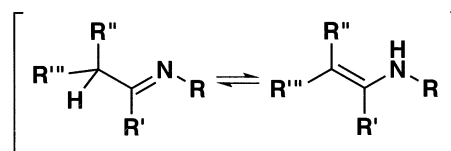
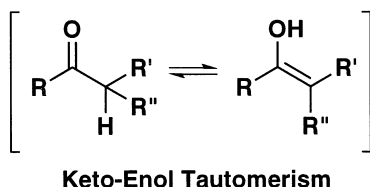
A high-resolution, double mass spectrometric (or MS/MS) technique that uses one mass spectrometer to separate mixtures of molecular ions into groups which upon entry into a second mass spectrometer are further fragmented and resolved into their characteristic mass spectra. Typically, the first mass spectrometer uses a chemical ionization method to impart kinetic energy with little or no fragmentation of each species; the second MS unit then relies on collisions of the parent ions with helium atoms to generate daughter ions which are characteristic and often predictable fragments of their respective parent ions.

TAU (τ)

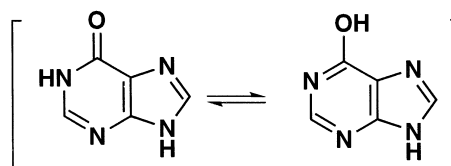
1. Symbol for relaxation time. 2. Symbol for a time constant. 3. Symbol for mean life.

TAUTOMERISM

A rapid and usually reversible internal isomerization between two or more structurally distinct compounds. The intramolecular process entails a heterolytic cleavage of a chemical bond followed by a recombination of the fragments, usually accompanied by the migration of a double bond or by ring opening and ring closing. The most common form of tautomerism involves the transfer of a proton (*i.e.*, prototropy or prototropic rearrangement). An example of this form of tautomerism is keto-enol tautomerism between a carbonyl group having an α hydrogen and its enol form:



Imine-Enamine Tautomerism



Lactam-Lactim Tautomerism

This reaction can be facilitated by both general acid and general base catalysis. In most cases, the equilibrium favors the keto form. For example, the percent enol content of acetaldehyde is only 1×10^{-3} while the enol percentage in acetone is 6×10^{-7} . Aqueous solutions of pyruvate and acetoacetate have less than 0.1% enol; however, the enol content of acetylacetone is 76.4%. The extent of enolization is greatly affected by the presence of double bonds or aromatic moieties that can be conjugated to the enol. In addition, the equilibria is affected by the nature of the solvent, by concentration, temperature, pH of the solution, and by the presence of metal ions. Thus, any perturbation of one or more of the factors can result in a displacement of the keto-enol equilibrium. This observation provides a tool in probing tautomeric specificity of particular proteins as well as the ability to assay enzymes that catalyze a keto-enol interconversion¹; for example, phenylpyruvate tautomerase² [EC 5.3.2.1] and oxaloacetate tautomerase³ [EC 5.3.2.2]. Following displacement of the equilibria, rates of ketonization or enolization can be followed by coupling the reaction to an enzyme that is specific for only one of the tautomeric species^{3,4}. **See** *Rearrangements; Anomeric Specificity; Valence Tautomerism*

¹I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.

²F. Blasi, F. Fragomele & I. Covelli (1969) *J. Biol. Chem.* **244**, 4864.

³R. G. Arnett & G. W. Kosicki (1969) *J. Biol. Chem.* **244**, 2059.

⁴G. W. Kosicki (1962) *Can. J. Chem.* **40**, 1280.

TELOMERASE

An enzyme that uses an RNA template to add DNA to the ends of chromosomes. Telomerase is normally active only in stem cells and those cells giving rise to sperm and egg, but telomerase also undergoes activation when cells become cancerous. In the latter case, telomerase action allows transformed cells to replicate without a limit, a process termed “immortalization”.

C. W. Greider (1996) *Ann. Rev. Biochem.* **65**, 337.

E. H. Blackburn (1992) *Ann. Rev. Biochem.* **61**, 113.

TEMPERATURE

A measure of the energy content or thermal content of a body or entity. A parameter that determines the directional transfer of heat to or from other bodies or entities. The SI temperature unit is the kelvin (abbreviated as K, and not °K).

TEMPERATURE CONTROL

Because enzymic reactions are so sensitive to variations in temperature, suitable care must be exercised such that the temperature of the system is constant and under adequate control. Ideally, the temperature-regulated water bath and/or circulator should reduce any variability to within less than $\pm 0.1^\circ\text{C}$. Before initiating any reaction kinetic measurement, all samples should be thermally equilibrated. This process may take several minutes, depending on the nature of the sample container and the volume of the sample.

When assays are conducted with a instrument such as a spectrophotometer, the sample compartment in the instrument should also be equilibrated to the same temperature. Typically, the control of temperature inside the sample compartment is less than that of the water bath or circulator. Many instruments now provide a temperature readout for the sample compartment that allows an investigator the opportunity to directly assess temperature control issues in the instrument.

If the assay temperature is significantly different than room temperature (*e.g.*, more than 10°C above or below), temperature control becomes a more serious concern. Simply mixing a reaction sample may perturb the sample temperature. Hence, after thermal equilibrium, handling

should be minimized. A wide variety of protocols have been utilized to achieve this end¹.

Ideally, the enzyme solution to be added to the reaction solution should be likewise thermally equilibrated prior to initiating the reaction. However, the thermal instability of many enzymes may require their storage in an ice bucket. An aliquot of the cold enzyme can then be brought up to temperature just prior to the initiation of the reaction. If the enzyme solution cannot be pre-warmed, then care has to be exercised in the analysis of the experimental data. For example, assume that an enzyme had a temperature dependency (Q_{10}) of two (*i.e.*, for every 10°C change in temperature, there is a twofold change in reaction rate). Thus, if a 0.1 mL aliquot of enzyme at 0°C is added to a reaction mixture of 3.0 mL at 30°C , the reaction velocity may be perturbed by as much as 20%. In such cases, the initial velocity should only be measured after the ongoing reaction has reached thermal equilibration.

¹R. D. Allison & D. L. Purich (1979) *Meth. Enzymol.* **63**, 3.

TEMPERATURE DEPENDENCE

Dixon and Webb¹ provided a useful list of various causes for the shape of the temperature effect seen in enzyme-catalyzed reactions: (1) effect on enzyme stability; (2) effect on the actual velocity of the reaction (especially on k_{cat}); (3) effect on affinity(ies) of the substrate(s) (*i.e.*, K_d values); (4) effect on the pH function of any or all components of the reactions (including the buffer); (5) effect on the affinity(ies) of enzyme effector(s); (6) an alteration in the rate determining (or rate contributing) step(s); (7) effect on the coupling enzymes of the assay; and (8) effect on physical properties (*e.g.*, solubility of substrates, particularly gas substrates such as O_2 and N_2 , dielectric constant of the solvent, *etc.*).

Clearly, many of these effects can be addressed through preliminary studies. Incubation of the enzyme at one temperature while assaying at a different temperature will address issues concerned with protein stability as well as effects of temperature on any coupling enzymes in the assay protocol. Nevertheless, temperature studies on each enzyme in a multienzyme system should be addressed individually at an early point in the investigation. Issues related to the effects on affinities can usually be minimized by assaying under saturating conditions. Care

should be exercised in this regard since, if the affinity of a substrate is altered significantly, a concentration that is saturating at one temperature may not be saturating at a different temperature. In a related issue, saturation may be difficult to achieve if the enzyme expresses substrate inhibition behavior at elevated substrate concentrations. Finally, since the pH may vary significantly with temperature and since pK_a values are temperature-dependent, a good temperature study of an enzyme is usually preceded by a pH study. At the very least, one should check the pH of the solution at different temperatures over the temperature range to be studied. **See** *Arrhenius Law; van't Hoff Relationship; Transition-State Theory*

¹M. Dixon & E. C. Webb (1979) *Enzymes*, Academic Press, New York.

TEMPERATURE DEPENDENCE OF KINETIC ISOTOPE EFFECTS

Observation of the temperature dependence of k_H/k_T and k_D/k_T and k_H/k_D kinetic isotope effects *via* Arrhenius plot is very useful in detecting and characterizing proton tunneling¹.

¹B. J. Bahnson & J. P. Klinman (1995) *Meth. Enzymol.* **249**, 373.

TEMPLATE CHALLENGE METHOD

A procedure used to assess the processivity of nucleic acid polymerases¹. In this method, the investigator perturbs the polymerization reaction proceeding on one template by the addition of a new template. **See** *Processivity*

¹W. R. McClure & Y. Chow (1980) *Meth. Enzymol.* **64**, 277.

TENSEGRITY

A model¹ for explaining how cells convert the mechanical signals (*i.e.*, the physical forces of gravity, hemodynamic stresses, and movement) into a chemical response. The model relies on recent experimental findings suggesting that cells use tensegrity architecture for their organization. Tensegrity predicts that cells are hard-wired for immediate response to mechanical stresses transmitted over cell surface receptors coupling the cytoskeleton to extracellular matrix. Mechanical signals are thought to be integrated with other environmental signals and transduced into a biochemical response through force-dependent changes in scaffold geometry or molecular mechanics.

¹D. E. Ingber (1997) *Annu. Rev. Physiol.* **59**, 575.

¹D. E. Ingber (1998) *Sci. Amer.* **278** (1), 48.

TERMINATION

1. The step¹⁻³ or steps in a chain reaction in which reactive intermediates are stabilized and rendered inactive, thus ending the chain reaction. For example, a common termination reaction is the combination of two radicals: *i.e.*, $R\cdot + R'\cdot \rightarrow R-R'$. Another type of termination reaction is disproportionation: *e.g.*, $2CH_3-CH_2\cdot \rightarrow$ ethane + ethene. 2. The quenching of a reaction. 3. The halting of any process.

¹S. J. Lapporte (1960) *Angew. Chem.* **72**, 759.

²M. J. Gibian & R. C. Corley (1973) *Chem. Rev.* **73**, 441.

³I. V. Khudyakov, P. P. Levin, & V. A. Kuzman (1980) *Russ. Chem. Rev.* **49**, 982.

TERNARY COMPLEX

A complex involving three species (such as the EAB complex formed in the ordered Bi Bi mechanism). **See** *Multisubstrate Mechanisms*

TER TER REACTION SYSTEMS

An enzyme-catalyzed reaction in which there are three substrates and three products. **See** *Multisubstrate Mechanisms*

TER UNI REACTION

An enzyme-catalyzed reaction in which there are three substrates and a single product. **See** *Multisubstrate Mechanisms*

TESLA

The SI unit for magnetic flux density or magnetic induction (symbolized by T) equal to a weber per square meter. One tesla is equivalent to 10^4 gauss.

TETRAHEDRAL INTERMEDIATE

An intermediate, often transient, appearing in a chemical or enzymatic reaction in which a carbon atom, which had been double-bonded (*i.e.*, in a trigonal structure) in a particular molecular entity, has been transformed to a carbon center having a tetrahedral arrangement of substituents. Tetrahedral intermediates of proteases have been stabilized with cryoenzymological techniques¹.

¹R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

TETRAHYDROPTEROYLTRIGLUTAMATE METHYLTRANSFERASE

This enzyme [EC 2.1.1.14], also known as 5-methyltetrahydropteroyltriglutamate:homocysteine *S*-methyltransferase and methionine synthase, catalyzes the reaction of 5-methyltetrahydropteroyltri-L-glutamate with L-homocysteine to produce tetrahydropteroyltri-L-glutamate and L-methionine. The reaction requires the presence of phosphate. The enzyme isolated from *E. coli* also requires a reducing system. **See** *N⁵-Methyltetrahydrofolate:Homocysteine Methyltransferase*

I. Saint-Gerons, C. Parsot, M. M. Zakin, O. Bârzy & G. N. Cohen (1988) *Crit. Rev. Biochem.* **23**, S1.

THERAPEUTIC RATIO

A ratio that characterizes the relative effectiveness of a pharmaceutical. It is equal to the maximally tolerated dose divided by the minimal curative or effective dose; or, to LD₅₀/ED₅₀.

THERMAL EQUILIBRATION

A necessary step for any initial rate study. Whenever practicable, the reaction solution should be preincubated at the final temperature. (This might not be possible for components that are unstable at that final temperature.) If enzyme is used to initiate the reaction, the enzyme stock solution should be briefly warmed. Occasionally a rapidly warmed enzyme may display hysteresis as a result of some temperature-dependent conformational change.

THERMAL LENSING

A method for determining the alteration of the refractive index of a medium as a result of the temperature rise in the path of a beam of coherent light absorbed by the medium. Thermal lensing can also occur with pigmented proteins, and this phenomenon can influence the accuracy of concentration gradient measurements in small aperture flow cuvettes as well as in ultracentrifugation.

Comm. of Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

THERMAL TRAPPING

The isolation of an intermediate or unstable species by the lowering of the temperature^{1,2}. The basic strategy is that one can “accumulate” otherwise unstable chemical species within potential energy wells along a reaction coordinate. **See also** *Cryoenzymology*

¹A. L. Fink & M. A. Geeves (1979) *Meth. Enzymol.* **63**, 336.

²D. L. Purich (1982) *Meth. Enzymol.* **87**, 3.

THERMODYNAMIC CONTROL

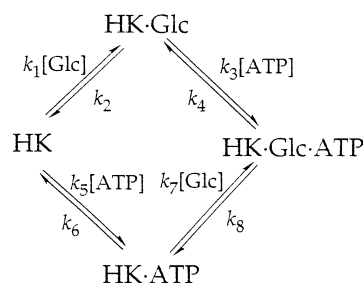
If a particular molecular entity(ies) participates in two or more parallel reactions and the proportion of the resulting products is determined by the relative equilibrium constants for the interconversion of reaction intermediates on or after the rate-determining step(s), then the more prevalent product is said to be thermodynamically controlled (*i.e.*, the more stable product will be the one formed in highest amounts). If the reactions are reversible and the system is allowed to go to equilibrium, the favored product is the thermodynamically controlled species. A synonymous term is equilibrium control. **See also** *Kinetic Control*

G. W. Klumpp (1982) *Reactivity in Organic Chemistry*, p. 36, Wiley, New York.

THERMODYNAMIC CYCLE

A closed cycle of ligand binding interactions or chemical reactions that can be fruitfully analyzed by taking advantage of thermodynamic relationships holding for branched pathways.

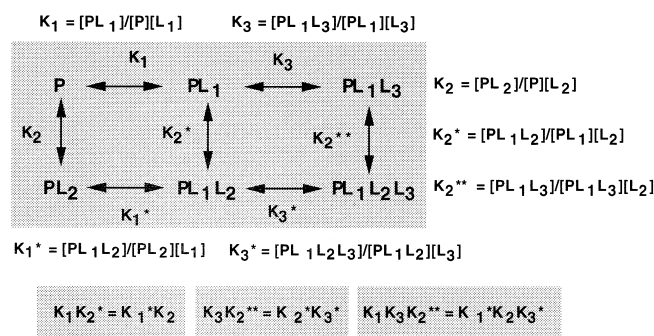
Consider the case of glucose (Glc) and ATP interactions with hexokinase (HK) in the absence of catalysis (which can be accomplished by omitting magnesium ion).



where $K_1 = k_2/k_1 = [\text{HK}][\text{Glc}]/[\text{HK}\cdot\text{Glc}]$, $K_2 = k_4/k_3 = [\text{HK}\cdot\text{Glc}][\text{ATP}]/[\text{HK}\cdot\text{Glc}\cdot\text{ATP}]$, $K_3 = k_6/k_5 = [\text{HK}][\text{ATP}]/[\text{HK}\cdot\text{ATP}]$, and $K_4 = k_8/k_7 = [\text{HK}\cdot\text{ATP}][\text{Glc}]/[\text{HK}\cdot\text{Glc}\cdot\text{ATP}]$. The changes in free energy from one enzyme form to another are independent of the path taken. Thus, $K_1K_2 = K_3K_4$, and knowledge of any three of the equilibrium constants allows one to determine the fourth.

Another excellent example is δ -chymotrypsin, a protease whose active conformation is maintained by a salt bridge

between the α -ammonium group on isoleucine-16 and the carboxylate anion of aspartate-194. Because pH-dependent changes in enzyme activity are linked to the maintenance or loss of this salt bridge, Fersht & Requena^{1,2} set out to determine how pH affects the enzyme's conformational equilibria. In a clever set of stopped-flow experiments in which the enzyme and the dye proflavin were rapidly combined, they took advantage of the fact that there is a burst in the 465 nm absorbance when the dye binds to the active conformation. This burst is then followed by a first-order activation attended by additional proflavin binding. By analyzing the kinetics of proflavin binding as a function of pH, these investigators constructed the following thermodynamic diagram for the making/breaking of this essential salt bridge. For an example of a thermodynamic cycle in the analysis of actin polymerization, **See** KINSIM/FITSIM.



Finally, in order to deal with the complexity of some ligand binding systems, one requires the use of more than a single thermodynamic cycle. While such a system is necessarily more complicated, the equilibrium relationships among dissociation constants allows one to dissect important features of the process.

¹A. R. Fersht & Y. Requena (1971) *J. Mol. Biol.* **60**, 279.

²A. R. Fersht (1971) *Cold Spring Harbor Symp.* **36**, 71.

THERMODYNAMIC EQUATIONS OF STATE

Expressions for internal energy, U , and enthalpy, H , in terms of pressure (P), volume (V), and absolute temperature (T):

$$(\partial U/\partial T)_T = -P + T(\partial P/\partial T)_V$$

$$(\partial H/\partial P)_T = V - T(\partial V/\partial T)_P$$

THERMODYNAMIC pK_a

The pK_a value of an ionizable group extrapolated to zero ionic strength.

THERMODYNAMICS, LAWS OF

Zeroth Law of Thermodynamics. If any two bodies are each in thermal equilibrium with a third body, then all three bodies are in thermal equilibrium with each other.

First Law of Thermodynamics. The total amount of energy within a closed system is constant (*i.e.*, the total energy of the system is conserved). Mathematically, this can be expressed as $\Delta U = q + w$ where ΔU is the change in internal energy, q is the heat transferred to the system, and w is the work done on the system (or, $dU = dq + dw$). Internal energy is a state function (*i.e.*, it is dependent only on the initial and final states and not on the path between those states). In addition, the validity of the first law means that perpetual motion machines are impossible. **See** Conservation of Energy

If nuclear reactions are to be considered, then the first law has to consider changes in mass associated with changes in energy. However, for normal chemical and biochemical processes, the change in mass is too small to be a factor.

Second Law of Thermodynamics. There have been numerous statements of the second law. To paraphrase Clausius: It is impossible to devise an engine or process which, working in a cycle, will produce no effect other than the transfer of heat from a colder to a warmer body. According to Caratheodory, the Second Law can be stated as follows¹: "Arbitrarily close to any given state of any closed system, there exists an unlimited number of other states which it is impossible to reach from a given state as a result of any adiabatic process, whether reversible or not".

A corollary to this law is that all reversible Carnot cycles, operating between the same two temperatures, must have the same efficiency. For a perfectly reversible cycle in which the only pressure, P , is uniform and external, then $dU = TdS - PdV$ where dU is the differential change in internal energy, T is the absolute temperature, dS is the differential change in entropy, and dV is the differential change in volume. This relation, which is a

combination of the first and second laws, is for systems of constant composition in which the only work done on or by the system is *PV* work. Another expression of the second law is that the entropy of a closed system increases with time. **See also** *Entropy*; *Caratheodory, Principle of*; *Carnot Cycle*; *Efficiency*; *Innate Thermodynamic Quantities*

Third Law of Thermodynamics. Also referred to as the Nernst heat theorem, this law states that it is impossible to reduce the temperature of any system, via a finite set of operations, to absolute zero. For any changes involving perfectly crystalline solids at absolute zero, the change in total entropy is zero (thus, $\Delta S_{0\text{ K}} = 0$). A corollary to this statement is that every substance, at $T > 0\text{ K}$, must have a positive and finite entropy value. The entropy of that substance is zero only at absolute zero when that substance is in pure, perfect crystalline form. **See** *Entropy*

¹J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, p. 147, Academic Press, New York.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Pathway of allosteric control as revealed by intermediate states of hemoglobin, **259**, 1; probes of energy transduction in enzyme catalysis, **259**, 19; macromolecules and water: probing with osmotic stress, **259**, 43; linkage of protein assembly to protein-DNA binding, **259**, 95; linkage at steady state: allosteric transitions of thrombin, **259**, 127; thermal denaturation methods in the study of protein folding, **259**, 144; kinetics of lipid membrane phase transitions: a volume perturbation calorimeter study, **259**, 169; tight binding affinities determined from thermodynamic linkage to protons by titration calorimetry, **259**, 183; calorimetric methods for interpreting protein-ligand interactions, **259**, 194; extracting thermodynamic data from equilibrium melting curves for oligonucleotide order-disorder transitions, **259**, 221; predicting thermodynamic properties of RNA, **259**, 242; thermodynamics and mutations in RNA-protein interactions, **259**, 261; melting studies of RNA unfolding and RNA-ligand interactions, **259**, 281; structural-perturbation approaches to thermodynamics of site-specific protein-DNA interactions, **259**, 305; thermodynamic parameters from hydrogen exchange measurements, **259**, 344; application of pressure to biochemical equilibria: the other thermodynamic variable, **259**, 357; molecular volume, **259**, 377; hydrostatic and osmotic pressure as tools to study macromolecular recognition, **259**, 395; sedimentation equilibrium as thermodynamic tool, **259**, 427; footprint phenotypes: structural models of DNA-binding proteins from chemical modification analysis of DNA, **259**, 452; low-temperature electrophoresis methods, **259**, 468; use of multiple spectroscopic methods to monitor equilibrium unfolding of proteins, **259**, 487; probing structural and physical basis of protein energetics linked to protons and salt, **259**, 512; evaluating contribution of hydrogen bonding and hydrophobic bonding to protein folding, **259**, 538; analyzing solvent reorganization and hydrophobicity, **259**, 555; simple force field for study of peptide and protein conformational properties, **259**, 576; probes for analysis of stability of different variants of aspartate aminotransferase, **259**, 590;

thermodynamic approaches to understanding aspartate transcarbamylase, **259**, 608; on the interpretation of data from isothermal processes, **259**, 628.

THERMOLYSIN

This enzyme [EC 3.4.24.27], also known as *Bacillus thermoproteolyticus* neutral proteinase, is a thermostable extracellular metalloendopeptidase containing zinc and four calcium ions. A member of the peptidase family M4, this enzyme catalyzes the hydrolysis of peptide bonds with a preference for Xaa—Leu > Xaa—Phe.

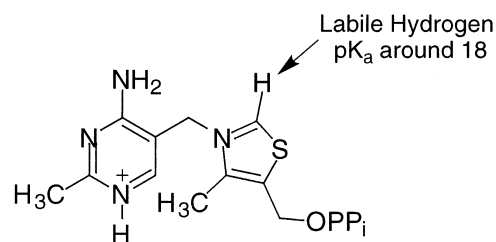
J. E. Coleman (1992) *Ann. Rev. Biochem.* **61**, 897.
D. W. Christianson (1991) *Adv. Protein Chem.* **42**, 281.
E. Lolis & G. A. Petsko (1990) *Ann. Rev. Biochem.* **59**, 597.
B. L. Vallee & A. Galdes (1984) *Adv. Enzymol.* **56**, 283.
J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

THETA (Θ)

1. Symbol (Θ) for characteristic temperature. 2. Symbol (θ) for degree of saturation of binding sites as defined in the Langmuir isotherm treatment for adsorption of a ligand onto a surface. **See** *Langmuir Isotherm*. 3. Symbol (θ) for plane angle. 4. Symbol (θ) for one of the space coordinates in the three-dimensional, spherical polar coordinate system. 5. Symbol (θ) for Celsius temperature.

THIAMIN: PHOSPHATES & ANALOGUES

A vitamin whose pyrophosphorylated form is an essential coenzyme in so-called α condensation and α cleavage reactions, whose cardinal feature is that the scissile bond always lies immediately adjacent to the carbonyl group.



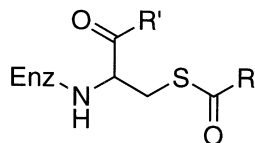
Selected entries from *Methods in Enzymology* [vol, page(s)]:

Acid dye method for the analysis of thiamin, **18A**, 73; electrophoretic separation and fluorometric determination of thiamin and its phosphate esters, **18A**, 91; catalytic polarography in the study of the reactions of thiamin and thiamin derivatives, **18A**, 93; preparation of thiamin derivatives and analogs, **18A**, 141; preparation of the mono- and pyrophosphate esters of 2-methyl-4-amino-5-hydroxymethylpyrimidine for thiamin biosynthesis, **18A**, 162; formation of the pyrophosphate ester of 2-methyl-4-amino-5-hydroxymethylpyrimidine by enzymes from brewers' yeast in thiamin biosynthesis, **18A**, 203; resolution, reconstitution, and other methods for the study of binding of thiamin pyrophos-

phate to enzymes, **18A**, 238; thiamin antagonists, **18A**, 245; enzymatic preparation, isolation, and identification of 2- α -hydroxyalkylthiamin pyrophosphates, **18A**, 259; an improved procedure for the determination of thiamin, **62**, 51; differential determination of thiamin and its phosphates and hydroxyethylthiamin and pyriothiamine, **62**, 54; fluorometric determination of thiamin and its mono-, di-, and triphosphate esters, **62**, 58; separation of thiamin phosphoric esters on Sephadex cation exchanger, **62**, 59; high-pressure liquid chromatography of thiamin, thiamin analogs, and their phosphate esters, **62**, 63; electrophoretic separation of thiamin and its mono-, di-, and triphosphate esters, **62**, 68; thiamin phosphate pyrophosphorylase, **62**, 69; synthesis of pyrimidine ^{14}C -labeled thiamin, **62**, 73; thiamin transport in *Escherichia coli*, **62**, 76; assay of thiamin-binding protein, **62**, 91; isolation and characterization of *Escherichia coli* mutants auxotrophic for thiamin phosphates, **62**, 94; a radiometric assay for thiamin pyrophosphokinase activity, **62**, 101; assay of thiamin pyrophosphokinase (ATP:thiamin pyrophosphotransferase) using anion-exchange paper disks, **62**, 103; affinity chromatography of thiamin pyrophosphokinase from rat brain on thiamin monophosphate-agarose, **62**, 105; enzymatic formation of thiamin pyrophosphate in plants, **62**, 107; preparation of thiamin triphosphate, **62**, 112; determination of thiaminase activity using thiazole-labeled thiamin, **62**, 113; membrane-bound thiamin triphosphatase, **62**, 118; transition-state analogs of thiamin pyrophosphate, **62**, 120.

THIOLESTER INTERMEDIATE

Thermodynamically unstable substances (structure below) that serve as enzymatic reaction intermediates and are typically formed by carboxylation (*i.e.*, carboxyl group transfer) of a sulfhydryl group on a cysteinyl residue of an enzyme.



Thiolester intermediates are formed in the glyceraldehyde-3-phosphate dehydrogenase and many CoA-linked enzymatic reactions.

THIOLS

Sulfhydryl groups play important roles in enzymatic reactions, serving in some cases as active site nucleophiles and in other instances as polar groups that stabilize subunit interactions in multisubunit proteins.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Assay with 2,2-dithiobisnitrobenzoic acid method, **233**, 381-382; pH effects, **233**, 384-385; formed by incubation with dithiothreitol, labeling, **233**, 409-410; protein thiol assay, **234**, 273-274; labeling, **233**, 414; reactivity with free radicals and reactive oxygen species, **233**, 405; reaction with ferrylmyoglobin, **233**, 196-197.

THREONINE DEHYDRATASE

This pyridoxal-phosphate-dependent enzyme [EC 4.2.1.16], also known as threonine deaminase and L-ser-

ine dehydratase, catalyzes the reaction of L-threonine with water to produce 2-oxobutanoate, ammonia, and water. The enzyme will also use L-serine as a substrate (thus, the activity of serine dehydratase).

H. E. Umbarger (1973) *Adv. Enzymol.* **37**, 349.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

THRESHOLD POTENTIAL

The membrane potential at an axon hillock below which an action potential will be generated.

THROMBIN

Thrombin [EC 3.4.21.5], also known as fibrinogenase, catalyzes the hydrolysis of peptide bonds, exhibiting preferential cleavage for the Arg—Gly peptide bond. The enzyme, a member of the peptidase family S1, activates fibrinogen to fibrin and releases fibrinopeptide A and B. Thrombin, formed from prothrombin, is more selective in peptide hydrolysis than trypsin or plasmin.

E. Di Cera, Q. D. Dang, Y. Ayala & A. Vindigni (1995) *Meth. Enzymol.* **259**, 127.

M. T. Stubbs & W. Bode (1993) *Thrombosis Res.* **69**, 1.

L. Lorand & K. G. Mann, eds. (1993) *Meth. Enzymol.*, vols. **222** and **223**.

W. Bode, D. Turk & A. Karshikov (1992) *Prot. Sci.* **1**, 426.

L. J. Berliner (1992) *Thrombin, Structure & Function*, Plenum Press, New York.

M. F. Scully (1992) *Essays in Biochem.* **27**, 17.

E. W. Davie, K. Fujikawa, K. Kurachi & W. Kisiel (1979) *Adv. Enzymol.* **48**, 277.

S. Magnusson (1971) *The Enzymes*, 3rd ed., **3**, 278.

THROMBOXANE A SYNTHASE

This heme-thiolate-dependent enzyme [EC 5.3.99.5] catalyzes the conversion of (5Z,13E)-(15S)-9 α ,11 α -epidioxy-15-hydroxyprosta-5,13-dienoate to (5Z,13E)-(15S)-9 α ,11 α -epoxy-15-hydroxythromba-5,13-dienoate. Thus, it converts prostaglandin H₂ into thromboxane A₂.

H. B. Dunford (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 195.

C. R. Pace-Asciak & W. L. Smith (1983) *The Enzymes*, 3rd ed., **16**, 543.

THYMIDINE 2'-HYDROXYLASE

This enzyme [EC 1.14.11.3], also known as thymidine:2-oxoglutarate dioxygenase, catalyzes the reaction of 2-deoxyuridine with α -ketoglutarate (or, 2-oxoglutarate) and dioxygen to produce uridine, succinate, and carbon dioxide. The enzyme, which also can act on thymidine, requires iron ions and ascorbate.

O. Hayaishi, M. Nozaki & M. T. Abbott (1975) *The Enzymes*, 3rd ed., **12**, 119.

THYMIDYLATE SYNTHASE

Thymidylate synthase [EC 2.1.1.45] reductively methylates 2'-deoxyuridine-5'-monophosphate to form 2'-deoxythymidine-5'-monophosphate in the following folate-dependent reaction: dUMP + N^5,N^{10} -methylene-tetrahydrofolate $\rightarrow$ dTMP + dihydrofolate.

Mechanistic and structural investigations, largely carried out by Santi's and Stroud's groups at the University of California, San Francisco, have provided important insights about the reaction mechanism of thymidylate synthase (TS), as well as the mode of inhibition of the antineoplastic agent fluorouracil¹. The ordered binding sequence of the substrates deoxyuridine monophosphate (dUMP) and methylenetetrahydrofolate ($\text{CH}_2\text{H}_4\text{folate}$) can be understood from the structures, where each forms a large part of the binding site for the other. The enzyme forms a complex with 5-fluoro-2'-deoxyuridine 5'-monophosphate (FdUMP), and the TS-FdUMP- $\text{CH}_2\text{H}_4\text{folate}$ complex shows a differential absorbance maximum at 326 nm, and is stable to SDS-PAGE. FdUMP is a slow, tight binding inhibitor of TS. The nature of covalently bound FdUMP has also been probed by ^{19}F nuclear magnetic resonance². Thermodynamic analysis³ of the binding of 5-FdUMP to thymidylate synthase over a range of temperatures supports the observation that this inhibitor binds to TS without producing profound conformational changes in the protein dimer. TS also catalyzes the dehalogenation of 5-bromo- and 5-iodo-2'-dUMP⁴. Wataya & Santi⁵ observed a secondary kinetic isotope effect in the cysteine-promoted dehalogenation of 5-bromo-2'-deoxyuridine, providing evidence for the transient formation of a 5,6-dihydropyrimidine intermediate in the TS reaction.

Thymidylate synthase also catalyzes an exchange⁶ of tritium of $[5\text{-}^3\text{H}]\text{dUMP}$ for water protons in the absence of $\text{CH}_2\text{H}_4\text{folate}$. The turnover number for this exchange reaction is about 1/45,000 that of dTMP formation, and the K_m is 1.2×10^{-5} M, about the same as the K_d of the enzyme-dUMP complex estimated by equilibrium dialysis. The exchange reaction provided compelling evidence that the enzymic reaction involves attack of an enzyme nucleophile on the 6 position of dUMP to pro-

vide a 5,6-dihydro-dUMP intermediate that becomes covalently bound to the enzyme.

Birdsall *et al.*⁷ examined the X-ray structures of ternary complexes of the wild-type TS in attempts to discern the structural cues that choreograph three steps along the reaction pathway: (a) binding of substrates; (b) orientation; and (c) release of products. Each complex was formed by diffusion of either the $\text{CH}_2\text{H}_4\text{folate}$ or its analog 10-propargyl-5,8-dideazafolate into crystals of thymidylate synthase with bound dUMP. A two-substrate/enzyme complex forms without further reaction. These investigators found that the imidazolidine ring is unopened, and the pterin ring of the $\text{CH}_2\text{H}_4\text{folate}$ binds at an unproductive "alternate" site. They suggested that pterin binding at this site is an initial interaction that precedes all catalysis. The structure of the enzyme complex with dUMP and 10-propargyl-5,8-dideazafolate showed the substrate and cofactor analog bound in orientations favorable for initiating catalysis and may resemble the productive complex. Another complex involving dTMP and 7,8-dihydrofolate revealed the geometry after catalysis and just prior to product release.

¹C. W. Carreras & D. V. Santi (1995) *Annu. Rev. Biochem.* **64**, 721.

²T. L. James, A. L. Pogolotti, Jr, K. M. Ivanetich, Y. Wataya, S. S. Lam & D. V. Santi (1976) *Biochem. Biophys. Res. Commun.* **72**, 404.

³L. Garcia-Fuentes, P. Reche, O. Lopez-Mayorga, D. V. Santi, D. Gonzalez-Pacanowska & C. Baron (1995) *Eur. J. Biochem.* **232**, 641.

⁴Y. Wataya & D. V. Santi (1975) *Biochem. Biophys. Res. Commun.* **67**, 818.

⁵Y. Wataya & D. V. Santi (1977) *J. Am. Chem. Soc.* **99**, 4534.

⁶A. L. Pogolotti, Jr., C. Weill, D. V. Santi (1979) *Biochemistry* **18**, 2794.

⁷D. L. Birdsall, J. Finer-Moore & R. M. Stroud (1996) *J. Mol. Biol.* **255**, 522.

The following are additional discussions of the molecular and physical properties of this enzyme.

D. R. Knighton, C.-C. Kan, E. Howland, C. A. Janson, Z. Hostomska, K. M. Welsh & D. A. Matthews (1994) *Nature Struct. Biol.* **1**, 186.

M. A. Ator & P. R. Ortiz de Montellano (1990) *The Enzymes*, 3rd ed., **19**, 213.

P. Reichard (1988) *Ann. Rev. Biochem.* **57**, 349.

C. Heidelberger, P. V. Danenberg & R. G. Moran (1983) *Adv. Enzymol.* **54**, 57.

S. J. Benkovic (1980) *Ann. Rev. Biochem.* **49**, 227.

T. W. Bruce, C. Garrett, Y. Wataya & D. V. Santi (1980) *Meth. Enzymol.* **64**, 125.

M. Friedkin (1973) *Adv. Enzymol.* **38**, 235.

J. I. Rader & F. M. Huennekens (1973) *The Enzymes*, 3rd ed., **9**, 197.

TIGHT-BINDING INHIBITOR KINETICS

The study and characterization of tight-binding inhibitors requires special techniques and protocols¹. Recall that an investigator typically varies the concentration of an inhibitor within a range bracketing the dissociation constant of that inhibitor. But, if an inhibitor has a very low K_d value, that concentration range could easily be close in magnitude to the concentration of the enzyme. In order for steady-state kinetics to be applicable to a system, not only must the substrate concentration be in great excess over the total enzyme concentration, but the enzyme concentration has to be much smaller than the concentration of any effector (in this case, an inhibitor). In such situations, different protocols have to be followed¹. Not only will the investigator have to measure initial rates at different concentrations of substrate and fixed concentrations of inhibitor, but it will also be necessary to look at the steady-state velocities as a function of total enzyme concentration (at fixed levels of the inhibitor). **See** *Henderson Plot*

¹J. W. Williams & J. F. Morrison (1979) *Meth. Enzymol.* **63**, 437.

TIGHT ION PAIR

An ion pair in which the constituent ions are not separated by a solvent or other intervening molecule. Tight ion pairs are also referred to as contact ion pairs. If X^+ and Y^- represent constituent ions, then a tight ion pair would be symbolized by X^+Y^- . An example of a tight ion pair would be the case in which an enzyme stabilizes a carbonium ion with juxtaposed negatively charged side-chain groups. **See** *Loose Ion Pair; Ion Pair; Solvent-Shared Ion Pair; Solvent-Separated Ion Pair*.

TIME

A reference frame that, while resisting unambiguous definition, serves to measure the progress/motion of chemical/physical processes. The SI unit for time is the second (symbolized as s). **See** *Time & Chronomals*

Time is readily sensed physiologically, and the so-called “flow of time”^{1,2} is measured by the regularity of motions and changes occurring within and around us. Plato’s interpretation of Heraclitus’ analogy³ of life to a river is that all things are in flux. We are immersed in a universe of processes that act as clocks, and even in solitude, we can still sense heartbeat, pulse, heat flow and the motions of other internal processes. Consciousness itself appears

to depend on the pace of coordinated depolarization/repolarization of interneurons within the larger context of the brain’s neural networks⁴. Channels opening, ions flowing down concentration gradients, channels closing, ion pump ATPases actively restoring transmembrane potentials—these are all subcellular processes serving as miniature clocks. Time can thus be described as the reciprocal of the frequency of motional change. The precision with which periodic change can be determined allows time to serve as a standard frame of reference for marking the progress of physicochemical processes. The *second* now corresponds to the duration of 9,192,631,770 periods of the radiation corresponding to the transition between two hyperfine levels of the cesium-133 atom in the ground state. This precision leads to the universal use of *dynamical time* as the independent variable in differential equations (*i.e.*, of the form dX/dr) to describe the time evolution of changes in location and energy. **See** *Circadian Rhythm*

¹The term “flow of time” actually implies the absurd: that we can write an equation for the change in time per unit time (*i.e.*, $d\Delta t/dt$). In this regard, our intuitive sense of time passing quickly relates more to our having been distracted from an awareness of time’s passage.

²“The besetting sin of philosophers, scientists, and, indeed all who reflect about time is describing it as if it were a dimension of space”. C. W. K Mundle (1967) in *The Encyclopedia of Philosophy*, **8**, p. 138, Macmillan, New York.

³“Upon those stepping in the same river, ever different waters flow”. This statement, attributed to Heraclitus of Ephesus, who was probably the first Grecian philosopher, survives only as a fragment of metrical verse torn from its original context.

⁴F. H. C. Crick (1994) *The Astonishing Hypothesis*, Scribner, New York. This fascinating book describes Crick’s quest for an understanding of the chemical basis of human consciousness.

TIME & CHRONOMALS

A chronomal is a dimensionless parameter [symbolized by I or $I(x)$] that is proportional to time. Chronomals are especially useful in dealing with diffusion, chemical reactions, and other related processes. One can choose to express the properties of such systems as t equal to $K_D I_D$, where K_D contains all the physical constants and has overall units of time, whereas I_D is a chronomal expressed in terms of the extent of reaction ξ . In many respects, the chronomal can be regarded as dimensionless time.

In the following example dealing with the kinetics of precipitation, we will use the parameter ξ , the extent of reaction to designate the amount of substance that has

undergone reaction to the amount of substance able to react. If the initial concentration is c_o at t_o , then

$$\xi = (c_o - c)/(c_o - s) \quad \text{or} \quad (c - s) = (c_o - s)(1 - \xi)$$

where s is the solubility of the substance. Now, if one treats the precipitate as a monodisperse set of spherical particles whose single radius r depends on time (*i.e.*, $r(t)$ changes as the reaction progresses), then

$$r(t) = r_1 \xi^{1/3}$$

Combining the last two equations with the rate law for precipitation [*i.e.*, $dr/dt = vD(c - s)/r$] and separating variables, we obtain an expression that can be written in form, $t = K_D I_D$, where $K_D = r_1^{2/3} vD(c_o - s)$ and I_D is the definite integral from $x = 0$ to ξ of $[x^{-1/3}(1 - x) - 1 dx]$. K_D contains all the physical constants and has units of time, whereas I_D is the chronomal. Nielsen presents a useful account of chronomals in the appendix to his treatment of crystallization and precipitation¹.

¹A. E. Nielsen (1964) *Kinetics of Precipitation*, pp. 135-138, Pergamon, Oxford Univ. Press.

TIME-CORRELATED SINGLE PHOTON COUNTING

The measurement of the time histogram of a sequence of photons with respect to a periodic event. This technique requires a time-to-amplitude converter. **See also** *Photon Counting*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

TIME-OF-FLIGHT MASS SPECTROMETRY

A mass spectrometer employing brief exposure to a pulse of electrons to impart kinetic energy on sample molecules as well as the use of a drift tube to separate molecules on the basis of molecular mass. Because the imparted kinetic energies are virtually identical before entering the drift tube, and because the path length from entry port to detector is the same for all ionic species, the time-of-flight (or TOF) depends solely on the mass of a particular species. For biological macromolecules exhibiting little tendency to vaporize, one may use matrix-assisted laser-desorbed ion TOF mass spectrometry (or simply MALDI-TOF MS). In this technique, macromolecules are mixed with a matrix substance that is placed on an inert target and transferred to the sample compartment of the mass spectrometer. Immediately upon laser photo-excitation of the target, the matrix va-

porizes so rapidly that the contained macromolecule(s) are likewise swept into the sample stream for electron bombardment and time-of-flight determination.

TIME-RESOLVED MACROMOLECULAR CRYSTALLOGRAPHY

A X-ray crystallographic method¹⁻³ for detecting the transient accumulation of intermediates in enzyme catalysis, protein folding, ligand-binding interactions, and other processes involving macromolecules. The approach is premised on the well documented retention of substantial reactivity of biological macromolecules, even in the crystalline state.

¹H. Bartunik (1983) *Nucl. Instrum. Methods* **208**, 523.

²J. Hajdu, K. R. Acharya, D. I. Stuart, D. Barford & L. M. Johnson (1988) *Trends in Biochem. Sci.* **13**, 104.

³K. Moffat (1989) *Ann. Rev. Biophys. Biophys. Chem.* **18**, 309.

TIME-RESOLVED SPECTROSCOPY

Spectra recorded at a series of time intervals following excitation or other perturbation.

TIME-RESOLVED X-RAY DIFFRACTION

A method¹⁻³ for analyzing the kinetics of chemical/physical processes by combining the advantages of a focused, monochromatic synchrotron X-ray beam with a suitable live time-imaging device, thereby permitting preferably continuous monitoring of changes in the diffracted X-rays. Caffrey³ discusses how this technique can be gainfully applied to investigate lipid phase transitions.

¹B. Batterman & N. W. Ashcroft (1979) *Science* **206**, 157.

²P. Eisenberger (1986) *Science* **231**, 687.

³M. Caffrey (1989) *Ann. Rev. Biophys. Biophys. Chem.* **18**, 159.

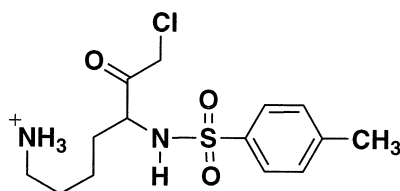
TIME-SCALE SEPARATION AND METABOLIC PATHWAYS CONTROL

Delgado *et al.*¹ recently demonstrated that time-scale separation is an effective way to localize metabolic control to only a few enzymes. They considered model pathways in which the eigenvalues of the Jacobian of the system are widely separated (*i.e.*, systems with time-scale separation). Their treatment assumes the system possesses a unique, asymptotically stable steady-state and that the reaction steps of the system under analysis are

well represented by linear kinetics around the steady state. While cells may produce many enzymes at far greater concentrations than needed to maintain a certain steady-state, they can achieve time-scale separation by controlling expression of only a few enzymes. The over-expressed enzymes catalyze what are termed “fast” reactions and lead to small response times of the system. For these “fast” reactions, metabolite control coefficients are small compared with “slow” reactions and do not effectively control overall flux. Nonetheless, at pathway branch points, “fast” reactions may be mutually competitive and result in significant control coefficients for fluxes to the branches.

¹J. Delgado & J. C. Liao (1995) *Biosystems* **36**, 55.

TLCK



The inhibitor tosyl-lysine chloromethyl ketone which reacts irreversibly with the active site serine hydroxyl group of trypsin. In the initial step of the inactivation process, TLCK's ϵ -amino group directs the inhibitor into the negatively charged specificity-conferring pocket where lysyl or arginyl side-chains normally are bound within trypsin's active site. Then, nucleophilic attack by the active-site serine hydroxyl leads to the formation of an ether linkage, accompanied by expulsion of chloride ion. **See** *Chymotrypsin*

TOLUENE DIOXYGENASE

This iron-sulfur oxygenase [EC 1.14.12.11] catalyzes the reaction of molecular oxygen with toluene and NADH to produce (1S,2R)-3-methylcyclohexa-3,5-diene-1,2-diol and NAD⁺. This reductase is an iron-sulfur flavo-protein (FAD) that contains ferredoxin. Ethylbenzene, 4-xylene, and some halogenated toluenes can likewise undergo conversion to the corresponding *cis*-dihydrodiols.

L. P. Wackett (1990) *Meth. Enzymol.* **188**, 39.

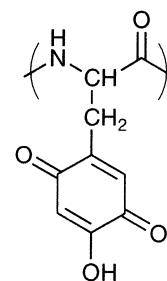
TONICITY

1. The tension or osmotic pressure of a solution; also, ionic strength, usually measured as weight percentage. Often the tonicity of a solution is presented as relative to some physiological solution (*e.g.*, blood plasma). **See** *Hypertonic; Hypotonic; Isotonic; Isotonic Buffers*

2. An isotope tracer method¹ for estimating the total amount of water within an organism.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, p. 190, Wiley-Liss, New York.

TOPAQUINONE



A trihydroxyphenylalanine residue (symbolized TPQ) that plays an essential cofactor role in catalysis of amine oxidases that use molecular oxygen and copper ions.

D. Wemmer, A. J. Smith, S. Kaur, D. Maltby, A. L. Burlingame, and J. P. Klinman (1990) *Science* **248**, 981.

D. Mu, K. F. Medzihradsky, G. W. Adams, P. Mayer, W. M. Hines, A. L. Burlingame, A. J. Smith, D. Cai, and J. Klinman (1994) *J. Biol. Chem.* **269**, 9926.

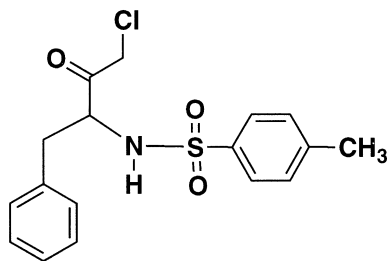
TOTAL ENERGY EXPENDITURE

The energy intake by an organism plus or minus any change in body mass or composition. Lifson and McClintock¹ introduced the use of doubly labeled water (²H₂O and H₂¹⁸O) to estimate energy expenditure. In this strategy, ¹⁸O is lost from the body as labeled water and carbon dioxide, while loss of the deuterium label allows one to measure turnover of the water pool. Wolfe² provides a lucid account of how the technique is applied, along with assumptions, strengths and weaknesses.

¹N. Lifson, W. S. Little, D. G. Levitt & R. M. Henderson (1975) *J. Appl. Physiol.* **39**, 657.

²R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, pp. 207-233, Wiley-Liss, New York.

TPCK



The active-site-directed inhibitor tosylphenylalanine chloromethyl ketone that specifically and irreversibly inhibits chymotrypsin. This chloroketone inhibitor relies on its toluene sulfonyl (or tosyl) group for binding into the aromatic binding pocket of chymotrypsin's active site. Inactivation occurs by alkylation of histidine-57 (pseudo-first order rate constant 0.2 min^{-1}). **See** *Chymotrypsin*

TRACEE

The unlabeled metabolite, drug, or other substance under study in a kinetic analysis of its biosynthesis/entry (appearance) and catabolism/clearance (disappearance). **See** *Compartmental Analysis*

TRACER

The labeled form of a metabolite, drug, or other substance used in a kinetic analysis of tracee biosynthesis/entry (appearance) and tracee catabolism/clearance (disappearance). **See** *Compartmental Analysis*

TRACER PERTURBATION METHOD

A enzyme kinetic technique, introduced by Britton and co-workers¹⁻⁴, that permits one to measure the equilibrium distribution of enzyme-bound substrate(s), intermediate(s), and product(s). In this procedure, radiolabeled substrate or product is initially permitted to react with enzyme for sufficient time to equilibrate. At thermodynamic equilibrium, all the different enzyme-bound species will be present at concentrations reflecting their stability relative to each other. One can then add a large excess of unlabeled substrate. Under this condition, any unbound or newly released radiolabel will mix with the large unlabeled pool of substrate or product, where it will undergo substantial reduction in its radiospecific activity. This dilution effectively reduces or eliminates any significant recycling of released radiolabel. One can then

sample the system as a function of time, and the rates and amounts of labeled substrate and labeled product depend on the kinetics of synthesis and/or interconversion of bound radiolabel to substrate and product, which upon release is unlikely to rebind.

¹H. G. Britton & J. B. Clarke (1972) *Biochem. J.* **110**, 161.

²H. G. Britton & J. B. Clarke (1972) *Biochem. J.* **130**, 397.

³H. G. Britton, J. Carreras & S. Grisolia (1971) *Biochemistry* **10**, 4522.

⁴H. G. Britton, J. Carreras & S. Grisolia (1971) *Biochemistry* **11**, 3008.

TRACER/TRACEE RATIO

A parameter used as a measure of isotope enrichment in stable isotope exchange and compartmental analysis experiments:

$$R = [r_{\text{sample}} - r_{\text{reference}}] / [r_{\text{blank}} - r_{\text{reference}}]$$

where r_{sample} , $r_{\text{reference}}$, and r_{blank} are the ion current ratios determined for an experimental sample, a reference sample, and a blank, respectively. The tracer/tracee ratio approximates specific activity (*i.e.*, [mol tracer]/[mol tracer + mol tracee]) when the tracer content is negligible relative to the tracee content. The tracer/tracee ratio R can be most directly determined using an ion ratio mass spectrometer to measure the ratio of ion currents for isotopomers such as $^{12}\text{CO}_2$ and $^{13}\text{CO}_2$ (at masses 44 and 45, respectively). **See** *Compartmental Analysis*

TRANSALDOLASE

This enzyme [EC 2.2.1.2], also known as dihydroxyacetone transferase and glycerone transferase, catalyzes the reversible reaction of sedoheptulose 7-phosphate with D-glyceraldehyde 3-phosphate to produce D-erythrose 4-phosphate and D-fructose 6-phosphate.

R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.

P. R. Levering & L. Dijkhuizen (1990) *Meth. Enzymol.* **188**, 405.

O. Tsolas & B. L. Horecker (1972) *The Enzymes*, 3rd ed., **7**, 259.

TRANSCARBOXYLASE

This enzyme [EC 2.1.3.1], also known as methylmalonyl-CoA carboxyltransferase, catalyzes the reaction of (*S*)-2-methyl-3-oxopropanoyl-CoA with pyruvate to produce propanoyl-CoA and oxaloacetate. The enzyme requires biotin, cobalt, and zinc as cofactors.

J. N. Earnhardt & D. N. Silverman (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 495.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

A. S. Mildvan, D. C. Fry & E. H. Serpersu (1991) in *A Study of Enzymes* (S.A. Kuby, ed.) vol. **2**, p. 105, CRC Press, Boca Raton.
 J. R. Knowles (1989) *Ann. Rev. Biochem.* **58**, 195.
 I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.
 H. G. Wood (1972) *The Enzymes*, 3rd ed., **6**, 83.

TRANSCRIPTION COMPLEX KINETICS

von Hippel¹ recently developed a model that integrates the structural, thermodynamic, and kinetic properties of the prokaryotic transcription complex that allows RNA polymerase to catalyze high-fidelity copying of a gene's DNA template strand into a cRNA transcript. This model allows one to understand how the transcription complex chooses between three competing pathways (e.g., moving forward along the template with attending elongation of the RNA chain, moving backward along the template with or without transcript shortening, or dissociating from the template). The choice between elongation and termination appears to favor elongation during the polymerization reaction, because the $\Delta G_{\text{release}}^{\ddagger} > \Delta G_{\text{forward}}^{\ddagger}$; however, when the polymerase arrives at a termination position, $\Delta G_{\text{release}}^{\ddagger} \approx \Delta G_{\text{forward}}^{\ddagger}$. To render the polymerase pathway kinetically accessible, the change in $\Delta G_{\text{release}}^{\ddagger}$ may be related to a massive destabilization of the transcription complex as it approaches terminators along the DNA template.

¹P. H. von Hippel (1998) *Science* **281**, 660.

TRANSCYTOSIS

The transport of a vesicle from one side of a cell to another. Typically, the vesicle is generated via endocytosis or phagocytosis and then exocytosed.

TRANSFERASES

A major class of enzymes that catalyze the transfer of a group or moiety from one compound to another. The groups being transferred can be one-carbon units such as methyl, hydroxymethyl, carbamoyl, or amidino moieties. Enzymes transferring aldehyde or ketonic groups such as transketolase are members of this class. Other examples include acyltransferases, glycosyltransferases, aminotransferases, phosphotransferases, and sulfotransferases.

TRANSFORMATION

Any conversion of reactant (substrate) into a particular product, irrespective of reagents or mechanisms involved. Transformations are distinct from reactions, be-

cause a full description of the latter would state or imply all the reactants and all the products.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

TRANSGLUTAMINASE

This enzyme [EC 2.3.2.13], also known as protein: glutamine γ -glutamyltransferase, catalyzes the calcium ion-dependent reaction of the side chain of a glutamyl residue in a protein with an alkylamine to generate an N^5 -alkylglutaminy residue in the protein and ammonia.

F. Tokunaga & S. Iwanaga (1993) *Meth. Enzymol.* **223**, 378.
 J. E. Folk (1983) *Adv. Enzymol.* **54**, 1.
 J. E. Folk & J. S. Finlayson (1977) *Adv. Protein Chem.* **31**, 1.
 J. E. Folk & S. I. Chung (1973) *Adv. Enzymol.* **38**, 109.

TRANSIENT SPECIES (or, Transient Chemical Species)

A relative term referring to an intermediate in a reaction or series of reactions which is short-lived under the given experimental conditions and assay procedures. Such species have a shorter half-life than persistent chemical species.

The term "metastable" has often been used to describe transient species. However, care should be exercised in such cases. Metastability relates to the thermodynamic stability of a molecule whereas the term "transient" refers to a kinetic property. Nevertheless, most transients are unstable relative to reactants and products. **See also** *Persistent*

TRANSIENT-STATE KINETIC METHODS

Any experimental technique that discloses the time-resolved behavior of a chemical/physical process. These approaches allow one to surmount the inherent limitations of steady-state and/or equilibrium kinetic measurements in the detection and quantification of species that comprise the internal equilibria of enzymic catalysis.

Johnson^{1,2} and Fierke & Hammes³ have presented detailed accounts of how rapid reaction techniques allow one to analyze enzymic catalysis in terms of pre-steady-state events, single-turnover kinetics, substrate channeling, internal equilibria, and kinetic partitioning. **See** *Chemical Kinetics; Stopped-Flow Techniques*

¹K. A. Johnson (1992) *The Enzymes*, 3rd ed., **20**, 1.

²K. A. Johnson (1995) *Meth. Enzymol.* **249**, 38.

³C. A. Fierke & G. G. Hammes (1995) *Meth. Enzymol.* **249**, 1.

TRANSITION COORDINATE

The normal reaction coordinate at the transition state corresponding to a vibration with an imaginary frequency. Motion along the transition coordinate will lead to either reactants or products¹. **See** *Reaction Coordinate*

¹IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

TRANSITION DIPOLE MOMENT

The amplitude of the oscillating electric or magnetic moment that is induced in a molecular entity by an electromagnetic wave.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

TRANSITION POLARIZATION

The direction of the transition moment in a molecular entity.

TRANSITION PROBABILITY FUNCTIONS

Relationships used in [¹⁸O]phosphate analysis to determine the probability that an enzyme's reaction with a phosphate containing *m* labeled oxygens in the presence of unlabeled water will result in the release of a phosphate that contains *n* labeled oxygens.

D. D. Hackney, K. E. Stempel & P. D. Boyer (1980) *Meth. Enzymol.* **64**, 60.

TRANSITION STATE

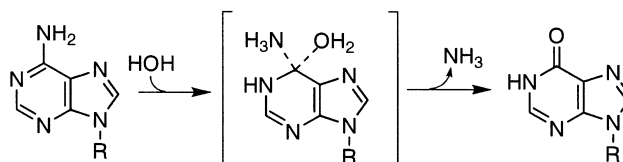
In terms of elementary reactions, a transition state (the assembly of atoms at the transition state is often called an activated complex) is that species having a more positive Gibbs free energy than the products or reactants of that elementary reaction. This species is represented by that assembly(ies) of atoms which, if placed in these geometries, energies, and bonds (both complete and partial) would have an equal probability of forming either the reactants or products of that elementary reaction. Note that the transition state could be represented by a set of states, each having the ability to form products or reactants. This set of states is characterized by the partition function for the transition state. The transition state is characterized by one and only one imaginary frequency.

The terms "early" and "late" transition states are qualitative descriptions related to concepts such as those advanced by the Hammond principle. An early transition

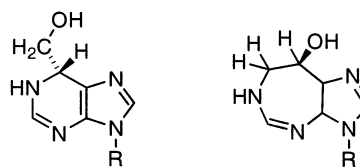
state is that assembly of atoms or moieties that closely resembles the reactant(s), such that only a relatively small reorganization will generate the reactant(s). Analogously, a late transition state more closely resembles the structure of the reaction product(s). **See** *Chemical Kinetics; Transition State Theory; Potential Energy Surface; Hammond Principle; Transition Structure*

TRANSITION-STATE ANALOGUES

A molecule designed to mimic the properties, structure, and/or geometries of the transition state of a particular reaction. Such compounds are often potent inhibitors of enzymes. **See** *Molecular Similarity*



Likely Transition State in the Adenosine Deaminase Reaction



NDHPR ($K_i = 10^{-5}$ M) Coformycin ($K_i = 10^{-11}$ M)

Figure 1. Adenosine deaminase reaction mechanism, indicating nucleophilic substitution reaction at position-6 of the purine ring. Note that the inhibitors NDHPR and coformycin also contain sp^3 hybridization in the position analogous to the carbon atom undergoing substitution in the adenosine deaminase reaction. Tight binding is thought to be a consequence of the complementarity of the enzyme for analogues mimicking the transition-state configuration.

As pointed out by Wolfenden^{1,2}, the classical Pauling-Haldane model of transition-state stabilization model for enzyme catalysis offers a means to rationalize the extraordinarily high affinities achieved by some substrate analogues. Using the example of the adenosine deaminase reaction (Fig. 1), we recognize that a key feature in this aromatic nucleophilic substitution reaction is the backside entry of a water molecule to attain loss of aromaticity as well as sp^3 hybridization at position-6 of the purine ring. Agents that can mimic the stereochemical and/or electrostatic arrangement achieved in the transition state are thought to possess a special binding advantage over typical substrates. This special advantage relates to the fact that such analogues are complementary

to the transition-state configuration of the enzyme, whereas substrates in the ground state are not. Thus, while a substrate must necessarily suffer some loss in binding energy as it is contorted along its path to reach the transition state, the same is not true for transition-state analogues.

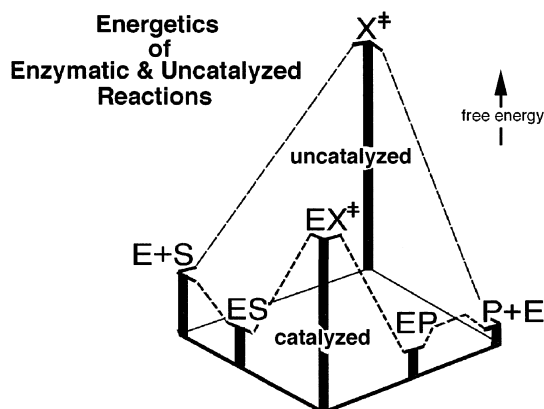


Figure 2. Energetics of enzymatic and uncatalyzed reactions. Conversion of substrate S directly to the transition state in the absence of enzyme is energetically more unfavorable than the corresponding conversion on the enzyme surface.

The energetics of enzymatic and their corresponding uncatalyzed reference reactions can be understood by the cyclic path that allows for substrate conversion to product by the uncatalyzed and enzymatic routes (Fig. 2). Note that the uncatalyzed reaction is characterized by a transition state that is far less stable than its enzymatic counterpart. Note also that the initial and final conditions are the same for either route, an absolute requirement for any catalyzed process (*i.e.*, no effect on the overall equilibrium constant).

Thermodynamic Cycle for Enzymatic and Noncatalyzed Reactions

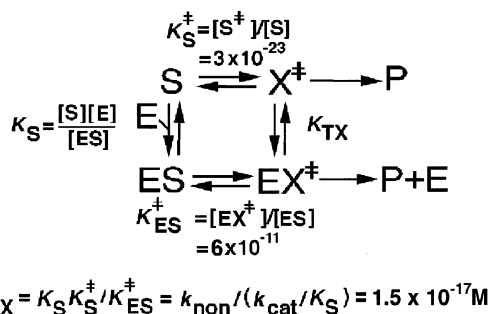


Figure 3. Thermodynamic cycle for enzymatic and noncatalyzed reactions, indicating that the limiting dissociation constant for binding of the transition-state $X^\ddagger$ to the enzyme to form $EX^\ddagger$.

In principle, one can never exactly duplicate the transition state, because transition state theory requires that such an intermediate species would disproportionate back to E·Substrate complex as well as proceed onward to E·Product complexes. However, the scheme shown in Fig. 3 permits one to estimate the maximal affinity that should be achievable if one were to approximate closely the electronic and stereochemical configuration of the enzyme and substrate in the transition state. An accurate estimation of K_{TX} requires detailed knowledge that the uncatalyzed reference reaction follows the same mechanism as the enzyme-catalyzed process. **See Enzyme Proficiency; Reference Reaction**

¹R. Wolfenden (1969) *Nature (London)* **223**, 704.

²A. Radzicka & R. Wolfenden (1994) *Meth. Enzymol.* **249**, 284.

TRANSITION STATES IN BIOLOGICAL CATALYSIS

Rather than adopting a purely fundamentalist viewpoint centering on strong reactant(s) binding of a transition-state geometry, Cannon *et al.*¹ recently offered a different perspective on enzyme rate enhancements. By re-examining the reaction coordinate, their analysis suggested that the ground state of the enzyme-substrate complex and the transition state are related by the mean force acting along the reaction path. Their treatment would suggest that catalysis is not easily resolvable into the effects of ground state destabilization or transition-state stabilization. The ability of the enzyme to preorganize the reaction environment may instead be a dominant factor, and gating of solvent water molecules may eliminate or minimize the inhibitory effects of water molecules on chemical reactivity.

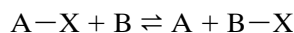
¹W. R. Cannon, S. F. Singleton & S. J. Benkovic (1996) *Nature Struct. Biol.* **3**, 821.

TRANSITION-STATE THEORY

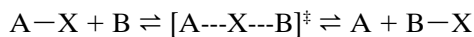
Theoretical descriptions of absolute reaction rates in terms of the rate-limiting formation of an activated complex during the course of a reaction. Transition-state theory (pioneered by Eyring¹⁻⁴, Pelzer and Wigner⁵, and Evans and Polanyi⁶) has been enormously valuable, and beyond its application to chemical reactions, the theory applies to a wider spectrum of rate processes (*e.g.*, diffusion, flow of liquids, internal friction in large polymers, *etc.*). Transition state theory assumes: (1) that classical mechanics can be used to calculate trajectories over po-

tential energy surfaces; (2) that reactants and activated complexes maintain equilibrium with each other and that the energetics of components follow Boltzmann distributions of energy; and (3) that reactions will occur once a certain point on the potential energy surface has been passed.

This theory assumes that the rate of a reaction at a given temperature is proportional to the concentration of an activated complex that is in equilibrium with the unactivated reactants. In proceeding from substrates to products, the reactants form an activated complex, also said to be in the transition state. As an example, consider the bimolecular reaction in the scheme below, in which the moiety X is being transferred.



In transition state theory, an activated complex occurs at an intermediary point prior to the formation of products:



The “double-dagger” symbol, $\ddagger$, is exclusively used to designate activated complexes or the transition state. This activated complex will only form when the reacting molecules have sufficient energy (*i.e.*, $E \geq E_a$). The activated complex is that configuration of interacting molecules at the maximum point in the potential energy barrier for a reaction progressing up a potential energy “hill”. In some reactions (*e.g.*, proton or electron transfer), the proton or electron may leak through a potential energy barrier by a process called tunneling. Tunneling is clearly significant in electron-transfer reactions, but it can also be appreciable in the transfer of protons⁷.

In transition-state theory, the absolute rate of a reaction is directly proportional to the concentration of the activated complex at a given temperature and pressure. The rate of the reaction is equal to the concentration of the activated complex times the average frequency with which a complex moves across the potential energy surface to the product side. If one assumes that the activated complex is in equilibrium with the unactivated reactants, the calculation of the concentration of this complex is greatly simplified. Except in the cases of extremely fast reactions, this equilibrium can be treated with standard thermodynamics or statistical mechanics⁸. The case of

extremely fast reactions introduces a small difference of about 10% between experimental rates and predicted rates. **See** *Chemical Kinetics; Collision Theory; Temperature Dependence; Q_{10} ; Unimolecular Reactions and Transition-State Theory; Transition-State Theory (Thermodynamics); Tunneling*

¹S. Glasstone, K. J. Laidler & H. Eyring (1941) *The Theory of Rate Processes*, McGraw-Hill, New York.

²H. Eyring (1935) *Chem. Rev.* **17**, 65.

³H. Eyring (1935) *J. Chem. Phys.* **3**, 107.

⁴W. F. K. Wynne-Jones & H. Eyring (1935) *J. Chem. Phys.* **3**, 492.

⁵H. Pelzer & E. Wigner (1932) *Zeit. Physik Chem.* **B15**, 445.

⁶M. G. Evans & M. Polanyi (1935) *Trans. Faraday Soc.* **31**, 875.

⁷R. P. Bell (1973) *The Proton in Chemistry*, 2nd ed., Cornell Univ. Press, Ithaca.

⁸R. D. Present (1959) *J. Chem. Phys.* **31**, 747.

TRANSITION-STATE THEORY (Thermodynamics)

Transition-state theory can also be expressed in thermodynamic terms¹. Since $K^{\ddagger}$ in the Eyring equation ($k = k_B T K^{\ddagger} / h = R T K^{\ddagger} / L h$ where k is the bimolecular rate constant, k_B is Boltzmann's constant, T is the absolute temperature, h is Planck's constant, L is Avogadro's constant, R is the universal gas constant, and $K^{\ddagger}$ is the equilibrium constant between the reactants and the transition state) is similar to a true equilibrium constant, let $\Delta G^{\ddagger}$ be the standard Gibbs free energy of activation between the reactants and the activated complex in the transition state. Strictly speaking, $\Delta G^{\ddagger} = -RT \ln [K^{\ddagger}(c^{\ominus})^{n-1}]$ where $c^{\ominus}$ is a standard-state concentration and n is the molecularity of the reaction. Thus, $K^{\ddagger} = e^{-\Delta G^{\ddagger}/RT} (c^{\ominus})^{1-n}$. However, standard states are, in general, unit concentrations, using whatever unit is being used to evaluate the rate constant. The bimolecular rate constant, in the Eyring equation ($k = (k_B T / h) K^{\ddagger}$), can be expressed as

$$k = \frac{k_B T}{h} e^{-\Delta G^{\ddagger}/RT} (c^{\ominus})^{1-n} = \frac{k_B T}{h} e^{\Delta S^{\ddagger}/R} e^{-\Delta H^{\ddagger}/RT} (c^{\ominus})^{1-n}$$

These parameters ($\Delta G^{\ddagger}$, $\Delta H^{\ddagger}$, and $\Delta S^{\ddagger}$) differ slightly from normal standard parameters in that the contribution of motion along the reaction coordinate toward the transition state is not included. The values are the difference in free energy, enthalpy, and entropy between 1 mole of activated complex and 1 mole of each reactant, all substances being at their standard-state concentrations (usually 1 M).

For a simple one-substrate, one-product enzyme-catalyzed reaction there are at least eighteen thermodynamic parameters that characterize the reaction coordinate profile. However, it should be noted that the Michaelis binary complex and the activated complex (*i.e.*, the Eyring complex) are not equivalent. An activated complex must form prior to the Michaelis complex, its formation governed by k_1 , the first bimolecular rate constant. In addition, at least one activated complex occurs between the ES and the EP binary complexes. Knowing the $\Delta H^\ominus$ and $\Delta H^\ddagger$ values of each step in an enzyme-catalyzed reaction, allows the investigator to draw a reaction coordinate diagram. Similar diagrams can be plotted using $\Delta G^\ominus$ and $\Delta G^\ddagger$ values or $\Delta S^\ominus$ and $\Delta S^\ddagger$ values. Dixon and Webb² have provided a table reviewing the different procedures for the determination of the various thermodynamic quantities for a simple one-substrate, one-product enzyme-catalyzed reaction. A truly thorough analysis necessitates the ability to separate each rate constant.

If an investigator studies temperature effects on K_m and V_{max} values of an enzyme-catalyzed reaction, then interpretation of the results depends on the precise definitions of these kinetic parameters³. If the rate-determining step in the reaction changes with temperature then either one or both of the plots with respect to K_m or V_{max} may be curvilinear. When analyzing binding behavior ideally, true K_d values (rather than K_m values) should be studied. In addition, the ionization state of the protein and/or substrates and effectors (including the buffer) may change with temperature as well. Often temperature studies should be preceded by a pH study. **See** Arrhenius Equation; Transition-State Theory; Gibbs Free Energy of Activation; Enthalpy of Activation; Entropy of Activation; Volume of Activation; Transition-State Theory in Solutions; Diffusion Control; Temperature Dependence; Entropy and Enthalpy of Activation, Interpretations

¹J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

²M. Dixon & E. C. Webb (1979) *Enzymes*, 3rd ed., Academic Press, New York.

³K. D. Gibson (1953) *Biochim. Biophys. Acta* **10**, 221.

TRANSITION STRUCTURE

The saddle point on a three-dimensional potential-energy surface, characterized by one negative force constant in the harmonic force constant matrix.

IUPAC (1994) *Pure and Appl. Chem.* **66**, 1077.

TRANSKETOLASE

This thiamin pyrophosphate-dependent enzyme [EC 2.2.1.1], also known as glyceraldehyde transferase, catalyzes the reversible reaction of sedoheptulose 7-phosphate with D-glyceraldehyde 3-phosphate to produce D-ribose 5-phosphate and D-xylulose 5-phosphate. The enzyme exhibits a wide specificity for both reactants. It also can catalyze the reaction of hydroxypyruvate with R—CHO to produce carbon dioxide and R—CH(OH)—C(=O)—CH₂OH. Transketolase isolated from *Alkaligenes faecalis* shows high activity with D-erythrose as the acceptor substrate.

R. L. Schowen (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 217.

J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

A. Schellenberger, G. Hübner & H. Neef (1997) *Meth. Enzymol.* **279**, 131.

Y. Lindqvist & G. Schneider (1993) *Curr. Opin. Struct. Biol.* **3**, 896.

R. Kluger (1992) *The Enzymes*, 3rd ed., **20**, 271.

C. J. Gubler (1991) *A Study of Enzymes* **2**, 117.

TRANSMISSION COEFFICIENT

A ratio in transition state theory (symbolized by κ) that represents the probability that the activated complex will go on to form product(s) rather than return to reactants. In most cases, this value is approximately one; however, if reactants do not obey the Boltzmann law or if the temperature is very high, then the coefficient can be less than one. **See** Transition-State Theory

TRANSMISSION DENSITY

Symbolized by D , the transmission density is the negative decadic logarithm of the transmittance; thus, $D = -\log \tau$. **See** Transmittance

TRANSMITTANCE

A dimensionless quantity, symbolized by τ or T , equal to the transmitted radiant power, P_{tr} , divided by the radiant power incident on the sample, P_0 ; thus, $\tau = P_{tr}/P_0$. It is a measure of the ability of a body, solution, entity, *etc.*, to transmit electromagnetic radiation. It is synonymous with transmission factor. **See also** Internal Transmittance; Transmission Density; Total Transmittance; Beer-Lambert Law; Absorption Spectroscopy

TRANSPORT NUMBER

A dimensionless number, symbolized by t (or by t^+ or t_+ for a cation and t^- or t_- for an anion), that is equal

to the fraction of the total current (of an electrolytic solution) carried by that ion. Thus, at infinite dilution $t_0^+ = \Lambda_0^+/\Lambda_0$ and $t_0^- = \Lambda_0^-/\Lambda_0$ where the subscripts refer to infinite dilution, Λ_0 is the molar conductance at infinite dilution and Λ_0^+ and Λ_0^- are the molar ionic conductances at infinite dilution. Transport numbers, also called transference numbers, can be measured by a number of means: *e.g.*, the Hittorf method or the moving boundary method. By knowing the molar conductance of an electrolytic solution, knowledge of the transport number enables one to determine the mobility of an ion.

TRAPPING

The binding, reaction, or interception of a reactive molecular entity or transitory intermediate in a reaction pathway to convert the substance to a more stable form and/or remove that substance from the system. Trapping may involve binding or reaction with another molecular entity or involve the alteration of some parameter (*e.g.*, thermal trapping)¹.

¹D. L. Purich (1982) *Meth. Enzymol.* **87**, 3.

TRIMOLECULAR

Trimolecular reactions (also referred to as termolecular) involve elementary reactions where three distinct chemical entities combine to form an activated complex¹. Trimolecular processes are usually third order, but the reverse relationship is not necessarily true. All truly tri- or termolecular reactions studied so far have been gas-phase processes. Even so, these reactions are very rare in the gas-phase. They should be very unlikely in solution due, in part, to the relatively slow-rate of diffusion in solutions. **See** *Molecularity; Order; Transition-State Theory; Collision Theory; Elementary Reactions*

¹J. W. Moore & R. G. Pearson (1981) *Kinetics and Mechanism*, Wiley, New York.

TRIOSE-PHOSPHATE ISOMERASE

This enzyme [EC 5.3.1.1], also known as triosephosphate mutase (TIM) and phosphotriose isomerase, catalyzes the interconversion of D-glyceraldehyde 3-phosphate and dihydroxyacetone phosphate (IUPAC: glycero-3-phospho-D-glyceraldehyde). As pointed out by Rose¹, this enzyme is chiefly responsible for the largely symmetrical conversion of the two three-carbon segments of glucose to lactate and for the nearly uniform distribution of ¹⁴C from pyruvate in the glucosyl units of liver glycogen. The

reaction favors dihydroxyacetone phosphate, DHAP, with an equilibrium constant of 22. This enzyme is extraordinarily efficient, displaying a specific activity of 6700 mmol/min/mg at saturating G3P concentrations². Like phosphoglucose isomerase, a *cis*-enediolate intermediate is a key catalytic feature³, and Alberty and Knowles⁴ used kinetic isotope effect measurements to analyze the energetics of TIM catalysis. In fact, this enzyme and proline racemase are among the extensively characterized enzymes in terms of the energetics of the catalytic reaction cycle⁵⁻⁷. **See** *Haldane Relation; Enzyme Energetics (the Case of Proline Racemase); Kinetic Isotope Effect; Affinity Labeling; Reaction Coordinate Diagram*

¹I. A. Rose (1969) in *Comprehensive Biochemistry* (M. Florkin & E. H. Stokes, eds.), vol. **17**, p. 126, Elsevier, Amsterdam.

²E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

³I. A. Rose (1982) *Meth. Enzymol.* **87**, 84.

⁴W. J. Alberty & J. R. Knowles (1976) *Biochemistry* **15**, 5631.

⁵J. R. Knowles (1991) *Nature* **350**, 121.

⁶R. C. Davenport, P. A. Bash, B. A. Seaton, M. Karplus, G. A. Petsko & D. Ringe (1991) *Biochemistry* **30**, 5821.

⁷J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.

TRIPLE-COMPETITIVE METHOD

A method used to determine primary intrinsic isotope effects¹. In this procedure, three differently labeled substrates are used to react with a labeled cosubstrate and the distribution of the labels in the products is measured.

¹D. B. Northrop (1982) *Meth. Enzymol.* **87**, 607.

TRIPLET STATE

A state having a total electron spin quantum number of one. **See** *Fluorescence*

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

TRIPLET-TRIPLET ANNIHILATION

A reaction between two species (*i.e.*, atoms or molecular entities), both of which are in triplet electronic states, resulting in at least two products, one of which is in its ground singlet state and the other in an excited singlet state. Delayed fluorescence often accompanies such processes. **See also** *Annihilation; Wigner Spin Conservation Rule; Delayed Luminescence*

TRIPLET-TRIPLET ENERGY TRANSFER

Energy transfer from a donor that is in an electronically excited triplet state to produce an electronically excited acceptor which is also in its triplet state.

TRIPLET-TRIPLET TRANSITIONS

Electron transitions in which both the initial and final states are triplet states.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

TRITIUM

A hydrogen isotope with a nucleus consisting of one proton and two neutrons (the nucleus is referred to as the triton). Tritium, a radioisotope symbolized by ^3H or T, decays by negative beta emission (0.01860 MeV) with a half-life of 12.32 years. The atomic weight of tritium is 3.01605 amu. It is frequently used in metabolic and kinetic experiments. The following decay data indicate the time followed by the fraction of original amount at the specified time: 0, 1.000; 1 month, 0.995; 2, 0.991; 3, 0.986; 4, 0.981; 5, 0.977; 6, 0.972; 7, 0.968; 8, 0.963; 9, 0.959; 10, 0.954; 11, 0.950; 1 year, 0.945; 2, 0.893; 3, 0.844; 4, 0.798; 5, 0.754; 6, 0.713; 7, 0.674; 8, 0.637; 9, 0.602; 10, 0.569; 11, 0.538; 12, 0.509.

Special precautions: The maximum range in air is 0.47 cm. Because no organ selectively concentrates tritiated metabolites, the entire body is considered to be the critical organ in terms of dose. The effective *biological* half-life in humans is around 10 days. This period can be shortened by increased water intake. Double gloving is sufficient, but changing the outer glove is recommended every 20 min when direct contact with gloves cannot be avoided.

TRITON

1. The name for a radioactive hydrogen ion. **See** *Tritium*
2. An abbreviation used for Triton X-100, a nonionic detergent that is used to solubilize membrane-bound proteins.

Triton X-100 has proved to be of great value in the surface dilution model^{1,2} for lipolytic enzyme action. In this experimental strategy, the surface concentration of phospholipid in mixed micelles is reduced by the addition of Triton as a neutral diluent, thereby increasing the average distance between phospholipids. This allows one to draw mechanistic inferences about the binding interactions of lipases and phospholipases with their lipid substrates¹.

¹T. G. Warner & E. A. Dennis (1975) *J. Biol. Chem.* **250**, 8044.

²R. A. Deems, B. R. Eaton & E. A. Dennis (1975) *J. Biol. Chem.* **250**, 9013.

TRYPSIN

This endopeptidase [EC 3.4.21.4], a member of the peptidase family S1, hydrolyzes peptide bonds at Arg—Xaa and Lys—Xaa. **See** *Chymotrypsin; Catalytic Triad; Acyl-Serine Intermediate*

D. R. Corey & C. S. Craik (1992) *J. Amer. Chem. Soc.* **114**, 1784.

A. A. Kossiakoff (1987) in *Biological Macromolecules & Assemblies* (F. A. Jurnak & A. McPherson, eds.) **3**, p. 369, Wiley, New York.

R. J. Coll, P. D. Compton & A. L. Fink (1982) *Meth. Enzymol.* **87**, 66.

J. S. Fruton (1982) *Adv. Enzymol.* **53**, 239.

B. Keil (1971) *The Enzymes*, 3rd ed., **3**, 250.

TRYPTOPHANASE

This enzyme [EC 4.1.99.1], also known as L-tryptophan indole-lyase, catalyzes the hydrolysis of L-tryptophan to generate indole, pyruvate, and ammonia. The reaction requires pyridoxal phosphate and potassium ions. The enzyme can also catalyze the synthesis of tryptophan from indole and serine as well as catalyze 2,3-elimination and β -replacement reactions of some indole-substituted tryptophan analogs of L-cysteine, L-serine, and other 3-substituted amino acids.

I. Behbahani-Najad, J. L. Dye & C. H. Suelter (1987) *Meth. Enzymol.* **142**, 414.

A. E. Braunstein & E. V. Goryachenkova (1984) *Adv. Enzymol.* **56**, 1.

A. E. Martell (1982) *Adv. Enzymol.* **53**, 163.

E. E. Snell (1975) *Adv. Enzymol.* **42**, 287.

L. Davis & D. E. Metzler (1972) *The Enzymes*, 3rd ed., **7**, 33.

TRYPTOPHAN SYNTHASE

This pyridoxal-phosphate-dependent enzyme [EC 4.2.1.20] catalyzes the reaction of L-serine with 1-(indol-3-yl)glycerol 3-phosphate to produce L-tryptophan and glyceraldehyde 3-phosphate. The enzyme will also catalyze (a) the conversion of serine and indole into tryptophan and water and (b) conversion of indoleglycerol phosphate into indole and glyceraldehyde phosphate.

K. A. Johnson (1992) *The Enzymes*, 3rd ed., **20**, 1.

E. W. Miles (1991) *Adv. Enzymol.* **64**, 93.

C. C. Hyde & E. W. Miles (1990) *Biotechnology* **8**, 27.

E. W. Miles, R. Bauerle & S. A. Ahmed (1987) *Meth. Enzymol.* **142**, 398.

E. W. Miles (1979) *Adv. Enzymol.* **49**, 127.

C. Yanofsky & I. P. Crawford (1972) *The Enzymes*, 3rd ed., **7**, 1.

TUBULIN-GTP HYDROLYSIS (Apparent Irreversibility)

The pathway of exchangeable-site GTP hydrolysis associated with bovine brain microtubule polymerization was

investigated using an assay of intermediate ^{18}O -positional isotope exchange reactions¹. Under a variety of conditions that influence the rate and extent of tubulin self-assembly, GTP hydrolysis proceeded without any evidence of multiple reversals that are characteristic of any reversible phosphoanhydride-bond cleavage. These results also accord with published findings² that ATP hydrolysis during actin polymerization fails to display intermediate exchange reactions. **See** *Microtubule Assembly Kinetics*

¹J. M. Angelastro & D. L. Purich (1990) *Eur. J. Biochem.* **191**, 507.

²M. F. Carlier, D. Pantaloni, J. A. Evans, P. K. Lambooy, E. D. Korn & M. R. Webb (1988) *FEBS Lett.* **235**, 211.

TUBULIN POLYMERIZATION ASSAYS

Microtubule assembly and disassembly results from the repetitive endwise gain or loss of tubulin dimers^{1,2}, and a number of useful procedures have been developed to study the kinetics of tubulin polymerization and microtubule depolymerization. These include³: (a) Turbidity measurements which take advantage of the fact that the turbidity of microtubule protein solutions is a direct measure of the polymer weight concentration; (b) Electron microscopy which allows one to directly measure polymer length changes in samples fixed at various times after polymerization or depolymerization is induced; (c) Viscosity which is based on the use of low-shear conditions to observe changes in the apparent viscosity of solutions containing microtubules; (d) Sedimentation which is based on the fact that large protein aggregates such as microtubules can be separated from much more slowly sedimenting tubulin dimers; (e) Flow birefringence which allows one to orient long rod structures under the influence of an imposed flow shear force; (f) Dark-field microscopy which is based on Tyndall's classical 19th century design of a "ultramicroscope" for detecting particles by their light scattering properties; (g) X-ray scattering methods based on the small-angle scattering of synchrotron radiation; (h) Filtration^{4,5} to separate polymeric and protomeric tubulin forms and to utilize the trapping of radioactively labeled GTP into the assembled microtubules; and (i) Biotinylated tubulin⁶ to directly visualize the addition of tubulin molecules on each end of microtubules after subsequent streptavidin binding and immunocytochemical detection. **See** *Microtubule Assembly Kinetics*

¹D. L. Purich & D. Kristofferson (1984) *Adv. Prot. Chem.* **36**, 133.

²D. L. Purich & J. M. Angelastro (1994) *Adv. Enzymol.* **69**, 121.

³F. Gaskin (1982) *Meth. Enzymol.* **85**, 433.

⁴R. B. Maccioni & N. W. Seeds (1978) *Arch. Biochem. Biophys.* **185**, 262.

⁵L. Wilson, H. P. Miller, K. W. Farrell, K. B. Snyder, W. C. Thompson & D. L. Purich (1985) *Biochemistry* **23**, 5254.

⁶D. Kristofferson, T. Mitchison & M. W. Kirschner (1986) *J. Cell Biol.* **102**, 1007.

TUBULIN:TYROSINE LIGASE

This enzyme [EC 6.3.2.25] catalyzes the ATP-dependent religation of L-tyrosine to the C-terminal L-glutamate residue in the detyrosinated $\alpha\beta$ -tubulin dimer, yielding ADP, orthophosphate, and the religated tubulin dimer. L-Phenylalanine and 3,4-dehydroxy-L-phenylalanine are also substrates. This is the only known C-terminal peptide bond-forming reaction involving a protein substrate.

See also *Arsenolysis*

N. L. Deans, R. D. Allison & D. L. Purich (1992) *Biochem. J.* **286**, 243.

J. Wehland, H. C. Schröder & K. Weber (1986) *Meth. Enzymol.* **134**, 170.

B. J. Terry & D. L. Purich (1982) *Adv. Enzymol.* **53**, 113.

TUNGSTOENZYMES

Enzymes that require tungsten as a metallic cofactor for catalytic activity. These enzymes include formate dehydrogenase (from *Clostridium thermoaceticum* and *Escherichia acidaminophilum*), acetylene hydratase from *Pleobacter acetylenicus*, aldehyde ferredoxin oxidoreductase (from *Pyrococcus furiosus*, *P. sp. ES-4*, and *Thermococcus sp. ES-1*), formylmethanofuran dehydrogenase (from *Methanobacterium wolfei* and *M. thermoautotrophicum*), carboxylic acid reductase (*Clostridium thermoaceticum* and *C. formicoaceticum*), formaldehyde ferredoxin oxidoreductase (*Thermococcus litoralis* and *Pyrococcus furiosus*), aldehyde dehydrogenase (from *Desulfovibrio gigas*), and glyceraldehyde-3-phosphate ferredoxin oxidoreductase (from *Pyrococcus furiosus*).

A. Kletzin & M. W. W. Adams (1996) *FEMS Microbiol. Rev.* **18**, 5.
P. A. Bertram, R. A. Schmitz, D. Linder & R. K. Thauer (1994) *Arch. Microbiol.* **161**, 220.

F. da Silva & R. J. P. Williams (1991) *The Biological Chemistry of the Elements: the Inorganic Chemistry of Life*, Clarendon Press, Oxford.

TURBIDITY

The apparent absorption of light resulting from scattering of incident radiation out of the transmitted light path. The intensity of light scattered in various directions from a beam of incident light passing through a solution of

macromolecules or other large particles is diminished, such that $I = I_0 e^{-\tau \lambda}$ where turbidity τ is the measure of decreased intensity I of the incident beam intensity I_0 per unit length, λ , of a solution. This expression is formally analogous to the Beer-Lambert Law for absorption of light. The increase in turbidity is an important kinetic tool in studying macromolecule polymerizations. Recalling that the Rayleigh ratio, R_θ , is an important parameter in light scattering experiments, the turbidity is related to the Rayleigh ratio, R_{90} , for N particles of uniform size per unit volume V at $\theta = 90^\circ$, by the relationship $\tau = (16\pi/3)R_{90}$. **See also** *Absorption Spectroscopy; Light Scattering; Actin Assembly Assays; Microtubule Assembly Kinetics; End-wise Depolymerization*

TURNOVER

1. The amount of material (*e.g.*, macromolecule or metabolite) metabolized or processed per unit time. 2. The balance between synthesis and degradation of a biomolecule. 3. One complete turn of a reaction cycle. **See** *Turnover Number; Protein Turnover; Pulse-Chase Experiments; Exponential Decay*

TURNOVER TIME

The period required for a pool of tracer to be replaced by newly biosynthesized substrate (also termed the tracee), corresponding to about five half-lives [*i.e.*, $1 - (0.5)^5$] or about 97% replacement of tracer by unlabeled substrate. **See** *Compartmental Analysis*

TWIDDLING

The random mechanochemical action of chemotactic bacteria, responsible for resetting their vectorial movement in gradients of chemotractant. Bacteria that experience a constant or increasing concentration of attractant twiddle infrequently, whereas those experiencing a drop in attractant concentration have a higher frequency of twiddling. Hence, those moving toward an attractant tend to continue in their migration, while those off-course use twiddling to reorient themselves. In their quantitative analysis, Dahlquist *et al.*¹ describe the importance of twiddling in making chemotaxis efficient.

Chen's laboratory^{2,3} used quasi-elastic light scattering to track migrating chemotactic bands of *Escherichia coli* in a buffer solution. The temporal development of the bacterial density profile is observed by the intensity of

scattered light as the band migrates through a fixed laser beam. They offered a model accounting for some aspects of the observed time evolution of the density profile. The microscopic motility characteristics of the *E. coli* in the band were simultaneously examined by photon correlation spectroscopy, and measured correlation functions yielded the spatial dependence of the half-width within the band. By analyzing the correlation function as a superposition of bacteria proceeding in a straight line and those engaging in twiddling motions, they obtain good agreement between their theoretical treatment and the measured angular dependence of the line shape. This effort helped them to estimate the fraction of twiddling bacteria in the center of the migrating band. **See** *Persistence*

¹F. W. Dahlquist, P. Lovely & D. E. Koshland, Jr. (1972) *Nature New Biol.* **236**, 120.

²M. Holz & S. H. Chen (1978) *Biophys. J.* **31**, 15.

³P. C. Wang & S. H. Chen (1981) *Biophys. J.* **36**, 203.

TWO-PHOTON EXCITATION

The excitation of a molecular entity due to absorption, successively or simultaneously, of two photons. If successively, the species that absorbs the second photon still has some of the excitation energy imparted by the first photon.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

TWO-PHOTON PROCESS

A process triggered by a two-photon excitation.

TWO-PROTONIC-STATE ELECTROPHILES

Compounds exhibiting different electrophilic reactivities depending on the compound's protonic or ionization state¹. Such molecules serve as excellent probes of active-site thiols and as reporter groups. **See also** *Affinity Labeling*

¹K. Brocklehurst (1982) *Meth. Enzymol.* **87**, 427.

TYPE 1,2,3 COPPER BINDING SITES

The copper-binding sites in copper-containing proteins are characterized by three distinct classes. In Type-1, or blue copper centers, the copper is coordinated to at least two imidazole nitrogens from two histidyl residues and one sulfur from a cysteinyl residue. Type-1 coppers have small copper hyperfine couplings and a strong visible absorption in the Cu(II) state. Type-2 or non-blue copper

sites contain copper that is bound mainly to imidazole nitrogens from histidyl residues. Type 3 copper centers are comprised of two spin-coupled copper ions, bound to imidazole nitrogens.

TYROSINE AMINOTRANSFERASE

This pyridoxal-phosphate-dependent enzyme [EC 2.6.1.5], also known as tyrosine transaminase, catalyzes the reaction of L-tyrosine with α -ketoglutarate (or, 2-oxoglutarate) to produce 4-hydroxyphenylpyruvate and L-glutamate. L-Phenylalanine can act as the substrate instead of tyrosine. In some systems, the mitochondrial enzyme may be identical with aspartate aminotransferase.

H. Hayashi, H. Wada, T. Yoshimura, N. Esaki & K. Soda (1990) *Ann. Rev. Biochem.* **59**, 87.

A. E. Braunstein (1973) *The Enzymes*, 3rd ed., **9**, 379.

TYROSINE KINASES, PROTEIN

The following are recent reviews on the molecular and physical properties of this enzyme [EC 2.7.1.112]. This enzyme [EC 2.7.1.112], also known as protein-tyrosine kinase and tyrosyl protein kinase, catalyzes the reaction of ATP with the hydroxyl group of a tyrosyl residue in a protein to produce ADP and the phosphotyrosyl residue.

B. M. Sefton & M.-A. Campbell (1991) *Ann. Rev. Cell Biol.* **7**, 257.

T. Hunter & B. M. Sefton, eds. (1991) *Meth. Enzymol.*, vols. **200** and **201**, Academic Press, San Diego.

J. Larner (1990) *Adv. Enzymol.* **63**, 173.

T. Hunter & J. A. Cooper (1986) *The Enzymes*, 3rd ed., **17**, 191.

M. F. White & C. R. Kahn (1986) *The Enzymes*, 3rd ed., **17**, 247.

T. Hunter & J. A. Cooper (1985) *Ann. Rev. Biochem.* **54**, 897.

D. A. Flockhart & J. D. Corbin (1982) *Crit. Rev. Biochem.* **12**, 133.

U

UBIQUINOL:CYTOCHROME c OXIDOREDUCTASE

This enzyme complex [EC 1.10.2.2], also known as cytochrome *bc*₁ and complex III, catalyzes the reaction of ubiquinol (QH₂) with two ferricytochrome *c* to produce ubiquinone (Q) and two ferrocytochrome *c*. The complex also contains cytochrome *b*-562, cytochrome *b*-566, cytochrome *c*₁, and a two-iron ferredoxin.

- J. V. Schloss & M. S. Hixon (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **2**, 43.
 A. Tzagoloff (1995) *Meth. Enzymol.* **260**, 51.
 U. Schulte & H. Weiss (1995) *Meth. Enzymol.* **260**, 63.
 H.-P. Braun & U. K. Schmitz (1995) *Meth. Enzymol.* **260**, 70.
 H. Schagger, U. Brandt, S. Gencic & G. von Jagow (1995) *Meth. Enzymol.* **260**, 82.
 U. Brandt & B. Trumpower (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 165.
 B. L. Trumpower & R. B. Gennis (1994) *Ann. Rev. Biochem.* **63**, 675.
 Y. Hatefi (1985) *Ann. Rev. Biochem.* **54**, 1015.
 M. Wikström, K. Krab & M. Saraste (1981) *Ann. Rev. Biochem.* **50**, 623.
 G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

UBIQUITIN-PROTEIN LIGASES

These enzymes [EC 6.3.2.19] catalyze the reaction of ATP with ubiquitin and a lysyl residue in a protein to produce a protein containing an *N*-ubiquityllysyl residue, AMP, and pyrophosphate (or, diphosphate). Ubiquitin is coupled to the protein by an isopeptide bond between the C-terminal glycine of ubiquitin and ϵ -amino groups of lysyl residues in the protein. An intermediate in the reaction contains one ubiquitin residue bound as a thiolester to the enzyme, and a residue of ubiquitin adenylate noncovalently bound to the enzyme.

- K. D. Wilkinson (1995) *Ann. Rev. Nutr.* **15**, 161.
 A. Herskko & A. Ciechanover (1992) *Ann. Rev. Biochem.* **61**, 761.

UDENFRIEND'S REAGENT

A mixture of ascorbic acid, oxygen, EDTA, and a ferrous salt that is used in a number of oxidation reactions,

including the hydroxylation of an aromatic carbon^{1,2}. **See also** *Fenton Reaction*

- ¹S. Udenfriend, C. T. Clark, J. Axelrod & B. B. Brodie (1954) *J. Biol. Chem.* **208**, 731.
²B. B. Brodie, P. A. Shore & S. Udenfriend (1954) *J. Biol. Chem.* **208**, 741.

UMBRELLA EFFECT

An effect observed with a number of compounds which have apparent chiral centers on elements other than carbon. For example, secondary and tertiary amines have a pyramidal structure in which the unshared pair of electrons is at the top of the pyramid. If the three substituents linked to the nitrogen are all different, one might suspect that the tertiary amine would give rise to optical activity and be resolvable. However, rapid oscillation of the unshared pair of electrons on one side of the nitrogen to the other (hence, pyramidal inversion) in effect causes interconversion of the two enantiomers and prevents resolution. If the nitrogen is at a bridgehead, this "umbrella effect" is inhibited and optical isomers can be isolated.

Other elements besides nitrogen can exhibit an umbrella effect: *e.g.*, phosphorus, sulfur, and arsenic. Fortunately, the effect is slow in most of these cases. Hence, one can prepare optically pure phosphoesters (such as, ¹⁶O¹⁷O¹⁸OP—OR). Thus, chiral analogs of ATP can be utilized to study enzyme stereochemistry. Similarly, chiral sulfones and sulfoxides have proved useful in assessing the topology of a protein's active site.

- J. B. Lambert (1972) *Topics Stereochem.* **6**, 19.

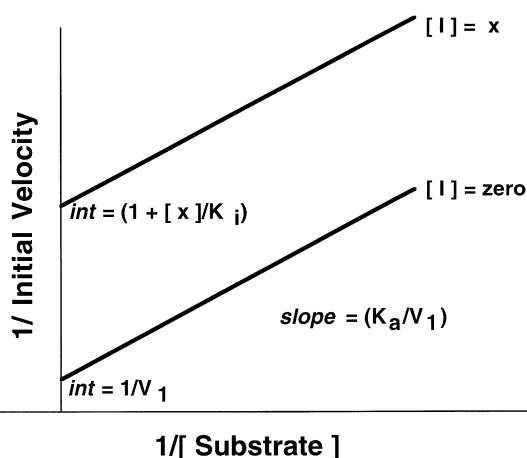
UNCOMPETITIVE INHIBITION

Inhibition of an enzyme-catalyzed reaction in which the inhibitor does not bind to the free, uncomplexed enzyme and does not compete with the substrate for the enzyme's active site^{1,2}. For a Uni Uni mechanism ($E + A \rightleftharpoons EX \rightarrow E + P$), an uncompetitive inhibitor would bind to

the EX binary complex (thus, $\text{EX} + \text{I} \rightleftharpoons \text{EXI}$), with a dissociation constant of K_{ii} (equal to $[\text{EX}][\text{I}]/[\text{EXI}]$). Both the steady-state and rapid equilibrium derivations yield an equation with the same general format:

$$v = \frac{V_{\max}[\text{A}]}{K_m + [\text{A}] \left(1 + \frac{[\text{I}]}{K_{ii}} \right)}$$

However, note that K_m is replaced with K_{ia} (the dissociation constant of A for the free enzyme) in the rapid-equilibrium equation. A standard double-reciprocal plot ($1/v$ vs. $1/[\text{A}]$) at different concentrations of inhibitor will yield a series of parallel lines. A vertical intercept vs. $[\text{I}]$ secondary replot will provide a value for K_{ii} on the horizontal axis. If questions arise as to whether the lines are truly parallel, one possibility is to replot the data *via* a Hanes plot ($[\text{A}]/v$ vs. $[\text{A}]$). In such a plot, the lines of an uncompetitive inhibitor intersect on the vertical axis.



Uncompetitive inhibition can also be a possibility in multisubstrate reactions. For example, in an ordered Bi Bi reaction, a competitive inhibitor with respect to the second substrate B, will act as an uncompetitive inhibitor with respect to the first substrate, A.

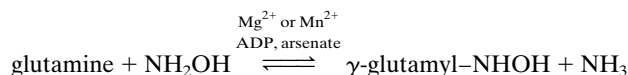
¹J. A. Todhunter (1979) *Meth. Enzymol.* **63**, 383.

²H. J. Fromm (1995) *Meth. Enzymol.* **249**, 123.

UNCONSUMED SUBSTRATE OR COFACTOR

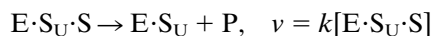
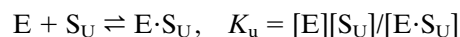
Any reactant that remains unchanged (or is rapidly regenerated) before and after (but not necessarily during) an enzymic reaction. These could include a metal ion, an activator (*e.g.*, calmodulin), a dissociable regulatory

subunit, a free-radical-containing subunit (*e.g.*, the tyrosyl radical-containing subunit of ribonucleotide reductase), or any unconsumed components. Examples of the last case are ADP and arsenate in the γ -glutamyltransferase reaction catalyzed by glutamine synthetase:



In this reaction, arsenate ester analogues of the γ -glutamylphosphate and ADP are likely to be formed in a manner analogous to that of the biosynthetic reaction; however, γ -glutamyl-arsenate and ADP-arsenate undergo rapid hydrolysis in each catalytic round, and the concentrations of ADP and arsenate remain unchanged. Polymeric substrates can also behave as unconsumed substrates. For example, glycogen contains n glucosyl units, and when serving as substrate for an exo-carbohydrase such as glycogen phosphorylase, the corresponding product with $(n - 1)$ glucosyl units is also deemed unaltered when n is large.

Unconsumed substrates are treated as “substrates” or essential activators in deriving rate equations and studying detailed mechanisms. Nonetheless, one must indicate whether an unconsumed substrate (U) remains bound to the enzyme or not (in this case, U also becomes an unaltered product) in the reaction scheme. In practice, unconsumed substrates are likely to be involved in all the typical multisubstrate kinetic mechanisms¹. Only one case is illustrated here, namely that the unconsumed substrate S_U activates catalysis when bound in a rapid-equilibrium ordered mechanism:



The initial rate law for this mechanism is:

$$\frac{1}{v} = \frac{1}{V_{\max}} + \frac{K_s}{V_{\max}[\text{S}]} \left\{ 1 + \frac{K_u}{[\text{S}_U]} \right\}$$

This rate equation is identical to that for a rapid equilibrium ordered addition bisubstrate mechanism (*i.e.*, a scheme where substrate A rapidly binds prior to the addition of the second substrate B). Huang¹ has presented the theoretical basis for mechanisms giving rise to

typical rapid-equilibrium ordered kinetic patterns, along with a helpful discussion on various sequential cases.

¹C. Y. Huang (1990) *Biochemistry* **29**, 158.

UNDERSHOOT

A temporary or transient decrease below a final steady-state value.

UNIMOLECULAR

Elementary reactions (also termed monomolecular reactions) that involve only a single entity in the formation of an activated complex. Unimolecular rate constants, k , are concentration-independent and are typically expressed in units of s^{-1} . Unimolecular reactions are expected to be first order (*i.e.*, $-dc/dt = kc$ where c is the concentration and t is time). Examples of unimolecular processes include radioactive disintegrations, isomerizations, disassociations, and decompositions. Reactions in solution are unimolecular only if the solvent is not covalently incorporated into the product(s).

An entire mechanism can be unimolecular, provided that the rate-limiting step is unimolecular. **See also** *Chemical Kinetics, Molecular; Order of Reaction; Elementary Reactions*

UNINTERESTING ENZYME

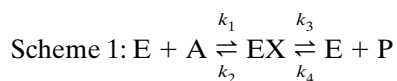
The null set. Just as there can be no uninteresting numbers in mathematics, each enzyme has the power to teach us fundamental *and* interesting facets of chemistry and biology.

UNIORT

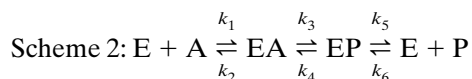
A system that transports one molecule or ion without concomitant transport of another molecule or ion. **See** *Membrane Transport*

UNI UNI ENZYME KINETIC MECHANISM

Many enzymes (especially isomerases, racemases, and epimerases) catalyze intramolecular reactions that can be described by a Uni Uni mechanism. These are typically written with one or two central complexes:



or



Both schemes yield the same generalized steady-state initial rate equation:

$$v = \frac{V_{\max,f} V_{\max,r} ([A] - [P]/K_{eq})}{(V_{\max,r} K_a + V_{\max,r} [A] + (V_{\max,f} [P]/K_{eq}))}$$

In Scheme 1, the rate parameters $V_{\max,f}$ and $V_{\max,r}$ are the maximum velocities in the forward and reverse direction, respectively (such that $V_{\max,f} = k_3 [E_{\text{total}}]$ and $V_{\max,r} = k_2 [E_{\text{total}}]$), K_a is the Michaelis constant for substrate A ($K_a = (k_2 + k_3)/k_1$), and K_{eq} is the equilibrium constant (equal to $k_1 k_3 / k_2 k_4$) and having the Haldane relationships $K_{eq} = V_{\max,f} K_p / (V_{\max,r} K_a) = K_{ip} / K_{ia}$ where K_p is the Michaelis constant for P (equal to $(k_2 + k_3)/k_4$) and K_{ia} and K_{ip} are the dissociation constants for A and P, respectively ($K_{ia} = k_2/k_1$ and $K_{ip} = k_3/k_4$). The determinants for the two enzyme forms are $[E] = k_2 + k_3$ and $[EX] = k_1[A] + k_4[P]$. In the absence of the product P (*i.e.*, in initial rate studies where velocities are observed with less than 10% substrate conversion), the expression above reduces to $v = V_{\max,f} [A] / (K_a + [A])$. For Scheme 2, the kinetic parameters are now: $V_{\max,f} = k_3 k_5 [E_{\text{tot}}] / (k_3 + k_4 + k_5)$; $V_{\max,r} = k_2 k_4 [E_{\text{tot}}] / (k_2 + k_3 + k_4)$; $K_a = (k_2 k_4 + k_2 k_5 + k_3 k_5) / (k_1 (k_3 + k_4 + k_5))$; $K_p = (k_2 k_4 + k_2 k_5 + k_3 k_5) / (k_6 (k_2 + k_3 + k_4))$; $K_{ia} = k_2/k_1$; $K_{ip} = k_5/k_6$; and $K_{eq} = k_1 k_3 k_5 / (k_2 k_4 k_6) = V_{\max,f} K_p / (V_{\max,r} K_a)$. The determinants for the enzyme forms for Scheme 2 are: $[E] = k_2 (k_4 + k_5) + k_3 k_5$; $[EA] = k_1 [A] (k_4 + k_5) + k_4 k_6 [P]$; and $[EP] = k_6 [P] (k_2 + k_3) + k_1 k_3 [A]$. **See also** *Michaelis-Menten Equation; Iso Uni Uni Mechanism; Rapid Equilibrium Assumption*

H. J. Fromm (1975) *Initial Rate Enzyme Kinetics*, Springer-Verlag, New York.

UNREACTIVE

The opposite of reactive; hence, a body or molecular entity fails to react (or reacts more slowly) in some reaction, under specified conditions, as does some standard. The term is not synonymous with stable (however, a stable entity may be more reactive than some reference species).

IUPAC (1979) *Pure and Appl. Chem.* **51**, 1725.

UNSTABLE

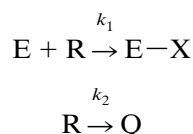
A body or molecular entity is said to be unstable with reference to another substance, if its molar Gibbs free energy is greater than the reference substance. This de-

scriptor is not synonymous with reactive or transient. **See** *Metastable*

UNSTABLE AFFINITY LABELING REAGENT

Some affinity labeling reagents react with water to undergo deactivation, and depending on the kinetics of the reagent deactivation compared to the enzyme's alkylation by the active affinity labeling substance, the kinetics of enzyme inactivation can appear to be anomalous. For example, the inactivation of horse alcohol dehydrogenase by diethylpyrocarbonate (also known as ethoxyformic anhydride) does not proceed by pseudo-first-order kinetics, and this would be indicated by the nonlinearity of a plot of the logarithm of remaining enzyme activity *versus* the period of exposure to diethylpyrocarbonate. Because this alkylating reagent has a half-life of only 20 min at pH 8 at room temperature, one must take its hydrolysis into account in defining the kinetics of its action on alcohol dehydrogenase.

The following scheme describes reagent R reacting with an enzyme as well as the reagent inactivation process:



where E, R, E-X, and Q are active enzyme, active reagent, inactivated enzyme, and inactivated reagent, respectively. The following differential equations describe the kinetics of the above reactions:

$$-dE/dt = k_1[E][R] \quad \text{and} \quad -dR/dt = k_2[R]$$

By integrating the second equation first, we get $[R] = [R]_{\text{initial}}e^{-k_2t}$; thus, we may substitute this expression for $[R]$ into the first differential equation to obtain the rate law that allows for enzyme inactivation by a reagent which itself is undergoing deactivation. After separating variables and integrating the combined expression, we obtain

$$\ln([E]_t/[E]_{\text{initial}}) = (k_1[R]_{\text{initial}}/k_2)[e^{-k_2t} - e^0]$$

or

$$[E]_t/[E]_{\text{initial}} = e - \{(k_1[R]_{\text{initial}}/k_2)(1 - e^{-k_2t})\}$$

The observed time course of an enzyme inactivation process can thus be fitted to this integrated rate law to

obtain the value of k_1 and k_2 , although in practice one can independently obtain a value for the latter in the absence of the enzyme.

UPCONVERSION

A nonlinear optical effect in which the frequency of light has increased.

UREA KINETICS

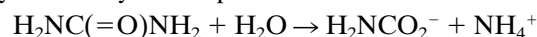
Wolfe¹ has presented an excellent description of the systematic application of stable and radioactive isotope tracers in determining the kinetics of urea production, urea recycling, and interorgan nitrogen transfer in living systems.

¹R. R. Wolfe (1992) *Radioactive and Stable Isotope Tracers in Biomedicine*, pp. 341-355, Wiley-Liss, New York.

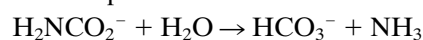
UREASE

Urease [EC 3.5.1.5], a protein containing two tightly bound nickel ions, catalyzes the hydrolysis of urea. Technically, the enzymatic phase proceeds up to the formation of carbamate:

Enzyme-Catalyzed Step:



Nonenzymatic Step:



Carbamate then degrades nonenzymatically under aqueous conditions to yield carbon dioxide and a second molecule of ammonia. **See** *Nickel-Dependent Enzymes*

M. A. Halcrow (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 506.

F. J. Reithel (1971) *The Enzymes*, 3rd ed., **4**, 1.

J. B. Sumner (1955) *Meth. Enzymol.* **2**, 378.

J. B. Sumner (1926) *J. Biol. Chem.* **69**, 435.

UV ENDONUCLEASES

Uvr(A)BC acts on a very broad spectrum of DNA damage, and in the damage recognition and incision steps, an intricate ATP hydrolysis-coupled process is employed rather than direct damage binding-incision utilized by most of the damage-specific repair enzymes.

A. B. Britt (1996) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **47**, 75.

L. Grossman & S. Thiagalingam (1993) *J. Biol. Chem.* **268**, 16871.

B. Van Houten (1990) *Microbiol. Rev.* **54**, 18.

B. Weiss & L. Grossman (1987) *Adv. Enzymol.* **60**, 1.

V

VALENCE-BOND METHOD

A procedure for obtaining an approximate wave function of a molecular orbital by treating the molecule as different canonical forms. The overall wave expression is thus the weighted sum of the wave equation for each of these canonical forms.

VALINE DEHYDROGENASE

This enzyme [EC 1.4.1.8] catalyzes the reaction of valine with NADP⁺ and water to produce 3-methyl-2-oxobutanoate, ammonia, and NADPH.

A. R. Clarke & T. R. Dafforn (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 1.

N. M. W. Brunhuber & J. S. Blanchard (1994) *Crit. Rev. Biochem. Mol. Biol.* **29**, 415.

VANADATE ION

The vanadium oxide VO₄³⁻ [conjugate base of the acid VO₃(OH)²⁻] that adopts trigonal bipyramidal geometry and acts as a transition-state mimic for in-line phosphoryl transfer enzymes. **See** Oxygen, Oxides, Oxo Anions

Vanadate can be prepared as follows: dissolve vanadium(V) oxide in 2.1 molar equivalents of 1 N NaOH; stir 2-3 days, until the yellow color becomes faint; dilute with water to a final concentration of 100 mM.

Vanadate-inhibited enzymes often reactivate on addition of EDTA, which forms high-affinity vanadate:EDTA complexes; if divalent cations must be chelated, use EGTA. Vanadate interacts with many other organic compounds and buffer salts, HEPES being a useful exception. Some commercial preparations of nucleotide triphosphates and carbamoyl-P contain vanadate, and one should ascertain that reagents are vanadate-free before use in kinetic and mechanistic studies. Vanadate also complexes with peroxide anion to produce pervanadate, a reagent that can oxidatively inactivate enzymes containing active-site thiols.

Vanadium in Biological Systems (N. D. Chasteen, ed.) Kluwer Academic Publishers, Netherlands.

A. Butler, M. J. Clague & G. E. Meister (1994) *Chem. Rev.* **94**, 625.
A. Tracey, M. Gresser & S. Liu (1988) *J. Am. Chem. Soc.* **110**, 5869.

VANADIUM-DEPENDENT ENZYMES

Enzymes requiring vanadium for catalytic activity. Perhaps the best studied of these are the vanadium-dependent nitrogenases [EC 1.18.6.1]. Other vanadium-dependent enzymes include vanadium haloperoxidase, vanadium chloroperoxidase, and vanadium bromoperoxidase. In the vanadium chloroperoxidase and bromoperoxidase reactions, the vanadium(V) is coordinated in a trigonal bipyramidal site to a histidyl residue, three nonprotein oxygens, and, presumably, to a hydroxide.

G. J. Leigh (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 349.

A. Butler (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 427.

VAN DER WAALS FORCES

Attractive or repulsive forces between molecular entities or groups within the same molecular entity (*i.e.*, both intermolecular and intramolecular) not due to bond formation or to electrostatic interactions of ions or ionic groups with one another or with neutral molecules. The origin of van der Waals forces is in electric polarization of uncharged atoms, groups, or molecules and includes dipole-dipole interactions, dipole-induced dipole interactions, and London forces (induced dipole-induced dipole interactions).

VAN'T HOFF RELATIONSHIP

Thermodynamic considerations led van't Hoff to conclude¹ that the equilibrium constant, K_{eq} , for a reaction (scheme below) is equal to the ratio of the rate constants



for that reaction, $K_{eq} = k_1/k_{-1}$. He also noted² that the effect of temperature on this equilibrium constant (at

constant pressure) was described by the following equation,

$$d \ln K_{eq}/dT = \Delta U^{\ominus}/RT^2$$

where R is the molar gas constant (*i.e.*, $8.31431 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ or $1.98722 \text{ cal} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$), T is the absolute temperature, and ΔU is the change in internal energy. The superscript symbol $\ominus$ (or $\circ$) denotes standard conditions of temperature, pressure, and ionic strength of an ideal solution at 1 M^{3-5} . In most biological systems, processes typically occur at constant pressure and usually involve only small changes in volume. Recall that the change in enthalpy, ΔH , is equal to the heat absorbed at constant pressure, q_p , when no work other than $P\Delta V$ work is being done: thus, $\Delta H = \Delta U + P\Delta V$ where P is the pressure and ΔV is the change in volume. For reactions in solutions, there usually is very little $P\Delta V$ changes when the pressure is low (*e.g.*, 1 atmosphere). Hence, $\Delta H^{\ominus}$ and $\Delta U^{\ominus}$ are very nearly equal. At substantially higher pressures, such as those found deep in the ocean, the $P\Delta V$ change can be a contributing factor. Substituting $\Delta H^{\ominus}$ into first equation and integrating yields $\ln K_{eq} = -(\Delta H^{\ominus}/RT) + \text{constant}$. If the thermodynamic parameters are to be measured under standard conditions, then $\Delta H^{\ominus}$ can be determined from a plot of $\ln K_{eq}$ as a function of T^{-1} (a van't Hoff plot), the slope of the line being equal to $-\Delta H^{\ominus}/R$. The vertical intercept will be equal to $\Delta S^{\ominus}/R$ (since $-\ln K_{eq} = \Delta G^{\ominus}/RT = (\Delta H^{\ominus}/RT) - \Delta S^{\ominus}/R$). **See** *Collision Theory; Arrhenius Law; Transition-State Theory; Temperature Dependence*

¹J. H. van't Hoff (1877) *Chem. Ber* **10**, 669.

²J. H. van't Hoff (English translation, 1896) *Studies in Chemical Dynamics*, Chem. Publ. Co., Easton, Penn.

³I. Mills (1988) *Quantities, Units and Symbols in Physical Chemistry*, Blackwell Sci. Publ., Oxford.

⁴IUPAC-IUPAB-IUB, Intercommission on Biothermodynamics (1985) *Eur. J. Biochem.* **153**, 429.

⁵Interunion Commission on Biothermodynamics (1976) *J. Biol. Chem.* **251**, 6879.

VAN'T HOFF'S LAWS

1. The osmotic pressure, at constant temperature, of a dilute solution is proportional to the concentrations of the dissolved substances; *i.e.*, $\Pi = RT \sum c_i$ where Π is the osmotic pressure, R is the molar gas constant, T is the absolute temperature and c_i is the molar concentration of species i . 2. In stereochemistry, all optically active molecules contain one or more multivalent atoms united

with other atoms such that there is an unsymmetrical arrangement in space around those multivalent atoms.

VARIANCE

A statistical measure of the dispersion of the distribution of the random variable, typically obtained by taking the expected value of the square of the difference between the random variable and its mean. The variance is the square of the standard deviation. **See** *Statistics (A Primer)*

VARIANCE RATIO

The ratio of two independent estimates of the same variance from a Gaussian distribution based on sample sites $(n + 1)$ and $(m + 1)$, respectively. This ratio, often symbolized by F , can be used to test a null hypothesis. **See** *Statistics (A Primer)*

VELOCITY

A term usually synonymous to rate of reaction in the field of enzyme kinetics. Rate is the preferred terminology. **See** *Rate of Reaction; Rate of Conversion*

VELOCITY COEFFICIENT

The rate of transformation of a unit mass of substance in a reaction.

VELOCITY, MACROSCOPIC

The macroscopic velocity is the sum of all microscopic velocities of an enzyme system which acts on a polymer at several different positions, thus yielding a number of different products from the same substrate. One good example is polysaccharide depolymerase¹.

¹J. D. Allen (1980) *Meth. Enzymol.* **64**, 248.

VENOM EXONUCLEASE

This enzyme [EC 3.1.15.1], also called venom phosphodiesterase, catalyzes the exonucleolytic cleavage of polynucleic acids (with a preference for a single-stranded substrate) in the 3' to 5' direction to yield 5'-phosphomononucleotides.

M. Laskowski, Sr. (1971) *The Enzymes*, 3rd ed., **4**, 313.

VESICLE TRANSPORT IN CELLS

The kinetic control of vesicle transport, from vesicle budding in the Golgi (and other origins) to fusion of the vesicle with the target membrane, is currently a vigorous area of research. The formation of COP-coated vesicles

(vesicles carrying newly synthesized proteins from Golgi cisternae) requires binding of a G-protein referred to as ARF (ADP-ribosylation factor). The ARF-GDP complex first binds to budding vesicle, followed by a GDP exchange. This event appears to trigger the binding of coatmer which causes the bud to “pinch” off from the Golgi. Hence, coatmer binding appears to drive budding. When GTP is hydrolyzed, ARF and coatmer are released and the vesicle is ready to fuse with the target membrane. Thus, the triggering of GTP hydrolysis allows the vesicle to fuse with its target. Just how all the steps are controlled and directed is a significant aspect of the study on this topic.

J. Ostermann, L. Orci, K. Tani, M. Amherdt, M. Ravazzola, Z. Elazar & J. E. Rothman (1993) *Cell* **75**, 1015.
J. E. Rothman (1994) *Nature* **372**, 55.
J. E. Rothman (1996) *Sci. Amer.* **274**, 70.

VIBRATIONAL COUPLING

Interaction between electronic and vibrational motions in a molecular entity.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

VIBRATIONAL REDISTRIBUTION

An intramolecular redistribution of energy among all the vibrational modes. For large molecules, this redistribution does not require collisions.

VIBRATIONAL RELAXATION

The loss of vibrational excitation energy by a molecular entity through energy transfer to the environment due to collisions.

Comm. on Photochem. (1988) *Pure and Appl. Chem.* **60**, 1055.

VIBRATIONAL SPECTROSCOPY

Any spectroscopic technique that directly provides information about the vibrational behavior of chemical substances.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Biomolecular vibrational spectroscopy, **246**, 377; Raman spectroscopy of DNA and proteins, **246**, 389; resonance Raman spectroscopy of metalloproteins, **246**, 416; structure and dynamics of transient species using time-resolved resonance Raman spectroscopy, **246**, 460; infrared spectroscopy applied to biochemical and biological problems, **246**, 501; resonance Raman spectroscopy of quinoproteins, **258**, 132.

VIBRONIC TRANSITION

A transition which is purely electronic or purely vibrational. The transition involves a change in both vibra-

tional and electronic quantum numbers and a change in both vibrational and electronic energy.

VICINAL

A term used to designate the geometric arrangement of moieties attached to two adjacent atoms within a molecule. An example of a vicinal diol would be ethylene glycol ($\text{HOCH}_2\text{CH}_2\text{OH}$) and a vicinal dihalide would be 1-bromo-2-chloroethane.

VINYLALETYL-CoA ISOMERASE

This enzyme [EC 5.3.3.3] catalyzes the interconversion of vinylacetyl-CoA and crotonoyl-CoA. 3-Methylvinylacetyl-CoA can also be utilized as a substrate.

D. J. Creighton & N. S. R. K. Murthy (1990) *The Enzymes*, 3rd ed. **19**, 323.

VINYL CATION (or Vinylic Cation)

A very short-lived carbocation of the form $\text{R}_2\text{C}=\text{C}^+\text{R}$ that has been proposed as an intermediate in a number of chemical mechanisms.

VIRTUAL TRANSITION STATE

An approach and construct^{1,2} used to understand isotope effects. The isotope effect observed in an enzyme-catalyzed reaction is a weighted average of several steps in the reaction. The transition state that one constructs from these studies is also a “weighted” average of several transition states: thus, the virtual transition state.

¹R. L. Schowen (1978) in *Transition States of Biochemical Processes* (R. D. Gandour & R. L. Schowen, eds.), p. 77, Plenum, New York.

²K. B. Schowen & R. L. Schowen (1982) *Meth. Enzymol.* **87**, 551.

VISCOSITY

A measure of the frictional resistance that a fluid in motion offers to an applied shearing force. Let F be the frictional force, and let S be the area of the interface between a stationary surface plane and a fluid with velocity dv/dr . Then the resisting or frictional force is given as:

$$F = \eta S dv/dr$$

where the constant η is termed the coefficient of viscosity. The SI units are $\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1}$, with the cgs unit (called poise) equal to one-tenth of the SI unit. The following are variously defined terms based on the measured solvent viscosity η_o and the solution viscosity η , where c is the concentration in g/dL: (1) Relative Viscosity = η/η_o ; (2) Specific Viscosity = $(\eta - \eta_o)/\eta_o$; (3) Reduced Viscosity = $(\eta - \eta_o)/\eta_o c$.

ity = $c^{-1}(\eta - \eta_0)/\eta_0$; and (4) Intrinsic Viscosity = $\lim_{c \rightarrow 0} [c^{-1}(\eta - \eta_0)/\eta_0]$.

VISCOSITY COEFFICIENT

A parameter equal to the value of the force per unit area required to maintain a unit relative velocity between two parallel planes a unit distance apart.

VITAMIN A GROUP

Metabolic derivatives of the fat-soluble vitamin retinol, which was first described in 1913. These metabolites play a central role in the visual process where the primary photoevent is attended by conversion of 11-*cis*-retinal to all-*trans*-retinal. The oxidized metabolite retinoic acid is an agonist in cell growth and proliferation.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Colorimetric estimation of vitamin A with trichloroacetic acid, **67**, 189; indirect spectrophotometry on vitamin A products: peak signal readout, **67**, 195; colorimetric analysis of vitamin A and carotene, **67**, 199; analysis of geometrically isomeric vitamin A compounds, **67**, 203; hplc of vitamin A metabolites and analogs, **67**, 220; gas chromatography, gas chromatography-mass spectrometry, and high-pressure liquid chromatography of carotenoids and retinoids, **67**, 233; separation of carotenoids on lipophilic Sephadex, **67**, 261; carotenoid biosynthesis by cultures and cell-free preparations of flavobacterium R1560, **67**, 264; synthesis and properties of glycosyl retinyl phosphates, **67**, 270; fluorescence assay of retinol-binding holoprotein, **67**, 282; purification of cellular retinol and retinoic acid-binding proteins from rat tissue, **67**, 288; cellular retinol-, retinal-, and retinoic acid-binding proteins from bovine retina, **67**, 296; isolation and purification of bovine rhodopsin, **67**, 301; quantitative analysis of retinal and 3-dehydroretinal by high-pressure liquid chromatography, **123**, 53; enzymatic synthesis and separation of retinyl phosphate mannose and dolichyl phosphate mannose by anion-exchange high-performance liquid chromatography, **123**, 61; separation and quantitation of retinyl esters and retinol by high-performance liquid chromatography, **123**, 68; separation of geometric isomers of retinol and retinoic acid in nonaqueous high-performance liquid chromatography, **123**, 75; use of enzyme-linked immunosorbent assay technique for quantitation of serum retinol-binding protein, **123**, 85; Fourier transform infrared spectroscopy of retinoids, **123**, 92; purification of interstitial retinol-binding protein from the eye, **123**, 102; quantification of physiological levels of retinoic acid, **123**, 112.

VITAMIN D GROUP

The vitamin and metabolic derivatives of a fat-soluble substance calciferol. These antirachitic vitamins are produced from $\Delta^{5,7}$ -unsaturated sterols upon irradiation with ultraviolet light. Population biologists have adduced strong evidence for the hypothesis that lighter skin color, the primary characteristic of race, is related to the lower solar radiance in northern climes. Vitamin D plays a central role in calcium metabolism, both in terms of

absorption of this metal ion and its subsequent role in bone mineralization.

Selected entries from *Methods in Enzymology* [vol, page(s)]:

Colorimetric determination of vitamin D₂ (calciferol), **67**, 323; analysis of vitamin D₂ isomers, **67**, 326; gas-liquid chromatography of vitamin D and analogs, **67**, 335; gas-liquid chromatographic determination of vitamin D in multiple vitamin tablets and their raw materials, **67**, 343; gas-liquid chromatographic determination of vitamin D, **67**, 347; thin layer-gas chromatographic determination of 25-hydroxycholecalciferol, **67**, 355; high-performance liquid chromatographic determination of vitamin D and metabolites, **67**, 357; HPLC of vitamin D metabolites and analogs, **67**, 370; mass fragmentographic assay of 25-hydroxyvitamin D₃, **67**, 385; determination of vitamin D and its metabolites in plasma, **67**, 393; assay of 24R,25-dihydroxycholecalciferol in human serum, **67**, 414; enzymic preparation of [³H]1,25-dihydroxyvitamin D₃, **67**, 424; vitamin D metabolites: extraction from tissue and partial purification prior to chromatography, **67**, 426; purification and properties of vitamin D hydroxylases, **67**, 430; 25-hydroxylation of vitamin D₃ in liver microsomes and their smooth and rough subfractions, **67**, 441; measurement of the chicken kidney 25-hydroxyvitamin D₃ 1-hydroxylase and 25-hydroxyvitamin D₃ 24-hydroxylase, **67**, 445; purification, characterization, and quantitation of the human serum binding protein for vitamin D and its metabolites, **67**, 449; competitive binding assay for 25-hydroxyvitamin D using specific binding proteins, **67**, 459; competitive protein binding assay for plasma 25-hydroxycholecalciferol, **67**, 466; steroid competition assay for determination of 25-hydroxyvitamin D and 24,25-dihydroxyvitamin D, **67**, 473; features of the receptor proteins for the vitamin D metabolites, **67**, 479; measurement of kinetic rate constants for the binding of 1 α ,25-dihydroxyvitamin D₃ to its chick intestinal mucosa receptor using a hydroxyapatite batch assay, **67**, 488; structural aspects of the binding of 1 α ,25-dihydroxyvitamin D₃ to its receptor system in chick intestine, **67**, 494; radioimmunoassay for chick intestinal calcium-binding protein, **67**, 500; purification of chick intestinal calcium-binding protein, **67**, 504; characteristics and purification of the intestinal receptor for 1,25-dihydroxyvitamin D, **67**, 508; a sensitive radioreceptor assay for 1 α ,25-dihydroxyvitamin D in biological fluids, **67**, 522; use of chick kidney to enzymatically generate radiolabeled 1,25-dihydroxyvitamin D and other vitamin D metabolites, **67**, 529; isolation and identification of vitamin D metabolites, **123**, 127; a new pathway of 25-hydroxyvitamin D₃ metabolism, **123**, 141; enzyme-linked immunoabsorption assay for vitamin D-induced calcium-binding protein, **123**, 154; measurement of 25-hydroxyvitamin D 1 α -hydroxylase activity in mammalian kidney, **123**, 159; quantitation of vitamin D₂, vitamin D₃, 25-hydroxyvitamin D₂, and 25-hydroxyvitamin D₃ in human milk, **123**, 167; 1,25-dihydroxyvitamin D microassay employing radioreceptor techniques, **123**, 176; assay for 1,25-dihydroxyvitamin D using rabbit intestinal cytosol-binding protein, **123**, 185; cytochrome assay for 1,25-dihydroxyvitamin D, **123**, 190; monoclonal antibodies as probes in the characterization of 1,25-dihydroxyvitamin D₃ receptors, **123**, 199.

VITAMIN D 25-HYDROXYLASE

The following are recent reviews on the molecular and physical properties of this liver enzyme which converts cholecalciferol (vitamin D₃) to 25-hydroxycholecalciferol.

H. F. DeLuca & C. Zierold (1998) *Nutr. Rev.* **56**, 54.

H. L. Henry & A. W. Norman (1984) *Ann. Rev. Nutr.* **4**, 493.

H. F. DeLuca & H. K. Schnoes (1983) *Ann. Rev. Biochem.* **52**, 411.
H. F. DeLuca & H. K. Schnoes (1976) *Ann. Rev. Biochem.* **45**, 631.

VITAMIN K-DEPENDENT γ -GLUTAMYL CARBOXYLASE

This enzyme catalyzes the vitamin K-dependent post-translational carboxylation of specific glutamic acid residues in prothrombin and other blood coagulation proteins to form 4-glutamylcarboxylate side chains.

P. Dowd, S.-W. Han, S. Naganathan & R. Hershline (1995) *Ann. Rev. Nutr.* **15**, 419.
J. W. Suttie (1993) *FASEB J.* **7**, 445.
K. J. Kotkow, D. A. Roth, T. J. Porter, B. C. Furie & B. Furie (1993) *Meth. Enzymol.* **222**, 435.
E. Cadenas, P. Hochstein & L. Ernster (1992) *Adv. Enzymol.* **65**, 97.
C. Vermeer (1990) *Biochem. J.* **266**, 625.
J. W. Suttie (1980) *Crit. Rev. Biochem.* **8**, 191.
J. Stenflo (1978) *Adv. Enzymol.* **46**, 1.

VITAMIN K EPOXIDASE

The following is a recent review on the molecular and physical properties of this enzyme.

R. E. Olson (1984) *Ann. Rev. Nutr.* **4**, 281.

VITAMIN K-2,3-EPOXIDE REDUCTASE

Vitamin-K-epoxide reductase (warfarin-sensitive) [EC 1.1.4.1] catalyzes the reaction of 2-methyl-3-phytyl-1,4-naphthoquinone with oxidized dithiothreitol and water to produce 2,3-epoxy-2,3-dihydro-2-methyl-3-phytyl-1,4-naphthoquinone and 1,4-dithiothreitol. In the reverse reaction, vitamin K 2,3-epoxide is reduced to vitamin K and possibly to vitamin K hydroquinone by 1,4-dithioerythritol (which is oxidized to the disulfide). Some other dithiols and butane-4-thiol can also act as substrates. This enzyme is strongly inhibited by warfarin.

Vitamin-K-epoxide reductase (warfarin-insensitive) [EC 1.1.4.2] catalyzes the reaction of 3-hydroxy-2-methyl-3-phytyl-2,3-dihydronaphthoquinone with oxidized dithiothreitol and water to produce 2,3-epoxy-2,3-dihydro-2-methyl-3-phytyl-1,4-naphthoquinone and 1,4-dithiothreitol. In the reverse reaction, vitamin K 2,3-epoxide is reduced to 3-hydroxy- (and 2-hydroxy-) vitamin K by 1,4-dithioerythritol (which is oxidized to the disulfide). The enzyme is not inhibited by warfarin.

R. E. Olson (1984) *Ann. Rev. Nutr.* **4**, 281.

VITAMIN K GROUP

This group of fat-soluble naphthoquinones contain isoprenoid side-chains of varying length and are important

factors in coagulation (hence, K for the German term Koagulation). Vitamin K plays a central role as a cofactor in the synthesis of γ -carboxyglutamic acid (Gla), and deficiency in this vitamin leads to blood-clotting disorders characterized by a much greater tendency for hemorrhage. Dicoumarol is a potent antagonist of vitamin K, and this agent is used clinically to prevent blood clotting.

Selected entries from *Methods in Enzymology* [vol., page(s)]:

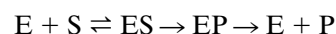
Spectrophotometric determination of the K vitamins, **67**, 125; spectroscopic determination of vitamin K after reduction, **67**, 128; polarographic determination of vitamin K₂ in aqueous solution, **67**, 134; photochemical-fluorometric determination of the K vitamins, **67**, 140; determination of vitamin K₁ in photodegradation products by gas-liquid chromatography, **67**, 148; assay procedure for the vitamin K₁ 2,3-epoxide-reducing system, **67**, 160; vitamin K-dependent carboxylase, **67**, 165; microsomal vitamin K-dependent carboxylase, **67**, 180; assay of coumarin antagonists of vitamin K in blood by high-performance liquid chromatography, **123**, 223; assay of K vitamins in tissues by high-performance liquid chromatography with special reference to ultraviolet detection, **123**, 235; analysis of bacterial menaquinone mixtures by reverse-phase HPLC, **123**, 251.

$V_{\max}$

Symbol for maximal velocity of an enzyme-catalyzed reaction, usually expressed as the molarity change in product per unit time (usually, second). $V_{\max}$ must not be confused with k_{cat} or specific activity; the former has dimensions of time^{-1} , and the latter is usually expressed as micromol product per unit time per milligram of protein. **See** *Michaelis-Menten Equation; Enzyme Rate Equations (1. The Basics)*

$V_{\max}/K_m$ (As Measure of Substrate Capture and Product Release)

Although $V_{\max}/K_m$ is traditionally treated as a first-order rate constant for enzyme reactions at low substrate concentration, Northrop¹ recently pointed out that $V_{\max}/K_m$ actually provides a measure of the rate of *capture* of substrate by free enzyme into a productive complex or the complexes *destined* to go on to form products and complete a turnover at some later time. His analysis serves to underscore the concepts (a) that any catalytic cycle must be characterized by the efficiency of reactant capture and product release, and (b) the Michaelis constant takes on meaning beyond that typically associated with affinity for substrate. Consider the case of an enzyme and substrate operating by the following sequence of reactions:



where k_1 is the rate constant for ES formation, k_2 is the rate constant for ES dissociation to E and S, k_3 is the constant for the conversion from ES to EP, and k_4 is the rate constant for EP breakdown. For the case where ES formation is faster than EP breakdown (*i.e.*, the product is “sticky” and its release is slower than bond-breaking/making), the maximal velocity $V_{\max}$ and $V_{\max}/K_m$ are:

$$V_{\max} = [\text{ES}]/(1/k_3 + 1/k_4) k_4[\text{E}_{\text{total}}]$$

$$V_{\max}/K_m = k_1 k_3 [\text{E}]/(k_2 + k_3)$$

If the substrate is also “sticky” (*i.e.*, $k_2 \ll k_3$), we arrive at the diffusion-controlled parameter $V_{\max}/K_m \approx k_1 [\text{E}_{\text{total}}]$. Northrop¹ defines $V_{\max}/K_m$ as the apparent rate constant for the *capture* of substrate into enzyme complexes that are *destined* to yield product(s) at some later time. Moreover, $V_{\max}$ is then defined as the apparent rate constant for the *release* of substrate from *captured complexes* in the form of free product(s). If one divides $V_{\max}$ by $V_{\max}/K_m$, one obtains K_m :

$$K_m = V_{\max}/(V_{\max}/K_m) = k_5/k_1 = k_{\text{release}}/k_{\text{capture}}$$

and K_m is the ratio of the rate constants for release and capture. As pointed out by Northrop, this approach allows one to liken $1/K_m$ to the substrate affinity, even though K_m is not purely a dissociation constant. **See Specificity Constant**

¹D. B. Northrop (1999) *Adv. Enzymol.* **72**, 12.

$V_{\max}$ -TYPE (or, V-TYPE) ALLOSTERIC SYSTEM

Most known allosteric enzymes rely on modulation of affinity, and are hence designated as *K*-type to signify changes in ligand binding affinity as reflected in changes in their apparent K_m values for substrates. Nonetheless, Monod *et al.*¹ anticipated that other enzymes would modulate their activity through changes in catalytic efficiency, and they used the descriptor “ $V_{\max}$ -type or *V*-type” to describe this less frequently observed form of allosteric behavior.

One excellent example^{1,3} of a $V_{\max}$ -type allosteric enzyme is *Escherichia coli* phosphoglycerate dehydrogenase (PGDH), a tetramer of identical subunits that catalyzes the formation of D-3-phosphohydroxypyruvate from D-3-phosphoglycerate in a reaction that uses NAD⁺ as a redox cofactor. This regulatory enzyme is allosterically controlled by serine. All available information suggests that the effects on the K_m for substrate are minor

and that enzyme activity depends on serine-induced changes in $V_{\max}$. Inhibition studies with numerous amino acids and L-serine analogues indicate that all three functional groups of serine are required for optimal interaction. Kinetic studies indicate there are a minimum of two serine-binding sites, although the crystal structure of PGDH with bound serine as well as direct serine-binding studies revealed four serine-binding sites. The serine-binding sites reside at the interface between regulatory domains of adjacent subunits. Two serine molecules bind at each of the two regulatory domain interfaces in the enzyme. When all four serines are bound, one observes nearly total inhibition; however, with one bound serine at each interface 85% inhibition is observed. Tethering of the regulatory domains by multiple H-bonds from serine to each subunit appears to prevent structural reorientation that is required in the catalytic cycle. These investigators inferred that part of the conformational change may involve a hinge (formed where two antiparallel beta-sheets are joined within the regulatory domain). Serine binding may prevent the enzyme from closing the cleft between the substrate-binding domain and the nucleotide-binding domain. If this conformational change is necessary for catalysis, then a plausible model emerges for $V_{\max}$ -type allosteric control based on serine-regulated movement of rigid domains about flexible hinges.

¹J. Monod, J. Wyman & J.-P. Changeux (1965) *J. Biol.* **12**, 88.

²G. A. Grant, D. J. Schuller & L. J. Banaszak (1996) *Protein Sci.* **5**, 34.

³D. J. Schuller, G. A. Grant & L. J. Banaszak (1995) *Nature Struct. Biol.* **2**, 69.

$V_{n,\max}$

The initial velocity of a cooperative enzyme for which the Hill coefficient is maximal.

VOLATILE BUFFERS

Chemical species which can be used as buffers and, due to their volatile nature (*i.e.*, relatively low boiling points), can be readily removed from a system (*e.g.*, by evaporation). Such buffers have proved to be quite useful in procedures which later require ion-exchange chromatography or electrophoresis (particularly high voltage electrophoresis). Examples of some common components of volatile buffer systems (with their corresponding boiling points and pK_a values [at 25°C]) would be formic acid (100.5°C; 3.75), acetic acid (118°C; 4.76), pyridine (115.5°C; 5.23), triethanolamine (335.4°C; 7.76), ammo-

nia (-33.35°C ; 9.25), ethanolamine (170.58°C ; 9.50), and trimethylamine (2.9°C ; 9.80).

A few common volatile buffer mixtures, along with their respective pH range are: pyridine-formic acid (2.3–3.5); trimethylamine-formic acid (or acetic acid) (3.0–6.0); triethanolamine-HCl (6.8–8.8); ammonia-formic acid (or acetic acid) (7.0–10.0). **See** *Buffers*

D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

VOLT

The SI unit for electric potential and for electromotive force (symbolized by V) equivalent to one joule per coulomb. It is the difference in electric potential needed for a one ampere current to flow through a resistance of one ohm.

VOLTAGE CLAMP

A circuit that uses a differential amplifier to maintain constant membrane potential by electronically balancing the ion channel current. This method allows the experimenter to analyze action potentials of excitable membranes resulting from an initial transient rise in sodium ion permeability followed by a transient rise in potassium ion permeability¹. The technique is especially valuable for studying kinetic properties of voltage-gated channels as well as voltage-dependent channels. **See** *Membrane Potential; Patch Clamp Methods*

¹A. L. Hodgkin & A. F. Huxley (1952) *J. Physiol.* **116**, 424; 472; and 497.

VOLUME OF ACTIVATION

A quantity (symbolized by $\Delta V^{\ddagger}$ or $\Delta^{\ddagger}V^{\ominus}$) derived from the pressure dependence of a reaction rate constant: $\Delta V^{\ddagger} = -RT(\partial \ln k / \partial P)_T$ where R is the molar gas constant, T is the absolute temperature, k is the reaction rate constant, and P is pressure¹. For this equation, the rate constants of all non-first-order reactions are expressed in pressure-independent units (e.g., molarity) at a fixed temperature and pressure.

In transition-state theory, this parameter represents the difference between the partial molar volume of the transition state ($V^{\ddagger}$) and the sums of the partial molar volumes of the reactants at the same temperature and pressure. Thus, $\Delta V^{\ddagger} = V^{\ddagger} - \sum(rV_R)$ where r is the order of

the reactant R and V_R is the partial molar volume. The net volume of activation for a bimolecular reaction is expected to be negative, whereas the volume of activation for a unimolecular reaction is expected to be zero or very small.

Moore² offers a concrete example in a problem asking the student to formulate an equation based on transition state theory to calculate the volume of activation per mole of a reaction that experiences a doubling in the bimolecular rate constant by an increasing pressure from 1 to 3000 atm at 298 K. Transition state theory requires that the activated complex remain in equilibrium with its reactants ($A + B = X^{\ddagger}$) where $\Delta G^{\ddagger} = -RT \ln K^{\ddagger}$. Recalling that $dG = VdP - SdT$, we can obtain the partial differential equation: $(\partial G^{\ddagger} / \partial P)_T = \Delta V^{\ddagger}$. Therefore, $(\partial \ln K^{\ddagger} / \partial P)_T = \Delta V^{\ddagger} / RT$. Again recalling that the reaction rate constant, $k_{\text{reaction}} = (k_B T / h) K^{\ddagger}$, we can obtain the expression $\ln(k_{\text{rxn}}(3000 \text{ atm}) / k_{\text{rxn}}(1 \text{ atm})) = (P_2 - P_1) \Delta V^{\ddagger} / RT$. Thus, $\ln 2 = (3000 - 1) \Delta V^{\ddagger} / (82.05 \times 298) = 5.7 \text{ cm}^3$. This exercise indicates that reaction rate is relatively insensitive to pressure changes if $\Delta V^{\ddagger}$ is small. **See** *Transition-State Theory Expressed in Thermodynamic Terms; Gibbs Free Energy of Activation; Enthalpy of Activation; Entropy of Activation*

¹IUPAC (1979) *Pure Appl. Chem.* **51**, 1725.

²W. J. Moore (1972) *Physical Chemistry*, 4th ed., p. 418, Wiley, New York.

VOLUME OF DISTRIBUTION

The volume or space over which a tracee is distributed within the body or structure. **See** *Compartmental Analysis*

VOLUME STRAIN

A dimensionless quantity for the change in volume, V , per unit volume. A synonymous term is bulk strain. An example of volume strain can be seen in bodies experiencing hydrostatic pressure. Volume strain is usually symbolized by θ : thus, $\theta = \Delta V / V_0$.

V SYSTEMS

Cooperative enzyme systems in which the presence of an allosteric effector results in an alteration of the V_{max} value of the system (as opposed to changes in the K_m value(s): K systems). **See** V_{max} -Type (or, V-Type) Allosteric System; Allostereism

W

WALDEN INVERSION

The inversion of stereochemical configuration at a chiral center in a nucleophilic substitution reaction¹⁻⁴. The uncatalyzed reaction is bimolecular: *i.e.*, it is an S_N2 reaction (in fact the Walden inversion was noted before the criteria for S_N2 reactions were presented). **See** S_N1 and S_N2 Reactions; Nucleophilic Substitution Reactions

¹P. Walden (1895) *Ber.* **28**, 1287 and 2766.

²C. A. Bunton (1966) in *Studies of Chemical Structure and Reactivity* (J. H. Ridd, ed.), pp. 73-102, McGraw-Hill, New York.

³H. A. Bent (1968) *Chem. Rev.* **68**, 587.

⁴D. P. G. Harmon (1970) *J. Chem. Ed.* **47**, 398.

WALL EFFECT

Any detectable effect on the reaction or behavior of a particular system by the interior wall of the container or reaction vessel. Because proteins can form high-affinity complexes with glass and plastic surfaces, one must exercise caution in the choice of reaction kinetic conditions. Wall effects can be discerned if one determines catalytic activity under different conditions that minimize or maximize contact of the solution with the container. In principle, an enzyme-catalyzed reaction should proceed at the same rate if placed in a capillary or a culture tube; however, contact with the wall is maximized in a capillary, and wall effects should be more prominent. Some investigators add bovine serum albumin to prevent adsorption of their enzyme onto the container's walls.

WASHING LABWARE

Biochemical experiments require glassware that is scrupulously free of contaminants that can alter the structure or activities of proteins and other macromolecules. Thus, washing must be more rigorous than that used in a typical organic or inorganic laboratory. Simple visual inspection can often detect if any residual detergent remains; evi-

dence of any remaining soap films can be readily detected with a pair of polarized sunglasses.

After cleaning with a phosphate-free detergent, labware should be rinsed extensively with deionized water and finally rinsed at least twice with doubly distilled water before drying. Avoid using paper towels, because lint and microscopic paper fibers often contain metal ions as well as acids/bases used in commercial treatment of paper products. Paper fibers also scatter light and interfere with spectrophotometric assays.

Once cleaned, glassware (and plasticware) should be covered or kept inverted. Even dust can be a troublesome contaminant.

Even scrupulously clean glass can become contaminated by fingerprints which often contain metal ions, amino acids, and oils. Heavy metal ions are present in the ink of permanent markers and can diffuse through the wall of certain plastic containers.

WASH-OUT

The loss of an isotopic label from one or more sites of a labeled compound in the absence or presence of an enzyme. Nonenzymatic tritium and deuterium wash-out occurs whenever a carbanion or ylide is sufficiently stable so as to promote proton release. Enzymatic tritium and deuterium wash-out also occurs when a carbanion is formed and the protonated base can undergo exchange with protons from the solution. The carbon-8 atom of many purines forms an ylide, and tritium or deuterium bound at this position can undergo facile exchange. For this reason, many tritium labeled nucleotides and nucleosides should be purified immediately prior to use. Another good example is proline racemase, which is thought to proceed by two-base catalysis through a transition state that has two thiols sandwiching a proline carbanion.

WATER

No single treatise can provide a sufficiently thorough account of the properties of water! Yet, the kinetics and thermodynamics of every biochemical process are linked to the molecular interactions of water with macromolecules, membranes, metabolites, anions, cations, protons and even electrons. For this reason, this handbook provides a brief overview of the structure and general properties of this most fascinating of all solvents. Where deemed appropriate, references are provided for further reading by those motivated to examine these topics at greater depth.

MOLECULAR STRUCTURE. The bond lengths and bond angle of water in its three forms (*e.g.*, H₂O, HDO, and D₂O) are well known, and Benedict *et al.*¹ gives the following bond lengths and angles: H₂O, 0.9572 Å and 104.52°; HDO, 0.9571 Å and 104.53°; and D₂O, 0.9575 Å and 104.47°. These are idealized values, and the bond lengths and angles are in constant flux. As pointed out by Eisenberg and Kauzmann², the near equivalence of bond lengths and angles for H₂O, HDO, and D₂O is consistent with the Born-Oppenheimer approximation that nucleus does not influence a molecule's electronic structure. (This fact is important in studies on kinetic isotope effects and solvent isotope effects.) In addition to its zero-point vibration, water molecules exhibit three vibrational modes: symmetric stretching along both O—H bonds, asymmetric stretching along only one O—H, and bending. The dipole moment of water is 1.83 debye.

HYDROGEN BONDING. The stability of hydrogen bonds ($\Delta G_{\text{formation}} \approx -5$ kcal/mol or -21 kJ/mol) can be compared to the weaker van der Waals interactions ($\Delta G_{\text{formation}} \approx -0.3$ kcal/mol or -1.26 kJ/mol) or covalent bonds ($\Delta G_{\text{formation}} \approx -100$ kcal/mol or -420 kJ/mol). Because hydrogen bonding is so extensive in the case of liquid water³, it is useful to consider some of the general features of hydrogen bonds. Three factors appear to be important in determining the strength of hydrogen bonds. First, the O—H bond's polarization determines (a) the ease with which a hydrogen atom can be shared by its neighboring electronegative oxygen atoms, and (b) the net electrical charge on the oxygen atoms that share the hydrogen. Of course, acids form stronger hydrogen bonds than water and alcohols be-

cause their O—H bonds are strongly polarized with greater net negative charge on the oxygen and more net positive charge on the hydrogen. This explains the propensity of acetic acid to form doubly hydrogen-bonded dimers in solid, liquid, and even vapor states. Second, the strongest hydrogen bonds are those most apt to have all three participating atoms arranged in a linear manner. Bent hydrogen bonds are considerably weaker. Third, the total interatomic distance between the two electronegative bonding partners also factors into bond strength, and the optimal interatomic distance for maximum bond overlap is around 3 Å.

Water is both a weak acid and a weak base, thereby promoting formation of good hydrogen bonding networks. Moreover, all of the above factors conspire to promote strong hydrogen bonding between water molecules. As discussed by Eisenberg and Kauzmann², the substantial effect of hydrogen bonding on the thermal energy of water is appreciated best when the thermodynamic properties of water are compared to those of other related liquids. They point out that the molal heats of vaporization of the related series of compounds (*i.e.*, H₂Te, H₂Se, and H₂S) decrease as their molecular weights increase. Water is the next lower molecular-weight homologue in this series; yet, water has a heat of vaporization that is more than twice that of hydrogen sulfide.

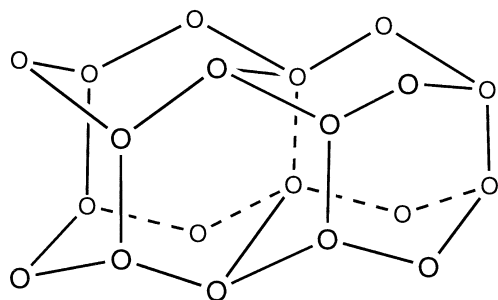
LIQUID WATER. Although there is still no universally accepted structure for liquid water, the ability to form strong hydrogen bonds is thought to dominate water structure in the liquid state. Indeed, there are also several models for the structure of liquid water that would suggest that the thermal energy of water relates to the hydrogen bonding equilibrium [written as: O—H···O = O—H + O] for which a shift to the right (a) is favored at elevated temperatures of the liquid, and (b) represents an increased configurational potential energy. While ice has a reasonably ordered structure involving hydrogen bonds, the fact that ice has a lower density than liquid water reminds us that H₂O molecules in liquid water are somehow more densely packed.

X-ray diffraction studies yield radial distribution function data⁴ which are dominated by the much greater scattering power of the more electron-rich oxygen atoms in water. These diffraction results tell us something about

how closely packed are these oxygen atoms with respect to each other. Such experimental measurements must be analyzed using structural⁵ or probabilistic⁶ models to retrieve structural information from the convoluted radial distribution data, and this complexity arises from the lack of any regular arrangements of molecules within a fluid. These molecules may form short-lived clusters that relentlessly form and break up. Indeed, no straightforward model yet accounts for either the maximal density of water at 4°C or water's substantial expansion as one proceeds to cool the liquid below that point.

Narten *et al.*⁴ arrived at 4.4 for the number of nearest neighboring H₂O molecules in solution. Furthermore, newer neutron diffraction studies⁷ suggest that each water molecule has 4.5 to 5.0 nearest neighbors. Nèmethy and Scheraga⁸ treated the radial distribution data as a composite of hydrogen bonding and non-hydrogen bonding populations. Bernal⁹ suggested that water molecules may form irregular networks of two classes of ring structures: the first comprising four-membered rings, as in ice-IV; and those made up of two closely positioned six-membered rings with bent hydrogen bonds allowing closer packing of non-hydrogen-bonded molecules.

Rather than interpreting the nearly four-coordinate structure of liquid water in terms of ice-like regions within the liquid, Symons^{10,11} envisages liquid water "as a three-dimensional network involving units fortuitously linked together, with four, five, six, and seven (membered) ring systems distinguishable but not of any particular significance because they must be haphazardly strung together". He suggests that water molecules form what he has termed as bifurcated hydrogen bonds where two OH groups share one lone pair or two lone pairs share an OH group. Furthermore, he suggests that the concentration of monomeric water molecules is likely to be quite low at room or physiologic temperatures.



Positions of Water Oxygens in Hexagonal Ice

Stillinger^{12,13} analyzed the structure of water by first considering hexagonal ice (illustrated above with solid lines indicating chemical bonds and dashed lines for hydrogen bonds). Starting with any oxygen atom, he points out that one can traverse many closed paths of contiguous hydrogen bonds by the simple rule that one always jumps to the nearest oxygen neighbor. Only polygonal paths with any even number of steps (or sides) offer a route that returns one to the origin, and this property is not observed with other hydrogen-bonded networks, such as that observed with HF or :NH₃. In the vapor phase, (H₂O)₂ dimers have linear hydrogen bonds with O-to-O interatomic distances of 2.98 Å, a value nearly 10% longer than the 2.74 Å for ice. Stillinger points out that quantum mechanical calculations support the linearity of hydrogen bonds, but they also suggest that rotations about those linear hydrogen bonds does not cost much in the way of lost stabilization energy. He finds appealing the ideas of Frank and Wen¹⁴ who view liquid water as a cooperative hydrogen bonding network, where there are chains of hydrogen bonds that are shorter and more stable than observed with (H₂O)₂ dimers. Stillinger's own computer modeling^{12,15} gives the following fractions of water molecules in liquid water (at 10°C) engaging in: zero H-bonds, ~4%; one H-bond ~18%; two H-bonds, ~36%; three H-bonds, ~32%; and 4 H-bonds, ~10%. This would suggest that the average water molecule in the liquid state makes 2.3 hydrogen bonds with its neighbors and that very few free water molecules (~4%) are present. By analyzing computer simulated molecular configurations of water molecules, Stillinger¹³ argued in favor of the following model for liquid water: "This network has a local preference for tetrahedral geometry, but it contains a large proportion of strained and broken (hydrogen) bonds. These strained bonds appear to play a fundamental role in kinetic properties, because their presence enhances prospects for molecules to switch allegiances, trading a bond here for one there and thus altering the network topology".

IONIZATION AND SELF-DIFFUSION. Water molecules readily dissociate to form protons and hydroxide ions, and the dissociation constant is:

$$K_{\text{water}} = k_d/k_r = [\text{H}^+][\text{OH}^-]/[\text{H}_2\text{O}]$$

The rate constant k_d for dissociation is about $2 \times 10^{-5} \text{ s}^{-1}$, and the reassociation rate constant k_r is about $1 \times$

$10^{11} \text{ M}^{-1}\text{s}^{-1}$ for the neutralization reaction¹⁶. The hydroxide ion and proton released in the dissociation process are strongly hydrated, and the hydrated proton can be represented in aqueous solution as $(\text{H}_2\text{O})_4\text{H}^+$. The lifetime of any intermolecular complex, whether it is $(\text{H}_2\text{O})_4\text{H}^+$ or other hydrogen-bonded clusters, is quite short, around 10^{-12} seconds. Moreover, a proton of a hydronium ion H_3O^+ or $(\text{H}_2\text{O})_4\text{H}^+$ can jump from one water molecule to another by virtue of the extensive hydrogen bonding in the liquid state. Indeed, this proceeds with diffusion-limited rate constants of about $1.1 \times 10^{10} \text{ M}^{-1}\text{s}^{-1}$ for protons (*i.e.*, $v = k[\text{H}_2\text{O}][\text{H}^+]$), or about three times faster than proton transfer between water and hydroxide ion.

Eigen and DeMaeyer¹⁶ summarized the following ionic properties of water (at 298 K): $K = [\text{H}^+][\text{OH}^-]/[\text{HOH}]$, $1.821 \times 10^{-16} \text{ M}$; $[\text{H}^+]$, $1.004 \times 10^{-7} \text{ M}$; $[\text{OH}^-]$, $1.004 \times 10^{-7} \text{ M}$; k_d , $2.5 \times 10^{-5} \text{ s}^{-1}$; k_a , $1.4 \times 10^{11} \text{ M}^{-1}\text{s}^{-1}$; proton transfer rate (for $\text{H}_2\text{O} + \text{H}_3\text{O}^+ \rightarrow \text{H}_3\text{O}^+ + \text{H}_2\text{O}$), $10.6 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$; proton transfer rate (for $\text{H}_2\text{O} + \text{HO}^- \rightarrow \text{HO}^- + \text{H}_2\text{O}$), $3.8 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$.

Biological fluids are more complicated still, because their water structure is strongly influenced by the presence of many biological and inorganic solutes. Collins^{17–19} has discussed how hydration of biomacromolecules is strongly influenced by the charge density and size of certain ions. He indicates that small ions of high charge density, known as kosmotropes, bind water molecules strongly relative to the strength of water-water interactions in the bulk solution. Another set of large monovalent ions, known as chaotropes, are of low charge density and bind water molecules weakly with respect to the strength of $\text{H}_2\text{O}-\text{H}_2\text{O}$ in bulk solution. Phosphates and carboxylates are the major intracellular kosmotropes that form highly soluble, solvent-separated ion pairs with the intracellular chaotropes: potassium ion, arginine's guanidinium group, histidine's imidazolium group, and lysine's ϵ -ammonium group. Accordingly, Collins¹⁷ has suggested that ion charge density may be regarded as a central determinant of structure and function in biological fluids. Lastly, the presence of macromolecular biological substances can also alter the fluid's viscosity and can restrict or promote certain interactions that are sensitive to the molecular crowding.

HYDROPHOBIC INTERACTIONS. These bonding interactions arise from the tendency^{20–23} of nonpolar side chains of amino acids (or lipids) to reside in the interior, nonaqueous environment of a protein (or membrane/micelle/vesicle). This process is accompanied by the release of tightly bound water molecules from these apolar side-chain moieties. The hydrophobic effect is thermodynamically driven by the increased disorder (*i.e.*, $\Delta S > 0$) of the system, thereby overcoming the unfavorable enthalpy change (*i.e.*, $\Delta H < 0$) for water release from the apolar groups.

Cantor and Schimmel²⁴ provide a lucid description of the thermodynamics of the hydrophobic effect, and they stress the importance of considering both the unitary and cratic contributions to the partial molal entropy of solute-solvent interactions. Briefly, the partial molal entropy (S_A) is the sum of the *unitary* contribution (S_A') which takes into account the characteristics of solute A and its interactions with water) and the *cratic* term ($-R \ln \chi_A$) which is a statistical term resulting from the mixing of component A with solvent molecules. The unitary change in entropy for transfer of a hydrocarbon side chain from water to the nonaqueous interior of a protein is positive. Hence, its contribution ($-T\Delta S$) to the Gibbs free energy favors transfer of a side chain to the protein's interior. Note that in the presence of denaturing agent (such as 8 M urea), a side chain exhibits a reduced tendency to reside in the apolar interior²⁵. About one-third to one-half of all amino acid residues in a protein have apolar side chains, thereby accounting for the cumulatively unfavorable effect of 8 M urea on protein folding.

Fersht and Serrano²⁶ recently described the systematic application of site-directed mutagenesis, together with calorimetry, to evaluate how specific side chains play roles in stabilizing protein structure. (**See also** *Hydrogen Bond in Molecular Recognition*). Many self-assembly processes are entropy driven, and Lauffer²⁷ proposed the release of water from apolar patches on the unpolymerized subunits as these patches interact with each other in the assembled state. In the case of microtubules, the warm-induced polymerization and cold-induced depolymerization are hallmarks of an entropy-driven process. Finally, although the common practices of enzyme purification are storage and processing at ice-water temperature, some enzymes are cold-inactivated. The expe-

rienced investigator will test the activity and structural properties of a newly characterized enzyme after storage at several different temperatures.

- ¹W. S. Benedict, H. H. Claassen & J. H. Shaw (1956) *J. Chem. Phys.* **24**, 1139.
- ²D. Eisenberg & W. Kauzmann (1969) *The Structure and Properties of Water*, Oxford Univ. Press, New York.
- ³L. Pauling (1960) *The Nature of the Chemical Bond*, 3rd ed., Cornell Univ. Press, Ithaca, N.Y.
- ⁴A. H. Narten & H. A. Levy (1969) *Science* **165**, 447.
- ⁵H. S. Frank (1972) in *Water: A Comprehensive Treatise* (F. Franks, ed.) pp. 515-543, Plenum Press, New York.
- ⁶A. Ben-Naim (1972) in *Water: A Comprehensive Treatise* (F. Franks, ed.) pp. 413-442, Plenum Press, New York.
- ⁷A. H. Narten (1972) *J. Chem. Phys.* **56**, 5681.
- ⁸G. Némethy & H. Scheraga (1962) *J. Chem. Phys.* **36**, 3382.
- ⁹J. D. Bernal (1964) *Proc. Royal Soc. London, Ser. A* **280**, 299.
- ¹⁰M. C. R. Symons (1975) *Philos. Trans. Royal Soc. London, Ser. B* **272**, 13.
- ¹¹M. C. R. Symons (1981) *Acc. Chem. Res.* **14**, 179.
- ¹²F. H. Stillinger (1975) *Adv. Chem. Phys.* **31**, 1.
- ¹³F. H. Stillinger (1980) *Science* **209**, 451.
- ¹⁴H. S. Frank & W. Y. Wen (1957) *Discuss. Faraday Soc.* **24**, 133.
- ¹⁵F. H. Stillinger & A. Rahman (1974) *J. Chem. Phys.* **74**, 3677.
- ¹⁶M. Eigen & L. DeMaeyer (1958) *Proc. Roy. Soc., Series A* **247**, 505.
- ¹⁷K. D. Collins (1997) *Biophys. J.* **72**, 65.
- ¹⁸K. D. Collins (1995) *Proc. Natl. Acad. Sci. U.S.A.* **92**, 5553.
- ¹⁹K. D. Collins & M. W. Washabaugh (1985) *Q. Rev. Biophys.* **18**, 323.
- ²⁰W. Kauzmann (1959) *Advances in Protein Chemistry* **14**, 1.
- ²¹C. Tanford (1962) *J. Amer. Chem. Soc.* **84**, 4240.
- ²²C. Tanford (1980) *The Hydrophobic Effect: Formation of Micelles and Biological Membranes*, 2nd ed., Wiley, New York.
- ²³F. M. Richards in *Protein Folding* (T. E. Creighton, ed.), pp. 1-58, Freeman, San Francisco.
- ²⁴C. R. Cantor & P. R. Schimmel (1980) *Biophysical Chemistry*, vol. **1**, pp. 281-288, Freeman, San Francisco.
- ²⁵P. L. Whitley & C. Tanford (1962) *J. Biol. Chem.* **237**, 1735.
- ²⁶A. R. Fersht & L. Serrano (1993) *Curr. Opin. Struct. Biol.* **3**, 75.
- ²⁷M. A. Lauffer (1975) *Entropy-Driven Processes in Biology*, Springer-Verlag, Berlin.

WATER, BOUND-

1. The water that is associated with a macromolecule, a colloid, or other substance such that it is not removable by simple filtration. 2. The water molecule(s) bound to the active site of an enzyme or to the surface of a protein.

WATER EXCHANGE RATES OF AQUA METAL IONS

Water molecules in the so-called coordination sphere of a metal ion complex can exchange with water molecules in the medium, and the rates depend largely on the nature of the metal ion and its electric charge, and to a lesser extent on the other coordinated substituents. The rate constants for substitutions of inner sphere water of various aqua ions were determined largely by Eigen's

group. Those with rate constants of 10^9 – 10^{10} s⁻¹ are K⁺, Rb⁺, Cs⁺, Ba²⁺, and Hg²⁺; 10^8 – 10^9 s⁻¹ are Li⁺, Na⁺, Ca²⁺, Sr²⁺, Cu²⁺, Cr²⁺, and Cd²⁺; 10^7 – 10^8 s⁻¹ are Y³⁺, Sc³⁺, La³⁺, Tb³⁺, Gd³⁺, and Zn²⁺; and 10^6 – 10^7 s⁻¹ are In³⁺, Fe²⁺, and Mn²⁺.

M. Eigen & K. Tamm (1962) *Z. Elektrochem.* **66**, 107.

F. Basolo & R. G. Pearson (1997) *Mechanism of inorganic reactions*, Wiley, New York.

WATER PURITY

The purity of the water used in kinetic and mechanistic studies should always be an issue of concern to an investigator. At a minimum level, all water used in a biochemical laboratory should be glass distilled. However, distillation is relatively ineffective in removing dissolved ionized gases and, although distillation will remove most dissolved ionized solids, not all are removed. It is best to combine glass distillation with one or more ion exchange resins. The American Society for Testing and Materials (ASTM) classifies ionically pure water as Type I water, if the maximum specific conductance is less than 0.06 micromho, less than 0.1 mg/L particulate matter, and pyrogen-free.

After purification, water should be stored in scrupulously cleaned glass containers to minimize metal ion contamination. If water or stock solutions are to be kept in plastic containers, then the containers should *not* be marked directly on the sides of the containers with marking pens commonly found in laboratories. Such commercial markers contain heavy metal ions which have been reported to permeate the walls of certain plastic containers.

As mentioned in the entry on *Substrate Purity*, ammonia is often a contaminant in many laboratories. It may be necessary to check for ammonia levels on a regular basis as ammonia can act as an inhibitor for many enzyme-catalyzed reactions acting on amine-containing substrates. **See** *Substrate Purity*

WATER (Temperature Effect on K_w)

If proton (or deuteron) concentrations and OH⁻ (or, OD⁻) concentrations are measured in molalities, m_H^+ (or, m_D^+) and m_{OH^-} (or, m_{OD^-}) respectively, then K_w is defined as the product of the molalities of the positively charged and negatively charged species. Since K_w is a thermodynamic constant, it is not surprising that it should vary with temperature. Defining $pK_w = -\log K_w$,

the following indicates the pK_w values for pure H_2O (and D_2O) at the listed temperature under saturating vapor pressure: 0°C, 14.94 (5.97); 5°C, 14.73 (15.74); 10°C, 14.53 (15.53); 15°C, 14.34 (15.32); 20°C, 14.16 (15.13); 25°C, 13.99 (14.95); 30°C, 13.84 (14.78); 35°C, 13.69 (14.62); 40°C, 13.54 (14.46); 50°C, 13.27 (14.18); 75°C, 12.71 (13.57); and 100°C, 12.26 (13.10). Notice that, because of this temperature effect, neutrality of pure water is pH 7.0 at only 25°C. At lower temperatures, neutrality occurs at pH values greater than 7.0; at temperatures higher than 25°C, neutrality occurs below pH 7.0. At normal human adult body temperature (37°C), pK_w is about 13.6. Hence, at 37°C, neutrality will take place at pH 6.8. For a solution of D_2O at 37°C, neutrality occurs at a pD value of about 7.28.

W. L. Marshall & E. U. Franck (1981) *J. Phys. Chem. Ref. Data* **10**, 295.

R. E. Mesmer & D. L. Herting (1978) *J. Solution Chem.* **7**, 901.

WATT

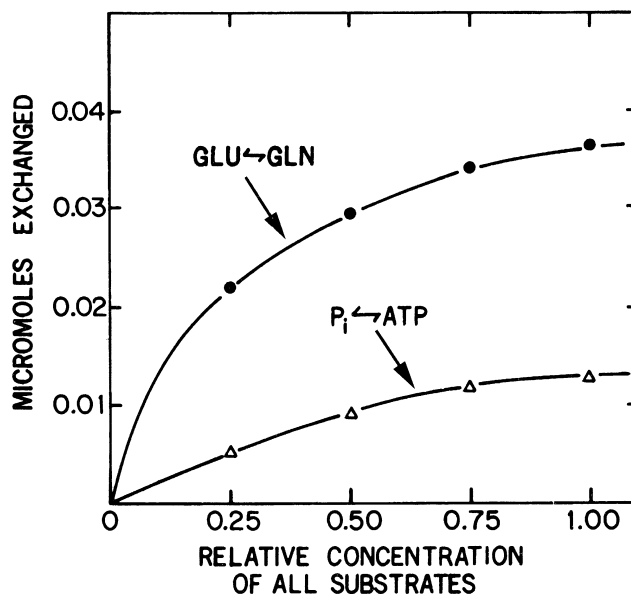
The SI unit of power or radiant flux (symbolized by W) equal to one joule per second (or, meter² per kilogram per second³). With respect to electric currents, the watt is the rate of work expressed by a current of one ampere and a potential difference of one volt. **See** *Power*; *Radiant Flux*

WEBER

The SI unit for magnetic flux (symbolized by Wb) equal to 10⁸ maxwells. The weber is the magnetic flux that produces an electromotive force of one volt, in linking a circuit of one turn, when the flux is reduced to zero at a uniform rate of one ampere per second. One weber is equivalent to one volt-second.

WEDLER-BOYER ISOTOPE EXCHANGE PROTOCOL

A very useful strategy¹ applied in equilibrium exchange studies to obviate problems associated with the formation of abortive complexes. Rates of isotope exchange are measured as a function of the absolute concentration of any one of the substrates or products (see figure below) under conditions such that the concentrations of all substrates and products are being varied in a constant ratio (*i.e.*, K_{eq}). This technique also facilitates detailed exchange kinetic studies of the effects of regulatory effectors on the exchange rates of enzymes. **See** *Isotope Exchange at Equilibrium*



Experimental demonstration that the *Escherichia coli* glutamine synthetase reaction operates by a random order of addition of substrates and a random order of release of products.

¹F. C. Wedler & P. D. Boyer (1972) *J. Biol. Chem.* **247**, 984.

WEIGHT

The force with which a body or entity is attracted to earth. The weight, W , is equal to the product of the mass, m , of the body and the acceleration of free fall in a vacuum (*i.e.*, $W = mg$). Since the earth is not a perfect sphere of uniform density, the weight of a body varies at various locales on the earth's surface. In practice, the mass of a body is determined by comparing its weight with a set of known standards: *i.e.*, $m_1/m_2 = W_1/W_2$.

WEIGHTING OF DATA

A statistical procedure that usually involves the multiplication of some quantity by a factor such that the result will be comparable to other quantities of the same type. For example, if a series of measurements of concentrations of a biomolecule are made, an investigator may wish to give greater "weight" to those values having the smallest associated error. **See** *Statistics (A Primer)*

Wilkinson¹ and Cleland² have pointed out the importance of proper weighting of initial rate data in the study of enzyme-catalyzed reactions. If the initial velocities vary by a factor of five or less, the weighting factor involves v_i^4 terms. If the velocities differ by a factor of ten or more, Cleland has suggested that the variance of

the velocities (*i.e.*, the weighting factor) be proportional to v_i^2 . **See** *Statistics (A Primer)*

¹G. N. Wilkinson (1961) *Biochem. J.* **80**, 324.

²W. W. Cleland (1979) *Meth. Enzymol.* **63**, 103.

WIGNER SPIN-CONSERVATION RULE

A rule that affects energy transfers in photochemical reactions, particularly photosensitization processes. The total electron spin (*i.e.*, the vectorial overall spin angular momentum of the system) does not change after the electronic energy transfer between an excited molecular entity and another molecular entity.

WOMACK-COLOWICK DIALYSIS METHOD

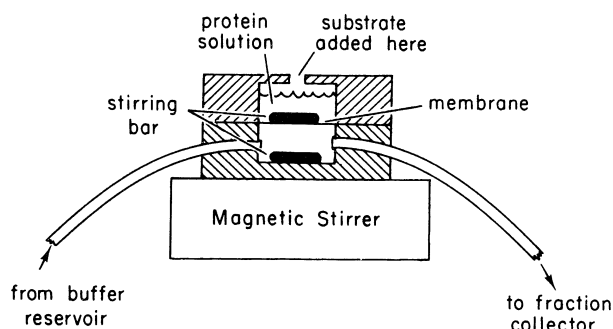


Diagram of the Womack-Colowick flow dialysis apparatus¹. Reprinted with permission of the *Journal of Biological Chemistry*.

A ligand binding technique that relies on the ability to monitor the concentration of uncomplexed ligand in equilibrium with a macromolecule by determining the ligand's rate of transport across a dialysis membrane. The technique¹ allows one to make successive additions of ligand to the upper chamber (see diagram below) containing the protein, and one can typically obtain suf-

ficient binding data within 20 min to construct a Scatchard plot for ligand binding at a single protein concentration. A miniaturized flow-dialysis cell has also been described².

¹F. C. Womack & S. P. Colowick (1973) *Meth. Enzymol.* **17**, 464.

²N. C. Brown & P. Reichard (1969) *J. Mol. Biol.* **46**, 39.

WORK

A measure of the mechanical or nonmechanical energy change corresponding to the action of some force through a distance, expressed in joules (with force in newtons multiplied by distance in meters) or ergs (with force in dynes multiplied by distance in centimeters). Other examples of work include: the product of ΔG and the number of moles of reactant; the product of surface tension and area; the product of pressure and volume; the product of electromotive force and electrical charge; and the product of magnetic field and magnetization.

Reversible work is a conceptual measure of the energy required to apply a force through a distance along a reversible path connecting states that are all in equilibrium with each other. To ensure that all intermediate states can attain equilibrium with each other, transit along the interconnecting path must proceed slowly. By contrast, irreversible work is achieved when a force is applied along a path at a pace that precludes equilibration of intermediate states, thereby hindering the full transfer of energy stepwise along that path. Reversible work represents the maximal work that can be achieved in a system, and this conceptual measure forms the basis for thermodynamic and statistical mechanical treatments of energy changes.

X

XANTHINE DEHYDROGENASE

This molybdenum-dependent enzyme [EC 1.1.1.204] catalyzes the reaction of xanthine with NAD^+ and water to produce urate and NADH. A variety of purines and aldehydes can act as substrates. Mammalian xanthine dehydrogenase can be interconverted to a form that has xanthine oxidase [EC 1.1.3.22] activity. The liver enzyme exists primarily in the dehydrogenase form, but can be converted into xanthine oxidase: (a) on storage at -20°C , (b) on treatment with proteolytic agents or organic solvents, or (c) by thiol reagents. Xanthine dehydrogenase can also be converted into xanthine oxidase by the action of enzyme-thiol transhydrogenase (glutathione disulfide) in the presence glutathione disulfide. In other animal tissues, the enzyme exists almost entirely as xanthine oxidase. **See** *Xanthine Oxidase*

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.
 N. E. Tolbert (1981) *Ann. Rev. Biochem.* **50**, 133.
 R. C. Bray (1980) *Adv. Enzymol.* **51**, 107.
 R. C. Bray (1975) *The Enzymes*, 3rd ed., **12**, 299.

XANTHINE OXIDASE

This enzyme [EC 1.1.3.22] catalyzes the reaction of xanthine with dioxygen and water to produce urate and hydrogen peroxide. Enzymatic activity requires iron, FAD, and molybdenum. Hypoxanthine and some other purines and pterins can act as substrates. Under some conditions, the product is mainly superoxide rather than hydrogen peroxide: thus, $\text{R}-\text{H}$ reacts with two dioxygen and water to produce $\text{R}-\text{OH}$, two H^+ , and two $\text{O}_2^{\cdot-}$ molecules. The *Micrococcus* enzyme can use ferredoxin as the acceptor substrate. The mammalian enzyme can be interconverted to xanthine dehydrogenase [EC 1.1.1.204]. **See** *Xanthine Dehydrogenase*

- B. A. Palfey & V. Massey (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 83.
 P. E. Baugh, D. Collison, C. D. Garner & J. A. Joule (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **3**, 377.
 C. Kisker, H. Schindelin & D. C. Rees (1997) *Ann. Rev. Biochem.* **66**, 233.
 S. C. Kundu & R. L. Willson (1995) *Meth. Enzymol.* **251**, 69.
 R. C. Bray (1975) *The Enzymes*, 3rd ed., **12**, 299.
 G. Palmer (1975) *The Enzymes*, 3rd ed., **12**, 1.

XI (x)

Symbol for extent or advancement (or extent of reaction) of a chemical reaction. ξ is the symbol for rate of conversion in which $\xi = d\xi/dt$; SI unit is $\text{mol}\cdot\text{s}^{-1}$ **See** *Extent of Reaction; Time & Chronomals*

XYLANASE

This enzyme [EC 3.2.1.8], also referred to as endo-1,4- β -xylanase and 1,4- β -D-xylan xylanohydrolase, catalyzes the endohydrolysis of 1,4- β -D-xylosidic linkages in xylans.

- G. Davies, M. L. Sinnott & S. G. Withers (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 119.
 M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.
 T. Hayashi (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 139.

XYLOGLUCAN SYNTHASE

This enzyme, officially known as 1,4- β -D-xylan synthase [EC 2.4.2.24], catalyzes the reaction of UDP-xylose with (1,4)- β -D-xylan_n to yield UDP and (1,4)- β -D-xylan_{n+1}. UDP-D-glucose is also a substrate.

- T. Hayashi (1989) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **40**, 139.

D-XYLONATE DEHYDRATASE

This enzyme [EC 4.2.1.82], also referred to as D-xyloaldonate dehydratase, catalyzes the conversion of D-xylo-aldonate to 2-dehydro-3-deoxy-D-xylonate and water.

- W. A. Wood (1971) *The Enzymes*, 3rd ed., **5**, 573.

XYLOSE ISOMERASE

This enzyme [EC 5.3.1.5], which requires magnesium ions, catalyzes the interconversion of D-xylose to D-xylu-

lose. This enzyme isolated from certain sources will also interconvert D-glucose and D-fructose.

C. I. F. Watt (1998) *Comprehensive Biological Catalysis: A Mechanistic Reference* **1**, 253.

I. A. Rose (1975) *Adv. Enzymol.* **43**, 491.

E. A. Noltmann (1972) *The Enzymes*, 3rd ed., **6**, 271.

α -XYLOSIDASE

The following is a recent review on the molecular and physical properties of this enzyme.

S. C. Fry (1995) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **46**, 497.

β -XYLOSIDASE

This enzyme [EC 3.2.1.37], also referred to as xylan 1,4- β -xylosidase and exo-1,4- β -xylosidase, catalyzes the hydrolysis of 1,4- β -D-xylans resulting in the removal of successive D-xylose residues from the nonreducing termini. Xylobiose can also be utilized as a substrate. The sheep liver enzyme has been reported to exhibit other exoglycosidase activities.

M. W. Bauer, S. B. Halio & R. M. Kelly (1996) *Adv. Protein Chem.* **48**, 271.

G. Mooser (1992) *The Enzymes*, 3rd ed., **20**, 187.

P. M. Dey & E. del Campillo (1984) *Adv. Enzymol.* **56**, 141.

D-XYLULOKINASE

This enzyme [EC 2.7.1.17], also known as xylulose kinase, catalyzes the reaction of ATP with D-xylulose to produce ADP and D-xylulose 5-phosphate.

W. L. Dills, Jr., P. D. Parsons, C. L. Westgate & N. J. Komplin (1994) *Protein Expr. Purif.* **5**, 259.

M. S. Neuberger, B. S. Hartley & J. E. Walker (1981) *Biochem. J.* **193**, 513.

L-XYLULOKINASE

This enzyme [EC 2.7.1.53], also known as L-xylulose kinase, catalyzes the reaction of ATP with L-xylulose to produce ADP and L-xylulose 5-phosphate.

J. C. Sanchez, R. Gimenez, A. Schneider, W. D. Fessner, L. Baldoma, J. Aguilar & J. Badia (1994) *J. Biol. Chem.* **269**, 29665.

R. L. Anderson & W. A. Wood (1962) *J. Biol. Chem.* **237**, 1029.

Y

YAGA-OZAWA TREATMENT

A graphical procedure for analyzing effects of two competitive inhibitors at the active site of an enzyme: v_0/v_i is plotted as a function of $([I_1] + [I_2])$ where v_0 is the initial-rate velocity in the absence of an inhibitor(s), v_i is the velocity in the presence of the inhibitor(s), and $[I_1]$ and $[I_2]$ are the two inhibitor concentrations^{1,2}. If there are no interactions between the two inhibitors, then a straight line will be obtained; if there is an interaction at the active site, curvature will be observed. **See** *Yonetani-Theorell Treatment*

¹K. Yagi & T. Ozawa (1960) *Biochim. Biophys. Acta* **42**, 381

²T. Yonetani (1982) *Meth. Enzymol.* **87**, 500.

YIELD COEFFICIENT (or, Coefficient of Yield)

A parameter used in the study of the growth and cultivation of microorganisms. It is equal to the mass of the amount of microorganism produced divided by the mass of the substrate consumed. The term “energy yield coefficient” is used if the ratio represents the yield of ATP produced per the theoretically possible yield.

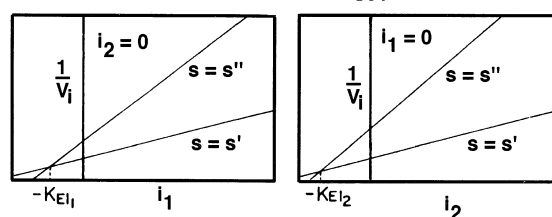
YLIDES

A class of compounds in which a positively charged atom from group V or VI of the periodic table (e.g., N, O, S, P, As, Se) is bonded to a carbon atom having an unshared pair of electrons. Whereas there is only one canonical form for nitrogen ylides ($R_3N^+-\ddot{C}R_2^-$), because of $p\pi-d\pi$ bonding, two canonical forms can be written for phosphorus (i.e., $R_3P=CR_2 \leftrightarrow R_3P^+-\ddot{C}R_2^-$) and sulfur ylides ($R_2S=CR_2 \leftrightarrow R_3S^+-\ddot{C}R_2^-$). A number of enzyme-catalyzed reactions have been reported to utilize ylide-based chemistry. For example, the ylide form of the

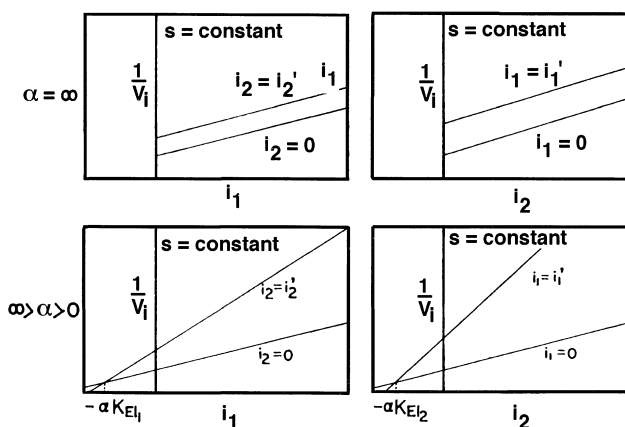
thiazolium ring on thiamin pyrophosphate is the nucleophile in pyruvate carboxylase. **See** *Carbanion*

YONETANI-THEORELL TREATMENT

Dixon Plot (to illustrate the analogy)



Yonetani-Theorell Plots



A graphical procedure for assessing the possible interaction of two different inhibitors binding at the active site of an enzyme. In this method, the investigator plots v_0/v_i as a function of the concentration of one of the two inhibitors at different, constant concentrations of the second substrate^{1,2}. Such plots provide a means for the calculation of the inhibitor interaction constant which provides the investigator with a tool for assessing the degree of overlap of subsites at an enzyme's active center.

¹T. Yonetani and H. Theorell (1964) *Arch. Biochem. Biophys.* **106**, 243.

²T. Yonetani (1982) *Meth. Enzymol.* **87**, 500.

Z

ZEATIN CIS-TRANS ISOMERASE

The following is a recent review on the molecular and physical properties of this enzyme.

A. N. Binns (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 173.

ZEATIN REDUCTASE

This enzyme [EC 1.1.1.242] catalyzes the reversible reaction of dihydrozeatin with NADP⁺ to produce zeatin and NADPH.

A. N. Binns (1994) *Ann. Rev. Plant Physiol. Plant Mol. Biol.* **45**, 173.

ZEOLITE

Highly crystalline, microporous substances comprised of an aluminosilicate framework wherein oxygen atoms join silicon and aluminum atoms located within tetrahedral units corresponding to [SiO₄]⁴⁻ and [AlO₄]⁵⁻. Zeolites were first discovered by Cronstedt¹ who recognized their ability to serve as boiling stones. Today, molecular sieves have many applications, including water removal from organic solvents, ion-exchange applications, adsorption of odorant molecules, and chemical catalysis.

In aqueous solutions, the high negative surface charge of a zeolite must be neutralized by binding counterions, such as Na⁺, K⁺, and Ca²⁺. The distribution of zeolite pore size can be modified, such that small molecules are included in the pores; those too large to diffuse into the pores are excluded. Structure types are named by a three-letter IUPAC code, based in part on the name of the zeolite first used to identify the specific type. They are also classified by pore size, framework density, and/or symmetry.

¹A. F. Cronstedt (1756) *Kongl. Svenska Vetenskaps Acad. Handlingar* **17**, 120.

ZERO ORDER

Referring to reactions in which the reaction velocity is independent of the reactant under consideration. For example, for the reaction $A + B \rightarrow C$, if the empirical rate expression is $v = k[B]$, the reaction is first order with respect to B but zero order with respect to A. *See Chemical Kinetics; Rate Saturation; Michaelis-Menten Equation*

ZERO-ORDER REACTIONS

Reactions in which the velocity (v) of the process is independent of the reactant concentration, following the rate law $v = k$. Thus, the rate constant k has units of M·s⁻¹. An example of a zero-order reaction is a Michaelis-Menten enzyme-catalyzed reaction in which the substrate concentration is much larger than the Michaelis constant. Under these conditions, if the substrate concentration is raised even further, no change in the velocity will be observed (since $v = V_{\max}$). Thus, the reaction is zero-order with respect to the substrate. However, the reaction is still first-order with respect to total enzyme concentration. When the substrate concentration is not saturating then the reaction ceases to be zero order with respect to substrate. Reactions that are zero-order in each reactant are exceedingly rare. Thus, zero-order reactions address a fundamental difference between order and molecularity. Reaction order is an empirical relationship. Hence, the term pseudo-zero order is actually redundant. All zero-order reactions cease being so when no single reactant is in excess concentration with respect to other reactants in the system.

Another example of zero-order kinetics is the rate of dissolution of encapsulated solutes restricted in the egress by passage through a small orifice in the capsule. If a soluble salt is added in addition to the encapsulated solute, one obtains an osmotically driven solute release system. *See also Order of Reaction; Molecularity; Michaelis-Menten Equation; First-Order Reaction*

ZERO POINT ENERGY

The residual energy (designated E_0) of a harmonic oscillator in the ground state. The Heisenberg Uncertainty Principle does not permit any state of completely defined position *and* momentum. A one-dimensional harmonic oscillator has energy levels corresponding to:

$$E_\nu = (\nu + 0.5)h\nu$$

where $\nu = 0, 1, 2, 3, \dots$, correspond to the vibrational energy levels, h is Planck's constant, and ν is the frequency. The zero-point energy occurs at the lowest energy state (*i.e.*, the ground state, where $\nu = 0$), and E_0 equals $0.5h\nu$. **See** Hooke's Law Spring; Kinetic Isotope Effects

ZETA (ζ)

Symbol for zeta potential.

ZIPPER MECHANISMS FOR DNA UNFOLDING

The sequential and often cooperative disassembly of double-helical structure, occurring whenever the sample temperature exceeds the so-called melting temperature (T_m) for a given segment of DNA. Because of the low concentrations of intermediate states lying between helix and coil structures, the helix-coil transition can be approximated as a two-state, all-or-nothing process. **See** DNA Unwinding; Kinetic Model for Small DNA

H. C. Spatz & D. M. Carothers (1969) *J. Mol. Biol.* **42**, 191.

D. Pörschke & M. Eigen (1971) *J. Mol. Biol.* **62**, 361.

Z VALUE

A calculated transition energy used to assess the polarity of a solvent. The solvent ionizing capability directly affects the position of a peak, easily measured, in the ultraviolet region of the spectrum of the complex of an iodide ion with 2-methyl- or 1-ethyl-4-carbomethoxypyridinium ion. Water has a Z value of 94.6, ethanol has a value of 79.6, dimethyl sulfoxide's value is 71.1, and benzene has a value of 54. A similar polarity scale, known as $E_T(30)$ values, is related to the Z value scale: $Z = 1.41E_T(30) + 6.92$. **See** Solvent Effects

E. M. Kosower (1958) *J. Amer. Chem. Soc.* **80**, 3253, 3261, and 3267.

E. M. Kosower, G. Wu & T. S. Sorensen (1961) *J. Amer. Chem. Soc.* **83**, 3147.

M. H. Abraham, P. L. Grellier, J. M. Abboud & R. M. Doherty (1988) *Can. J. Chem.* **66**, 2673.

M. J. Kamlet, J. M. Abboud & R. W. W. Taft (1981) *Prog. Phys. Org. Chem.* **13**, 485.

J. Shorter (1982) *Correlation Analysis of Organic Reactivity*, Wiley, New York.

C. Reichardt (1988) *Solvents and Solvent Effects in Organic Chemistry*, 2nd ed., VCH, New York.

ZWITTERGENTS

Zwitterionic detergents or surfactants. **See** Detergents; Micelle

ZWITTERION

A molecular entity containing oppositely charged acid-base groups. An example is $^+\text{H}_3\text{NCH}(\text{CH}_3)\text{COO}^-$, which is by far the predominant species of alanine at neutral pH. The term "zwitterion" is a German word designating the presence of an "inner salt" or an internal ionic species within a molecule. Most biochemistry textbooks indicate that zwitterions are electrically neutral, lacking any overall net charge. This is only true for a substance containing one positively charged and one negatively charged ionizable group. Polyvalent molecules may have zwitterionic forms with overall net charge. For glutamic acid, one zwitterionic species ($^+\text{H}_3\text{NCH}(\text{CH}_2\text{CH}_2\text{COOH})\text{COO}^-$) is in fact neutral at low pH; the other is a monoanion present at higher pH ($^+\text{H}_3\text{NCH}(\text{CH}_2\text{CH}_2\text{COO}^-)\text{COO}^-$).

A zwitterion must not be confused with an ylide, which has opposite electric charges residing on adjacent atoms. The term "dipolar ion," while often used in place of zwitterion, is not considered by IUPAC to be an acceptable synonym, and should be avoided.

The relative concentrations of zwitterionic forms and the uncharged species is well illustrated in the case of glycine¹⁻⁴. We begin by considering the relationship between macroscopic and microscopic constants. Titration experiments provide information on the proton dissociation constants; at 25°C and extrapolated to zero ionic strength, $K_1 = 4.45 \times 10^{-3}$ ($\text{p}K_1 = 2.35$) and $K_2 = 1.66 \times 10^{-10}$ ($\text{p}K_2 = 9.78$). The first macroscopic constant represents the dissociation of a proton from the fully protonated species ($^+\text{H}_3\text{NCH}_2\text{COOH}$) to form either the zwitterion ($^+\text{H}_3\text{NCH}_2\text{COO}^-$) or the neutral form ($\text{H}_2\text{NCH}_2\text{COOH}$). Similarly, the second macroscopic constant represents the dissociation of a proton from either the zwitterion or the uncharged species to form

the anionic species, $\text{H}_2\text{NCH}_2\text{COO}^-$. To determine the relative abundance of the zwitterion and the uncharged species, the corresponding microscopic ionization equilibria must be evaluated. Note that these macroscopic constants are related to the microscopic events by the following expressions:

$$K_1 = \frac{([\text{H}_3\text{NCH}_2\text{COO}^-] + [\text{H}_2\text{NCH}_2\text{COOH}])([\text{H}^+])}{[\text{H}_3\text{NCH}_2\text{COOH}]}$$

$$K_2 = \frac{[\text{H}_2\text{NCHCOO}^-][\text{H}^+]}{[\text{H}_3\text{NCH}_2\text{COO}^-] + [\text{H}_2\text{NCH}_2\text{COOH}]}$$

Microscopic, or intrinsic, equilibrium constants are

$$k_1 = \frac{([\text{H}_3\text{NCH}_2\text{COO}^-][\text{H}^+])}{[\text{H}_3\text{NCH}_2\text{COOH}]}$$

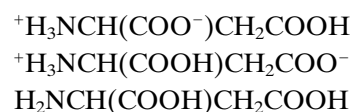
$$k_2 = \frac{[\text{H}_2\text{NCH}_2\text{COOH}][\text{H}^+]}{[\text{H}_3\text{NCH}_2\text{COOH}]}$$

$$k_3 = \frac{[\text{H}_2\text{NCH}_2\text{COO}^-][\text{H}^+]}{[\text{H}_3\text{NCH}_2\text{COO}^-]}$$

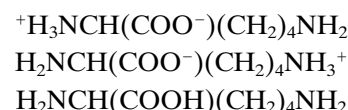
$$k_4 = \frac{[\text{H}_2\text{NCH}_2\text{COO}^-][\text{H}^+]}{[\text{H}_2\text{NCH}_2\text{COOH}]}$$

Note that $K_1 = k_1 + k_2$ and $K_2 = [k_3^{-1} + k_4^{-1}]^{-1}$. The four microscopic constants are thermodynamically interdependent, such that: $k_1 k_3 = k_2 k_4$. The $\text{p}K_a$ value for the amino group of glycine ethyl ester was experimentally determined to be 7.73 at 25°C; for glycine methyl ester, the value is 7.66^{3,4}. If one assumes that this dissociation constant (an average value of 7.70) is identical to the dissociation constant of the amino proton from glycine, then $\text{p} k_2 = 7.70$ (i.e., $k_2 = 2.00 \times 10^{-8}$). Thus, $k_1 = K_1 - k_2 \approx 4.45 \times 10^{-3}$. Combining the expression for K_2 above and the thermodynamic interrelationship of the microscopic constants, one finds that $k_3 = K_2(k_1 + k_2)/k_1 = K_1 K_2/k_1 \approx 1.66 \times 10^{-10}$. Hence, k_4 equals $k_1 k_3/k_2$, or about 3.69×10^{-5} . From the definitions for the microscopic dissociation constants, $k_1/k_2 = k_4/k_3 = [\text{H}_3\text{NCH}_2\text{COO}^-]/[\text{H}_2\text{NCH}_2\text{COOH}] \approx 2.23 \times 10^5$. This ratio is pH-independent, and the zwitterion form predominates over the uncharged species at all pH's. Similar calculations can be made with other monoaminomonocarboxylic acids. Although there is some variability in the ratio for other common amino acids, the zwitterion is by far the major form of these amino acids.

For compounds with three $\text{p}K_a$ values, a similar but somewhat more demanding analysis can be made³. The three $\text{p}K_a$ values for aspartic acid are 1.99, 3.90, and 9.90, whereas the $\text{p}K_a$ value for the diethyl ester is 6.5. The following compounds



are present in solution in the approximate proportions 28,000:1600:1, respectively. For lysine's isoelectric structures



the relative proportions are 3.2×10^5 : 1.8×10^6 : 1, respectively.

Because dissociation constants vary with temperature, the zwitterion/uncharged ratio may also vary with temperature⁵. For glycine⁶, the temperature dependencies are $\text{dp}K_1/\text{dT} = -0.0020$ and $\text{dp}K_2/\text{dT} = -0.025$.

Finally, the zwitterion is not the major form for all "amino acids." For both *o*-aminobenzoic acid (anthranilic acid) and *p*-aminobenzoic acid, the uncharged species is the more prevalent form³ whereas the zwitterion/uncharged ratio for *m*-aminobenzoic acid is about 2.31. This behavior arises primarily from the low $\text{p}K_a$ values for the aromatic ring's amino group (e.g., the $\text{p}K_1$ and $\text{p}K_2$ values for anthranilic acid are 2.05 and 4.95 and the $\text{p}K_a$ value for the ester is only 2.09).

¹L. Ebert (1926) *Z. Physik. Chem.* **121**, 385.

²J. T. Edsall & M. H. Blanchard (1933) *J. Amer. Chem. Soc.* **55**, 2337.

³J. T. Edsall (1943) in *Proteins, Amino Acids and Peptides as Ions and Bipolar Ions* (E. J. Cohn & J. T. Edsall, eds.), pp. 75-115, ACS Monograph Series #1, Reinhold Publ. Corp., New York.

⁴J. T. Edsall & J. Wyman (1958) *Biophysical Chemistry*, vol. I, Academic Press, New York.

⁵R. M. C. Dawson, D. C. Elliott, W. H. Elliott & K. M. Jones (1986) *Data for Biochemical Research*, 3rd ed., Clarendon Press, Oxford.

⁶D. D. Perrin & B. Dempsey (1974) *Buffers for pH and Metal Ion Control*, Chapman and Hall, London.

ZYMOGEN

A precursor or storage form of an enzyme, sometimes used synonymously with proenzyme, in which the precursor is converted to the active (or more active) enzyme

by proteolysis. This conversion can involve one or several hydrolytic steps. If more than one step occurs, the intermediate form(s) may have partial activity. For example, prochymotrypsin (or, chymotrypsinogen) is converted to active α -chymotrypsin following three proteolytic cleav-

ages and involves the formation of an active intermediate referred to as π -chymotrypsin. In some cases, the polypeptide products produced by proteolytic activation do separate from each other; in other instances, they remain associated in the final mature protein.

WORDFINDER

Use this Wordfinder to take fullest advantage of the text and reference material contained in this **HANDBOOK**.

This Wordfinder comprises all listed source words (**CAPITALIZED**) as well as other subheadings, keywords, or synonyms (written in lower case). Each of the latter is immediately followed by the recommended source entry (**CAPITALIZED & ITALICIZED**).

Note that many biochemical terms and enzymes (*e.g.*, ACAT, ATP, NADH, *etc.*) as well as computer soft-

ware programs (*e.g.*, KINSIM, SAAM, FITSIM) are always capitalized. To avoid confusion, look for a comma following the entry: source entries stand alone and are not followed by a comma; the presence of a comma indicates that the entry is not a source word.

Many source words are also cross-referenced, and these source words (again **CAPITALIZED & ITALICIZED**) appear below each entry.

A

A, B, C, . . . P, Q, R, . . .

CLELAND NOMENCLATURE

AB INITIO MOLECULAR-ORBITAL CALCULATIONS

ABM-1 & ABM-2 SEQUENCES IN ACTIN-BASED MOTORS

ABORTIVE COMPLEXES

DEADEND COMPLEXES

ENZYME REGULATION

FRIEDEN DILEMMA

INHIBITION

ISOTOPE EXCHANGE AT EQUILIBRIUM

ISOTOPE TRAPPING

LACTATE DEHYDROGENASE

LIGAND EXCLUSION MODEL

NONPRODUCTIVE COMPLEXES

PRODUCT INHIBITION

SUBSTRATE INHIBITION

Abortive complexes in isotope exchange experiments,

ABORTIVE COMPLEXES

Abortive complex interaction,

ISOTOPE EXCHANGE AT EQUILIBRIUM

ABSCISSA

ABSOLUTE CONFIGURATION

CONFIGURATION

CAHN-INGOLD-PRELOG SYSTEM

CORN RULE

DIASTEREOMERS

ENANTIOMERS

(R/S)-CONVENTION

Absolute rate theory,

TRANSITION-STATE THEORY

ABSOLUTE TEMPERATURE

ABSOLUTE UNCERTAINTY

RELATIVE UNCERTAINTY

ABSOLUTE ZERO

ABSORBANCE

BEER-LAMBERT LAW

ABSORPTION COEFFICIENT

ABSORPTION SPECTROSCOPY

ABSORBED DOSE

ABSORPTION

ABSORPTION COEFFICIENT

BEER-LAMBERT LAW

ABSORBANCE

MOLAR ABSORPTION COEFFICIENT

ABSORPTION SPECTROSCOPY

ABSORPTION SPECTROSCOPY

ABSORPTIVITY

ABSORBANCE

ABSORPTION COEFFICIENT

BEER-LAMBERT LAW

ABSORPTION SPECTROSCOPY

ABSTRACTION REACTION

Abzyme,

See preferred descriptor CATALYTIC

ANTIBODIES

ACAT,

ACYL-CoA:CHOLESTEROL ACYLTRANSFERASE

ACCELERATION

CATALYTIC RATE ENHANCEMENT

CATALYTIC PROFICIENCY

EFFICIENCY FUNCTION

Acceptor,

FLUORESCENCE

DONOR NUMBER

LEWIS ACIDITY

Acceptor Number,

DONOR NUMBER

LEWIS ACIDITY

PHYSICAL ORGANIC CHEMISTRY TERMINOLOGY

ACCRETION

ACCURACY

PRECISION

Accuracy of measurement,

NOISE

ACETALDEHYDE DEHYDROGENASE (ACETYLATING)

2-(Acetamidomethylene)succinate,

METHYLHYDROXYPYRIDINECARBOXYLATE DIOXYGENASE

2-(ACETAMIDOMETHYLENE)SUCCINATE HYDROLASE

Acetate,

LACTATE MONOOXYGENASE

LEUCINE KINETICS

ACETATE KINASE

ACYL-PHOSPHATE INTERMEDIATE

NUCLEOSIDE-5'-TRIPHOSPHATE REGENERATION

ACETATE KINASE (PYROPHOSPHATE)
 Acetate kinase reduction,
BOROHYDRIDE REDUCTION
 ACETAZOLAMIDE
 Acetic acid, autoprotolysis constant,
AUTOPROTOLYSIS
 ACETOACETATE DECARBOXYLASE
 Acetoacetate decarboxylase reduction,
BOROHYDRIDE REDUCTION
 ACETOLACTATE SYNTHASE
 Acetone,
MEERWEIN-PONDRORFF-VERLEY REDUCTION
 Acetylcholine dissociation from acetylcholine
 receptor,
DISSOCIATION KINETICS
 ACETYLCHOLINESTERASE
 Acetylcholinesterase reaction,
ACYL-SERINE INTERMEDIATE
 ACETYL-CoA
 Acetyl-CoA:arylamine *N*-acetyltransferase,
ARYLAMINE ACETYLTRANSFERASE
 ACETYL-CoA CARBOXYLASE
 ACETYL-CoA SYNTHETASE
 ACETYLENE MONOCARBOXYLATE HY-
 DRATASE
 N-ACETYLGALACTOSAMINE-4-SULFATE
 SULFATASE
 N-ACETYLGALACTOSAMINE-6-SULFATE
 SULFATASE
 α -*N*-Acetyl-D-galactosaminidase,
 α -GALACTOSIDASE
 N-ACETYLGALACTOSAMINIDE SIALYL-
 TRANSFERASE
 N-Acetylglucosamine,
LACTOSE SYNTHASE
LYSOZYME
MAPPING SUBSTRATE INTERACTIONS
USING KINETIC DATA
 N-Acetyl-D-glucosamine 1,6-bisphosphate,
 as activator,
PHOSPHOACETYLGLUCOSAMINE
MUTASE
 N-Acetylglucosamine 2-epimerase,
N-ACYLGLUCOSAMINE 3-EPIMERASE
 N-ACETYLGUCOSAMINE KINASE
 N-Acetyl-D-glucosamine 1-phosphate,
PHOSPHOACETYLGLUCOSAMINE
MUTASE
 N-Acetyl-D-glucosamine 6-phosphate,
PHOSPHOACETYLGLUCOSAMINE
MUTASE
 N-ACETYLGUCOSAMINE-6-PHOSPHATE
 2-EPIMERASE

N-ACYLGLUCOSAMINE-6-PHOSPHATE
2-EPIMERASE
N-Acetylglucosamine-phosphate mutase,
PHOSPHOACETYLGLUCOSAMINE
MUTASE
 Acetylglucosamine phosphomutase,
PHOSPHOACETYLGLUCOSAMINE
MUTASE
 N-ACETYLGUCOSAMINE-6-SULFATE SUL-
 FATASE
 α -*N*-ACETYLGUCOSAMINIDASE
 β -*N*-ACETYLGUCOSAMINIDASE
N-Acetyl-D-glucosaminyl-lipopolysaccharide,
LIPOPOLYSACCHARIDE N-ACETYLGU-
COSAMINYLTRANSFERASE
N-Acetylglucosaminyl residues,
LIPOPOLYSACCHARIDE N-ACETYLGU-
COSAMINYLTRANSFERASE
N-ACETYLGUTAMATE SYNTHASE
GLUTAMATE ACETYLTRANSFERASE
N-ACETYL- γ -GLUTAMYL-PHOSPHATE RE-
 DUCTASE
 β -*N*-ACETYLHEXOSAMINIDASE
 O-ACETYLHOMOSERINE (THIOL)-LYASE
N-Acetylglucosamine synthase,
LACTOSE SYNTHASE
N-Acetylmuramic acid,
LYSOZYME
 3-Acetyl-NAD⁺ as an alternative substrate,
ABSORPTION SPECTROSCOPY
N-ACETYLNEURAMINATE LYASE
N-Acetylneuraminyl-D-galactosyl-*N*-acetyl-D-
 galactosaminyl-(*N*-acetylneuraminyl)-D-
 galactosyl-D-glucosylceramide,
SIALYLTRANSFERASE
 α -*N*-Acetylneuraminyl-2,6- β -D-galactosyl-1,4-
N-acetyl- β -D-glucosamine,
SIALYLTRANSFERASE
 α -*N*-Acetylneuraminyl-2,6- β -galactosyl-1,4- β -
 D-glucosylceramide,
SIALYLTRANSFERASE
*N*²-ACETYLORNITHINE AMINOTRANS-
 FERASE
N-ACETYLORNITHINE DEACETYLASE
 Acetyl-phosphate,
NUCLEOSIDE-5'-TRIPHOSPHATE REGEN-
ERATION
PHOSPHATE ACETYLTRANSFERASE
PYRUVATE OXIDASE
 O-Acetylserine sulphydralase,
ACETYL SERINE (THIOL)-LYASE
 O-ACETYLSERINE (THIOL)-LYASE
 Acetylserotonin methyltransferase,

HYDROXYINDOLE O-METHYLTRANS-
FERASE
 Acetyl transacetylase,
FATTY ACID SYNTHETASE
 ACHIRAL
CHIRALITY
 Achirotopic,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
 ACID
LEWIS ACID
LEWIS BASE
BRØNSTED THEORY
 Acid anhydrides,
ACYL-PHOSPHATE INTERMEDIATE
 ACID-BASE EQUILIBRIUM CONSTANTS
DEGREE OF DISSOCIATION
HENDERSON-HASSELBALCH EQUATION
 Acid/base groups in enzymes,
pH EFFECTS ON ENZYMIC CATALYSIS
 Acid β -D-glucosidase,
GLUCOSYLCERAMIDE β -D-GLUCO-
SIDASE
 Acid hydrolysis,
METAL ION COORDINATION REAC-
TIONS
 ACIDITY
DEGREE OF DISSOCIATION
HENDERSON-HASSELBALCH EQUATION
 ACID-BASE EQUILIBRIUM CONSTANTS
BRØNSTED THEORY
LEWIS ACID
ACIDITY FUNCTION
LEVELING EFFECT
 ACIDITY FUNCTION
 ACID-LABILE SULFIDES
 ACID PHOSPHATASE
 ACONITASE
 ACONITATE DECARBOXYLASE
 ACONITATE Δ -ISOMERASE
 Acridine dye intercalation in DNA,
SCATCHARD PLOT
 ACROSIN
 ActA,
ABM-1 & ABM-2 SEQUENCES IN ACTIN-
BASED MOTORS
ACTIN-BASED BACTERIAL MOTILITY
 ACTIN ASSEMBLY ASSAYS
 ACTIN ASSEMBLY KINETICS
ACTIN FILAMENT CAPPING PROTEIN
ACTIN FILAMENT SEVERING PROTEIN

ACTIN CROSS-LINKING/BUNDLING PROTEIN	ACTIVATOR	Acylation,
ACTIN ATPase REACTION (APPARENT IRREVERSIBILITY)	ACTIVATION	AFFINITY LABELING
ACTIN-BASED PATHOGEN MOTILITY	ALLOSTERIC EFFECTOR	Acyl carrier protein,
ABM-1 AND ABM-2 SEQUENCES IN ACTIN-BASED MOTORS	LINKED FUNCTIONS	HOLO-[ACYL CARRIER PROTEIN] SYNTHASE
Actin capping proteins,	LONDON-STECK PLOT	MALONYL-CoA:ACP-TRANSACYLASE
ACTIN FILAMENT CAPPING PROTEIN	Active enzyme crystallography,	Acyl carrier protein synthase,
Actin cross-linking/bundling,	SOLVENT-TRAPPED INTERMEDIATES	HOLO-[ACYL CARRIER PROTEIN] SYNTHASE
ACTIN REGULATORY PROTEINS	ACTIVE SITE	ACYL-CoA:CHOLESTEROL O-ACYLTRANSFERASE
ACTIN FILAMENT CROSS-LINKING/BUNDLING PROTEIN	LAMBDA (or Λ -) ISOMERS OF METAL ION-NUCLEOTIDE COMPLEXES	ACYL-CoA DEHYDROGENASES
ACTIN FILAMENT CAPPING PROTEIN	LOCK-AND-KEY MODEL OF ENZYME ACTION	ACYL-CoA DESATURASE
ACTIN FILAMENT CROSS-LINKING/BUNDLING PROTEIN	LOW-BARRIER HYDROGEN BONDS: ROLE IN CATALYSIS	DESATURASES
ACTIN FILAMENT GROWTH (Polymerization Zone Model)	YAGA-OZAWA PLOT	ACYL-CoA:RETINOL O-ACYLTRANSFERASE
ABM-1 & ABM-2 SEQUENCES IN ACTIN-BASED MOTORS	YONETANI-THEORELL PLOT	Acyl-CoA synthetase,
ACTIN FILAMENT LATERAL BINDING PROTEIN	INDUCED FIT MODEL	LONG-CHAIN FATTY ACYL-CoA SYNTHETASE
ACTIN FILAMENT SEVERING PROTEIN	Active-site-directed irreversible enzyme inhibitor,	ACYL ENZYME INTERMEDIATES
ACTIN FILAMENTS (MECHANICAL PROPERTIES)	AFFINITY LABELING	ACTIVE SITE TITRATION
ACTIN MONOMER SEQUESTERING PROTEIN	Active site residues,	SERPINS (INHIBITORY MECHANISM)
ACTINOMETER	AFFINITY LABELING	N- ACYLGLUCOSAMINE 2-EPIMERASE
Actin severing proteins,	ACTIVE-SITE TITRATION	N-ACYLGLUCOSAMINE-6-PHOSPHATE 2-EPIMERASE
ACTIN REGULATORY PROTEINS	ACETYLCHOLINESTERASE	2-Acylglycerol acyltransferase,
ACTION	AFFINITY LABELING	2-MONOACYLGLYCEROL ACYLTRANSFERASE
ACTION POTENTIAL	DIISOPROPYLFLUOROPHOSPHATE	Acylglycerol lipase,
MEMBRANE POTENTIAL	ACTIVE TRANSPORT	MONOACYLGLYCERIDE LIPASE
ACTION SPECTRUM	ION PUMPS	1-Acyl-sn-glycerol 3-phosphate,
Activated cluster,	MEMBRANE TRANSPORTERS	LYSOPHOSPHOLIPASE D
ACTIN FILAMENT GROWTH (Polymerization Zone Model)	$\Delta G_{\text{diffusion}}$	ACYLGLYCERONE-PHOSPHATE REDUCTASE
Activated complex,	ACTIVITY	1-ACYLGLYCERO-3-PHOSPHATE O-ACYLTRANSFERASE
TRANSITION STATE	SPECIFIC ACTIVITY	1-Acylglycerophosphocholine,
Activated complex theory,	TURNOVER NUMBER	LECITHIN-CHOLESTEROL ACYLTRANSFERASE
TRANSITION-STATE THEORY	OPTICAL ACTIVITY	1-Acylglycerophosphocholine, formation of,
ACTIVATION	ACTIVITY COEFFICIENT	PHOSPHOLIPASE A ₂
LIPID ACTIVATION	CHEMICAL POTENTIAL	2-Acylglycerophosphocholine, formation of,
LONDON-STECK PLOT	FUGACITY	PHOSPHOLIPASE A ₁
ARRHENIUS EQUATION	LATENT ACTIVITY	1-Acyl-sn-glycero-3-phosphoethanolamine,
ACTIVATION, CALMODULIN-DEPENDENT	BIOMINERALIZATION	LYSOPHOSPHOLIPASE D
Activation energy,	ACTIVITY COEFFICIENT	Acyl halide reduction to alcohols (or aldehydes),
ENERGY OF ACTIVATION	ACTIVITY	BOROHYDRIDE REDUCTION
ARRHENIUS EQUATION	DEBYE-HÜCKEL LAW	Acyl hydroxamate,
MARCUS EQUATION	BIOMINERALIZATION	ACYL-PHOSPHATE INTERMEDIATE
MARCUS RATE THEORY	ACTOMYOSIN ATPase	ACYL-LYSINE DEACYLASE
Activation energy barrier,	MYOSIN	N-ACYLMANNOSAMINE KINASE
QUANTUM MECHANICAL TUNNELING	ACYL-ACYL-CARRIER-PROTEIN Δ^9 -DESATURASE	ACYLNEURAMINATE CYTIDYLYLTRANSFERASE
	Acyl-adenylate,	
	LUCIFERASE	
	ACYLAL INTERMEDIATE	
	ACYLAMINOACYL-PEPTIDASE	

N-ACYLNEURAMINATE 9-PHOSPHATASE	Adenosine 5'-diphosphate-2',3'-dialdehyde, MALAPRADE REACTION (PERIODATE OXIDATION)	NOREPINEPHRINE N-METHYLTRANSFERASE
N-ACYLNEURAMINATE 9-PHOSPHATE SYNTHASE	ADENOSINE KINASE	PHOSPHATIDYLETHANOLAMINE N-METHYLTRANSFERASE
Acyloxyacyl hydrolase, LIPOPOLYSACCHARIDE DEACYLATING ENZYME	Adenosine 5'-monophosphate, AMP	PUTRESCINE N-METHYLTRANSFERASE
ACYL PHOSPHATASE	Adenosine 5'-triphosphate, ATP	rRNA (ADENINE-6-)-METHYLTRANSFERASE
ACYL-PHOSPHATE-HEXOSE PHOSPHOTRANSFERASE	Adenosine 5'-triphosphate-2',3'-dialdehyde, MALAPRADE REACTION (PERIODATE OXIDATION)	rRNA (GUANINE-1-)-METHYLTRANSFERASE
ACYL-PHOSPHATE INTERMEDIATE D-ALANINE-D-ALANINE LIGASE	S-Adenosylhomocysteine, MAGNESIUM-PROTOPORPHYRIN METHYLTRANSFERASE	S-ADENOSYLMETHIONINE DECARBOXYLASE
MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA	METHYL TRANSFER REACTIONS USING S-ADENOSYLMETHIONINE	S-Adenosylmethionine decarboxylase reduction, BOROHYDRIDE REDUCTION
Acyl phosphate reduction, BOROHYDRIDE REDUCTION	NOREPINEPHRINE N-METHYLTRANSFERASE	S-Adenosylmethionine:guanidinoacetate, N-METHYLTRANSFERASE GUANIDINOACETATE METHYLTRANSFERASE
ACYL-SERINE INTERMEDIATE ACTIVE SITE TITRATION	S-ADENOSYLMETHIONINE LYASE	S-ADENOSYLMETHIONINE HYDROLASE
Acyl transfer from a peptidyl-tRNA, PEPTIDYL TRANSFERASE	S-ADENOSYLMETHIONINE NUCLEOSIDASE	S-Adenosylmethionine:sterol methyltransferase, Δ^{24} -STEROL METHYLTRANSFERASE
ADAIR CONSTANTS	METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	S-Adenosylmethionine synthetase, METHIONINE ADENOSYLTRANSFERASE
ADAIR EQUATION	S-ADENOSYL-L-METHIONINE	ADENYLATE CYCLASE (or, ADENYL CYCLASE)
ALLOSTERISM	METHYL TRANSFER REACTIONS	Adenylate energy charge, ENERGY CHARGE
COOPERATIVITY	S-Adenosyl-L-methionine as a substrate, S-ADENOSYLMETHIONINE DECARBOXYLASE	ADENYLATE ISOPENTENYLTRANSFERASE
KOSHLAND-NEMETHY-FILMER MODEL	S-ADENOSYLMETHIONINE HYDROLASE	ADENYLATE KINASE (or, MYOKINASE)
Adair model, COOPERATIVITY	1-AMINOCYCLOPROPANE-1-CARBOXYLATE SYNTHASE	METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS
Ada methyltransferase, DNA REPAIR METHYLTRANSFERASES	CATECHOL O-METHYLTRANSFERASE	ENERGY CHARGE
β -Addition of water, CROTONASE SUPERFAMILY	COBALAMIN ADENOSYLTRANSFERASE	5'-Adenylic acid aminohydrolase, AMP AMINOHYDROLASE
ADDITION REACTION	COLUMBAMINE O-METHYLTRANSFERASE	ADENYLOSUCCINATE LYASE
Additive feedback inhibition, FEEDBACK INHIBITION	DNA (CYTOSINE-5-)-METHYLTRANSFERASE	ADENYLOSUCCINATE SYNTHETASE
ADDITIVITY PRINCIPLE	GLYCINE METHYLTRANSFERASE	Adenylyl cyclase, ADENYLATE CYCLASE
ADDUCT	HISTAMINE N-METHYLTRANSFERASE	ADENYLYLSULFATE-AMMONIA ADENYLYLTRANSFERASE
COMPLEX	HOMOCYSTEINE METHYLTRANSFERASE	ADENYLYLSULFATE KINASE
LEWIS ADDUCT	HYDROXYINDOLE O-METHYLTRANSFERASE	ADENYLYLSULFATE REDUCTASE
MEISENHEIMER ADDUCT	LYSINE 2,3-AMINOMUTASE	ADENYLYLSULFATE SULFOHYDROLASE
Adenine, METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	MAGNESIUM-PROTOPORPHYRIN METHYLTRANSFERASE	Adhesion, ACCRETION
ADENINE DEAMINASE	METHIONINE ADENOSYLTRANSFERASE	BIOMINERALIZATION
ADENINE NUCLEOTIDE TRANSLOCASE	N ^F -METHYLTETRAHYDROFOLATE: HOMOCYSTEINE METHYLTRANSFERASE	Adiabatic compression and dilatation in a sound wave, CHEMICAL KINETICS
ADENINE PHOSPHORIBOSYLTRANSFERASE	mRNA (GUANINE-7-)-METHYLTRANSFERASE	ADIABATIC PROCESS
Adenosine 3',5'-bisphosphate synthesis, 3'-PHOSPHOADENYLYL SULFATE SULFOHYDROLASE	NICOTINAMIDE N-METHYLTRANSFERASE	ISOTHERMAL PROCESS
ADENOSINE DEAMINASE		HEAT CAPACITY
Adenosine 5'-diphosphate, ADP		

ADP, ABORTIVE COMPLEXES ADENYLATE KINASE MAGNESIUM ION (INTRACELLULAR)	AEQUORIN FURA-2 Aerobic deamination of activated primary amines, REDOX-ACTIVE AMINO ACIDS Affinity chromatography of NAD ⁺ , NICOTINIC ACID: ANALOGUES AND CO-ENZYMES AFFINITY LABELING AGARITINE γ -GLUTAMYLTRANSFERASE γ -GLUTAMYLTRANSPEPTIDASE Aggregation, ENZYME CONCENTRATION EFFECTS MICELLES "AGIRE," COMPUTER ALGORITHMS & SOFTWARE AGMATINE DEIMINASE AGONIST RECEPTOR-AGONIST ACTION Agostic, PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE AICAR transformylase, PHOSPHORIBOSYLAMINOIMIDAZOLE-CARBOXAMIDE FORMYLTRANSFERASE L-Alanine, ALANINE AMINOTRANSFERASE γ -GLUTAMYL TRANSPEPTIDASE L-SELENOCYSTEINE β -LYASE D-ALANINE-D-ALANINE LIGASE ALANINE AMINOTRANSFERASE ALANINE:GLYOXYLATE AMINOTRANSFERASE D-ALANINE AMINOTRANSFERASE D-AMINO ACID AMINOTRANSFERASE β -ALANINE AMINOTRANSFERASE ALANINE DEHYDROGENASE ALANINE:GLYOXYLATE AMINOTRANSFERASE SERINE:PYRUVATE AMINOTRANSFERASE ALANINE AMINOTRANSFERASE D-ALANINE HYDROXYMETHYLTRANSFERASE ALANINE RACEMASE β -ALANOPINE DEHYDROGENASE D-ALANYL-D-ALANINE PEPTIDASE D-Alanyl-D-alanine synthase, ARSENOLYSIS β -ALANYL-CoA AMMONIA LYASE O-ALANYLPHOSPHATIDYLGLYCEROL SYNTHASE	ALBERRY DIAGRAM ALBUMIN ALCOHOL DEHYDROGENASE ALCOHOL DEHYDROGENASE (QUINOLINE-DEPENDENT) ALCOHOL OXIDASE ALCOHOL SULFOTRANSFERASE Alcoholysis, PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE ALDEHYDE DEHYDROGENASE ALDEHYDE HYDRATION ALDEHYDE OXIDASE ALDEHYDE OXIDOREDUCTASE ALDOSE REDUCTASE Aldehyde reduction to alcohols, BOROHYDRIDE REDUCTION ALDOLASE Aldolase reduction, BOROHYDRIDE REDUCTION ALDOL CONDENSATION ALDOSE 1-EPIMERASE ALDOSE REDUCTASE ALGINATE LYASE ALKALINE PHOSPHATASE, ATP & GTP DEPLETION ALKANE HYDROXYLASE 1-Alkenylglycerophosphocholine acyltransferase, PLASMALOGEN SYNTHASE ALKENYLGLYCEROPHOSPHOCHOLINE HYDROLASE ALKENYLGLYCEROPHOSPHOETHANOLAMINE HYDROLASE 1-ALKYL-2-ACETYLGLYCEROL CHOLINE PHOSPHOTRANSFERASE 1-ALKYL-2-ACETYLGLYCEROPHOSPHOCHOLINE ESTERASE Alkylation, AFFINITY LABELING S-ALKYLCYSTEINE LYASE S-Alkylcysteine sulfoxide lyase, ALLIIN LYASE ALKYLDIHYDROXYACETONE-PHOSPHATE SYNTHASE ALKYLGLYCEROL KINASE 1-ALKYLGLYCEROPHOSPHOCHOLINE O-ACYLTRANSFERASE Alkylglycerophosphoethanolamine phosphodiesterase, LYSOPHOSPHOLIPASE D N-Alkylimino ester reduction, BOROHYDRIDE REDUCTION
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1-ALKYL-2-LYSOGLYCERO-3-PHOSPHOCHOLINE O-ACETYLTRANSFERASE	MEERWEIN-PONDRORFF-VERLEY REDUCTION	β -AMINOISOBUTYRATE AMINOTRANSFERASE
ALLANTOICASE	Aluminum isopropoxide,	5-AMINOLEVULINATE AMINOTRANSFERASE
ALLANTOINASE	MEERWEIN-PONDRORFF-VERLEY REDUCTION	5-AMINOLEVULINATE DEHYDRATASE
ALLENE OXIDE CYCLASE	AMADORI REARRANGEMENT	5-AMINOLEVULINATE SYNTHASE
ALLENE OXIDE SYNTHASE	"AMBER,"	AMINOMALONATE DECARBOXYLASE
ALLIIN LYASE	COMPUTER ALGORITHMS & SOFTWARE	(R)-3-Amino-2-methylpropionate-pyruvate aminotransferase,
ALLOPHANATE HYDROLASE	"AMBER/OPLS,"	β -AMINOISOBUTYRATE AMINOTRANSFERASE
All-or-none cooperativity,	COMPUTER ALGORITHMS & SOFTWARE	5-Aminopentanoate,
HILL EQUATION & PLOT	AMBIDENT NUCLEOPHILE	PROLINE REDUCTASE
ALLOSE KINASE	ω -AMIDASE	AMINOPEPTIDASE
ALLOSTERY	AMIDOPHOSPHORIBOSYLTRANSFERASE	See also specific aminopeptidases
ADAIR EQUATION	Amine dehydrogenase,	AMINOPEPTIDASE P
COOPERATIVE LIGAND BINDING	METHYLAMINE DEHYDROGENASE	Aminopropyltransferase,
HILL EQUATION & PLOT	AMINE OXIDASES	SPERMIDINE SYNTHASE
KOSHLAND-NEMETHY-FILMER MODEL	AMINE SULFOTRANSFERASE	Ammonia as a substrate or product,
MONOD-WYMAN-CHANGEUX MODEL	Amino acid acetyltransferase,	ADENINE DEAMINASE
NEGATIVE COOPERATIVITY	N-ACETYLGLUTAMATE SYNTHASE	ADENOSINE DEAMINASE
POSITIVE COOPERATIVITY	D-AMINO ACID AMINOTRANSFERASE	ω -AMIDASE
HEMOGLOBIN	D-ALANINE AMINOTRANSFERASE	D-AMINO ACID OXIDASE
ALLOTOPIC EFFECT	D-AMINO ACID DEHYDROGENASE	L-AMINO ACID OXIDASE
Allowed electronic transitions,	L-AMINO ACID DEHYDROGENASE	AMP DEAMINASE
JABLONSKI DIAGRAM	D-AMINO ACID OXIDASE	ASPARAGINASE
ALLYLIC SUBSTITUTION REACTION	L-AMINO ACID OXIDASE	ASPARAGINE SYNTHETASE
ALPHA (α)	ω -AMINO ACID-PYRUVATE AMINOTRANSFERASE	CARBAMOYL PHOSPHATE SYNTHETASE
ALPHA-ADDITION (α -ADDITION)	AMINO ACID RACEMASE	GLUTAMATE DEHYDROGENASE
ALPHA CLEAVAGE (α -CLEAVAGE)	Amino acids, peptides & proteins,	GLUTAMINASE
ALPHA EFFECT	BIOCHEMICAL NOMENCLATURE	GLUTAMINE SYNTHETASE
ALPHA-ELIMINATION (α -ELIMINATION)	AMINO ACID TURNOVER KINETICS	γ -GLUTAMYL TRANSPEPTIDASE
ALPHA EXPULSION	AMINOACYLASE	GLYCINE CLEAVAGE COMPLEX
ALPHA PARTICLE	AMINOACYL-tRNA HYDROLASE	LEUCINE DEHYDROGENASE
Alternant,	Aminoacyl-tRNA synthetase,	LYSINE DEHYDROGENASE (α -DEAMINATING)
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	EDITING MECHANISMS	LYSYL OXIDASE
ALTERNATIVE HYPOTHESIS	AMINOACYL-tRNA SYNTHETASES	METHANE MONOOXYGENASE
STATISTICS (A PRIMER)	2-AMINOADIPATE AMINOTRANSFERASE	METHIONINE γ -LYASE
ALTERNATIVE PRODUCT INHIBITION	2-AMINOADIPATE 6-SEMIALDEHYDE DEHYDROGENASE	METHYLAMINE DEHYDROGENASE
ABORTIVE COMPLEXES	4-Aminobutanal, synthesis of,	3-METHYLASPARTATE AMMONIA-LYASE
ALTERNATIVE SUBSTRATES	PUTRESCINE OXIDASE	N-METHYLGLUTAMATE SYNTHASE
COMPETITIVE INHIBITOR	4-AMINO BUTYRATE AMINOTRANSFERASE	UREASE
ABORTIVE COMPLEXES	α -AMINO- ϵ -CAPROLACTAM RACEMASE	Ammonia, autoprotolysis constant,
MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA	1-AMINOCYCLOPROPANE-1-CARBOXYLATE SYNTHASE	AUTOPROTOLYSIS
MEMBRANE TRANSPORT	2-AMINO-4-HYDROXY-6-HYDROXYMETHYLDIHYDROPTERIDINE PYROPHOSPHOKINASE	Ammonium ion,
ENERGY OF ACTIVATION	5-Amino-4-imidazole carboxamide ribonucleotide transformylase,	See Ammonia as a substrate or product
Q_{10}	PHOSPHORIBOSYLAMINOIMIDAZOLE-CARBOXAMIDE FORMYLTRANSFERASE	AMOUNT-OF-SUBSTANCE
ARRHENIUS EQUATION		CONCENTRATION
ALTRONATE DEHYDRATASE		AMP
Aluminum,		ADENYLATE KINASE
IONIC RADIUS		AMP DEAMINASE
Aluminum alkoxides,		

ASPARAGINE SYNTHETASE	Angiotensinogenase,	ANTIBONDING ORBITAL
CARNOSINE SYNTHETASE	RENIN	MOLECULAR ORBITAL
LEUCYL-tRNA SYNTHETASE	Angle strain,	BONDING ORBITAL
LONG-CHAIN FATTY ACYL-CoA SYNTHETASE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	ORBITAL
METHIONYL-tRNA SYNTHETASE	ANGULAR MOMENTUM	Antichymotrypsin,
NITROGENASE	8-Anilino-1-naphthalene sulfonate,	SERPINS (INHIBITORY MECHANISM)
PYRUVATE, ORTHOPHOSPHATE DIKINASE	FLUORESCENCE	Anticlinal,
PYRUVATE, WATER DIKINASE	ANION EFFECTS ON pH-RATE DATA	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
AMP DEAMINASE	pH STUDIES	ANTICOOPERATIVITY
Amperometry	ACTIVATION	LIGAND BINDING COOPERATIVITY
MEDIATED AMPEROMETRIC BIO-SENSOR	OPTIMUM pH	NEGATIVE COOPERATIVITY
AMPHIPHILIC	Anion radical,	Antiperiplanar,
AMPHIPROTIC	RADICAL ANION	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
PROTOGENIC	ANISOTROPIC	ANTIPORT
Amphiprotic molecular entities,	ISOTROPIC	UNIPORT
AUTOPROTOLYSIS	Anisotropic diffusion coefficients,	SYMPORT
Amphiprotic solvent,	DIFFUSION WITHIN CELLULAR/EXTRA-CELLULAR SPACES	MEMBRANE TRANSPORT
AUTOPROTOLYSIS	ANNULATION (and ANNELATION)	Anti-Stokes shift,
AMPHOTERIC	Annulene,	STOKES SHIFT
AMPLIFICATION	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	Anti-streptavidin antibody,
ENZYME CASCADE KINETICS	ANOMER,	BIOTIN AND DERIVATIVES
AMPLITUDE	FISCHER PROJECTION	AO reductase,
AMP NUCLEOSIDASE	EPIMERS	NAD(P)H DEHYDROGENASE (QUINONE)
AMYLASES	DIASTEREOMERS	ap,
LIMIT DEXTRAN	ANOMERIC EFFECT	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
γ -Amylases,	ANS,	Ap ₄ A & Ap ₅ A,
GLUCOAMYLASE	FLUORESCENCE	ADENYLATE KINASE
Amylo-1,6-glucosidase,	ANTAGONIST	Apical,
DEBRANCHING ENZYME	ANTARAFACIAL & SUPRAFACIAL MIGRATIONS	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
AMYLOSUCRASE	ANTARAFACIAL & SUPRAFACIAL REACTIONS	Apical and equatorial ligand interchange,
Anaerobic fermentation rate,	ANTARAFACIAL & SUPRAFACIAL REACTIONS	PSEUDOROTATION
MEYERHOF OXIDATION QUOTIENT	ANTHOCYANIDIN SYNTHASE	APICOPHILICITY
ANALOG COMPUTER SIMULATION	ANTHRANILATE 1,2-DIOXYGENASE	APOACTIVATOR
COMPUTER SIMULATION; KINSIM/FITSIM	ANTHRANILATE HYDROXYLASE	Apolar side-chain interactions,
Analogues of naturally occurring peptide bonds,	ANTHRANILATE PHOSPHORIBOSYLTRANSFERASE	HYDROPHOBIC EFFECT
PEPTIDOMIMETIC COMPOUND	ANTHRANILATE SYNTHASE	Apparent equilibrium constant,
Ananain,	ANTI	ATP, THERMODYNAMICS OF HYDROLYSIS
BROMELAIN	ANTIAROMATICITY	Apparent reaction order for microtubule nucleation,
ANAPLEROTIC	AROMATIC	MICROTUBULE ASSEMBLY KINETICS
ANATION	ANTI-ARRHENIUS BEHAVIOR	APROTIC
ANCHIMERIC ASSISTANCE	ARRHENIUS EQUATION	A PRIORI PROBABILITY
NEIGHBORING GROUP MECHANISM	ARRHENIUS PLOT, NONLINEAR	Aps kinase,
ANCHOR PRINCIPLE	Antibiosis,	ADENYLYLSULFATE KINASE
ANGIOTENSIN CONVERTING ENZYME	D-ALANINE-D-ALANINE LIGASE	APYRASE
Angiotensin-forming enzyme,	ANTIBODY-HAPTEN INTERACTIONS	AQUACOBALAMIN REDUCTASE
RENIN		AQUAMOLALITY
		CONCENTRATION

AQUATION	ENERGY OF ACTIVATION	Assembly-induced GTP hydrolysis,
HYDRATION: WATER	Q_{10}	MICROTUBULE ASSEMBLY KINETICS
α -L-ARABINOFURANOSIDASE	ARRHENIUS PLOTS (NONLINEAR)	Assimilatory nitrate reductase,
D-ARABINOKINASE	ARRHENIUS EQUATION & PLOT	NITRATE REDUCTASE
ARABINONATE DEHYDRATASE	Q_{10}	Assistance,
ARABINOSE ISOMERASE	TRANSITION-STATE THEORY	NEIGHBORING GROUP MECHANISM
ARABINOSE-5-PHOSPHATE ISOMERASE	VAN'T HOFF RELATIONSHIP	ASSOCIATION
Arachidonate 5-lipoxygenase,	Arrhenius treatment,	ASSOCIATION CONSTANT
5-LIPOXYGENASE	CHEMICAL KINETICS	DISSOCIATION CONSTANT
ARACHIDONYL-CoA SYNTHETASE	Arsenate,	STABILITY CONSTANT
Arene oxide intermediate,	ARSENOLYSIS	Association/dissociation rate constants, de-
NIH SHIFT	OXYGEN, OXIDES & OXO ANIONS	termination,
ARGINASE	ARSENOLYSIS	FLUORESCENCE
L-Arginine,	ARYLALKYLAMINE N-ACETYLTRANS-	ASTACIN
D-NOPALINE SYNTHASE	FERASE	ASYMMETRIC INDUCTION
NITRIC OXIDE SYNTHASE	ARYLAMINE N-ACETYLTRANSFERASE	ASYMMETRY PARAMETER
Arginine carboxypeptidase,	Arylamine sulfotransferase,	ASYMPTOTE
CARBOXYPEPTIDASE N	AMINE SULFOTRANSFERASE	ATCase,
ARGININE DECARBOXYLASE	ARYLFORMAMIDASE	ASPARTATE CARBAMOYLTRANSFERASE
ARGININE DEIMINASE	ARYLSULFATASES	ATMOSPHERE
Arginine:glycine amidinotransferase,	ARYLSULFOTRANSFERASE	ATOM PERCENT EXCESS
GLYCINE AMIDINOTRANSFERASE	ASCORBATE-CYTOCHROME b_5 RE-	TRACER/TRACEE RATIO
ARGININE KINASE	DUCTASE	COMPARTMENTAL ANALYSIS
ARGININE MONOOXYGENASE	ASCORBATE OXIDASE	ISOTOPE EXCHANGE KINETICS
ARGININE RACEMASE	ASCORBATE PEROXIDASE	ATOMIC MASS UNIT
L-Arginine synthesis,	ASPARAGINASE	ATOMIC ORBITAL
OCTOPINE DEHYDROGENASE	ASPARAGINE AMINOTRANSFERASE	ATOMIZATION
ARGININOSUCCINATE LYASE	ASPARAGINE SYNTHETASE	ATP
ARGININOSUCCINATE SYNTHETASE	1-Aspartamido- β -N-acetylglucosamine ami-	ATPase, Ca^{2+}
ARISTOLOCHENE SYNTHASE	dohydrolase,	ATPase, F_1F_o ,
Arogenate dehydrogenase,	ASPARTYLGLUCOSAMINIDASE	F_1F_o ATPase
CYCLOHEXADIENYL DEHYDROGENASE	ASPARTATE AMINOTRANSFERASE	BINDING CHANGE MECHANISAM
AROMATIC	ASPARTATE AMMONIA-LYASE	CF_1CF_o ATPase
ANTIAROMATICITY	ASPARTATE CARBAMOYLTRANSFERASE	ATPase, H^+/K^+
HÜCKEL'S RULE	ASPARTATE α -DECARBOXYLASE	ATPase, Na^+/K^+
AROMATIC AMINO ACID AMINOTRANS-	ASPARTATE β -DECARBOXYLASE	ATPase reaction,
FERASE	ASPARTATE KINASE	ION PUMPS
AROMATIC AMINO ACID DECARBOX-	D-ASPARTATE OXIDASE	ATP CITRATE LYASE
YLASE	ASPARTATE RACEMASE	ATP-dependent enzymes,
AROMATIC AMINO ACID-GLYOXYLATE	ASPARTATE-SEMIALDEHYDE DEHYDRO-	ACETATE KINASE
AMINOTRANSFERASE	GENASE	ACETYL-CoA CARBOXYLASE
Aromatic amino acid hydroxylases,	Aspartate transcarbamylase,	ACETYL-CoA SYNTHETASE
PHENYLALANINE HYDROXYLASE	ASPARTATE CARBAMOYLTRANSFERASE	N-ACETYLGUCOSAMINE KINASE
TYROSINE HYDROXYLASE	ASPARTIC PROTEASES	ACTIN ATPase
TRYPTOPHAN HYDROXYLASE	β -ASPARTYL-N-ACETYLGUCOSAM-	ACTOMYOSIN ATPase
ARRHENIUS CONSTANT	INIDASE	N-ACYLMANNOSAMINE KINASE
ARRHENIUS EQUATION & PLOT	β -ASPARTYLDIPEPTIDASE	ADENINE NUCLEOTIDE TRANSLOCASE
ARRHENIUS EQUATION & PLOT	ASPARTYL-tRNA SYNTHETASE	ADENOSINE KINASE
BOLTZMANN DISTRIBUTION	AMINOACYL-RNA SYNTHETASES	ADENYLATE KINASE (MYOKINASE)
COLLISION THEORY	Aspergillopepsin,	ADENYLYLSULFATE KINASE
TEMPERATURE DEPENDENCY,	PROTEINASE A	D-ALANINE-D-ALANINE LIGASE
TRANSITION-STATE THEORY		

ALKYLGLYCEROL KINASE	FMN ADENYLYLTRANSFERASE	PHOSPHOFRUCTOKINASE
ALLOPHANATE HYDROLASE	10-FORMYLTETRAHYDROFOLATE SYN- THETASE	3-PHOSPHOGLYCERATE KINASE
ALLOSE KINASE	FRUCTOKINASE	5-PHOSPHOMEVALONATE KINASE
AMINOACYL-tRNA SYNTHETASES	GALACTOKINASE	PHOSPHORYLASE KINASE
2-AMINO-4-HYDROXY-6-HYDROXY- METHYLDIHYDROPTERIDINE PYRO- PHOSPHOKINASE	GENTAMICIN 2"-NUCLEOTIDYLTRANS- FERASE	POLYNUCLEOTIDE 5'-HYDROXYL KINASE
APYRASE	GERANOYL-CoA CARBOXYLASE	POLYPHOSPHATE KINASE
D-ARABINOKINASE	GLUCOKINASE	PROPIONYL-CoA CARBOXYLASE
ARACHIDONYL-CoA SYNTHETASE	GLUCURONOKINASE	PROPIONYL-CoA SYNTHETASE
ARGININE KINASE	GLUTAMINE SYNTHETASE	PROTEASE La
ARGININOSUCCINATE SYNTHETASE	γ-GLUTAMYL-CYSTEINE SYNTHETASE	PYRIDOXAL KINASE
ASPARAGINE SYNTHETASE	GLUTATHIONE SYNTHETASE	PYROPHOSPHOMEVALONATE DECAR- BOXYLASE
ASPARTATE KINASE	GLYCERATE KINASE	PYRUVATE CARBOXYLASE
ASPARTYL-tRNA SYNTHETASE	GLYCEROL KINASE	PYRUVATE DEHYDROGENASE KINASE
ATPase, Ca ²⁺	GLYCERONE KINASE	PYRUVATE KINASE
ATPase, F ₁ F ₀	GMP SYNTHETASE	PYRUVATE, WATER DIKINASE
ATPase, H ⁺ /K ⁺	GUANYLATE KINASE	RESTRICTION ENZYMES
ATPase, Na ⁺ /K ⁺	HEXOKINASE	RHODOPSIN KINASE
ATP CITRATE LYASE	HISTONE KINASE	RIBOFLAVIN KINASE
ATP:GLUCOSE-1-PHOSPHATE ADENYL- YLTRANSFERASE	β-HYDROXY-β-METHYLGLUTARYL-CoA REDUCTASE KINASE	RIBONUCLEOTIDE REDUCTASE
ATP PHOSPHORIBOSYLTRANSFERASE	5-HYDROXYMETHYLPYRIMIDINE KINASE	RIBOSEPHOSPHATE PYROPHOSPHO- KINASE
BENZOATE-CoA LIGASE	HYPOTAUROCYNAMINE KINASE	RNA POLYMERASES
BIOTIN HOLOCARBOXYLASE SYN- THETASE	INOSINE KINASE	SELENOPHOSPHATE SYNTHETASE
CARBAMOYL PHOSPHATE SYNTHETASE I and II	INOSITOL-1,3,4,6-TETRAPHOSPHATE 5-KINASE	SHIKIMATE KINASE
CARNOSINE SYNTHETASE	INOSITOL-3,4,5,6-TETRAPHOSPHATE 1-KINASE	SULFATE ADENYLYLTRANSFERASE
CHAPERONES	INOSITOL-1,4,5-TRISPHOSPHATE 3-KINASE	TUBULIN:TYROSINE LIGASE
CHOLINE KINASE	LEUCYL-tRNA SYNTHETASE	UBIQUITIN-PROTEIN LIGASES
CHOLOYL-CoA SYNTHETASE	LONG-CHAIN FATTY ACYL-CoA SYN- THETASE	D-XYLULOXOKINASE
COBALAMIN ADENOSYLTRANSFERASE	LUCIFERASE	ATP-dependent protease La, PROTEASE La
4-COUMAROYL-CoA SYNTHETASE	5,10-METHENYLTETRAHYDROFOLATE SYNTHETASE	ATP-dependent racemization, PHENYLALANINE RACEMASE
CREATINE KINASE	METHIONINE ADENOSYLTRANSFERASE	ATP-dependent serine proteinase, PROTEASE La
CTP SYNTHETASE	METHIONYL-tRNA SYNTHETASE	ATP depletion, ATP & GTP DEPLETION
CYTIDYLATE KINASE	β-METHYLCROTONYL-CoA CARBOX- YLASE	ATP:GLUCOSE-1-PHOSPHATE ADENYLYL- TRANSFERASE
2-DEHYDRO-3-DEOXYGLUCONOKINASE	MEVALONATE KINASE	ATP & GTP DEPLETION
DEHYDROGLUCONOKINASE	MYOSIN and MYOSIN ATPase	ATP hydrolysis-dependent conformational change, ION PUMPS
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DEOXYADENYLATE KINASE	NAD ⁺ SYNTHETASE	ATP-regenerating enzymes, ENERGY CHARGE
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(DEOXY)NUCLEOSIDE MONOPHOS- PHATE KINASE	NUCLEOSIDE DIPHOSPHATE KINASE	ATP sulfurylase, SULFATE ADENYLYLTRANSFERASE
DEOXYTHYMIDINE KINASE	5-OXOPROLINASE	
DEPHOSPHO-CoA KINASE	PHENYLALANINE RACEMASE	
DETHIOBIOTIN SYNTHASE	1-PHOSPHATIDYLINOSITOL 4-KINASE	
DIACYLGLYCEROL KINASE	PHOSPHATIDYLINOSITOL-4-PHOSPHATE 5-KINASE	
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CERAMIDASE	CRYPTAND	Chirality sense, <i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>
CERAMIDE:CHOLINEPHOSPHOTRANSFERASE	Cheletropic reaction, <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	CHIRAL METHYL GROUPS
CEREBROSIDE SULFATASE	CHELEX	Chiral purity, <i>STEREOCHEMICAL PURITY</i>
CERULOPLASMIN <i>FERROXIDASE</i>	CHEMICAL BOND	Chiroptic/chiroptical, <i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>
CF ₃ CF ₃ ATPase	Chemical exchange, <i>MAGNETIZATION TRANSFER</i>	CHI-SQUARE DISTRIBUTION
cGMP-DEPENDENT PROTEIN KINASE	CHEMICAL EXCHANGE EFFECT ON NMR SPECTRUM	CHITINASE
cGMP PHOSPHODIESTERASE	CHEMICAL FLUX	CHITIN DEACETYLASE
CHA METHOD <i>RAPID EQUILIBRIUM ASSUMPTION</i>	CHEMICAL KINETICS	CHITIN SYNTHASE
<i>ENZYME RATE EQUATIONS (2.)</i>	Chemically induced dynamic nuclear polarization (CIDNP), <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	CHLORAMPHENICOL ACETYLTRANSFERASE
Chain initiation, <i>INITIATION</i>	CHEMICAL POTENTIAL <i>CHEMICAL THERMODYNAMICS</i>	CHLOROPEROXIDASE
<i>CHAIN REACTION</i>	<i>DEBYE-HÜCKEL MODEL</i>	CHLOROPHYLLASE
<i>TERMINATION</i>	<i>GIBBS FREE ENERGY</i>	Chloroplast, <i>PHOTOSYNTHESIS</i>
CHAIN PROPAGATION	<i>ATP, THERMODYNAMICS OF HYDROLYSIS</i>	Chloroplast ATP synthase, <i>CF₃CF₃ ATPase</i>
CHAIN REACTION	CHEMICAL RESCUE	
CHAIN RULE		

CHOLERA TOXIN <i>See also</i> ADP-RIBOSYLATION	Chymotrypsin hydrolysis of <i>p</i> -nitrophenyl ethylcarbonate or <i>p</i> -nitrophenyl acetate, <i>BURST KINETICS</i>	CLOSTRIPAIN C.M.C.
Cholesterol acyltransferase, <i>ACYL-CoA:CHOLESTEROL O-ACYL-TRANSFERASE</i>	CHYMOTRYPSIN-LIKE SERINE PROTEASE SUPERFAMILY	CMP- <i>N</i> -acetylneuraminate, <i>SIALYLTRANSFERASE</i>
Cholesterol turnover, <i>LIPID TRACER KINETICS</i>	CIDNP (Chemically induced dynamic nuclear polarization), <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	CMP- <i>N</i> -acetylneuraminate- β -galactoside α -2,6-sialyltransferase, <i>SIALYLTRANSFERASE</i>
CHOLESTERYL ESTER HYDROLASE	Cine-substitution, <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	CMP- <i>N</i> -ACETYLNEURAMINATE- β -GALACTOSIDE SIALYLTRANSFERASES
CHOLINE ACETYLTRANSFERASE		COACTIVATOR
CHOLINE DEHYDROGENASE		COBALAMIN ADENOSYLTRANSFERASE
CHOLINE KINASE		Cobalamin-dependent enzymes, <i>AQUACOBALAMIN REDUCTASES</i>
CHOLINE OXIDASE		COBALAMIN ADENOSYLTRANSFERASE
Choline phosphatase, <i>PHOSPHOLIPASE D</i>	<i>trans</i> -Cinnamate, <i>PHENYLALANINE AMMONIA-LYASE</i>	ETHANOLAMINE AMMONIA-LYASE
CHOLINE-PHOSPHATE CYTIDYLYLTRANSFERASE	<i>trans</i> -CINNAMATE 4-MONOOXYGENASE	GLYCEROL DEHYDRATASE
CHOLINEPHOSPHOTRANSFERASE	CIRCADIAN RHYTHM	LEUCINE 2,3-AMINOMUTASE
Cholinesterase, <i>BUTYRYLCHOLINE ESTERASE</i>	CIRCE EFFECT	β -LYSINE 5,6-AMINOMUTASE
CHOLINE SULFATASE	CIRCULAR DICHROISM SPECTROSCOPY	3-METHYLASPARTATE AMMONIA-LYASE
CHOLINE SULFOTRANSFERASE	Cisoid conformation, <i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>	2-METHYLENEGLUTARATE MUTASE
CHOLOYL-CoA SYNTHETASE	CITRAMALATE LYASE	<i>L</i> -METHYLMALONYL-CoA MUTASE
CHONDROITIN 4-SULFOTRANSFERASE	CITRAMALYL-CoA LYASE	<i>D</i> -ORNITHINE 4,5-AMINOMUTASE
CHONDROSULFATASES	CITRATE LYASE	PROPANEDIOL DEHYDRATASE
CHORISMATE MUTASE	CITRATE SYNTHASE	COBALAMINS AND COBAMIDES (B ₁₂)
CHORISMATE SYNTHASE	<i>L</i> -Citrulline, <i>ORNITHINE TRANSCARBAMOYLASE</i>	Cobalt(III) nucleotide complex, <i>EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES</i>
Chromatin self-assembly, <i>NUCLEOSOME CORE PARTICLE SELF-ASSEMBLY</i>	<i>NITRIC OXIDE SYNTHASE</i>	COBALT NUCLEOTIDES
CHROMIUM-51 (⁵¹ Cr)	Citrulline phosphorylase, <i>ORNITHINE TRANSCARBAMOYLASE</i>	COCOONASES
Chromium ion, <i>IONIC RADIUS</i>	CLAISEN CONDENSATION	Coefficients in binomial expansion, <i>PASCAL'S TRIANGLE</i>
Chromium(III)-ATP, <i>SCREW SENSE OF METAL-NUCLEOTIDE COMPLEXES</i>	Class (A) metal ion, <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	Coenzyme A-dependent enzymes, <i>ACETALDEHYDE DEHYDROGENASE (ACETYLATING)</i>
Chromium(III) nucleotide complex, <i>EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES</i>	Class (B) metal ion, <i>PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE</i>	ACETYL-CoA ACETYLTRANSFERASE (or, THIOLEASE)
CHROMIUM-NUCLEOTIDES	CLATHRATE	ACETYL-CoA ACYLTRANSFERASE
CHROMOGENIC SUBSTRATE <i>ABSORPTION SPECTROSCOPY</i> <i>FLUORESCENCE SPECTROSCOPY</i>	<i>CAGE COMPOUND</i>	ACETYL-CoA:ACP TRANSACYLASE
CHROMOPHORE	CLELAND NOMENCLATURE	ACETYL-CoA CARBOXYLASE
CHYMASE	Clinal,	ACETYL-CoA SYNTHETASE
CHYMOPAPAIN	<i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>	<i>N</i> -ACETYLGUTAMATE SYNTHASE
CHYMOSIN (or, RENNIN)	Closed catalytic cycle, <i>HYPERCYCLE</i>	ACYL-CoA:CHOLESTEROL ACYLTRANSFERASE
CHYMOTRYPSIN <i>ACYL-SERINE INTERMEDIATE</i> <i>AFFINITY LABELING</i> <i>SERPINS (INHIBITORY MECHANISM)</i>	CLOSED SYSTEM	ACYL-CoA DEHYDROGENASES
CHYMOTRYPSIN C	Close-packing of polypeptide chains in globular proteins, <i>PACKING DENSITY</i>	ACYL-CoA DESATURASE
	<i>Clostridium oedematiens</i> β - and γ -toxins, <i>PHOSPHOLIPASE C</i>	ACYL-CoA:RETINOL ACYLTRANSFERASE
	<i>Clostridium welchii</i> α -toxin, <i>PHOSPHOLIPASE C</i>	1-ACYLGLYCEROL-3-PHOSPHATE ACYLTRANSFERASE
		1-ACYLGLYCEROL-3-PHOSPHO-CHOLINE ACYLTRANSFERASE
		β -ALANYL-CoA AMMONIA LYASE
		5-AMINOLEVULINATE SYNTHASE
		ARACHIDONYL-CoA SYNTHETASE

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 ARYLAMINE ACETYLTRANSFERASE
 ATP CITRATE LYASE
 BENZOATE-CoA LIGASE
 BIOTIN HOLOCARBOXYLASE SYNTHETASE
 BUTYRYL-CoA DEHYDROGENASE
 CARBON MONOXIDE DEHYDROGENASE
 CARNITINE ACETYLTRANSFERASE
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 CARNITINE PALMITOYLTRANSFERASE
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 CHOLINE ACETYLTRANSFERASE
 CHOLOYL-CoA SYNTHETASE
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 CYSTEINE S-CONJUGATE N-ACETYLTRANSFERASE
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 GERANOYL-CoA CARBOXYLASE
 GLUCOSAMINE ACETYLTRANSFERASE
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 3-KETOACID CoA-TRANSFERASE
 α -KETOBUTYRATE SYNTHASE
 α - and β -KETOGLUTARATE SYNTHASES

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 3-METHYLGLUTACONYL-CoA HYDRATASE
 METHYLMALONYL-CoA DECARBOXYLASE
 METHYLMALONYL-CoA EPIMERASE
 D -METHYLMALONYL-CoA HYDROLASE
 L -METHYLMALONYL-CoA MUTASE
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 DEHYDROASCORBATE REDUCTASE
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 2-DEHYDRO-3-DEOXY-D-GLUCONATE 5-DEHYDROGENASE
 2-DEHYDRO-3-DEOXY-D-GLUCONATE 6-DEHYDROGENASE
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 2-Dehydro-3-deoxy-D-pentonate aldolase,
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5-Dehydroshikimate,
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 Deoxycytidylate kinase,
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 3-DEOXY-D-manno-OCTULOSONATE ALDOLASE
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 2'-Deoxynucleoside,
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 2'-Deoxynucleoside 5'-monophosphate,
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 Diffusion and replacement of molecules
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DIHYDROLIPOAMIDE DEHYDROGENASE	BENZIMIDAZOLE PHOSPHORIBO- SYLTRANSFERASE	"DISMAN,"
DIHYDRONEOPTERIN ALDOLASE		COMPUTER ALGORITHMS & SOFTWARE
DIHYDRONEOPTERIN TRIPHOSPHATE PYROPHOSPHOHYDROLASE	6,7-Dimethyl-8-(1-D-ribityl)lumazine, RIBOFLAVIN SYNTHASE	DISPROPORTIONATION
DIHYDROOROTASE	DIMINISHED ISOTOPE EFFECT	Disrotatory,
DIHYDROOROTATE DEHYDROGENASE	Dinitrogenase,	PHYSICAL ORGANIC CHEMISTRY NO- MENCLATURE
DIHYDROOROTATE OXIDASE	NITROGENASE	DISSOCIATION
DIHYDROPTERIDINE REDUCTASE	Dinitrogenase reductase,	DISSOCIATION CONSTANT
DIHYDROPTEROATE SYNTHASE	NITROGENASE	DISSOCIATION CONSTANTS (Metal-Nucleo- tide Complexes)
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Dihydroxyacetone phosphate,	Dioxygen,	Dissociation rate of an enzyme-substrate complex,
RHAMNULOSE-1-PHOSPHATE AL- DOLASE	OXYGEN, OXIDES & OXO ANIONS	ISOTOPE TRAPPING
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DIHYDROXYACETONE SYNTHASE	DIPEPTIDYL-PEPTIDASE I	FRACTAL REACTION KINETICS
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DIHYDROXYBIPHENYL 1,2-DIOXYGENASE	Diphosphate,	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
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2,3-DIHYDROXYINDOLE 2,3-DIOXYGENASE	2,3-Diphosphoglycerate formation, PHOSPHOGLYCERATE MUTASE	PHYSICAL ORGANIC CHEMISTRY NO- MENCLATURE
3,4-DIHYDROXYPHENYLACETATE 2,3-DIOX- YGENASE	Diphosphoglyceromutase,	DISTORTION, FREE ENERGY OF DISTRIBUTION
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AROMATIC AMINO ACID DECARBOX- YLASE	Diphosphoinositide kinase,	DITHIOTHREITOL
DIHYDROXYPTHALATE DECARBOXYLASE	PHOSPHATIDYLINOSITOL-4-PHOSPHATE 5-KINASE	Disulfide rearrangements in proteins,
2,5-DIHYDROXYPYRIDINE 5,6-DIOXY- GENASE	(R)-5-Diphosphomevalonate, as substrate, PYROPHOSPHOMEVALONATE DECAR- BOXYLASE	PROTEIN DISULFIDE ISOMERASE
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 DNase,
 DEOXYRIBONUCLEASES
 DNase digestion,
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 DOMAIN
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 Donor,
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 Dose, absorbed ionizing radiation,
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 DOSE, LETHAL
 Dose-response curve,
 CONCENTRATION-RESPONSE CURVE
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 DPN hydrolase,
 NAD⁺ GLYCOHYDROLASE
 Drug,
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 Drug absorption, quantitative models for,
 PHARMACOKINETICS
 DRUG ACTION,
 DIFFUSION OF LIGAND TO RECEPTOR
 DRUG RECEPTOR
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 ACTION
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 DRUG ACTION (OPERATIONAL MODEL)
 Drug binding to receptor,
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 PHARMACOKINETICS
 Drug excretion rates,
 PHARMACOKINETICS
 Drug metabolism/physiology,
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 THYMIDYLATE SYNTHASE
 Dual substituent-parameter equation,
 PHYSICAL ORGANIC CHEMISTRY NO-
 MENCLATURE
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 DYNAMIC EQUILIBRIUM
 Dynamic instability,
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Dynamic instability of microtubules,
 MICROTUBULE ASSEMBLY KINETICS
 Dynamic light scattering,
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 DYNAMICS
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 Eclipsing strain,
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 EC Nomenclature system
 ENZYME COMMISSION NOMEN-
 CLATURE
 EC number,
 ENZYME COMMISSION NOMEN-
 CLATURE
 EDITING
 EDITING MECHANISMS
 EDTA
 Effective concentration,
 EFFECTIVE MOLARITY
 DEBYE-HÜCKEL TREATMENT
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 EFFECTOR
 MODIFIER
 ACTIVATOR
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 EFFICIENCY FUNCTION
 EF HAND
 Eigen,
 CHEMICAL KINETICS
 EIGENFUNCTION
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 ION COMPLEXATION

EINSTEIN	Electron movements across the anode-to-cathode junction,	ACTIN ASSEMBLY KINETICS
EINSTEIN-SUTHERLAND EQUATION	NOISE	MICROTUBULE ASSEMBLY KINETICS
DIFFUSION	ELECTRON-NUCLEAR DOUBLE RESONANCE (ENDOR)	ELONGATION (No Change in Polymer Number Concentration)
Eisenthal-Cornish-Bowden plot,	Electron-pair donor (or Lewis base),	"EM" algorithm,
DIRECT LINEAR PLOT	NUCLEOPHILE	COMPUTER ALGORITHMS & SOFTWARE
ELASTASE	ELECTRON SINK	EMBRYO
ELASTICITY	ELECTRON SPIN RESONANCE	EMISSION
ELASTICITY COEFFICIENT	ELECTRON TRANSFER	EMISSION SPECTRUM
ELECTRIC CHARGE	MARCUS EQUATION	ACTION SPECTRUM
ELECTRIC POTENTIAL	ELECTRODE KINETICS	EXCITATION SPECTRUM
ELECTROMOTIVE FORCE	Electron transfer mechanism,	FLUORESCENCE
ELECTROACTIVE SPECIES	OUTER SPHERE ELECTRON TRANSFER REACTION	Empirical rate equations,
ELECTROCATALYSIS	ELECTRON TRANSFER REACTIONS	CHEMICAL KINETICS
Electrocyclic reaction,	ELECTRON TRANSPORT	Enantioconvergence,
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	ELECTROPHILE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
ELECTRODE KINETICS	ELECTROPHILICITY	Enantioenriched,
ELECTROFUGES	NUCLEOPHILE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
Electrolyte behavior,	ELECTROPHILIC CATALYSIS	Enantiomer,
DEBYE-HÜCKEL TREATMENT	ELECTROPHILICITY	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
Electromeric effect,	NUCLEOPHILICITY ELECTROPHILE	Enantiomer excess/Enantiomeric excess,
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	ELECTROPHILIC SUBSTITUTION REACTION	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
ELECTROMOTIVE FORCE	Electrophoretic mobility,	Enantiomeric ratio,
ELECTRIC POTENTIAL	OFFORD PLOT	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
NERNST EQUATION	Electrostatic attraction/repulsion,	Enantiomerically enriched/Enantioenriched,
ELECTRON	DEBYE-HÜCKEL TREATMENT	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
ELECTRON ACCEPTOR	ELECTROSTATIC BOND	Enantiomerically pure/Enantiopure,
ELECTRON AFFINITY	ELECTROSTATIC SURFACE POTENTIAL	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
Electronation,	ELECTROSTRICTION	ENANTIOSELECTIVE REACTION
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	ELECTROTAXIS	ASYMMETRIC INDUCTION
Electron attachment,	ELECTROVALENT BOND	ENCOUNTER COMPLEX
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	ELEMENTARY CHARGE	ENCOUNTER-CONTROLLED RATE
ELECTRON DENSITY	ELEMENTARY REACTION	DIFFUSION CONTROL FOR BIMOLECULAR COLLISIONS
CHARGE DENSITY	Elementary reaction stoichiometry,	ENDERGONIC PROCESS
ELECTRON DENSITY FUNCTION	MOLECULARITY	ENDO- α (or β)-N-ACETYLGALACTOSAMINIDASE
ELECTRON DONOR	CHEMICAL KINETICS	ENDO- β -N-ACETYLGLUCOSAMINIDASE
ELECTRONEGATIVITY	UNIMOLECULAR	ENDOENZYME
ELECTRONEUTRALITY PRINCIPLE	BIMOLECULAR	ENDO- β -GALACTOSIDASE
Electronic transitions,	TRANSITION-STATE THEORY	Endo-1,4- β -glucanase,
JABLONSKI DIAGRAM	ELEMENTARY REACTION	CELLULOSE
FLUORESCENCE	Element effect,	Endo-1,4- β -mannanase,
ABSORPTION SPECTROSCOPY	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	MANNANASE
$n \rightarrow \pi^*$ TRANSITION	Element of chirality,	ENDONUCLEASE
$n \rightarrow \sigma^*$ TRANSITION	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	DEOXYRIBONUCLEASES
$\pi^* \rightarrow \pi^*$ TRANSITION	ELIMINAND	RESTRICTION ENZYMES
Electron microscopy,	β -Elimination of water,	
ACTIN ASSEMBLY ASSAYS	CROTONASE SUPERFAMILY	
	Elongation,	

ENDONUCLEASE III	RADIATIVE ENERGY TRANSFER	VOLUME OF ACTIVATION
ENDONUCLEASE IV	SPECTRAL OVERLAP	Entropy change for ligand binding,
ENDONUCLEASE V (PYRIMIDINE DIMER)	ENHANCER	CALORIMETRY
ENDONUCLEASE VII	ENOLASE	ENTROPY OF ACTIVATION
ENDO-OLIGOPEPTIDASE A	ENOLASE SUPERFAMILY	TRANSITION-STATE THEORY (THERMO-
Endopeptidase,	Enolate anion generated by nucleophilic aro-	DYNAMICS)
NEUROLYSIN	matic addition,	GIBBS FREE ENERGY OF ACTIVATION
ENDOPEPTIDASE Clp	CROTONASE SUPERFAMILY	ENTHALPY OF ACTIVATION
Endopeptidase La,	Enolate anions,	VOLUME OF ACTIVATION
PROTEASE La	ENOLIZATION	ENTROPY AND ENTHALPY OF ACTI-
ENDOPOLYGALACTURONIDASE	ENOLIZATION	VATION
ENDOPOLYPHOSPHATASE	2-Enolpyruvate-phosphate,	Entropy, rate of change in,
Endoskeleton and exoskeleton,	NUCLEOSIDE-5'-TRIPHOSPHATE REGEN-	NONEQUILIBRIUM THERMODYNAMICS:
BIOMINERALIZATION	ERATION	A PRIMER
ENDOTHERMIC PROCESS	5-ENOLPYRUVYLSHIKIMATE-3-PHOSPHATE	ENTROPY TRAP MODEL
ENTHALPY	SYNTHASE	Envelope conformation,
EXOTHERMIC	ENOYL-ACP REDUCTASE	STEREOCHEMICAL TERMINOLOGY,
ENDOGONIC	FATTY ACID SYNTHETASE	IUPAC RECOMMENDATIONS
ENDOERGIC	ENOYL-CoA HYDRATASE (or, CROTONASE)	Environmental noise,
ENDOTHIAPEPSIN	ENOYLPYRUVATE	NOISE
Endo-1,4- β -xylanase,	Enrichment,	ENZYME
XYLANASE	ISOTOPE ENRICHMENT	Enzyme activation,
END PRODUCT	ATOM PERCENT EXCESS.	BASAL RATE
End-product regulation,	Ensemble average fluctuation with time,	Enzyme binding interaction with polymeric
CUMULATIVE FEEDBACK INHIBITION	CORRELATION FUNCTION	substrate,
End rule,	ENTATIC STATE	PROCESSIVITY
N-END RULE	ENTERING GROUP	ENZYME CASCADE KINETICS
END-TO-END ANNEALING	ENTEROPEPTIDASE	Enzyme catalytic model,
End-wise depolymerization,	ENTEROTOXIN	ORBITAL STEERING
DEPOLYMERIZATION (END-WISE)	ENTHALPY	ENZYME CATALYSIS
Eneamine,	ENTHALPY-DRIVEN REACTION	ENZYME COMMISSION NOMENCLATURE
ENOLIZATION	ENTHALPY OF ACTIVATION	ENZYME CONCENTRATION EFFECTS
ENE REACTION	TRANSITION-STATE THEORY (THERMO-	ENZYME DEPLETION (Apparent Noncom-
Energetics of biological macromolecules,	DYNAMICS)	petitive Inhibition)
THERMODYNAMICS OF BIOLOGICAL	TRANSITION-STATE THEORY	Enzyme electrodes,
MACROMOLECULES	GIBBS FREE ENERGY OF ACTIVATION	BIOCATALYTIC ELECTRODE
ENERGY BARRIER	ENTROPY OF ACTIVATION	ENZYME ENERGETICS (The Case of Proline
ENERGY CHARGE (or, ADENYLATE EN-	VOLUME OF ACTIVATION	Racemase)
ERGY CHARGE)	ENTROPY	ENZYME EQUIVALENT WEIGHT
Energy level diagram,	GIBBS FREE ENERGY	ENZYME INHIBITION
FLUORESCENCE	ENTHALPY	Enzyme isomerization,
ENERGY OF ACTIVATION	THERMODYNAMICS	RAY-ROSCELLI PLOT
ARRHENIUS EQUATION	NONEQUILIBRIUM THERMODYNAMICS:	TRACER PERTURBATION METHOD
LEAST MOTION, PRINCIPLE OF	A PRIMER	ENZYME-LINKED IMMUNOSORBENT
MARCUS EQUATION	ENTROPY AND ENTHALPY OF ACTIVA-	ASSAYS (ELISA)
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TRANSITION-STATE THEORY	TRANSITION-STATE THEORY	Enzyme purification,
Q ₁₀	TRANSITION-STATE THEORY (THERMO-	ENZYME PURITY
ENERGY TRANSFER	DYNAMICS)	ENZYME PURITY
FÖRSTER RESONANCE EXCITATION	TRANSITION-STATE THEORY IN SOLU-	ENZYME RATE EQUATIONS (1. The Basics)
TRANSFER (FRET)	TIONS	FROMM'S METHOD FOR DERIVING EN-
	ENTROPY OF ACTIVATION	ZYME RATE EQUATIONS

NET RATE CONSTANT METHOD	KINETIC ISOTOPE EFFECT	Excluded volume effects in cytoplasm, MOLECULAR CROWDING
ENZYME RATE EQUATIONS (2. Derivation of Initial Velocity Equations)	EQUILIBRIUM PERTURBATION METHOD	Excluding kinetic reaction mechanisms, HALDANE RELATION
ENZYME RATE EQUATIONS (3. Derivation of Isotope Exchange Rate Equations)	KINETIC ISOTOPE EFFECT	EXERGONIC
Enzyme reaction intermediate, ACYL-PHOSPHATE INTERMEDIATE	SOLVENT ISOTOPE EFFECT	EXOCYTOSIS
Enzyme regulation, ABORTIVE COMPLEXES	EQUILIBRIUM THERMODYNAMICS (Measur- able Quantities)	EXOENZYME
ACTIVATION	ERG	EXOERGIC
ALLOSTERISM	ERGODIC	Exo- β -fructosidase, FRUCTAN β -FRUCTOSIDASE
COOPERATIVITY	ERGODIC PROCESS	EXO-MALTOHEXAOHYDROLASE
HYSTERESIS	ERGOTHIONASE	EXO-MALTOTETRAOHYDROLASE
INHIBITION	D-Erythrose 4-phosphate, RIBULOSE-5-PHOSPHATE 3-EPIMERASE	EXONUCLEASE
ISOSTERY	Erythrose-4-phosphate epimerase, RIBULOSE-5-PHOSPHATE 3-EPIMERASE	EXONUCLEASE III
PRODUCT INHIBITION	Escherichia coli glutamine synthetase, ENZYME CASCADE KINETICS	EXONUCLEASE V
SUBSTRATE INHIBITION	GLUTAMINE SYNTHETASE	EXOPOLYGALACTURONIDASE
Enzymes, BIOCHEMICAL NOMENCLATURE	Essential amino acid residues in catalysis, AFFINITY LABELING	EXOPOLYPHOSPHATASES
Enzymes, major classes of, ENZYME COMMISSION NOMEN- CLATURE	ESTERASES	Exoribonuclease II, RIBONUCLEASE II
ENZYME STABILITY	CHOLESTERYL ESTER HYDROLASE	Exo- α -sialidases, SIALIDASES
ENZYME-SUBSTRATE COMPLEX	ESTER HYDROLYSIS MECHANISMS	EXO-SITE
Enzymes with oligomeric/polymeric sub- strate subsites, SHIFTED BINDING	ETA (η)	Exoskeleton, BIOMINERALIZATION
EPIMERIZATION	ETHANOLAMINE AMMONIA-LYASE	EXOTHERMIC
EPITAXY	ETHANOLAMINE KINASE	ENDOTHERMIC
EPOXIDE HYDROLASE	CHOLINE KINASE	EXERGONIC
trans-EPOXYSUCCINATE HYDROLASE (or, HYDRATASE; or, TARTRATE EPOX- IDASE)	ETHANOLAMINE N-METHYLTRANSFERASE	EXOERGIC
EPSILON (ϵ)	Ethanolamine phosphate cytidyltransferase, CTP:ETHANOLAMINEPHOSPHATE CYTI- DYLTRANSFERASE	EXOTOXIN A
EQUIL SOFTWARE, BIOMINERALIZATION	ETHANOLAMINEPHOSPHOTRANSFERASE	EXPERIMENTAL DESIGN OF INITIAL RATE ENZYME ASSAYS
EQUATIONS OF STATE	ETHANOLAMINE-SERINE BASE-EXCHANGE ENZYME	COUPLED ENZYME ASSAYS
EQUI-EFFECTIVE DOSE/MOLARITY RATIO	Ethanol, autoprotolysis constant, AUTOPROTOLYSIS	CONTINUOUS VS. SINGLE-POINT ASSAYS
EQUILIBRIUM	Ethoxyformylation of proteins, DIETHYL PYROCARBONATE	PROGRESS CURVE
EQUILIBRIUM CONSTANT	Ethylene-forming enzyme, 1-AMINOCYCLOPROPANE-1- CARBOXYLATE OXIDASE	INITIAL RATE CONDITION
ASSOCIATION CONSTANT	ETIOCHOLANOLONE SULFATASE	SUBSTRATE PURITY
DISSOCIATION CONSTANT	Evolution of enzyme catalysis, INTERNAL EQUILIBRIA	ENZYME PURITY
MASS ACTION RATIO	EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES	WATER PURITY
EQUILIBRIUM CONSTANT DETERMINA- TIONS	Exchange of protein-bound nucleotide, NUCLEOTIDE EXCHANGE FACTORS	SUBSTRATE STABILITY
EQUILIBRIUM DIALYSIS	Excited singlet state, JABLONSKI DIAGRAM	ENZYME STABILITY
EQUILIBRIUM ENZYMES IN METABOLIC PATHWAYS	EXCITED STATE	MIXING TIME
Equilibrium exchange, ISOTOPE EXCHANGE AT EQUILIBRIUM	Excited state lifetime, PHASE MODULATION FLUORESCENCE	INITIAL RATE CONDITION
EQUILIBRIUM ISOTOPE EFFECT		QUENCHING
ISOTOPE EFFECT		EXPONENTIAL
		EXPONENTIAL BREAKDOWN
		Extended Debye-Hückel equation, DEBYE-HÜCKEL TREATMENT
		EXTENSIVE PROPERTY
		EXTENT OF REACTION
		RATE OF CONVERSION

RATE OF REACTION
VELOCITY
External compensation,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
External magnetic field strength effect on
molecule,
PERMEABILITY
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ABSORPTION SPECTROSCOPY
Extraneous and random disturbances,
NOISE
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Extrinsic fluorescence,
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Eyring treatment,
CHEMICAL KINETICS
E/Z CONVENTION
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F

F-430
Fab fragments,
ANTIBODY
FACILITATED TRANSPORT
MEMBRANE TRANSPORT
F-actin,
ACTIN ASSEMBLY KINETICS
BIOCHEMICAL SELF-ASSEMBLY
FAD-dependent enzymes,
ACYL-CoA DEHYDROGENASES
ADENYLYLSULFATE REDUCTASE
ALCOHOL OXIDASE
AMINE OXIDASES
D-AMINO ACID DEHYDROGENASE
L-AMINO ACID DEHYDROGENASE
D-AMINO ACID OXIDASE
L-AMINO ACID OXIDASE
D-ASPARTATE OXIDASE
BENZOATE 1,2-DIOXYGENASE
BERBERINE BRIDGE ENZYME
CHOLINE OXIDASE
CYCLOHEXANONE MONOOXYGENASE
CYTOCHROME P-450 REDUCTASE
DIHYDROLIPOAMIDE DEHYDROGENASE

DIHYDROOROTATE OXIDASE
DIMETHYLANILINE MONOOXYGENASE
FERREDOXIN:NADP⁺ REDUCTASE
GLUTATHIONE REDUCTASE
D-2-HYDROXY ACID DEHYDROGENASE
3-HYDROXYBENZOATE 6-MONOOXY-
GENASE
4-HYDROXYBENZOATE 3-MONOOXY-
GENASE
IMIDAZOLEACETATE HYDROXYLASE
ISOVALERYL-CoA DEHYDROGENASE
KYNURENINE 3-HYDROXYLASE
D-LACTATE DEHYDROGENASE (CYTO-
CHROME)
LYSINE MONOOXYGENASE
MALATE DEHYDROGENASE
MELILOTATE HYDROXYLASE
5,10-METHYLENETETRAHYDROFOLATE
REDUCTASE
5,10-METHYLENETETRAHYDROFOLATE:
tRNA (URACIL-5-)-METHYLTRANS-
FERASE (FADH₂ OXIDIZING)
METHYLHYDROXYPYRIDINECARBOXY-
LATE DIOXYGENASE
N⁵-METHYLTETRAHYDROFOLATE:
HOMOCYSTEINE METHYLTRANS-
FERASE
NADH-CYTOCHROME b₅ REDUCTASE
NADH DEHYDROGENASE
NADH DEHYDROGENASE, OTHER THAN
COMPLEX I
NITRATE REDUCTASE
NITRITE REDUCTASE
PHENOL HYDROXYLASE
PROLINE DEHYDROGENASE
PUTRESCINE OXIDASE
PYRUVATE OXIDASE
SALICYLATE 1-MONOOXYGENASE
SUCCINATE DEHYDROGENASE
SULFITE REDUCTASE
XANTHINE OXIDASE
Falling ball viscometry,
ACTIN POLYMERIZATION ASSAYS
FARADAY'S CONSTANT
NERNST EQUATION
FARADAY'S LAWS
FARNESYL-DIPHOSPHATE FARNESYL-
TRANSFERASE
Farnesyl pyrophosphate synthetase,
GERANYLTRANSTRANSFERASE
Farnesyltransferases,
PROTEIN FARNESYLTRANSFERASES
FARNESYLTRANSTRANSFERASE

Fast-atom bombardment (Fab) mass spec-
troscopy,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
Fast reaction technique,
FLASH PHOTOLYSIS
Fat absorption (intestinal),
LIPID TRACER KINETICS
FATTY ACID SYNTHASE
Fatty acid tracer kinetics,
LIPID TRACER KINETICS
Fatty acyl-CoA desaturases,
DESATURASES
Fatty acyl-CoA synthetases,
LONG-CHAIN FATTY ACYL-CoA SYN-
THETASE
MEDIUM-CHAIN FATTY ACYL-CoA SYN-
THETASE
FEEDBACK INHIBITION
CUMULATIVE FEEDBACK INHIBITION
Feedback inhibitor,
CUMULATIVE FEEDBACK INHIBITION
FEEDBACK INHIBITION
FEED-FORWARD ACTIVATION
"FELIX,"
COMPUTER ALGORITHMS & SOFTWARE
FeMo-COFACTOR
FENTON REACTION
UDENFRIEND'S REAGENT
Fermentation,
MEYERHOF OXIDATION QUOTIENT
FERREDOXIN
RESONANCE RAMAN SPECTROSCOPY
Ferredoxin-dependent enzymes,
ADRENODOXIN
BENZENE 1,2-DIOXYGENASE
FERREDOXIN:NADP⁺ REDUCTASE
GLUTAMATE SYNTHASE
HYDROGENASE
6-HYDROXYNICOTINATE REDUCTASE
 α -KETOBUTYRATE SYNTHASE
 α -KETOGLUTARATE SYNTHASE
4-METHOXYBENZOATE MONOOXY-
GENASE
NITRITE REDUCTASE
NITROGENASE
PYRUVATE SYNTHASE
RUBREDOXIN:NAD⁺ REDUCTASE
STEROID 11 β -HYDROXYLASE
SULFITE REDUCTASE
TUNGSTOENZYMES

UBIQUINOL:CYTOCHROME c OXIDORE- DUCTASE	CHEMICAL KINETICS	PHASE MODULATION FLUORESCENCE
FERREDOXIN:NADP ⁺ REDUCTASE	First-order relaxation,	FLUORESCENCE LIFETIME IMAGING MICROSCOPY
Ferric ion,	FLUORESCENCE	Fluorescence phosphate assay,
IONIC RADIUS	FISCHER PROJECTION	ORTHOPHOSPHATE CONTINUOUS ASSAY
Ferricytochrome b ₅ reduction,	Fischer-Rosanoff Convention,	FLUORESCENCE RECOVERY AFTER PHOTBLEACHING
NADH-CYTOCHROME b ₅ REDUCTASE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	Fluorimeter,
FERROCHELATASE	"FITSIM" COMPUTER PROGRAM	FLUORESCENCE
FERROMAGNETIC	KINSIM/FITSIM	FLUOROACETATE/FLUOROCITRATE
Ferromagnetic substances,	FLAGELLAR ROTATION	Fluorocitrate
PERMEABILITY	Flagpole,	FLUOROACETATE/FLUOROCITRATE
MAGNETIC SUSCEPTIBILITY	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	Fluorometric assay for avidin and biotin, BIOTIN AND DERIVATIVES
Ferrous ion,	FLASH PHOTOLYSIS	Fluorophore,
IONIC RADIUS	CHEMICAL KINETICS	FLUORESCENCE
FERROXIDASE	Flash vacuum pyrolysis (FVP),	"FLUSYS,"
[2Fe-2S] CLUSTER	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	COMPUTER ALGORITHMS & SOFTWARE
[4Fe-4S] CLUSTER	FLAVANONE 3 β -HYDROXYLASE	FLUX
F ₁ F ₀ ATPase and H ⁺ -TRANSPORTING ATPase	FLAVINS AND DERIVATIVES	NONEQUILIBRIUM THERMODYNAMICS: A PRIMER
F ₁ F ₀ ATPase mechanism,	Flavokinase,	RADIANT ENERGY FLUX
BINDING CHANGE MECHANISM	RIBOFLAVIN KINASE	RADIANT POWER
Fibrils, pathological brain-derived,	FLAVONE SYNTHASE	FLUX CONTROL IN METABOLIC PATHWAYS
PRION PLAQUE FORMATION	FLAVONOL SYNTHASE	FLUX-GENERATING STEPS
FICIN	Flicker,	Fluxional,
FICK'S LAWS OF DIFFUSION	NOISE	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
DIFFUSION	Flip-flop of lipids across membrane bilayers,	FLUX RATIO
TRANSPORT PROCESSES	PHOSPHOLIPID FLIP-FLOP	Flux through a pathway
Fiducial group,	"FLOG,"	METABOLIC CONTROL ANALYSIS
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	COMPUTER ALGORITHMS & SOFTWARE	METABOLONS
FIELD EFFECT	Flow birefringence,	META-MODEL
INDUCTIVE EFFECT	ACTIN ASSEMBLY ASSAYS	FMN,
"FIRM,"	Fluctuating system,	L-LACTATE DEHYDROGENASE (CYTOCHROME)
COMPUTER ALGORITHMS & SOFTWARE	CORRELATION FUNCTION	LACTATE MONOOXYGENASE
FIRST DERIVATIVE TEST	Fluctuation correlation,	LUCIFERASE
First law of thermodynamics,	FLUORESCENCE	4-METHOXYBENZOATE MONOOXY- GENASE
CONSERVATION OF ENERGY	FLUCTUATIONS	RIBOFLAVIN KINASE
THERMODYNAMICS, LAWS OF	Fluorouracil,	FMN ADENYLYLTRANSFERASE
First-order kinetics,	THYMIDYLATE SYNTHASE	FMN-dependent enzymes,
ISOTOPE EXCHANGE AT EQUILIBRIUM	FLUORESCENCE (1. General Principles)	CYTOCHROME P-450 REDUCTASE
First-order process/reaction,	FLUORESCENCE (2. Time-resolved Methods)	DIHYDROOROTATE OXIDASE
CHEMICAL KINETICS	FLUORESCENCE (3. Resonance Energy Transfer)	FMN ADENYLYLTRANSFERASE
First-order rate behavior,	Fluorescence changes,	GLUTAMATE SYNTHASE
AUTOPHOSPHORYLATION	ACTIN ASSEMBLY ASSAYS	GLYCOLATE OXIDASE
FIRST-ORDER REACTION KINETICS	FLUORESCENCE CORRELATION SPEC- TROSCOPY (FCS)	(S)-2-HYDROXY-ACID OXIDASE
ORDER OF REACTION	FLUORESCENCE DECAY	L-LACTATE DEHYDROGENASE (CYTOCHROME)
HALF-LIFE	FLUORESCENCE	
SECOND-ORDER REACTION	FLUORESCENCE DEPOLARIZATION	
ZERO-ORDER REACTION	Fluorescence lifetime,	
MOLECULARITY		
MICHAELIS-MENTEN EQUATION		

LUCIFERASE	5-Formyltetrahydrofolate cyclo-ligase,	FREE RADICAL
NADPH DEHYDROGENASE	5,10-METHENYLTETRAHYDROFOLATE	TRANSIENT CHEMICAL SPECIES
NITRATE REDUCTASE	SYNTHETASE	PERSISTENT
PROTOPORPHYRINOGEN OXIDASE	10-FORMYLTETRAHYDROFOLATE DE-	RADICAL
SARCOSINE DEHYDROGENASE	FORMYLASE	BIRADICAL
SULFITE REDUCTASE	10-FORMYLTETRAHYDROFOLATE DEHY-	Free rotation,
F _o F ₁ -ATPase (and RELATED PROTON-	DROGENASE	STEREOCHEMICAL TERMINOLOGY,
TRANSPORTING ATPases)	FORMYLTETRAHYDROFOLATE HY-	IUPAC RECOMMENDATIONS
Folic acid,	DROLASE	FREQUENCY
BIOCHEMICAL NOMENCLATURE	10-FORMYLTETRAHYDROFOLATE SYN-	FRET,
FOLYLPOLY- γ -GLUTAMATE SYNTHASE	THETASE	FLUORESCENCE
FOOTPRINTING	FOSTER-NIEMANN PLOTS	FRIEDEN'S DILEMMA
FORCE	ISO MECHANISMS	FRIEDEN'S METHOD FOR TER-REACTANT
FORCE EFFECTS ON MOLECULAR	FOURIER TRANSFORM	ENZYMIC KINETICS
MOTORS	FOURIER TRANSFORM IR/PHOTOACOUS-	FROMM'S METHOD FOR DERIVING EN-
Force-field calculations,	TIC SPECTROSCOPY TO ASSESS SEC-	ZYME RATE EQUATIONS
PHYSICAL ORGANIC CHEMISTRY	ONDARY STRUCTURE	FROMM'S METHOD FOR TER-REACTANT
NOMENCLATURE	Four substrate enzyme reactions,	ENZYMIC KINETICS
FORCING FUNCTION	QUADREACTANT SYSTEMS	FRONTIER ORBITALS
FORMALDEHYDE DEHYDROGENASE	"Fourth" law of thermodynamics,	FROST-SCHWEMER METHOD
FORMALITY	ADDITIVITY PRINCIPLE	1,2- β -FRUCTAN:1,2- β -FRUCTAN 1- β -FRUC-
FORMAMIDOPYRIMIDINE GLYCOSIDASE	RACTAL	TOSYLTRANSFERASE
Formate acetyltransferase,	CHAOS THEORY	FRUCTAN β -FRUCTOSIDASE
PYRUVATE FORMATE LYASE	Fractal dimensionality of diffusion,	β -Fructofuranosidases,
FORMATE DEHYDROGENASE	DIFFUSION, REDUCTION IN DIMENSION-	INVERTASE
FORMATION CONSTANT	ALITY	FRUCTOKINASE
LIGAND BINDING	RACTAL REACTION KINETICS	D-Fructose,
METAL ION COMPLEXATION	Fractional saturation,	SORBITOL DEHYDROGENASE
STABILITY CONSTANT	ADAIR EQUATION	FRUCTOSE-1,6-BISPHOSPHATASE
BIOMINERALIZATION	FRACTIONATION FACTOR	FRUCTOSE-2,6-BISPHOSPHATASE
DEGREE OF DISSOCIATION	Fraction of active enzyme,	D-Fructose 2,6-bisphosphate,
Formic acid, autoprotolysis constant,	BURST KINETICS	6-PHOSPHOFRUCTO-2-KINASE
AUTOPROTOLYSIS	FRAGMENTATION	Fructose-1,6-bisphosphate aldolase,
FORMIMINOGLUTAMASE	FRANCK-CONDON PRINCIPLE	ALDOLASE
FORMIMINOGLUTAMATE DEIMINASE	FLUORESCENCE	D-Fructose 1,6-bisphosphate, synthesis of,
Formiminoglutamate formiminotransferase,	JABLONSKI DIAGRAM	PHOSPHOFRUCTOKINASE
GLUTAMATE FORMIMINOTRANSFERASE	FRAP,	PYROPHOSPHATE FRUCTOSE-6-PHOS-
Formiminoglycine formiminotransferase,	FLUORESCENCE RECOVERY AFTER PHO-	PHATE 1-PHOSPHOTRANSFERASE
GLYCINE FORMIMINOTRANSFERASE	TOBLEACHING	D-Fructose 6-phosphate, interconversion
5-FORMIMINOTETRAHYDROFOLATE	Free energy diagrams for enzymes	with D-glucose 6-phosphate,
CYCLODEAMINASE	REACTION COORDINATE DIAGRAM	PHOSPHOGLUCOSE ISOMERASE
Formyl-CoA synthesis,	ENZYME ENERGETICS	Fructose-6-phosphate,2-kinase/fructose-2,6-
OXALYL-CoA DECARBOXYLASE	POTENTIAL-ENERGY SURFACES	phosphatase,
N-Formylglutamate transformylase,	TRANSITION-STATE THEORY	BIFUNCTIONAL ENZYME
GLUTAMATE FORMYLTRANSFERASE	ARRHENIUS EQUATION	D-FUCONATE DEHYDRATASE
FORMYLGLYCINAMIDE RIBONUCLEOTIDE	VAN'T HOFF RELATIONSHIP	FUCOSE-1-PHOSPHATE GUANYLYLTRANS-
AMIDOTRANSFERASE		FERASE
FORMYLGLYCINAMIDINE SYNTHETASE		α -L-FUCOSIDASE
FORMYLMETHANOFURAN DEHYDRO-		FUCOSYLTRANSFERASES
GENASE		FUCULOSE-1-PHOSPHATE ALDOLASE
FORMYLMETHANOFURAN:5,6,7,8-		FUMARASE
TETRAHYDROMETHANOPTERIN		
FORMYLTRANSFERASE		

FUMARATE REDUCTASE

FURA-2

CALCIUM ION INDICATOR DYES

METAL ION COMPLEXATION

FURIN

G ΔG of ATP hydrolysis,

ATP, THERMODYNAMICS OF HYDROLYSIS

 ΔG° (Relation to K_{eq})

ACTIVITY COEFFICIENTS

ADDITIVITY PRINCIPLE

BIOCHEMICAL THERMODYNAMICS

CHEMICAL POTENTIAL

EQUILIBRIUM CONSTANTS

HESS'S LAW

INNATE THERMODYNAMIC QUANTITIES

MOLECULAR CROWDING

THERMODYNAMICS, LAWS OF

THERMODYNAMIC CYCLE

THERMODYNAMIC EQUATIONS OF STATE

G-actin,

ACTIN ASSEMBLY KINETICS

 $\Delta\Delta G$ AS AN INDEX OF COOPERATIVITY

GALACTARATE DEHYDRATASE

GALACTITOL 2-DEHYDROGENASE

Galactocerebrosidase,

GALACTOSYL CERAMINIDASE

GALACTOKINASE

GALACTONATE DEHYDRATASE

GALACTOSE 1-DEHYDROGENASE

GALACTOSE OXIDASE

GALACTOSE-1-PHOSPHATE URIDYL-TRANSFERASE

Galactose-6-sulfate sulfatase,

N-ACETYLGALACTOSAMINE-6-SULFATE SULFATASE

 α -GALACTOSIDASE β -GALACTOSIDASE

GALACTOSIDE FUCOSYLTRANSFERASES

GALACTOSIDE SIALYLTRANSFERASES

 β -D-Galactosyl-1,4- β -D-glucosylceramide, SIALYLTRANSFERASE

"GAMESS,"

COMPUTER ALGORITHMS & SOFTWARE

GAMMA (γ)

Gas-phase acidity,

PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE

Gas-phase basicity,

PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE

Gauche,

STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS

Gauche effect,

STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS

GAUSS

"GAUSSIAN",

COMPUTER ALGORITHMS & SOFTWARE

GDP-Fucose synthase,

FUCOSE-1-PHOSPHATE GUANYLYL-TRANSFERASE

GDP-MANNOSE 4,6-DEHYDRATASE

GDP-MANNOSE DEHYDROGENASE

GDP-MANNOSE:DOLICHYL PHOSPHATE MANNOSYLTRANSFERASE

GDP-MANNOSYLTRANSFERASES

GEAR ALGORITHM

STIFFNESS OR STIFFNESS INSTABILITY

Gel-liquid crystalline phase transition,

CALORIMETRY

GELSOLIN

Geminate recombination,

PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE

GENERAL ACID CATALYSIS

SPECIFIC ACID CATALYSIS CATALYSIS

GENERAL BASE CATALYSIS

GENERAL BASE CATALYSIS

SPECIFIC BASE CATALYSIS

CATALYSIS

GENERAL ACID CATALYSIS

GENETIC CODE

GENTAMICIN 2"-NUCLEOTIDYLTRANSFERASE

GENTISATE 1,2-DIOXYGENASE

Geometric constraints on diffusion,

FRACTAL REACTION KINETICS

Geometric inhibitor,

D-ALANINE-D-ALANINE LIGASE

Geometric isomerism,

STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS

GEOMETRIC MEAN, RULE OF THE

GERANOYL-CoA CARBOXYLASE

Geranyl-diphosphate cyclase,

BORNYL PYROPHOSPHATE SYNTHASE

Geranyl diphosphate synthase,

DIMETHYLALLYLTRANSTRANSFERASE

Geranylgeranyl diphosphate synthase,

FARNESYLTRANSTRANSFERASE

Geranylgeranyltransferases,

PROTEIN GERANYLGERANYLTRANSFERASES

GERANYLTRANSTRANSFERASE

GIBBERELLIN 2 β -HYDROXYLASEGIBBERELLIN 3 β -HYDROXYLASE

GIBBS FREE ENERGY

BIOCHEMICAL THERMODYNAMICS

ENDERGONIC

EXERGONIC

ENTHALPY

ENTROPY

HELMHOLTZ ENERGY

THERMODYNAMICS

GIBBS FREE ENERGY DIAGRAM

POTENTIAL ENERGY PROFILE

ENZYME ENERGETICS

GIBBS FREE ENERGY OF ACTIVATION

TRANSITION-STATE THEORY (THERMODYNAMICS)

GIBBS-HELMHOLTZ EQUATION

GIBBS FREE ENERGY

GLOBAL ANALYSIS

1,4- α -Glucan branching enzyme,

BRANCHING ENZYME

GLUCAN ENDO-1,3(4)- β -GLUCANASEGLUCAN ENDO-1,2- β -GLUCOSIDASEGLUCAN ENDO-1,3- β -GLUCOSIDASEGLUCAN ENDO-1,6- β -GLUCOSIDASEGLUCAN 1,3- β -GLUCOSIDASE1,4- α -Glucanotransferase,

DEBRANCHING ENZYME

1,4- α -Glucan:orthophosphate α -glucosyltransferase,

GLYCOGEN PHOSPHORYLASE

1,3- β -GLUCAN SYNTHASE

GLUCARATE DEHYDRATASE

D-Glucitol,

SORBITOL DEHYDROGENASE

GLUCOAMYLASE

GLUCOCEREBROSIDASE

GLUCOKINASE

GLUCONATE DEHYDRATASE

Gluconeogenesis tracer kinetics,

GLUCOSE TRACER KINETICS

GLUCOSAMINATE AMMONIA-LYASE
 GLUCOSAMINE N-ACETYLTRANSFERASE
 GLUCOSAMINE-6-PHOSPHATE ISOMERASE
 GLUCOSAMINE-6-PHOSPHATE SYNTHASE
 Glucose as a substrate or product,
 GLUCOKINASE
 HEXOKINASE
 LACTASE
 LACTOSE SYNTHASE
 LEVANSUCRASE
 MALTASE
 MALTOSE PHOSPHORYLASE
 MAPPING SUBSTRATE INTERACTIONS
 USING KINETIC DATA
 XYLOSE ISOMERASE
 GLUCOSE OXIDASE
 GLUCOSE-6-PHOSPHATASE
 ISOTOPE EXCHANGE (*Reactions Away
 From Equilibrium*)
 Glucose-1-phosphate adenyltransferase,
 ATP:GLUCOSE-1-PHOSPHATE ADENYL-
 YLTRANSFERASE
 GLUCOSE-6-PHOSPHATE DEHYDRO-
 GENASE
 α -D-Glucose 1-phosphate, interconversion
 with α -D-glucose 6-phosphate,
 PHOSPHOGLUCOMUTASE
 D-Glucose 6-phosphate, interconversion with
 D-fructose 6-phosphate,
 PHOSPHOGLUCOSE ISOMERASE
 Glucose-6-phosphate isomerase,
 PHOSPHOGLUCOISOMERASE
 D-Glucose 6-phosphate synthesis,
 6-PHOSPHO- β -GLUCOSIDASE
 GLUCOSE-1-PHOSPHATE URIDYLYLTRANS-
 FERASE
 GLUCOSE TRACER KINETICS
 Glucosylceramide β -D-glucosidase,
 GLUCOCEREBROSIDASE
 GLUCURONATE-1-PHOSPHATE URIDYLYL-
 TRANSFERASE
 GLUCURONATE-2-SULFATE SULFATASE
 β -GLUCURONIDASE
 GLUCURONOKINASE
 GLUCURONOLACTONE REDUCTASE
 GLUCURONOSYLTRANSFERASES
 GLUTACONYL-CoA DECARBOXYLASE
 Glutamate acetyltransferase,
 N-ACETYLGUTAMATE SYNTHASE
 GLUTAMATE N-ACETYLTRANSFERASE
 GLUTAMATE N-ACETYLTRANSFERASE
 N-ACETYLGUTAMATE SYNTHASE

D-GLUTAMATE:D-AMINO ACID AMINO-
 TRANSFERASE
 Glutamate-amino acid transaminases,
 See specific aminotransferase
 Glutamate as a substrate or product,
 GLUTAMATE DECARBOXYLASE
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Intrinsic isotope effects on enzyme-catalyzed reactions, NORTHROP ISOTOPE EFFECT METHOD	EXTERNAL ION-PAIR RETURN HIDDEN RETURN	(R)-2-METHYLMALATE DEHYDRATASE NICOTINATE DEHYDROGENASE NITRATE REDUCTASE NITRITE REDUCTASE PHENYLALANINE MONOOXYGENASE PROLYL 3-HYDROXYLASE PROLYL 4-HYDROXYLASE PROTocatechuAte 3,4-DIOXYGENASE PROTocatechuAte 4,5-DIOXYGENASE RIESKE IRON-SULFUR PROTEIN RUBREDOXIN RUBREDOXIN:NAD ⁺ REDUCTASE SUCCINATE DEHYDROGENASE SULFITE REDUCTASE SULFUR DIOXYGENASE THYMIDINE 2'-HYDROXYLASE XANTHINE OXIDASE
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LINEAR STRAIN	LIPOPOLYSACCHARIDE N-ACETYLGLU- COSAMINYLTRANSFERASE	Long-chain acid anion, LONG-CHAIN ALDEHYDE DEHYDRO- GENASE LONG-CHAIN FATTY ACYL-CoA SYN- THETASE
Line formula, PHYSICAL ORGANIC CHEMISTRY NO- MENCLATURE	Lipopolysaccharide core, LIPOPOLYSACCHARIDE N-ACETYLGLU- COSAMINYLTRANSFERASE LIPOPOLYSACCHARIDE GALACTOSYL- TRANSFERASE LIPOPOLYSACCHARIDE GLUCOSYL- TRANSFERASE	Long-chain acyl-CoA, LONG-CHAIN ACYL-CoA DEHYDRO- GENASE LONG-CHAIN FATTY ACYL-CoA SYN- THETASE
LINE-SHAPE ANALYSIS	LIPOPOLYSACCHARIDE GALACTOSYL- TRANSFERASE	LONG-CHAIN ACYL-CoA DEHYDRO- GENASE
Lineweaver-Burk plot, DOUBLE-RECIPROCAL PLOT ENZYME KINETIC EQUATIONS (1. The Basics)	LIPOPOLYSACCHARIDE GLUCOSYL- TRANSFERASE	Long-chain aldehyde, LONG-CHAIN ALDEHYDE DEHYDRO- GENASE
LINKED FUNCTIONS ALLOSTERISM BASIC REGULATORY KINETICS BOHR EFFECT COOPERATIVITY FEEDBACK EFFECTORS LE CHATELIER'S PRINCIPLE HEMOGLOBIN ADAIR EQUATION	LIPOPOLYSACCHARIDE GLUCOSYL- TRANSFERASE	LONG-CHAIN ALDEHYDE DEHYDRO- GENASE
LINOLEATE ISOMERASE	LIPOPOLYSACCHARIDE GLUCOSYLTRANS- FERASE	LONG-CHAIN FATTY ACYL-CoA SYN- THETASE
Linoleoyl ester, LECITHIN-CHOLESTEROL ACYLTRANS- FERASE	LIPOPROTEIN LIPASE LIPASES	Longitudinal strain, LINEAR STRAIN
LIPASES LANGMUIR TROUGH AND BALANCE LIPOPROTEIN LIPASE MICELLES	LIPOXYGENASE 5-LIPOXYGENASE 8-LIPOXYGENASE 12-LIPOXYGENASE 15-LIPOXYGENASE <i>Listeria monocytogenes</i> , ACTIN-BASED BACTERIAL MOTILITY ACTIN FILAMENT GROWTH (Polymeriza- tion Zone Model)	Long-lived luminescence, PHOSPHORESCENCE JABLONSKI DIAGRAM
LIPID ACTIVATION	Lithium, IONIC RADIUS	LONG-RANGE LOOPS
Lipid mobilization kinetics, LIPID TRACER KINETICS	Local isolation, method of, NONEQUILIBRIUM THERMODYNAMICS (A Primer)	LOOSE ION PAIR ION PAIR SOLVENT-SEPARATED ION PAIR SOLVENT-SHARED ION PAIR TIGHT ION PAIR
Lipid phase state, LIPID PHASE TRANSITION KINETICS	Local maxima in a continuous and differenti- able function, FIRST DERIVATIVE TEST	LOSSEN REARRANGEMENT
LIPID PHASE TRANSITION KINETICS	Local minima in a continuous and differenti- able function, FIRST DERIVATIVE TEST	LOW-BARRIER HYDROGEN BONDS (Poten- tial Role in Catalysis) HYDROGEN BONDS WATER
LIPID TRACER KINETICS	LOCK-AND-KEY MODEL OF ENZYME ACTION INDUCED-FIT MODEL	Low-density lipoproteins, LIPOPROTEIN LIPASE
Lipid transfer across membrane bilayers, PHOSPHOLIPID FLIP-FLOP	LIGAND-INDUCED CONFORMATIONAL CHANGE	Lowest-lying excited state, FLUORESCENCE
Lipoamide, PYRUVATE DEHYDROGENASE	LOG-RESPONSE CURVE FOR ENZYME COOPERATIVITY	Low-k _{cat} GTPase in tubulin polymerization, MICROTUBULE ASSEMBLY KINETICS
LIPOIC ACID AND DERIVATIVES		LUCIFERASE
Lipolysis, LANGMUIR TROUGH AND BALANCE MICELLES		D(-)-Luciferin, LUCIFERASE
		D(-)-Luciferyl adenylate, LUCIFERASE

LUMEN
 Lumichrome,
 RIBOFLAVINASE
 Luminal calcium ion,
 ION PUMPS
 LUMINESCENCE
 FLUORESCENCE
 Luminometer,
 LUCIFERASE
 Luminous flux,
 LUMEN
 LYASES
 ENZYME COMMISSION NOMENCLATURE
 LYATE ION
 LYONIUM ION
 LEVELING EFFECT
 Lysine,
 LYSINE 2,3-AMINOMUTASE
 LYSINE 6-AMINOTRANSFERASE
 LYSINE DECARBOXYLASE
 LYSINE DEHYDROGENASE (α -DEAMINATING)
 LYSINE MONOOXYGENASE
 LYSOPINE DEHYDROGENASE
 ZWITTERION
 D-Lysine,
 D-LYSINE 5,6-AMINOMUTASE
 LYSINE 2,3-AMINOMUTASE
 β -LYSINE 5,6-AMINOMUTASE
 D-LYSINE 5,6-AMINOMUTASE
 LYSINE 6-AMINOTRANSFERASE
 Lysine ϵ -aminotransferase,
 LYSINE 6-AMINOTRANSFERASE
 LYSINE DECARBOXYLASE
 LYSINE DEHYDROGENASE (α -DEAMINATING)
 LYSINE 2-MONOOXYGENASE
 D- α -Lysine mutase,
 D-LYSINE 5,6-AMINOMUTASE
 Lysolecithin,
 LYSOPHOSPHOLIPASE
 MICELLES
 Lysolecithinase,
 LYSOPHOSPHOLIPASE
 2-Lysophosphatidylcholine,
 LYSOPHOSPHOLIPASE
 LYSOPHOSPHOLIPASE
 LYSOPHOSPHOLIPASE D
 Lysopine,
 LYSOPINE DEHYDROGENASE

LYSOPINE DEHYDROGENASE
 D-Lysopine synthase,
 LYSOPINE DEHYDROGENASE
 LYSOZYME
 KINETIC ISOTOPE EFFECT
 LYSYL ENDOPEPTIDASE
 LYSYL HYDROXYLASE
 LYSYL OXIDASE
 Lysyl residue,
 LYSYL HYDROXYLASE
 LYSYL OXIDASE
 LYSYLTRANSFERASE
 L-Lysyl-tRNA^{lys},
 LYSYLTRANSFERASE
 Lysyl-tRNA:phosphatidylglycerol transferase,
 LYSYLTRANSFERASE
 α -LYTIC PROTEINASE
 CHYMOTRYPSIN
 CATALYTIC TRIAD
 D-Lyxose,
 LYXOSE ISOMERASE
 MANNOSE ISOMERASE
 LYXOSE ISOMERASE
 Lyxose ketol-isomerase,
 LYXOSE ISOMERASE
 D-Lyxosides,
 MANNOSIDASES

M

MACROMOLECULAR SEQUENCE ANALYSIS (Computer Methods)
 MACROSCOPIC CONSTANTS
 MICROSCOPIC CONSTANTS
 ZWITTERION
 MACROSCOPIC DIFFUSION CONTROL
 MICROSCOPIC DIFFUSION CONTROL
 Magic acid,
 PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
 Magnesium carbonate (MgCO₃),
 SOLUBILITY PRODUCT
 Magnesium hydroxide (Mg(OH)₂),
 SOLUBILITY PRODUCT
 Magnesium ion,
 ENERGY CHARGE
 IONIC RADIUS
 LAMBDA (or Λ -) ISOMERS OF METAL ION-NUCLEOTIDE COMPLEXES
 METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS
 XYLOSE ISOMERASE

MAGNESIUM ION (Intracellular)
 ENERGY CHARGE
 METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS
 BIOCHEMICAL THERMODYNAMICS
 Magnesium oxalate dihydrate (MgC₂O₄·2H₂O),
 SOLUBILITY PRODUCT
 Magnesium phosphate (Mg₃(PO₄)₂),
 SOLUBILITY PRODUCT
 Magnesium protoporphyrin,
 MAGNESIUM-PROTOPORPHYRIN O-METHYLTRANSFERASE
 MAGNESIUM-PROTOPORPHYRIN O-METHYLTRANSFERASE
 Magnesium protoporphyrin monomethyl ester,
 MAGNESIUM-PROTOPORPHYRIN O-METHYLTRANSFERASE
 MAGNETICALLY ANISOTROPIC GROUP
 Magnetic balance,
 MAGNETIC SUSCEPTIBILITY
 Magnetic dipoles,
 QUADRUPOLE
 Magnetic equivalence,
 PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
 Magnetic field,
 LARMOR PRECESSION
 MAGNETICALLY ANISOTROPIC GROUP
 MAGNETOTACTIC
 Magnetic field strength,
 MAGNETIC SUSCEPTIBILITY
 Magnetic flux,
 MAXWELL
 Magnetic flux density,
 PERMEABILITY
 MAGNETIC SUSCEPTIBILITY
 MAGNETIC SUSCEPTIBILITY
 PARAMAGNETIC
 HEMOGLOBIN
 DIAMAGNETIC
 Magnetization,
 MAGNETIC SUSCEPTIBILITY
 MAGNETIZATION TRANSFER
 MAGNETOTACTIC
 Magnitude of k_H/k_D ,
 QUANTUM MECHANICAL TUNNELING
 MALAPRADE REACTION (Periodate Oxidation)
 (R)-Malate,
 MALATE DEHYDROGENASE

(S)-Malate, ABORTIVE COMPLEXES MALATE DEHYDROGENASE MALATE SYNTHASE (S)-2-METHYLMALATE DEHYDRATASE	Maltose, MALTASE MALTOSE 1-EPIMERASE MALTOSE PHOSPHORYLASE MALTOSE PHOSPHORYLASE MALYL-CoA LYASE MAMILLARY MODEL CATENARY MODEL COMPARTMENTAL ANALYSIS	MANNOSE ISOMERASE α -D-Mannose 1-phosphate, MANNOSE-1-PHOSPHATE GUANYLYL-TRANSFERASE
Malate condensing enzyme, MALATE SYNTHASE MALATE DEHYDROGENASE ABORTIVE COMPLEXES	Mandelate, MANDELATE 4-MONOOXYGENASE MANDELATE RACEMASE	D-Mannose 6-phosphate, MANNOSE-6-PHOSPHATE ISOMERASE MANNOSE-1-PHOSPHATE GUANYLYL-TRANSFERASE
Malate dehydrogenase (acceptor), MALATE DEHYDROGENASE	L-Mandelate 4-hydroxylase, MANDELATE 4-MONOOXYGENASE	D-Mannose 1-phosphate, interconversion to D-mannose 6-phosphate, PHOSPHOMANNOMUTASE
Malate dehydrogenase (decarboxylating), MALATE DEHYDROGENASE	MANDELATE 4-MONOOXYGENASE MANDELATE RACEMASE	MANNOSE-6-PHOSPHATE ISOMERASE
D-Malate dehydrogenase (decarboxylating), MALATE DEHYDROGENASE	Mandelonitrile, MANDELONITRILE LYASE	D-Mannose residues, MANNOSIDASES
Malate dehydrogenase (NADP ⁺), MALATE DEHYDROGENASE	MANDELONITRILE LYASE	MANNOSIDASES
Malate dehydrogenase (oxaloacetate-decarboxylating), MALATE DEHYDROGENASE	Manganese(II) carbonate (MnCO ₃), SOLUBILITY PRODUCT	D-Mannosides, MANNOSIDASES
Malate dehydrogenase (oxaloacetate-decarboxylating, NADP ⁺), MALATE DEHYDROGENASE	Manganese(II) hydroxide (Mn(OH) ₂), SOLUBILITY PRODUCT	1,4- β -D-Mannosidic linkages, β -MANNANASE
MALATE SYNTHASE	Manganese ion, GLUTAMINE SYNTHETASE MANGANESE PEROXIDASE	MANOMETRIC ASSAY METHODS EUDIOMETER
MALDI-TOF MS, TIME-OF-FLIGHT MASS SPECTROMETRY	METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS PYRUVATE, WATER DIKINASE IONIC RADIUS	Manometry, MANOMETRIC ASSAY METHODS
Maleate, MALEATE ISOMERASE	Manganese(II) oxalate dihydrate (MnC ₂ O ₄ ·2H ₂ O), SOLUBILITY PRODUCT	MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA
MALEATE ISOMERASE	MANGANESE PEROXIDASE PEROXIDASE	MARCUS EQUATION MARCUS RATE THEORY
4-Maleylacetoacetate, MALEYLACETOACETATE ISOMERASE	β -MANNANASE MANNOSIDASES	MARKOV CHAIN
MALEYLACETOACETATE ISOMERASE	Mannan endo-1,4- β -mannosidase, β -MANNANASE	Markovnikoff rule, PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
Maleylpyruvate, MALEYLACETOACETATE ISOMERASE MALEYLPYRUVATE ISOMERASE	Mannans, β -MANNANASE	MARQUARDT ALGORITHM COMPUTER ALGORITHMS & SOFTWARE
MALEYLPYRUVATE ISOMERASE	Mannase, MANNOSIDASES	MASS-ACTION RATIO
Malic dehydrogenase, MALATE DEHYDROGENASE	D-Mannonate, MANNONATE DEHYDRATASE	Mass-action ratio, determination, ISOTOPE EXCHANGE AT EQUILIBRIUM
Malic enzyme, MALATE DEHYDROGENASE	MANNONATE DEHYDRATASE	MASSIEU FUNCTION HELMHOLTZ ENERGY PLANCK FUNCTION
Malonyl-[acyl-carrier-protein], MALONYL-CoA:ACYL-CARRIER-PROTEIN TRANSACYLASE	D-Mannonate hydrolase, MANNONATE DEHYDRATASE	MASS SPECTROMETRY
MALONYL-CoA:ACYL-CARRIER-PROTEIN TRANSACYLASE	D-Mannose, MANNOSE ISOMERASE	Matrix of biominerals, BIOMINERALIZATION
Malonyl-coenzyme A, MALONYL-CoA:ACYL-CARRIER-PROTEIN TRANSACYLASE	MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA	MATRIX ISOLATION MAXIMUM LIKELIHOOD METHOD
MALTASE α -GLUCOSIDASE		Maximum velocity, MICHAELIS-MENTEN EQUATION
		MAXWELL MAXWELL'S THERMODYNAMIC RELATIONS
		Mean activity coefficient, DEBYE-HÜCKEL TREATMENT
		Mean lifetime, LIFETIME
		Mean residence time, MEAN TRANSIT TIME

MEAN TRANSIT TIME	Membrane transport in organelles, MEMBRANE TRANSPORT	Metal ion exchange times, LAMBDA (or Λ -) ISOMERS OF METAL ION-NUCLEOTIDE COMPLEXES
COMPARTMENTAL ANALYSIS		
MECHANISM-BASED INHIBITOR	Menadione reductase, NAD(P)H DEHYDROGENASE (QUINONE)	Metal ion-ligand complex, METAL ION COMPLEXES (DETERMINING STABILITY CONSTANTS)
MEDIATED AMPEROMETRIC BIOSENSOR	MEPRIN	
BIOSENSOR		
MEDIUM	Mercaptoethanol, 3-MERCAPTOPYRUVATE SULFURTRANS- FERASE	Metal ion-nucleotide complexes, LAMBDA (or Λ -) ISOMERS OF METAL ION-NUCLEOTIDE COMPLEXES
Medium effect, SOLVENT EFFECTS		METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS
MEERWEIN-PONDORFF-VERLEY REDUCTION	3-Mercaptopyruvate, 3-MERCAPTOPYRUVATE SULFURTRANS- FERASE	
MEGA-	3-MERCAPTOPYRUVATE SULFURTRANS- FERASE	Metal ion oxidation state, MIXED VALENCY
MEISENHEIMER ADDUCT		METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS
TWO-PROTONIC STATE ELECTROPHILES		
Melilotate,	Mercuric ion, MERCURY(II) REDUCTASE	Metal ligand complexation, LIGAND FIELD SPLITTING
MELILOTATE HYDROXYLASE		METALLO-BIOCHEMICAL METHODS
MELILOTATE HYDROXYLASE	Mercuric reductase, MERCURY(II) REDUCTASE	Metalloproteins, LASER-FLASH KINETIC ANALYSIS
Melilotate 3-monooxygenase, MELILOTATE HYDROXYLASE		METAL ION CATALYSIS
MELTING	Mercury, MERCURY(II) REDUCTASE	METALLOTHIONEINS
Melting curve, MELTING	Mesaconase, (S)-2-METHYLMALATE DEHYDRATASE	Metal-nucleotide complex, EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES
Melting point, MELTING	Mesaconate, 3-METHYLASPARTATE AMMONIA-LYASE	
Membrane,	Mesaconate hydratase, (S)-2-METHYLMALATE DEHYDRATASE	Metal oxides, OXYGEN, OXIDES & OXO ANIONS
LIPID ACTIVATION		META-MODEL
MEMBRANE TRANSPORT	MESO FORM (or, MESO ISOMER)	METAPHOSPHATE
Membrane disruption, LATENT ACTIVITY	Meso isomer, MESO FORM (or, MESO ISOMER)	ACYL PHOSPHATES
Membrane electric potential, MEMBRANE POTENTIAL	MESOLYTIC CLEAVAGE	OXYGEN, OXIDES & OXO ANIONS
Membrane fusion, LIPID PHASE TRANSITION KINETICS	Metabolic activity of an individual organism, BASAL RATE	METASTABLE STATE
Membrane lipid asymmetry, PHOSPHOLIPID FLIP-FLOP	METABOLIC CONTROL ANALYSIS TOP-DOWN CONTROL ANALYSIS	JABLONSKI DIAGRAM
Membrane permeability, PERMEABILITY	Metabolic flux, METABOLIC CONTROL ANALYSIS	TRANSIENT CHEMICAL SPECIES
PERMEABILITY CONSTANT	Metabolite compartmentation, MAMILLARY MODEL	
MEMBRANE POTENTIAL	METABOLONS	Metastable triplet state, JABLONSKI DIAGRAM
ACTION POTENTIAL	SUBSTRATE CHANNELING	Metavanadate, OXYGEN, OXIDES & OXO ANIONS
DEPOLARIZATION	Metal complexes, METAL ION COORDINATION REACTIONS	METHANE MONOOXYGENASE
GOLDMAN EQUATION		Methanol, autoprotolysis constant, AUTOPROTOLYSIS
NERNST EQUATION	METAL ION CATALYSIS	METHANOL DEHYDROGENASE
RESTING POTENTIAL	METAL ION COMPLEXES (Determining Stability Constants)	5,10-METHENYLTETRAHYDROFOLATE CYCLOHYDROLASE
THRESHOLD POTENTIAL	METAL IONS IN NUCLEOTIDE-DEPENDENT REACTIONS	5,10-METHENYLTETRAHYDROFOLATE SYNTHETASE
PATCH-CLAMP TECHNIQUE	METAL ION COORDINATION REACTIONS	METHIONINE ADENOSYLTRANSFERASE
Membrane protein dynamics, FLUORESCENCE RECOVERY AFTER PHOTOBLEACHING	WATER	METHIONINE γ -LYASE
MEMBRANE TRANSPORT	WATER EXCHANGE RATES OF AQUA METAL IONS	METHIONINE SULFOXIDE REDUCTASE
ION PUMPS		Methionine synthase, N ⁵ -METHYLTETRAHYDROFOLATE: HOMOCYSTEINE METHYLTRANSFERASE

METHYL TRANSFER REACTIONS USING S-ADENOSYLMETHIONINE	2-METHYL-BRANCHED-CHAIN ACYL-CoA DEHYDROGENASE	Methylitaconate, METHYLITACONATE Δ -ISOMERASE
Methionine:tRNA ligase, METHIONYL-tRNA SYNTHETASE	2-Methyl-4-carbomethoxypyridinium ion, Z VALUE	METHYLITACONATE Δ -ISOMERASE
Methionyl-tRNA ^{Met} , METHIONYL-tRNA FORMYLTRANSFERASE	β -METHYLCROTONYL-CoA CARBOXYLASE,	(R)-2-Methylmalate, (R)-2-METHYLMALATE DEHYDRATASE
METHIONYL-tRNA SYNTHETASE	3-Methylcrotonyl-coenzyme A β -METHYLCROTONYL-CoA CARBOXYLASE	(S)-2-Methylmalate, (S)-2-METHYLMALATE DEHYDRATASE
METHIONYL-tRNA FORMYLTRANSFERASE	2-Methyleneglutarate, 2-METHYLENEGLUTARATE MUTASE	(R)-2-METHYLMALATE DEHYDRATASE
METHIONYL-tRNA SYNTHETASE	2-METHYLENEGLUTARATE MUTASE	(S)-2-METHYLMALATE DEHYDRATASE
AMINO-ACYL tRNA SYNTHETASES	Methylene-interrupted polyunsaturated fatty acids, LIPOXYGENASE	2-Methylmaleate, (R)-2-METHYLMALATE DEHYDRATASE
Method of continuous variation, JOB PLOT	2-Methylene-3-methylsuccinate, 2-METHYLENEGLUTARATE MUTASE	Methylmalonate semialdehyde, METHYLMALONATE-SEMIALDEHYDE DEHYDROGENASE
4-Methoxybenzoate, 4-METHOXYBENZOATE MONOOXYGENASE	5,10-Methylenetetrahydrofolate, 5,10-METHYLENETETRAHYDROFOLATE DEHYDROGENASE	METHYLMALONATE-SEMIALDEHYDE DEHYDROGENASE (ACYLATING)
4-METHOXYBENZOATE MONOOXYGENASE (O-DEMETHYLATING)	5,10-METHYLENETETRAHYDROFOLATE REDUCTASE	(R)-Methylmalonyl-CoA, METHYLMALONYL-CoA EPIMERASE
3-Methyladenine, 3-METHYLADENINE-DNA GLYCOSYLASE	5,10-METHYLENETETRAHYDROFOLATE: tRNA (URACIL-5-)-METHYLTRANSFERASE (FADH ₂ OXIDIZING)	L-METHYLMALONYL-CoA MUTASE
7-Methyladenine, 3-METHYLADENINE-DNA GLYCOSYLASE	SERINE HYDROXYMETHYLTRANSFERASE	(S)-Methylmalonyl-CoA, METHYLMALONYL-CoA DECARBOXYLASE
3-METHYLADENINE-DNA GLYCOSYLASE	THYMIDYLATE SYNTHASE	METHYLMALONYL-CoA EPIMERASE
N-Methyl-L-alanine, N-METHYL-L-ALANINE DEHYDROGENASE	5,10-METHYLENETETRAHYDROFOLATE DEHYDROGENASE	D-METHYLMALONYL-CoA HYDROLASE
N-METHYL-L-ALANINE DEHYDROGENASE	5,10-METHYLENETETRAHYDROFOLATE REDUCTASE	PROPIONYL-CoA CARBOXYLASE
Methylamine, N-METHYL-L-ALANINE DEHYDROGENASE	5,10-METHYLENETETRAHYDROFOLATE: tRNA (URACIL-5-)-METHYLTRANSFERASE (FADH ₂ OXIDIZING)	METHYLMALONYL-CoA DECARBOXYLASE
METHYLAMINE DEHYDROGENASE	2-Methylfumarate, (S)-2-METHYLMALATE DEHYDRATASE	METHYLMALONYL-CoA EPIMERASE
RESONANCE RAMAN SPECTROSCOPY	3-METHYLGLUTACONYL-CoA HYDRATASE	D-METHYLMALONYL-CoA HYDROLASE
TOPAQUINONE	3-Methylglutaconyl-coenzyme A, β -METHYLCROTONYL-CoA CARBOXYLASE	L-METHYLMALONYL-CoA MUTASE
Methylamine-glutamate methyltransferase, N-METHYLGLUTAMATE SYNTHASE	3-METHYLGLUTACONYL-CoA HYDRATASE	4-Methyl-2-oxopentanoate, LEUCINE AMINOTRANSFERASE
N-Methyl-4-aminobenzoate, 4-METHOXYBENZOATE MONOOXYGENASE	N-Methyl-L-glutamate, N-METHYLGLUTAMATE SYNTHASE	LEUCINE DEHYDROGENASE
β -Methylaspartase, 3-METHYLASPARTATE AMMONIA-LYASE	N-METHYLGLUTAMATE SYNTHASE	LEUCINE KINETICS
L-threo-3-Methylaspartate, 3-METHYLASPARTATE AMMONIA-LYASE	3-Methylguanine, 3-METHYLADENINE-DNA GLYCOSYLASE	2-Methyl-3-oxopropanoate, METHYLMALONATE-SEMIALDEHYDE DEHYDROGENASE
β -METHYLASPARTATE-GLUTAMATE MUTASE	7-Methylguanine, 3-METHYLADENINE-DNA GLYCOSYLASE	(R)-2-Methyl-3-oxopropanoyl-coenzyme A, METHYLMALONYL-CoA EPIMERASE
3-METHYLASPARTATE AMMONIA-LYASE	γ -Methyl- γ -hydroxy- α -ketoglutarate, γ -METHYL- γ -HYDROXY- α -KETOGLUTARATE ALDOLASE	L-METHYLMALONYL-CoA MUTASE
β -METHYLASPARTATE-GLUTAMATE MUTASE	γ -METHYL- γ -HYDROXY- α -KETOGLUTARATE ALDOLASE	(S)-2-Methyl-3-oxopropanoyl-coenzyme A, METHYLMALONYL-CoA DECARBOXYLASE
Methylaspartate mutase, β -METHYLASPARTATE-GLUTAMATE MUTASE	METHYLHYDROXYPYRIDINECARBOXYLATE DIOXYGENASE	METHYLMALONYL-CoA EPIMERASE
METHYLATION (As a Mechanistic Probe)		N-Methylphenylethanolamine, NOREPINEPHRINE N-METHYLTRANSFERASE
		N-Methylputrescine, synthesis of, PUTRESCINE N-METHYLTRANSFERASE
		5-Methyltetrahydrofolate, 5,10-METHYLENETETRAHYDROFOLATE REDUCTASE
		N ⁵ -METHYLTETRAHYDROFOLATE:

HOMOCYSTEINE METHYLTRANSFERASE	MICHAELIS CONSTANT (APPARENT)	Millipore filtration assays,
N ⁵ -METHYLTETRAHYDROFOLATE: HOMOCYSTEINE METHYLTRANSFERASE	MICHAELIS-MENTEN EQUATION	ACTIN ASSEMBLY ASSAYS
TETRAHYDROPTEROYLTRIGLUTAMATE METHYLTRANSFERASE	L'HÔPITAL'S RULE	MINI-CHAPERONES
METHYLTETRAHYDROMETHANOPT-ERIN:COENZYME M METHYLTRANSFERASE	MICHAELIS CONSTANT	Mini-chromatography column,
Methylthioadenosine,	MICHAELIS-MENTEN KINETICS	RAPID BUFFER EXCHANGE
METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	PROGRESS CURVE ANALYSIS	MINIMAL LETHAL DOSE
Methylthioadenosine nucleosidase,	UNI UNI MECHANISM	MINIMAL STRUCTURAL CHANGE, POSTULATE OF
METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	ZERO-ORDER REACTIONS	MOLECULAR REARRANGEMENT
METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	MICHAELIS-MENTEN KINETICS	REARRANGEMENT STAGE
S-ADENOSYLHOMOCYSTEINE NUCLEOSIDASE	MICHAELIS-MENTEN EQUATION	MITOCHONDRIAL INTERMEDIATE PEPTIDASE
5-Methylthio-D-ribose,	UNI UNI MECHANISM	Mitochondrial oxidation-reduction state,
METHYLTHIOADENOSINE NUCLEOSIDE HYDROLASE	ENZYME RATE EQUATIONS (1. The Basics)	NADH/NADPH SURFACE FLUORESCENCE IN LIVING TISSUES
Methyltransferases,	Michaelis pH function,	MITOGEN-ACTIVATED PROTEIN KINASE
METHYL TRANSFER REACTIONS USING S-ADENOSYLMETHIONINE	pH EFFECTS ON ENZYMIC CATALYSIS	MITOGEN-ACTIVATED PROTEIN KINASE KINASE
METHYL TRANSFER REACTIONS USING S-ADENOSYLMETHIONINE	MICRO-	MIXED-FUNCTION OXIDASE
Mevaldate,	Microelectrodes,	CYTOCHROME P450
MEVALDATE REDUCTASE	MAGNESIUM ION (INTRACELLULAR)	Mixed kinetic order,
MEVALDATE REDUCTASE	MICROENVIRONMENT	CHEMICAL KINETICS
(R)-Mevalonate,	Microinjection of fluorescently labeled tubulin into cells,	MIXED-TYPE INHIBITION
MEVALDATE REDUCTASE	MICROTUBULE ASSEMBLY KINETICS	MIXED VALENCY
MEVALONATE KINASE	Microscopic constants,	MIXING TIME
Mevalonate-5-diphosphate decarboxylase,	MACROSCOPIC CONSTANTS	Mobile charge carrier in water,
PYROPHOSPHOMEVALONATE DECARBOXYLASE	ZWITTERION	CHEMICAL KINETICS
MEVALONATE KINASE	MICROSCOPIC DIFFUSION CONTROL	Möbius aromaticity,
Mevalonic acid 5-pyrophosphate anhydridecarboxylase,	MACROSCOPIC DIFFUSION CONTROL	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
PYROPHOSPHOMEVALONATE DECARBOXYLASE	MICROSCOPIC REVERSIBILITY	MODELS
MEYERHOF OXIDATION QUOTIENT	CHEMICAL REACTION	MODIFIER
Micellar catalysis,	DETAILED BALANCING, PRINCIPLE OF	EFFECTOR
MICELLES	CHEMICAL KINETICS	ACTIVATOR
Micelle, light scattering from,	MICROTUBULE ASSEMBLY KINETICS	INHIBITOR
ABSORPTION SPECTROSCOPY	BIOCHEMICAL SELF-ASSEMBLY	MODULATION
MICELLES	ACTIN ASSEMBLY KINETICS	MOLAR ABSORPTION COEFFICIENT
Micellization,	HEMOGLOBIN-S POLYMERIZATION	ABSORPTION SPECTROSCOPY
MICELLES	MICROTUBULE CYTOSKELETON AS "SENSISOME"	BEER-LAMBERT LAW
MICHAELIS COMPLEX (Distal Motions Affecting ES Formation)	Microtubule depolymerization,	MOLAR GAS CONSTANT
MICHAELIS CONSTANT	DEPOLYMERIZATION, END-WISE	BOLTZMANN CONSTANT
MAPPING SUBSTRATE INTERACTIONS USING KINETIC DATA	MICROTUBULE TREADMILLING	Molecular bonding and conformation,
	MICROTUBULE ASSEMBLY KINETICS	SHIELDING
	Migrating chemotactic bacteria,	MOLECULAR CROWDING
	TWIDDLING	Molecular diffusion through large pores,
	MIGRATION	DIFFUSION OF MOLECULES INTO A PORE
	MOLECULAR REARRANGEMENT	MOLECULAR DYNAMICS
	Migratory aptitude,	MOLECULAR ENTITY
	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	MOLECULAR HITCH-HIKING
	Migratory insertion,	MOLECULARITY
	PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE	CHEMICAL KINETICS

UNIMOLECULAR
 BIMOLECULAR
 TRANSITION-STATE THEORY
 ELEMENTARY REACTION
 MOLECULAR MECHANICS CALCULATIONS
 MOLECULAR ORBITALS
 MOLECULAR REARRANGEMENT
 MOLECULAR SIMILARITY
 Molecular stoichiometry of an elementary reaction,
 CHEMICAL KINETICS
 Molecules of product formed per absorbed quantum,
 QUANTUM YIELD
 MOLE FRACTION
 CONCENTRATION
 METHOD OF CONTINUOUS VARIATION
 MOLYBDENUM COFACTOR (MoCo)
 Molybdenum-dependent reactions,
 ALDEHYDE OXIDASE
 MOLYBDOPTERIN
 NITRATE REDUCTASE
 NITROGENASE
 SULFITE OXIDASE
 XANTHINE DEHYDROGENASE
 MOLYBDOPTERIN
 MONOD-WYMAN-CHANGEUX MODEL
 MONOGLYCERIDE LIPASE
 MONO-ISO MECHANISMS
 ISOMERIZATION MECHANISMS
 MONO-ISO PING PONG BI BI REACTION SCHEME
 Monomer-polymer equilibrium,
 PROTEIN POLYMERIZATION
 ACTIN ASSEMBLY KINETICS
 MICROTUBULE ASSEMBLY KINETICS
 MONOMER-POLYMER EXCHANGE KINETICS
 Monomer sequestering proteins,
 ACTIN REGULATORY PROTEINS
 MONOMOLECULARITY
 Monomolecular surface coverage,
 BIOMINERALIZATION
 MONOPHENOL MONOOXYGENASE
 Monophosphatidylinositol phosphodiesterase,
 1-PHOSPHATIDYLINOSITOL PHOSPHODIESTERASE
 Monosialoganglioside sialyltransferase,
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Morphine as a peptidomimetic,
 PEPTIDOMIMETIC COMPOUND
 MÖSSBAUER SPECTROSCOPY
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 MU (μ)
 MUCONATE CYCLOISOMERASE
 MUCONOLACTONE Δ -ISOMERASE
 MUCROLYSIN
 Multicenter bond,
 PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
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 PHARMACOKINETICS
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 ENZYME CASCADE KINETICS
 Multidimensional NMR,
 NUCLEAR MAGNETIC RESONANCE
 Multielectronic atom interactions with the nucleus,
 SHIELDING
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 Multiple attack enzyme,
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 MULTIPLE DEAD-END INHIBITION
 MULTIPLE-TURNOVER CONDITIONS
 MULTIPLICATIVE MODEL
 MULTIPLICITY
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 MULTISTAGE REACTIONS
 MULTISUBSTRATE (or GEOMETRIC) ANALOGUE
 TRANSITION-STATE ANALOGUE
 MULTISUBSTRATE MECHANISMS
 MUSCLE MASS DETERMINATION BY ISOTOPIC TRACERS
 MUTAROTASES
 ANOMERS
 FISCHER PROJECTION
 ALDOSE 1-EPIMERASE
 MUTAROTATION KINETICS
 MYELOBLASTIN
 MYELOPEROXIDASE
 MYOGLOBIN OXYGENATION
 MYOSIN ATPase
 FORCE EFFECTS ON MOLECULAR MOTORS

N

$n \rightarrow \pi^*$ TRANSITION
 ABSORPTION SPECTROSCOPY

$n \rightarrow \sigma^*$ DELOCALIZATION
 ABSORPTION SPECTROSCOPY
 HYPERCONJUGATION
 RESONANCE
 $n \rightarrow \sigma^*$ TRANSITION
 ABSORPTION SPECTROSCOPY
 ν ,
 NU
 NaBH₄, NaB²H₄, and NaB³H₄,
 BOROHYDRIDE REDUCTION
 NAD⁺,
 ADP-RIBOSYLATION
 NICOTINAMIDE COENZYMES (SPECTRAL PROPERTIES)
 NADase,
 ISOTOPE EXCHANGE (REACTIONS AWAY FROM EQUILIBRIUM)
 NAD⁺ GLYCOHYDROLASE
 NAD⁺-dependent dehydrogenases,
 ABSORPTION SPECTROSCOPY
 NAD⁺-dependent enzymes,
 L-AMINO ACID DEHYDROGENASE
 AQUACOBALAMIN REDUCTASES
 BENZOATE 1,2-DIOXYGENASE
 BETAIN-ALDEHYDE DEHYDROGENASE
 CDP-GLUCOSE 4,6-DEHYDRATASE
 DEHYDROASCORBATE REDUCTASE
 2-DEHYDRO-3-DEOXY-D-GLUCONATE 6-DEHYDROGENASE
 2,4-DIAMINOPENTANOATE DEHYDROGENASE
 DIHYDROLIPOAMIDE DEHYDROGENASE
 DIHYDROURACIL DEHYDROGENASE
 DIODOPHENYLPYRUVATE REDUCTASE
 DNA LIGASES
 FORMALDEHYDE DEHYDROGENASE
 FUMARATE REDUCTASE
 GALACTITOL 2-DEHYDROGENASE
 GDP-MANNOSE DEHYDROGENASE
 GLUTAMATE DEHYDROGENASE
 GLUTAMATE SYNTHASE
 GLYCERALDEHYDE-3-PHOSPHATE DEHYDROGENASE
 GLYCERATE DEHYDROGENASE
 GLYCEROL-3-PHOSPHATE DEHYDROGENASE
 GLYOXYLATE REDUCTASE
 HISTIDINOL DEHYDROGENASE
 HOMOISOCITRATE DEHYDROGENASE
 HOMOSERINE DEHYDROGENASE
 HYDROGEN DEHYDROGENASE
 3-HYDROXYACYL-CoA DEHYDROGENASE

HYDROXYBUTYRATE DEHYDROGENASES	HYDROXYLAMINE REDUCTASE	GLUTAMATE DEHYDROGENASE
β -HYDROXYPROPIONATE DEHYDROGENASE	4-HYDROXYPHENYLACETATE 3-MONO-OXYGENASE	GLUTAMATE γ -SEMIALDEHYDE DEHYDROGENASE
20-HYDROXYSTEROID DEHYDROGENASE	IMIDAZOLEACETATE HYDROXYLASE	GLYCERALDEHYDE-3-PHOSPHATE DEHYDROGENASE
IMIDAZOLEACETATE HYDROXYLASE	NADH-CYTOCHROME b_5 REDUCTASE	HOMOSERINE DEHYDROGENASE
IMP DEHYDROGENASE	NADH DEHYDROGENASE	3-HYDROXYACYL-CoA DEHYDROGENASE
myo-INOSITOL 2-DEHYDROGENASE	NADH DEHYDROGENASE, OTHER THAN COMPLEX I	15-HYDROXYPROSTAGLANDIN DEHYDROGENASE
myo-INOSITOL-1-PHOSPHATE SYNTHASE	NADH PEROXIDASE	HYDROXYSTEROID DEHYDROGENASES
ISOCITRATE DEHYDROGENASE	NITRATE REDUCTASE	ISOCITRATE DEHYDROGENASE
3-ISOPROPYLMALATE DEHYDROGENASE	NADH-CYTOCHROME b_5 REDUCTASE	β -KETOACYL-ACP REDUCTASE
LACTATE DEHYDROGENASE	NADH DEHYDROGENASE	α -KETOGLUTARATE SEMIALDEHYDE DEHYDROGENASE
LEUCINE DEHYDROGENASE	NADH DEHYDROGENASE (Other Than Complex I)	LYSOPINE DEHYDROGENASE
LONG-CHAIN ALDEHYDE DEHYDROGENASE	NADH dehydrogenase (quinone), NADH DEHYDROGENASE, OTHER THAN COMPLEX I	MALATE DEHYDROGENASE
LYSINE DEHYDROGENASE (α -DEAMINATING)	NADH/NADPH SURFACE FLUORESCENCE IN LIVING TISSUES	MERCURY(II) REDUCTASE
MALATE DEHYDROGENASE	NADH oxidation, NADH PEROXIDASE	N-METHYL-L-ALANINE DEHYDROGENASE
5,10-METHYLENETETRAHYDROFOLATE DEHYDROGENASE METHYLMALONATE-SEMIALDEHYDE DEHYDROGENASE	NADH PEROXIDASE	5,10-METHYLENETETRAHYDROFOLATE DEHYDROGENASE
NAD ⁺ GLYCOHYDROLASE	NAD ⁺ KINASE	NAD(P) ⁺ TRANSHYDROGENASE
NAD ⁺ KINASE	NAD ⁺ nucleosidase, NAD ⁺ GLYCOHYDROLASE	NICOTINATE DEHYDROGENASE
OCTANOL DEHYDROGENASE	NADP ⁺ , NICOTINAMIDE COENZYMES (SPECTRAL PROPERTIES)	D-NOPALINE SYNTHASE
OCTOPINE DEHYDROGENASE	NADP ⁺ -dependent enzymes, N-ACETYL- γ -GLUTAMYL-PHOSPHATE REDUCTASE	SHIKIMATE DEHYDROGENASE
PHENYLALANINE DEHYDROGENASE	ACYL-CoA DEHYDROGENASES	SULFITE REDUCTASE
POLY(ADP-RIBOSE) SYNTHETASE (or, SYNTHASE)	ACYLGLYCERONE-PHOSPHATE REDUCTASE	VALINE DEHYDROGENASE
PREPHENATE DEHYDROGENASE	ALDEHYDE DEHYDROGENASE	ZEATIN CIS-TRANS ISOMERASE
PYRIDOXAL 4-DEHYDROGENASE	ALDOSE REDUCTASE	NADPH DEHYDROGENASE
Δ^1 -PYRROLINE-5-CARBOXYLATE DEHYDROGENASE	2-AMINOADIPATE 6-SEMIALDEHYDE DEHYDROGENASE	NAD(P)H DEHYDROGENASE (QUINONE)
RUBREDOXIN:NAD ⁺ REDUCTASE	ASPARTATE-SEMIALDEHYDE DEHYDROGENASE	NADPH-dependent enzymes, ACYL-ACYL-CARRIER-PROTEIN Δ^9 -DESATURASE
SERINE DEHYDROGENASE	N^5 -(CARBOXYETHYL)ORNITHINE SYNTHASE	ACYL-CoA DEHYDROGENASES
SORBITOL DEHYDROGENASE	CORTISONE β -REDUCTASE	ADRENODOXIN
XANTHINE DEHYDROGENASE	7-DEHYDROCHOLESTEROL REDUCTASE	ANTHRANILATE HYDROXYLASE
NAD ⁺ formation, NADH PEROXIDASE	2-DEHYDRO-3-DEOXY-D-GLUCONATE 6-DEHYDROGENASE	AQUACOBALAMIN REDUCTASES
NAD ⁺ GLYCOHYDROLASE	3-DEHYDROSPHINGANINE REDUCTASE	N^5 -(CARBOXYETHYL)ORNITHINE SYNTHASE
NADH, NICOTINAMIDE COENZYMES (SPECTRAL PROPERTIES)	DIAMINOPIMELATE DEHYDROGENASE	trans-CINNAMATE 4-MONOOXYGENASE
NADH-dependent enzymes, AQUACOBALAMIN REDUCTASES	FERREDOXIN:NADP ⁺ REDUCTASE	CYCLOHEXANONE MONOOXYGENASE
BENZENE 1,2-DIOXYGENASE	10-FORMYLTETRAHYDROFOLATE DEHYDROGENASE	CYTOCHROME P-450 REDUCTASE
DEHYDROASCORBATE REDUCTASE	GLUCOSE-6-PHOSPHATE DEHYDROGENASE	DIHYDROFOLATE REDUCTASE
GLUTAMATE SYNTHASE	GLUCURONOLACTONE REDUCTASE	DIMETHYLANILINE MONOOXYGENASE
GLYOXYLATE REDUCTASE		FARNESYL-DIPHOSPHATE FARNESYLTRANSFERASE
3-HYDROXYBENZOATE 6-MONOOXYGENASE		FATTY ACID SYNTHASE
		GLUTATHIONE REDUCTASE
		HEME OXYGENASE
		3-HYDROXYBENZOATE 6-MONOOXYGENASE
		4-HYDROXYBENZOATE 3-MONOOXYGENASE

<i>β</i> -HYDROXY- <i>β</i> -METHYLGLUTARYL-CoA REDUCTASE	LIGAND-INDUCED CONFORMATIONAL CHANGE	[¹⁴ C]Nicotinamide riboside,
HYDROXYSTEROID DEHYDROGENASES	COOPERATIVITY	NICOTINIC ACID: ANALOGUES AND COENZYMES
KYNURENINE 3-HYDROXYLASE	ALLOSTERISM	Nicotinate,
5,10-METHYLENETETRAHYDROFOLATE REDUCTASE	NEIGHBORING GROUP MECHANISM	NICOTINATE DEHYDROGENASE
NAD(P)H DEHYDROGENASE (QUINONE)	ANCHIMERIC ASSISTANCE	NICOTINATE PHOSPHORIBOSYLTRANSFERASE
NITRIC OXIDE SYNTHASE	NEIGHBORING-GROUP PARTICIPATION	NICOTINATE DEHYDROGENASE
PHENOL HYDROXYLASE	INTRAMOLECULAR CATALYSIS	Nicotinate hydroxylase,
PROTOCHLOROPHYLLIDE REDUCTASE	SYNARTETIC ACCELERATION	NICOTINATE DEHYDROGENASE
SQUALENE SYNTHASE	N-END RULE	NICOTINATE MONONUCLEOTIDE ADENYL- YLTRANSFERASE
SULFITE REDUCTASE	PROTEIN TURNOVER KINETICS	Nicotinate-nucleotide adenylyltransferase,
NADPH diaphorase,	NEPRILYSIN	NICOTINATE MONONUCLEOTIDE ADE- NYLYLTRANSFERASE
NADPH DEHYDROGENASE	NERNST EQUATION	NICOTINATE-NUCLEOTIDE DIMETHYLBEN- ZIMIDAZOLE PHOSPHORIBOSYL- TRANSFERASE
NADP[³ H], preparation of stereospecific forms,	ELECTRON-TRANSFER REACTIONS	NICOTINATE PHOSPHORIBOSYLTRANS- FERASE
NICOTINIC ACID: ANALOGUES AND CO- ENZYMES	HALF-REACTIONS	Nicotinate D-ribonucleotide,
NAD(P)H:quinone reductase,	NET RATE CONSTANT METHOD FOR DE- RIVING ENZYME EQUATIONS	NICOTINAMIDE MONONUCLEOTIDE AD- ENYLYLTRANSFERASE
NAD(P)H DEHYDROGENASE (QUINONE)	Neuraminidases,	NICOTINATE-NUCLEOTIDE DIMETHYL- BENZIMIDAZOLE PHOSPHORIBO- SYLTRANSFERASE
NADP ⁺ synthesis from NAD ⁺ ,	SIALIDASES	NICOTINAMIDE PHOSPHORIBOSYL- TRANSFERASE
NAD ⁺ KINASE	NEUROLYSIN	QUINOLINATE PHOSPHORIBOSYL- TRANSFERASE
NAD(P) ⁺ TRANSHYDROGENASE	Neurotensin proteolysis,	NICOTINIC ACID: ANALOGUES AND COENZYMES
NAD(P) ⁺ transhydrogenase (B-specific),	NEUROLYSIN	NIH SHIFT
NAD(P) ⁺ TRANSHYDROGENASE	Neutral density filter,	Nitrate conversion to nitrite,
NAD ⁺ pyrophosphorylase,	ATTENUANCE FILTER	NITRATE REDUCTASE
NICOTINAMIDE MONONUCLEOTIDE AD- ENYLYLTRANSFERASE	Newman projection,	NITRATE REDUCTASE
NAD ⁺ synthesis,	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	NITRENE
NICOTINAMIDE MONONUCLEOTIDE AD- ENYLYLTRANSFERASE	NEWTON	NITRENIUM ION
NAD ⁺ synthesis from deamido-NAD ⁺ ,	NEWTON-RAPHSON METHOD	Nitric oxide,
NAD ⁺ SYNTHETASE	N-H EXCHANGE	OXYGEN, OXIDES & OXO ANIONS
NAD ⁺ SYNTHETASE	NHOMO,	NITRIC OXIDE SYNTHASE
NANO-	PHYSICAL ORGANIC CHEMISTRY NO- MENCLATURE	AUTOINHIBITION
NANOSECOND ABSORPTION SPEC- TROSCOPY	NICKEL-DEPENDENT ENZYMES	Nitrile aminohydrolase,
NARCISSISTIC REACTION	Nickel ion,	NITRILE HYDRATASE
NATURAL ABUNDANCE	IONIC RADIUS	NITRILE HYDRATASE
ISOTOPE EXCHANGE KINETICS	NICOTINAMIDASE	NITRITE REDUCTASE
COMPARTMENTAL ANALYSIS	Nicotinamide,	Nitrogen analogues of carbenes,
Natural lifetime,	NAD ⁺ GLYCOHYDROLASE	NITRENE
RADIATIVE LIFETIME	NICOTINAMIDE N-METHYLTRANS- FERASE	NITROGENASE
NDP-hexose pyrophosphorylase,	NICOTINAMIDE PHOSPHORIBOSYL- TRANSFERASE	NITROGEN FIXATION
NUCLEOSIDE TRIPHOSPHATE:HEXOSE- 1-PHOSPHATE	NICOTINAMIDE COENZYMES (SPECTRAL PROPERTIES)	Nitroxide spin-labeled phospholipids,
NUCLEOTIDYLTRANSFERASES	NICOTINAMIDE N-METHYLTRANSFERASE	PHOSPHOLIPID FLIP-FLOP
NDPK,	NICOTINAMIDE MONONUCLEOTIDE ADE- NYLYLTRANSFERASE	NOISE
HALF-REACTIONS	Nicotinamide nucleotide transhydrogenase,	
NEGATIVE COOPERATIVITY	NAD(P) ⁺ TRANSHYDROGENASE	
LIGAND BINDING	NICOTINAMIDE PHOSPHORIBOSYLTRANS- FERASE	
KOSHLAND-NEMETHY-FILMER MODEL		

Nonadditivity of entropies and free energy for weak interactions, <i>ADDITIVITY PRINCIPLE</i>	D-NOPALINE SYNTHASE	<i>RITCHIE EQUATION</i>
Nonaprenyl-diphosphate synthase, <i>trans-OCTAPRENYLTRANSTRANSFERASE</i>	NOREPINEPHRINE N-METHYLTRANSFERASE	<i>ELECTROPHILE</i>
Nonbonded interactions, <i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>	NORMAL DISTRIBUTION <i>STATISTICS (A Primer)</i>	NUCLEOPHILIC CATALYSIS
NONCOMPETITIVE INHIBITION <i>MIXED-TYPE INHIBITION</i>	NORMAL ERROR CURVE <i>STATISTICS (A Primer)</i>	NUCLEOPHILIC COMPETITION
Noncompetitive inhibition, limiting case of, <i>MIXED-TYPE INHIBITION</i>	NORMALITY (or, NORMAL CONCENTRATION)	NUCLEOPHILICITY <i>ALPHA EFFECT</i> <i>ELECTROPHILICITY</i>
NONCOVALENT INTERACTIONS <i>ALLOSTERIC INTERACTION</i> <i>BINDING INTERACTION</i> <i>BINDING ISOTHERM</i> <i>BIOSENSOR</i> <i>COOPERATIVE LIGAND BINDING</i> <i>EQUILIBRIUM CONSTANT</i> <i>EQUILIBRIUM DIALYSIS</i> <i>HUMMEL-DREYER METHOD</i> <i>LINKED FUNCTION THEORY</i> <i>KLOTZ PLOT</i> <i>MACROSCOPIC CONSTANT</i> <i>MICROSCOPIC CONSTANT</i> <i>MOLECULAR CROWDING</i> <i>SCATCHARD PLOT</i>	NORTHROP ISOTOPE EFFECT METHOD <i>KINETIC ISOTOPE EFFECTS</i>	Nucleophilicity of hydrazines and hydroxylamines, <i>ALPHA EFFECT</i>
NONEQUILIBRIUM THERMODYNAMICS (A Primer)	NOTEBOOK ENTRIES	NUCLEOPHILIC SUBSTITUTION REACTION
Non-first-order process, <i>NOYES EQUATION</i>	NOYES EQUATION	Nucleophilic substitution reactions on phosphorus, <i>CHIRAL ATP</i> <i>PSEUDOROTATION</i>
NONHYPERBOLIC KINETICS	NU (ν)	Nucleoside 2',3'-cyclic phosphodiester, <i>RIBONUCLEASE (RNase)</i>
Noninteger reaction orders in elementary reactions, <i>FRACTAL REACTION KINETICS</i>	NUCLEARITY	NUCLEOSIDE DEOXYRIBOSYLTRANSFERASE
NONLINEAR INHIBITION <i>CLELAND RULES FOR DEAD-END INHIBITION</i> <i>ABORTIVE COMPLEX</i>	NUCLEAR MAGNETIC RESONANCE <i>LARMOR PRECESSION</i> <i>LIGAND BINDING ANALYSIS</i> <i>LINE-SHAPE ANALYSIS</i> <i>LOW-BARRIER HYDROGEN BONDS: ROLE IN CATALYSIS</i> <i>MAGNESIUM ION (INTRACELLULAR)</i> <i>MAGNETIZATION TRANSFER</i>	NUCLEOSIDE DIPHOSPHATASE
Nonlinear kinetics of drug bioavailability, <i>PHARMACOKINETICS</i>	Nuclear pores, <i>DIFFUSION OF MOLECULES INTO A PORE</i>	NUCLEOSIDE DIPHOSPHATE KINASE <i>HALF-REACTIONS</i>
NONLINEAR LEAST SQUARES ANALYSIS	Nucleation, <i>ACTIN ASSEMBLY KINETICS</i> <i>BIOCHEMICAL SELF-ASSEMBLY</i> <i>BIOMINERALIZATION</i> <i>PRION PLAQUE FORMATION</i>	Nucleoside 5'-diphosphate phosphotransferase, <i>NUCLEOSIDE DIPHOSPHATE KINASE</i>
Nonmetal oxides, <i>OXYGEN, OXIDES & OXO ANIONS</i>	Nucleation as a highly cooperative process, <i>MICROTUBULE ASSEMBLY KINETICS</i>	Nucleoside diphosphokinase, <i>NUCLEOSIDE DIPHOSPHATE KINASE</i>
Nonprocessivity, <i>RANDOM SCISSION KINETICS</i>	Nucleation center, in microtubule self-assembly, <i>MICROTUBULE ASSEMBLY KINETICS</i>	Nucleoside monophosphate, <i>NUCLEOSIDE DIPHOSPHATASE</i>
Nonproductive reversible complexes, <i>ABORTIVE COMPLEXES</i>	Nucleation of polymerization, <i>HEMOGLOBIN-S POLYMERIZATION</i>	NUCLEOSIDE PHOSPHOTRANSFERASE
Nonproductive substrate binding, <i>SHIFTED BINDING</i>	Nucleic acid dynamics, <i>NUCLEAR MAGNETIC RESONANCE</i>	NUCLEOSIDE TRIPHOSPHATE: HEXOSE-1-PHOSPHATE NUCLEOTIDYLTRANSFERASES
NONRADIATIVE DECAY	NUCLEASE S1	Nucleoside triphosphate reaction with a hexose 1-phosphate, <i>NUCLEOSIDE TRIPHOSPHATE HEXOSE-1-PHOSPHATE NUCLEOTIDYLTRANSFERASES</i>
	NUCLEATION <i>ACTIN ASSEMBLY KINETICS</i> <i>MICROTUBULE ASSEMBLY KINETICS</i> <i>PROTEIN POLYMERIZATION KINETICS</i>	NUCLEOSIDE-5'-TRIPHOSPHATE REGENERATION
	NUCLEIC ACID RENATURATION KINETICS	NUCLEOSOME CORE PARTICLE SELF-ASSEMBLY
	Nucleic acid structure, <i>BASE STACKING</i>	3'-NUCLEOTIDASE
	NUCLEOFUGE <i>ELECTROFUGE</i> <i>NUCLEOPHILE</i>	5'-NUCLEOTIDASE
	NUCLEOPHILE	NUCLEOTIDE EXCHANGE FACTORS
		Nucleotides and nucleic acids, <i>BIOCHEMICAL NOMENCLATURE</i>
		NUMERICAL COMPUTER METHODS
		NUMERICAL INTEGRATION <i>STIFFNESS</i> <i>GEAR ALGORITHM</i> <i>NUMERICAL COMPUTER METHODS</i>

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OCCAM'S RAZOR
 OCCUPANCY THEORY OF DRUG ACTION
 OCTAHEDRAL COORDINATION
 OCTANOL DEHYDROGENASE
trans-OCTAPRENYLTRANSTRANSFERASE
 OCTOPINE DEHYDROGENASE
 D-Octopine synthase,
 OCTOPINE DEHYDROGENASE
 OFFORD PLOT
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 Old yellow enzyme,
 NADPH DEHYDROGENASE
 2',5'-OLIGOADENYLATE SYNTHETASE
 OLIGOMERIZATION
 OMEGA (Ω or ω)
 OMPTIN
 One-dimensional diffusion,
 DIFFUSION, REDUCTION IN DIMENSION-
 ALITY
 One-dimensional diffusion in the linear in-
 terval,
 DIFFUSION, REDUCTION IN DIMENSION-
 ALITY
 One-step reaction systems,
 CHEMICAL KINETICS
 One-substrate mechanism,
 RANDOM UNI BI MECHANISM
 OPEN SYSTEM
 Opposite-end assembly and disassembly,
 MICROTUBULE TREADMILLING
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 OPTICAL ACTIVITY
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 Optically active substance,
 OPTICAL ROTATION
 Optically labile,
 STEREOCHEMICAL TERMINOLOGY,
 IUPAC RECOMMENDATIONS
 OPTICAL ROTATION
 OPTICAL TWEEZERS
 FORCE EFFECTS ON MOLECULAR
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 OPTICAL YIELD
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 OPTIMAL pH
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ORDERED BI BI ENZYME MECHANISM
 ISOTOPE EXCHANGE AT EQUILIBRIUM
 ORDERED BI BI THEORELL-CHANCE MECH-
 ANISM
 ORDERED BI TER MECHANISM
 ORDERED BI UNI MECHANISM
 ORDERED BI UNI UNI BI PING PONG
 MECHANISM
 ORDER OF REACTION
 MOLECULARITY
 CHEMICAL KINETICS
 FIRST-ORDER REACTIONS
 RATE CONSTANTS
 ORDERED ON-RANDOM OFF BI BI MECH-
 ANISM
 ORDERED TER BI MECHANISM
 ORDERED TER TER MECHANISM
 ORDERED UNI BI MECHANISM
 Oriented diffusion,
 BIOMINERALIZATION
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 ORNITHINE TRANSCARBAMOYLASE
 D-ORNITHINE 4,5-AMINOMUTASE
 ORNITHINE AMINOTRANSFERASE
 ORNITHINE DECARBOXYLASE
 Ornithine carbamoyltransferase,
 ORNITHINE TRANSCARBAMOYLASE
 Ornithine:ketoacid aminotransferase,
 ORNITHINE AMINOTRANSFERASE
 Ornithine:oxo-acid aminotransferase,
 ORNITHINE AMINOTRANSFERASE
 ORNITHINE TRANSCARBAMOYLASE
 Orotate,
 OROTATE PHOSPHORIBOSYLTRANS-
 FERASE
 OROTATE PHOSPHORIBOSYLTRANS-
 FERASE
 Orotidine 5'-phosphate,
 OROTATE PHOSPHORIBOSYLTRANS-
 FERASE
 OROTIDYLATE DECARBOXYLASE
 Orotidine-5'-phosphate decarboxylase,
 OROTIDYLATE DECARBOXYLASE
 Orotidine-5'-phosphate pyrophosphorylase,
 OROTATE PHOSPHORIBOSYLTRANS-
 FERASE
 OROTIDYLATE DECARBOXYLASE
 Orotidylic acid phosphorylase,
 OROTATE PHOSPHORIBOSYLTRANS-
 FERASE
 Orthophosphate,
 OXYGEN, OXIDES & OXO ANIONS

Orthophosphate as substrate or product,
 ACETATE KINASE (PYROPHOSPHATE)
 ACETYL-CoA CARBOXYLASE
 ACID PHOSPHATASE
 ACTOMYOSIN ATPase
 ACYL PHOSPHATASE
 ASPARTATE-SEMIALDEHYDE DEHYDRO-
 GENASE
 ATPases
 CARBAMOYL PHOSPHATE SYNTHETASE
 I and II
 CELLOBIOSE PHOSPHORYLASE
 CHORISMATE SYNTHASE
 COBALAMIN ADENOSYLTRANSFERASE
 3-DEHYDROQUINATE SYNTHASE
 DEOXYCYTIDYLATE HYDROXYMETHYL-
 TRANSFERASE
 3-DEOXYOCTULOSONATE 8-PHOS-
 PHATE SYNTHASE
 DETHIOBIOTIN SYNTHASE
 DIHYDROFOLATE SYNTHETASE
 DIPHOSPHOMEVALONATE DECARBOX-
 YLASE
 FOLYLPOLY- γ -GLUTAMATE SYNTHASE
 FORMYLGLYCINAMIDINE SYNTHETASE
 FRUCTOSE BISPHOSPHATASES
 GLUTAMINE SYNTHETASE
 γ -GLUTAMYL CYSTEINE SYNTHETASE
 γ -GLUTAMYL KINASE
 GLUTATHIONE SYNTHETASE
 GLYCERALDEHYDE-3-PHOSPHATE DEHY-
 DROGENASE
 GLYCOGEN PHOSPHORYLASE
 HISTIDINOL-PHOSPHATE PHOSPHATASE
 myo-INOSITOL 1-MONOPHOSPHATASE
 INOSITOL POLYPHOSPHATE PHOSPHA-
 TASES
 MALTOSE PHOSPHORYLASE
 5,10-METHENYLTETRAHYDROFOLATE
 SYNTHETASE
 β -METHYLCROTONYL-CoA CARBOX-
 YLASE
 MYOSIN and MYOSIN ATPase
 NAD⁺ SYNTHETASE
 3'-NUCLEOTIDASE
 5'-NUCLEOTIDASE
 ORTHOPHOSPHATE CONTINUOUS
 ASSAY
 PHOSPHATE ACETYLTRANSFERASE
 PHOSPHATIDYLGLYCEROL PHOSPHATE
 PHOSPHOHYDROLASE
 PHOSPHOENOLPYRUVATE CARBOXYKI-
 NASE (PYROPHOSPHATE)

PHOSPHOENOLPYRUVATE CARBOXYLASE	Oxidative phosphorylation, index of, P/O RATIO	DIMETHYLANILINE MONOOXYGENASE
PHOSPHORYLASE PHOSPHATASE	OXIDOREDUCTASES	DOPAMINE β -MONOOXYGENASE
PHOSPHOTYROSINE PROTEIN PHOSPHATASE	ENZYME COMMISSION NOMENCLATURE	FLAVANONE 3 β -HYDROXYLASE
POLYNUCLEOTIDE PHOSPHORYLASE	Oxocarbonium ion intermediate, KINETIC ISOTOPE EFFECTS	GALACTOSE OXIDASE
PROPIONYL-CoA CARBOXYLASE	5-OXOPROLINASE	GENTISATE 1,2-DIOXYGENASE
PROTEASE La	Oxyanion intermediate, CROTONASE SUPERFAMILY	GIBBERELLIN HYDROXYLASES
PURINE NUCLEOSIDE PHOSPHORYLASE	Oxygen binding, BOHR EFFECT	GLYCOLATE OXIDASE
PUTRESCINE CARBAMOYLTRANSFERASE	Oxygen binding to hemoglobin, ADAIR EQUATION	HEME OXYGENASE
PYRUVATE CARBOXYLASE	HEMOGLOBIN	HOMOGENTISATE 1,2-DIOXYGENASE
PYRUVATE DEHYDROGENASE PHOSPHATASE	Oxygen consumption, QUOTIENT, RESPIRATORY	HYDROCARBON MONOOXYGENASE
PYRUVATE OXIDASE	Oxygen (dioxygen) as a substrate, ACYL-ACYL-CARRIER-PROTEIN Δ^9 -DESATURASE	3-HYDROXYANTHRANILATE 3,4-DIOXYGENASE
RIBOSE-5-PHOSPHATE ADENYLYLTRANSFERASE	ACYL-CoA DESATURASE	HYDROXYBENZOATE MONOOXYGENASES
SELENOPHOSPHATE SYNTHETASE	ALKANE HYDROXYLASE	4-HYDROXYPHENYLACETATE 3-MONOOXYGENASE
STARCH PHOSPHORYLASE	AMINE OXIDASES	4-HYDROXYPHENYLPYRUVATE HYDROXYLASE
SUCCINYL-CoA SYNTHETASE	D-AMINO ACID OXIDASE	IMIDAZOLEACETATE HYDROXYLASE
SUCROSE PHOSPHORYLASE	ANTHRANILATE 1,2-DIOXYGENASE	INDOLEAMINE 2,3-DIOXYGENASE
TUBULIN:TYROSINE LIGASE	ANTHRANILATE HYDROXYLASE	myo-INOSITOL OXYGENASE
ORTHOPHOSPHATE CONTINUOUS ASSAY	ARGININE MONOOXYGENASE	JUGLONE MONOOXYGENASE
Orthovanadate,	ASCORBATE OXIDASE	KYNURENINE 3-HYDROXYLASE
OXYGEN, OXIDES & OXO ANIONS	D-ASPARTATE OXIDASE	LACCASE
Osmolarity,	BENZENE 1,2-DIOXYGENASE	LACTATE MONOOXYGENASE
ISOTONIC BUFFERS	BERBERINE BRIDGE ENZYME	LIPOXYGENASE
OSTWALD RIPENING	CAMPOR 5-MONOOXYGENASE	5-LIPOXYGENASE
OSTWALD SOLUBILITY COEFFICIENT	CATECHOL 1,2-DIOXYGENASE	8-LIPOXYGENASE
OUTER SPHERE ELECTRON TRANSFER REACTION	CATECHOL 2,3-DIOXYGENASE	12-LIPOXYGENASE
METAL ION COORDINATION	CATECHOL OXIDASE	15-LIPOXYGENASE
Out-isomer,	CHOLINE OXIDASE	LUCIFERASE
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	trans-CINNAMATE 4-MONOOXYGENASE	LYSINE MONOOXYGENASE
OVERSATURATION	COPROPORPHYRINOGEN OXIDASE	LYSYL HYDROXYLASE
OXALOACETATE DECARBOXYLASE	CYCLOHEXANONE MONOOXYGENASE	MANDELATE 4-MONOOXYGENASE
Oxaloacetate, synthesis in gluconeogenesis,	CYSTEAMINE DIOXYGENASE	MELILOTATE HYDROXYLASE
PYRUVATE CARBOXYLASE	CYTOCHROME P-450 REDUCTASE	METHANE MONOOXYGENASE
PHOSPHOENOLPYRUVATE CARBOXYKINASE (PYROPHOSPHATE)	DESATURASES, FATTY ACYL-CoA	4-METHOXYBENZOATE MONOOXYGENASE
OXALYL-CoA DECARBOXYLASE	DIHYDROOROTATE DEHYDROGENASE	METHYLHYDROXYPYRIDINECARBOXYLATE DIOXYGENASE
Oxidation,	2,3-DIHYDROXYBIPHENYL 1,2-DIOXYGENASE	MONOPHENOL MONOOXYGENASE
ELECTRODE KINETICS	3,4-DIHYDROXYPHENYLACETATE 2,3-DIOXYGENASE	NITRIC OXIDE SYNTHASE
Oxidation/reduction reaction,	2,5-DIHYDROXYPYRIDINE 5,6-DIOXYGENASE	PEPTIDYL GLYCINE α -AMIDATING MONOOXYGENASE
NERNST EQUATION	2,3-DIHYDROXY-9,10-SECOANDROSTA-1,3,5(10)-TRIENE-9,17-DIONE 4,5-DIOXYGENASE	PHENOL HYDROXYLASE
Oxidative addition,		PHENYLALANINE MONOOXYGENASE
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE		PLASMANYLETHANOLAMINE Δ^1 -DESATURASE
Oxidative coupling,		PROLYL 3-HYDROXYLASE
PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE		PROSTAGLANDIN H ₂ SYNTHASE
		PROTOCATECHUATE 3,4-DIOXYGENASE
		PROTOCATECHUATE 4,5-DIOXYGENASE

PROTOPORPHYRINOGEN OXIDASE
 PUTRESCINE OXIDASE
 PYRUVATE OXIDASE
 QUERCETIN 2,3-DIOXYGENASE
D-RIBULOSE-1,5-BISPHOSPHATE CAR-
 BOXYLASE/OXYGENASE
 SALICYLATE 1-MONOOXYGENASE
 SARCOSINE OXIDASE
 STEROID HYDROXYLASES
 SULFITE OXIDASE
 SULFUR DIOXYGENASE
 THYMIDINE 2'-HYDROXYLASE
 TOLUENE DIOXYGENASE
 XANTHINE OXIDASE
 OXYGEN ELECTRODE FOR MONITORING
 HEMOGLOBIN OXYGENATION
 OXYGEN, OXIDES & OXO ANIONS
 Oxygen, photo-induced incorporation,
 PHOTO-OXYGENATION
 OXYGEN RADICALS
 Oxygen transport in erythrocytes,
 DIFFUSION OF OXYGEN WITHIN RED
 CELLS
 Oxygen uptake,
 MEYERHOF OXIDATION QUOTIENT
 Ozone,
 OXYGEN, OXIDES & OXO ANIONS

P

PACEMAKER REACTION
 PACKING DENSITY
 Palitzsch buffers,
 ISOTONIC BUFFERS
 PANTOTHENIC ACID, COENZYME A, AND
 DERIVATIVES
 PAPAIN
 Paracrystals,
 BIOMINERALIZATION
 PARALLEL FIRST-ORDER REACTIONS
 Parallel line kinetics,
 PHI VALUES IN BISUBSTRATE MECHA-
 NISMS
 INITIAL RATE ENZYME ASSAYS
 PARALLEL REACTIONS
 See specific type of parallel reaction
 PARAMAGNETIC
 PERMEABILITY
 MAGNETIC SUSCEPTIBILITY
 DIAMAGNETIC
 ELECTRON SPIN RESONANCE

PARTIAL COMPETITIVE INHIBITION
 COMPETITIVE INHIBITION
 PARTIAL DERIVATIVES (Basic Properties)
 Partial exchange reactions,
 SUBSTRATE SYNERGISM
 PARTIAL INHIBITION
 PARTIAL COMPETITIVE INHIBITION
 INHIBITION
 PARTIAL MOLAR VOLUMES
 DALTON'S LAW OF PARTIAL PRESSURES
 CONCENTRATION
 MOLECULAR CROWDING
 PARTIAL PRESSURE
 DALTON'S LAW OF PARTIAL PRESSURE
 CONCENTRATION
 PARTIAL RATE FACTOR
 SELECTIVITY FACTOR
 Partition and diffusion coefficients relative to
 membrane thickness,
 PERMEABILITY COEFFICIENT
 PARTITIONING FACTOR
 PARVULIN
 PASCAL
 PASCAL'S TRIANGLE
 PATCH CLAMP
 Pauling's postulate transition state stabili-
 zation,
 ENTROPY TRAP MODEL
 MOLECULAR SIMILARITY
 TRANSITION-STATE INHIBITORS

pCa
 pD
 Pectate,
 PECTINESTERASE
 Pectin demethoxylase,
 PECTINESTERASE
 PECTINESTERASE
 Pectin methoxylase,
 PECTINESTERASE
 Pectin methylesterase,
 PECTINESTERASE
 Peltier cooling devices, noise reduction in
 PM tubes,
 NOISE
 Penefsky centrifuged-column procedure,
 RAPID BUFFER EXCHANGE
 PENICILLOPEPSIN
 Pentacovalent phosphorane intermediates,
 PSEUDOROTATION
 Pentavalent intermediate, in phosphotrans-
 fer reactions,
 PSEUDOROTATION

Pentose-5-phosphate 3-epimerase,
 RIBULOSE-5-PHOSPHATE 3-EPIMERASE
 PEPSINS and PEPSINOGENS
 Peptide bond(s),
 PEPTIDOMIMETIC COMPOUND
 PEPTIDE METHIONINE SULFOXIDE RE-
 DUCTASE
 PEPTIDOMIMETIC COMPOUND
 PEPTIDYL-ASPARTATE ENDOPEPTIDASE
 Peptidylglycine,
 PEPTIDYL GLYCINE α -AMIDATING
 MONOOXYGENASE
 PEPTIDYL GLYCINE α -AMIDATING
 MONOOXYGENASE
 Peptidylglycine 2-hydroxylase,
 PEPTIDYL GLYCINE α -AMIDATING
 MONOOXYGENASE
 Peptidylglycine monooxygenase,
 PEPTIDYL GLYCINE α -AMIDATING
 MONOOXYGENASE
 Peptidylprolyl *cis-trans* isomerase,
 PROLYL ISOMERASE
 Peptidylprolyl isomerase,
 PROLYL ISOMERASE
 PEPTIDYLTRANSFERASE
 Percent diastereoisomer excess,
 STEREOCHEMICAL TERMINOLOGY,
 IUPAC RECOMMENDATIONS
 Percent enantiomer excess,
 STEREOCHEMICAL TERMINOLOGY,
 IUPAC RECOMMENDATIONS
 Percolation clusters,
 FRACTAL REACTION KINETICS
 PERIODATE CLEAVAGE
 MALAPRADE REACTION
 Periodicity,
 CIRCADIAN RHYTHM
 Period of sustained chemical behavior or bio-
 logical response,
 PERSISTENCE TIME
 Periplanar,
 STEREOCHEMICAL TERMINOLOGY,
 IUPAC RECOMMENDATIONS
 Periselectivity,
 PHYSICAL ORGANIC CHEMISTRY NO-
 MENCLATURE
 PERMEABILITY
 MAGNETIC SUSCEPTIBILITY
 PERMEABILITY COEFFICIENT
 PERMISSIVE ACTIVE SITE
 PERMUTATIONS AND COMBINATIONS
 STATISTICS (A Primer)
 LINKED FUNCTIONS

PEROXIDASE	PHENYLALANINE RACEMASE	PHOSPHATIDYLGLYCEROL PHOSPHATE PHOSPHOHYDROLASE
<i>HORSERADISH PEROXIDASE</i>	PHENYLMETHYLSULFONYL FLUORIDE	Phosphatidylglycerophosphatase,
OVOPEROXIDASE	<i>CHYMOTRYPSIN</i>	<i>PHOSPHATIDYLGLYCEROL PHOSPHATE</i> <i>PHOSPHOHYDROLASE</i>
MYELOPEROXIDASE	<i>PROTEASE INHIBITOR "COCKTAILS"</i>	Phosphatidylglycerophosphate,
Perpendicular effect,	Phenylpyruvate,	<i>PHOSPHATIDYLGLYCEROL PHOSPHATE</i> <i>PHOSPHOHYDROLASE</i>
<i>PHYSICAL ORGANIC CHEMISTRY</i> <i>NOMENCLATURE</i>	<i>PHENYLALANINE DEHYDROGENASE</i>	1-PHOSPHATIDYLINOSITOL 4-KINASE
PERSISTENCE TIME	<i>PREPHENATE DEHYDRATASE</i>	PHOSPHATIDYLINOSITOL-4-PHOSPHATE 5-KINASE
<i>TWIDDLING</i>	PHENYLPYRUVATE SYNTHASE	Phosphatidyl-1D-myo-inositol,
PERSISTENT	PHI (ϕ or Φ)	<i>PHOSPHATIDYLINOSITOL SYNTHASE</i>
<i>PHYSICAL ORGANIC CHEMISTRY</i> <i>NOMENCLATURE</i>	PHI VALUES IN BISUBSTRATE MECHANISMS	1-Phosphatidyl-1D-myo-inositol 4,5-bisphosphate synthesis,
<i>TRANSIENT CHEMICAL SPECIES</i>	<i>INITIAL RATE ENZYME ASSAYS</i>	<i>PHOSPHATIDYLINOSITOL-4-PHOSPHATE</i> <i>5-KINASE</i>
PERTUSSIS TOXIN	Phosphate,	1-Phosphatidyl-1D-myo-inositol 4-phosphate synthesis,
Phalloidin,	<i>ORTHOPHOSPHATE CONTINUOUS</i> <i>ASSAY</i>	<i>PHOSPHATIDYLINOSITOL-4-PHOSPHATE</i> <i>5-KINASE</i>
<i>ACTIN ASSEMBLY KINETICS</i>	<i>OXYGEN, OXIDES & OXO ANIONS</i>	1-Phosphatidyl-1D-myo-inositol 4-phosphate synthesis,
PHARMACOKINETICS	PHOSPHATE ACETYLTRANSFERASE (Phosphotransacetylase)	<i>1-PHOSPHATIDYLINOSITOL 4-KINASE</i>
PHASE	PHOSPHATE CONTINUOUS ASSAY	1-PHOSPHATIDYLINOSITOL PHOSPHODIESTERASE
PHASE MODULATION FLUORESCENCE	<i>ORTHOPHOSPHATE CONTINUOUS</i> <i>ASSAY</i>	Phosphatidylinositol phospholipase C,
PHASE-TRANSFER CATALYSIS	PHOSPHATE DIESTER HYDROLYSIS	<i>1-PHOSPHATIDYLINOSITOL PHOSPHODIESTERASE</i>
pH dependence in enzymic catalysis,	Phosphate ester,	PHOSPHATIDYLINOSITOL SYNTHASE
pH EFFECTS ON ENZYMIC CATALYSIS	<i>ARSENOLYSIS</i>	Phosphatidyl-N-methylethanolamine,
pH EFFECTS ON ENZYMIC CATALYSIS	PHOSPHATE MONOESTER HYDROLYSIS	<i>PHOSPHATIDYLETHANOLAMINE</i> <i>N-METHYLTRANSFERASE</i>
pH ELECTRODE	Phosphatidase,	PHOSPHATIDYL-N-METHYLETHANOLAMINE N-METHYLTRANSFERASE
pH electrode correction for D ₂ O,	<i>PHOSPHOLIPASE A₂</i>	O-sn-Phosphatidyl-L-serine,
pD	3-sn-Phosphatidate,	<i>PHOSPHATIDYLSERINE SYNTHASE</i>
Phenethylamine,	<i>PHOSPHATIDATE PHOSPHATASE</i>	PHOSPHATIDYLSERINE DECARBOXYLASE
<i>PHENYLALANINE DECARBOXYLASE</i>	PHOSPHATIDATE CYTIDYLYLTRANSFERASE	PHOSPHATIDYLSERINE SYNTHASE
PHENOL HYDROXYLASE	PHOSPHATIDATE PHOSPHATASE	Phosphinate inhibitor,
Phenol 2-monooxygenase,	Phosphatidolipase,	<i>D-ALANINE-D-ALANINE LIGASE</i>
<i>PHENOL HYDROXYLASE</i>	<i>PHOSPHOLIPASE A₂</i>	PHOSPHOACETYLGLUCOSAMINE MUTASE
Phenylalaninase,	Phosphatidylcholine 2-acylhydrolase,	Phosphoacylase,
<i>PHENYLALANINE MONOOXYGENASE</i>	<i>PHOSPHOLIPASE A₂</i>	<i>PHOSPHATE ACETYLTRANSFERASE</i>
Phenylalanine,	Phosphatidylcholine, hydrolysis of,	Phosphoadenylylsulfatase,
<i>PHENYLALANINE AMINOTRANSFERASE</i>	<i>PHOSPHOLIPASE D</i>	<i>3'-PHOSPHOADENYLYL SULFATE SULFOHYDROLASE</i>
<i>PHENYLALANINE AMMONIA-LYASE</i>	PHOSPHATIDYLCHOLINE:SERINE O-PHOSPHATIDYLTRANSFERASE	3'-Phosphoadenylylsulfate hydrolysis,
<i>PHENYLALANINE DECARBOXYLASE</i>	Phosphatidyl-N-dimethylethanolamine,	<i>3'-PHOSPHOADENYLYL SULFATE SULFOHYDROLASE</i>
<i>PHENYLALANINE DEHYDROGENASE</i>	<i>PHOSPHATIDYL-N-METHYLETHANOLAMINE</i> <i>N-METHYLTRANSFERASE</i>	3'-PHOSPHOADENYLYL SULFATE SULFOHYDROLASE
<i>PHENYLALANINE MONOOXYGENASE</i>	Phosphatidylethanolamine,	Phosphoanhydride,
<i>PHENYLALANINE RACEMASE</i>	<i>PHOSPHATIDYLETHANOLAMINE</i> <i>N-METHYLTRANSFERASE</i>	<i>ACYL-PHOSPHATE INTERMEDIATE</i> <i>ENERGY CHARGE MODEL</i>
PHENYLALANINE AMINOTRANSFERASE	<i>PHOSPHATIDYLSERINE DECARBOXYLASE</i>	6-Phospho-2-dehydro-D-gluconate and NAD(P)H,
AROMATIC AMINO ACID AMINOTRANSFERASE	PHOSPHATIDYLETHANOLAMINE N-METHYLTRANSFERASE	<i>6-PHOSPHOGLUCONATE DEHYDROGENASE</i>
PHENYLALANINE AMMONIA-LYASE	Phosphatidylglycerol,	
<i>DEHYDROALANINE</i>	<i>PHOSPHATIDYLGLYCEROL PHOSPHATE</i> <i>PHOSPHOHYDROLASE</i>	
<i>BOROHYDRIDE REDUCTION</i>		
PHENYLALANINE DECARBOXYLASE		
PHENYLALANINE DEHYDROGENASE		
Phenylalanine(histidine) aminotransferase,		
<i>PHENYLALANINE AMINOTRANSFERASE</i>		
PHENYLALANINE HYDROXYLASE (Phenylalanine Monooxygenase)		

Phosphodeoxyribomutase, PHOSPHOPENTOMUTASE	3-Phospho-D-glycerate, D-RIBULOSE-1,5-BISPHOSPHATE CAR- BOXYLASE/OXYGENASE	PHOSPHOPENTOMUTASE
PHOSPHODIESTERASES See also specific enzyme	PHOSPHOGLYCERATE MUTASE	Phosphopentose isomerase, RIBOSE-5-PHOSPHATE EPIMERASE
Phosphodiester hydrolysis, EXONUCLEASES	3-PHOSPHOGLYCERATE DEHYDRO- GENASE	PHOSPHOPROTEIN PHOSPHATASES
Phosphoenolpyruvate, PHOSPHOENOLPYRUVATE CARBOX- YLASE	3-PHOSPHOGLYCERATE KINASE	Phosphopyruvate carboxylase, PHOSPHOENOLPYRUVATE CARBOXY- KINASE
PYRUVATE KINASE	PHOSPHOGLYCERATE MUTASE	PHOSPHOENOLPYRUVATE CARBOXYKI- NASE (PYROPHOSPHATE)
PYRUVATE, WATER DIKINASE	Phosphoglycerate phosphomutase, PHOSPHOGLYCERATE MUTASE	PHOSPHORESCENCE
PYRUVATE, ORTHOPHOSPHATE DIK- INASE	Phosphoglyceromutase, PHOSPHOGLYCERATE MUTASE	JABLONSKI DIAGRAM
PHOSPHOENOLPYRUVATE CARBOXY- KINASE	3-Phospho-D-glyceroyl phosphate, 3-PHOSPHOGLYCERATE KINASE	FLUORESCENCE
PHOSPHOENOLPYRUVATE CARBOXYKI- NASE (Pyrophosphate)	PHOSPHOGLYCERIDE:LYSOPHOSPHATIDYL- GLYCEROL ACYLTRANSFERASE	5-Phospho- α -D-ribose 1-diphosphate, RIBOSEPHOSPHATE PYROPHOSPHO- KINASE
PHOSPHOENOLPYRUVATE CARBOXYLASE PHOSPHOENOLPYRUVATE CARBOXYKI- NASE (Pyrophosphate)	Phosphohexose isomerase, PHOSPHOGLUCOSE ISOMERASE	NICOTINAMIDE PHOSPHORIBOSYL- TRANSFERASE
PHOSPHOENOLPYRUVATE CARBOXY- KINASE	Phosphohexomutase, PHOSPHOGLUCOSE ISOMERASE	NICOTINATE PHOSPHORIBOSYLTRANS- FERASE
Phosphoenolpyruvate carboxytransphosphor- ylase, PHOSPHOENOLPYRUVATE CARBOXYKI- NASE (PYROPHOSPHATE)	3-Phosphohydroxypyruvate formation, 3-PHOSPHOGLYCERATE DEHYDRO- GENASE	Phosphoriboisomerase, RIBOSE-5-PHOSPHATE EPIMERASE
Phosphoenolpyruvate, interconversion with 3-phosphonopyruvate, PHOSPHOENOLPYRUVATE MUTASE	PHOSPHOKETOLASE	5'-Phosphoribosyl-5-amino-4- imidazolecarboxamide, PHOSPHORIBOSYLAMINOIMIDAZOLE- CARBOXAMIDE FORMYLTRANS- FERASE
PHOSPHOENOLPYRUVATE MUTASE	PHOSPHOLIPASE See specific enzyme	PHOSPHORIBOSYLAMINOIMIDAZOLECAR- BOXAMIDE FORMYLTRANSFERASE
PHOSPHOENOLPYRUVATE PHOSPHATASE	MICELLE	PHOSPHORIBOSYL-AMP CYCLOHY- DROLASE
Phosphoenolpyruvate phosphomutase, PHOSPHOENOLPYRUVATE MUTASE	LIPASE	5-Phosphoribosyl-AMP, formation of, PHOSPHORIBOSYL-ATP PYROPHOS- PHATASE
Phosphoenolpyruvate synthase, PYRUVATE, WATER DIKINASE	INTERFACIAL CATALYSIS	5-Phosphoribosyl-AMP, hydrolysis of, PHOSPHORIBOSYL-AMP CYCLOHY- DROLASE
Phosphoenolpyruvate synthesis, PHOSPHOENOLPYRUVATE CARBOXY- KINASE	PHOSPHOLIPASE A ₁ MICELLE	PHOSPHORIBOSYLANTHRANILATE ISO- MERASE
PYRUVATE, WATER DIKINASE	PHOSPHOLIPASE A ₂	5-Phosphoribosyl-ATP, hydrolysis of, PHOSPHORIBOSYL-ATP PYROPHOS- PHATASE
PHOSPHOFRUCTOKINASE ATP & GTP DEPLETION	PHOSPHOLIPASE B	PHOSPHORIBOSYL-ATP PYROPHOS- PHATASE
6-PHOSPHOFRUCTO-2-KINASE	PHOSPHOLIPASE C	N ¹ -(5-Phospho- α -D-ribosyl)-5,6- dimethylbenzimidazole, NICOTINATE-NUCLEOTIDE DIMETHYL- BENZIMIDAZOLE
6-Phospho-D-galactose, 6-PHOSPHO- β -GALACTOSIDASE	PHOSPHOLIPASE D	PHOSPHORIBOSYLTRANSFERASE
6-PHOSPHO- β -GALACTOSIDASE	PHOSPHOLIPID FLIP-FLOP	5'-Phosphoribosyl-5-formamido-4- imidazolecarboxamide, PHOSPHORIBOSYLAMINOIMIDAZOLE- CARBOXAMIDE FORMYLTRANS- FERASE
β -D-Phosphogalactoside galactohydrolase, 6-PHOSPHO- β -GALACTOSIDASE	PHOSPHOMANNOMUTASE	N-(5'-PHOSPHORIBOSYLFORMIMINO)-5- AMINO-1-(5'-PHOSPHORIBOSYL)-4-
PHOSPHOGLUCOMUTASE	(R)-5-Phosphomevalonate, as substrate, 5-PHOSPHOMEVALONATE KINASE	
6-PHOSPHOGLUCONATE DEHYDRO- GENASE	5-PHOSPHOMEVALONATE KINASE	
PHOSPHOGLUCOSE ISOMERASE	3-Phosphonopyruvate, interconversion with phosphoenolpyruvate, PHOSPHOENOLPYRUVATE MUTASE	
6-PHOSPHO- β -GLUCOSIDASE	(R)-4'-Phosphopantothenate, as substrate, PHOSPHOPANTOTHENOYL-CYSTEINE SYNTHETASE	
	(R)-4'-Phosphopantothenoyl-L-cysteine, PHOSPHOPANTOTHENOYL-CYSTEINE SYNTHETASE	
	PHOSPHOPANTOTHENOYL-CYSTEINE SYN- THETASE	

IMIDAZOLECARBOXAMIDE ISOMERASE	Photochemistry,	$\pi \rightarrow \pi^*$ TRANSITION
5'-Phosphoribosyl-N-formylglycinamide, PHOSPHORIBOSYLGLYCINAMIDE FORMYLTRANSFERASE	QUANTUM YIELD	ABSORPTION SPECTROSCOPY
5'-Phosphoribosylglycinamide, as substrate, PHOSPHORIBOSYLGLYCINAMIDE FORMYLTRANSFERASE	PHOTOCHROMISM	$\pi \rightarrow \sigma^*$ TRANSITION
PHOSPHORIBOSYLGLYCINAMIDE FORMYLTRANSFERASE	PHOTO-CROSS-LINKING	ABSORPTION SPECTROSCOPY
5'-Phosphoribosylglycinamide trans-formylase, PHOSPHORIBOSYLGLYCINAMIDE FORMYLTRANSFERASE	PHOTOAFFINITY LABELING	PICO-
Phosphoribosyl pyrophosphate synthetase, RIBOSEPHOSPHATE PYROPHOSPHOKINASE	PHOTODIMERIZATION	PICORNAIN
Phosphoribulose epimerase, RIBULOSE-5-PHOSPHATE 3-EPIMERASE	PHOTOISOMERIZATION	PING PONG BI BI MECHANISM
PHOSPHORUS-32 (^{32}P)	PHOTOLYSIS	ISOTOPE EXCHANGE AT EQUILIBRIUM
PHOSPHORYLASE KINASE	FLASH PHOTOLYSIS	MULTISITE PING PONG BI BI MECHANISM
PHOSPHORYLASE PHOSPHATASE	PHOTOREACTIVE CAGED COMPOUNDS	HALF REACTION
PHOSPHORYLATION STATE RATIO	Photomultiplier,	SUBSTRATE SYNERGISM
6-Phosphoryl-IMP, ADENYLOSUCCINATE SYNTHETASE	FLUORESCENCE	Ping-pong versus sequential mechanisms, PHI VALUES IN BISUBSTRATE MECHANISMS
Phosphoryl-phosphinate adduct, D-ALANINE-D-ALANINE LIGASE	NOISE	INITIAL RATE ENZYME ASSAYS
PHOSPHOSERINE AMINOTRANSFERASE	RAPID SCAN SPECTROSCOPY	Pitzer strain,
3-PHOSPHOSHIKIMATE 1-CARBOXYVINYLTRANSFERASE	PHOTON COUNTING	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
Phosphotransacetylase, PHOSPHATE ACETYLTRANSFERASE	TIME-CORRELATED SINGLE PHOTON COUNTING.	Planar chirality,
PHOSPHOTRANSFERASE SYSTEM	FLUORESCENCE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
PHOSPHOTYROSINE PROTEIN PHOSPHATASE	PHOTOOXIDATION	PLANCK FUNCTION
PHOSPHOPROTEIN PHOSPHATASES	PHOTOXYGENATION	GIBBS FREE ENERGY
PHOTOACOUSTIC SPECTROSCOPY	PHOTOREDUCTION	MASSIEU FUNCTION
FOURIER TRANSFORM IR/PHOTOACOUSTIC SPECTROSCOPY TO ASSESS SECONDARY STRUCTURE	PHOTOXYGENATION	PLANCK'S CONSTANT
PHOTOAFFINITY LABELING	PHOTOPHYSICAL PROCESS	PLASMALOGEN SYNTHASE
AZIDO PHOTOAFFINITY REAGENTS	PHOTOREACTIVATION	PLASMANYLETHANOLAMINE Δ^1 -DESATURASE
AFFINITY LABELING	PHOTOREACTIVE CAGED COMPOUNDS	PLASMIN
Photobleaching, FLUORESCENCE RECOVERY AFTER PHOTOBLEACHING	PHOTOLYSIS	Plasmon resonance (SPR),
Photocathode fluctuations, NOISE	PHOTOREDUCTION	BIOSENSOR
Photochemical deactivation, QUENCHING	Photosensitive precursors, PHOTOREACTIVE CAGED COMPOUNDS	^{31}P MAGNETIC RESONANCE OF CONTRACTILE SYSTEMS
Photochemically produced singlet oxygen, PHOTOOXYGENATION	PHOTOSYNTHESIS	NUCLEAR MAGNETIC RESONANCE
PHOTOCHEMICAL REACTION	PHOTOSYNTHETIC REACTION CENTER	pMg
	PHOTOSYNTHETIC WATER OXIDATION	PMSF,
	PHOTOSYSTEM I	PHENYLMETHYLSULFONYL FLUORIDE
	pH profile for inactivation, AFFINITY LABELING	Point group,
	pH-RATE PROFILE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
	pH EFFECTS	POINT-OF-CONVERGENCE METHOD
	pH STAT	POINT OF INFLECTION
	pH VALUE	POISE
	pH* VALUE	POISSON DISTRIBUTION
	Phylloquinone reductase, NAD(P)H DEHYDROGENASE (QUINONE)	STATISTICS (A Primer)
	PHYSICAL ORGANIC CHEMISTRY TERMINOLOGY (IUPAC Recommendations)	Poisson statistics,
	6-PHYTASE	COUNTING STATISTICS
	PHYTOENE SYNTHASE COMPLEX	STATISTICS (A Primer)
	GERANYLGERANYL DIPHOSPHATE GERANYLGERANYLTRANSFERASE	POLAR EFFECT
	PI (π)	INDUCTIVE EFFECT
	π -BOND (PI-BOND)	FIELD EFFECT
	π -Orbital interactions in nucleic acids, BASE STACKING	

POLARITY	Positional isotope exchange rate laws, <i>NET RATE CONSTANT METHOD</i>	PREEQUILIBRIUM <i>MICHAELIS-MENTEN KINETICS</i>
POLARIZABILITY		PREEXPONENTIAL FACTOR <i>ARRHENIUS EQUATION</i>
POLARIZATION	POSITIVE COOPERATIVITY <i>LIGAND BINDING COOPERATIVITY</i> <i>$\Delta\Delta G$ AS INDEX OF COOPERATIVITY</i> <i>HILL EQUATION AND PLOT</i> <i>MONOD-WYMAN-CHANGEUX MODEL</i> <i>KOSHLAND-NEMENTHY-FILMER MODEL</i> <i>ADAIR EQUATION</i> <i>LINKED FUNCTIONS</i> <i>HEMOGLOBIN</i> <i>SCATCHARD PLOT</i> <i>COOPERATIVITY</i> <i>ALLOSTERISM</i> <i>HYSTERESIS</i> <i>ASPARTATE CARBAMOYLTRANSFERASE</i> <i>NEGATIVE COOPERATIVITY</i>	COLLISION THEORY <i>TRANSITION-STATE THEORY</i> <i>ENTROPY OF ACTIVATION</i>
Polarization of chemical bond, <i>FIELD EFFECT</i>		PRENYL-PROTEIN-SPECIFIC ENDOPEP- TIDASE
Polarized light interaction with matter, <i>OPTICAL ROTATION</i>		PREORGANIZATION <i>PODAND</i> <i>SPHERAND</i> <i>HOST</i> <i>GUEST</i> <i>CROWN ETHER</i> <i>MARCUS THEORY</i>
Polyadenylate polymerase, <i>IRREVERSIBLE POLYMERIZATION</i>		PREPHENATE DEHYDRATASE PREPHENATE DEHYDROGENASE
POLY(ADP-RIBOSE) SYNTHETASE		Pressure, <i>PASCAL</i>
Poly(A) polymerase, <i>IRREVERSIBLE POLYMERIZATION</i>		PRESSURE-JUMP METHOD <i>CHEMICAL KINETICS</i>
POLYGALACTURONASE		PRE-STEADY STATE PHASE
POLYMER FLEXING AND EXTENDING		Prigogine, <i>NONEQUILIBRIUM THERMODYNAMICS: A PRIMER</i>
Polymeric substrate binding, <i>SHIFTED BINDING</i>		PRIMARY PHOTOCHEMICAL PROCESS <i>PRIMARY PHOTOPRODUCT</i>
POLYMERIZATION	POSTTRANSLATIONAL MODIFICATION OF PROTEINS	PRIMARY PHOTOPRODUCT
<i>QUASI-EQUIVALENCE</i>	POTENTIAL ENERGY <i>ENERGY</i> <i>KINETIC ENERGY</i>	PRIMITIVE CHANGE <i>ELEMENTARY REACTION</i>
<i>ACTIN ASSEMBLY KINETICS</i>	Potential-energy barrier, <i>HYDROGEN TUNNELING</i> <i>QUANTUM MECHANICAL TUNNELING</i>	Principle of detailed balance, <i>CHEMICAL KINETICS</i>
<i>ACTIN-BASED MOTILITY</i>	POTENTIAL ENERGY PROFILE <i>REACTION COORDINATE</i> <i>POTENTIAL ENERGY SURFACE</i> <i>GIBBS FREE ENERGY DIAGRAM</i>	Principle of microscopic reversibility, <i>NONEQUILIBRIUM THERMODYNAMICS: A PRIMER</i>
<i>MICROTUBULE ASSEMBLY KINETICS</i>	POTENTIAL ENERGY SURFACE <i>ENZYME ENERGETICS</i> <i>REACTION COORDINATE</i> <i>GIBBS FREE ENERGY DIAGRAM</i> <i>POTENTIAL ENERGY PROFILE</i> <i>MORE O'FERRALL-JENCKS DIAGRAM</i> <i>KINETIC ISOTOPE EFFECTS</i>	Principle of parsimony, <i>OCCAM'S RAZOR</i>
<i>IRREVERSIBLE POLYMERIZATION</i>	Power-compensated DSC, <i>DIFFERENTIAL SCANNING CALO- RIMETRY</i>	PRION PLAQUE FORMATION
<i>CRITICAL CONCENTRATION</i>		PROBABILITY <i>STATISTICS (A Primer)</i> <i>PERMUTATIONS AND COMBINATIONS</i>
<i>BIOCHEMICAL SELF-ASSEMBLY</i>	PRECISION <i>ACCURACY</i> <i>STATISTICS (A Primer)</i>	PROBABILITY DENSITY FUNCTION
<i>PROCESSIVITY</i>	Precision of measurement with electronic de- vices, <i>NOISE</i>	PROBABLE ERROR <i>NORMAL ERROR CURVE</i> <i>STATISTICS (A Primer)</i>
POLYMERIZATION KINETICS (Irreversible)	PRECUSOR <i>SERIES FIRST ORDER REACTION</i> <i>PULSE CHASE EXPERIMENTS</i>	Probable error in a single measured value, <i>NORMAL ERROR CURVE</i>
<i>IRREVERSIBLE POLYMERIZATION</i>	Predictor-corrector algorithm, <i>STIFFNESS</i>	Probability of tunneling, <i>HYDROGEN TUNNELING</i>
Polymerization time course, <i>IRREVERSIBLE POLYMERIZATION</i>	PREDISSOCIATION	PROCESSIVITY <i>RANDOM SCISSION KINETICS</i>
Polymer length redistribution, <i>ACTIN ASSEMBLY KINETICS</i>		
POLYNUCLEOTIDE 5'-HYDROXYL-KINASE		
POLYNUCLEOTIDE PHOSPHORYLASE		
Polyol dehydrogenase, <i>SORBITOL DEHYDROGENASE</i>		
POLYPHOSPHATE GLUCOKINASE		
POLYPHOSPHATE KINASE		
Polypodal attachment, <i>BIOMINERALIZATION</i>		
Polyribonucleotide synthase (ATP), <i>RNA LIGASE</i>		
POLYSACCHARIDE DEPOLYMERASE		
Polytopal rearrangement, <i>STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS</i>		
POOL SIZE <i>COMPARTMENTAL ANALYSIS</i>		
P/O RATIO		
PORPHOBILINOGEN DEAMINASE		
POSITIONAL ISOTOPE EXCHANGE ("PIX")		

PROCESSIVITY OF KINESIN ATPase	Prolyl residue isomerases, <i>PARVULIN</i>	PROTEIN FARNESYLTRANSFERASES
PROCHIRALITY	Prompt emission, <i>FLUORESCENCE</i>	PROTEIN GERANYLGERANYLTRANS- FERASES
<i>CHIRALITY</i>		Protein interior, water exclusion from, <i>HYDROPHOBIC EFFECT</i>
<i>CHIRALITY PROBES</i>	PROPAGATION	PROTEIN KINASE
PROCOLLAGEN N-PEPTIDASE	<i>CHAIN REACTION</i>	PROTEIN KINASE C
PRODUCT	<i>CHAIN TRANSFER</i>	Protein-ligand interactions, <i>NUCLEAR MAGNETIC RESONANCE</i>
PRODUCT-DETERMINING STEP	<i>INITIATION</i>	"Protein only" hypothesis for prion plaque formation, <i>PRION PLAQUE FORMATION</i>
PRODUCT DEVELOPMENT CONTROL	<i>TERMINATION</i>	PROTEIN OXIDATION (In Aging & Disease)
Product formation,	Propanal, synthesis of, <i>PROPANEDIOL DEHYDRATASE</i>	PROTEIN PRENYLTRANSFERASES
<i>INITIAL RATE PHASE</i>	PROPANEDIOL DEHYDRATASE	PROTEIN-PROTEIN RECOGNITION KI- NETICS
Product inhibition experiments, practical limi- tations on,	Propanoyl-CoA, formation of, <i>PROPIONYL-CoA SYNTHETASE</i>	Protein structure and dynamics, <i>NUCLEAR MAGNETIC RESONANCE</i>
<i>PRODUCT INHIBITION STUDIES</i>	Propionate:CoA ligase, <i>PROPIONYL-CoA SYNTHETASE</i>	PROTEIN SYNTHESIS INHIBITORS
PRODUCT INHIBITION STUDIES	PROPIONYL-CoA CARBOXYLASE	PROTEIN TURNOVER
<i>ABORTIVE COMPLEXES</i>	PROPIONYL-CoA SYNTHETASE	<i>N-END RULE</i>
PRODUCTION	PRO-PROCHIRAL	<i>LEUCINE KINETICS</i>
<i>COMPARTMENT ANALYSIS</i>	<i>PROCHIRALITY</i>	<i>PROTEASE La</i>
PRODUCT RULE	<i>CHIRALITY</i>	PROTocatechuate 3,4-DIOXYGENASE
Profilin,	<i>pro-R GROUP</i>	PROTocatechuate 4,5-DIOXYGENASE
<i>ABM-1 & ABM-2 ACTIN-BASED MOTOR</i> <i>SEQUENCES</i>	<i>pro-S GROUP</i>	2-Protocatechuoylphloroglucinolcarboxylate, <i>QUERCETIN 2,3-DIOXYGENASE</i>
<i>ACTIN-BASED BACTERIAL MOTILITY</i>	PROSTACYCLIN SYNTHASE	PROTOchlorophyllide REDUCTASE
PROGRESS CURVE ANALYSIS	PROSTAGLANDIN E: 9-KETO (α)-RE- DUCTASE	PROTOGENIC
<i>CHEMICAL KINETICS</i>	PROSTAGLANDIN H, PROSTAGLANDIN D ISOMERASE	<i>PROTOPHILIC</i>
<i>KINSIM/FITSIM</i>	PROSTAGLANDIN H, PROSTAGLANDIN E ISOMERASE	<i>AMPHIPROTIC</i>
Progress curve with respect to product inhi- bition,	PROSTAGLANDIN H ₂ SYNTHASE	α -Proton abstraction from thioesters, <i>CROTONASE SUPERFAMILY</i>
<i>FOSTER-NIEMANN PLOTS</i>	Prostereoisomerism, <i>CHIRALITY</i>	PROTON AFFINITY
PROLINE DEHYDROGENASE	<i>CHIRALITY PROBES</i>	Proton exchange in base pairs, <i>BASE PAIRING</i>
PROLINE DIPEPTIDASE	<i>PROCHIRALITY</i>	Proton flow to ATP synthesis/hydrolysis, <i>BINDING CHANGE MECHANISM</i>
PROLINE IMINOPEPTIDASE	PROSTHETIC GROUP	PROTON INVENTORY TECHNIQUE
<i>cis,trans</i> -Proline isomerization,	PROTEASE ACCESSIBILITY (As Probe of Pro- tein Structure)	α -Proton lability in racemization of phenyl- alanine, <i>PHENYLALANINE RACEMASE</i>
<i>PROLYL ISOMERASE</i>	PROTEASE INHIBITOR "COCKTAILS"	Proton shuttle, <i>SHUTTLE GROUP</i>
Proline-limited folding,	PROTEASE La,	1,3-Proton transfer, <i>CROTONASE SUPERFAMILY</i>
<i>PARVULIN</i>	Protein aggregation, in Creutzfeldt-Jakob disease, <i>PRION PLAQUE FORMATION</i>	PROTON TRANSFER IN AQUEOUS SO- LUTION
PROLINE RACEMASE	PROTEINASE A	Proton-translocating ATP synthase/hy- drolase, <i>BINDING CHANGE MECHANISM</i>
<i>ENZYME ENERGETICS (The Case of Pro- line Racemase)</i>	Proteinase inhibitor, <i>SERPINS (INHIBITORY MECHANISM)</i>	
PROLINE REDUCTASE	PROTEIN COMPLEMENTATION	
D-Proline reductase, borohydride reduction, <i>BOROHYDRIDE REDUCTION</i>	<i>DOMAIN SWAPPING</i>	
D-Proline reductase (dithiol), <i>PROLINE REDUCTASE</i>	PROTEIN DISULFIDE ISOMERASE	
D-Proline, synthesis of, <i>PROLINE REDUCTASE</i>	PROTEIN-DISULFIDE REDUCTASE (Gluta- thione)	
L-Proline to D-proline interconversion, <i>PROLINE RACEMASE</i>		
PROLIPOPROTEIN SIGNAL PEPTIDASE		
PROLYL 3-HYDROXYLASE		
PROLYL 4-HYDROXYLASE		
PROLYL ISOMERASE		

PROTOPHILIC	AMINO ACID RACEMASE	<i>L</i> -SELENOCYSTEINE β -LYASE
PROTOGENIC	2-AMINOADIPATE AMINOTRANSFERASE	SERINE DEHYDRATASE
AMPHIPROTIC	1-AMINOCYCLOPROPANE-1-CARBOXYLATE SYNTHASE	SERINE PALMITOYLTRANSFERASE
PROTOPORPHYRINOGEN OXIDASE	5-AMINOLEVULINATE AMINOTRANSFERASE	O-SUCCINYLMOMOSERINE (THIOL)-LYASE
Pseudo-asymmetric carbon atom, STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	5-AMINOLEVULINATE DEHYDRATASE	THREONINE DEHYDRATASE
Pseudo-axial, STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	5-AMINOLEVULINATE SYNTHASE	TRYPTOPHANASE
PSEUDO BUFFERS	ARGININE DECARBOXYLASE	TYROSINE AMINOTRANSFERASE
PSEUDOCATALYSIS	ARGININE RACEMASE	Pyridoxal 5'-phosphate, synthesis of, PYRIDOXAL KINASE
PSEUDOHALIDES	AROMATIC AMINO ACID AMINOTRANSFERASE	PYRIDOXAMINE:PYRUVATE AMINOTRANSFERASE
Pseudoisomeric forms, BIOCHEMICAL THERMODYNAMICS ATP, THERMODYNAMICS OF HYDROLYSIS	ASPARTATE AMINOTRANSFERASE	PYRIDOXINE & PYRIDOXAL (Analogues and Derivatives)
PSEUDOROTATION	ASPARTATE β -DECARBOXYLASE	4-Pyridoxolactone, synthesis of, PYRIDOXAL 4-DEHYDROGENASE
PSI (ψ)	BRANCHED-CHAIN AMINO ACID AMINOTRANSFERASE	PYRIMIDINE DIMER DNA GLYCOSYLASE
PTERIDINES, ANALOGUES & FOLATE CO-ENZYMES	CYSTATHIONINE β -LYASE	Pyroglutamase (ATP-hydrolyzing), 5-OXOPROLINASE
PULLULANASE	CYSTATHIONINE β -SYNTHASE	PYROGLUTAMYL PEPTIDE HYDROLASE
PULSE-CHASE EXPERIMENTS	CYSTEINE CONJUGATE β -LYASE	PYROLYSIS
N-END RULE	CYSTEINE LYASE	PYROPHOSPHATASE, INORGANIC Pyrophosphate, OXYGEN, OXIDES & OXO ANIONS
PULSE RADIOLYSIS	DIAMINOPIMELATE DECARBOXYLASE	Pyrophosphate as a substrate or product, ACETATE KINASE (PYROPHOSPHATE)
PURINE NUCLEOSIDASE	DIIODOTYROSINE AMINOTRANSFERASE	ACYLNEURAMINATE CYTIDYLYLTRANSFERASE
PURINE NUCLEOSIDE PHOSPHORYLASE	GLUCOSAMINATE AMMONIA-LYASE	ADENINE PHOSPHORIBOSYLTRANSFERASE
PUTRESCINE CARBAMOYLTRANSFERASE	GLUTAMATE DECARBOXYLASE	ADENYLATE CYCLASE
PUTRESCINE <i>N</i> -METHYLTRANSFERASE	GLUTAMATE FORMIMINOTRANSFERASE	ALKALINE PHOSPHATASE
PUTRESCINE OXIDASE	GLUTAMINE AMINOTRANSFERASE	AMIDOPHOSPHORIBOSYLTRANSFERASE
Putrescine synthesis, ORNITHINE DECARBOXYLASE	GLYCINE ACETYLTRANSFERASE	AMINOACYL-tRNA SYNTHETASES
P-VALUE	GLYCINE AMINOTRANSFERASE	ANTHRANILATE PHOSPHORIBOSYLTRANSFERASE
Pyramidal inversion, STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	GLYCINE DECARBOXYLASE	ARACHIDONYL-CoA SYNTHETASE
PYRAZOLE	GLYCINE:OXALOACETATE AMINOTRANSFERASE	ARGININOSUCCINATE SYNTHETASE
ALCOHOL DEHYDROGENASE	GLYCINE SYNTHASE	ASPARAGINE SYNTHETASE
Pyridine nucleotide transhydrogenase, NAD(P) ⁺ TRANSHYDROGENASE	HISTIDINE DECARBOXYLASE	ATP:GLUCOSE-1-PHOSPHATE ADENYLYLTRANSFERASE
PYRIDOXAL 4-DEHYDROGENASE	HISTIDINOL-1-PHOSPHATE AMINOTRANSFERASE	ATP PHOSPHORIBOSYLTRANSFERASE
PYRIDOXAL KINASE	HOMOCYSTEINE DESULFHYDRASE	BENZOATE-CoA LIGASE
Pyridoxal-phosphate-dependent enzymes, N ² -ACETYLMORNITHINE AMINOTRANSFERASE	erythro- β -HYDROXYASPARTATE DEHYDRATASE	BIOTIN HOLOCARBOXYLASE SYNTHETASE
O-ACETYLSERINE (THIOL)-LYASE	KYNURENINASE	BORNYL PYROPHOSPHATE SYNTHASE
ALANINE AMINOTRANSFERASE	KYNURENINE AMINOTRANSFERASE	CARNOSINE SYNTHETASE
ALANINE:GLYOXYLATE AMINOTRANSFERASE	LEUCINE AMINOTRANSFERASE	CHOLINE-PHOSPHATE CYTIDYLYLTRANSFERASE
ALANINE RACEMASE	LYSINE 2,3-AMINOMUTASE	CHOLOYL-CoA SYNTHETASE
S-ALKYLCYSTEINE LYASE	LYSINE 6-AMINOTRANSFERASE	COBALAMIN ADENOSYLTRANSFERASE
ALLIIN LYASE	LYSINE DECARBOXYLASE	4-COUMAROYL-CoA SYNTHETASE
	METHIONINE γ -LYASE	
	ORNITHINE AMINOTRANSFERASE	
	PHENYLALANINE DECARBOXYLASE	
	PHOSPHATIDYLSERINE DECARBOXYLASE	
	PHOSPHOSERINE AMINOTRANSFERASE	
	PYRIDOXAMINE:PYRUVATE AMINOTRANSFERASE	

CTP:ETHANOLAMINEPHOSPHATE CYTIDYLYLTRANSFERASE
 DEOXYCYTIDYLATE HYDROXYMETHYLTRANSFERASE
 DIHYDROPTEROATE SYNTHASE
 DIMETHYLALLYLTRANSTRANSFERASE
 DNA LIGASES
 FARNESYLTRANSTRANSFERASE
 FUCOSE-1-PHOSPHATE GUANYLYLTRANSFERASE
 GALACTOSE-1-PHOSPHATE URIDYLYLTRANSFERASE
 GENTAMICIN 2"-NUCLEOTIDYLTRANSFERASE
 GERANYLTRANSTRANSFERASE
 GLUCOSE-6-PHOSPHATASE
 GLUCOSE-1-PHOSPHATE URIDYLYLTRANSFERASE
 GMP SYNTHETASE
 GTP CYCLOHYDROLASE II
 GUANYLATE CYCLASE
 HEXOSE-1-PHOSPHATE GUANYLYLTRANSFERASE
 HYPOXANTHINE (GUANINE) PHOSPHORIBOSYLTRANSFERASE
 LEUCYL-tRNA SYNTHETASE
 LONG-CHAIN FATTY ACYL-CoA SYNTHETASE
 LUCIFERASE
 MANNOSE-1-PHOSPHATE GUANYLYLTRANSFERASE
 METHIONINE ADENOSYLTRANSFERASE
 METHIONYL-tRNA SYNTHETASE
 NAD⁺ SYNTHETASE
 NICOTINAMIDE MONONUCLEOTIDE ADENYLYLTRANSFERASE
 NICOTINATE PHOSPHORIBOSYLTRANSFERASE
 NUCLEOSIDE TRIPHOSPHATE HEXOSE-1-PHOSPHATE
 NUCLEOTIDYLTRANSFERASES
 trans-OCTAPRENYLTRANSTRANSFERASE
 OROTATE PHOSPHORIBOSYLTRANSFERASE
 OXALYL-CoA DECARBOXYLASE
 PHOSPHATIDATE CYTIDYLYLTRANSFERASE
 PHOSPHOENOLPYRUVATE CARBOXYKINASE (PYROPHOSPHATE)
 PHOSPHORIBOSYL-ATP PYROPHOSPHATASE
 PROPIONYL-CoA SYNTHETASE
 PYROPHOSPHATASE, INORGANIC
 PYROPHOSPHOMEVALONATE DECARBOXYLASE

PYRUVATE, ORTHOPHOSPHATE DIKINASE
 QUINOLINATE PHOSPHORIBOSYLTRANSFERASE
 REVERSE TRANSCRIPTASE
 RIBITOL-5-PHOSPHATE CYTIDYLYLTRANSFERASE
 RIBOSEPHOSPHATE PYROPHOSPHOKINASE
 RNA LIGASE
 RNA POLYMERASE
 RUBBER *cis*-POLYPRENYLCISTRANSFERASE
 SQUALENE SYNTHASE
 SULFATE ADENYLYLTRANSFERASE
 UBIQUITIN-PROTEIN LIGASES
 PYROPHOSPHATE FRUCTOSE-6-PHOSPHATE 1-PHOSPHOTRANSFERASE
 PYROPHOSPHOMEVALONATE DECARBOXYLASE
 Δ^1 -PYRROLINE-5-CARBOXYLATE DEHYDROGENASE
 Δ^1 -PYRROLINE-5-CARBOXYLATE REDUCTASE
 1-PYRROLINE-4-HYDROXY-2-CARBOXYLATE DEAMINASE
 PYRUVATE CARBOXYLASE
 PYRUVATE DECARBOXYLASE
 PYRUVATE DEHYDROGENASE
 PYRUVATE DEHYDROGENASE KINASE
 PYRUVATE DEHYDROGENASE PHOSPHATASE
 Pyruvate:ferredoxin 2-oxidoreductase, PYRUVATE SYNTHASE
 PYRUVATE FORMATE LYASE
 PYRUVATE KINASE
 NUCLEOSIDE-5'-TRIPHOSPHATE REGENERATION
 PYRUVATE, ORTHOPHOSPHATE DIKINASE
 PYRUVATE OXIDASE
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 PYRUVATE SYNTHASE
 PYRUVATE, WATER DIKINASE

Q

Q₁₀
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 QUADRUPOLE

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 Quantitative structure-activity relationships, HANSCH'S CORRELATION ANALYSIS
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 Quasi-equatorial, STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS
 QUASI-EQUILIBRIUM ASSUMPTION
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 QUERCETIN 2,3-DIOXYGENASE
 QUINOLINATE PHOSPHORIBOSYLTRANSFERASE
 Quinone reductase, NAD(P)H DEHYDROGENASE (QUINONE)
 Quinones, BIOCHEMICAL NOMENCLATURE
 QUIN-2 TETRAACETOXYESTER
 FURA-2
 QUOTIENT FUNCTION
 QUOTIENT, RESPIRATORY
 QUOTIENT, SYNERGISM (Q_{syn})

R $R_{0.9}$, $S_{0.9}/S_{0.5}$

RACEMASES

Racemic conglomerate,

STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS

RACEMIC MIXTURE (or Racemate)

RACEMIZATION

RAD

RADIAN

RADIANCE

RADIANT ENERGY

RADIANT (or, ENERGY) FLUX

Radiant power,

RADIANCE

Radiationless transfer,

JABLONSKI DIAGRAM

RADIATIONLESS TRANSITION

FLUORESCENCE

RADIATIVE LIFETIME

RADICAL (or, FREE RADICAL)

RADICAL CENTER

RADICAL ION

Radioactive decay,

CHEMICAL KINETICS

RADIOACTIVE SUBSTANCES (General Pre-
cautions)

RADIOCARBON DATING

COSMOGENIC DATING

RADIOLYSIS

PULSE RADIOLYSIS

Radiometric technique,

PULSE-CHASE EXPERIMENTS

RAMAN SCATTERING

RANDOM AB/QR, ORDERED C/P TER TER
MECHANISMRANDOM AC/PR, ORDERED B/Q TER TER
MECHANISMRANDOM BC/PQ, ORDERED A/R TER TER
MECHANISM

RANDOM BI BI MECHANISM

MULTISUBSTRATE MECHANISM

RANDOM BI BI DUAL-THEORELL-CHANCE
MECHANISM

MULTISUBSTRATE MECHANISM

Random Bi Bi ternary complex mechanism,

ISOTOPE EXCHANGE AT EQUILIBRIUM

RANDOM BI UNI MECHANISM

RANDOM BI UNI BI PING PONG
MECHANISM

RANDOM ERROR

STATISTICS (A Primer)

RANDOM EXPERIMENT

STATISTICS (A Primer)

RANDOMIZATION

STATISTICS (A Primer)

RANDOM MECHANISMS

MULTISUBSTRATE MECHANISMS

Random number table,

STATISTICS (A Primer)

RANDOM-ON/ORDERED-OFF BI BI MECH-
ANISM

RANDOM SCISSION KINETICS

RANDOM TER TER MECHANISM

RANDOM UNI BI MECHANISM

RANDOM UNI UNI BI BI PING PONG
MECHANISM

RANDOM VARIABLE

STATISTICS (A Primer)

RANDOM WALK

DIFFUSION

PERSISTENCE TIME

TWIDDLING

RAPID BUFFER EXCHANGE

Rapid equilibrium assumption,

QUASI-EQUILIBRIUM ASSUMPTION

RAPID EQUILIBRIUM MECHANISMS

Rapid gel filtration of biomacromolecules,

RAPID BUFFER EXCHANGE

RAPID MIXING

QUENCHED FLOW EXPERIMENTS

Rapid quench experiments,

QUENCH-FLOW EXPERIMENTS

Rapid reaction kinetics,

PULSE RADIOLYSIS

Rapid reaction techniques,

CRYOENZYMOLOGY

PULSE RADIOLYSIS

Rapid removal of enzyme stabilizing agents,

RAPID BUFFER EXCHANGE

RAPID SCAN SPECTROSCOPY OF FAST RE-
ACTIONS

RAPID-START COMPLEX

Rate coefficients exhibiting temporal
memory,

FRACTAL REACTION KINETICS

RATE CONSTANT

CHEMICAL KINETICS

RATE-CONTRIBUTING STEP

RATE-CONTROLLING STEP

RATE-DETERMINING STEP or RATE-LIM-
ITING STEP

KINETIC ISOTOPE EFFECTS

RATE-DETERMINING STEP or RATE-LIM-
ITING STEP

Rate enhancement,

CATALYSIS

CHEMICAL KINETICS

ORBITAL STEERING

Rate enhancement observed in two-phase re-
action,

PHASE-TRANSFER CATALYSIS

RATE LAW

CHEMICAL KINETICS

Rate-limiting step in inner-sphere coordi-
nation,

EIGEN-TAMM MECHANISM

Rate-limiting step in metabolic pathway,

PACEMAKER REACTION

RATE OF APPEARANCE

CHEMICAL KINETICS

RATE OF DISAPPEARANCE

CHEMICAL KINETICS

Rate of intracellular synthesis and/or turn-
over of a metabolite,

PULSE-CHASE EXPERIMENTS

RATE OF REACTION

CHEMICAL KINETICS

Rate processes in aqueous solutions,

CHEMICAL KINETICS

RATE SATURATION BEHAVIOR

RAY-ROSCELLI TREATMENT

ISOMERIZATION

ISO MECHANISMS

RE-

REACTANCY

Reactant conversion into its mirror image,

NARCISSISTIC REACTION

REACTING BOND RULES

REACTING ENZYME CENTRIFUGATION

REACTION COORDINATE DIAGRAM

POTENTIAL ENERGY DIAGRAM

SADDLE POINT

ENZYME ENERGETICS (The Case of Pro-
line Racemase)

REACTION CYCLE

Reaction intermediate,

ACYL-PHOSPHATE INTERMEDIATE

PERSISTENT

TRANSIENT CHEMICAL SPECIES

REACTION MECHANISM

CHEMICAL KINETICS

Reaction order of nonenzymic reaction,

CHEMICAL KINETICS

NOYES EQUATION

REACTION PROGRESS CURVE FOR UNSTABLE ENZYMES	REGULATION OF ENZYME ACTIVITY	Retro-aldol condensation, ALDOL CONDENSATION
PROGRESS CURVE ANALYSIS	RELATIVE BIOLOGICAL EFFECTIVENESS	Reverse micelle, INVERTED MICELLE
REACTION RATE/VELOCITY	RELATIVE DENSITY	REVERSE TRANSCRIPTASE
CHEMICAL KINETICS	RELATIVE SUPERSATURATION (RS or RSS)	VIRAL POLYMERASES
Reactions at interfaces,	BIOMINERALIZATION	Reversible adiabatic change, ADIABATIC PROCESS
FRACTAL REACTION KINETICS	RELAXATION	Reversible process, ENTROPY
Reaction schemes,	CHEMICAL KINETICS	α -L-Rhamnoase, α -L-RHAMNOSIDASE
CHEMICAL KINETICS	RELAXATION PERIOD FOR PRECIPITATION	L-Rhamnose, RHAMNOSE ISOMERASE
Reactions in understirred solutions,	BIOMINERALIZATION	RHAMNOSE ISOMERASE
FRACTAL REACTION KINETICS	RELAXATION TIME	α -L-RHAMNOSIDASE
Reaction velocity,	FAST REACTION TECHNIQUES	α -L-Rhamnosides, α -L-RHAMNOSIDASE
ENZYME KINETIC EQUATIONS (1. The Basics)	CORRELATION FUNCTION	L-Rhamnulose, RHAMNOSE ISOMERASE
REACTIVATION KINETICS (Temperature Effects)	Removal of enzyme/protein stabilizing agents, RAPID BUFFER EXCHANGE	L-Rhamnulose 1-phosphate RHAMNULOSE-1-PHOSPHATE ALDOLASE
REAGENT PURITY	RENATURATION	RHAMNULOSE-1-PHOSPHATE ALDOLASE
SUBSTRATE PURITY	Renaturation of nucleic acids, NUCLEIC ACID RENATURATION KINETICS	RHIZOPUSPEPSIN
WATER PURITY	RENIN	PEPSIN AND PEPSINOGEN
ENZYME PURITY	Rennin,	RHO (ρ)
Receptor complex,	CHYMOSIN	$\rho\sigma$ -Equation (Rho-Sigma Equation), HAMMETT EQUATION
AGONIST	REORGANIZATION ENERGY	RHODANESE
Reconceptualization of biochemical thermodynamics,	INTRINSIC BARRIER	Rhodium(III) nucleotide complex, EXCHANGE-INERT METAL-NUCLEOTIDE COMPLEXES
ATP, THERMODYNAMICS OF HYDROLYSIS	MARCUS EQUATION	RHODOPSIN
Rectangular hyperbola,	Repeatability, REPRODUCIBILITY	RHODOPSIN KINASE
ENZYME KINETIC EQUATIONS	REPLOTS	RHO-VALUE (ρ -Value) HAMMETT EQUATION
REDOX-ACTIVE AMINO ACIDS	Reporter group, FLUORESCENCE	Ribitol, RIBOFLAVINASE
TOPAQUINONE	Residual activity, BASAL RATE	Ribitol dehydrogenase, ABORTIVE COMPLEXES
REDOX POTENTIAL	RESOLUTION	D-Ribitol 5-phosphate, RIBITOL-5-PHOSPHATE CYTIDYL-TRANSFERASE
Redox reactions,	Resonance hybrid, DEWAR STRUCTURES	RIBITOL-5-PHOSPHATE 2-DEHYDROGENASE
HALF-REACTIONS	KEKULÉ STRUCTURES	RIBITOL-5-PHOSPHATE CYTIDYL-TRANSFERASE
NERNST EQUATION	RESONANCE RAMAN SPECTROSCOPY	RIBITOL-5-PHOSPHATE 2-DEHYDROGENASE
REDUCED CONCENTRATION	RAMAN SPECTROSCOPY	RIBITOL-5-PHOSPHATE 2-DEHYDROGENASE
Reduced dimensionality of diffusion,	Resonance stabilization, DEWAR STRUCTURES	4-(1-D-Ribitylamino)-5-amino-2,6-dihydroxypyrimidine, RIBOFLAVIN SYNTHASE
DIFFUSION, REDUCTION IN DIMENSIONALITY	KEKULÉ STRUCTURES	
REDUCED MASS	RESONANCE STABILIZATION ENERGY	
HOOKE'S LAW SPRING	RESPIRATORY BURST OXIDASE	
KINETIC ISOTOPE EFFECTS	NADPH OXIDASE	
REDUCING AGENT (or Reductant)	RESPIRATORY EXCHANGE RATIO	
REDUCTION	Restricted rotation, STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	
REDUCING AGENTS,	RESTRICTION ENZYMES	
BOROHYDRIDE REDUCTION	DEOXYRIBONUCLEASES	
REDUCTIO AD ABSURDUM	Retinoids, BIOCHEMICAL NOMENCLATURE	
REDUCTION		
ELECTRODE KINETICS		
OXIDATION		
REFERENCE REACTION		
CATALYTIC PROFICIENCY		
REGIOSELECTIVITY		

Riboflavin,
RIBOFLAVINASE
RIBOFLAVIN KINASE
RIBOFLAVIN SYNTHASE
 RIBOFLAVINASE
 RIBOFLAVIN KINASE
 RIBOFLAVIN SYNTHASE
 RIBONUCLEASE (or RNase)
PSEUDOROTATION
DIETHYL PYROCARBONATE
 RIBONUCLEASE F
 RIBONUCLEASE II
 RIBONUCLEASE III
 Ribonuclease inactivation,
DIETHYL PYROCARBONATE
 RIBONUCLEASE IV
 RIBONUCLEASE P
 Ribonuclease, pancreatic,
RIBONUCLEASE (or RNase)
 Ribonucleoside synthesis,
3'-NUCLEOTIDASE
5'-NUCLEOTIDASE
 RIBONUCLEOTIDE REDUCTASE
 D-Ribose 5-phosphate,
RIBOSE-5-PHOSPHATE EPIMERASE
RIBOSEPHOSPHATE PYROPHOSPHO-
KINASE
 RIBOSE-5-PHOSPHATE ADENYLYLTRANS-
 FERASE
 RIBOSE-5-PHOSPHATE EPIMERASE
 D-Ribose 1-phosphate, interconversion to
 D-ribose 5-phosphate,
PHOSPHOPENTOMUTASE
 Ribose 5-phosphate isomerase,
RIBOSE-5-PHOSPHATE EPIMERASE
 RIBOSEPHOSPHATE PYROPHOSPHO-
 KINASE
 Ribozyme,
CATALYTIC RNA
ENZYME
 D-Ribulose 1,5-bisphosphate,
D-RIBULOSE-1,5-BISPHOSPHATE CAR-
BOXYLASE/OXYGENASE
 D-RIBULOSE-1,5-BISPHOSPHATE CARBOXYL-
 ASE/OXYGENASE
 D-Ribulose 5-phosphate,
RIBITOL-5-PHOSPHATE 2-DEHYDRO-
GENASE
RIBULOSE-5-PHOSPHATE 3-EPIMERASE
 L-Ribulose 5-phosphate,
RIBULOSE-5-PHOSPHATE 4-EPIMERASE
 RIBULOSE-5-PHOSPHATE 3-EPIMERASE

RIBULOSE-5-PHOSPHATE 4-EPIMERASE
 RIESKE IRON-SULFUR PROTEIN
 Ring reversal (or ring inversion),
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
 Ring structures, role in microtubule as-
 sembly,
MICROTUBULE ASSEMBLY KINETICS
 RITCHIE EQUATION
 RNA, exonucleolytic cleavage,
PHOSPHODIESTERASES
 RNA LIGASE
 RNA POLYMERASES
EDITING MECHANISMS
REPLICASE
 RNA stability,
DIETHYL PYROCARBONATE
 ROENTGEN
 ROENTGEN-EQUIVALENT-MAN (rem)
 Rosanoff convention,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
 Rotamer,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
 Rotary engine,
BINDING CHANGE MECHANISM
 Rotational barrier,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
 Rotational catalysis by ATP synthase,
BINDING CHANGE MECHANISM
 ROTATIONAL CORRELATION TIME
CORRELATION FUNCTION
 ROTATIONAL DIFFUSION
FLUORESCENCE
 rRNA (ADENINE-6-)-METHYLTRANSFERASE
 rRNA (GUANINE-1-)-METHYLTRANSFERASE
 R_s
 RUBBER *cis*-POLYPRENYLCISTRANSFERASE
 Rubidium ion,
IONIC RADIUS
 Rubisco,
D-RIBULOSE-1,5-BISPHOSPHATE CAR-
BOXYLASE/OXYGENASE
 RUBREDOXIN
 RUBREDOXIN:NAD⁺ REDUCTASE
 RUNGE-KUTTA ALGORITHM
NUMERICAL COMPUTER METHODS
NUMERICAL INTEGRATION
STIFFNESS
GEAR ALGORITHM
 R_v

S

$S_{0.5}$
 $S_{0.9}/S_{0.1}$ (or, R_s)
 SAAM
 Sacrificial hyperconjugation,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
 SADDLE POINT
 SALICYLATE 1-MONOOXYGENASE
 SAMPLE SPACE
RANDOM EXPERIMENT
STATISTICS (A Primer)
 Sample turbidity,
FLUORESCENCE
 SAPONIFICATION
 Sarcoplasmic reticulum calcium pump,
ION PUMPS
 SARCOSINE DEHYDROGENASE
 SARCOSINE OXIDASE
 SATURATION
 Saturation function,
LIGAND BINDING
COOPERATIVE LIGAND BINDING
 SATURATION KINETICS
 Saturation ratio,
RELATIVE SUPERSATURATION
 SCALAR COUPLING RELAXATION
 Scatchard equation,
SCATCHARD PLOT
 SCATCHARD PLOT
 SCAVENGER
 SCHIFF BASE (or, Schiff's Base)
 Schiff base reduction,
BOROHYDRIDE REDUCTION
 SCHRÖDINGER EQUATION
QUANTUM MECHANICAL TUNNEL-
LING
 SCINTILLATORS
 SCISSILE BOND
 Scrambling,
ISOTOPIC SCRAMBLING
POSITIONAL ISOTOPE EXCHANGE
 SCREW SENSE OF METAL-NUCLEOTIDE
 COMPLEXES
 Sea shell formation,
BIOMINERALIZATION
 SECOND
 SECONDARY ISOTOPE EFFECT
KINETIC ISOTOPE EFFECT

Secondary kinetic isotope effect, KINETIC ISOTOPE EFFECT	SENSITIVITY	SHIKIMATE DEHYDROGENASE
Second law of thermodynamics, THERMODYNAMICS, LAWS OF ENTROPY CARATHEODORY, PRINCIPLE OF CARNOT CYCLE EFFICIENCY	Sensitivity of measurement, NOISE	SHIKIMATE KINASE
Second-order autocatalytic reaction, AUTOCATALYSIS	SEPARATION OF VARIABLES	Shikimate 3-phosphate, SHIKIMATE KINASE
Second-order rate constant, ENCOUNTER-CONTROLLED RATE	SEQUENTIAL MECHANISMS	Shot noise, NOISE
SECOND-ORDER REACTION	MULTISUBSTRATE MECHANISMS	SHUTTLE GROUP
CHEMICAL KINETICS	Sequential versus ping-pong mechanisms, PHI VALUES IN BISUBSTRATE MECHANISMS	SI-
ORDER OF REACTION	INITIAL RATE ENZYME ASSAYS	SIALIDASES (or Neuraminidases)
NOYES EQUATION	Series first-order processes, CHEMICAL KINETICS	O-SIALOGLYCOPROTEASE
MOLECULARITY	SERIES REACTIONS	SIALYLTRANSFERASE
AUTOCATALYSIS	CHEMICAL KINETICS	SIDE REACTIONS
FIRST-ORDER REACTION	Serine	SIGMA (σ)
SEDIMENTATION EQUILIBRIUM ULTRACENTRIFUGATION	SERINE DEHYDRATASE	σ -ADDUCT (Sigma-Adduct)
SEDIMENTATION VELOCITY ULTRACENTRIFUGATION	SERINE DEHYDROGENASE	σ -CONSTANT (Sigma-Constant)
Seeded assembly, POLYMERIZATION	SERINE HYDROXYMETHYLTRANSFERASE	$\sigma \rightarrow \sigma^*$ TRANSITION
ELONGATION (No Change in Polymer Number Concentration)	SERINE PALMITOYLTRANSFERASE	ABSORPTION SPECTROSCOPY
SELECTION RULE	Serine deaminase	SIGMATROPIC REARRANGEMENT
Selective protein degradation rates, N-END RULE	SERINE DEHYDRATASE	Sigmoidal dose-response, ENZYME CASCADE KINETICS
Selenium donor protein, SELENOPHOSPHATE SYNTHETASE	SERINE DEHYDROGENASE	SIGNAL AMPLIFICATION
L-SELENOCYSTEINE	Serine esterase inhibitor, irreversible, PHENYLMETHYLSULFONYL FLUORIDE	ENZYME CASCADE KINETICS
L-SELENOCYSTEINE β -LYASE	SERINE HYDROXYMETHYLTRANSFERASE	SIGNAL PEPTIDASE, EUKARYOTIC MICROSOMAL
L-SELENOCYSTEINE β -LYASE	SERINE PALMITOYLTRANSFERASE	SIGNAL PROCESSOR
L-SELENOCYSTEINE	Serine protease inhibitor, irreversible, PHENYLMETHYLSULFONYL FLUORIDE	Signal-to-noise ratio, NOISE
Selenophosphate, SELENOPHOSPHATE SYNTHETASE	Serine proteinase inhibitor, SERPIN	SIGNIFICANT FIGURES
SELENOPHOSPHATE SYNTHETASE	Serine/threonine-specific protein phosphatase, PHOSPHOPROTEIN PHOSPHATASES	PRECISION
SELF-ABSORPTION	SERPINS (INHIBITORY MECHANISM)	Simple diffusion through a membrane, PERMEABILITY COEFFICIENT
Self-activating process, AUTOCATALYSIS	L-SERYL-tRNA ^{Ser} SELENIUM TRANSFERASE	Simplest explanation or model, OCCAM'S RAZOR
Self-ordering reactants, FRACTAL REACTION KINETICS	L-SELENOCYSTEINE	Simplified treatment of enzyme kinetics, NET RATE CONSTANT METHOD
Self-phosphorylation process, AUTOPHOSPHORYLATION	Severing protein, CYTOSKELETAL PROTEIN POLYMERIZATION	Simulation, analysis, and modeling program, SAAM
SELF-PROTECTION MECHANISM	SHIELDING	Single-electron transfer mechanism (SET), PHYSICAL ORGANIC CHEMISTRY NOMENCLATURE
SELF-QUENCHING	CHEMICAL SHIFT	Single-ion activity coefficients, DEBYE-HÜCKEL TREATMENT
Self-replicating elements, HYPERCYCLE	SHIFTED BINDING	SINGLE-POOL MODEL FOR METABOLITE COMPARTMENTATION
Self-unmixing of reactants, FRACTAL REACTION KINETICS	SHIFTING SPECIFICITY MODEL FOR ENZYME CATALYSIS	SINGLE-STEP REACTION
	Shigella, ACTIN-BASED BACTERIAL MOTILITY ACTIN FILAMENT GROWTH (Polymerization Zone Model)	Single-stranded RNA and DNA substrates, NUCLEASE S1
	Shikimate, SHIKIMATE DEHYDROGENASE SHIKIMATE KINASE	Singlet dioxygen, OXYGEN, OXIDES & OXO ANIONS
		SINGLET MOLECULAR OXYGEN

- Singlet oxygen,
OXYGEN, OXIDES & OXO ANIONS
- SINGLET STATE
JABLONSKI DIAGRAM
- SINGLET-TRIPLET ENERGY TRANSFER
- SINGLE-TURNOVER CONDITIONS
- Site-directed inhibitor,
AFFINITY LABELING
- Site-site interaction,
COOPERATIVITY
- Site-specific enzyme modification,
AFFINITY LABELING
- SITE-SPECIFIC ISOTOPE FRACTIONATION
IN BIOCHEMICAL MECHANISMS
- Site-specific labeling,
AFFINITY LABELING
- SI UNITS
- Skew,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
- Slit widths,
FLUORESCENCE
- SLOPE
- SLOPE-INTERCEPT EQUATION
- SLOPE REPLOTS (or, Slope Secondary Re-
plots)
- SLOW-BINDING INHIBITOR
D-ALANINE-D-ALANINE LIGASE
- Slowest enzymic step in metabolic pathway,
PACEMAKER ENZYME
- Slow partial exchange reactions,
SUBSTRATE SYNERGISM
- SLOW TEMPERATURE-JUMP METHODS
DEPOLYMERIZATION, END-WISE
- SLOW, TIGHT-BINDING INHIBITOR
- SMOLUCHOWSKI TREATMENT FOR AG-
GREGATION
- S_N1 and S_N2 reactions,
KINETIC ISOTOPE EFFECTS
- S_N DISPLACEMENT REACTIONS
SOLVOLYSIS
- S/N ratio,
NOISE
- SODIUM-22 (²²Na)
- Sodium ion,
IONIC RADIUS
- Sodium, potassium ATPase,
ATPases
- Solanesyl-diphosphate synthase,
trans-OCTAPRENYLTRANSTRANSFERASE
- Solid-state NMR of inhibited complex,
D-ALANINE-D-ALANINE LIGASE
- Solubility,
BIOMINERALIZATION
- SOLUBILITY PRODUCT
- Solubility product constant,
BIOMINERALIZATION
- Solute-solvent interactions,
HYDROPHOBIC EFFECT
- SOLUTION
- SOLVATION
HYDRATION ATMOSPHERE
CHEMICAL KINETICS
- Solvatochromic relationship,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- Solvatochromism,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- SOLVENT-ACCESSIBLE SURFACE AREA
- Solvent acidity function,
BUNNETT-OLSEN EQUATIONS
- Solvent cage,
CHEMICAL KINETICS
SOLVENT-TRAPPED INTERMEDIATES
- SOLVENT ISOTOPE EFFECT
KINETIC ISOTOPE EFFECT
- Solvent parameter,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- SOLVENT TRAPPING OF ENZYME INTERME-
DIATES
- SOLVOLYSIS
- Solvophobicity parameter,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- SOMO
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- SONICATION
- SORBITOL DEHYDROGENASE
- L-Sorbose,
SORBITOL DEHYDROGENASE
- Speciation,
BIOMINERALIZATION
- SPECIFIC ACTIVITY
ISOTOPE DILUTION
- SPECIFIC CATALYSIS
GENERAL ACID CATALYSIS
GENERAL BASE CATALYSIS
- SPECIFICITY
SPECIFICITY CONSTANT
MAPPING SUBSTRATE INTERACTIONS
USING KINETIC DATA
- SPECIFICITY CONSTANT
- Spectrophotofluorimeter design,
FLUORESCENCE
- Spin adduct,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- Spin counting,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- Spin density,
PHYSICAL ORGANIC CHEMISTRY NO-
MENCLATURE
- SPIN LABEL
ELECTRON SPIN RESONANCE
- Spin multiplicity,
PHOSPHORESCENCE
JABLONSKI DIAGRAM
- Spin-spin coupling,
NUCLEAR MAGNETIC RESONANCE
- SPONTANEOUS PROCESS
- Spontaneous resolution,
STEREOCHEMICAL TERMINOLOGY,
IUPAC RECOMMENDATIONS
- SQUALENE SYNTHASE
- "S—S rearrangase,"
PROTEIN DISULFIDE ISOMERASE
- STABILITY CONSTANT
METAL ION COMPLEXATION
STABILITY CONSTANT DETERMINATION
BY A LINEAR PLOT METHOD
- STABILITY CONSTANT DETERMINATION
BY A LINEAR PLOT METHOD
- STABILITY, PRODUCT
SUBSTRATE STABILITY
- Stabilization,
ENZYME STABILITY
- STANDARD CURVE
BEER-LAMBERT LAW
ABSORPTION SPECTROSCOPY
- STANDARD DEVIATION
STATISTICS (A Primer)
- Standard deviation of the data set,
NORMAL ERROR CURVE
STATISTICS (A Primer)
- Standard error curve,
NORMAL ERROR CURVE
STATISTICS (A Primer)
- Standard error of the mean,
NORMAL ERROR CURVE
STATISTICS (A Primer)
- STAPHYLOCOCCAL NUCLEASE
- STARCH PHOSPHORYLASE

STARCH SYNTHASE	<i>pro-R GROUP</i>	STURGILL-BILTONEN METHOD
STATE FUNCTION	<i>pro-S GROUP</i>	Suberyldicholine dissociation from acetylcholine receptor,
STATIONARY STATE	Stereoisomerism,	<i>DISSOCIATION KINETICS</i>
STEADY STATE	STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	SUBSITE MAPPING
STATISTICAL ANALYSIS OF ENZYME RATE DATA	Stereoisomers,	SUBSTITUENT
STATISTICS (A Primer)	DIASTEREOMERS	SUBSTITUTION REACTION
STATISTICAL FACTOR	STEREOSELECTIVITY	S _N REACTIONS
STATISTICS (A Primer)	STEREOSPECIFICITY	Substitution reactions at a saturated carbon,
Statistical mechanics,	STERIC-APPROACH CONTROL	ALPHA EFFECT
ENTROPY	STERIC HINDRANCE	SUBSTRATE
STATISTICS (A Primer)	Steric hindrance effect on ligand binding,	SUBSTRATE-ASSISTED CATALYSIS
STEADY STATE	SCATCHARD PLOT	SUBSTRATE CHANNELING
Steady-state approximation for polymer nucleation,	Stern layer,	Substrate contaminated with the enzyme,
MICROTUBULE ASSEMBLY KINETICS	BIOMINERALIZATION	BASAL RATE
STEADY-STATE ASSUMPTION	STEROID 11 β -HYDROXYLASE	Substrate cycling,
ENZYME KINETICS	STEROID 17 α -HYDROXYLASE	BIFURCATION THEORY
NUMERICAL INTEGRATION	STEROID 21-HYDROXYLASE	Substrate depletion,
Steady-state processes,	Steroids,	INITIAL RATE PHASE
NONEQUILIBRIUM THERMODYNAMICS (A Primer)	BIOCHEMICAL NOMENCLATURE	Substrate dissociation rate,
Stepwise depolymerization,	STEROID SULFOTRANSFERASE	ISOTOPE TRAPPING
DEPOLYMERIZATION, END-WISE	STICKY SUBSTRATES	STICKY SUBSTRATES
STEPWISE REACTION	ISOTOPE TRAPPING	Substrate-induced conformational change,
MULTISTAGE REACTION	STIFFNESS (or Stiffness Instability)	INDUCED FIT MODEL
STEREOCHEMICAL FIDELITY	GEAR ALGORITHM	SUBSTRATE INHIBITION
STEREOCHEMICAL TERMINOLOGY (IUPAC Recommendations)	Stoichiometric coupling,	ABORTIVE COMPLEX FORMATION
STEREOCHEMISTRY OF HYDRIDE TRANSFER TO/FROM NADH	ION PUMPS	LACTATE DEHYDROGENASE
STEREOCHEMISTRY OF HYDROGEN TRANSFER TO/FROM FADH ₂	STOICHIOMETRIC NUMBER	LEE-WILSON EQUATION
STEREOCHEMISTRY OF ISOMERASES INVOLVING PROTON TRANSFER	Stoichiometry of elementary reactions,	Substrate ionization, effect on enzyme catalysis,
Stereoconvergence,	CHEMICAL KINETICS	pH EFFECTS ON ENZYME CATALYSIS
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	MOLECULARITY	SUBSTRATE OXIDATION KINETICS
Stereodescriptor,	UNIMOLECULAR	Substrate-product ratio,
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	BIMOLECULAR	ISOTOPE EXCHANGE AT EQUILIBRIUM
STEREOELECTRONIC	TRANSITION-STATE THEORY	SUBSTRATE PURITY
Stereogenic unit (or Stereogen),	ELEMENTARY REACTION	SUBSTRATE STABILITY & HANDLING
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	STOKE'S SHIFT	SUBSTRATE SYNERGISM
Stereoheterotopic,	STOPPED-FLOW MASS SPECTROMETRY IN ENZYME KINETICS	SUBTILISIN
STEREOCHEMICAL TERMINOLOGY, IUPAC RECOMMENDATIONS	STOPPED-FLOW TECHNIQUES	Substituent or moiety bound to a molecular entity,
Stereoheterotopic face of a trigonal atom,	Stopping a chemical reaction,	RADICAL (or FREE RADICAL)
RE-	QUENCHING	Subunit assembly in multisubunit proteins,
SI-	STRAIN	BIOCHEMICAL SELF-ASSEMBLY
Stereoheterotopic groups in a prochiral atom,	Streak camera method,	Subzero enzymology,
	FLUORESCENCE	CRYOENZYMOLGY
	Streptavidin,	SUCCESSOR COMPLEX
	BIOTIN AND DERIVATIVES	Succinate,
	STREPTOKINASE	SUCCINYL-CoA SYNTHETASE
	Stress,	O-SUCCINYLMOMOSERINE (THIOL)-LYASE
	PASCAL	SUCCINATE DEHYDROGENASE
	Strong cooperativity,	
	BINDING CHANGE MECHANISM	

Succinyl-CoA,
SUCCINYL-CoA SYNTHETASE
 SUCCINYL-CoA SYNTHETASE
 O-Succinyl-L-homoserine,
O-SUCCINYLBOMOSERINE (THIOL)-LYASE
 O-SUCCINYLBOMOSERINE (THIOL)-LYASE
 SUCRASE
 SUCROSE PHOSPHORYLASE
ARSENOLYSIS
 SULFATE ADENYLYLTRANSFERASE
 SULFATE ADENYLYLTRANSFERASE (ADP)
 Sulfhydryl-disulfide interchange,
PROTEIN DISULFIDE ISOMERASE
 SULFIDES
 Sulfite,
RHODANESE
 SULFITE DEHYDROGENASE
 SULFITE OXIDASE
 SULFITE REDUCTASE
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 Surface plasmon resonance (SPR),
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RANDOM BC/PQ, ORDERED A/R TER TER MECHANISM
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TER UNI REACTION MULTISUBSTRATE MECHANISMS	Thermodynamics of protein unfolding, INNATE THERMODYNAMIC QUANTITIES	THYMIDYLATE SYNTHASE
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TRACER PERTURBATION METHOD	Transition-state mimic, D-ALANINE-D-ALANINE LIGASE	TRANSMISSION DENSITY TOTAL TRANSMITTANCE
TRACER/TRACEE RATIO COMPARTMENTAL ANALYSIS	TRANSITION STATES IN BIOLOGICAL CATALYSIS	BEER-LAMBERT LAW ABSORPTION SPECTROSCOPY
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 TUNGSTOENZYMES

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 RADICAL (or Free Radical)
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 Vicinal hydroxyl groups,
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 VITAMIN D GROUP
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 $V_{\max}$ -TYPE (or, V-TYPE) ALLOSTERIC SYSTEM
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Volume change of charge neutralization,
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WEIGHTING OF DATA
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XANTHINE OXIDASE
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D-Xylose 5-phosphate,
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 α -XYLOSIDASE
 β -XYLOSIDASE
D-XYLULOKINASE
L-XYLULOKINASE
X-Pro dipeptidase
 PROLINE DIPEPTIDASE
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 SOLUBILITY PRODUCT
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CARBONIC ANHYDRASE
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 GENASE
CARBOXYPEPTIDASES
CARNOSINASE
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Zinc ion,
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